TRƯỜNG ĐẠI HỌC KIẾN TRÚC HÀ NỘI
KHOA CÔNG NGHỆ THÔNG TIN

TÀI LIỆU NGHIÊN CỨU CHUYÊN NGÀNH

TỔNG HỢP 16 TÀI LIỆU NGUỒN CHUẨN HÓA

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Banner Advertising Effectiveness: An Experimental Study

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Banner advertising is the most popular form of advertising in the internet, which promotes products and services in websites. Banner is a graphic image that is designed to catch the attention of the targeted audience. Banner advertisements are embedded into web pages and when an internet user visits a particular website this ad is presented virtually as the web page loads. Banner advertising is used to a) increase website traffic, b) enhance product sales, c) inform targeted audiences of new product and services or sales and discounts and d) create brand awareness. For companies banner advertising is cost-effective promotes company's image, generates sales leads and provides the opportunity to trace or track its effectiveness. Moreover, banner advertising can engage a target group with the company by using attractive designs (colors, graphics, logos and texts). Due to its importance, banner advertising has attracted significant research attention. Studies have attempted to explain its effectiveness as expressed by website visits, advertising attitudes, purchase intentions and actual internet purchase behavior. Others have examined the elements that make this tool more efficient and attractive such as consumer characteristics, interactivity, exposure frequency, embedded content and design elements. In line with previous research, the scope of our study is to evaluate the effectiveness of banner advertising by investigating the role of product type and content elements. Specifically we examine the effect of congruity (between the advertising copy and the image of the advertised product) and the type of advertising copy. In order to achieve the objectives of our study we conducted two experiments with a 2(congruency: congruent, incongruent) x 2(type of advertising copy: emotional, informational) x 3(product involvement: low, medium and high) design. Experiment 1 was run with three unknown brands and experiment 2 with three well known brands. Overall 30 people participated in the first experiment and 20 in the second. In the first experiment, a two-way analysis of variance (ANOVAs) indicated no statistically significant interaction between congruency and advertising copy type on attitude toward the ad ($F=.021$, $p<.885$). On the contrary, in the second experiment, the interaction between the two variables was significant ($F=6.757$, $p<.012$). It appears that a congruent emotional copy ($M=3.53$) leads to more positive attitude toward the ad than an incongruent emotional copy ($M=2.60$), while an incongruent informational copy (M-3.40) results in more positive attitude toward the ad than a congruent

informational copy (3.07). The level of involvement exerts no effect on the interaction between the two variables (experiment 1: $F=1.321$, $p<.273$ and experiment 2: $F=.018$, $p<.982$). The managerial and theoretical implications of the study are discussed and compared with earlier work on banner advertising effectiveness.

Designing for Clarity

Graphic Design Tips for Non-Graphic Designers

Welcome to Designing for Clarity: Graphic design tips for non-designers.

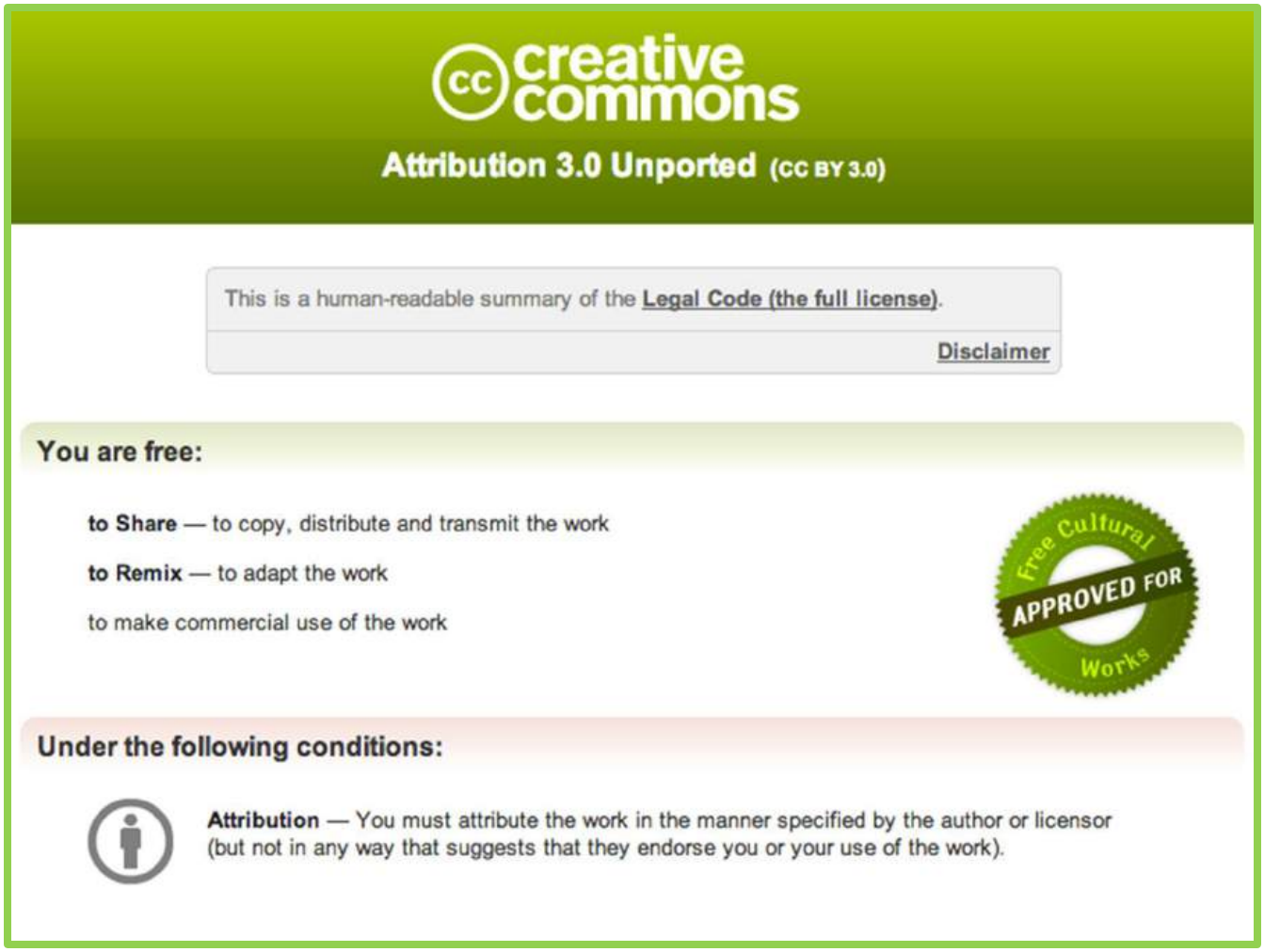
This session is an overview of practical, “use on Monday”, design advice and tips for people without a graphic design background. Obviously, the intended audience is people who are graphic design beginners, but there’s also value for people who have experience with graphic design and are looking for ways to explain these concepts to beginners.

Created by:

Bianca Woods

Instructional Designer/Technologist at BMO Financial Group

Former art teacher



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I'd like to start with a story about why design and clarity matter. It's a story about hats.

Last fall I took a hat making class, not for anything vital, just for fun. I had some sewing experience, but no hat making experience in the slightest. The course description didn't indicate I needed anything other than that, so I thought I'd be fine. Unfortunately, the class instructor wasn't used to teaching beginners, and it was her first time teaching the class. So wouldn't you know it, the instruction handout we received for our first assignment looked something like this:

Instructions

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Now, I'm using placeholder text instead of the actual instructions I received to avoid copyright issues. This is Lorem Ipsum text. It's gibberish that sort of looks like Latin that's used to block out where you'll eventually put your real text. It gives you a sense of how a layout with text will look, but it's all nonsense words.

I'll be honest though, the instructions were pretty much as incomprehensible as this fake Latin. It was just blocks and blocks of text. No categories, no numbering of steps, no diagrams: just text. You might hope that each paragraph was it's own step, but you'd be disappointed. Each paragraph was made of a number of steps which, to a classroom of beginners, just made it all the more easy to accidentally skip or misread steps. I'll admit, this was deeply intimidating and not easy to follow.

So, as a class, we begged for more clarity. The instructor finally agreed and made one small design change that made it substantially easier to follow:

Instructions

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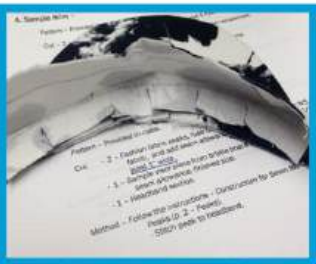
She numbered the steps.

That was it, just one simple design change and the assignment was substantially clearer and easier to complete.

But what if she had made a few more basic changes?

Instructions

1



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Add a few more design elements and you have a substantially clearer set of instructions.

The reason I'm sharing this story is to show just how big a difference using a few simple design rules can make. Careful design and layout is incredibly effective at simplifying your message and making it understood easily.

This is valuable in any field, but particularly so in ours. If there are design rules that can help to better understand content, and even feel less intimidated, frustrated, or just plain bored by the experience, then we should absolutely take advantage of them.

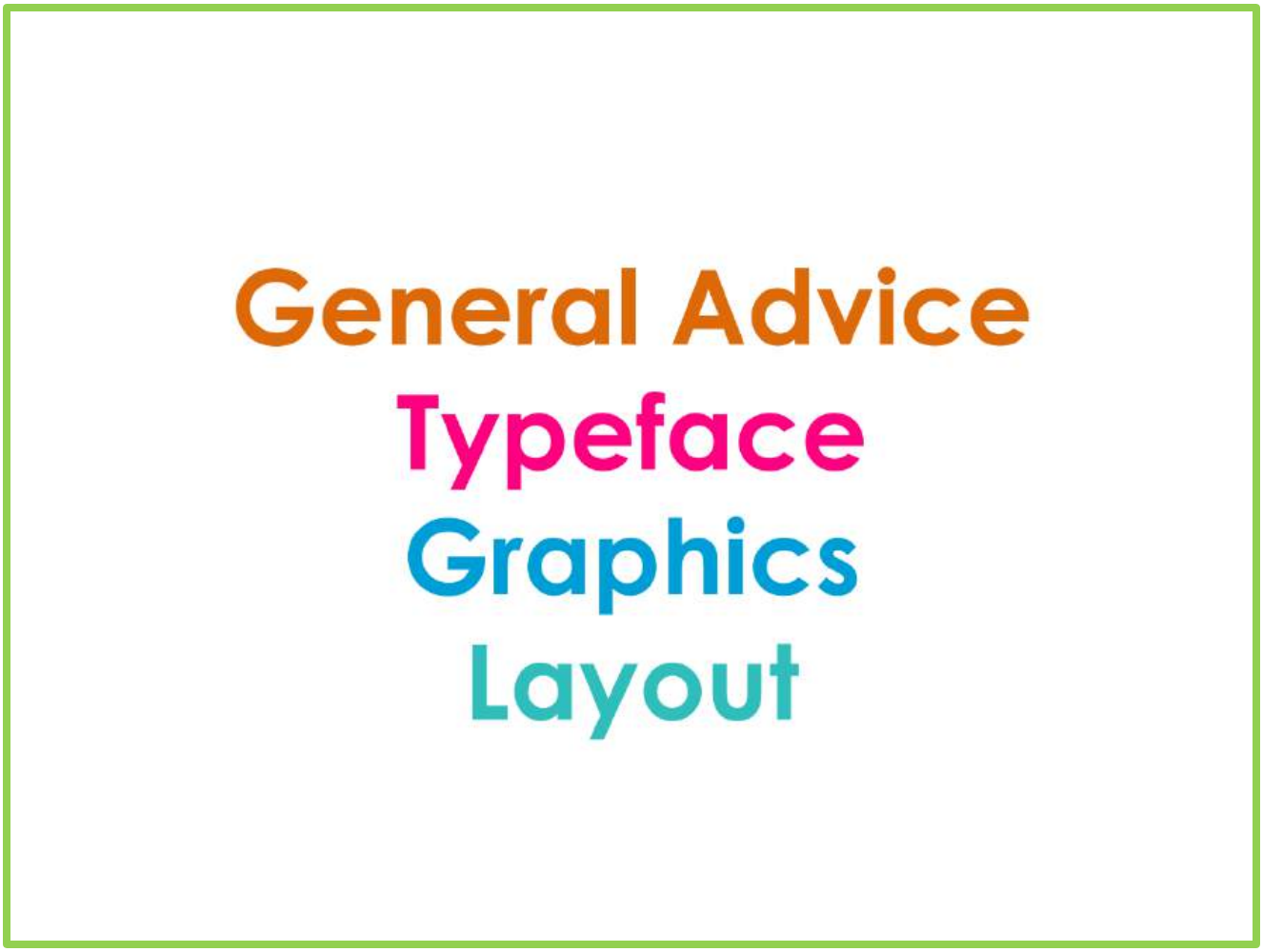
“But I’m not a designer,” you say.

That’s okay. Basic design is a skill that anyone can learn with a bit of practice. And that’s what we’re going to begin in this session.

Today I’m going to walk you through some basic, “use on Monday” design advice. That’s right, we’re skipping theory and diving right into things you can use immediately. These practical tips can be used to take any project, from a simple handout to a smartphone app, and make it easier for others to understand and interact with. Consider this your design training wheels.

The advice in this session isn’t going to make you a graphic designer overnight, but it will give you some basic design rules to use that you can use immediately much trouble and then continue to build on as you become more and more comfortable with design.

A pre-warning: in some cases I’m going to give you rules that you’ve seen designers break over and over again. That’s because the advice you’re getting today is the simplified, un-risky basics. They’re hard to mess up, even as a beginner. It’s not that I’m advocating for never taking any risks, but when you’re getting started it’s more effective to stick to some basic rules you know work every time.



Let's get started. We're going to break down tips for clear and concise design into 4 areas:

- General advice
- Typeface
- Graphics
- Layout

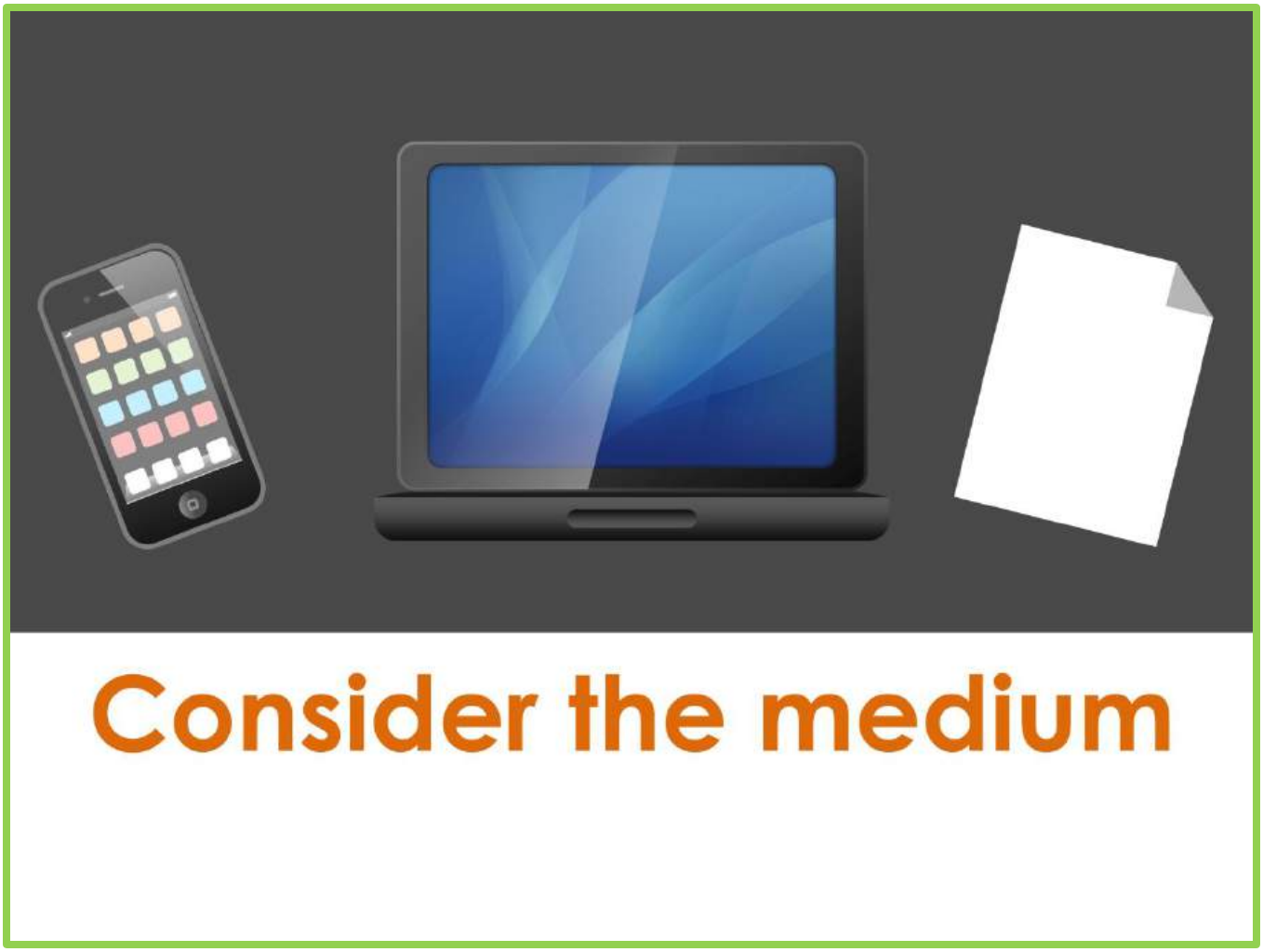
Let's start with some basics that should come at the beginning of any project, no matter what you're designing.

Consider your audience

You need to start by considering your audience.

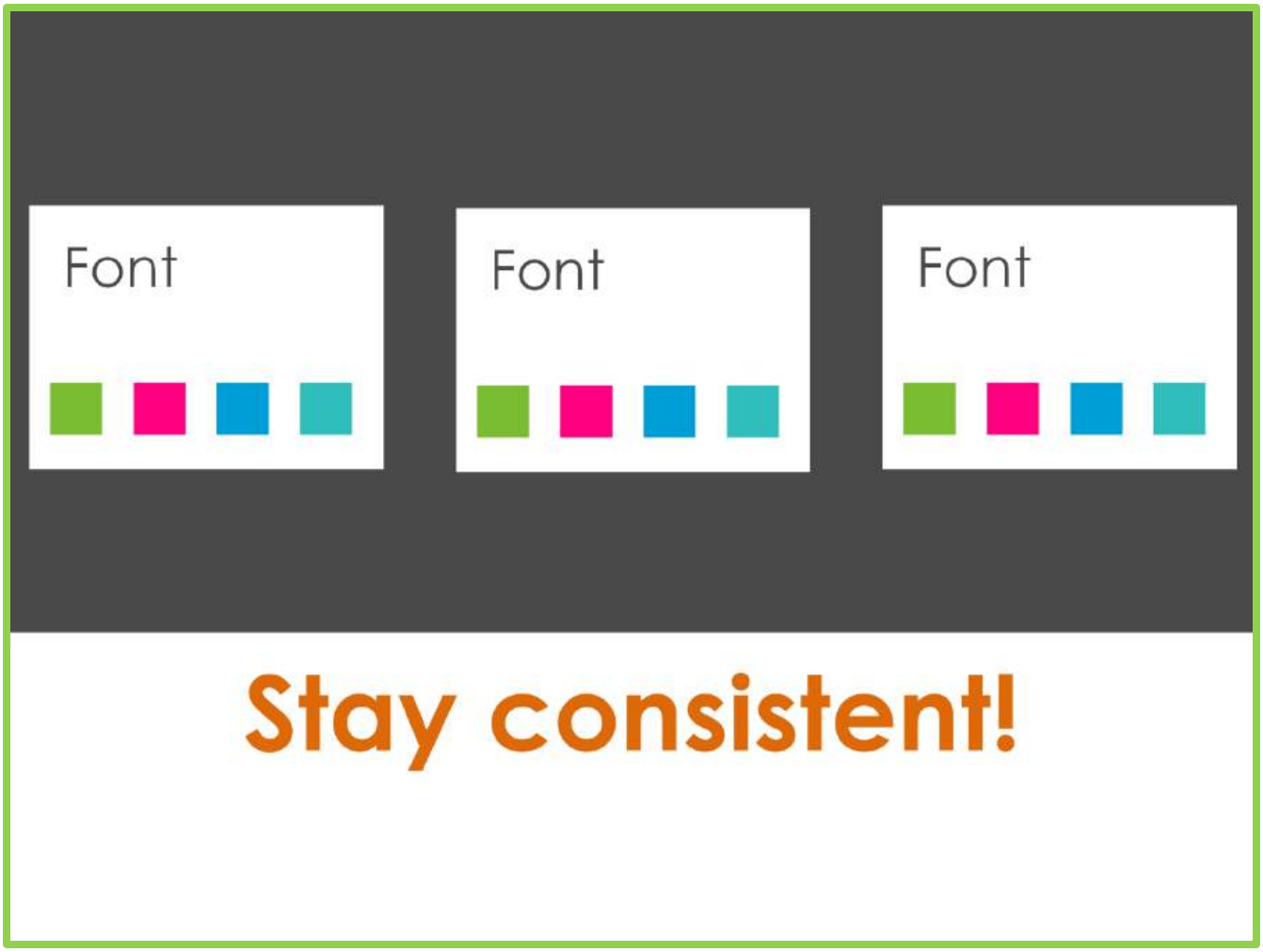
What makes up “good design” isn’t the same for everyone. That’s why it’s important to begin any design project, honestly any project in general, by thinking of who the final audience is. For instance, if you’re designing an instruction manual for experts, your audience may appreciate having more information on each page to make it easier to search through quickly. If you’re designing that same document for beginners, though, a great deal of information on each page might be intimidating.

Let the needs and preferences of your audience guide all of the design decisions you make.



Next, consider the medium (or mediums) your project will be delivered in. Think about it, something that's designed to look good as a large poster isn't going to look great as a PDF on a smartphone.

There are real strengths and limitations to every way we display information, so clear design means taking these into account at the beginning of the process.



Finally, stay consistent. When you make a design choice, use that same choice throughout the entire project, and even on related materials if you can. This makes all of your items look like they belong together. It's also easier on your audience.

Think about it. If, say, you use a particular font and colour for a subtopic title on the first page of a textbook, your audience is going to expect that pattern throughout the entire book. We're humans. We like patterns. If you change it half way through, it's incredibly confusing. Keep things simple by being consistent.

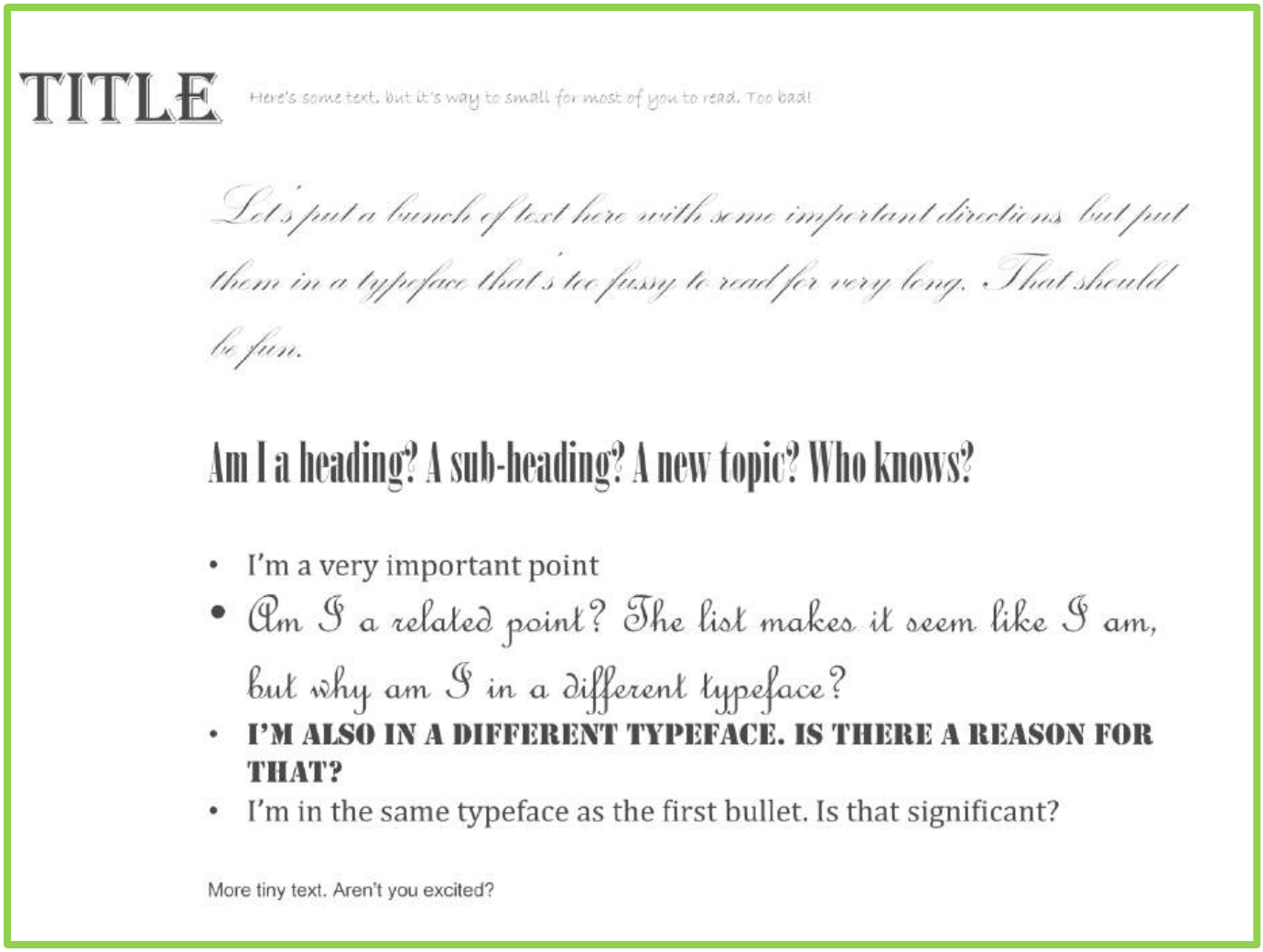
Now that we've got the general advice down, let's get a bit more specific and turn to the actual lettering that you use in your work: typeface.

So what exactly is a typeface? Simply put, a typeface is a set of characters that share common design features. Right now you're looking at seven different examples of typefaces.

If you're not a graphic designer, you're probably more used to calling typefaces "fonts". Technically, that term is incorrect, but no one here is going to kick you out for using it.

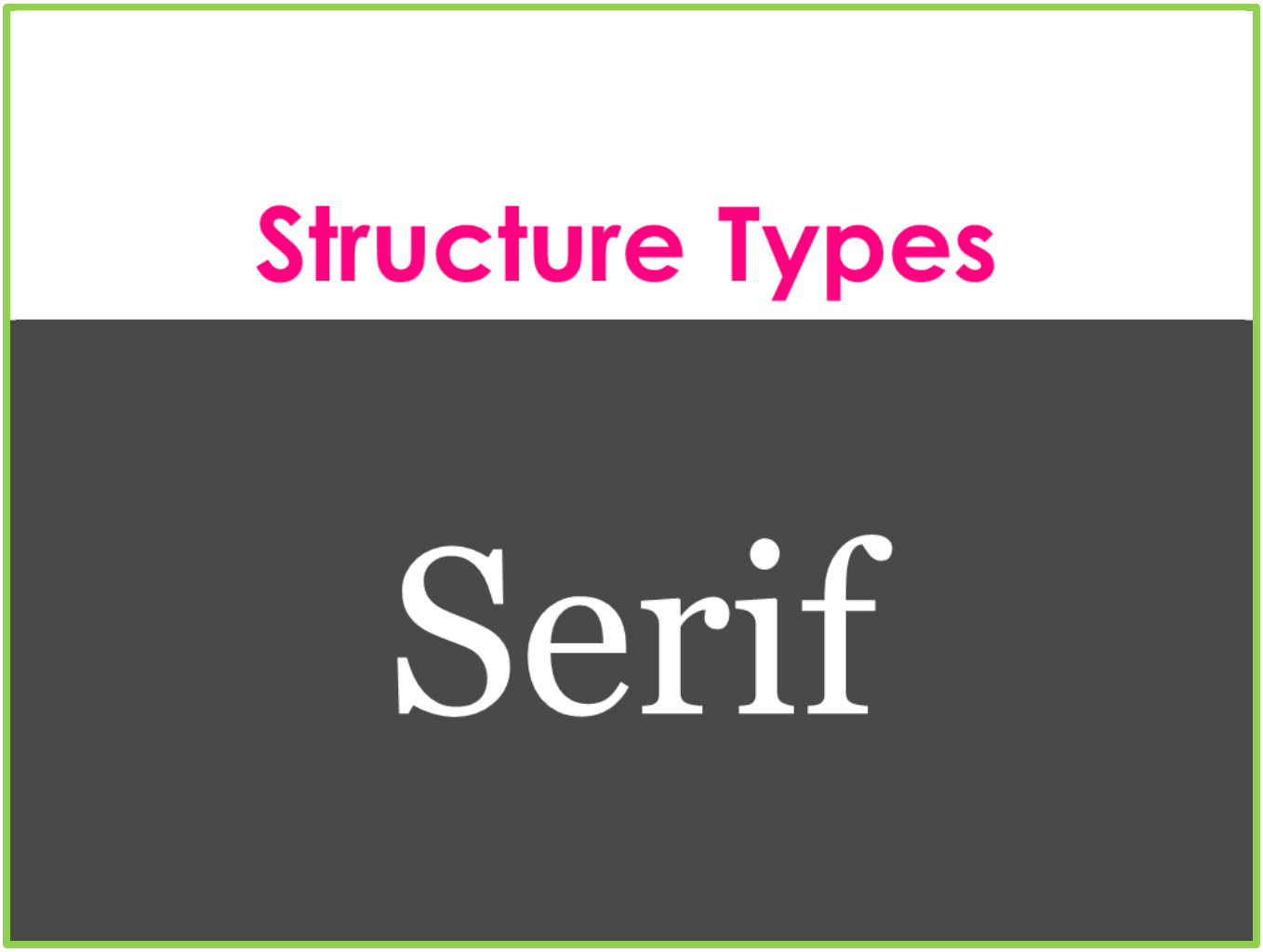
What's important to notice here is that each typeface has it's own unique aesthetic, or flavour. The look and feel of a typeface gives a very specific impression, and you can use this in order to influence how people perceive what you've designed. For instance, the thin and simple lines of Century Gothic come across as sophisticated, Cooper Black's rounded ends makes it feel a bit more casual, and Broadway is instantly reminiscent of old plays and musicals.

Make sure the typeface you choose reinforces the image you're trying to convey.

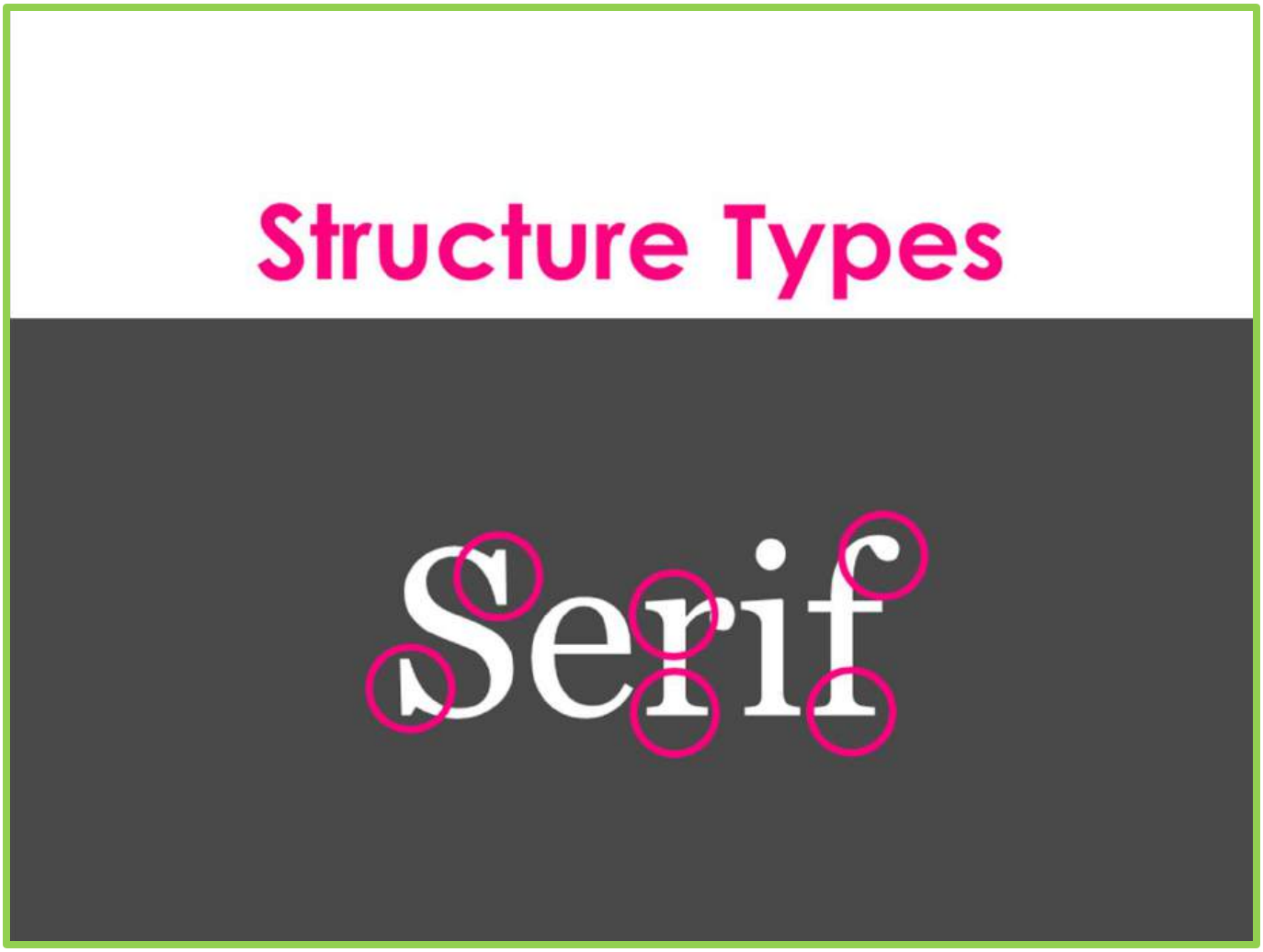


Typeface is unfortunately one of those things that people sometimes go a bit crazy on when they're trying to make something look more "designed". I'm sure you've all seen documents that look like this. It's chaotic, hard to read, and not particularly professional looking. There are some easy rules, though, that can help you avoid a mess like this.

Structure Types

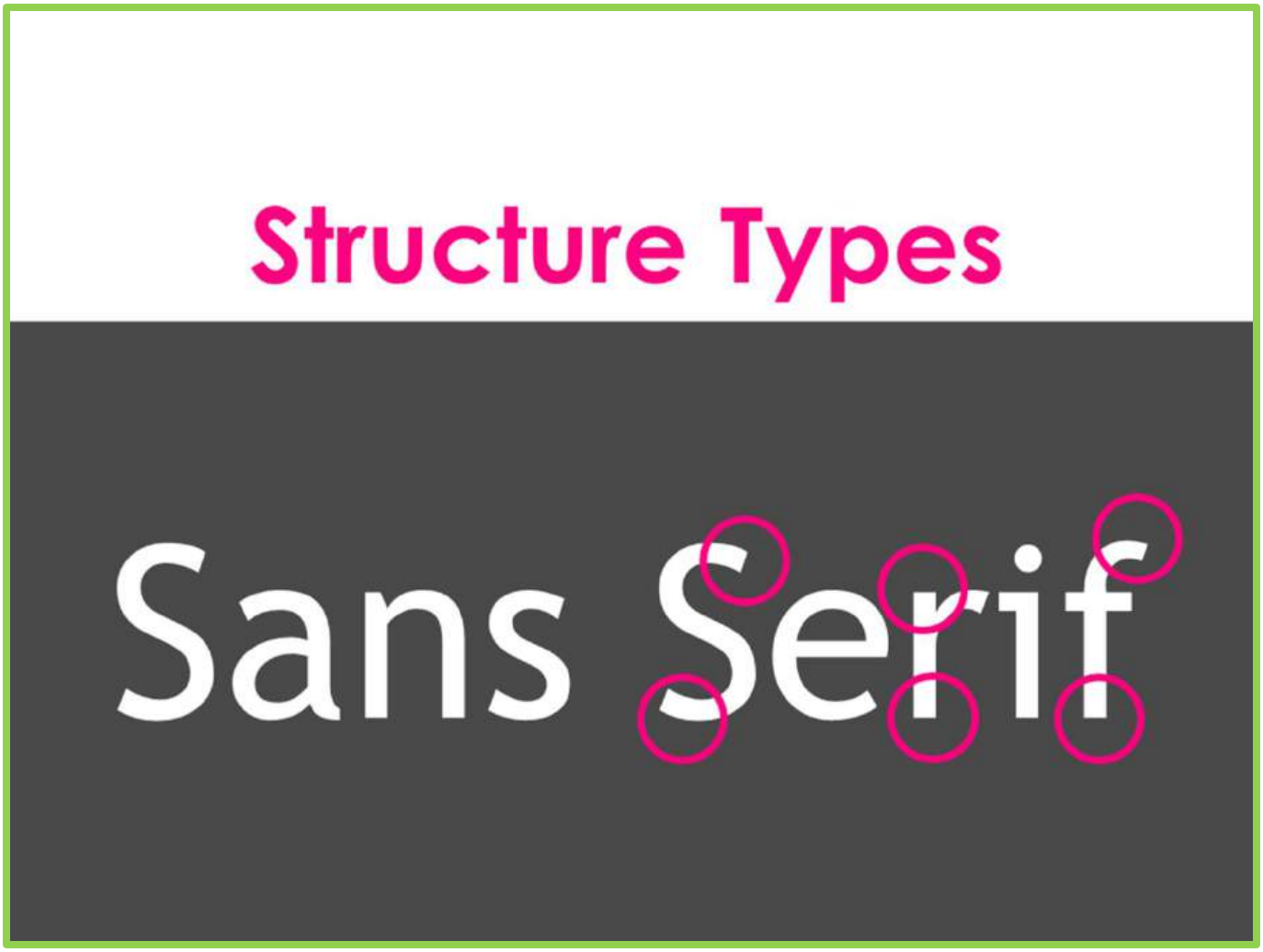


Typefaces themselves are organized into a few broad groups based on their design features. For the sake of this session I'm keeping it simple by putting everything into 3 larger groups. This is the point where anyone who knows anything about typography will insist that there are more than just 3 categories. They're right. But when you're starting out, sorting typefaces into just these three broad structure categories will help you make better decisions about which to use:



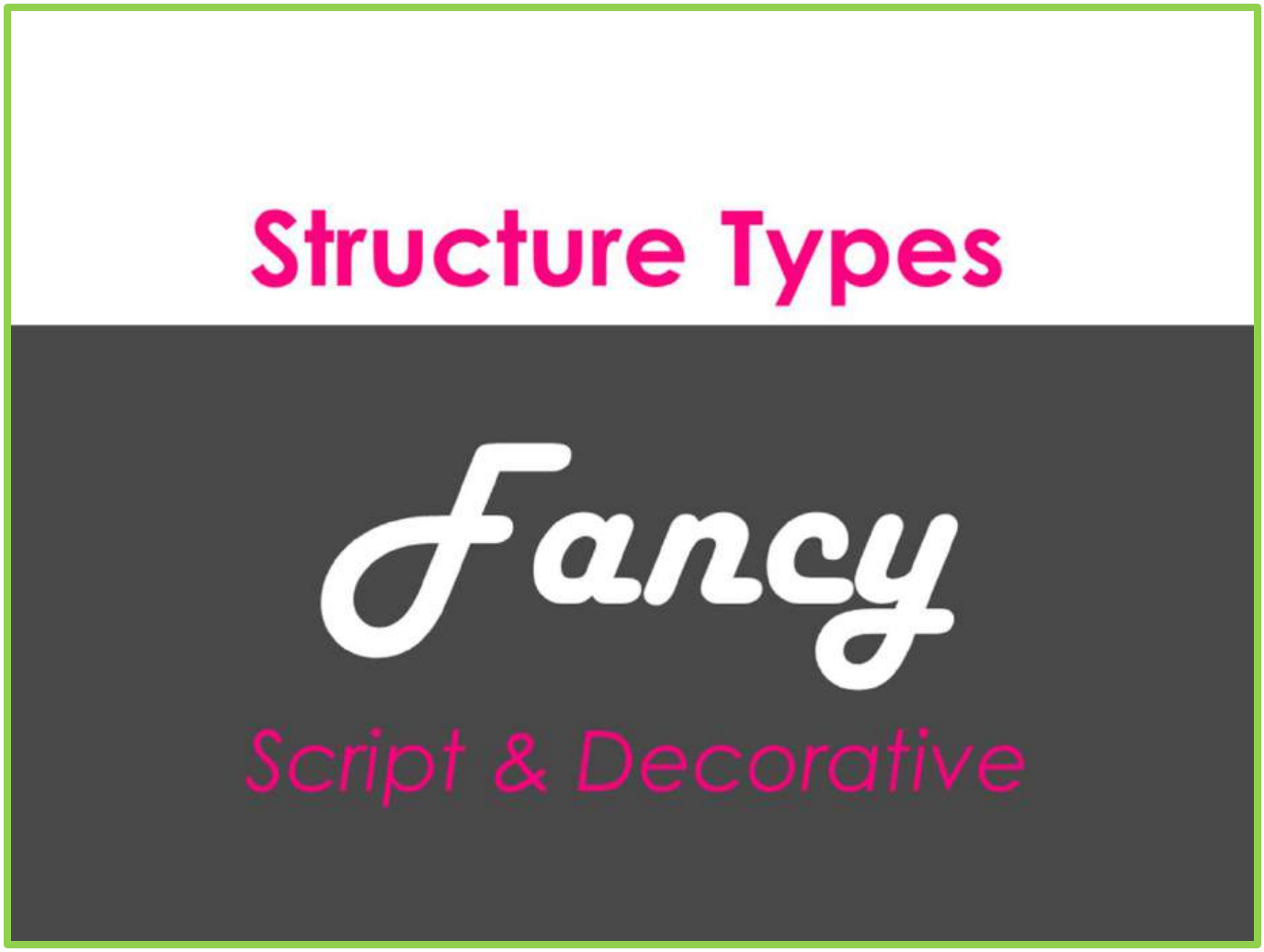
First we have serif typefaces. What defines a serif typeface?

It has simple-looking letters with these caps on the ends of each character called, you guessed it, serifs. Serif typefaces are typically thought of as the easiest to read when printed, although that fact is up for debate.

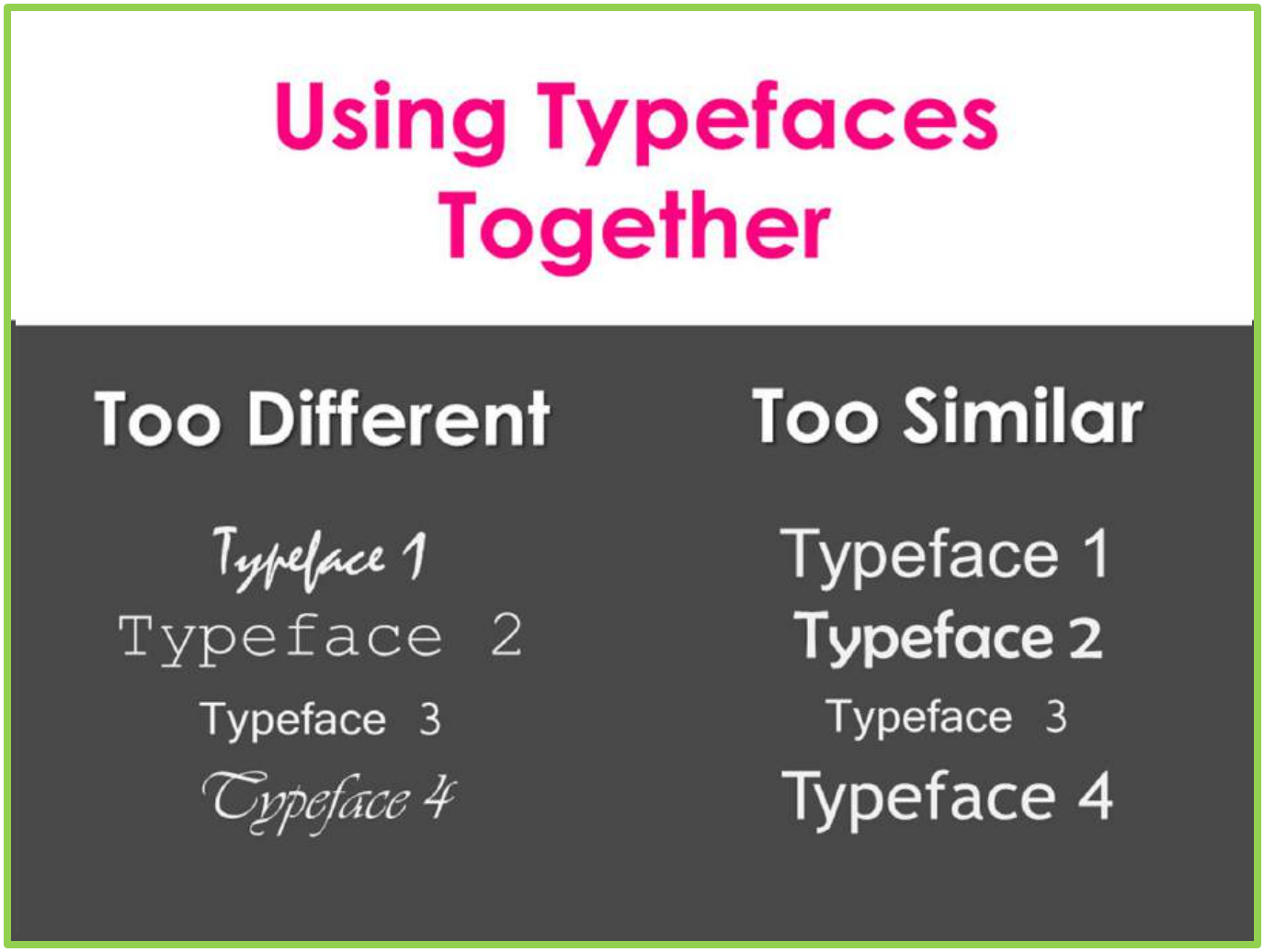


Next up is sans serif. A sans serif typeface is recognizable because..

While it also has simple letter design, each character *doesn't* have serifs. That's easy enough to remember, especially if you know any French. Because sans serif letters have a simpler design, they're typically thought of as the easiest category of typeface to read for digital projects.



Finally, we have a very broad category that, for the sake of this presentation, we're going to call "Fancy". This category consists of typefaces that are highly stylized or look like script. They typically have a more decorative flair than your serif and sans serif letters, but they're also not nearly as readable. Because of this it's good to use them sparingly, such as for titles or headings.



Mixing a large number of wildly different typefaces together just looks chaotic. Mixing very similar typefaces from the same structure type also looks odd.

That's why when you're starting out an easy rule is to choose just 2 typefaces for your entire project: a serif and a sans serif. Use one for titles or headings, and the other for your body text and you'll have a safe combination nearly every time.

Typeface size is typically measured in points. Size will dictate what people look at first. The larger something is, the more people will perceive it as important.

What you want to do is use size to your advantage to organize your content.

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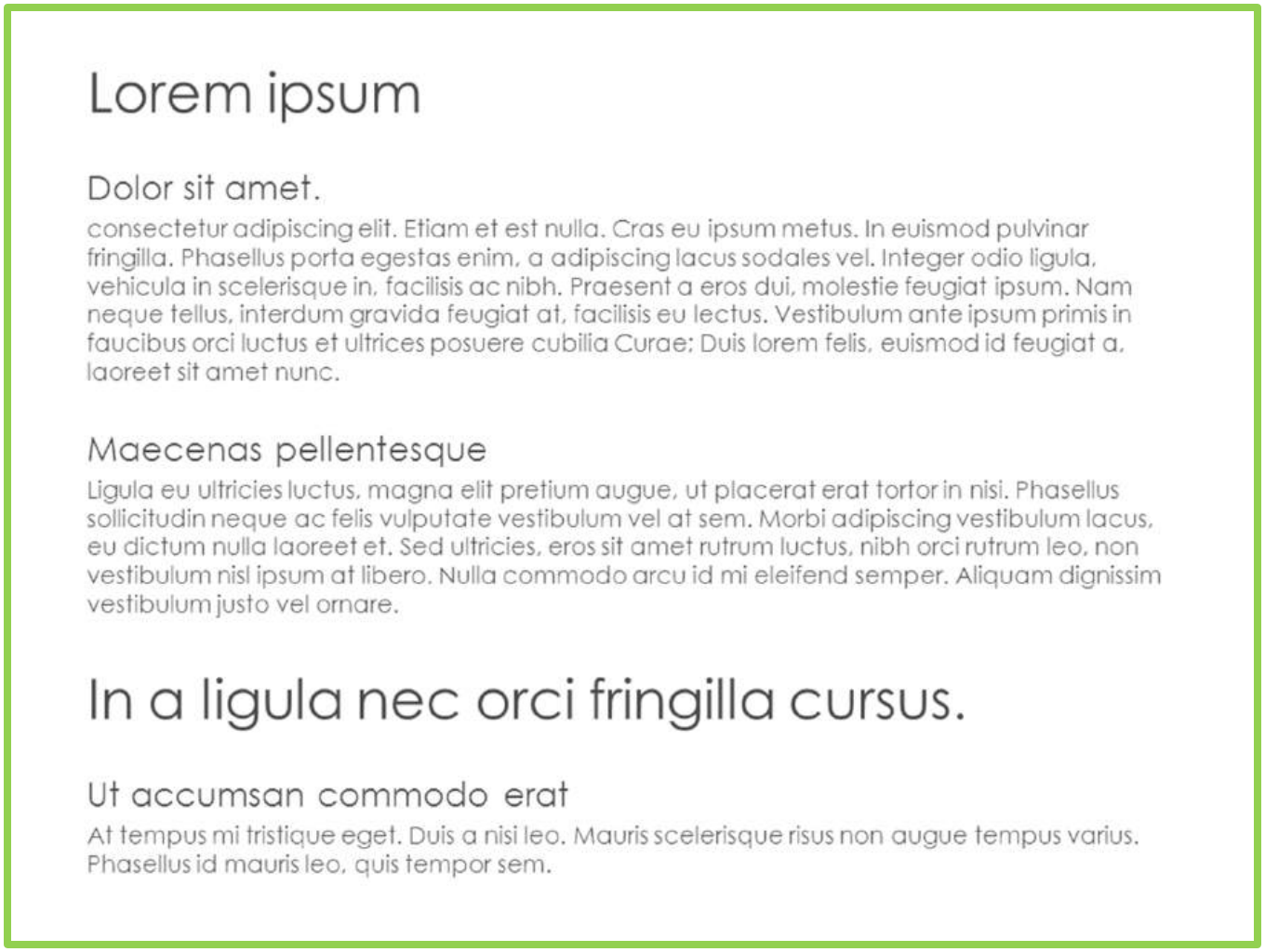
At tempus mi tristique eget. Duis a nisi leo. Mauris scelerisque risus non augue tempus varius. Phasellus id mauris leo, quis tempor sem.

For instance, here's some text that's vaguely organized in layout. You can sort of see what might be headers and titles, but you have to work at it a bit.



Now here's the exact same text and the exact same layout, but I've increased the size of the headings and subheadings.

You'll notice that I'm using a consistent size for the headings, as well as a different, but also consistent size for the subheadings. This consistency helps the brain find patterns.

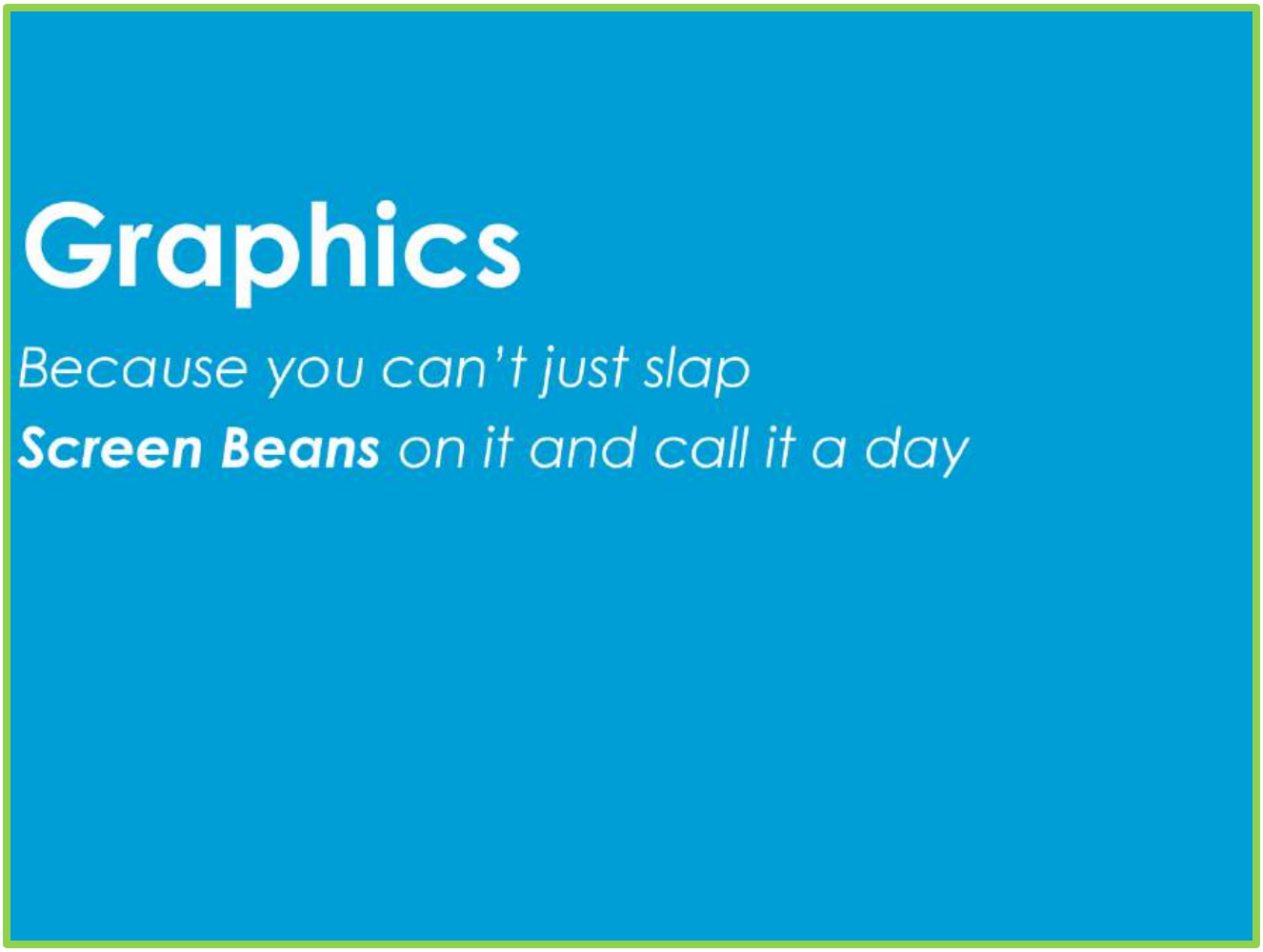
When you're trying to use typeface size to your advantage, be sure to make the font sizes *noticeably* different. If you use similar font sizes, such as 12 point for the regular text and 14 point for the title, people will have a much harder time noticing the difference, if they even notice it at all. To use size well, you should be able to instantly see the size difference.

Since I'm telling you not to use more than two typefaces, you may worry about things looking sort of boring. Don't worry: you can mix things up a bit with weight. Weight refers to how wide or skinny the lines that make up each letter are. Thankfully many professional typefaces have multiple weights, as shown here in BMO's Dax font. Even if you're just using the default typefaces that come with your software, you can get different weight by using the Bold function.

Because you're essentially using variations on a theme, you can use multiple weights on one document without it looking messy. The only thing you want to keep in mind is, once again, consistency. Much like size, keep how you use each weight consistent. If you decide to use a bold versions of a typeface for one heading, use it for all headings.

Finally, some typefaces come with a lot of emotional baggage, and because of this you'll want to avoid using them. There's two reasons for that. First, there's overexposure. Typefaces like Times New Roman and Arial are so ubiquitous that people are tired of seeing them. They seem tired and old to people.

The other reason is aesthetics. Some typefaces, like Comic Sans and Papyrus, have been deemed just plain ugly by the design community. That judgment is easily passed on to anything you make using them, so it's best to avoid them completely.



Now that we've got the basics of typeface down, let's move to the other thing you immediately see when you look at a project: graphics.

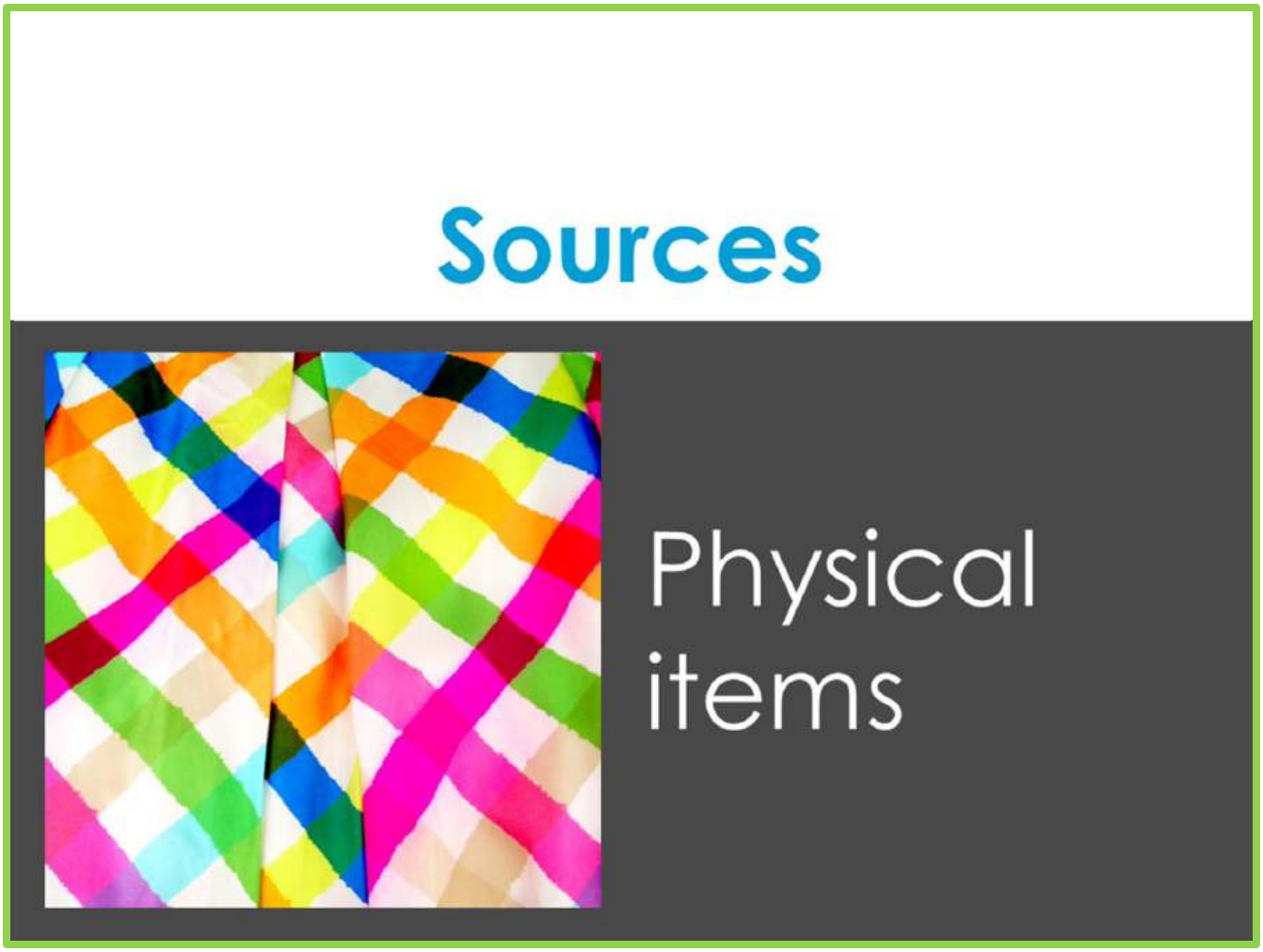
What we're trying to avoid is this sucker here. Tired clip art, random graphics, tiny graphics, text colours that interfere with readability, no noticeable colour scheme. I'd like to pretend I haven't seen things that look this bad, but that would sadly be a lie.

The secret to cleaning this up is to just keep things simple, which actually makes things a lot easier for you in the long run.

First we should talk about colour.

Like typefaces, if you use a bunch of different colours haphazardly it ends up looking messy, and sometimes even nauseating. The best way to keep your colours in order is to use a set colour palette. That's a small selection of specific colours that you use consistently throughout your entire document. For example, the rectangles on this screen represent the colour palette I used for this presentation.

Select a few colours that look good together, all used consistently, and you're set. There are going to be some people here that feel very comfortable putting together colour combinations on their own, but for everyone else there are some great shortcuts.



Here are some easy ways to put together a colour palette that just works:

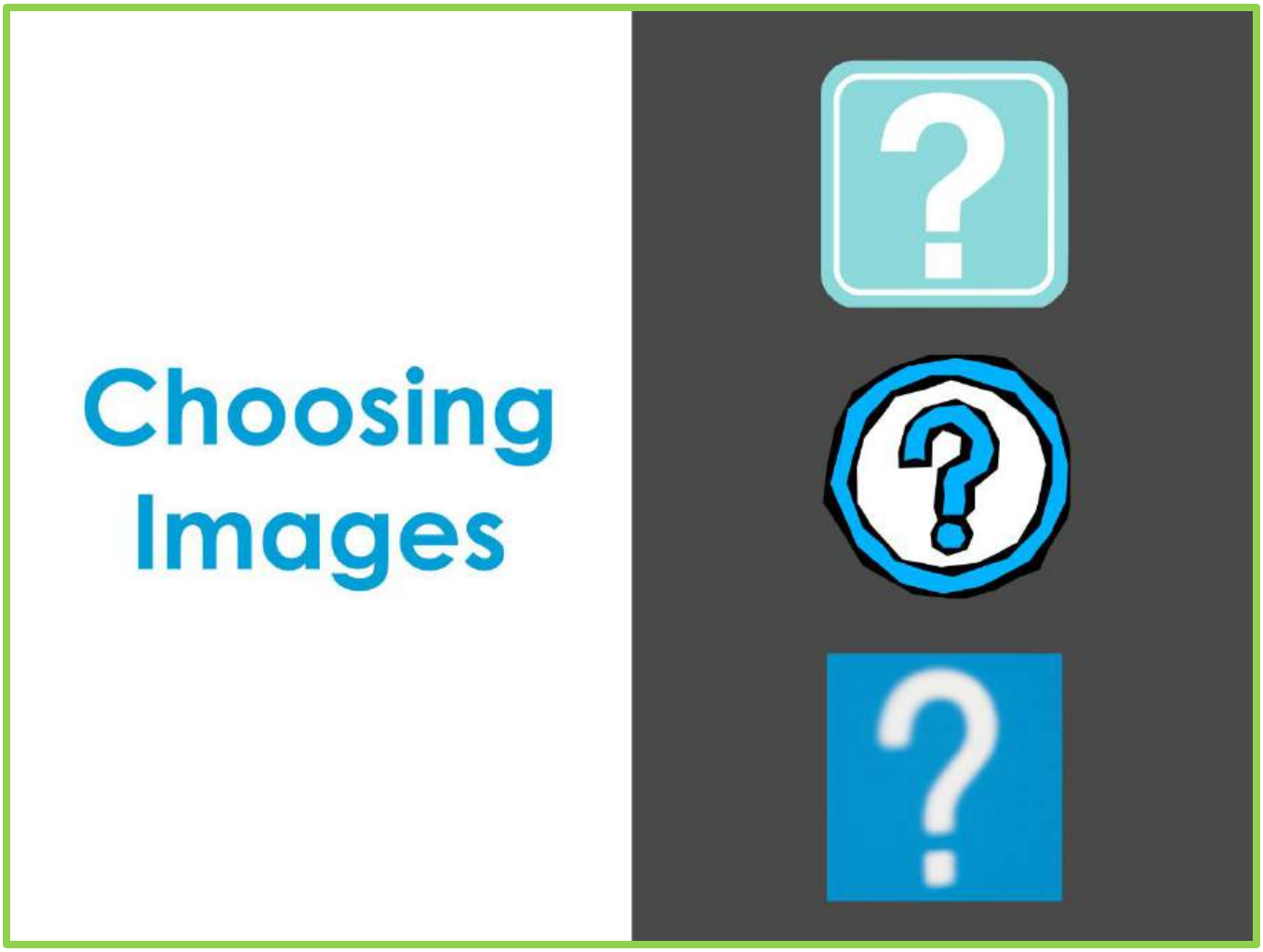
One option is to look at things around you for ideas. For those of you who have company logos or websites, that's a good place to start if you want colours that reinforce your company image. If you want to branch out more though, there are sources for ideas everywhere. For instance, I got my initial idea for the colour scheme in this presentation from one of my favourite skirts. Pick something that seems well designed, be it a photo, book cover, cereal box, or your laptop bag.

Sources

Pre-Created Palettes

Another option is to take advantage of pre-created palettes. These are, as it sounds, colour schemes that other people have put together for you. If you use any Microsoft Office products you'll likely be familiar with the colour scheme sets that come standard with the software. They're actually pretty decent. Another option, though, is websites that curate colour palettes. The image here is from a site called Design Seeds. The designer for this site selects a photo for inspiration and then puts together colour palettes based around the dominant colours in the image. Sites like this can be a great way of finding colour sets put together by experts.

Finally, you can always try a palette generator website. On these sites you choose an image that has colours you think work well with your project, upload it to the site, and the site quickly generates a palette based on predominant colours from the image.



Moving on from colour, we now need to look at actually *choosing* images.

The first thing you need to consider is quality. There are a number of ways an image can be low quality.

There's quality of the image file itself. A small image stretched large eventually looks blurry. If an image looks blurry at the size you need it to be, you should always replace it with one that's crisp.

Also, taking an image and warping it to fit a space also lowers its quality. You're better off cropping the image to make it the right size or, once again, choosing an image that's already a similar size to the area you're trying to fill.

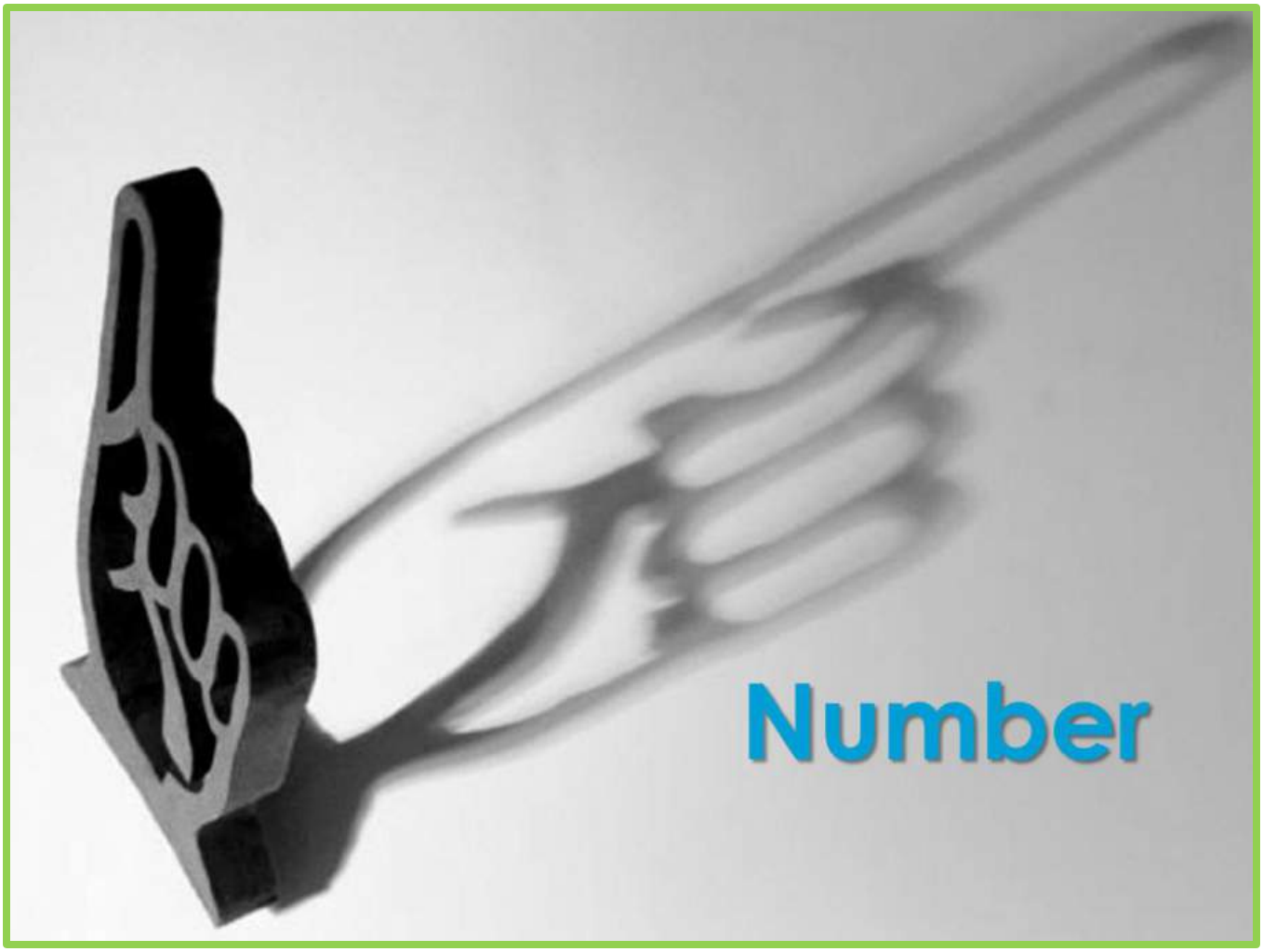
Next, there's also the quality of the subject matter. An image that doesn't have a strong connection to your subject matter, or any connection at all, just gets in the way of clarity. I have yet to think of a situation in which this particular image, a personal favourite of mine from the Microsoft Office images library, would ever have a strong connection to a topic. I'm open to suggestions though.

Image Quality

Finally, don't forget that an image that's overused loses impact and should be avoided. Things like these Screen Beans guys or pictures of people shaking hands in front of a globe are so overused that they'll appear trite or hokey to your audience. Avoid them at all costs.

You also want to consider the number of images you use.

In most cases, less is more.



It’s better to have one fantastic picture than a bunch of okay ones. It also looks more focused and sleek to the viewer. Magazine layouts or advertising posters can be a great example of this.

Size

The size of the images you use is also important. People are often afraid of letting the image take over a large area of a document, so they keep them tiny like this example.

If you've picked a good image, it will make more of an impact if you give it a large amount of space. One again, look to magazines and ads to see this at work.

Sources

Stock Photo Sites

You know that high quality images are important, but where can you go to find cheap and easy sources for them?

Stock photo sites are a good place to start. They're relatively cheap ways to purchase the rights to professionally created images. They've also got fantastic search engines that allow you to look for specific image types, layouts, and even colour schemes. The trick to using these sites is to be very picky about the images you choose. You want the image you pick to be deeply relevant to your topic and not seem lifeless or phony. It's also important to avoid images that people have seen time and time again. Since many people will just jump on the first image in a search that works for them, completely skipping the first page of search results can be a good way to find stock images that aren't overused.

Sources

Another option, especially if you're on a shoestring budget, is to use free images. This doesn't mean going to Google Image search and taking photos from there. It means finding images that the owner has licensed people to use for free:

- Flickr has a great search function that allows you to search for images that have been made available under a Creative Commons license. If you're not familiar with the term, Creative Commons is a flexible type of license that a content creator can use to share content freely, but with a few established restrictions.
- Stock image sites often have free images of the week.
- Graphic designers and community groups like E-Learning Heroes also commonly give out free design resources.

In all these cases you want to double check what the licensing requirements are before you use them. For instance, many Creative Commons licenses require that you credit the image owner.

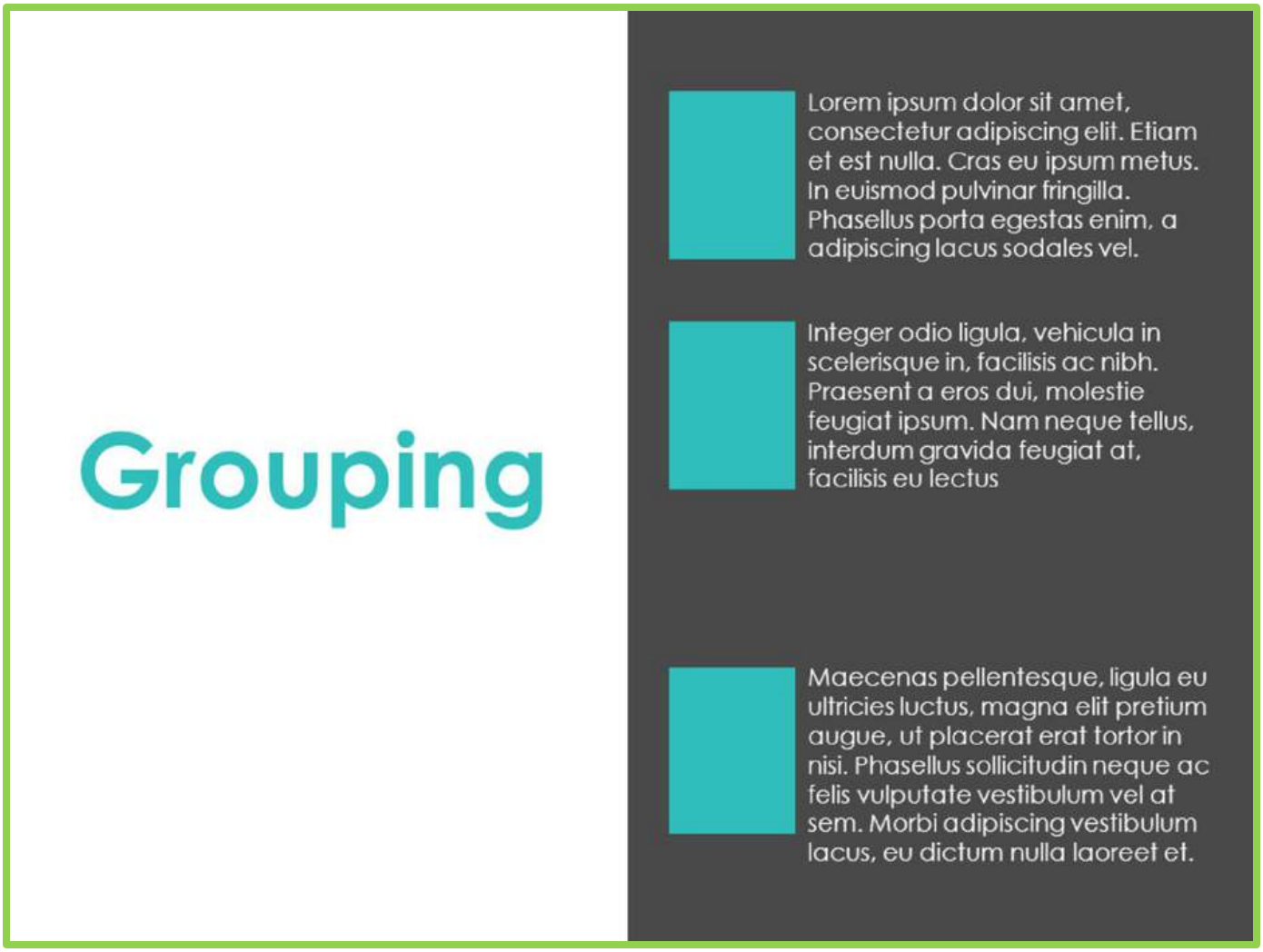
Sources

If you can't find just the right image elsewhere, don't forget that you have a great deal of tools available to make your own graphics. Anyone with a smartphone and a few apps has the tools to create decent images with a bit of practice. The two I tend to use the most are Instagram and Camera+. Another unsung hero is PowerPoint 2010. There are some amazing hidden functions that allow you to easily put together professional looking vector graphics quickly and then export them as graphics for use anywhere. The iPhone image on this screen was actually created entirely with PowerPoint.

Finally, there's layout: how exactly you arrange items in your project. This includes text, images, videos, or other visual aspects.

When laying out a project, you want to be aware of how people perceive grouping. Things that are close together are unconsciously linked. The layout on the right here doesn't have any noticeable grouping as it stands now.

I can change that by clustering elements of this design together.

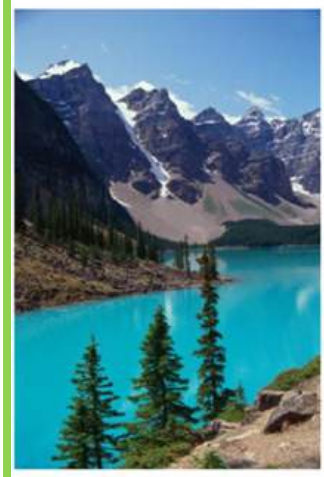


Now each block of text looks linked to the image on its left. Also, the first two blocks of text seem more connected to each other than the last block.

Be purposeful about your grouping and use it to your advantage.

Whitespace is the blank space in a design. Whitespace is vital to a clear design, so it's important to not be afraid of leaving space, or "room to breathe", in between your design elements.

Title Text



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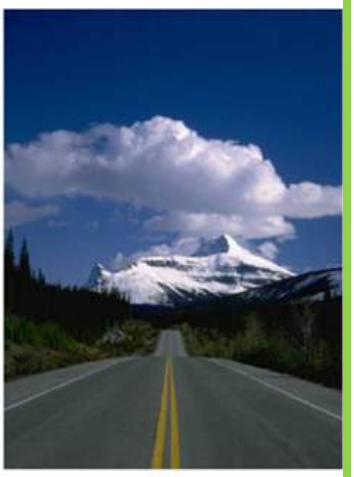

A layout that's all squished together may fit more information on an individual screen or page, but it's much harder to read or interact with the content with when everything is so close together.

Title Text

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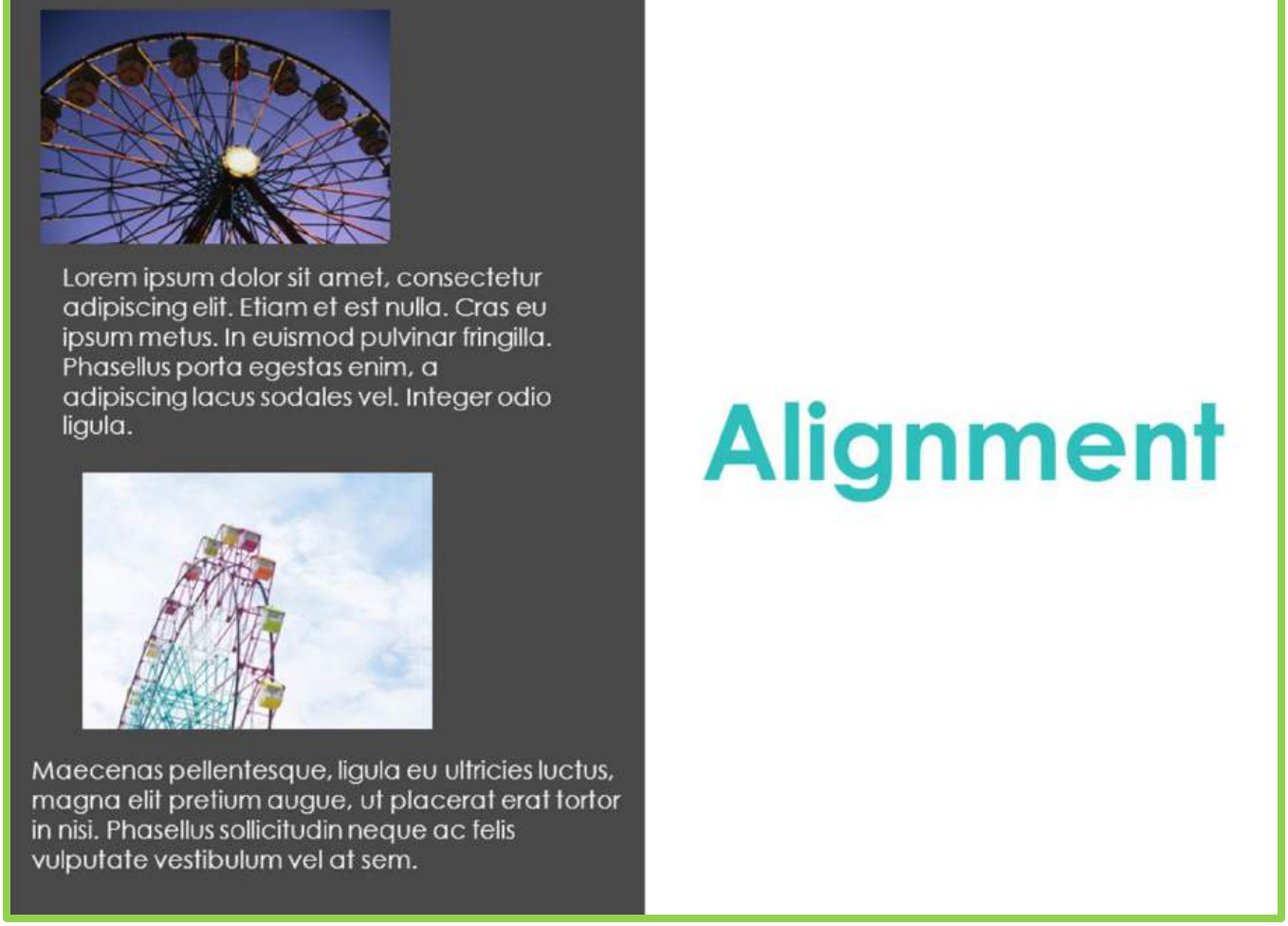
It's much clearer to a reader when there's space between design elements.

Now, you'll notice that adding in more whitespace meant I had to move some content off screen. This will make a longer document in the end, but that's okay. In most cases it's better to have a slightly longer document that's well spaced and easy to read than a shorter document that's cramped and hard to read.



You also want to think about how things line up, or alignment. Alignment of images and/or text also helps make your project look polished. The images and text on the left here aren't aligned, so they look a bit off even to an untrained eye.

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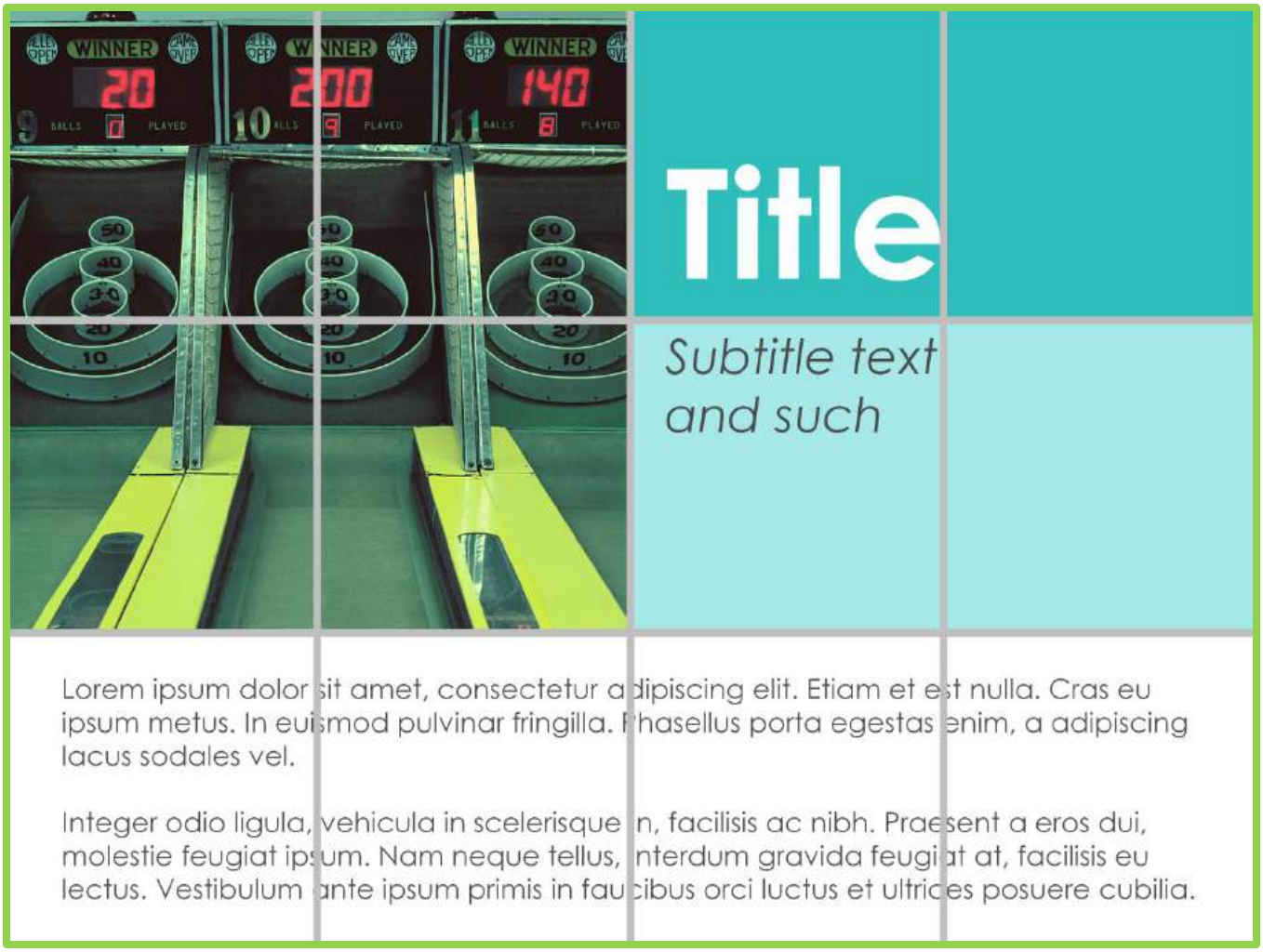
Alignment

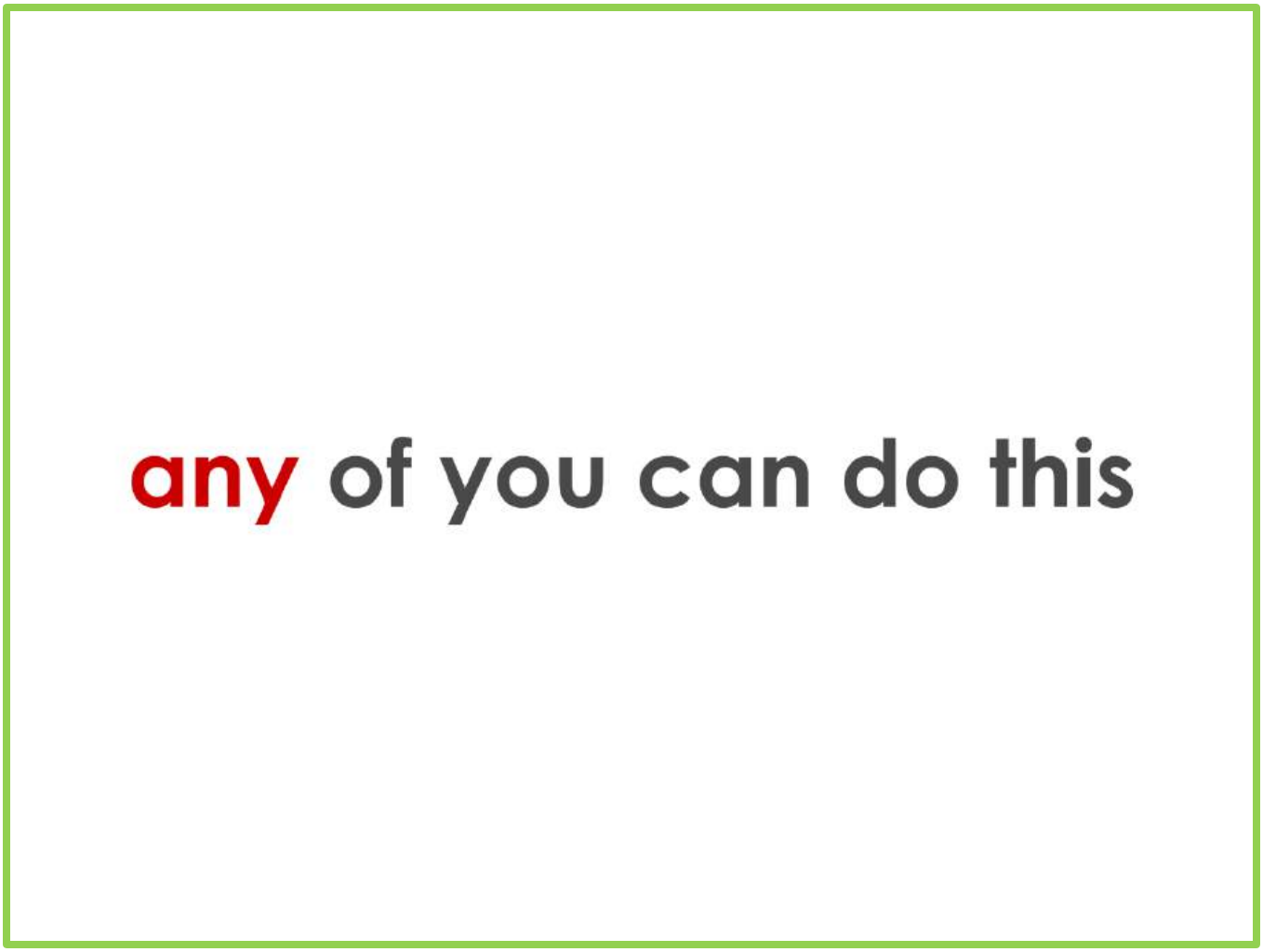
Aligning design elements, in this case I've lined up the left edge of text and images, makes everything seem connected and organized. Thankfully if you're using a recent version of Office or iWork the software actually has alignment features built in. When you move an object around a page, the software will actually alert you when the edge of an image or text box is aligned with other elements on the page. Take advantage of this to help tidy up your project.

If you're unsure of how to start to laying a project out in the first place, try using a grid.

Doing this is quite simple. Start by breaking the space you're using for the project out into an even grid (a 3x4 grid like this is a good place to start).



Then fit your design elements into one or more cells. As you can see, this builds a clean and sleek layout that keeps you from having too many, or two few, design elements in your project.



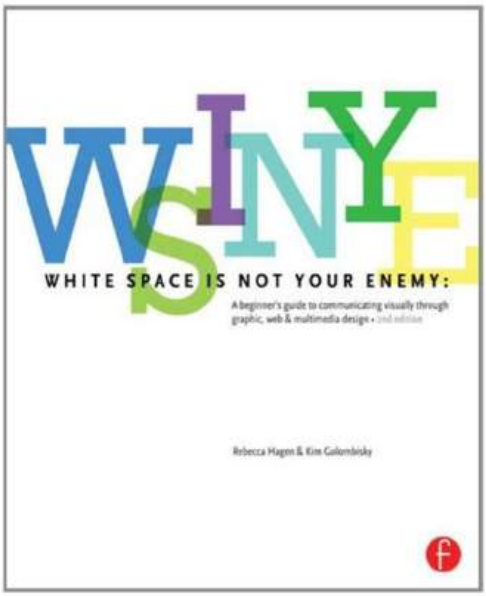
So that's the very basics of clear and concise design. The thing I'm hoping you've gotten out of this experience is that every single one of you is capable of using these tips to make your work look cleaner and communicate more effectively.

What's next?

Because knowing is half the battle

Now, eventually you're all going to have these basic ideas mastered. Because of this, I thought I'd also a few suggestions for resources you can use to build your design skills even further.

Getting Started

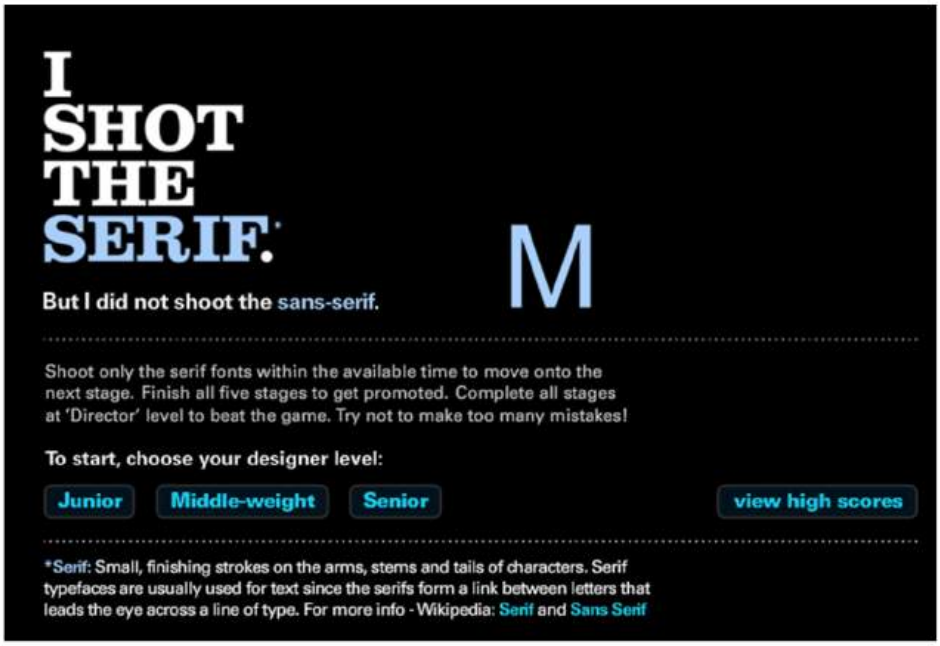


White Space is Not Your Enemy

Kim Golombisky
Rebecca Hagen

For general advice that's still aimed at beginners, try the book "White space is not your enemy" by Kim Golombisky and Rebecca Hagen. This book goes into even more detail about the topics I talked about today and is a great way to keep learning about the essentials of design without being overwhelmed.

Typeface



Web and iOS

When it comes to typeface, I've got two amusing web games for you.

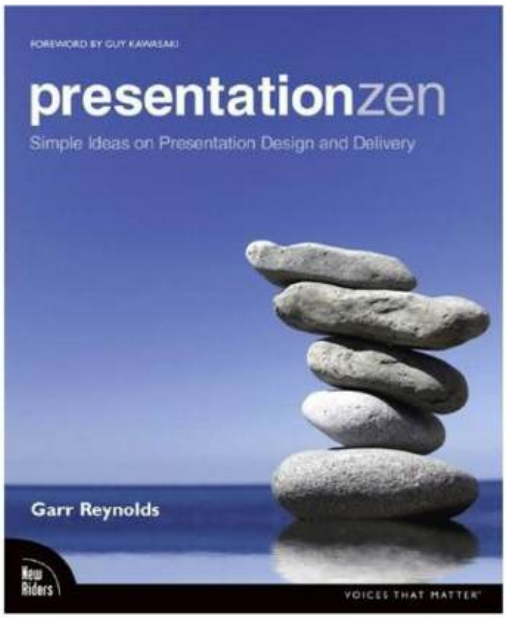
First is "I shot the serif" which is available to play online or through an iOS app. It challenges you to look at a screen full of different letters in different typefaces, and then "shoot" only the letters in a serif typeface. It's great for wrapping your brain around the difference between serif and sans serif.

Typeface

Web

Once you're feeling ready to move beyond the simple "one serif, one sans-serif" typeface pattern, you might want to play with TypeConnection: a "dating game" for matching typefaces together successfully. The site allows you to pick a typeface, learn a bit more about its "personality", and then set it up with another typeface that's compatible with it. The games gives you a lot of guided feedback, so it's perfect for learning more about why certain typefaces work, or don't work, with each other.

Graphics

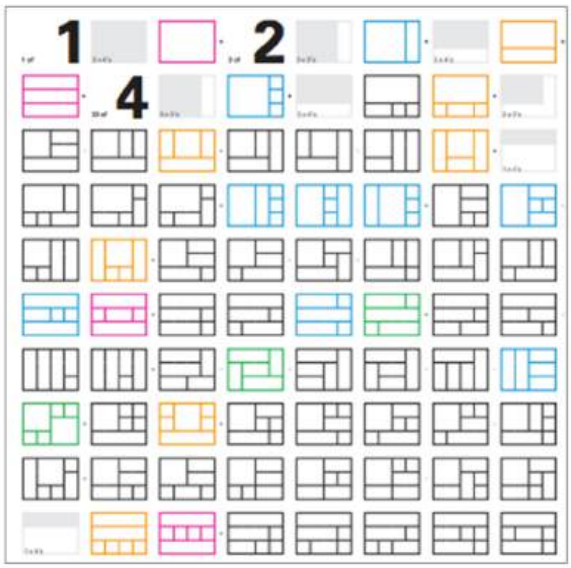


Presentation Zen

Garr Reynolds

For graphics, I'd recommend Presentation Zen by Garr Reynolds. It's not actually a book about graphics specifically, it's actually about presentations, but it has some fantastic insight into simplifying your graphics that can honestly be applied to any design project.

Layout



The 892 unique ways to partition a 3 x 4 grid

Dubberly Design Office

For layout, I suggest checking out a PDF of **The 892 unique ways to partition a 3 x 4 grid**. It literally shows every possible variant of partitioning this type of grid, which is incredibly helpful for when you're stumped at how you want to layout a page or screen. This graphic here is just one tiny corner of the full document.

Finally, here's the link I promised to the additional resources. It has a copy of this deck, along with the presenters notes, links to all the materials mentioned in this presentation, and more resources I thought you'd find helpful: **http://biancawoods.weebly.com/ls-2013.html**

I'm also constantly blogging and tweeting about learning, design, and technology, so feel free to check me out on Twitter or my blog.

Blog
http://e-geeking.blogspot.ca/

Twitter
@eGeeking

BrandMaster

The complete guide to banner advertising

How Display Ads is a game changer for your marketing team

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WHY SHOULD YOU READ THIS EBOOK?

As a marketer you are facing the daily challenges of delivering a relevant brand and message to your market. It requires skills, resources and time you don't have and you are falling behind.

Creating dynamic content is not only draining your time, it is killing your budget. But you know the importance and now you are looking for a solution.

In this eBook we'll guide you through banner advertising and how Display Ads from BrandMaster is the answer to your worries.

The intention is to provide you with know-how on great possibilities that is in front of you and how investing in technology is beneficial in the long run.

Do you need banner advertising?

Relevant advertising and consumer behavior information is important regardless of the type of business you are in or the size of your company. Consumers are already expecting it, and the demand will only grow in the future.

However, the extent of your usage of banner advertising will vary. A B2C brand is more likely to have a higher rate on this type of communication than a B2B brand. So the question is: Can your brand benefit from banner ads? If at least three of the following statements are true for your brand, you should consider investing in Display Ads.

Your ads require many different ad formats

Display Ads reduce the time spent on editing and making numerous ad versions. Instead, your creative team can focus on producing new great content.

You struggle to keep up with the market changes

Display Ads help you take ad relevancy to a complete new level. You can create ads and have them go live in a few minutes.

Protecting your brand is important to you

Display Ads let you regain control over your ads. With Display Ads, you can quickly produce content for a variety of channels without compromizing your brand identity.

The trends among your target groups requires personalized communication

Display Ads let you communicate with specific target groups at the right time and place in the customer journey.

Outsourcing costs are draining your budget

If it takes you days to fix a problem in your digital ads, you may lose customers and it is definitly not good for your budget. With Display Ads, you can easily change your ads and see them go live immediately. No skills requried.

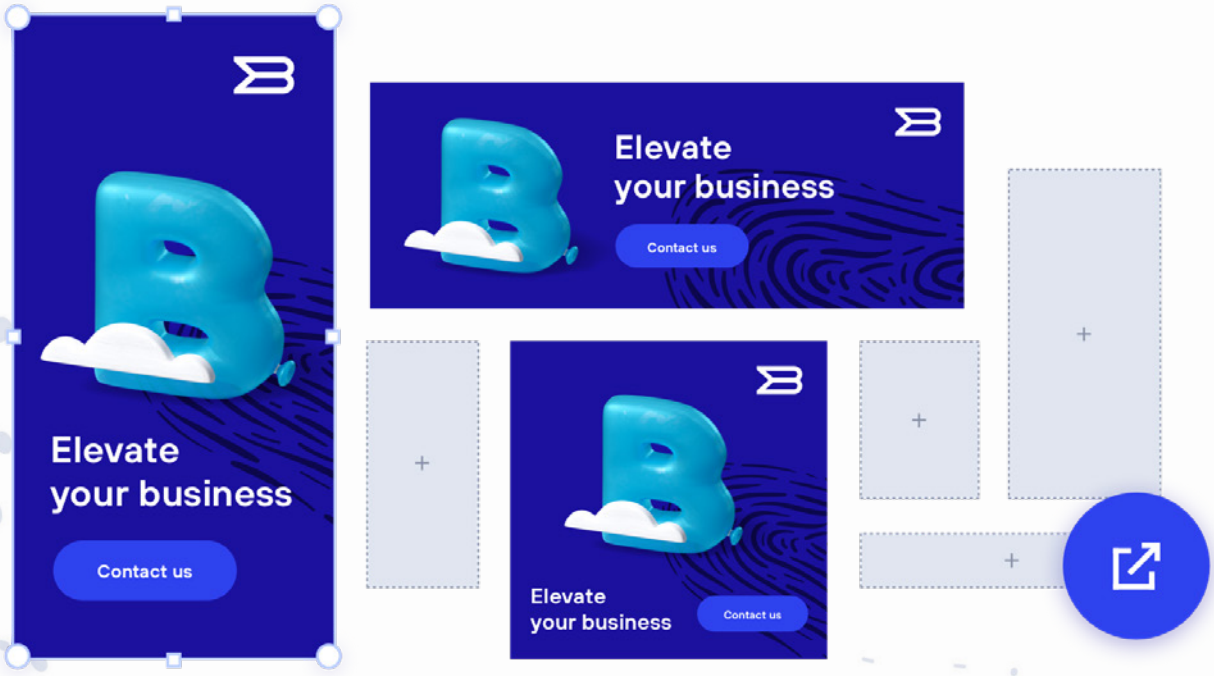
What is Display Ads from BrandMaster?

Display advertising, banner ads or dynamic content is content that changes and is presented based on the reader's preference, demographic and personal information. The content can be tailored to the consumer's specific needs and can, for example, be displayed on your website, landing pages or in emails.

You see it daily in online stores or social networks when the website recommends products based on your browsing or purchase history, but Display Ads can do so much more. For starters, it can be broadcasted on external channels.

Now, if you have any experience with creating this type of content, you know how time consuming it is. And unless you are a skilled designer, you need external help.

When distributing ads across a variety of channels, your design team or agency often have to create custom designs, spending valuable time on cropping and adapting, rather than focusing on the message that needs to be delivered. Also, it is nearly impossible to change the ad content once they have been distributed, which, when you think about it, is a pretty bad strategy



Say goodbye to bad strategies

With Display Ads, your designers will create ads in multiple formats at the same time, and the ads will be updated automatically as soon as you make changes. This way, you can easily spot if the design is not working for certain formats and correct it. As a result, your team can spend more time creating great, customized content and less time on the nitty gritty stuff.

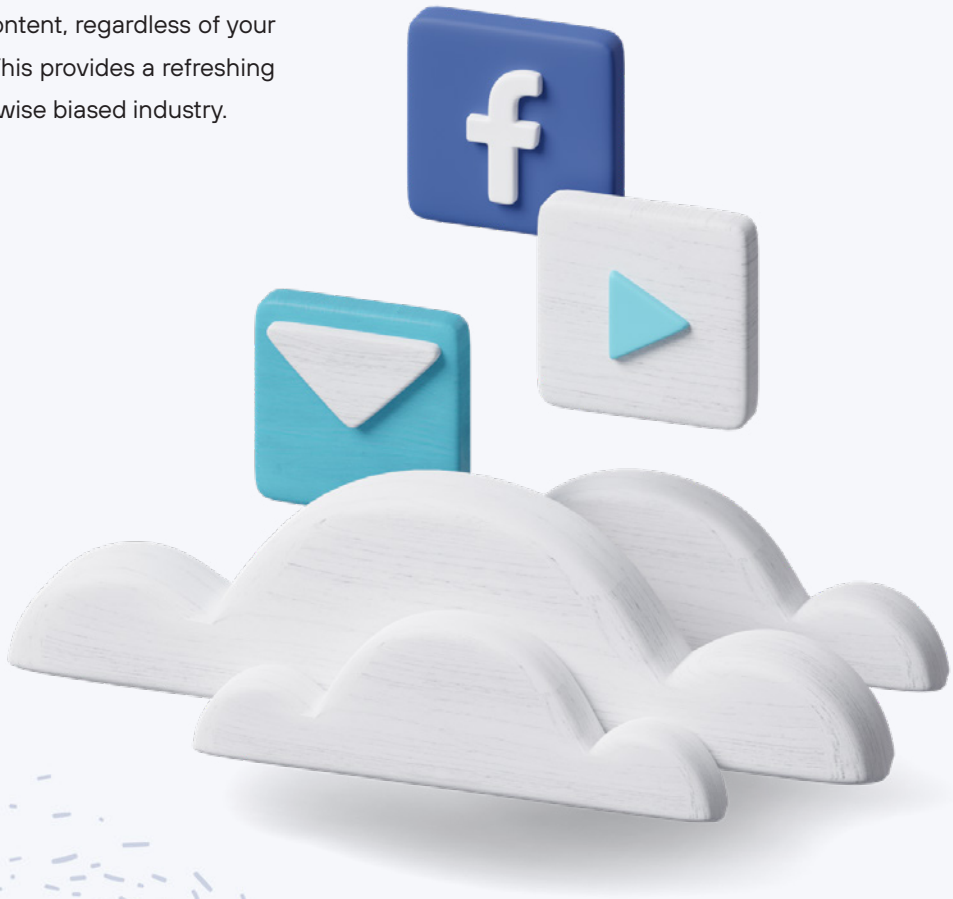
Independent tool

Compared to 3rd party agencies or media houses. this application is delivered independently, and the only motive is to give you the tools you need to create great relevant content, regardless of your chosen media channels. This provides a refreshing independence in an otherwise biased industry.

Cloud-based service

Creating banner ads happens in the cloud. When your creative team has finalized the ad content, they deliver a living file to the distribution company. This is the key to always providing relevant and updated information. The same goes for your website, where cookies track consumer behavior and use that information to send relevant emails and offers.

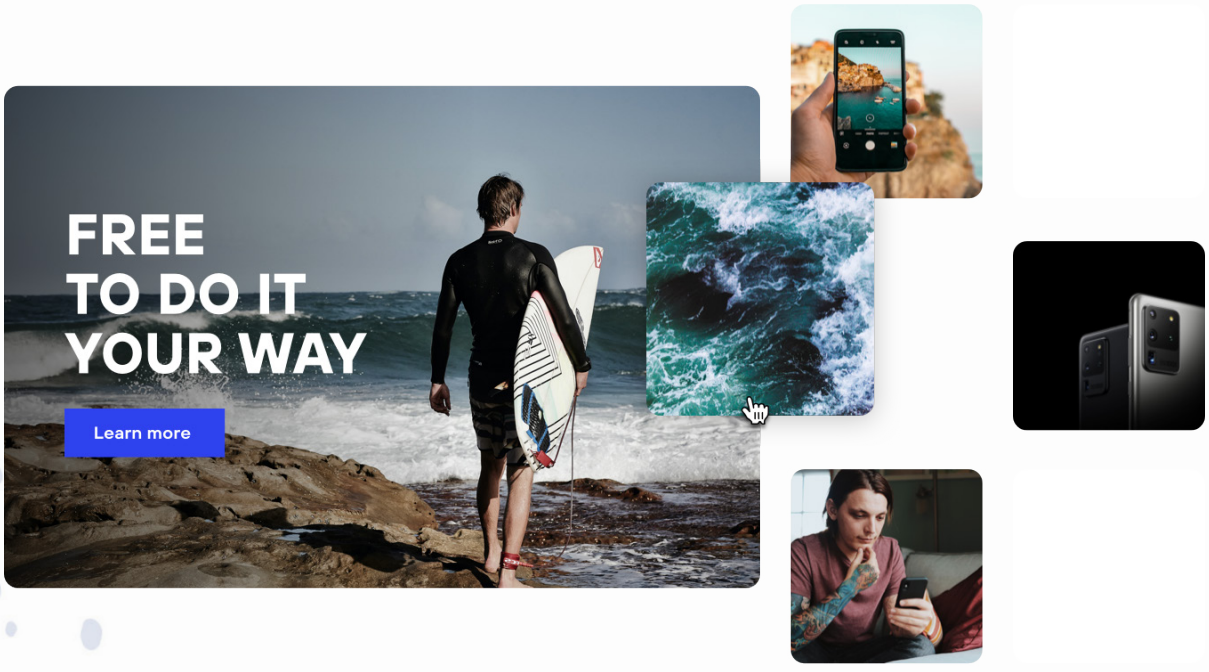
The right digital infrastructure and workflows must be in place for Display Ad creation to be cost effective, and your team must be comfortable working in the cloud.



Put your brand in front of a receptive audience

Display Ads enhances the distribution of relevant information. The ability to edit content across channels in a matter of minutes will enable your brand to keep up with current events, respond to social phenomenas, interact with the consumers, create engagement and counter competing offers.

Using Display Ads to deliver relevant content to your prospective consumers will give you a huge competitive advantage in today's market, but this kind of marketing strategy will not be optional for long. The consumers will come to expect it, and if marketers do not deliver, even more people will turn to ad-block or develop banner blindness.



Common misconceptions

A tool like this is still relatively new to the advertising industry, and there are a few common misconceptions that should be straightened out immediately.

This technology will replace designers

Designers will in no way become redundant. Instead, they will spend more time creating amazing designs and content, and less time editing templates and existing designs.

The software is too expensive

You know what is expensive – pushing out irrelevant ads to an undefined target group suffering from banner blindness. It is also extremely time consuming to redo a whole ad campaign because you discover a typo, or you figure out the image does not work. This quickly accumulates to an unnecessary high media spending.

Social relevancy will hurt my brand

Not every company believes it is necessary to take advantage of current news; it depends on the type of business. However, if you decide in favor of it, your company is bound to benefit from it, provided you do it the right way.

Naturally, there is the right way and wrong way to respond to social happenings. Do not be afraid to explore different approaches and consult professionals. The advantage of Display Ads is that you can change the content in a heartbeat if the campaign falls flat on its face. Standing passively on the sideline falls under the "wrong way" of marketing.

Why you need Display Ads

Let us get one thing straight: relevant advertising and consumer behavior information is important regardless of the type of business you are in or the size of your company. As we mentioned before, consumers are already expecting it, and the demand will only grow in the future.

The only thing that is up for debate is whether you need Display Ads software.

What's in it for you?

What exactly can you achieve with the help of Display Ads? The point of any marketing strategy should always be to increase sales, but when you commit to banner ad creations, you achieve some additional benefits.

Relevancy

We have already briefly touched upon the importance of staying relevant, but let us make it abundantly clear: Pushing out ads to an undefined target group is not only annoying for the recipients, it is also a total waste of your time and resources.

That is not to say you will not still catch an occasional fish despite being completely indifferent to important factors like the tide and choice of bait, but that will not be the case in the future.

Consumers are exposed to hundreds of commercials every day, and their behavior and expectations can change drastically in a short amount of time. You will only be able to catch the consumer's attention by delivering relevant content, and Display Ads is a solution to that.

Brand protection

We often make use of several marketing channels at the same time to reach our prospective clients. With so many ads going live at once, it is hard to keep track of how they are displayed across different channels.

An ad which has the picture and text half out of the frame looks neither enticing nor safe. Sloppy ads are likely to damage the credibility of your brand. Display Ads let you regain control over your live ads and ensure brand safety.

Social engagement

Since the birth of social media, the gap between brand and consumers has greatly shrunk. Satisfied and engaged consumers happily promote their favorite brands by posting pictures and reviews, while dissatisfied people write openhearted status updates.

Social engagement is invaluable. It can provide you with vital consumer information and build customer loyalty, thus strengthening the reputation of your brand. But be aware, not every breaking news story is appropriate for marketing purposes. Consider first how you will contribute to the lives of your target audience and make sure your team is competent enough to handle social media. In the (un)likely event of your social media involvement going horribly wrong, you can regain control by quickly changing the content with Display Ads.

Display Ads, a game changer

First of all, information about consumers and where and how to reach them should be part of your marketing plan. Know your audience, know how they engage and where to reach them. And be prepared for fast market changes.

The content creator will become a technologist

Company roles will have to change. Content creators have to adopt a technological approach. They must stay updated on the newest marketing technology, be able to analyze the performance of different campaigns and take action accordingly.

Change in marketing planning

A static marketing plan set for the whole year will not be sufficient when working with dynamic content. Your marketing team must be able to act fast and adapt to changes in the market. Consequently, the marketing plan must be operational and accessible at all times.

Remember that, as technology advances, you will gain access to masses of information about prospective clients. Using this information strategically is crucial to achieve success.

What will this look like in practice? Firstly, you will no longer allocate resources to specific marketing channels for a whole year. You will still set overall goals and keep a budget as before, but you will also be ready to change channels and strategies in real time, to adapt to changes in consumer behavior and technology.

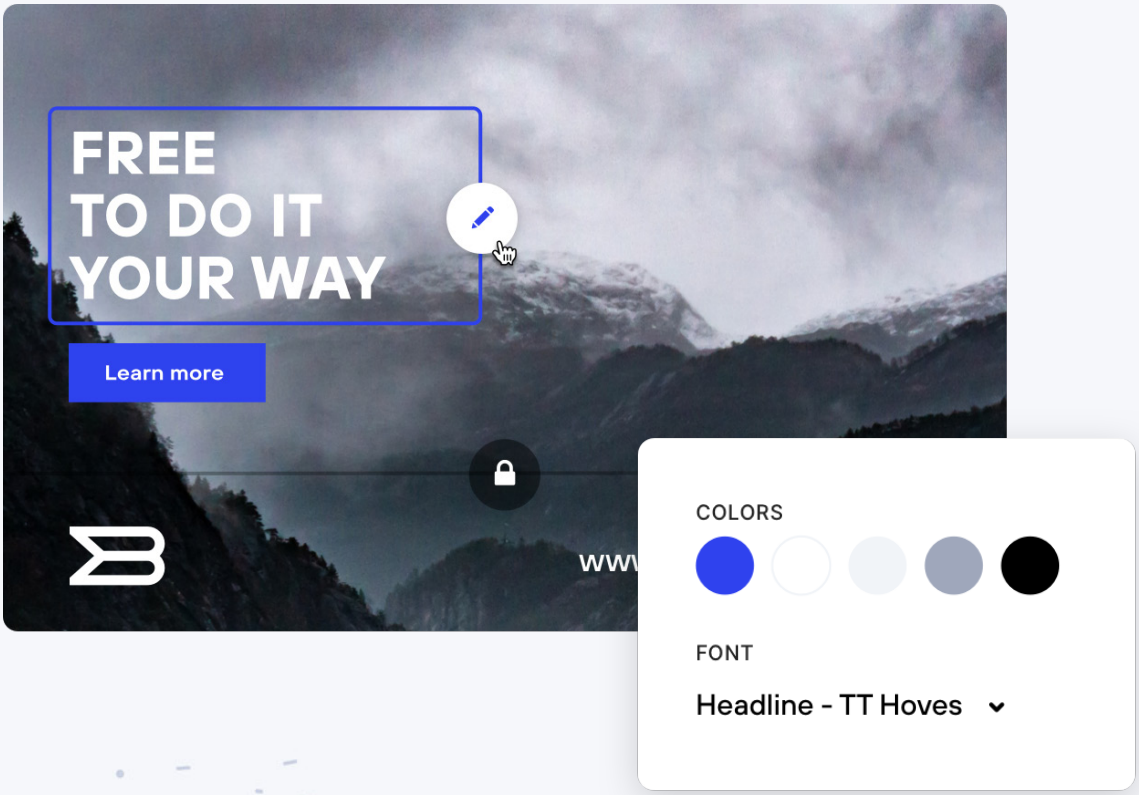
Secondly, you will have to set up a system that lets you update the marketing plan in the cloud and make it accessible for your team. You should be able to keep track of how your campaigns are performing in real time so that you can make adjustments accordingly.

In-house competence is within reach

The company as a whole will no longer work in separate silos with separate goals. You will instead strive to create a culture of teamwork and knowledge exchange by gathering all the employees with relevant competence in the same room as marketing.

For example, it is crucial that the web developers know what marketing wants to achieve with a new web page, and it is equally important that marketing gains insight into the developing process.

It is not enough to simply create content; you must possess the information concerning conversion tracking, analysis, social media and distribution channels. The time of out-sourcing is over, the future is in-house.



Get started with Display Ads

Before you get started with your Display Ads tools you should do an assessment of your company and brand needs and map up everything to make sure your Display Ads application delivery according your needs from day one.

1. **What type of dynamic content is needed?**
2. **What channels requires your presence?**
3. **How often do you plan to publish new content?**
4. **Do you have a design team who can create the creatives or do you need an external agency?**
5. **What type of ad formats do you need?**
6. **What type of content and assets is required?**

Once you have mapped out your needs it will be easier to map up the template needs and get a establish a tool that will advance your content creation from day one.

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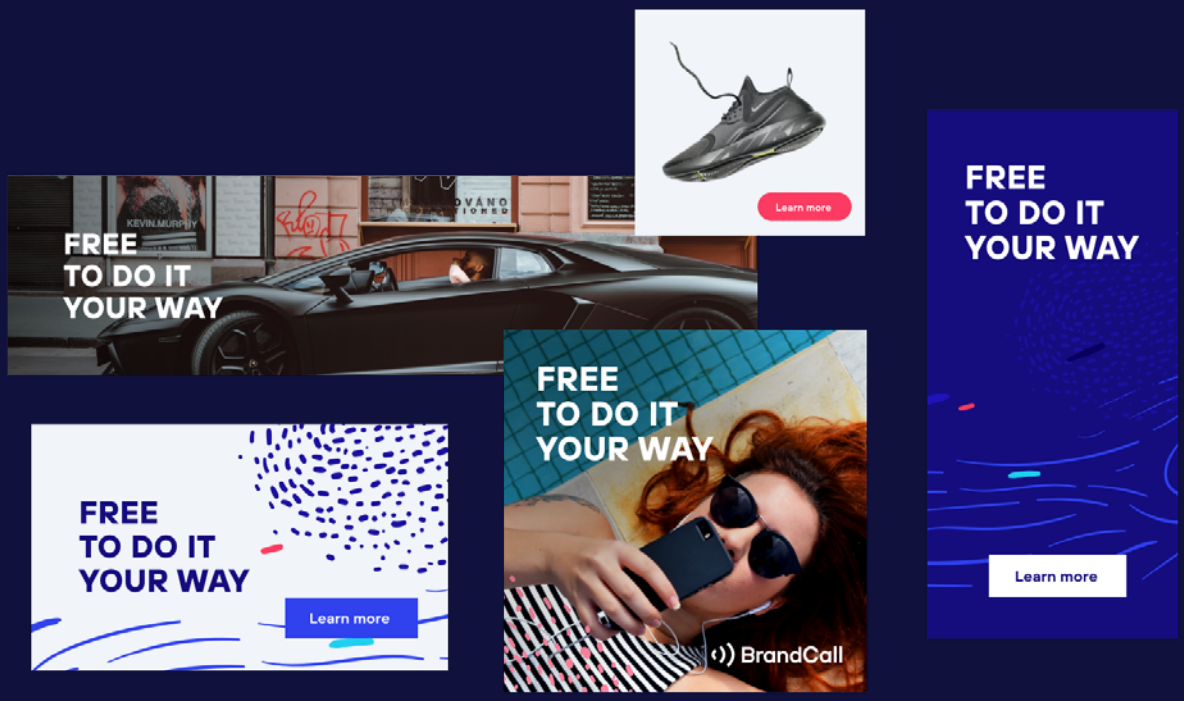

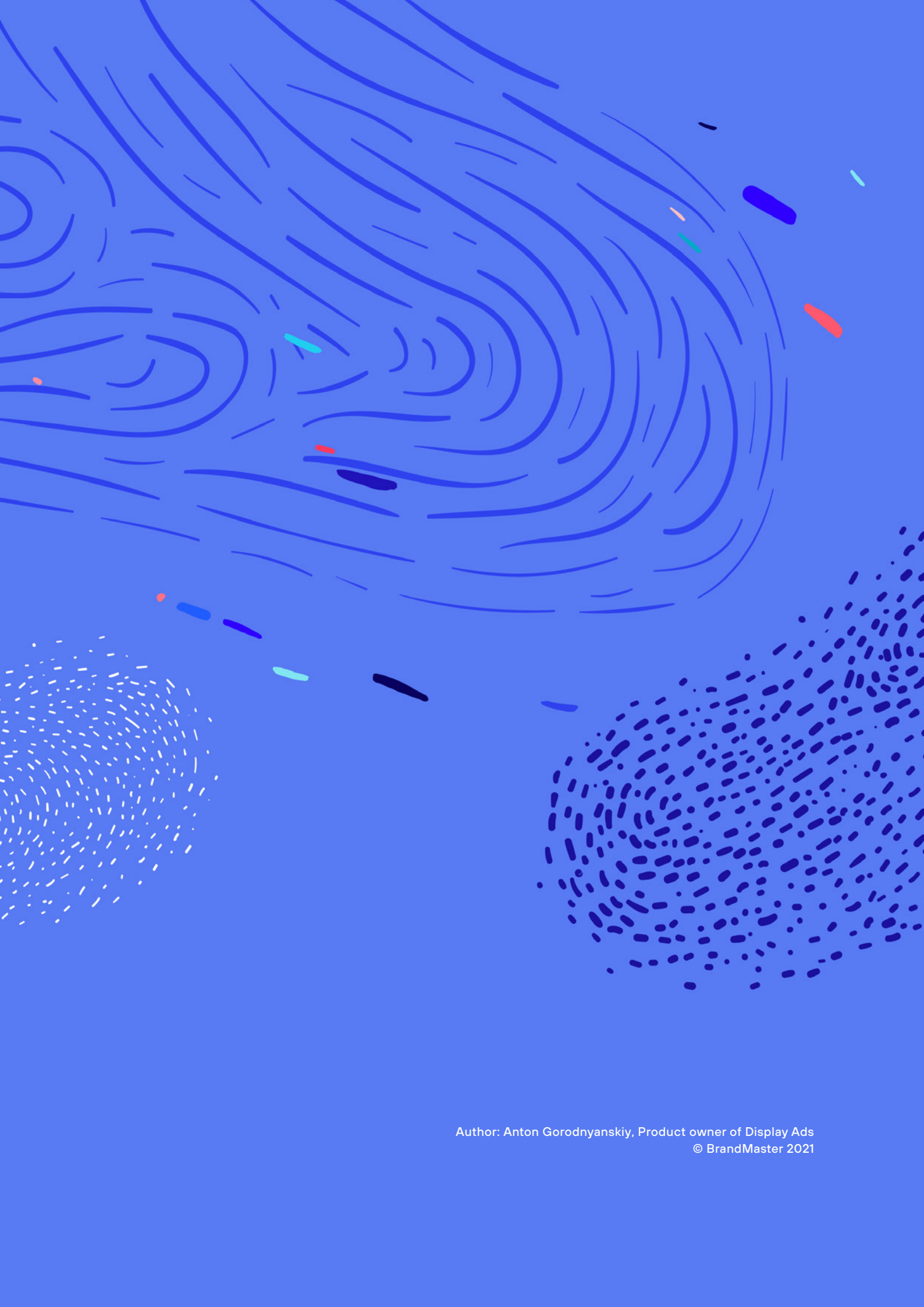

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Graphic Design and Print Production Fundamentals

Graphic Design and Print Production Fundamentals

Graphic Communications Open Textbook Collective

Wayne Collins, Alex Hass, Ken Jeffery, Alan Martin, Roberto Medeiros, and Steve Tomljanovic

BCCAMPUS
VICTORIA, B.C., CANADA

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Introduction

Ken Jeffery

Figure I.1 Car graphics are an example of modern day print design

On any given day, you can look around your surroundings and come in contact with print design. Information comes to you in many forms: the graphics on the front of a cereal box, or on the packaging in your cupboards; the information on the billboards and bus shelter posters you pass on your way to work; the graphics on the outside of the cup that holds your double latte; and the printed numbers on the dial of the speedometer in your car. Information is communicated by the numbers on the buttons in an elevator; on the signage hanging in stores; or on the amusing graphics on the front of your friend's T-shirt. So many items in your life hold an image that is created to convey information. And all of these things are designed by someone.

Figure I.2 Times Square has many examples of print design

Traditionally referred to as graphic design, communication design is the process by which messages and images are used to convey information to a targeted audience. It is within this spectrum that this textbook will address the many steps of creating and then producing physical, printed, or other

imaged products that people interact with on a daily basis. Design itself is only the first step. It is important when conceiving of a new design that the entire workflow through to production is taken into consideration. And while most modern graphic design is created on computers, using design software such as the Adobe suite of products, the ideas and concepts don't stay on the computer. To create in-store signage, for instance, the ideas need to be completed in the computer software, then progress to an imaging (traditionally referred to as printing) process. This is a very wide-reaching and varied group of disciplines. By inviting a group of select experts to author the chapters of this textbook, our goal is to specifically focus on different aspects of the design process, from creation to production.

Each chapter begins with a list of Learning Objectives, and concludes with Exercises and a list of Suggested Readings on the Summary page. Throughout, key terms are noted in bold and listed again in a Glossary at the end of the book.

In Chapter 1, we start with some history. By examining the history of design, we are able to be inspired by, and learn from, those who have worked before us. Graphic design has a very rich and interesting heritage, with inspirations drawn from schools and movements such as the Werkbund, Bauhaus, Dada, International Typographic Style (ITS), as well as other influences still seen in the designs of today.

Figure I.3 Johannes Itten was a designer associated with the Bauhaus school

We now work in an age where the computer has had an influence on the era of Post Modernism. Is this a new age? Are we ushering in an era unseen before? Or are modern-day designs simply a retelling of the same tropes we have seen for hundreds of years?

Chapter 2 follows with a discussion about the design process. Contrary to what we tend to see in popular television shows and movies where advertising executives are struck with instant, usable, and bold ideas, design strategies are seldom insights gained through such a sudden outburst of inspiration. The design process is a deliberate, constructive, and prescriptive process that is guided by specific strategies. For example, before any piece of designed communication can be started, some very detailed research needs to be performed. This happens well before any graphic design or layout software is opened on a computer. Designing is a form of problem solving, where a system is created to communicate a specific

and targeted message. The design process is the way that a designer breaks the problem into discrete creative activities. First is an exploration of what is trying to be achieved. Facts are gathered about the problem, and the problem itself is often defined very specifically. The idea phase is where brainstorming and ideation occurs, often without judgment, as a way to gather as many different ideas and directions as possible. From this, solutions are evaluated, both for their perceived impact on the target audience and for their perceived effectiveness in portraying the desired message. Finally, all of this information is distilled into an accepted solution. Designers do not sit around waiting for ideas to just happen; they follow a process in order to make it happen.

Figure I.4 The golden ratio is a constant that appears in nature

Chapter 3 presents the most important and necessary design elements required for effective graphic layout and design. When designing a layout, the designer cannot just ‘throw’ all of the information onto the page. Design is a thoughtful process that makes use of many different skills to create a design that is both appealing and legible. We discuss the grid in its many forms, including different types of grid such as the ITS grid, the golden ratio, and even strategies for using no grid at all. Space is an important design element, with different items on the page requiring more or less area to be effective. We also talk about the density, or ‘colour’ of type on the page, along with a number of different typographical conventions for making the most of the collection of words on the layout.

In Chapter 4, we begin to move along in the production process and discuss some of the more physical attributes of design. And one of the most important topics in creating printed products is that of colour. It is a complex part of the design process, affecting how an image is transmitted to the eye, how the colours are perceived, and what makes one thing look different from another, even if it is the same colour. Have you ever printed something on your home printer only to be disappointed that it doesn’t look like it did on your computer screen? Highly detailed systems of colour management are put in place to mitigate these differences.

As we proceed toward creating printed output, Chapter 5 is where it all starts to come together. In the print process, this stage is called prepress. Prepress is where all the design work is translated from a file on the computer in front of you into a form that can be ‘printed’ onto a given surface. Imagine the requirements for creating not just one copy of a design, but thousands! This is a very important step, and if mistakes or production hurdles are not discovered and overcome at this step, then the project can end up being very costly for all parties involved, from the designer, to the printer, to the client. This chapter deals with topics such as preflight, imposition, separations, platemaking, and considerations for other print and finishing processes.

Chapter 6 is a comprehensive look at how all of this design work will result in a finished product. The many ways that a design can be printed are varied and complex, but having some knowledge about how the print process works will help to create a more successful project. Is it going to be printed on a box, or on a billboard? How many copies are needed: one or one million? These and many more decisions influence how a product will be produced. This chapter outlines some of the more popular printing technologies, along with industry standard procedures for working with them. Suggestions for choosing the right paper (or other types of **substrates**) are also made along with best practices for working with colour on the printed page.

Chapter 7 rounds out this textbook with a look at online technologies and how they affect, and are affected by, the printed word. We examine online web-to-print solutions and their contribution to bridging the process from graphic design to printed work. We also highlight other considerations such as branding and digital file resolution strategies. As the world has moved into an Internet-connected, always-on compendium of information, print remains a vital, relevant, and important part of the media mix. Effective communication campaigns make the most of *all* opportunities that media design and, in particular, print design can offer.

The goal of this text is to bridge the disciplines of communication design and print production to form a concise, accessible compendium outlining the design process in this modern, computer-driven age. While it is common, or perhaps easy, to surmise that graphic design is solely a computer-driven pursuit, when we take a step back, and look at the entire process, we see that computer-aided design is only one part of a larger picture. And by including this larger domain in our studies, we can truly gain an appreciation for the influences and strategies needed to be successful in this field.

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Chapter 1. Design History

1.1 Introduction

Alex Hass

Learning Objectives

- Identify the unique attributes of major modern graphic design styles, beginning with William Morris. The design styles discussed will be those that have a presence or an influence in our current visual culture:
    - Morris
    - Werkbund
    - Bauhaus
    - Dada
    - International Typographic Style (ITS)
    - Late Modern
    - Post Modern
- Evaluate the influence of past design styles on one another
- Explain the influence of culture on major modern graphic design styles
- Identify the cross-cultural influences of visual culture that impacted graphic design style
- Identify the technological influences that affected and advanced graphic design

Industrial Revolution Overview

The Craftsman

Before the Industrial Revolution (1760-1840 in Britain) most aspects of design and all aspects of production were commonly united in the person of the craftsman. The tailor, mason, cobbler, potter, brewer, and any other kind of craftsman integrated their personal design aesthetic into each stage of product development. In print, this meant that the printer designed the fonts, the page size, and the layout of the book or broadsheet; the printer chose (even at times made) the paper and ran the press and bindery. Unity of design was implicit.

Typography in this pre-industrial era was predominantly used for books and broadsheets. The visual flavour of the fonts was based on the historic styles of western cultural tradition — roman, black letter, italic, and grotesque fonts were the mainstay of the industry. Typography was naturally small scale

— needed only for sheets and pages — and was only large when it was chiseled into buildings and monuments.

Technological Shift

The Industrial Revolution radically changed the structure of society, socially and economically, by moving vast numbers of the population from agrarian-based subsistence living to cities where manufacturing anchored and dominated employment and wealth. Agrarian-based society was tied to an aristocracy overseeing the land and controlling and directing production through the use of human labour. In contrast, urban production, though still very much in need of human labour (female and child labour in particular was in huge demand), was dominated by the mechanized production of goods, directed and controlled by industrialists instead of the aristocracy. The factories were powered initially by steam, and eventually by gasoline and electricity. These new manufacturing models were dominated by an engineering mentality that valued optimization of mechanical processes for high yields and introduced a compartmentalized approach to production.

Design and Production Separate

The design process was separated from the production-based process for a number of reasons. Primary was the efficiency-oriented mindset of the manufacturers who were focused on creating products with low unit costs and high yield outcomes, rather than on pleasing aesthetics or high-quality materials. Design process is time consuming and was considered unnecessary for each production stage of manufactured goods.

Manufactured products were intended for the working and middle classes, and high-quality output was not a goal. These products were never intended to vie for the attention of the upper classes — enticing them away from the services and bespoke products of the craftsman (a contemporary example is Tip Top Tailors attracting Savile Row customers). Rather, they supplied common people with goods they had not been able to afford before. This efficient line of thinking created the still existing equation of minimal design plus low material integrity equalling low-cost products.

Design, rather than being a part of each step of production (implicit in the craftsman's approach), was added for form development and when a product needed more appeal for the masses — usually during the later stages of production through decorative additions. Design was now directed by the parameters and constraints of the manufacturing process and its needs.

Advertising Emerges

Despite low product standards, the high quantities and low costs of manufactured goods "stimulated a mass market and even greater demand" (Meggs & Purvis, 2011, p. 127). The historic role of graphic design for broadsheets and books expanded at this point to include advertising. Each company and product needed exposure to sell these manufactured products to the mass market — no earlier method of promotion could communicate to this number of people.

The design aesthetic of these times was relatively untouched by stylistic cohesion or design philosophy. Industrialists used a pastiche of historic styles that aspired to make their products look more upscale,

but did not go as far as to create a new visual language. This was a strategy that made sense and has since been repeated (consider early computer design aesthetics). Usually, when a new medium or communication strategy is developed (advertising in print and the posters of the Industrial Revolution), it uses visual and language styles that people are already familiar with, and introduces a new way to deliver the message. Too much change alienates, but novelty of delivery works by adding a twist on the shoulders of an already familiar form.

Font Explosion

In addition to its new role in promoting products to the mass market, graphic design moved forward with an explosion of new font designs as well as new production methods. The design of fonts had earlier been linked to the pragmatic and cultural objectives of producing books and broadsheets. With large format posters and numerous other print components, text needed to do much more than represent a phonetic symbol. Innovations in production affected — perhaps infected — printers with the pioneer spirit of the times, and all products and their potential were examined and re-evaluated. This attitude naturally included the function and design of fonts and the methods used to reproduce them. Text was often the only material used to promote its subject and became integral to a visual communication. Jobbing printers who used either letterpress or lithographic presses pushed the boundaries of both, competing with each other by introducing innovations and, in turn, pushing artists and type foundries to create more products they could use. An entirely new font category, slab serif — sometimes called Egyptian — was created. Thousands of new fonts emerged to meet the demand of the marketplace.

Photography

In addition to font development, the Industrial Age also contributed the photograph and ultimately its use in books and advertising. Photography (for print design) was originally used as a research tool in developing engravings, but this was costly and time consuming. Numerous inventors searched for ways to integrate photography into the press process since the early years of its development in the 1830s. Photo engraving eventually arrived in 1871 using negatives and plates. From that time forward, photography has been used to conceptually and contextually support the communication of graphic design in its many forms.

1.2 William Morris and the Arts & Crafts Movement

Alex Hass

Conditions and Products of the Industrial Age

The Arts & Crafts movement emerged in the second half of the 19th century in reaction to the social, moral, and aesthetic chaos created by the Industrial Revolution. William Morris was its founder and leader. He abhorred the cheap and cheerful products of manufacturing, the terrible working and living conditions of the poor, and the lack of guiding moral principles of the times. Morris "called for a fitness of purpose, truth to the nature of the materials and methods of production, and individual expression by both artist and worker" (Meggs & Purvis, 2011, p. 160). These philosophical points are still pivotal to the expression of design style and practice to this day. Design styles from the Arts & Crafts movement and on have emphasized, in varying degrees, either fitness of purpose and material integrity, or individual expression and the need for visual subjectivity. Morris based his philosophy on the writings of John Ruskin, a critic of the Industrial Age, and a man who felt that society should work toward promoting the happiness and well-being of every one of its members, by creating a union of art and labour in the service of society. Ruskin admired the medieval Gothic style for these qualities, as well as the Italian aesthetic of medieval art because of its direct and uncomplicated depiction of nature.

Many artists, architects, and designers were attracted to Ruskin's philosophy and began to integrate components of them into their work. Morris, influenced by his upbringing in an agrarian countryside, was profoundly moved by Ruskin's stance on fusing work and creativity, and became determined to find a way to make it a reality for society. This path became his life's work.

Pre-Raphealite Brotherhood

Morris met Edward Burne-Jones at Exeter College when both were studying there. They both read extensively the medieval history, chronicles, and poetry available to them and wrote every day. Morris published his first volume of poetry when he was 24, and continued to write and publish for the rest of his life. After graduation, Morris and Burne-Jones tried a few occupations, and eventually decided to become artists. Both became followers of Dante Gabriel Rossetti who founded the Pre-Raphealite brotherhood that was based on many of Ruskin's principles. Morris did not last long as a painter, eventually finding his design vocation while creating a home for himself and his new wife (Rosetti's muse and model).

Discovering the lack of design integrity in Victorian home furnishings and various additional deficiencies in other aspects of home products, he chose to not only design his home, but all its furniture, tapestries, and stained glass.

Morris & Co.

In 1860, Morris established an interior design firm with friends based on the knowledge and experiences he had in crafting and building his home. He began transforming not only the look of home interiors but also the design studio. He brought together craftsmen of all kinds under the umbrella of his studio and began to implement Ruskin's philosophy of combining art and craft. In Morris's case, this was focused on making beautiful objects for the home. The craftsmen were encouraged to study principles of art and design, not just production, so they could reintegrate design principles into the production of their products. The objects they created were made and designed with an integrity a craftsman could feel proud of and find joy in creating, while the eventual owner would consider these products on par with works of art (an existing example is the Morris chair). The look of the work coming out of the Morris studio was based specifically on an English medieval aesthetic that the British public could connect to. The English look and its integrity of production made Morris's work very successful and sought after. His organizational innovations and principled approach gained attention with craftsmen and artisans, and became a model for a number of craft guilds and art societies, which eventually changed the British design landscape.

William Morris and the Kelmscott Press

Morris's interest in writing never waned and made him acutely aware of how the book publishing industry had been negatively affected by industrialization. One of his many pursuits included the revitalization of the book form and its design components through the establishment of the Kelmscott Press. The press was created in 1888 after Morris, inspired by a lecture about medieval manuscripts and incunabula publications, began the design of his first font, Golden, which was based on the Venetian roman face created originally by Nicolas Jenson.

In his reinterpretation of this earlier font, Morris strove to optimize readability while retaining aesthetic integrity — in the process reviving interest in font design of earlier periods. Morris used this font in his first book, *The Story of Glittering Plain*, which he illustrated, printed, and bound at his press. The design approach of this publication and all others Kelmscott produced in its eight years was based on recreating the integrated approach and beauty of the incunabula books and manuscripts of the medieval period. All aspects of the publication were considered and carefully determined to create a cohesive whole. The press itself used hand-operated machinery, the paper was handmade, and the illustrations, fonts, and page design were all created and unified by the same person to make the book a cohesive, beautiful object of design. Morris did not wholly reject mechanization, however, as he recognized the advantages of mechanical process. He considered, redesigned, and improved all aspects of design and production to increase physical and aesthetic quality.

Kelmscott Press produced over 18,000 volumes in the eight years of its existence and inspired a revival of book design on two continents. In addition, Morris inspired a reinterpretation of design and design practice with his steadfast commitment to Ruskin's principles. Future generations of designers held to Morris's goals of material integrity — striving for beautiful utilitarian object design and carefully considered functionality.

1.3 Deutscher Werkbund

Alex Hass

In the early years of the 20th century, the German Hermann Muthesius returned to Germany from England with Morris's Arts & Crafts concepts. Muthesius published the *The English House* in 1905, a book wholly devoted to the positive outcomes of the English Arts & Crafts movement. Muthesius was a sometime cultural ambassador, possibly an industrial spy, for Germany in England. His interest in the Arts & Crafts movement was not based on returning German culture to the romantic values of an earlier pre-manufacturing era. He was focused on infusing the machine-made products of Germany with high-quality design and material integrity. Muthesius believed manufacturing was here to stay. He was one of the original members of the state-sponsored Deutscher Werkbund — an association that promoted the union of art and technology. The Werkbund integrated traditional crafts and industrial mass-production techniques, and put Germany on a competitive footing with England and the United States. Its motto "Vom Sofakissen zum Städtebau" (from sofa cushions to city-building) reveals its range.

Design Embraces the Manufacturing Process

Peter Behrens and Henry van de Velde were also part of the original leadership, and with Muthesius developed the philosophy of *Gesamtkultur* — a cohesive cultural vision where design was the driving force of a wholly fresh, man-made environment. Every aspect of the culture and its products was examined and redefined for maximum use of mechanization in its production. The new visual language of *Gesamtkultur* was a style stripped of ornament in favour of simplicity and function. All areas of cultural production were affected by this new philosophy — graphic design, architecture, industrial design, textiles, and so forth — and all were reconfigured and optimized. Sans serif fonts dominated the reductive graphic design style as did standardization of sizes and forms in architecture and industrial design. Optimization of materials and mechanical processes affected every area. Germany embraced this new philosophy and visual style for its simplicity and exactness. In 1919, Walter Gropius, a modernist architect whose work was inspired by Werkbund ideals, was finally successful in opening a school he called the Bauhaus (in Weimar where artists, industrialists, and technicians would develop their products in collaboration). These products would then build a new future for German exports by virtue of their high level of functional utility and beauty.

1.4 Bauhaus

Alex Hass

The Bauhaus philosophy has become famous for its integrated approach to design education; “it precipitated a revolution in art education whose influence is still felt today” (Whitford, 1995, p. 10). Most art colleges and universities still base much of their foundational curriculum on its fundamental ideas.

The Bauhaus school was founded with the idea of creating a ‘total’ work of art in which all arts, including architecture, would eventually be brought together. The first iteration of the school brought together instructors from all over Europe working within the latest art and design styles, manufacturing ideologies, and technologies. An example of this new teaching style can be found in its first-year curriculum. This foundation year exposed all students to the basic elements and principles of design and colour theory, and experimented with a range of materials and processes. This allowed every student the scope to create projects within any discipline rather than focus solely on a specialty. This approach to design education became a common feature of architectural and design schools in many countries.

In addition to its influence on art and design education, the Bauhaus style was to become a profound influence upon subsequent developments and practices in art, architecture, graphic design, interior design, industrial design, and typography.

The school itself had three iterations in its 14-year run. With each iteration, the core concepts and romantic ideals were modified and watered down to work within the realities of the difficult Nazi culture. When the school was finally closed by its own leadership under pressure from the Nazi-led government, most of the faculty left the country to teach in less difficult circumstances and continued to spread Bauhaus precepts all over the world. Many of its artists and intellectuals fled to the United States. Because the Bauhaus approach was so innovative and invigorating, the institutions that were exposed to the Bauhaus methodology embraced its principles. This is why the Bauhaus had a major impact on art and architecture trends in Western Europe, the United States, and Canada.

Later evaluation of the Bauhaus design philosophy was critical of its bias against the organic markings of a human element, an acknowledgment of “… the dated, unattractive aspects of the Bauhaus as a projection of utopia marked by mechanistic views of human nature” (Schjeldahl, 2009, para. 6). And as Ernst Kállai proposed in the magazine *Die Weltbühne* in 1930, “Home hygiene without home atmosphere” (as cited in Bergdoll & Dickerman, 2009, p. 41).

The very machine-oriented and unadorned aesthetic of the Bauhaus refined and evolved, eventually informing the clean, idealistic, and rigorous design approach of the International Typographic Style.

1.5 Dada

Alex Hass

> Dada does not mean anything. We read in the papers that the Negroes of the Kroo race call the tail of the sacred cow: dada. A cube, and a mother, in certain regions of Italy, are called: Dada. The word for a hobby-horse, a children's nurse, a double affirmative in Russian and Rumanian, is also: Dada. (Tzara, 1992)
> – Tristan Tzara, *Dada Manifesto*

Dada was an artistic and literary movement that began in 1916 in Zurich, Switzerland. It arose as a reaction to World War I, and the nationalism and rationalism, which many thought had brought war about. Influenced by ideas and innovations from several early avant-gardes — Cubism, Futurism, Constructivism, and Expressionism — its influence in the arts was incredibly diverse, ranging from performance art to poetry, sculpture, and painting, to photography and photographic and painterly collage.

Dada's aesthetic, marked by its mockery of materialistic and nationalistic attitudes, became a powerful inspiration for artists and designers in many cities, including Berlin, Paris, and New York, all of which generated their own groups. The movement radically changed typographic ideals and created fresh approaches to text. Unburdened of its rules and conventions, type was allowed to become expressive and subjective. The poetic output of the group was fresh and different, and needed its typography to be as expressive and innovative as its content. Dada, in combination with aspects of Constructivist and Suprematist typography, balanced the cultural discipline created and applied to typography by other streams of contemporary design like the Bauhaus. This movement in particular advanced typography as a medium of its own. It promoted the use of typography as an art material that could be manipulated by artists and designers expressively and without preordained rules and structural principles.

> Words emerge, shoulders of words, legs, arms, hands of words. Au, oi, uh. One shouldn't let too many words out. A line of poetry is a chance to get rid of all the filth that clings to this accursed language, as if put there by stockbrokers' hands, hands worn smooth by coins. I want the word where it ends and begins. Dada is the heart of words. (Ball, 1996)
> – *Hugo Ball's manifesto*, read at Zunfthaus zur Waag on July 14, 1916

1.6 International Typographic Style

Alex Hass

International Typographic Style (ITS), also known as the Swiss Style, emerged in Switzerland and Germany in the 1950s. ITS became known for design that emphasized objective clarity through the use of compositional grids and sans serif typography as the primary design material (or element).

Guiding Principles

ITS was built on the shoulders of the 'less is more' ideal of the German Werkbund and the Bauhaus school. But its pioneers pursued ideologies that had much more depth and subtlety. Ernst Keller, whose work in design spanned over four decades, brought an approach to problem solving that was unique. His contribution to design was in defining the problem. For Keller, the solution to a design problem rested in its content. Content-driven design is now a standard practice. Max Bill, another pioneer, brought a purist approach to design that he had been developing since the 1930s. He was instrumental in forming Germany's Ulm School of Design, famous for its ITS approach. The school introduced Greek rhetorical devices to amplify concept generation and produce greater conceptual work, while the study of semiotics (creating and understanding symbols and the study of sending and receiving visual messages) allowed its design students to understand the parameters of communication in a more scientific and studied way. At this time, there was also a greater interest in visual complexity. Max Huber, a designer known for his excellent manipulation of presses and inks, layered intense colours and composed chaotic compositions while maintaining harmony through the use of complex grids that structured and unified the elements. He was one of many designers who began using grids in strategic ways. ITS design is now known for its use of anchored elements within a mathematical grid. A grid is the "most legible and harmonious means for structuring information" (Meggs & Purvis, 2011, p. 355). Visual composition changed in many ways due to the grid. Design was already moving toward asymmetrical compositions, but now even the design of text blocks changed — from justified text to aligned flush left, ragged right. Fonts chosen for the text changed from serif fonts to sans serif, a type style believed to "express the spirit of a more progressive age" by early designers in the movement. Sans-serif typefaces like Helvetica, Univers, and Akzidenz Grotesk were favoured because they reflected the ideals of a progressive culture more than traditional serif fonts like Times or Garamond. ITS balanced the stabilizing visual qualities of cleanliness, readability, and objectivity with the dynamic use of negative space, asymmetrical composition, and full background photography.

Photography

ITS did not use illustrations and drawings because of their inherent subjectivity. Photography was preferred because of its objective qualities, and was heavily used to balance and organically complement the typography and its structured organizational grid. Often the photograph sat in the background with the type designed to sit within it; the two composed to strengthen each other to create a cohesive whole. ITS refined the presentation of information to allow the content to be understood clearly and cleanly,

without persuading influences of any kind. A strong focus on order and clarity was desirable as design was seen to be a “socially useful and important activity … the designers define their roles not as artists but as objective conduits for spreading important information between components of society” (Meggs & Purvis, 2011, p. 355).

Josef Müller-Brockmann, another one of its pioneers, “sought an absolute and universal form of graphic expression through objective and impersonal presentation, communicating to the audience without the interference of the designer’s subjective feelings or propagandistic techniques of persuasion” (Schneider, 2011). Müller-Brockmann’s posters and design works feature large photographs as objective symbols meant to convey his ideas in particularly clear and powerful ways.

After World War II, international trade began to increase and relations between countries grew steadily stronger. Typography and design were crucial to helping these relationships progress — multiple languages had to be factored into a design. While clarity, objectivity, region-less glyphs, and symbols were essential to communication between international partners, ITS found its niche in this communicative climate and expanded beyond Switzerland, to America.

ITS is still very popular and commonly used for its clarity and functionality. However, there is a fine line between clean and simple, and simply boring. As the style became universal, its visual language became less innovative and was perceived to be too restrictive. Designers wanted the freedom to be expressive, and the culture itself was moving from cultural idealism to celebratory consumerism. ITS can be a very successful design strategy to adopt if there is a strong concept binding all of the design components together, or when there is a vast amount of complexity in the content and a visual hierarchy is needed to calm the design to make it accessible.

1.7 Late Modern | New York Style

Alex Hass

Late Modernism encompasses the period from the end of World War II to the early 21st century. Late Modernism describes a movement that arose from and reacted to trends in ITS and Modernism. The Late Modern period was dominated by American innovations spurred on by America's new-found wealth. The need for more advertising, marketing, and packaging was matched by a new mood in the culture — a mood that was exuberant and playful, not rigid and rule-oriented.

Late Modern was inspired by European avant-garde immigrants. These immigrants found work in design and quickly introduced Americans to early modern principles of an idealistic and theoretical nature. American design at this point had been pragmatic, intuitive, and organic in composition. The fusion of these two methodologies in a highly competitive and creative climate produced design work that was original in concept, witty, and provocative and, as personal expression was highly prized, full of a variety of visual styles. Paul Rand is one of the great innovators of this style. Rand was adept at using ITS when its rules and principles were called for, but he was also very influenced by European art movements of the times. In his work, he fused the two and made works that were accessible, simple, engaging, and witty. His work was inspirational, but his writing and teaching were as important, if not more, to redefining the practice of design. He restructured the design department at Yale and published books on design practice informed by ITS principles, softened by wit, and espoused the value of the organic look of handmade marks. As a result, artists and designers began to merge organic shapes with simple geometry.

The look of graphic design also changed through advancements in photography, typesetting, and printing techniques. Designers felt confident in exploring and experimenting with the new technologies as they were well supported by the expertise of the print industry. Designers began to cut up type and images and compose directly on mechanical boards, which were then photographed and manipulated on the press for colour experimentation. As well, illustration was once again prized. Conceptual typography also became a popular form of expression.

Push Pin Studios

An excellent example of this expansive style can be found in the design output of New York's Push Pin Studios. Formed by Milton Glaser and Seymour Chwast, Push Pin was a studio that created innovative typographic solutions — I♥NY— brand identities, political posters, books, and albums (such Bob Dylan's album *Dylan*). It was adept at using and mixing illustration, photography, collage, and typography for unexpected and innovative visual results that were always fresh and interesting as well as for its excellent conceptual solutions. The influence of Push Pin and Late Modern is still alive and has recently experienced a resurgence. Many young designers have adopted this style because of its fresh colours, fine wit, and spontaneous compositions.

1.8 Post Modern

Alex Hass

By the early 1970s, the idealistic principles of Modernism were fading and felt flat and lifeless. Pluralism was again emerging as people craved variety as a reaction to the reductivist qualities that modernism espoused.

Punk

In the late 1970s in Britain, Australia, and parts of the United States, a youthful rebellious culture of anger and distain arose against the establishment. In many ways, the design language of Punk echoed the Dadaist style, though Punk was anchored with a pointed, political message against the tyranny of society and the disenfranchisement of youth. A use of aggressive collages, colours, and experimental photography were its hallmarks. These free-form, spontaneous design works incorporated pithy tag lines and seethed with anger in a way that Dada work never attempted to achieve. Punk actively moved away from the conformities of design, and was anti-patriotic and anti-establishment. Punk established the do-it-yourself (DIY) ethos and stylized it with the angry anti-establishment mood of the mid 1970s, a time of political and social turbulence. DIY style was considered shocking and uncontrolled. However, the influence on design has been far reaching and subsequently widely emulated.

Jamie Reid, a pioneer of the Punk style, developed the visual signature look for the Sex Pistols and many other punk bands. His personal signature style was known for a collaged ‘ransom note’ typography that became a typographic style of its own. Reid cut letters out of newspapers and magazines, and collaged them together to be photographed. By doing this, he could see what he was creating as he went along, trying out different font styles and sizes and seeing the results instantly. Treating type as if it were a photograph also freed him from the restrictions of typesetting within a structured grid and allowed him to develop his ideas and concepts as he created. This unguided, process-free approach to design became a part of the Post Modern experimentation that was to come.

When Punk first exploded in the 1970s, it was deemed a youthful rebellion. In actuality, it was one of the many forms of visual expression that manifested as part of the Postmodernist movement that began as a reaction to the rigid restrictions of Modernism.

Early Post Modernism

Early Swiss Post Modern design was driven by the experimentations and teachings of Wolfgang Weingart who taught at the Basel School of design in Basel, Switzerland. Weingart was taught ITS by the masters of the style, Emil Ruder and Armin Hofmann at the Basel School. But once he became an instructor there, he questioned the “value of the absolute cleanliness and order” (Meggs & Purvis, 2011, p. 465) of the style. He experimented vigorously with breaking all typographic and organizational rules to see what the effect on the audience would be. He invigorated typography with energy and in turn

changed the viewer's response to the visual information. Instead of a simple fast reading, the reader now faced dynamic complexity free of any rules or hierarchies. The viewer was now compelled to spend more time with a design piece to understand its message and parse the meaning of its symbolism.

One of his American students, April Greiman, brought this new design language back to California with her and heavily influenced the youth culture there. David Carson, a self-taught designer working in the surf magazine world, took the ideas of the style and adopted them to his own typographic experiments in the surfing magazines he designed. For Carson, Post Modern design reflected the free spirit of the surf community.

Post Modernism is actually an umbrella term for many visual styles that came about after the 1980s. They are unified by their reaction to Modernism's guiding principles — particularly that of objectivity. A key feature of Post Modern design is the subjective bias and individual style of the designers that practise it. Additional defining stylistic characteristics can be summarized in the idea of 'de-construction.' The style often incorporates many different typefaces breaking every traditional rule of hierarchy and composition. Visual organization becomes more varied and complicated with the use of layers and overlapping. The use of image appropriation and culture jamming is a key feature. Dramatic layouts that do not conform to traditional compositions are another common characteristic. A traditional grid is not used to organize the layout of the elements, making composition look 'free-style.' Other organizational systems for the elements developed — axial, dilatational, modular, and transitional systems created a fresh way to organize the information. The combination of multiple geometric shapes layered with photographs created depth that worked well on the computer monitor — now a component of contemporary society.

Post Modernism is still in use today, though selectively. The chaos created by our technological advancements needs to be balanced with the ease of accessing information. The Apple brand is a good example of a contemporary design approach that feels fresh and current, while delivering massive amounts of information in a clean and simple way. The Post Modern methods of built-in visual difficulty are less welcome in our data-saturated culture.

1.9 Summary

Alex Hass

The technological revolution of the 1990s brought the mobile phone and computer to every home and office and changed the structure of our current society much as manufacturing in the 1800s changed Britain and the Western world. As with the Industrial Revolution, the change in technology over the last 20 years has affected us environmentally, socially, and economically. Manufacturing has slowly been moved offshore and replaced with technology-based companies. Data has replaced material as the substance we must understand and use effectively and efficiently. The technological development sectors have also begun to dominate employment and wealth sectors and overtake manufacturing's dominance. These changes are ongoing and fast-paced. The design community has responded in many novel ways, but usually its response is anchored by a look and strategy that reduce ornament and overt style while focusing on clean lines and concise messaging. The role of design today is often as a way-finder to help people keep abreast of changes, and to provide instruction. Designers are once again relying on established, historic styles and methods like ITS to connect to audiences because the message is being delivered in a complex visual system. Once the technological shifts we are experiencing settle down, and design is no longer adapting to new forms of delivery, it will begin to develop original and unique design approaches that complement and speak to the new urban landscape.

Questions to consider after completing this chapter:

1. What design principles do Dada and Punk have in common?
2. What influence does ITS have on Post Modern design?
3. What influence does ITS have on current design practice?
4. How did World War II influence design education?
5. How did Morris and the Arts & Crafts movement help to create the Bauhaus design philosophy?
6. How did technology influence early German design?
7. How does technology influence contemporary design practice?

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Chapter 2. Design Process

2.1 Introduction

Alex Hass

Learning Objectives

- Explain the role of communication design in print and media
- Describe how the creative process relates to strategic problem solving
- Contrast how the creative process relates to the design process
- Define critical phases of the design process
- Discover how project research helps to define a communication problem
- Give examples of brainstorming techniques that generate multiple concepts based on a common message
- Learn about metaphors and other rhetorical devices to generate concepts
- Explore how concepts translate into messages within a visual form

Communication Design and The Design Process

The practice of graphic or communication design is founded on crafting visual communications between clients and their audience. The communication must carry a specific message to a specific audience on behalf of the client, and do so effectively — usually within the container of a concept that creates context and builds interest for the project in the viewer.

See an illustrated model of the design process here: A Model of the Creative Process

Overview of the Design Process

The process of developing effective design is complex. It begins with research and the definition of project goals. Defining goals allows you to home in on precisely what to communicate and who the audience is. You can then appropriately craft the message you are trying to communicate to them. Additional information regarding how to deliver your message and why it's necessary are also clarified in the research stage. Often the preferred medium becomes clear (i.e., web, social media, print, or advertising) as does the action you want your audience to take. Asking a millennial to donate to a cause is a good example. Research reveals that transparency of donation use, donor recognition, and ease of making the donation are vital to successfully engaging a millennial audience (Grossnickle, Feldmann, White, & Parkevich, 2010). Research also reveals that millennials resist negative advertising, so the

message must be crafted in positive terms that are anchored to a realistic environment (Tanyel, Stuart, & Griffin, 2013). Knowing this information before the concept development begins is vital to crafting a message that will generate the response your client needs. Critiquing and analysis allow you to evaluate the effectiveness of the design approach as it develops through the stages of an iterative process.

In order to design visual materials that communicate effectively, designers must understand and work with the syntax of visual language. Meaning is expressed not only through content but through form as well, and will include both intellectual and emotional messages in varying degrees.

Developing Concepts into Design Solutions

Designers are responsible for the development of the creative concepts that express the message. A **concept** is an idea that supports and reinforces communication of key messages by presenting them in interesting, unique, and memorable ways on both intellectual and emotional levels. A good concept provides a framework for design decisions at every stage of development and for every design piece in a brand or ad campaign. An early example of this is the witty and playful ‘think small’ Volkswagen Beetle (VW) advertising campaign of the 1960s. By amplifying the smallness of its car in a ‘big’ car culture, VW was able to create a unique niche in the car market and a strong bond between the VW bug and its audience (see Figure 2.1).

Figure 2.1 Volkswagen Beetle

When you implement solutions, you put concepts into a form that communicates effectively and appropriately. In communication design, form should follow and support function. This means that what you are saying determines how you say it and in turn how it is delivered to your audience. Design is an **iterative** process that builds the content and its details through critiquing the work as it develops. Critiquing regularly keeps the project on point creatively and compositionally. Critiquing and analysis allow you to evaluate the effectiveness of the whole design in relation to the concept and problem. The number of iterations depends on the skill of the designer in developing the content and composition as well as properly evaluating its components in critique. In addition, all of this must occur in the context of understanding the technologies of design and production.

As you begin to build and realize your concepts by developing the content, the elements, and the layouts, you must apply compositional and organizational principles that make sense for the content and support the core concept. Compositional principles are based on psychological principles that describe how human beings process visual information. Designers apply these principles in order to transmit meaning

effectively. For example, research has shown that some kinds of visual elements attract our attention more than others; a designer can apply this knowledge to emphasize certain parts of a layout and give a certain element or message importance. These principles apply to all forms of visual materials, digital media, and print.

When dealing with text, issues of legibility and readability are critical. Designers organize information through the use of formal structures and typographic conventions to make it easier for the viewer to absorb and understand content. The viewer may not consciously see the underlying structures, but will respond positively to the calm clarity good organization brings to the text.

Media Attributions

- Volkswagen Beetle by IFCAR © Public Domain

2.2 Design Research and Concept Generation

Alex Hass

Defining Design Problem Parameters

Many designers define communication design as a problem-solving process. (The problem/opportunity is how to deliver information effectively to the desired audience.) The process that takes the designer from the initial stages of identifying a communication problem to the final stage of solving it covers a lot of ground, and different models can be used to describe it. Some are very complicated, and some are simple. The following sections break the design problem-solving process into four steps: (1) define, (2) research, (3) develop concepts, and (4) implement solutions.

2.3 Define

Alex Hass

Step 1: Define the Communication Problem

The inventor Charles Kettering is famously quoted as saying “a problem well-stated is half-solved.” Clearly the first step in any design activity is to define the communication problem properly. To do this, you will need to meet with clients to establish initial goals and objectives.

Here are some of the questions you should ask:

- What is the business of the client; what products or services does the client offer?
- What are the client’s long-term business goals? (What does the client want its business to have accomplished in 5 or 10 years?)
- What is the purpose of the project? What does the client hope to achieve with it? (The goals of a specific project are usually narrower than overall long-term business goals, but should fit within the larger picture.)
- What are the performance criteria that will be used to evaluate whether project goals are met?
- Who is the target audience?
- What is the client’s message to this audience?
- How does this project fit in with existing corporate materials?
- Does this piece require more than one format or medium?
- What corporate guidelines (if any) must be adhered to?
- Are illustration, photography, or any other special services required?
- Are there any special or unusual considerations around this project?
- What quantity is needed (for print)?
- What distribution method will be used (for print)?
- What is the budget?
- Who will approve the project? Will that person be available for sign-off when required?

Good planning at the beginning can make a project run smoothly and without surprises. Don’t assume anything; both the designer and the client should listen closely to each other and ask plenty of questions. Keep in regular communication, document discussions, and ensure that you have written confirmation of decisions.

2.4 Research

Alex Hass

Step 2: Conduct Research

Gather and analyze information What else do you need to know? The information you collected in the first stage is just a starting point — now you need to do more research in order to fine-tune your goals and process. Check every assumption, ask more questions, and add detail.

Research practices may involve:

- Competitor analysis: analyzing the competition to see what they do and determine their strengths and weaknesses
- **Ethnographic** research: observing user behaviour and culture
- Site research: observing and understanding the strengths and weaknesses of a space to optimize the effectiveness of the design experience you will be creating; site research is necessary to any design project that is situated in a built environment
- Marketing research: analyzing behaviour in terms of consumer practices, including demographic profiling (grouping people based on variables such as age/income/ethnicity/ location to create profiles generally describing their thinking/behaviour)
- User testing: measuring the ability of the product or service to satisfy users' needs
- Co-creation: inviting end-users to **brainstorm** solutions with the design team before the concept phase of design begins

Incorporating Research into the Design Process

Research should be a part of all design process, but what kind of research is done, and who does it, will be determined by the scope and budget of the project. Some information may be publicly available, for example, through corporate publications or previously published marketing studies or market data, but a design company may need to partner with a research firm in order to do targeted in-depth research.

At the very least, design research should include:

- A literature review (gathering and reviewing all existing material that is relevant to your subject)
- Collected details (existing materials, corporate guidelines) of your client's business and the services the client offers
- Information on the **target audience** (What do they want? need? expect?)
- Analysis of competitors (Who are they? how are they different? how are they the same? how

do they advertise or make information available?)

- Estimates and technical advice from subcontractors (e.g., printers)

Some things to consider:

- Is a full design audit required? Much like a SWOT analysis, which assesses strengths, weaknesses, opportunities, and threats, a design audit applies the same stringent methodology to analyzing your competitors' visual presence in the marketplace.

> A graphic design audit is a fantastic and relatively easy way to get a clear picture of how your competitors are perceived, what key messages they are communicating and how you look when placed alongside them. It's also a valuable exercise that informs you about the type of communication your customers are receiving on a regular basis from your key competitors. (Clare, 2006)

- What are the implications of the audience profile in relationship to the project goals?
- What is the most appropriate means to communicate with this audience (i.e., what media and marketing tools should you use)?
- How do the goals of this project fit into your client's long-term goals?
- Is your client's message what actually needs to be communicated in order to further the client's business goals?

Research takes time and can cost money, but in the larger picture will save time and money by helping to focus the direction of the design process. It also helps you provide justification for your proposed communication solutions to your client. Remember that all research must be carefully documented and raw sources saved and made available for future reference.

Now that you have gathered all the information, it's time to craft the design problem into a well-defined, succinct statement.

A Problem Well-stated is Half-solved

The writer Mark Levy, in his article A Problem Well-stated is Half-solved, developed six steps you can take to state a design problem so its solutions become clearer:

1. State the problem in a sentence. A single sentence forces you to extract the main problem from a potentially complex situation. An example of a problem statement: "We need to increase revenue by 25%."
2. Make the problem statement into a question. Turning the problem statement into a question opens the mind to possibilities: "How do we increase revenue by 25%?"
3. Restate the question in five ways. If you spin the question from a variety of perspectives, you'll construct new questions that may provide intriguing answers.
   For instance, try asking: "How could we increase revenue by 25% in a month?" "How could we increase it by 25% in an hour?" "How could we increase it by 25% in a minute?" "What could we stop doing that might cause a 25% revenue increase?" "What ways can we use our existing customer base to affect the increase?"
4. Give yourself thinking quotas. An arbitrary production quota gives you a better shot at coming up

with something usable, because it keeps you thinking longer and with greater concentration.
When I asked you to "Restate the question five ways," that was an example of an arbitrary quota. There's nothing magical about five restatements. In fact, five is low. Ten, or even a hundred, would be far better.

5. Knock your questions. Whatever questions you've asked, assume they're wrong-headed, or that you haven't taken them far enough.
You might ask, "Why do we need an 25% increase at all? Why not a 5% increase? A 500% increase? A 5,000% increase? What other things in the business might need to change that would be as important as revenue?"
6. Decide upon your new problem-solving question. Based on the thinking you've already done, this step may not even be necessary. Often, when you look at your situation from enough angles, solutions pop up without much more effort.

However, if you still need to pick a single question that summarizes your problem, and none seems perfect, force yourself to choose one that's at least serviceable. Going forward is better than standing still.

Now you can start brainstorming.

Concept Mapping

A good way to begin the process of research and problem definition is to write down everything that you already know about your subject. This brainstorming can be done in a linear way by developing lists, or in a non-linear way, popular with designers, called *concept mapping*. Concept mapping is a non-linear approach that allows a designer to see what is known and what still needs to be researched. Concept mapping is also used to generate concepts and to create associations and themes.

W5 + 1

The first step is to take a sheet of paper and write a central title or topic in the centre. Then surround this central idea with information gathered by answering the following questions, based on the 5 Ws (who, what, where, why, and when), plus one more, how:

- What are you trying to communicate? (the problem)
- Why must communication occur? (what is its purpose?)
- Who is the target audience?
- Where will communication take place? (in what medium and location?)
- When will communication take place?
- How will you implement the concept?
- What if? (what would be ideal?)

Once you've added all the information you have at hand, you will see any assumptions and gaps in that information, and you can begin specific directed research to create a larger, more objective picture.

Here is an example of a concept map (See Figure 2.2). To see a concept map that details the scope of visual communication.

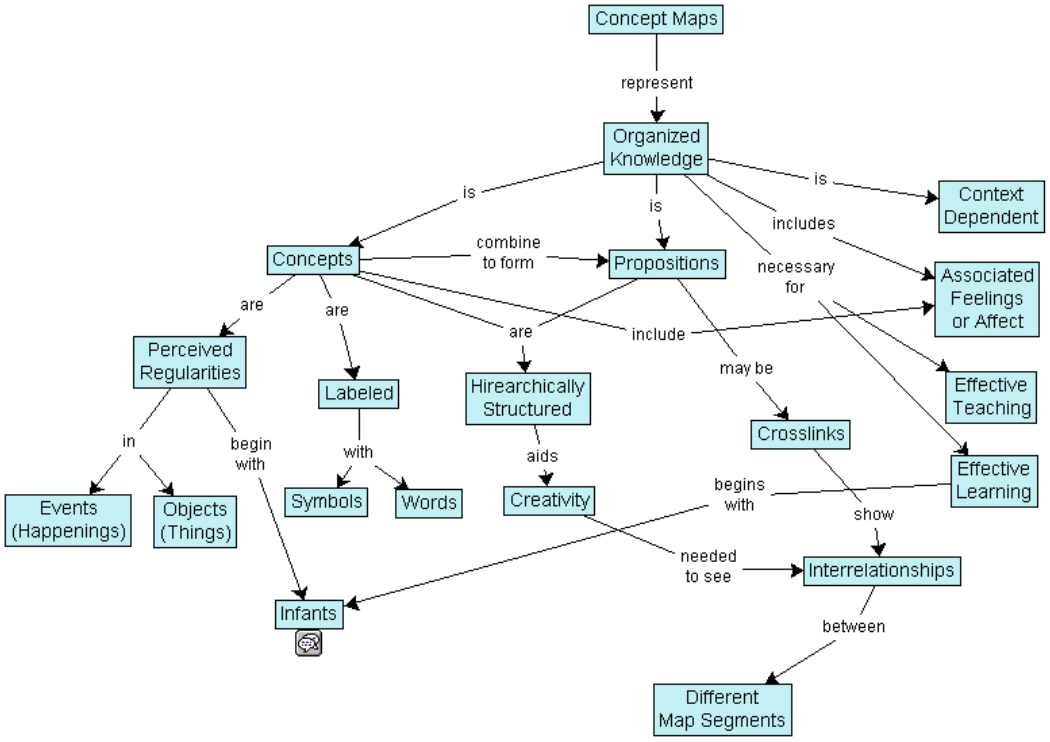

Figure 2.2 Example of a concept map

You can use the information in a concept map to generate other themes and concepts for your project. For example, in the concept map above, you could develop another theme by highlighting in yellow all information from the 1970s. This would reveal the parameters of design practice in the 70s and would additionally reveal what has been added and changed in design practice since.

Text Attributions

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2.5 Develop Concepts

Alex Hass

Step 3: Developing Concepts

Concept development is a process of developing ideas to solve specified design problems. The concepts are developed in phases, from formless idea to precise message in an appropriate form with supportive visuals and content. Once you have done your research and understand exactly what you want to achieve and why, you are ready to start working on the actual design. Ideally, you are trying to develop a concept that provides solutions for the design problem, communicates effectively on multiple levels, is unique (different and exciting), and stands out from the materials produced by your client's competitors.

Generate, test, and refine ideas

A good design process is a long process. Designers spend a great deal of time coming up with ideas; editing, revising, and refining them; and then evaluating their results every time they try something. Good design means assessing every concept for effectiveness.

The design process looks roughly like this:

- Generating a concept
- Refining ideas through visual exploration
- Preparing rough layouts detailing design direction(s)
- Setting preliminary specifications for typography and graphic elements such as photography, illustration, charts or graphs, icons, or symbols
- Presenting design brief and rough layouts for client consideration
- Refining design and comprehensive layouts, if required
- Getting client approval of layouts and text before the next phase

Developing Effective Concepts

A concept is not a message. A concept is an idea that contextualizes a message in interesting, unique, and memorable ways through both form and design content.

A good concept reinforces strategy and brand positioning. It helps to communicate the benefits of the offer and helps with differentiation from the competition. It must be appropriate for the audience, facilitating communication and motivating that audience to take action.

A good concept provides a foundation for making visual design decisions. For example, Nike's basic

message, expressed by its tagline, is “Just Do It.” The creative concept Nike has used since 1988 has been adapted visually in many ways, but always stays true to the core message by using images of individuals choosing to take action.

> “It was a simple thing,” Wieden recalls in a 2009 Adweek video interview in which he discusses the effort’s genesis. Simplicity is really the secret of all “big ideas,” and by extension, great slogans. They must be concisely memorable, yet also suggest something more than their literal meanings. Rather than just putting product notions in people’s minds, they must be malleable and open to interpretation, allowing people of all kinds to adapt them as they see fit, and by doing so, establish a personal connection to the brand (Gianatasio, 2013).

A good concept is creative, but it also must be appropriate. The creativity that helps develop effective, appropriate concepts is what differentiates a designer from a production artist. Very few concepts are up to that standard — but that’s what you should always be aiming for.

In 1898, Elias St. Elmo Lewis came up with acronym AIDA for the stages you need to get consumers through in order for them to make a purchase. Modern marketing theory is now more sophisticated, but the acronym also works well to describe what a design needs to do in order to communicate and get people to act.

In order to communicate effectively and motivate your audience, you need to:

A — attract their attention. Your design must attract the attention of your audience. If it doesn’t, your message is not connecting and fulfilling its communication intent. Both the concept and the form must stand out.

I — hold their interest. Your design must hold the audience’s interest long enough so they can completely absorb the whole communication.

D — create a desire. Your design must make the audience want the product, service, or information.

A — motivate them to take action. Your design must compel the audience to do something related to the product, service, or information.

Your concept works if it makes your audience respond in the above ways.

Generating Ideas and Concepts from Concept Mapping

You can use the information in a concept map to generate additional concepts for your project by reorganizing it. The concept tree method below comes from the mind-mapping software blog (Frey, 2008)

1. Position your design problem as the central idea of your mind map.
2. Place circles containing your initial concepts for solving the problem around the central topic.
3. Brainstorm related but non-specific concepts, and add them as subtopics for these ideas. All related concepts are relevant. At this stage, every possible concept is valuable and should not be judged.

4. Generate related ideas for each concept you brainstormed in step 3 and add them as subtopics.
5. Repeat steps 3 and 4 until you run out of ideas.

Applying Rhetorical Devices to Concept Mapping

After you have placed all your ideas in the concept map, you can add additional layering to help you refine and explore them further. For example, you can use rhetorical devices to add context to the concepts and make them come alive. **Rhetoric** is the study of effective communication through the use and art of persuasion. Design uses many forms of rhetoric — particularly metaphor. If you applied a metaphor-based approach to each idea in your concept map, you would find many new ways to express your message.

Rhetorical Devices Appropriate for Communication Design

Allusion is an informal and brief reference to a well known person or cultural reference. In the magazine cover linked below, an allusion is used to underline the restrictive nature of the burqa, a full body cloak worn by some Muslim women, by applying it to Sarah Jessica Parker, an actor whose roles are primarily feminist in nature. (Harris, 2013)

Follow the link to see an example: Marie Claire Cover

Amplification involves the repetition of a concept through words or images, while adding detail to it. This is to emphasize what may not be obvious at first glance. Amplification allows you to expand on an idea to make sure the target audience realizes its importance. (Harris, 2013)

Follow the link to see an example: Life's too short for the wrong job Marketing Campaign

Analogy compares two similar things in order to explain an otherwise difficult or unfamiliar idea. Analogy draws connections between a new object or idea and an already familiar one. Although related to simile, which tends to employ a more artistic effect, analogy is more practical; explaining a thought process, a line of reasoning, or the abstract in concrete terms. Because of this, analogy may be more insightful. (Harris, 2013)

Follow the link to see an example: WWF Lungs Before It's Too Late

Hyperbole is counter to understatement. It is a deliberate exaggeration that is presented for emphasis. When used for visual communication, one must be careful to ensure that hyperbole is a clear exaggeration. If hyperbole is limited in its use, and only used occasionally for dramatic effect, then it can be quite attention grabbing.

Follow the link to see an example: Final Major Project by Mark Studio

A written example would be: *There are a thousand reasons why more research is needed on solar energy.*

Or it can make a single point very enthusiastically: *I said "rare," not "raw." I've seen cows hurt worse than this get up and walk.*

Hyperbole can be used to exaggerate one thing to show how it differs from something to which it is being compared: *This stuff is used motor oil compared to the coffee you make, my love.*

Hyperbole is the most overused rhetorical device in the world (and that is no hyperbole); we are a society of excess and exaggeration. Handle it like dynamite, and do not blow up everything you can find (Harris, 2013).

Metaphor compares two different things by relating to one in the same terms commonly used for the other. Unlike a simile or analogy, metaphor proposes that one thing is another thing, not just that they are similar (Harris, 2013).

Follow the link to see an example: Ikea Bigger Storage Idea

Metonymy is related to metaphor, where the thing chosen for the metaphorical image is closely related to (but not part of) that with which it is being compared. There is little to distinguish metonymy from synecdoche (as below). Some rhetoricians do not distinguish between the two (Harris, 2013).

Follow the link to see an example: London Logo

Oxymoron is a paradox presented in two words, in the form of an adjective and noun (“eloquent silence”), or adverb-adjective (“inertly strong”), and is used to impart emphasis, complexity, or wit (Harris, 2013). See Figure 2.3 for another example.

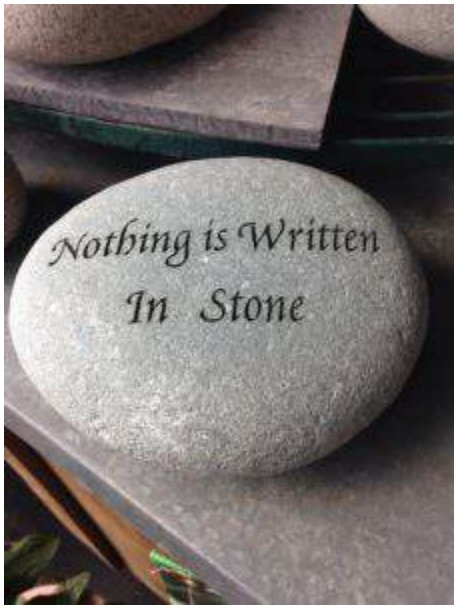

Figure 2.3 Example of an oxymoron

Personification attributes to an animal or inanimate object human characteristics such as form, character, feelings, behaviour, and so on. Ideas and abstractions can also be personified. For example, in the poster series linked below, homeless dogs are placed in environments typical of human homelessness (Harris, 2013).

Follow the link to see an example: Manchester Dogs’ Home Street Life

Simile is a comparison of one thing to another, with the two being similar in at least one way. In formal

prose, a simile compares an unfamiliar thing to a familiar thing (an object, event, process, etc.) known to the reader. (Harris, 2013)

Follow the link to see an example: Strong Handle Billboard

Synecdoche is a type of metaphor in which part of something stands for the whole, or the whole stands for a part. It can encompass many forms such that any portion or quality of a thing is represented by the thing itself, or vice versa (Harris, 2013).

Follow the link to see an example: A Global Warming Poster

Understatement deliberately expresses a concept or idea with less importance as would be expected. This could be to effect irony, or simply to convey politeness or tact. If the audience is familiar with the facts already, understatement may be employed in order to encourage the readers to draw their own conclusions (Harris, 2013).
For example: *instead of endeavouring to describe in a few words the horrors and destruction of the 1906 earthquake in San Francisco, a writer might state: The 1906 San Francisco earthquake interrupted business somewhat in the downtown area.*

Follow the link to see an example: Nike's Just Do It

An excellent online resource for exploring different rhetorical devices is "A Handbook of Rhetorical Devices" (Harris, 2013). The definitions above have been paraphrased from this site.

Developmental Stages of Design

No design work should ever be done without going through an iterative development process in which you try out different ideas and visual approaches, compare and evaluate them, and select the best options to proceed with. This applies to both form and content.

The development of the concept starts with brainstorming as wide a range of ideas as possible, and refining them through a number of development stages until you are left with those that solve the communication problem most effectively.

The development of graphic forms starts with exploring a wide range of styles, colours, textures, imagery, and other graphic devices and refining them through development stages until you are left with those that best reinforce the concept and message.

The development process starts with **thumbnails** and works through rough layouts and comprehensives to the final solution. Thumbnails are small, simple hand-drawn sketches, with minimal information. These are intended for the designer's use and, like concept maps, are visuals created for comparison. These are not meant to be shown to clients.

Their uses include:

- Concept development and visualization of ideas
- Preliminary evaluation of content (they allow you to sift and sort ideas quickly and effectively)

- Preliminary evaluation of form (value studies, compositional studies, potential placement of elements)
- Note-taking (a tool to record verbal or visual information quickly and accurately)

Quantity is very important in thumbnails! The idea is to get as many ideas and options down as possible. Designers typically take one of two approaches when they do thumbnails: they either brainstorm a wide range of ideas without exploring any of them in depth, or they come up with one idea and create many variations of it. If you use only one of these approaches, force yourself to do both. Brainstorm as many ideas as possible, using a mix of words and images. The point here is the quantity of ideas — the more the better. Work fast and don't judge your work yet.

Once you have a lot of ideas, take one you think is good and start exploring it. Try expressing the same idea with different visuals, from different points of view, with different taglines and emotional tones. Make the image the focal point of one variation and the headline the focal point of another. The purpose here is to try as many variations of an idea as possible. The first way of expressing an idea is not necessarily the best way, much like the first pancake is not usually the best.

After you've fully explored one idea, choose another from your brainstorming session and explore it in the same way. Repeat this with every good idea.

Roughs are exactly that — rough renderings of thumbnails that explore the potential of forms, type, composition, and elements of your best concepts. Often a concept is explored through the development of three to five roughs. These are used to determine exactly how all of the elements will fit together, to provide enough information to make preliminary evaluation possible, and to suggest new directions and approaches.

The rough:

- Uses simple, clean lines and basic colour palettes.
- Accurately renders without much detail (the focus is on design elements, composition, and message)
- Includes all of the visual elements in proper relationship to each other and the page

Comps are created for presenting the final project to the client for evaluation and approval. The comp must provide enough information to make evaluation of your concept possible and to allow final evaluation and proofing of all content.

The comp:

- Is as close as possible to the final form and is usually digital
- May use final materials or preliminary/placeholder content if photographs or illustrations are not yet available

Hand-drawn or Digital?

Comps might be hand-drawn when you are showing a concept for something that doesn't yet exist, such

as a product that hasn't been fabricated, a structure that hasn't been built, or to show a photographer how you want material to be laid out in a photograph that has not yet been taken. Although you could create these comps digitally, it's often more cost effective to create a sketch.

Designers sometimes create hand-drawn comps in order to avoid presenting conceptual work that looks too finished to a client, so they will not be locked into a particular approach by the client's expectations.

Even in this digital age, you should draw all thumbnails by hand (using pen, pencil, or tablet) for the following reasons:

- You don't have to make time-wasting decisions that you shouldn't be making at this early stage (e.g., what typeface should I use? what colour should this be?)
- It's much faster than doing it digitally.
- Work done on a computer tends to look finished and professional, and this can trick you into thinking an idea is better than it is.
- The technology of a tool tends to define the way it is used. If you are using a computer, you will tend to come up with solutions that can be executed only on a computer, and that limits your creative options. For example, would you think of creating an illustration from coloured paper if you were using the computer?
- Hand-drawn sketches provide a paper trail that shows your concept development process and can be presented in case studies to reveal your entire design process in a more personal and engaging way.

Media Attributions

- nothing-is-written-in-stone-527756_1920 by SBM © Public Domain

2.6 Implement Solutions

Alex Hass

Step 4: Solution Implementation

In this step, we are ready to select the final concept options and carry their application through to completion in producing the final design(s). This part of the process requires that you know how to work with photographers and illustrators, as well as with people in production technologies — primarily, programmers and printers. You may also require project management skills. You should also put a process in place so your final solutions can be evaluated for their effectiveness. Did they work? Did they achieve their goals?

There are many components that require attention during the production phase:

Production and Implementation

- Copy placement and preparation of layouts from approved text
- Liaison with suppliers and subcontractors
- Completion of photography, illustration, charts/graphs, icons/symbols
- Ongoing client liaison for proofreading and corrections
- Scanning and electronic preparation of images (black and white, duotones/tritones, colour); may include colour correction and/or digital manipulation
- Preparation of electronic files in line with press/prepress/web requirements
- Supervision of all prepress materials (final files and proofs)
- Organization, maintenance, and archiving of all digital materials related to the job

Production Supervision

- Discuss production options with client, solicit quotes, and select printer/programmer
- When contract is awarded, liaise with production services to discuss and refine project details
- Prepare or review production specifications
- Liaise with client and production to check proofs
- Oversee production to ensure quality control
- Follow up after production work is complete

Evaluation

Every step of a project should be evaluated in terms of the goals you have defined. Two fundamental questions about every design decision you make are:

- What does this accomplish?
- How does what is accomplished help to meet the project goals?

After the original design challenge has been defined, evaluate every stage of the process in that context. It's surprisingly easy to stray off track when you're designing. If you find yourself designing something brilliant, but it doesn't communicate what it should to the right audience, then all that brilliance is wasted.

Communication

Whether they are in print or multimedia, all design works are intended to communicate to a specific audience, and the design must support that function. All concepts must be evaluated with that end in mind. For example:

- Does the work communicate the key message(s) and support the client's goals?
- Does the work effectively integrate images, design, and text (form and content) to support that communication; create an overall 'look'; make the piece work as a unified whole with no distractions?
- Is the piece physically easy to read and/or understand?
- Do the design choices amplify material (subject matter, mood) in the text?
- Is the piece appropriate to the audience? (children, youth, adults, seniors have particular interests and needs)

Economic Efficiency

- What is possible and most effective within the budget?
- Will this method attract the desired audience/buyer?

Design and Materials

- Are the design choices compatible with technological requirements for production?
- For print materials, is there efficient and economical use of paper?
- Will the materials chosen support the intended use and method of distribution?

2.7 Summary

Alex Hass

Communication design can be described as a problem-solving process that can be broken into four steps: (1) define, (2) research, (3) develop concepts, and (4) implement solutions. Research should be a part of all design process determined by the scope and budget of the project. Concept mapping is a non-linear approach that outlines what is known, what is needs, creates associations and themes, and helps generate ideas. Good design takes time that involves generating and assessing concepts. Time is also spent editing, revising, refining , and evaluating ideas.

In conclusion, defining the design process is complicated as it has many stages and involves many steps at each stage. Complicating it further is the reality that every project is unique in its parameters, goals, time period, and participants. This chapter is meant to facilitate the beginning of how you define your individual design process by basing it on general guidelines. If you are still developing an understanding of your personal design strengths and weaknesses, allow extra time for each stage and track your time for each stage. You'll soon discover if you fall into the category of a brainstorming, conceptual, or project development type. Conceptual designers find it easy to develop multiple concepts, but less easy to take the steps to develop them to their full potential. Project development types are the opposite — finding concepts hard to create, but developing projects quite easy. Allow extra time to discover which category you fall into and also to develop strengths in your weaker area. As you gain experience developing design projects, you will start to personalize your design process and be able to estimate how long it takes with a fair degree of accuracy. This will help you to estimate project design costs more accurately and gauge the steps needed to bring a project to a successful conclusion.

Questions to consider after completing this chapter:

1. How does communication design work within the constraints of print and media?
2. How does the creative process relate to strategic problem solving?
3. How is the creative process related to the design process?
4. What are the critical phases of the design process?
5. How does project research help to define a communication problem?
6. What are some examples of brainstorming techniques that generate multiple concepts based on a common message?
7. How does using a metaphoric device generate concepts?
8. How do concepts translate into messages within a visual form?

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Chapter 3. Design Elements, Design Principles, and Compositional Organization

3.1 Introduction

Alex Hass

Learning Objectives

- Utilize basic design principles relating to visual composition
- Define design terminology pertaining to form
- Describe organizational systems and core principles for layout grids
- Differentiate between typographic categories
- Establish a visual hierarchy within a layout
- Express ideas using the principles of composition and form

Communication design is essentially the crafting of a message meant for a specific section of the public. This written message is infused with meaningful and relevant visual components. The composition of these components should amplify, clarify, and enhance the message for the viewer. To assist in making sound design choices, a designer applies principles of composition and principles of organization to the design elements selected for a project.

Understanding how to utilize the fundamentals of design elements, principles, and composition is necessary to be able to confidently move through the stages of the design development process and build a project from the initial design brief to the final published design work.

Definitions from various design sources about what comprises a design element are consistent for the most part, but defining design principles is not as consistent and varies from one text to the next. Marvin Bartel's (2012) definitions of these categories are both simple and on point. He defines a visual element as any "basic thing that can be seen," and a design principle as a method for "arranging things better." Also included in this chapter are organizational systems that can focus and direct the overall direction a composition will take.

3.2 Visual Elements — Basic Things That Can be Seen

Alex Hass

> Point, line, and plane are the building blocks of design. From these elements, designers create images, icons, textures, patterns, diagrams, animations, and typographic systems. (Lupton & Phillips, 2014, p. 13)

Figure 3.1 Design using points, lines, planes

Point

A point is a precise position or location on a surface. In purely mathematical terms, a point marks a set of coordinates — it has no mass at all. In this objective definition, a point is essentially a place. Visually, a point is a dot and therefore the basic building block of every variation of line, texture, and plane.

Subjectively, the term *point* has a lot of power. *Point* can direct attention, be the focus of attention, create emphasis, and cut through veiled information. The compositional term *focal point* brings the objective and subjective together by being the first place the eye is drawn to in a composition and usually contains the most important piece of visual communication.

Line

Figure 3.2 Lines (by Ken Jeffery)

A line is the second most basic element of design — a line is a collection of points arranged in a linear manner (see Figure 3.2). A line connects two points, or traces the path of a movement. A line can be actual or implied — for instance, as a composition of two or more objects in a row. Lines in nature act as defining planes — examples are a horizon or the silhouette of a forest against the sky. Long straight lines do not often occur in nature, and therefore when they are present, they tend to dominate the landscape visually. Natural settings are usually parsed by the eye into shorter sequences of curved or straight lines and organic shapes.

When made by the hand, a line is created by the stroke of a pencil, pen, brush, or any mark-making tool. These lines can be thin or wide, and are expressive and distinct, reflecting the texture of the tool used to make them. Lines can create a plane (a shape) by being clustered together or by defining a shape. If the line is thickened, it changes and becomes a plane. When lines are made digitally, they can acquire many of the same qualities possessed by hand-drawn lines through the application of effects.

Plane

Figure 3.3 Planes

Like lines, planes (shapes) can be organically made or they can be geometric, as in the example shown in Figure 3.3. A plane is a flat surface that has defined borders. “A line closes to become a shape, a bounded plane” (Lupton & Phillips, 2014, p. 38). Planes are excellent compositional tools for clustering visual elements into visual fields. A plane can also act as a separating device and allow the viewer to see that one section of information is not linked to another.

In design software, a vector graphic is a shape created by defining its parameters with a line, and then filling it with a solid or textured fill. Grids help to create and define typographic planes that float or interact with solid planes of image, texture, or colour. In the physical world, everything is composed of shapes that are either two- or three-dimensional. How you choose to organize and arrange the planes in your photograph, your illustration, or your design will structure the composition and determine not only how the elements intersect with one another but also how the viewer interacts with the composition.

Colour

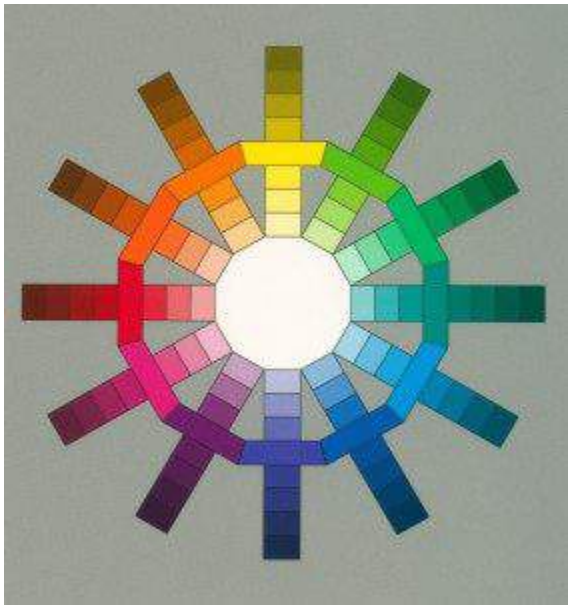

Figure 3.4 Colours

Graphic design has evolved over the last two centuries from a craft that designed text and images primarily in black and white for books and broadsheets, to a craft that works with full colour in analog and digital media and on every kind of substrate. Controlling and effectively using colour to support communication is now more important than it has ever been. Both media and advertising have become very sophisticated over the last few decades and are adept at creating exciting, sensuous, and energetic environments that are crafted with the skillful use of colour and texture. The public, in turn, has absorbed these unprecedented levels of image saturation with a variety of outcomes. One is an expectation that the visual palette match and enhance the message. A second outcome is a high expectation for strong and authentic visuals of places or objects. A third outcome is a cultural nostalgia for earlier looks created by various devices. Examples like 8-bit graphics or 1950s Kodachrome both possess unique colour and texture palettes and have properties the public can discern. When one of these nostalgic colour palettes is applied to an image, it adds another layer of meaning to the work, and that meaning has to make sense for the viewer.

The explosion of tools for making and sharing digital photography and graphics also reveals how good the general public has become at crafting visuals with relevant atmosphere and texture. The bar has been raised very high with colour use in contemporary times, and understanding colour basics is an absolute necessity.

RBG and CMYK Colour Spaces

Given that design and colour are united in every project, it is important to realize that there are two colour systems, and often a project needs to work in both. Digital media works in the additive colour system, and its primary colours are red, green, and blue (RGB). In this system, the absence of colour equals black, while combining all colours results in white. RGB is the colour system of visible light (see

Figure 3.5). This light system is called *additive* because the three primaries together create all the hues in the spectrum.

Subtractive colour is the system needed for print media, and its primary colours are cyan, magenta, yellow, and black (CMYK), as shown in Figure 3.5. In CMYK, the absence of colour equals white, while combining all colours creates black. Both of these systems have many overlapping colours but their colour spheres are not exactly the same. Understanding where the overlaps e xist and where they don't correspond is vital to the success of a project. If your print materials cannot be replicated on screen, you will have a major design problem that will have to be corrected. Always choose colours that will work in both systems.

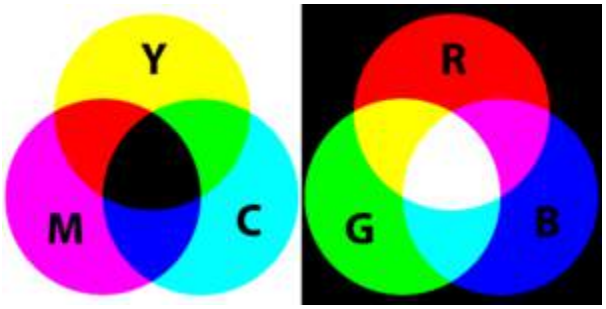

Figure 3.5 Primary colours for the additive and subtractive colour schemes

Environment is another aspect of colour choice that is very important. Both the natural world and the world within the screen vary from moment to moment and screen to screen. Colours are affected and influenced by the amount of atmospheric light available to them as well as by the colours in contact with the object they are viewing. Texture also changes our perception of a colour as does the brightness or darkness around it.

However much a designer hopes to define the parameters of a colour palette, there will always be unknown factors influencing the palette on the viewers' end. Create a palette that is focused enough to create the right atmosphere and energy level for your project, but one that doesn't rely too heavily on a specific colour. Careful, considered colour use will help define a message and create a mood that supports the composition and concept of a design work. Always create a palette that will work with both colour systems and also be robust enough to work in less than optimal environmental circumstances.

Negative Space

Negative space, which is also called *white space*, is the visually quiet area that surrounds the active area of a composition (see Figure 3.6). It is also referred to as figure/ground, and has a very important role in composition as it shapes the visual perception of the subject. Without negative space, there is no positive space — the effect is similar to seeing a polar bear in a snowstorm. Negative space is often thought of as as passive and unimportant, but the active elements or 'figure' are always perceived in relation to their surroundings by the mind of the viewer. The composition of the negative space frames and presents the active elements in a flat or dynamic way. If the surrounding area is busy with many other elements, the focal point loses its power because the elements all have a similar visual value. The works of Gustav Klimt exhibit this quality.

Figure 3.6 Example of negative or white space

If, on the other hand, the work is balanced and the negative space is active, it brings energy to the form and its space. The focal point or figure increases its visual power because there is contrast for the eye. Another way to look at this is to see that the range or gamut of visual activity is increased and therefore the experience is more satisfying to the eye.

When designers play with reducing or confusing positive and negative space, they create ambiguity. Ambiguity creates tension, which increases the interest of a composition to the viewer and also increases the visual energy of a design. There are three types of figure/ground relationships.

Stable figure/ground is the most common type. The positive element is clearly separate and defined against its negative space. A good example of this is text blocks in magazines or books.

Reversible figure/ground is the second type and is found in most of the work of M.C. Escher. Both the positive and negative space delivers ‘active’ information that feels equal to the eye and therefore creates a toggling effect in the viewer. One shape is comprehended while the other acts as its negative space, then the opposite happens and the negative space becomes meaningful and its opposite becomes the neutral ‘holding’ space.

Ambiguous figure/ground creates a confusing lack of focal point. The eye searches for a dominant visual ‘starting point’ in the composition but can’t find one. Often this creates energy, and if the effect is compelling, it invites the viewer to stay with the work for a long period of time, absorbing all of the visual information.

Figure 3.7 FedEx express truck

Designers often utilize figure/ground in the crafting of symbols, wordmarks, and logos because of its capacity to create meaning with the space surrounding a mark. An excellent example of figure/ground is the FedEx wordmark (see Figure 3.7). The negative space needed to define the letterforms also augments their meaning by creating a forward pointing arrow. In print design, negative space can also allude to what is outside the frame and makes the field of the page or poster larger that it physically is. On a static or moving screen, negative space has the ability to change the flow of time, to introduce a break, or to create space around an important point.

Composing strong figure/ground tension is an excellent skill to acquire for designers of any media. Crafting white space eventually becomes as important to a designer as selecting the words and the elements of a project. Composing the negative spaces of a composition will allow you to vary visual emphasis of the elements, and control and increase the visual energy overall.

Texture

Figure 3.8 Example of texture

Texture is a visual and a tactile quality that designers work with (see Figure 3.8). Texture is used both in composition and also on the printed substrate or media space. Designers create textures for their projects with anything at hand. A texture can be made with typography, generated in raster or vector software like Photoshop or Adobe Illustrator, or by using a camera and capturing elements in the material world.

Using texture thoughtfully will enhance a visual experience and amplify the context for the content. Often adding texture adds visual complexity and a bit of visceral depth to a two-dimensional design

project. It can also tie one piece of design to another, or become a defining element of a brand or a series of communications.

The tactile aspect of a design work comes into play with the choices we make for the substrate we print on. The surface can be smooth or rough, glossy or matte, thick or thin, translucent or opaque, paper, plastic, concrete, metal, wood, or cloth. Paper can even have two or more of these qualities if we augment the original look of the paper with layers of varnish that reverse the tactile effect of the substrate. Often the choice of substrate is most effective if it is sympathetic to or contrasts with the concept and content of the piece. The choice of substrate texture affects how the viewer perceives the content — both physically and optically. Glossy substrates often feel sophisticated, hard, and cold. They are imbued with a sense of precision because the ink sits on top of the surface of the paper and retains almost all of its original integrity. A textured matte paper feels organic, accessible, and warm because the ink is partially absorbed by the paper, and is therefore influenced by and fused to its softer characteristics.

Pattern is part of the element of texture, but because of its special ability to hold content that is meaningful, and its long and significant cultural history, it deserves a special mention. All patterns can be reduced to dot and line and are organized by a grid system of some kind. Their ‘flavour’ is a reflection of the culture and time they come from and of the materials that created them. Patterns can be a subtle addition to the content of any design work. A pattern can be created using a relevant graphic (like a logo) or repeated multiple times, or it can support the organizational principles developed by the designer in a decorative way; for example, if a grid is based on the square and the texture of the pattern is also based on the square.

When the pattern is seen as a whole, its individual components melt away and lose their identity to the larger field of the pattern. This ability to focus on a pattern in multiple ways creates a second purpose for the graphic element (such as a circle, a square, a logo, or symbol) the designer has used. In modern design practice, pattern is an opportunity to augment the clean and simple material surfaces we work with and ornament a page or a website with a relevant texture.

Typography

Figure 3.9 Typography

Typography is the medium of designers and the most important element we work with (see Figure 3.9).

Typography not only carries a message but also imbues a message with visual meaning based on the character of a font, its style, and its composition. Words are meaningful in and of themselves, but the style and composition of words tells a reader you are serious, playful, exciting, or calm. Typography is the tonal equivalent of a voice and can be as personal or as general in flavour.

Typography traditionally has two functions in most design projects. One function is to call attention to or to 'display' the intent of a communication. This function is called titling or display typography and it is meant to call attention to itself. The second function is to present the in-depth details of a communication within a text block. This function requires a different typographic approach — one that is quiet and does not call attention to itself. Instead, it is intended to make the content accessible and easy to read.

Font Categories

There are many ways to categorize and subcategorize type. This overview discusses the seven major historical categories that build on one another. Serif fonts comprise four of these categories: humanist, old style, transitional, and modern. Italics, first designed in the 1500s, have evolved to become part of a font 'family' and were at one time a separate category. They were initially designed as independent fonts to be used in small pocket books where space was limited. They were not embraced as text fonts, but were considered valuable for adding emphasis within a roman text and so became part of the set of options and extensions a font possessed. The trajectory of use is the opposite for the sans serif category. Sans serif fonts have historically been used for display only, but in the 20th century, they became associated with the modern aesthetic of clean and simple presentation and have now become very popular for text-block design. Egyptian or slab serif fonts can be used as either display or text depending on the characteristic of the font design.

Blackletter

Figure 3.10 Example of Blackletter type

Blackletter was the medieval model for the first movable types (see Figure 3.10). It is also know as

Block, Gothic, Fraktur, or Old English. The look of this font category is heavy and dark. The letterforms are often condensed and put together tightly in a text block creating a dark colour (tone) for a page — between 70% and 80% grey. To put the tone in context, the usual tone of a modern text page is between 55% and 70% grey. The look of the letterforms makes it hard to read the page, because legibility was not their first function as it is today. The beauty of the font and the form of the book was the primary goal for early publications. Books were considered to be objects of wealth and beauty, not solely as a means to convey information.

Humanist

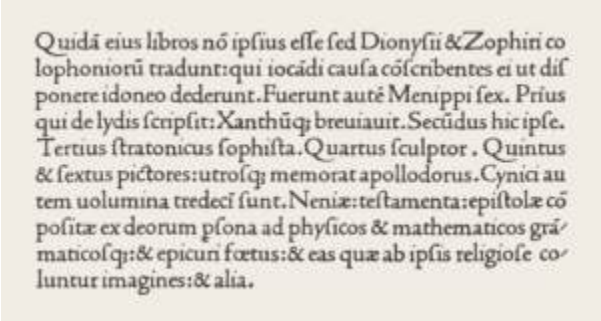
Quidã eius libros nõ ipſius eſſe ſed Dionyſii & Zophiri co
lophoniorũ tradunt: qui iocãdi cauſa cõſcribentes ei ut diſ
ponere idoneo dederunt. Fuerunt autẽ Menippi ſex. Prius
qui de Iydis ſcripſit: Xanthũq; breuiauit. Secũdus hic ipſe.
Tertius ſtratonicus ſophiſta. Quartus ſculptor. Quintus
& ſextus pictores: utroſq; memorat apollodorus. Cynici au
tem uolumina tredecĩ ſunt. Neniæ: teſtamenta: epiſtolæ cõ
poſitæ ex deorum pſona ad phyſicos & mathematicos grã
maticoſq;: & epicuri fœtus: & eas quæ ab ipſis religioſe co
luntur imagines: & alia.

Figure 3.11 Example of Humanist type

Humanist fonts are also referred to as Venetian, because they were developed in and around Venice in the mid-15th century (see Figure 3.11). Their design was modelled on the lighter, open serif letterforms and calligraphy of the Italian humanist writers. The designers strove to replicate many of the characteristics found in this writing style, including multiple variations of a glyph (letterform) that a written document possessed. For instance, a font could have up to 10 different lowercase a's to set a page with. Humanist types were the first roman types. Though they were much easier to read and lighter on the page than blackletter, they still created a visually dark and heavy text block in contrast to the fonts we have become accustomed to. Humanist fonts have little contrast between the thick and thin strokes — the strokes are usually heavy overall. The x-height of a humanist font is small compared to contemporary fonts, and this impedes quick comprehension and legibility. Humanist fonts are not often used for these reasons, though they are well respected because they are the original model so many other fonts are based on. It is important to remember that these fonts were a perfect match to the earliest printing technologies and that those presses could not have printed our light and delicate fonts. Fonts have evolved alongside the technological advancements of the printing industry.

Examples of humanist fonts include Jenson, Centaur, Verona, Lutetia, Jersey, and Lynton.

Old Style

Figure 3.12 Example of Old Style type

Old style fonts, also known as Garalde fonts, are the next leap in font design, and their stylistic developments were driven by the technological advancement of presses and the improved skills of punchcutters (see Figure 3.12). Font designers began to explore the possibilities of their medium — both the metal of the punches and the abilities of the presses and their papers. The letterforms became more precise, their serifs more distinct. The contrast of the stroke weights was also increased, and the presses held true to the design and didn't distort them. The aim of these new fonts ceased to be about replicating the look of handwriting and more about refining the letterforms to create a lighter overall tone.

Examples of old style fonts include Goudy Old Style, Granjon, Janson, Palatino, Perpetua, Plantin, and Sabon.

Transitional

Figure 3.13 Example of Transitional type

A few centuries later, font design was again refined, and this time the impetus came from France and the Enlightenment movement. Fonts were created along the rationalist principles of the times. The strokes were contrasted further with very thick main strokes and very thin sub-strokes, and the serif, which capped the stroke, did not use bracketing (the rounding underneath the intersection of the two strokes). The letterforms took on a look that implied they were constructed mathematically and anchored within a grid. These new fonts broke with humanist and old style tradition and ceased to reference calligraphy.

Examples of transitional fonts include Baskerville, Bookman, Fournier, and Joanna (see Figure 3.13).

Modern

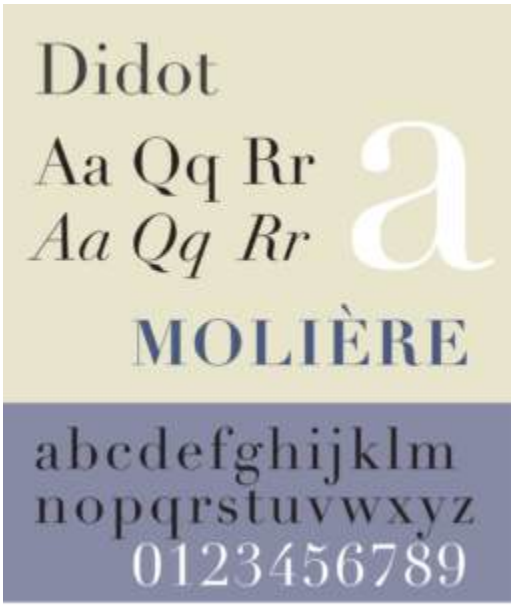

Figure 3.14 Example of Modern type

Modern fonts are also known as Didones and take the contrast started by the transitional fonts much, much further (see Figure 3.14). Bodoni is an excellent example font as nearly everyone can bring to mind the extreme contrast of its thick and thin strokes. The Frenchman Didot and the Italian Bodoni were the first to bring this design style to the public. Its major attributes align with the Romantic period's aesthetics.

> Romantic letters can be extraordinarily beautiful, but they lack the flowing and steady rhythm of the Renaissance forms. It is that rhythm which invites the reader to enter the text and read. The statuesque forms of Romantic letters invite the reader to stand outside and look at the letters instead. (Bringhurst, 2004, p. 130)

The major characteristics of modern fonts are extreme contrast between thick and thin strokes, clean, unbracketed, hairline serifs, and a completely vertical axis. These fonts have an almost mechanical look because of their precise, sharp, and clean appearance. They also possess an elegance that complrments the time period they emerged in. Modern fonts are often used as display fonts and can sometimes be used for text, though very carefully.

Examples of modern fonts include Fenice, Zapf Book, New Caledonia, Bodoni, and Didot.

Egyptian

Figure 3.15 Example of Egyptian type

Egyptian is also known as slab serif, square serif, or mechanical (see Figure 3.15). This category of font was created in England in the 1880s — a design expression of the industrial revolution. The category was named Egyptian because of the popularity of all things Egyptian after Napoleon's return from a three-year Egyptian expedition. The name of the style has nothing to do with any element of Egyptian culture. The style was created initially for display copy, but over the centuries, fonts like Clarendon have become popular for setting text blocks because they contain the quality of objectivity and yet still feel traditional.

Examples of Egyptian fonts include Officina Sans and Officina Serif, Clarendon, and every typewriter font.

Sans Serif

Figure 3.16 Example of Sans Serif

Sans serif fonts have existed since ancient times, but it was only in the late 19th century that font designers began to consider removing serifs and letting the letterforms stand on their own (see Figure 3.16). These fonts were initially considered appropriate only for titling and display purposes, and only became text fonts in the hands of the 20th-century modernists. The first sans serif forms were created on the early humanist and old style calligraphic forms, but eventually the forms were influenced by objective modernist principles and geometry.

Examples of sans serif fonts include Univers, Helvetica, and Akzidenz-Grotesk.

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3.3 Compositional Principles — Strategies for Arranging Things Better

Alex Hass

> We have many words for the frustration we feel when an interface isn't directing us to what we need to know. Loud, messy, cluttered, busy. These words. . . express our feeling of being overwhelmed visually by content on a screen or page. We need them to express how unpleasant a user experience it is to not know where to direct our attention next. (Porter, 2010, para 1)
>
> If everything is equal, nothing stands out. (Bradley, 2011)

The proper composition of visual elements generates not only visual stability, it enhances mood through composition and generates order that prevents visual chaos. Designers use compositional rules in their work to make the reader enter their work and experience a design environment that is calm yet exciting, quiet yet interesting. A magazine designer, for example, creates a grid and applies an order to the typographic elements creating a comprehensible hierarchy. This design system is interpreted in different ways, in pages and spreads, issue after issue. If the organizational system is versatile and planned with thought and depth, it can be used to produce unique and exciting layouts that remain true to the rules determined for the overall system initially designed. Organizational principles create a framework for design without determining the end results.

Compositional rules can be used to generate content as well as organize it. The Bauhaus artist and designer Laszlo Moholy-Nagy created a series of paintings by calling in a set of instructions to a sign painter using the telephone. Here is his account of the experience, written in 1944:

> In 1922 I ordered by telephone from a sign factory five paintings in porcelain enamel. I had the factory's color chart before me and I sketched my paintings on graph paper. At the other end of the telephone, the factory supervisor had the same kind of paper divided in to squares. He took down the dictated shapes in the correct position. (It was like playing chess by correspondence). (Moholy-Nagy, 1947, p. 79)

Designing visual elements into a strong composition is a complex endeavour on its own, but increasingly designers are being asked to create vast compositional systems that other people will implement. Much like Laszlo Moholy-Nagy, designers need to be able to make strong compositional systems and also convey how their systems work, how to apply their rules, and how to apply them so they retain a relevant freshness.

Alignment

Figure 3.17 Alignment

Alignment refers to lining up the top, bottom, sides, or middle of a text, composition, or grouping of graphic elements on a page. Often a design composition includes a grid where the alignment of text blocks is dictated by the design of the columns of the grid (see Figure 3.17).

Typographically, horizontal alignment includes flush left (also called left justified or ragged right), flush right (also called right justified or ragged left), centred, and fully justified. Vertical alignment in typography is usually linked to baseline alignment. A baseline grid exists in digital software that is meant for layout of type and is the invisible line where font characters sit.

Contrast

Contrast is a visual device that increases the special character of both elements that have been paired. Contrast assists composition by creating focal points, and adds energy and visual range to a composition. Using contrast enables us to distinguish the qualities of one object by comparing differences with another. Some ways of creating contrast among elements in the design include the use of contrasting colours, sizes, and shapes. Johannes Itten, a design instructor and artist at the Bauhaus focused his research on the concept of contrast in both composition and colour. Itten's list of contrasts can be applied to both the composition and the atmosphere of a design work. His list includes these pairings: large/small, hard/soft, thick/thin, light/heavy, straight/curved, continuous/intermittent, much/little, sweet/sour, pointed/blunt, light/dark, loud/soft, black/white, strong/weak, diagonal/circular. No design makes use of only one kind of contrast, but usually one dominates the others.

Colour Contrast

Johannes Itten also worked with contrast in his seminal theory of colour and determined that there are seven kinds of contrast.

1. *Contrast of hue* occurs when a hue or colour is separated by being outlined in black or white lines. White lines weaken the 'strength' and appearance of the colour and the colours around the white lines seem darker. In contrast, a black line around a colour strengthens the appearance of the colour, while the colours around the black lines appear to be lighter.
2. *Light-dark contrast* is the contrast between light values and dark values.
3. *Cold-warm contrast* refers to the contrast between cool and warm colours. Warm colours are the red, orange, and yellow colours of the colour wheel, while cool colours are blue, green, and purple.
4. *Complementary contrast* is the contrast between colours directly opposite each other on the colour wheel.
5. *Simultaneous contrast* occurs between two colours that are almost complementary. One colour is one section to the left or right of the complementary colour of the other.
6. *Contrast of saturation* refers to the contrast between intense colours and tertiary or muted colors. Muted colours appear duller when placed next to intense colours, and intense colours appear more vivid when next to a muted colour.

7. *Contrast of extension* refers to the contrast between the area of one colour and another. Different areas of one colour are needed to balance another.

For text, contrast is achieved by using varied colours, serif and sans serif, type styles that are not often paired, or type in place of an image. As contrast in elements diminishes, the elements begin to feel similar, and the level of visual interest decreases.

Emphasis

A focal point in a composition draws the eye to it before the eye engages with the rest of the visual information. This is called *emphasis* and is achieved by making a specific element gain the attention of the eye. Emphasis is created in graphic design by making only one focal point and clearly emphasizing it by placing the elements on the page in positions where the eye is naturally drawn to the proper entry into the work. Designers rely on additional compositional principles to support the hierarchy of a composition such as contrast, repetition, or movement.

Designers use emphasis to assist viewers in identifying the relative importance of each element in a composition. Emphasis is strongly linked to visual hierarchy. Both emphasis and visual hierarchy create order for the viewer, allowing the eye to see the first element of importance, then the second, then the third, and so on. Graphic elements gain or lose emphasis by changing in size, visual intensity, colour, complexity, uniqueness, placement on the page, and relationship to other elements.

Movement

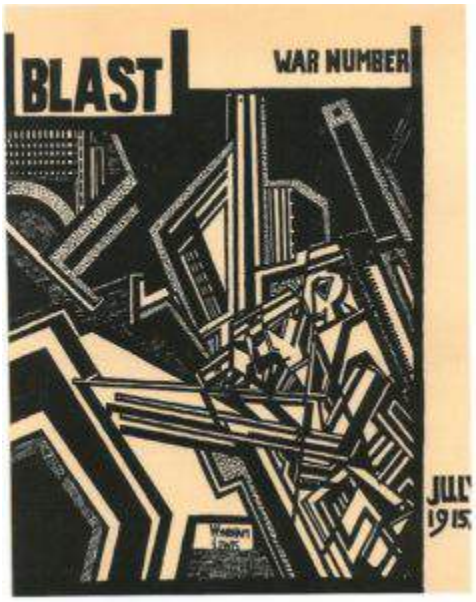

Figure 3.18 Example of movement

Movement is made by creating visual instability — like motion in a photograph that blurs the image, as shown in the example in Figure 3.18. Creating the illusion of movement photographically or artistically is not difficult because a blur translates into movement in the mind of the viewer. However, it is not

the only option for a designer. A composition can also achieve movement if the graphic elements are arranged in a way that directs the eye to move in a specific direction — usually by creating a diagonal that takes the eye up to the right corner (forward motion) or down to the left corner (backward motion). Movement can also be created using overlapping planes that imply depth and distance by becoming progressively smaller and lighter in tone (mimicking depth). Using typography as a visual medium is also an option. Overlapping the text blocks and/or sentences effectively creates both depth and movement (though it destroys legibility). David Carson is a designer who often uses this technique to create movement in his work.

Scale

Varying scale (size) is one of the major tools in the designer's toolbox. Changing scale is important on two levels. The first is purely compositional — a composition needs variety in the size of its elements to be dynamic and effective. If all the elements have the same visual weight, the composition will be flat. Another aspect to varied scale is conceptual. If a design visually distorts the size relation of one element to another, the viewer is instantly engaged in discovering why. This is a great method to engage the viewer and add a twist to the message embedded in the design. A great example of this is the 'think small' ad campaign of the 1960s for Volkswagen Beetle.

The series is witty and engaging and plays on how we perceive size. This distortion is witty and playful, and presents smallness as desirable. Subtle scale differences do not make much visual impact, but large ones are very dramatic. The concept and context of a project should determine the relationship of scale differences for a composition. Large differences in scale are suited to dramatic and energetic design content, while smaller differences in scale are appropriate for professional and institutional content.

Proximity and the Gestalt Theory of Visual Relationships

Proximity of elements is part of Gestalt theory, which is a framework of spatial relationships developed in the 1920s by the German psychologists Max Wertheimer, Wolfgang Kohler, and Kurt Koffka. The term Gestalt means *unified whole*, and points to the underlying conceptual structure of this framework. Gestalt works because the mind seeks to organize visual information. A composition created using Gestalt principles predetermines how each of the elements within it interacts with the others spatially. In this system of relationships, close proximity of objects, regardless of shape, size, or content, indicates a connection. There are six basic Gestalt principles: (1) similarity, (2) continuation, (3) closure, (4) proximity, (5) figure/ground, and (6) symmetry and order.

Similarity

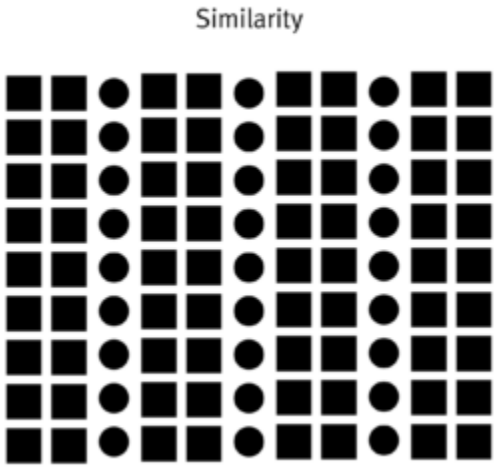

Figure 3.19 Similarity

When visual elements have a similar shape or look as one another, a viewer will often connect the discrete components and see a pattern or grouping (see Figure 3.19). This effect can be used to create a single illustration, image, or message from a series of separate elements. Similarity of medium, shape, size, colour, or texture will trigger a sense of similarity. The sense of grouping will be strengthened or weakened by increasing or decreasing the commonality of the individual elements.

Continuation

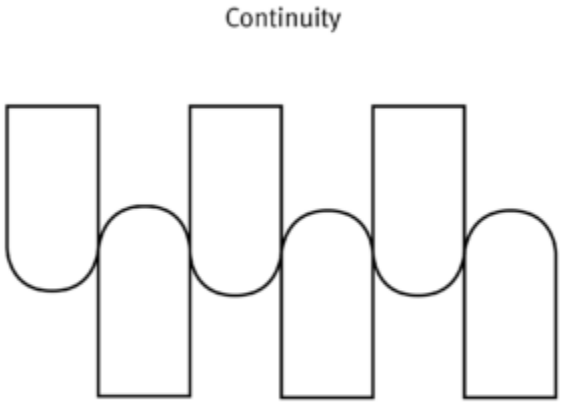

Figure 3.20 Continuity

Continuation is the tendency of the mind to see a single continuous line of connection rather than discrete components (see Figure 3.20). The eye is drawn along a path, line, or curve, as long as there is enough proximity between objects to do so. This tendency can be used to point toward another element in the composition, or to draw the eye around a composition. The eye will continue along the path or direction suggested by the composition even when the composition ends, continuing beyond the page dimensions.

Closure

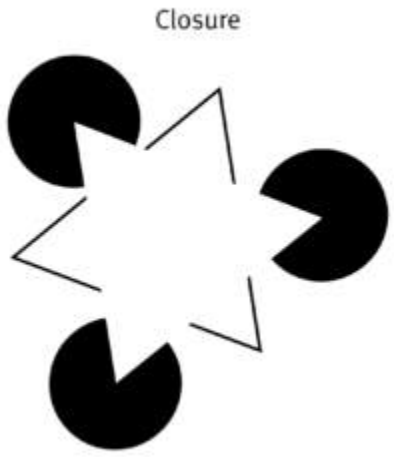

Figure 3.21 Closure

Closure is a design technique that uses the mind's tendency to complete incomplete shapes (see Figure 3.21). The principle works if the viewer is given enough visual information to perceive a complete shape in the negative space. In essence, the mind 'closes' a form, object, or composition. In the example above , the triangle is formed by the viewer's mind, which wants to close the shape formed by the gaps and spaces of the adjacent circles and lines. The partial triangle, outlined in black also hints at the missing shape.

Proximity

Figure 3.22 Proximity

Proximity is an arrangement of elements that creates an association or relationship between them (see Figure 3.22). If individual elements are similar, they will probably be perceived first as a whole and second as discrete components. If, like the example above, some of the components form to create a large ‘whole,’ similar elements positioned away from the main shape will also be associated with the large shape. In this case, the viewer interprets them as falling off or away from the main shape. The shapes used do not have to be geometric to create the effect of proximity. Any components have a strong commonality in shape, colour, texture, size, or other visual attribute can achieve proximity. Proximity can also be achieved with dissimilar shapes and textures if cleverly and conceptually composed.

Figure/Ground

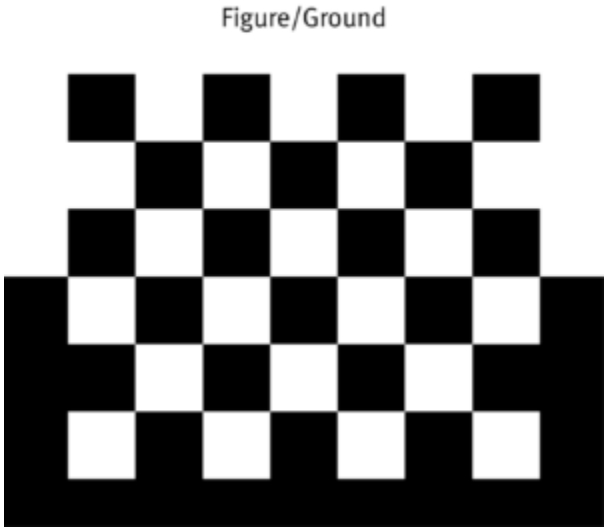

Figure 3.23 Figure/Ground

Figure/ground was discussed earlier, but it is part of Gestalt theory, so we present it here again. This principle describes the mind's tendency to see as two different planes of focus, information in both positive and negative space (see Figure 3.23). It works if those spaces are suggestive enough in their composition.

Symmetry and Order

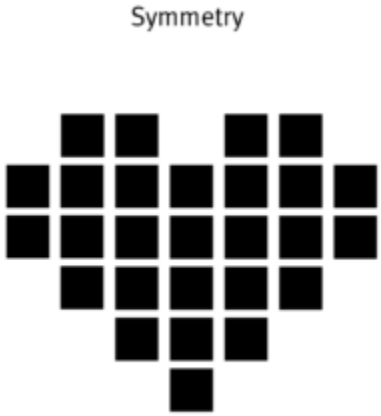

Figure 3.24 Symmetry

Symmetry and order follow the premise that a composition should not create a sense of disorder or imbalance (see Figure 3.24), because the viewer will waste time trying to mentally reorder it rather than

focus on the embedded content. The photographic example in Figure 3.25 is composed symmetrically and allows the viewer to concentrate on the figure in the centre. Achieving symmetry in a composition also gives the composition balance and a feeling of harmony.

Figure 3.25 Example of symmetry and order

Rhythm

Rhythm is integral to the pacing of a design composition and is also necessary for creating a pattern, as used in the example in Figure 3.26. The pacing of a repeating motif or element at regular or irregular intervals within a design determines the energetic quality of a composition; it also creates a consistent and unifying backdrop for the introduction of new elements.

Rhythm is the effect produced in a magazine or book by varying the placement of elements within the grid structure. The changes in the density of elements and visual tones of the spreads translate into a rhythmic visual energy as the energy of each page grows or shrinks. Rhythm is the glue that connects one page to the next; it reveals recurrent themes and creates movement, tension, and emotional value in the content. When viewers understand the rhythm of a book, a magazine, or a website, they will also appreciate the variations that break with or punctuate the rhythm and create interest, change, or tension.

Figure 3.26 Example of rhythm

Repetition

Repetition creates visual consistency in page designs or in visual identities, such as using the same style of headline, the same style of initial capitals, and the same set of elements, or repeating the same basic layout from one page to another (see Figure 3.27).

Excessive repetition, however, creates monotony. This usually leads to viewer boredom and dull, uninteresting compositions for the designer. Be sure to create a design system that allows the repetitions within it to be lively and interesting page after page. The example above uses a simple set of rules, but because the rules allow for colour and compositional changes, each discrete component is as interesting on its own as it is within the whole. If you cannot avoid excessive repetitions, try to add some visual breaks and white spaces where the eyes can rest for a while.

Figure 3.27 Example of repetition

Balance

Balance and symmetry are important design qualities because they are deeply embedded in human DNA. Because our bodies are symmetrical, we have a strong association and satisfaction with centred, symmetrical design. Balancing visual elements compositionally calms the tensions and grounds the design (see Figure 3.28). This is important if you wish to convey a sense of stability to the viewer. When we look at a design, we use our innate sense of what constitutes ‘right balance’ to assess its stability. If that stability is missing, we feel tension, which can counteract the core of the message. Centred design compositions work very well for stable, security-inspiring content, but what about content that demands attention, or tension, or excitement?

When a centred (or stable) composition is not desirable, developing an asymmetrical composition is the best strategy. Asymmetry has been explored in graphic design for the last 150 years, and designers continue to discover new strategies that feel fresh. Asymmetry has no empirical rules but is guided by balancing the distribution of main elements around the space of a composition in an unexpected way. Contrast and counterpoint are the main tools of composition in asymmetry — large shapes balance small ones; intense colours balance neutrals. Creating asymmetrical design is not easy because there are no firm rules to follow, but it is exciting to create and exciting to see for exactly the same reason.

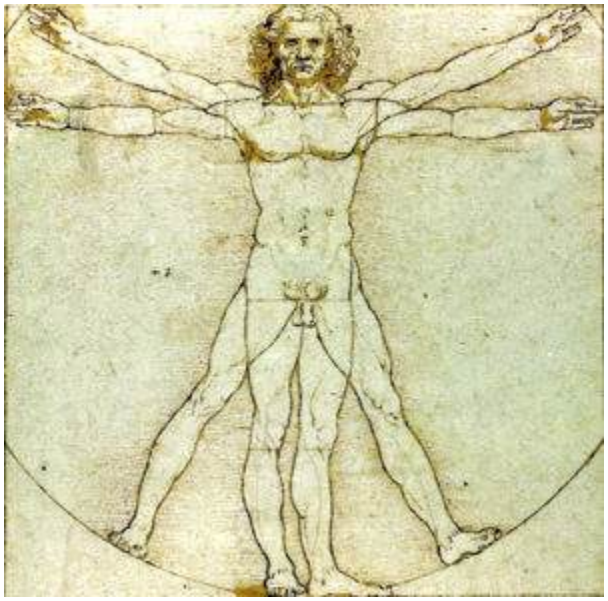

Figure 3.28 Example of balance

Hierarchy: Dominance and Emphasis

Simply put, hierarchy is applying an order of importance to a set of elements. Hierarchical order is apparent in every facet of our lives and is a defining characteristic of our contemporary culture. Hierarchy can be very complex and rigorous — an instruction manual is a good example of this. It can also be uncomplicated and loose. Hierarchy in composition is conveyed visually through variations of all the elements — size, colour, placement, tonal value, and so on (see Figure 3.29).

Figure 3.29 Example of hierarchy

Graphic design does not always embrace hierarchy. There are some messages that are more suited to visual anarchy and chaos (Punk design is a good example). These projects often connect to an audience

by experimenting with, and breaking free from universal rules of visual structure. It is important is to match the structure of the composition to the needs of the project.

Typographic hierarchy is very important in design. A body of text is made more comprehensible by imposing order through a system of titles, subtitles, sections, and subsections. Hierarchy is created when the levels of the hierarchy are clear and distinguishable from one another. Subtle signs of difference are not effective. Typography acts as a tonal voice for the viewer, and must create clear variation in tone, pitch, and melody.

Hierarchy is usually created using similarity and contrast. Similar elements have equality in typographic hierarchy. Dominant and subordinate roles are assigned to elements when there is enough contrast between them. The bigger and darker an element is, the more importance it has. Smaller and lighter sizes and tones imply lesser importance.

Every hierarchy has a most important level and a least important level. The elements that fall between the two are ranked according to size and position. However, if you subdivide the text with too many levels, the contrast between different levels will blur their differences in the hierarchical order.

A good strategy to follow with text design is to apply three levels of typographic hierarchy.

Title

The function of a title is to attract the reader to the content of the text block. Often, the title is visually ‘flavourful’ and possesses a strong visual dynamic and energy.

Subtitle

Second-level typography gives the reader the ability to distinguish between types of information within the text block. This level of type includes subheads, pull quotes, captions, and anything else that can help detail and support understanding of the text-block information.

Text block

The text block is the content. As opposed to the ‘display’ function of the title and subtitle, the function of the text block is to make the content legible and easy to digest visually. Readers should be able to decide if they want to read this level based on primary (title) and secondary (subtitle) type levels.

Typically, a typographic hierarchy will convey information from general to specific as it progresses from title to text block. The general points presented in the title will be the most important and will be seen by most everyone. Think of how a newspaper is scanned for interesting news items: If readers are interested in the title, they may choose to read more detailed and in-depth information in the associated text block.

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3.4 Organizational Principles

Alex Hass

Compositional organization is complex, but even more so when applied to typography. Typography is a complicated medium to work with as it contains two levels of information (display and content), and requires its components to be read in proper sequence with proper emphasis, good legibility, and strong contrast to the substrate. Many elements need to be organized to allow the reader a seamless experience when reading the content. Designing with type requires adept handling of the hierarchy, refining and designing the display elements for focal emphasis and also refining the quiet details of the text block so it sits perfectly and quietly in its space.

Think of these organizational systems as ‘large picture’ constraints. Constraints (rules) allow a designer to focus on the other aspects of a project. Designers make myriad decisions about concept, style, visuals, form, font, size, spacing, colour, placement, proportion, relationships, and materials. When some factors are determined in advance, the designer is able to spend time with the other parts of the project. A well-defined constraint can free up the thought process by taking some decisions off the table. The following eight organizational systems cover composition for type (but can also be applied to general composition), including the traditional ordering system of the grid.

Grid

A grid is a network of lines that structure the placement of elements and create relationships between them. A grid divides a design space into vertical and horizontal divisions. The grid is a bridge between a design rationale and the beginning of implementation for each project, converting a concept into a structured space. It is an exceptional tool for composing, arranging, and organizing every kind of visual element. The grid usually works invisibly in the background, but it can become an active, visible element as well. Designers use grids in a variety of ways. They can be very disciplined about adhering to their grid structure from the beginning of a project, or use it as a starting point for composition and order.

Grid systems create a formal composition in comparison to more casual compositional approaches like transitional or random structures. Grids are often used in publication and web design because they introduce consistency and guide hierarchy. Consistent margins and columns create an underlying structure that unifies the multiple pages of a document or website, and makes the layout process more efficient.

The plan for the grid comes from the content and concept of the design project. The objective in creating a grid is to set up the relationships between elements in a way that stays true to the concept. For instance, if your publication is a book of poetry, the grid must have generous amounts of negative space and generous leading. If, on the other hand, your publication is a daily newspaper, the spacing relationships cannot be so generous, and have to clearly show which article relates to which image. Hierarchy of information must be very clear as well, and should reveal which news item is most important and which is least important. A well-made grid will naturally allow the designer generous scope for variation in

image style, text size, and graphic style. Often, a grid that is complex allows for some freedom where the designer can introduce a new element or effect.

A grid activates the entire surface of a project by making all of it available for active elements. It helps create both stable symmetrical and dynamic asymmetrical compositions. By breaking down space into smaller units, grids encourage designers to leave some areas open rather than fill up the whole page.

Types of Grids

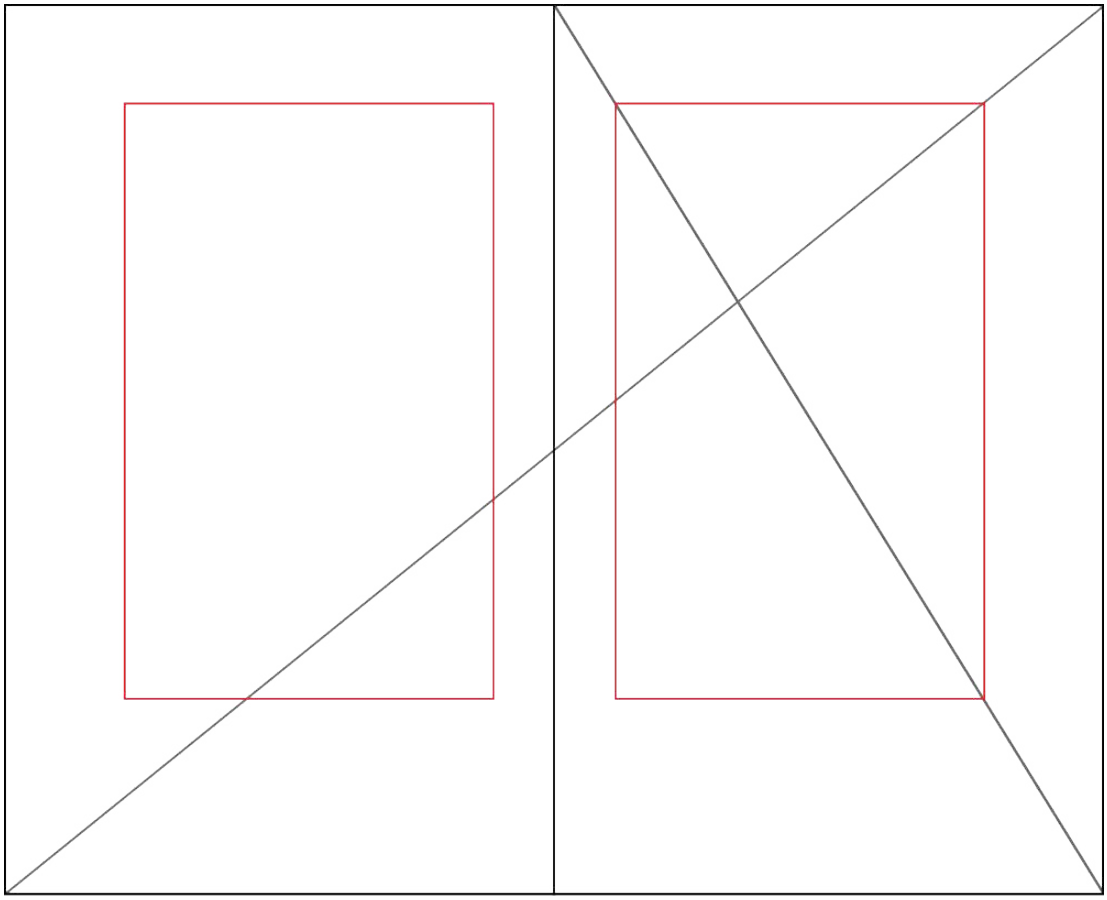

Figure 3.30 The golden section

The golden section is also known as the golden ratio, golden mean, or divine proportion, and it is found in mathematics, geometry, life, and the universe — its applications are limitless (see Figure 3.30)

The golden section is a ratio — a relationship between two numbers — that has been applied as an organizational system in art, design, and architecture for centuries. Expressed numerically, the ratio for the golden section is 1 : 1.618. The formula for the golden section is a : b = b : (a+b). In other words, side a is to side b as side b is to the sum of both sides.

Graphic designers use the golden section to create grids and layouts for websites and books. Photographers use it to compose the focal point of an image and also to compose the elements found in an image.

Single-Column Grid

Figure 3.31 Single-column grid

A single-column grid is an excellent approach if the content a designer is working with is formatted in a simple manner (see Figure 3.31). Content that is appropriate for a single-column grid consists of main text for the text block, a few levels of display type, possibly some images, and finally page numbers.

The main column of this style of grid must sit properly on the page, held in place by the negative space that surrounds it. To determine the right amount of negative space on the top, bottom, and sides of the page, a designer usually considers facing pages as a spread. In books and magazines, the two-page spread, not the individual page, is the main unit of design. The designer determines the right amount of negative space on the top and bottom, gutter (inside margin), and outside edge. The spread is often symmetrical, and the pages mirror one another.

Multi-Column Grid

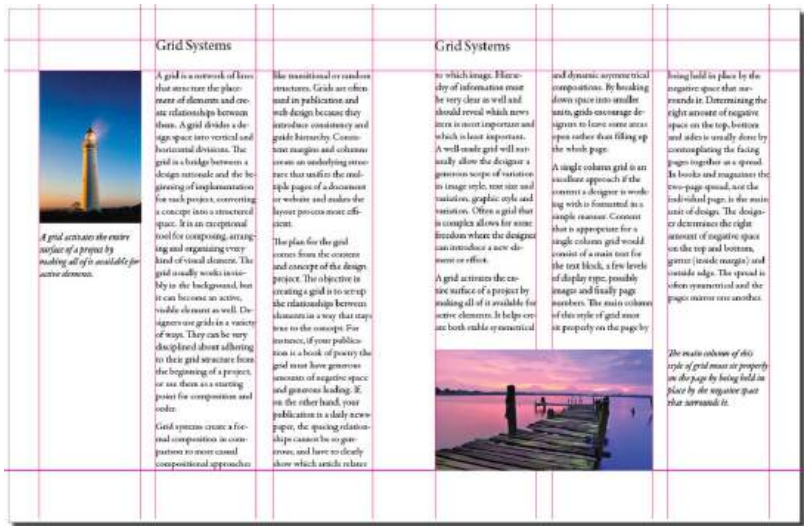

Figure 3.32 Multi-column grid

When a designer creates a grid for a document that is complicated, he or she may use multi-column grids because they allow for a complex hierarchy and provide more options for integrating text and visuals (see Figure 3.32). The more columns you create, the more flexible your grid will be. You can use the grid to articulate the hierarchy of the publication by creating zones for different kinds of content. The columns act as visual units that can be combined or kept separate. A photo can span several columns or be reduced to the width of only one. A text can also occupy a single column or span several.

Hang Lines

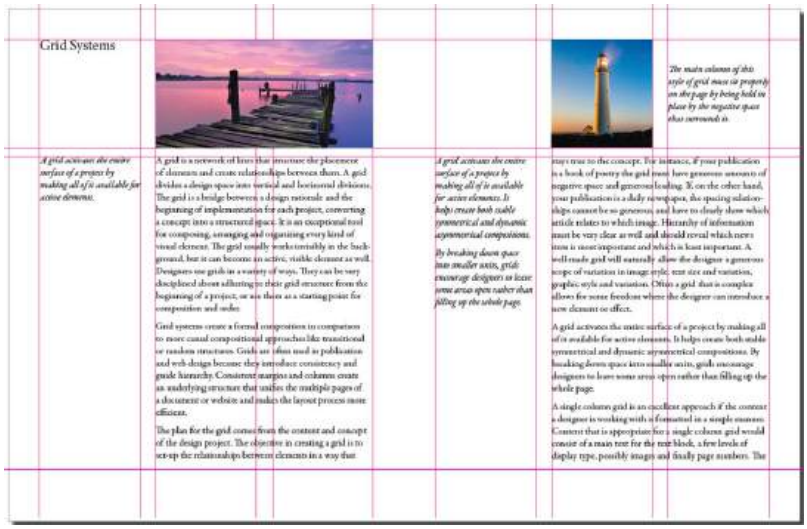

Figure 3.33 Hang lines

In addition to creating vertical columns in a grid, you can also divide the page horizontally. Often, a designer determines the placement of hang lines (see Figure 3.33) by applying the rule of thirds

(breaking up the horizontal plane into three equal parts). This compartmentalization allows the designer to reserve certain sections for images and others for the text block.

Modular Grid

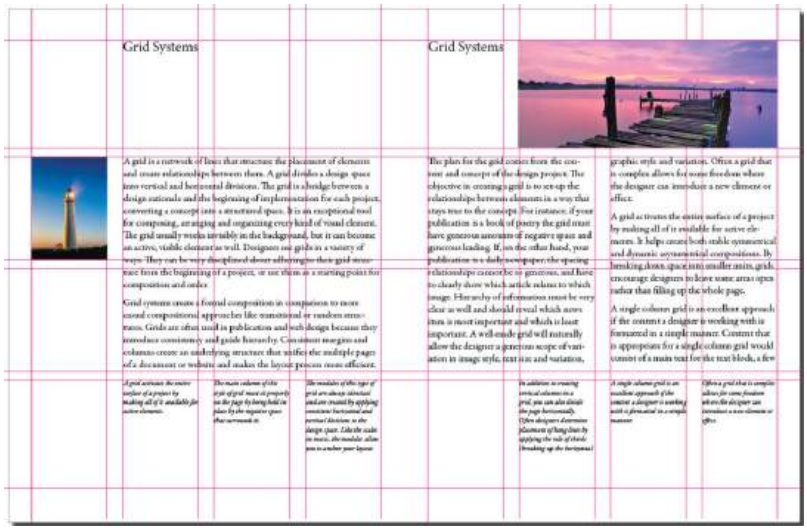

Figure 3.34 Modular grid

The modules of this type of grid are always identical and are created by applying consistent horizontal and vertical divisions to the design space. Like the written notes in a musical score, the modules allow you to anchor your layout elements and typography to a specific rhythm. With a modular grid, the horizontal guidelines are tied to the baseline grid that governs the whole document. Baseline grids serve to anchor most of the elements to a common leading (typographic line spacing). See Figure 3.34.

Baseline Grid

A baseline grid is the horizontal grid that determines where all of the type will sit. You can also use it to determine the placement and edges of your visual and graphic elements. To create a baseline grid, determine the right font, size, and leading for your text block, then go to your baseline grid settings (found with the other grid preferences) and change the baseline grid default (usually 12 pt) to the leading you will be using in your text block.

Axial

The axial system has a simple premise — all elements are arranged on either side of an axis or line. You can centre the axis itself in the composition or, for a more energetic asymmetrical composition, place the axis off centre to either the right or left. This compositional strategy creates a dynamic negative space on the opposite side. To create a more complex composition, designers often employ an axial system combined with another — like the radial or dilatational system (see below). They may also use double-axis compositions with the axes either parallel to each other, or intersecting to create a strong focal point. There are many instances of the axial system in nature — tree trunks, roots, and vines are good examples.

Like these organic examples, an axis does not need to be a straight line — it can be curved, zigzag, or circular.

Modular

Modular organization is a compositional method that utilizes rigour (by constraining the shape) and freedom from structure (modules can be any size and placed anywhere in the space). Modules can also be uniform and contained within a structure (a grid). A module is a fixed element used within a larger system or structure. For example, a pixel is a module that builds a digital image.

Bilateral

The bilateral system is based on mirrored symmetry and is therefore both classic and ubiquitous. Because of its predictability, it is a challenge for designers to work with. Nature exhibits many examples of bilateral composition — the bodies of mammals, the points of a snowflake, and the fractal symmetry of plants are all quickly understood, appreciated, and then dismissed by the viewer. To create a composition based on the bilateral system, a designer must make some part of the composition unusual. The designer can achieve this by moving the axis to a diagonal, off-centre location, which allows the negative space on either side of the bilateral composition to be varied. A second method is to introduce a double axis: the designer uses two columns of bilateral information and varies the size of each.

Radial

The radial system takes its name from the sun — all elements are arranged like rays coming from a central focal point. This is a dynamic compositional strategy as it references dynamic action. Examples of the radial form from the natural world, such as explosions, flowers, spiders, stars, and so on, are all exciting and dynamic. Much like it is difficult to handle the natural objects, reproducing a radial composition is not that easy. There are problems with legibility unless type is very carefully placed and scaled. Every line of type starts and ends in a different place, so continuity is also hard to control. For example, a designer may take a traditional approach so the text reads from top to bottom, or an inverse approach so the text reads from bottom to top. Arranging the text on either side of centre may also be effective. It is important to try placing the type in different positions and in different relationships until it works with the composition and is easy to read.

As in the organizational systems we have discussed, designers can add radial points for a more complex composition or combine a radial system with one that adds stability, such as a grid, axial, or modular system.

Dilatational

Dilatational systems mimic the look of still water when a pebble is dropped into it, creating rings of greater and greater size as they move away from the centre. Like the radial system, this composition has

a strong focal point, but unlike the radial system, the composition creates rings, not rays, that expand from the centre. Other examples of this system are the iris of the eye or representations of sound waves.

Random/Spontaneous

Creating a composition that does not follow any compositional principle is not as easy as it sounds. Finding examples of randomness is also one of the most difficult exercises for design students. Random design does not follow any rule, method, direction, or pattern. If a project calls for randomness, it is best to start with materials that are conducive to spontaneity like Jackson Pollock's paint throws. Allow the elements that fall to organize themselves naturally — eventually, a dynamic and fresh composition should emerge. Random compositions exhibit visual qualities that are not patterned, aligned, or horizontal. Instead, they tend toward compositions that exhibit overlapping, cropping, angling, and textures.

Transitional

The transitional typographic system is defined by the layering of lines of text into informal textured planes and shifting tonal bands. The shapes and bands created with this layering approach are not aligned with one another, and create an overall organic atmosphere. This visual approach is often used in Post Modern design where the clear legibility of text is not as important as the visual atmosphere of the design. Text planes in Post Modernist works point the viewer to the main theme of the message rather than articulate the message in clean, concise text arrangements.

Compositions using the transitional approach have a light, airy look that abstractly imply cloud formations or wood grain patterns rather than solid concrete shapes created by using the grid or axial systems. A transitional composition has lively and active negative space, and can create an excellent ground for a vital focal point if it is in sharp contrast to the rest of the composition.

Attributions

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3.5 Summary

Alex Hass

Exploring the design possibilities that the organizational systems discussed in this chapter possess is an endless undertaking. Once these systems are innately understood individually, a designer can begin to play with layering two systems within one design project. Combining contrasting systems often works well. For instance, an axial system combined with a radial system tempers the axial system's linear focus and anchors and diffuses the rays emanating from the radial shapes. A grid combined with a dilatation system gives the composition both vertical and horizontal structure that is softened by the rounded shapes. Organizational systems give the designer ways to distribute words or images within a structure while allowing negative space into the centre of the design space.

Compositional strategies are design constraints. The definition of a design constraint is to apply or impose limitations on the elements or design of a system. The compositional strategies (systems) discussed above are in fact design constraints, and they should be understood as parameters that assist the designer in the design process rather than as restraints that limit the designer's creativity. Parameters are necessary in every visual system. Applying a visual organizational system also allows the designer to focus on the message and the details of a design project rather than on the structure of the composition that holds the work together. Visual systems create visual unity.

Questions to consider after completing this chapter:

1. Name the design principle that distorts realistic relationships for visual effect and emphasis.
2. Name the three building blocks of design that pertain to form.
3. Describe the eight organizational systems that apply to typography.
4. What are two typographic categories?
5. How many levels of visual hierarchy are needed for hierarchy to exist?

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Suggested Readings

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Chapter 4. Colour Management in the Graphic Technologies

4.1 Introduction

Alan Martin

Learning Objectives

- Define the relationship between the observer, the illuminant, and the object in colour appearance
- Describe the physics of light and colour
- Recognize how the eye-brain system affects colour perception
- Explain the components of the CIE colorimetric system, specifically the Lab colour space and Delta E measurements
- Differentiate between device specific and device independent colour models
- Describe the role of colour management in achieving consistent appearance between proofing cycles and final printed production
- Define an ICC profile and describe its application in digital imaging
- Use a spectrophotometer to capture colour data from industry standard targets as the first step in profile creation
- Create ICC colour profiles for standard colour transformations
- Calibrate, profile, and colour manage an LCD monitor
- Describe the application of display, input, and output colour profiles in the electronic prepress workflow
- Apply colour profiles in a variety of industry-standard applications to achieve desirable results for graphic reproduction
- Combine source and destination profiles into a single device link profile for device specific colour transformations
- Apply colour management with ICC or device link profiles in the colour control module of Prinergy's Process Plan for proofing output
- Combine surplus spot colours into a single separation for successful printing-plate production

A knowledgeable application of a colour-managed **workflow** is required for the successful and predictable generation of all colour content in print production and related graphic technologies.

This process requires familiarity with the fundamental colour science that drives the practical steps of **colour profile** creation and colour profile use in our industry-standard software applications. From colour science, we can progress to an examination of the tools and detailed steps required for profile creation and successful profile use.

4.2 Colour Science

Alan Martin

The Colour Event

The first challenge in dealing with colour in graphic reproduction is to learn to think of colour as an *event* rather than as an attribute or characteristic of a particular object.

Colour is an event because it requires the participation of three components at a particular point in time to take place. In addition to the object, we require a light source and an observer. Only with the interaction of these three things — object, light, and observer — can we have a colour event or experience.

We need some help from three branches of science, physics, physiology and psychology, to understand how the object, light, and observer contribute to the colour event. If you like memory aids, you can use the acronym POLO to remind you of the three science P's and the object, light, and observer.

Object

The object and light fall under the domain of physics, while we need both physiology and psychology to describe the observer's role in the colour event.

The object's role is to interact with light, and the object can either reflect light back from its surface or transmit light through itself. Reflectance and transmission are the two potential interactions. The majority of objects are opaque, so most of the time we are dealing with the reflection of light. If an object is semi-opaque, and transmits a portion of light, we refer to it as translucent.

Light

Visible light is a tiny sliver of the total electromagnetic spectrum. The **electromagnetic spectrum** contains all forms of energy, ranging from kilometre-long radio waves at one end and progressing in shortening wavelengths down through microwaves, infrared waves, ultraviolet waves, X-rays, and finally, gamma waves with wavelengths of a subatomic dimension (see Figure 4.1).

Visible light is nestled in-between the infrared and ultraviolet range (see Figure 4.2). The progression from longest to shortest wavelength is from red (following infrared) to violet (preceding ultraviolet) in the 700 to 380 nanometre (millionths of a metre) wavelength distribution.

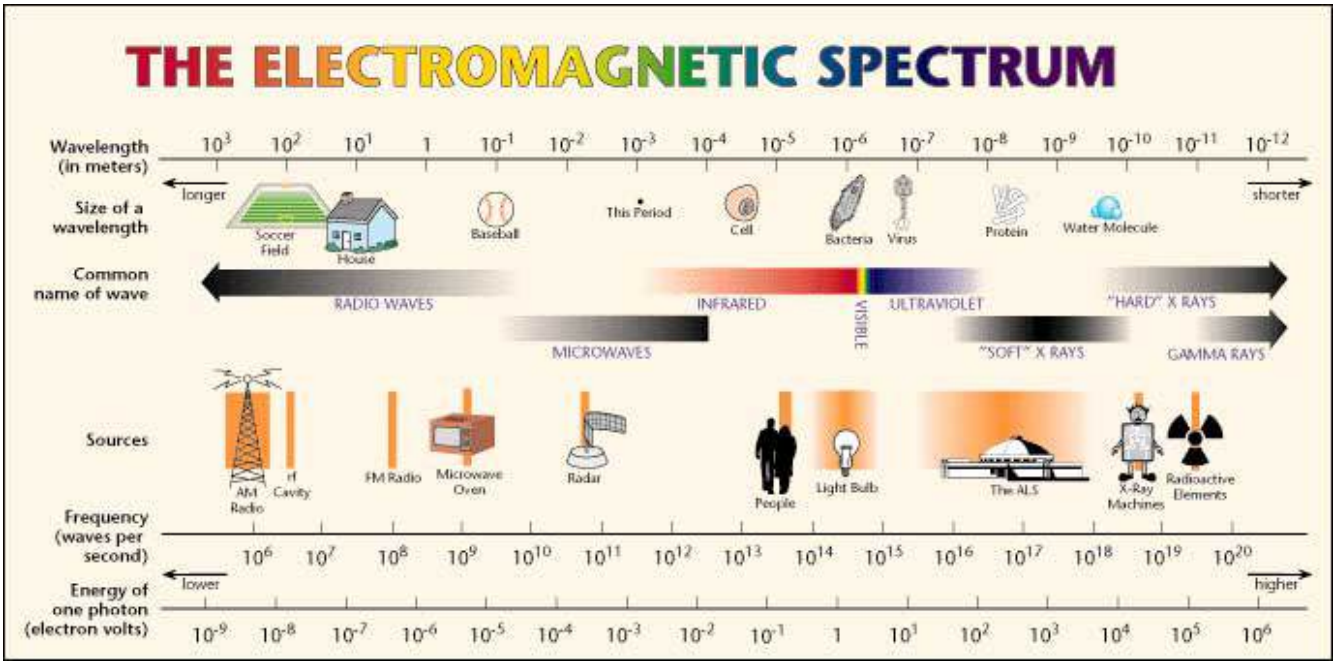


Figure 4.1 The Electromagnetic Spectrum

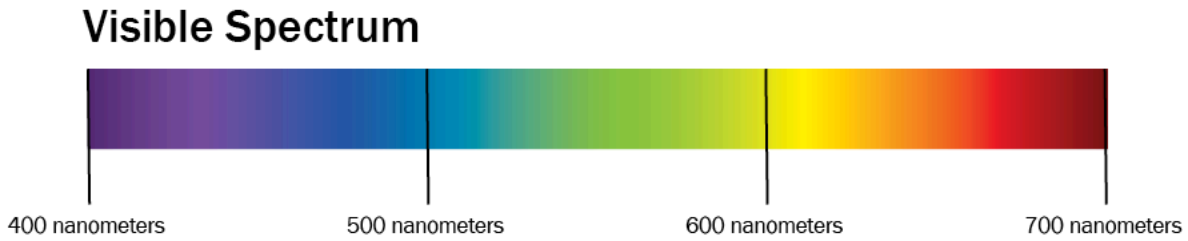


Figure 4.2 Visible Spectrum

We describe the **temperature** (relative warmness to coolness) of light in degrees Kelvin. Typical daylight ranges from 5000 to 6500 degrees Kelvin. We use the labels D50 and D65 to indicate daylight-viewing conditions at these temperature points.

Observer

The greatest complexity of the colour event occurs in the interaction with the observer. The science of physiology, the study of the human body's functions, provides half the story. Psychology, which provides insights about the function of the mind, completes the tale.

We begin with how our eyes, our optic systems, respond to light. Trichromacy and opponency are the key concepts.

Trichromacy

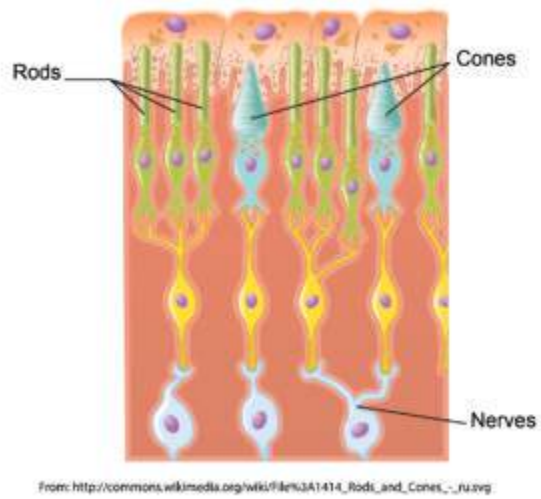


Figure 4.3 Rods and cones (adapted by Ken Jeffery)

We call it ‘visible’ light because it is the portion of the electromagnetic spectrum that our eyes are sensitive to. The two types of receptors in our eyes are cones and rods (see Figure 4.3). The cones respond to colour and the rods come into play in low-light situations when we see only varying shades of grey. There are three types of cones, each sensitive to approximately one-third of the visible spectrum. We characterize those segments of the spectrum as red, green, and blue, and this three-colour or trichromatic response by the cones is where all colour experience begins. Every colour we perceive comes from mixing varying volumes of the red, green, and blue signals from the three types of cones in our eyes.

The Additive Primaries

We refer to the *red*, *green*, and *blue* colour set (**RGB**) as the **additive primaries**. When we combine or add all three of these, we get the entire spectrum and, thus, white light. This is the primary colour set involved whenever we talk about the transmission of light, such as the display on a computer monitor, a tablet, or from a projector. For this reason, red, green, and blue are also referred to as the **transmissive primaries**.

The Subtractive Primaries

What happens when we project two of the three additive primaries on top of each other? This is the same as removing or subtracting one of the additive primaries from white light. Let’s start with red and green. Though not at all intuitive, if you have any experience with mixing paint or inks, the combination of red and green light produces yellow. Remember that we are adding light to light, so the production of a

brighter colour is to be expected. Continuing on: combining green and blue gives us a light blue that we call cyan, while the combination of red and blue produces magenta.

Since each of these colours is produced by subtracting one of the additive primaries from the full complement of white light, we refer to this colour set of *cyan*, *magenta*, and *yellow* (**CMY**) as the **subtractive primaries**. Each of the subtractive primaries acts as a filter for its complementary colour in the additive primary colour set. Cyan absorbs all red light, reflecting only green and blue. Magenta absorbs all green light, returning only red and blue; while yellow absorbs all blue light and reflects back only red and green. What colour would you see if you shone green light on a magenta patch?

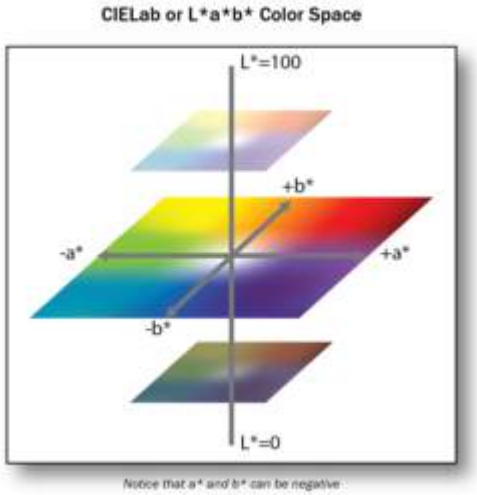
Just as we can produce any colour sensation in the transmission of light by mixing the appropriate quantities of red, green, and blue, we can produce the corresponding colour sensation when we put ink on paper by absorbing the necessary portions of the visible spectrum so that only the required amounts of red, green, and blue are reflected back. This is how cyan, magenta, and yellow work as our primary colours in the printing environment, and why we also call them the **reflective primaries**.

Opponency

The second half of the role that our human physiology plays in the observer's part of the colour event is the concept of opponency. Our eyes' tristimulus response (a response to the red, green, and blue portions of incoming light) is the input, but the interpretation occurs when we map that input to a location in a colour space determined by three axes of opposing sensations. We have a built-in colour map where we define our colour perception by identifying the perceived colour based on its degree of greenness to redness, blueness to yellowness, and darkness to lightness.

These three pairs of opposites — green-red, blue-yellow, dark-light — are the fundamental guide posts we use to position any colour we perceive on our internal colour map. These opponent colour pairs are exclusive colour entities, occupying opposing ends of the range of our interpretation. Unlike a yellowish-orange or a reddish-purple, we cannot imagine a colour having the properties of red and green or blue and yellow at the same time.

Lab Colour Space



Once the opponent nature of colour interpretation was understood, colour scientists were able to create a model colour space based on the opposing pairs. This is the **Lab colour space** (see Figure 4.4). The Lab variation of interest to us is officially called *CIELAB,* and all references in this textbook to *Lab* will mean *CIELAB.* Additionally, references to *L*, *a*, and *b* in this textbook are equivalent to the *L**, *a**, and *b** units of the CIELAB colour space. Each of the opposing pairs provides one axis of this three-dimensional colour space. L is the axis for darkness to lightness; a is the axis for greenness to redness; and b is the axis for blueness to yellowness. By providing a value for each of the L, a, and b attributes, a colour is accurately and uniquely located in the colour space. The tremendous utility of the Lab colour space is that it allows for the mathematical description of a colour in a non-ambiguous and meaningful way.

Psychology of Colour Perception

We come to the last of our three science P's: psychology. After the trichromatic response is passed to opponent interpretation in the physiology of our optic systems, the final engagement of colour perception occurs in our minds. This interaction complicates and potentially confounds our objective measurements, so it is critical to be aware of the typical issues that the filter of our mind brings to the arena of colour perception.

Colour Constancy

Colour constancy is how our mind adjusts our colour perception to discount or remove the effects of an overall colour cast due to a coloured illuminant. If we see someone wearing a yellow sweater illuminated by a bluish cast of light, we still 'see' a yellow sweater, even though we have no trouble identifying the colour in the sweater as green if it is isolated from the scene. In our mind, we remove the blue

constituents of all colours in the scene that we assume are coming from the tint in the light source. This behaviour is also known as **chromatic adaptation**.

The effect of **adjacency** is very similar to colour constancy. A colour placed next to a light colour appears darker than when that same colour is placed next to a dark colour (see examples in Figures 4.5 and 4.6). We make adjustments to our interpretation based on our assessment of the environment.

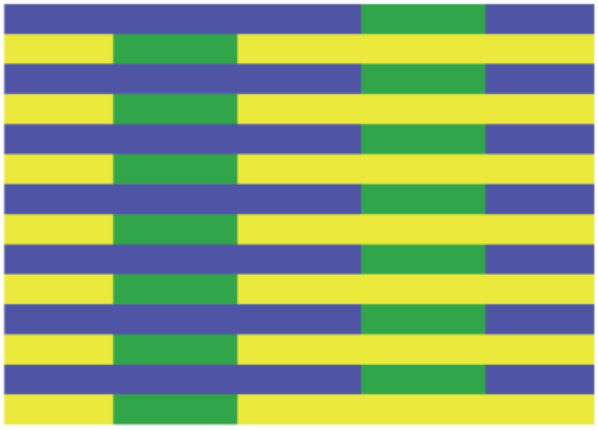


Figure 4.5 Both greens are the same colour

Figure 4.6 Both reds are the same colour

The effect of colour constancy provides a very important lesson in judging our success in colour matching: it is more important to preserve the overall colour relationships in our image than to focus on individual colour accuracy.

Memory Colours

In our mind's eye, not all colours are created equal. Due to their historical importance to our survival, we pay special attention to certain colours. Flesh tones, the blue of the sky, and the greens of grass are known as memory colours due to the additional weight they have in our hierarchy of colour.

We need to give these memory colours a priority when we evaluate our colour management efforts. If these key colours aren't right, then everything will look wrong.

The significant impact of our mind's contribution to colour perception enforces the requirement to take colour matching beyond the raw numbers we can extract from the physics and physiology of light's interaction with an object and our optic systems. The psychological components such as colour constancy and memory colours can only be accommodated by human intervention in a colour management system.

Media Attributions

- 1414_Rods_and_Cones_modified-01-01 by Kaidor, adapted by Ken Jeffrey © CC BY-SA (Attribution ShareAlike)
- Lab colour space by Ken Jeffrey
- green-yellow by Ken Jeffrey
- colours in the middle by Ken Jeffrey

4.3 Measuring Devices

Alan Martin

We measure light to provide the data needed to manage colour in a graphic production environment. There are three ways to measure light and three corresponding tools available to take those measurements: densitometer, colorimeter, and spectrophotometer.

Densitometre

To measure only the volume of light, we use a densitometer. The **densitometre** provides a known volume of light and then records what remainder of that light is returned to the device. A transmissive densitometer records how much light gets through a semi-transparent material such as camera film, and a reflective densitometer measures how much light has bounced back. The majority of densitometers in the print environment are reflective.

How does measuring the volume of light help us? Maintaining a consistent thickness of ink in printing is a very good way to control consistency and quality, and measuring the amount of light absorbed by the ink is a very accurate indicator of ink thickness.

Since our eyes have to function over a very wide range of brightness, we have a non-linear response to increasing volumes of light. That means it takes approximately 10 times the amount of light for us to experience one step in our perception of brightness. To match this behaviour of our eyes, the density scale is based on powers of 10, with each larger whole number representing one-tenth the volume of light of the preceding number. A density reading of 1.0 means that 1/10 of the original light has been reflected back. This is a typical reading for a process Yellow patch in offset lithographic printing. A density reading of 2.0 indicates that 1/100 of the original light is returned, while a density reading of 3.0 shows only 1/1000 coming back. Black ink is usually in the 1.7 density range, with cyan and magenta at 1.3 to 1.4.

Scanning or hand-held densitometers are typically found in the viewing station by a press. Densities are recorded when the printed sample matches the desired result and then ongoing adjustments to maintain the target densities keep the printing on target.

Colorimetre

Colorimetres mimic the three-colour response of our eyes by using red, green, and blue filters to measure the amount of light present in each third of the spectrum. They have built-in software to calculate Lab values based on what volume of red, green, and blue is returned from a sample. Colorimeters are particularly useful for calibrating and profiling monitors. Some well-known examples of colorimeters are the X-Rite ColorMunki or i1 Display devices.

Spectrophotometre

Spectrophotometres measure slices of the spectrum to produce a spectral ‘map’ of the light reflected back from a sample. Spectrophotometers are typically more expensive than densitometers and colorimeters but are employed because they can more accurately do the jobs of both devices. They work by recording the light at specific wavelengths over the wavelength range of visible light, and then by converting this spectral data to colorimetric and densitometric values.

While we are talking about measuring spectral values, it is important to note that we do not depend on identical spectral values to achieve matching colour experiences. Different spectral values can trigger the same volume of colour signals in our optic system and lead to matching colour perception. In fact, we depend on this phenomenon in graphic production in order for proofing devices to simulate the colour output of a printing press or for any two devices to be colour aligned. The ability of the CMYK (cyan, magenta, yellow, black) process colour set to mimic most of the colours in the world is also based on the fact that we can achieve a colorimetric match without having identical spectral values.

4.4 Lab Colour Space and Delta E Measurements

Alan Martin

The CIE (Commission Internationale d'Eclairage or International Commission on Light) is a scientific body formed by colour scientists in the 1930s that has provided much of the fundamental colour knowledge we possess today. Three core definitions provided by the CIE are the standard observer, the Lab colour space, and Delta E measurements. The latter two are particularly important for colour management.

The Lab Colour Space Revisited

In section, 4.2, we mentioned the Lab colour space as a natural outgrowth of understanding the function of opponency in human vision. It's comprised of three axes: *L* represents darkness to lightness, with values ranging from 0 to 100; *a* represents greenness to redness with values of -128 to +127; and *b* represents blueness to yellowness also with values from -128 to +127.

Notice that there are no negative values on the L axis as we can't have less than zero light, which describes absolute darkness. The L axis is considered **achromatic** meaning without colour. Here we are dealing with the volume rather than the kind of light. In contrast, the a and b axes are **chromatic,** describing the colour character and the type of light.

The standard two-dimensional depiction is of only the a and b axes, with a as the horizontal axis and b as the vertical axis. This places *red* to the right, green to the left, *blue* at the bottom, and *yellow* at the top. If you found our previous mnemonic aid of POLO helpful, you can use RGBY to remember the colour pairs. For correct placement, remember that *red* is on the *right*, and *blue* is on the *bottom*.

Colours are more neutral and grey toward the centre of the colour space, along the L axis. Imagine that equivalent values of the opposing colours are cancelling each other out, reducing the saturation and intensity of those colours. The most saturated colours are at the extremes of the a and b axes, in both the large positive and negative numbers. For a visual depiction of the Lab colour space, open the ColorSync application found in the Utilities folder of any Macintosh computer and view one of the default profiles such as Adobe RGB.

Now it's time to explore the practical application of this colour map for the comparative analysis of colour samples. We can't make any progress in evaluating our success in colour matching unless we have a frame of reference, some yardstick to determine how much one colour sample is similar or different from another. That yardstick is the Delta E measurement.

Delta E

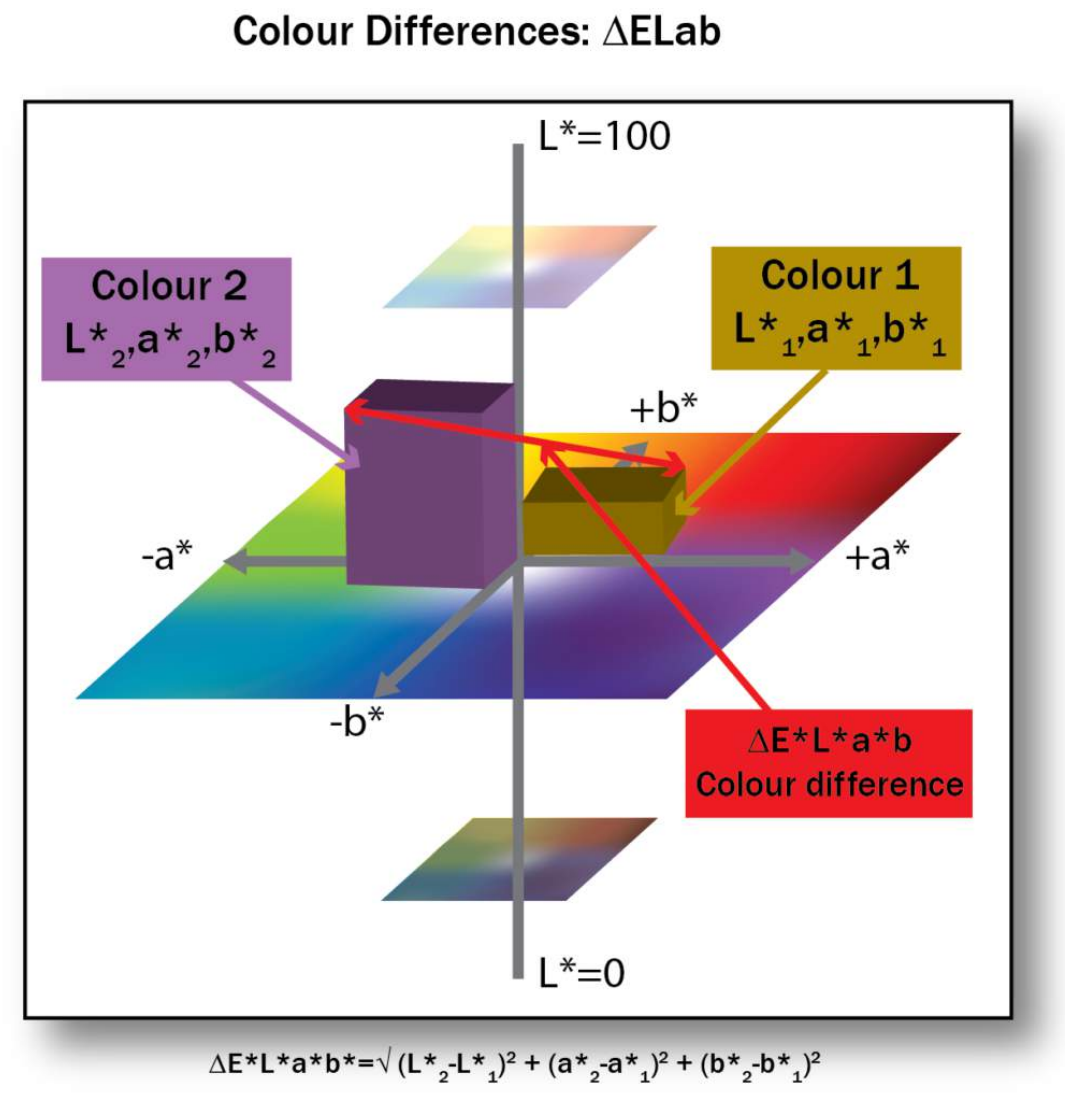


Figure 4.7 Lab Colour Space

Delta, the fourth letter of the Greek alphabet and symbolized as a triangle, is used in science to indicate difference. **Delta E** is the difference between two colours designated as two points in the Lab colour space. With values assigned to each of the L, a, and b attributes of two colours, we can use simple geometry to calculate the distance between their two placements in the Lab colour space (see Figure 4.7).

How do we do that? It looks a lot like the formula used to determine the long side of a right triangle that you may remember from high school geometry. We square the difference between each of the L, a, and b values; add them all together; and take the square root of that sum. Written out as a formula it looks a little more daunting: $\sqrt{(L_1 - L_2)^2 + (a_1 - a_2)^2 + (b_1 - b_2)^2}$.

Let's try a simple example to see what we get. Colour 1 has a Lab value of 51, 2, 2 and Colour 2 is 50, 0, 0 (right at the centre of the colour space):

$L_1 - L_2 = 51 - 50 = 1$, and$1 \times 1 = 1$, so our first value is 1.

$a_1 - a_2 = 2 - 0 = 2$; and $2 \times 2 = 4$, so our second value is 4.

$b_1 - b_2 = 2 = 0 = 2; 2 \times 2 = 4$, so the third value is also 4.

Add them together: $1 + 4 + 4 = 9$; and take the square root: $\sqrt{9} = 3$.

The Delta E (difference) between our two colours is 3. Could we detect that difference if we were viewing those two colours? Probably just barely. The minimum Delta E for seeing a difference is about 2. Smaller differences can normally be detected in neutral colours (such as our samples), while more saturated colours require a slightly larger Delta E. A Delta E of 4 is the upper threshold for acceptable machine repeatability or consistency.

Delta E provides a value indicating the overall difference between two colours. It does not provide any colour-related data such as which colour is lighter/darker, redder/greener, more blue/more yellow. To understand how the colours are different, we have to evaluate the comparative L, a, and b differences independently.

Experimentation over time has come to show that conventional Delta E is about 75% accurate in showing the difference we see between two colours. Delta E numbers exaggerate the differences in yellows and compress our perceptual distance between blues. To improve on the representation of our interpretation of colour difference, scientists have produced a modified formula known as Delta E(94).

Delta E(94)

Delta E(94) is a modified formula that provides about 95% accuracy in correlation to human perception of colour differences. Here it is in all its splendour:

$$\delta E*_{94} = \sqrt{\left(\frac{\delta L^*}{k_L S_L}\right)^2 + \left(\frac{\delta C^*_{ab}}{k_C S_C}\right)^2 + \left(\frac{\delta H^*_{ab}}{k_H S_H}\right)^2}$$

where:

$S_L = 1$

$S_C = 1 + 0.045 C_{ab}$

$S - H = 1 + 0.015 C_{ab}$

$K_L = K_C = K_H = 1$ (for reference conditions)

$C_{ab} = C^*_{ab}(\text{standard}) \text{ OR} \sqrt{C_{ab,1} C_{ab,2}}$

$\delta H^*_{ab} = \sqrt{\delta {E^*_{ab}}^2 - \delta {L^*}^2 - \delta {C^*_{ab}}^2}$

You can see that it is still the comparison of three values: L, C, and H, where C and H are produced by applying modifying factors to the original Lab values to compensate for perceptual distortions in the colour space. Each difference is squared and the root taken of the sum of the squares, just as in the original Delta E formula.

There is an additional refinement in Delta E(2000) where further adjustments are applied to blue and neutral colours and compensations for lightness, chroma, and hue. This is a much smaller correction than the change from Delta E to Delta E(94).

The good news is that you don’t need to understand or remember any of the details of these equations. Just remember that these are real numbers measuring actual plotted distances between colour samples. Delta E represents the distance between two points in the Lab colour space. Delta E(94) and Delta E(2000) are enhancements, providing improved numbers that more closely match our perception of colour differences.

Media Attributions

- Delta E in Lab Colour Space by Ken Jeffrey

4.5 Working with a Spectrophotometer to Read Standard Colour Charts for Output Profiling

Alan Martin

Armed with our fundamental concepts in colour theory, we can apply these principles to the physical process of colour management. The practical application to print production requires a procedure for measurement, colour profile generation, and the correct use of profiles in the manufacturing process. Let's begin with measurement and discuss the working components of a standard graphic arts spectrophotometer and the colour charts we would use it with.

X-Rite i-One (i1) Pro Spectrophotometer

The i1 Pro is one of the most common hand-held spectrophotometers used in the graphic reproduction industry. It can also be mounted in the iO base for automated scanning. As described in section 4.3, the **spectro** works by recording spectral data from incremental slices of the range of wavelengths included in visible light. To do this properly, the spectro must calibrate its white point to establish the baseline for interpretation. It does this by reading the white tile supplied in the baseplate that comes with the spectro. Each baseplate is uniquely matched to a specific spectrophotometer and marked with a serial number that corresponds to its spectro. Make sure you confirm that you have the correct baseplate that matches the serial number on your spectro. When used commercially, the manufacturer recommends that a spectro be returned for factory recalibration every two years. The packaging will include a certificate indicating the expiry date for the current calibration.

The spectro may also be identified (on both the serial number tag and surrounding the light-emitting aperture) as a UV-cut. This indicates it has an ultraviolet filter, which acts to remove the impact of fluorescence from optical paper brighteners. If you have access to more than one spectro device, be sure that any related measurements are done consistently either with or without the filter. Best practice is to use exactly the same instrument for any series of measurements.

The USB cable provides power to the spectro and so should be plugged directly into the computer and not into a peripheral such as a keyboard. Additional typical accessories for the spectro include a weighted strap and screw-in base for hanging the spectro against the screen of a monitor and a proof mount base with sliding rail for the spectro to read printed colour targets.

Colour Charts or Targets

You will typically be dependent on the colour management software application that you have chosen to produce a pdf file of the colour chart that your spectro can read. While in the software, you select a reading device (such as the X-Rite i1 Pro or i1 iO) from the list of devices that the software supports and then provide the dimensions for your output device. The choice of reading device will determine the

size of colour patches and page format, and the size of output will define how many pages are ganged to a sheet. When prompted, name the pdf with a clear identifier (output device and date or equivalent) and save it.

Once you have the pdf, use it for any direct output, such as a proofer, or to make a plate for your printing press so that the colour chart can be printed. In all cases, it is critical that no colour management be applied to the output device for the production of the chart so that the natural base state of the device is captured in the colour target produced. It is also essential that the proofer or press be in an optimal operating state so that the output is an accurate reflection of the device's capabilities. This may require a calibration process for a proofer or standard maintenance procedure on the press.

There are several colour chart standards that you should be aware of. The chart produced by your colour management software will likely be one of these or a slight variation thereof. The original standard is the IT8.7/3, composed of 928 colour patches. This was followed by the ECI 2002, which has 1,485 colour samples. There is now an update to the IT8, the IT8.7/4, which has extended the colour sampling to 1,617 patches. The larger number of patches provides a more detailed snapshot of the colour capability of the device that you are profiling. Of course, it takes more time to read the greater number of samples, so the newer sets are more manageable with automated reading devices such as the iO table. If you are reading with a hand-held spectro, choose the IT8.7/3 or smaller patch set if it is offered by your colour management software. The other trade-off between larger and smaller numbers of colour patches lies in the smoothness of colour transitions. Fewer data points mean less interpolation and potentially smoother curves in colour modulation. Be aware that it will require some experimentation on a device-by-device basis to determine the ideal intersection of accuracy and smooth interpretation for your measurement data.

The colour charts come in two standard patterns of organization: random and visual. Random is exactly that, and comprises the vast majority of charts that are produced for measuring. The colour swatches are distributed to provide optimal contrast between adjacent patches to aid the spectrophotometer that is reading them in distinguishing where one patch ends and the next begins. Having the colours scattered over the sheet also generates relatively even distribution of cyan, magenta, yellow, and black ink which leads to better performance from the press. The visual pattern places the colour blocks in a logical and progressive sequence so it's easy to see the chromatic ranges that are covered. Some scanning spectrophotometers can read the visual arrangement.

Measuring Your Colour Chart

Once you have used the pdf to generate the printed sample, you have to measure the sample. For charts from proofers, it is critical to wait a minimum of 30 to 90 minutes after the sample has been produced before measuring in order for the colour to stabilize. To create a measurement file, you need three things: colour management software; a target from that software; and a measuring instrument supported by the software. After connecting your measuring device to the computer (remember to connect directly to a USB port on the computer, not to a peripheral such as a keyboard), enter the measuring step of your colour management software. You will need to point to the correct measurement chart that was used, which can be easily identified if you have named it distinctively, and confirm the right measuring device is indicated. If you are getting errors once you begin measuring and can't proceed, the typical culprit is the selection of an incorrect colour chart or incorrect measuring device.

When you begin the measurement process, there are a few option prompts you may have to respond to. Some software allows for the averaging of multiple press sheets to produce an improved characterization of the device. This software can scan the colour bars placed at the bottom of the sheet and indicate which sheets are the best candidates for measuring. If you have chosen to do multiple sheets, then you will have to record the sheet number on each of the pages that you cut from the full press sheet in order to enter the sheet number correctly as you carry on measuring.

Proofing devices are stable enough that it is not necessary to average multiple sheets. You may still be cutting out single pages from a larger sheet (depending on the size of your proofing device) and should label the individual pages to stay organized. You can skip any of the prompts that deal with choosing multi-sheet options.

Once past the initial options, you will be prompted to calibrate the measuring instrument. Make sure the i1 Pro is correctly seated on its base plate and push the button. After successful calibration, you will be instructed to mount page 1 of your colour chart and begin measuring. For hand-held measuring with the i1 Pro, use the white plastic proof mounting base it came with. Secure your page under the spring clip at the top of the mounting base, positioning it so that the sliding clear plastic guide the spectro rides on can reach the first row of colour patches. The clear plastic guide has a horizontal opening that the head of the spectro nestles into to traverse the horizontal row of colour swatches. There is a cradle for the base of the spectro to rest on that slides horizontally across the guide. The spectro must begin on white paper to the left of the colour patch row, and finish on white paper to the right of the row. If your target has a semi-circle printed to the left of its colour rows, use it to position the chart under the scanning opening of the plastic guide. The left semi-circle on the opening in the guide should align with the printed semi-circle on the chart. If there is no printed semi-circle on the chart, allow ¼ – ½ inch of white space showing in the opening to the left of the first colour patch.

With your chart properly placed in the mounting base, position the spectro all the way to the left on the plastic guide with its base in the sliding cradle and reading aperture in the cut-away channel. Press the button on the left side of the spectro and keep it pressed. Wait one second while the white of the paper is read (there may be a beep to indicate this is done), slide the spectro smoothly and evenly to the right until you reach white paper past the colour patches, and then release the button. You should be rewarded with a prompt to scan row B. Slide the entire plastic guide down one row, move the spectro all the way back to the left, and do it all over again! If you receive an error, clear the prompt and try scanning the same row again. Success at hand-held scanning comes with a little bit of practice and a Zen approach. Your degree of success will be inversely proportional to the amount of anxiety you feel. The techniques that contribute to getting it right are smooth and even passage of the spectro; consistent pressure (make sure you are not pushing down on the head); white paper at either end; and keeping the button pressed from start to finish.

After you receive a couple of errors on the same row, the software may switch to individual patch reading. In this case, the prompt will now say “Scan patch B1” instead of the previous “Scan row B.” You must position the read head over each individual patch and press the spectro’s button when prompted on screen. After completing one row of individual patches, you will be returned to full row scanning mode. This procedure allows the measurement process to go forward when the software is having trouble detecting colour boundaries in a particular row.

Having completed all rows on all pages (which may take some time), you will be prompted to save the

measurement file. Some colour management software will take you directly to the additional settings required to define a profile and then proceed with the profile creation.

4.6 The Measurement File's Role in Colour Profiling

Alan Martin

It's important to remember that the measurement file is not a profile! It provides the basic information about the colour behaviour of the device but needs some critical additional instructions and the processing of the colour management software in order to produce a colour profile that can be used for colour transformations. The measurement file is a snapshot in time of device behaviour and depends on the appropriate calibration of the device prior to the chart's creation for its accuracy.

Content of the Measurement File

```
CGATS.5
ORIGINATOR  "1.1.6.3.717"
DESCRIPTOR  "ColorFlow MeasurementData File"
CREATED     "December 21, 2010 12:24:30 PM PST"
#Filename   measurements5475167630861979 48.cgt
#Rows 26
#Columns    19
#This chart generated from averaging several files.  This header extracted
from the first file
NUMBER_OF_FIELDS 51
BEGIN_DATA_FORMAT
SAMPLE_ID   CMYK_C      CMYK_M      CMYK_Y      CMYK_K      XYZ_X XYZ_Y XYZ_Z
      LAB_L LAB_A LAB_B D_RED D_GREEN     D_BLUE      D_VIS 380_NM
      390_NM      400_NM      410_NM      420_NM      430_NM      440_NM
      450_NM      460_NM      470_NM      480_NM      490_NM      500_NM
      510_NM      520_NM      530_NM      540_NM      550_NM      560_NM
      570_NM      580_NM      590_NM      600_NM      610_NM      620_NM
      630_NM      640_NM      650_NM      660_NM      670_NM      680_NM
      690_NM      700_NM      710_NM      720_NM      730_NM
END_DATA_FORMAT
NUMBER_OF_SETS    1976
BEGIN_DATA
A01   100   0     0     0     14.9593     24.264      56.2451     56.3512
      -43.1895    -51.288     1.60652     0.381318    0.166744    0.739
      0.7006      0.3927      0.3573      0.4336      0.5074      0.5794
      0.6549      0.7106      0.7356      0.7434      0.7396      0.7272
      0.7008      0.6495      0.5669      0.456       0.3352      0.2187
      0.1228      0.0686      0.0436      0.0327      0.026       0.022
      0.0214      0.0227      0.0256      0.0322      0.0417      0.0476
      0.0458      0.0404      0.0327      0.0274      0.0312      0.0457
```

Figure 4.8 Example of measurement file

We can open the measurement file in a text editor and examine its structure to understand what information it contains (see Figure 4.8). The first section is header info and provides some basic

information about the file. This is followed by a key to the colour data that follows. The actual colour information comprises a row and column ID to identify the specific colour patch and then the numerical data that goes with the patch.

The key information is contained in the second section where the data format is laid out. This indicates what colour spaces are captured in the data for each colour patch. The first four values are CMYK; the next three are **XYZ** (a device independent colour space like Lab); the next three are Lab; then RGB; and finally there is a wavelength measurement for each of the 10 nanometer slices of the visible spectrum (380-730 nanometers; violet to red).

The CMYK values for each patch are fixed in the software, coming from either the IT8 or ECI specification. These CMYK values have to remain constant because they are our point of reference for all devices. When we read the patch with the spectro, it uses the spectral information (each of the 10 nanometer slices) to calculate the XYZ and Lab values that describe the colour appearance of the colour swatch. By matching the measured Lab value to the predetermined CMYK value supplied in the original pdf, we have the raw material to build a translation table describing the device's colour behaviour. Once we have this look up table (LUT), we can predict the Lab equivalent for any CMYK value produced on the device. Correspondingly, when we are given the Lab values from another device that we want to match, we can produce the appropriate CMYK values to provide that colour appearance.

Device Dependent Versus Device Independent Colour Spaces

We've defined CMYK, RGB, and Lab colour spaces, and we've seen how the first step in colour profiling is to measure output from a device to define a relationship between the CMYK device dependent and the Lab device independent numbers. Establishing this relationship between device dependent (RGB or CMYK) values and device independent (Lab) values is a fundamental component of the colour management process.

We call the CMYK and RGB colour spaces **device dependent** because the results we get from specific RGB or CMYK values depend entirely on the device producing them. When we tell a device to produce "50% cyan," we are saying, "Give me half of as much cyan as you can produce." Since the capacity and colour appearance of cyan from any two devices will not be the same, it should not be surprising that a specification of "half that much" will also produce colour events that do not match. Similarly, the RGB values on a monitor or projector simply specify some proportion of the full red, green, and blue signals that the device can produce. Since there is no common starting point between two monitors in terms of what a full red, green, or blue signal is, then providing the same RGB values to those two monitors will in no way provide an opportunity to generate the same colour appearance.

For RGB devices, RGB values simply identify the volume of signal for each channel. For printers, proofers, and presses the CMYK percentages dictate what proportion of pigments are deposited. The numbers associated with specific RGB and CMYK colours only have colour meaning when attached to a particular device. There is no inherent consistency between any two devices based on providing the same RGB or CMYK values.

So if the device dependent colour spaces don't give us any consistency or control of colour between devices, where can we turn? Enter the **device independent** colour spaces of Lab and XYZ. We

spoke of Lab earlier as the three-dimensional model colour science has produced to map the way we perceive colour. The Lab colour space is device independent because we do not depend on the values associated with the output of a specific device to enumerate the colour. Lab values are calculated from spectral readings in a controlled environment so that they define a consistent colour experience for any circumstance and from any device. Device independent colour is the Rosetta stone of colour management that allows us to translate from the unique dialect of colour behaviour on one device into a universal language and then back to the specific colour dialect of a different device and maintain the colour meaning.

The device independent colour spaces are known as profile connection spaces (PCS) since they provide this gateway service between the device dependent colour spaces of individual devices. Lab and XYZ are the two PCS allowed in the **ICC** specification that defines profile creation. From the examination of the measurement file, we can see how it provides the first step in establishing and tabulating a relationship between the device dependent values of a particular device's output and the corresponding device independent values that characterize the actual colour appearance associated with the output.

We've talked about the measurement file as the gateway to our profile creation, so let's see what steps remain to get us there.

4.7 Profile Creation

Alan Martin

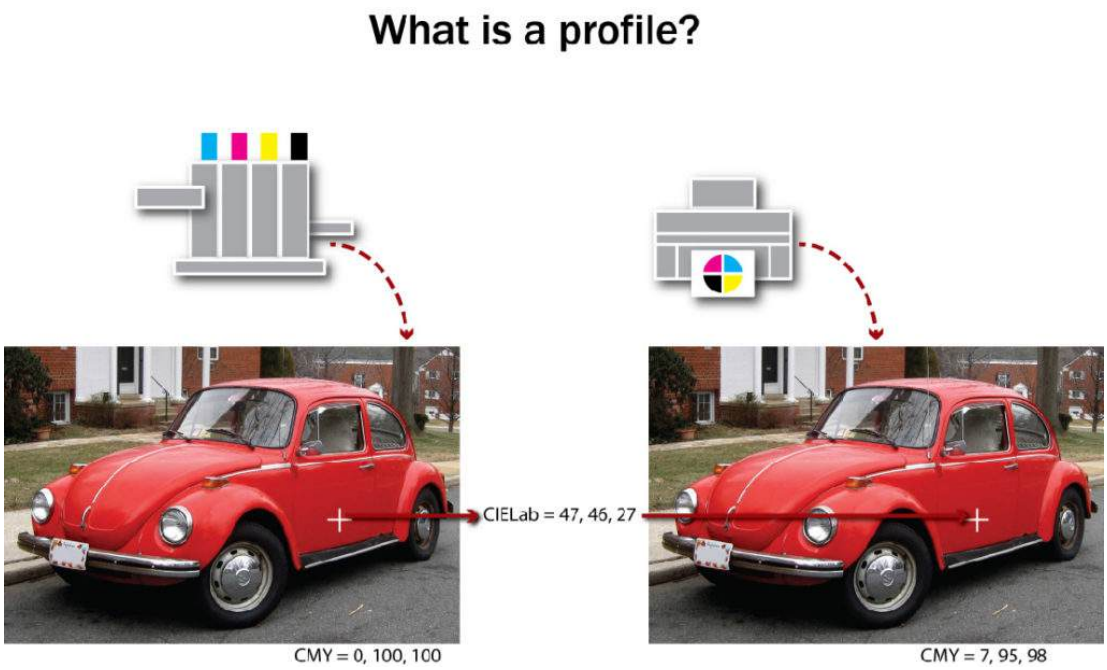


Figure 4.9 what is a profile

The measurement file contains the raw data of the relationship between the device dependent colour space of the device and the device independent colour space of the profile connection spaces (PCS). There are two additional parameters we have to provide along with this data in order for the colour management software to produce an output profile: total ink limit and black generation (see Figure 4.9).

Total Ink Limit

Total ink limit is a percentage, usually between 240% and 400%, that represents the upper threshold that the sum of the four process inks can total. Why would this change? Each device, in combination with its particular substrates (papers) and ink sets, has a different capability in terms of how much ink can be deposited. A coldset web newspaper press might be close to the lower end of 240%, while a high-end inkjet can support 100% from each of cyan, magenta, yellow, and black, and have a 400% rating. This total ink limit setting affects the colour distribution in darker colours. As the total ink limit decreases, dark colours require a higher proportion of black ink relative to CMY to stay within the ink limit percentage. With an ink limit of 360%, when black is at 90%, we still have room left for 90% of cyan, magenta, and yellow (4 x 90 = 360). But if the ink limit is 240%, with black at 90%, we can't exceed 150% for the sum of the other three (150 + 90 = 240). Since cyan leads in a grey balance mix, we might have 60% cyan, 45% magenta, and 45% yellow (60 + 45 + 45 = 150). As the ink limit drops, black has to do more of the work of providing darkness for any heavy colour mix.

Black Generation

We also have to provide instruction on how the black component of the CMYK mix will be created. Remember that our subtractive primaries are CMY and that, theoretically, they provide all the colours in the visible spectrum. Where does K (black) come in? To bridge the gap between theory and real world performance, black ink does some very important things:

- It compensates for the spectral shortcomings of CMY inks: our inks are not perfect representations of what cyan, magenta, and yellow colours should be.
- It eliminates registration problems in black type: if there were no discreet black ink, every time we wanted to print black type we would have to stack cyan, magenta, and yellow on top of one another and make sure they fit perfectly.
- It helps us achieve easier neutrals: we can use black to replace the grey component of colours reducing the amount of CMY required to stay in balance on the press to provide consistent neutral tones.
- It provides cost savings: black ink is cheaper than coloured ink.
- It increases contrast: black's greater density extends the tonal range of CMYK printing and improves the appearance of the final printed piece.

Since black is an add-on to our primary CMY colour set, we must provide instructions for its creation. Typically, we specify the black start point, maximum black, and black strength.

- Black start point: The point at which black enters colour mixes (range of 0% to 50%). If the black start point is 20%, then tones from white to 20% will carry CMY only.
- Maximum black: The upper allowable percentage of black ink used in the K separation (range 70% to 100%).
- Black strength: The relative quantity of black versus cyan, magenta, and yellow in the neutral grey component of colours (range 5 to 75%). As the number increases, colours can contain more black.

Black strength settings are also referred to as grey component replacement (GCR) values in colour separations. These may be set as percentages or simply as light, medium, and heavy, where more equals a larger proportion of black taking the place of CMY in neutral tones. GCR is the successor to UCR (under colour removal), which only moved exactly equivalent amounts of CMY to the black channel. UCR separations exhibit a 'skeletal' black, where there is a small amount of content in the black separation. With GCR, and as the black strength increases, more and more content moves into the black channel.

Final Processing

With the measurement file available, and total ink limit and black generation specified, processing can start in the colour management software and an output profile created. It will take from two to five minutes for all the calculations to complete. You may be prompted to specify whether to save the profile in the system or user location, or both. The system location will make the profile available to

all users of the computer, while the user location will restrict it to the current user. Depending on your permissions at the computer, you may not be able to write into the system location. On an Apple OS X machine, the default location for the system is System/Library/ColorSync/Profiles and for the user, Users/(userID)/Library/ColorSync/Profiles.

Viewing Your Profile

The Mac provides a built-in tool for viewing a profile or comparing two profiles. From the Utilities folder, launch ColorSync. Select the second tab from the left and you'll see a selection window on the left and a display pane on the right. The selection window contains a list of all ColorSync aware locations on your computer. Tip open the available folders and browse to find the profile you wish to view. Select a profile and a three-dimensional representation of its **gamut** in the Lab colour space appears on the right. Use your cursor to rotate the profile through all three of its axes. From the drop-down triangle in the display pane, choose Hold for comparison. The current profile is ghosted back, and you can select a second profile to display at full strength on top of it so you can compare the relative gamuts of the two profiles.

Profile Limitations

Remember that your profile represents the behaviour of the device for particular imaging parameters such as a specific substrate, resolution (screen ruling), and ink set. If any of those parameters are significantly altered, it requires a new profile to accurately represent the colour results. For real-world applications, profiles representing related groups of parameters are employed. One profile might represent all uncoated stock at 100 to 133 lines per inch, a second for coated stock at 150 to 175 lines per inch, and a third for a high saturation ink set on glossy stock with high resolution stochastic screening.

Media Attributions

- what is a profile by IFCAR adapted by Ken Jeffrey

4.8 Beyond Output Profiling: Display, Input, and Device Link Profiles

Alan Martin

Up to this point, we have focused exclusively on the output profile in our discussion of profiling. This makes sense, since this is the predominant profile we are concerned with in graphic production. Output profiles characterize output devices such as printers, proofers, and presses, but there are other devices that we have to manage in the full spectrum of a colour-managed workflow, and these require two additional classes of **ICC profiles**: display and input.

Display profiles capture the colour characteristics of monitors, projectors, and other display devices. Input profiles characterize devices that capture images such as scanners and digital cameras.

Display Profiling

You may hear this class of profile referred to as *monitor profiling*, but the more accurate designation is *display profiling* to acknowledge the inclusion of components beyond the monitor such as the video card and video drivers. Though less commonly profiled, this class of profile encompasses digital projectors as well.

In preparation for display profiling, the cardinal rule of thumb is to make whatever adjustments we can in the actual monitor. Any software adjustments to the VideoLUT (the look up table stored on the video card) reduce the operating range of the monitor and limit the spectrum of the display. With the predominance of LCD monitors, this means that the brightness or white luminance is the only hardware adjustment available. If you see reference to black level or colour temperature settings, this hearkens back to CRT monitors where these were actual hardware settings. For LCD monitors, these are software controls. For an LCD, all light comes from the backlight, which is a fluorescent array behind a diffuser, so the only monitor control is the brightness of this backlight.

Display profile software typically combines calibration and profiling. A setting called the vcgt (video card gamma type) tag in the display profile can download settings to the VideoLUT on the video card and change monitor behaviour. This is an unusual deviation from the standard protocol in colour management where the profile never alters the behaviour of the device. Calibration is used to optimize device function and characterization or profiling captures a description of device behaviour. Normally, the application of a profile should not have any influence on the device function.

Before calibration, it’s essential to warm up an LCD monitor for 30 to 90 minutes. Check the full brightness. If the monitor has aged to the point where it can’t achieve adequate brightness, then it should be abandoned as a candidate for profiling. Set the standard refresh rate and resolution that will be used on the monitor. If these are changed after profiling, then the profile cannot be considered accurate. Clean the screen with an appropriate gentle cleaner.

When you begin the profiling software, you will be prompted to identify your instrument. Colorimeters are often provided in display profiling packages, but most software works with standard

spectrophotometers (spectros), such as the i1 Pro.
The recommended settings to enter for the set-up phase are:

- White point: D65 (6500 K)
- Gamma: 2.2

The setting 6500 K is usually close to the native white point of an LCD monitor. You can choose Native White Point if you feel that 6500 is too far from the actual white point of your monitor. Gamma is the tone reproduction curve of the monitor. The setting 2.2 typically provides the smoothest gradients in monitor display.

Next is the choice of a patch set from small, medium, and large options. This determines the number of colour swatches that will be projected on screen for the instrument to read. The trade-off is between calibration time and colour range. Start with the small patch set and see if you are happy with the results.

To start this process, make sure the i1 is on its base plate for the instrument calibration step and then suspend the spectro in the monitor mounting strap on the monitor. The weight at one end of the strap hangs behind the monitor to counterbalance the spectro, and the i1 Pro locks into the mounting plate at the other end of the strap to keep it in place on the monitor screen. The reading aperture of the spectro should be approximately in the centre of the screen. Tip the monitor back very slightly to help the spectro sit flat of the front of the LCD.

When you tell the software to proceed, it begins projecting a series of colour swatches that the colorimeter or spectro records. As you did to produce the measurement file for your output profile, you are building a comparative table of known device values (the RGB swatches projected on the screen), with device independent values (Lab readings from the spectro) that describe their appearance. This may take from three to ten minutes. During this process, make sure that no screen saver becomes active and you keep the mouse out of the scanning area and. If you leave before the process is completed, check that the spectro is properly positioned when you return.

Once the colour patches are done, you will be prompted to name and save the profile. Make a habit of naming your profile with the date so its age can be easily checked. Saving display profiles is similar to saving output profiles, where the user chooses system and user options. With display profiles, there is no value in saving previous versions. All you are interested in is the current state of the monitor.

To see the active profile on a Mac, choose System Preferences/Displays/Color. The active profile will be highlighted at the top of the list. There is a check box toggle limiting the list so only profiles that are known to be associated with the monitor show.

Input Profiles

As mentioned, we need input profiles when we capture images. There are predominantly two types of devices associated with image capture: scanners and digital cameras. The fundamental concept in producing an input profile is that RGB device values scanned or photographed from a target are matched to device independent Lab values either provided by the target vendor or measured from the target itself.

For input profile creation, the targets always consist of two parts: a physical sequence of colour patches and a target description file (TDF) with the profile connection space (PCS) values for the swatches. The TDF accuracy varies from individually measured targets (done by you or a specialty vendor) at the high end to averaged samples from a batch run (an economical alternative).

As with output profiling, there are standard scanner target formats. We have IT8.7/1 for transmissive originals (film transparencies like slides) and the IT8.7/2 for reflective (photo print paper). These targets are available from a variety of vendors and allow you to match the material of the target to the material you will be scanning. If you will be scanning Kodachrome slides, you will want a Kodachrome IT8/7.1 target. Conversely, if your originals are Fuji photo prints, then you will want an IT8/7.2 target on the matching Fuji photo print paper.

The X-Rite ColorChecker targets are commonly used for digital cameras. There is the original ColorChecker with 24 tiles and the later digital ColorChecker SG with 140 colour tiles. The larger target can be used for initial set-up and there is a mini version of the original ColorChecker that will work in most photo shoots for an ongoing reference check.

Though scanners and digital cameras both fall into our domain of input profiling, they have some very different characteristics that we have to take into account when preparing to produce a useful profile. As with output profiling, we need to calibrate the device by stabilizing and optimizing its performance prior to capturing its colour behaviour. In order to stabilize, we need to understand the potential variables that the device presents. Scanners have a controlled light source and stable filters and typically have the option for extensive software intervention. In contrast, cameras have stable filters and moderate software controls but have the potential for hugely variable lighting conditions. The excessive variability of outdoor lighting limits useful profile creation to interior and in-studio camera work. If the lighting conditions can be controlled adequately in the studio, then colour-accurate capture can take place and colour accuracy can be maintained in the production work that follows.

Stabilizing a scanner’s performance comes from turning off any automatic adjustments for colour correction:

- White and black point setting
- Removing colour casts
- Sharpening

If you can’t turn these off, then the scanner is likely not a good candidate for profiling. Optimize the scanner’s behaviour with an output gamma setting of 2.6 to 3.0.

Stabilizing a camera’s performance comes from the appropriate lighting and capture settings. Use even lighting and full highlight and shadow exposure for target capture. For colour temperature, use grey balance for camera back devices, and white balance for colour filter arrays (CFA). Optimize the camera’s bit depth retention with gamma 1.0 for raw profiling.

With calibration complete, it’s time to capture the target. For a scanner:

- Mount straight
- Mask open bed areas

- Scan as high-bit tiff
- Open in Photoshop (beware of any automated conversions or profile assignments) to rotate, crop, and spot

Comparatively, for the digital camera:

- Light evenly
- Capture square
- Open in Photoshop (beware of any automated conversions or profile assignments) to rotate, crop, and spot

With the digital image in hand, we're ready for the input profile creation. Measure the commercial target with the spectro if you are not using a supplied target description file. Launch your colour management software and you will be prompted to identify the target image and the corresponding target description file. The profiling software reads the RGB values from the scanned or captured image, and the software processes the target description file and RGB measurement file to produce the input profile. File-saving options are very similar to what we have previous described for output and display profiles.

Device Link Profiles

Device link profiles are most closely related to the output class of profiles. A device link profile combines two output profiles to provide the specific conversion instructions between two particular devices. It provides the opportunity to maintain black and other separation purity (i.e., what begins as black only in the source colour space emerges as black only in the destination colour space) by removing the need for passing the colour transformation through the PCS. To define a device link, we identify a source and destination profile to our colour management software, specify the rendering intent, and provide details on how constrained the re-separation should be. By avoiding the passage into and back out of the PCS, we can very strictly control the parameters of the colour conversion. The options for conversion are:

- Full re-separation — Complete re-separation. Solid colours in the original file may not remain solid. The black generation parameters that you specify are used, which may result in using less chromatic ink and more black ink.
- CMYK integrity — All colour builds can be adjusted. The relative amount of black versus CMY will be preserved in content processed through the device link.
- Black purity only — Any colours other than the black channel (solid K, K-greys) can be adjusted.
- Colour and black purity — The same as fully constrained, but solid colours can be reduced to a tint.
- Fully constrained — Any colour made with only one or two inks will not have other inks added. Solid (100% tints) primaries and secondaries are not affected and remain solid.
- Ink optimizing — A proprietary term in the ColorFlow colour management software for applying a full re-separation with a heavy grey component replacement (GCR) algorithm.

Colour management software used to be required to preview the results of applying a device link. With the last few versions of Adobe Photoshop, a device link option has been added to the advanced dialog window of the Color Conversion menu, making device link previewing much more accessible. Currently, Photoshop only supports CMYK to CMYK device links. It does not support RGB to CMYK device links. Another alternative for viewing the results of applying a device link is to generate a virtual proof (VPS) in Kodak Prinergy with the device link specified. For details, see section 4.11 in this chapter.

With this extraordinary level of control, why don't we use device links for every colour conversion? The truth is, that with our gain in managing the colour conversion process, we sacrifice an even greater degree of flexibility. The premise of colour management and the use of profiles is that we do not have to generate a unique profile for each pairing of devices. With the power of the PCS gateway to provide the device independent colour description, we only need a single profile for each colour condition of a device and any two profiles can be positioned on either side of the PCS to provide a pathway for the colour conversion.

Where it does make sense to go to the extra trouble of generating a device link profile is a situation where a specific pairing of two devices is used over and over again, such as a proofer for a particular press condition, or to keep two presses in a shop matched for their colour output.
If we process an image from RGB to CMYK at the beginning of our production process, we gain the stability of having the image in our known CMYK space, but we surrender the flexibility of converting to the optimal CMYK space at final output. For final stage or late-binding conversion, we are dependent on the RIP environment for managing the calculations between the profile pair (see Section 5.2). A device link provides additional security in the conversion process by reducing the variability that can come with the processing application input that is part of a profile pair transformation.

4.9 A Review of the Profile Classes

Alan Martin

We've now touched on the four types or classes of profiles: display, input, output, and device link. What traits do they have in common? They are:

- Specified by ICC-defined file formats
- Contain colour tables
- Have device colour values associated with device independent colour values (PCS)
- Use a measuring device (spectrophotometer) and targets for creation
- Require a specified rendering intent
- Have standard OS library locations

There are also some unique characteristics for each profile class that help define the role they play in the overall colour management process. The display class of profiles:

- Have no separate, tangible target: the device 'is' the target
- Can affect device behaviour
- Are mostly integrated with device calibration

The input class of profiles:

- Are unidirectional: A to B table only (device values to PCS)
- Are able to exclude any external measuring (with a supplied TDF)
- Ensure that the target's job is tell us how the device 'sees' the target

Output (and device link) class of profiles:

- Use CMYK (versus RGB for display and input)
- Have black handling settings
- Have the largest and most complex colour tables
- Ensure the target's job is to tell us how the device 'makes' the target
- Provides preview capability for upstream editing

We also discussed the functions of a profile in the colour equation. The two functions are source and destination. A source profile is a profile used to convert device dependent RGB or CMYK information to a device independent colour space (Lab). A destination profile is a profile used to convert information from a device independent colour space (Lab) into a device dependent colour space of an output device

(RGB or CMYK). You can think of the source profile as the colour reference or the place from which our desired colour appearance comes. The destination profile describes the location where we are producing colour in the current pairing of devices. If we want a proofer to simulate the colour behaviour of a particular press, the press's profile is the source (the desired colour behaviour) and the profile for the proofer is the destination (our colour output).

Do not confuse a profile's class with its function; they are independent and separate characteristics. A particular output profile can assume the function of source in one colour equation and then turn around and assume the role of destination in a second. Take the example of the press above, where the press profile acted as the source. If we have multiple presses in the printing plant, and we have another press that is our master press for colour behaviour, the press profile that acted as a source for the proofer output will now function as a destination profile to align that press to the colour behaviour of the master press (the master press profile is the source profile in this pairing).
Profiles enable the two key processes of any colour managed workflow: clear communication of colour meaning and the transformation of device values to maintain colour appearance.

4.10 The Components and Purpose of a Colour Management System

Alan Martin

Our primary goal in colour management is to provide a consistent perceptual experience. As we move from device to device, within the limits of the individual device's colour gamut, our interpretation of the colour event should be the same.

As we've discussed, but it's worth repeating, we achieve that goal in two fundamental steps:

1. We give colour its meaning through device independent colour values correlated to a specific device's RGB or CMYK numbers.
2. We transform our destination device's specific values to match the perceived colour definitions of our source.

The Components

We have spoken at great length about colour profiles, but there are three additional pieces required to enact a colour-managed workflow: the profile connection space (PCS), a colour management module (CMM), and rendering intents.

The PCS provides the device independent colour definitions to which device values can be matched. The ICC specification allows the XYZ or Lab colour spaces for the PCS.

Profiles provide the look up table for the conversion of device values to PCS and vice versa. Any conversion requires two profiles (or a device link into which two profiles have been merged).

The CMM is the software engine that actually does the conversion. It lives in a product like Adobe Photoshop or Kodak Prinergy, where file processing takes place. The CMM provides two major benefits: it reduces profile size by doing the interpolation in colour calculations and compensates for shortcomings in the Lab colour space. The CMM does the heavy lifting by providing the computation of the conversion. For consistency, it is best to keep to a common CMM as much as possible. The Adobe Color Engine or ACE is the CMM seen referenced in the Color Setup dialog window of the Adobe Creative products.

Rendering intents are the instructions for dealing with out-of-gamut colours (see Figure 4.10). They are user-selectable controls to define colour rendition when you move colour from one device to another. There are four types of rendering intents: perceptual, saturation, relative colorimetric, and absolute colorimetric. Each intent represents a different colour conversion compromise, resulting in a different gamut mapping style.

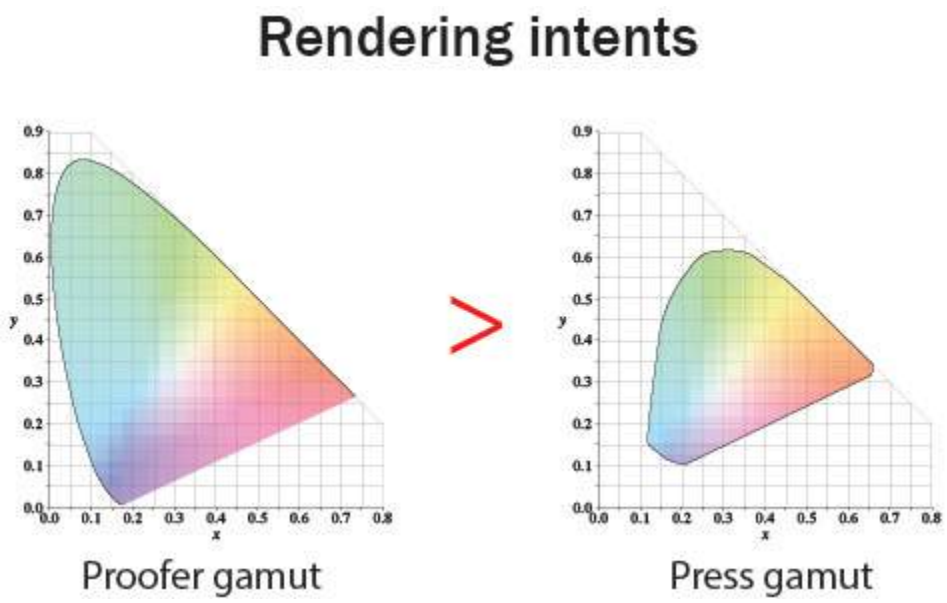


Figure 4.10

Perceptual and saturation intents use gamut compression, where the overall gamut space is adjusted. Relative and absolute colorimetric intents use gamut clipping, where colour matching is maintained throughout the available gamut, and out-of-gamut colours are moved to the available extremes of the destination gamut.

The perceptual intent is used mainly for RGB to CMYK transformations, which are typically image conversions. Since we are moving from a larger gamut (RGB) to a smaller gamut (CMYK), it makes sense to employ a rendering intent that preserves the overall relationship rather than one that emphasizes one-to-one colour matching within the gamut.

The saturation intent is the least relevant for colour-managed workflows. When you use this intent, colour saturation is preserved as much as possible at the expense of hue and luminance. The result is a bad colour match, but the vividness of pure colours is preserved. This intent is usually used for documents such as presentations, charts, and diagrams, but not for graphic arts jobs.

The two colorimetric intents, relative and absolute, are closely related. They are both used for CMYK to CMYK conversions where the gamuts of source and destination are closely matched or the destination is larger (typical of a proofer compared to the press it is matching). They emphasize exact colour matching for in-gamut colours and clip out-of-gamut colours.

The only difference between the two colorimetric rendering intents is in white point handling. The absolute intent pays attention to the colour of white in the source and reproduces that in the destination. Think of newspaper printing where the whitest colour is the paper that has a dull and beige tone. With an absolute rendering intent, a proofer matching the newspaper would add that beige colour to all of the white areas of the proof. This matching of white destination to white source colour is not usually necessary due to the chromatic adaptation or colour constancy that we discussed earlier. We have a built-in mechanism for adjusting to judge the overall colour relationship independent of the appearance of white. For this reason, the relative colorimetric intent is used most of the time and the white of the destination is not adjusted to simulate the white point of the source.

With all of the pieces of the colour management process clearly delineated, we can put them to use in our standard graphic arts workflow applications.

Attribution

Figure 4.10
Image modified from: Comparison of some RGB and CMYK colour gamut by BenRG and cmglee is used under a CC BY SA 3.0 license.

4.11 Applying Colour Management in the Adobe Creative and Kodak Prinergy Software

Alan Martin

Colour management comes into play at two primary points in the print production workflow: during file creation with authoring tools like the Adobe Creative applications (Photoshop, InDesign, Illustrator), and then when the file is processed for output with a workflow software program such as Kodak Prinergy. Let's examine the details in these most widely used software tools to provide concrete examples.

Colour Set-up in the Adobe Creative Applications

The primary tool for colour management in the Adobe products is the Color Settings dialog under the Edit menu. Fortunately, these settings can be shared across all of the Adobe applications to coordinate a consistent delivery of colour strategy. Define your settings in Photoshop, as this is the application with the largest number of options, to guarantee that all possible options have been set to your choices.

Launch Photoshop and, from the Edit menu, choose Color Settings. There are three main sections to the dialog window: Working Spaces, Color Management Policies, and Conversion Options. Change the Settings option above the Working Spaces panel to North American Prepress 2. This applies a set of defaults that are optimal for a print production workflow.

Working Spaces is Adobe's name for default profiles. These are the profiles that will be used if no other information is available. If you open a file that is *untagged* (the terminology for a document that has no profile attached to it), the profile listed for the colour space matching the file will be *assumed* and used as long as the file is open. It will not persist with the file once the file is closed. If you create a new file in the application, the profile listed will be *assigned* and the profile reference will move forward with the file.

Let's review and clarify the terminology associated with describing the status of a colour profile relative to a particular document or file. A file that has a profile is referred to as *tagged* while one without profile is *untagged*. A tagged document can have one of two relationships with its colour profile. The colour profile can be *embedded* or *assigned*. An embedded profile is actually written into the file content. This increases the file size, but guarantees that the desired profile will be available. For an assigned profile, only a reference to the profile is contained in the document. File size is reduced, but access to the profile depends on the application and environment processing the object. You can think of an *assumed* profile as a temporary assignment that will only last as long as the file is open.

For Working Spaces options, the RGB default of Adobe RGB (1998) that comes with the North American Prepress 2 setting is a perceptually even RGB space, which makes it better for editing and a good choice. The CMYK field is where you should choose a profile specific to the final output device if it is known. The SWOP profile is a reasonable fallback and is commonly used as the industry standard

for generic work. Be aware that choosing SWOP will constrain the gamut to the capability of a web offset print condition.

The list of profiles available for selection comes from the various ColorSync aware folders that we have previously discussed. Priority is given to profiles in the Library/Application Support/Adobe/Color/Profiles/Recommended folder, and these profiles are listed first. If you have a profile that you wish to give prominence to for selection, place it in this folder.

The Color Management Policies subsection controls behaviour when opening or creating documents and when moving objects between documents. There are three options for each of the available colour space settings: Off, Preserve Embedded, or Convert to Working.

The choice of Off is the most misleading, because we can't actually turn colour management off: there is always an assumed profile if no other information is presented. With Off, copy and paste of an object moves tint values: a 50% cyan value from the originating document lands as 50% cyan in the destination document.

Preserve Embedded does what it says and maintains what's in place. New documents use the Working Space profile and become tagged. An untagged file assumes the working space profile but stays untagged. If you copy and paste native RGB objects, they are converted. If you copy and paste native CMYK objects, the tint values are maintained.

Our final choice for colour management policy is potentially the most dangerous. Convert to Working converts tagged documents using the existing profile as a source profile and the Working Space profile as the destination. If you do not have the Alert check boxes ticked, this can happen without your awareness. For an untagged document, it assumes the Working Space profile. Copy and pasting RGB or CMYK objects always converts to preserve appearance (changes the tint values).

After reviewing the choices, the recommendation for Color Management Policies is Preserve Embedded, and make sure all Alert boxes are checked. This allows you to confirm that any action is what you actually want before proceeding.

The last section of the Color Settings dialog window is the Conversion Options. The Engine option refers to the colour management module (CMM) that will be used for calculations in the colour conversions. The default choice of the Adobe Color Engine is good for maintaining consistency. Here we have as well the Rendering Intent entry, which will function as a default unless an alternate intent is specified in any dialog. Relative Colorimetric is a reasonable choice unless you know that almost all of your conversions will be RGB to CMYK for which Perceptual is the appropriate intent option. Always check Use Black Point Compensation. This maps the black point source to the black point destination, avoiding any clipping or flattening of the darkest colours and maintains the full dynamic range.

Now that we have all of our working parameters correctly defined, we can OK the Color Settings dialog and look at the basic mechanism for using colour profiles in the Adobe Creative applications. There are two actions we can invoke from the Edit menu to apply colour profiles: Assign Profile and Convert to Profile.

Assign Profile allows us to select a source profile for the open document. This action will replace the existing profile for a tagged document or provide a new reference for an untagged file. To have the

association persist, you must save the file and check the option to embed the current colour profile. Assigning a profile will change the onscreen appearance of the file but not alter any of the file's tint values (colour data). We are simply instructing the computer on how the original CMYK or RGB values would look if they are coming from our new source (described in the look up table of the new colour profile).

Convert to Profile is an immediate and irreversible (once the file is saved) transformation of the document's tint values. The assigned profile is used as a source and the user selects a destination profile and rendering intent. The conversion takes place and the file now has new RGB or CMYK numbers as a result. If you use the advanced dialog to specify a device link profile, then the currently assigned source profile is ignored for the calculations since the device link contains information for both the source and destination conditions.

Both the Assign and Convert dialog windows come with a Preview check box that will allow you to toggle between the before and after state to visually validate your choices and experiment with the effects of choosing different rendering intents.

Assessing the Effect of a Colour Profile

Once the profile is applied, what should we be looking for? Both on screen and when comparing hard-copy (printed) samples, there are specific areas of the image that should be checked. There is also industry-specific language used in describing colour appearance that it is helpful to be familiar with. The areas of the image to pay special attention to are saturated colours, flesh tones, neutrals, and the highlights. Proof and print sample sheets will have four or five images that emphasize these areas along with tone ramps in the process and overprint colours. Focusing on these areas of interest will make it easiest to identify variation when checking for colour matching.

The terminology that is often employed is *colour cast*, to indicate a shift toward a particular colour; *heaviness*, to suggest excessive tone (particularly in the highlights); *dirty*, to specify too much complementary colour resulting in greying; and *flat*, to describe a lack of contrast and/or saturation. Knowing the terminology will help you understand the comments your co-workers may make and will help remind you of the types of analysis you should be doing.

Additional Colour Tools in Adobe Acrobat

In addition to the fundamental profile handling procedures described above, there are several powerful and useful colour tools in Adobe Acrobat that can be used once you have exported your desktop file to a PDF. These are found among the Print Production Tools in the Acrobat Tools menu. Two are of particular note: Convert Colors and Output Preview.

Convert Colors allows you to convert colour spaces, such as changing RGB content to CMYK. It also enables transforming *spot colours* (such as Pantone) to CMYK. In addition, if the file incorrectly contains multiple instances of a spot colour that should all appear together on the same printing plate (i.e., Pantone 283 and s*pecial blue*), they can be linked to behave as a single entry on the colour palette.

Output Preview does not apply any changes to the file, but is an extraordinarily powerful review

mechanism. It enables you to confirm that colour elements in the file are as they should be before you commit to the expensive step of actual output. With Output Preview, you can turn individual separations on and off to check overprints and knockouts; check the separation list to confirm which elements are attached to each separation; identify the colour space of each object; and even highlight any area that exceeds your threshold for total ink coverage.

Profile Use in Kodak Prinergy

The final topic in our exploration of colour management in graphic technologies is an example of the application of colour management in a print production workflow. We'll use one of the predominant workflow applications, Kodak Prinergy, as a model.

The processing instructions for file handling in Prinergy are contained in a *process template*. Input files are 'refined' to PDF in the Prinergy workflow and an important portion of the refining process is the instructions relating to colour conversions. In addition, we have process templates for output both to proof and final output. These templates also contain colour control options. For both *refine* and *output* process templates, the colour management information is contained in the Match Colors box of the Color Convert subsection.

Prinergy offers a comprehensive colour management application of its own called ColorFlow. There is a check box in Color Convert to turn on ColorFlow awareness and pass all of the colour management decisions to what has been predefined in ColorFlow. Discussing the structure and functional logic of ColorFlow is beyond the scope of this text. To use Prinergy as a more generic workflow example, we'll uncheck the ColorFlow option and turn off the integration.

The standard location to store profiles for use by Prinergy is \Prinergy\ CreoAraxi\data\ICC-Profiles. Profiles are not immediately available from this location in Prinergy's drop-down menus, but can be browsed to if placed in job-specific subfolders.

Let's look at the Match Colors section of the refine process template. With ColorFlow integration turned off, the entry fields under Assign Input Device Conditions become available. If you check Override Embedded Profiles, then profiles that are checked on in the Assign Input Device Conditions section will replace all existing profiles in the files being processed. Notice that there is a very granular level of control, with independent entries for CMYK images, CMYK graphics, RGB images, and RGB graphics. If you specify a device link profile, it will override any tagged profile whether or not Override Embedded Profiles is checked.

Convert to Destination is where you indicate the destination profile. This will only be used for instances not covered by a device link in the assign section. Remember that any device link contains both source and destination information. Beneath the Convert box is a box for selecting rendering intents. Entries are only available for the options that were checked on under Assign Input Device Conditions. The default entry is Perceptual since most input files will require image conversions from larger colour gamuts.

For output process templates in Prinergy, the options are very similar to what has been described above. We typically use colour management for output to proofs, either single page or imposed. There is a separate ColorFlow section to uncheck to enable the more traditional colour management approach. In the Match Colors box, Match Colors in Page Content must be checked on for any colour profiles to

be applied. Also, you must select one of the following radio buttons–As Defined Below, If Not Set in Refining, or Exactly as Defined Below– so you will be able to choose a profile in the Input Device Conditions box. Since all content will now be in the CMYK space from the refine process template, there are no longer independent options for RGB, graphics and images: only one source or device link can be specified. The rendering intent is entered in the Rendering Intent field. The destination profile (usually the proofer profile) goes in the Device Condition box. Once again, this destination profile will be ignored if a device link has been chosen for the source.

By following the steps described, the workflow application user can produce predictable and accurate colour reproduction results through both the file processing and output steps of any print production workflow.

4.12 Summary

Alan Martin

In essence, colour management is a translation process. The tools and precepts of colour science give us the lexicon to populate the dictionaries of each of the devices we use in our graphic technology workflow. Our device's dictionary is the colour profile we generate with the help of a colour target, an instrument to measure the target and colour management software to process the measurements.

Since the device's dictionary (colour profile) defines the specific RGB or CMYK values of that device in a device independent colour language, we have a conduit between any two devices. By aligning the device independent values between the two devices, we can translate from the device dependent values of our source to the new RGB or CMYK values at our destination to provide a common colour experience.

With an understanding of how to configure our standard graphic arts applications to effectively use our device's profiles, we can make successful colour translations happen in our real-world production scenarios. Successful colour translations provide the improved productivity that comes with predictability and confidence in process control.

Questions to consider after completing this chapter:

1. What are the three components of the colour event?
2. How does visible light relate to the electromagnetic spectrum?
3. How does the zone theory of optical systems resolve the apparent incompatibility of trichromacy and opponency?
4. What is the importance of the Lab colour space?
5. Define Delta E.
6. Which of RGB, CMYK, and Lab are device dependent colour spaces, and which are device independent colour spaces?
7. How does a spectrophotometer work?
8. What is the purpose of a colour target?
9. What Mac OS utility can be used to view colour profiles?
10. How does a measurement file relate to an ICC profile?
11. What is the vcgt tag in a display profile?
12. Name the common characteristics of output, display, and input profiles.
13. What characteristics are unique to each of output, display, and input profiles?

14. Working Spaces serve what purpose in Adobe's Color Setup dialog?
15. Which Acrobat tool allows for the merging of spot colours?
16. How do we make a device link profile?
17. Where is a device link profile used in Prinergy's output process template?

References

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Suggested Readings

Fraser, B., Murphy, C., & Bunting, F. (2004). *Real world color management* (2nd ed.). Berkeley, CA: Peachpit Press.

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Chapter 5. Pre-press

5.1 Introduction

Wayne Collins

Learning Objectives

- Explain why raster image processing requires so much data for print imaging
- Compare resolutions required for digital media and print media
- Compare and contrast the positive and negative attributes between using process and spot colours
- Discuss why Pantone colours are more accurate on a printed swatch than on screen.
- List a number of different industry standard spot colour systems
- Describe trapping issues that occur when adjacent colours are imaged independently
- Analyze different imaging technologies for trapping requirements
- Interpret how black ink is to be used in overprint and boost situations
- Define transparency within the context of prepress workflow management
- Differentiate between flattening transparency on the desktop, or at raster image processing
- Describe the most common press sheet imposition styles
- Analyze different binding styles to select the correct imposition required
- Identify opportunities for nesting multiple images to save materials
- Explain the importance of preflight within the context of pre-press workflow

North America's fifth-largest manufacturing sector is graphic communications technologies. We can become aware of just how huge this industry is by listing all the manufactured images we see in our day. Your list might include the morning paper or magazine you read, the graphic on the side of the bus you ride to work, and the labels on the grocery shelf where you select your evening meal. Increasingly, more of the graphics that are driving that massive industry are produced with computer graphics software on personal computers. Most of the graphics software used to create the images for reproduction is designed to create images for electronic media — primarily the Internet. Computer graphics designers are not aware of, or concerned with, optimizing their designs for the manufacturing process they are driving. This problem is a root cause of less profitability in most sectors of the graphic communications industry. To tackle this problem, we must become aware of all that happens to a computer graphic from the time it leaves the picture created on the computer screen to the image on the label on the package on the grocery shelf, or the photograph on the side of a bus.

We must first distinguish between traditional pre-press technologies and the pre-imaging processes that are relevant in today's graphic communications industry. Pre-press processes are different from

the way we process images for electrophotographic imaging or imaging with an inkjet engine. We must also distinguish between preparing images for a lithographic press and a flexographic press. Electrophotography and inkjet are growing technologies used to produce customized — or individualized — communications materials. Lithography and flexography are used to manufacture mass-produced media products. These four imaging processes are the core imaging technologies that reproduce 90% of the images produced in the graphic communications industry.

Many graphic designers are not aware of what must happen to the computer graphics they produce in order to ready them for manufacturing reproduction. Their experience is limited to hitting 'command P' and their computer graphic magically transforming the illuminated masterpiece on their Apple Cinema Display, to the disappointing rendition that appears on the tray of their inkjet printer. Most of the pre-imaging processes are automated in software functions that are built into the print driver, so people are not aware of how a computer graphic must be prepared for an imaging device. Since more and more of the images produced through inkjet, electrophotography, lithography and flexography start their lives as computer graphics, it is important to understand these pre-imaging processes to properly design computer graphics for the manufacturing process.

This chapter will analyze six pre-imaging processes in detail, and describe how they are altered to prepare computer graphics differently for each of the four imaging technologies. We will refer back to the computer graphic design/creation process to outline how graphics could be altered so they can be more effectively reproduced with each imaging technology. This is the missing link in the graphic communications business in today's marketplace. Designers create computer graphics in software that is increasingly designed for electronic image creation. They do not realize that the same graphic they created for a home page on the Internet should not be used for the cover of a book. They email the image to a lithographic print production facility and the pre-press department of that facility does hand springs trying to alter the image to work on their sheet-fed presses. This adds time and cost to the job that is usually buried. The designer never gets feedback on how the design could be altered to be more effective for lithographic production.

When pre-press was a computer-to-film process, there were two important factors that ensured designers got this critical feedback. The software for computer graphic production was specialized for print creation and content could be photographed or computer-generated and combined on film. Computer graphic designers knew their image was only going to be used for the cover of a book and created it appropriately. They also had to submit their computer graphic to a graphic communications production facility that was separate from the lithographic print facility. If there were extra costs incurred to prepare the computer graphic for a lithographic press, the designer was informed and invoiced for the extra work the image preparation entailed. So the designers were working with computer graphic software that would not let them create imagery that was not appropriate for print production, and if they did dream up an image that did not work well, they were immediately informed of the extra costs they were incurring.

In the 21st-century marketplace, all graphics that drive our four primary imaging technologies are created on the computer. Computer graphics software is designed to create effects for images that will stay in the electronic media: web, broadcast, digital film, and hand-held communication technologies. Pre-imaging processes are either automated or a part of the print manufacturing business and usually considered the painful part of feeding the print machinery that no one wants to talk about. So computer graphic designers drive software that lets them create outrageous images for imaging reproduction manufacture. They are less concerned about the 'print' part of a media campaign, and manufacturers

are hesitant to inform them that their designs incurred extra costs to reproduce. We can contribute to a solution to this problem by analyzing all of the pre-imaging processes for each type of reproduction manufacture and link them back to the computer graphic design software.

We will examine six pre-imaging processes:

- Raster image processing (RIP) technologies that are common to all four manufacturing processes
- Colour management for repeatability, as a part of the RIP process
- Trapping to lithographic and flexographic specifications
- Transparency, which is a visual effect that has a great impact on imaging
- Imposition for pre-RIP and post-RIP for media utilization
- Preflight analysis and automation for computer file creation

5.2 Raster Image Processing

Wayne Collins

The raster image processor (RIP) is the core technology that does the computational work to convert the broad range of data we use to create a computer graphic into the one-bit data that drives a physical imaging device. Let's examine the creation of a single character of the alphabet, or **glyph**. A font file delivers PostScript language to the RIP that describes a series of points and **vector** curves between those points to outline the letter A. The RIP has a matrix grid at the resolution of the output device and computes which spots on the grid get turned on and which are turned off to create the shape of that letter A on the output device. The spots on the grid can only be turned on or off — which is how binary data is encoded — either as 0 or 1. The grid then acts as a switch to turn a mechanical part of the imaging engine on or off.

With computer-to-plate technology for lithographic printing plate production, a laser is used to expose an emulsion on a printing plate. Most plate-setters have a resolution of 2,000 to 3,000 lspi (laser spots per inch). The RIP calculates all the spots that must be turned 'on' to create the graphic that will be imaged on the printing plate. If the image fills a typical sheet-fed press, it is (30 inches x 3,000 lspi) x (40 inches x 3,000 lspi) = 1.08 trillion, which takes 10 gigabytes of computer memory to store and transfer. A printing plate for flexographic print production is created by turning a laser on and off at a slightly lower resolution. An inkjet printer uses the same RIP process to deliver the same one-bit data to each inkjet nozzle for each colour of ink in the printer. Most inkjet engines have a resolution between 600 and 1,200 spots per inch — so the matrix grid is smaller — but if it is an eight-colour printer, the data for all eight nozzles must be synchronized and delivered simultaneously. An electophotographic (Xerox) printer usually has a resolution similar to an inkjet printer and utilizes a similar RIP process to change a grid of electrostatic charges to positive or negative on an electrostatic drum that is the maximum media size the machine can image. Each colour in the printer has a separate **raster image** that charges the drum in the right spot to attract that colour of toner to that exact location. The data for each colour must be synchronized for simultaneous delivery. The data must refresh the charge on the drum after each print in order to pick up new toner. That is a very important fact to remember when we talk about personalizing print with variable data later in this chapter.

This basic understanding of RIP's place in a computer graphic workflow is essential to understanding how to prepare files for, and manage, RIP resources. It is also essential in solving some of the common problems we see in various RIPs. When we compare the two mass production imaging technologies, lithography and flexography, to the personalized imaging technologies, electrophotography and inkjet, we can identify some core similarities. In lithography and flexography, a high-powered laser is used to alter a physical emulsion that is durable and finely grained enough to let the laser image a spot that is one three-thousandth of an inch without affecting the spot of equal size beside it. We can reliably image that spot in a serif of a glyph set in one point type or a hair on a face in a photo that is imaged with a 5 micron frequency modulated (FM) screening pattern. The mass production technology assures us that the first print will be identical to the millionth print.

The raster grid of one-bit data that the RIP produces must be delivered to the imaging drum or the inkjet

nozzle for every image that is produced with an inkjet printer or an electrophotographic engine. This is what allows us to make every image different and personalize it for the person we are delivering the image to. It also makes the process slower and less reliable for mass production. The RIP produces a lower resolution raster grid, so the detail in photos and letter shapes is not as precise. We can have a RIP discard data if we have too much detail for the raster grid it is producing. The RIP does not do a good job of interpolating more data to produce additional detail in a photo or graphic shape if that information is missing to begin with.

That brings us to examining the resources that a RIP must have to produce a perfect raster for every graphic shape it renders, and for every colour being reproduced. The resource a RIP consumes is data. In the graphic communications industry, we should all wear T-shirts that say 'Pigs for data!' just to distinguish us from our media colleagues who are producing computer graphics for electronic media. If we think of a RIP as an auto assembly line we are feeding with parts, in the form of files in different data formats, it will help us understand how to make a RIP more efficient. If we feed too many parts into the assembly line, it is easier to throw some parts away than it is to stop and recreate a part that is missing. If we feed the assembly line with five times as many parts needed to make a car, it is still more efficient to throw parts away than it is to stop and recreate a missing part.

If we apply this analogy to image resolution, we can point to examples where designers regularly repurpose images from a web page to use on a book cover or poster print. The web page needs to deliver the photo across a network quickly and only needs to fill a typical computer screen with enough detail to represent the photo. A typical photo resolution to do that properly is 72 pixels per inch. Now remember that the raster grid for a lithographic printing press that will print the book cover is 3,000 lspi. Our RIP needs much more data than the web page image contains! Most of the photos we are reproducing today are captured with electronic devices — digital cameras, phones, scanners, or hand-held devices. Most store the data with some kind of compression to reduce the data the device has to store and transfer. Those efficiencies stop at the RIP though, as this computational engine has to decompress the data before applying it to the graphic page it is rasterizing. It is like breaking a steering wheel down to wires, bolts, and plastic sleeves that efficiently fit into a one-inch-square shipping package, and putting this 'IKEA furniture' steering wheel onto an auto production line for the assembler to deal with in two-point-two minutes!

On the other hand, we can capture a digital photo at 6,000 pixels per inch (ppi) and use it on a page scaled to half the original dimension. That is like packing a finished steering wheel in 10 yards of bubble wrap and setting it on the assembly line in a wooden shipping crate! So it is important for designers to pay attention to the resolution of the final imaging device to determine the resolution that the RIP will produce from the graphic files it is processing.

Halftone Screening

It is important to stop here for a discussion about halftone screening that a RIP applies to photographs and graphics to represent grey levels or tonal values in a graphic element. We described how the RIP makes a grid of one-bit data, but graphics are not just black and white — they have tonal values from 0% (nothing) printing to 100% (solid) printing. If we want to render the tonal values in-between in half percent increments, we need 200 addresses to record the different values. Computer data is recorded in bits, two values (on and off), and bytes, which are eight bits strung together in one switch. The number

of values a byte can record is 256 — the number of combinations of on and off that the eight bits in the byte can express. A computer records a byte of data for each primary colour (red, green, and blue — RGB) for each detail in a photo, as a *pixel* (picture element), which controls the phosphors on electronic imaging devices. A RIP must convert the eight-bit RGB values into the four primary printing ink colours (cyan magenta, yellow, and black — CMYK). There are two distinct steps here: (1) conversion from RGB to CMYK continuous tone data (24 bit RGB to 32 bit CMYK); and (2) continuous tone to one-bit screening algorithms. We have to be in the output colour space before we can apply the one-bit conversion. It converts the eight-bit tonal values into one-bit data by dividing the area into cells that can render different sizes and shapes of dots by turning spots on and off in the cell. A cell with a grid that is 10 laser spots wide by 10 laser spots deep can render different 100 dot sizes (10 x 10), from 1% to 99%, by turning on more and more of the laser spots to print. If we think back to the plate-setter for lithographic platemaking, we know it is capable of firing the laser 2,000 to 3,000 times per inch. If the cells making up our printing dots are 10 spots square, we can make dot sizes that have a resolution of 200 to 300 halftone screened dots in one inch. A RIP has screening (dot cell creation) algorithms that convert the data delivered in RGB pixels at 300 pixels per inch into clusters of laser spots (dots) for each printing primary colour (CMYK).

This description of how a RIP processes photographic data from a digital camera can help us understand why it is important to capture and deliver enough resolution to the RIP. It must develop a detailed representation of the photo in a halftone screened dot that utilizes all of the laser spots available. The basic rule is: Required PPI = 2 x lines per inch (LPI) at final size. So if you need to print something at 175 lines per inch, it must have a resolution of 350 pixels per inch at the final scaled size of the reproduction. Use this rule if you are not given explicit direction by your print service provider. You can use a default of 400 ppi for FM screening where lpi is not relevant.

WYSIWYG

It is important to know that each time we view a computer graphic on our computer screen, it is imaging the screen through a RIP process. The RIP can change from one software program to another. This is why some PDF files look different when you open them in the Preview program supplied with an Apple operating system than they do when opened in Adobe Acrobat. The graphics are being processed through two different RIPs. The same thing can happen when the image is processed through two different printers. The challenge is to consistently predict what the printed image will look like by viewing it on the computer screen. We use the acronym **WYSIWYG** (what you see is what you get) to refer to imagery that will reproduce consistently on any output device. Designers have faced three significant challenges in trying to achieve WYSISYG since the advent of desktop publishing in the early 1980s.

The first challenge was imaging typography with PostScript fonts. The second was colour managing computer screens and output devices with ICC profiles. The third and current challenge is in imaging transparent effects predictably from one output device to another. Font problems are still the most common cause of error in processing client documents for all imaging technologies. Let’s look at that problem in depth before addressing the other two challenges in achieving WYSIWYG.

Font Management

The development of the PostScript computer language was pioneered by Adobe in creating the first device independent font files. This invention let consumers typeset their own documents on personal computers and image their documents on laser printers at various resolutions. To achieve WYSIWYG on personal computer screens, the font files needed two parts: screen fonts and printer fonts. Screen fonts were bitmaps that imaged the letter shapes (glyphs) on the computer screen. Printer fonts were vector descriptions, written in PostScript code, that had to be processed by a RIP at the resolution of the printer. The glyphs looked significantly different when imaged on a 100 dpi laser printer than they did on a 600 dpi printer, and both were quite different from what graphic artists/typographers saw on their computer screen. That was not surprising since the shapes were imaged by completely different computer files — one raster, one vector — through different RIP processors, on very different devices. Many graphic designers still do not realize that when they use Adobe type font architecture they must provide both the raster screen font and the vector PostScript font to another computer if they want the document that utilizes that font to process through the RIP properly. This was such a common problem with the first users of Adobe fonts that Microsoft made it the first problem they solved when developing TrueType font architecture to compete with Adobe fonts. TrueType fonts still contained bitmap data to draw the glyphs on a computer screen, and PostScript vector data to deliver to a RIP on a print engine. The TrueType font file is a single file, though, that contains both raster and vector data. TrueType fonts became widely distributed with all Microsoft software. Microsoft also shared the specifications for TrueType font architecture so users could create and distribute their own fonts. The problems with the keeping screen font files with printer font files went away when graphics creators used TrueType fonts.

The quality of the fonts took a nose dive as more people developed and distributed their own font files, with no knowledge of what makes a good font, and what can create havoc in a RIP. Today, there are thousands of free TrueType fonts available for downloading from a multitude of websites. So how does a designer identify a good font from a bad font? The easiest way is to set some complicated glyphs in a program like Adobe InDesign or Illustrator and use a ‘convert to outlines’ function in the program. This will show the nodes and bezier curves that create the glyph. If there are many nodes with small, straight line segments between them, the font may cause problems in a RIP. Remember that PostScript was meant to be a scalable device independent programming language. If the poorly made glyphs are scaled too small, the RIP has to calculate too many points from the node positions and ends up eliminating many points that are finer than the resolution of the raster image. On the other hand, if the glyph is scaled too large, the straight lines between points make the smooth curve shapes square and chopped-looking. These fonts are usually created by hand drawing the letter shapes, scanning the drawings, and auto tracing them in a program like Illustrator. The ‘convert to outlines’ test reveals the auto tracing right away, and it is a good idea to search out another font for a similar typeface from a more reputable font foundry.

Another good test is to look at the **kerning** values that are programmed into the font file. Kerning pairs are glyph shapes that need the space between them tightened up (decreased) when they appear together. A good font usually has 600 to 800 kerning pair values programmed into its file. The most common pair that needs kerning is an upper case ‘T’ paired with a lower case ‘o’ (To). The ‘o’ glyph must be tucked under the crossbar of the T, which is done by programming a negative letter space in the font file to have less **escapement** when the imaging engine moves from rendering the first shape to when it starts imaging the second shape. If we set the letter pair, and put the curser in the space between them, a

negative kerning value should appear in the kerning tool. If no kerning value appears, the font is usually a poor one and will cause spacing problems in the document it is used in.

Another common problem occurred when combining Adobe Type 1 fonts with TrueType fonts in the same document. Adobe was the creator of the PostScript programming language, and although it was easy enough to copy its code and create similar fonts, Adobe has maintained fairly tight control over licensing the PostScript interpreting engines that determine how the PostScript code is rendered through a raster image processor. The RIP stores the glyph shapes in a font file in a matrix that can be speedily accessed when rendering the glyphs. Each glyph is assigned an address in the matrix, and each font matrix has a unique number assigned to it so that the RIP can assign a unique rendering matrix. Adobe could keep track of its own font identification numbers but could not control the font IDs that were assigned to TrueType fonts. If a TrueType font had the same font ID number as the Adobe Type 1 font used in a document, the RIP would establish the glyph matrix from the first font it processed and use the same matrix for the other font. So documents were rendered with one font instead of two, and the glyphs, word spacing, line endings, and page breaks were all affected and rendered incorrectly. For the most part, this problem has been sorted out with the creation of a central registry for font ID numbers; however, there are still older TrueType font files out there in the Internet universe that will generate font ID conflicts in a RIP.

Adobe, Apple, and Microsoft all continued to compete for control of the desktop publishing market by trying to improve font architectures, and, as a result, many confusing systems evolved and were discarded when they caused more problems in the RIPs than they solved. There is a common font error that still causes problems when designers use Adobe Type 1 fonts or TrueType fonts. Most of these fonts only have eight-bit addressing and so can only contain 256 glyphs. A separate font file is needed to set a bold or italic version of the typeface. Some page layout programs will allow the designer to apply bold or italic attributes to the glyphs, and artificially render the bold or italic shapes in the document on the computer screen. When the document is processed in the RIP, if the font that contains the bold or italic glyphs is not present, the RIP either does not apply the attribute, or substitutes a default font (usually Courier) to alert proofreaders that there is a font error in the document. The line endings and page breaks are affected by the error — and the printing plate, signage, or printout generated becomes garbage at great expense to the industry.

To solve this problem, Adobe actually cooperated with Microsoft and Apple in the development of a new font architecture. OpenType fonts have unicode addressing, which allows them to contain thousands of glyphs. Entire typeface families can be linked together to let designers seamlessly apply multiple attributes such as condensed bold italic to the typeface, and have the RIP process the document very closely to what typesetters see on their computer screen. PostScript is also the internal language of most page layout software, so the same OpenType font files are used to rasterize the glyphs to screen as the printer’s RIP is using to generate the final output. There can be significant differences in the RIP software, but many font issues are solved by using OpenType fonts for document creation.

One common font error still persists in the graphic communications industry that acutely underlines the difference between creating a document on a single user’s computer but processing it through an imaging manufacturer’s workstation. Designers usually own a specific set of fonts that they use for all the documents they create. The manufacturer tries to use the exact font file each designer supplies with the document. The problem once again involves the font ID number, as each font file activated in an operating system is cached in RAM memory to make the RIP-to-screen process faster. So the font

files the manufacturer receives can be different versions of the same font created at different times, but assigned the same font ID number. For example, one designer uses a 1995 version of Adobe's Helvetica typeface and another uses a 2015 version, but the two typefaces have the same font ID number. The manufacturer's operating system will not overwrite the first font matrix it cached in RAM, so it is the first font file that renders the document on screen and will be sent down to the RIP. Usually, there are few noticeable changes in the glyph shapes. But it is common for font foundries to adjust kerning values between letter pairs from one version to the next. So if a manufacturer has the wrong version of the font file cached in RAM, a document can have line-ending changes and page reflows. This is a hard error to catch. There are programs and routines the imaging manufacturer can implement to clear the RAM cache, but many times, more 'garbage' is generated before the problem is diagnosed. Modern PDF creation usually includes the production of a uniquely tagged font subset package that only contains the glyphs used in the document. The unique font subset ID avoids the potential for font ID conflicts.

Managing fonts on a single user computer has its own challenges, and Apple has included Font Book with its operating systems to help manage fonts in applications on an Apple OS. Adobe offers Typekit with its latest Creative Cloud software to provide greater access to a wide variety of typefaces from a reliable foundry. Third-party font management programs like Suitcase Fusion also help graphic artists manage their fonts for repurposing their documents effectively. It is still the responsibility of individual operators to know how to use the fonts in their documents. They should also make sure that the fonts are licensed and packaged to deliver to other computer systems so that they can drive many different RIPs on a wide variety of output devices.

5.3 Colour

Wayne Collins

The second challenge with implementing WYSIWYG for electronic documents that are imaged on substrates is managing colour expectations. Chapter 4 discussed the challenges of colour management from the perspective of how we see colour, measure it, and manage output devices with ICC profiles. In this chapter, we will explore colour management from the perspective of how we recognize and manage the ICC profiles that are embedded in client documents. We will also explore the preflight issues in managing spot colours in documents. This will also lead us to a discussion of trapping for lithography and flexography.

To design with colour in computer graphics software, we must understand how the software generates the colour values. Page layout and illustration software usually have several systems for creating colour on a page. The colour settings or colour preferences attached to a file can change from one document to another or in the same document restored from one computer to another. If a designer uses the RGB colour model to specify the colours in a document, the colours on the monitor can change depending on the translations done to the colour settings. This is a major turning point for designers creating documents intended to stay in the electronic media. No one pays much attention to how a particular colour of red is rendered from one web browser to another. Designers pay more attention to how the colours interact relative to one another in web-page documents. It is only when we image a computer graphic on a substrate that we must pay attention to rendering the exact hue of red from one device to another. Coca-Cola has very exact specifications for the red used in its documents that tie into its brand recognition. So designers for documents intended for imaging on substrates must use colour models that are proven to render exactly the same results from one output device to another.

Pantone Colours

This is a lofty ideal that the graphic communications industry aspires to. There are systems in place that are proven to render very accurate results. There are challenges in understanding the systems, and one wrong step in the process and accuracy is destroyed. The process starts with how a designer chooses colours in a document. The most-used system for choosing accurate colours was created by the Pantone company. Pantone has developed a library of ink recipes that are published as swatch books. A designer can buy a printed book of a library of colours that matches an electronic library that can be imported into computer software programs. Designers compare their on-screen rendering of a colour to the printed sample swatch. If a designer is developing a corporate identification package with logos that use Pantone 123 and Pantone 456, the designer can be assured that the documents he or she creates will be imaged with inks that have similar spectral values to the swatch books used to choose the colour. I say similar, because the swatch books fade over time, and the substrates the books are printed on don’t usually match all the substrates a corporate logo is imaged on.

It is also important to realize that the Pantone library was created to mix pigments for **spot colour** inks rather than **process colour** inks. Spot colours are mixed independently and must each be applied

to the substrate independently. Process inks are only the four primary colours: cyan, magenta, yellow, and black. Process inks are transparent and are intended to be combined by halftone screening different percentages on a substrate to render any colour in the Pantone library. Spot colour inks are more opaque and are intended to be applied to a substrate one at a time, through distinctly separate printing units. Since most colour photography is colour separated to render the photo in only the four primary process inks, most documents are created intending to convert the spot colours to process colours. They can be imaged with the photographs in the document. A designer must know how many colours the output device is capable of when deciding which colours will remain as spot colours and which will be converted to CMYK process colours. Most inkjet and electrophotographic devices are only capable of imaging with the four process colours. Some lithographic presses have extra printing units that can print spot colours, so six- and eight-colour presses commonly print the four process colours and two (or four) extra spot colours in one pass. It is not uncommon to have 10- and 12-colour flexographic presses that image no process colours but use 12 spot colours. This is because, historically, flexo plates cannot consistently reproduce very fine halftone dots reliably. This is changing with the development of high-definition plating technology, so we are seeing more photographic content produced well on flexographic presses. Flexography is primarily used in the packaging industry where spot colours are very closely tied to brand recognition in retail outlets. A designer must be very aware of the imaging technology used to reproduce a document when deciding which colours will remain spot colours.

This is where the next round of challenges begins when preflighting (assessing) documents for imaging to substrates. If design elements stay as spot colours, it is a simple process to maintain the spot colour on the output device and to image with the appropriate ink or toner. Some software will not maintain the spot colour in a document easily in some situations. Usually, the problem comes with applying gradients to spot colours. It is very easy to introduce a median colour value on a spot colour gradient that is simulated with a process colour value. The screen version displays a nice smooth gradient that looks like what the designer intended to create. When imaging on a substrate, the gradient will have to be broken down into individual colours: from the solid spot colour to a value of CMYK and back to spot colour. It is very hard to recognize this by viewing the document, or even a composite PDF file. Viewing separated PDF files, or using a ‘separations’ tool in Acrobat, will show the problem before it gets to a printing plate.

There are also colour problems associated with nested files generated in different software. For example, if we create a magazine page with a headline colour named “PMS 123,” add a logo created in Adobe Illustrator with type in a colour named “Pantone 123,” and insert a PDF ad created in Apple’s Pages layout with a border specifying “PANTONE 123,” then even though they are all the same, colour-separating software will generate three separate spot colour plates for that page. The spot colours have to be named exactly the same and come from the same library. Some modern workflows include aliasing rules that will match numbered PMS colours to try to alleviate the problem. Colour libraries can be a problem as well, especially if our software allows the library to convert the spot colour to a process colour. The same colour library in two different versions of Adobe’s Creative Suite software can generate a different process colour recipe for the same Pantone colour. This is not a problem if all the document elements are created within one software package and all spot colours are converted to process colours. The problem arises when a designer places a graphic file from other software on a page with the same colour elements. A logo created in an older version of Adobe Illustrator will use that colour library to look up process colour recipes that can be very different from the recipes in a recent colour library used in Adobe’s InDesign software. So all the Pantone orange colours in a document are supposed to look the same, but do not because the spot colour to process colour conversion has not been

done with the same library. The problem becomes worse when we combine files from different software vendors, as designers often have to do when building a document. It is common these days to bring together graphics created in Microsoft Office and Apple software and generate a final document with Adobe InDesign. The best way to create consistent documents for reproduction is to specify a common CMYK colour value that will print reliably on the output device.

Pantone also publishes a swatch book that shows the difference between the swatches run as spot colour ink mixes, and the same swatch printed as halftone screen builds of process inks. This is a designer's most valuable tool for specifying process ink recipes. It also illustrates that many Pantone colours cannot be simulated very well using halftone screen values of the four process inks. It is very apparent that very vibrant orange, purple, and green Pantone spot colours are not achievable with process inks. There are systems like Hexachrome for colour separations that use more than just CMYK inks to extend the gamut of the Pantone colours that can be reproduced. There are also more and more inkjet and electrophotographic engines that will use extra spot colours to extend the colour range of the device beyond CMYK. The businesses that employ those devices usually know they are unique in the marketplace and have developed marketing tools to help designers use those capabilities successfully.

Accuracy in Design

If we reflect back to the concept of WYSIWYG for a moment, we can use the Pantone selection process to illustrate the challenge very well. If we ask a designer to choose colours for a document based on computer screen displays, we know that the RGB or HSL values they can select will be far too vibrant for reproduction with any imaging engine. To set proper expectations for WYSIWYG, we ask the designer to calibrate a monitor and select the proper output profiles to tone down the screen view and set more realistic expectations. We also ask that a print designer use printed swatch books to select from a library of specified colours and assign realistic CMYK process colour values to her or his colour palette. If those steps are followed, there is a very reasonable chance that the process will achieve WYSIWYG. However, it can break down in a few places. The spot colour swatch books set expectations about colours that cannot be achieved with process inks. When a mixture of spot colours and process inks are used, it is difficult to display both on the same computer screen with reliable colour. Graphics files can originate in different software with different libraries using different process colour recipes for the same Pantone colours.

There are also many spot colour libraries to choose from, and designers don't know when to use each library. We have described why the Pantone library is a North American standard, and some of its limitations. There are other design communities in the world that use spot colour libraries that are included as choices in graphic creation software tools. There are almost as many spot colours to choose from as there are free fonts files to download from the Internet. Spot colour classification has led to thousands of discrete colours being given unique names or numbers. There are several industry standards in the classification of spot colour systems. These include:

- Pantone, the dominant spot colour printing system used in North America and Europe.
- Toyo, a spot colour system common in Japan.
- DIC colour system guide, another spot colour system common in Japan.
- ANPA, a palette of 300 colours specified by the American Newspaper Publishers Association for spot colour usage in newspapers.

- GCMI, a standard for colour used in package printing developed by the Glass Packaging Institute (formerly known as the Glass Container Manufacturers Institute, hence the abbreviation).
- HKS is a colour system that contains 120 spot colours and 3,250 tones for coated and uncoated paper. HKS is an abbreviation of three German colour manufacturers: Hostmann-Steinberg Druckfarben, Kast + Ehinger Druckfarben, and H. Schmincke & Co.
- RAL is a colour-matching system used in Europe. The RAL Classic system is mainly used for varnish and powder coating.

The guiding principle for using any of these spot colour systems is to check that the manufacturer of the reproduction is using that ink system. The Trumatch library is quickly gaining favour as a tool for colour selection. That library of spot colours has been developed to be exactly replicated with process colour halftone screening. There are no spot colours a designer can choose from that library that cannot be reproduced well with standard process inks. As more computer graphics are being produced on digital imaging devices that only use CMYK, this colour library is becoming the choice for cross-platform or multi-vendor media publications.

5.4 Trapping

Wayne Collins

Trapping can be a very complex procedure in pre-imaging software for certain imaging technologies. It is an electronic file treatment that must be performed to help solve registration issues on certain kinds of press technologies. Generally, if a substrate has to move from one colour unit to another in the imaging process, the registration of one colour to another will not be perfect. That mis-registration must be compensated for by overlapping abutting colours. As soon as two colours touch in any two graphic elements we must create a third graphic element that contains both colours and overlaps the colours along the abutment line. That third element is called a *trap line* and can be generated many different ways that we will review.

Electrophotography

First let's look at the differences between the four most common imaging technologies and determine where and why we need to generate these trap lines. Electrophotography, or toner-based digital printers, generally use only process colours. Each time an electrostatic drum turns, it receives an electrical charge to attract the toner colour it is receiving. The drum keeps turning until all colours of all toners are on the drum, and then all colours are transferred to the substrate at one time. There is no chance for mis-registration between the cyan, magenta, yellow, and black toners as they are imaged at the resolution of the raster generated by the RIP, and the placement of the electronic charge for each colour can be adjusted until it is perfect, which makes it stable from image to image.

Lithography

Let's compare electrophotography to the lithographic print process. In lithography, a printing plate is generated for each colour and mounted on a plate cylinder. The plates are registered by manually turning wrenches to hold plate clamps, so the plate-mounting procedure can generate registration errors. Each colour printing unit is loaded with a separate ink, a plate that has been imaged to receive that ink, and a blanket that offsets the image from the plate before it transfers it from the blanket to the substrate. This is another mechanical transfer point that can cause registration errors. Most high-end lithographic presses have servo motors and cameras that work together to adjust for mechanical registration errors as the press runs. The substrate must travel from one printing unit to the next, and it is here that most registration errors occur. There are slight differences in the substrate thickness, stability, lead (or gripper) edge, and a different rate of absorbing ink and water that cause slight mis-registration. Also, consider that most sheet-fed litho presses are imaging around 10,000 sheets per hour, and we are only talking about movements of one-thousandth of an inch. On most graphic pages, however, the naked eye can see a mis-registration of one-thousandth of an inch, so the process must be compensated for. The solution is generating trap lines to a standard for lithography of three one-thousandths of an inch. This trap line allowance in abutting colours allows for mis-registrations of two-thousandths of an inch that will not show on the final page.

Inkjet

Inkjet is the next imaging technology we must assess and compare. The print heads on all inkjet machines are mounted on the same unit travelling on the same track. Each ink is transferred one after the other and the substrate does not move after receiving each colour. It is like electrophotography in that mis-registration between print heads can be adjusted electronically, and once in register remain stable for multiple imaging runs on the same substrate. If the substrate is changed between imaging, the operator must recalibrate to bring all colours into registration, and ensure the placement of abutting colours is perfect and no compensation is needed. As a result, no trapping will be needed for most inkjet imaging processes.

Flexography

Flexography is the fourth imaging technology we need to assess. This technology has the most points where mis-registration can occur. The printed image must be raised on the plate to receive ink from an anilox roller that can deliver a metered amount of ink. The computer graphic must be reduced (or flexed) in only one direction around the plate cylinder. A separate printing plate is developed for each colour and mounted on a colour unit that includes an ink bath, anilox roller, doctor blade, and a plate cylinder. The substrate travels from one print unit to the next on a continuous web that is under variable amounts of tension. If a graphic has a lot of white space around it, the substrate can be pushed into the blank space and cause distortion and instability in the shape and pressure of the raised inked image on the substrate. Flexography is used to image the widest range of substrates, from plastic films to heavy corrugated cardboard. This process absolutely needs trap lines generated between abutting colours. Standard traps for some kinds of presses can be up to one point (1/72 of an inch, almost five times our standard litho trap). Graphic technicians need to pay particular attention to the colour, size, and shape of the trap lines as much as to the graphic elements. In most packaging manufacturing plants, there are pre-imaging operators that specialize in creating just the right trapping.

Let’s examine some of the ways these traps can be generated. The simplest way is for a graphic designer to recognize that he or she is designing a logo for a package that will be imaged on a flexographic press that needs one-point trap lines generated for all abutting colours. The designer isolates the graphic shapes that touch and creates overprinting strokes on those graphic elements that contain all colours from both elements. That doesn’t even sound simple! (And it’s not.) It becomes even more complicated when the graphic is scaled to many different sizes on the same package or used on many different packages. So most designers do not pay attention to creating trap lines on the graphics they create and leave it to the manufacturer to create trapping for the specific documents on the specific presses they will be reproduced on.

There is specialized software that analyzes a document, determines where abutting colours are, and generates the tiny graphic lines as a final layer on top of the original graphic. This is done before the document goes to the RIP so it is raster-image processed at the same time as the rest of the document. Most RIPs process PDF files these days, and there are specialized plug-ins for Adobe Acrobat that will analyze a document, generate trap lines, and let an operator examine and edit the thicknesses, shape, and colour of those lines. It takes a skilled operator to examine the extra trap lines and determine if they are appropriate for the press they are going to be printed on. Press operators also need to determine the

trap values of their inks. This refers to the ability of one printing ink to stick to another. Inks vary in viscosity depending on the density and types of pigments they are carrying. The trap characteristics and transparency of a printing ink are part of what determines the printing order in which they are applied to the substrate. For example, a process primary yellow ink is very transparent and will not stick (trap) well if printed on top of a heavy silver metallic ink. The metallic silver is thick and very opaque, so it will hide everything that it overprints. A graphics technician must generate trap lines for a graphic that has metallic silver abutting to a process yellow shape. The technician will increase (spread) the shape of the yellow graphic to go under the abutment to the silver. The silver shape will not be altered, and when it overprints, the yellow ink will stick to and hide the yellow trap line shape. The best analogy, we have heard is from a press person — the peanut butter sandwich analogy. We know the jelly sticks to the peanut butter and the peanut butter will not stick to the bread if the jelly is spread first. If a press person does not know the trap values of the inks, he or she can make as big a mess of the press sheet as an upside-down peanut butter and jelly sandwich makes on the front of your shirt! For this reason, trapping should be left to the specialists and is usually applied to a final PDF file before it is sent to a RIP. Ninety percent of trap lines for lithographic and flexographic imaging reproduction are generated automatically by specialized trapping software. Operators are trained to recognize shapes and colour combinations that will cause problems on the press. They will custom trap those documents with the Acrobat plug-ins we talked about earlier.

Special Consideration for Black

There is one trapping combination that should be considered and applied to all four imaging technologies. It is the way that black ink is handled in the document and applied on the imaging device. Most type is set in black ink, and much of it overprints coloured backgrounds. In all four imaging technologies, black is the strongest ink and hides most of what it overprints. It is still a transparent ink and most process black ink is more dark brown than the rich dark black we love to see in our documents. If the size of the black type or graphic is large enough, we will be able to see the black colour shift as it overprints stronger or weaker colours under it. Graphic designers should pay attention to setting whether black type or graphics overprint the background, or knock out the background to print a consistent black colour. A useful rule of thumb is that type above 18 points should be knocked out and boosted. Raise this threshold for very fine faces such as a script where larger point sizes can overprint, and reduce it for excessively heavy fonts like a slab serif. If the graphic is large enough, it should also be ‘boosted’ with other process colours.

The way we handle black ink or toner deserves special consideration in all four imaging technologies. Black is a supplemental colour to the three primary process colours. It is intended to print only when the other three colours are all present in some kind of balance. In all imaging technologies, we must watch that our **total ink coverage** does not approach 400%, or 100% of each ink overprinting the other inks in the same place. This is usually too much ink or toner for the substrate to absorb. As a result, it will not dry properly and will offset on the back of the next sheet, or bubble and flake off the media in the fuser. Therefore, we must pay attention to how our photographs are colour separated, and how we build black in our vector graphics.

Colour Separations

When colour separating photographs, we can build in an appropriate amount of GCR or UCR to give us the right total ink coverage for the imaging technology we are using for reproduction. UCR stands for under colour removal. It is applied to very dark shadow areas to remove equal amounts of cyan, magenta, and yellow (CMY) where they exceed the total ink limit. For example, in sheet-fed lithography, a typical total ink limit is 360. In areas that print 100% of all four colours, UCR will typically leave the full range black separation, and remove more and more CMY the deeper the shadow colour is. A typical grey balance in shadows may be 95% cyan, 85% magenta, and 85% yellow. Including a 100% black, that area would have a total ink coverage of 365. Other imaging technologies have different total ink limits, and these can vary greatly from one substrate to another within an imaging technology. An uncoated sheet will absorb more ink than a glossy coated sheet of paper and so will have a different total ink limit.

GCR stands for grey component replacement, and it is intended to help improve grey balance stability in a print run and save on ink costs. GCR can affect far more colours than UCR as it can be set to replace equal amounts of CMY with black all the way into the highlight greys. This is particularly useful in technologies like web offset newspaper production. Grey balance is quickly achieved in the **make-ready** process and easily maintained through the print run. Black ink for offset printing is significantly cheaper than the other process colours, so there are cost savings for long runs as well. GCR is used in photos and vector graphics produced for other imaging technologies as well. Any process where grey balance of equal values of the three primary colours could be an issue is a smart place to employ GCR.

You may be wondering how we can check the shadow areas of every photo we use. These GCR and UCR values can be set in ICC profiles by linking the shadow and neutral Lab values to the appropriate CMYK recipes. When the ICC profile is applied for a given output device, the shadows get the proper ink limits, and the grey tones get the prescribed amount of black, replacing CMY values.

Keylines

Black keylines, or outline frames for photos, are common in many documents. This is another place where a document should have trapping software applied for every imaging technology. Outline strokes on graphics can also have a ‘hairline’ setting, which asks the output device to make the thinnest line possible for the resolution of the device. This was intended for in-house studio printers where the resolution is 300 dpi — so the lines are 1/300th of an inch. But the same command sent to a 3,000 lspi plate-setter will generate a line 1/3000th of an inch, which is not visible to the naked eye. These commands must be distinguished in PostScript and replaced with lines at an appropriate resolution — trapping and **preflight** software will do this.

Knock outs

The use of solid black backgrounds is becoming more popular in documents, which can cause problems in reproduction with all imaging technologies. The first problem is with filling in details in the graphic shapes that are knocked out of the solid black background. Fine type serifs, small registered or trademark symbols, or the fine hairlines mentioned above will all fill in and be obliterated when imaged. The

problem is multiplied when we boost the black colour by adding screened values of cyan, magenta, or yellow to the colour block. When white type or graphics knock out of these background panels, any slight mis-registration of any colour will leave a halo of that colour in the white type or graphic. This problem can also be solved with trapping software. Essentially, the trapping engine outlines all the white type with a small ‘black only’ stroke that knocks out the process colour that boosts the black, making the white type fatter in that colour separation. This ‘reverse trapping’ works well when applied to the four imaging technologies we have been examining: lithography, flexography, electrophotography, and inkjet.

5.5 Transparency

Wayne Collins

The biggest challenge in reproducing computer graphics on output devices in today's marketplace is dealing with transparency in graphic files. This truly emphasizes the importance of WYSIWYG in proofing for the graphic communications industry. We must first emphasize that page layout graphic software is not developed for producing documents for mechanical reproduction. This software prioritizes the creation of documents for viewing on electronic media; they are created on a computer screen for viewing on a computer screen. We have reviewed some of the issues with rasterizing vector shapes consistently, and reliably representing colour from one device to another. Viewing a graphic with three-dimensional transparent elements is significantly different on an illuminated medium where the light is transmitted compared to an opaque medium where the light is reflected. It is very hard to judge how the transparent effects will translate from one to another. There is room for the same kind of collaborative research in this realm, as there was in developing OpenType font architecture and ICC profiles.

The problems in WYSIWYG production for transparency fall in two categories. The first problem is setting expectations so a designer can make a reasonable prediction of how the document will look when imaged on a given media. The second problem is the sheer proportions of the computational processes we are asking of a RIP. PostScript is a three-dimensional language that allows a creator to stack and prioritize elements on a page. The RIP can literally 'throw away' raster data that is knocked out by graphic elements that completely cover the elements behind. If those elements have to show through the foreground elements by 20%, the RIP must hold much more raster data in physical memory addresses. Many times, data is lost if there are not enough addresses available for the computations, and this can change from one processing of the document to the next.

Designers can employ strategies at each level of document creation to manage these problems. The first strategy is to use layers well in document creation. By isolating different effects on separate layers, it becomes easier to isolate and edit the transparent effects when they don't produce the desired results in the final output. The layers can be included in a PDF file of the document, and this allows the possibility of relatively quick editing in PDF editing software closer to the output stage. This can be a completely different working style for some graphic artists. If we start with the premise that the computer screen representation of the document is NOT good WYSIWYG and will probably need editing, then we can justify working with layers more to isolate effects. We can organize design elements on layers after creation — when we are fine-tuning the effects. Usually, this is a good technique when creating many elements on several page dimensions. Designers can review their documents and decide if there are distinct dimensional levels, as page elements are pushed further into the background to pull other page elements forward. A simple example is a book cover for a retrospective, with pictures from four distinct decades. The photos and type from each decade can be set on distinct layers, and transparent values of 25%, 50%, 75%, and 100% can be set for each layer. The screen will render one version of the document, and the printer will render another. It is easier to fine-tune the four layer levels of transparency than to go back and set new transparency levels for dozens of individual page elements.

Another strategy that must be considered for processing multiple transparent page elements is allowing the page layout software to raster the page elements, so it sends raster data to the RIP. This technique treats the transparent elements, such as a photograph on the page, and allows the creator to choose the resolution of the raster. Care must be taken here to ensure overlapping vector elements will raster at the same resolution in the RIP. Let's say we have a type block that crosses a photo on the page, but it is transparent to let the photo show through the type. If we rasterize the transparent type at 300 ppi — the resolution of the photo — it will be significantly different from the raster of the vector type at the RIP, which might be 3,000 lspi for some plate-setters. The letter shape will be 10 times thicker over the photo, and that will be VERY noticeable if the type crosses the photo in the middle of the glyph. The solution is to make sure to raster the transparent type at 3,000 ppi to match the plate-setter raster. This makes the PDF file very large because it contains lots of raster data. But this solution is also a disadvantage because it does not allow late-stage editing of the transparent values in the PDF file. The advantage is that the transparency elements will have better WYSIWYG, process more consistently in multiple RIPs, and use less RIP resources in processing.

It is very important to be aware of the transparent elements you are creating in a document. It is not always apparent when using effects, plug-ins, or effects filters available in page layout software. Using a bevel or emboss effect, or a simple drop shadow, makes that page element use transparent routines in the RIP. Programs like Adobe InDesign let designers view all the transparent elements on a page. Designers should examine each one to decide if it should be rasterized before RIP-ing or at the RIP. This is a good point at which to decide if transparent elements can be grouped, or organized, on common layers. It is also a good point to decide how the transparent element contributes to the design, and how critical the level of transparency, or WYSIWG value, is in the overall design. In the retrospective book cover design referred to above, WYSIWYG is very important in communicating the message of the book and getting predictable results.

Transparent elements can be rasterized at the page layout stage, the PDF creation stage, and at the RIP stage for the final output device. Adobe Acrobat also has a tool to view transparent elements in a PDF file. It is important for a designer to compare the transparent elements in the PDF to those in the page layout software. The primary concern is that the elements rasterized in the PDF are no longer editable, so it is critical that the levels are right to create the desired overall effect. It is also important for a preflight operator to view the transparent elements in a PDF file to check what the RIP will have to process and to make sure the computational resources are available. If there are processing errors in the final output, they are most likely to occur in rendering the transparent objects. Viewing the transparent elements on a page in Acrobat should provide a mental checklist for the operator when she or he views the final output.

Communication Is Key

The graphic communications industry still has collaborative work to do to make the processing of transparent elements on a page more predictable and repeatable. It is important for designers to understand the problems they can be creating for a RIP, especially for output on an extremely high-resolution device like a plate-setter for waterless lithography. It is also important for operators who are managing documents with lots of transparency to be aware of the checkpoints in a document, and to know when there is not adequate WYSIWYG for the transparent elements on a page. Good questions for all stakeholders to ask when processing a document that relies on many transparent elements are:

- Where are the transparent elements?
- Did they process correctly?
- Is anything missing in the layers that should show through the transparency?
- Are there transparency values that can be adjusted to optimize the overall effect?

Let’s review the primary tools for reproducing transparent page elements in a document. We can utilize layers in a document for setting common transparency values. We should view all transparent elements in a document before and after creating a PDF file. There are several stages to rasterizing the transparent elements. The earlier we rasterize them, the less editable the document becomes, and the more consistent the final output will be. We are creating a larger file to process when we rasterize transparent elements early. Much less computational resources are required at the RIP, and the more predictable our final output will be. When managing late-stage processing of transparency, we must be aware that what we are viewing on a computer screen is not necessarily a good representation of the final output. Graphic artists at all levels of production must pay attention to the transparent areas of a document to check for accuracy.

5.6 Imposition

Wayne Collins

Imposition of individual graphics page files serves two primary purposes. The first, and perhaps most important purpose, is to utilize media and manufacturing equipment with the most economic efficiencies. The second is to add what has historically been referred to as ‘furniture’ to the manufactured sheet to control processes. We will discuss both priorities for each primary imaging technology we are examining in this book. There is also a range of equipment capabilities for each technology that affects how documents are imposed. There are a few ways to impose the files either pre-RIP or post-RIP. We will also look at ways of imposing in graphic creation software and in specialized imposition software.

The first technology we will look at is electrophotography, where imposition is perhaps the most underutilized. Electrophotographic, or Xerox-type copiers are usually used for short-run lengths with demands for instant turnaround. Duplexing is the simplest type of imposition, but there are four choices for how to orient the back of a single page on the front. The duplexing style can be specified in the print driver, in the PDF file, or in the RIP. Most small printers will turn on duplexing, rather than image the fronts, turn the printed paper over the right way, and image the back of the sheet. Fewer will take the time to utilize the machine and media capabilities to step and repeat an image two up on a larger sheet to half the run time. Yet, as a manufacturing process, electrophotography is the slowest technology for image reproduction, and the most in need of saving time. There are simple rules of automation that can be programmed in a RIP to automatically impose if a run length is over 100 copies. For example, letter-size documents are the most often imaged on this type of equipment. If run lengths of more than 100 copies were imposed two up on a tabloid sheet, it would halve the run time and open up more imaging time on the machine. This introduces another process — cutting the sheets in half before final delivery. Management can determine how well the imaging engine run time is utilized and when it is efficient to have the same operator cut printed sheets in half. Making that management decision requires a knowledge of workflow options for efficiency. Those efficiencies are the primary purpose of implementing imposition software.

Using a ‘step and repeat’ or ‘duplexing’ imposition of single-page file is the simplest example of imposing files for electrophotographic workflows. More and more copiers have capabilities to fold and bind the finished product ‘inline’ in one continuous process. This process is driven by imposing the single-page files in the correct order as they are processed in the RIP, so they image in the proper spot on the media to fold, and bind together in the correct order.

Imposing page files for binding styles usually follows two types of machine capabilities: saddle stitching and perfect binding. **Saddle stitching** is a binding process that folds the media after it is imaged on both sides, stacks the printed folded sheets one inside the other, and applies staples to the spine of the book. The other dominant style of book binding built into copiers is **perfect binding**. Media is imaged on both sides and folded, but the folded sheets are stacked on top of each other, glue is applied, and a cover sheet is wrapped around the glued book to encase the spine. The pages have to be imposed in a completely different order on the printed sheets. The first and last sheets of a saddle-stitched book are imaged on the

same sheet, whereas a perfect-bound book has the first pages imaged on the same sheet, and last pages on separate sheets of media.

There are many options for folding a sheet of substrate before binding it together. The options increase the larger a sheet is. An imposition must account for the preferences that are best practices for the specific machines involved. If we look at sheet-fed lithography first, we can identify some common best practices that can also apply to toner-based electrophotography and inkjet. We shall leave an examination of imposition for flexography to discuss nested dieline shapes for packaging applications.

Imposition standards are based on workflows for standard-sized pages, primarily letter-sized pages measuring 8½” x 11″. We speak of *two up* devices that can generally image substrates up to a maximum size of 12″ x 18″ or 13″ x 19.” Two up devices can accommodate two letter-sized pages plus bleeds, grip, marks, and colour bars — sometimes referred to as furniture on an imposed sheet. These four elements all serve a purpose in page reproduction manufacturing that we shall define later. *Four up* devices generally accommodate imaging substrates up to 20″ x 29″ to image four letter-sized pages. Forty-inch devices are referred to as *eight up* and image on a maximum sheet size of 30″ x 40″, which can image eight letter-sized pages and the furniture mentioned above.

There are four common styles of imposition for eight up devices: sheet-wise, work and turn, work and tumble, and cut and stack.

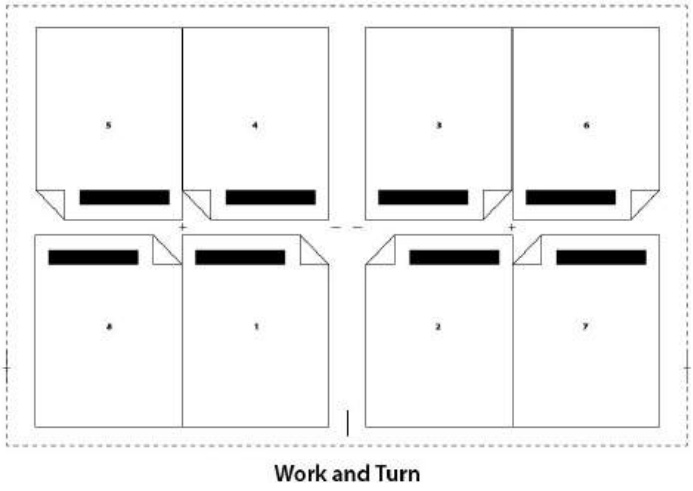

Figure 5.1 work and turn(a)

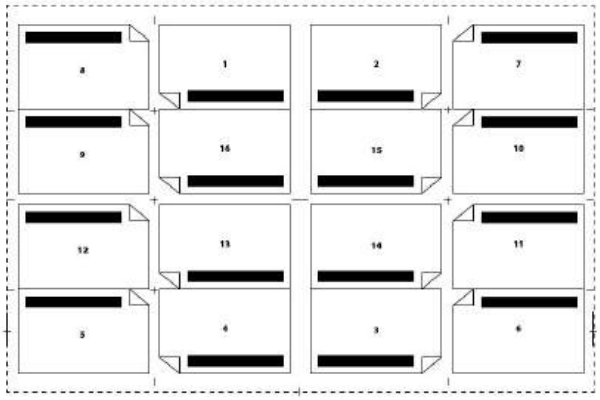

Figure 5.2 work and turn(b)

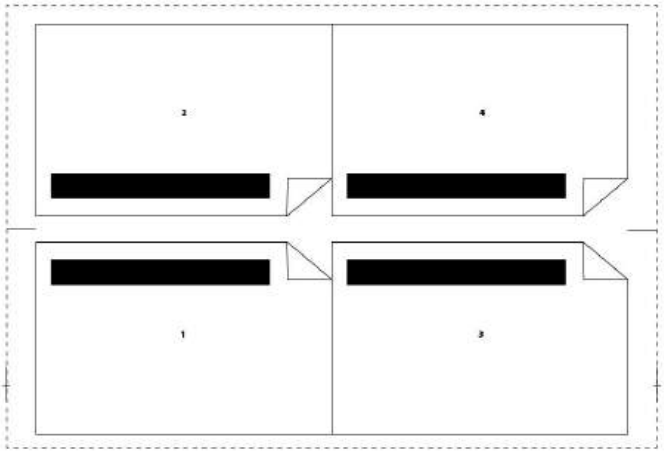

Figure 5.3 work and tumble

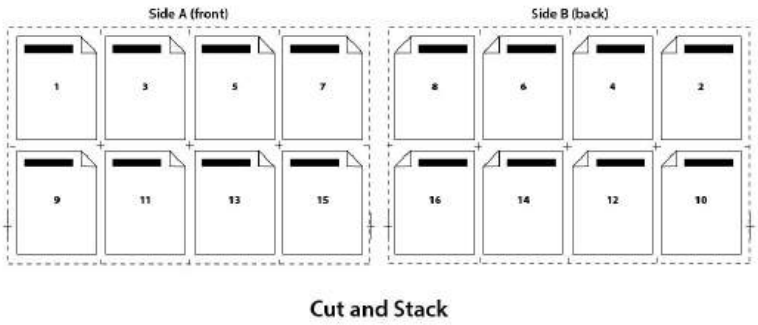

Figure 5.4 cut and stack

Sheet-wise impositions image the fronts of all pages on one side of the sheet and impose all the backs on a separate set of plates for a press run that will back up all the sheets. Work and turn imposes the fronts on one half of the sheet and the backs on the other half, with the axis running perpendicular to the grip of the sheet (see Figures 5.1 and 5.2). Work and tumble imposes all fronts on the bottom half of a sheet and backup images on the top half of the sheet (see Figure 5.3). The sheets are flipped halfway through the press run, with the axis parallel to the grip edge of the sheet. Cut and stack imposes the pages so full press sheets can be collated, and the collated sheets cut and stacked in order on top of each other to make a final book (see Figure 5.4).

Lithographic web offset presses have imposition orders that depend on how wide the web of paper is, and how many web rolls and half rolls will be brought together before folding on a first former, and cutting on a second former. There are many options for configuring a web-fed litho press, depending on the number of pages in a publication. Usually, the entire publication is printed and folded by running the stack of web paper together and folding it in half over a former.

Imposition has to account for creep and bottling when imposing for thicker publications. *Creep* pushes the image on a page closer in to the spine the further the page is toward the spine, by the width of the thickness of the publication at the stapled, spine edge. *Bottling* skews the image on a page to account for the skewing of web rolls of paper that are folded in very thick signatures. The thicker the folded signature of a bound book, the more skewing takes place, which should be accounted for in the 'Bottling' value in an imposition.

Imposition for inkjet mediums is usually done when the image is rasterized. The RIP will store the raster image and nest several raster images together to fill the dimensions of the media being imaged. This is usually set as an automated function in the RIP, and is tied to the size and cost of the media being used. When imaging very low resolution images on very low cost media, the manufacturer is usually more concerned with the speed of the machine than the utilization of the media. If an expensive media is being used, the automatic imposition will be utilized in the RIP to save media. Often inkjet images are not square, and the media will be die cut or cut with a router after imaging. The RIP can be set to

impose the images so the shapes nest inside each other. This is usually outside of the automatic features for imposition in a RIP, and requires operator intervention. In that case, the imposition operator must know the die cutting or router processes very well to make sure the imaged media can be cut out even though it is nested with another image.

This nesting of images to be die cut after imaging is most prevalent in flexographic printing for packaging. Most package or label shapes are not square and the media is expensive. The imposition function becomes very important for preparing flexographic plates. Nesting the die cut shapes for several packages together on a continuous roll of media takes very specialized software and a highly skilled operator. There are many variables to consider, including media thickness, ink coverage, die shapes, glue releases, and image floor on the flexo plate. Flexo imaging for packaging generally takes more understanding of CAD software and the construction of the final three-dimensional product. Imposition operators must know the structural requirements as well as the press limitations to nest together several package images on the same press run.

The final consideration for all impositions in all imaging technologies is the computer resource requirements for the RIP. We usually require an imaging engine to raster a single document, and proof it one page at a time through a proofing device. When we impose the same document with many other pages in completely different orientations, sometimes RIP processing errors can occur. Fonts drop out, and more commonly, transparent elements do not process properly. This is another checkpoint to make sure the imposed image matches the proof of the single page. It is essential to discuss preflighting for print at this point to establish where the routine checkpoints in document processing should be.

Media Attributions

- work and turn 01 by Ken Jeffrey
- work and turn 02 by Ken Jeffrey
- work and tumble by Ken Jeffrey
- cut and stack-03 by Ken Jeffrey

5.7 Preflight

Wayne Collins

We have covered quite a few parameters that must be considered when preparing a computer graphic for manufactured image reproduction. The parameters shift with different substrates and imaging technologies. The task of checking a computer graphic document in preparation for the manufacturing process is called preflight. Most graphics are created by designers who are working separately from the manufacturer. In some cases, preflight preparation is the responsibility of the designer, or graphics creator, and in some cases, it is the responsibility of the manufacturer. Some manufacturers charge extra if they have to correct a graphics file that is not prepared properly for their imaging process. Most do not charge extra for preflighting, trapping, or imposing a file for their imaging process. Problems occur when all parties believe a file is prepared properly, but it causes errors in the RIP process. A poor font, improper colour separations, and transparency settings that drop out layers on a page are problems that occur most often. This is when time, materials, and money are wasted, and critical media campaign deadlines are missed. Preflight tries to catch the problems before they reach the RIP.

Designers or graphics creators can purchase separate preflight software that will generate reports about a PDF or PostScript file before they submit it to a manufacturer. The most popular dedicated preflight software is from Markzware and is called FlightCheck. There are also a few other companies that are popular in the marketplace. Enfocus bundles its preflight software with a suite of PDF editing tools called Pitstop. The Adobe Creative Suite has preflight functions built into its software suite. Adobe InDesign is the page layout software of choice for creating multi-page documents such as brochures, pamphlets , or books. The preflight module in InDesign will generate a report that can be included with the packaged contents a designer should provide to a manufacturer. The report will list important information about the InDesign document, such as that found in the list below. Adobe Illustrator also has a built-in preflight tool, and Adobe Acrobat has preflight tools that designers should use to analyze their PDF files before submitting them to a RIP.

Various industries have set PDF standards. The magazine publishing industry, for example, developed a PDF/X standard to ensure PDF/X files can be written only if they meet a set of specifications that are common for lithographic magazine production. Other manufacturing processes adopted the standards if they were appropriate for their imaging technology and PDF workflows.
Most preflight software checks for the following elements in an electronic document:

- File format
- Colour management
- Fonts
- Spot colour handling
- Page structure
- Thin lines
- Black overprint

- Trapping

5.8 Summary

Wayne Collins

This chapter has looked at computer graphic creation through the lens of a manufacturer that must reproduce the electronic image on a substrate. The image must be processed through a RIP that drives a laser, or other imaging technology, to transfer pigments to that substrate. There are unique variables that must be considered in preparing the computer graphic for the reproduction process. We have explored routines for processing vector data such as fonts through a RIP, spot colour handling, trapping, and imposition. The next chapter will look at each of the imaging technologies in more depth.

Questions to consider after completing this chapter:

1. Describe six pre-imaging file analysis processes that should be considered when developing a computer graphic for reproduction manufacture.
2. Describe four major imaging technologies that utilize computer graphics to image on different substrates.
3. Describe the difference between raster data and vector data when creating a computer graphic file.
4. Compare the raster resolution of the data for a typical lithographic plate-setter compared to the resolution of a typical inkjet device.
5. How many addressable values can be recorded in an eight-bit byte of computer data?
6. What does the acronym WYSIWYG stand for?
7. How many kerning pairs are present in a ‘good’ font file?
8. What colour matching library has been developed exclusively for process colour printing inks (CMYK)?
9. What two printing processes must have trapping applied to computer graphics files before making printing plates?
10. What can a page layout artist do to a graphics file if the transparent elements on the page are dropping out or not processing in the RIP?

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Chapter 6. Imaging

6.1 Introduction

Roberto Medeiros

Learning Objectives

- Describe digital printing methods and their differences
- List the various inks used in inkjet printing and their characteristics
- Identify the key components of electrophotography
- Explain the seven steps of the electrophotographic process
- Describe the differences between toner types and how they affect imaging
- Evaluate the suitability of a paper for a project based on its characteristics
- Convert between paper basis weights and grammage
- Describe the key differences between page description languages
- Acknowledge the historical significance of Postscript in desktop publishing
- Explain the differences between PDF/X versions
- Describe the function of a RIP in a DFE
- Explain why calibration is critical in electrophotography
- Describe the key component in variable data printing
- Identify key benefits of open standard VDP formats

Digital printing can be defined as the reproduction of an image or document onto a substrate, directly from an electronic file, without the use of a fixed image plate. Traditional printing transfers an image permanently onto a fixed image plate, whereas digital printing transfers the image temporarily onto a photoconductive cylinder, called a drum, or directly onto the substrate itself. Printing in this manner provides some unique capabilities that sets it apart from traditional print methods. There is virtually no set-up or make ready, finishing tasks can be accomplished inline, and each sheet can have unique content, which makes this printing method ideal for publication printing, short print runs, or highly dynamic content.

The two most common digital printing methods in use today are electrophotographic (toner based) and inkjet (ink based). Both technologies are used in a wide range of printing devices from small desktop printers to large high-volume, high-speed digital presses. The term *digital press* is often used to describe commercial digital printers. In the past, speed was the determining factor of this designation. Today, we have specific criteria published by Idealliance, a not-for-profit member organization that develops standards and best practices for the digital media supply chain (Idealliance, n.d.). Apart from

speed, colour accuracy to meet a specification and consistency over long print runs are key parts of Idealliance's certification process.

For the purposes of this text, we will focus on digital printers and presses used in commercial printing rather than on consumer or office printers.

6.2 Inkjet

Roberto Medeiros

Inkjet printing is a type of digital imaging where drops of ink are jetted onto the substrate in a very precise pattern from a nozzle. This nozzle, also called the print head, is required to be very precise and accurate, which is a challenge when you consider that the goal is to get many thousands of tiny drops of ink to land exactly where needed on the printed surface. Over time, inkjet technology has become more advanced, allowing greater resolution, more accurate colour, and overall, finer visual fidelity to the original. The most common method of inkjet printing for commercial purposes is called drop-on-demand (DOD). This type of inkjet print head only fires each individual droplet of ink when needed (on demand) and comes in two types, thermal or piezoelectric (see Figure 6.1). Accuracy in DOD inkjet printing is achieved by keeping the print head close to the surface being printed (substrate) as the velocity of the jetted ink is low.

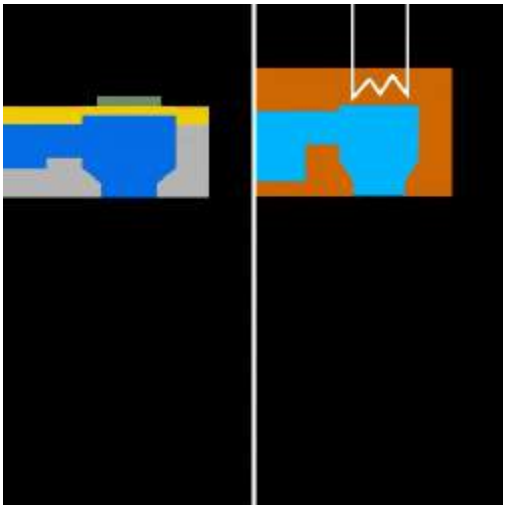


Figure 6.1 Piezoelectric head on the left, thermal on the right

Thermal Inkjet

In a thermal print head, each nozzle contains a special reservoir that is bounded by a heating element. When current is passed through the heating element, it causes the ink to expand rapidly, ejecting out of the nozzle to land on the substrate in a given position. The print head is made up of a matrix of many of these chambers, and each print head is connected to a different colour of ink. As the ejected ink leaves the chamber, fresh ink is drawn into the reservoir by surface tension and the vacuum created by the previous drop of ink leaving.

Thermal inkjet is most common in household and consumer grade inkjet printers. A major benefit to using thermal printhead technology is the relatively inexpensive print head. Since each colour printed requires a separate print head, and some print devices can contain eight or more colours of ink, thermal

technology keeps the initial cost of the device low and reduces replacement costs when a print head fails, or is damaged.

Piezoelectric Inkjet

Piezoelectric (piezo) print heads also use a tiny reservoir to hold a droplet of ink. However, unlike thermal printheads, piezo heads contain a small flexible membrane, or diaphragm, that moves up and down to squirt the ink out of the print nozzle. The pressure caused by the flexing of the piezo material is very precise, allowing a drop, or multiple drops, to strike the substrate accurately. Similar to thermal, the print head is made up of a matrix of a number of these individual nozzles. And by using multiple print heads, multiple colours are possible.

Piezoelectric is more common in commercial and large-format printing applications, although there are a few consumer grades of printers that use piezo as well. Piezo is more accurate, and because the ink in the chamber doesn't have to be vaporized to form the droplets of ink, piezo can print with a wider variety of inks such as aqueous, ultraviolet, and latex.

Types of Ink

Inkjet printing has become more advanced not only in the mechanics of how the print heads work, but also in the variety and usage of different types of ink. Below are some common types of ink and a brief explanation of how they might be used.

Aqueous Ink

Aqueous ink, as discussed earlier, is a water-based ink. This type of ink is used in consumer printers using thermal technology, but can also be used in commercial piezo printers as well. Aqueous is well suited to thermal technology because the formulation of the ink allows it to vaporize in the print head for expulsion onto the paper. The water component of the ink, however, also contributes to its greatest drawback: the susceptibility of the finished printed piece to run or smear if it gets wet. Many users of desktop printers in their homes have been disappointed when they take their printed pages outside in the rain. Even a few drops of water can cause the ink to run and bleed into the paper.

In commercial uses, aqueous inkjet is well known for colour fidelity and quality, but the finished piece has to be protected from moisture. These types of print products would most likely only be used indoors, mounted behind glass, or with a laminated plastic layer on top. There are water-resistant coatings that can be sprayed onto a finished product, but even then, you would not want to leave it outside for an extended period of time. Aqueous ink is a common choice for art prints.

Ultraviolet Inkjet

Ultraviolet (UV) ink is a type of **energy-cured ink** that stays wet until it is bombarded with ultraviolet radiation. This UV radiation is commonly projected onto the freshly printed surface by means of a special high-intensity light bulb. Once the UV rays hit the ink, a special molecular process is triggered,

causing the chains of molecules in the ink to bond and solidify instantly. UV ink does not dry from exposure to air, nor from heat. Once the UV ink has been cured, however, it is very solid and quite durable.

For commercial use, UV inks tend to be popular for outdoor uses such as banners and signage. Indoor signage is commonly printed using UV as well because of its durability and rub resistance. Since UV inks dry instantly, they can be removed from the printer and handled much sooner. UV inks sit mostly on top of the surface of the substrate, and because of their solid bond are more prone to cracking if bent or folded. UV is not a good choice of ink where flexibility of the substrate is required.

Latex Inkjet

Latex ink is a newer formulation that has exploded onto the inkjet printing scene in the last few years. Latex inks are water based and cure primarily through heat, but more importantly, they are not subject to moisture damage once cured. This is because the pigment is carried by the latex molecules, and once the latex has bonded to the substrate, the pigment stays intact. Latex printed products dry immediately and are ready to use as soon as they come off the print device.

Latex inks are used in many commercial applications, particularly where outdoor durability and flexibility are needed. One of the many common uses of latex inkjet printing is in imaging car wraps. A car wrap is a flexible adhesive material that is printed flat, then stretched or wrapped around the contours of a vehicle, usually for marketing or advertising purposes. Figure I.1 in the introduction of this textbook shows an example of a car wrap. Because of this flexibility, latex printed signage can also be adhered to rougher surfaces such as wood or brick. The popularity of latex can be attributed to a previously popular inkjet formulation that is solvent based.

Solvent Inkjet

Because of the rise in popularity of latex ink over the last few years, there has been a great decline in the use of solvent inkjet inks. Formerly, this was the type of ink needed for flexible, durable printing. Solvent inks are formulated using solvent as a carrier for the pigment, and as the solvent dries, the pigment remains bonded to the substrate. A big concern for the use of solvent-based printing is the release of volatile organic compounds, or VOCs. These VOCs are released into the atmosphere during the printing and drying of solvent prints, and have to be vented outdoors so as not to pollute the workspace. Even newer eco-friendly inks still release VOCs, albeit at a lower level. Some areas have environmental laws that restrict the release of pollutants into the air (United States Environmental Protection Agency, 2000). Customers often complain about the smell of solvent prints, particularly when used indoors. Because of this, solvent inkjet is primarily chosen for outdoor uses such as large-format signage, banners, and car wraps. Solvent can be very economical, and while the quality isn’t as sharp as UV or aqueous, it is excellent for very large projects that will be viewed from even a moderate distance. Pressure on the solvent ink market comes because most of these uses can now be achieved with latex inks as well, and the industry has seen a divergence between companies that still use solvent or eco-solvent inks and those that are switching to latex.

Media Attributions

- Micro_Piezo_Comparison by Javachan © CC BY-SA (Attribution ShareAlike)

6.3 Electrophotography

Roberto Medeiros

Electrophotography (also known as xerography) is a complex process commonly used in copiers and faxes, as well as in digital printers. It is an imaging technology that takes a digital file and utilizes a photoreceptor, light source, electrostatic principles, and toner to produce the printed output. Before this process was used for digital printing, it was extensively used in analog copiers where a lamp illuminated the page being copied, and then a series of mirrors reflected the page directly onto the surface of a drum. Digital copiers replaced the direct light path with a sensor that converts the analog image into digital information, then a laser or an **LED** array writes the image onto the drum. Many digital printers today are based on the same platform as digital copiers. The technology has seen many improvements over the years, but the electrophotographic process at its core remains relatively unchanged.

Photoreceptor

The photoreceptor is commonly referred to as a drum. It is a cylinder coated with a material that becomes conductive when exposed to light. Areas that are not exposed have a high resistance which allows these areas to hold the electrostatic charge necessary for the process.

Light Source

Light sources used in digital printing include LED arrays or, more commonly, lasers. VCSEL (vertical cavity surface emitting laser) is an advanced type of laser used in the most current digital presses in the market. A VCSEL array can position its beam with high accuracy (addressability) for optimal clarity, resolution, and image positioning. This makes it ideally suited for a digital press.

Electrostatic Principles

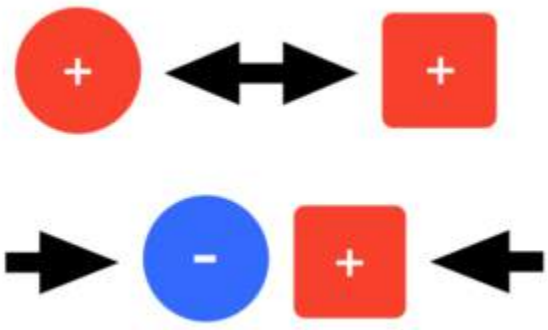


Figure 6.2 Like charges repel each other while opposite charges are attracted

To understand electrophotography, we must first understand some basic electrostatic principles. When certain materials come in contact then separate from each other, these materials can become electrically charged. Rubbing these materials together can increase this effect. This is called the **triboelectric effect.** Static electricity buildup on your clothes in a dryer or from rubbing a balloon on your hair are examples of the triboelectric effect. Charges can have either a positive or negative polarity. Like charges repel each other while opposite charges are attracted, in much the same way as the polarities in magnets (see Figure 6.2).

These properties are at the core of the technology and are utilized in almost every stage of the digital imaging process.

Toner Basics

Toner is a very fine, dry powder medium used in the electrophotographic or xerographic process. It is composed primarily of a resin and includes pigment, wax, and process-enhancing additives. The term *xerography*, in fact, is derived from the Greek words *xeros*, 'dry' and *graphia*, 'writing,' reflecting how toner rather than ink is used in the imaging process. Toner particles become electrically charged when stirred or agitated through a triboelectric effect. The composition of the toner not only contributes to its imaging characteristics but to its ability to maintain and control its charge properties. The shape of the toner also is a factor in its charging capability. This electrical charge is what allows the toner to be precisely manipulated throughout the process.

There are two basic types of toner production, pulverized and chemical (Figure 6.3). Pulverized toner was commonly used in earlier digital printers and is manufactured by successive compound mixing and grinding steps until the desired consistency and size is achieved. The resulting toner particles are irregular in size and shape and typically average around 6.2 to 10.2 microns in size. Pulverized toner produces good results, up to 600 dpi resolution; however, a consistent size and shape along with a smaller particle size is required to produce better clarity and detail at higher resolutions.

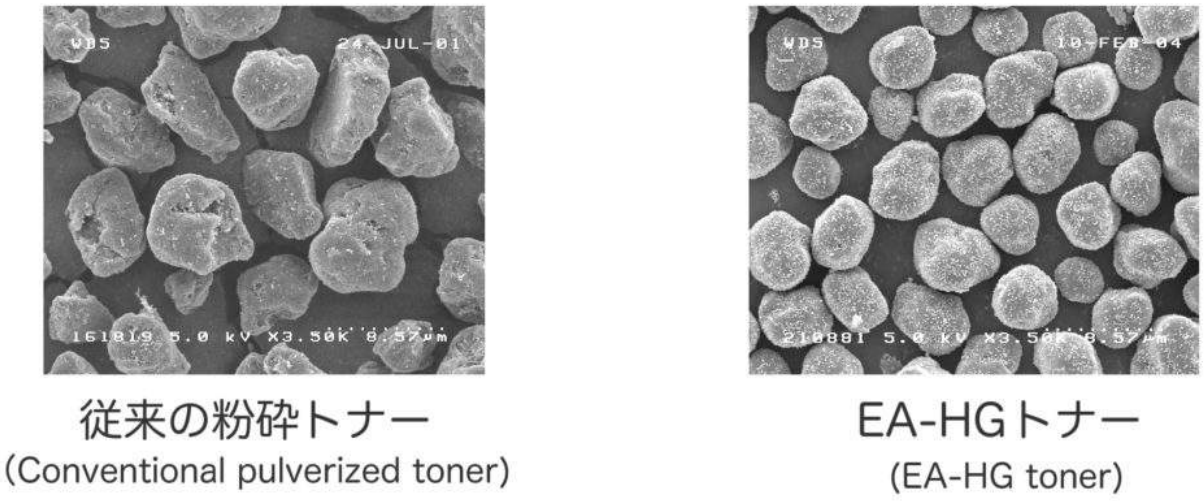


Figure 6.3 Two basic types of toner production

Chemical toners were introduced later to overcome those limitations and are in common use today. Each manufacturer has its own process for creating this type of toner and unique names as well. Xerox's EA toner, Ricoh's PxP toner, and Konica Minolta's Simitri toner are all examples of chemical toners. As the name suggests, chemical toners are created through a process of building or 'growing' the particle chemically. This process allows for the precise control of the shape and size of the toner particle (under 5 microns in some cases), resulting in higher definition and resolution capabilities. Resolutions of 1,200 dpi and 2,400 dpi are possible largely due to the use of this type of toner. Other benefits include much lower energy consumption, both in the manufacturing process and printing process, as well as narrower particle size and charge distributions.

Here is a YouTube video of how chemical toner is made: https://youtu.be/852TWDP61T4

Dry toner comes in two forms: mono component and dual component. Both rely on magnetic iron or iron oxide particles to 'hold' the charged toner on a magnetic roller. Mono component toners incorporate the magnetic material in the composition of the toner particle itself where dual component toners have the magnetic material mixed together with the toner but as separate components. This mixture is called developer.

ElectroInk

ElectroInk is a unique form of toner used in HP Indigo digital presses. The toner comes in the form of a paste and is mixed internally in the press with imaging oil, a lightweight petroleum distillate. This type of toner is considered a liquid toner as the particles are suspended in the liquid imaging oil, but still uses an electrophotographic process for imaging. One of the important advantages of this type of toner is its particle size. ElectroInk toner particles are 1 to 2 microns, significantly smaller than the smallest dry toner particle. At this size, a dry toner would become airborne and would be very difficult to control. The toner and oil suspension achieves higher resolutions, uniform gloss, sharp image edges, and very thin image layers. A thin image layer allows the toner to conform to the surface of the substrate, producing a consistent look between imaged and non-imaged areas. A drawback of this toner, however,

is that substrates may need to be pre-treated in order for the toner to adhere properly. There are substrates available for use specifically on HP Indigo digital presses, but typically these are more expensive or may not be compatible with other printing methods. Some Indigo presses are equipped with a pre-treating station that expands substrate compatibility extensively and even surpasses that of other forms of digital printing.

Nanography

Nanography is a very new and exciting print technology currently in development by the creator of the Indigo digital press, Benny Landa. It borrows some of the same concepts used in the Indigo but with a different approach to the implementation of these. The technology centres around NanoInk, a breakthrough ink with pigment sizes in the tens of nanometers. In comparison, pigments found in good-quality offset inks are in the 500 nanometre range. Colorants intensify and ink density increases at this microscopic level, thereby expanding the ink's colour gamut considerably. The ink uses water as a carrier instead of imaging oil making it more cost effective and eco-friendly. Billions of ink droplets are jetted onto a heated blanket, not directly onto the substrate as in inkjet printing. The ink spreads uniformly on the blanket and the water quickly evaporates leaving only an ultra-thin (approximately 500 nanometres), dry polymeric film. This film transfers completely onto the substrate on contact and produces a tough, abrasion-resistant image. This print technology can be used with almost any substrate without pre-treatment and, due to its minuscule film thickness, does not interfere with the finish. Whether high gloss or matte, the ink finish matches that of the substrate. Although the technology is poised to revolutionize the print industry, the first press to use it is currently in beta testing. You can find the latest news and more information on nanography on this webpage: http://www.landanano.com/nanography

Media Attributions

- Electrostatic principles by Roberto Medeiros
- EA-HG_toner by Fuji Xerox Co., Ltd

6.4 Electrophotographic Process

Roberto Medeiros

The electrophotographic process consists of seven stages (see Figure 6.4). For the purpose of this text, we will be describing the process using a negatively charged dry toner. The process is the same for a positive toner except the polarity would be reversed in each stage.

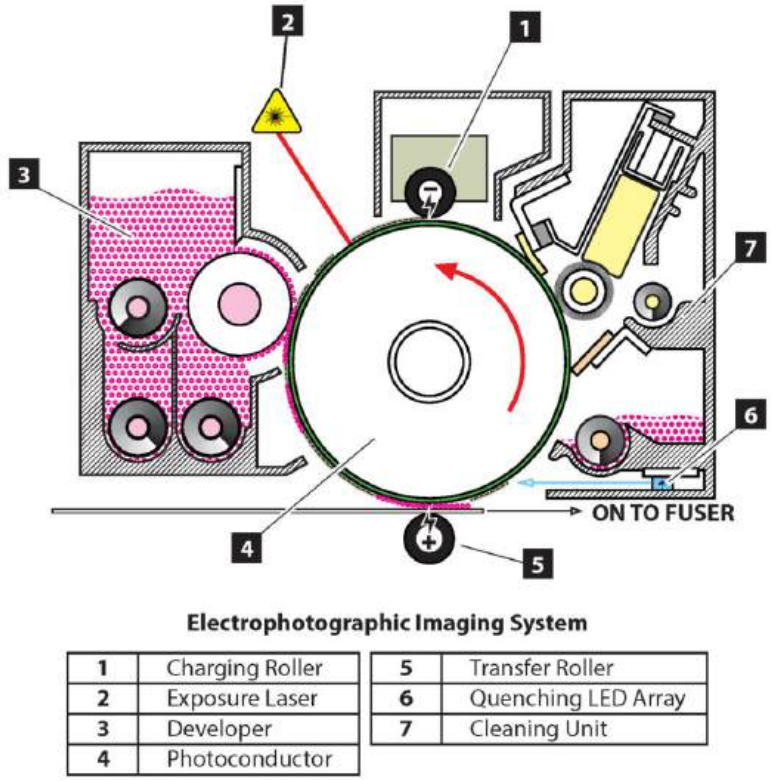


Figure 6.4 Electrophotographic Imaging System

Charging

In the first stage, a high negative voltage of approximately -900 volts is provided to a charge roller (see Figure 6.5). The voltage used varies by manufacturer and model. The charge roller applies a uniform layer of negative charge to the surface of the drum. The resistivity of the unexposed photosensitive drum coating allows the charge to remain on the surface.

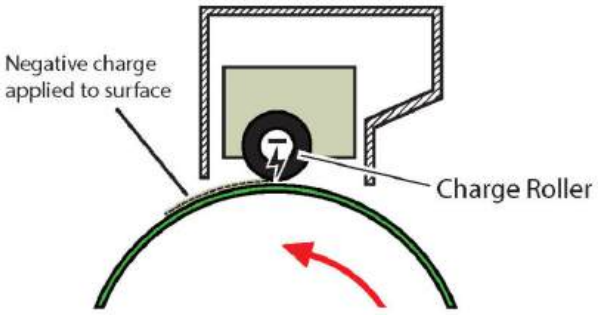


Figure 6.5 Charge roller

Exposure

A laser is used to write the image onto the charged surface (see Figure 6.6). Because the photosensitive coating on the drum becomes conductive when exposed to light, the charges on the surface of the drum exposed to the laser conduct to the base layer, which is connected to a ground. The result is a near zero volt image and a negative background. This is known as the latent image.

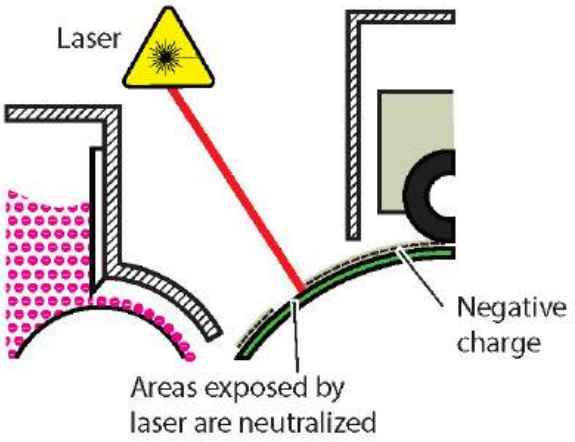


Figure 6.6 Exposure

Development

Many digital printers and presses use a dual component development system (see Figure 6.7). The developer is a mixture of non-magnetic toner and a magnetic carrier. As the developer is stirred and the particles rub up against each other, a triboelectric charge is generated between the them. The toner becomes negatively charged while the carrier becomes positive. The opposite charges cause the toner to be attracted to the carrier. A magnetic development roller holds the mostly iron carrier in alignment with magnetic lines of force forming a magnetic brush. This magnetic brush in turn ‘carries’ the attracted toner to the surface of the drum. A high negative bias is applied to the development roller repelling the

toner onto the drum. The toner is attracted to the areas of the drum exposed by the laser, which, being close to zero volts, is much more positive than the negatively charged toner. In this way, the latent image is developed. As the carrier remains on the development roller, it continues to attract toner from the hopper to maintain the optimal concentration on the magnetic brush.

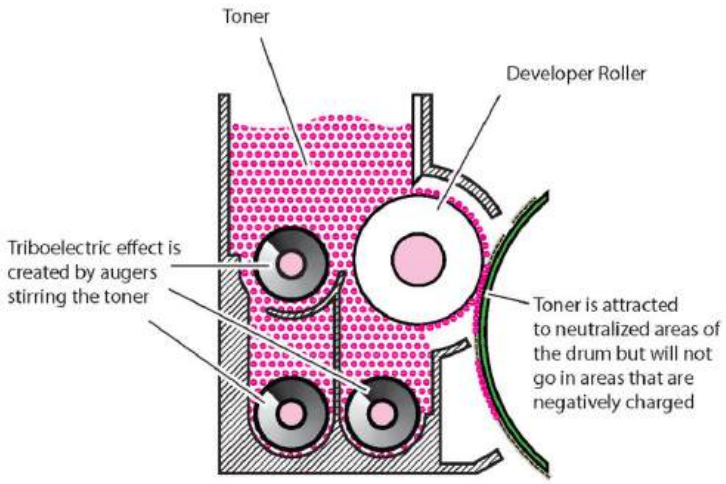


Figure 6.7 Development

Transfer

A sheet of paper or substrate passes between the drum and a transfer charge roller that has a high positive voltage applied to it (see Figure 6.8). The negatively charged toner of the developed latent image on the drum is attracted to the more positive transfer roller and adheres to the sheet in-between. The charge applied to the back of the sheet causes the paper to cling to the drum. A high negative voltage is applied to a discharge plate immediately after the transfer charge roller to aid in the separation of the sheet from the drum. The curvature of the drum along with the weight and rigidity of the sheet also aid in the separation.

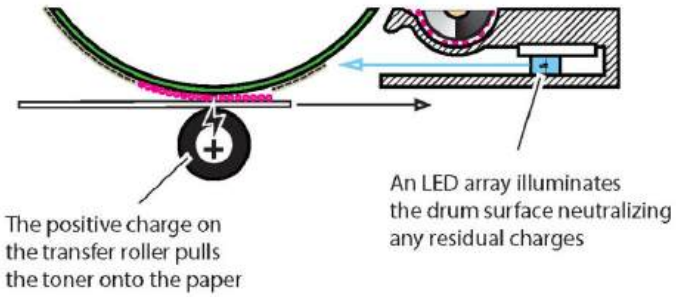


Figure 6.8 Transfer

A more advanced method of transfer utilizes an intermediate transfer belt system. This is most common on colour digital presses where four or more colours are transferred onto the belt before transferring the complete image onto the sheet. Charge rollers beneath the belt, under each drum, pull off the developed **latent images** of each separation directly onto the belt. In the transfer stage, a transfer charge roller beneath the belt applies a negative charge to push the toner onto the sheet. A second roller, directly beneath the first on the other side of the belt, applies pressure keeping the paper in contact with the belt and aiding in transfer for more textured stocks. The lower roller may have a small positive charge applied to it or may be grounded. Some systems can also alternate the charge applied to the transfer charge roller to further aid toner application onto textured substrates.

After this stage, the sheet moves on to fusing where the toner permanently adheres to the substrate. The next two stages described below are post-imaging steps that are necessary to prepare the drum surface for the next print cycle.

Cleaning

After the transfer stage, some toner may be left behind on the surface of the drum. If left there, the background of each successive print would slowly become darker and dirtier. To prevent this, a cleaning blade removes any residual toner from the drum’s surface (see Figure 6.9). Some systems will recycle this toner back to the developing unit, but mostly the waste toner is collected in a container for disposal.

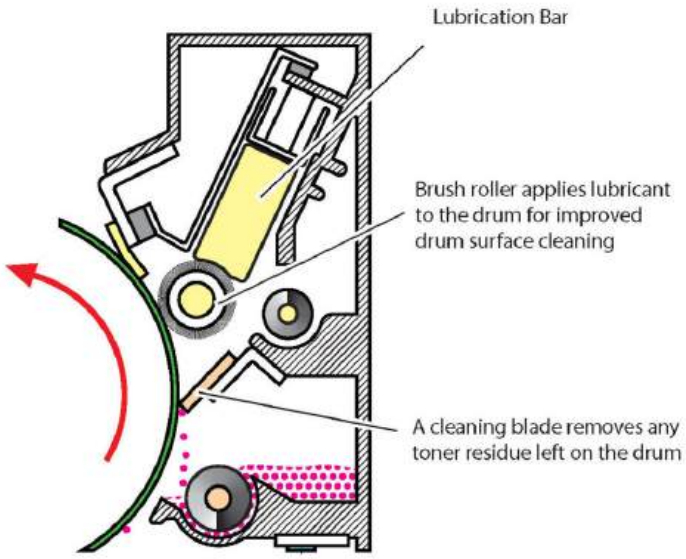


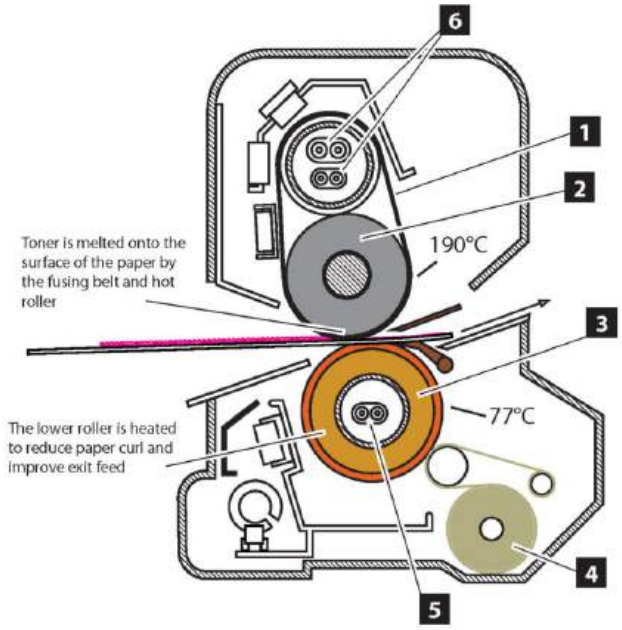
Figure 6.9 Cleaning

Erasing

In this stage, an LED array exposes the length of the drum, bringing this area of the drum to near zero volts. This prepares the drum surface for the charging stage of the next print cycle.

Fusing

This is the final stage in the electophotographic process. The fusing mechanism, or *fuser*, consists of a heat roller, a pressure roller, and cleaning mechanism (see Figure 6.10). Toner is composed mostly of resin. When the toner is heated by the heat roller and pressure applied by the complement pressure roller, it melts and is pressed into the fibres of the sheet. The toner is never absorbed by the paper or substrate but rather is bonded to the surface. A negative charge is applied to the heat roller or belt to prevent the toner from being attracted to it and the cleaning section removes any toner or other contaminates that may have remained on the heat roller. Heat may also be applied to the pressure roller (at a much lower temperature) to prevent the sheet from curling.



Fuser

1	Fusing Belt	4	Cleaning Belt
2	Hot Roller	5	Lower Heater
3	Pressure Roller	6	Upper Heaters

Figure 6.10 Fusing

Along with the transfer stage, fusing can be greatly affected by the paper or substrate used. The thicker and heavier the sheet, the more heat it absorbs. Because of this, these sheets require higher temperatures so there is sufficient heat remaining to melt the toner. Insufficient heat can cause the toner to scratch off easily or not bond at all. Too much heat can cause moisture in the substrate to evaporate quickly and get trapped beneath the toner causing tiny bubbles that prevent the toner from sticking wherever they occur. This issue is seen more on thinner stocks that do not absorb as much heat. Too much heat can also cause toner residue to stick to the heater roller and deposit it on subsequent sheets.

The heat roller can heat up quite quickly but may take much longer to cool down. This can cause delays in producing work that switches between different paper weights. To combat this, some devices use a thin belt that can be both heated and cooled quickly in place of the heater roller. In some cases, a cooling mechanism is also employed further mitigating the cooling lag.

Media Attributions

- EP Imaging System – Complete by Roberto Medeiros
- EP Imaging System – Charging by Roberto Medeiros)
- EP Imaging System – Development by Roberto Medeiros
- EP Imaging System – Transfer by Roberto Medeiros

- EP Imaging System – Cleaning by Roberto Medeiros
- EP Imaging System – Fusing by Roberto Medeiros

6.5 Paper Basics

Roberto Medeiros

When talking about substrates used in printing, paper is usually what comes to mind. Paper is made most commonly from wood fibre. Today, many papers also have some percentage of recycled fibre as well as fillers and other additives. These all contribute to the quality of the paper itself and to the quality of the printed output. It's important to understand some basic attributes of paper as they all have a direct impact on imaging processes and results.

Formation

Formation refers to the distribution of fibres, fillers, and additives in paper and how evenly they come together. When you hold a sheet up to a strong light source and look through it, the mix of dark and light areas are the result of formation. The more uniform the formation, the less mottling is observed in the paper. Papers with uniform formation accept inks and toners more evenly, have reduced print mottling, and enhance clarity.

Opacity

In strict terms, opacity is the degree to which light is prevented from travelling through a paper. In practical terms, it's how well a paper prevents the image on the backside of a sheet showing through to the front. This is measured on a scale from 1 to 100, where 100 is completely opaque. **Opacity** can be increased with fillers, pigments, or even coatings. In general, a thicker paper, coloured paper, or coated paper is more opaque than its counterparts. Opacity values are very important when projects require thinner paper stocks and both sides of the sheet are being printed.

Basis Weight and Grammage

When looking at the label on a ream of paper used in North America, you usually see two weight designations: the **basis weight**, designated in pounds (#) and the equivalent **grammage**, in grams per square metre (g/m^2 or gsm). In most of the world, grammage is primarily used. In North America, the basis weight is more common. Grammage is simply how many grams per square metre paper weighs. No other factors are represented by this designation. So we can deduce that the higher the grammage, the thicker or denser the sheet. Basis weight is the weight of 500 sheets of paper at a specific size, known as the 'parent' sheet size, which varies based on the historical use of the specific paper. To understand this better, let's examine two different basis weights.

Cover basis weight is based on a 20″ x 26″ parent sheet. So 500 sheets of 80# cover (the # symbol is used to indicate pounds) at the parent sheet size weighs 80 pounds. Likewise, 500 sheets of 80# text at the text-weight parent sheet size of 25″ x 38″ also weighs 80 pounds. This can be very confusing as a cut

sheet of letter (8.5″ x 11″), 80# text, is much thinner than the same size of 80# cover. Table 6.1 shows common basis weights, parent sheet sizes, and typical uses.

Table 6.1 Paper weights, sizes, and uses

Basis Weight	Parent Sheet Size	Typical Use
Bond	17″ x 22″	Historically used as writing paper and typically uncoated. Standard office paper is 20# bond, while colour prints are more commonly done on 24# or 28# bond due to the need for higher opacity.
Cover	20″ x 26″	Used for paperback book covers, business cards, post cards. Business cards have typically been 100# cover, but have been trending toward higher weights of 110# and 120#.
Text	25″ x 38″	Used for magazines and posters. Relatively thin sheets with higher opacity. Magazines typically use a coated text weight paper for both the cover and the body. Typical weights are 70# to 100#.
Index	25.5″ x 30.5″	Used for index cards and tab stock. Tab stocks are typically uncoated 90# index.

Although basis weight is used as the primary weight on a paper label and description, a digital press will typically use grammage to define the weight property when assigning a paper to a tray. Paper weight is one of the key characteristics that affect many parameters on the digital press, including how much vacuum strength is used for feeding, how much charge is required to transfer toner to paper, and how much heat is required to maintain a consistent fusing temperature to bond toner to the paper, among others. Entering the wrong values for the paper weight can cause paper misfeeds, poor image quality, or toner not adhering to the paper. Using grammage simplifies data entry and avoids errors due to incorrect basis weight selection for the numeric weight value. It may, however, require one to do a conversion calculation if only basis weight is provided. The following conversion factors can be used to do these calculations.

Conversion Factors:

Bond (lbs.) x 3.7606 = gsm
Cover (lbs.) x 2.7048 = gsm
Text (lbs.) x 1.4805 = gsm
Index (lbs.) x 1.8753 = gsm

Grain Direction

In the paper manufacturing process, a slurry of fibre travels over a high-speed mesh conveyor belt that is oscillating side to side. This action and movement causes the fibres to interlace and develop a predominant alignment along the direction of movement. This predominant alignment of the fibres is called **grain direction**. Short grain refers to fibres running parallel to the short dimension of the sheet, and, conversely, long grain refers to fibres running parallel to the long dimension of the sheet.

It is important to keep grain direction in mind when choosing a paper for a project. You need to consider the print process and binding or finishing method you will use, as choosing the wrong grain direction can produce poor results or may be incompatible with the printing method you have chosen. Sheet fed offset lithography papers are often long grain and are most common. Digital presses require the grain to run perpendicular to the feed direction in order to feed properly and make the sharp turns typically found in a digital press. In this case, most sheets are fed into the press with the short edge first therefore requiring short grain paper. When folding is required, folds that run parallel to the grain will be smooth and sharp while folds that run across the grain will be ragged, and the fibres on the top of the sheet may pull apart. Toner used in digital printing bonds to the surface of the paper and does not penetrate. Folding across the grain will cause the toner to break apart where the fibres separate.

The second or underlined dimension of the sheet will indicate the direction of the grain. For example, 18″ x 12″ is a short grain sheet, and 12″ x 18″ is long grain. If the underline method is used, short grain would be 12″ x 18″ and long grain would be 12″ x 18“. If the dimensions are not noted or the sheet is not in its original packaging, grain direction can be determined by folding the sheet along both dimensions. As noted previously, a fold that runs parallel to the grain will be smooth and sharp while a fold that runs across the grain will be ragged. You can also gently bend the paper in either direction. The bend running in the direction offering the least resistance is the grain direction.

Caliper

Caliper, unlike grammage and basis weight, is a measure of thickness. The most common measurement used in North America is thousandths of an inch, designated as points (common for paper) or mils (common for synthetic paper). This terminology can be confusing, however, as points can also refer to 1/72 of an inch when referring to font size, line thickness, and dimensions on a page. Mils can be confused with millimetres as well. A common misconception is that points and mils can be converted to grammage or basis weight. This is not true. The caliper can vary depending on the coatings or finish. In general, a rougher finished stock will have a higher caliper than the same weight of a smooth stock. Coatings can be heavier than paper fibre so coated paper can have a smaller caliper than the same weight of an uncoated counterpart. A process called calendaring, which irons the paper between two highly polished chrome rollers, improves smoothness and printability but also reduces the caliper without changing the weight of the paper.

Brightness and Whiteness

Brightness and **whiteness** define the optical properties of paper and differ mainly in how they are measured. Whiteness measures the reflective properties of the paper across the entire visible spectrum of light (defined by CIE). In other words, it defines how white the paper is. A perfect reflecting, non-fluorescent white material measures 100 whiteness. Brightness also measures the reflective properties of paper, on a scale of 1 to 100, but specifically in the blue area of the spectrum at a principal wavelength of 457 nanometres and 44 nanometres wide (defined by TAPPI and ISO standards). This wavelength coincides with lignin absorption. Lignin is what binds the cellulose fibres in wood and pulp and gives it its initial dark brown colour. The more bleaching done to the pulp, the more lignin is removed, and the higher the blue reflectance and therefore brightness. In most parts of the world, paper whiteness

measurement is used; however, in North America, most papers use brightness measurement instead. Some papers have brightness values that exceed 100. This is due to the addition of fluorescent whitening agents (FWAs), which return additional blue light when exposed to UV light. The same is true for whiteness, as papers with higher blue reflectance levels tend to have higher whiteness levels.

Finish

Finish defines the look and feel of the paper's surface and can be achieved during the paper-making process (on-machine) or after (off-machine). On-machine finishes are achieved by the application of a pattern onto the paper by a marking roller while it is still wet. Examples of on-machine finishes are smooth, vellum, laid and felt (see Table 6.2). Off-machine finishes are accomplished with rollers that press the pattern into the paper after it has been made. Off-machine finishes are also known as embossed finishes. Linen, stipple, and canvas are examples of these; Table 6.3 gives a description of each.

Table 6.2 On-machine finishes

On-machine Finishes	Description	Typical Uses
Smooth	Paper is passed through various calendaring rollers, producing a finish that is uniform, flat, and smooth to the touch.	Ideal for general digital printing and copying as toner is applied to the surface and does not penetrate the fibres.
Vellum	A consistent eggshell appearance that is not quite as smooth as smooth finish but has a velvety feel. Not to be confused with the substrate called vellum, which is translucent.	Used most commonly for book paper.
Laid	Consists of a series of wide-spaced lines (chain lines) and more narrowly spaced lines (laid lines), which are at 90 degrees to the chain lines.	Used for letterhead, reports, presentations.
Felt	A felt-covered roller is used to produce this finish. The appearance resembles that of felt.	Used for letterhead, reports, presentations.

Table 6.3 Off-machine finishes

Off-machine Finishes	Description	Typical Uses
Linen	A cross-hatch pattern resembling linen fabric.	Used for personal stationery, letterhead, fine-dining menus, business cards.
Stipple	A fine bump texture that resembles the painted surface of a wall.	Used where a subtle uneven texture is desired.
Canvas	Simulates the surface of canvas.	Used for art prints or where a 'painted' appearance is desired.

Coated papers have calcium carbonate or china clay applied to their surface. The coating fills in the spaces between the fibres on the paper's surface, resulting in a smoother finish. The amount of coating and calendaring produces different finishes and gloss appearance. Examples of coated finishes are matte, dull, satin, silk, and gloss, described in Table 6.4.

Table 6.4 Coated finishes

Coated Finish	Description	Gloss Level
Matte	Roughest surface of coated paper. Very flat, no lustre, no glare, no calendaring applied.	None
Dull	Smoother surface than matte. No luster, no glare, minimal calendaring.	Very low
Satin	Smooth and soft to the touch. Slight lustre, low glare, light calendaring.	Medium low
Silk	Smooth and silky to the touch. Low lustre, low glare, light calendaring.	Moderate
Gloss	Smooth and slick. Shiny, high calendaring.	High

Cast coated paper has a very high gloss finish on the front side and is uncoated and rough on the back. The high gloss finish is created by applying a heated chrome roller to the coated surface to quickly dry it while moisture is released through the uncoated back of the sheet. Calendaring is not used, allowing the back surface to be rough and ideally suited for labels. Cast coated paper holds ink well, but the toner used in digital printing may not adhere to it.

6.6 Page Description Languages

Roberto Medeiros

Many page description languages (PDL) exist today; however, Printer Command Language (PCL) and PostScript are the most common and widely adopted. Each has its strengths, weaknesses, and jobs for which it is best suited. It is important to understand these differences and choose the method best suited to your particular printing requirements.

PCL

PCL is a page description language developed by Hewlett-Packard (HP) and was originally used on HP impact and inkjet printers. PCL 3 was the first version to be used with a laser printer, the HP LaserJet, released in 1984, around the same time PostScript was introduced. The goal of PCL was to have an efficient printer control language that could be implemented consistently across HP's printer line. Simple commands and functionality would not require expensive print controllers, making it very attractive for utility-level printing. Many other printer manufacturers implemented PCL for this reason. Commands are embedded at the beginning of the print job and set the parameters for the printer to use for the job. These commands remain set until a new value is assigned for the command or the printer is reset. If the printer does not support a specific command, it ignores it.

When colour laser printing became available, PCL 5c was developed with similar goals. New commands were added to the existing command set, as was the case with all the predecessors, to add support for colour printing. This ensured backwards compatibility while minimizing development. When it came to colour, HP's goal was to have colour on the printed page look the same as what was displayed on screen. There were many challenges to achieving this, so print quality adjustments were included to give users the ability to fine-tune the output. With the emergence and widespread adoption of the sRGB standard to define and describe colour on a display, the PCL colour command set could be simplified by adopting this standard for colour printing. Thus, HP's goal could be achieved without the complexity and overhead of a full colour management system. Operating systems and applications, for the most part, have standardized how they display colour in sRGB, so this approach is the simplest way to achieve acceptable colour between display and print. PCL is most appropriate for general office use where a simple, low-cost print device that produces good quality colour is expected. It is not suitable, however, for a colour critical or print production environment where precision and full colour management is required.

PostScript

PostScript is a page description and programming language developed by Adobe that describes text, graphics, and images, and their placement on a page, independent of the intended output destination. The code created is in plain text that can be written and examined with a basic text editor. The output itself can either be to a printer, display, or other device possessing a PostScript interpreter, making it

a device independent language. The interpreter processes the PostScript instructions to create a raster image the device can render. The interpreter is often referred to as a RIP or raster image processor for this reason. It is possible to write valid PostScript code from scratch, but it is impractical as page composition applications can either generate the PostScript code directly or can utilize a print driver, which can convert the page to the PostScript language.

Since PostScript is a general-purpose programming language, it includes many elements that you wouldn't associate specifically with printing such as data types (numbers, arrays, and strings) and control primitives (conditionals, loops, and procedures). It also has an interesting feature called *dictionary*, which stores information in a table consisting of a collection of key and value pairs. The values can be entered into the dictionary while the keys are used to reference the information needed. These features made possible documents that acted like an application which could generate pages dynamically from data directly on the printer itself. These printer-based applications were stored temporarily in memory or permanently in the printer's hard drive, and triggered by a command in the print stream. These capabilities made variable data printing possible using PostScript and are still being used today for that purpose.

The first printer to use PostScript was the Apple LaserWriter in 1985. The same day that Apple announced the LaserWriter, Aldus Corporation announced PageMaker, a page layout application developed to take advantage of the Apple Macintosh computer's GUI (graphical user interface) and the PostScript PDL. This series of events is considered by many as the genesis of the desktop publishing revolution. In fact, the term *desktop publishing* is attributed to the founder of Aldus Corporation.

PDF

Portable document format (PDF) is one of the most popular file formats for displaying and printing documents. When this format was released by Adobe in 1993, it shared many of the same concepts and components of PostScript. But where PostScript was designed primarily to provide device independent consistency in print output, PDF was focused on maintaining the visual appearance of a document onscreen, independent of the operating system displaying it. Over the years, PDF has expanded to more specific-use specifications for engineering, archiving, health care, universal access, and printing.

PDF/X is a branch of PDF and an ISO standard that deals specifically with print. It was developed by the Committee for Graphic Arts Technologies Standards (CGATS). Table 6.5 shows the evolution of the standard.

Table 6.5 Evolution of PDF
Data source: Adobe Systems Inc, 2008, p. 4

Preset	Compatibility	Settings	Usage
PDF/X-1a: 2001	Acrobat 4/PDF 1.3	• Convert RGB colour to CMYK (spot colors allowed) • Transparency flattened	PDF/X-1a ensures that the files are ready for print production—fonts are embedded, colours must be CMYK or spot, layers and transparency are flattened. Note that there is no minimum resolution required for PDF/X.
PDF/X-1a: 2003	Acrobat 5/PDF 1.4		
PDF/X-3: 2002	Acrobat 4/PDF 1.3	• Leave RGB and CIELab color unchanged (profiles allowed) • Transparency flattened	PDF/X-3 has all the benefits of PDF/X-1a plus it allows colour-managed workflows.
PDF/X-3: 2003	Acrobat 5/PDF 1.4		
PDF/X-4: 2008	Acrobat 7/PDF 1.6	• Leave RGB and CIELab colour unchanged (profiles allowed) • Live (unflattened) transparency • Layers allowed	Has all the benefits of PDF/X-3 plus it allows live (unflattened) transparency and layers for versioning. Print workflows based on the Adobe PDF Print Engine will be able to process PDF/X-4 jobs natively, without flattening artwork or converting to PostScript.
PDF/X-4p: 2008	Acrobat 7/PDF 1.6		Use PDF/X-4p when a required ICC profile is unambiguously identified and supplied separately.

Submitting documents for print using one of these standards is highly recommended as it eliminates many of the causes of print issues and is a more reliable method for graphics file exchange.

Digital Front End

Digital front end (DFE) describes the combination of hardware and software that drives and manages a print device. Hardware is often custom built for this specific purpose and may have proprietary video interfaces that connect directly to the print engine. An operating system serves as the base for the software components of the DFE and is often Microsoft Windows based or a Linux or Unix variant. Although the Windows running on a DFE is much the same as its desktop counterpart, Linux- and Unix-based systems are often custom distributions that are compiled specifically for the DFE.

One of the key components of a DFE is the raster image processor (RIP). The RIP refers to the software component that interprets the PDL and performs the function of rendering or rasterizing the complete instructions into an image or raster the print engine will reproduce. The term RIP is often used interchangeably with DFE. This may have been accurate in the past when a DFE really only performed the ripping function and little else. Modern DFEs, however, do much more. In fact, a DFE may contain

multiple RIPs, and within those RIPs they can utilize the multi-threading and processing power of modern computer hardware and operating systems to process many pages or channels simultaneously. PostScript has been the defacto PDL in digital printing for many years but with the development of the PDF/X print standards and the subsequent release of the Adobe PDF Print Engine (APPE), a native PDF RIP, many DFEs now include both PostScript and APPE as their available RIP engines.

ICC-based colour management workflow may be part of the RIP process or can be an independent component of the DFE. Different elements within a print file get processed through their respective channels in the colour management system. Common channels include CMYK, RGB, black, and named colours. The idea is to convert all colour elements into the colour gamut of the print engine's colorants/paper combination. The conversion process can be complicated, but the basic concept is device dependent source colour spaces (CMYK, RGB, etc.) are converted to a device independent colour space, referred to the profile conversion space (PSC), then from the PSC to the output gamut defined in the output ICC profile. The idea is to take the source 'recipe' and define the visual appearance of it first. That is why it needs to convert to device independent colour space, which defines the visual appearance of colour. Once the visual appearance is defined, the 'recipe' for the specific output can be calculated.

Systems that support named or spot colour rendering follow a similar process. The named colour is located in a look up table. The name must match perfectly, including punctuation, spaces, and case. Each named colour is defined in a device independent colour space, typically Lab. There is no calculation in this step. The last step is the same; Lab values are then converted via the output profile.

There is one more calculation applied before passing the information through to the printer. Electrophotography is affected by rapid changes in humidity and temperature. The electrophotographic process relies on components that do become less effective over time and use. These factors all affect the colour output. Calibration should be performed on a regular basis to compensate for these variables. Calibration is a process of creating a correction curve to maintain consistent and repeatable print output. This correction curve is applied right after the conversion to output profile, ensuring output is consistent with what was defined in the output profile itself.

In order to maintain all these aspects of the DFE, an intuitive and user-friendly interface is critical. The user interface includes many components. Here is where you would configure the DFE, find the status of the print device and consumables, view and process jobs and job queues, and examine and export job histories and logs. Many WYSIWYG tools are accessed via the user interface, such as those for workflow, imposition, complex job composition, paper libraries, spot colour refinement, the launching of external tools, and even interfaces into other systems such as web2print. DFEs are becoming more powerful and perform more than just the traditional RIP functions. As the digital print industry continues to evolve, DFEs will be called on to perform more duties and functions. User interfaces will need to evolve as well to maintain usability and stay intuitive.

6.7 Variable Data Printing

Roberto Medeiros

Variable data printing, or VDP, refers to a special form of digital printing where document content is determined by entries in a record or data set and can be highly personalized. Varied text, graphics, and images are typical content elements, but layout, element positioning, and even document choice are just some of the other variables. Because the content on the printed page is constantly changing, it would not be feasible to produce this type of print product with traditional offset lithography or with any other process that requires a fixed image plate. Electrophotographic and ink jet printing are ideally suited for this type of printing as each page is imaged individually.

VDP can take many forms. Transactional documents like invoices and statements are probably the oldest form of VDP, but these have evolved to include marketing or informational content. This is known as trans-promo or **trans-promotional**. A mail merge is a simple form of VDP where a static document has data elements added directly to it. Each record in the data set produces one document. Another VDP form is when you enter the record manually or upload a simple text-based data table, which then fills the content of a template. This method is typically found in web2print solutions and produces items such as business cards, where the layout, fonts, and required elements can be predetermined and the content based on the data entered. More advanced VDP solutions may include campaign management tools, workflow management, two-dimensional barcode generation, image-based font technology, and integration into external systems such as databases, email, web2print solutions, data cleansing, or postal optimization solutions.

One of the core purposes of VDP is to increase response rate and, ultimately, conversions to the desired outcome. In order to accomplish this, it is critical that the content presented is relevant and has value for the intended audience. Today, there are massive amounts of data available on customers and their behaviour. Analyzing and understanding customer data is essential to maintaining a high degree of relevancy and engagement with the customer.

VDP can be broken down into six key components: data, content, business rules, layout, software, and output method. Each component can vary in complexity and capability and may require advanced software solutions to implement. However, even the most basic tools can produce highly effective communications.

Data

Data used for VDP can be simply thought of as a table or data set. Each row in the table is considered a single record. The columns are the fields used to describe the contents of the record. Some examples of columns or fields would be first name, last name, address, city, and so on. The simplest and most common form of representing this table is by using a delimited plain text format like comma separated value (CSV) or tab delimited. The *delimiter* separates the columns from one another and a new line represents a new row or record in the table. Here is an example of CSV data:

```
"FirstName","LastName","Gender","Age","FavQuotes"

"John","Smith","M","47","Do or do not, there is no try."

"Mary","Jones","F","25","Grey is my favourite colour."
```

The first row contains the row headers or what the fields represent and is not considered a record. You'll notice that each field is separated by a comma but is also enclosed within quotes. The quotes are text qualifiers and are commonly used to prevent issues when the delimiting character may also be in the contents of the field as is the case with the first record above. Many VDP applications support more advanced, relational databases like SQL, but a query must be performed to extract the data to be used, which ultimately results in the same row and column record structure. The data must be attached or assigned to the document in the page layout or VDP application.

Content

Content refers to elements displayed on each page. This would include text, graphics, and images, both static and dynamic. Dynamic content uses placeholders, typically named by the column headers of the data, to mark the position of the element and reference the data in the specific column of the current record. When the document is rendered, the placeholder is replaced by the record data element.

"Dear <<FirstName>>…" becomes "Dear John…" when the document is rendered for the first record and "Dear Mary…" for the second record, and so on. A complete document is rendered per record in the dataset.

Business Rules

Business rules are one of the key elements that make VDP documents highly useful. They can be thought of as a series of criteria that are checked against the data to determine what gets displayed on the page. They can also be used to manipulate the data or filter out relevant content. In almost every case, some level of scripting is required. Advanced VDP solutions have built-in scripting capability, utilizing either common scripting language such as VBScript or JavaScript, or a proprietary scripting language that is only applicable in that specific application. If the page layout tool you are using to create your VDP document does not have scripting capability, you can apply business rules to data beforehand in a spreadsheet application like Microsoft Excel or even Google Sheets.

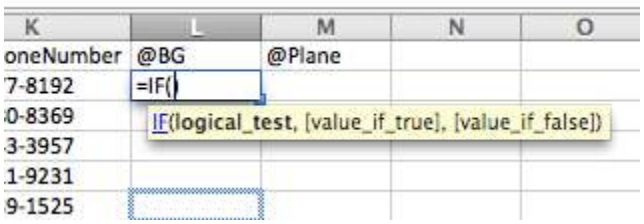


Figure 6.11 Logical test

One of the most common methods for implementing a business rule is using a conditional or IF statement comprising a logical test, an action for a 'true' result, and an action for a 'false' result (see Figure 6.11).

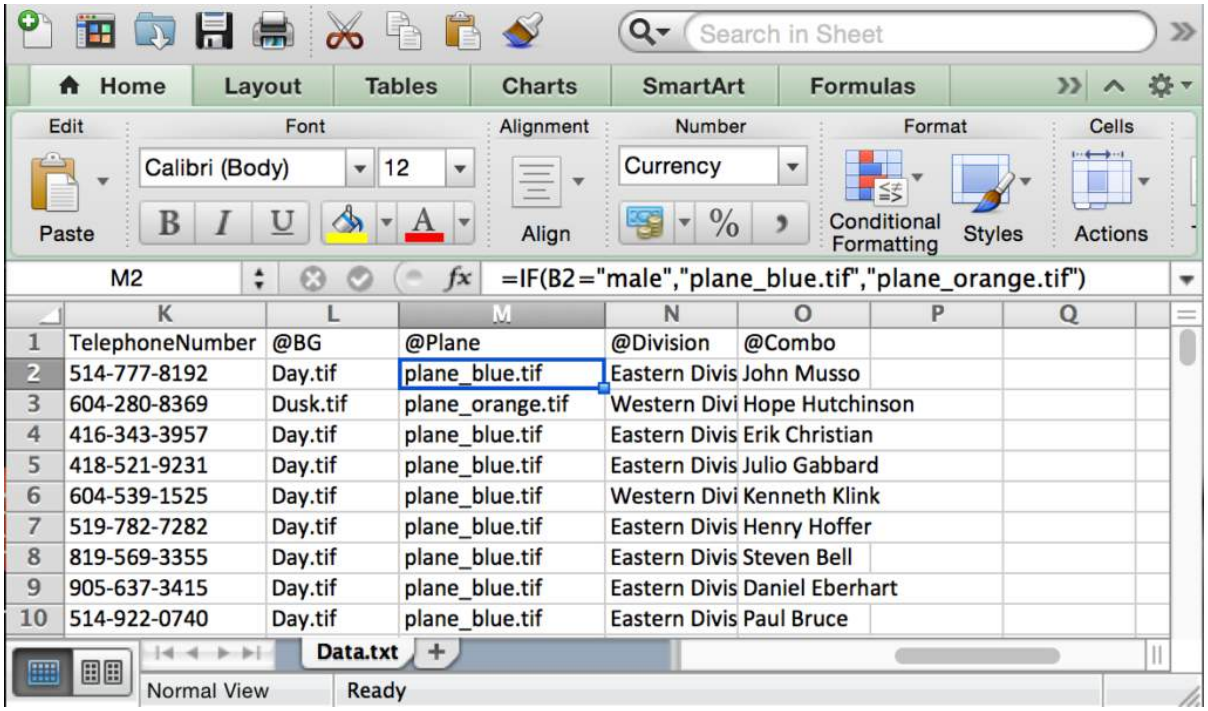
The logical_test means that the answer will be either true or false. In this case, we want to change our graphics based on gender.
In plain English, you may say:

```
"IF gender is male, THEN use plane_blue.tif, or ELSE use plane_orange.tif"
```

In scripting, it would look something like this:

```
IF(Gender="male","plane_blue.tif","plane_red.tif")
```

When doing this in a spreadsheet, you would enter the script in a cell in a new column. The result of the script is displayed in the cell, not the script itself. This new column could now be used to specify the content to be displayed in your layout application. In dedicated VDP applications, the script is attached to the object itself and is processed and displayed in real time.



	K	L	M	N	O	P	Q
1	TelephoneNumber	@BG	@Plane	@Division	@Combo		
2	514-777-8192	Day.tif	plane_blue.tif	Eastern Divis	John Musso		
3	604-280-8369	Dusk.tif	plane_orange.tif	Western Divi	Hope Hutchinson		
4	416-343-3957	Day.tif	plane_blue.tif	Eastern Divis	Erik Christian		
5	418-521-9231	Day.tif	plane_blue.tif	Eastern Divis	Julio Gabbard		
6	604-539-1525	Day.tif	plane_blue.tif	Western Divi	Kenneth Klink		
7	519-782-7282	Day.tif	plane_blue.tif	Eastern Divis	Henry Hoffer		
8	819-569-3355	Day.tif	plane_blue.tif	Eastern Divis	Steven Bell		
9	905-637-3415	Day.tif	plane_blue.tif	Eastern Divis	Daniel Eberhart		
10	514-922-0740	Day.tif	plane_blue.tif	Eastern Divis	Paul Bruce		

Figure 6.12 Plane blue

The "@Plane" column was added to dynamically change a graphic based on the contents of the cell in the "Gender" column (B2) (see Figure 6.12).

Business rules can also be applied to the VDP workflow. In this case, the workflow application or component can manipulate the data before applying it to a document, or it can select the document or destination to be used for the record and much, much more.

Layout

When working with variable data documents, there are special layout considerations you should be aware of. Because word lengths will change per record, there needs to be sufficient space to accommodate the largest and smallest record, and prevent **oversetting** while maintaining the desired visual appearance. This challenge is compounded by the prolific use of **proportional fonts**. Character widths differ with each letter so word lengths will vary even when the number of characters is the same. This can also force a paragraph to reflow onto another page and change the number of pages in the document. Additional scripting may be required to handle reflow scenarios. Some applications use special copy-fitting algorithms to dynamically fit text into a defined area. The use of tables for layout purposes can also be helpful. Because we are dealing with dynamically generated documents, we may also want to vary the images. Using images with a consistent size and shape make it easier to work with. Transactional documents, such as statements and invoices, extensively use numbers. Most fonts, including proportional ones, keep numbers mono-spaced. In other words, every number character occupies the same amount of space. This is important because, visually, we want numbers to be right justified and lining up vertically in columns with the decimal points aligned. There are, however, some fonts that do not follow this common practice. These fonts may be suitable for use in a paragraph but are not for displaying financial data.

Software

Software that can generate a data-driven document is required for variable data printing. In the early days of VDP, there weren't many choices for designers. It was common practice to hand code VDP in PostScript, since it was both a programming language and a PDL. Applications like PageMaker and Illustrator were PostScript design applications but lacked VDP capabilities. Applications like PlanetPress emerged as dedicated PostScript VDP applications. Today, designers have a wide variety of software available for creating VDP. There are three basic VDP software types: a built-in function within a page layout or word-processing software, a third-party plug-in, or a dedicated VDP application.

Microsoft Word, for example, has a mail merge function but does not have the ability to vary images, just text. Adobe InDesign has the data merge function, which is basically a mail merge but includes the ability to vary images as well. In both these examples, business rules would be applied to the data prior to using it in these applications.

There are a number of plug-ins available for InDesign that are very sophisticated. These leverage the extensive page layout capability of InDesign while adding scripting and other VDP specific capabilities. XMPie and DesignMerge are examples of these types of plug-ins. FusionPro is another plug-in based VDP product, and while it does have an InDesign plug-in, it only uses this to allocate variable text and image boxes in the layout. Business rules and specific content are applied in its complement plug-in for Adobe Acrobat.

PlanetPress and PrintShop Mail are examples of dedicated applications that combine both page layout and VDP functions. Although they are very strong in VDP functionality, they sometimes lack the sophistication you'd find in InDesign when it comes to page layout. These particular applications have recently moved from PostScript-based VDP to a more modern HTML5 and CSS (cascading

style sheets) base, making it easier to produce and distribute data-driven documents for multi-channel communications.

Output Method

The ‘P’ in VDP stands for printing and is the main output method we will discuss here. However, document personalization and data-driven communications have evolved to also include email, fax, web (PURL, personalized landing page), SMS text messaging, and responsive design for various mobile device screen sizes. With the emergence of quick response **(QR) codes**, even printed communications can tap into rich content and add additional value to the piece. In order to take advantage of these additional distribution and communications channels, a workflow component is often employed.

A key element for optimized print output of VDP documents is caching. This is where the printer’s RIP caches or stores repeatable elements in print-ready raster format. This means the RIP processes these repeating elements once, and then reuses these preprocessed elements whenever the document calls for them. This does require a RIP with enough power to process large amounts of data and resources with support for the caching scheme defined in the VDP file but, ultimately, allows the printer to print at its full rated speed without having to wait for raster data from the RIP.

There have been many proprietary VDP file formats that have striven to improve performance of VDP over the years, but the industry is moving rapidly toward more open standards. PODi, a not-for-profit consortium of leading companies in digital printing, is leading the way with two widely adopted open VDP standards. These standards are PPML (Personalized Print Markup Language) and PDF/VT (portable document format/variable transactional).

PPML

PPML, first introduced in 2000, is a device independent **XML**-based printing language. There are two types of PPML: *thin* and *thick*. Thin PPML is a single file, with the .ppml extension, containing all the instructions necessary for producing the VDP document. It does include caching instructions; however, all resources such as fonts or images are stored externally of the file. The path to these resources is defined in the RIP and retrieved during the rendering process of the document. Thin PPML is ideal for in-house VDP development where resources may be shared by multiple projects. These files are extremely small, however, and network speed and bandwidth may affect performance and are more difficult to implement if using an external print provider. Thick PPML is a .zip file containing all the required resources (fonts, images, instructions, etc.). This format makes the file highly portable and easy to implement on the print device, but it has a larger file size when compared to thin PPML. RIPs that support the PPML format can import the .zip file directly. Regardless of the type used, PPML benefits from exceptional performance, an open standard, open job ticketing support (JDF), and overall reduced file size. To generate PPML, an advanced VDP solution is required.

PDF/VT

PDF/VT is a relatively new international standard (ISO 16612-2) that has a lot of potential. It is built off

the PDF/X-4 standard, benefiting from its features, such as support for transparency, ICC-based colour management, extensive metadata support, element caching, preflighting, and much more. In short, PDF/VT includes the mechanisms required to handle VDP jobs in the same manner as static PDF printing allows print providers to use a common workflow for all job types, including VDP. Many of the latest releases of advanced VDP solutions already support PDF/VT as well as many DFE manufacturers.

For more information on PPML and PDF/VT, please refer to the PODi website at: http://www.standards.podi.org

Media Attributions

- plane_blue by Roberto Medeiros

6.8 Summary

Roberto Medeiros

Digital printing encompasses a number of technologies that each has unique characteristics, strengths, and applications. Digital imaging may even be the only print method able to produce a certain type of work, as is the case with VDP. Paper is also a major factor in the success of a project. Your paper choice can convey a message or set a tone as much as the content printed on it. Having a good understanding of technology and paper fundamentals can go along way when making choices for producing your print project.

Questions to consider after completing this chapter:

1. All xerography can also be called electrophotography, but not all electrophotography can be called xerography. What key element validates this statement?
2. What are the four key components in electrophotography?
3. How does toner acquire its charge?
4. What is the difference between paper brightness and whiteness?
5. Which PDLs support an ICC colour-managed workflow?
6. Which PDF/X standard leaves layers and transparency live?
7. Why are data content and business rules critical in VDP?

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Chapter 7. Web2print

7.1 Introduction

Steve Tomljanovic

Learning Objectives

- Describe web2print and how it benefits the supplier-customer relationship
- Differentiate between the different business models of web2print implementation
- Explore the economic impact of implementing web2print
- Research target markets that would benefit from online selling processes
- Define business to business (B2B) and business to consumer (B2C) sales models
- Examine variable templates and identify their components
- Discuss how print on demand works and how it affects the sales cycle
- Give examples of print workflows and management information system (MIS) integration
- Discuss how web2print business opportunities might affect return on investment

As modern modes of communication are constantly changing, the print industry has had to become very competitive. Successful print companies have evolved beyond producing only printed products; today, they also provide other services geared to an electronic marketplace. With the ever-growing number of people going online, customers have adapted to having instant information (Poon & Swatman, 1995), which printers can now provide. One such service is web2print. Customers benefit from the ease of ordering print anywhere around the world, with quicker turnaround times than traditional methods. Print companies benefit by building a larger customer base, streamlining workflows, and reducing production costs — all while staying competitive.

Print has traditionally been a service requiring many human touch-points, but with advances in business communication over the Internet, the dynamics of custom manufacturing are changing, particularly with respect to the ordering process (Shim, Pendyala, Sundaram, & Gao, 2000). Putting an effective online service in place involves strategically selecting suitable products and services. Online ordering is not ideal for every print product, but rather for specific products that are unnecessarily labour intensive. Products that were once very labour intensive, such as business cards, can now be ordered online. Customers enter their information directly into a template on screen while they sit in their own offices. A print-ready PDF file enters an automated folder in the print company (called a hot folder) and moves into the print workflow, allowing for a fully automated process called *lights-out prepress*, which saves on labour and allows for greater profits. Web2print provides a client company a value-added feature, while improving workflow process for the print company.

Technology Is the Key

Web2print completely automates the print ordering process by integrating online orders with print production. Using the Internet automates order entry and provides time savings to both the customer and the company.

> Electronic commerce offers the possibility of breakthrough changes: changes that so radically alter customer expectations that they redefine the market or create entirely new markets (Gunasekaran, Marri, McGaughey, & Nebhwani, 2002, p. 195).

This technology allows businesses to interact with customers in a new way in addition to more traditional forms of ordering such as email, file transfer protocol (**FTP**), phone, fax, and face-to-face meetings. While web2print removes the need for more traditional ways of ordering, it does not replace them. Some customers may prefer these ways because they may not know, or be comfortable with, the online options that are increasingly available to them. It is advantageous to a company to inform and educate customers, encouraging them to evolve their buying habits. Traditionally, purchasing involved gaining knowledge about print companies through their sales reps, selecting the appropriate company, and trusting that the sales rep would ensure the product was delivered on time and for the best price. Today, many customers use search engines to obtain information about and decide on which print company to use. Using web2print ensures that the production time frame and the price match what a customer is looking for.

A printing company has to work with and implement many complex and interconnected systems in order to generate a printed product. From the inception of a new item to its delivery to the end-user, there are many opportunities for streamlining. Web2print creates efficiencies at the beginning of the print process, and those benefits trickle all the way through to imaging. The ultimate goal is for the data collected from the customer to proceed directly to the raster image processor (**RIP**) that drives the production process. A print company that takes advantage of this type of automation is positioning itself for smoother and more cost-effective print runs.

7.2 E-commerce for Print Manufacturing

Steve Tomljanovic

E-commerce is by definition the buying and selling of goods online. Print companies are turning to e-commerce to target the ever-growing number of people who want self-service and a more convenient way to purchase products in their busy lives. Today, many customers are online and the number of users purchasing goods through the Internet is growing substantially (Statistics Canada, 2014). The ability to offer an effective e-commerce service for ordering can put a print company ahead of its competitors. This approach challenges traditional thinking that a printer is only a bricks-and-mortar or face-to-face business. As more and more businesses move online to satisfy their customers' procurement needs, so must print companies (Supply Management, 2014).

A print company should transition with its customers to stay ahead of the competition. E-commerce allows a customer to order any product 24/7 without having to leave the office or pick up the phone. Providing online ordering has other benefits besides increased sales. It can help a company gain more knowledge about its customers and track purchasing trends. As well, online ordering allows print companies to offer customers additional products and services that may not come up in discussions during traditional forms of print procurement.

Not all business can be conducted online, as the more complex a project is the more a company will benefit from offering its services and expertise in traditional face-to-face communications. However, the ability to stay ahead of the competition by using Internet technologies helps solidify a company's relationship with its customers. For example, even if customers are buying products offline, they are likely doing their research online first. Therefore, a company can use the valuable data generated during this research to greatly improve its ability to evaluate and manage its business. When customers place orders, companies can analyze the frequency of those orders, their customers' buying trends, and which products are more successful than others. Data-gathering is the most important advantage online ordering has over traditional offline ordering. Using an e-commerce system allows a print company to learn more about what, when, and how customers are ordering. Online technologies enable a company to broaden the products and services it provides more than ever before.

Templated Variable Data Printing

Templated variable data technology allows a customer to produce a product using a template-driven order-entry system. These templates are used most frequently through an e-commerce store. A variable template allows the user to control the information made available on a product and see the real-time end result. Variable data entry is an ideal solution for direct mailers, promotional flyers, event posters, and stationery. Variable data allows for the custom printing of one-of-a-kind pieces that are unique to the targeted market. The potential for variable data is endless and is only limited by the imagination behind a design.

Variable data is not limited to digital presses, as many lithography presses can incorporate designs with

the simple swap of a plate. It is common for companies ordering large quantities of business cards to print blank shells on which they can later imprint a person's information. A print company obtains this information by having the customer use a variable template, which is then uniquely created to ensure all branding standards are consistent. This allows any employee of the client company to order business cards and stay true to the brand. The person fills out his or her name, title, and phone and email contact information, then views the result on a soft-proof. Then the person simply submits the card through a shopping cart system, eliminating the need for multiple communications, and making production efficient by having fewer hands touch the project.

The Benefits to a Print Company

Web2print has multiple benefits for a print company. Software can automate tasks, eliminating the need for staff to touch a project, which makes the service profitable. Allowing the customer to create a print-ready PDF is an asset, not a loss of control, as it allows the customer to assume responsibility for any typos or errors. Web2print also makes the production process much faster and efficient by automating time-consuming steps, and most importantly, helps build solid rapport and customer loyalty.

Once a PDF is ordered, a job ticket (called a *docket*) is automatically generated containing all of the order specifications, including pricing. The PDF then travels to a hot folder where it is accessed by the imposition software, which imposes it onto a press sheet. The press sheet either goes straight to the digital press to be printed or to the plate setter if it is being printed by conventional lithography. The whole process is extremely efficient and takes considerably less time to complete than having staff continually involved as the PDF travels through their departments.

The Benefits to the Customer

Web2print is all about customer satisfaction. That should be the top priority when a print company creates customized online storefronts and variable templates. Customers no longer have to call or physically meet a sales rep, as they now have the ability to order their printed materials at any time of the day and from anywhere. Today, many customers do not sit behind a desk for their entire workday, and establishing an online service allows a company to target those on-the-go customers who need to order print.

Companies can track orders in real time, which helps them predict future buying trends. This is especially beneficial to customers wanting to print blank items, otherwise known as *shells*. Using this information, a company can help a customer determine how many shells to print in the next run.

Business Models: Licensed or Subscribed Software

Web2print services come in two primary types: licensed software and subscribed software. Licensed software allows a print company to own the technology by paying an upfront fee; subscription-based software typically requires a company to pay a monthly or yearly fee to use the software.

Licensed software strategies typically represent a large cash outflow up front. The company can then use

the software to create as many portals or implementations as it wishes. While the software is supported through a licence, it is expected that the company will maintain its own web presence, and therefore may require highly trained web developers or programmers. The outlay of expense is gradually recouped over time through increased user implementation.

The subscription model, also referred to as **SaaS** (software as a service) reduces a company's support and maintenance costs up front. Saas allows adding new print products to move forward more quickly because the need to support the same version throughout computers internally is removed. All subscribers operate on the same version at the same time, and are upgraded regularly at the same time. Because the Internet is constantly evolving, the SaaS model is flexible and better aligned to move with it. This business model contributes to a company's return on investment (**ROI**) as well. Since a company typically doesn't pay a large sum of money upfront for the software, it can easily budget for payments based on monthly usage. The company only pays for the services it uses. This business model builds a partnership between the print company and its SaaS vendor, and a positive ROI is beneficial to both parties in the partnership because it keeps both vendor and subscriber focused on mutual success.

7.3 Web2print Strategies and Goals

Steve Tomljanovic

Evaluating Strategies and Setting Goals

Print companies must have clear strategies and goals to ensure continued success when implementing web2print. The first step is to evaluate the type of sales they make. There are two basic types of sales a print company makes: business to business (**B2B**) and business to consumer (**B2C**). It is very common for a printing company to serve a primarily B2B customer base; however, since B2C requires a vastly different storefront, this decision needs to be made early in the process of implementing web2print.

Once a print company determines the type of storefront its customers need, it should research the three basic types of service: print on demand (**POD**), variable data printing (**VDP**), and static warehoused items. By analyzing its target market, a print company can determine which of these services customers will use most. Once the print company chooses a software vendor that can provide the most suitable storefront, only then can it decide on the specific services to offer each of its customers.
Therefore, to be successful, a print company must:

- Know the target market
- Choose an appropriate vendor and storefront
- Make plans to add new customers to the system by setting goals
- Choose the types of products to offer to each customer based on need

Know the Target Market

Every print company has a different customer base, and thus serves a different market. A print company must analyze the customers it serves to determine exactly what its target market is. The biggest mistake print companies make when committing to the purchase of an online ordering system is not researching the technology in relation to their target market. Print companies should choose the system that best suits their needs and benefits their customers. There are hundreds of vendors and products with thousands of features, so print companies need a strategy to ensure they can maximize their return on investment (ROI) while providing the best possible services to their specific, targeted customer base.

Choosing a Digital Storefront and Variable Software

Since not all vendors of e-commerce systems are the same, print companies need to exercise due diligence in making their choice of vendor. They should analyze their own internal workflow to ensure they find a vendor that best meets their specific needs. As well, print companies should determine what their employees' strengths are and ensure the appropriate staff are hired to accommodate online needs.

Staff involved in the implementation and operation of an e-commerce ordering system need a basic knowledge of many web-based programming languages in order to give them a good grasp of the back-end coding necessary to build and maintain the online system. Every online ordering system uses its own method of coding to create its storefronts and templates, so having previous programming knowledge is a major asset.

Many companies such as IBM, HP, Creo, and EFI are building platforms to provide VDP service to print companies. The software these companies provide creates a web2print workflow. This includes the internal processes needed to print a job, as well as a client-facing website, from which customers can order. It is important to understand the benefits of every digital storefront as they all offer different options. Digital storefronts must provide a simple ordering process for the customer while being very robust and efficient for the print company. Selecting the order quantity and displaying pricing should be simple and not confusing for the end-user. Customizing VDP products or ordering POD or warehoused items should be simple and quick. The ability to split products for multiple-shipping destinations should also be considered.
Selecting a storefront that can be integrated into a management information system **(MIS)** to streamline orders from customization to invoicing is beneficial. The ability to have customers approve or reject orders placed by others is also beneficial, as it allows for an extra review to ensure order information is correct.

To ensure they make appropriate choices, print companies request copies of documentation from a software provider to see what they need to learn if they plan to be a self-service user of the software. They request references and ask how the software provider handles support, system outages, and upgrade development to get a sense of how other users perceive the company. Print companies attend demonstrations of the product and give specifics on what they want to hear beyond the generic sales pitch. Print companies also seek specific information about short- and long-term product upgrades, which gives them a chance to glimpse the software company's strategic vision and how the product might develop in the future.

Other Considerations Before Purchasing

Print companies take other considerations into account before purchasing software.

Usability: If they have current B2B customers, print companies ask them to test the software before committing to a purchase. If these end-users have difficulty using the software, then it is not the right choice. If print companies have B2C customers, they ask someone without any print knowledge or experience to test the product. Testing against online competitors to see how the software compares is another way print companies assess the usability of a product. They also research customer feedback.

Partnership compatibility: The relationship between a print company and a software provider is a partnership, not just a sales interaction. Print companies are in frequent contact with their software provider to solve technical difficulties, arrange training, or add improved services. Therefore, determining if the software provider will make a compatible partner is important. Print companies don't rely solely on what the sales rep tells them; they try to get a sense of the software provider's team by calling the support desk and talking to customer service. This helps print companies determine how well they will be treated and whether the software provider's staff are knowledgeable.

Features: Assessing software features is usually part of the decision-making process. Print companies generally want to know the following before purchasing:

- How easily will customers understand and use the software's features in a self-service situation?
- Was the software built to support the print industry or first created for some other use and applied to the print industry? If the former, are the features transferable?
- Do the features allow set up and administration of the site, creation of B2B storefronts, and product development. Do they enable the print company to add variable elements, create users, and take orders without relying on the software provider?

An important tip when choosing software technology is to not put too much emphasis on the number of features offered. Features tend to constantly change, and more does not necessarily mean better. While software product development tends to centre on adding more features, it is not necessarily adding more value. If a feature is added to a product, but is never used by customers, it is possible that the feature did nothing more than add complexity to the ordering process. Such a feature may result in discouraging customers from using the system, or placing future orders.

Starting with a New Customer

One way to introduce a new customer to web2print is to build a single item for them. This allows the customer to learn the ordering process and the print company to learn and incorporate the customer's products into a production workflow. A workflow should be designed to be as automated as possible, from order entry to production to invoicing. New workflows should include sufficient time to allow a customer to test the variable templates to extremes, entering best- and worst-case scenarios to ensure the template can perform in all situations without errors. Only once the initial workflow has been proven to be efficient should more products be added to the storefront. This ensures that both the customer (external activity) and the print company (internal activity) are confident enough in the workflow to handle more orders.

Setting Goals and Site Testing

Printing companies should allow time to educate their customers in all steps of the process when launching an e-commerce system or when adding a new variable-template-driven item. The easiest way to meet customer expectations is to involve them in the development process, regularly inviting feedback and eliciting suggestions for improvement. Customer satisfaction is important, so a company must ensure that it takes client feedback seriously, incorporating customer input to improve the service process. As the site is being developed, both the programmer and the customer need to rigorously test new products and templates to ensure they are completely satisfied long before allowing product ordering. It is common for a programmer to envision how a template will behave, while the customer intends it to behave in a different way. Often a customer has expectations that the programmer may not have foreseen. Once the entire site, including products and templates, has been developed, it still isn't ready. A testing phase or pilot period is necessary to find any other bugs or shortcomings that may be more easily discovered once real data is being used. Implementing a pilot period before an official launch

of the full workflow also allows everyone to learn how the system will impact them, exposes potential workflow issues (which can arise in the many steps between ordering and invoicing), and allows the customer to provide final feedback.

Most important to keep in mind is that the system only works when customers use it. They will often find opportunities during the pilot period to suggest where the process can be improved, as unforeseen problems are discovered only after people start using a new system or variable template. Often these user-experience issues can prevent adoption of the system by the customer. As well, customers may fall back to the more familiar method of traditionally ordering print if they do feel comfortable using the new system. Including the customer in the entire process allows for the greatest chance of success, and is the best way to ensure the success of the site.

Choosing the Right Type of Products

Before setting out to create products, a print company should determine whether it is a variable template, a print-on-demand piece, or a warehoused item. Other key information needed is the name of the product and the communication intent (i.e., Is the piece promotional or educational? What audience is it intended to reach? How knowledgeable is this audience?). Print companies also need to know whether the product will be ordered regularly or be a one-time communication. It is important to choose the right products before the development phase begins. It is common for a product to be almost completely programmed before it is discovered that another similar product would have been more appropriate. Below are explanations of the three most common types of products, followed by a list of more specific options.

Variable Templates

Variable templates contain all the necessary information for a customer to customize and soft-proof a print order. This usually results in the creation of an automated, print-ready PDF, which is generated while the customer is still online.

A PDF of the design is created containing variable fields assigned for every element. Coding is then applied to each field to determine how the template will behave under given circumstances, such as during customization. For example, coding can force a name to be upper case or email to be lower case. Coding can also be used to upload custom images or force phone numbers to use hyphens (e.g., 604-123-4567) instead of dots (e.g., 604.123.1234). Coding is critical for keeping a customer’s brand consistent, so regardless of who creates an order, all products will be formatted consistently and have the same look.

Deciding which VDP software or plug-in is more appropriate and how it interacts with the digital storefront is important. VDP software comes in the form of third-party applications such as XMPie or is accessed online through a self-hosted dashboard.

Print on Demand

POD products are the opposite of VDP products. POD allows the customer to order a static product to

be printed and shipped. POD products do not require customization and are printed using a completed file uploaded by the customer or stored in the system by the programmer.

Warehousing

Storefronts can act as an inventory management system for any products that can be warehoused. These products can be ordered online using the same process as a POD item. Each product has a real-time inventory count associated with it, which updates after every order. Notifications can be sent about low product inventory levels, reminding a customer that a product needs to be replenished. Inventory counts benefit customers by showing their buying patterns, which helps them to effectively determine future quantities.
Below are other examples of different types of products that can be ordered online:

- *Ad hoc print:* an online print product where the customer provides the content during the ordering process via file upload, such as brochures, flyers, and newsletters.
- *Ad hoc business documents:* an online print product where the customer provides the content during the ordering process via file upload, such as training manuals, presentations, and reports.
- *Ad hoc oversize:* An online print product where the customer provides the content during the ordering process via file upload, such as posters, signs, and banners.
- *Static print product:* An online print product where the content is stored in a catalogue and printed on demand after ordering, such as sales sheets, flyers, and white papers.
- *Inventory product:* An online print product where the content is stored in a catalogue and pulled from existing inventory after ordering.
- *Digital publishing:* An online product where the final product is a downloadable PDF instead of a printed product, such as white papers, personalized sales materials, and presentations.
- *Kit:* An online print product where the customer can buy a basket of goods contained in a single item.
- *Promo product:* A set of products that are branded with a logo for use in marketing or promotional programs, such as mugs, baseball hats, and pens.
- *Integrated campaign:* A product that combines multiple-marketing channels to create an integrated campaign a customer can use to track metrics when launching a new product or sales promotion.
- *Photo product:* An online print product using uploaded photos or photos from online collections, such as photo books, photo cards, and photo calendars.
- *Quote request:* An online print product used to request a quote for a print job.

7.4 Implementation and Workflow Considerations

Steve Tomljanovic

Implementation and Deployment

By this point, the workflow strategy has been created, the customer has been included in discussions about its goals, and the print company has created some sample products and print items for the customer to test. As well, the print company and customer have completed a pilot period and identified unforeseen workflow issues. What remains is the final step of making the site live.

Making the site live involves 'turning it on' to accept orders from the entire user base. If the above steps have been completed properly, there should be very few issues.

Continuous Assessment

Even after a storefront has been launched, it is not considered complete. There should always be a system of continuous assessment in place to respond to customer feedback and correct any errors as the orders start coming in. Even after the site is live, the programmer should navigate the storefront to ensure its usability, and place a test order to ensure no issues arise for the customer during the ordering process. Also of consideration is a post-order assessment, where the internal processes in the printing company are evaluated for completeness and efficiency, as outlined below.

Workflows and Automation

Orders should enter an automated workflow, creating a seamless transition while bypassing several departments. Once an order has been placed, the appropriate staff are notified to fulfill it. If a VDP product was customized, then a print-ready PDF should automatically be uploaded to a hot folder. At this point, either an automated system or a prepress operator reviews the file for print standards and imposes it on the print template. These files can then be automatically produced on a digital press or be sent to the plate setter to be prepared for litho printing. Throughout every step of the process, email notifications should be sent to appropriate staff so they can fulfill the order, and to the customer so they can be kept informed of anything related to the order such as invoices and product shipping.

MIS Integration

It is beneficial to select a storefront suitable for integration into a management information system (MIS) to streamline orders from customization to invoice. Integration is a connection between two systems that enables the exchange of data. The information is automatically entered into an electronic docket, which is a database that collects and maintains customer information, products ordered, shipping

information, and billing information automatically. When integrating two systems, it is important to note which system is the master data holder and which is the subscriber to that data. Only one digital system should 'hold' the data, whereas all the other systems access the same database. In a print environment with a functioning print MIS system, it is the MIS system that should be considered to be the master in every case. The MIS system collects orders from everywhere, not just the orders placed through storefronts online. The web2print system pushes data into the MIS system and subscribes to the master data stored and managed in that system, such as pricing and job specifications. This can be challenging because the web2print software and the print MIS system are often provided by separate vendors, which can prevent a smooth exchange of data.

Web2print is one of many secondary, or subscriber, connections into a printer's business and production workflow. Web2print should serve its main purpose, which is to capture orders in the most efficient manner while maintaining a competitive edge for a print company's sales team. Orders must be transitioned seamlessly and smoothly into the production workflow and MIS system so they can be treated like any other order, whether they were placed online or by traditional means. In this way, web2print is regarded as only one of many business opportunities that bring sales to a print company.

Analyzing the ROI

When making any business decision, investment must be weighed against return. This is known as return on investment, or ROI. Moving a business online to accept orders is a serious business decision. Web2print can be a worthwhile investment and understanding how to measure ROI before investing in a vendor's software is important.

Typically, in a print company, the estimate for the actual printing process is very well defined. The estimating department can provide detailed analysis of all of the costs associated with printing a specific printed product. Where web2print differs, however, is in the costs of capturing the sale and in streamlining the process in the print shop. For example, there are specific increased costs in running a web2print site online. If the system was paid for as a one-time licence, then the total cost must be amortized over the life of the licence, and each print order shares a small part of that overall cost. Some SaaS systems, on the other hand, charge a piece rate or a monthly fee. These are easier to incorporate into the costs of the job. On the savings side, there are processes within the print company that are made more efficient, so an analysis of cost savings can be made there as well. However, print companies should not fall into the trap of thinking that just because a print job can be completed in a shorter time, it is automatically cheaper to produce. In order to assess the total ROI, only real costs that affect the print product's profitability should be assessed.

Timing is important when calculating ROI because a printer must determine when to invest money based on an expected return of that investment. Purchasing or building an online system is not automatically going to generate revenue. It is likely that the print company could invest thousands of dollars very quickly before the system provides any value in return. There is a human aspect to this as well. Sales professionals are still critical for driving new customer sales to the web and finding new online opportunities, both of which will help improve the return on the initial investment.

Systems with monthly payments are sometimes better for new online ventures, as they do not require a huge investment upfront. Up-front payments force a print company to give away all monetary leverage

in a single transaction, and while they might be more cost-effective when serving large numbers of customers, they can do serious financial damage in the short term.

7.5 Summary

Steve Tomljanovic

Web2print is the online connection between a print company and its customers, and the technology should help to solidify this relationship, not hinder it. Print companies offer their services online in response to their customers' needs and buying trends. As web2print becomes more integrated into a print company's day-to-day business, it becomes a main channel for interacting with a customer. A key to the strategy for implementing web services is involving the customer as much as possible, since the customer's use and acceptance of the ordering portal is critical for its success. Print companies should research the types of products and services that will be helpful to customers in the specific target markets they serve, and not add too many products too quickly. Print companies must analyze the types of products their customer needs, and plan how a streamlined workflow will create efficiencies in its operations. Finally, a pilot phase to assess both accuracy of the storefront and user experience is important. To ensure continued customer satisfaction, print companies should be prepared to make ongoing improvements once the site goes live. System integration with print companies' internal processes is also ongoing, as efficiencies and production enhancements are realized. The print industry continues to evolve and a successful implementation of a web2print portal will help print companies keep up with this evolution and stay in front of the competition.

Questions to consider after completing this chapter:

1. How would you describe web2print and the technology involved?
2. How would you describe e-commerce and how web2print utilizes it?
3. What are the benefits of using web2print to a company and it customers?
4. What are the strategic steps in creating a web2print system?
5. What types of products can be offered through a web2print system?
6. In what ways can a web2print system be integrated into a production workflow?

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Glossary

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The Graphic Communications Open Textbook Collective comprises the following authors.

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1.01	June 6, 2019	Updated the book's theme.	The styles of this book have been updated, which may affect the page numbers of the PDF and print copy.
2.00	July 18, 2019	Entire book revised for accessibility.	Accessibility remediation: • Image descriptions added. • Link text edited to be descriptive. • Added an Accessibility Statement • Headings added. Other changes: • Added List of links by chapters for print users. • Added pop up features for glossary. • Formulas formatted in LaTeX
2.01	July 30, 2019	The following changes were part of a project to standardize BCcampus-published books.	• Added additional publication information • Updated copyright and attribution information • Added ISBNs • Renamed "About the book" to "About BCcampus Open Education" and updated the content • Updated the book's cover

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- A Handbook of Rhetorical Devices: https://www.virtualsalt.com/rhetoric.htm
- A Model of The Creative Process: http://www.dubberly.com/concept-maps/creative-process.html
- “A Problem Well-stated is Half-solved” by Mark Levy: http://www.levyinnovation.com/a-problem-well-stated-is-half-solved/
- Concept Tree Method: https://mindmappingsoftwareblog.com/concept-tree/
- Example of Allusion – the magazine cover: http://2.bp.blogspot.com/_6YrCK0O5xBw/SeDYHUDq3tI/AAAAAAAAAFs/Paovl5_nz04/s1600/eyes.jpg
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- Example of Hyperbole: http://n-tiffany0811fmp.blogspot.ca/2011/02/mark-studio.html
- Example of Metaphor: http://media.iqads.ro/2009/09/ikea-bigger-storage-ideas-full.jpg
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- Example of Synecdoche: http://40.media.tumblr.com/tumblr_lfthryFqOH1qaz1ado1_500.jpg
- Example of Understatement: http://3.bp.blogspot.com/-qobPI4uVlcI/UGQUXup0rtI/AAAAAAAAAEU/d3icG0PcEsA/s320/1486396940.jpeg
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Chapter 3. Design Elements, Design Principles, and Compositional Organization

- “Think Small” Campaign: https://en.wikipedia.org/wiki/Think_Small

Chapter 6. Imaging

- Digital Press System Certification: https://www.idealliance.org/certification/digital-press-certification
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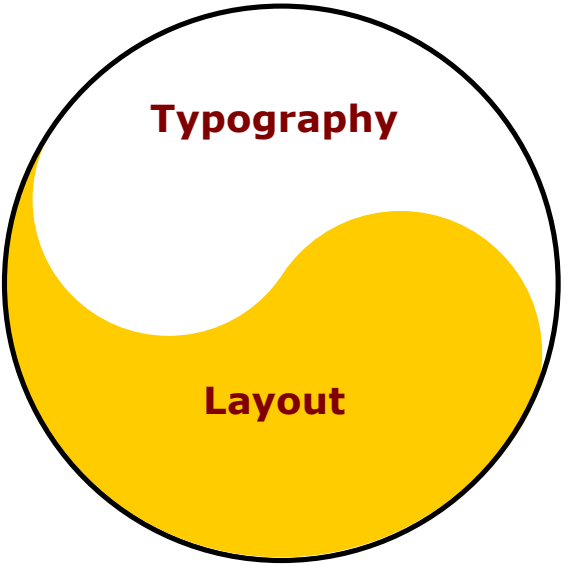

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Information Design 4–Graphic Design

Yin and yang, or yin-yang, is a concept used in Chinese philosophy to describe how seemingly opposite forces are interconnected and interdependent, and how they give rise to each other. Many natural dualities, such as life and death, light and dark, are thought of as physical manifestations of the concept. Yin and yang can also be thought of as complementary forces interacting to form a dynamic system in which the whole is greater than the parts. In information design, theory and practice is an example where *the whole is greater than the parts*.

In this book drawings and photos are my own, unless other information.

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Preface

Information design is a multi-disciplinary, multi-dimensional, and worldwide consideration with influences from areas such as language, art and aesthetics, information, communication, behaviour and cognition, business and law, as well as media production technologies.

Since my retirement I have edited and revised sections of my earlier books, conference papers and reports about information design, message design, visual communication and visual literacy. The result is this series of six books:

Information Design 1–Message Design
Information Design 2–Text Design
Information Design 3–Image Design
Information Design 4–Graphic Design
Information Design 5–Cognition
Information Design 6–Predecessors & Pioneers

These books include definitions, selected results from research, theoretical considerations, as well as practical guidelines for message design. The intended reader is especially interested in research and theory related to message design, and the design of information materials and information sets.

Tullinge, Sweden
Rune Pettersson, Ph.D.
Retired Professor of Information Design

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Organized structure

Although we may not think about it, the practice of graphic design is as old as recorded history. We see the results of graphic design every day in books, magazines, packages, papers, posters, symbols, and many other products.

Graphic design

A generally accepted view is that graphic design may be described as the art and craft of bringing a functional, aesthetic, and organized structure to different kinds of texts and illustrations. Graphic design is a *process* (verb) as well as a *result* (noun) of that process. Traditional graphic design is a kind of "all purpose-design" used in the production of various media. Modern visual graphic design has its roots in the rational, functional aesthetics that evolved in traditional graphic design over the centuries for the print media, and are now used in industrial design, as well as in architecture.

Purpose for graphic design

The purpose of work with graphic design is to find a *suitable presentation of the message* with respect to the intended receiver, medium, and economical situation. A well-designed book appears as a "unified whole." All design elements serve to enhance the content.

Melin (1999, 2011) found that it is more likely that graphically complex texts will be read than "plain" texts. And it takes less time to read a graphically complex text than a "plain" text. Readers often react in a positive way to texts with good typography and good illustrations. The overall visual look shapes

users' perceptions of consistency and user satisfaction in browsing tasks (Ozok & Salvendy, 2000).

Within a given area, such as a page in a book, a computer screen, or a projected image, we may alter the presentation of text (headings, running text, captions, lists), pictures, tables, and the background (margins, ornaments, page number, space).

When text and graphics are organised into meaningful semantic clusters, this makes it easier for readers to chunk the content (Kahn, Tan & Beaton, 1990). Visual grouping of text enables readers get a sense of the overall structure (Tullis, 1997), and help readers remember content (Niemela & Saarinen, 2000). How the content is grouped may influence the readers' first impressions of the content (Lindgaard et al., 2006). Readers will better remember the content and make fewer errors. Careful integration of words and pictures engage people more effectively than words or pictures alone (Sadoski & Paivio, 2001).

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures, will impair the understanding of the message. Active voice, clarity, comprehensibility, consistency, legibility, precision, readability, reading value, simplicity, and structure are the key concepts in information design. Hiebert (1998) saw graphic design as interface design (p. 9):

> In the largest sense, all graphic design serves as an interface. It is the filter that facilitates communication between users and products, places, processes, information, and services. This interface may have the purpose of description or persuasion. It may be in a process of one-way presenta-

> tion or two-way discourse (interactive). It has a functional rather than a decorative purpose.

The role of graphic design is functional. According to Bull (1999) there are currently many debates as to what the role of a designer is. Terms like "Visual Communicator" and "Information Architect" have arose. Many of these debates have grown out of many designers' misuse of technology and the computer. Bull concluded (p. 54):

> In our fast-paced, drive-thru oriented society, the graphic design community often suffers from the public perception of what I have coined the 'Kinko's Mentality'. This is where anyone with a computer is legitimized as a designer and solutions to your problem can be given form based on a solution to another problem. And of course, it all perpetuates the idea of the designer as just someone who just dresses things up.

The study of rational, functional aesthetics, as well as effective and efficient layouts for all media makes graphic design interesting for information design scholars. Graphic design is a tool with which we can manipulate the raw materials – words in different typefaces, sizes, styles, empty space, illustrations, colour, paper and ink, and the final number of pages – to achieve the best possible communications between people.

Waller (1980) provided a list of codifying rules for graphic language and functions of the typographic organization of text in a book. Waller discussed two main categories (I) Rhetorical functions, and (II) Access functions (p. 246):

I. Rhetorical functions

1. About the argument.

   - Summarization (title, summary).
   - Introduction (foreword, preface, introduction).
2. Within the argument
   - Emphasis (underlining, italics, etc.).
   - Transition (headings, space, etc.).
   - Bifurcation (alternative options, parallel texts, interpolation sections).
3. Extra to the argument
   - Substantiation (footnotes, appendices, references).
   - Addenda (apologia, acknowledgements, etc.).

II. Access functions

4. About the book.
   - Overviews (contents list, abstract).
   - Definitives (glossary, index).
   - Identifiers (title, author, style).
5. Within the book.
   - Locators (topical headings, typographic signalling).
   - Descriptors (functional headings, captions).
6. Extra to the book.
   - Study guidance (recommended reading, exercises).

Graphic design objectives

Any graphical message should be *legible* for the intended audience. (It should also be *readable*, and well *worth reading*.) In graphic design the main objective is to provide functional, aesthetic, and organised structure to all kinds of information sets. The intended individual information interpreters might be seen as “readers.” They may develop new views, relaxation, emotions, awareness, attention, and understanding. In the writing of *graphic design objectives* it may be an advantage to use verbs

like *find, identify, read,* and *recognise*. These verbs all denote observable behaviour. A few examples of performance objectives in graphic design may be:

- For a table: 100% of the users should be able to find the time for departure of the train to x.
- For a package: 100% of the buyers should be able to read the text on the package without any difficulty.
- For a non-fiction book: 100% of the readers should be able to read the text in the book without any difficulty.
- For a logotype: 60% of the readers should be able to identify a new logotype within six weeks.

Of course we have to decide the actual numbers, with respect to percent and allowed time, in each specific case.

Graphic design processes

Around 1850 the sculptor Horatio Greenough wrote, “form follows function”. The architect Louis Sullivan made this phrase famous in 1896. However, Mijksenaar and Westendorp (1999, p. 34) concluded that architects, engineers and designers rarely follow this rule. Many theories of design processes may have been devised with three-dimensional products in mind. However Adams (1999, p. 4) argued that there is no reason not to apply these design processes to their two-dimensional relatives.

Some examples of design processes

Several processes for problem solving are related to design. For example Shadrin (1992:29) discussed problem solving based on the following seven steps or “constants” as a system for problem solving:

- Design activity (What is the problem?)

- Analysis (What is the purpose and function of the design?)
- Historical reference (How was it done before?)
- Visual communication (How can I communicate my idea?)
- Skills (What skills do I need for this design?)
- Technology (How will the design or product be made?)
- Evaluation (Is this the best solution I can come up with?)

Roozenburg and Eekels (1995) presented a design process with error elimination step by step. They noted the following steps:

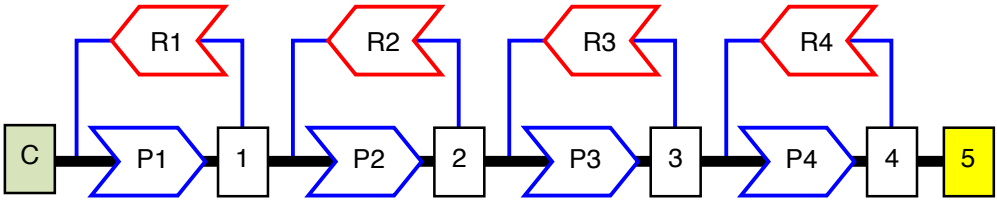
- Problem definition
- Analysis
- Criteria
- Synthesis
- Preliminary design
- Simulation
- Expected properties
- Evaluation (based on Criteria)
- Value of the design
- Decision (feedback to Synthesis and Analysis)
- Acceptable design

Roozenburg and Eekels stress the development of criteria to be used in judging the success of a resulting design. The functioning of a potential design solution is assessed in a simulation. The observed properties of the assessed solution are used to infer the properties that we may expect from the final product. Bull (1999) noted the following steps in a graphic design process (p. 53):

- Problem definition.
- Research.
- Conceptualization.

- Prototyping.
- Testing.
- Implementation.
- Documentation.

Adams (1999, p. 4) described how the writer and the designer are working together in some manner using a mental model of the user and of the way in which the communication will be used. Within this mental model words, verbal structures, possibly illustrative pictures, and indeed the whole design concept are constructed. When the user approaches an information product, he or she brings a history of interaction with that type of information product. The user will be of a social and cultural background, will have a certain education and ability level and will have a certain motivation regarding the use of the product.



The creative message and information design processes include four different production (P) and review activities (R). The production activities are analysis and production (P1) of synopsis (1), production (P2) of draft (2), production (P3) of script (3), and production (P4) of original 4) and master (5). C = commission.

My own "message design and information design model" includes the following four process activities: analysis and synopsis, production of draft, production of script, and production

of original and master. Each activity includes a design sub-process, activity documentation, and a review process. Main message design tools will include text (printed and spoken), symbols, pictures (drawings and photos), typography and lay-out, light and light effects, sound and sound effects.

Often a team of people with skills in the different areas are working together in a *design team*. The task may be too overwhelming for a single person.

(See the book *Information Design 1–Message Design*.)

Reduction

According to Mullet and Sano (1995, p. 38) the most fundamental design technique is reduction. Wherever possible the designer should remove insignificant elements in layout, pictures, and texts in order for significant design elements to be clearly noticed. An elegant design must be reduced to its essential elements and each element reduced to its essential form. As an example a good symbol for a sign is simple and clear. It has an optimal colour, dimension, form, and size (Barlow & Wogalter, 1991; Dewar, 1999; Wogalter, 1999).

A text on a poster, and on a screen, shall be bold enough (Mayer, 1993; Ormrod, 1989; Pettersson, 1993; Wileman, 1993), and large enough (Ormrod, 1989; Pettersson, 1993; Wileman, 1993). Too small or too large lettering will impair reading. The text shall also have good readability (e.g. Kirkman, 2003, 2005; Klare, 1985; Lipton, 2007; Mackiewicz, 2004; Pettersson, 1989; Young, 1989).

It is generally accepted that good design is simple, bold, and direct. It ensures that significant design elements will be noticed by removing insignificant elements wherever possible. Most people read instructional materials selectively. Readers

rarely, if ever, begin at the beginning and read straight through to the end. Usually we use a combination of browsing, reading headings, looking at illustrations and captions, reading certain parts carefully, skimming others, and avoiding some parts completely.

Information layout

Information layout differs from a *decoration layout,* in which purely aesthetic aspects dominate. When illustrations in textbooks are not relevant to the prose contents, they do not facilitate understanding. On the contrary, illustrations can actually have a negative effect on reading comprehension and prose learning. Therefore illustrations should not be used only for decoration in information and learning materials. (See the section *Image functions in teaching and learning* in the book *Information Design 3–Image Design.*)

Today many visuals are too complicated and would communicate better if designers valued simplicity over decoration. Aesthetically pleasing visuals may deceive the learners about their instructional values. There are many situations where colour and typographic elements can be used for decoration. However, a decorative use of colour or typography should never be mixed with the intended use to provide clear structure, simplicity and hierarchy. It must always be very clear and easy for the receiver to understand when colour and typography are used for decoration and when the use is meant to have some cognitive importance.

History of graphic design

People have been designing, planning and executing information sets, information materials, and messages in all times. Mullet and Sano (1995, p. 9) noted that whereas *art* strives to express fundamental ideas and perspectives on the human condition, *design* is concerned with finding the representation best suited to the communication of specific information.

The Egyptians were the first culture to produce illustrated manuscripts and wall decorations that combined their writing system with illustrations. In the fifth century, the Greeks introduced the concept of symmetry. This was based on their observations about the natural world. Later, Roman artists developed symmetric arrangements of letters and graphic elements. For centuries monks produced hand-drawn manuscripts in their monasteries. They paid careful attention to the selection and placement of each graphic element, and they tried to maintain a consistent style throughout every single work.

Step by step

With the invention of the printing press the publisher or art director employed assistants to design typefaces for good legibility and high aesthetic value (Meggs, 2005). At this point, more care could be given to typography, illustrations and graphic design. Book pages were often decorated with borders and various kinds of ornaments as natural parts of the final design. Artistic and aesthetic values of each historical period have been dominating in traditional graphic design.

Since the time of Gutenberg graphic design has developed as new printing devices were invented. For centuries words and pictures were separated by less developed technology. There

was no method to print text and pictures together with good quality until the lithographic process was invented.

Originally all lithographs were printed from a flat, lithographic limestone on a flatbed press. The image and the text are drawn on the flat surface of the stone with a greasy crayon, an oily wash, or with India ink. After a chemical treatment the parts of the stone without the image and text elements are susceptible to water. The printing ink is transferred to the paper from the surface with the image and the text. Lithography was invented in 1798 and has been used for printing of lithographic art, but also posters, placards, cards, advertisements, etc.

By 1834 specially treated zinc plates began to replace the heavy stones. From 1860 to 1900, lithography was commonly used for art reproductions, posters, greeting cards, and business cards. The use of photomechanical metal plates in the early 1900s made the technique of hand transferring from stones obsolete. Lithography expanded the range of graphic design.

With the invention of the half-tone screen it was possible to print photographs as half-tone pictures. This information is converted to the raster-dots that are employed in the printing of the picture. The printed dots vary in size from nothing or a very small dot to gradually increasing dot-sizes. Smaller dots give the impression of light grey areas in the image, and larger dots give the impression of darker tones. The number of raster-dots defines the quality of the final printed image.

The first colour photographs were reproduced in 1881. Advances in photoengraving and halftone techniques allowed the regular use of photographs in print media by World War I. Because of the modern computer the methods for presenting photographs are changing radically. Halftone screens can now

be simulated with computer programmes that sidestep the entire photoengraving process.

Today the graphic designer has even more freedom than the monks during the Middle Ages. Now, it is possible to combine words and pictures at will, in effective and efficient layouts for all media. Modern visual graphic design has its roots in the rational, functional aesthetics that evolved in traditional graphic design over the centuries for the print media, and in industrial design, as well as architecture.

According to Baggerman (2000, p. 9) possibly the most important thing to keep in mind is that there's no recipe that's going to make your design appropriate for all applications and users. It helps to think of any designed piece as an interface, be it a book, shampoo bottle, retail store, or Web site.

Modern graphic design

Today graphic designers work in "persuasion design," as well as in "instruction design," and in "information design." Graphic design is a natural part of these design areas. Graphic designers are responsible for typography and layout in information and learning sets. The graphic designers may also produce the final master for printing.

At the beginning of the twentieth century avant-garde designers, from the modern art movement, found inspiration from efficient, functional, interesting, and powerful new machinery. The avant-garde designers used visual forms that were fitting for the new and modern world. They explored asymmetrical layout, functionalism, geometric typefaces, hierarchy, minimalism, serial design, universality, and white "empty" space. Modern graphic design emerged out of the modern art movement.

The history of graphic design reveals that the manipulation of visual structure has always been fundamental to our thinking about how communication works (Davis, 2012, p. 22). During several decades graphic design was gradually consolidated into a design profession. The theoretical base for graphic design was developed from avant-garde movements such as the Russian artistic and architectural philosophy *Constructivism*, the Dutch artistic movement *de Stijl*, and the German school *Bauhaus* that combined crafts and fine arts. Bauhaus was famous for the new approach to design. After World War II many art schools worked with influences from these early movements and practices. In the middle of the twentieth century graphic design became a profession in its own right.

Design scholars like György Kepes (1944), Rudolf Arnheim (1954), and Donis Dondis (1973) wrote textbooks with design principles based on abstract painting and Gestalt psychology. All of them have a focus on *visual perception*. For these authors design is an abstract and formal activity. A theory of design that isolates visual perception from linguistic interpretation encourages indifference to cultural meaning (Lupton & Miller, 1999, p. 62).

Kepes (1944) argued that visual communication is universal and international. Visual communication knows no limits of tongue, vocabulary, or grammar. This book was used as a college textbook, and had thirteen printings, in four languages.

Arnheim (1954) described picture perception as a matter of responding to basic forms such as gestalt laws. An important point is that visual perception includes the same behaviours that we commonly consider only as matters of cognition or thinking.

Dondis (1973) discussed the use of several pairs of oppositions as techniques for visual communication. A few examples are: Balance – Instability, Simplicity – Complexity, and Transparency – Opacity. These oppositions present the graphic designer with effective means of making expressive visual communication.

From the 1950s/1960s graphic designers abandoned the old avant-garde ideals, and used formal methods. They worked with restricted typography in layouts based on strict grids. The new *International Typographical Style*, also known as the *Swiss Style*, favoured values of asymmetric layouts, cleanliness, legibility, neutrality, objectivity, photographs rather than drawings, rationality, and sans-serif typefaces.

According to Horn (1999, p. 25–26) information design is experiencing a variety of tensions. Graphic designers "learn in art school to worship the gods of Style and Fashion, Novelty, Impact and Self-expression." Technical communication people "worship the gods of Clarity, Precision, Legibility, Comprehensibility, and (often) Simplicity."

In the book *"Graphic design theory Readings from the field"* (Armstrong, 2009) twenty-four graphic designers explores the aesthetic and social purposes of design practice. These authors present what they think about what they experience is going on in the world of design discourse. Topics range from Bauhaus, over postmodernism and social responsibility, to Internet.

Armstrong (2009, p. 97) noted that digital technology fundamentally transformed graphic design "as one millennium ended and another began." The old avant-garde issues of authorship, social responsibility, and universality were reborn within society's newly decentralized network structure. In-

dustry-standard software and restrictive web protocols formed a new universal graphic language, while the subjective shift expressed in New Wave and postmodern design instilled a revived sense of agency among designers.

Providing clarity, providing emphasis, and *providing unity* are three of the *Functional principles* in information design (Pettersson, 2010). Information materials should be as clear, simple, unambiguous and transparent as possible. We should avoid unusual typefaces, as well as fonts that are too small or too large. Typeface and font size must be adapted to meet the limitations of the medium, and the technical production. The most important elements in an information material may be emphasized to enhance attention and perception. Emphasis may be used to *attract*, *direct* and *keep* attention. Information materials should have an "overall coherence and togetherness." Inconsistencies in information materials may confuse the receivers.

Research-based design

According to Lester (1995, p. 168) the designer W. A. Dwiggins was the first to use the term *graphic design*. This was in 1922. During his career he worked on more than 300 book designs. Although we may not think about it, the practice of graphic design is as old as recorded history, and we see the results of graphic design every day. We see books, cards, exhibitions, magazines, newsletters, packages, papers, posters, signs, tickets, and many other products. Often graphic design is thought of with regard only to the print medium, but it is also used in several other media. Hurlburt (1981, p. 22) defined graphic design in the following way: "Graphic design is an umbrella term that covers a broad range of printed and projected images. Its

three principal functions are to persuade, to identify, and to inform."

Advertising is generally regarded as the means for persuasion, although at times its purpose may also be to identify or to inform. Corporate identification focus on identity, but corporate literature will often inform, and corporate advertising may also have a persuasive purpose. In package design the function is also to protect the content in the package.

Hightower (1989, p. 7) noted that graphic design is a ubiquitous presence in our daily lives that can engage and inform us or simply add to the visual morass of contemporary culture. Important and unimportant messages are graphically communicated throughout the day.

According to Friedman (1989, p. 10) the taint of commerce has relegated graphic design to the status of "second class" discipline in the academic realm. In order to better this low status, the discipline needs to adopt more theory. However, at present there seems to be few theoretical foundations for graphic design. Much of the work is apparently based on intuition, and some on fundamental principles.

According to Zwaga, Boersma and Hoonhout (1999, p. xix) researchers active in a field of applied behavioural science implicitly expect that designers concerned with the usability of their products will eagerly pick up the fruits of scientific investigation and use them in their designs.

Many guideline documents have been published over the years. However, to the disappointment of ergonomists and behavioural scientists designers do not read their handbooks, or their scientific journals, and they are not prepared to apply experimental data to their work. Often designers are not even aware that this guideline information exists. The efforts were

wasted because they did not understand what designing was all about. However, Dillon (1994) found that designers seek further guidelines and usability standards, and that they are prepared to apply guideline information.

Adams (1999, p. 19) concluded that designing of good information-giving material is a difficult task, but it is one that can be assisted greatly by appropriate usability testing. Wogalter (1999) noted that if the testing shows that a number of people do not understand the message, or worse, misunderstand it, then the material must be redesigned and tested again – until it is understood by the intended audience. Users of information products still largely blame themselves when faced with poor design of any kind. Thus a push for research-based information design must come from the information designers. In each case members of the specific user group may be invited to evaluate preliminary texts and sketches for drawings and photographs.

Aesthetic principles

Aesthetics is usually regarded as a branch of philosophy, along with epistemology, ethics, logic, and metaphysics. It is the youngest branch of philosophy with its own name. Aesthetics implies a hierarchical judgement in which the elements of any compositions, and their organisations, are considered more or less appealing to the senses by virtue of their ability to create higher emotional, intellectual, or moral appreciations in people. The group "*Aesthetic principles*" is one of the four groups of message design principles. This section includes *Aesthetic proportion principle*, and *Harmony principle*.

Aesthetic proportion principle

Basically proportion is a mathematical concept. However, the concept *aesthetic proportion principle* is very much a subjective concept. It is related to the appropriate and pleasing relations between elements in information and learning materials. Aesthetic proportion deals with aesthetic aspects of information sets.

We may all have different ideas of what we find beautiful and rewarding, and what we find boring, disturbing, distracting or ugly. When a design is *out of proportion* it is lacking appropriate relations. It may be disproportionate, exaggerated or overemphasized. Aesthetic designs are perceived as easier to use than less-aesthetic designs (Lidwell et al., 2010, p. 20). The information designer may:

- Find out receiver preferences of aesthetic proportions.
- Be careful using proportions according to the "golden section."

- Never mix a decorative use of colour with cognitive importance.

This section includes the following two sub-sections 1) fine art, and 2) information aesthetics.

Fine art

Throughout history all cultures have produced musical, verbal as well as visual art. Fine art is usually assumed to mean the visual arts, like architecture, painting, and sculpture. The impulse to create, to recognise and to generate order, seems to be universal.

Different reasons

One of the earliest human artistic acts was to spend more time shaping tools than was functionally necessary. Palaeolithic Neanderthal man carefully decorated their spearheads. Some theorists argue that the origin of human creativity can be found in these early crafts of making tools and weapons.

Throughout history people have produced art for different reasons, such as religious devotions, special commemorations, adornments, and also for personal expressions. Art has also been created on many scales, from huge buildings and monuments, large murals and paintings to small jewellery and miniature paintings. The broadest generalisation may be that the visual arts are spatial rather than temporal. Music and literature must be experienced serially in time. The visual arts must be experienced in space. Space is an illusion in painting. Here an indication of three dimensions in two dimensions is rendered by conventions.

However, the conventions vary during different periods and in different places. In many situations it is not at all possible to

"understand the meaning of fine art." To some degree, the physical materials that artists use will influence the properties of the resulting artwork. As an example, a fresco painting on a wall has visual properties that are different from the properties of an oil painting on canvas.

Divine proportion

In fine art classical formats are based on the *divine proportion* or the principle of the *golden ratio*. This is an irrational number is calculated from a line that is divided into two segments in a certain way. The ratio of a line (a+b) to the larger segment (a) is the same as the larger segment (a) is to the shorter segment (b). Thus (a+b)/a = a/b (Livio, 2002, p. 3).

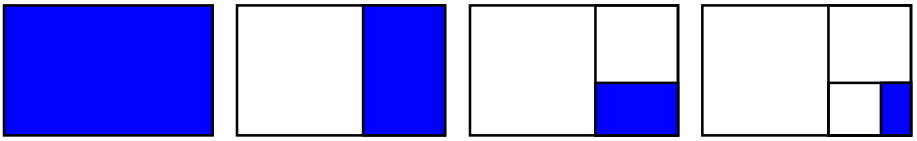
According to the Divine Proportion (a+b) is to a as a to b.

This mathematical relationship forms an irrational number, i.e. a number that never ends. Around 300 B.C. Euclid of Alexandria, in Egypt, provided the first mathematical definition of the *golden ratio* (Livio, 2002, p. 3). (With ten decimals the *golden quota* is 1.6180339887.) In the early 20th century the American mathematician Mark Barr named this irrational number "phi" in honour of the Greek Sculptor Phidias (Livio, 2002, p. 5). Historians believe that Phidias lived circa 490–430 B.C.

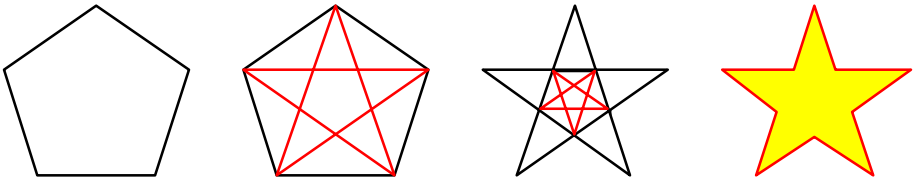
The principle of the golden ratio is comparable to the well-known "Fibonacci numbers": 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, and so forth. In this sequence any term after the first two is the sum of the previous two terms. This property is a close ap-

proximation of the golden quota (8/5 = 1.6). A *golden rectangle* therefore has sides of approximately the same proportions, 8/5. Any such rectangle is enlarged by being multiplied by 1.62, and reduced by being multiplied by 0.62.

Traditionally, the golden rectangle is considered aesthetic in the western world (Arnheim, 1974; Berndal & Frigyes, 1990; Bringhurst, 2004; Koblanck, 1999; Moriarty, 1991; Pettersson, 2002). For 2,500 years the principle of the golden ratio has been used in art and architecture to create harmonious proportions. The golden ratio is a well-known standard format for fine art, flags, symbols and more. During the nineteenth century the golden ratio was much used in painting academies.



When we divide a rectangle according to the golden ratio by adding a square, the rest of the original rectangle (the blue surface) gets the golden ratio. In theory we can repeat this process forever; in reality though, there are always physical limits to this.



When diagonal lines are drawn in a pentagon, several examples of the golden ratio emerge. At the same time "phi" is the property between the diagonal and one side in the pentagon. The five-point star is used in many ways.

The *aesthetic value* of a message is how the intended receivers perceive it with respect to its beauty. Material with a (sufficiently) pleasing form has greater potential for conveying a particular message than does non-aesthetic material. With the invention of the printing press the publisher, or the art director, employed assistants to create typefaces for good legibility and high aesthetic value. In this new situation, more care could be given to typography, illustrations, and layout. As previously noted book pages were often decorated with borders and various kinds of ornaments as natural parts of the final design. Artistic and aesthetic values of each historical period have been dominating in traditional graphic design. Since the time of Gutenberg the art and practice of graphic design has developed step by step, as new printing devices were invented and introduced.

Aesthetically pleasing visuals may not be of great instructional value. "The value of different types of visual illustrations is not a valid assessment of instructional effectiveness that is, aesthetically pleasing visuals may not be of great instructional value" (Dwyer & Dwyer, 1989, p. 122). It is, however, possible that aesthetically pleasing information material will be noticed and used better than material without any aesthetic qualities (Holmes, 1993). Malamed (2009, p. 203) noted that when viewers look at both pleasant and unpleasant pictures, they consistently demonstrate an emotional reaction indicated by pronounced brain activity that does not occur when they look at neutral pictures.

Information aesthetics

Norman (2002) argued that *visual attractiveness* is important for the cognitive domain since the affective domain is highly related to human cogitation.

Lidwell et al. (2010, p. 20) found that aesthetic designs are perceived as easier to use than less-aesthetic designs. Information aesthetics deal with aesthetic aspects of information sets. In some situations art may be used for information, and information may in some situations be classified as art. A good example of this is the brightly coloured posters designed by artists like Jules Chéret (1836-1932), and Henri de Toulouse-Lautrec (1864-1901).

(See the book *Information Design 6–Predecessors & Pioneers*.)

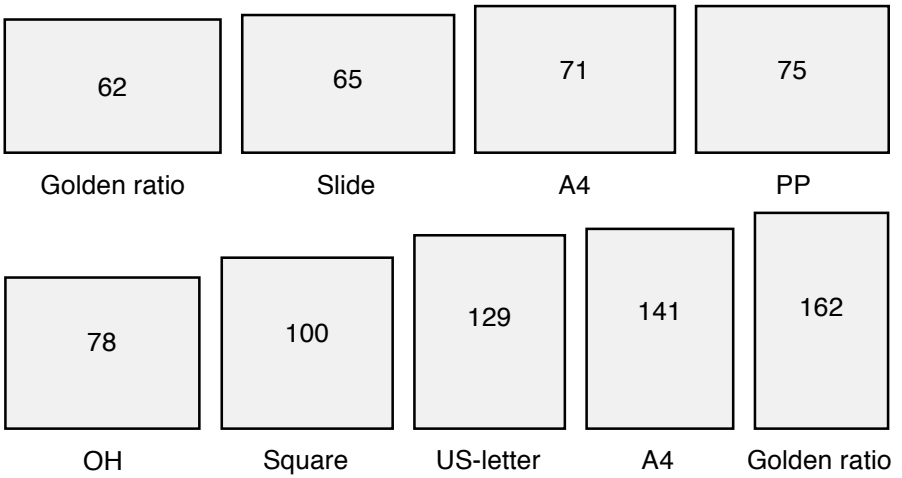
Chéret is often called both "the father of modern advertising" and "the father of the poster". He produced about 1,200 posters. Toulouse-Lautrec is often called "the king of posters". A register of his complete works shows in all 350 lithographs, of which about 30 are posters. The 27-year-old Toulouse-Lautrec became famous over a night, when his poster "Moulin Rouge: La Goulue" was put on walls and advertising pillars all over Paris in October, 1891.Toulouse-Lautrec revolutionized the art of posters and gave the commercial poster the status as an independent art form. Jules Chéret, Henri de Toulouse-Lautrec and many other artists who worked with commercial posters did not follow the old principle of the Golden Ratio.

A format index

It is difficult to compare pages and pictures, especially in different media and in different sizes. To objectively compare formats of pages, papers, and different kinds of images and pictures I

have developed a simple *format index*. It is calculated as the height/width x 100. Regardless of the size of the individual artefacts all wide formats get index numbers below 100. All square formats get index 100, and all vertical formats get index values over 100.

The format index can be used to compare pages and sheets of paper (as a *page format index,* or a *paper format index*). When we use standard A-series papers horizontally they get index 71. When we use the same papers vertically they get index 141. When we use US-letter papers horizontally they get index 77. When we use these papers vertically they get index 129.



This illustration shows some index values.

The format index can be used to compare pictures and projected images (as a *picture format index*). The horizontal golden ratio format has index 62, and the vertical golden ratio format has 162. A projected slide has index 65, a PP 75, and an OH 78. The image area on a horizontally held iPad Air2 has index 75, and when it is held vertically the index is 133.

Study of book pages

For many years designers and fine artists have been using, and are still using, the proportions of the golden ratio in their work. It can be used in paintings, page formats, or picture formats in print products, e.g. in books. Furthermore, handbooks in typography and graphic design recommend the use of the golden ratio. In her book *Creative Advertising*, Moriarty (1991, p. 237) noted that:

> Most quality books and magazines use page sizes that are close to a 3:5 ratio. The amount of type on the page relative to the overall page size is roughly 3:5. The proportions of the page margins to the text area are also 3:5.

The Penguin publishing house has used the page format 111x180 mm (3:4.9) for more than half a century (Bringhurst, 2004, p. 157).

In order to study if, or and to what degree, the golden ratio is a good principle for design of harmonious book pages we carried out a study with 126 university students from Sweden and 44 from USA (Pettersson & Strand, 2006). Subjects were asked to fold a white paper, size 45x45 cm, to a book page that they felt was the most harmonious. There was no time limit.

Results showed that there is a large variety in the values of what a harmonious book page is. *On average the most harmonious book page is 22.7 cm high and 16.6 cm wide.* The mean area is 382 cm^2. Very few subjects created book pages according to the principle of the golden ratio. There was no notable difference between the perceptions of women when it comes to a harmonious format on a book page. In this study the mean format index was 137 for men and 135 for women. These format indexes are close to regular paper, and far away from the golden

ratio. The study showed no significant difference in values of subjects in the two countries. It was concluded that the golden ratio is *not a good principle* when it comes to design of book pages.

Study of headings

According to Berndal and Frigyes (1990) the golden ratio has been used in the past to estimate suitable levels for headlines in a document. The size of body type is multiplied with 1.62, and then rounded off. If the body type is ten Didot points (=10.7 pica points), and there are four levels for headlines in the document, the following sizes are adequate: 10, 16, 26 and 42 Didot points.

However, the use of the golden ratio in art and design seem to be based on tradition rather than on facts. In one experiment I studied how subjects perceived different headings. In the fall of 2002, students at Mälardalen University in Sweden were asked to carefully study and then rank six variants of page layouts (Pettersson, 2003, p. 45). Subjects were asked to carefully study the six variations and rank them. There is a relatively large spread in their perceptions. All the six layout variants were ranked according to the alternative "best," or "second best" by at least one person. Taken all together two clear groups emerge, one group that is considered "better," and one group that is considered "not as good."

To the "better group" belongs: classic book typography (average ranking 2.5), the x-law with normal body type (average ranking 2.6) and my model for study books (average ranking 2.8). These three variants are considered clearly better than the ones that belong to the "not as good" group.

The “not as good” group consists of the X-law with bold text type (average ranking 3.8), and the two layout variants that are based on the principle of the golden ratio (average ranking 4.6; and 4.7). The principle of the golden ratio apparently gave too much difference between the type sizes. Also note that the two variants with bold headlines belong to the ”not as good” group.

Harmony principle

The *harmony principle* is one of the two aesthetic principles in information design (Pettersson, 2010). Harmony describes the effect of the pleasing interaction, and combination of elements to form a consistent and orderly whole of all elements in a design. Certain design elements look good when they are placed together and when they interact in a final design. Other design elements may look ugly and be distracting. Harmony in design can be said to be a pleasing arrangement and combination of elements to form a consistent and orderly whole. We need to find a good vertical as well as a good horizontal balance on a page.

Kandinsky wrote (1912/1977 p. 47): “never has there been a time when it was more difficult than it is today to formulate a complete theory, ...” Here Kandinsky was referring to a theory of harmony and a firm artistic basis. This is still difficult, and not only with reference to art, beauty and harmony-but also with reference to design. The information designer should:

- Develop standard templates for graphic design.
- Use standard templates for graphic design.
- Find balance between the design elements.

The term *harmony* may be used in all design disciplines to mean that the design decisions, and the design elements fit to-

gether. Graphic designers and information designers often use the term harmony in discussions on typography and layout. There is harmony in information material when all design elements fit well together and form harmonious relationships. Harmony is often closely related to *unity* (Wileman, 1993). *Balance* and *contrast* are important aspects of harmony (Mullet & Sano, 1995).

This section includes three sub-sections: 1) dichotomy, 2) harmony in typography, and 3) harmony in colour.

Dichotomy

In the Greek mythology there is a perpetual struggle between two sets of opposing forces or ideals, the *Apollonian and Dionysian dichotomy*. Apollo and Dionysus were both sons of Zeus. Apollo was the god of common sense, distance, dreams, healing, reason, self-control, and the sun. Dionysus was the god of wine, god of emotion, excess, ecstasy, passion, rage, and wildness. The Apollonian ideal celebrates human creativity through reason and logical thinking. The Dionysian ideal is based on chaos and appeals to the emotions and instincts.

The Apollonian and Dionysian philosophical dichotomy is commonly associated with the German philosopher Friedrich Nietzsche (1844-1900). The visual arts share the same features as Apollon, while music is the essence of Dionysus. Paglia (1990) used this dichotomy as the basis in her *theory of art and culture*. The Apollonian is light and structured, and associated with males. The Dionysian is dark and chthonic, dwelling within or under the earth, and associated with females.

Harmony in typography

Harmony in typography will be achieved when there is good relationship between the individual elements in the design and the "wholeness". A balanced typography gives an impression of quality and credibility.

Contrast is the difference between the brightest and the dimmest parts of a picture or of a text. Regardless of the colour and colour-contrast effects contrast should be clear and differentiate image elements from one another.

Contrast in typography may be achieved by using different fonts, styles, sizes and colours. Improper contrast is far too common. It is quite often a more or less even shade of grey or chromatic colours. Different hues may have the same value, and as a result almost no contrast. Another common problem is the use of too small symbols and small type. Text must be large enough. The difference in resolution in different media is very important. We tend to order impressions that form natural opposites, thereby reinforcing one another, in groups.

Balance is the sum of all the elements, the verticals and horizontals, the darks and lights that make up the design. Man has an intuitive sense of balance. Information material should display good balance, in a manner, which is interesting but not disturbing or distracting. Balance can be formal or informal. Formal balance has total symmetry and it is felt to be static and harmonious. It may, however, also be boring.

Composition can be used to direct the viewers (Wileman, 1993, p. 93). Informal balance contributes to a feeling of dynamism (Fleming & Levie, 1978; Pettersson, 1993). It may attract attention to a specific picture, to a part of a text or to the entire information material. However, imbalance and inconsis-

tent use of colours, graphics, or typography, have been found to reduce learning (Bradshaw, 1996, 2003).

Harmony in colour

Historical colour theories have included principles used to create harmonious colour combinations in architecture and painting. Perceived relationships between different pure colours have been visually represented with colours displayed on a circle, a "colour wheel." Harmonious combinations of pure colours will look good together. Colour combinations that are considered especially pleasing are called *colour harmonies* or *colour chords*. The artist may use 1) any two colours that are opposite each other in a colour wheel, 2) any three colours that are equally spaced and forming a triangle on the colour wheel, or 3) any four colours forming a square or a rectangle on the colour wheel. A colour wheel shows relationships between primary colours, secondary colours, and tertiary colours.

The "primary colours" are red, yellow and blue. The "secondary colours" are green, orange and purple, created by mixing two primary colours. Mixing of primary and secondary colours gives the six "tertiary colours." In theory it should be possible to mix all colours. However dyers, painters, and printers preferred pure pigments to primary colour mixtures, because the mixtures were too dull.

Colour and typographic elements can be used for decoration. However, it must always be very clear and easy to understand for the receiver when colour and typography is used for decoration, and when the use is meant to have some cognitive importance.

Aesthetic theories

Aesthetics is usually regarded as a branch of philosophy, along with epistemology, ethics, logic, and metaphysics. It is the youngest branch of philosophy with its own name. Aesthetics implies a hierarchical judgement in which the elements of any compositions, and their organisations, are considered more or less appealing to the senses by virtue of their ability to create higher emotional, intellectual, or moral appreciations in people.

Aesthetic theories that are based on perception favour sensation over intellect, favour seeing over reading, favour universality over cultural differences, and favour physical immediacy over social mediation (Lupton & Miller, 1999, p. 62).

The group *art and aesthetic disciplines* includes disciplines such as aesthetics, architecture, art history, computer graphics, film, fine art, iconography, iconology, illustration, music, painting, photography, and sculpture. At present the *aesthetics theory for ID* includes, but is not limited to, the following two fields of knowledge: 1) beauty theories, and 2) colour theories.

Beauty theories

Aestheticians discuss the *philosophy of art* from various perspectives, such as: definitions of art, experiences of art, interpretations of art, metaphysics of art, and production of art. Philosophers have made many unsuccessful attempts to define beauty. Three art theories concerning the nature of beauty are: 1) the formal theory, 2) the emotional theory, and 3) the relational theory.

Aestheticians value art for its originality and expressiveness. Its focus is on individual artefacts crafted through the manual and aesthetic virtuosity of the artist. Design, in contrast,

is valued for its fitness to a particular user and to a particular task (Mullet & Sano, 1995, p. 8). While a painter or a sculptor can choose any imaginable shape, a designer is limited by the function of the thing being designed. Of course many designers want to provide aesthetic experiences where possible, but the design aesthetic is always related to the intended function of the information products intended for widespread distribution and use.

This field of knowledge includes three areas of knowledge: 1) philosophy of beauty, 2) philosophy of art, and 3) aesthetics and usability.

Philosophy of beauty

The philosophy of beauty recognises aesthetic phenomena outside of the arts. These aesthetic phenomena can be found in nature, and in non-artistic cultural areas such as mathematics, morality, and science. The philosophy of beauty is concerned with fine arts only insofar as art may be perceived as beautiful. Philosophers have made many unsuccessful attempts to define beauty. However, there is much more to art than beauty. In fact in many situations art may have little, or nothing to do with beauty.

Until the eighteenth century, scholars regarded the study of beauty as the main problem of aesthetics. Since then aestheticians have devoted less effort to the *philosophy of beauty* than to the *philosophy of art*. Today it is also highly relevant to discuss *aesthetics and usability*.

Iconologia

The Italian art historian and scholar Cesare Ripa (c. 1560–c. 1645) published *Iconologia,* a didactic encyclopaedia in Rome

1593. A second edition was published in Rome in 1603, this time with 684 concepts and 151 woodcuts. The book was extremely influential in the 17th and 18th centuries and published in another eight Italian editions, and eight editions in other languages (Manning, 2002).

For more than three hundred years experts in art history all over Europe used *Iconologia* as their prime source of knowledge when they discussed description, identification, interpretation of the content of classical art. Art historians had learned, and knew the "true" language of art. However this was, and still is, not true for people in general. The general public lack these necessary frames of references.

(Further see the book *Information Design 6–Predecessors & Pioneers*.)

Judgement of beauty

One of the effects of art is the feeling that some things are beautiful and other things are ugly. In 1790 Immanuel Kant defended the validity of a "subjective universality" as a "universal voice" of the imagination through which beauty became known. He asserted that the "judgement of beauty" is "subjective," and defined aesthetic apprehension as "taste," an a priori judgement separate from cognition and morality. Aesthetic criticism implies a judgement that utilises analysis, synthesis, evaluation, and feeling in the understanding of beauty.

Before Kant, the common assumption was that beauty designated some objective feature of things. Most of the earlier theories of beauty declared that beauty was a complex relation between parts of a whole. Some philosophers called this relation "harmony". From the time of the Greeks, a common assumption was that beauty applied not only to art. Beauty manifested itself

in cultural institutions and moral character as well as in natural and artificial objects. Philosophers sometimes established very firm rules about what artists should create and what people should like. Many of these rules have been abandoned.

Psychological pleasure derived from an artwork comes from the similarity of the artist's values and sense of life to one's own values. Admiration of an artwork comes from the viewer's evaluation of the artist's skill, style, and integrity. An individual can dislike the values, the sense of life, or the theme of an artwork, but can admire the artist's skill or style.

Philosophers have made many unsuccessful attempts to define beauty. In the 18th century philosophers agreed that beauty could not be defined in terms of the qualities shared by all beautiful objects. Three theories concerning the nature of beauty are 1) the formal theory, 2) the emotional theory, and 3) the relational theory. The formal theory locates beauty in the qualities of objects. According to this theory an object may be considered beautiful when it has integrity, unity, proportion, and splendour. However, the emotional theory identifies beauty with the mental responses of the audience, the listeners and the viewers, and their aesthetic experiences. According to the relational theory beauty includes aspects from the other two theories.

Instructional value

Aesthetically pleasing visuals may *not* be of great instructional value. Dwyer and Dwyer (1989, p. 122) found that:

> The value of different types of visual illustrations is not a valid assessment of instructional effectiveness, that is, aesthetically pleasing visuals may not be of great instructional value.

It is, however, quite possible that aesthetically pleasing information material will be *noticed*, and then actually *used* in a better way than material without any obvious aesthetic qualities.

Malamed (2009, p. 203) noted that when viewers look at both pleasant and unpleasant pictures, they consistently demonstrate an emotional reaction indicated by pronounced brain activity that does not occur when they look at neutral pictures.

Information design may vary a lot in style and quality. Edward Tufte (1983, 1990, 1997) has provided information design with results from pioneering studies on how information materials used for communication can be both beautiful, and at the same time useful. His concepts of *chartjunk* and *data-to-ink ratio* are useful contributions to information design. In newspapers attractive information graphics, with "infotainment" values, attract more readers (Holmes 1993).

Philosophy of art

Alexander Gottlieb Baumgarten, a leading German philosopher, introduced the term *aesthetics* in 1735. However, a large number of philosophers from Plato to the present day have discussed "the philosophy of art." Seward Barry (1994) defined *aesthetics* broadly as an "appreciation of the beautiful," and more narrowly as a "philosophy of art, its creative sources, forms, and effects." Aestheticians try to understand the arts in broad and fundamental ways.

The philosophy of art includes traditional, philosophical theories of art: for example: Aristotelian, empiricist, existentialist, feminist, idealist, Marxist, phenomenological, Platonic, postmodernist, and rationalist aesthetics theories.

Such theoretical positions inform, but are also tested by, critical and interpretive articles about particular types or exam-

ples of artworks. The information designer may focus attention upon the visual arts – as opposed to dance, literature, music, and theatre. Philosophers have encountered difficulties in framing a theory of "aesthetic perception" and, more importantly, of the remarkable variety of visual arts. We can discuss the philosophy of art from various perspectives, such as: art and design, definitions of art, experiences of art, interpretations of art, metaphysics of art, and production of art.

Art and design

Aestheticians study all the arts, from all countries, and from all periods of history, in relation to their cultural, physical, and social environments. They try to organise knowledge of art in systematic ways. Aestheticians value art for its originality and expressiveness. Its focus is on individual artefacts crafted through the manual and aesthetic virtuosity of the artist.

Art is valued for its originality and its expressiveness. Focus is on individual artefacts crafted through the manual and aesthetic virtuosity of the individual artist. Design, in contrast, is valued for its fitness to a particular user and to a particular task (Mullet & Sano, 1995, p. 8). While a painter or a sculptor can choose any imaginable shape, a designer is limited by the function of the thing being designed. Many designers want to provide aesthetic experiences where possible, but the *design aesthetic* is always related to the intended function of the information products intended for widespread distribution and use.

Definitions of art

A number of philosophers have made unsuccessful attempts to define art. Usually definitions of art aim at establishing a set of characteristics applicable to all kinds of fine arts, as well as the differences that set them apart. After some hundreds of years of

discussions aestheticians have not yet agreed upon a definition of art. Some say that it is impossible to define art.

Art criticism may be defined as the process of judging the aesthetic qualities of visual art, mainly painting, sculpture, and architecture, but also craft. This has become a specialised field, but judgements about works of art have appeared since ancient times. In the Western world, reflection on art began with the philosophers of ancient Greece. Plato discussed proportion as the source of beauty, and imitation, as the primary mode of art. Aristotle identified different kinds of imitation. Xenocrates wrote about painting and sculpture, and the ideal synthesis of imitation and proportion.

Experiences of art

It is a common belief that a person's response to art is a mystical experience that has no basis in reality and serves no practical purpose. But the opposite may very well be true. A positive response to art is a phenomenon of reality that may reflect a person's most important values. Another belief is that art is entirely subjective and cannot be evaluated on an objective basis.

Discussions about how people experience art have been dominated by theories devised in the 18th century. Some philosophers still think of the typical experience of art as *contemplative and disinterested*, different from everyday concerns. A few modern aestheticians have stressed the continuity between aesthetic experience and everyday experience. The social sciences, such as anthropology, archaeology, and sociology, help aesthetics researchers explain the forms that art has taken in various cultures, and how the arts are related to economic activities, government, religion, and science. Recent theories of post structuralism and interpretation theory propose strategies

of critical evaluation that incorporate multiple viewpoints and accept a basic indeterminacy in meaning.

Interpretations of art

In aesthetics one area of study is concerned with the study of the interpretations of art, how people criticise, enjoy, and use art. What happens in the minds of people when they listen to music, look at paintings, or read poetry? Knowledge of psychology helps the aestheticians to understand how people act, desire, feel, hear, imagine, learn, see, and think, in relation to art and aesthetic experiences.

One question is whether strictly deductive reasoning based on premises descriptive of the artworks can back evaluative judgements. According to some opinions judgments are merely expressions of personal preferences. Thus these opinions cannot be considered true or false. Another basic question is whether conflicts over interpretations of a work of art can be settled by facts about the work, or whether more than one interpretation of a specific work is possible or not.

For the media theorist Marshall McLuhan *Cubist art* required "instant sensory awareness of the whole" (McLuhan, 1964, p. 13). With Cubism one could not ask about the content/message in the artwork, but rather consider the artwork in its entirety.

Metaphysics of art

Aestheticians ask questions like these. What access does art give people to reality? What kind of entity is a work of art? Works of art may be physical objects, such as sculptures. However, not all works of art are physical objects. A dance and a theatrical performance may be as aesthetically relevant as any physical object. A flat painting can represent spatial depth and fast move-

ments. What the painting represents may often seem more aesthetically relevant than its physical dimensions.

Some philosophers have concluded that works of art represent *mental entities*, such as visions and dreams. Other philosophers have noticed that artists may express their attitudes, emotions, and personality traits in their art, and have concluded that art works belong in a category with non-verbal communications rather than with physical objects. Still other philosophers argue that works of art exist only in the minds of their creators and of their audiences. The question whether art can provide knowledge of, or insight into, reality is as old as philosophy itself. Plato argued that art has the power to represent only the appearances of reality. The opposite position is common among modern philosophers, artists, and critics.

Since art can reflect powerfully emotional values to the beholder, art can be appreciated, enjoyed, and loved for those values. The artwork, however, is an extension of the artist and thus can never be spiritually possessed or owned by anyone else, even though the physical ownership of artefacts as well as copyrights can be transferred or sold.

Production of art

Wassily Kandinsky (1866-1944) was an influential Russian modernist, and pioneering painter and art theorist. He is considered by many to be the father of abstract art and a leader in the movement to free art from the strict bonds of tradition. According to Kandinsky art is the expression of the spiritual atmosphere of a certain period. Art evolves from the culture that inspires artistic expression. The foundation of forms, the harmony of colours, and the principle of art is an “inner necessity,” or a “right of the artist” to unlimited freedom. Art is born from

the inner necessity of the artist. Kandinsky taught at the Bauhaus school of art and architecture from 1922 until the Nazis closed it in 1933. He then moved to France. In 1912 Kandinsky argued (1912/1977 p. 1): “

> Every work of art is the child of its age and, in many cases, the mother of our emotions. It follows that each period of culture produces an art of its own which can never be repeated.

In 1925 Kandinsky published his thoughts of the role of the line, point, and other key elements of non-objective painting. A point, a small bit of paint on the canvas, is neither a geometric point nor a mathematical abstraction. It has colour and a simple or complex shape. A point can be isolated, or it can resonate with other points or lines on the canvas. A horizontal line corresponds with the ground. A vertical line corresponds with height, and offers no support. A diagonal line is un-stable.

Main questions about the production of art deal with creativity, imagination, and the role of innate ability in any artistic production. Ancient and medieval philosophers assumed the same model for producing fine art and crafts. They had no conception that the two are distinct. The present distinction between the production of fine art and crafts emerged in Western culture after the renaissance. Today most aestheticians assume that something is unique about producing fine art.

Aesthetics and usability

We could argue that the very moment that shape exceeds functionality is the point at which usability is compromised. However, we may also argue that art and aesthetic concerns in gen-

eral serve a basic human need. We may say that artistry makes a thing more usable and useful by way of making it special.

Interface design

Dreilinger (1993, p. 6) wrote about usability and aesthetics as the yin and yang of interface design. Norman (1998) asserted that if everyday design were ruled by aesthetics, life might be more pleasing to the eye but less comfortable; if ruled by usability, it might be more comfortable but uglier. The aesthetic properties of a design contribute to its ability to offer a different perspective on the world. Paying attention to the aesthetic of any design is more important than simply making things look pretty.

When the Apple Macintosh first appeared on the market people in the business world assumed that aesthetics have no useful function beyond making something look nice. Business people were accustomed to the IBM PC. They assumed that the aesthetically pleasing Macintosh interface could not possibly be as useful as the less attractive text-based interface of the PC. Aesthetic properties of a design make the product more personal, more intimate, and more special.

According to Baggerman (2000, p. 11) all design elements in interface design should serve a purpose. It is the same for information design. In information design functional properties are always more important than aesthetic properties. However, it may be an advantage if information materials also look good.

Use of material

In many areas of design the use of *material* is a very important factor. It represents many different aspects of aesthetic as well as functional qualities. As an example warning signs must be clear and easily noticed in bad and degraded conditions such as

fog, weak illumination, and smoke (Lerner & Collins, 1983). Some warning signs need adequate reflectance and good lighting equipment (Wogalter, 1999). In the area of wayfinding, or wayshowing, the right decision may be to paint lines in different colours on the floor in a hospital, or use reflecting materials in signs. When messages are printed the use of paper, or plastic, is of vital importance. The material has to be "right" for the situation.

Perceptual aesthetics

Unlike the classical tradition perceptual aesthetics seeks meaning through the process of becoming or developing according to nature through perceptual dynamics (Seward Barry, 1994). The concept of "perceptual aesthetics" provides us with a unified approach to art that reveals a cross-pollination of perceptual insights and aesthetic insights. A perceptually based approach to art provides a simple and more basic system of aesthetic judgement than do approaches based on morality, contemporary manners, and taste of style. Perceptual aesthetics is based on Gestalt psychology and its understanding of the perceptual process. Perceptual aesthetics seeks to understand how art may be structured. Here simplicity regularity, and symmetry provides the foundation from which to judge the effectiveness of composition in art.

Aesthetic experience in instructional design

In 1934 the American educational reformer John Dewey (1859–1952) argued that aesthetic is a quality of experience that "possesses internal integration and fulfilment" (Dewey, 2005a, p. 46). Dewey emphasized that aesthetic experience is an everyday

life experience, not something that takes place just on a special occasion.

According to Parrish (2005) Dewey's thoughts from the 1930s brings insights to *instructional design*. These insights enhance instructional practices (Uhrmacher, 2009). Instructional designers frequently point to their affiliations with other design disciplines and look to them as useful analogues of their practice (Parrish, 2005). Aesthetic aspects are of high importance to both designers and end-users in many design disciplines.

Parrish (2009) suggested that "an instrumental view of learning may consider only the immediately measurable outcomes of a learning experience, particularly its impacts on cognition, behavior, or performance" (p. 513). However, aesthetic, cultural, emotional, political, and social qualities of experiences have more than immediate rewards. Parrish (2009) provided principles that contribute to developing the aesthetics of instructional design:

- Designing clear phases of the learning experience: Challenge, engagement, and Conclusion/Resolution.
- Learners are made active, not passive recipients of information.
- Designing instruction according to concrete activities not to obscure constructs.
- Designing the setting of learning in a way that all the content and activities of the lesson become integrated in coherent manner.
- Being a role model of an active learner while being careful about the principles above.

For Parrish, Uhrmacher, and many others aesthetics is far more than beauty. The opposite of aesthetic experience is boredom, disengaged habitual behaviour, or imposed labour.

Colour theories

Colour has probably always been considered as important for people. The *Blombos Cave* in South Africa has rich deposits from 100,000–70,000 years ago (Henshilwood et al. 2009). Here red ochre pieces have been deliberately engraved or incised with abstract geometric designs, maybe the oldest known human "artwork."

At the beginning of the 7th century pope Gregorius the Great (about 540–604) had said that pictures are used in churches so that those who cannot read at least can look at the walls and understand what they cannot read in books (Piltz, 2007, p. 128; Sandquist Öberg, 2007, p.171). Gregorius the Great presented his insights and views in a new and simple manner (Hill, 2008, p. 169).

From about 1250 *Biblia Pauperum* consisted of a collection of about 50 colourful hand-painted loose pages (Cornell, 1925). The purpose of *Biblia Pauperum* was to recount the teachings of the Bible to the illiterate, who were at that time in the majority among the inhabitants in most countries.

The *colour theory* and the colour principles that *Leon Battista Alberti* published in *Della Pittura* (On Painting) have influenced the use of colour in the visual arts. Classical painters like Leonardo da Vinci, Michelangelo, Raphael, Peter Paul Rubens, and Rembrandt, worked with colourful compositions with great sense of colour and light. Leonardo da Vinci set out his beliefs on colour theory in his *Treatise on Painting*, which was not published until 1651. He wrote that black and white was

indeed colours, and he assigned white, yellow, green, blue, red, and black as the simple or primary colours (Anderson Feisner, 2006, p. 13).

Historical colour theories have included principles used to create harmonious colour combinations in architecture and painting. This field of knowledge includes three areas: 1) colour wheels, 2) colour systems, and 3) colour for information.

Colour wheels

Perceived relationships between different pure colours have been visually represented with colours displayed on a circle, a "colour wheel." Harmonious combinations of pure colours will look good together. Colour combinations that are considered especially pleasing are called *colour harmonies* or *colour chords*. The artist may use 1) any two colours that are opposite each other in a colour wheel, 2) any three colours that are equally spaced and forming a triangle on the colour wheel, or 3) any four colours forming a square or a rectangle on the colour wheel. A colour wheel shows relationships between primary colours, secondary colours, and tertiary colours.

Every historical colour wheel is a visual representation of a *specific colour theory* (Anderson Feisner, 2006). In 1704, Sir Isaac Newton made the first colour wheel. He split white sunlight into red, orange, yellow, green, cyan, and blue.

In 1810 Johann Wolfgang von Goethe created a colour wheel showing the psychological effect of each colour. The "primary colours" are red, yellow and blue. The "secondary colours" are green, orange and purple, created by mixing two primary colours. Mixing of primary and secondary colours gives the six "tertiary colours." In theory it should be possible to mix all colours. However dyers, painters, and printers preferred

pure pigments to primary colour mixtures, because the mixtures were too dull.

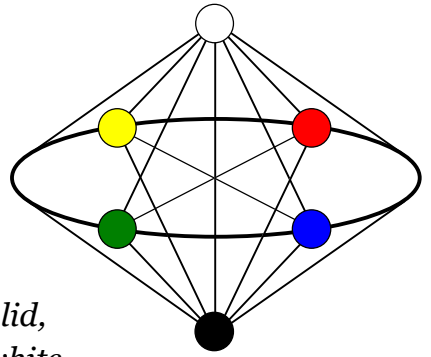
Colour systems

There are many theories about how perception of colours actually works. Colour can be described in aesthetical, physical, physiological, psychological, and technical terms. Hue, value, and saturation describe what we see when we look at pictures as well as the real world. Intensity, purity and wavelength are physical dimensions. The relationship between brightness, hue, lightness and saturation is very complicated. Colour is reflected in and through light. There is nothing yellow in a banana. It's all in how it is perceived. For practical use in art and in industry several different systems providing numerical indexes for colour have been developed.

The Natural Colour System

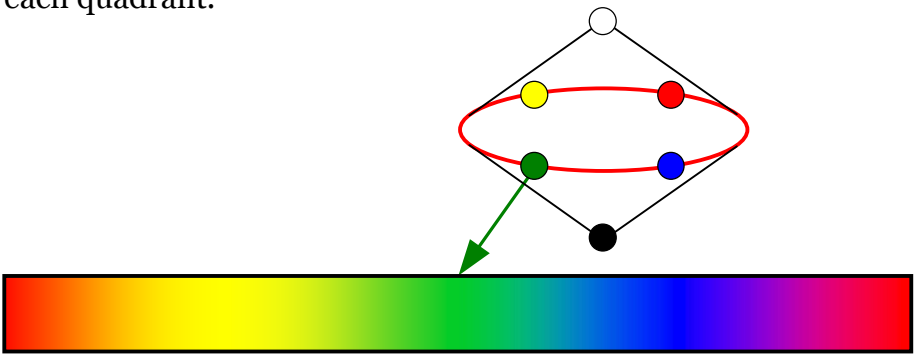
In 1925 Hering based his "natural system" on man's natural perception of colour that presupposes two pairs of chromatic colours blocking each other, red/green and blue/yellow. Hering's "natural system" is the principle for the *Natural Colour System* (NCS), developed during the 1970s in the Swedish Colour Center Foundation in Stockholm (Hård & Sivik, 1981).

From a perceptual point of view, we perceive six colours as "pure." Black and white are achromatic colours. Yellow, red, blue, and green are chromatic colours. These six colours are called elementary colours. All colours that are not pure elementary colours have a varying degree of resemblance to several elementary colours. Thus every possible colour can be described with a specific location in a three-dimensional model, a twin cone, called the "NCS Colour Solid."



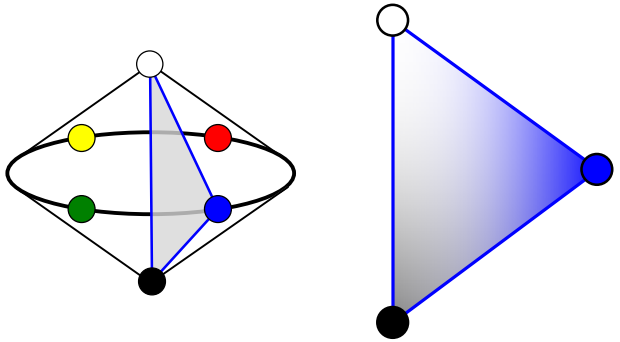
The NCS Colour Solid with the six elementary colours. Yellow, red, blue, and green are all located on the circumference of the Colour Circle. The Colour Triangle is any vertical sector through half of the NCS Colour Solid, such as, e.g., white–blue–black–white.

The chromatic elementary colours yellow, red, blue, and green are all located on the circumference of the *Colour Circle*. One hundred steps, thus describing the hue of a colour, can divide each quadrant.



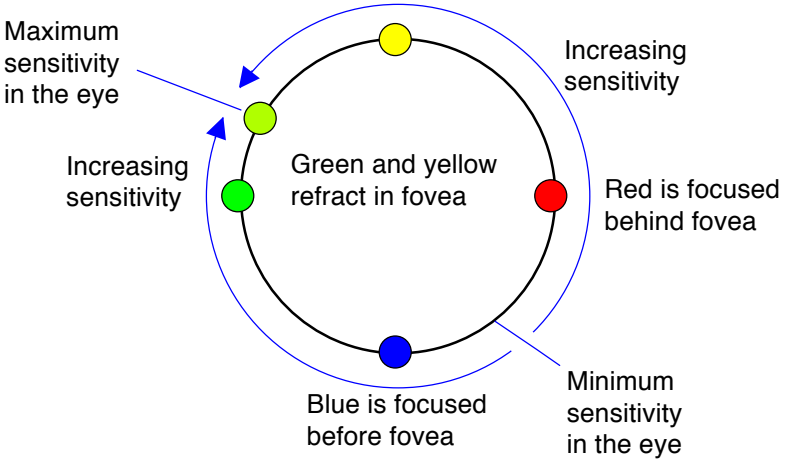
This figure illustrates a cut and opened circumference of the colour circle, here starting with red and ending with red.

The *Colour Triangle* is any vertical sector through half of the NCS Colour Solid. It is used to describe the nuance of a colour, i.e., its degree of resemblance to white, black, and the pure chromatic colour of the hue concerned (chromaticness).



There are many thousands of colour triangles. In this example the figure illustrates the colour triangle ending in blue.

When we want to describe a colour using the colour triangle and the colour circle, it is done in the following sequence: blackness, chromaticness, and hue. For example, a colour of 10 percent blackness, 80 percent chromaticness, and with a hue of Y70R will have the notation 1080-Y70R.



The NCS Colour Circle combined with information on the eye's sensitivity.

Some other colour systems

The NCS places emphasis on qualitative variation in the colour sensation whereas the Munsell System is based on equally spaced visual scales. Both systems are based on surface colours. The *Munsell Colour System* (MCS) was introduced in 1905 and it has been modified several times. The system consists of fixed arrays of samples that vary in hue, lightness (here called value), and saturation (here called chroma). The value scale ranges from white to black with nine steps of grey. Forty equal steps in a circle represent hue. The value and the hue are related to each other by a maximum of sixteen "saturation steps."

There are many theories about how perception of colours actually works. In 1807, Young proposed a tri-chromatic colour vision system. In 1924, Young's theory was formalized by von Helmholz, who proposed hypothetical excitation curves for three kinds of cones in the retina, sensitive for red, green, and blue.

In the *Hue Lightness Saturation System* (HLS), the hues are arranged as circles on the outside of a double cone resembling the NCS Colour Solid (Murch, 1983). Hue specifications start with blue at 0° and then follow the spectral order around the circle. Lightness and saturation are defined as percentages from 0 to 100. The HLS system is easy to use for colours on the surface of the model. However, colours inside the model are difficult to define. As in the Munsell- and NCS-systems, brightness creates problems.

The *Hue Value Saturation System* (HVS) is a model that is rather similar to the NCS-system but it utilizes another coding (Samit, 1983). Here value is defined as the relative lightness. White has full value and black has no value at all.

There are many other colour systems, colour wheels, and colour circles. Today printers who use modern *subtractive colour methods* use magenta, yellow, cyan, and black as primaries. Colour scientists often use the *additive primaries*, red, green and blue (RGB). These colours are used for anything light-based visual displays, like computer monitors, and television screens. People who are specially trained can use the RGB proportions as a colour description system. However, this is not possible for people in general.

In web design colours are defiend with a six-digit hexadecimal number or triplet, a *hex value/hex triplet*. Colours can be specified in the format #RRGGBB, where RR, GG, and BB are the hexadecimal values for red, green and blue values of the colour. The values varies from zero to maximum 255 for each component.

Colour for information

Colour is regularly used in printed materials, not only in illustrations, but also in the text itself. Colour can be used to clarify the structure of the text and to make learning easier. Certain parts of the text may be printed with colours or printed on top of backgrounds in different colours. Black type has good contrast to many light background colours. The legibility will always be affected when there is insufficient contrast between the type and the background.

From many experiments, it is clear that people prefer colour in visuals. To some extent colour is a language of its own. Colour enhances the attention and perception of a visual message. If people like the contents in a picture, they like them even more when the visual is presented in colour. Advertising is

known to be much more effective when visuals are in colour than in black and white.

We can use colour to clarify the structure of a text, group elements, indicate meaning, and to make learning easier. Colour can be used as an important and a successful part of information design (Bradshaw, 2001; Dwyer, 1971, 1978; Hannafin & Peck, 1988; Lipton, 2007; Moriarty, 1991; Muter & Marrutto, 1991; Pettersson, 1989; Scharff, Hill & Ahumada, 2000; Winn, 1993; Wogalter, 1999).

Inconsistent and improper use of colour can be distracting, fatiguing, and upsetting, and it can actually produce negative results and reduce learning. It should also be remembered that some people are red-green colour blind and they perceive these hues as grey.

(See the book *Information Design 3–Image Design* regarding *Use of colour*, and the book *Information Design 5–Cognition* regarding *Colour blindness*.)

Contributions for ID

The *aesthetics theory for ID* contributes to information design with valuable facts, practices, principles, and theoretical approaches. Some examples of important concepts are: aesthetic design, additive colour, aesthetic proportions, aesthetically pleasing, architecture, balance, beauty, colour chords, colour circles, colour harmonies, colour systems, colour theories, colour wheels, composition, decoration, divine proportion, fine art, golden ratio, golden rectangle, golden section, harmony principle, illustration, material, philosophy of art, philosophy of beauty, photography, primary colours, proportion, quality, secondary colours, subtractive colour, and tertiary colours.

Here are some facts, hypotheses, and postulates based on the aesthetics theory for ID.

- Aesthetic designs are perceived as easier to use than less-aesthetic designs.
- Aesthetic value is the aggregate effect of all picture variables.
- Colour can be described in aesthetical, physical, physiological, psychological, and technical terms.
- Colour enhances the attention and perception of a visual message.
- Colour wheels are visual representations of colour theories.
- Every historical colour wheel is a visual representation of a specific colour theory.
- For the receiver it must always be absolutely clear if colour is used for decoration, or if it has some cognitive importance.
- Graphic form, text, and visuals should always work together to fulfil information design objectives.
- Imbalance and inconsistent use of colours, graphics, or typography, have been found to reduce learning.
- Informal balance contributes to a feeling of dynamism.
- People in different cultures may have varying opinions of harmony.
- Proportions according to the “golden section” are not always appropriate in information design.

Functional principles

The group "*Functional principles"* includes six message design principles: *Defining the problem, Providing structure, Providing clarity, Providing simplicity, Providing emphasis,* and *Providing unity*. Guidelines that are based on these six principles will assist the information designer to design information materials that are well suited for the intended receivers.

(See the book *Information Design 1–Message Design* for information regarding *Defining the problem*, *Information Design 2–Text Design* for information regarding *Providing structure,* and *Information Design 3–Image Design* for more information regarding *Legibility of pictures*.)

Providing clarity

The *legibility of a graphical message* is determined by the technical design of texts and pictures, that is, their *clarity*. The information designer will have to make the content stand out clearly from the background. In general information materials should be as clear, simple, unambiguous and transparent as possible. We should avoid unusual typefaces, as well as fonts that are too small or too large. We read words in a text as "pictures," not letter by letter. Typeface and font size must be adapted to meet the limitations of the medium and technical production. A message has good legibility if it is easy to read, and if the reader can easily see and distinguish all different parts of the message.

Good legibility is always economically advantageous, whereas poor legibility is a costly business. Good models make the production of documents simple and inexpensive. In my view it is not economical to cram too much information on a

page. It is better to edit the text and reduce its bulk, and thereby increase its legibility.

Legibility can be measured rather objectively and its quality is assessable whether we understand the content of the message or not. Dissatisfaction with the execution of a message may also cause dissatisfaction with the content of the message (Pettersson, 1989). Therefore the information designer has to consider the legibility of text printed on paper, displayed and projected on screens, as well as legibility of pictures, legibility of layout, legibility of symbols, legibility of numerals, and legibility of colours.

Legibility of text

Depending on the purpose, we can read a text in several ways. An active reader makes good use of the structure embedded in the text. The preface, the table of contents, the headings, as well as the captions and the illustrations provide an overview of the content in the whole book.

The concept "legibility of text" refers to a text's external properties (Lipton, 2007; Pettersson, 1993; Williams & Tollet, 1998). These are properties such as letter size, inter-line distance, line length, the distance between letters, the number of letters per line, the distance between words, headings, the subdivision into paragraphs, headings in the margin, the layout, colour of the printing ink and paper, the paper quality, etc. These different external properties have not been found to have a drastic effect on legibility as long as the text is presented within the framework of variation normally found in contemporary books.

Furthermore legibility refers to production and material quality, environmental conditions, room lighting and tempera-

ture, noise level, et cetera. Principles for legibility are presented in the following sub-sections: *Legibility of print media, Legibility of text on wall charts, Legibility of text on computer screens,* and *Legibility of projected texts.*

Legibility of print media

A printed text in books, handouts, reports and other printed documents must have good legibility. Therefore the information designer will have to:

- Use clear, direct, simple and transparent typography.
- Use a common typeface, between nine and twelve Pica points, for continuous text in a book, a pamphlet, or a report.
- Restrict the number typefaces and only use a few per information material.

See the two chapters *Typography* and *Layout* for more information.

Legibility of projected texts

In verbal presentations, many of the overhead transparencies, slides, filmstrips, and projected computer presentations consist mainly, or sometimes only of text. Here lettering must be considered carefully in order to guarantee good legibility for all listeners. In preparing the material the information designer should:

- Use no more than six rows of six words in each image, set in a linear typeface, with characters large and bold enough.
- Maintain a good contrast between foreground and background.
- Avoid graduated and tonal background fills.

Before the presentation the presenter will need to reduce room illumination and clean slides, lenses, and screens. During the presentation it is important to really project the images in focus and on the screen, preferably horizontally. (See the chapter *The elements of graphic design* for more information.)

Legibility of text on wall charts

A wall chart must have good legibility. The information designer should:

- Set text bold and large enough, adjusted to the reading distance.
- Use lower case letters and avoid all-capital printing for running text.
- Restrict the number typefaces.

Posters and wall charts shall be read from some distance. Therefore text should be large enough (Ormrod, 1989; Pettersson, 1993; Wileman, 1993), and bold enough (Mayer, 1993; Pettersson, 1993; Wileman, 1993, p. 79). Too small or too large lettering will impair reading. The text on a poster or a wall chart may often have to be ten times larger in size than a text in a book or on a print out. Text should be set in lower case letters, because all-capital printing has been shown to markedly reduce the speed of reading (Henney, 1981; Poulton and Brown, 1968). Since the texts on posters and wall charts should be short it may be a good idea to use a sans serif typeface like Arial or Helvetica. If so the running text will need some extra space between the lines (Collier & Cotton, 1989).

Legibility of text on screens

Compared with traditional graphic presentations, a presentation of information on visual displays such as television sets and

computer terminals is very limited. Still, information may be presented in many different ways. The design may vary with respect to spatial organization like directive cues, colours, columns, headings, justification, lines, scrolling text, spacing, and twinkling characters or words. A text on a computer screen must have good legibility. The information designer should:

- Use typefaces designed for screen display.
- Use black text on a white or yellow background.
- Avoid the use of all capital letters.

(See the two chapters *The elements of graphic design,* and *Layout* for more information.)

Legibility of pictures

A message has good legibility if it is easy to read, and if the reader can easily see and distinguish all different parts of the message. Legibility can be measured rather objectively and its quality is assessable whether we understand the content of the message or not. Dissatisfaction with the execution of a message may also cause dissatisfaction with the content of the message. Therefore the information designer has to consider the legibility of text printed on paper, displayed and projected on screens, as well as legibility of pictures, legibility of layout, legibility of symbols, legibility of numerals, and legibility of colours. The information designer should:

- Use picture elements that are bold and large enough.
- Use a style guide for picture elements in schematic pictures.
- Set words in images and pictures bold and large enough to read.

(See the book *Information Design 3–Image Design* for more information in these areas.)

Legibility of layout

An “informative layout” must have good legibility. Therefore the information designer should:

- Use standard page sizes with standard grids for pre-planning of pages.
- Use a clear and simple layout.
- Use arrows, bullets, lines, and symbols in various colours; also margin notes, repetition, and space to highlight relevant information.

(See the chapter *Layout* for general information.)

Legibility of symbols

The use of symbols has a long tradition and various symbols can be used to aid communication. The information designer should:

- Use distinct colours and simple graphical elements to design symbols that will function in any size.
- Design solid figures with a distinct contrast to the background.
- Use characters and graphical elements that are bold, distinct and large enough.

(See the book *Information Design 3–Image Design* for more information.)

Legibility of numerical values

Numerical data and information can be presented in tables and in graphs. A table in an information material must have good legibility. Therefore the information designer should:

Tables

- Use type between 8 and 12-point size for table cells.
- Use vertically oriented tables.
- Use rounded off numbers.

Graphs

- Compare lengths of variables to show their relationships.
- Compare areas of variables to show their parts of a whole.
- Use actual figures in graphs when accuracy is needed.

(See the chapters *The elements of graphic design* and *Layout* for more information.)

Legibility of maps

Maps must have good legibility. Therefore the information designer should:

- Use bold and distinct symbols in a consistent size.
- Restrict the number of typefaces and complexity of patterns.
- Provide distinct contrast in form and dimensions.

(See the book *Information Design 3–Image Design* for more information.)

Legibility of colour

Colour can be used to clarify the structure of a text. Certain parts of the text may be printed with colours or printed on top of backgrounds in different colours. Colours used in informa-

tion materials must have good legibility. Therefore the information designer should:

- Use a light or a dark background colour appropriate to the content, and then use a colour with good contrast for the figure or text.
- Make sure that differences between colours are clear and obvious.
- Combine colours with shape in warning signs.

Colour is regularly used in printed materials, not only in illustrations, but also in the text itself. The most legible combinations of print colours are black or dark brown text on a light yellow background (Pettersson, 1989; Pettersson et al., 1984a). Black type on a white background gives the highest and most comfortable contrast for sustained reading (Bradshaw, 2001; Lipton, 2007; Muter & Marrutto, 1991; Scharff, Hill, & Ahumada 2000). Other combinations may attract more attention but are less legible and, thus, require larger type. The legibility will always be affected when there is insufficient contrast between the type and the background. There are strong cultural differences in interpreting the meanings of colour.

Here black type on a yellow background gives good legibility. The orange text on the same background gives poor legibility.

The background colour of a computer screen should be "fairly light" or "fairly dark," depending on the content. The text

displayed on a screen should have an opposite ("fairly dark" or "fairly light") colour (Bradshaw, 2001). The most legible colour combination is black text on a white or yellow background (Pettersson et al., 1984a).

Unfortunately, red and green are often used as discriminating colours in symbols and in warning signs. Since many colour-blind people perceive red and green as grey, colour can only be used to code the information redundantly. Colour may be combined with shape, and position, or with both, which is often seen in traffic signs.

Providing emphasis

The most important elements in information material may be emphasized to enhance attention and perception. A dark dot in a light field and a jog in a straight line are two good examples of emphasis. Emphasis may be used to *attract*, *direct* and to *keep* attention. Typography and layout will better show the structure and the hierarchy of the content in the information material when important parts are emphasised. The information designer should:

- Use specific elements for emphasis.
- Use clear contrasts for emphasis.
- Use variables like complexity, directionality, exaggerated features, humour, isolation or motion for emphasis.

It is possible to provide emphasis in information material with the help of a number of specific design elements. Generally speaking highlighting cues and emphasis in a message will result in attention to that message. We should, however, never overuse any accenting techniques because if we do they may completely lose their meanings and their power to emphasize

(Dwyer, 1978; Hartley, Bartlett, and Branthwaite 1980; Benson, 1985; Bausell and Jenkins 1987).

See the section *Emphasis* in book 3 *Image Design* for emphasis in pictures and symbols.

Emphasis in text

There are a several possibilities to emphasize elements in typography. We can use boldface, colour, italics, key words in red, light against dark, small against large, and underlining key words. In order not to confuse the readers, it is important to establish a consistent system for how to signal emphasis.

When used purposefully, contrast can reveal the structure of the content helping readers see relationships among the parts (Ivory, Sinha and Hearst, 2001). Typographic cuing generally refers to the use of bold or italic type or underlining to signal the important ideas in a text. There is little doubt that cuing does work well in drawing attention to the cued material (Glynn, Britton and Tillman, 1985). The consensus is that readers are more likely to remember cued ideas than un-cued ideas (Hartley, 1987). In an experiment with 300 subjects Cisotto and Boscolo (1995) found that the use of paragraph headings improved learning. However, underlining of relevant information did not have the same effect. Underlining did not improve learning.

Headings should always be relevant and identify the subject matter. The purposes of headings are to attract the attention of the readers, make the subject matter readily apparent, and indicate the relative importance of different items in the document. To avoid too large masses of text, it is a good idea to divide the text into sections, subsections, and paragraphs. Headings on different hierarchic levels will provide the readers with refer-

ence point and help them to organize information cognitively for better retention and recall. Headings set in different type versions aid comprehension of the text content (Jonassen, 1982).

Normally boldface and italics will not be used for continuous text. Italic print is read more slowly than regular type and is also disliked by many readers (Tinker, 1965). However, the information designer can decide to use boldface and italics for emphasis of parts of a text (Lipton, 2007; Mayer, 1993; Tinker, 1965).

The use of underlining and all capital letters should be restricted to headings and titles, if they are used at all. Usually bold and italics are quite sufficient. Underlining in the middle of a sentence makes the lower line more difficult to read (Isaacs, 1987). Shadow and outline letters should be avoided.

There are a few other possibilities to emphasize paragraphs in texts, such as adjunct question to relevant information (Mayer, 1993) and statements of objectives for emphasis (Briggs & Wager, 1989; Mayer, 1993).

Emphasis in layout

Most people read instructional materials selectively. Readers rarely, if ever, begin at the beginning and read straight through to the end of a document. Usually we use a combination of browsing, reading headings, looking at illustrations, reading captions, reading certain parts carefully, skimming others, and avoiding some parts completely. Many readers will only spend time on a limited amount of information in a newspaper. It is known that elements like headings, photos, drawings, and information graphics attract attention and often are entry point into a page. Size and placement of such elements influence how

the reader will actually read the page. Many readers may jump over too large pictures and never look at them at all.

The competition for our attention is usually very fierce in commercial arts and in advertising. Thus discontinuity is often used intentionally to attract and even to hold attention of the viewers. The intended message may be hidden within verbal or visual puns, within metaphors, satires, parodies, or within humour. In these cases, designers break the traditional rules or guidelines of instructional message design. It might also be possible to deliberately use the unexpected to attract attention to instructional materials.

There are a several possibilities to emphasize elements in typography and layout. The information designer can use a number of "tools," such as:

- Areas of colour (Mijksenaar, 1997).
- Areas of shading (Mijksenaar, 1997; Pettersson, 1989).
- Arrows (Beck 1984; Hartley, 1987; Jonassen & Kirschner, 1982; Lamberski & Dwyer, 1983; Mayer, 1993; Pettersson, 1989; Winn, 1989;).
- Black type on a white background (Muter & Marrutto, 1991; Scharff, Hill & Ahumada, 2000).
- Bold against standard (Pettersson, 1989).
- Boldface (Hartley, 1987; Jonassen & Kirschner, 1982; Mayer, 1993).
- Boxes (Mijksenaar, 1997).
- Bullets (Mayer, 1993).
- Circles or ovals around objects (Pettersson, 1989).
- Clear contrasts (Pettersson, 1989).
- Colour (Winn, 1993; Wogalter, 1999).
- Colour against no colour (Pettersson, 1989).

- Colour coding (Bradshaw, 2003; Hannafin & Peck, 1988)
- Complexity (Pettersson, 1989).
- Contrast (Muter & Marrutto, 1991; Pettersson, 1989; Scharff, Hill & Ahumada 2000).
- Detail against no detail (Pettersson, 1989).
- Directionality (Pettersson, 1989).
- Headings (Hartley, 1987; Jonassen & Kirschner, 1982; Mayer, 1993).
- Highlight relevant information (Mayer, 1993).
- Icons (Mayer, 1993).
- Illustrations (Mijksenaar, 1997).
- Imbalance (Fleming & Levie, 1978).
- Irregular design (Fleming & Levie, 1978, 1993)
- Italics (Mayer, 1993; Mijksenaar, 1997; Pettersson, 1989).
- Key words in red (Fleming & Levie, 1978).
- Larger font (Mayer, 1993).
- Light against dark (Pettersson, 1989).
- Lines (Mijksenaar, 1997).
- Logos (Mijksenaar, 1997).
- Margin notes (Mayer, 1993).
- Repetition (Mayer, 1993).
- Shaded areas against plain backgrounds (Pettersson, 1989).
- Small against large (Pettersson, 1989).
- Symbols (Mijksenaar, 1997).
- Tonal areas against plain backgrounds (Pettersson, 1989).
- Underlining key words (Fleming & Levie, 1978; Mayer, 1993).
- Unexpected design (Fleming & Levie, 1978, 1993)
- Unstable design (Fleming & Levie, 1978, 1993).
- Variation out of context (Pettersson, 1989).

- White space (Hartley, 1987; Jonassen & Kirschner, 1982; Mayer, 1993).

Italics give emphasis to a word or a group of words. In order not to confuse the readers, it is important to establish a consistent system for how to signal emphasis. Use italics for emphasis sparingly; too many italicised words may reduce the emphasis effect.

Providing unity

Aesthetic theories that are based on *perception* favour sensation over intellect, favour seeing over reading, favour universality over cultural differences, and favour physical immediacy over social mediation (Lupton & Miller, 1999, p. 62). Aesthetics treats the conditions of sensuous perception and aims to establish general principles of art and beauty, and of proportion and harmony.

Unity suggests an "overall togetherness" in the information material. It is more or less the opposite to emphasis. According to some authors unity is the fabric about which the entire information structure with its text elements and pictures is interwoven. In a design that lacks unity the different elements compete for attention. It is chaos and it creates confusion for the receivers. However, the systems for desktop publishing make it possible for the individual author to integrate words and visuals to aid communication.

All kinds of inconsistencies in an information material may confuse the receivers. It may become unnecessarily complicated for them to interpret and understand the intended content of the message. There is a close relationship between guidelines aimed at providing unity and guidelines aimed at providing

harmony. To provide unity in information material the information designer can:

- Use style and terminology in a consistent way in each specific information material.
- Use layout and typography in a consistent way.
- Use highlighting techniques in a consistent way.

To get maximum impact from a picture it should be introduced in the text between the appropriate paragraphs. Each picture should also have its own caption. Substantial research has clearly shown that learning efficiency is much enhanced when words and visuals interact and supply redundant information (Levie & Lentz, 1982; Levin & Lesgold, 1978). The improvement sometimes exceeds sixty percent and averages thirty percent.

Many authors have found that pictures should be put as close to the relevant running text as possible (Benson, 1985; Braden, 1983; Clark & Lyons, 2004; Haber & Hershenson, 1980; Hartley & Burnhill, 1977a; Lidman & Lund, 1972; MacDonald-Ross, 1977; Mayer, 1993; Mayer & Sims, 1994; Mayer et al., 1995; Moreno & Mayer, 2000; Pettersson, 1989, 1993; Schriver, 1997; Wright, 1982). Illustrations in textbooks are often "forgotten" by students as well as teachers; therefore it is important for editors and information designers to clearly instruct the learners to make good use of the pictures (Hannus, 1996; Peeck, 1993, 1994; Reinking, 1986).

Visuals designed to complement oral and printed instruction should both have a high correlation with the message they are attempting to support, and need to be designed to provide the learning environment with specific types of stimuli needed by particular types of learners, to achieve special educational objectives.

Wileman (1993, p. 93) provided the following checklist with questions for evaluation of visuals with reference to unity:

1. Are the visual elements (words, pictures, graphic design elements) well laid out?
2. Does the margin unify or add to the composition of the visual?
3. Does the entire message (words and images) fill the screen?
4. If words are used, is it clear which words go with what pictures?
5. Does the composition direct the viewers to the centre of attention?
6. Does the composition support what is being taught?

We may distinguish between three types of unity. *Pragmatic unity* is that the characteristics of the viewer can work for or against recognition of an image. *Semantic unity* is the possibility to identify an image. *Syntactic unity* is an acknowledgement that an image exists. A minimum requirement is that the bounds of each image should be discernible.

Typography

Typography is the art and technique of arranging type in order to make the message visible and it provides a large number of possibilities to make the structure in a document clear (Benson, 1985; Tinker, 1963, 1965). The graphic designer will select typefaces, point sizes, line length, and line spacing and alter the presentation of headings, running text, captions, lists, tables, and ornaments.

Typography determines the clarity of the message and has been noted to have the capacity to influence brand perception (Childers & Jass, 2002, p. 104) and consumer motivation (McCarthy & Mothersbaugh, 2002, p. 664). Graphic designers use typography in different ways. Over time, we have become accustomed to associating certain styles of typographic design with certain kinds of content and tone of voice (Moys, 2011).

Any typographic feature that reduces ability to readily recognize and discriminate individual letters within words and words within larger text units will decrease legibility (McCarthy & Mothersbaugh, 2002, p. 670). “Novelty” or “incongruity” may have the inverse effect of generating interest in advertising (McCarthy & Mothersbaugh, 2002), and this is certainly also true in information design.

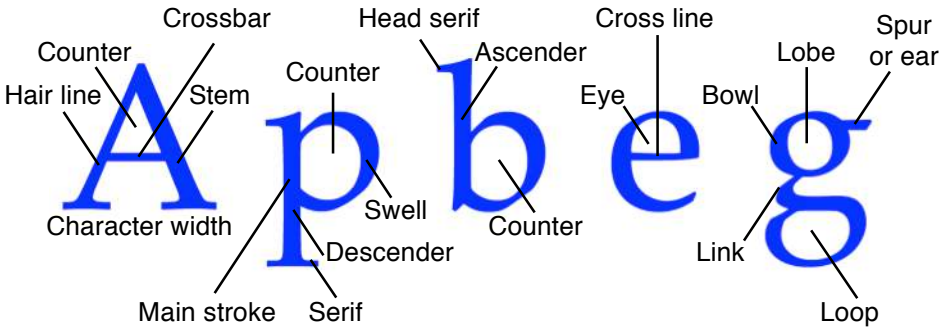
Type

The words we need in books and other graphical media use *type*. The term *type* refers to all characters that are used in printing. It may be hard to imagine how people lived 500 years ago without type. At that time all texts were copied by hand or they were printed from wood blocks. Today printers use thousands of type styles. A type style makes somewhat the same im-

pression on the reader that a speaker's voice makes on the listener. Some type may suggest strength. Others may suggest delicacy.

Design of characters

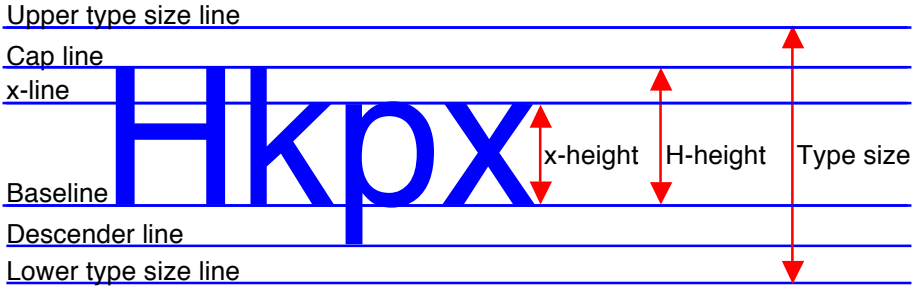
Individual characters can be designed in many different ways (Strizver, 2006). Many characters are based on geometrical shapes like the circle, oval, rectangle, square, and triangle. Circles and triangles need to have a slightly larger height (3–4%) than a corresponding square.



The terminology in typography is based on hot metal type composition. Here are some examples.

The differences between many typefaces are subtle. It is not always possible to see the differences without special training. Typefaces can be identified by looking for classic traits, such as the shape of specific parts of lower-case letters. A complete assortment of characters of the same style and size is called a *font* or a *font of type*. Thus, each typeface appears in many different fonts. However, in computer manuals the word font is sometimes used to mean typeface, which is of course confusing.

Imaginary lines are used in the positioning of letters. The baseline is the main point of reference in typography. The x-height is the vertical space occupied by lowercase letters (excluding ascenders and descenders). The H-height or cap height is the height of capital H. The point size or type size is the vertical space allowed for any character of a typeface.



Imaginary lines are used in the positioning of letters. Helvetica 72 point is used in this illustration.

When designing characters for digital environments, it is important to consider how to compensate for the limitations in fonts that are designed for traditional printing. All fine lines in characters present huge challenges for adaptation to screens, since they only represent a small percent of a pixel (Josephson, 2008; Bias et al., 2010). Digital-text developers improve text for low-resolution computer displays by font design, font hinting, and rendering technologies (Bias et al., 2010, p. 746). *Font hinting* refers to editing typeface outlines to available pixels. *Rendering technologies* aim to give screen text the clarity of printed text.

The lowercase letters of a and g are designed both in “one storey, or single storey, letter forms” and in “two storey letter

forms." Walker and Reynolds (2003) discussed whether 6-year-old children find serif text with "infant character forms" (e.g. variants of 'a' and 'g') easier or more difficult to read than "adult character forms." The results showed that children in the test group could read specially-designed texts set in Gill and Century, with and without infant characters, equally well.

One storey "infant character forms" (left) and "two storey adult character forms" (right).

Typefaces

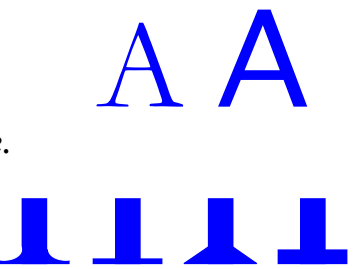
The term *typeface* refers to the general outline, the face, personality, and shape of the individual characters. During the little more than 500 years of western printing history, probably more than 60,000 typefaces have been designed (Mijksenaar, 1997). It is easy to combine them in many ways. Most of these typefaces are, however, limited in their usefulness. Differences are often subtle. It is not always possible to see the differences without special training. The distinctive details and the explicit forms of words may facilitate word recognition.

Lupton (2010, p 13) argued that: "Typefaces are an essential resource employed by graphic designers, just as glass, stone, steel, and countless other materials are employed by architects."

Experts employ several systems for classification. Classification systems may be based on chronology, on evolution, and on various elements of letter shapes. Most systems are incomplete and more or less confusing. The system that has enjoyed the largest general favour divides typefaces into four main

classes: 1) Roman type style. 2) Sans serif type style. 3) Script type style. 4) Black letter type style. Within these classes groups of type designs with important similarities form “type families.”

Roman type style with serifs (left) and sans serif type style (right). In this example we can compare Garamond and Geneva of the same size. There are four main types of serifs: bracketed serif, hairline serif, wedge serif, and slab serif.



Roman type style

The *Roman type style* includes most of the typefaces used in modern printing. These typefaces have *serifs*, finishing strokes normally at the top and at the bottom of the main strokes of letters. Serifs are not just put there for decoration. They help us distinguish between characters. Serif typefaces are often considered to make it easier to follow text lines and thus be easier to read than sans serif typefaces (Tinker, 1963), except for small letter sizes (Benson, 1985; Braden, 1985; Mackiewicz 2004). Visual psychophysical studies in adult readers have shown that serifs have little effect on legibility (Arditi & Cho, 2005).

Most typefaces in books for children are sans serif. Walker and Reynolds (2003) studied whether reading books for young children should be set in serif or sanserif type. Their results suggested that there is little significant difference in children's reading performance when Century is compared with Gill.

Serifs on characters make it easier to follow the horizontal lines with text. However this part of a word is very hard to read.

The upper parts of letters shape the images of words. Thus this part of the word is rather easy to read.

Serifs are terminal strokes, normally at the top and the bottom of the main strokes of letters in a Roman type style. This typeface is called Garamond.

Typefaces are often named after the designers who created them or after the printers who first used them. Common examples of serif typefaces are Baskerville, Berling, Bodoni, Bookman, Caslon, Century Old Style, Garamond, New Century Schoolbook, Palatino, and Times New Roman.

The French printer Nicolas Jenson worked in Italy. He perfected the first Roman style type already in 1470. In France Claude Garamond created the Garamond-style about 1540. In England William Caslon designed the Caslon-style in 1734; and John Baskerville created the Baskerville-style about 1757. In Italy Giambatista Bodoni designed a new kind of Roman type

style in 1768. For more than 200 hundred years the Bodoni-style has been called modern or modern type. Classical newspaper typefaces are Century Old Style, Ionic, Excelsior, Times New Roman, and Corona. They were all created for the hot-metal print and the rotary letterpress printing.

Sans serif type style

The *Sans serif type style* has no serifs on the characters. These typefaces provide uniform weight when there are less-than-optimal reading conditions and they are often used for headings, labels in pictures, diagrams, captions and tables (Benson, 1985; Lipton, 2007; Mackiewicz, 2004; Pettersson, 1993). Williams and Tollet (1998) suggests we should use sans serif type to improve legibility. Sans serif type can be used successfully for the body text in books, pamphlets and reports.

Common examples of sans serif type are Avant Garde, Futura, Geneva, Gill, Helvetica, Optima, Univers, and Venus. Helvetica typefaces may be the most widely used among the sans serif typefaces in the world today (Collier & Cotton, 1989). Subjects perceive san serif fonts as clean, modern and technical (Brumberger, 2003).

Serif fonts as well as sans serif fonts have been designed specifically for use on computer screens. However, sans serif fonts are still regarded more legible for screen reading than serif fonts (Josephson, 2008).

Some other typefaces

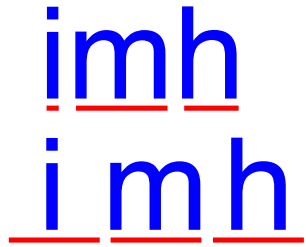
Script type style looks somewhat like modern handwriting that is carefully executed with a brush dipped in India ink. The individual characters are joined together. It is not possible to use script type for whole words in upper case, or capital letters. Examples are Constance, Palace Script, and Zaph Chancery.

Black letter type style resembles old German manuscript handwriting. Black letter type is difficult to read and rarely used in the USA. Examples are Fraktur, Rotunda, Schwabacher, and Textura.

A more modern classification scheme is the *serif-evolution system*. This system provides eight main classes: Venetian, Old Style (Dutch-English and French), Transitional, Modern, Contemporary (sans serif and square serifs), Black letter, Scripts, and Decorative letters.

Most typefaces are proportionally spaced. Then different letters are assigned different spacing in accordance with their individual sizes and shapes. However, on most typewriters and some printers the typefaces are mono-spaced. In such typefaces all the letters have the same amount of space. The letter "i" takes the same space as the letter "m". Proportionally spaced type is usually easier to read than mono-spaced type.

Note the differences in character space, especially for the letters i and m. These characters are set in 40 point Helvetica.



Size of type

Too small or too large lettering impairs reading. We read words in a text as patterns or even as a series of "word pictures," not letter by letter (Hallberg, 1992; Ingvar and Hallberg, 1989). Wilkins et al. (2009) found that the size and design of the typeface in textual material for children aged 7–9 (in the UK) may impair their speed of reading and their comprehension, and

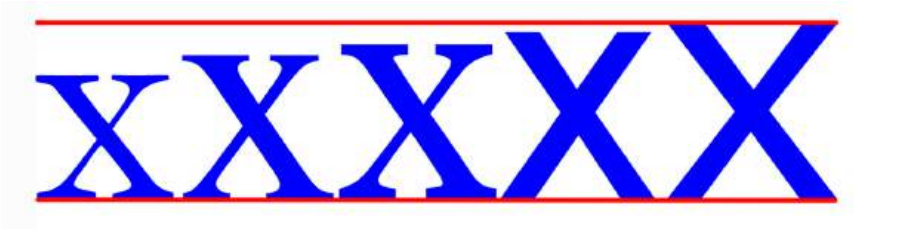
measurement of reading attainment. One study compared the speed with which sample sentences were comprehended. The sentences were printed in Arial font with an x-height of 4.2 mm (22pt) and 5.0 mm (26 pt) respectively. The sentences were verified 9% more quickly when presented in the larger typeface. Two studies compared the font Sassoon Primary with the font Verdana and showed that Verdana was read and searched more quickly.

Our perception of size is always relative. According to White (1987), some 10-point type looks gigantic and some minuscule. It all depends on the design of the face and its x-height. Letters can be varied in size and even by type to differentiate different types of data (Hartley, 1987). To achieve optimum legibility it is known that letter size must be adjusted to the visual format and the reading distance.

Height of character

The *x-height* is the vertical distance equal to the height of the lowercase letters such as x (without ascenders and descenders). Higher x-heights allow for smaller typeface size, which is just as legible but still is space saving and economical. This is important in the production of newspapers.

According to Tinker (1963) text smaller than nine points is too small to be considered legible. Text that should be read in a continuous manner should be set between nine and twelve Pica points (Benson, 1985; Bernard & Mills, 2000; Braden, 1983, 1985; Haber and Haber 1981; Tinker, 1963). A twelve point normal type is often suitable. Even after reduction in the printing process to ten point type, the text will exhibit very good legibility.



Height and width vary in different typefaces of the same size. This figure includes Garamond (left), Times New Roman, Georgia, Arial and Verdana (right). The differences in the shape and size of the characters will influence legibility.

In early readers for children the x-height is typically about 4 mm. Over the course of five years the x-height decreases to an "adult size" (about 2 mm). Hughes and Wilkins (2000) have argued that the typeface becomes too small too early in life.

In its *Publication Manual* the American Psychological Association (1996, p. 151) recommends a minimum type size of eight points in figures. The maximum size is 14 points. According to Lenze (1991) font size above twelve points slows the reading process because the reader must examine each letter individually to recognize it. Thus, we can conclude that type should be "large enough". A larger type size in a single column is preferable to a smaller type size in a double column layout. But the text on a display poster should be at least three to five times as large. A poster might require text that is at least ten times as large since it is designed to be read from a distance.

Measurement systems

Traditionally the vertical height of letters is designated in typographical *points*. However the size of such a point is somewhat different in various parts of the world. There are three basic

measurement systems used for traditional typesetting, and one main system for desktop publishing.

- The *Pica system* is used in USA and Great Britain.
  Here 1 pica = 12 points = 12 pts = .1660 inch = 4.22 mm.
  1 pt = .3516 mm (72,27 pts/inch)
- The *Didot system* is used in most parts of Europe (except Great Britain and France).
  Here 1 Cicero = 12 points = 12 p = .1780 inch = 4.511 mm.
  1 p = .3759 mm (67.6 pts/inch)
- The *Mediaan system* is used in France.
  Here 12 points = .1649 inch = 4.205 mm.
  1 pt = .3504 mm (72.5 pts/inch)
- *Desktop publishing systems* use a version of the Pica system.
  Here 1 pica = 12 points = 12 pts = 1/6 inch = .1667 inch= 4.23 mm.
  1 pt = .3528 mm (72 pts/inch)

In some situations these differences may be significant.

This is the width of 100 *Pica* points
This is the width of 100 *Didot* points
This is the width of 100 *Mediaan* points
This is the width of 100 *DTP Pica* points

This is the width of 25 millimetres
This is the width of 1 inch (25.4 mm)

Initial letters

Sometimes initial letters are larger than the running text. Initial letters may be lowered as “drop caps” or raised as “raised initial letters”. Because of the possibility to create fancy initial letters

in desktop systems, this medieval practice has become common again (Stokstadand Cothren, 2011). Misanchuk (1992) argues that fancy initial letters are not at all likely to help readers.

Different impressions

Traditional hot metal type is cast in sizes ranging from 4 points to 144 points. Photo-typesetting machines and systems for desktop publishing can produce even larger characters. Sizes of type that are in common use are 6, 7, 8, 9, 10, 11, 12, 13, 14, 18, 24, 30, 36, 42, 48, 60, and 72 points. Characters should obviously neither be too small, nor too large. In the first case we cannot read them at all. Legibility is very poor. In the latter case we can only have a few words on each line.

Too small or too large lettering impairs reading. As an example, a text on a poster should be at least five times larger than a corresponding text in a book. Newspaper print is usually eight or nine points. In books ten to twelve points are common. The following pages include five examples of a paragraph of text in five size versions: Georgia 8, 10, 12, 14, and 16 point type.

Georgia 8. Infodidactics is the methods used for teaching the various aspects of information design. The huge spread among the different disciplines makes information design an interesting, but also a complex area of research and teaching. Understanding is the goal of all scientific enterprise. When we understand a subject matter we are able to explain phenomena and predict new phenomena.

Georgia 10. Infodidactics is the methods used for teaching the various aspects of information design. The huge spread among the different disciplines makes information design an interesting, but also a complex area of research and teaching. Understanding is the goal of all scientific enterprise. When we understand a subject matter we are able to explain phenomena and predict new phenomena.

Georgia 12. Infodidactics is the methods used for teaching the various aspects of information design. The huge spread among

the different disciplines makes information design an interesting, but also a complex area of research and teaching. Understanding is the goal of all scientific enterprise. When we understand a subject matter we are able to explain phenomena and predict new phenomena.

Georgia 14. Infodidactics is the methods used for teaching the various aspects of information design. The huge spread among the different disciplines makes information design an interesting, but also a complex area of research and teaching. Understanding is the goal of all scientific enterprise. When we understand a subject matter we are able to explain phenomena and predict new phenomena.

Georgia 16. Infodidactics is the methods used for teaching the various aspects of information design. The huge spread among the different disciplines makes information design an interesting, but also a complex area of research and teaching. Understanding is the goal of all scientific enterprise. When we understand a subject matter we are able to explain phenomena and predict new phenomena.

As we can see from these examples different size versions of a text provide different impressions. In information design it is important to find a good balance between our need for a general overview and a good legibility of individual words. We need a balance between “not to large” and “not to small.” In books, magazines, and newspapers larger type may be used for head-

ings. Below are some examples of the word “information” in five larger size versions: Helvetica 18, 24, 28, 36 and 48 point type.

information 18

information 24

information 28

information 36

information 48

Which type sizes do you prefer for different headings in a book and in a technical report?

Information	Information
Information	Information

Fine lines in Times (upper line) may sometimes almost “disappear.” It is better to use a sans serif type, like Helvetica, when the text is white on a black background (bottom line). Both examples are set in 14 points.

Often text is printed with black ink. However text may also be printed in a number of colours. This is especially easy in documents for the Internet. The use of colour will influence im-

pressions as well as legibility. White text on a black background should usually be larger than traditional black text on a white background. Fine lines in a Roman type style may sometimes almost “disappear.” It is better to use a sans serif type when the text is white on a black background.

See the sub-section *Legibility of colour* in the chapter *Functional principles,* and the section *The message on the screen* in the chapter *Screen typography*.

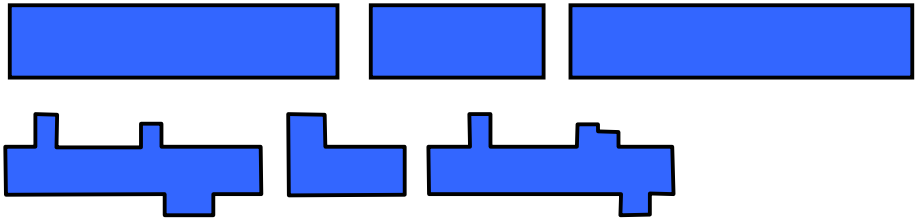
Stylistic variation of type

With respect to line thickness, inclination, and width characters in a typeface can be drawn in many different versions. The major examples of stylistic variation of regular text are **bold**, *italic*, and ***bold italic***. The visual weight and visual impression varies a lot. It may be compared with physical weight. A small and extra bold and expanded character may be seen as heavier than a large, light and condensed character.

A typeface may be available as light condensed, light, light expanded, bold condensed, bold, bold expanded, regular condensed, regular, regular expanded, extra bold condensed, extra bold, and extra bold expanded. With respect to inclination a typeface may be designed in italic letter style versions. A typeface may also be available as outlined, in-lined, and shadowed.

Regular type is easier to read than uncommon type. Boldface or italics should normally not be used for continuous text. Italic print is read more slowly than regular type and is also disliked by many readers (Tinker, 1965). However, boldface and italics may be used for emphasis. Italics may be used for small parts in a running text, for headings, and for captions. Bold type may be used for headings.

Different style manuals have different recommendations for when italic and bold type versions should and shouldn't be used. The use of underlining and all capital letters should be restricted to headings and titles, if they are used at all. Usually bold and italics are quite sufficient. Make type big enough to stand out from the background and heavy enough to be visible (Lipton, 2007). Underlining in the middle of a sentence makes the lower line more difficult to read (Isaacs, 1987). Shadow and outline letters should be avoided.



Words in all capital letters have similar shapes and profiles. The three words "DESIGN FOR CLARITY" (at the top) cover rather similar areas. However written with lower case letters (at the bottom) the words "design for clarity" look quite different. (This illustration is based on Futura.)

Reading speed can be substantially reduced when italic text is used for continuous prose (Foster and Bruce, 1982). Extensive use of italic text in continuous prose will impair their functions to emphasize (Glynn, Britton and Tillman, 1985).

The most common type versions are easier to read than uncommon type versions. We should avoid odd type versions like outline, shadow, and reverse type like white type on black background. Boldface and/or italics should not be used for running text but rather for emphasis of important parts of the text. Ty-

pography can be used to modulate a message. Lund (1999) reviewed research literature on the legibility of seriffed and sans serif type and concluded that the research is not only inconclusive, but it is mostly of too poor a quality to be relied upon. Other aspects, like lighting condition, and size of type are more important for legibility. For continuous reading, the typeface must be comfortable to read for longer periods of time.

We should remember, though, that content is more important than form. A text will retain its content even when type design is changed. Graphic design can be and should be used to build consistency and aid communication. Glynn, Britton and Tillman (1985) reviewed studies on the effect of "typographic cueing" on learning. Typographic cueing generally refers to the use of bold or italic type or underlining to signal the important ideas in a text. There is little doubt that cueing does work in drawing attention to the cued material. The consensus is that readers are more likely to remember cued ideas than un-cued ideas (Hartley, 1987).

The angle of the thickest part in a curved letter is usually described as either diagonal or vertical. It can also be horizontal. This design feature is called *stress of a typeface*. For example the typeface Sabon has diagonal stress, and Century Schoolbook has vertical stress. Stress reminds of the visual properties of writing. It is easy to see the angle of stress by drawing a line through the thinnest points of the letter o. There is no obvious stress in a sans-serif typeface with uniform strokes.

Typeface personalities

Several scholars have discussed the physical characteristics that lead to "perceived personas" and "typeface personalities" of typefaces. According to Hassett and Curwood (2009, p. 272) the

use of font is a mode of communication that serves as a semiotic resource. Research indicates that the presentation of texts affects the perception of the contents of the message.

Krause (2007) and Samara (2007) have offered different conceptual categories in their taxonomies of fonts, including moods, concepts, energy, elegance, technology and specific areas. Machin (2007, p. 87) noted that "letterforms themselves have become more important as part of the overall meaning of composition and have themselves become more graphic and iconic." Typographical features like colour, flourish, formality, framing, size, slant, and weight focus on elements of characters as well as overall composition. Strategies that support how readers interpret multimodal texts are evolving (Albers, 2008; Anstey & Bull, 2006; Serfini, 2010; Serafini and Clausen, 2012).

Several typefaces elicit strong emotions from groups of viewers. One example is Comic Sans. This typeface was designed for Microsoft based on lettering from comic books, but it has been used in many other situations. Depending on your point of view, Comic Sans is fun, breezy, silly or vulgar and lazy – and many graphic designers and aesthetes just hate it (Steele, 2009).

Viewers judge the material based on the visual attributes versus the meaning of the text (Childers & Jass, 2002, p. 95). To ensure that the intended message is understood, document designers must consider how typography and text present a united message.

There are many different and subjective opinions regarding the personalities of typefaces. With *serif type* it is easy to follow text lines (Tinker, 1963). *Sans serif type* is seen as clean, modern and technical (Brumberger, 2003).

Light type has been described as delicate, gentile, and feminine (Brumberger, 2003), as difficult to read and unprofessional (Mackiewicz, 2005), and as insubstantial and timid (Van Leeuwen 2006). *Moderate weight type* has been regarded as professional (Mackiewicz, 2005).

Bold type has been seen as aggressive, masculine and strong (Brumberger, 2003), as difficult to read and unprofessional (Mackiewicz, 2005), as assertive, daring, domineering, overbearing, solid and substantial (Van Leeuwen, 2006), as daring, overbearing, stable and substantial (Machin, 2007).

Van Leeuwen (2006) described *narrow and condensed type* as precise, economical, cramped, and restrictive. Machin (2007) described them as cramped and un-assuring. *Expanded type* was regarded positive light, and providing room to breathe (Van Leeuwen, 2006).

Cursive type fonts as well as *script fonts* are seen as elegant (McCarthy and Mothersbaugh, 2002), or organic, personal, and formal (Van Leeuwen, 2006). *Angular type* conveys cultural meanings. It is abrasive, brisk, controlled, decisive, harsh, masculine, modern, rational, and technical (Van Leeuwen, 2006). *Curvature* in type is perceived as fluid, gradual, maternal, natural, organic, smooth, and soft (Van Leeuwen, 2006).

According to Van Leeuwen (2006) a *flattened type* has heaviness, inertia, self-satisfaction, and solidity. A *vertical type* has instability, lightness, and an up-wards aspiration. Mackiewicz (2005) regarded *imperfection* in type as friendly. Abe Pittman (2012) concluded that regardless of the ability to communicate technical features of a typeface's anatomy, viewers share an emotional reaction. When selecting typefaces it is important to consider the amount of text, purpose, intended audience and medium.

Van Leeuwen (2006, p. 142) concluded that a new typography has emerged. Typography is no longer a humble craft and an "abstract art" in the service of the written word. It is a means of communication in its own right and it is spearheading innovation in graphic design.

Moys (2011) studied how typography and layout in a selection of magazine feature spreads and covers influenced reader's judgments. When it came to typeface participants in the study "were more likely to discuss its treatment than its choice: whether something was in bold, capitals, italics, colour, had a drop shadow or other effects, seemed to influence its distinctiveness far more than the choice of typeface." Moys concluded that "typographic voice" is conveyed through a range of variables. It is not just a question of establishing a typographic or visual mood. Typography plays a role in how a document is seen to address its readers and position itself.

Selection of typefaces

The decision of which typeface to select should rest largely on the purpose and on the intended audience for the document (Benson, 1985; Black, 1990). Using the correct typeface for the job is based on solving a problem and meeting a need.

Lenze (1991) noted that *private documents* may invite the use of ornate and stylish looking fonts. *Professional documents*, however, require maximum legibility (Benson, 1985; Pettersson, 1989; Tinker, 1963).

Advertisements must be noticed. Therefore, it is important to choose a graphical form that arouses interest. Thus, unusual typefaces can be useful and stimulate attention, entice the reader to look at the pictures and begin reading the text.

When it comes to *information materials*, the graphical form should basically be simple and "transparent" and not arouse any special interest or attention for its own part. The graphical form should not be too exciting or provocative, nor should it be patently dull. McDougall and Hampton (1990, p. ix) concluded that design and layout should enhance visual and verbal content by making it appealing and understandable. It should not be excessively ornamental, thus calling attention to itself.

To encourage readers to pay attention to relevant information, text designers should help the reader to control his or her cognitive processes during learning (Mayer, 1993). To achieve optimum legibility it is known that:

- Common typefaces are easier to read than uncommon typefaces (Benson, 1985; Paterson & Tinker, 1932; Spencer, Reynolds & Coe, 1974; Tinker, 1963, 1965).
- Serif typefaces are often considered to be easier to read than sans serif typefaces, except for small letter sizes (Tinker, 1963).
- Running text should have a "normal" combination of upper and lower case letters. All-capital printing has been shown to markedly reduce the speed of reading (Henny, 1981; Poulton & Brown, 1968; Tinker, 1965).
- Without sufficient white space around words in all capitals, the words will become too difficult to read (Kinney & Showman, 1967).

The different typefaces have their individual characteristics. They vary in space and in their visual appeal. Some are more legible and some are less legible. In the USA children usually learn to read using textbooks with serif typeface like New Cen-

tury Schoolbook. In European countries it is not unusual that sans serif typefaces are used in textbooks. Children in China learn to read the complicated Chinese characters. And we can find equally good readers in all these countries. As a matter of fact, the reader should never become too conscious of the typeface.

Type can be created in a variety of styles. A change in type style can signal a change of purpose, a new section, or another degree of importance. Typographical techniques can alert learners to such things as main ideas, important concepts, rules, sections, subsections, and more. However, multiple type styles on a page tend to be confusing rather than facilitating. The decision of which font or which fonts to use should rest largely on the purpose and audience of the document (Benson, 1985).

Systems for desktop publishing are now common. Software that is easy to handle give the layman the opportunity to combine verbal and visual messages. This technological revolution enables us to do anything we have in mind to do on a page or on a screen. The mechanical processes of typesetting and makeup do no longer restrict us. Unfortunately, most desktop publishers know little about typography, graphic design, and information design. With the new systems it is almost too easy to manipulate a text and make use of all the possibilities to change the appearance of a page.

It is far too easy to create confusion or even a complete mess-up. As White (1987) put it: “The distinguished art and craft of typography developed over centuries by sensitive craftsmen/artists cannot be handed over to an indifferent typist keyboarding a machine (however marvellous the technology) with results of equal excellence and stature. The reader feels the difference in terms of the piece’s ease of reading, charm, and

comfort." It is wrong economy to cut back on investment in quality, believing the readers won't know the difference. Winn (1993) noted that in text, attention is drawn to words or passages that stand in contrast to the rest of the body of the text.

Baggerman (2000, p. 69) pointed out that designers make a frequent mistake when they choose typefaces according to their own tastes instead of according to the needs of the users.

Some common typefaces

A "good" typeface might actually be one that is more or less "invisible" to us as readers (Goldenberg, 1993; Scieszka, 1998; Melin, 2011). We shouldn't be too concerned with the design of the typeface. The typefaces in common use are all more or less equally legible (Paterson and Tinker, 1932; Tinker, 1963, 1965). However, readers are likely to have strong preferences about the aesthetics of typefaces. One important aspect of the design of a typeface is the perceived size of the individual characters. Some type looks big and some type looks small. This depends on the design of the characters and their x-height.

Next to the x-height, the shaping of the various letters is important. The counters are important for printing and also for the recognition of words. In desktop publishing all parts of the letters must be clear and distinct. Clear contrasts increase the legibility, uniformity reduces it. A typeface can influence our appraisal of a printed message. Thus, it might be important to make a choice so that typography reinforces the message.

As mentioned earlier, thousands of fonts are available. In the practical work with graphic design for information, we need only a few typefaces in different versions. Times New Roman was created for the London newspaper The Times. Today Times

New Roman is available in most computers and used in a large number of documents.

Common font families like Baskerville, Berling, Bookman, Garamond, New Century Schoolbook, and Palatino can be used successfully for the body text in books. Modern newspaper typefaces include Stone (1987), Swift (1985), Gulliver (1990), and Utopia (1989). These are all serif typefaces. Like Times New Roman they have large x-heights and good legibility.

The following pages present some examples of the same text segment rendered with different typefaces.

Georgia

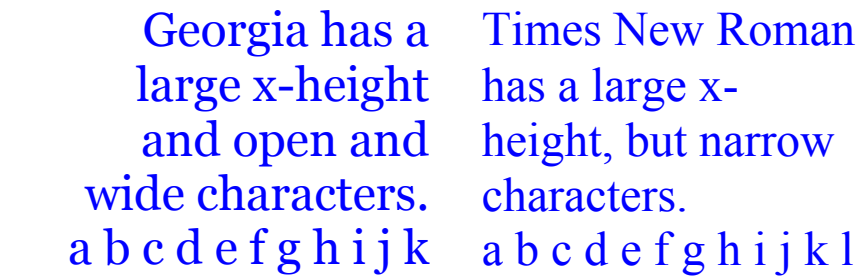
The type designer Matthew Carter created the two typefaces *Georgia* and *Verdana* for the Microsoft Corporation. They were released in 1996.

Georgia is a transitional serif typeface specifically designed for clarity. It is similar to Times New Roman, but it is larger at the same point size. Georgia has a large x-height and open and wide characters. Thus Georgia has good legibility on computer screens, even at small sizes. Times New Roman is narrow with a more vertical axis. Serifs in Georgia are almost horizontal, more blunt and wide. The Georgia typeface is very common for running text in documents at the Internet.

Georgia: abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSVWXYZ,.;:!?”()/&%+-=><*1234567890

Infology is the science of verbal and visual presentation and interpretation of information. On the basis of man’s prerequisites, infology encompasses studies of the way a verbal and visual representation should be designed in order to achieve optimum communication between sender and receiver. Infology models

contain both theoretical (descriptive) elements and normative (prescriptive) elements.



In this illustration Georgia is compared with Times New Roman in the same type size.

Helvetica

The Swiss typeface designer Max Miedinger created *Helvetica* in 1957. Helvetica has endured over time as an example of a clean, durable, malleable, rational, and universal typeface. This all-purpose font is everywhere, particularly in commercial use representing all kinds of brands. It is the ultimate design of the Bauhaus–Swiss–International schools. Helvetica is designed with simple, striking lines in a compact way. Like Times, Helvetica has been developed in many typefaces and font versions. Helvetica typefaces are very useful for captions, headings, tables, and titles, but they might be hard to read in running text. The similarities between upper case I and lower case l cause real problems in words like “Illustration” (See the illustration at *Verdana*). Running text in Helvetica will need extra space between the lines. Helvetica is often used for reader slides, overhead transparencies and in business graphics. According to Collier and Cotton (1989) Helvetica typefaces may be the most

widely used sans serif typefaces in the world today. In this book Helvetica has been used for labels and captions within figures.

Helvetica: abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOP QRSVWXYZ,.;:!?”()/&%+-=><*1234567890

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Times New Roman

The British type designer Stanley Morrison created the original *Times New Roman* for Monotype in 1932, to be used in the London newspaper *The Times*. He worked together with Victor Lardent, an advertising designer and draftsman at The Times The mission was to produce a new, first-class and easy-to-read typeface with very good legibility even in small sizes for the newspaper. The production methods for The Times newspaper were far above the world standard at that time. Times New Roman was designed for hot-metal print and the rotary letterpress printing. Times New Roman combines the classic elegance of typefaces for books with modern demands for economy of space. Several versions of Times New Roman have been developed. Typefaces and fonts belonging to “the Times family” are all easy to read. They are probably the most widely used serif typefaces in the world. Times typefaces are used in newspapers and magazines as well as in books. However, Times New Roman

cannot be successfully used on computer screens, and laser printers cannot print the fine lines in small letter sizes.

Times New Roman: abcdefghijklmnopqrstuvwxyzABCDEFGHIJK LMNOPQRSVWXYZ,.;:!?”()/&%+-=><*1234567890

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Verdana

The type designer Matthew Carter created *Verdana* as a humanist sans-serif typeface for computer screens. It was specifically designed for clarity. Verdana has no serifs, large x-height as well as open and wide characters. The spacing between characters is loose. In contrast to Helvetica similarly shaped characters have emphasized distinctions. Thus Verdana has good legibility on computer screens, even at small sizes.

Comparisons between Helvetica, Georgia and Verdana.

Typeface	*Digit 1*	*Lowercase l*	*Uppercase I*
Helvetica	1	l	I
Times	1	l	I
Verdana	1	l	I

Verdana is very useful for titles, headings, captions, and tables of various kinds but it might be hard to read in running text. However, in 2009 the international furniture company IKEA

changed the typeface in its catalogue from the classical typeface Futura to the modern Verdana. The company expressed a desire to unify its branding between print and web media.

Verdana: abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLM NOPQRSVWXYZ,.;:!?”()/&%+-=><*1234567890

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Paper and ink

In all graphical media the quality of the paper and ink is of vital importance for the final result. The paper has its specific bulk, finish, grain, texture, and weight. It may be stiff or floppy. It may be glossy and shining or matte and maybe dull. It may be thick or thin. Thin lines, in serifs on small type sizes, may require a harder paper than normal lines and larger type. The printing technique used will affect the type of paper needed. The quality of ink is also an important factor in graphical communication.

All these factors affect legibility and our reading comfort and also our perceived value of the product. The degree of contrast between the colour of the ink and the colour of the paper should be optimum. For text printed in black all paper surfaces are equally legible if they have a reflectance of at least 70%. The most legible combination is a black text on a light yellow back-

ground. In a normal reading situation, black print on white paper is over 10% more efficient than white on black (Pettersson, 1991).

Cost effective typography

If a document is to have many readers who will try to read and understand it during working hours, the cost of reading the document will be the greatest expense it incurs. Even though it may be expensive to produce information, it usually costs even more to store, access, and use it. Therefore, the more people who will partake of certain information, the greater the total cost will be. Because the cost of reading is closely linked to the type of material in the document and to the various groups of readers it targets, there will be great opportunities to reduce the total costs and save money by presenting the information in a suitable fashion.

To facilitate our work with cost effective typography and graphical form, we may need a number of guidelines, or rules of thumb. In an international organization such as a multinational corporation advice and guidelines may be developed according to the following criteria and requirements:

1. The text must exhibit good legibility. Because the typography should provide ease of reading without being “visible” (that is, without arousing notice), we must avoid unusual typefaces.
2. Our point of departure should primarily be the reader's psychological capacity for perception, and secondly the potential performance of the technical equipment at hand.
3. It should be as easy as possible for technical writers and editors to handle templates for graphical forms. Thus, these

people should be given the necessary training to handle computer programs and the like where these templates are available.

4. Size ratios in typography and layout should be based on conventions of typography and on research findings in perception psychology.
5. The typography and layout should be acceptable in many countries, even though national tastes may vary widely.
6. The typography should be adapted to the technology, that is, it should look good when printed on a laser printer, and when copied on an ordinary office copying machine. Look for, and adopt to the week part in the production chain.
7. The typefaces should be available as standard selections in computers and laser printers all over the world. Even so, we have to accept the fact that printouts made by different printers will not be identical in appearance.
8. The typography should usually be "economical". It should be possible to accommodate a great deal of information in a limited space without a crowded appearance.
9. We should choose typefaces with high x-height for good legibility, even when small type sizes are used.
10. The typography and layout should produce good results on standard paper. In Europe, the standard paper size is A4 (210 x 297 millimetres), whereas in the USA, US letter (216 x 279.5 millimetres) is the standard. We deal with these differences by varying the margins on the page.
11. The dimensions of pages and length of lines are given in centimetre or millimetres. Type sizes are given in "points," the typographical unit of measurement. Note that there are different systems in USA and Europe. In the Pica system 12 points = 4.2333 mm. In the Didot system 12 points = 4.511

mm. In the Mediaan system 12 points = 4.205 mm. The most common programs for desktop publishing use the Pica system, giving points as “pts”.

12. It should be possible to print a document and insert the pages directly into a loose-leaf binder. This means that right-hand and left-hand pages should have the same appearance, basically a right-hand page layout.
13. If a document is to be reproduced using conventional technology, it should be possible to use printouts of it from a laser printer or from a phototypesetter as a basis for direct reproduction. When A4 is reduced to 80.4%, the result conforms to the conventional printing format, G5 (169 x 239 millimetres).
14. Before conventional printing, the layout of right-hand and left-hand pages may easily be changed.
15. All documents might be coded in accordance with the SGML standard, so that it is easy to use the information in different ways, and in different formats. Sometimes other standards may be used (like HTML, and XML).
16. Reduction through successive refinement is the best way to reach clarity. To create an elegant solution, anything that is not essential to the communication task has to be removed.

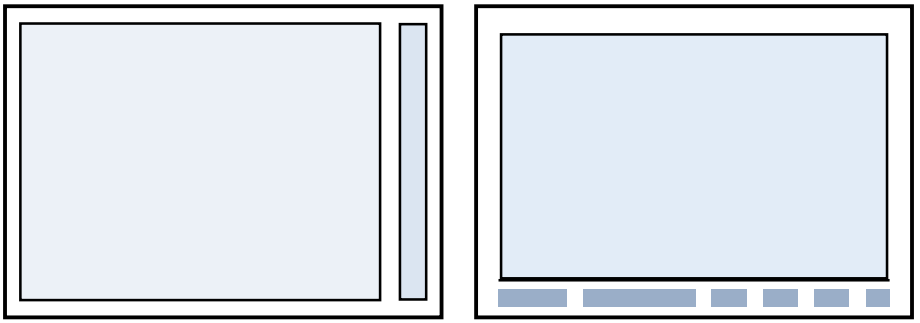
Projected typography

In audio-visual instruction and in audio-visual materials, such as overhead transparencies, slides, filmstrips, and computer-based presentations, lettering must be considered carefully in order to improve legibility. Type must be large as well as bold enough to see. In verbal presentations, many of the projected images consist only of text. In written documentation, this type of information should be worked into the body of text instead.

(See the book *Information Design 2–Text Design* for more information about oral presentations and the use of projected images.)

OH and PP

According to the standard for OH, overhead transparencies, SIS 62 23 01 SD A4L, the actual plastic film is 210 x 297 mm. The picture surface is 190 x 245 mm, which is a height-to-width ratio of 1:1.289. Administrative data are put in the 35 mm right margin. The top margin is 15 mm, bottom margin 5 mm and left margin 17 mm. In some organizations there is a need for more administrative data, such as logotype and/or name of the organisation, name of author, document name, security classification, revision, publication date and document number. These data are put in the bottom margin. Here the picture surface may be 170 x 270 mm, which is a height-to-width ratio of 1:1.588. In PowerPoint the height-to-width ratio is 1:1.328. The size of the image can be adjusted to the situation on the computer screen.



The layout to the left is the OH-standard SIS 622301 SD A4L. The layout to the right is used in some organizations. This layout can also be used for PP and other similar systems.

Many speakers use text transparencies, or "word-visuals," containing key words in attempting to supply an overview, clarify, reinforce, and summarize complicated arguments. These texts must then be brief and concise. They must also be easy to read. For good legibility, characters should be large, distinct, and bold, not less than six millimetres high when projected in a room the size of a normal classroom. Larger rooms require larger character sizes.

Normal text, containing both uppercase and lower-case letters, is easier to read than texts using only uppercase letters. Overhead transparency texts should usually consist of black characters on a light background. Text transparencies are a useful adjunct for a speaker but are sometimes very boring to an audience. In verbal presentations, many of the pictures consist only of text. For good legibility of projected text we should consider the following aspects:

- Use no more than six rows of six words each in one overhead transparency. The line distance should consist of the size of the typeface plus 15 to 20% of extra white space.
- Maintain a good contrast between foreground and background. Good colour combinations are yellow – black, and white – blue.
- Avoid graduated and tonal background fills.
- Overhead transparencies should preferably be horizontally presented.
- Overhead transparencies should be projected in a correct way and not in any way distorted.
- Text should have maximum legibility. It may be a good idea to use a linear typeface such as Helvetica or Verdana; for ex-

ample 36 and 30 points for headings, 24 points for the body text, and 18 points for texts within illustrations.
- Text must also be legible to spectators who are sitting at the rear of the hall. The minimum size of letters in overhead transparencies is six millimetres (Helvetica or Verdana 18 points).
- Restrict stylized and fancy typefaces to opening slides.
- Edit the text into sections that are easily read.
- Be consistent in the use of uppercase and lowercase letters.
- Text should usually be left justified.
- White – yellow, red – green, and red – blue are all very bad colour combinations.
- A good general guideline for reader slides is to use a light or a dark background colour appropriate to the content, and then use a colour with good contrast for the text.
- Put the necessary identifications data on each transparency.
- Check the spelling.

In written documentation, this type of information should be worked into the body of the text instead. Basically the guidelines for overhead transparencies are also relevant for PowerPoint presentations, and other similar systems.

Slides

A slide's subject should appear against a black background. Text slides should only be used to a limited degree. Texts should be light (white or yellow) on a dark background (black or dark blue). Many colour combinations are hard or even impossible to read on a screen.

According to the standard for 5 x 5 slides, SIS 62 23 03, the individual part of the film is 24 x 36 mm. The picture surface is

22,5 x 34,5 mm, which is a height-to-width ratio of 1:1.533, which is quite close to the golden mean (1:1.618). The actual frame is 5 x 0 cm.

In verbal presentations, many of the pictures consist only of text. We should consider the following aspects for good legibility:

- Use no more than six rows of six words in each image, set in a linear typeface, with characters large and bold enough.
- Maintain a good contrast between foreground and background. Good colour combinations are yellow – black, and white – blue.
- Avoid graduated and tonal background fills.
- Slides should preferably be horizontally presented.
- 15 – 20 words on a slide is maximum for effective communication.
- Letters should be medium to medium-bold.
- Lettering height should be no less than one 25th of the height of the artwork to be transferred to film.
- Upper-case letters are often used in slides. Be consistent in the use of uppercase and lowercase letters.
- Restrict stylized and fancy typefaces to opening slides.
- Edit the text into sections that are easily read.
- Text may be centred.
- White – yellow, red – green, and red – blue are all very bad colour combinations.

Screen typography

Compared with traditional graphic presentations, a presentation of information on visual displays such as television sets and computer terminals is very limited. Still, information may be presented in many different ways.

Visual displays

Visual displays can be built in many ways. A colour television set, an advanced computer terminal, and a liquid crystal display all have different characteristics. A television set is built to be used at a distance of more than 120 centimetres. A computer terminal, however, is built to be used at a distance of 60 centimetres and has a much better picture resolution than a television set. Our perception of text and pictures on visual displays are, of course, to a large extent dependent on the quality of the screen. A videotex terminal simply cannot reproduce a simple illustration like a pie chart, since the resolution is only a total of about 5,000 graphical elements.

Computer terminals are often built with a screen resolution of 72–82 dots per inch. Today, an ordinary television image consists of about 250,000 image points, or picture elements, which vary with respect to both grey scale and colour information. High Definition Television (HDTV) uses 1,125 scanning lines and can contain five or six times more information than the present NTSC standard colour television system, with 525 lines. (The European PAL-system has 625 lines.) HDTV developments of flat PAL 625 screens will also give increased technical possibilities to present text and pictures.

However, visual displays will continue to have a resolution far below a printed page. Some laser printers produce documents with 1,200 dots per inch horizontally, and 600 dots per

inch vertically. Phototypesetters produce 1,250–2,600 dots per inch. This is enough to create very small and close raster dots and to produce very fine lines in the final print.

Colour displays

In discussions on technology, colour is related to measurable amounts of light. In 1931 an international body called the Commission International de l'Eclairage (International Commission on Illumination), or CIE, defined standards of light and colour. In this context colour primaries are the basic colour stimuli used for the synthesis of any colour, by addition or subtraction. Additive combinations of a very limited amount of radiation are used for colour synthesis of a range of colours in a cathode ray tube (CRT) and in a visual display unit (VDU).

A colour CRT is a vacuum tube, enclosing one or three electron guns for generating beams of electrons, a system for focusing the beam to produce a spot of visible light at the point of impact on the phosphorous screen, and for electric field deflection of the beam, suitable deflection electrodes. The thousands of small phosphorous dots are grouped into threes, called triads, with one dot emitting radiation that appears red, one dot emitting radiation appearing to be green, and the third emitting radiation appearing to be blue. Red, green, and blue are called the "three primaries," RGB. One lumen of white is given by 0,30 red + 0,59 green + 0,11 blue. Any two primary colours may be mixed to produce other colours. Red and green added can produce a range of hues around yellow. Green and blue produce a range centred on blue-green, while red and blue mixtures produce a red-blue range. The total number of colours that can be produced in a CRT depends upon the number of steps or grey levels obtainable for each phosphorous dot. Advanced systems

are capable of producing up to 256 simultaneously visible colour stimuli chosen from a palette of 16 million. However, in most cases only a few colour stimuli are needed at the same time.

The uncertainties in the coordinates of colours are rather large as a consequence of the heterogeneous distribution and efficiency of the phosphorous over the screen, the defects in electron beam convergence, and the departures of the relations between the values of the colour signals and the digital counts. One consequence of additive combinations in colour television is that characters presented in white (the three-colour combination) are less sharp than those presented in yellow, blue-green, or red-blue (all two-colour combinations). In a similar way, the latter colours are less sharp than red, green, and blue (pure colours). Sometimes colour rims may be seen at the characters with two- or three-colour combinations. The additive combination starts in dark adding light to produce colour. Thus, another consequence of additive combinations is that secondary colour stimuli will always appear brighter than the primaries.

Luminance is a photometric measure of the amount of light emitted by a surface (lumen/steradian/sq.m.). Radiance is a radiometric measure of light emitted by a surface (watt/steradian/sq.m.). It should be noted that neither luminance nor radiance is the equivalent of brightness, which is the experienced intensity of light (bright-dull). In colour displays it is very difficult to distinguish brightness from lightness (white-black). When the signal to the display is increased, the brightness of the total screen is increased. If a signal to a specific part on the screen is increased, the lightness of the area is increased compared to the total screen.

It is usually possible in a CRT to adjust the luminance, the hue, and the saturation. Like brightness and lightness, hue and saturation are also psychological dimensions. Hue is the basic component of colour corresponding to different wavelengths. Saturation is most closely related to the number of wavelengths contributing to a colour sensation. We should always remember that the production of colour, by additive or subtractive methods, has nothing to do with the actual perception of colours.

In a CRT some 8,000 to 20,000 volts of tension are required to form an image on the screen. Screens with several colours even require up to 30,000 volts. The electrostatic field of the CRT is positively charged. Thus the person sitting in front of the screen is negatively charged and a strong field is created. The field affects the movement of dust particles in the air. Since the majority of the particles are positively charged, they are attracted to the operator. The rate of deposit can reach 10,000 particles per square millimetres of skin an hour. Thus skin and eyes become irritated. People suffering from allergies can experience extreme discomfort. The electrostatic field can be eliminated with the help of a grounded filter, mounted on the screen.

The large magnets used to focus and direct the electron beam on the screen create the electromagnetic field. The electromagnetic radiation consists of two components of which one can be shielded. It is not clear as to whether this radiation affects the adult human body unfavourably. However, it is known that this radiation might be injurious to unborn children when pregnant women work too many hours in front of the screen.

Interface design

A human-computer interface is a communications channel between the user and the computer. An interface includes both 1) physical components, and 2) conceptual components. Physical components are devices for input and output of information. Input devices include microphone, touch panels, joysticks, speech recognizers, eye trackers, and data gloves. Output devices include visual displays and sound or speech synthesizers. Conceptual components include command languages, menus, and typography and screen layout.

Interface design is interdisciplinary. It may involve facts from psychology, computer science, information science, engineering, education, and communications. The information processing model of cognition is used in cognitive psychology. It establishes that: 1) our working memory is limited to five to seven "chunks" of information; 2) our attention must frequently be refreshed; and 3) recalling information requires more cognitive effort than recognizing information. Novices and casual users prefer menus to command languages, recognizing an option is easier than remembering a command. Touch panels in information kiosks and graphic displays in most video games overcome many psychological limitations. They share the “load” between physical and cognitive activity. Immediate feedback and easily reversibility invite user exploration.

The psychological theory of mental models, or schemas, has also been applied to interface design. Users develop mental models for objects, events, and ideas. These internal representations are incomplete and often inaccurate, but they help people deal with their systems.

A common approach in interface design is to define metaphors that links existing user knowledge to the functions of the

system. The desktop metaphor may be the best known example. However, a more fundamental metaphor is the screen as a scroll of paper. Metaphors are useful, but they can also constrain the user's view of the system. Concern with ease of learning can eventually interfere with skilled use.

Research in psychology has led to a number of principles for better interface design. Perhaps the most basic principle is that an interface should be designed around the needs of the user rather than added on after a system has been completed, thus serving the constraints imposed by the system. Marchionini (1991) provided the following seven design principles:

1. The interface should compensate for human physical and cognitive limitations whenever possible. However, the interface should be "transparent," not getting in the way of the user's actions or impeding his or her progress. The interface itself should not overload the user with complexity or unnecessary "bells and whistles" that interfere with or distract from the task at hand.
2. The physical components of the interface should be ergonomically designed, taking into account the comfort and health of the user as well as his or her special needs and characteristics. For example, a touch panel design for a word processing program demands far too much arm movement for lengthy sessions, but serves quite nicely in an information kiosk of a shopping mall when positioned to be touchable by users of various heights.
3. The interface should be consistent. For example, selection methods, positioning of important text and buttons, text fonts and styles, and window layout and management should be consistent in all parts of an interface.

4. Non-command interaction styles such as direct manipulation and menus are preferable to command languages, although the expert user should be given “type ahead” capability to quickly move through layers of menus.
5. The interface should handle errors by providing simple and concise error messages that assist the user in recovery and future avoidance.
6. The interface should support reversible actions (e.g., the UNDO capability in many systems).
7. The interface should be subjected to usability testing early in the design process and as each iteration of the product evolves.

Computer systems are becoming increasingly interactive, and this trend will continue as new interfaces are developed. Interactivity will be supported by new input and output devices that take fuller advantage of the many communication channels humans employ. The development of interfaces that support multiple input and output devices in parallel will be important. For example an interface that accepts voice and gesture concurrently will give richer control to users who must move about while controlling systems. Likewise, video and sound output together provide a more powerful communication channel for information flow. Intelligent agents are also under development. Agents can be assigned specific tasks by the user and then sent out to execute those tasks.

The message on the screen

The message on the screen may consist of text, numeric data, and visuals. The layout of the message is important for our perception.

Text on the screen

The quality of visual displays is important for our perception. Colour as well as blank space are essentially free and might be used to increase legibility. Experiments with 11,000 judgments of perceived reading efforts of text on visual displays (Pettersson et al., 1984a) showed that colours presented on colour displays are ranked in the same order as surface colours in traditional print media. Blue was most popular. The best combination is black text on a white or yellow background. Black was also the best background colour, because it has good contrast to most text colours. However, the background colour of a computer screen should be "fairly light" or "fairly dark," depending on the content (Bradshaw, 2001). The text displayed on a screen should have an opposite ("fairly dark" or "fairly light") colour. When the screen resolution is average or poor, sans serif is more legible (Bernard et al., 2001).

A presentation of text on a visual display depends on the type of characters used, the design of the information, the background, and also the content. The characters may vary with respect to font, size, lowercase and uppercase letters, colour, and contrast to the background. Legibility of the text depends on the execution of the individual character and the possibility for each one to be distinguished from all others. People generally perform better with text on paper than with electronic text (Dillon and McKnight, 1990).

The most important consideration when working with typography and colour for computer screens is to achieve an appropriate contrast between text and its background (Carter, Day and Meggs, 2007). They wrote (p. 80): "It has long been considered that black type on a white background is the most legible (combination). While this combination remains an excellent

choice, other alternatives may offer equal if not improved legibility due to improved digital and printing technologies, and the fact that colour is a relative phenomenon. ... Generally, all legibility guidelines related to working with colour and type in print apply also to type appearing on a computer screen." It is also important to use typefaces specially designed for screen display, such as Georgia, Trebuchet and Verdana (Bias et al., 2010; Hoffman, White & Aquino, 2005; Josephson, 2008).

Subjects generally prefer reading text on paper to reading electronic text on a screen (Dillon & McKnight, 1990; Wright & Lickorish, 1983). People perform slower and less accurate on instructional tasks when they have to read from computer screens and not from paper (Wright & Lickorish, 1983). Proofreading of text from print on paper is 20-30% faster than proofreading from computer screens (Gould & Grischowsky, 1984). A lot of work has been conducted to create legible characters. Knave (1983) has given guidelines for the creation of characters. A minimum of ten to twelve raster lines per character is required. When characters are built by dots in a dot matrix, the characters will be round or square and not elongated. A dot matrix of seven by nine dots is often regarded as a minimum. The height of the characters should be a minimum of four millimetres for a viewing distance of sixty centimetres.

Subjects dislike fast scrolling text on computer screens (Burg et al., 1982; Kolers, Duchnicky & Ferguson, 1981). For maximum legibility on a computer screen double spaces should be used between lines in a continuous text (Grabinger, 1989; Kolers, Duchnicky & Fergusson, 1981). Blinking and flashing text can be used as an accenting (Rambally & Rambally, 1987).

General design rules should be employed also in the design of screen displays. Thus material should be arranged and dis-

played so that it is easy to read, from top to bottom and from left to right. A visual display design may vary with respect to spatial organization like headings, length of lines, justification, spacing, number of columns, number of colours at the same "page," and directive cues like colour coding, twinkling characters or words, and scrolling text. Experiments (Pettersson et al., 1984a) with 11,000 individual judgments of perceived reading efforts of text on visual displays were concluded as follows: Colours presented on colour displays seem to be ranked in the same order as surface colours in traditional print media. Blue was most popular.

- When text is shown on a visual display, there is no easily read colour combination. About thirty-five of one hundred and thirty-two combinations are acceptable.
- The best text colour is black, which causes good contrast to most background colours.
- The best combination is black text on a white or yellow background.
- A text can be easy to read in any colour, provided the background is carefully selected.
- The best background colour is black, which has good contrast to most text colours.
- Reading efforts of colour combinations are independent of the sex of the subjects.
- There was no difference between colour blind (red-green) users and users with normal vision.

Inverse writing in various colours within a text may be used to achieve emphasis. Other possibilities may be a box around a paragraph or a change in font or size of letters. Blank space in printed material increases cost, since more paper is required.

Thus, it is not used often. However, colour as well as blank space on a visual display is essentially free and might be used to increase readability. Full text screens in several colours are difficult to read and quite annoying. What about double spaced lines and/or spaces between columns? As stated above, it is possible to make rather clear statements with respect to the use of colours. However, it is not as easy to give guidelines for the other variables.

Studies of attitudes to various variables (Pettersson et al., 1984b) in the presentation of text on visual displays showed that subjects dislike fast scrolling text. This was also found by Kolers, Duchnicky and Fergusson (1981), and by Burg et al. (1982). Subjects seem to dislike more than three or four text colours on the same "page". They seem to consider colour coding and/or twinkling text to be a good way to show that something is especially important.

Subjects also seem to agree that text in uppercase letters is harder to read than normal text. According to Kolers, Duchnicky and Fergusson (1981), and Grabinger (1989) we should use double spaces between lines in a continuous text for maximum legibility. Attitudes are indifferent to a few design variables. Thus text on every second line does not seem to make it easier to read than text on all lines. Higher characters do not seem to be easier to read than standard characters. Half lines do not seem to be better than full lines. A two-column layout does not seem to be better than a one-column layout.

Other research suggests that margins should neither be narrow nor very wide. Rambally and Rambally (1987) showed that blinking and flashing of a text on a computer screen is an accenting technique that will grab the attention of the user.

The background may vary with respect to colour and brightness. Good combinations of text and background colours always have a good contrast. Optimum contrast is often found to be 8:1 to 10:1. Most subjects prefer a positive image, i.e., dark text on light background with a minimum refresh rate of 70 Hz.

CLEA-research (Pettersson, 1984) concerned with perceived reading efforts of text on visual displays and altering colours of the actual equipment found that the close context is really important for the perceived reading effort. The colour of a terminal should be rather subdued. The best of ninety combinations were black text on a white screen with a dark grey terminal, closely followed by the context colours black, white, and light grey. It was also found that it is an advantage when the context colour is the same as the colour of either the colour of the text or the background on the screen. The combination of context and text/background colours must match against each other. If they clash the reading effort increases. Further experiments (op. cit.) with altering ambient light levels showed that this is of no or very limited importance for the perceived reading effort.

Numeric data on the screen

Computer graphics hardware and software have become widely available. In advertisements it is often stated that business graphics communicate the information effectively, thus being very useful. However, in real-life situations graphics often tend to be very poorly designed. Thus they may fail to improve the communication. Sometimes bad design might even make communication difficult or even impossible. Bertin (1967), Cossette (1982), McCleary (1983), and Pettersson (1983, 1989) all discussed the importance of individual design variables in visual

language. However, these discussions are all based on research on traditional print media. Ehlers (1984) points out problems of legibility in business graphics. According to him, direction and pattern and texture of graphic elements appear to be important factors as well as colour and size.

The CLEA-laboratory studies of attitudes to different variables in the presentation of information on visual displays (Pettersson et al., 1984b) showed that subjects consider it easy to see the difference between vertical bars as well as between horizontal bars.

Further experiments by Fahlander and Zwierzak (1985) have shown that the greater the difference is between the colour in a graphic presentation, the more distinct is our perception of the border between the colour spaces. On white background the following colour combinations are suitable to use: black combined with yellow, yellow-red, red, blue, green or the mixtures of red-blue (magenta), blue-green (cyan), and green-yellow. On black background the following colour combinations are suitable to use: white combined with yellow-red, red, blue, or the mixtures of red-blue. On white as well as on black background the following combinations are suitable to use: yellow combined with red and blue, red combined with blue-green or green-yellow, red-blue combined with green or green-yellow.

Following this study Azoulay and Janson (1985) found that some colours used in business graphics have much higher aesthetic values than others. Blue, red, and green are liked the most. In an effort to find some more detailed knowledge about our perception of business graphics, two comprehensive experiments were carried out with respect to relationships between variables and parts of a whole (Pettersson & Carlsson,

1985). The findings, based on more than 2,300 individual assessments, were conclusive in the following points:

- Graphical information is good in conveying a survey of a situation.
- When relationships between variables are presented, comparisons of lengths give the best results.
- When parts of a whole are presented, comparisons of areas can be used as well.
- Design of graphic elements is important to consider. Most available patterns are probably less good. Patterns should be subdued and not disturbing.
- Colours like blue, red, and green are liked very much but they do not improve our possibility of reading the message accurately.
- Different parts in graphic figures should have about the same luminance and radiance. The true differences between areas can be hard to see when shaded differently.
- When accuracy is needed, graphical information should be combined with actual figures.

"Choosing the Right Chart" ISSCO (1981) supplies 21 practical guidelines and pointers on effective chart design such as: "Make bars and columns wider than the space between them". In a study Ek and Frederiksen (1986) used the CLEA-equipment to find out about effective chart design. Forty subjects assessed the difference in size between two bars in a bar chart with six bars. The bars had one of three possible widths (1/60, 1/30, and 1/15 of the screen width). The distance between the bars had one of six possible values, from zero to more than twice the bar width. The bar charts were produced and displayed at random and al-

ways presented with blue bars on black background. The findings, based on 3,600 individual assessments, showed:

- The bar width has no influence on our perception of size.
- The space between bars has no influence on our perception of size.

This study confirmed earlier findings (with the perception of vertical lines). It can be concluded that we can make screen design according to aesthetic appeal. Finally, it may be stated that it is extremely easy to convey misleading information about statistical relationships by using misleading illustrations. Those who are serious in their work should seek to avoid these mistakes.

Visuals on the screen

Our perception of visuals on visual displays are of course to a large degree dependent on the quality of the screen, especially when pie charts are used. European videotex terminals simply cannot reproduce a pie chart since the graphics resolution is only about 5,000 graphical elements. An ordinary television image consists today of about 250,000 image points or picture elements which vary with respect both to grey scale and colour information. Hayashi (1983) reported on the development of High Definition Television (HDTV) in Japan. HDTV uses 1125 scanning lines and can contain five or six times more information than the present NTSC standard colour television system with 525 lines. HDTV developments of flat plasma screens will also give increased technical possibilities for better perception of the visual information.

Windows on computer displays

General design rules should be employed also in the design of screen displays with information that is intended to be read on the screen rather than printed out on paper. Thus, material should be arranged and displayed so that it is easy to read: from top to bottom and from the left to the right.

For good legibility text in windows on a computer display may be set in 14 point Helvetica or New Century Schoolbook with 16–18 point line spacing, and an additional six points of space after each paragraph. The line length may be up to 120 millimetres wide. A window heading may be set in 18 points bold Helvetica with 24 point line spacing and an additional six points of space after the paragraph. For text in illustrations, I recommend 10–12 point Helvetica. Blank space in printed material increases cost, since more paper is required. Thus, it is not often used. However, colour as well as blank space on a visual display are essentially free and might be used to increase legibility and readability.

The layout of an "ideal window for a computer screen" should be adapted to the need of showing both separate video pictures and video sequences. An ordinary television screen corresponds with a 60 to 90-millimetres-wide picture on a computer screen. Thus the window should be at least 90 millimetres wide. The layout of the window may be adapted to well-known picture sizes, such as: overhead transparencies in accordance with SIS, with the height-width ratio of 1:1,29, television (1:1,32), standard film (1:1,33), European A and G series for paper (1:1,41), Cinema U and IMAX (1:1,44), the effective picture area of a slide (1:1,53), the golden section (1:1,62), Vistavision Europe (1:1,85), Showscan (1:2,18), and Cinemascope (1:2,35).

The layout of an "ideal window for a computer screen" should be adapted for easy transfer of traditional, and page related electronic documents into an electronic document. Moving pictures between different formats and different media should be as easy as possible: documents on paper – documents on computer screens – overhead transparencies – slides – video. It is a good idea to only use a limited number of document templates with agreed standards for typography and layout. The agreed layout limits the size of pictures. The picture width is the most critical; it is often limited by the text column width. The picture height is limited by the column height, usually the height of the paper reduced with upper and lower margins. Sometimes, there are problems with the picture widths, but almost never with the picture heights. The window height should be adapted to the normal text height, and the height of the most common pictures including captions.

In search of an "ideal window for a computer screen" a summary of all these criteria, the lowest common denominator, results in an information window (143 x 190 millimetres) with a text and picture area, an "information area," (120 x 160 millimetres). This active text and picture area has the same side ratio as standard film (1:1,33), and almost the same as a television screen (1:1,32).

Each information material may be regarded as a reel of film in a feature film, with film-frames instead of pages in a book. Contrary to the situation with film, we can with the help of hypertext links move as we wish, from frame to frame in the complete information material. The window is designed from the side ratio 3:4 (1:1,32). This ratio is the basis for the size of the hypertext keys (8 x 6 millimetres), for the distance between the buttons (three millimetres), for the placement of the buttons

(six millimetres from the information area and three millimetres from the borders), and for the placement of the identification texts (six millimetres under the information area and three millimetres above the border). The space above the information area is totally nine millimetres. Below the information area, to the left, there are a number of buttons for hypertext-linked steps to each section (level) in the information material. To the right there are a number of hypertext tools: go to the first page, go one page back, go one page ahead, go to the previous page, go to the page with information on originators, go to list of pictures, go to page with copyright information, go to page with a list of illustrations, go to page with an index and hypertext links to the complete material, go to the page with help and explanations, and close the document.

Above the information area there is a field with administrative information, such as the company name, the title of the information material, the document number, and the current number of the "frame" or page. This information only has a limited value in the electronic document, where all the administrative information may be put together in the last frame. However, this information is important when printing one or more frames. On an A4 paper there is space enough to print one frame. The window may be used for one column (160 millimetres) and also for two columns (40 + 120 millimetres). The header levels number two and three may start to the very left of the window, that is, in the narrow column. Small pictures, symbols and notes may also be placed here. The full width of the window may be used for large pictures.

In one design experiment as much as 93% of all the pictures in a major system description project could be moved directly into such an electronic document. The texts in the pictures,

however, needed to be enlarged from nine to ten, or preferably to a twelve point linear typeface such as Helvetica. However, the information in an information material may well be divided so that the text is shown in one window and all the pictures in another, similar window on the computer screen.

The picture height will be decisive for the use of pictures. There will be some "empty" space both to the left and to the right of the picture. After appropriate enlargement, according to an enlargement table, both line width and texts usually need to be revised. If we want to work with text only in a window, and present all the pictures in a separate window next to the text window, we may use a narrower variation for the text. Such a "text window" may measure 143 x 150 millimetres, and have an "active" text area of 120 x 120 millimetres. This window will only contain a text column, with a column width of 120 millimetres.

In WWW the individual user is able to define the size of the display window at will. To a large degree the user can also decide on his or her own preferred typography. Although the typographic possibilities are limited in HTML-documents (compared with SGML-documents) it is still possible to provide a clear structure and make it possible to navigate within the document as well as between different documents. A consistent use of size and weight in headings is important. This should be combined with a consistent use of empty space to further enhance clarity.

Keeping a list of contents with links to chapters or sections in the current document (or group of documents) in the "list of contents area" makes it possible to maintain an understanding of the structure of the present document (or group of documents). In the "text and picture area" it is possible to succes-

sively view all the available information. A specific area for administrative information can be placed at the bottom of the text and picture area. This information will be printed when the document is printed.

Computer print-outs

In visual displays, an additive mixture of red, green, and blue produces the colour image. However, in the production of hard copies, computer printouts as well as in painting, printing and also colour photography a subtractive method of combining inks, dyes and pigments is used. Most colours can be generated in printing with the use of yellow, cyan (blue-green) and magenta (red-blue). Together these primaries produce black. However, pure black is often included as a fourth printing colour because the three primaries that produce the best chromatic colours usually do not produce the best black. Subtractive systems begin with a white surface. Colours darken as more wavelengths are absorbed. (However, in some colour hard copy printing, both additive and subtractive colour combinations can occur.)

The difference in the production of colours creates some problems. It should also be remembered that colour coding will lose its meaning when monochrome printers, as well as displays, are used. There are many hard copy systems and possibilities to make computer printouts for text as well as for pictures.

In a study (Pettersson et. al., 1984c) printouts from line, matrix, and daisy wheel printers were used. These eight samples were numbered and showed at random to 40 subjects who judged their perceived reading efforts of the different texts. The results showed that:

- Only the two daisy wheel printouts and a normal font, normal mode matrix printout cause little reading effort and thus are easy to read and quite acceptable.
- Reading efforts of print-outs are independent of the sex of the subjects

It was concluded from this study and the previous findings (Pettersson et. al., 1984a) that a text presented in a good colour combination on a visual display was easier to read than printouts from several printers used. Today, however good quality laser printers are often used.

Layout

A finished layout is the practical result of the work in the graphic design process. The purpose of this work is to find a suitable presentation for the content with respect to the receiver, the subject matter, the medium, and the overall financial situation. Within a given area – such as a page in a book, a poster, or a label – the graphic designer may arrange and distribute text, pictures (drawings and photographs), and the background (margins, space, patterns, and designs without any significant picture elements). Layout provides a large number of possibilities to make the structure in a document clear (Benson, 1985; Tinker, 1963, 1965). The graphical form should help the reader to benefit from the contents of a document.

Paper size

Traditionally, a large number of different paper sizes were defined for large sheets of paper. Local paper sizes were defined by the name of the sheet and by the number of times it had been folded. Different paper sizes were used for different purposes. The aspect ratio varies a lot in many standards for paper sizes.

Today there is one widespread international ISO standard and a local standard used in North America. Paper sizes affect cards, envelopes, printed documents, stationary, and writing paper.

All *ISO paper sizes* are based on a single aspect ratio of the square root of two, approximately 1:1.4142. The base for the system is the A0 paper. An A0 (A zero) paper has an area of 1 m^2, with the sides 841 and 1189 millimetres (33.1 in × 46.8 in). The *A-series* can be traced back to a French law from 1798 for certificate papers.

Successive *A series* paper sizes are defined by halving the preceding paper size along the larger dimension. The system with scaling is a significant advantage. The most frequently used paper size is A4 210 by 297 millimetres (8.3 in × 11.7 in). A standard A4 sheet made from standard 80 grams per m^2 paper weighs 5 grams.

Size A0 is 841 × 1189 mm, A1 594 x 841, A2 420 × 594, A3 97 × 420, A4 210 × 297, A5 148 × 210, A6 105 × 148, A7 74 × 105, A8 52 × 74, A9 37 × 52, and A10 26 × 37 mm. The number after A accounts for the number of folds.

The *B series* is less common than the A series. The area of a B series sheet is the geometric mean of successive A sheets. The B-series is widely used in the printing industry. Many posters use B-series paper or a close approximation, such as 50 cm × 70 cm.

Size B0 (B zero) is 1000 × 1414 mm, B1 707 × 1000, B2 500 × 707, B3 353 × 500, B4 250 × 353, B5 176 × 250, B6 125 × 176, B7 88 × 125, B8 62 × 88, B9 44 × 62, B10 31 × 44 mm.

The *C series* is used only for envelopes. The area of a C series sheet is the geometric mean of the areas of the A and B series sheets with the same numbers. A letter that is written on A4 paper fits inside an envelope in size C4.

Size C= (C zero) is 917 × 1297 mm, C1 648 × 917, C2 458 × 648, C3 324 × 458, C4 229 × 324, C5 162 × 229, C6 114 × 162, C7 81 × 114, C8 57 × 81, C9 40 × 57, and C10 28 × 40 mm.

In Sweden the ISO system of A, B, and C formats is supplemented by adding D, E, F, and G formats. G5 (169 × 239 mm) and E5 (155 × 220 mm) are popular for printing dissertations.

The most commonly used paper sizes in North America are called *Ledger, Legal, Letter*, and *Tabloid*. A ledger size paper is

432 x 279 mm (17 x 11 in). A legal size paper is 215.9 x 355.6 mm (8.5 x 14 in). A letter size paper is 215.9 × 279.4 mm (8½ × 11 in). Letter size is also known as American Quarto. A tabloid size paper is 279 x 432 mm (11 x 17 in). Note that a ledger size paper and a tabloid size paper have the same size, but ledger is vertically oriented and tabloid is horizontally oriented.

Arch is a series of paper sizes used in architecture. In this series the aspect ratios vary, 4:3 and 3:2 are common. Five successive paper sizes are defined by halving the preceding paper size along the larger dimension. Here the aspect ratio varies between 3:4 and 2:3.

Arch A is 9 x 12 in (229 x 305 mm), Arch B 12 x 18 in (305 x 457 mm), Arch C 18 x 24 in (457 x 610 mm), Arch D 24 x 36 in (610 x 914 mm), Arch E 36 x 48 in (914 x 1219 mm), Arch E1 30 x 42 in (762 x 1067 mm), Arch E2 26 x 38 in (660 x 965 mm), and Arch E3 27 x 39 in (686 x 991 mm).

Page composition

An “empty” page can be considered as an available area or space. This area may be and should often be used in different ways. In all printed matter, space can be used to convey the structure of the information. The information can be grouped in various ways. There may be plenty of white space on a printed page. Headings, margins and “empty” space can be used to aid communication when used in a consistent way (Hartley & Burnhill, 1977a, 1977b).

Learners are most able to build connections between verbal and visual representations when text and illustrations are actively held in memory at the same time. This can happen when text and illustrations are presented in close connection on the same page in a book, or when learners have sufficient experi-

ence to generate their own mental images as they read the text (Mayer et al., 1995).

According to Baggerman (2000, p. 64) most designers don't consider print design in terms of its interface. However, since desktop publishing has revolutionised the options for page layout, the professional designer must use the interface of print design in order to connect directly with the user.

Page size

In traditional as well as in desktop publishing the available paper defines the possible page sizes. The use of standard page sizes can aid communication. Hartley and Burnhill (1977b) are strong advocates of international standard paper sizes. Their own experimental materials are printed on A4 paper. We may choose from different kinds of page formats. A quadratic format is static. It is often considered uninteresting and it is seldom used. Wide formats correspond with our vision. In printed materials wide formats may be used to present pictures in large sizes. Tall formats are very suitable for presentation of printed text and are the most common format for books. Classical formats are based on the proportions of the golden section, 3:5, 5:8, 8:13, 13:21, 21:34, etc. Each new number in the scale is the sum of the previous two numbers. Many experiments have shown that there is an optical centre on a page.

In our daily life we often use A4 paper – with a width-to-length ratio of 1:1.41 – for copied/printed matter. The text area (the portion of the page that will contain text and pictures) should be large enough, in order for the information to fit in. To satisfy a requirement for single-sided printouts that can be inserted into a loose-leaf binder, while at the same time being able to make double-sided printouts that can be bound in book form,

right-hand and left-hand pages should be identical. In a traditionally printed book, however, we can use different inner and outer margins, and adapt the page layout for even more convenient reading.

If one wants to fill the whole width of the page with text, the text must be written in a large type, with ample distance between each line. It is preferable to have narrower lines than that. If pictures are used, it might be suitable to allow them to fill the entire text area from margin to margin. Necessary administrative information, the chapter title, and the page number may be provided in the lower or upper margins.

In documents with only text, or with text and very few pictures, two columns of text may function very well on an A4 page. In such a layout, a column width of 80 millimetres with 12 points text is quite satisfactory. However, in documents that have plenty of pictures, two or more columns will pose some difficulties and demand a good deal of extra work. As long as a document is subject to recurring amendment, supplementation and other updating measures, a layout with only one column is preferable.

Text face

The appearance of a book page is governed by the fact that Western readers begin at the top left of a page and read to the right, one line of text at a time until they get to the bottom of the text column. Often the main part of the page is the text-face. Depending on the page size the page may be used for one or more columns of text and pictures. Hartley and Burnhill (1977a, 1977b) have done extensive research on psychological research on typography based on basic principles of typographic deci-

sion-making. They worked with the following three main principles:

1. Use typographic space in a consistent way in order to convey the structure of the information.
2. Use standard page sizes.
3. Use grids for pre-planning of pages.

By grouping headings, paragraphs, illustrations, and captions the designer aids communication. An “empty” line might separate paragraphs, two lines subsections, and three or four lines sections. In this process the principles from the Gestalt theory can be utilized. When we use space to group graphic components, we employ the “proximity principle” or the “proximity law”. When we use a consistent type to signal a particular kind of graphic component, we employ the “similarity principle”. When we use grid systems these are based on the “closure principle,” together with the “continuity principle.”

The classic model for page design

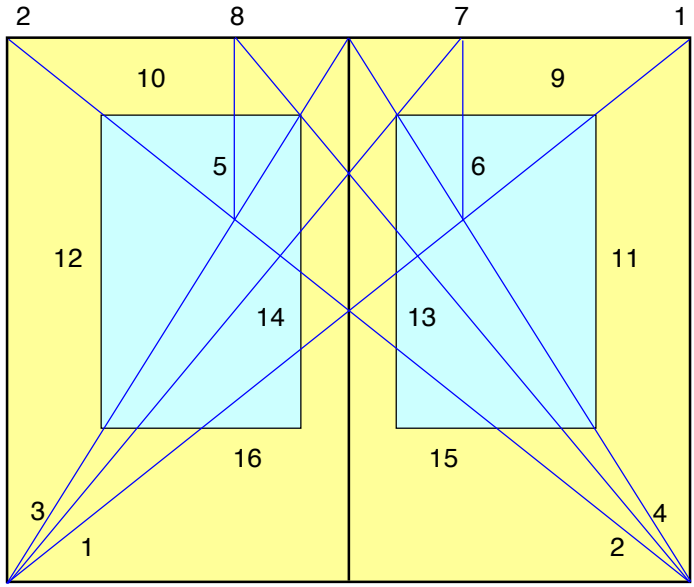
White or “empty” space on a page can contribute to the creation of a harmonious and functional product. An old method of deciding the appropriate sizes for margins on a two-page spread is based on geometry:

1. Draw a diagonal from the lower left corner of the left-hand page to the upper right corner of the right-hand page.
1. Draw a diagonal from the lower right corner of the right-hand page to the upper left corner of the left-hand page.
2. Draw a diagonal from the lower left corner on the left-hand page to the upper right corner on the same page.

3. Draw a diagonal from the lower right corner on the right-hand page to the upper left corner on the same page.
4. Draw a perpendicular to the top of the page from the point where the two diagonals cross on the left page.
5. Draw a perpendicular to the top of the page from the point where the two diagonals cross on the right page.
6. Draw a line from the point where the perpendicular meets the top of the right page, down to the lower left corner of the left page.
7. Draw a line from the point where the perpendicular meets the top of the left page, down to the lower right corner of the right page.
8. Draw a line to the right, parallel to the page edge, from the point where diagonal 4 and line 7 cross, to diagonal 1. This line will establish the top margin on the right page.
9. Draw a line to the left, parallel to the page edge, from the point where diagonal 3 and line 8 cross to diagonal 2. This line will establish the top margin on the left page.
10. Draw a line, parallel to the page, from the point where diagonal 1 and line 9 cross, down to diagonal 4. This line will establish the right margin on the right page.
11. Draw a line, parallel to the page, from the point where diagonal 2 and line 10 cross, down to diagonal 3. This line will establish the left margin on the left page.
12. Draw a line, parallel to the page gutter, from the point where line 7, line 9 and diagonal 4 cross. This line will establish the left margin on the right page.
13. Draw a line, parallel to the page gutter, from the point where line 8, line 10 and diagonal 3 cross. This line will establish the right margin on the left page.

14. Draw a horizontal line from the point where diagonal 4 cross line 11 to the left. This line will establish the bottom margin on the right page.
15. Draw a horizontal line from the point where diagonal 3 cross line 12 to the right. This line will establish the bottom margin on the left page.

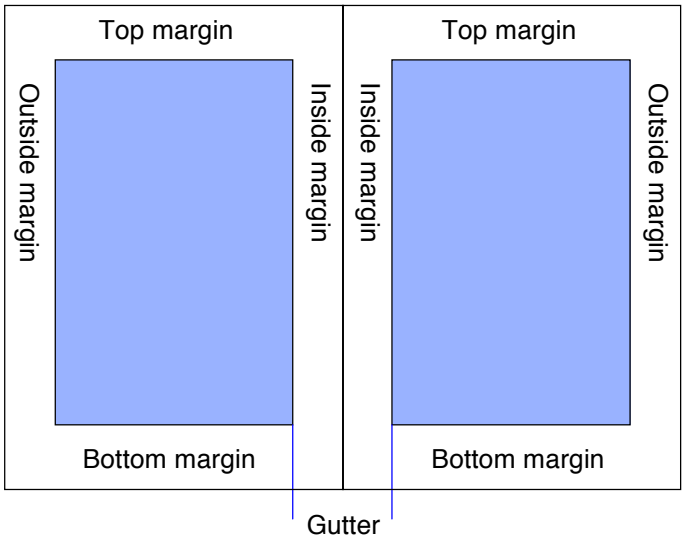
As can be seen in the illustration on the next page this method creates large margins, with the text covering only one-third of the page. West (1987) called this model “ideal” and “classic”.



This is an illustration of the construction of the classic model for page design.

Margins

Margins have three main functions. They provide space for comments, headings, illustrations, page numbers and personal notes. They provide space for fingers to hold a document while reading it. They make it possible to make notes.



The classic model for page design creates very large margins.

Surrounding of white space

The text-face is surrounded by margins. There are four margins; a header (or top margin) and footer (or bottom margin), an inner, and an outer or outside margin. Taylor (1960, p. 112) noted, "a surrounding of white space will bring attention to any object being surrounded". According to Waller (1987), the observations on which the Gestalt theory is based form a basic part of the graphic designer's craft knowledge. These principles might be seen as relatively inflexible perceptual rules that act as a fundamental constraint for the typographer alongside such con-

ventional rules as the left-to-right direction of the writing system.

Footers may appear at the bottom of every page. They should provide information that will help the reader navigate in a document. Footers often carry page numbers. There may also be footnotes.

Headers, or *running heads*, may appear at the top of every page. They should provide information that will help the reader navigate in a document. In some books every left-hand page have the title of the book, and every right-hand page have the name of the chapter. Headers may also carry page numbers.

Size of margins

A general guideline may be to make all margins one inch on a standard page (Bradshaw and Johari, 2000). However, there are a number of exceptions to this guideline and margins. Usually margins differ in size.

According to Hartley (1985, 1994), Lichty (1989, 1994) and Misanchuk (1992) readers expect margins to occupy 40–50% of a standard 8 1/2" x 11" page, although this amount can be reduced in professional or scholarly texts. Some guidelines call for even larger margins (Burns, Venit, and Hansen, 1988).

Guidelines for margins vary a lot (Ander, 2003; Bergström, 1998; Berndal & Frigyes, 1990; Bohman & Hallberg, 1985; Hellmark, 2000a, 2000b; Koblanck, 1999; Lohr, 2003; Misanchuk, 1992; Pettersson et al., 2004; White, 1983). Recommendations for top margins are often between 14 – 35 millimetres, for right margins 6 – 53 millimetres, for the bottom margins 14 – 34 millimetres and for the left margins 8 – 28 millimetres.

With very thick books, the part of the page that is bound will have a section that we cannot see. Thus, thick books need to

have wide inner margins. This part of the page is called the gutter. On a two-page spread the right side of the left page is the gutter, and the left side of the right side is the gutter.

Page numbers

In all kinds of reference materials, it is important to have page numbers, folios. The reader can find information in the material by using the table of contents and/or the index. Page numbers should be clear and easy to find. There are many possibilities for placement of the page numbers.

Page numbers can be put to the left, in the middle, or to the right in the header or in the footer. Page numbers are sometimes also put in the margin. In products with large page sizes, it is usual to put page numbers in the header. In products with small page sizes, it is quite common to put page numbers in the footer. Usually readers are likely to look for page numbers in the margin at the bottom of the page (Lipton, 2007, p. 134). However, this should not to be considered a rule. The important thing is consistency.

Page numbers indicate arbitrary divisions of the text. For technical reasons many books contain more than one series of page numbers. For example, technical manuals and instructional materials often use a separate numbering series for each chapter. Then a single chapter can be updated without reprinting the whole manual. The running head includes the name of the publication, the issue date, or other header or footer style material.

Grid systems

Look at different books, reports, magazines, and papers and study the variety of page sizes and page designs. In many books

the page has only one column, which is very good for undisturbed and easy reading. However, a wide page has to be divided. Too wide lines impair reading. Book pages with two columns are quite common. Even three columns can be used, e.g., in dictionaries and reference books. Pages with four or more columns are very rare. One-column layouts are the most used layouts in instructional materials. It is easy to work with, especially when materials may have to be revised frequently. Davies, Barry, and Wiesenberg (1986) suggest that one, two, or three columns per 8 1/2" x 11" page are reasonable for reports, papers, and other similar documents.

In magazines it is common that different sections have their individual page designs. One section may have three columns, another section four columns, and a third section five columns. Pages with six or more columns are very rare. In newspapers it is common that a page has six, seven, or eight columns. Newspaper pages with four or five as well as with nine or ten columns may also be used. Pages with eleven or more columns are very rare. It is very easy to create complicated typography and page layout. However, simplicity and consistency are very important. Complexity usually interferes with communication. Readers should not be forced to search on the page for headings and page numbers. Typographic grids may be used to ensure that space and print are used consistently. It is important that the printed page provide a reliable frame of reference from which the learner can return without confusion. By dividing a page into smaller sections, a specific topic may be presented in a variety of combinations of grid sections. A grid establishes a structure to build upon. There are different kinds of grid systems. We can distinguish between one-column grids, multicolumn grids, modular grids, and irregular grids. Many books

and other documents use a standard page with a fixed text-face, a live area. Type and other graphic elements occupy the text-face. The grid establishes the location of the text-face, the inner and outer margins, the header, and the footer on each page.

Traditionally graphic designers work with grid sheets printed in light blue ink. The “non-repro” blue ink will not show up on films, printing plates, or on the final print. The grid is used to ensure that type and graphic elements are pasted on correct locations. In electronic page layout systems the software can create electronic grids. They are called master pages. A master page is shown on the screen but does not print. In columnar grids the text flows from one column to another. Dictionaries and telephone directories are examples of simple columnar grids. In parallel columnar grids the text flows in columns that are related horizontally. This system is sometimes found in multilingual manuals with the same text in different languages.

The modular grid is the classical Swiss grid. It is based on regular rows as well as columns. Often a modular grid system is used for “blocked grids”. For example, it is possible to use columns with different widths to present texts and illustrations. Usually “topic frames” are rectangular. The purpose is to create clear visual gestalts. Many tabloid newspapers use irregular grids. The topic frames are often not rectangular. Editorial stories are mixed with advertising items. The purpose of irregular grids is actually to prevent clear visual gestalts. The reader can never get an understanding of the page at only one glance.

Effective grid-based design requires both careful planning and a willingness to adjust to the content itself when this is needed. Fitting the content to the grid should be seen as a way of regularizing the information and increasing its internal consistency in order to improve communication.

Oppositions

By tradition page layout is based on symmetrical page design. Centred as well as justified texts are examples of symmetrical layout. Symmetry is axial balance. It is logical and simple to design. Symmetrical page design is static and may be perceived as boring. An asymmetrical layout can be based on contrasts of size, strength, shape, area, or colour. Unjustified text flushed left or right are examples of asymmetrical layout.

Symmetrical as well as asymmetrical layouts can have a good balance. The favouring of uniformity in the use of design elements is *regularity*. The opposite, called *irregularity*, emphasizes the unusual and unexpected. A layout may be built on simplicity using few and simple elements and simple forms.

The opposite strategy is based on complexity. Dondis (1973) discussed the use of several pairs of oppositions as techniques for visual communication. These oppositions are:

Balance	–	Instability
Symmetry	–	Asymmetry
Regularity	–	Irregularity
Simplicity	–	Complexity
Unity	–	Fragmentation
Economy	–	Intricacy
Understatement	–	Exaggeration
Predictability	–	Spontaneity
Activeness	–	Stasis
Subtlety	–	Boldness
Neutrality	–	Accent
Transparency	–	Opacity
Consistency	–	Variation
Accuracy	–	Distortion

Flatness	–	Depth
Singularity	–	Juxtaposition
Sequential	–	Random
Sharpness	–	Diffusion
Repetition	–	Episodic

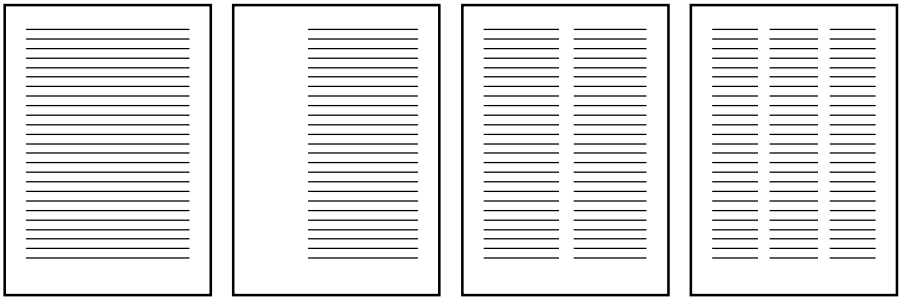
These different visual design techniques overlap and reinforce meaning. They present the graphic designer with effective means of making expressive visual communication.

Text layout

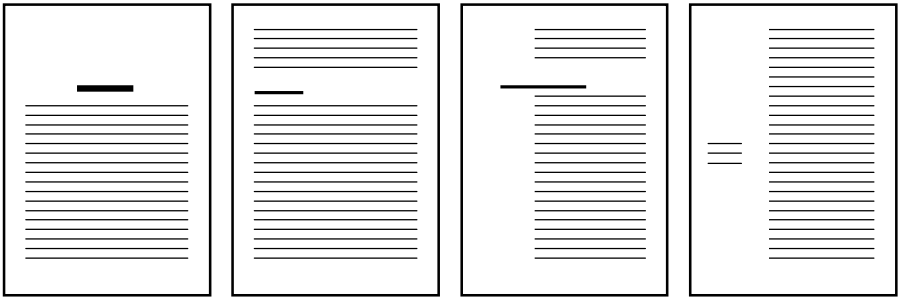
Lupton and Miller (1999, p. 33) started their chapter about *Period Styles* in this classical way:

GREEKANDLATINMANUSCRIPTSWEREUSUALLYWRITTEN
WITHNOSPACEBETEENWORDSUNTILAROUNDTHENINT
HCENTURYADALTHOUGH.ROMAN.INSCRIPTIONS.LIKET
HE.FAMOUS.TRAJANCOLUMN.SOMETIMES.SEPARATED.
WORDS.WITH.ACENTERED.DOT. ...

Today graphic designers use typographic variation to present the content in a text in a clear way. Reference materials, such as telephone books and dictionaries are examples of highly structured information. Here, a carefully thought-out, functional layout can facilitate the reader's ability to find the desired information quickly, easily, effectively, and reliably.



There are vast possibilities of text design. The whole page may be used for one column of text. The left or the right margin may be wide to leave space for notes. The text-face may also be divided into two or more columns.



The text may start down the page in each new chapter. Sub-headings are often located within the text column, but they may poke out in the margin. There may also be printed notes in the margin.

Text must have a typography that facilitates its legibility. Headings, sub-headings, main text, captions, boxes, summaries, etc., must be clearly distinguished from one another. For reports, and similar documents, it is reasonable to use one, two, or three columns on the page (Hartley & Burnhill, 1977a, 1977b). Consistent use of columns will help to establish a regular pattern throughout a project (Lipton, 2007, p.120).

In typography a “widow” is the first line of a paragraph alone at the bottom of a column. Sometimes a widow refers to the last line of any paragraph with only a few words. The first line of a paragraph should not, as a rule, begin on the last line of a text column. Similarly, the last words in a paragraph should not be placed alone at the top of a new page or a new column. Many word processing programs automatically eliminate awkward phenomena like these. An “orphan” is the last line of a paragraph when it is alone at the top of a column.

In some documents lines need to have numbers. Non-arbitrary numbering systems include the numbering of lines where line endings are meaningful. This might be the case in computer programs, in dictionaries, and in texts used for linguistic analysis. Non-arbitrary numbering systems also include the numbering of paragraphs and the numbering of headed sections.

Justified or unjustified text?

A text may be justified or unjustified. An unjustified text can be flushed left, centre justified, or flushed right. There are some advantages and some disadvantages with each system. The following pages include four examples of the same text fragment in the four different versions: justified text, centre justified text, flushed left text, and flushed right text.

Justified text

A number of authors argue that *justified text* (sometimes called right-justified text or full text) is aesthetically pleasing and that it is easier for people to read lines of the same length than reading lines with markedly varying right-hand ends (Lang, 1987;

Lichty, 1994). Justified text is commonly used for running text in books, magazines, and newspapers.

To achieve a justified text the technical system inserts extra space between words and between characters. According to Kleper (1987). The practice of having text justified is a tradition in the publishing industry that originated with the scribes who copied text by hand before the invention of movable type. At that time paper was expensive and the scribes attempted to put as much text as possible onto every page, filling each line completely.

The blue text fragment below is justified

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum communication between a sender and a group of receivers. Infology models contain both theoretical (descriptive) elements as well as normative (prescriptive) elements.

Burns, Venit, and Hansen (1988) pointed out that the use of metal type required that the right edges of type align in the page form so that the type could be locked into place. Until recently most publishers had regarded anything other than justified text as unprofessional. Lang (1987) argued that justified text is aesthetic, it serves to define the right margin, and it is familiar to the readers. Also Lichty (1989) argued that it is easier to read lines of the same length than lines with markedly varying right-hand ends. According to Machin (2007) justified text may be seen as authoritative, efficient and formal.

Hartley (1985) and Hartley and Burnhill (1977a) argued that justified text is not a good idea for instructional materials. The variable spacing between words as well as the use of hyphenation makes reading less smooth and more difficult.

When justified text is set in too short lines, there will be "rivers of space" between words, or characters spaced out to fill the lines. Justification should be avoided with narrow columns in multi-columnar layouts (Davies, Barry, and Wiesenberg, 1986). Misanchuk (1992) claims that there are no good arguments at all for using justified text.

Centre justified text

Symmetrical centre justified texts may be seen as formal, grave and momentous ((Machin, 2007). However, without symmetry centre justified texts may be seen as energetic, informal and playful. Symmetrical centre justified texts are often used for menus, quite often used for poetry, and sometimes used for short captions and tables of contents. In films and in television programs the participants are usually listed centre justified.

This blue text fragment below is centre justified

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum communication between a sender and a group of receivers. Infology models contain both theoretical (descriptive) elements as well as normative (prescriptive) elements.

Flushed left text

Today flushed left text, or *unjustified text*; with a "ragged–right–hand" edge is commonly used for running text in books, magazines, and some newspapers. A century ago flushed left text was restricted to poetry. At that time it was considered very odd and peculiar and most publishers regarded the use of anything other than justified text as unprofessional. According to Machin (2007) flushed left text may be seen as less formal and more relaxed than justified text.

Some authors argue that flushed left text is a much better choice than justified text (Hartley, 1994; Misanchuk, 1992). The exact spacing between letters and between all the words in unjustified text retains the optimal spacing between letters and between words and so keeps the visual rhythm constant. This aids reading, especially for young, inexperienced and poor readers (Gregory and Poulton, 1970; Zachrisson, 1965).

This blue text fragment below is flushed left

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum communication between a sender and a group of receivers. Infology models contain both theoretical (descriptive) elements as well as normative (prescriptive) elements.

Trollip and Sales (1986) experimented with justified and flushed left text. Results indicated significant increase in reading time for groups reading justified text. No differences in comprehension were detected. Poor readers have difficulty

reading justified text (Gregory and Poulton, 1970; Zachrisson, 1965). However, whether the text is justified or unjustified causes no significant difference in search time and comprehension of the information for advanced readers (Hartley, 1987).

However, many readers find justified text much more aesthetically pleasing than flushed left text. They may even feel that the ragged right is ugly and repulsive.

Flushed right texts

Flushed right texts can be used for captions that are positioned to the left of the pictures, and for tables of contents. This is, however, only possible when the line length is short. Since flushed right texts demands more effort to read they suggest more confidence than justified texts (Machin, 2007).

This blue text fragment below is flushed right

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum communication between a sender and a group of receivers. Infology models contain both theoretical (descriptive) elements as well as normative (prescriptive) elements.

Regardless of justification system the ends of sentences should be determined by syntax rather than by an idea of a set width of the line (Hartley, 1980; Bork, 1982).

Line length

It is easy to change the *length of lines*. According to Tinker (1963), readers tend to dislike both very short and very long line

lengths. Tinker made extensive studies of typography. He worked with characters in sizes of nine to twelve Pica points and recommended ten to twelve words per line. This results in a line length of eight to ten centimetres. There are, however, several other recommendations of line length.

West (1987) suggested that the line length should be 35–40 characters. Quite often the optimum line length seems to be about 1 1/2 alphabets – 42 characters (Pettersson, 1989; Walker, 1990). This is nine to eleven centimetres with optimum character size, ten to twelve points, at a normal reading distance. Lipton (2007) suggested 40–50 characters and Parker (1988) 50 characters. Burns, Venit, and Hansen (1988), and Zwaga, Boersma and Hoonhout (1999) suggested up to 60 characters. Miles (1987) suggested 60–65 characters. A text column may be widened up to 120–130 millimetres to accommodate more text, and still be easy to read for an experienced reader. It is quite clear that too wide lines impair reading.

In my opinion the maximum line length should not be much more than 60–70 characters. This is except for books intended for highly skilled readers. The optimum line length should be found for each individual purpose and each audience. However, costs often force people to use more characters on each line, so that the total number of pages can be reduced. The longer the lines the wider the vertical space between them needs to be (Waller, 1987).

The length of a line will affect reading speed (Duchnicky and Kolers, 1983). Different kinds of publications should use different line lengths. The optimum line length should be found for each individual purpose and audience. The width of a line of type is traditionally measured in picas or Cicero. A pica, as well as a Cicero, equals 12 points. As systems for desktop publishing

become more common, line lengths will be measured in inches or centimetres instead of picas or Cicero's.

The line length may vary considerably. The longer the line is, the larger the type size should be. The shorter the line is, the smaller the type size can be. If justified text is set in lines too short, you get rivers of space between words, or characters spaced out to fill. Following on the next pages are examples of different line lengths with the same type size; too short lines, short lines, optimum line length, and long lines. All examples are given with unjustified as well as with justified lines. All examples use the same rules for hyphenation. In all examples the shortest prefix as well as the shortest suffix is set to three characters in UK English.

However, Zwaga, Boersma and Hoonhout (1999, p. xxii) argued that there is no such thing as the "best" column width or line length. The typographic features of printed matter should be chosen in relation to the requirements of the material to be printed. Line length can be relevant in the context of material consisting of sentences grouped in paragraphs, but as less important when there are also components such as tables, lists, diagrams and mathematical formulae. In these cases the constraints of a narrow column might be a disadvantage.

Too short lines 1

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and

Too short lines 2

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and

Short lines 1

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures,

Short lines 2

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures,

Optimum line length 1
Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures, will impair the understanding of the message. Active voice, clarity, comprehensibility, consistency, legibility, precision, readability, reading value, simplicity, and structure are the key concepts in information design. Any graphical message should be legible, readable, and well worth reading for the intended audience.

Optimum line length 1
Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures, will impair the understanding of the message. Active voice, clarity, comprehensibility, consistency, legibility, precision, readability, reading value, simplicity, and structure are the key concepts in information design. Any graphical message should be legible, readable, and well worth reading for the intended audience.

Long line length 1

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures, will impair the understanding of the message. Active voice, clarity, comprehensibility, consistency, legibility, precision, readability, reading value, simplicity, and structure are the key concepts in information design. Any graphical message should be legible, readable, and well worth reading for the intended audience.

Long line length 2

Producers of information and learning materials can facilitate communication, and the learning processes of the readers. Complicated language, in both texts and pictures, will impair the understanding of the message. Active voice, clarity, comprehensibility, consistency, legibility, precision, readability, reading value, simplicity, and structure are the key concepts in information design. Any graphical message should be legible, readable, and well worth reading for the intended audience.

As we can see from these examples, short lines cannot handle justified setting very well. The distance between the words is too long. We get white "rivers" of space in the text column. A line of 40 to 50 characters or "strokes" results in a line length of 75 to 90 millimetres in a book or a paper. A text column may be widened up to 120 millimetres to accommodate more text, and still be easy to read for an experienced reader. This column is 120 millimetres.

Interline distance

The "interline distance," "interline spacing," "line space," or "vertical spacing" is the vertical distance between the baselines in a text. In Times an 11-point text may be set on a 13-point line. This is written as 11/13, and read as "eleven on thirteen." The term "leading" refer to the extra space between the lines, the

”line–to–line” spacing. In this case the interline distance is 13 points, and thus the leading is two points. The leading may vary considerably between different kinds of texts. In this column the text is set with Georgia 10/16. Georgia needs more space than Times.

The term *leading* is derived from the days of hot metal type when thin strips of lead were placed between lines of type to provide line spacing. However, today the word *leading* is sometimes used to measure the vertical distance between the baselines in a text.

Extra space is important for legibility. As previously noted the longer the lines, the larger the vertical distance should be. The reader needs to be able to find the next line without any trouble. In general, the opinion is that as the line length is increased, the need for more leading and larger type increases (Lipton, 2007; Misanchuk, 1992; Pettersson, 1989). Here the x-height is important. Typefaces with small x-heights manage well with less leading than typefaces with large x-heights.

For maximum legibility of the running text in a book a leading should be between one to three points when text size and line length are optimal (Benson, 1985; Hartley, 1987; Kleper, 1987; Lichty, 1989; Lipton, 2007, p. 124; Pettersson, 1989, Tinker, 1963). Texts on wall charts and overhead transparencies need more space between the lines. Children and inexperienced readers need more leading than experienced readers. Typefaces with small x- heights manage well with less leading than typefaces with large x-heights.

Here are six examples of a text fragment with ten short lines. Georgia 12/10 is compared with 12/12. Georgia 12/14 is compared with 12/16, and 12/18 is compared with 12/20. For

better and easier comparisons the left example in each pair is flushed right and the second example is flushed left.

Georgia 12/10

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

Georgia 12/12

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

Georgia 12/14

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

Georgia 12/16

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

Georgia 12/18

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

Georgia 12/20

Infology is the science of verbal and visual presentation and interpretation of messages. On the basis of man's prerequisites, infology encompasses studies of the way a combined verbal and visual representation should be designed and produced in order to achieve optimum

It is obvious that there should be a little extra space between the lines in order to facilitate reading. Thus, a line with 12 point type needs an interline distance of at least 2 points (Times) or 3–4 points (Georgia). Generally speaking, one can use the type size plus 15–30% to determine this ratio.

Space

Space is an important tool in typography (Hartley, 1985, p. 27): "It is space that separates letters from each other. It is space (with punctuation) that separates phrases, clauses and paragraphs from each other; and it is space (with headings and sub headings) that separates subsections and chapters from one another." Consistent spacing in a document will help the readers to:

- Increase the rate of reading because they are more able to see redundancies.
- Access the more personally relevant pieces of information.
- See the structure of the document.

Space between words

Space between words varies in each line depending on the actual words and the actual line length. First the computer system adds word spacing and then, if the space between words becomes too excessive, the system will add letter spacing.

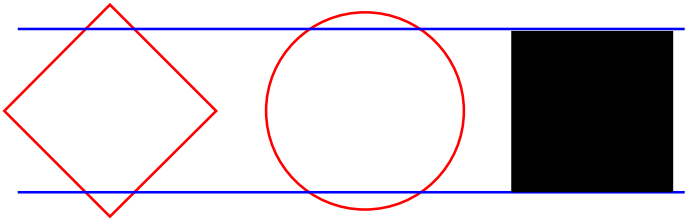
The distance between words shall be smaller than the distance between lines, and larger than the distance between characters. According to Walker and Reynolds (2004) a word space that is too wide or too narrow can ruin the design of a font. Space between elements should be used as a legibility tool (Lipton, 2007, p. 122). Text with a generous amount of space within it is rated as "easier" and "more interesting" than text that has a

more solid appearance. For comfortable reading it must be easy to distinguish between words. The distance between words should be relatively small. When the text has optimal spacing we can keep the reading rhythm constant.

Wendt (1979) inserted additional space between the constituents of the sentences to better convey the phrase structure of the text but found no difference in learning efficiency when compared with traditional text in one or in two columns.

Space between letters

The amount of space between letters in a text varies in each line depending on the actual words and the actual line length. Space between letters in text should not be too loose, or too tight. Visual design should be based on perceptual, rather than on physical phenomena. Compensation for shortcomings of our human vision is often required. Acute and rounded shapes need to be enlarged and extended in order to appear to be of equal height compared with a square. Visual alignment depends on careful optical adjustment to compensate for differences in shape of the element being aligned.



Acute and round shapes appear to be too low compared with a rectangle or a square.

Walker and Reynolds (2004) showed school children four texts with varying letter spacing and four texts with varying

word spacing. Results showed that more children noticed differences in letter spacing than differences in word spacing.

The distance between characters is sometimes too long. This is especially true for headlines in capitals. In traditional hand typesetting people used thin strips of metal, from 0.5 point upwards, for "correct" spacing of type.

In computer systems "empty space" is added automatically between the words and between individual letters when the text is set with justified lines. In photo typesetting it is possible to automatically and systematically increase the distance between characters in order to expand the text. It is also possible to reduce the distance between characters in order to condense the text. This is also possible in some systems for desktop publishing.

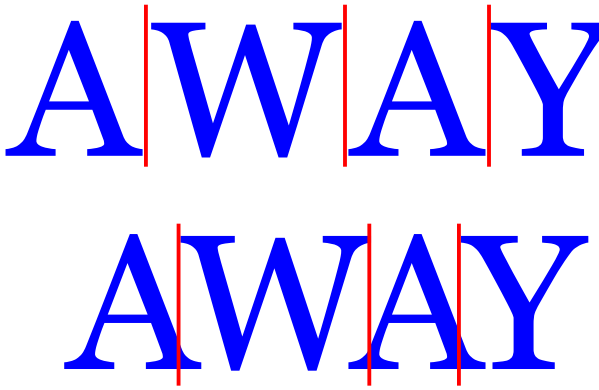
Kerning

In photo typesetting, and in some systems for desktop publishing, *kerning* is used to individually correct the distance between characters to achieve a better type. For example, when a capital A and a capital V are set without kerning, there is too much space between the letters. The A and V have slanted shapes, and the space between these letters is exaggerated.

With kerning, selected pairs of letters can be pushed together and overlap to create a better optical visual spacing between the letters. In optical spacing we need to equalize the area rather than the distance between characters. In typography, the tightest letter spacing is reserved for adjacent characters with curved edges or horizontally projected strokes. The widest spacing is allotted to character pairs with adjacent vertical strokes.

Hewson (1988) noted that it isn't worthwhile kerning any type under 18 points. Kerning is important for headings in

books, handouts, pamphlets, reports and other printed documents, and also for texts on OH transparencies, PP presentations and wall charts.



There may be too much space between the letters in large fonts (upper example). Here it is necessary to apply kerning (lower example) and correct the distances between characters to achieve a more legible type.

Hyphenation

Lines can be broken according to different principles. According to one principle, lines can be broken only between words. According to another principle, lines can be broken also within words. This can be done phonetically or according to etymology. In mechanical word breaks, lines are broken at the most convenient point, regardless of meaning. It is known that poor readers have difficulty reading hyphenated text (Gregory and Poulton, 1970; Lichty, 1989; White, 1983). Misanchuk (1992) provides guidelines for hyphenation.

Rules
Vertical and horizontal straight lines that are used in typography and layout are called rules. Horizontal lines can be used to separate sections in a text and rows in a table. Vertical lines can be used to separate columns of text on a page or columns in a table. It is; however, often better to use white space as a separating device. Rules wider than 12 points are called bars or bands. Sometimes horizontal bars have type in them. Rules may be in colour.

Headings

Headings, or *headlines*, attract the attention of the readers, enhance the hierarchic structure, make the subject matter readily apparent, and indicate the relative importance of items.

Show structure
Headings should be used to make the subject matter readily apparent, and indicate the relative importance of different items in the document (Cisotto and Boscolo, 1995; Jonassen, 1982; Mayer, 1993). Composition, intentional use of size, space, and placement of headings enhance the hierarchic structure in information material and help the readers to get the message (Jonassen, 1982; Wileman, 1993, p. 88). Headings should always be relevant and identify the subject matter. The purposes of headings are to attract the attention of the readers, make the subject matter readily apparent, and indicate the relative importance of items. Headings in large type may be printed in colour. In order to increase the contrast it is a good idea to use larger as well as bolder type when headings are printed in colour.

Headings on different hierarchic levels will provide the readers with reference points and help them organize information cognitively for better retention and recall.

To achieve a clear structure we can use a combined numbering and lettering system (Jonassen, 1982). Main points in a text are traditionally labelled with Roman numerals (I, II, III, IV etc.). Sub-points of the first degree are traditionally labelled with capital letters (A, B, C, D etc.). Second-degree sub-points are traditionally labelled with Arabic numerals (1, 2, 3, 4, etc.). The labelling hierarchy is I., A., 1., a., (1), (a). In Europe it is common to use a numbering system only (e.g., 1, 1.1, 1.2, 1.3, 1.4; 2, 2.1, 2.2, 2.3, 2.4; 3, 3.1, 3.2, 3.3, 3.4 etc.).

Size and style

We can vary the type size and/or use different type versions. According to a rule of thumb, called the "Rule of X's," the height of the uppercase X of a smaller typeface should be the same as the height of the lower-case x of a larger typeface in a hierarchy. For a ten point running text headings may be 10, 14, 18 and 24 points. For a twelve point running text headings may be 12, 16, 22 and 28 points.

Numbering and lettering systems can be combined with typographic cueing of headings. Headings set in different type versions aid comprehension of the material. We can use a special typeface for headings, for example a sans serif typeface like Avant Garde, Futura, Gill, or Helvetica.

Headings set in different type versions aid comprehension of the text content (Jonassen, 1982; Mayer, 1993). However, such a mix appears strange to some European readers. In Europe there are national differences concerning the use of typefaces with and without serifs. Scandinavian newspapers, for ex-

ample, often use Modern Style typefaces, whereas England and France prefer Transitional Style or Old Style typefaces. In Germany they have serif as well as sans-serif typefaces.

Placement of headings

The use of space and the actual placement of a heading can be used to enhance the hierarchic structure (Jonassen, 1982). Headings shall be placed above and close to the following text. This distance shall be smaller than the distance to the previous paragraph (Pettersson, 1993; Lipton, 2007). Sometimes headings are indented.

We need to provide space enough for the headings in a text. The amount of white space surrounding headings should correspond to the heading hierarchy. A major heading needs to have more space than a minor heading.

The distance between the heading and the text below the heading should always be smaller than the distance between the heading and the previous text (the text above). The heading "belongs with its own text". See the two examples on the next page. In each case the total space for the heading is the same but the positions of the headings are different. Distinguishing headings with extra white space makes a document easier to search and easier to read. See the two examples on next page.

Example 1

A major heading needs to have more space than a minor heading We need to provide space enough for the headings in a text.

Good placement

The distance between the heading and the text below the heading should always be smaller than the distance

Example 2

A major heading needs to have more space than a minor heading We need to provide space enough for the headings in a text.

Bad placement

The distance between the heading and the text below the heading should always be smaller than the distance

Paragraphs

There are a few possibilities to emphasize paragraphs. Paragraphs may be indented and/or separated with extra space for emphasis.

Paragraph indents

An indent is the distance between the lines in a text and the text-face, usually the right part of the left margin. Paragraphs may be indented and/or separated with extra space. *First line indents* are the distance between the beginning of the first line in a paragraph and the left margin. They are often used for the running text in books. There are also *negative indents, hanging indents, nested indents* and *full paragraph indents*. It is considered to be bad typography to allow an indent on the bottom line of a page. To avoid this it might be possible to edit the text until it fits.

First line indents

In order to emphasize the beginning of paragraphs in running text the first lines may be indented and start with empty spaces. The width of an em dash is commonly used for paragraph indentation. This width is equal to the height of the type. So in 12 point Georgia an em dash is 12 points wide. However, other widths are also common. The fragment below demonstrates a first line indent, here marked with a blue rectangle.

The width of the first line indent in this paragraph is marked with a blue rectangle. In this book first line indents are used in all paragraphs, with the exception of paragraphs following headers and certain citations. Headings may be indented.

According to Tinker (1963) indenting of the first sentence of each paragraph improves legibility by 7% for both single and double line spacing. According to Frase and Schwartz (1970) indenting the first sentence of each paragraph improves comprehension of printed materials. Indenting of every sentence will, however, slow down the reading speed.

Negative indents

The indent on the first line of a paragraph may also be directed back into the left margin (or sometimes even forwarded into the right margin). This is called a *negative indent* and may be used for reference lists, numbered lists and lists with bullets, and other characters like stars, squares etcetera. The fragment below of a paragraph demonstrates a first line indent.

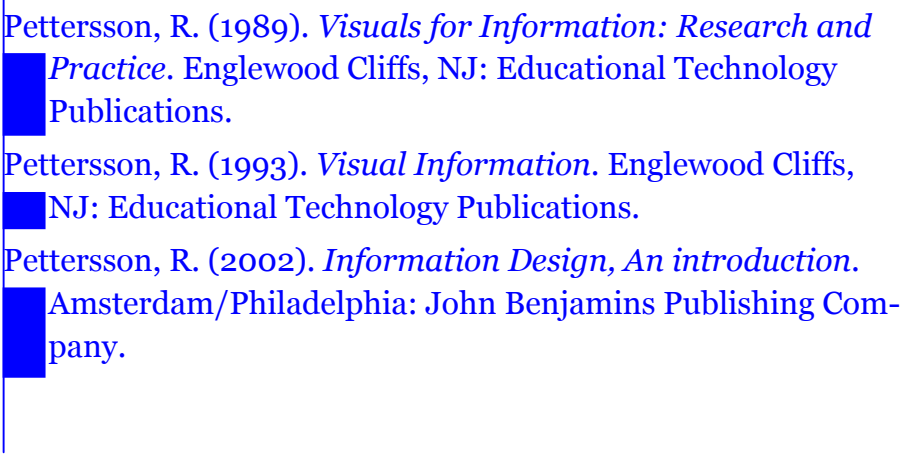
The first line in this paragraph starts in the left margin. This is an example of a negative indent, here marked with an “empty” space. The following lines will align with the margin. Headings may have negative indents.

Negative indents may also be used in texts where it is very important to draw attention to each new paragraph. However, as noted above, indenting of every sentence will slow down the reading speed.

Hanging indents

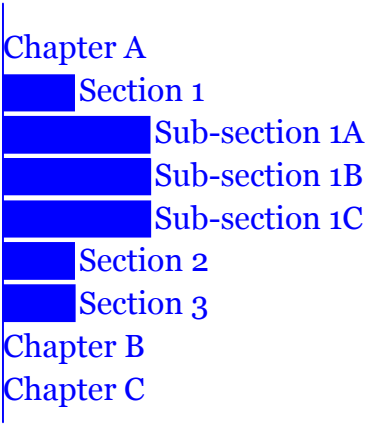
Hanging indents, or *outdentions*, are often used in reference lists, numbered lists and lists with bullets, and other characters like stars, squares etcetera. All lines, except the first lines, begin with an empty space. In contrast to texts with negative indents all the texts always remain within the text column. Indents make it easy to find single references.

The fragment below of a ”reference list” demonstrates hanging indents, here marked with blue rectangles.



Nested indents
Nested indents are a form of indention in which each subsequent indent is set relative to the previous indent, and not relative to a margin. Nested indents may be used to graphically show the relationship between chapters, sections, sub-sections and paragraphs within a text. Tables of contents often have nested indents to show each successive level.

The fragment below of a ”tables of contents” demonstrates nested indents, here marked with blue rectangles.



Full paragraph indents
Long quotations should be distinctly separated from the rest of the text. Quotations are often indented, sometimes also on the right side. Full paragraph indents may also be used for lists and tables. See the section *Quotations*.

Extra space
Readers prefer small text paragraphs to big ones. Often it is quite easy to divide the text in hierarchic and natural parts, portions, or sections. Natural breaks emphasized by typography are helpful. Providing “extra white space” between paragraphs and

between larger portions of the text provides valuable cues to the learners that a new section or a new type of activity follows (Waller, 1987). The extra space between paragraphs may vary, depending on the material.

Marks

The end of a sentence should be determined by syntax rather than by a set width of a line (Hartley, 1980; Bork, 1982). There are usually no problems with continuation of text in books. The text fills up page after page, and chapter after chapter. However, in magazines and other periodical publications, it is sometimes necessary to use special *continuation marks*. These marks may be arrows or triangles and references to page numbers. In these situations it might also be a good idea to use *terminal marks* after the last paragraph in the article. Such a mark is often a circle or a square, filled or unfilled. Sometimes the terminal mark is the initials or the signature of the writer.

Tables

Numeric data can be used to illustrate several different situations, such as chronological changes in single quantities or sets of quantities, parts of a whole, and relationships between two or more variables.

There are different ways to present numeric data. In print media numeric data can be presented in text, as figures and digits, in tables, or in various graphical formats. No more than one or two items of numeric data should be presented in prose form. Tables are often, but not always, a good solution. Tables may show the maximum of amount of information in the minimum amount of space.

Many people find tables confusing and difficult to understand (Wright, 1968). Tables should be structured and compact for easy accessibility. Sometimes a diagram or a graph may be a better choice.

Vertically oriented tables

The following three examples, Table 1, 2 and 3, demonstrate how vertically oriented tables may look. In these examples the data that are used has no interest at all. Table-titles are often, but not always, put above the actual table. They include descriptions of the table contents. However, in these examples the table-titles only comment on the design of the table below.

Table 1. This is a vertically oriented table. Ruling in the table is thin.

Products	*Sales in $*		
	2012	*2013*	*2014*
Product 1	100	120	110
Product 2	200	180	210
Product 3	200	220	210
Product 4	300	320	350
Product 5	400	420	410

Table 2. This is a vertically oriented table without ruling.

Products	*Sales in $*		
	2012	*2013*	*2014*
Product 1	100	120	110
Product 2	200	180	210
Product 3	200	220	210
Product 4	300	320	350
Product 5	400	420	410

Table 3. This is another vertically oriented table without any ruling. However, in this case the columns are very wide. The large distance between the numerical data in the columns makes it hard to read the table below.

Products	*Sales in $*		
	2012	*2013*	*2014*
Product 1	100	120	110
Product 2	200	180	210
Product 3	200	220	210
Product 4	300	320	350
Product 5	400	420	410

Readers prefer vertically oriented tables rather than horizontally oriented tables. In vertically oriented tables it is easy to see the target entries, and then quickly find the data in the table cells to the right (Wright, 1968; Wright and Fox, 1972; Ehrenberg, 1977). It is easy for us to compare data “side by side.”

Horizontally oriented tables

The following three examples, Table 4, 5 and 6, demonstrate how horizontally oriented tables may look. As in the previous examples of tables the data has no interest at all and the table-titles only comment on the design of the tables.

Table 4. This is a horizontally oriented table. Ruling in the table is thin.

Sales in $	*Products*				
	Product 1	*Product 2*	*Product 3*	*Product 4*	*Product 5*
2012	100	200	200	300	400
2013	120	180	220	320	420
2014	110	210	210	340	410

Table 5. This is a horizontally oriented table without ruling.

Sales in $	*Products*				
	Product 1	*Product 2*	*Product 3*	*Product 4*	*Product 5*
2012	100	200	200	300	400
2013	120	180	220	320	420
2014	110	210	210	340	410

Table 6. This horizontally oriented table is not only very ugly, it is also very confusing and provides very poor legibility. Please note that all the data are the same as in previous tables.

Sales in $	Products				
	Product 1	**Product 2**	Product 3	*Product 4*	Product 5
2012	100	200	200	300	400
2013	120	180	**220**	*320*	420
2014	110	*210*	210	340	410

Horizontally oriented tables are harder to use and more difficult to understand than vertically oriented tables (Wright, 1968; Wright and Fox, 1972). It is complicated to compare data "up and down."

Good tables

Tables that communicate the quantitative aspects of data are effective only when the data are arranged so that their meaning is *obvious at a glance* (Ehrenberg, 1977). After deciding what data to present but before design a table, we need to consider that (a) rounded-off values may display patterns and exceptions more clearly than precise values, (b) a reader can compare numbers down a column more easily than across a row, and (c) column and row averages can provide a visual focus that allows the reader to inspect the data easily.

According to Tinker (1963) the type used in tables should be no smaller than 8-point, and no larger than 12-point (Tinker, 1963; Wright and Fox, 1972, p. 241). Like pictures, all tables

should be integrated with the text content, in the flow of text. In a text, tables should preferably be put between paragraphs, and not forced to break paragraphs and thus disturb the reading. Sans-serif typefaces are usually used in tables. Good examples are Helvetica and Franklin.

In *friendly graphs* (Tufte, 1983; 1990), words are spelled out, they run from left to right (in western societies), and data are explained. Elaborately encoded shadings, cross-hatching, and colours are avoided. Colours are easy to distinguish, type is clear and precise, and is done in upper and lower case with serifs. In *unfriendly graphs*, abbreviations abound, words run in many directions. Graphics are repellent and cryptic with obscure coding. The design is insensitive to colour-deficient viewers Red and green are used for essential contrasts, and type is clotted and in all capitals in sans serif.

Wright and Fox (1972) offer the following guidelines for the construction of tables for the general public and non-professional audiences (p. 241):

- All the information the learner will need should be presented in the table. That is, the learner should not be required to interpolate, combine entries, draw inferences, or otherwise manipulate the contents of the table in order to determine the correct answer. Rather, the learner should only be required to scan the list to find the correct target entry.
- Type size used in the table should be between 8 and 12 point.
- Items should be arranged vertically in the table rather than horizontally (i.e., the list of target entries should be vertical).
- Items within columns should be grouped and separated from other groups by either white space or rules (lines) in

order to facilitate reading without accidentally moving to another row. Groups should contain no more than five items.

- Redundant abbreviations of units should not be included within the table entries [although they should be included in the column or row headings].
- 'Landmarks' or sub-headings – certain target entries in the table highlighted by using bold type or a larger font (e.g., every tenth entry, starting with 10, 20, 30, etc.) as an intended aid to locating entries – should not be used, since they appear to be ineffective and possibly confusing.
- It is useful to have adjacent columns printed in different fonts or styles to distinguish between them. That is, there is less chance of erroneous reading of the table if the column of target entries is in normal text (for example) and the column of associated answers is in boldface text.
- Related pairs of items in adjacent columns should be spaced closely together (i.e., the eyes should not have to traverse a great distance between the target entry and the associated answer).
- Whenever possible, columns should be arranged so that the target entries are to the left of the answers.

Ehrenberg (1977) provides some more general guidelines for the construction of tables for the general public and non-professional audiences:

- Numbers should be rounded off to no more than two significant figures to facilitate learners' making comparisons.
- Averages of rows and columns (as appropriate) should be given to facilitate learners' making comparisons of individual cell entries to them.

- Put the most important comparisons into columns (rather than rows), as columns make for the easiest comparisons.
- Numbers in rows or columns should be arranged in some meaningful order whenever possible (e.g., increasing or decreasing).

Bigwood and Spore (2003, 2013) discussed how to present charts, numbers, and tables for easy communication. They provided the following six rules of plain figures:

- Put figures in a logical order, often from the largest to the smallest.
- Add focus to figures. *Arithmetic means* and *medians* make it easier to see exceptions and patterns in the data. *Percentages* show proportions. *Totals* provide overviews.
- Keep numbers to be compared close. This proximity aids comparisons.
- Round figures for clarity. In many situations exact numbers, with four decimals, does not matter. It may often be a good idea to use variable rounding and convert exact numbers to two, or three, effective digits, regardless of the size of the number.
- Provide written summaries in the text of the charts, numbers, and tables.
- Use layout to guide the eye, and avoid distracting and unnecessary decorative elements that interfere with the content in graphs.

Table 7. This vertically oriented table provides good legibility. However, sometimes more lines are needed.

Products	*Sales in $*	
	2013	*2014*
Product 1	100	120
Product 2	200	180
Product 3	200	220
Product 4	300	320
Product 5	400	420

There are obviously many ways to design even a simple table. It is easy to agree with the *Publication Manual of the American Psychological Association* (1996, 121) where the American Psychological Association concludes "that an author's thoughtful preparation makes the difference between a table that confuses and one that informs the reader."

Captions

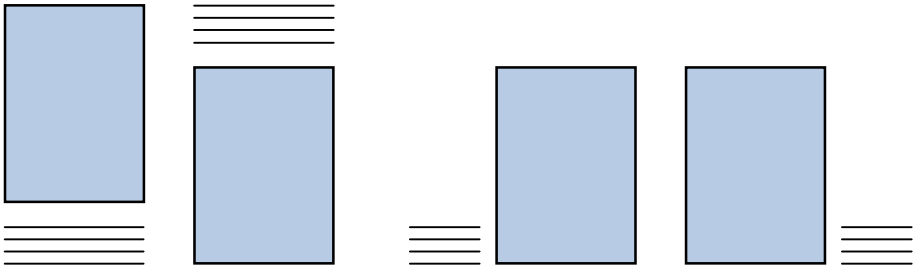
Most pictures are capable of several interpretations until anchored to one by a caption (Barthes, 1977). McDougall and Hampton (1990, p. ix) concluded that photographs nearly always need a partnership with words that will confirm, clarify and reinforce their messages.

Placement of caption

Each picture should have its own caption, unless two pictures or a series of pictures are closely together. A caption may be placed in many ways. The caption should always be located close to the

picture. Readers usually expect to find the captions beneath the pictures. However, captions can also be placed above, to the left, or to the right, of the picture, but never inside the picture frame. Use flushed right text if you want to put the caption to the left of the picture. Use flushed left text if you want to put the caption to the right of the picture. When the caption is put above or beneath the picture, we can use justified text or flushed left text and sometimes even centred text. Sometimes triangles or arrows pointing from the caption towards the picture are used, especially when pictures are gathered together many on a page, e.g., in a magazine.

Placing the caption inside the picture makes it harder for people to read the picture as well as the text. Nor should the caption be located on some other page, as happens far too often in reference books and textbooks used in schools.



Here are different ways to place captions. Readers usually expect to find captions beneath the picture as in the first example to the left.

Typography of caption

The caption should have a different typographic size or even a different typeface so it can be easily distinguished from the main text. The captions should not be in negative form in a colour picture since the slightest misalignment in printing makes

the captions extremely difficult to read. Never make the caption type larger than the main text. The title of the caption could be printed in boldface.

(See the book *Information Design 2–Text Design* for further information about captions.)

Quotations

Sometimes quotations are included in a text. Short quotations can be integrated into the body text, but long quotations should be distinctly separated from the rest of the text. Quotations are often indented, sometimes also on the right side. Any quotation that is used must have the source of reference cited, in keeping with common rules of writing. Below is an example of a short text segment (blue) with an embedded quotation (here green). The *full paragraph indent* is shown with a blue rectangle.

On occasion, unfortunately, communication does not seem to function. This may depend on insufficient information, but it may also be because we have difficulty reaching each other. By way of example, the following text is cited as a warning in *The Technical Writer's Handbook* by Young (1989, p. 206):

> Conditional symbolic modified single-digit arithmetic using optical content-addressable memory logic elements: Conditional symbolic modified signed-digit arithmetic operators.

Esoteric and impenetrable jargon like this can be perceived as incomprehensible by the uninitiated. Since readers who do not know the code are left out, its use poses the risk of 'one-way-only' communication, and has no value outside of initiated circles.

In this example, the body text and the quotation are set with 12 point Georgia with 16 point line spacing. Quite often the quotation is set in a smaller type. Between the text and the quotation, there are spaces that each corresponds to half a line, in this case eight points. Blank space (blank lines) like these can be used in various contexts to provide space above and below, for example, pictures, tables, and lists. In fact, we may use “empty lines” for vertical separation of text elements rather than printed lines of any kind. In this book the firsts paragraphs below quotations and examples are not indented.

Lists

Putting information into list form can facilitate comprehension (Hartley, 1987; Frase and Schwartz, 1970). There are several types of lists used in technical documentation. A book, for example, usually contains a table of contents, a subject index, and a bibliography. It may even contain lists of pictures and tables, or of authors or other persons. The text in the book may have various lists of characteristics, criteria, requirements, or prospects. Within a body text, there may be various lists of characteristics, criteria, items, requirements, persons, and products. Bullets and numbers are used for items of the first degree. Hyphens are used for lists within such a list. In a numbered list, the various items are ordered according to their importance, unlike the unnumbered list. In an unnumbered list items may be ordered alphabetically, or not at all. It is also possible to use upper case letters or upper case roman figures for items of the first degree, numbers in Arabic figures for the second degree, bullets for the third degree, and hyphens for the fourth degree.

Table of contents

The table of contents may have justified or unjustified text. The important thing is that the structure of the content in the document is clearly visible to the reader.

The table of contents immediately follows heading level one, ("Contents") and may be set in 12 point normal type with 14 point line spacing, left-aligned, and without any extra space before each entry. Chapter designations may be set in bold face, while section designations usually are set in normal typeface. Page-numbers follow each text on the same line. They may also be flushed right.

When a table of contents is extensive, it may be rendered in two columns. References to page numbers are often shown for heading levels one, two and three, with indents for each successive level. Heading levels four and more are usually not shown in a table of contents. This would impair the possibility to get an overview and understanding of the overall structure.

Bibliographies

References to cited works are usually put in a reference list at the end of a chapter or at the end of a book. When each chapter has its own bibliography, the heading "Bibliography" or "References" may be set as a heading 2 in the document. When there is only one bibliography for a factual study book, heading level 1 is used.

Bibliographies are often voluminous and may take up a great deal of space. Since the reader will usually peruse such a list to find only certain entries, they should be made easy to find. Therefore, the text should be distinct even though it is relatively compact. One common bibliography may be a good

idea when several references are the same in more than one chapter.

In a printed book bibliographies may be set in ten point type with 11 point line spacing, with 12 point negative indents. Above them, there may be 2 or 3 points of space, but no extra space is needed below. Indents make it easy to find single references.

A bibliography may be laid out to cover the full width of the column. If the bibliography is very extensive we should divide it into two columns that are equal in size, even if the body of the text in the document is arranged in one wide column.

The subject index

A well-constructed subject index, "Index," is often an indispensable tool for quickly finding information in books. As is the case with bibliographies, the readers are usually looking only for certain information when they refer to a list of this kind.

A subject index may be set in three 50-millimetres-wide columns, in ten point type on 11 point lines, with 12 point indents. It is not necessary to have extra space between the lines. However, there may be a half a line of space between groups of words that begin with successive letters of the alphabet. The title of the subject index may be set as a header 1.

Lists of items

Within a body text, there may be various lists of characteristics, criteria, items, requirements, persons, and products. Bullets and numbers are used for items of the first degree. Hyphens are used for lists within such a list. In a numbered list, the various items are ordered according to their importance, unlike the unnumbered list. In an unnumbered list items may be ordered alphabetically, or not at all. It is also possible to use upper case

letters or upper case roman figures for items of the first degree, numbers in Arabic figures for the second degree, bullets for the third degree, and hyphens for the fourth degree.

Notes

Notes tend to fray the reading fabric. If notes are required, they may be collected in a numbered list at the end of each chapter immediately before the bibliography, and arranged like a list of items with numbers instead of characters to mark them. Occasional notes and brief references may be placed as *footnotes* at the bottom of the page, or sometimes in the left or right margin or at the end of the chapter or article. Usually a smaller type size or a different typeface is used for footnotes. A horizontal line can separate the footnotes from the running text. When there are two or more columns on a page footnotes should be placed at the end of the document

The title pages

A book begins with a title page with the title of the book, the name(s) of the writer(s), and the name of the publisher. On the second page, the imprint page, administrative information such as the name of the document, a document number, a security classification, revision or version number, and date of publication may be provided at the top of the page. Information about the ISBN- or the ISSN-number and the copyright holder are usually provided at the bottom of the imprint page. There are usually no pagination on the title and imprint pages.

On the third page in a book, there is the *preface*. The header “Preface” may be set as heading one. A preface will usually not exceed two pages. Page three is usually the first page with pagination. On the fifth page, or the first odd page after page five, the table of contents starts. The header “Contents”

may be set as header one. However, in e-books intended for reading on screens the table of contents may start on page four.

The pages with preface and table of contents are often paginated with lower-case roman figures. The book is numbered with Arabic figures.

Miscellaneous

This section contains short notes on a limited number of items in graphic design.

Abstract, a short summary of a paper put in the beginning, below the author lines and above the body text.

Achromatic colours are black and white.

Active space or *positive space* in a visual is the part representing different objects. Space has no meaning in itself, but it may be used to separate or bring together different picture elements.

Agate lines are vertical measurement of space, there are fourteen agate lines to an inch.

Artistic layout may please the individual artistic graphic designer, but may have no relation to the content of the message.

Ascender. The part of a lower case letter (b, d, f, h, k, l) that ascend above its x-height. These letters may be taller than the cap height.

Author lines are the lines at the beginning of a paper with the name of the author or the authors. It is often centred and set in smaller type than the heading.

Balance is the sum of all the elements, the verticals and horizontals, the darks and lights, that make up the design.

Bleed is text and/or pictures that extend beyond the trim edge of the printed page.

Blowup is an enlarged picture.

Bold type, or boldface, members of a family of type in which letters are heavy, **bold**.

Borders are ornamental rules around the edge of a page or a page component.

Boxes.

Specially important messages like summaries can be boxed-in to gain special attention. If the lines are shorter the effect is even more obvious. Boxes usually have thin lines. Boxes may sometimes be filled with a background colour.

Bullets (•) are commonly used in lists of items in "point form" without numbers. Bullets are more powerful than hyphens (-) or asterisks (*).

Camera-ready is a final printout of a page with text and pictures that is ready for reproduction, and also or art that is ready for reproduction.

Cap height or H-height. The size of capital letters in a typeface, expressed as a percentage of the point size. (See illustration in chapter 2.)

Capital letters, *caps*, are harder to read than a combination of upper and lower case letters.

Caps is short for capital letters.

Check box. Check boxes (☐) can be used in instructional materials when learners may make check marks when they have finished assignments.

Chop-mark is the symbol of a printer specializing in printing of fine arts, such as etchings, etc.

Chromatic colours are blue, green, red and yellow.

Colour-key is a process for obtaining process colours on film by exposure to light.

Colour-spacing. Positioning letters so that balances of negative and positive spaces are pleasing.

Column is a vertical segment of a printed page.

Column rules are vertical lines between two columns.

Condensed. Horizontally narrow type designs.

Continuous-tone is a complete range of gray, from black to white.

Contrast is the result from differences along a common visual dimension that can be observed between various elements in a design.

Copy is usually the text in a design, but also all the elements of a finished design.

Credits. In books it is quite common to give credits to artists and photographers in a special “List of illustrations”. In magazines and in newspapers credits are usually put next to the ac-

tual illustration, often in a vertical position. Credits to the author are usually in the form of a byline.

Cutoff rules are horizontal lines that separate unrelated items above and below the line.

Cyan, the particular blue used in four colour process printing.

Dash. With the typewriters we had to create a dash (—) with two hyphens (--). With desktop publishing and typography this is not necessary. In 12 point Georgia the em dash is 12 points wide. In desktop publishing we can create the em dash or em rule (—), the en dash or en rule (–), as well as the hyphen (-). The em dash is used to indicate a break in thought. There should be no space before or after the em dash. The en dash is used between numerals (e.g., pp. 33–45 and as a minus sign (e.g., –12).

Descender is the part of some lower case letters (g, j, p, q, y) that descend below the baseline.

Didot system, a system for measuring size in typography, 12 points = 4.511 mm.

Differentiation. It is essential to be able to differentiate elements from one another.

Disinformation is the opposite of information design.

Dot. A dot is a full stop (.) to mark the end of a complete sentence.

Drop caps are lowered initial letters.

Dropout is a halftone in which no dots fall on white areas.

Dust cover or jacket is the paper wrapper of a hardbound book.

Ellipsis. An ellipsis (...) is normally used to indicate that a part of a text is missing in a quotation.

Em is the square of the body of type. See Dash.

Extended. Horizontally wide type designs.

Face is the printing surface of type; also, any particular style of type.

Fill-in blanks. Fill-in blanks (_) can be used in assignments, and in lists of various kinds, for people to mark their choices.

Folder. A folder consists of a single sheet, usually printed on both sides, and folded two or more times.

Font of type is a complete assortment of characters of the same style and size. However, in computer manuals the word *font* is sometimes used to mean typeface, which is confusing.

Footers. When present, footers appear at the bottom of every page. Like headers, footers may carry page numbers.

Formal balance has total symmetry and it is felt to be static and harmonious. It may, however, also be boring. Composition can be used to direct the viewers.

Glossy is a reproduction proof of type.

Graphical media include products such as almanacs, catalogues, children's books, dictionaries, educational materials, encyclopaedias, forms, graphic symbols, handbooks, indexes, maintenance information, manuals, maps, non-fiction books,

plans, product descriptions, reference books, reports, textbooks, and tickets.

Graphics editors work on schematic pictures that are delivered by subject matter experts and writers. They produce finished schematic pictures.

Grid, a geometric pattern used as a base for layout.

Grouping. By grouping headings, paragraphs, illustrations, and captions, the designer aids communication.

Gutter. The margin at the binding edge of a page, or inside edge of unbound pages.

H-height, or *cap height*, is the size of capital letters in a typeface, expressed as a percentage of the point size.

Hairline. A hairline is the finest line in metal type. Hairlines are often used to divide columns.

Halftone. A halftone is a reproduction in which tones have been photographed through a screen. The picture is divided into dots. Size of dots governs darks and lights.

Hanging indent, or *outdentation,* is the reverse of indentation. Examples are lists with bullets.

Harmony describes the effect of the pleasing interaction of all the design elements in a design.

Headers. When present headers, or running heads, appear at the top of every page. Like footers, headers may also carry page numbers.

Hot type, *hot lead typesetting, hot metal typesetting, mechanical typesetting, metal type*, refers to 19th-century technologies for typesetting text. Hot melted type metal is injected into moulds. The resulted characters are used for printing ink onto paper.

Interest. Scale, contrast, and colour may add visual interest to a design.

Interline distance is the vertical distance from one baseline to another baseline in a text.

Introductions to chapters in a book or to articles in a magazine or a newspaper are very often set in different type than the following running text. Sometimes the introduction is set with larger type size, sometimes in italic or bold typefaces.

Italic type, *italic*, or *italics*, are members of a family of type in which letters slant to the right. Italics are designed to complement the normal vertical design. See the section *Stylistic variation of type*.

Jacket or *dust cover* is the paper wrapper of a hardbound book.

Jim dashes are short cut-off rules that separate unrelated items above and below the line.

Kern (verb) manually adjusting the space between characters to achieve better legibility, (noun) the amount of space added or subtracted in the process. See the section *Spacing*.

Leading, spacing between lines in a text. See the section *Interline distance*.

Letterform is the design of individual characters of a typeface.

Letterhead is the top of a designed writing paper.

Letterspace (verb) to add extra space between letters; (noun) the amount of extra space added.

Ligatures. Ligatures are letters combined to new characters. Examples are Æ (A+E) and Œ (O+E).

Line length is the horizontal measure of a line.

Line space is the vertical distance from one baseline to another baseline. See the section *Interline distance*.

Lines. Various lines are often used for decoration to make a more aesthetically pleasing or artistic product. However, lines can also be used to aid communication.

Horizontal lines can separate sections or paragraphs or hold them together. Horizontal black lines are usually one to four points. When horizontal lines are printed in other colours they may be wider.

When two or more columns are used on a page, vertical lines can be used to clearly separate the text blocks from one another. By tradition vertical lines are thin, usually half a point or one point.

Linotype is a machine for setting hot type line by line.

Lithography is a printing process based on the fact that oil and water don't mix.

Lower case is the un-capitalized "small" letters of the alphabet (a, b, c etcetera).

Magenta is the particular red used in process printing with four colours.

Margins are the areas surrounding the text and pictures on a printed page. See the section *Text layout*.

Modern type is a typeface with thin, flat serifs. See the section *Typefaces*.

Moiré is an undesirable pattern in a picture that occurs by re-screening halftone copy.

Monotype is a machine for setting hot type letter by letter.

Negative space is the space between and around pictures, and also between the elements within a picture.

Oblique is an "artificial italic." It is formed by electronically slanting a vertical typeface. Italics, however, are specially designed.

Offset is a kind of lithographic printing in which ink is transferred to a rubber blanket, then to the paper.

Open ended boxes are boxes with rules only at the top and bottom.

Ornaments. Various ornaments and patterns can be used to separate different sections in a text. In instructional materials they are often used to mark specific activities. They can also be used for decoration, to make a more aesthetically pleasing or artistic product. Dingbats is a special PostScript font in desktop publishing systems

Outdentation, or *hanging indent,* is the reverse of indentation. Examples are lists with bullets.

Point, is used for measurement of letter size, see the section *Size of type*.

Pointers are lines combining labels with elements in a picture.

Primary colours, red, yellow, and blue; all others can be mixed from these.

Process colours are magenta, cyan, yellow and black. These are the inks used in four-colour process printing.

Quotation marks, ‘ ’ “ ” « », are used according to the different standards in different languages.

Rough is a preliminary layout.

Rules are vertical and horizontal straight lines used in typography and layout.

Run-arounds. When text is “wrapped” around an irregularly shaped illustration this is known as a “run-around.” Run-arounds almost never represent any improvements and should be avoided.

Sans serif is a typeface without serifs.

Scale describes the relative size or magnitude of a given design element in relation to other elements and to the design as a whole.

Separation. Use of separation devices such as bullets, numbers, and letters facilitate recall of information in list form.

Serifs are the terminal strokes, normally at the top and bottom of the main strokes of letters in a Roman type style.

Shaded letters have strong three-dimensional quality by use of heavy shadows on one side of main stroke.

Small caps or small capitals are an alternative set of capitals used for text setting. These SMALL CAPITALS are smaller than standard CAPITALS.

Special designs. Chemical, mathematical, astronomical, and medical signs are important for use in these subject matters.

Spreads are two facing pages in books, etc.

Subscript is a character set lower than the body of text, like the figure 2 in x_2. See *Superscript*.

Summaries. Important messages like summaries can be boxed-in to gain special attention. If the lines are shorter, the effect is even more obvious. Boxes usually have thin lines. Boxes may sometimes be filled with a background colour.

Superscript is a character set higher than the body of text, like the figure 2 in x^2. See *Subscript*.

Symbols. Various symbols can be used to aid communication in a book or a magazine. An example is a mark for continuation (>). Many ornaments may be used as symbols.

Title Case, each word begins with a capital letter, is sometimes used in titles and in figure texts.

Type. The term *type* refers to all characters that are used in printing. See the section *Typefaces*.

Upper case is a term for capital letters of the alphabet (A, B, C etcetera).

x-height is the vertical distance equal to the height of lowercase letters such as x (without ascenders and descenders).

Layout of text and pictures

A typical page in a telephone directory is almost filled with text. There are no pictures, and margins are narrow. The text may cover 90 percent of the page. In dictionaries the text covers 75-80 percent, and in specialist books and factual study books the text often covers 50-60 percent of the page. In pure literature and children's books, the text area is often even smaller than that. When a picture covers the entire page, there is no space for text or margins. The third extreme is an empty page.

Different layouts

A picture without a caption has no or almost no informational value. A picture is too ambiguous on its own. A picture caption must describe the picture and guide the reader to the interpretation the informer wished to convey to the reader. Both words and pictures may possess an emotive force that is not easily foreseen. A number of different value judgments may slip into a text when the purpose of the text is merely to supply information.

The reader's emotions may be aroused by seemingly insignificant details in a visual or nuances in the wording of a text. So sufficient effort must be invested in the editing of both texts and pictures. Interest can be focused on the central message in pictures through careful picture selection and editing, primarily by means of cropping, which is, deleting non-essential portions. Sometimes different visuals benefit from being presented in a group. This may be the case for a photograph and an explana-

tory drawing, or several photographs or drawings forming a mini-series or related picture sequence.

When texts and visuals are collected for informative pages and spreads, "message transmission" must be a central consideration. As previously noted this kind of "information layout" differs from a "decoration layout" in which purely aesthetic aspects are allowed to predominate.

The preparation and production of maps is called cartography. The product of a cartographer's efforts is a mathematically defined depiction of a reality based on measurements. Maps describe reality and shed light on a number of conditions, such as terrain, political subdivisions, the prevalence of certain types of soil, minerals, etc. The utilization of variations in shape and colour creates map symbols that provide a picture of the reality they represent. A carefully processed map contains more information per square inch than any other form of printed information.

In newspaper layout many different messages have to be communicated. The problem is to communicate a series of disconnected messages of infinitely varying significance within a limited space, time, and economy with a recognizably consistent style for each section of the paper. Many readers will only spend time on a limited amount of information in a newspaper. It is known that elements like headlines, photos, drawings, and information graphics attract attention and often are entry points into a page. Size and placement of such elements influence how the reader will read a page. It is possible that many readers may jump over too large pictures and never look at them at all. In magazine layout visuals are often very large.

Quite often the visuals have no or a very limited contact with the text. Here White (1987) asks for less typographic

amusement and more serious character building. Fiction layout certainly does not work very well in instructional message design. However, even in traditional non-fiction publishing, it is often very hard to access the information.

Picture placement has very often been based on aesthetic rather than instructional characteristics. The appearance of elements on the page can provide powerful cues to the reader. Thus, visuals should be located in close proximity to the parts of the text in which pictured motifs are discussed. Visuals plus their captions should preferably be inserted into the text between two paragraphs, not in the middle of a paragraph, so they do not disrupt reading rhythm.

Captions and visuals are often piled up in different areas and sometimes even on different pages. Here, an effort should really be made to achieve the best possible interplay of words and visuals. Visuals interact with the captions, other visuals, headings, running text, tables, and maps on a page and on a spread in a book, a magazine, or a newspaper. The layout of the whole spread should be attractive. However, the mere use of a lot of pictures is not at all enough. In an artistic layout, the message may actually be effectively hidden. The message is more effectively conveyed by an informative layout.

The basics for instructional message design could be summarized in the following: Select Arrangement, Balance and Colour to maximize Dynamism, Emphasis, Fidelity and Graphic Harmony.

Balance in design

In nature balance is normal. A design should usually display “good balance”. Elements of the design should fit together in an aesthetic and harmonious relationship in a manner that is in-

teresting but not distracting. Man has an intuitive sense of balance. When a single element is too large or too small, too light or too dark, too prominent or too indistinct, the entire design will suffer from this imbalance. However, good balance is something subjective. Different people may have quite different opinions on any aesthetic issues.

A composition is balanced when the visual weight of graphical elements on either side of the centre of balance are approximately equal. As with a physical balance, lighter elements can balance heavier elements if their size or value is increased, or if they are moved farther away from the centre of balance. It is also possible to move the heavier elements closer to the centre of balance. Mullet and Sano (1995, 51) noted that altering even a single attribute of one part in a complex composition could have a significant impact on the balance, the unity, and ultimately the harmony of the whole. When a single element is too large or too small, too light or too dark, too prominent or indistinct, the entire design suffers.

Imbalance creates an uncomfortable feeling in the reader and should often be avoided. Imbalance, however, can be used to attract attention within a picture or within a material. Unexpected, irregular, and unstable design will attract attention. As soon as instability is introduced in a design the result is a provocative visual expression. The eye will struggle in order to analyze the relationships and the balance within the picture.

Balance can be formal with total symmetry or informal. Formal balance is felt to be static and harmonious. Informal balance contributes to a feeling of dynamism. However, the mind needs stimulation and surprise. Contrast and imbalance can dramatize a design and attract attention. Several artists use

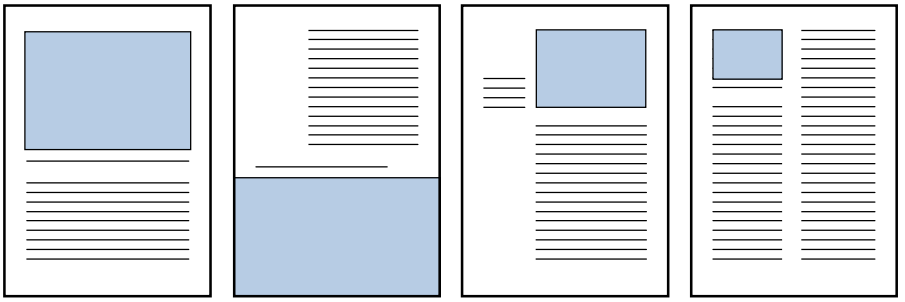
a visual strategy, such as combinations of dark and bright, large and small, round and square, to sharpen meaning.

If the lines used as pointers to elements in a picture are heavier than the lines in the picture itself this will create noise and clutter in the illustration. Pointers should be light in comparison with the lines in a picture.

Image placement

Pictures are frequent in most factual study books. A picture may be placed on a page in many different ways. It is usually, but not necessarily, adjusted to the width of the text column. A page with more than one column has more possibilities for placement of pictures. Usually, pictures are placed in a column of text, and are centred. As a rule, they may be placed where they best serve the presentation. Very small pictures can be placed in the narrow column. On the other hand, wide pictures may cover the entire text area of the page. Too many pictures will confuse the layout and reduce the chances of the reader getting involved in the text. Too few pictures appear to increase the size of the body text.

A picture may expand beyond the width of the column and cover the margins and sometimes the whole page. “Bleed” (covering the entire page, with no margins) may be used constructively and creative to expand the impact of important attention-getting images. We should avoid too much text within the pictures. The text that is absolutely necessary (that is, key words) may be set in nine point Helvetica with ten-point line spacing. Single words that are especially important or have a comprehensive function may be shown in bold face. In English texts, key words usually begin with capital letters.



A few examples of pages with text, pictures and their captions.

As previously noted many authors have noted that in materials for information a picture should be located as close to the relevant text passage as possible. It is usually a good idea to put pictures between the appropriate paragraphs in the text to get maximum impact (Pettersson, 1989, 1993). Pictures that are put within a paragraph will interfere with the reading of the text. Above the picture, there should be at least one blank line, if, of course, the picture is not at the top of the page, in which case the upper margin will provide sufficient empty space.

A picture can be tilted on the page. Tilting of a picture may draw special attention to it. Pictures on odd-numbered pages attract more attention than pictures on even pages. In newspapers and magazines it is quite often a correlation between placement of pictures and perceived importance. The higher up on a page a picture is placed, the more important it is considered to be (White, 1987). Many visuals have a built in "direction," e.g., a person on a photograph may be looking to the left. Such a picture should usually be placed on the right page for the person to look into the gutter and not out of the book or the magazine.

People who have not learned to read or write do not necessarily look at pictures in the order that has been intended by the

designer. Therefore it often proves helpful, as messages are being tested, to ask several groups of people to arrange the individual message into a sequence that seems most logical to them (Zimmermann and Perkin, 1982; Wileman, 1993, p. 105). This is a way to better get to know the intended audience.

Image framing

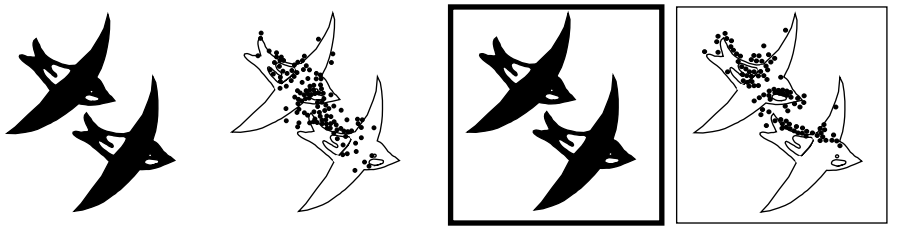
A frame or box around an image, or sometimes around an illustration and text, may have different functions. Image framing can be functional and/or attentional. A frame will separate the image from the surrounding context and draw special attention to information within the frame. In a newspaper, framing is a way of helping the readers to combine the corresponding text and pictures on the page.

Interplay of visuals

In many situations it is a good idea to use more than one visual to be able to convey information. We can use image pairs or sequences of images.

Image pairs

In many situations it is a good idea to use pairs of visuals in which one is true-to-life, such as a photograph, and the other represents an analytical representation, such as a simple line drawing. The analytical visual makes it easier for us to understand the content, and the realistic visual enables us to believe in the content. So the two visuals should be closely linked in a carefully thought-out relationship. The actual size of two pictures may reflect a natural relationship of scale, but it may also reflect a relationship of importance.

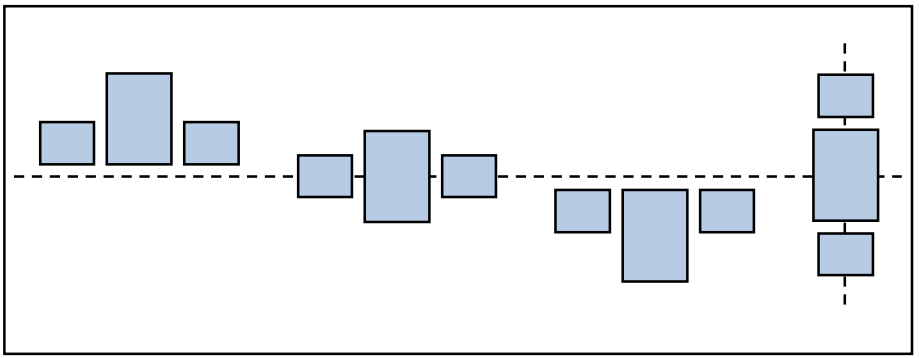


Seventeen subjects viewed a drawing of two flying House Martins. The recorded fixations have been plotted on the outlines to the right. Nine subjects viewed the version without a frame and made 117 fixations. Eight subjects viewed the frame version and made 98 fixations. Note that there are no fixations on the border in this case.

To enhance communication it is very good if there is a logical relationship between pictures placed next to each other. This relationship may be based on consistency or on continuity.

Image sequences

Usually picture editors seek a single picture that best sums up a situation. However, sometimes it is necessary to divide a message content into a series of visuals. The amount of details can be great or the content can include a certain period of time. Time scales and charts provide a reference in time and space. A sequence of pictures can be used to explain a development over time. It can be used in instructions. A sequence of pictures telling parts of the same story can be used to hold a chapter together to enhance the depth dimension in a printed material. A series of pictures can approximate the impression of a motion picture sequence. Several consecutive enlargements of a specific part will help the reader to understand the detailed structure of an object.



This example shows four ways of arranging a group of three pictures with reference to (dotted) "lines of balance". (Captions are not visible in this example.)

Understanding of a series of pictures is dependent upon the ability to recognize that the object or person in each frame is the same. For an exhibition, pictures and prints can be put on a wall or on a board in many different ways. Balance in the layout can be achieved by putting all pictures on a joint base line or bottom edge. Balance can also be achieved by putting all pictures centred on a "central" line. A third possibility is to put all the pictures on the wall with a joint upper edge. When we want to have two rows of pictures on the wall we can achieve balance with reference to a central line or cross or with reference to joint bottom and upper edges and joint side edges. A fourth possibility is to put all the pictures in a vertical sequence.

The picture area index

Evans, Watson and Willows (1987) discussed the use of visuals in Canadian textbooks. They analyzed 11,236 textbook pages. 8,304 pages had illustrations. Like Willows, Borwick and Hayvren (1981) they used two different methods of measuring visu-

als/pictures/illustrations. They calculated an "index of frequency," that is the number of pages with any visuals in percent of all pages (except title pages and indexes). The average index of frequency was 74% (8,304 pages with illustrations is 74% of all 11,236 pages). In lower grades almost all pages had illustrations (95-99%). For higher grades the index of frequency went down considerably.

In textbooks, however, many pages have more than one picture. When visuals are used for information the "actual number of illustrations" seem to be much more important than the number of pages with illustrations. A "picture index," defined as the average number of pictures for 100 pages, will no doubt be a better measure and more suited for international and intercultural comparisons of textbooks (Pettersson, 1990). It is possible to calculate both the index of frequency and the picture index from the data given by Evans et al. (1987) in their Appendices B and C. As seen from these data the index of frequency and the picture index really give complementary information about the textbooks. It is easy to see how the character of these textbooks changes from the lower to the higher grades within each group of subject matter.

Evans, Watson and Willows (1987) also indexed the sizes of illustrations according to the percentage of the page occupied by the illustrations. Illustrated pages were mentally divided into quarters and each illustration coded as occupying up to 25% of the page, 26 to 50% of the page, 51 to 75% of the page, or more than 75% of the page. For lower grades illustrations usually covered more than half of the page size. For higher grades the average picture size usually decreased.

Metallinos et al (1990) mentally divided the page into six equal parts, sixths, and indexed the sizes of illustrations in

secondary school geography textbooks. The most common size was 1/6 in Australia and Japan, and 2/6 in Greece, Sweden and the US. They defined the "average picture size" as the average part of a page covered by pictures.

It is however not enough to get information about the sizes of the pictures. We can also use an index showing how the available printed area in a book is divided between text and pictures. Thus the "picture area index" is defined as the average percentage of the text-face utilized for pictures. A book without any pictures will obviously have a picture area index of 0 (zero). When the index is 50 half of the total text-face in the book is covered with pictures. A picture area index of 100 leaves no room at all for any text in the book (except for texts printed in the actual pictures).

Some samples among textbooks for the 9–year Swedish comprehensive school showed that 60-80% of the text-face was covered by illustrations in textbooks used for elementary level (year 1.3). For intermediate level (year 4–6) the percentage was 50-70, and for senior level (year 6–9) it was 30-60, with a big difference between various textbooks (Pettersson, 1991).

Too be able to judge the quality of "mental indexing" of picture sizes and the possibility to use such data for calculations of picture area indexes the following study was done. One chapter in each of three textbooks were used and the picture area indexes were calculated with three different methods. (Melin and Pettersson studied these three textbooks with respect to captions, 1991).

Exact measurement

For each textbook the text-face was measured and calculated. Then all the pictures were measured. The heights and widths

were measured in millimetres. The areas of the pictures were calculated and added up to a total picture area. Then the picture area index was calculated. In the table below these results are called "the real picture area index".

Estimation with quarters

Based on the method used by Evans, Watson and Willows (1987) the text-faces (and not the pages) in each book were mentally divided into quarters and each illustration coded as occupying 0–1/4, 1/4–2/4, 2/4–3/4, and 3/4–4/4 of the text-face. (It is of course also easy to use a grid, drawn on a transparent foil.) Some pictures may actually be larger than the text-face. Such pictures are coded in the 3/4–4/4 category. The number of pictures within each category was multiplied by the theoretical mean size for each group (that is 12.5%, 37.5%, 62.5%, and 87.5% respectively). The added total picture area was then divided by the total text-face resulting in an "estimated picture area index".

Estimation with sixths

Based on the method used by Metallinos et al (1990) the text-faces (and not the pages) in each book were mentally divided into sixths and each illustration coded as occupying 0–1/6, 1/6–2/6, 2/6–3/6, 3/6–4/6, 4/6–5/6, and 5/6–6/6 of the text-face. (It is of course also easy to use a grid, drawn on a transparent foil.) Some pictures may actually be larger than the text-face. Such pictures are coded in the 5/6–6/6 category. The number of pictures within each category was multiplied by the theoretical mean size for each group (that is 8.8%, 25.0%, 41.6%, 58,3%, 75.0%, and 91,5% respectively). The added total picture area was then divided by the total text-face resulting in an "estimated picture area index".

Calculation results

To be able to compare the three methods the differences between the "real picture area index" and the two "estimated picture area indexes" were calculated. As we can see in the table below these differences are small with both methods.

Estimation with quarters resulted in a mean difference of 2.3%, and estimation with sixths resulted in a mean difference of .4%. These results indicate that the method of "Estimation with quarters" is "good enough" for practical work. This method is very quick and easy and far less labour-intensive than the laborious method of exact measurement.

The picture area index (%)

The picture area index	*Book 1*	*Book 2*	*Book 3*
Number of pictures in chapter	16.0	37.0	53.0
Number of pages	21.0	23.0	26.0
The real picture area index	26.9	32.6	26.5
Estimated picture area index, quarters	25.0	29.9	32.2
Difference 1	1.9	2.7	5.7
The real picture area index	26.9	32.6	26.5
Estimated picture area index, sixths	27.0	34.0	30,5
Difference 2	0.1	1.4	4.0

It would be interesting to make a collection of picture area index-data for "typical textbooks" used in different countries. At the same time one should also calculate the "picture index", defined as the average number of pictures for 100 pages (Pettersson, 1990).

Also see the discussion on the verbal and visual area diagram in the section *Text, picture, and background* in the chapter *Information graphics*.

Graphic symbols

Taking up only a very small amount of space, symbols can provide or convey information, equivalent to one or more sentences of text. Eco (1976) suggested that the verbal equivalent of an iconic sign is not a word but a phrase or indeed a whole story. This is, of course, also the case with the large number of Chinese kanji-characters, designating different words or sometimes phrases.

Use of graphic symbols

The use of graphic symbols has a long tradition. Functional, instructive graphic symbols are actually older than words. They are found in every culture however primitive. In specific areas symbols are a supplement to all languages to help create better and faster understanding. Symbols first appeared as paintings or carvings on caves and stone walls as early as 50,000 BC, with the first depiction of humans dating back about 11,000 years (Dewar, 1999). There is also a need for various symbols in modern societies.

Since 1974, the American Institute of Graphic Arts (AIGA) and the United States Department of Transportation (DOT) have collaborated to a symbol system that will help travellers navigate their way through airports, bus and train stations, large international events, and unfamiliar public places. This system, 2013 consisting of 50 symbols, is designed to guide people, regardless of age or culture, to where they are trying to go or what they are trying to reach.

Groups of graphic symbols

I distinguish between two main categories of representations, (I) figurative representations, and (II) non-figurative representations (Pettersson, 2002). Figurative representations include two groups, 1) *visuals* and 2) *graphic symbols*. The group graphic symbols include three subgroups 1) pictorial symbols, 2) abstract symbols, and 3) arbitrary symbols.

Pictorial symbols

Pictorial symbols (or representational symbols) are "image related" and simplified pictures. Pictorial symbols resemble the objects they represent. They can be characterized as silhouettes, shadows, or profiles with no surface detail. A traffic sign with a silhouette of a locomotive, to denote a railroad crossing, is an example of a pictorial symbol. In the design process, some pictorial symbols may be successively simplified into abstract graphic symbols.

Abstract symbols

Abstract graphic symbols can look like the objects they represent but have less detail than pictorial symbols. Good abstract graphic symbols are intuitive and we should be able to understand their meaning.

The modern symbols typically found in airports and in travel guides are intended to convey generalities of the same order of abstractness as words. Their characteristic graphic neutrality is perhaps the most significant aspect of their invention by the Isotype Institute (Neurath, 1936).

In athletic contests, like the Olympic games, abstract graphic symbols are often used to denote the different sports. Good abstract graphic symbols are intuitive and we should be able to understand their meaning.

Arbitrary symbols

Some figurative symbols are arbitrary symbols. They are invented and constructed out of the designer's imagination. Usually arbitrary graphic symbols have no resemblance at all to the objects or the ideas they represent. Many are based only on the use of geometric shapes and colours. Many signposts and some traffic signs are examples of signs with arbitrary symbols. Arbitrary symbols are unambiguous by convention. We agree and decide on their meaning. Just as new terms have to be learned when we begin to study a new topic; we have to learn arbitrary graphic symbols. Every motorist has to pass a test in order to get a driver's licence.

Cochenour, Rezabek and Westhoff (1998) studied the interpretations of 12 arbitrary graphic symbols. In this case, 96 respondents showed a wide disparity in their interpretations, with as many as 51 different categories of meaning for a single symbol. The average number of no-meaning responses was 13.

Visual terms

Quirk, Greenbaum, Leech, and Svartvik (1985) noted that pictograms most reliably can substitute for words in "block language" – single-word captions, headings, and labels – as distinct from sentenced language. Graphic symbols may be intended to convey generalities of the same order of abstractness as verbal terms. In some cases we can see graphic symbols as *visual terms*.

Symbols may be visually more distinctive than text. Their syntax and semantics may be simpler. Image perception is rapid, virtually "instantaneous". Reading and comprehending the equivalent message in words takes much more time, and may fail if it is hard to read the text. So symbols permit rapid

reading and comprehension, and they may require less time and effort for learning. This is important in numerous situations, e.g., in traffic, in industry, and in aviation.

Objectives

There are many reasons to use symbols. Graphical symbols can be used effectively in manuals and on screens to help readers to quickly recognize and identify a specific message. Symbols can be used to create an overview and provide a holistic perspective. This property is utilized in maps and informative signs as well as in catalogues and project reports.

Graphic symbols may be used to identify information, illustrate spatial and geographic position, illustrate size relationships, navigate in databases, provide a holistic perspective, recognize information, and represent an organization, a service, or a product. Graphic symbols may supply information and supply instructions. Graphic symbols have evolved to the point of universal acceptance in such areas as music, mathematics, and in many branches of science.

Pettersson (1985, 1987) concluded that perceived image content often is different from intended image content. Even simple pictures and also symbols may cause many different associations. A given set of basic picture elements and symbols can be combined to form completely different images. Moriarty and Sayre (1993) studied intended and perceived advertising meanings. They also found a high level of disagreement between intended and perceived messages. More than half of the responses were different from those intended, and expected by the message creators. Symbols must be meaningful, legible, learnable, memorable and used consistently (Dewar, 1999).

Interpretation of meaning

The meaning of a symbol is seldom easy to guess (Pettersson, 1989, 1993). Griffin and Gibbs (1993) and Olmstead (1999) found that graphic symbols were interpreted in many different ways. Sometimes only a few persons will understand the intended meaning of a symbol. Thus, the sender will always have to supply explanations for the symbols used in any specific situation.

Symbols in business presentations

Griffin et al. (1994) studied how international business people interpreted and understood commonly used clip art symbols. The symbols were taken from a Harward Graphics software package for business presentations. Based on 4.530 opinions expressed by 302 subjects in Japan, Sweden, Tanzania and USA, regarding 15 of the symbols in the software package, Griffin et al. made the following two conclusions:

- There are many ways to interpret symbols. Very few people share the same understanding of any given symbol.
- There are strong cultural differences in interpreting the meanings of symbols.

Four *verbal symbols* resulted in many interesting observations. Symbols in this category were either understood or not understood at all. There were 90% appropriate answers from the subjects in the USA and 80% from the subjects in Sweden. For Tanzania and Japan, the corresponding numbers were 27% and 29% respectively. The rank ordering of the mean numbers of appropriate answers for seven pictorial symbols was 80% for USA, 69% for Sweden, 60% for Tanzania, and 48% for Japan.

The remaining four *abstract symbols* were the most difficult symbols for all audiences to comprehend. Here definitions differed strongly by symbol and by country. In this study, it was common for the subjects to give several meanings for each symbol. For example, the 81 Swedish subjects gave 20 different interpretations of a symbol with the shape of a star.

Telecom symbols

Nilsson and Lindqvist (1995) examined a random selection of telecom publications. They browsed about 1,000 pages. From this material they selected 100 pictures and registered about 350 symbols. This study concluded that:

- A certain meaning is explained with several different symbols.
- A certain symbol has several different meanings.

Within the Swedish R&D-company Ellemtel, several people designed their own graphic symbols, which they used in various combinations in schematic pictures in their own documents. In a sample of documents no less than 29 different "telephone-pictures" were used. Some of the telephone-pictures were well-drawn schematic pictures, some were pictorial symbols, and some were abstract symbols. All these symbols were replaced by two graphic abstract symbols for telephones, passive, and active to be used in schematic pictures.

Sepulchral symbols

Cochenour and Rezabek (1997) studied the interpretations of 21 sepulchral symbols. Respondents showed a wide disparity regarding their ability to understand the meanings of the symbols. Nineteen percent of the responses indicated that no meaning at

all was conveyed, while 31% were considered as "understanding of the intended meaning".

It was concluded that symbols carry a variety of meanings for different people and when used alone as a means of visual communication cannot be expected to always convey a simple and direct meaning.

International communication

At the end of his book on writing for science and technology, Kirkman (1992) noted the possibility of using symbols in international communication (p. 155): "Perhaps, since use of words causes so much difficulty in international communication, we should abandon words wherever possible, and use icons instead."

Kirkman used the term "icon" in the same meaning as I use the term "symbol" here. Symbols may very well become more and more common in international communication. Kirkman commented (p. 156): "I have no doubt that we shall gradually have to include more and more icons in our presentations of information, especially in our onscreen presentations."

Many pictograms are culturally biased (Mangan, 1978) and thus arbitrary to those from other cultures. For example, when using a guidebook with symbols, we often have to look them up in a key in much the same way as we look up unfamiliar words in a dictionary. Their iconic origins may only become apparent after we are aware of their intended meaning (Waller, 1987). Baron (1981) reported that iconicity is a surprisingly unimportant factor in the learning of sign languages for the deaf, autistic, or mentally retarded.

Warnings

Many symbols are used for warnings. Wogalter (1999, p. 94) concluded that warnings should contain certain elements:

- A signal word such as “Danger” and “Caution” that enables people to recognize that the message is a warning, that a hazard is present, as well as providing information on the hazard level (with “Danger” signalling more serious and probable injury than “Caution”);
- A description of the hazard, e.g. in the case of a no diving sign, a statement such as “Shallow water” provides information about the specific danger involved;
- A description of the consequences that could occur if the person fails to obey the warning’s directions, e.g. “You can be permanently paralysed”;
- The directions or instructions, i.e., the specific actions that should or should not be done, e.g. “No diving”.

Traffic signs

Traffic signs and signposts are various devices with symbols, placed along, beside, or above a highway, roadway, pathway, or any other route in order to guide and help with directions, regulate the traffic and warn for dangers. The symbols may also be painted on the streets, and sometimes on the walls of houses. The traffic may include many kinds of motor vehicles, bicycles, pedestrians, equestrians, and other travellers.

Road signs and signposts may be divided into four main groups: 1) warning signs, 2) mandatory signs, 3) prohibitory signs, 4) information signs, and the STOP sign. Today road signs are manufactured in different materials such as aluminium, galvanized sheet iron, and flat or canned panel. The

symbols on the signs may be painted, printed or duplicated in other ways. The motifs are designed in different ways even if they carry the same or similar messages.

In order to be effective a warning must reach the intended audience and make them adopt the desired behaviour. Several steps are needed here. Initially any warning, and any symbol, must attract and capture the attention of the intended audience, the persons who need the information. The message must be legible at the appropriate distance, and must often be legible when seen for a short period of time under bad lighting conditions. A driver on a highway may only have a second or two to read a signpost. Then the message in the warning must be mentally processed and understood correctly by the intended audience. The action to be taken should be immediately obvious. Furthermore the message in the warning must be able to motivate the audience to comply with the desired behaviour.

Wogalter pointed out that warnings should have properties that allow them to be seen in degraded conditions such as low illumination, smoke or fog. In addition warnings should be adequately lit by direct light or by backlighting and/or have good reflectance so that they are visible under reduced-light conditions.

The United States Department of Transportation (2002) says that roadway signs in the United States increasingly use symbols rather than words to convey their messages. Thus it is important that these symbols maintain a consistency in colour, layout, position, shape, space, and text.

Project "crossing road signs"

In communication senders want to communicate messages or make information sets available to receivers or rather interpret-

ers. In the case of warning for and information about pedestrian crossing road signs the city authorities are the senders. The *warning sign* for a pedestrian crossing convey a simple, but important message to the *motorists*: "Look out and slow down, there is a pedestrian crossing ahead."

The warning sign is usually triangular, sometimes with soft corners. The actual symbol consists of a wide line forming a red triangle on a light background. Graphical elements, dots, lines, and areas, are combined to form a simple figurative and abstract representation of a person. Some graphical elements may also represent street lines showing a pedestrian crossing.

A warning sign and an information sign (from Germany and Lithuania). These symbols are classified as abstract.

The *information sign* is usually rectangular, sometimes with soft corners. This particular sign is from Lithuania. The triangle is white. The four graphical elements representing a person and the three graphical elements representing street lines are black. The person appears to be running to the right. This symbol is classified as abstract. In Lithuania there are also signs with the person moving to the left. Graphical elements, dots, lines, and areas, are combined to form a simple figurative representation of a person. Some graphical elements may also represent street lines showing a pedestrian crossing

An information sign about a pedestrian crossing convey slightly different messages for two different groups of receivers. The message to the *motorists* and other road-users is: "Look out

and slow down, pedestrians are told to cross the street here." The message to the *pedestrians* is: "This is the place where you should cross the street."

In several locations this information sign is replaced by traffic lights. Traffic lights convey more distinct and "sharp" instructions to motorists as well as to pedestrians than pedestrian crossing warning and information road signs. Here the instructions may be expressed as: Drive now!, Do not drive!, Walk now!, and Do not walk! respectively. The design of traffic lights may vary substantially. However, design of traffic lights is not included in this study, and not the sign explaining that children may be playing alongside the road as well as on the road.

I have studied 52 different pedestrian warning and information crossing road signs from 32 countries (Argentina, Australia, Austria, Belgium, Chile, China, Cyprus, Czech Republic, Estonia, France, Germany, Ghana, Iceland, India, Ireland, Italy, Jordan, Lithuania, Luxembourg, Malta, Mexico, Monaco, Norway, Peru, Poland, Portugal, South Africa, Spain, Sweden, Switzerland, Turkey, and USA). Some of these road signs were brought to Stockholm for a public exhibition in 1998. The graphic designer Anders Körling put together the exhibition *The World Walks in Stockholm* as an event during *Stockholm '98, Stockholm – Cultural City of Europe 1998*.

I have taken photographs of the crossing road signs at this exhibition. I have also taken photographs during my travels in various countries. In addition, people have provided me with photographs of crossing road signs in a few cases. The pictures presented in this report have been scanned and retouched using Adobe Photoshop™. The design of pedestrian warning and information crossing road signs wary with respect to shape, colours, size, and number of graphic elements in the symbols.

Pedestrian warning and information crossing road signs wary with respect to their shape, colours, size, and the design of the figurative representations. Differences in material and in the construction of the signs and signposts are not discussed here.

Shape

Pedestrian crossing warning and information road signs vary in shape in different parts of the world. In Europe warning signs are triangular, and information signs are rectangular. In other parts of the world the information sign may be rhombic or circular. On the signs from Argentina and Australia (below) the pictorial symbols are black on yellow backgrounds without indications of street lines.

The rhombic sign is from Argentina. The circular sign is from Australia. These symbols are classified as pictorial.

Colours

Pedestrian crossing warning and information road signs vary in colour in different parts of the world. In Europe warning signs usually have white, black and red colours. In Sweden warning signs are yellow, red and black. In many countries the information signs are blue, white and black. In some countries the colours are white and blue; white, yellow and black; or yellow and black. The figurative representations are usually black, on a white, blue or yellow ground.

Size

Pedestrian crossing warning and information road signs vary in size in accordance with different situations. In this sample the smallest sign is 40 cm (Cyprus) and the largest is 102 cm (Argentina). Most signs are between 55 and 75 cm. Thus the impression of the pictures printed here corresponds to the impression from viewing the real signs on a distance of five to seven metres.

Mr. Walker

The symbol on pedestrian crossing warning and information road signs varies to a large extent. Almost all are different with respect to their design. The person in the symbol is always "Mr. Walker". Nowhere is the figure representing a woman (Jofs, 1998). (However, in 2010 Sweden introduced Mrs. Walker.) In this study Mr. Walker is an abstract graphic symbol (27), or a pictorial graphic symbol (25).

As would be expected, there are no arbitrary graphic symbols in this sample of symbols. In one case (Australia) the sign only shows the lower parts of the legs and the feet. For some reason most figures on the signs (44) are crossing the street to the left. The remaining figures (8) cross in the other direction, to the right. Estonia and Lithuania have both versions of signs.

Most figures (42) seem to be walking across the street over to the other side, but some (9) appear to be running. In one case (Mexico) the figure appears to be standing and waiting for a possibility to cross the street. In two other symbols from Mexico the person appears to be walking. Some symbols are very clear and distinct; others are "blurred" by a large number of graphical elements.

The person is an abstract graphic symbol on the left warning sign from Malta, and a pictorial graphic symbol on the middle sign from Switzerland. In both cases the person is apparently walking to the left. The number of graphical elements varies to a large degree. On the sign to the right, from France, there are a total of 21 graphical elements. However, Mr. Walker consists of only one graphical element.

The numbers of graphical elements that are used to build the Mr. Walker figure vary (1–4), as the total number of graphical elements (2–21) in the symbols. The painted street lines form a graphic world of their own. The number varies to a large degree. In some cases the street lines may actually confuse people. Some signs have no street lines. I don't know to what extent the number of street lines varies in different countries. In Sweden there are four graphical elements on the symbols representing the lines painted on the streets. The number of painted lines may vary.

Message and symbol

There are probably pedestrian crossing road signs in most cities, at least where there are cars. In accordance with international conventions and national legislations on road signs and signals, most road signs have a similar design. For example, the official manual for signing in the United States is the Manual of Uniform Traffic Control Devices. Warning signs are triangular.

Information or instructional signs are rectangular or rhombic. Prohibitory signs and mandatory signs are round, but with different colours. However, the symbols on the signs may differ in several respects. Every country has its own version, or rather versions, because some road signs differ between regions. Some of these signs may be old versions, waiting to be replaced.

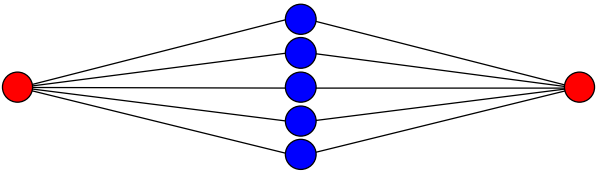
There are several requirements that must be fulfilled for road signs to be effective communicators of information. The designs of the symbols, as well as the contexts in which the signs appear are important factors for our perception of the message. The abstract or pictorial graphic symbol on the sign must have good legibility and be easy to read from a distance. Obviously pedestrian crossing road signs must be placed in such a way that the motorists as well as the pedestrians can see them. Signs should be placed as necessary for safety and proper regulation of traffic. However, the use of too many signs within a given location severely reduces the effectiveness of each individual sign at that specific location. There is a distinct risk that we will not see, or pay attention to, some of the signs.

It is quite obvious that traffic symbols are designed in many different ways, even when they convey the same or similar information to people who see them. This may not seem to be an important issue, since people have to learn the meaning of all-important symbols within their own society. However, with respect to the fact that international travel seems to increase all the time, and more people visit different countries, it would probably be an advantage to have a “world standard” for a set of basic symbols, including traffic signs. This would probably reduce the number of misunderstandings and traffic accidents.

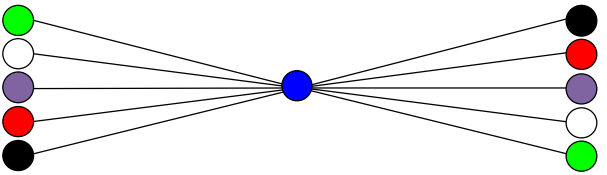
The above study supports the assumptions noted in the introduction. We may conclude that:

1. A specific message may be communicated to the receiver/s or interpreter/s with several different symbols.
2. A specific symbol may be used to communicate several different messages.
3. People have to learn the meaning of all-important symbols within their own society.

The first two points may also be visually represented in the following two schematic pictures:



A specific message (left red circle) may be communicated to the receiver/s or interpreter/s (right red circle) with several different symbols (middle blue circles) as long as people learn the intended meanings of the symbols.



A specific symbol (middle blue circle) may be used to communicate several different messages (left circles) to the receiver/s (right circles) as long as people learn the intended meanings of the symbol.

With respect to the design and the use of symbols it may further be concluded that:

1. Receivers may interpret symbols in many ways.
   - Few people share the same understanding of any given symbol.
   - People can usually not guess the meaning of symbols.
   - There are strong cultural differences in interpreting the meanings of symbols.
2. The senders will always have to supply explanations for symbols.
   - Symbols should be used in a consistent way.
   - A symbol must always have the same meaning within a specified context.
3. A good symbol is designed so it can be used in many different situations and in many contexts. A good symbol:
   - Is simple
   - Is clear
   - Has optimal size
   - Has good contrast in form, dimension, and colour
4. Graphical symbols may be intended to convey generalities of the same order of abstractness as verbal terms. In some cases we can see graphical symbols as visual terms. Graphical symbols may be used to:

   - Create an overview
   - Identify information
   - Illustrate position
   - Illustrate size relationships
   - Navigate in databases
   - Provide a holistic perspective
   - Recognize information
   - Represent an organization, a service, or a product
   - Supply information or instructions

Design of graphic symbols

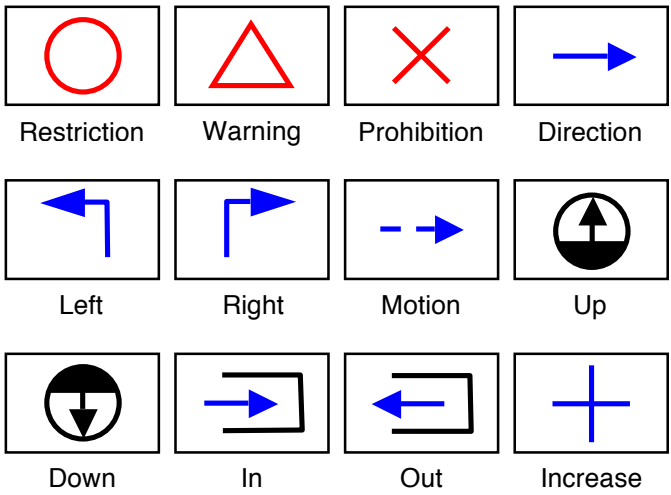
With the increase in international travel and trade, there is a growing need to communicate with people who do not understand the language of the country they are in. The use of symbols is one of the most common ways to deal with this situation. However, there may also be some disadvantages with symbols. Symbols may be less efficient than text in conveying abstract, as well as detailed information.

Dewar (1999) pointed out that the specific criteria for individual symbols or sets of symbols depend on their application. Legibility distance is essential in the case of traffic signs and many building signs, but not for symbols on maps or consumer products. There is a need to have as much uniformity as possible across different information systems.

When designers in different countries are working on similar problems they may create different solutions and different designs. In each case, they may have clear intentions and objectives with his or her information sets. However, it is always up to the interpreter/s to conceive or misconceive information, to use or not use it, to use or misuse it. It is reasonable to assume that 1) a specific message may be communicated to the receiver/s or interpreter/s with several different symbols. It is also reasonable to assume that 2) a specific symbol may be used to communicate several different messages. If the assumptions above are correct it would indicate that 3) we will normally have to learn the intended meaning of symbols.

Symbols are often composed of simple graphical elements, such as lines, circles, ovals, squares, rectangles, triangles, or combinations thereof. Distinctively shaped letters are often utilized in modern symbols. Regular, simple, geometrical figures are identified more quickly than complex ones. Keates

(1982) noted that discriminatory responses to map symbols depend on contrast in form, dimension, and colour. The problem of discrimination is generally more critical in monochrome maps, in which only contrasts in form and dimensions are possible for lines and small symbols.



Here are some basic symbols that appear and reappear in many different situations in various countries throughout the world.

The graphic symbol should be simple and clear with a distinct contrast to the background. A clear and stable figure to ground articulation is essential (Dewar, 1999; Easterby, 1970). The figure ("foreground") should be stable, i.e. spontaneously organised as one unit. This is achieved by close boundaries, appropriate line thickness and any other graphical means that help the visual system to organise the figure as one unit. It is most appropriate to use silhouette (side) views of certain components such as vehicles (Dewar, 1999). The graphic symbol

should be as symmetrical as possible. It should appear in an optimal size.

A good symbol is designed so it can be used in many different situations and in many contexts. For example, the McDonald's M or "golden arch" is designed to work in every conceivable size, from a few millimetres high in a brochure to more than six feet high in outdoor signs. It is often an advantage that a symbol is iconic, that is, that it looks like the real thing it represents. Then it may be intuitive to the users. In technical and scientific systems and documentation, hardware may be symbolised by squares and rectangles or the like. Softer forms, such as circles or ovals, may represent software.

Graphic symbols often make use of bright colours to intensify their meaning – in fact in some instances a change of colour creates a diametric change of meaning. Common hues are pure yellow, red, blue, green, white and black, or combinations of the same. Colour creates instant impact. It becomes a vital part of the first impression created. Industry employs colour coding in many areas. The countless wires in a complex cable are instantly traced by their hue; the colours of knobs and buttons on vehicles and machinery signify what they control (Dreyfuss, 1972).

However, since many people are colour-blind colour can only be used to code the information redundantly. Colour may be combined with shape or position. Complementary colours contrast and provide a warm – cool effect. Colours may have many different meanings; some common meanings and their attention getting values are listed in the table on next page.

Colours and their attention getting values.

Hue	*Meaning*	*Attention*
Red	Alarm, danger, financial loss, fire, heat, stop, unsafe, warning	Good
Orange	Warning	Good
Yellow	Caution, danger, hazard, oil, slow, warning	Good
Green	Go, normal state, safe, satisfactory	Poor
Light blue	Advisory, aerated water, cool	Bad
Dark blue	Advising, untreated water	Poor
Magenta	Alarm state	Good
White	Advisory, stem	Poor
Black	Financial gain	Poor

Warnings should have high contrast relative to the background (Barlow & Wogalter, 1991; Sanders & McGormick, 1993). They should have large, legible bold-faced alphanumeric characters (Wogalter, 1999). Legibility of symbols can be enhanced with the application of a few simple guidelines:

- Use realistic figures rather than abstract forms.
- Make important figures and characters large.
- Eliminate unnecessary elements.
- Use solid figures, not outline figures.
- Maximize separation between features in symbols.
- Maximize the luminance between a symbol and its background.
- Maximize the colour contrast between a symbol and its background.

According to Keates (1982), the use of colour on maps introduces a large number of variables that can enhance contrast, and therefore extend the number of perceptual differences that can be employed in discrimination. The effect is to aid legibility, and therefore to increase the total range of information which the map can present. Shape and colour components are often used for designating a link or relationship between groups of messages. The recognition of geographical features is much enhanced when areas are differentiated by hue. At the same time, complex colour arrangements may raise problems in discrimination, so that although multi-colour maps enlarge the graphic possibilities, they also increase the probability of errors in the judgment of discrimination.

According to Keates (1982), the most common case of quantitative judgment on maps occurs in the use of proportional symbols, that is, point or line symbols constructed to represent specific quantities. Cochenour, Lee and Wilkins (1995) provided the following design guidelines for functional image maps on the World Wide Web (p. 172):

- Use simple graphics.
- Use smaller rather than larger image sizes.
- Clearly define hot spots.
- Use unified rather than scattered groupings.
- Choose icons to accurately represent information.
- Design image maps to model the structure of information.
- Use a minimal number of layers.
- Limit the number of choices.
- Include site-specific help.

Before beginning to design new symbols, it is a good idea to consider employing already existing symbols. Standardisation

bodies like ISO (International Standards Organisation) and IEC (Commissioin Electrotechnique Internationale) officially recommend many symbols. There are a large number of existing symbols for different areas. Symbols are employed in different media. They are static and immutable in graphical media and may be more changeable in computer-based media. As noted in the introduction designers in different countries may be working on similar problems, and they will often create different designs. Most, if not all, countries have traffic signs. In order to study similarities and differences in the design of a highly restricted message I have analyzed pedestrian warning and information signs. This study is presented in the following section.

Information graphics

A particular message is sometimes presented through the purposeful integration of text, pictures, and graphic design into a clearly delineated and structured area, i.e., a functioning whole. (However, nowadays this term is used in a broader sense.) The result of this work is a product called information graphics, infographics, or just “graphics”. Infographics are widely used in newspapers and newscasts (Lamb et al., 2014; Smiciklas, 2012).

Examples of confusing, misleading, and ineffective graphics are everywhere. The three essential elements of good information graphics are rich content, inviting visualizations, and sophisticated execution (Wong, 2010, p. 14). *Rich content* brings meaning to the graphic. *Inviting visualizations* interprets the content and highlights the essence of the information for the reader. *Sophisticated execution* brings the content and the graphics to life.

Graphics

Information graphics are informative and may be entertaining. They aid communication, enabling better understanding and comprehension. Information graphics are attention-getters when they appear on a page in a newspaper. They may improve readability and increase retention. In the past, information graphics were produced by hand, a tedious work process. Today most information graphics are produced with computers. Ideas can be tested in less time and good solutions may be found. In the case of newspapers, information graphics can be the key to attract new readers and to hold on to old readers.

Most visualizations originating from quantitative data include axes or scales, coordinate systems, or glyphs that use spe-

cific data values as an input (Schroeder & Martin, 2005; Tversky, 2011). The appearance of a glyph corresponds directly to data as a result of the principled mapping of data variables to visual features such as position, size, shape, and colour (Heer, Bostock, & Ogievetsky, 2010, p. 67).

Information graphics provide the reader with a rapid and easily grasped overall view of a message and are therefore highly suitable as an introduction to and summary of a subject. However, conventionally illustrated text is better for analysis, discussion, and study of details. So information in graphical media can utilize text, pictures, information graphics, and graphical design in conveying its message. We should note that the word "graphics" can be used for completely different concepts:

1. One or more art forms in which copies can be made on paper or the same original.
2. Activity involving the printing of the written word.
3. Integrated presentation of text, pictures, and graphical design, in, e.g., the daily press, information graphics.
4. The technique of presenting data in the form of figures on a video display screen.

Wainer (2009, p. 31) observed: "graphical representation has been shown repeatedly over the past two hundred years to be perhaps the best way to communicate complex technical information to an intelligent, lay audience."

Modern information graphics has its origin in the tradition of making posters and advertisements. These visual forms of communication spring from a past in older times, when pictures and texts were combined into information on circus and theatre posters, and on signs on inns. The oldest posters were created at

the end of the 14th century, when printing on paper started in Europe. Leaflets with text and simple pictures were used by booksellers and travelling theatre companies and for political agitation. The posters' roots are traced back to antique Athens, where the City Fathers put up notices with regulations.

On the 15 September 1982, a new type of daily paper was published in the USA. From a central editorial office in Arlington, Virginia, the produced pages were sent via a satellite to printing houses in eleven cities, where USA TODAY was printed. As early as after one year, the daily edition was more than one million copies. Today, the paper is widely spread in the USA and in several other countries. The paper is as fragmented as the news on television, and is often being criticized for being too superficial.

USA TODAY is the paper of the television generation. It is divided into four separate sections: news, money, sports, and life. Each section has its own cover story and a large number of easily read mini-notices. The headings are effective, and the text is often heavily edited. The notice has replaced the article. There are many photographs, drawings, diagrams, and tables in the paper. USA TODAY introduced modern information graphics, and has been imitated all over the world. It is the most copied paper in modern times.

Baer and Vaccara (2010) provided a range of interesting case studies, from print projects to interactive and environmental information systems.

According to Lankow (2012) a test of the efficacy of an infographic is based on the first-century architect, author, and engineer Marcus Vitruvius Pollios principles of good design. We need to measure the *utility*, the *soundness*, and the *beauty* of infographics. *Utility* refers to whether the infographic meets the

designer's objectives or not. *Soundness* refers to whether the information presented in the infographic is complete, correct, and valuable to the viewer. *Beauty* refers to whether or not the design of the infographic is appealing and appropriate.

Objectives

According to Malamed (2011) the goal with information graphics is to re-present data in such a way that the intended audience is able to quickly grasp the content primarily by sight. Infographics are a popular visual approach to deliver abstract, complex, and dense messages in small areas (Lamb et al., 2014).

In order to facilitate an audience's engagement any infographic must exhibit appearance and explanation (Stone & Hall, 1997). Visual appeal attracts viewers' attention. The explanatory power of the infographic makes complex subject matter apprehensible and intelligible.

Information graphics can be classified according to different criteria, such as objectives, medium, size, and time for production (Pettersson, 1993). One specific graphic may very well belong to several of these groups. We can produce information graphics in order to achieve several different objectives.

Content graphics or everyday graphics are used for information about the content in packages, e.g., with food. Decorative and artistic graphics can also be found in both newspapers and television. Even though informative graphics can be both decorative and aesthetically attractive, the informative function always predominates. So purely decorative and artistic graphics are not information graphics. Explanatory graphics depict the ways things were, are, or will be, for example, the weather. These graphics range from simple drawings to complex combinations of drawings, maps, and photographs.

Expo graphics are used at exhibitions and trade fairs. A subject matter is presented using verbal and visual technique and the real objects. The graphical information may aid understanding of how the real objects can be used. Instruction graphics are used for instructions, e.g., in instructional manuals. Instruction graphics may deal with how to use, e.g., a machine or how to prepare, e.g., a meal step by step. Locating graphics are used to give the physical location of an event or of an object. Based on one or more maps, movements of an object can be explained. Maps are often also included in other kinds of graphics.

News graphics are used to convey all kinds of news that are fit to see rather than to read as a printed story. News graphics are found in, e.g., newspapers and some magazines, and also on television. Presentation graphics depict facts and are often used for different types of statistical tabulations. A graph or chart can be integrated into a symbolic image to heighten impact and identify the subject. Presentation graphics are often used for "business presentations". They are often called business graphics. Signal graphics are small-scale graphics used to add impact and visual relief to a text.

We encounter information graphics in many different media. In graphical media (books in particular), information graphics are often referred to as lexivision or lexi-visual information. In the corresponding manner, lexigraphy is a special version of infography for lexi-visual presentation of text, pictures, and information graphics in graphical media and computer media. Graphics are called reference book graphics in reference books and dictionary graphics in dictionaries and encyclopaedias.

There are a number of different types of information graphics, and we will probably see more types in the future (Lankow et al., 2012; Marcel, 2014).

Time for production

The American national daily newspaper USA TODAY was a pioneer in the field of news graphics and has acquired imitators in many different countries. “News graphics” is a summarizing designation for several somewhat differing forms of information graphics. Due to available time for production, news graphics can be divided into several groups.

Business graphics are found in computer programs and newspapers. Business graphics is a general designation for information graphics that present economic and statistical data, for example: “the production of crude oil over the past five years”.

Daily graphics is generally produced against tight deadlines. News must be published in the next edition or in the next TV news slot.

Feature graphics is a general designation for information graphics that describe more timeless subjects, such as popular science. Here, the producer may have several weeks or even months to create the copy and acquire the photographs.

Planned graphics is the designation for information graphics that the editors of news graphics have a few days to produce. This provides more opportunities for checking facts and more carefully thought-out execution.

Weather graphics are information graphics describing what the weather has been like and how it is likely to be according to available forecasts. The colourful weather graphics in

USA TODAY have inspired a large number of dailies to introduce information graphics.

George-Palilonis and Spillman (2013) asked 53 editors about their attitudes and practices about use of infographics. Most editors supported production of infographics. However, the availability of time and in-house expertise, as well as cost effectiveness influenced their decisions.

Everyday graphics

Today, we can study some of the classical posters from last century in art books and in museums, such as the Musée des Arts Décoratifs in Paris. It is possible – and even plausible – that some of the information graphics of today will find their way into art and design museums at the turn of this century or at the beginning of the 21st Century. (See *Information Design 6 – Predecessors & Pioneers*.)

In several respects, everyday graphics differs from other types of information graphics. Contrary to the producer of news graphics, but in conformity with the producer of posters, the producer of everyday graphics often has time enough to plan and design the messages in an optimal way. However, the space is often very limited for everyday graphics. The amount of the article and the size of the product decide the space available for the design of the verbal and visual message. In practice, however, there is the same space for about the same amount of information on a small label as on a normal-sized poster. When we read a label of 4x4 cm at normal reading distance (30-40 cm), it corresponds to a poster of 100 x 100 cm when reading it at a distance of 7-10 metres. When it comes to reading posters, the most common way of reading is at a distance.

Just as posters, everyday graphics have short and concentrated messages, expressed in a clear and distinct way. We read everyday graphics close at hand, but the retinal picture is about the same as the retinal picture of a poster. Usually, we do not take much interest in a label or a package. The same applies as for posters: it must not take too much time to perceive the message in everyday graphics. Just one glance should be enough to perceive the contents of a package. Does the bag contain sugar, salt, or pepper? Is there raspberry jam or strawberry jam in the jar? Is the content poisonous, or harmful for us in any other way? In several respects almost the same conditions apply when taking in information printed on a poster and on a label or a lid.

When it comes to small packages, the message still has to be designed in order to fit in all the necessary information. Sometimes, there is a national law on how to describe the contents in detail. For some products, there may also exist different types of instruction, enclosed inside the package. As producers of posters, the producers of everyday graphics are usually totally anonymous to the consumers. In other forms of information graphics, however, the names of the originators are often mentioned, and thus known. Everyday graphics may have different purposes. It may be used to instruct, to inform, and to tempt a presumptive customer to buy the product in question.

Instructing

Sometimes, schematic pictures are used to show how something is to be performed in several steps, in a practical, simple, quick, and safe way. One example is how to open a vacuum package with cheese, and another example is how to put together several separate parts in a box into a chair or a table. Symbols are often used. An arrow, for example, may be used to show where to

open a package and to explain in which order and in which way to join different parts.

Informing

Everyday graphics informs on the contents of the package, and the characteristics of the article or product. When it comes to provisions, for example, there is a legislation in some countries that sanctions that some types of data must be on the package. This may be information on the contents of the product, day of manufacture, keeping qualities, producer, and place of manufacture. Also trade agreements and practice within different trades may demand that a certain kind of information – for example, the price of a certain package and the price per kilogram – is presented to the customer/user.

Often, a large and lifelike colour *picture* – a photograph or a realistic drawing – gives the reader direct information or an association on the type of contents in the package. The picture is often reinforced with a *heading* in a large size of type and a clear typeface, often in upper-case letters. There is often a short *text* with detailed information describing the characteristics of the article or product.

The *declaration of contents* may be presented as a compact table or a short text. These texts are often set in a small size of type and printed in a colour with little contrast to the background. It always takes much more time to read the text than to read the picture.

Two examples of everyday graphics printed directly on the lids on individual portions of strawberry jam (life-size). The products are from Denmark (left) and Germany (right).

A *graphic symbol*, such as an arrow, will often show where to get hold of the lid and the direction where to pull, in order to open the package. Often a text and a *logotype* show the *name of the company* that produced the jam. The *address* to the company may be small. There may also be information about the *weight* of the content as well as an *identification code*. Figures written with an ink-jet printer show when the jam was packaged and/or a "best-before date."

Everyday graphics with clear pictures may also work internationally, with people that do not know the language in question. Here, the context is very important. The pictures of oranges, strawberries, raspberries, or peaches on individual portions of jam serve as a good guidance of the people who queue at the hotel's breakfast buffet. On several occasions, I have wit-

nessed how people after a quick glance on different packages make their choices. Obviously, just a few eye fixations on a package are enough to tell the contents.

An example of everyday graphics printed directly on the lid on two individual portions from a company in the USA, with strawberry jam and orange marmalade (life-size).

Tempting to buy

Often, the purpose of everyday graphics is to tempt consumers to buy a certain product in competition with other, more or less similar and equally good, products. Regarding everyday graphics, the "sender" often has the possibility to turn to experts for help to design the message. Experienced marketing people and publicity expert's work closely together to produce verbal and visual messages that function well. The text, pictures, and graphical design of everyday graphics are, therefore, given the final design with great care and consideration.

Everyday graphics may be both aesthetically attractive and decorative. There are also examples of everyday graphics that

are not successful. A label or package are often a part of a series: a part of a carefully prepared “whole”. Products from one company may be kept together with the help of a common logotype or symbol, and with similar graphical design for the choice of typeface and size of type, the placing of texts, and the use of pictures, colours, and decorative patterns. The intention is that a specific graphical profile will make it easier for the receiver/customer/reader to quickly and easily find articles and products from the same supplier.

Tempting to buy? Here is an example of an elegant and free and easy design from France (left), and an example of a less successful design from Belgium (right).

The illustration above contains a life-size example of an elegant and free and easy design of a lid on an individual portion of strawberry jam from a French company. The semicircular text contains information in several languages on the contents. The three strawberries are printed in colours that look “warmed by the sun” and naturally yellow-orange-red. There is

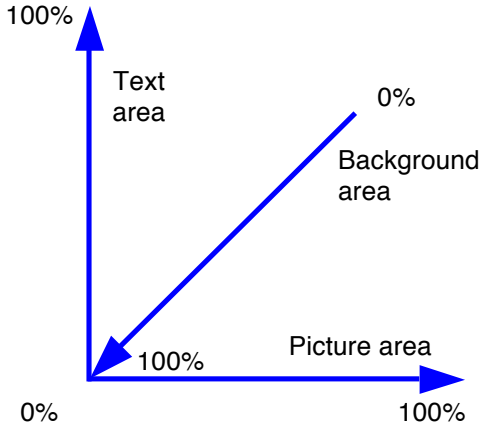
no doubt about the contents or about the trademark. Compare with the life-size example of a less successful design of a lid on an individual portion of strawberry jam, from a Belgian company. Here the message “strawberry jam” is clear, but the berries do not look appetizing enough. The colours are not successful and the trademark is very dominating.

Text, pictures, and background

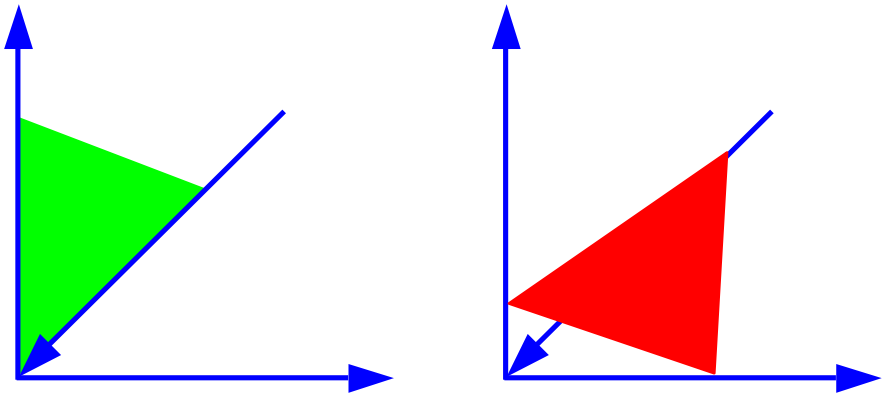
Within a given area – such as a page in a book, a poster, or a label – the designer may distribute text, picture, and background (margins, space, and patterns and designs without any significant picture elements). A typical page in a telephone directory is almost filled with text. There are no pictures, and the margins are narrow. Here, the text may cover 90 percent of the page. In dictionaries the text covers 75-80 percent, and in specialist books and factual study books the text often covers 50-60 percent of the page. In pure literature and children’s books, the text area is often even smaller than that. When a picture covers the entire page, there is no space for text or margins. The third extreme is an empty page.

In order to easily compare the relationship between text area, picture area, and background area in different graphical verbal and visual messages, I have developed a “verbal and visual area diagram”. This diagram has three axes: text area, picture area, and background area. The three axes are graded from zero to 100 percent. The text and picture axis have a common starting-point, and there is a right angel (90 degrees) between them. The background axis is situated at a 45-degree angel to both the other axis, and it is graded from the outside towards the starting point of the other axis. Consequently, where

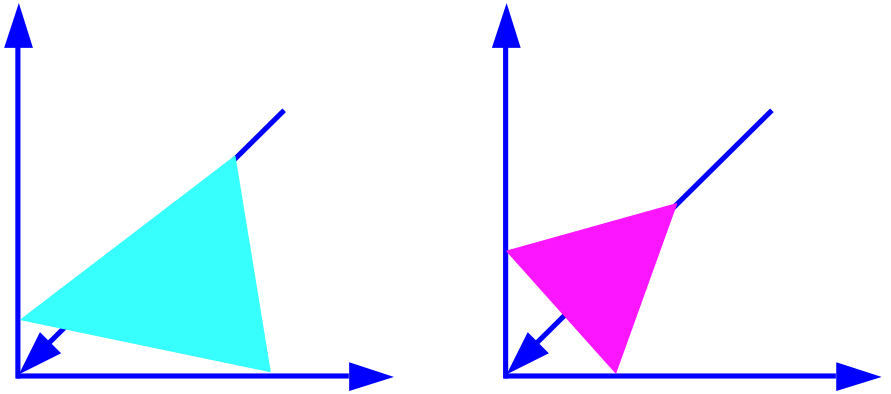
the text and picture areas are zero, 100 percent is background/emptiness.



This is the structure for the verbal and visual area diagram. The three axes are graded from zero to 100 percent.



Visual area diagrams for a fiction book without illustrations (green), and an illustrated book for children (red) may look like these two diagrams.



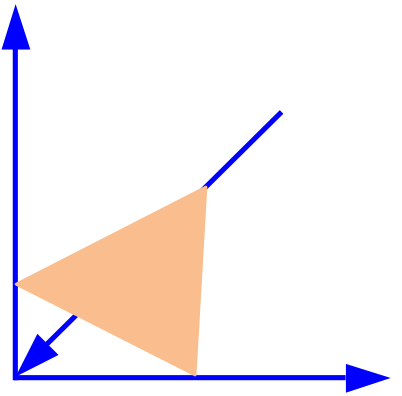
Here are the verbal and visual area diagrams on the mean values of four posters to the left (picture area 68%, text area 15%, background area 17%), and on the mean values of fourteen lids on jam packages to the right (picture area 30%, text area 33%, background area 37%).

The areas for text, picture, and background have been calculated by putting a transparency with a grid over enlarged copies (300 percent) of the everyday graphics, and over reduced copies of some posters. By counting the number of squares for each category, the percentage distribution of text area, picture area, and background area is easily calculated. When text is printed on a picture, the area is equally divided between the two categories. Text logotypes are regarded as text. Symbols are regarded as pictures. Depending on the shape and size of the lids, the number of squares varied between 180 and 270. For the posters, the number of squares varied between 345 and 450. Consequently, a single square always represents less than a percent of the area – often less than half a percent.

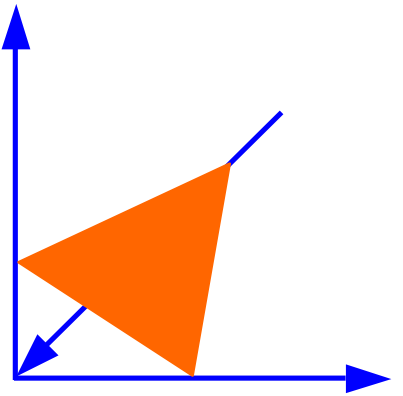
In the artful posters exemplifying this chapter, the picture area strongly dominates the areas for both text and background. This is shown in a very clear way in a verbal and visual area diagram. On the lids of the jam packages there is much more balance between the text area, picture area, and background area. Area diagrams on the mean values of the fourteen information graphics products in this analysis show great correspondence with a page in a book containing the same amount of text, pictures, and background.

However, among the information graphics products that have been studied, there are some examples with a similar distribution of the area as on the posters. We may consider today's everyday graphics as a development of the posters from the turn of the last century. In several respects, the two groups have similar functions, and there are similarities in their designs. A reason for differences in the designs between the two groups may be the great difference between the products to be "sold".

Chéret's advertising for aperitif, books, and foodstuff looks more like today's everyday graphics than the posters in this study. Here, the division of the area is very similar to the French package, "Confiture de fraises". Everyday graphics usually have a very good balance between text, picture, and background. It is probably a good idea to employ this balance also in the production of other kinds of information graphics, as well as in the production of learning materials.



Dubonnet Aperitif is a typical poster (116 x 83 cm) designed by Jules Chéret, 1895. To the right is the verbal and visual area diagram for this poster, (text area 25%, picture area 47%, background area 28%).



Here is an everyday graphic printed directly on the lid on an individual portion of jam from France. The verbal and visual area diagram is shown to the right (text area 33%, picture area 47%, background area 20%).

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INKSCAPE MANUAL

This is a free, user-editable, open source software manual.

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About Inkscape

Inkscape is an **open source** drawing tool for creating and editing **SVG** graphics. More than just a text **vector editor**, Inkscape provides a **WYSIWYG** interface for manipulation of **vector images**, allowing the artist to express himself freely. While other free and proprietary software exists with similar capabilities, Inkscape provides an interface to directly manipulate the underlying SVG code, which allows one to be certain that the code is in compliance with **W3C** standards. Since the beginning of its development, the Inkscape project has been a very active, providing stability for the current software and growth of capacities in the future.

Like other drawing programs, Inkscape offers creation of basic shapes (such as ellipses, rectangles, stars, polygons and spirals) as well as the ability to transform and manipulate these basic shapes by rotation, stretching and **skewing**.

Inkscape also offers functionality to manipulate objects more precisely by adjusting **node points** and curves. These functions are indispensable to useful drawing software, and allow the advanced artist to freely create what he imagines.

The properties of objects can either be manipulated individually and precisely through the **XML** editor, or more generally in an intuitive fashion by input devices such as mice, pen tablets or even touch screen.

In addition, Inkscape allows one to insert text and **bitmaps** (such as **PNG**, another W3C recommended bitmap image format) into an image as well as perform some basic editing functions on them. If further bitmap editing is required, other tools may be used (such as the **GIMP**) on images before importing them or after. In fact, if a linked bitmap is edited in another program, Inkscape will reflect these changes once the SVG is reloaded.

All of these characteristics make Inkscape a model drawing application, especially considering its flexibility and many other capabilities. Its strict compliance with the W3C SVG standards allow excellent portability of images to many applications and platforms on which these applications are used.

About SVG

Those who work with graphics for internet use are familiar with the problems tied to publication of images on the web. Traditionally, **bitmap** images (such as **JPG** or **GIF**) have been the only option for use in such documents, with the disadvantage that these images are either too large for quick transfer or, if they are small or highly compressed to reduce file-size, of poor quality.

As a solution to this problem, **Macromedia** created the **Flash** image format. While Flash satisfactorily solved the main problems inherent to bitmap images, there has been discontent for some users that the common **vector format** for the web is dependent solely on Macromedia for development of the file format and software. In order to address this discontent and provide an open option for vector graphics, the **W3C** created the **SVG** file format, making a freely usable vector format available to everyone.

Most image files are only able to be read by specific software that renders the image. SVG, however, is described in **XML** and **CSS**, and its files can be opened and edited in any **ASCII** text editor. While it is possible to create SVG images in this manner, it is highly unproductive and unintuitive. SVG editors and renderers have the ability to easily open and manipulate SVG files without a special interpreter.

Objectives of the SVG Format

The advantages of SVG are the same as for any vector image: high-quality images that are smooth and crisp ability to resize the image to any dimensions without diminishing quality, which is impossible with bitmap images. The SVG standard also defines animation, and with a little use of **Javascript**, one can make SVG interactive. Finally, since SVG is written in XML, it is possible to create graphics based on data that is stored in other XML-based formats, such as graphs, charts and maps. Despite its benefits, there is a lack of usable software to create and edit SVG files and take full advantage of its capacities; for this reason, SVG is not as usable at the moment as Flash.

The Current State of SVG Software

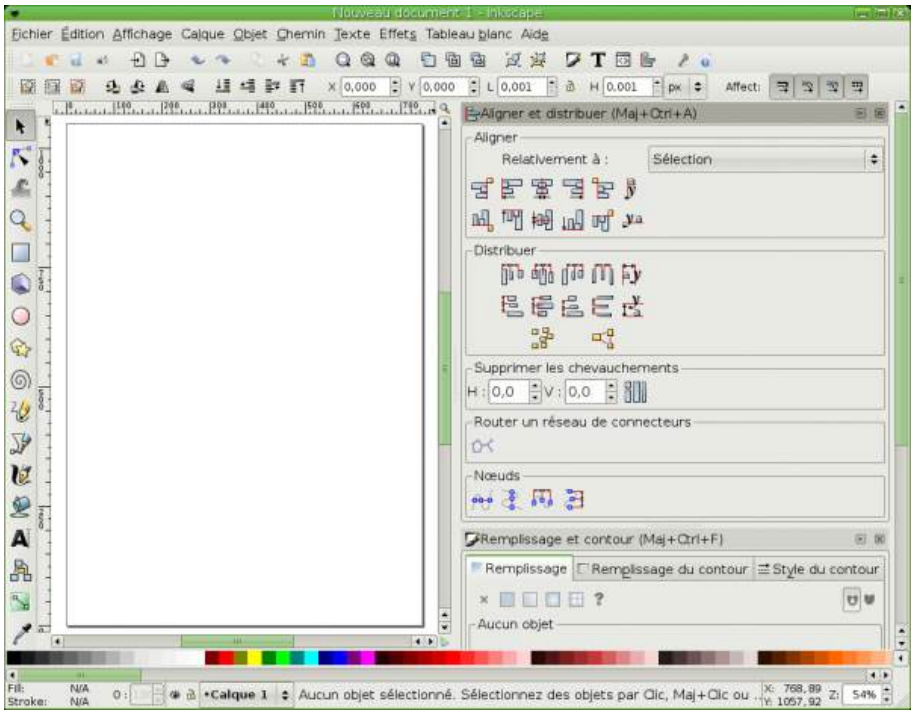
Creation of SVG files can be accomplished with several software applications today, both free and proprietary: Inkscape, Sketch/Skencil, sK1, Karbon14, xfig, Adobe Illustrator, Corel Draw, Xara, and any ASCII text editor.

Currently, although SVG is not well supported by most web browsers, Mozilla (Firefox, Netscape) and other browsers (such as Safari, Konqueror) support a basic subset of SVG, and there are plugins available for Internet Explorer (i.e. Renesis) which support most of the SVG standard. Amaya has good support for SVG display, including animations, and can also be used as a basic editor.

The Batik toolkit is a very useful tool for SVG display, and is often used as a reference for checking SVG implementations.

Inkscape Interface

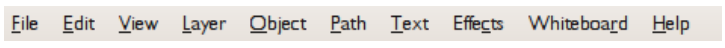
The **Inkscape** interface is constituted of elements which are designed to make work simple, harmonious and contextual. It is composed principally of a single window in which drawings are created and manipulated. Within the window are particular components which it is important to identify in order to easily navigate the software.



We may divide it into eight major areas:

1. the **Menu** (at the top of the window)
2. the **Commands Bar**
3. the **Toolbox**
4. the **Tool Controls Bar** (also called just **Controls Bar**)
5. the **Canvas**
6. the **Rulers**, **Guides** and **Grids**
7. the **Palette**
8. the **Status Bar**

The Menu



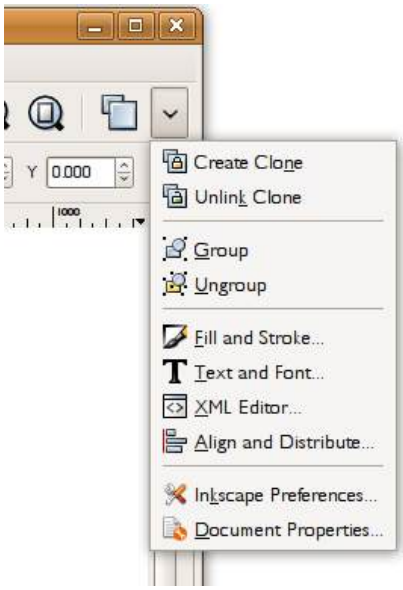
As in most **GTK** applications, the Inkscape Menu contains the essential functions of any program, those which concern the application itself: New, Open, Save, Export, Quit, etc. The functions relating to drawing are also present in the Menu.

The Commands Bar

The **Commands Bar** is located at the top of the workspace directly underneath the Menu. It contains icons which are shortcuts to commands which are otherwise accessible from the menus or shortcut key commands.

It also contains other controls for manipulating the document and drawing objects. For example, from the Commands Bar you can open a new or existing document, print, import an image, undo previous commands, zoom, open the dialog to adjust document properties, etc. It is possible to see all the functions by hovering over each one and reading the **tooltips**.

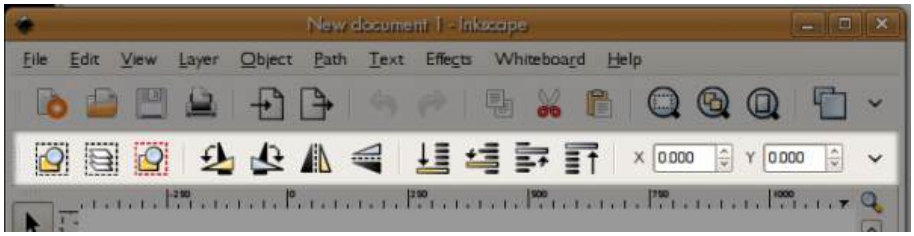
There may be an arrow on the right side of the Commands Bar pointing down which you can click to reveal any command shortcuts that were not able to fit on the bar due to monitor size or resolution settings.



The Toolbox and Tool Controls Bar

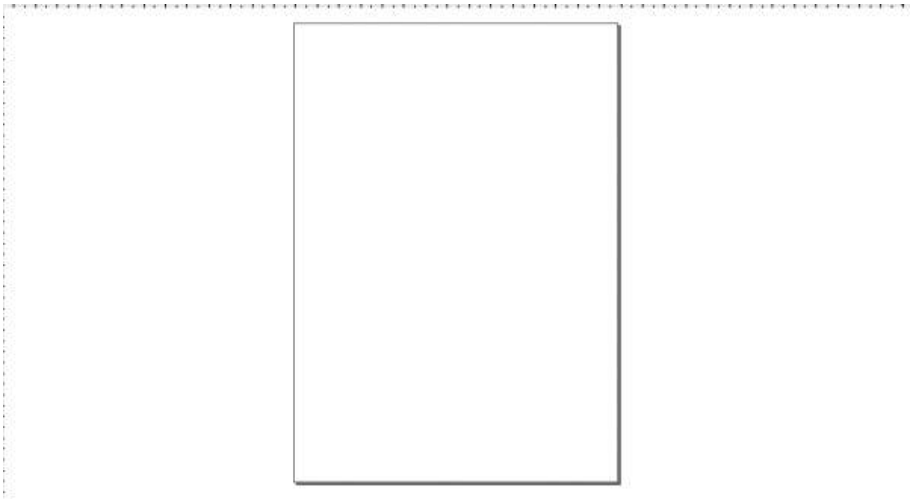
The **Toolbox**, consisting of vertically aligned buttons located on the left of the window, is Inkscape's main editing control. It contains the basic set of drawing utilities, in particular for creating and editing **shapes**. There are controls for geometric shapes as well as **free-form** shapes and lines, text, and **fills** (colors and **gradients**).

Located directly under the Commands Bar is the **Tool Controls Bar**.



When each tool is selected in the Toolbox, the Tool Controls Bar changes to show particular options associated with that tool. Depending on context, some of these options affect the selected **object** while some take effect only when drawing a new object; others can affect either existing or new objects.

The Canvas



The **Canvas** is the main workspace, and is the most central and important part of the interface, since it is here that the drawing is created and viewed. It is located in the middle of the window and is represented as a blank "page" with open space around it. By default, there is a Ruler above and a Ruler to the left of the Canvas which is set to measure in **pixels** (the standard **SVG** unit), but these defaults (*ruler visibility* and *unit*) can be adjusted in Document Properties.

While the "page" defines the boundaries of a document intended for certain media (print, export, etc.), an SVG is not limited to the page boundaries. In fact, the page border and shadow can be made invisible in the Document Properties. Some artists will prefer to use a particular page boundary and use the white space as "scratch paper"; others will prefer not to be limited by page boundaries.

Rulers

The Rulers are graduated lines placed on top and left of the canvas. The first is called "horizontal" and the second "vertical". Graduations represent distances and are expressed in units that can be set in the *Units option* of the *Page tab* of the *File > Document Preferences.*

When the mouse is over the canvas, two triangles appear in the rulers to show its X and Y coordinates, relative to the page's bottom left corner. Those coordinates are also displayed in the Status Bar (at the bottom of the document window) on the left, near the Zoom Control.

Note : In SVG, coordinates begin at the bottom left of the document like in **Cartesian geometry**.
Ctrl + R is a quick way to hide or display the Rulers. One can also do that with the *View>Show/Hide>Rulers*

Guides

Guides are user-defined 'magnetic' lines. Using Guides makes object alignment easy even with the mouse. To use Guides, click and drag from the Rulers to the point where the Guide is to be inserted and release. Clicking and dragging from the horizontal Ruler produces a horizontal Guide. Clicking and dragging from the vertical Ruler produces a vertical Guide.

How to use

Moving Guides

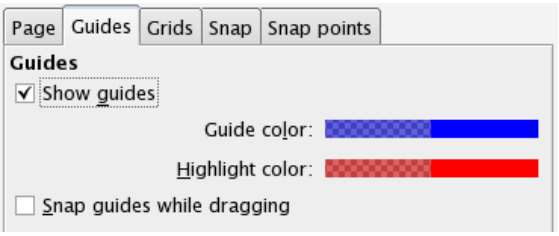
When the Selector Tool **F1** is active, passing the mouse over a Guide will change its color to red. Then, click and drag the Guide where you want.

Deleting guides

To delete a guide, just drag it to the appropriate Ruler with the Selector Tool **F1**

Guide Visibility

To make Guides invisible, without deleting them, select *View* > *Guides* from the Menu Bar. The keyboard shortcut for toggling Guide visibility is **Shift** | (hold shift and press the pipe - | - key, which is usually paired with the backslash key.)



File > *Document properties* let define if Guides should be displayed as default and change the color both of the Guide itself and for the highlight when the mouse passing over.

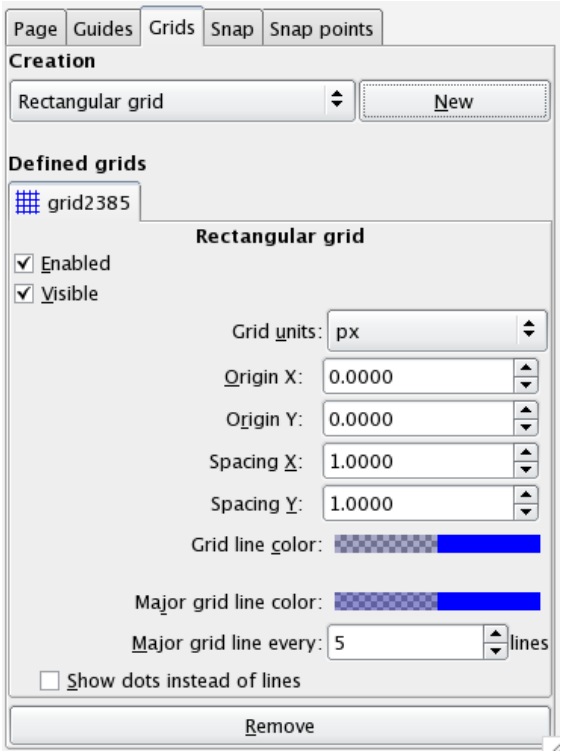
Guides are also often used with snapping that makes it much more easier to place object on the canvas, especially for precise or technical drawings. In this case just check the *Snap guides while dragging* checkbox.

Grids

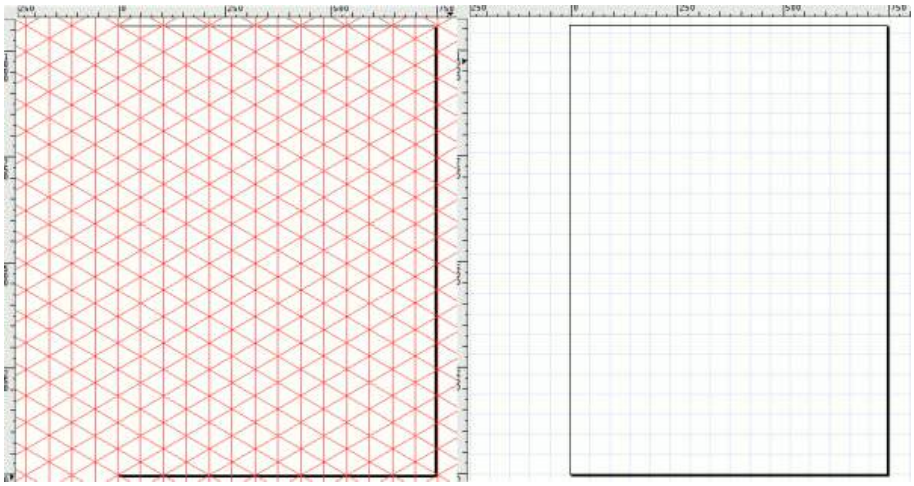
Instead of using lots of Guides, it can be useful to activate Grids. Do this with the *View* > *Grid* menu or press # (**Shift** + **3**).

There are of 2 types : rectangular and **axonometric**. They can be defined in the window from the *Document Properties* > *File* menu. Most commonly used is the rectangular Grid which is made of vertical and horizontal lines.

Axonometric Grids allow the user to define any kind of angled Grid which can be interesting for technical or architectural drawings.



Here is an example of standard axonometric Grid.



How to Use

To chose between one of those, just drop down the list in the document properties and click the new button. A new tab is created within the main one (one can define several Grids for a single document). Then define the units you would like to use and both the Origin point and the distance between to lines of the Grid. When on Axonometric Grids another option to define the angle is available.

Enabled

The user will use or not this Grid in the document

Visible

The user will see or not the grid on the canvas. This is the default value for that grid. But if the *View* > *Grid* is uncheck, the Grid won't be visible on the Canvas even if *Visible* is checked here.

Grid Units

Many commonly used units are available from mm, to feet and px. Choose the one that best suits your needs. If no special needs, keep the default px.

Origin X and Y

Define the beginning point of the Grid. Usually set to '0' (zero) it is useful to change if an offset is needed especially to define margins from the Canvas side.

Spacing X and Y

Defines the space between to lines of the Grid. These spaces can be different for horizontal and vertical lines so that the Grid pattern can be set to any kind of rectangle.

Angle X and Y

Only available for axonometric Grids, lets define the angle of the Grid lines.

Grid line color

Default color for the Grid is blue, but this can be changed here. There are two kinds of line. The most often used is the Grid line, but the major Grid line helps to evaluate the distance especially when the grid spacing is short and that many lines are displayed. In this case, one can define a different color for each, and set the frequency of major grid line , usually 5 or 10.

Show dots instead of lines

Since lines can overload the screen, it can be uneasy to work with Drawing Tools. It can be done here.

Swatches

Swatches is a quick way to apply color on shapes. It is display at the bottom of the Canvas, or in a window by *View* > *Swatches* (**Shift** + **Ctrl** + **W**)

How to Use

To find the color you like, just scroll the swatch line and choose. Yo can change the color by another preset by clicking the triangle at the right of the bar and choose one.

To apply a color in a shape as a fill color, just click on a color after selecting one or more shapes.

To apply the color on the stroke, press **Shift** while clicking and it's done.

Status Bar

Status Bar is the bottom-most of Inkscape interface. It includes (from left to right) :

- Color indicator for the object
- Quick layer selector
- Help message area
- Mouse Coordinate indicator
- and finally a zoom factor in which one can right the factor he wants to use.

Working with files

Create a New Document

Document creation is an important step for working. This function allow access to a new blank page for drawing. The *File* > *New* menu gives access to numerous predefined formats.

During new document creation, a new instance of the **Inkscape** window is opened with its own menu and own bars. There is no dialog that asks you to set the page properties. Instead the document default **parameters** are used. The default is a **portrait A4** page, though this may be changed by creating a new **default template**.

Default settings are stored in a **template** in your directory `share` (on Linux, most often `/usr/local/inkscape/share/templates`). The template name `Default.svg` could be replaced by a customised file that contains your own options like size, border, zoom, **metadata** and so on...

Using

You can create a new file in several ways including:

- By using the File Menu*: File* > *New*
- By using the keyboard shortcut*:* **Ctrl + N**

Additional Information

To change the document size after it's created, go to *File* > *Document preferences*.

Opening a Document

Instead of creating a new file you may wish to open an existing **SVG** document. This process can be useful for:

- modifying an existing document;
- getting some part of a document to reuse it for another one;
- analysing the method used to create a picture, especially by viewing the code in the Inkscape **XML source code** editor;
- exporting the document in a new **format**.

Usage

This command can be called by several ways including:

- By the File m=Menu*: File->Open*
- By the keyboard shortcut: **Ctrl + O**

Additional Information

- Documents are always opened in a new window. This way, the work on newly opened documents is totally independent from concurrently open documents.
- The closing of one document window does not close others.
- It is possible to exchange objects from one document to another by **copy/paste** functions.

Warning : this menu allows optimum opening only for SVG documents. This operation doesn't modify the type of the document. The opened formats are the same as the imported ones.

Saving files

As with document opening or creation, the saving of your file is an essential action, allowing you to finish the production later or to share the document with others.

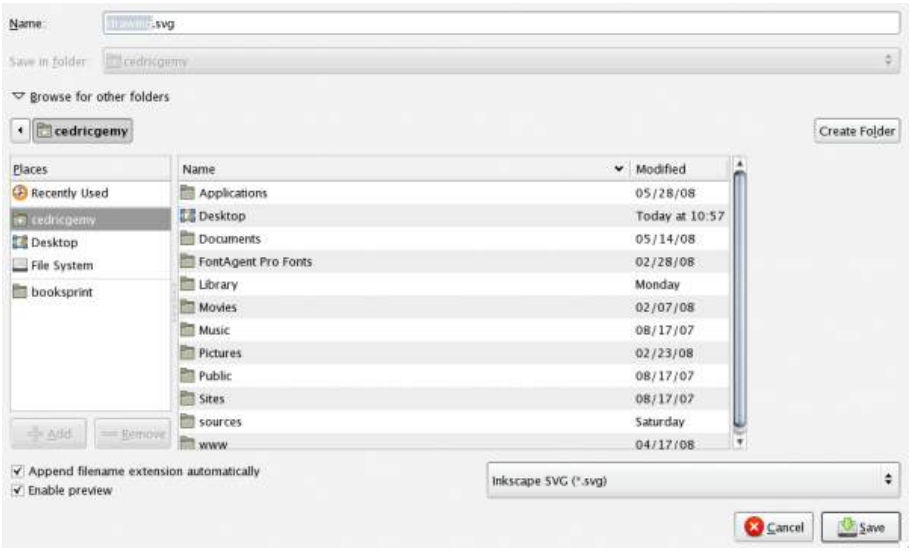
Two functions are traditionally used for saving:

- *Save* - modify existing document by saving changes. This operation should be done often, such as when the document reaches a new important step or is finished. Save modifies only the file it was last saved as.
- *Save as...* save under a different file name. The save as method allows you to save several versions of the same document by changing the document name (by adding an incremental version number, for example). Any new file should be immediately saved with this function to identify it.

How to use

Call this by using File > *Save* or File > *Save as* ...

or pressing **Ctrl+S** or **Ctrl+Shift+S**



Parts

The main area contains two parts. The right and huge one is the actual directory list of files. The left one gives a quick access to standard directories and to user-defined favourites too. To add a favourite just select it on the right area on click the *ADD* button.

Type

Lets you define the **file format** for the saving process. Most commonly used in Inkscape is SVG but one may use another one for special purposes. To have more information about file formats supported by Inkscape, read below.

Name

Just write the name of your new file here. You can add the extension (i.e. .svg, .pdf ...) so that you won't have to choose it the the format list.

Plain SVG is the most standard SVG. Use it if you don't know with what kind of SVG editor the file will be read on. It's also the best for interoperability and export path to **Gimp** or **Blender**.

Inkscape SVG (the default one) is to be considered as the most comprehensive as it keeps every detail of your drawing, including Inkscape specific stuff like **live shapes**, **layers**... This format may not be fully readable by other editors.

Formats

.svg (native Inkscape)

Inkscape default format, that keeps every shapes as easily editable as possible.

.svg (plain)

SVG recommended format, fully compliant with **W3C**'s spec. In this format many of the shapes (especially **primitives**) will be transformed to paths.

.svgz (compressed)

Compressed SVG with **gzip** compression. Low file size for quicker downloading or uploading on the web.

.pdf

Exchange format created by **Adobe**. Inkscape's PDF is 1.4 only, and needs to be improved.

.svg (Adobe Illustrator 9+ , Adobe Illustrator)

Adobe Illustrator file format, with its specification. For those who have to work with **proprietary software** users.

.outline

text outline format

.xaml

eXtensible Application Markup Language, developed by **Microsoft** to defined **Windows Vista** Graphical Interface.

.png

Raster image format recommended by W3C with **alpha** support.

.bmp

Raster image basic format. Produces very heavy pictures.

.wbmp

Wireless Bitmap format aimed for 2-bits (BW) encoding especially for early mobile applications.

.ico, .cur

Microsoft Windows format for cursor and icons.

.jpg, .jpeg

Raster image format commonly used for photos on the internet. many camera do save pictures in **JPEG** format too.

.pnm, .pgm, .pbm, .ppm

Portable Anymap, Portable greymap, Portable bitmap, Portable Pixmap : Raster image format

.ras

Raster image format used by Sun Microsystems computers, typically created on a Unix workstation; supports 1, 8, 24, and 32 bits per pixel; can be uncompressed or compressed using RLE compression; recognized by most image viewing programs.

.tiff

Raster image format especially made for professional printing process.

.xpm, .xbm

Raster image format used on **X11** server for icons.

.tga, .targa

High level Raster image format frequently used in 3D-design for textures or alpha support.

.pcx

Old raster image format used mainly in the '80 and '90s. JPEG and **PNG** are now recommended instead.

.ps (Postscript), .eps (encapsulated postscript), .epsi (encapsulated postscript interchange)

Printers main language. Commonly used since early '80s to late '90s for raster and vector mixing. Now mainly replaced by PDF.

.dia (dia)

Dia (software) editors diagrams

.ggr (Gimp Gradient)

GIMP gradient file format

.ani (animated)

Commonly used for animated cursors, especially on windows

.tex

LateX file format

.odg

Openoffice.org Draw (software), not fully compatible

.dxf

Standard format for technical drawings. Used to exchange with software as **Autocad**.

.pov

Pov **ray** file format for scripted 3D **rendering**.

.xcf

Gimp's file format file layer kept.

Selector Tool

The **Selector Tool** is used to select, position and transform objects on the **Canvas** with the mouse or other input device.

How to Use

Click once on an object with the **Selector Tool** to select it. The object will be framed with a bounding box (a black, dashed line) and scale **handles** will appear. Click again on the same object and the scale handles will change to rotation and skew handles. If the object is part of a group, the group will be selected, and dragging the object handles will transform the group. Double-click an object with the Selector Tool and the tool will change to the appropriate tool to edit the object (i.e., if you double-click an ellipse, the Ellipse Tool will be activated, etc.).

Selecting Objects

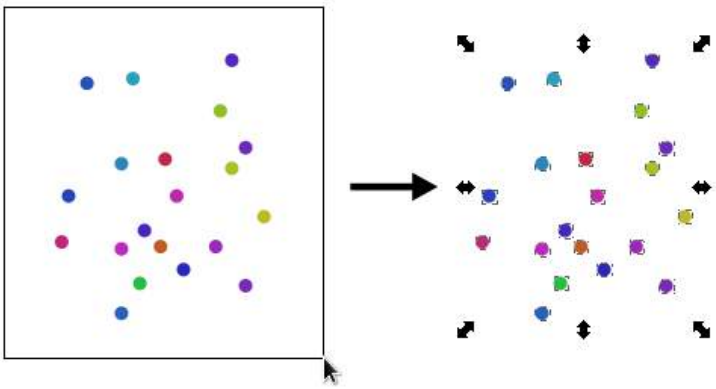
Just click any object once to select it.

Adding Objects to and Removing Objects from Selection

Shift+Click objects to add them to the current selection or to remove them from the selection.

Rubberband Selection

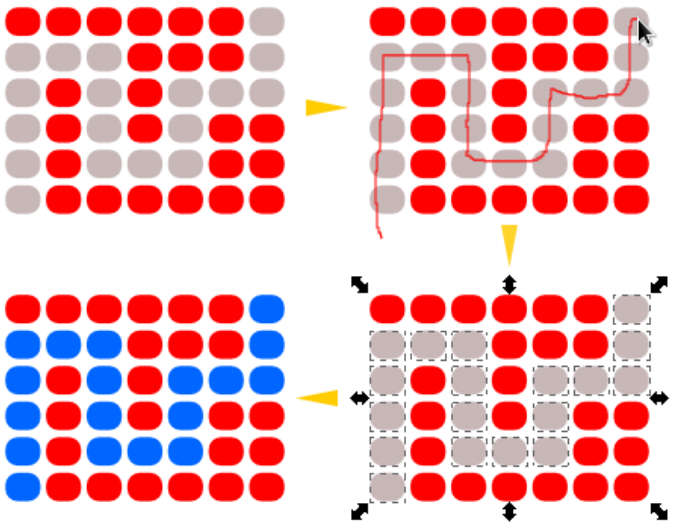
Rubberband selection is made by clicking on emptly canvas space (or over locked objects which are unselectable) and dragging the rectangular "rubber band" over several objects (i.e. click at one place and keep the button pressed while moving the mouse). **Shift+Click** will start the Rubberband Selection over objects as well.



Touch Selection

Touch Selection allows you to select objects by drawing a freehand path across them. This mode is very convenient in situations where you need to select objects so intermingled that selecting them by the other methods is too difficult or tedious.

If you are drawing a rubberband rectangle, press **Alt** to switch it to the touch mode. The rectangle will disappear and a red **touch path** will be shown instead. When dragging from an empty space, you can press first **Alt** and then start to drag to get the touch mode (note that your selection must be empty, otherwise **Alt** dragging will move the selected objects instead).



To start a touch selection from a point over an object, or to add to existing selection by touching, press **Shift+Alt** and then start to drag.

Inverting Object Selection

Use **!** to invert the selection to all unselected objects within the current layer; use **Alt+!** to invert the selection to all unselected objects within all unlocked layers.

Moving Objects

To move an object or an object group with the mouse, click on the object and hold the mouse button while dragging to the new location.

Dragging an object or several objects while holding **Ctrl** enables you to keep them aligned on an axis using the snap options.
To move objects **precisely** with the Selector Tool, you may use either of two methods:

Pressing the keyboard's cursor arrows moves selected objects 2px in the direction of the arrow. The default step can be changed in Inkscape Preferences.

Entering the coordinates into the **Tool Controls** bar X and Y coordinate controls will position the bottom left corner of the selection's bounding box at precisely the indicated coordinates.

Transforming Objects

There are two modes within the Selector Tool which are used to transform objects: *scale* and *rotate/shear* modes. You can switch between *scale* and *shear/rotate* modes with the mouse or keyboard:

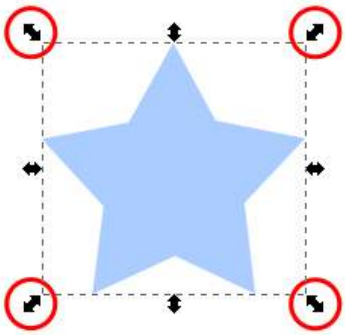
Activate *scale* mode by selecting an object with only one click (if it is unselected) or by clicking it once (if it is already in *rotate/shear* mode). Activate *rotate/shear* mode by selecting and object with a double-click (if it is unselected) or by clicking it once (if it is in *rotate* mode).

Switch between modes by pressing **Shift+S** while using the Selector Tool.

Scaling

Scaling an object resizes it vertically, horizontally or both. To scale and object, activate the *scale* mode.

To resize a selection in *scale* mode, drag the handles at the corners or press **<** or **>** (the factor for key scaling can be set in Inkscape Preferences).



You may also scale a selection by adjusting the parameters of the *width* and *height* boxes on the Tool Controls bar

Lock the width and height proportion with the lock toggle on the Tool Controls bar.

Hold **Ctrl** while dragging if proportion constraint is needed.

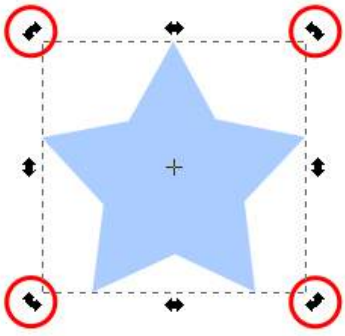
Hold **Shift** to use the center point of the object as the center of the transformation.

Dragging the scale handles with **Alt** scales the selection by an integer factor, i.e. up to **2**, **3**, **4**, etc. times the original size or down to **1/2**. **1/3**, **1/4**, etc. of the original size either horizontally or vertically (or both). ** In some Linux distributions, you may need to adjust your meta-key settings to make this work.*

Rotating

Rotating a selection is done in *rotate/shear* mode, and makes the selection turn around its axis point or **rotation center**.

To rotate the object with the Selector Tool, drag the arrow handles which are found at the corners or press **[** or **]**.



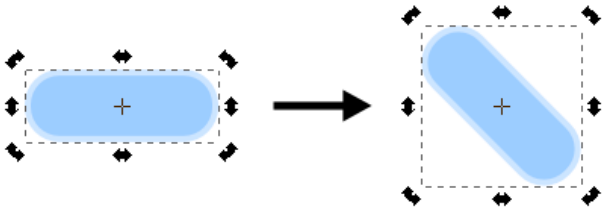
Hold **Ctrl** while rotating to constrain rotations to 15 degree increments.

Rotation Center

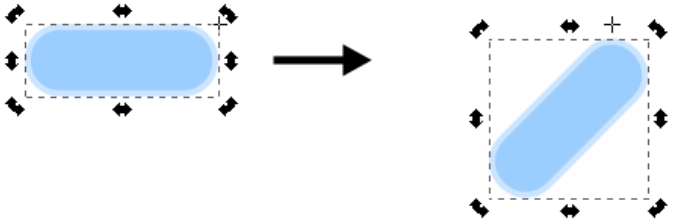
The **Rotation Center** of the selected object specifies the point around which to rotate the object. It looks like a crosshairs and may be placed anywhere on the canvas, not just within the **bounding box** of the object. When several objects are selected, they use the rotation center of the **first selected object**. If the first object does not have center set (i.e. if it's in a default central position), then several objects will rotate around the geometric center of their common bounding box

To move the rotation center of an object, click and drag it to the desired spot. It will snap to the bounding box of the object as well as other **snap points** as set in the Preferences.

Compare rotation around center which is in it default position



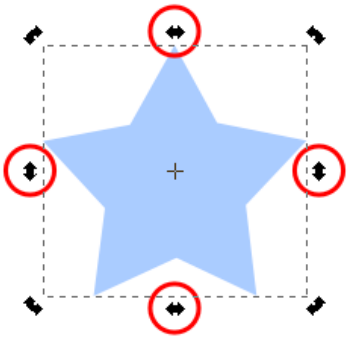
with rotation around center which is moved to upper right corner of a shape:



Shift+click on the rotation center resets it back to the center of the object's box.

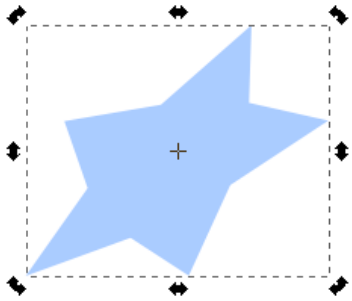
Skewing or Shearing

Skewing or shearing objects can be done in *rotate/shear* mode as well. This transformation shifts the parallel bounding edges of the selection in opposite directions so that the selection is warped diagonally.



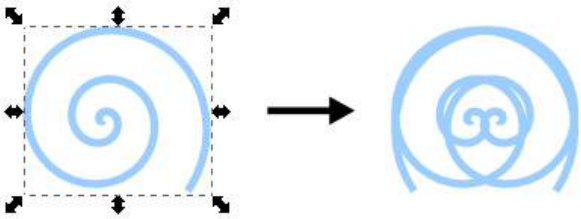
To skew the object with the Selector Tool, just drag the arrow handles placed at the middle of each sides. Horizontal handles will skew horizontally, meaning that horizontal line will keep

preserved and that only vertival lined will be affected

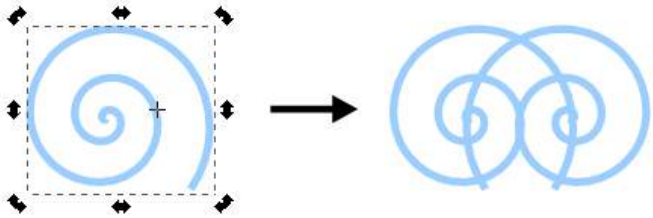


Mirroring

Mirroring flips a selection either vertically or horizontally. Flipping a selection while in *scale* mode makes it flip within its bounding box, so that the bounding box remains fixed.



However, in *rotate/shear* mode mirroring happens about an (imaginary) vertical/horizontal axis through the rotation center:



To mirror an object vertically, select Object->Flip Vertical or pressV.

To mirror an object horizontally, select Object->Flip Horizontal or press H.

There are also two buttons in the options bar that can do the same.

Scaling of Stroke Width, Rectangle Corners, and Fills

There four preferences that control whether or not to **scale stroke widths**, **scale rectangle corners**, **transform gradient fills**, and **transform pattern fills** with the object, represented by **four toggle buttons** in the Tool Controls Bar.

Tips

1. The Transform Window (Object->Transform or **Shift+Ctrl+M**) can be used for precise transformations. Changes
2. Press **Esc** to deselect, cancel selection, or cancels drag or transformation.
3. **Ctrl+Click** or **Shift+Ctrl+Click** will select objects within groups from outside that group.
4. **Select Under**: in Selector, **Alt+click** selects the object at cursor which is below (in z-order) the currently selected object at cursor; if the bottom object is reached, **Alt+click** again selects the top object. Thus, several Alt+**clicks** will cycle selection through the z-order stack at the click point. Combining **Alt** with **Ctrl** ("select in groups") and **Shift** ("add to selection") works, too. Note that on Linux, many window managers steal **Alt+click** by default; reconfigure your WM so you can use **Alt+click** in Inkscape.
5. **Drag Selected**: in Selector, **Alt+Drag** moves the currently selected object(s) no matter where you start the drag, unlike regular drag that first selects the object under cursor. This is convenient for dragging objects that are behind other objects in z-order. On Linux, you may need to disable dragging the window with **Alt** in your WM if you want to use "drag selected".

Node Tool

The Node tool has an essential role in Inkscape as it is used to select and manipulate **nodes** so as to be able to precisely modify the shape of **paths** or curves. In addition to allowing manipulating of node position, node handles extend from the node when it is selected which define the direction of the path segments originating from that node.

How to Use

Add Node

Creates a new node between two selected nodes. You can also double-click on the path at the place you want the node to be created.

Subtract Node

Removes selected nodes and joins adjacent nodes. It is also possible to press **Ctrl+Del** or **Ctrl+Backspace**.

To preserve the shape of the path better, either select the node and press Del or Bksp, or else**Ctrl+Alt+Click** on the node.

Join Nodes

Combines two end nodes into one node on a continuous path, moving both nodes to an average middle point; **Shift+J** can also be used (hovering the cursor over one node will preserve its position so that only the other node is moved).

Joing Nodes with New Segment

Adds a path segment between to open nodes.

Split Path Between Two non-Endpoint Nodes

Deletes the path segment between two selected nodes, leaving the nodes open (or unconnected).

Break Path at Selected Nodes

Splits a single node into two nodes in the same position. The nodes can then be moved apart.

Make Selected Nodes Corner (Convert to Cusp)

Changes one or more selected nodes into **cusp nodes** or angled/corner nodes. Paths on cusp nodes have an angle rather than an arch, and node handles can be moved independently of each other.

Make Selected Nodes Smooth

Changes one or more selected nodes into smooth nodes. Also available by **Shift+S**. Paths passing through smooth nodes have a continuous arch through the node point, but each handle of the node can have different lengths.

Make Selected Nodes Symmetric

Changes one or more selected nodes into **symmetric** nodes. Also available by **Shift+Y**. Paths passing through symmetric nodes have a continuous arc throught the node point, and handles have the *same* length.

Make Selected Segments Lines

Changes one or more selected segments (two adjacent nodes) into a straight path.

Make Selected Segments Curves

Changes one or more selected segments (two adjacent nodes) into a curved path. Also available by Shift+U.

Convert Selected Object to Path

Converts an object that is not already a path, like a **Live Shape** or a text object, to path. (This command effectively creates "outline" text, removing dependency on installed fonts.)

Convert Selected Object's Stroke to Paths

Creates an outline of a path, creating parallel combined paths separated by the width of the stroke.

Show the Bezier Handles of Selected Nodes

Toggle whether the node's handles are displayed while editing.

Show Next Path Effect Parameter for Editing

Cycles the display of **LPE** controls, often indicated by a red or green path or specific handle types.

Invert Node selection

The **!** key inverts node selection in the current subpath(s) (i.e. subpaths with at least one selected node); **Alt !** inverts in the entire path. (This is similar to how these keys work in Selector, with current subpath(s) instead of the current layer.)

Hotkeys

Multiple select

Shift+Click to select several nodes in succession.

Near selection

Clicking on a selected path selects the two nodes closest to the click point (on either side). **Shift+Click** adds or removes these two nodes to the node selection (when only one path is selected; otherwise **Shift+Click** works as in Selector).

Adding nodes

Double-click or **Shift+Alt+Click** anywhere on the selected path (even if it is under other objects) creates a new node at the click point, without changing the shape of the path.

Tab selects the next node
Shift+Tab selects the previous node

Other

Rotate (direction of handles) by pressing **[** or **]** keys and scale (length of handles) by **< or > keys**. **Ctrl+** any of these keys will affect only the left or right node handle (according to whether the left or right Ctrl key is pressed).

Rotating objects or control handles of a node with **Ctrl** restricts rotation to 15 degree increments. **Dragging** nodes may be restricted to horizontal/vertical with **Ctrl** and to the directions of the node's handles (or there perpendiculars) with **Ctrl+Alt**. **Dragging a node's control points** with **Alt** locks the length of the handle, and with **Shift**, rotates the other handle by the same angle. When *several* nodes are selected, pressing **<** or **>** scales, **[** or **]** rotates the selected nodes **as if they were an â objectâ** , around the center of that object. So, for example, in a single-path silhouette portrait, you can select the nodes of the nose and rotate/scale the nose as a whole without breaking the path into pieces. Pressing **Alt** with these keys gives pixel-sized movement depending on zoom, the same as in Selector. Also, you can press **h** or **v** to flip the selected nodes horizontally or vertically.

Ctrl+click toggles cusp/smooth/symmetric, **Ctrl+Alt+click** deletes a node

Tips

1. When an object is selected handles appear making it possible to handle the shape of the object in an intuitive and precise way.
2. You can switch the not-yet-finalized (red) segment of the path being drawn from curve to line (**Shift L**) or back to curve (**Shift U**).
3. **Esc** deselects, cancels selection, and cancels drag or transformation of any kind (so far only in selector and node edit). Arrows, **Ctrl+a**, and **Tab / Shift+Tab** act on nodes in node editor exactly as they do on objects in Selector. The first **Escape** or empty-space click deselects any selected nodes,

the second one deselects the selected object removing the node display.

4. The Node Tool selects objects regardless of grouping. This means you don't need to switch to selector for a **Ctrl+click** if you want to edit a grouped path.
5. You can reverse the direction of selected path(s) either via Path > Reverse or by pressing 'r' in node tool (useful for markers and combining paths).
6. **Ctrl+click** on a node handle retracts that handle back to its node.
7. If a node does not show one or both handles (i.e. they are retracted), you can **drag a handle out** by dragging away from the node **with Shift.**
8. Pressing **Ctrl+A** with some nodes selected will select not all nodes in the path but all nodes in the subpath(s) containing the selected node(s). To select all nodes in the path unconditionally, either deselect any nodes before pressing Ctrl+A, or use *Select all in all subpaths* (**Ctrl+Alt+A**) in Node tool.
9. Edit the selected path by dragging directly on the path.
10. if you press Shift before starting to drag, you always get a node selection rubberband rectangle
11. When you **switch the type** of the selected node to Smooth or Symmetric by pressing **Shift+S / Shift+Y**, you can now **preserve the position** of one of the two handles by hovering your mouse over it, so that only the other handle is rotated/scaled to match.
12. You can **grow or shrink node selection** by hovering the mouse pointer over a node and using **mousewheel** (up = grow, down = shrink) or the keys **PageUp** (grow) and **PageDown** (shrink). *Growing* adds the closest unselected node to the selection; shrinking deselects the farthest selected node. There are two modes that differ by how the closest/farthest nodes are chosen:

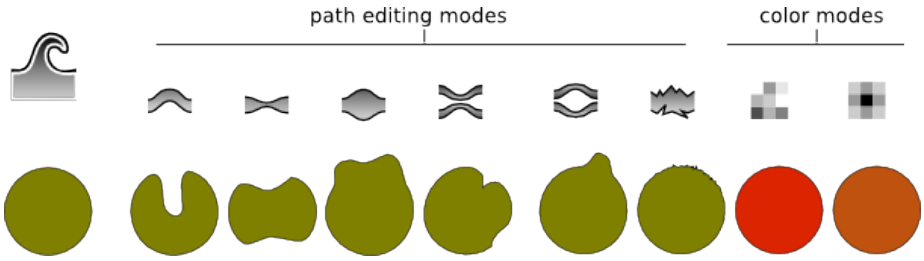
    ◊ **Spatial selection** (mousewheel, **PageUp / PageDown**): distances to nodes are measured directly, regardless of which subpath a node belongs to.

    ◊ **Linear selection** (Ctrl+**mousewheel**, Ctrl+PageUp/Ctrl+PageDown): node distances are measured *along the path*, and only the nodes belonging to the same subpath as the hovered node are considered (i.e. other subpaths are never selected).

13. This technique is convenient for quickly selecting an area in a complex path starting from a center - for example, for node sculpting.
14. If any of the nodes in the currently selected path is mouseovered, then horizontal/vertical flipping ('H' and 'V' keys), stepwise rotation ('[' and ']' keys) and scaling ('<' and '>' keys) now all use this specific node as center/axis. If there is no mouseovered node, the center of the bounding box is used instead (as is currently the case unconditionally). Nodes that are covered by one of their handles are also detected as mouseovered.
15. Two entry fields are available on the Tool Controls bar which allow precise editing of the coordinates of selected nodes.

Tweak Tool

The Tweak Tool is an exciting way to edit drawings which largely blurs the distinction between **vector** and **raster** editing. Instead of meticulously selecting some objects and then performing an action on the selection, you can select *all* objects (or all objects you are interested in) and apply the Tweak Tool's **brush** to smoothly and naturally change the shape or style of *only those objects* (or parts thereof) *that the brush touches*.



The area of the tool's action - its *brush* - is marked by an orange-colored circular outline that moves with your mouse cursor. However, that area actually has no sharp boundaries; the power of the tool's action falls off gradually, following a smooth bell-shaped profile. This makes the tool act softly and smoothly.

The tool will work on any number of selected objects; for example, you can select all **Ctrl + A** and "smear" your entire drawing by **Push** mode or paint it by **Color Paint** mode. You can also apply it to groups of objects; it will go into groups and act on individual objects inside groups. If you're trying to use it without anything selected, it will remind you by a statusbar message to select some objects.

How to Use

Width

The width of the tool's brush, in the range from 1 to 100, can be changed by the **Width** control in the tool's controls bar above the **canvas**. You can also change width by **Left** and **Right** arrow keys (same as in the **Calligraphy Tool**) at any time (including during action) as well as **Home** and **End**. Also, as in Calligraphy Tool, the visible width of the brush is independent of zoom; simply zooming in or out is often easier than adjusting the width if you want to cover a smaller or larger area of the drawing.

Force

The next control is **Force** which adjusts the power of the action, also in the range from 1 to 100. You can also change width by **Up** and **Down** arrow keys at any time (including during action).

If you have a pressure-sensitive tablet and your "Use pressure" button on the right-hand end of the controls bar is on, then the force will also depend on how hard you actually press your pen into your tablet, changing in the range from zero to whatever you set in the Force control. If all you have is a mouse, then the force will be constant but still settable by the Force control.

Path editing modes

The Tweak tool has a number of **modes**, selectable by toggle buttons in the tool's Controls bar and by keyboard shortcuts. Some of these modes change the shapes of **paths** while others affect the colors of objects. All these modes share the **Width** and **Force** controls but

otherwise are quite different. Let's look at the path editing modes first.

Unlike the Node tool, to edit paths with the Tweak tool you don't need to worry about where the nodes of a path are and how to manipulate them. You just apply the tool's brush to any point, and the selected paths at that point will reshape smoothly and naturally - as if made of soft jelly - regardless of where its nodes lie. If applied to a shape or text object, the tool converts them to paths automatically.

While not very useful for technical drawings, tweaking paths will be indispensable for artistic uses of **Inkscape** - cartoons, drawings, sketches, anime, etc. This new functionality is somewhat similar to the tools such as "Pucker" and "Bloat" in the latest versions of **Adobe Illustrator**.

There are currently six path editing modes in the Tweak Tool: **Push**, **Shrink**, **Grow**, **Attract**, **Repel**, and **Roughen**.

Push

This default mode of the tool, **Push**, **Shift + P** simply displaces the part of the path under the cursor in the direction of the drag. The path behaves like soft jelly, bending and bulging smoothly and naturally. It's an easy way to produce various irregular, lifelike, handmade-looking shapes starting from something as simple as an ellipse or a calligraphic stroke. For parallel-stroke hatching (engraving) done in the Calligraphy tool, pushing is an easy way to bend, pinch, or curve the entire hatching uniformly.

Shrink and Grow

The **Shrink** (**Shift + S**) and **Grow** (**Shift + G**) are two opposite modes that move each point of a path in a direction perpendicular to the path's surface at the point, either inwards (Shrink) or outwards (Grow). This is similar to the **Inset** and **Outset** commands, except that the Tweak Tool can act on a part of a path instead of the whole path.

For example, the visible lightness/darkness of an engraving hatching may not exactly correspond to your artistic intention. Also, the ends of Calligraphy pen strokes are often far from ideal - they may be too blunt or have unsightly bends or blobs. This is where the Tweak Tool may help. Select all the strokes in a hatching pattern and apply a light Shrink action where you want the lines to become thinner (and the hatching to become lighter), up until total disappearance. If you press hard, shrinking works as an eraser, so you can easily clean the strokes' ends to make them thin, sharp, and uniform. Conversely, applying Grow makes strokes wider (i.e. the hatching becomes darker).

Of course, shrinking and growing are useful not only for calligraphic strokes. Same as with Push, with Shrink and Grow you can **sculpt** any path, spawning smooth treacle-like appendages with Grow and carving holes with Shrink. Unlike the "node sculpting" mode in the Node tool, however, this does not require adding new nodes to the shape.

Attract and Repel

The **Attract** (**Shift+ A**)and **Repel** (**Shift + E**) modes work by moving each affected point on a path towards (Attract) or from (Repel) the cursor point. In some cases this may look similar to Shrink and Grow, but the difference is that shrinking/growing moves paths perpendicularly to the path in each point, whereas attracting/repelling moves them to or from the cursor regardless of the path shape. These modes are similar to the Pinch effect in AI; you can use them for various central-symmetric distortions in parts of your paths.

Roughen

The **Roughen** (**Shift +R**) mode does exactly this: roughens the edge of the path without changing its overall shape. Slight roughening simply makes the edge crooked and uneven; strong roughening tears and explodes the edge into random blobs and splotches. Note that this operation, especially with high Fidelity, adds a lot of nodes which increases the size of your SVG document and may slow down Inkscape considerably. In particular, pushing, shrinking, or growing of a roughened path becomes much slower and more difficult, so it's recommended to finalise the overall shape of a path first and roughen it, if necessary, only as the final step.

Fidelity

Any tweaking of a path slightly distorts the entire path, including even those parts that you didn't touch. These distortions are similar to those that a Simplify command produces. The **Fidelity** value (also in the range from 1 to 100, default is 50) allows you to control the amount of these distortions. With a higher fidelity, the distortions are less noticeable, but the path may end up having a lot of nodes which inflates up the SVG size and slows down Inkscape.

The best value of Fidelity depends on the nature of your artwork. If you're sculpting an amorphous blob, you can do with low fidelity of about 20. If, however, you are pushing or inflating a text string (as a single path) and want the letters outside the distorted area to remain crisp and clean, you will need to raise fidelity to 80 or more.

Color editing modes

The **Color Paint** (**Shift + C**) and **Color Jitter** (**Shift + J**) modes, unlike the path editing modes, change the colors of objects instead of their shapes. Yet they share enough common features with the path editing modes to be part of the same tool: These modes also use a circular soft-edged brush controlled by the Width and Force parameters on the Controls bar and affected by the pen pressure (if you have a pressure-sensitive tablet).

- **Color Paint** applies the style of the tool to the selected objects under the brush. The style of the tool is visible in the style swatch at the rightmost end of the tool's control bar; it can be changed by clicking on the color palette or by any other style assignment command, such as Fill and Stroke dialog. (**Note**: unlike all other tools, in Tweak tool in Color Paint mode you cannot assign style directly to selected objects; any style-setting command changes the tool's style instead.)
  The fill from the tool's style applies to the fills of the painted objects, and the stroke applies to the strokes. If the tool's style has no fill or no stroke, it won't affect fills or strokes, correspondingly. For example, if you want to color the fills of objects blue but leave their strokes untouched, assign blue fill to the tool's style (just click blue on the palette) but set its stroke to None (middle-click the Stroke swatch in the statusbar). Similarly, master opacity in the tool's style affects master opacities of the touched objects (if the O channel is on, see below).
  This mode allows you to literally paint over objects, shifting their colors towards the target style of the tool. For example, if you paint with yellow fill over a blue-filled object, the object will become greenish blue, then green, then yellowish green, and end up being exactly the yellow color you're painting with. This speed of this gradual transition depends on both Force parameter and pen pressure; also, objects touched by the periphery of the brush are less affected than those hit by the brush centre. Overall, using this tool is very similar to a soft brush in a raster editor such as Gimp or Photoshop.

- **Color Jitter** mode does not apply any color, but instead jitters (randomises) the colors of the objects it touches. The force of the action determines how strong is the randomisation, i.e. how far the colors deviate from the original values. This mode does not use the tool's style.

Both modes work on flat fills and gradients; for gradients, the tool takes into account not only the position of the entire object with gradient, but also the position of each gradient stop relative to the brush. This means that, for example, you can change the blue color only in an object filled with blue-red gradient simply by painting over its blue end with a brush small enough to not touch the red. (Note that color tweaking does not create gradients on objects that used flat color before, but only adjusts existing gradients in the drawing.)

Channels

Color Paint and Color Jitter honor the **Channels** control. This control comprises the four buttons: **H**, **S**, **L**, and **O**, which allow you to turn on and off the tool's action on the object's hue, saturation, lightness, and opacity, correspondingly. For example, if you want to raise the saturation of some part of your drawing without changing the hue, select some maximum-saturation color (e.g. pure red) and turn off all Channels buttons except **S**. Similarly, you can replace the hues without affecting saturation or lightness (only **H** pressed), or lighten/darken all colors without changing their hues and saturation (only **L** pressed). Pressing **O** allows you to apply the master opacity from the tool's style to the master opacity of objects (but not fill or stroke opacity).

Usage notes

Color painting with Tweak tool is similar, but not exactly analogous to bitmap painting. Even though the tool itself works as a soft brush, it still applies its color to vector objects, which behave as vector objects usually do. For example, if you want to change the tint of the face in your drawing, and if a hand in the drawing is part of the same object as the face, that hand will change its tint too even if it's located far from the point you are painting. (We foresee a "fracture" command in one of the next versions of Inkscape which will help you turn a monolithic object into a mosaic of small fragments that will be then easy to paint with Tweak tool.) Still, even with this limitation, color painting is a novel way of dealing with vector drawings which allows you to quickly and intuitively make adjustments which would be awkward and slow with traditional approach.

Drawings containing patterns or scatterings of small independent objects are best suited for color painting with Tweak tool. Examples include:

- freehand drawings with Calligraphy pen, consisting of many separate strokes;
- gradient meshes imported from Adobe Illustrator files (Inkscape renders these meshes as lattices of small polygons; while there's no direct support for gradient meshes in Inkscape yet, color painting on such lattices is almost as good);
- text converted to paths and with Break Apart command applied so that each letter is a separate path;
- patterns made with the Tile Clones command; note that you need to unset the fill and/or stroke on the original object and use the Color tab to assign some initial color to the clones - this will make them paintable with the Tweak tool without unlinking.

Moreover, color tweaking can be useful for compositions with a few objects or even for single objects. Unlike all other color selection methods, painting with the Tweak tool implements the *color mixing* metaphor which is much more familiar to traditional artists than RGB sliders or even the color wheel. For example, start with a rectangle of pure blue color; then, pick different colors by Color Paint and apply light touches with minimum Force and minimum pen pressure: add a little green, a little brown, a little yellow, etc. until you have the

exact hue you need. Similarly, you can whiten or blacken any hue by admixing white or black.

You can also use color tweaking to add a tint, darken/lighten, saturate/desaturate, or color jitter your entire drawing. Just select all in all layers, zoom out, choose a large brush width so it covers all of the drawing, and apply a little color tweaking (with minimum Force) that will therefore affect all visible objects.

Hotkeys

- **W**, **Shift + F2**: switch to the Tweak Tool
- **Shift +P**: switch to the Push mode
- **Shift+S**: switch to the Shrink mode
- **Shift+G**: switch to the Grow mode
- **Shift+A**: switch to the Attract mode
- **Shift+E**: switch to the Repel mode
- **Shift+R**: switch to the Roughen mode
- **Shift+C**: switch to the Color Paint mode
- **Shift+J**: switch to the Color Jitter mode
- **Left, Right, Home, End**: change width
- **Up, Down**: change force
- **mouse drag**: act on selected path in the current mode
- **Ctrl+mouse drag**: temporarily switch to Shrink (while **Ctrl** is down)
- **Shift+Ctrl+mouse drag**: temporarily switch to Grow (while **Shift+Ctrl** is down)

Zoom Tool

Zoom Tool is a great way to navigate on the **canvas**. It can be accessed via several ways that are very completementary and comfortable. Each zoom action is saved in a special history so that one can easily be zooming and unzooming without difficulties. All zooming operations via mouse preserve the point under the cursor (no centering).

How to use

Use this tool by:

- *clicking in the toolbox :*
- *pressing :* **F3**

  or : **Ctrl Alt Mouse wheel**

  or : Shift **Middle Mouse Button**

Zooming

Zooming is just set the zoom factor to a higher level so that one can work more precisely. Pressing **+** does the same.

Unzoom

Unzooming can be used to have a larger view of the drawing or area. Pressing **-** can unzoom too.

Zoom to 1:1

Just displays the drawing at real pixel size. This way a banner drawing (for example) which is 468 pixels wide, will be 468 pixels on screen. If the Inkscape window is smaller, some part of the drawing may be invisible.

Zoom to 1:2

Just displays the drawing at half the real pixel size. This way a Banner drawing which is 468 pixels wide, will be 234 pixels on screen. If the Inkscape window is smaller, some part of the drawing may be invisible.

Zoom to 2:1

Just displays the drawing at twice the real pixel size. This way a Banner drawing which is 468 pixels wide, will be 936 pixels on screen. If the Inkscape window is smaller, some part of the drawing may be invisible.

Adjust selection to window

The selecect area will fill the entire Inkscape window.

Adjust drawing to window

The existing current drawing will be displayed to fit the window size.

Adjust page to window

Best to have a complete overview of the page and work on a layout, makes the whole page fit the window size depending on the orientation. Pressing **5** does the same action.

Adjust page width to window

Same as before except the whole page may not be displayed, especially if the page is portrait oriented. Gives a better result than previous with landscape drawing. Press **6** to activate it with keyboard.

Previous zoom

Come back to the previous zoom factor, as saved in the zoom history. Inkscape now preserves the history of zoom settings for each document. The ` key restores previous zoom; pressing ` repeatedly will guide you through all the zoom settings you've used in this session. **Shift** + ` moves forward in the zoom history.

Next zoom

Goes back to the next zoom factor, as saved in the zoom history, especially when previous zoom has jus be used.

Hotkeys

Shift is diminishing zoom factor.

Middle click zooms in, **Shift** + **middle click** zooms out (in addition to **middle button + drag** which pans canvas).

Tips

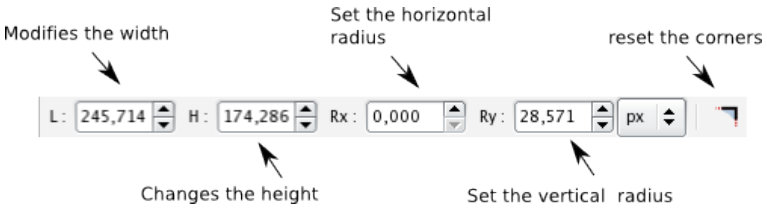
1. By default, rotating the mouse wheel scrolls the canvas vertically and Ctrl+wheel zooms in and out. Now, if you turn on the **Mouse wheel zooms by default** checkbox in the Scrolling tab of the Inksape Preferences dialog, this behavior is reversed: mouse wheel zooms without Ctrl and scrolls with Ctrl. This new mode should be familiar for users of AutoCAD and CorelDraw.
2. Now you can use **Shift+middle button drag**, in any tool, to zoom into an area. This works the same as simple drag in Zoom tool, but is faster because it does not require switching away from your current tool. Together with **middle button drag** (panning), **middle button click** (zoom in) and **Shift+middle button click** (zoom out), this completes the set of canvas navigation shortcuts available in any tool or context.
3. In the Zoom tool, right mouse button always zooms out instead of calling the context menu

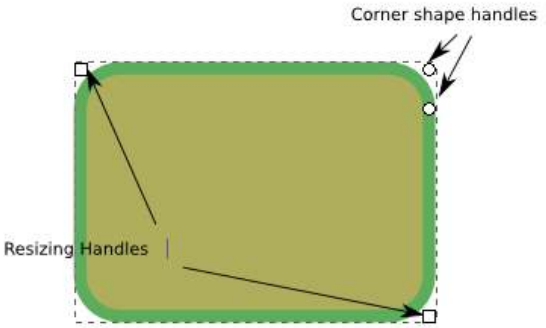
Rectangle Tool

With the **Rectangle Tool** R, one can draw rectangles and squares in the document.

Rectangles have two resize handles in the opposite corners (in addition to the rounding handles in the third corner). Dragging the resize handles with **Ctrl** snaps the rectangle so that either its width, height, or proportion is preserved. **Shift Click** on a rounding handle removes rounding; **Ctrl Click** makes rounding radii equal.

How to Use



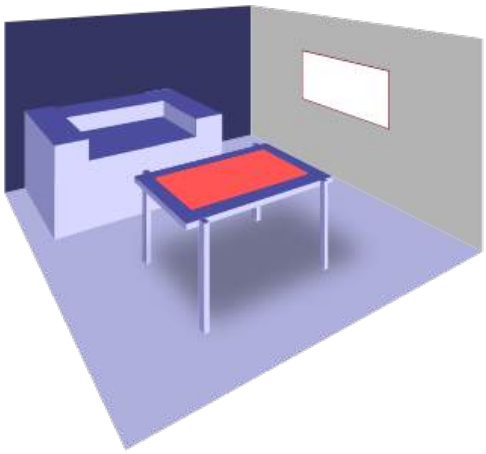


Hotkeys

Ctrl constraints rectangle sides so that they have the same value, creating a square.
Shift Draws the shape from its center and not from the upper left corner.

3D Box Tool

Inkscape is a two-dimensional drawing tool. However, very often it is used to create drawings or objects that appear three-dimensional. The **3D box tool** helps you create such drawings by automating the most common operation: creating a box in a given **3D** perspective. The tool automatically ensures that all sides of the box lie on the corresponding perspective lines.



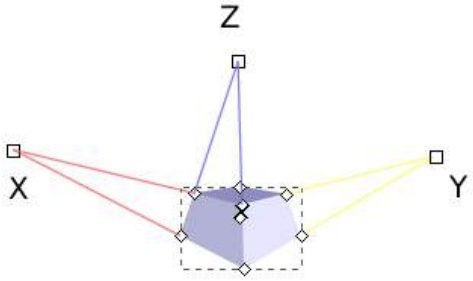
Basic Usage

In order to represent an object in 3D (it's emulated 3D, so it''s rather **2.5D**), the 3D box tool deals with **axes** (XYZ), **planes** (XY, XZ, YZ), **parallel lines** (PL), **converging lines** and **vanishing points** (VP).

Each dimension can lie on either parallel or converging lines. These lines are drawn in different colors, so that users could distinguish one line from another:

- red lines stand for **X direction** (width);
- blue lines stand for **Y direction** (height);
- yellow lines stand for **Z direction** (depth).

If lines are converging, they have a vanishing point â a point where they cross. Here is a sample to illustrate how it works:

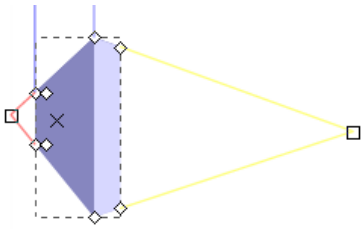


X is the vanishing point for X plane

Z is the vanishing point for Z plane

Y are parallel lines for Y plane

Here is what happens when you keep moving borders of X and Z planes closer to vanishing points:



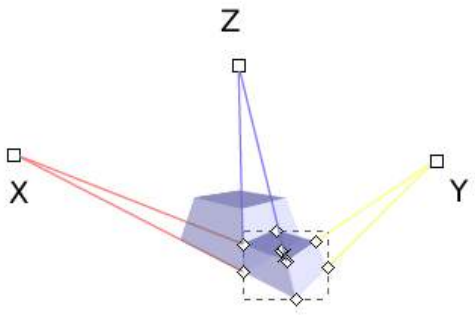
X and Z planes nearly converged, but Y is more or less the same.

By default Inkscape makes X and Z lines converging, because it represents usual perspective.

Note : Parallel lines do not necessarily have to be 90 degrees to X axis â changing the angle will somewhat skew the object.

Drawing a 3D box

Press **X** or **Shift+F4** to switch to the 3D Box Tool. Start dragging cursor on canvas. You can use **Shift**+drag without releasing the mouse button to extrude in Z direction.



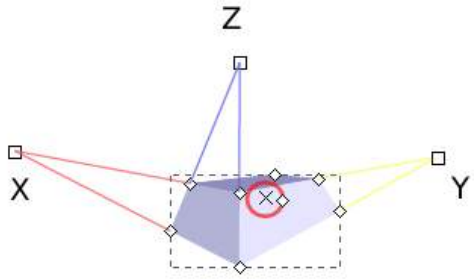
If you need to draw another 3D box with same vanishing point, just keep the previous box selected and start drawing a new one:

Editing a 3D box

Adjust any of its 3 dimensions by handles. The 4 handles on the front X/Y side resize it, the four ones on the back X/Y side expand/contract the box in Z direction. With **Shift**, the functions of handles on front and back sides are reversed. With **Ctrl**, the side-resizing handles snap to the coordinate axes or diagonals.

Moving a 3D box

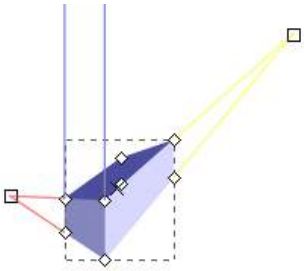
You can move a 3D box in perspective by dragging its center marked as an "X" character; without modifiers, movement occurs within the XY-plane (press **Ctrl** to constrain the movement to the directions of the coordinate axes or diagonals), with **Shift** the box moves parallel to the Z-axis.



Handling vanishing points

Adjust the vanishing points of a perspective by dragging them on the canvas (see below) or toggling their states; all boxes sharing this perspective are affected by a vanishing point move.

When several boxes are selected, all vanishing points of their associated perspectives are shown on the canvas. If vanishing points of different perspectives coincide, they are combined in a single "dragger". Moving this dragger moves all the vanishing points simultaneously and transforms the associated boxes accordingly. Note that some non-selected boxes may also be reshaped if their perspectives share the same vanishing point. Pressing **Shift** while moving the dragger can be used to only transform the selected boxes, separating their perspectives from the non-selected ones'. On the other hand, when a vanishing point being dragged comes close enough to another one, both snap together and are combined in a single dragger.



Rotating perspective lines

The various kinds of parentheses, namely **[ ]**, **( )**, **{ }**, can be used to rotate infinite (i.e., parallel) perspective lines in X-, Y-, and Z-direction, respectively. Closing parentheses rotate clockwise and opening parentheses rotate counterclockwise. The angle of rotation is taken from the preferences. Pressing **Alt** reduces the amount of rotation to 1 screen pixel.

Changing mode of perspective lines

When the 3D box tool is active, **Shift+X**, **Shift+Y**, and **Shift+Z** toggle perspective lines in the corresponding direction between "converging" and "parallel".

Tips

Inside 3D Box tool

In SVG, a 3D box is represented as a **group** (svg:g) with a special extension attribute (in Inkscape's namespace); this group contains the 6 quadrilateral **paths** representing the sides of

the box. Only the 3D box tool treats this object as a box; for all other tools it is just a group, so you can select any of the paths by Ctrl+click, apply any style to it, delete it, etc. You can of course transform the entire box or any face in it using Selector or Node tools.

Ellipse

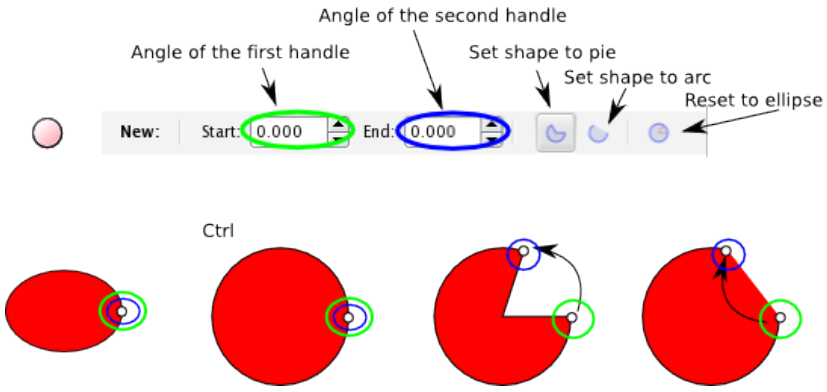
With the **Ellipse tool**, the **Inkscape** artist can draw an ellipse, circle, or arc. As a **Live Shape**, the object created with the Ellipse tool can be converted from an ellipse to an arc, and can be transformed into any elliptical shape.

How to Use

When the ellipse is first drawn with the Ellipse tool, there are three edit handles, *top*, *left* and *right*.

The top and left square handles control the horizontal and vertical dimensions of the shape.

The right round handle edits the arc of the shape. Dragging the right handle with the Ellipse tool in either direction, horizontal or vertical, will start an arc. When the tool is moved inside the bounds of the shape (the whole ellipse) the arc will be open. When the tool is dragging the arc and is moved outside the bounds of the shape, the arc will be closed and create a pie shape.



Tips

1. Hold Ctrl while dragging the arc control to enable snapping at angles as specified in the Inkscape Preferences window.
2. Hold Shift while drawing to draw the shape with its centre at the cursor's point of origin in order to keep position static during the drawing.
3. Hold Ctrl while drawing to keep the bound of the ellipse always touching the cursor - otherwise the cursor indicates the intersection of the x and y bound coordinates.

Star Tool

Stars and polygons are special shapes, difficult to draw by hand while maintaining a regular shape. The **Star Tool** enables stars and regular polygons to be drawn easily in the document. This tool is very convenient even though it may be less often used than the **Rectangle Tool** and **Ellipse Tool**.

Inkscape stars are **Live Shapes**, and can thus be infinitely modified after creation, with **handles** or tool control **parameters**. By modifying some of the following parameters, this tool can be used to draw triangles and other regular polygons with any number of sides. The included deformation methods allow for some amazing shapes, all while keeping the powerful aspects of easy editability of an **SVG Primitive**.

How to Use

To switch to the Star Tool, you can either

- click the Toolbox icon
- or press * (Shift+8)

Options

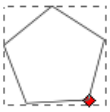
The tool options can be accessed in the **Tool Controls** bar

Drawing a Regular Polygon or Star

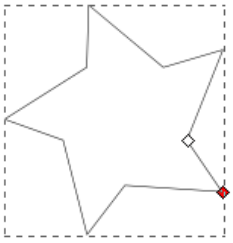
The left icons allow respectively to use the **Regular Polygon Mode** or the **Star Mode** .

The next polygon/star will be created in the **Selected Mode**. An object's mode and shape will be changed if it is selected while one of these icon is clicked.

In Regular Polygon Mode, the shape has no spokes. Only one handle is available, allowing modification of the size and orientation of the polygon.



In Star Mode, spokes start from regular polygon corners. There are two handles - the red handle is the same as for polygons, allowing modification of size and orientation of the object; the white handle at an interior spoke angle allows simultaneous modification of all the spokes' shapes and sizes.



Corners

You can define the number of corners and spokes of the shape by adjusting the corners parameter - it can either be specified before drawing a polygon or dynamically modified after selecting an existing polygon. Corner counts are limited from 3 to 1024.

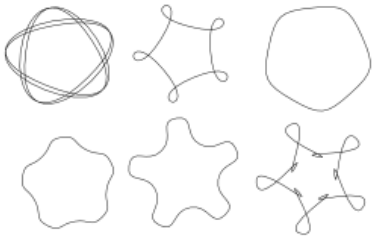
Spoke Ratio

Only accessible in Star Mode, it allows you to define the ratio of spoke length between the star core and its spoke tips. This parameter range is from 0.0 to 1.0.

Rounded

This setting allows you to round the corners of the shape in both Polygon Mode and Star Modes of this tool. The range value is from -10.0 to 10.0. Press Shift and drag either handle to modify the rounding of the shape on-canvas.

Note that dragging each handle gives different results because the degree of rounding is proportional to the angle of displacement from the center of the shape - thus a drag of the middle handle will create a greater degree of rounding than a same-length drag of the outer handle.
The following image shows some rounding examples, with regular polygons on the first line and stars at the second. There are infinitely more possibilities than shown here.



There is a **Spirograph Tool** (*Effects* > *Render* > *Spirograph*) which can produce similar effects and much more.

Randomised

The **Randomised** option of Star Tool displaces the inner and outer points of the spokes to random coordinates around the center of the star. The range value runs from -10.0 to 10.0. Near 0.0, the randomness of the displacement ratio low, and it increases as the parameter approaches -10.0 or 10.0.

You can also change the power of the randomised effect on-screen by pressing Alt and dragging the handle.

Default

Reset shape parameters to defaults.

Additional information

Polygons are always drawn with their geometric centers at the cursor origin - this cannot be changed

Ctrl - Constrain the star shape to follow some angles. The cursor move allow to modify angle by regular increment. The increment value can be defined in the general preferences window: *File* **>** *Inkscape Preferences...* **> Steps tab**.

Spiral

The **Spiral Tool** is one of the special class of tools that create **Live Shapes**. With the Spiral tool, shapes of spiral arcs can be drawn.

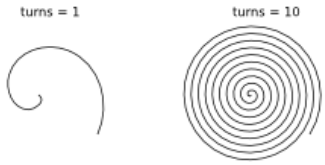
How to use

Call the Spiral Tool with its icon on the Toolbox or by pressing **F9** or **I**.

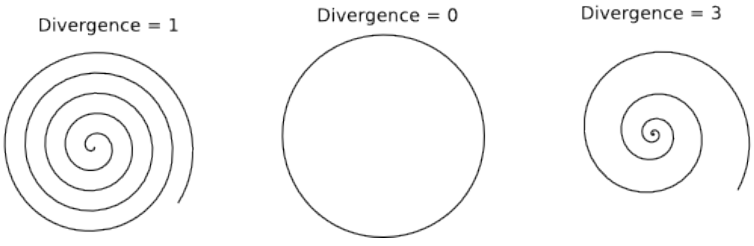
The spiral drawn with the tool has two handles, one on either end of the shape's path. Clicking and dragging the inner-most handle with the spiral tool will change the interior radius of the arc, and the handle on the outter-most end of the path edits the number of turns of the spiral.

There are **four options** in the Tool Controls bar which can be used to further manipulate the spiral:

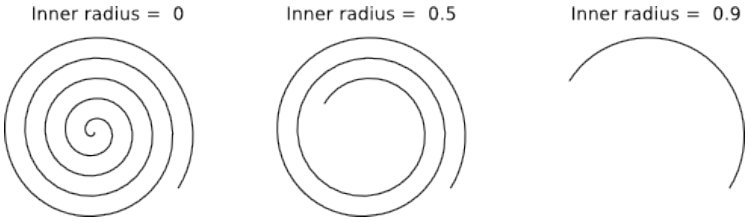
- **Turns** adjusts the number of rotations the spiral takes around its center. Increasing the turns will decrease the width between successive spiral paths, since this option does not increase the dimensions of the spiral.



- **Divergence** tightens the spiral either toward its center (as if it were being wound tighter) or toward the exterior. When the divergence is increased above 1, it makes the outer part of the spiral looser, or widens the gap between its outermost paths. The reverse is true for values below 1.



- **Inner Radius** does the same as dragging the inner handle. The inner radius parameter is a relation or fraction representing the distance from the exact center of the spiral to the beginning of the inner spiral path relative to the whole spiral radius. When the inner handle has been moved from its position at the center of the spiral, the radius becomes more than 0. When this is true, growing the spiral path by dragging the outer handle will decrease the inner radius, showing its relative nature.



- The **Broom** icon will clear all parameters to the defaults set at the spiral's creation.

Key Commands

- **Ctrl `drag`** constrains the rotation angle to 15 degree increments

Outer handle:

- **Shift `drag`** scales the spiral and ;rotates it
- **`Alt drag`** keeps the radius static while increasing or decreasing the turns around the center

Inner handle:

- **`Alt drag`** vertically adjusts divergence
- **`Alt click`** resets divergence
- **`Shift click`** moves the inner handle to the center

Pencil Tool

With the **Pencil Tool** (**P** or **F6**), the **Inkscape** artist creates freehand paths by drawing directly on the canvas in the desired curves. Inkscape evaluates the line or shape drawn by the user and produces **nodes** to form the path. After the path is drawn, the nodes of the path can be edited with the **Node Tool**, like other paths.

How to use

Choose the Pencil Tool and just click and drag the mouse to draw the line. By default it has no **fill**, but this can be set by any means (swatch or Fill and Stroke dialog). The line can also be set with **stroke** properties and colors in the Fill and Stroke dialog.

It is possible to close the line drawn while returning towards the initial point. When the mouse is close to this point, the point changes color to red to specify that a release of the mouse at this moment closes the shape.

Tips

These tools can create **single dots** by **Ctrl + click** on the canvas. This creates a small circle filled with the current stroke color. The radius can be set in the Preferences of the respective tools (it is specified as a multiple of the current stroke width). **Shift + Ctrl + click** creates a dot twice the specified size, and **Alt + Ctrl + click** varies the size of the created dots randomly.

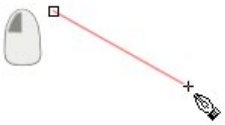
Bezier with Pen Tool

Bezier Curves, available in the Pen Tool, enable you to draw smooth curves with precision.

How to use BÃ©zier

To create a Bezier curve in Inkscape, click on the butto in the Toolbox, or press **b** or **Shift + F6**.

Left click to create the first node.



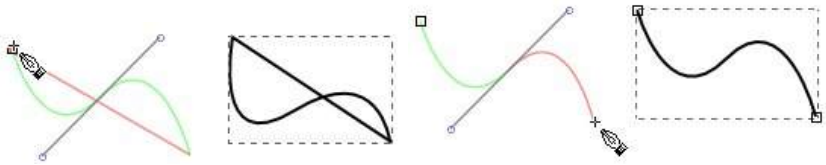
To draw a segment, just click again further where you want the segment to end. Inkscape draws immediately a straight line between these two points.

Delete a Segment

To erase the last segment/node, press **del**.

Draw a Curve

If you want to draw a curve, just keep the button pressed after clicking and just drag to make curve control point appear. They are symetrically placed to make a perfect smooth curve so that you also need to move only one **node**. To stop drawing the actual curve you can either click the first node of the global curve (if you want it to be a closed shaper), or press **Enter** or **double-click.**



Continue a Path

To continue a path that has been previously drawn, just press **b** to activate the pen and click on either end and just go on.

A pen made path can be continued with the pencil **F6** if needed .

Tips

A **right-click** closes the shape but doesn't add any new point. It then becomes a path that you can edit with any other tool, especially the Node Tool **F2** .

Further editing of any curve can be accomplished with the **Node Tool**.

Calligraphy Tool

The **Calligraphy Tool** uses dynamic drawing techniques that apply simple filters to the cursor place and motion. The SVG "stylus" transforms as if it were a physical stylus or brush, depending on its mass, speed, orientation and friction. (Some of these parameters are affected only by use of an input device such as a drawing tablet.) By changing these parameters, various types of "strokes" can be made. This makes the Calligraphy tool excellent for drawing more natural, smooth and consistent strokes, particularly when using a pen tablet or similar input device.

The Calligraphy Tool does not draw a single path line like the freehand tool, but a whole filled shape. This is not a live shape, like rectangles and stars, but an arbitrary shape consisting of node paths. Being comprised of multiple nodes, calligraphy strokes can be modified by other path tools, for example the *Node* and *Tweak* tools. Also, like all arbitrary SVG shapes, calligraphy paths have strokes at their edges, so they are affected by the usual *Fill* and *Stroke* settings.

How to Use

The Calligraphy Tool can be called by clicking its icon in the Toolbox or by pressing **C** or **Ctrl+F6**.

Calligraphy shapes are drawn in the same manner as any other shape - just click on the canvas and drag.

Options

The Calligraphy Tool has many options available in the Tool Controls bar which allow the artist to create very specific types of strokes.

Width

Used to set the basic width of the line. This basic width is then automatically modified depending on the other values (such as pressure of the tablet pen) and settings (such as the toggle setting of *Input Device Pressure*).

Warning: Calligraphy stroke width is relative to the current view and zoom factor.

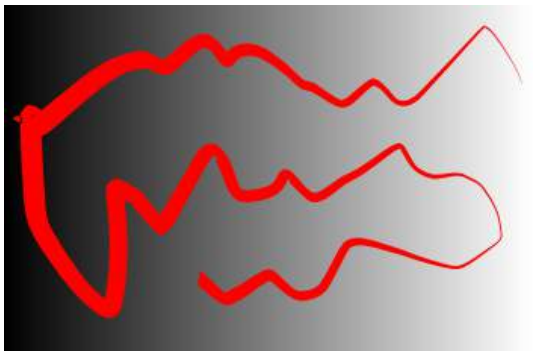
Input Device Pressure

When on, **Calligraphy** uses pen tablet pressure to affect such values as stroke width.

Trace Lightness to Width

Trace Lightness to Width adjusts the width of the stroke to the lightness of objects behind it. In the background objects, white translates into the minimum stoke width (1) and black translates to the maximum (which is set by the Width parameter). This works with both bitmap and vector images and allows the artist to not only hatch over an imported bitmap

image or any drawing, but to do so automatically reproducing the highlights and shades of the background with your strokes becoming lighter and heavier as needed. This can work alone or in combination with pressure sensitivity, depending on whether the "Use pressure" button is also toggled.



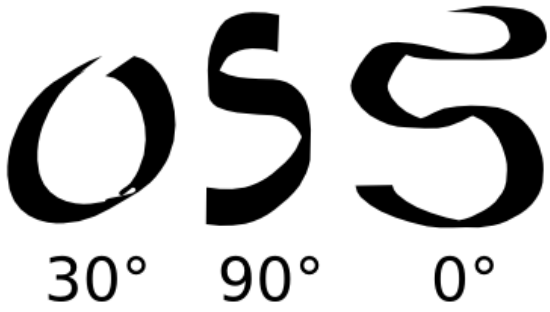
Thinning

Thinning is a way to modify the width of the calligraphy stroke dynamically according to the speed at which the stroke is made. This value enables the calligraphy tool to emulate true ink flow from a pen or brush. For instance, dragging a stroke at a uniform speed will create a mostly uniform stroke width, while increasing speed will decrease width, and decreasing speed will increase width (to a degree). The higher the thinning value is, the more the stroke will be thinned.

Some examples are shown below. Notice that a negative thinning results in a thicking. When set to 0.0, the line keeps its width with uniformity.

Angle

The **Angle** setting is used to emulate a stylus type of writing instrument. Angle will affect the direction at which the stroke creates its thinnest part, just like a calligraphy pen. Values can be set from -90Â° to 90Â°. When set to 0, the hair line is horizontal; to 90, vertical.



Tilt to angle

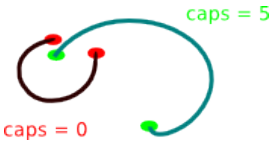
When activated, Angle is modified relatively to tilt of the tablet pen.

Fixation

Fixation changes the way the Angle width follows the calligraphic path. When set to 0.0, Angle is set always perpendicularly to the path so that the width looks nearly the same all along the path (as if the stylus were rotated constantly in the direction of the stroke). When set to 1.0, Angle is set to adjust to stroke direction most strictly (as if the stylus were kept exactly in the same direction at all times as a machine might be able to do). A setting of a little less than 1.0 (such as 0.9) will most closely follow natural hand movement, like using a real stylus.

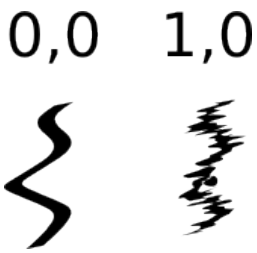
Caps

Caps determines how the line ends. At 0, the end caps will be drawn flat. Increasing the value will create elliptical end caps, and the higher the value the longer the ellipses will be. The max value is 5.00.



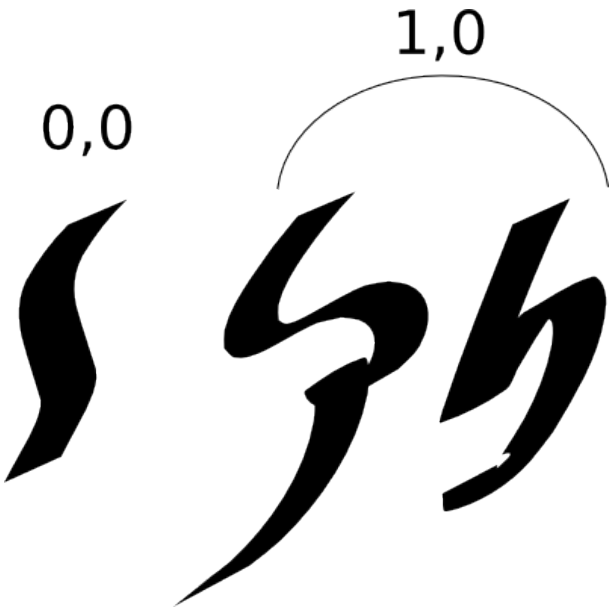
Tremor

Tremor affects jitteriness of the stroke. It can be set from 0.0 to 1.0. When set to 0, the line is the most regular.



Wiggle

Wiggle is a kind of randomization on drawn curves, making a sort of "bumpy" stroke. It generates these curves or "bumps" regularly, and can help to draw some nice typographic shapes, though the result is hardly predictable.



Mass

Mass affects how quickly the stroke follows the cursor. A heavier mass makes the stroke slower and increases smoothness or regularity of the stroke. Values can be set from 0.0 to 1.0. When set to 0.0 the path just follows the mouse as normal. When set to 1.0, the drawing of the stroke is very slow.

Default

Default resets all Calligraphy Tool Controls settings to defaults as defined in Preferences.

Drawing

Adding a New Stroke to a Calligraphy Object

Press **Shift** to add a new calligraphic line to those that are selected, keeping all strokes together as a single object.

Tracking a Shape

One of the most common operations in line engraving is **hatching** (or sometimes *cross-hatching* when several hatching grids cross): filling a space with many parallel straight or variously curved lines (usually of varying width to represent a gradual shading). You could try to achieve a similar effect with e.g. path interpolation (blending), but it is rather cumbersome and limited; manual drawing of hatch lines, on the other hand, is tedious and nearly impossible to do uniformly. Now Inkscape provides "assisted hatching" by **tracking a guide path**, allowing you to hatch quickly and uniformly and at the same time giving you sufficient manual control over the process.

First, select the **guide path** that you will track. It may be another calligraphic stroke, any path or shape, or even a letter of a text object. Then switch to Calligraphic pen, select the desired parameters (line width, angle, fixation etc.) and, before starting to draw, press Ctrl. You will see a gray **track circle** centered at your mouse pointer and touching the closest point on the selected guide path. (If you have no guide path selected, a statusbar message will tell you to select it.)

The Now move your mouse close to the guide path, so that the track circle radius is equal to the desired spacing of your hatch pattern, and start drawing along the guide path. At that moment, the radius of the circle gets locked; now the circle slides along the guide path - and the actual stroke is drawn by the center of the tracking circle, *not* by your mouse point. As a result, you are getting a smooth stroke going parallel to the guide path and always at the same distance from it.

When the stroke is ready, release your mouse button (or lift your tablet pen) but **do not let go of Ctrl** because, as long as you have it pressed, the tool remembers the hatch spacing you set when you started drawing. Now, you have just created a new stroke and, as usual with Inkscape tools, it gets selected instead of what was selected before. In our case, this means that the newly drawn stroke itself becomes the new guide path. Next, you can draw a second stroke along the first one, then a third one along the second, etc. Eventually you can fill any desired space with uniform hatching.

Alternatively, if you uncheck "Select new path" in the Calligraphy tool preferences, newly created strokes will not be selected, so your original guide path will be kept selected. In this mode, Inkscape will increase the tracking distance after each created stroke so that you can create uniformly spaced hatching by tracking a single guide path.

The attachment to the guide path is not absolute. If you stray your mouse pointer far enough from the guide path, you will be able to tear it off (the track circle turns from green to red) and move freely. This is intentional; this feature allows you, for example, to continue drawing a stroke past the end of a guide stroke, thus making your hatching cover a wider area than the initial guide path. Special care is taken to make such tearing off as smooth as possible and to suppress violent jerks, but this is not always possible; the general advice is to not try to hatch too fast. If jerking and unintended tearoffs still bother you, try increasing the Mass parameter.

Also, special code is in place to prevent flipovers - accidental jumps to the other side of the guide path. Brief flipovers are suppressed, but if you intentionally go over to the other side and stay there, eventually Inkscape will obey and your tracking stroke will also flip over to follow you.

Tracking a guide also allows some slight feedback by gradually changing the tracking distance in response to your drawing behavior. Thus, if you're consistently trying to draw closer or farther from the guide than the current tracking distance, the distance will correspondingly decrease or increase, so you will get a hatching that is slightly spacing in or out. (The effect is very slight, however, so as not to become a nuisance.) Also, note that since

tracking follows the edge of the stroke, strokes of varying width (such as those tracing background, see below) will result in gradual bending of the hatching pattern as you proceed

Engraving

Combined with the **Ctrl** key, the Calligraphy Tool is able to simulate the ancient art of line engraving. Traditional engraving is a very labour-intensive process, and while for a long time it was the only practical way of reproducing lifelike images in black-and-white print, about a century ago it was almost completely displaced by automatic halftone screens in industrial process. However, line engravings have their characteristic charm, and are often still used for certain art applications.

Bucket Fill Tool

The **Bucket Fill Tool** is simple - it fills in unfilled areas with color. Being a **vector tool**, however, **Inkscape's** Paint Bucket actually creates a new **path** that "fills in" the area in which you clicked.

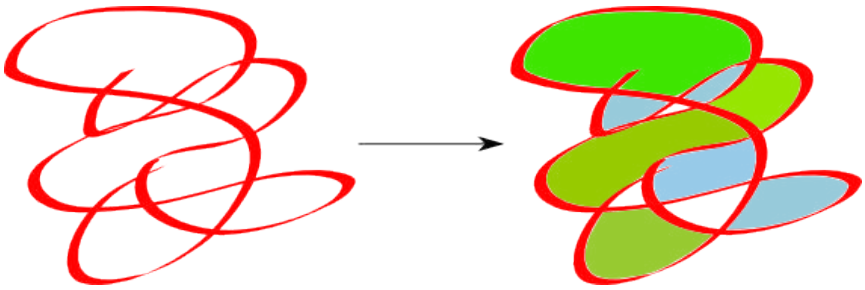
It is important to note that the tool's operation **perceptual**, not geometric. That is, when looking for the boundaries around the point you clicked, it takes for such boundaries any *visible* color changes. This means that filling will stop at **gradients**, blurs, and even the color boundaries in imported **bitmaps**, but will ignore any paths or other objects that are fully (or almost fully) transparent or for any other reason do not stand out from the background. In short, it will work exactly as if you were filling a **rasterized** version of your image in a bitmap editor like **Photoshop** or **GIMP** - but will give you a **vector object** to work with.

Internally, the tool works by performing a bitmap-based **flood fill** on a **rendered** version of the visible **canvas**, then tracing the resulting fill using **potrace** and placing the traced path into the document.

It places the rendered path onto the current layer, so you can have a **layer** on top (for example, "Inks") and select the layer below ("Colors") and do the fills so that they always appear below the Inks.

Because the tool operates in this way, you can, for example, scan a pencil sketch, import the bitmap into Inkscape, and quickly fill all its **cells** with colors without tracing the bitmap first. This is a very convenient and interactive way of digitizing your paper drawings, making the traditional bitmap tracing unnecessary in many cases.

The **resolution** of the bitmap image used to perform the trace is dependent upon your **current zoom level** -- the more 'zoomed in' to an area that you are, the higher the resolution of the bitmap-based flood fill. So, if you have a fill that is too imprecise, has rough corners, or doesn't go where it is supposed to go, just undo it (CTRL + z), 'zoom in' closer and repeat filling from the same point. Conversely, if the fill leaks out through a small gap, zoom out to make the gap less visible and fill again (or use the **auto gap** closing parameter - see below).



How to use

The **Paint Bucket Tool** works fairly intuitively: click in any area bounded on all sides and it will fill it with color - or, in reality, a path which can be filled and adjusted like any arbitrary path.

Style

Like all object-creating tools, the Paint Bucket may use the **last-set style** for the **objects** it creates (this is the default), or it can use its own **fixed style**. You can switch between these modes on this tool's page in *Inkscape Preferences* (Ctrl+Shift+P). As in all other tools, the **style swatch** on the far right of the **Controls Bar** shows the style that will be used for the next fill object you create.

Controls

In the **Tool Controls** bar: Paint Bucket's perceptual fill can use either all visible colors or specific **color channels**. Using the **Fill by** drop-down list, you can restrict the fill algorithm to one the following channels:

- Red
- Green
- Blue
- Hue
- Saturation
- Lightness
- Alpha

The **Threshold** (in per cent units) controls how large the color difference must be at a point (compared to the initial click point) to stop the fill. 'Zero tolerance' means only an area which is strictly the same color will be filled; the larger the tolerance, the easier it will be for the fill to leak into adjacent different-color areas. The default value is 10%.

Using the **Grow/shrink by** parameter, you can control the amount of **inset/outset** to be applied to the created fill path. Setting a positive outset causes fill paths to be larger than the filled bitmap area (good for eliminating **anti-aliasing** errors), while setting a negative outset causes the path to be smaller. This works much the same as the Outset and Inset path commands, except it's done automatically after every fill.

With the **Close gaps** parameter, you can make the Paint Bucket tool ignore any gaps in the area boundaries that would normally cause the fill to spill out of the desired area. There are four settings to auto gap:

- None
- Small (close gaps up to 2 pixels in size)
- Medium (4 pixels)
- Large (6 pixels)

Note : setting this parameter to anything other than **None** may slow down Inkscape noticeably while filling large areas.

Hotkeys

The tool's **shortcuts** are:

- **Single click** performs filling from the click point.
- **Shift+click** performs filling from the click point and then unions the resulting path with the selected path. This way, if your first attempt did not fill in all of the desired area, you can Shift+click the remaining corner to fill it in separately and combine the result with the result of the previous fill.
- **Ctrl+click** on an object simply changes that object's fill to the current fill color of the tool, and **Shift+Ctrl+click** changes the **stroke** to the current stroke color.
- **Click+drag** performs filling from all of the points that you pass while dragging (you will see your path visualized by a red line). From each point, the fill spreads to its neighbours with the colors similar to that point - in other words, it's like clicking with this tool at each point of the drag path and joining the results. This lets you easily fill an area occupied by a gradient or blur - just drag from the darkest to the lightest points in the area you want to fill.

- **Alt+click** and drag works similarly to simple drag, except from each point of the drag path, the fill spreads to the neighbours (if any) with the colors similar to the *initial point* (the point where you started the drag). This lets you fill a series of similarly-colored yet separated areas (for example, multiple cells in a cartoon) by starting the drag in one of those areas, and **Alt**+dragging the tool through all the other areas.

Text tool

The **Text Tool** makes it possible to write text in an SVG drawing.

How to Us

To invoke the tool, press **Shift + Ctrl + T**. Alternatively there is a Text menu which allows you to change the aspect of the characters or their behavior with respect to other elements of the drawing. Text Tool uses several short cut keys to modify typographic elements.

There are two ways to use the **Text Tool** :

1. Select the Text Tool and click on the **canvas** to create a simple **line field**. It will grow with the written text.
2. Click and drag to draw a rectangle this becomes a **text box** : this frame will contain the text and will be the limit for the word wrapping. The text box will not grow with the text, so if there is too much text for the frame, some letters will simply not be displayed. This frame can be manually resized by dragging the handle at the bottom right corner. The **Flow into Frame** command of the **Text** menu makes it easy to use any shape as a **text container**.

Special Characters

- **Ctrl +Space** inserts on **non-breaking space**

- **Ctrl U** allows to use **Unicode encodings** within the document. To find a Unicode value refer to a Unicode character map program or reference.
    - ♦ To insert an arbitrary Unicode character, type **Ctrl + U**, then the hexadecimal code, then **Enter.** For example, type **Ctrl U 2 0 1 4 Enter** for an **em-dash**; **Ctrl + U** a 9 + **Enter** for a **copyright sign**.
    - ♦ To stay in Unicode mode after inserting the character, press **Space** instead of **Enter**
    - ♦ Press Esc or another **Ctrl U** to cancel Unicode mode without inserting the character.

Tips

- If you're not sure all your text is not visible in the frame, use the Text and Font window to read it all.

Connector Tool

The **Connector tool** (**Ctrl+F2** or the **o** key) draws lines between **objects** that stay connected to other objects as they are manipulated. Any object may be marked as an object *to avoid*, which causes connectors to automatically route around the object. This is helpful for creating technical illustrations like flowcharts.

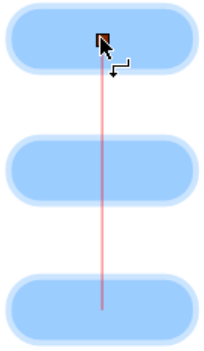
How to Use

Connecting

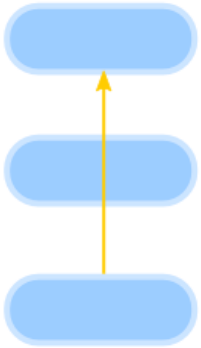
A new **connector** can be drawn by clicking and dragging from any point on the canvas. Connectors can also be created with two clicks, rather than click-and-drag, if this is preferred. In this case, click once **on an empty point on the canvas** to begin drawing the connector, then move the mouse to the new connector's target point and then click again to finalize the connector. Single clicking on a canvas object selects/deselects that object, just as with other tools. Usually connectors are drawn *from* an existing object:

Connection point handles are shown while the mouse cursor is hovering over a non-connector object in the connector tool. Currently they are shown only at the center of objects. When creating a connector, if the connector is started or ended over a connection point then the connector will be **attached** to that object. From then on the connector will be automatically rerouted whenever the attached object is moved.

Connectors attached to objects are currently drawn to the **bounding box** of those objects. It is planned that they will be drawn instead to the edges of objects.



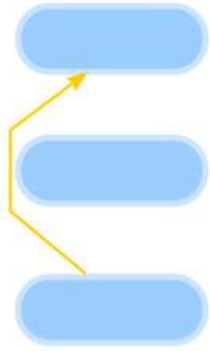
The connector is finalized, when the mouse is released:



The **Make connectors avoid selected objects** button marks all objects in the selection as "avoided", causing all current and future connectors to automatically avoid these objects.

The **Make connectors ignore selected objects** button marks all objects in the selection as "ignored", causing all current and future connectors to completely ignore these objects. This is the default for all canvas items, i.e., no objects are automatically routed around by default.

Here the middle object is avoided:



By default, the Connector tool will not attach connectors to text objects. A checkbox in the Connector preferences controls this setting.

Rerouting

A selected connector shows two **endpoint handles**. By clicking and dragging these, the connector can be rerouted and attached/detached from objects.

Connectors moved as part of a selection will stay attached to other objects in the selection, rather than becoming detached from them.

Arranging

The **margins around avoided shapes** used for autorouting connectors can be adjusted via the "Spacing" control on the controls bar.

The **Remove Overlaps** button to move the selected objects enough that they don't overlap each other. A minimum spacing between the boundaries of objects can be specified. Together with the automatic layout tool, described above, this should be a significant addition to Inkscape's usability for diagramming. Removing overlaps is different from the "Unclump" button in that the former is completely deterministic and guarantees removing overlaps on the

first application, but is not concerned with visual perceptive distances between objects. Unclumping, on the other hand, attempts to equalize perceptive distances between objects and can be applied repeatedly for gradual effect.

Automatic Diagram Layout: this function is available in the **Align and Distribute** dialog and performs automatic layout of diagrams involving a network of shapes and connectors. Edges are treated as if they are springs such that the distance between nodes will be proportional to the path length â number of connectors â between them. Disconnected components (where not every shape is connected) will be arranged around the circumference of a circle.

Gradients

The **Gradient Tool** allows the progressive passage of one color to another (or multiple others) in an object. It can be used on any **shape**, closed or opened, as a **fill** or a **stroke** (applied separately). Any number of selected objects can simultaneously display **handles** and **direction lines** for the **linear** and **radial gradients** in their fills or strokes. You can drag these handles directly in the drawing, to interactively adjust gradient positions.

The gradient can be set in the Fill and in the Stroke Paint dialog to replace of any other type of coloration.

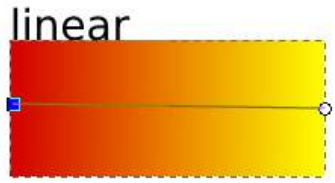
How to use

To invoke, either click the Gradient Tool icon or press **Ctrl +F1**.

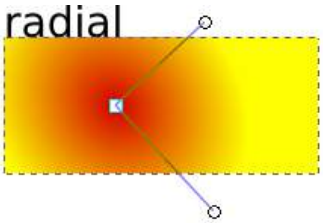
Types of gradients

There are two types of gradients:

1. **Linear gradients** transition the color change from one point to another in a straight line. Just click where the gradient should begin, hold and drag, and release where it should end. Two handles now appear: the handle of the beginning is square and the end is a circle. The two handles can now be used to change the color (select each one and choose the color for it from the fill and stroke dialog) or move and change the gradient direction.



2. **Radial gradients** work as colored circles with the color transition beginning at the center. Three handles are available : the square one is the center of the gradient, and the circle ones change the radius of the gradient. When the radius is the same for both circle handle, the gradient is perfectly circular. To move the placement of a radial gradient in the object, drag the square handle.



Once a gradient is created it is automatically saved in the drop-down list of the Gradient Tool Options bar so that it can be easily reused on other shapes.

More colors

To have more than two colors on a gradient, just double-click on the gradient base path to make a new handle (stop) appear. **Click** in the **swatch** to set the color. You can move this handle to adjust the position of this stop. To delete a stop, just press the**Backspace** key.

Selecting multiple stops

More than one gradient stop can be selected at a time. Shortcuts for working with multiple stop selections are generally modeled on the Node tool.

- Add a stop to the selected stops by **Shift+click**.
- Press **Ctrl+A** to select all stops in the selected objects.
- **Shift+drag** around stops to add them to selection.

Multiple selected stops:

- Can be moved together by **mouse drag** or by **arrow keys**. For example, creating a linear gradient, then press **Ctrl+A** to select all stops and use arrow keys to move the entire gradient as a whole.
- Can be deleted at the same time by pressing **Del**.

An always up-to-date description of the current handle selection is provided in the statusbar in the Gradient tool, including the number of selected handles (and the type of the single selected handle), as well as the total number of handles and selected objects.

Editing intermediate stops

Intermediate stops in gradients can be added, deleted, and edited right on canvas.

Stops can be added by **double clicking** or by **Ctrl+Alt+Click** on the gradient line. Also, you can **drag-and-drop** a color from the palette onto the gradient line to create a new stop with this color. Dropping a color on an existing stop changes the color of that stop.

When two or more adjacent stops are selected, pressing **Ins** adds stops in the middles of all selected stop intervals.

Intermediate stops can be **mousedrag**ged or moved by **arrow keys** along their gradient line, within the limits of the adjacent unselected stops (or end handles).

- Dragging with **Ctrl** moves the selected stops snapping them to 1/10 fractions of the available range.
- Dragging with **Alt** moves the selected stops depending on how close each one is to the stop being dragged, using a smooth bell-like curve similar to the node sculpting feature in Node tool. This makes it easy to approximate different gradient profiles; for example, if you have a two-stop gradient that you want to shape according to a curve profile, select both ends of the gradient, press **Ins** a few times to add a number of intermediate nodes, then **Alt+drag** a node in the middle to smoothly profile the gradient.

Stops can also be moved by **arrow keys** with all the regular modifiers (**Shift** for 10Ã movement, **Alt** for pixel-size movement at the current zoom, **Shift+Alt** for 10 pixels movement at the current zoom).

Stops can be deleted by **Ctrl+Alt+Click** on a stop or by the **Del** key for all the selected stop(s).

- When you delete an end stop, the nearest intermediate stop becomes the new end stop of the gradient (without moving â i.e., the gradient span becomes shorter).

- When you delete an end stop and there are no intermediate stops, the object will be painted with a solid fill taken from the color & opacity of the remaining stop.

Pressing **Ctrl+L** with some intermediate stops selected attempts to *simplify* the selected portion of the gradient, removing those stops that can be removed without too much change in the way the gradient looks. In particular, new stops created by double-clicking or pressing Ins initially do not change the appearance of the gradient, so if you press **Ctrl+L**, all redundant stops that weren't moved or repainted since creation will be deleted.

Automatic duplication of gradients

When copy/pasting or duplicating an object with gradient, it automatically gets a **copy** of the original gradient, so modifying it does not affect the source object's gradient anymore .

However, to accommodate the needs of users who rely on sharing the same gradient definition across objects, this behavior can be optionally suppressed. The **Prevent sharing of gradient definitions** checkbox on the Misc tab of Inkscape Preferences is by default checked; if you uncheck it, Inkscape does not automatically copy gradient definitions for new objects, which means that copy/pasting, duplicating, pasting style, and explicit assignment of a gradient to an object via the Gradient tool controls results in a shared gradient definition, so that changing the colors or mid-stop positions of the gradient on one object (but not changing the coordinates of the end handles) affects all other objects that share the same definition.

Dropper Tool

The **Dropper Tool** is used to select an **object**'s **fill** or **stroke** color by **sampling** the color of an area of the **canvas**. The color selected is the single point at the center of the cross at the end of the Dropper Tool icon.

How to Use

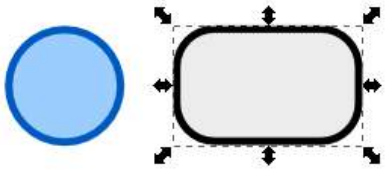
Dropper is a very simple, yet powerful tool. Select an object to which you want to apply a new color and press **F7** or **D** to switch to Dropper. All you need to do is just click an area with desired color to apply new color to fill, or use **Shift** to apply this color to stroke.

The shortcut **D** can be used to *toggle* (not just switch to) the Dropper tool â much like space is used to toggle the Selector Tool. That is, pressing **D** a second time switches back to the tool used before.

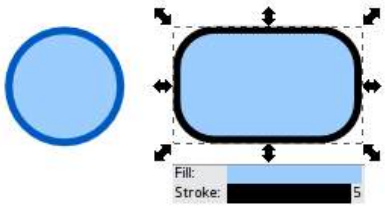
There are cases when you will want to change Dropper's options in the Tool Controls bar:

The left icon stands for picking **alpha channel**. The right icon stands for setting alpha channel.

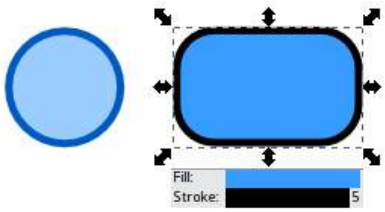
Suppose you have an object selected and, using Dropper, click on an object which has blue transparent fill (#389bff7f).



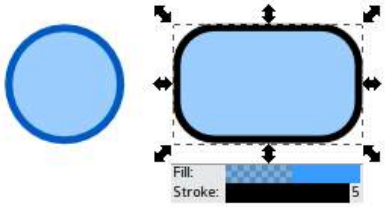
If the "Pick alpha" checkbox is off, the selected object will get the fill color #9ccdffff (i.e. faded-out blue) and opaque:



If the "Pick alpha" checkbox is on, but "Set alpha" is off, the selected object will get the fill color #389bffff (bright blue) and opaque:

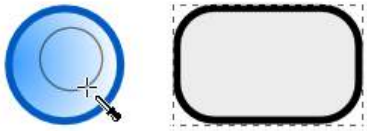


If both "Pick alpha" and "Set alpha" are on, the selected object will get the fill color #389bff7f (pale blue) and half-transparent, i.e. fully match the fill style of the object from which the color was inherited:

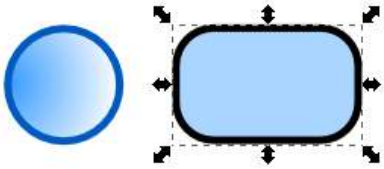


Note that in no situation can Dropper change the *master opacity* of the selected object(s) (only the fill/stroke opacity), although it can pick it just as it does any other kind of opacity.

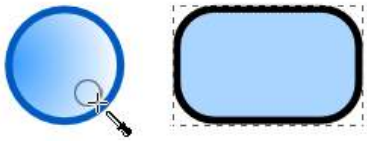
There are cases also, when you might want to apply an average color of an area, for example, an average color of an object with **gradient** fill. Here is how to do it. Select the unpainted object, switch to Dropper and start dragging it from the central point of an area with the average color you want to pick:



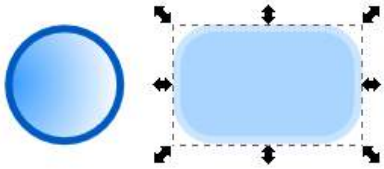
You get:



Same can be done for stroke:



And you get:



Hotkeys

Alt+Click picks the inverse of the color at point (works with dragging and **Shift** too).

Tips

You can drag to pick an average color from the object itself, if it has a gradient fill. You can use it for setting color of stroke as average color of this object's gradient fill.

Path modification Effects

Modify Path Effects work on a selection of any number of **paths** and modify them in various ways without creating new paths or deleting existing ones.

Available effects

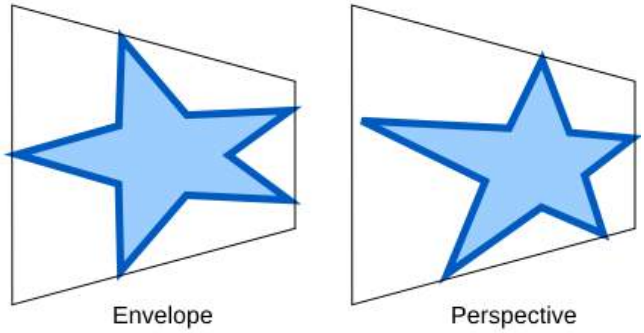
Envelope and Perspective

Maps **bounding box** of a path to **quadrilateral** to achieve effect of a perspective. These two effects are quite similar, but differ in level of natural look of the desired effect.

Here is how to use it:

1. Draw your to-be-distorted path
2. Draw and position a four **node** path (**envelope frame**).
3. Place the to-be-distorted path inside/above the envelope frame.
4. Select the path you wish to distort first and then add to selection the envelope frame. The original position of the four nodes is considered to be clockwise around the bounding box of the path to distort beginning in the upper left corner.
5. Apply the effect.

Here is the difference:



Add Nodes

Adds nodes to the selected paths. Each segment of the selected path is subdivided into ceil (Length/Max) equal length segments. Lengths are measured in **SVG User Units** calculated from the path data and do not take into account any transforms.

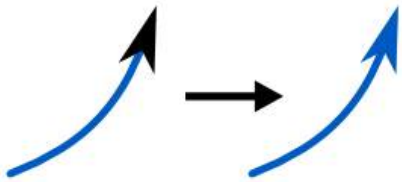
Parameters for example below:

- Division method: by number of segments
- Maximum segment length (px): 5
- Number of segments: 2

Color Markers to Match Stroke

Currently available SVG 1.1 Full specification doesn't allow using the same color for the path and its markers. This extension is a temporary workaround to fix that (until the new improved

SVG specification is out). Just change the stroke color of your path and call this effect to recolor its markers to match.



Flatten Beziers

Flattens paths in the current selection, approximating each path with a **polyline** whose segments meet the specified criteria for flatness.

Parameters for example below:

- Flatness:10

Fractalize

Replaces each segment of the selected path by a crooked line, subdivided to the given depth, with randomly displaced nodes.

Available options are:

Parameters for example below:

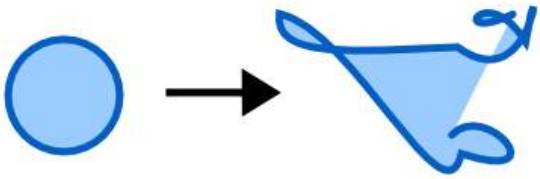
- Subdivisions: 6
- Smoothness: 4.0

Jitter nodes

Randomly shifts nodes (and, optionally, **node handles**)

Parameters for example below:

- Maximum Displacement in X: 50.0
- Maximum Displacement in Y: 50.0
- Shift nodes: checked
- Shift node handles: checked
- Use normal distribution: checked

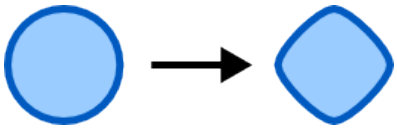


Straighten Segments

Straightens curved segment to a user defined level (in percents), using of two behaviours (**1** stands for rounded corners, **2** stands for less rounded corners).

Parameters for example below:

- Percent:50
- Behavior: 1

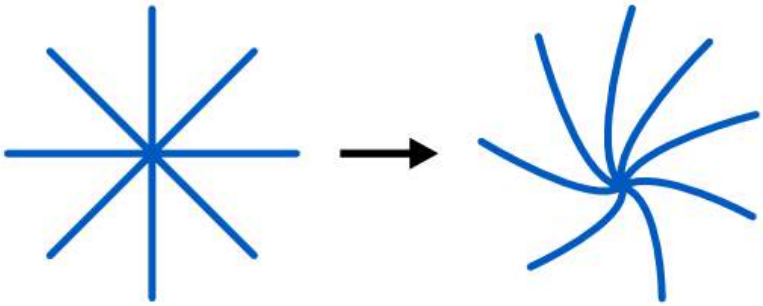


Whirl

Twists the selected paths around the specified center point.

Parameters for an example below:

- Amount of whirl: 15
- Rotation is clockwise: checked



Styling Text

Many modifications can be done to **Text**. Some are available in **Text Option Bar**, but some aren't. Here is a little overview of the most useful options.

Text Selection

- **Ctrl** with arrows **left** and **right** : moves the beam word by word

- **Shift** with arrows **left** and **right** : selecting glyph by glyph
- **Ctrl Shift** with arrows **left** and **right** : selection by words
- **Double click** : select the word
- **Triple click** : select the line
- **Shift** + **Home** : select from the beginning of the line
- **Shift** + **End** : select until the end of the line
- **Ctrl** + **Shift** +**Home** : select from the beginning of the text
- **Ctrl** + **Shift** + **End** : select until the end of the text

Text Aspect

The Tool Controls Bar provides several options such as :

- Font families
- Font size
- Bold style
- Oblique/Italic style
- Alignment

Hot Keys

- **Ctrl +B** and **Ctrl + I** applies **bold** and *italic* to the selected text.
- **Kerning** : **Alt** with arrows **left** and **right** increase or decrease the space between caracters at the end or beginning ofthe selection. You can also use **Alt + >**, **Alt <**, **Shift +Alt + >**, or **Shift + Alt + <** to change the letter spacing more significantly. Use the Remove manual kerns menu of the Text menu if your are not satisfied with it.
- Pressing **Alt + [**, **Alt +]** rotates precisely the letters; **Ctrl** + **[**, **Ctrl** + **]** do 90Â° rotation at once.

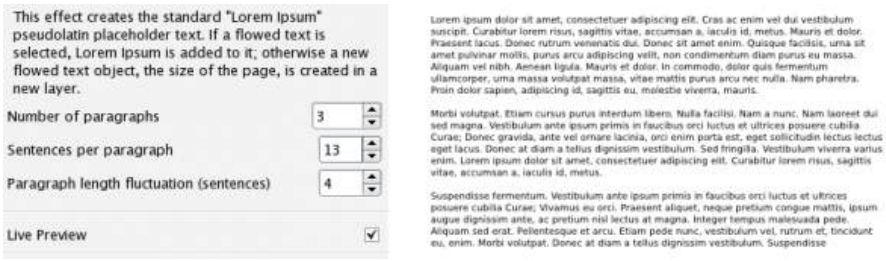
- **Alt arrows up and down** change the vertical position of the selected text relatively to the baseline.

Text Effects

Writing text often requires the repetition of the same operations. The *Effect* > *Text* menu can do some of them automatically.

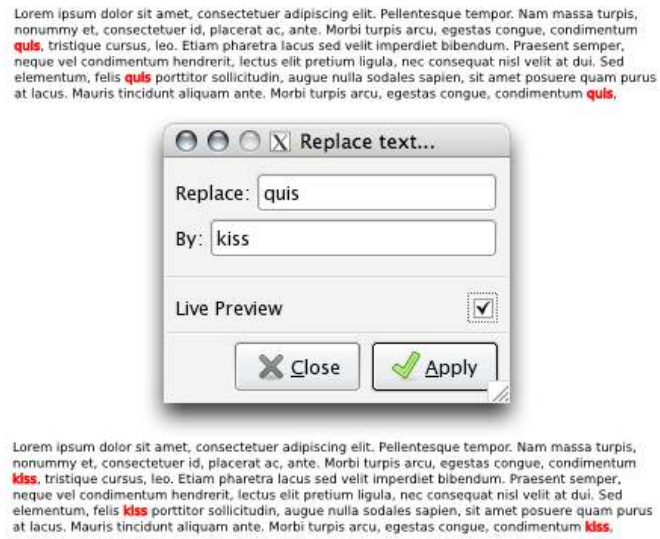
Lorem Ipsum Example

In the example below a standard latin text taken from Cicero is insterted into a text frame. This is useful in cases when a text placeholder is required (e.g. webdesign).



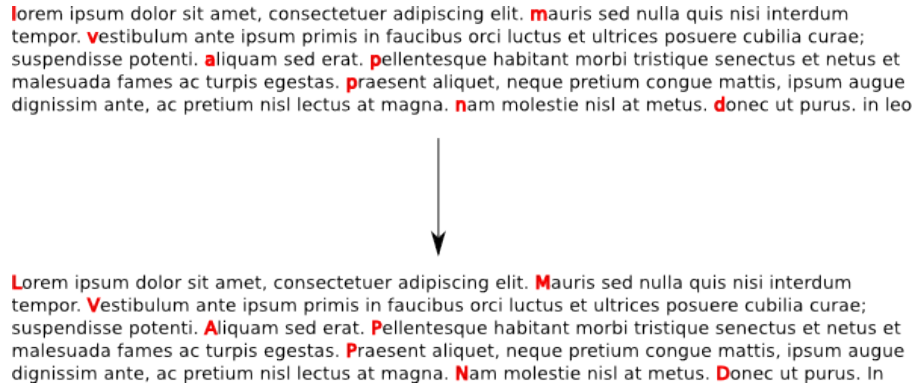
Replace Text

Replace Text simply replaces the text written in the first field of the dialog by the text written in the second one.



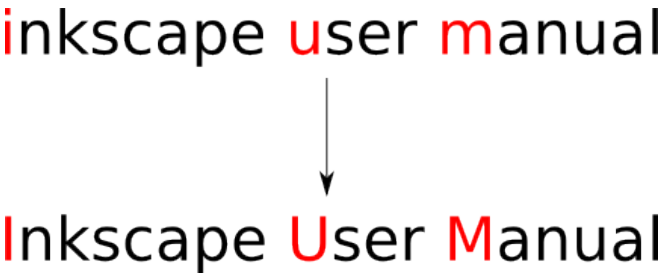
Sentence Case

Sentence Case replaces lower case characters by capitals ine the beginning of every sentence.



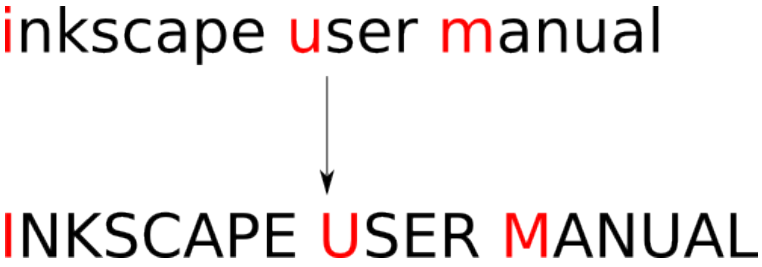
Title Case

Capitalizes first letter of each word.



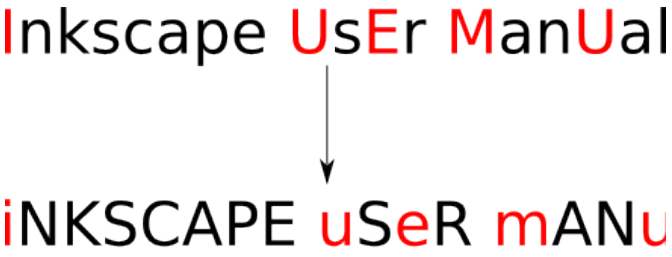
UPPER CASE

Capitalizes all letters.



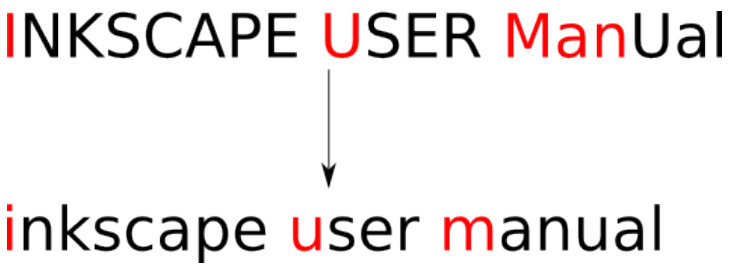
fLIP cASE

Reverses letter case, so that all capitals become lower case, and all lower case letters become upper case letters.



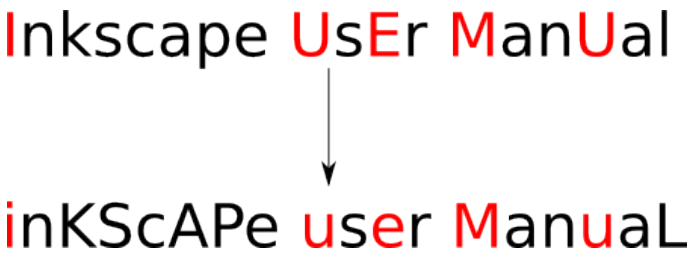
lower case

All the letters become lower case.



rANdOm CasE

Arbitrarily toggles letter case in the whole text.



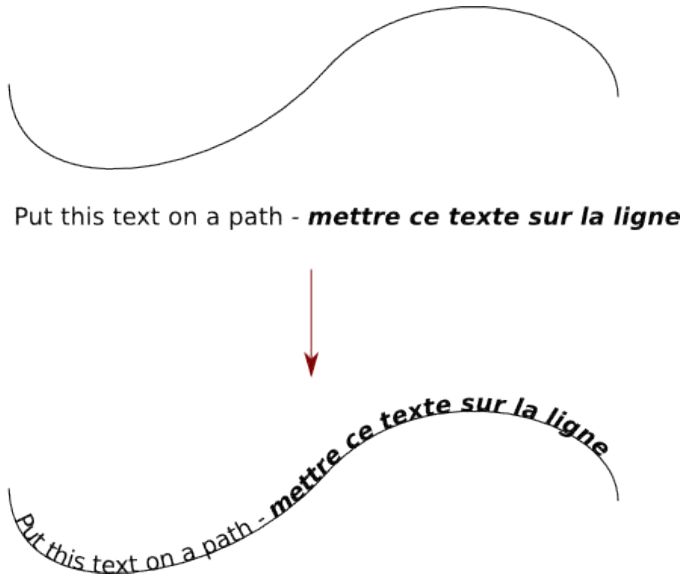
Text and Paths

Text and path can be complementary elements that work together. Some options of the Text menu are very interesting.

Put on Path / Remove from Path

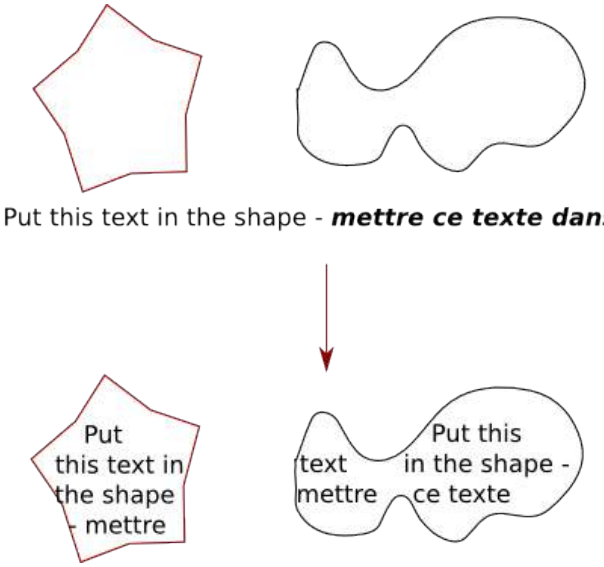
Inkscape has support for the SVG element <textPath>. That allows putting a text on a path so that it follows its shape. Both text and path remain fully editable (including kerns and letterspacing in text).

Select the path and the text and use Text->Put on Path. When you move the path, its attached text moves with it; however, you can move the text away from its path or transform it without losing the link. The Text->Remove from Path command converts a text-on-path into a regular text object.



Flow into Frame / Unflow from Frame

The Flow into frame **Alt+W** command of the Text menu takes any selected text and puts it into the selected shape. Word-wrapping is automatically done so that the text fits the shape as precisely as possible. Manual adjustment can still be done on the text after this. To separate the text from the path, just use Unflow or press **Shift+Alt+ W**.


Put this text in the shape - **mettre ce texte dans la forme**
Put
this text in
the shape
- mettre
text
mettre
Put this
in the shape -
ce texte

Copy, Clone and Duplicate

Introduction

The artist may have need, at some point in his work, to replicate an object he has created for any number of purposes. Possibly he is populating a scene with bubbles or blades of grass which will all look exactly alike, or else he may want to create a very similar object which has some minor modifications.

Fortunately, Inkscape offers several tools to replicate objects. Each one has advantages in disadvantages, of course, which will be defined also by the purpose of the replicant object.

A strict *copy* of an object repeats the object on screen and within the xml code, and the new object is entirely separate from the original. This means that changes can be made to the replicant indifferently to the original, and changes made to the original do not affect the replicant either. This is done with either the **Copy** or **Duplicate** commands.

A *clone* is merely a repetition of the original on screen and in the code. Most modifications made to the original, like fill and transparency, are transferred to the replicant as well, and only a few modifications, such as transform and position, may be made to the replicant directly (some of these only under certain circumstances). Clones are created with the **Clone** command.

How to Use

Copy

The **Copy** command (Ctrl + C or *Edit* > *Copy* or the command bar button) makes an exact copy of the selection in Inkscape's memory; the **Paste** command (Ctrl + V or *Edit* > *Paste* or the command bar button) puts an exact copy under the cursor. The last selection to be copied can be pasted an indefinite number of times. It is also possible to paste the object wherever it is needed, for instance on an other layer, without the trouble of having to reposition the copy after its creation.

Duplicate

The **Duplicate** command (Ctrl + D or *Edit* > *Duplicate*) allows you to copy and paste a selection automatically with one command. The new object or selection is in exactly the same position as the original and is selected (instead of the original), ready for further manipulation.

Clone

The **Clone** command (Alt + D or *Edit* > *Clone*) allows you to create a copy of the selection that is linked to the original, also known as its *parent*. The clone's shape, style (fill and stroke, transparency) and dimensions are linked to its parent. Thus, editing the parent's transparency (etc.) will also affect the clone's transparency.

You can clone multiple object by grouping them first.

A clone can be a parent to another clone. Any modifications to the original will affect all clones down the hierarchy of clones.

The *fill* and *stroke paint* of a clone is the same as its parent's *unless the* ***parent*** has "Unset paint" activated for either in the *Fill and Stroke* dialog. Unsetting *fill* will make the parent's fill black as well as the clone's, but the clone's fill can then be reset while the parent's cannot; unsetting *stroke paint* will make the stroke disappear so that the clone's *stroke* and *stroke paint* can be adjusted while the parent's cannot (it will remain invisible

unless paint is reset for the parent).

Unlink clone

Unlink clone (Alt + Shift + D or *Edit* > *Clone* > *Unlink clone*) separates a clone definitively from its parent, effectively making it only a copy. After that you can apply changes to the parent without changing the copy.

Select original

Use **Select original** (Shift + D or *Edit* > *Clone* > *Select original*) to find the parent of a selected clone. This is very useful when you have lost your parent.

Clipping paths and masks

Introduction

Clip paths, masks, and patterns are greatly useful additions to an artist or designers' tool set. A common property that all three share is the concept of boundaries. Clip paths and masks bound the contents of the items they contain, whereas patterns are only visible inside the bounds of the object they are applied to (in all three cases cropping visibility).

These methods do not delete parts of your objects, they merely tell the SVG renderer not to display them. This non-destructive editing makes it so that we can also release objects from from their bounded areas as well.

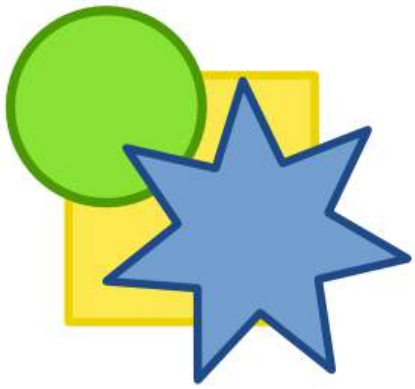
There are benefits to working with clip paths or masks combined with either individual or grouped objects.

Use of patterns fills is addressed in a dedicated chapter.

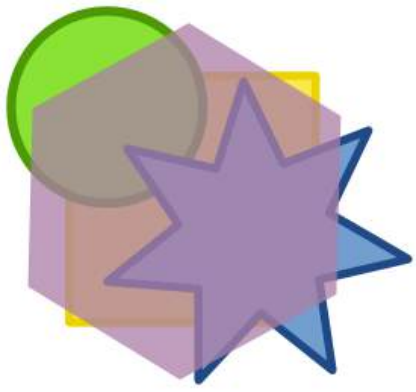
Clipping

The clip function in Inkscape allows an object or a group of objects to be clipped (cropped) so that only parts of the elements are visible. Clipping can be applied to any object (including groups, layers, bitmaps, etc.)

The easiest way to understand clipping is to see it in action. Take the following assortment of objects:

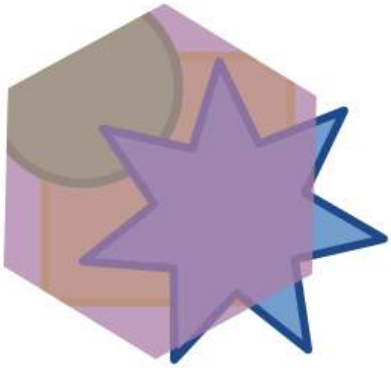


Now, place another object on top of all the others. This is what we are using for our clipping object. Note that the Z-order of the clipping object is important, as the default behavior in Inkscape takes the top-most object as the clipping object. Also note that the style (fill, stroke, opacity etc.) of the object is irrelevant, all that is used when clipping is the shape of the object.



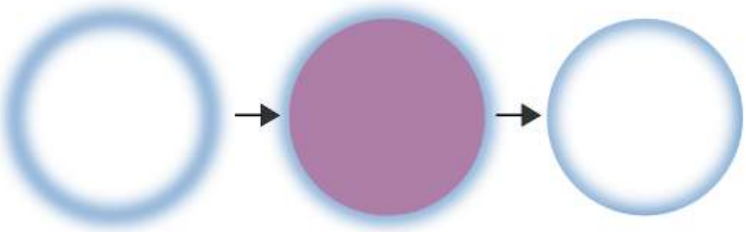
The next step is to select all the objects that you want clipped and the clipping object. Then in the menus, go to Object > Clip > Set. The result is:

Since this is all non-destructive editing, it means that we can also release objects from a clip. Do this by selecting the object that you wish to release, and in the menus go to Object > Clip > Release. In the following example the blue star object was selected, and the clip was released:

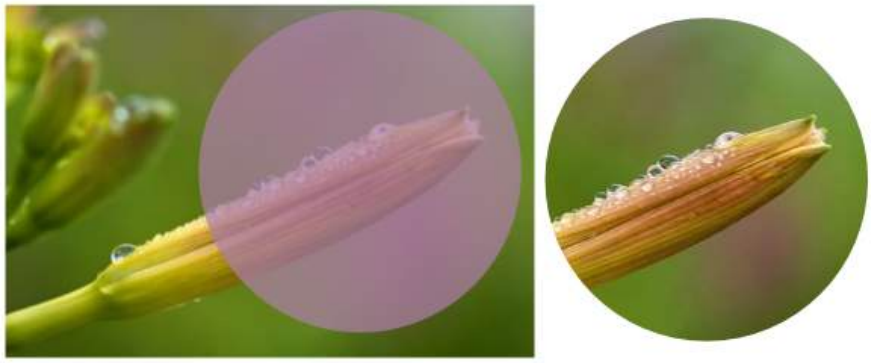


So the purple clipping object has now returned, and the blue star object is no longer clipped. Note that the other two objects are still clipped, as the clip was applied to 3 seperate objects. These clips still apply and can be released if needed. Because clipping paths can be applied to any object, if you wanted to treat the 3 objects as a whole, you would group them, then apply the clip to the group.

Clipping can be done on any object, even objects that have had filters applied to them. Below is an example of using clipping to create a bubble effect on a circle:



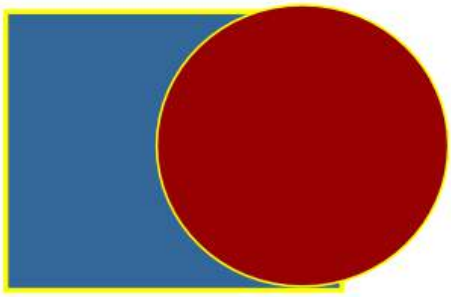
A clip can also be done on bitmaps that are embedded into or linked from an SVG:

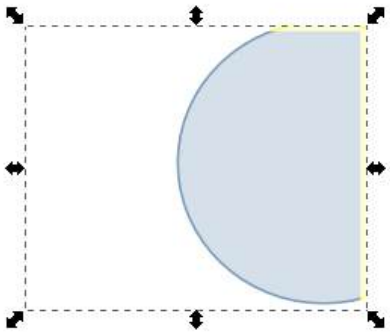


Masking

Like clip paths, masks allows an object or a group of objects to be visually cropped. The difference is that the mask's black or transparent areas become fully transparent in the masked object; mask's opaque white areas become fully opaque; and all intermediate colors translate into intermediate levels of opacity in the masked object. This allows you to apply, for example, arbitrary transparency gradients to objects.

In the following example you will see both the masks and the items they're applied to. As you'll notice, masks offer the ability for you to texture objects, give them depth, and other things that clip paths can't provide:



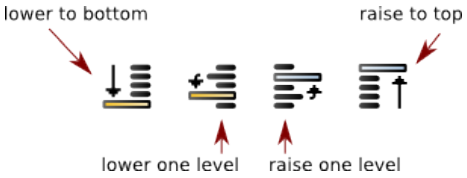


Introduction to Positionning

Every new **object** is placed at a higher level of **z-order**. It is impossible to have two or more objects at the same level. If there is a need to change the **stack**, use some Object menu entries or **Selector Tool F1** options.

If layers are defined, the stack changes occur in the layer.

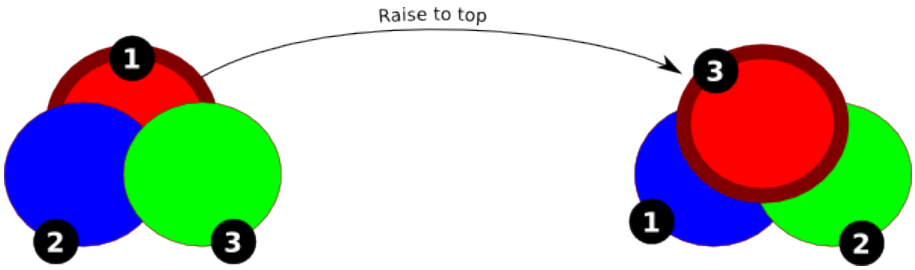
When the **Selector Tool F1** is on, this option appears in the Tool Controls bar.



Raise to Top

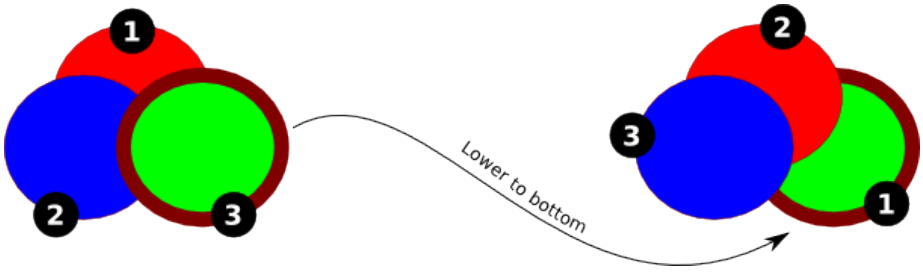
Raise to top (**Home**) command puts any selected object at the top, above all other objects in the drawing.

SVG places any newly created object at top. However it is impossible to have 2 objects at the same stack-level. Some **shapes** or **paths** may be masked by these new objects.



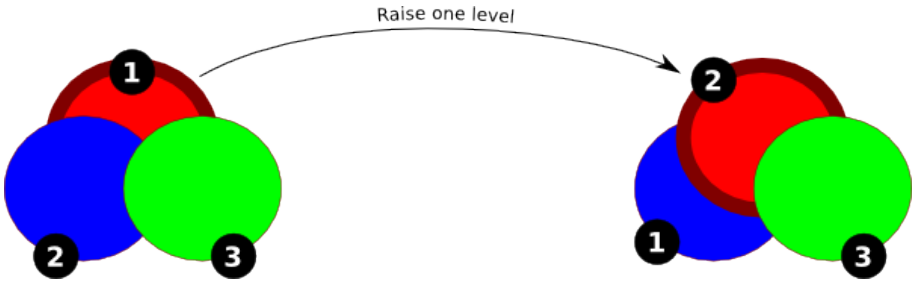
Lower to Bottom

Lower to Bottom puts all selected objects at the bottom, below all other objects of the drawing.



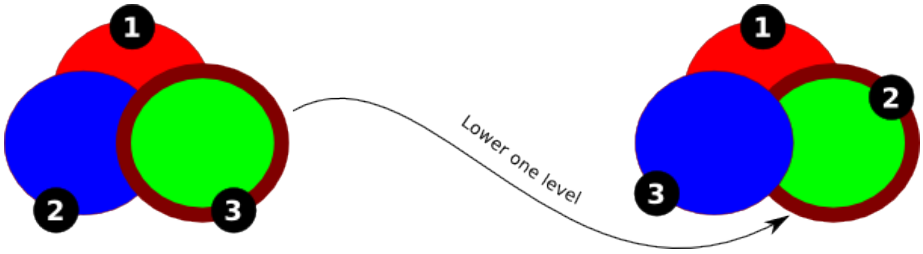
Raise

The Raise (PgUp) command changes the stack of any selected object so that its z-order is just changed by one level and goes over only one other.



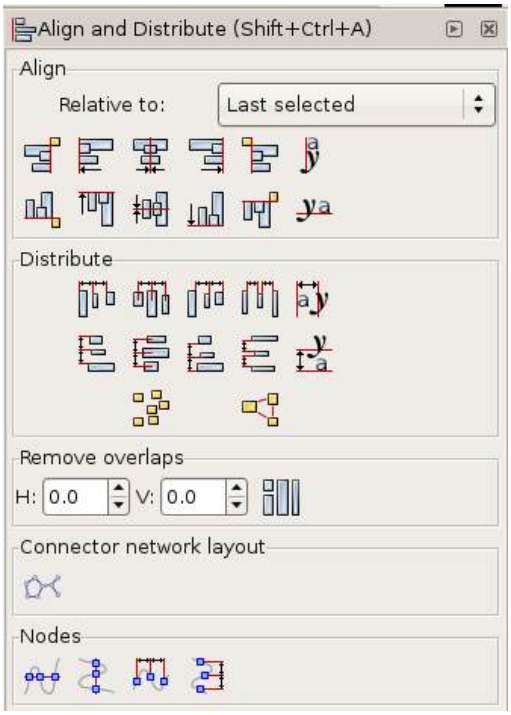
Lower

The Lower (PgDn) command changes the stack of any selected object so that its z-order is just changed by one level and goes below one other.



Align and Distribute

Align and Distribute moves multiple selected objects to the same axis and distributes them equally relative to a selected anchor object. The Align and Distribute dialog offers many ways of applying these functions to object selections.



How to Use

The Align and Distribute dialog can be called by pressing **Shift+Ctrl+A**, selecting *Object* > *Align and Distribute*, or clicking the icon on the Toolbar:

Make sure your objects are selected. Depending on which object you wish to have as the **anchor** and which *relative to* setting you have selected, order of selection will be important.

"Relative to"

Objects can be aligned and distributed relative to:

- the last or first selected
- the biggest or smallest item
- the page
- the drawing (all objects within the document)
- the selection of objects

Align

To align selected objects, click one of the *align* icons:

- Align right sides of objects to left side of anchor
- Align left sides
- Center on vertical axis
- Align right sides
- Align left sides of objects to right side of anchor
- Align baseline anchors of text vertically
- Align bottoms ob objects to top of anchor
- Align tops

- Center on horizontal axis
- Align bottoms
- Align tops of objects to bottom of anchor
- Align baseline anchors of texts horizontally

Distribute

To distribute selected objects, click one of the *distribute* icons:

- Distribute left sides equidistanly
- Distribute centers equidistanly horizontally
- Distribute right sides equidistanly
- Make horizontal gaps between objects equal
- Distribute baseline anchors of text horizontally
- Distribute tops euidistanly
- Distribute centers equidistanly vertically
- Distribute bottoms equidistanly
- Make vertical gaps between objects equal
- Distribute baseline anchors of text vertically
- Randomize centers in both dimensions
- Unclump objects ; try to equalize edge-to-edge distances

Remove overlaps

You can also define:

- Minimum horizontal gaps between objects,
- Mimimum vertical gaps,
- Move objects as little as possible so that their bounding boxes do not overlap

Connector network layout

If you are using the connector tool and need to distribute the connected objects, use this button to rearrange the selected connectors.

Nodes

There are 4 buttons to organize nodes in path. Select the nodes with Node Tool and click one of the *node align* buttons:

- Align selected nodes horizontally
- Align selected nodes vertically
- Distribute selected nodes horizontally
- Distribute selected nodes vertically

Layers

Layers are a type of object group within an **SVG** document. As the name indicates, they are like stacked slices of the image which can be stacked, arranged, added and removed. In addition, layers can be locked and made invisible to ease editing of objects within other layers of the document.

Layers can be very useful for several purposes:

- **Arranging** sets of **objects** by **z-order** (or **stacking**);
- **Grouping** objects which should be made visible/invisible or locked/unlocked together;
- Setting background layers which are visible but not selectable;
- etc (limited only by your imagination)

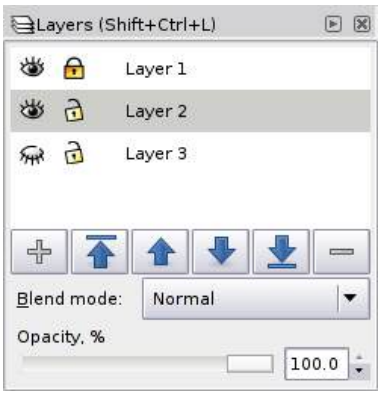
The Layers Widget on the Status Bar

The quickest and easiest way to work with layers is to use the layers **widget** on the **statusbar**.

Once you have created multiple layers in your drawing, you can select a layer easily from the combo box. You can also **hide / unhide** or **lock / unlock** the current layer from this widget.

The Layers Dialog

The *Layers* dialog controls all layer functions - adding new layers, **renaming**, **deleting**, **arranging**, **toggling** visibility and **edit-locking,** and setting **Blend** mode and **opacity**.



Blend Mode and Opacity

You can choose to apply a Blend mode to layers, which makes all objects in a layer blend together as if they were one object - i.e., the two objects within the same layer will not blend with each other when layer blend modes are used. The available modes are the common **Multiply**, **Screen**, **Darken**, and **Lighten**.

Opacity for an entire layer operates in a similar manner - objects within the layer have their opacity set as if they were a single object, so that one will not show through the others. Only objects from layers beneath will show through the affected layers.

How to Use

To open the *Layers* dialog, press **Ctrl+Shft+L** or select *Layer>Layers*.

Adding Layers

New documents created by **Inkscape** using the default template have only one layer. It is simple to create new layers which can be used right away.

To create a new layer, click "Create a new layer" on the *Layers* dialog.

Note : Although it is possible to have the same name for multiple layers (since each is given a different ID automatically), this is not the SVG standard. It is good practice to name each layer differently.

Renaming Layer

To rename a layer, **double-click** on the layer name in the *Layers* dialog or **right-click** the layer and select Rename Layer.

Selecting Layers

To select a layer (in order to make it active), simply **click** on the layer once in the *Layers* dialog or select it from the layers widget on the status bar.

Moving a Selection Between Layers

To move the current selection to the previous or next layer press **Shift+PgUp** or **Shift+PgDn** .

Moving Layers

To raise a layer one level press **Shift+Ctrl+PgUp** or click the "raise the current layer" arrow in the Layers dialog.

To lower a layer one level press **Ctrl** or click the "lower the current layer" arrow in the *Layers*

To raise a layer to the top press **Shift+Ctrl+Home** or click the "raise the current layer to the top" arrow in the *Layers* dialog.

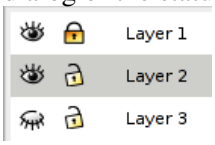
To lower a layer to the bottom press **Shift+Ctrl+End** or click the "lower the curent layer to the bottom" arrow in the *Layers* dialog.

Deleting Layers

To delete a layer, it must be selected. Click the "delete the current layer" button in the *Layers* dialog.

Locking Layers

To lock or unlock layers, click on the lock icon to the left of the layer name in the *Layers* dialog or the status bar. The icon changes to reflect the current edit-lock status.

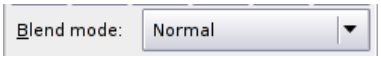


Hiding Layers

To make a layer invisible or visible, click on the eye icon to the left of the layer name in the *Layers* dialog or the status bar. The icon changes to reflect the current visibility status.

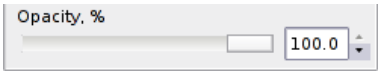
Blend Mode

To change the blend mode, click on the blend mode dropdown in the *Layers* dialog and select a new mode.



Opacity

To change the opacity of a layer, move the opacity slider in the Layers dialog to the right or left.



Grouping

Sometimes several **objects** need to be manipulated together in a **group** rather than separately. For example, if you want to keep the position of several objects relative to each other but still be able to grab and move those objects around the **canvas** quickly and easily. You might also want to adjust one setting for several objects at the same time, such as **fill** or **transparency**.

A group is treated as a *single* object, and for most operations it can be manipulated as such. For example, groups can be "stacked", meaning that you can create groups made of groups or groups other objects. (This can be helpful, but it can also be very confusing if it becomes complicated.)

Groups cannot, however, be created between groups; in other words, you cannot select one object from two groups and create a new group without destroying or modifying the original groups.

How to Use

Creating and Releasing Groups

Groups are created by selecting multiple objects and pressing **Ctrl+G or** clicking the *Group* button on the Commands Bar.

Groups are release by selecting the group and pressing **Shift+Ctrl+G** or clicking the *Ungroup* button on the Commands Bar.

Working with Objects in a Group

Individual objects within a group can be manipulated as normal by "entering" the group.

To enter a group, **double-click** the group. You may then select any object in the group individually to work with it.

Selecting Objects within a Group (from outside the group)

Objects may be selected from within groups for manipulation without entering or releasing the groups. This is very useful if you need to quickly change only one member of a group, yet leave the group intact.

To select an object within a group without entering that group, press **Ctrl** and **left-click** the object.

To select multiple objects within groups (they can exist withing various groups), press **Shft+Ctrl** and **left-click** the objects.

Adding Objects to a Group

In order to add objects to a group, you can use several methods, each having its own benefits and drawbacks:

1. Select the group, ungroup, add the object to the selection, then regroup.
2. Enter the group and draw a new object.
3. Copy or cut the object from the canvas, enter the group, paste the object.

Introduction to Styling

Objects in **SVG** can have various types of **styles** applied to them, and **Inkscape** offers multiple approaches to applying these styles. Unfortunately, this makes the discussion of styling objects fairly complex. This chapter aims to simplify the discussion as much as possible while revealing the possibilities of Inkscape and SVG drawings. Accordingly, this introduction will focus on explaining the types of object styling and a little bit about alternative ways of applying styles.

Fill

Fill is the visual content within the **path** borders of an SVG object. Each SVG object has some kind of fill (except diagram connectors).
Note : Even **bezier** and **freehand** lines have fill, although by default it is set to transparent.

There are three basic types of fill:

1. **Solid Fills** - invariant color fills the object
2. **Gradient Fills** - are a combination of two or more colors (or transparencies). A gradient begins with the first color and blends smoothly into the next, etc. Gradients can either be **linear** or circular.
3. **Pattern Fills** - are comprised of repeating images which can be selected from a preset or defined by the user. They can be either paths and **shapes** or **bitmaps**.

Stroke Paint

Stroke Paint is the visual content following the path borders of an SVG object.
Each SVG object can have a stroke set or unset. Stroke Paint is basically the same as fill in that it can consist of solid colors, gradients or patterns. **Stroke Width** can be defined as well.

Stroke Width

Stroke Width is the thickness setting of an object's Stroke Paint.
The center of an object's stroke follows the path of the object. For example, if the stroke is 10 pixels wide, the stroke will be drawn 5 pixels in opposite directions on either side of the path.

An interesting point: when stroke paint has partial transparency, the object's fill will show through the inner half of the stroke paint but not the outer half, since the path stops at the center of the stroke and bounds the object's fill. This can either make for interesting drawing possibilities or frustrated artists. This is by SVG specification design.

Application of Styles

Here is a description of methods of applying styles.

Copy Styles from Other Objects

If you want to copy styles from one object to another, you can do so easily by copying the object as normal (press **Ctrl+C**) and then paste the style by pressing **Shft+Ctrl+V**.

Copy Dimensions from Other Objects

There are six options for copying dimensions from one object to a new object. They are found in *Edit > Paste Size*

Paste Size - paste both the copied *height and width* to the selected object(s) so that the whole selection has the same dimensions as the original

Paste Width - paste only the copied *width* to the selected object(s) so that the whole selection has the same width as the original

Paste Height - paste only the copied *height* to the selected object(s) so that the whole selection has the same height as the original

Paste Size Separately - paste the copied *height and width* to *each selected object individually* (when multiples are selected) so that each separate object within the selection has the same height and width as the original

Paste Width Separately - paste the copied *width* to *each selected object individually* (when multiples are selected) so that each separate object within the selection has the same width as the original

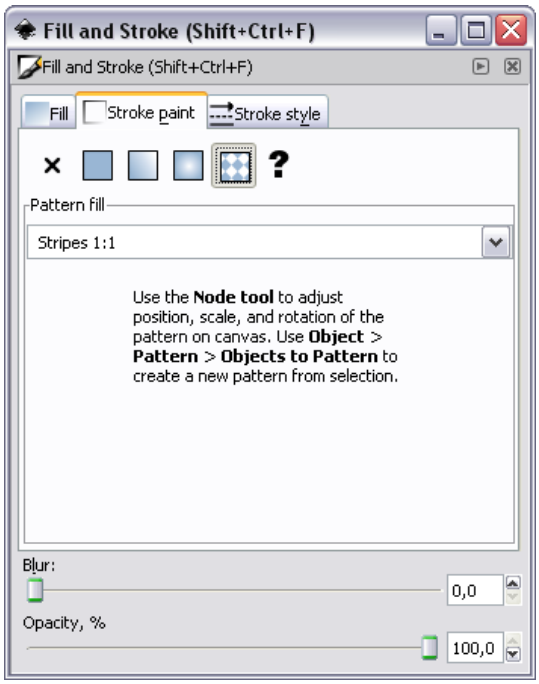
Paste Height Separately - paste the copied *height* to *each selected object individually* (when multiples are selected) so that each separate object within the selection has the same height as the original

Pattern Fill

An object can be filled with a pattern. You can create your own pattern or use an existing (pre-set) pattern.

How to use

Select the object with the Selector Tool and open the Fill and Stroke dialog. Go to the Fill or Stroke tab and click the *pattern fill* icon.



You can choose the pattern you want from the selection box (path-based, such as stripes, dots, polka dots, checkboard, wavy, camouflage, and ermine, or bitmap-based, such as sand, clothes and old paint).

Object to pattern

To create your own pattern, select the object from which the pattern will be created. Then select Object > Pattern > Object to Pattern or pressAlt+I. Your object will dissapear from the canvas and appear on the list of patterns.

To do the reverse, select Object > Pattern > Pattern to Object or press Shift+Alt+I.

If the object filled with pattern is transformed, the pattern will be also. If you don't want the pattern to be transformed, uncheck "Transform patterns" from the Transforms tab in the Preferences dialog.

Stroke Style

Object **strokes** or **outlines** can have various styles applied to them from the Stroke Style panel on the Fill and Stroke dialog. (**Stroke paint** is applied from the Stroke Paint dialog and is not covered in this chapter.)

How to Use

Activate the Fill and Stroke dialog by pressing Shift+Ctrl+F or selecting *Object* > *Fill and Stroke*.

A brief list of stroke style options follows:

- **Width** controls how wide the stroke is and your desired measurement unit type.
- **Join** controls the type of joint - **miter** (angled corners), **round** (rounded corners) or **bevel** (flat corners).
- **Miter limit** sets the maximum length of the miter.
- **Cap** controls the type of cap that is on the end of an open path - **butt cap** (flat), **round cap** (half-circle) or **end cap** (flat but extended).
- **Dashes** controls the dash pattern and its offset from the primary node.
- **Start Markers**, **Mid Markers** and **End Markers** set a marker selection from the dropdown (there is a pre-set selection of arrow-heads, etc). Mid Markers places only one marker at the middle of the path.

Creating New Markers

Create your object on the canvas, select it and then select Object > Objects to marker. Your marker will appear in the selection box for markers.

If you are putting markers on a path and want to match them to the color of the path, select Effects > Modify Path > Color Markers.

Color Swatches

Introduction

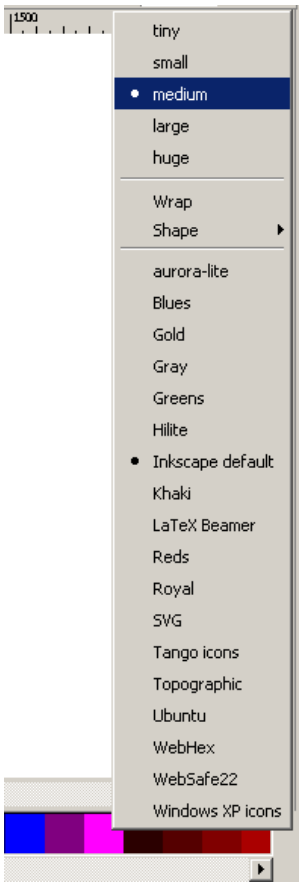
The **swatch palette** allows you to easily apply colors to objects or paths on your canvas. Click a color in the palette to change the fill color of the current selection and press **Shift + Click** to change the stroke color of the current selection. Alternatively, you can right click a color in the palette to revel an option box to set the fill or the stroke.

Mousing over a color will reveal the specific color's given name (if any).

How to Use

Customize the Swatch Palette

Clicking the arrow to the right of the color swatch palette reveals different options for customizing how it is displayed.



Swatch Size

The first several options (Tiny, Small, Medium, Large, Huge) determine the height of the palette and the amount of screen real estate that it takes up.

Small Option

Huge Option

Wrap

If you have the swatch size set to tiny, selecting â wrapâ from the options will break the palette into two lines.

Shape

This allows you to customize the shape of each swatch in the palette to allow for more or less swatches to be visible.

Tall Option

Wide Option

Libraries

The last set of options is a list of all the available swatch libraries â choosing one of these will replace the current palette with the chosen palette.

Create a Custom Swatch Library

Swatch Libraries are stored in the share/palettes folder of Inkscape and are .gpl files that can be shared with The Gimp. Each of the files is a simple text file that follows a standard format:

```
GIMP Palette
Name: aurora-lite
Columns: 5
# The Best Swatch Ever
207 171 213 Purple
77 129 113 Dark Green
144 205 187 Lite Green
154 138 93 Lite Brown
129 117 Dark Brown
```

To create your own palette, create a new file in that directory and populate the file in that same format.

GIMP Palette â This needs to stay as is for it to be recognized by Inkscape

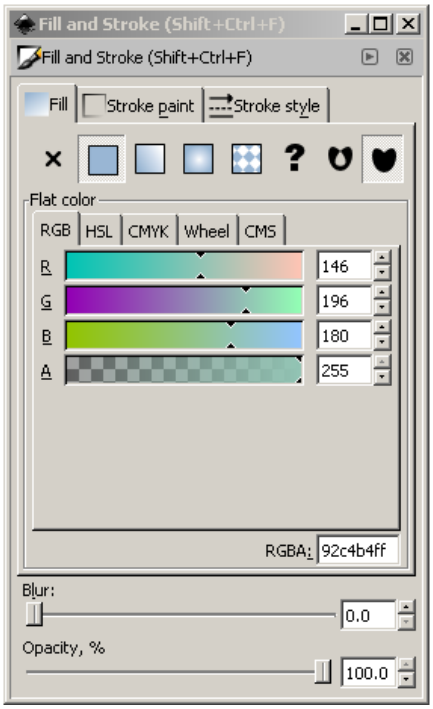
Name: Any Name â This is a custom name for the palette that will appear in the options when selecting the palettes

Columns: 5 â The number of colors in the palette

#Add your comment here - This is for adding in an comment

207 171 213 Purple - After the heading of the file, each color is listed with its RGB values followed by a name for the color.

The RGB Color can be pulled from the Fill and Stroke palette:



Live Path Effects

Live path effects (not to be confused with **Extension Effects** or **SVG Filters**) are a new way to **non-destructively modify path and shape objects**. Path Effects affect the path data of an object but not its **style**. The original path is preserved and can be edited directly on **canvas**, and the path effect applied to it will be updated live.

In the version 0.46 several path effects that are analogous to the corresponding extension effects (such as **Path along Path effect** and **Pattern along Path** that replaces the extension of the same name) are included. The most important advantage of path effects is that they are, indeed, live â you can still edit the original path and the effect will update in real time (unlike the extension effects which were one-time one-way transformations).

Details about operation

The following schematic tries to explain how LPE work.

```
original style  ------------>  output style
original path   -->  LPE  -->  output path
                       ^
                       |
                  parameters
```

The original style and path are from the path that the effect is applied on. The output is what is visible on screen. What is very important to notice is that **output style equals original style**.

The parameters can be paths, numbers, points, text, in principle anything.

Applying effects

Path effects are applied through the Path Effects dialog. This is opened from the Path menu, or by pressing **Ctrl +Shift +7**. This dialog is also used for controlling the effect's parameters and for removing effects.

When a path with a path effect applied is selected, the **statusbar** description gives details, for example "**Path (4 nodes, path effect)**".

There is a special **Paste Path Effect** command **Ctrl+ 7** that can be used to copy effects from one path to another.

Editing effect parameters

When switching to the **Node Edi**t tool **F2**, the **original path** can be edited. The original path is shown as a red helper path. Normal path operations, like **simplify**, still work.

Some parameters of these effects can be edited on-canvas. For example, path parameters can be node-edited, by pressing the **edit on-canvas** button in the **Path Effects** dialog. Press **7** to cycle through the different on-canvas editable parameters. This way, one can edit the parameters without opening the Path Effects dialog. The statusbar tells the name of the parameters that is currently being shown.

Available effects

Currently available live path effects are:

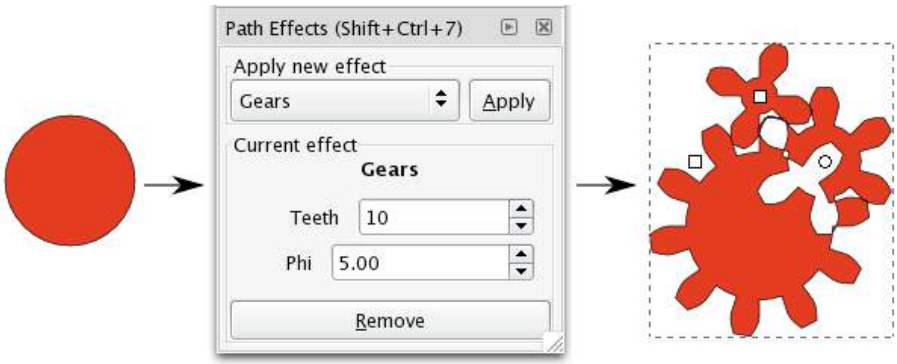
- Bend
- Pattern along path
- Stitch subpaths
- Gears

Development of new effects

One of the goals of the **Summer of Code** project was to make it easy to create new effects. There is a framework that greatly simplifies effect implementation; very little code is needed to get the effect hooked into **Inkscape**. This leaves valuable time for the actual effect to be implemented. See the http://wiki.inkscape.org/wiki/index.php/MakingLivePathEffects wiki page for an explanation of how to get started with your own effect!

Gears

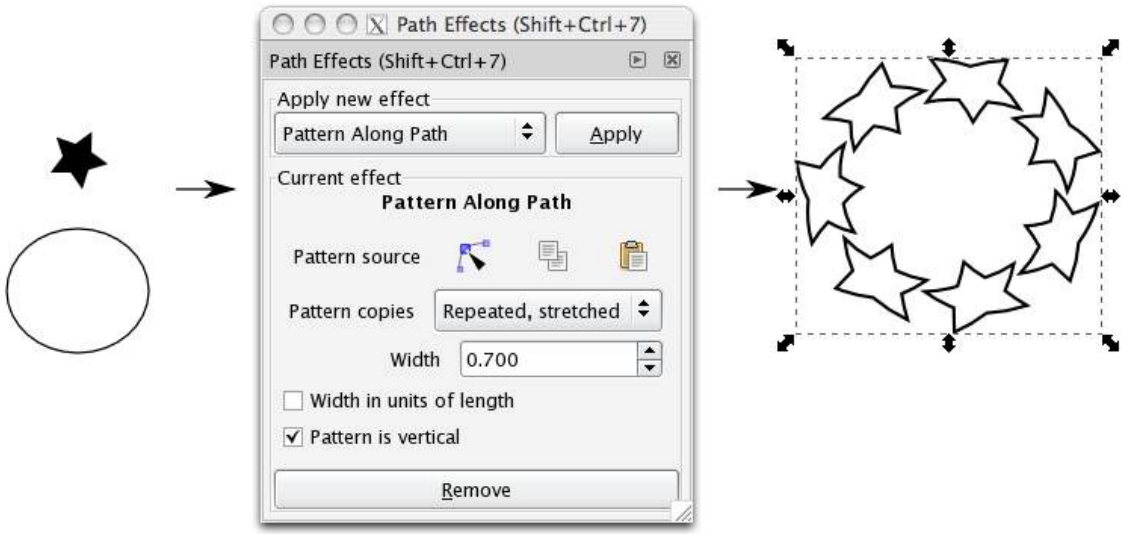
The **Gears** effect is a toy effect. It generates a chain of interconnected gears from the path that has the effect applied to it. The nodes of the path define the centers of the gears. The first 3 nodes are special; the first defines the start angle of the chain, the second defines the center of the first gear and the third knot specifies the radius of the first gear. That is, to create a chain of 2 gears, you will need a path with 4 nodes; for 3 gears, 5 nodes, and so on.

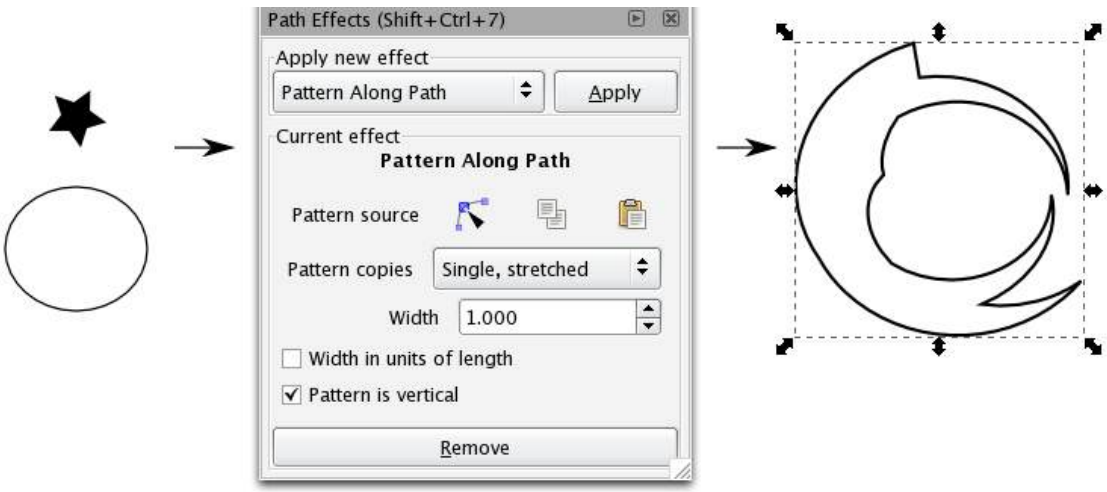


Pattern along path

The **Pattern along Path** effect can curve a path along another path. When this effect is applied to path A (called *skeleton*), another path B (called *pattern*) can then be passed as a parameter. The result is that path B is bent along path A. With the node edit tool, path A can be changed **on-canvas** and the result is **updated live**. This provides a direct equivalent of "vector brushes" or "skeletal strokes" features in other vector editors.

In the effect's control panel in the Path Effects dialog, you can select how many copies of the pattern are attached (either **single** or **repeated**) and whether the pattern is **stretched** to fill the skeleton path. You can also choose the pattern for the selected skeleton [either directly or] by **pasting** it from clipboard (that is, you select and copy to the clipboard the pattern, then select the skeleton, apply the Path along path effect, and paste the pattern). The **Scale width** parameter allows you to change the width of the pattern applied to the path.





Introduction to SVG Filter Effects

Basics

SVG filters are different from e.g. GIMP filters in a sense that what counts in GIMP for a filter in SVG is a filter primitive. And a filter in SVG can contain multiple primitives. For example, an SVG filter that creates an effect of fire consistes of *Turbulence*, *Color Matrix* and *Gaussian blur* primitives. SVG filters are also non-destructive in its nature and canbe applied to both vector and bitmap objects of an SVG document.

What should be taken into consideration all the time is that SVG Filters always deal with bitmap representation of vector graphics. A filter either makes a momentary snapshot of all data below it, or instantly reapplies changes to underlying bitmap representation of an image when it changes.

Filter primitives

Inkscape supports the following SVG filter proimitives:

- The **feBlend** filter primitive gives us image blending modes, like in many image manipulation programs. These modes are screen, multiply, darken and lighten. There's a caveat, though: when blending an object against an semi-transparent background, the background will be accumulated twice, resulting in thicker objects under the bounding box of blended object. This is a limitation of current version of SVG format, not a bug in Inkscape.

- The **feColorMatrix** filter primitive applies a matrix transformation to colour of each rendered pixel. This allows for effects like turning object to grayscale, modifying colour saturation and changing colour hue.

- The **feComposite** filter primitive composites two images using one of the Porter-Duff blending modes (described in paper Compositing Digital Images by T. Porter and T. Duff, published in SIGGRAPH '84 Conference Proceedings, Association for Computing Machinery, Volume 18, Number 3, July 1984) or the aritmetic mode described in SVG standard. Porter-Duff blending modes are essentially logical operations between the images. For example, xor mode shows the areas, where either one of the objects is, but not the areas where both of the objects are. Arithmetic mode lets you specify coefficients k1-k4 for blending equation (result colour) = k1 * (first input colour) * (second input colour) + k2 * (first input colour) + k3 * (second input colour) + k4.

- The **feConvolveMatrix** lets you specify a Convolution to be applied on the image. Common effects created using convolution matrices are blur, sharpening, embossing and edge detection. There's a fairly good explanation and some example matrices at www.gamedev.net/reference/programming/features/imageproc/page2.asp. Note that while gaussian blur can be created using this filter primitive, the special gaussian blur primitive is faster and resolution-independent.

- Filter primitives **feDiffuseLighting** and **feSpecularLighting** create lighting maps for the object in input image. SVG doesn't have concept of third dimension, so these filters use alpha channel of input image as a height map: the more opaque given point in input image is, the nearer spectator it is considered to be. There exists an example for using these in Inkscape distribution, in share/examples/lighting_effects.svg.

- The **feDisplacementMap** filter primitive displaces the pixels in the first input using the second input as a displacement map, that shows from how far the pixel should come from. Classical examples are whirl and pinch effects, that can be found in most image manipulation programs and even in some screensavers, where this kind off effect is moving around screen, twisting desktop beneath it.

- The **feFlood** filter primitive fills its region with a given color and opacity. It can be used as an auxiliary tool, usualy in combination with other filter primitives, in order to facilitate some common color handling operations.
- The **feGaussianBlur** filter primitive allows natural blurring any Inkscape objects: paths, shapes, groups, text, images. Gaussian blur enables a wide range of photorealistic effects: arbitrarily shaped shades and lights, depth of field, drop shadows, glows, etc. Also, blurred objects can be used as masks for other objects to achieve the "feathered mask" effect.

  The **feImage** filter primitive allows using external images as part of filtering chain. For example, one could use external image as a displacement map for feDisplacementMap or as a height map for lighting effects. Note that while SVG standard allows using other parts of the SVG file in this filter primitive, the current Inkscape implementation only allows external images.

- The **feMerge** filter primitive composites several temporary images inside the filter primitive to a single image. It uses normal alpha compositing for this. This is equivalent to using several feBlend primitives in 'normal' mode or several feComposite primitives in 'over' -mode.

- The **feMorphology** filter primitive provides erode and dilate effects, that are common in image manipulation programs. With erode, darker and more transparent areas spread to lighter and more opaque areas, whereas with dilate lighter and more opaque areas spread to darker and more transparent areas. For single-colour objects, this basically means, erode makes the object thinner and dilate makes it thicker.

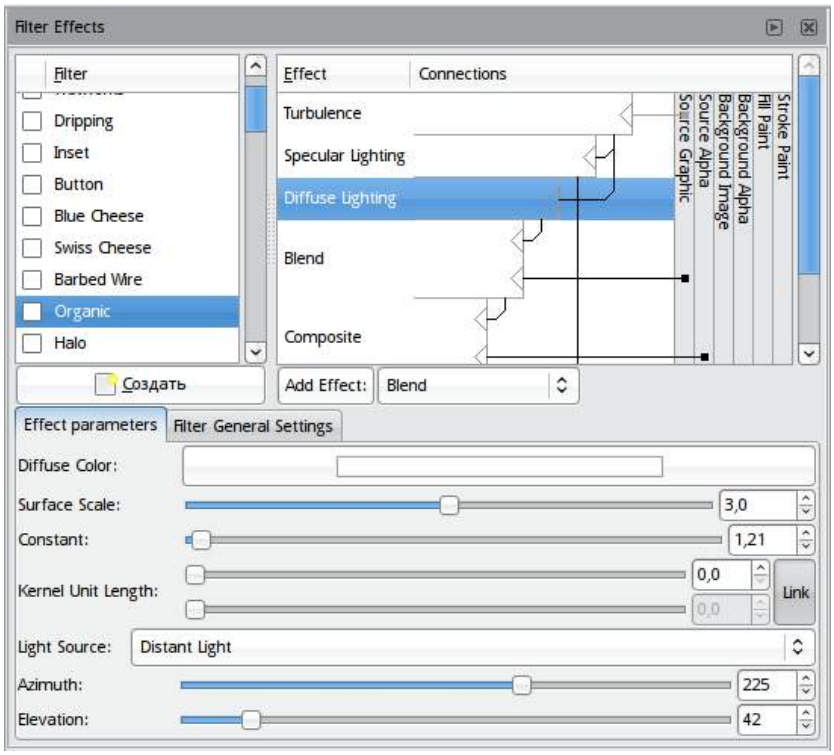
- The **feOffset** filter primitive offsets the image by an user-defined amount. For example, this is useful for drop shadows, where the shadow is in a slightly different position than the actual object.

- The **feTurbulence** filter primitive renders Perlin noise. This kind of noise is useful in simulating several nature phenomena like clouds, fire and smoke and in generating complex textures like marble or granite.

Filter UI

Parts of the dialog

Creating and modifying filter effects is done in a dedicated dialog Object>Filter Effects....



The list at the left of the dialog displays all filters currently in the document.

Managing filters

- New filters can be added with the **Add** button beneath the list
- Right-clicking a filter for the pop-up menu allows duplicating or removing a filter.
- Double-clicking a filter will apply it to all selected objects
- A black dot is placed next to whatever filter is applied to the selected objects. If more than one filter is in use by selected objects, an unfilled dot is used instead.

Managing filter primitives

The second list, at the left of the dialog, displays the **filter primitives** that are contained within the currently-selected filter.

- New primitives can be added by selecting the primitive type from the combo box beneath the list, and then pressing the **Add** button.
- Right-clicking a primitive for the pop-up menu allows duplicating or removing a primitive.
- Primitives can be rearranged by clicking and dragging any filter in the list.
- When a filter is selected, the **Settings** group at the bottom of the dialog will change to display the attributes available for that primitive. Changing a setting results in an immediate update to the document.
- The "in" and "in2" attributes for filters that support them are not shown in the **Settings** group. These input connections are displayed graphically in the list, under the **Connections** column.
    - ♦ Inputs for a particular filter are displayed as triangles. Depending on the primitive type, there may be one or two inputs (or more for Merge primitives.) Connections can be created by clicking on a triangle and dragging.
    - ♦ There are six standard input types that can be used for any primitive input; *Source Graphic*, *Source Alpha*, *Background Image*, *Background Alpha*, *Fill Paint*, and *Stroke Paint*. These are displayed vertically on the far right of the list. Click and drag from an input triangle to one of the standard inputs to connect them.

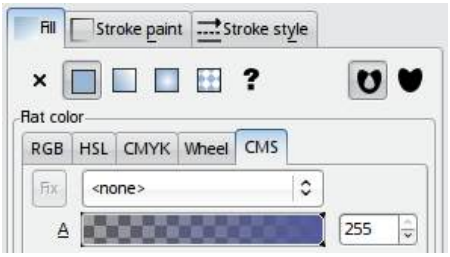
- Primitives can also be connected to other primitives by clicking an input triangle and dragging upwards to another primitive. A primitive can only be connected to one higher up the list.
- Single-clicking on an input triangle will unset it, returning it to the default. If it is on a *Merge* primitive, the input will be deleted.
- Merge inputs have an empty input at the end. Dragging a connection from this input will add a new input to the primitive.

Color Management

Calibrated SVG Color including CMYK

Inkscape supports color-managed color definitions that use a colorspace other than sRGB (for example Adobe RGB, or calibrated CMYK colors). In the SVG file, this is done using the optional "icc-color(...)" paint components as described in section 11.2 "Specifying paint" of the SVG 1.1 specification. A fallback sRGB value will be used for non color managed workflows. This allows using of calibrated color spaces, including using CMYK values that are preserved across applications.

The CMS color selector tab allows these colors to be edited.



Display Adjustment

Color Management tab in Inkscape Preferences dialog provides options for enabling display adjustment:

- **Display Profile**. Here you set an ICC file for you calibrated and profile display.
- **Retrieve profile from display**. On X11-based systems (i.e. Unix and Mac OSX) use of ICC Profiles In X Specification (or XICC) can be enabled. Support for version 0.2 of this specification has been implemented. Enabling this option by choosing to retrieve profiles from the display will switch Inkscape to using profiles attached to screens at runtime. These allow display adjustment to be changed on the fly, and to be set/cleared per-display. This is especially helpful for a multi-display configuration.

  When XICC support is enabled, windows will adjust to the proper profile as they are moved across monitors. Also, as the windows are moved onto monitors with no profile attached, the adjustment toggle will become disabled. When the windows are moved onto screens that do have profiles, the toggle will become enabled.

- **Display Rendering Intent**. You can choose between **Perceptual**, **Relative Colorimetric**, **Saturation** and **Absolute Colorimetric**. Perceptual intent prevents from gamut clipping by preserving relationships between color. Saturation intent preserves saturation and best suits for illustrations with indexed color palette like logos. Relative colorimetric intent maps original white color to color of the paper and translates other colors accordingly. Absolute colorimetric intent tries to produce closest match, but suits mostly spot color workflows.

It has to be noted that display adjustment is enabled and disabled for each Inkscape's window. This allows for simultaneous viewing of adjusted and unadjusted views of a single document by using multiple windows. There is a toggle at the bottom-right corner of the scrollbars that allows for turning on and off display adjustment. The toggle will have a disabled state to provide visible feedback when no profile is set.

Proofing

This section in Color Management tab allows defining settings for an output device such as printer:

- **Simulate output on screen** option enables softproofing, that is â the document will look exactly as it will be outputted to printer.
- **Mark out of gamut colors**. All devices have a unique set of colors (gamut) they can reproduce, be it a display or a printer. This option enables marking the colors that cannot be reproduced in destination color space witha user defined color. By default neutral grey color is used.
- **Device profile**. This is an ICC profile for the output device (usually, a printer). For Europe the **Euroscale Uncoated v2** profile is recommended to be used by default, and for USA â **U.S. Web Coated (SWOP) v2** or **U.S. Sheet-fed Coated v2**.
- **Device Rendering Intent**. You can choose between Perceptual, Relative Colorimetric, Saturation and Absolute Colorimetric.
- **Black Point Compensation**. BPC's primary function is to map dynamic range of an image to dynamic range of an output device, so that contrast us not lost. It is usually used for publications with a lot of full color graphics like photos. Note that BPC is used only with relative colorimetric rendering intent and should be disabled for other rendering intent types.
- **Preserve Black**. Makes Inkscape do its best to provide same level of blackness in the output.

Operation Systems Support

Đ¡olor management is supported in Linux and Mac OS X only inthe current stable version of Inkscape.

Creating ICC Profiles

For creating ICC profiles with free software LProf and ArgyllCMS are recommended.

Generate Template

The only available **extension** to create a **template** right now is **Perfect-Bound Cover** which modifies the document to create a cover for perfect-bound books using US size and paper weight measurements. This extension will resize the document to include the width, height, spine width, and bleed measurements that are provided to the extensions, so it should be the first operation done before designing.

Ideally, you want to know the **PPI** of the paper stock youâ ll be using. **Average caliper size** or **point size** (which is **the caliper size times 1000**) are just as good. Your printer should be able to tell you any of these measurements. If not, youâ ll have to guess. Or find another printer. If you donâ t know the PPI, caliper, or point size of the paper your printer is using, but do know **the type of paper and its weight**, you can estimate PPI or caliper using the charts on the **Case Paper** (http://www.casepaper.com/calc_chart_caliper.htm) or the **Micro Format** (http://www.paper-paper.com/weight.html) site.

For instance, letâ s say your printer uses **20# bond paper**, a pretty standard paper weight, to print the interior pages of your 200 page book. Most likely, the caliper size of this paper is **.004**, but again, *verify this with your printer before proceeding*. Calculate the PPI by diving the number 2 by the caliper size:

2/.004 = 500

The PPI of this paper is **500**, which means that, for **every 500 pages** in your book, the spine will be **1 inch thick**.

Now, with the PPI, you can calculate the thickness of the spine, without the allowance for the cover. The size of the spine for the interior pages will be:

200/500 = .4 inches

Next, do the same for the cover. Letâ s say your cover is going to be printed on 9pt paper stock. Points to caliper is a really easy conversion â **divide the points by 1000** to get .009, then calculate:

2/.009 = 222

Since the cover is four â pagesâ (outside and inside covers), the additional inches you need to add to the spine will be:

4/222 = .018

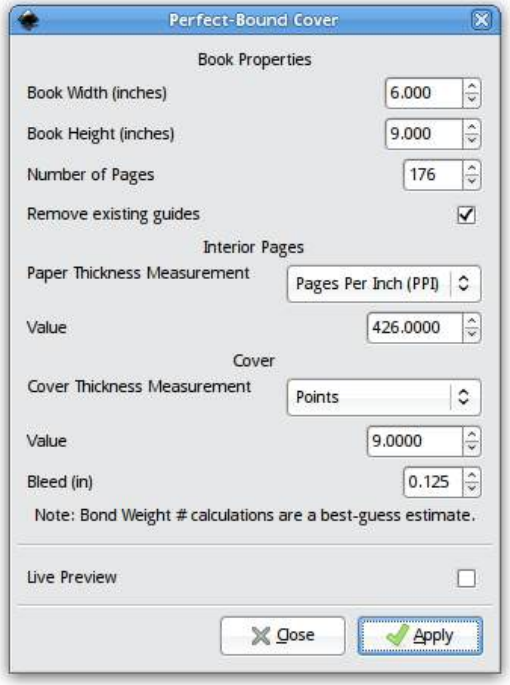
So your final spine width in inches for a 200 page book printed on 20# bond with a cover that is printed on 9pt stock is:

(200 / (2/.004)) + (4 / (2/.009)) = .418

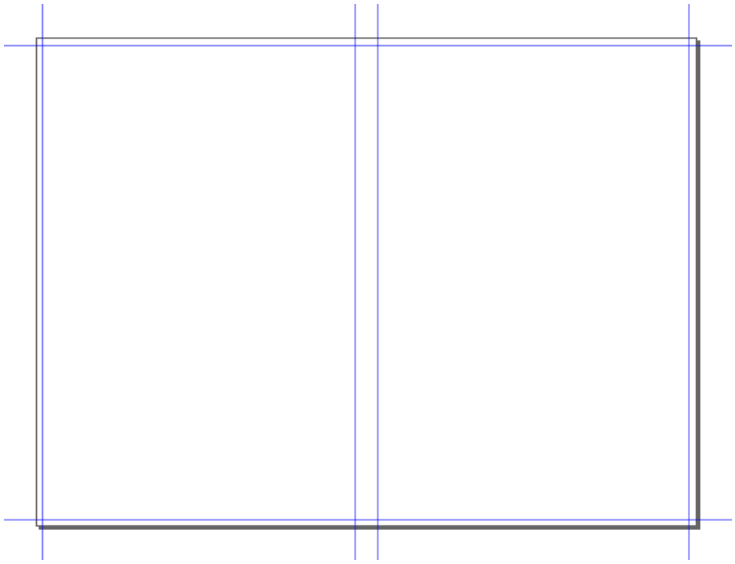
.4 + .018 = .418 inches

All this is different in metric.

Let's take an example. You have a 6â ³x9â ³ book with 176 pages to be printed on paper with a PPI of 426 and the cover â on 9pt stock. You also need 1/8â ³ bleeds. Hereâ s how you would set up the document:



And here is how the final document looks:

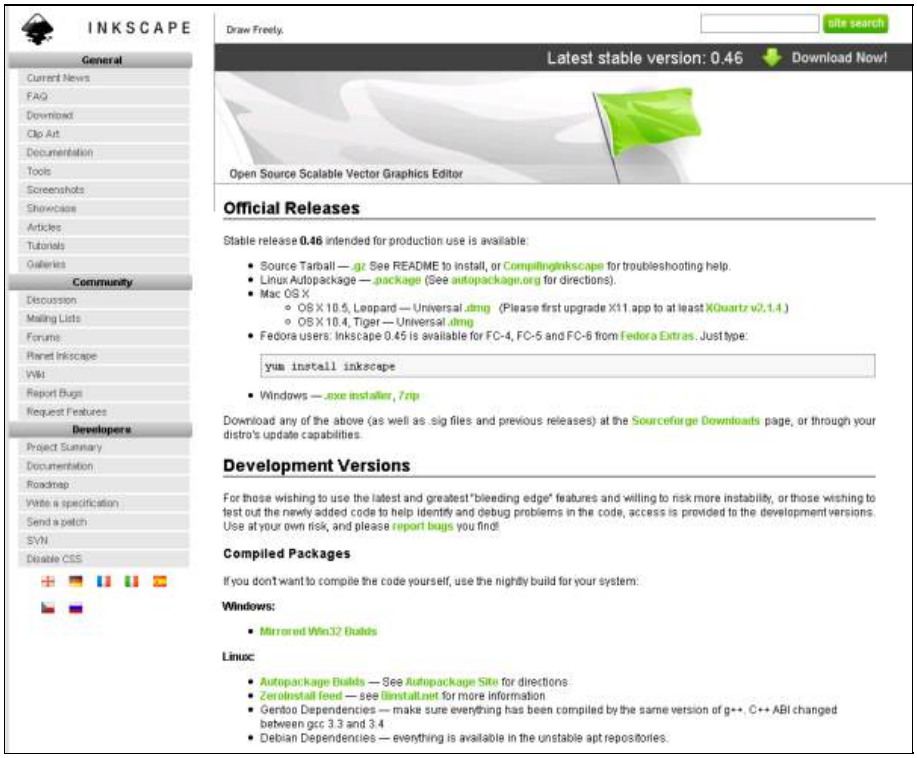


You can also use the **Live Preview** option to see what you get.

Installing Inkscape on Windows

Software name : Inkscape
Homepage : http://www.inkscape.org/
Software version used for this installation : 0.46
Operating System use for this installation : Microsoft Windows (2000)
Recommended Hardware : 200 Mhz processor (CPU) minimum

Installing Inkscape is relatively easy. You need only an internet connection and browser. First visit the download page of Inkscape : http://www.inkscape.org/download/

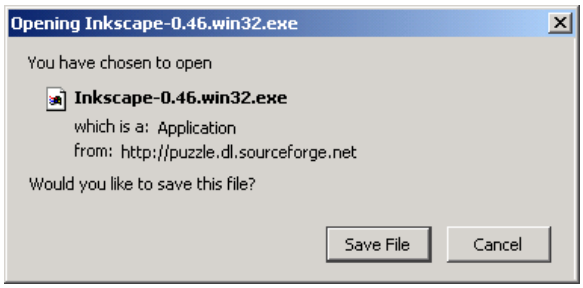


Now scroll down to the Official Releases' section to the link that says 'Windows' :

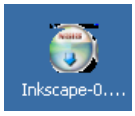
- Windows — .exe installer, 7zip

We will choose the '.exe installer' so click on this. You will be redirected to a 'SourceForge' page. This is where the installation files are hosted :

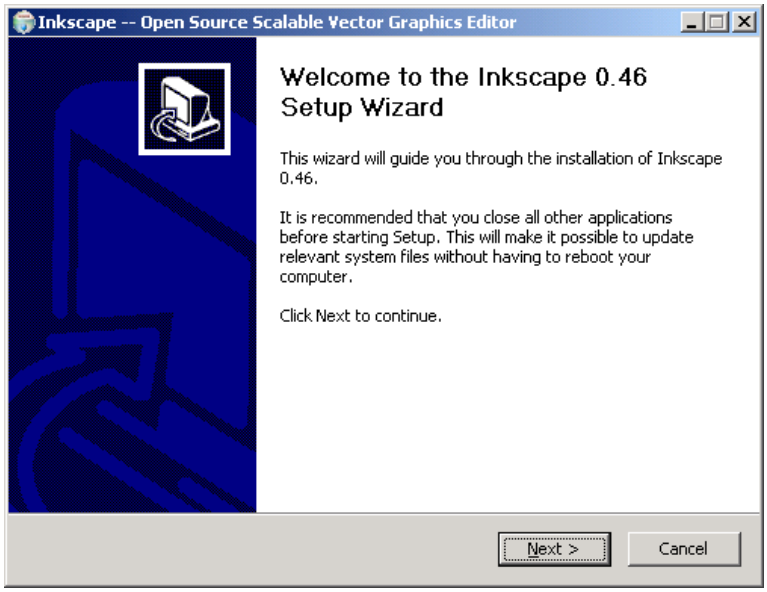
The download will begin automatically. You chould see a dialog like this (it may look different according to the browser you are using):



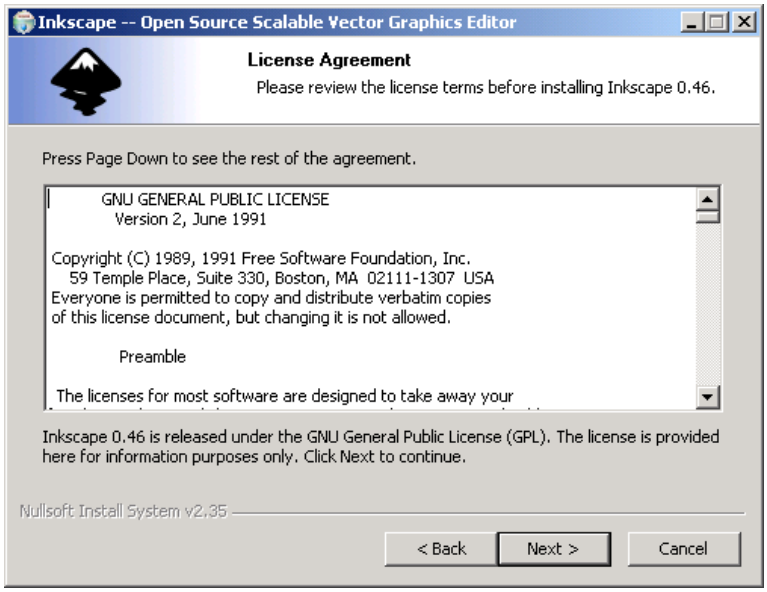
Press 'Save File'. Your installation file will start downloading. You need to know where the file is downloading to. If you know where this is then you need to find the file on your computer. You should see an icon like this :



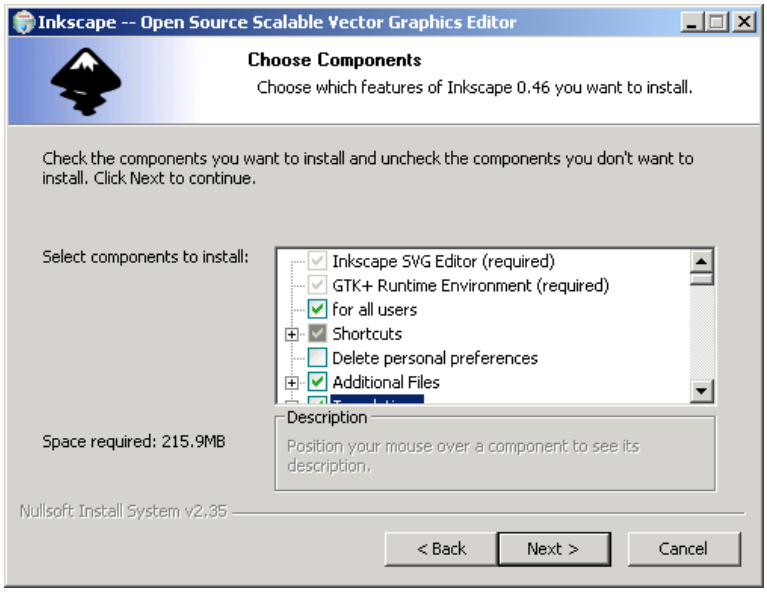
Now you need to double-click on tis icon and the installation process will begin.



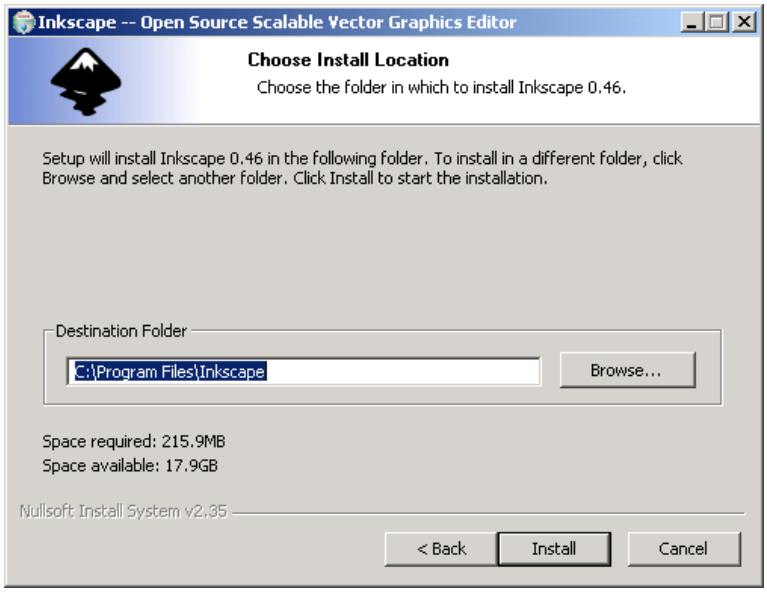
You can click through the above introduction screen.



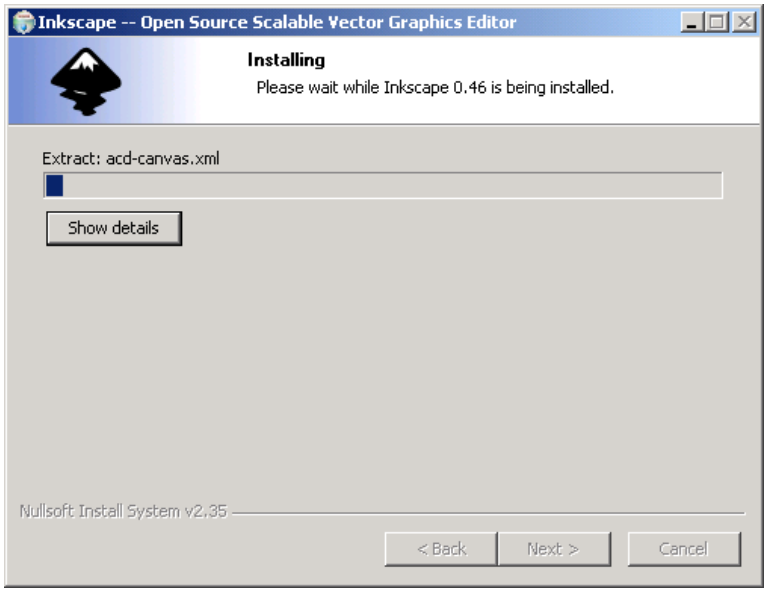
The above is just the license agreement. If you don't agree to it then press cancel but this means you will not be able to install the software. If you wish to install the software click 'Next >' :



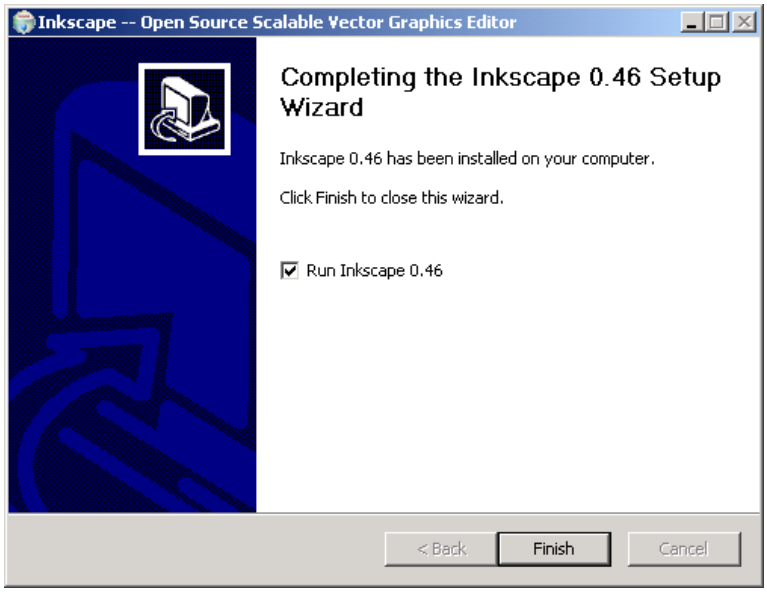
The entire installation size is listed here as 215.9MB. If this is too big then you may wish to turn off some of the boxes with green arrows. The items with green arrows are optional, the best saving would be to turn of all the Translations (saves 45MB) - however this means you will not be able to use Inkscape in any language other than english. When you finished with the options click 'Next >' :



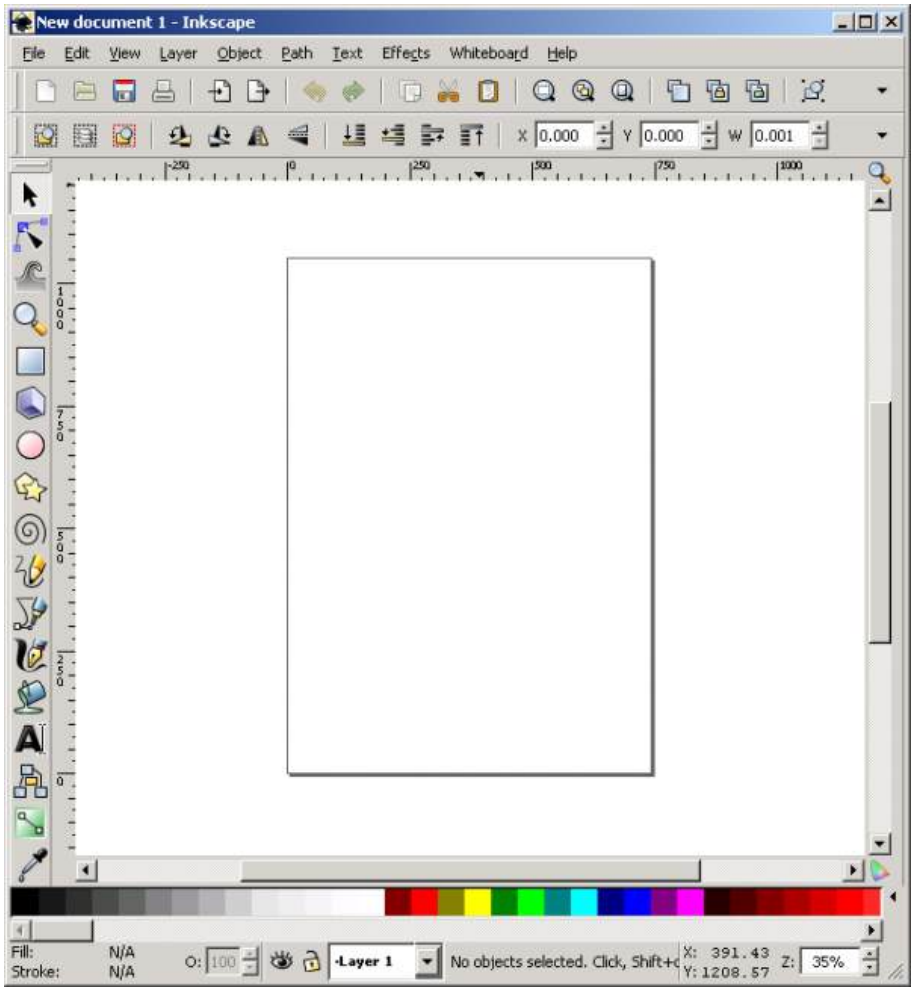
The above window just tells you where Inkscape will be installed. Unless you have good reason to change the above settings then it is better to leave it as it is and press 'Install'. Then the installation window appears and reports the progress:



When the install has finished you will see the following:



To run Inkscape just leave the 'Run Inkscape 0.46' and press 'Finish'. Inkscape then appears:


New document 1 - Inkscape
File Edit View Layer Object Path Text Effects Whiteboard Help
X 0.000 Y 0.000 W 0.001
Fill: N/A
Stroke: N/A
O: 100
Layer 1
No objects selected. Click, Shift+c
X: 391.43
Y: 1208.57
Z: 35%

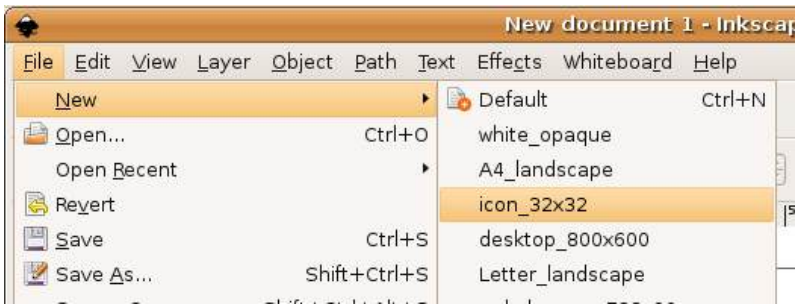
Create a Basic Icon

Inkscape is an excellent tool for creating icons. Generally speaking icons are required in different formats so that makes **SVG** an excellent format as it is a scalable graphics format. This means you can shrink or enlarge the graphic to the required size without losing any quality.

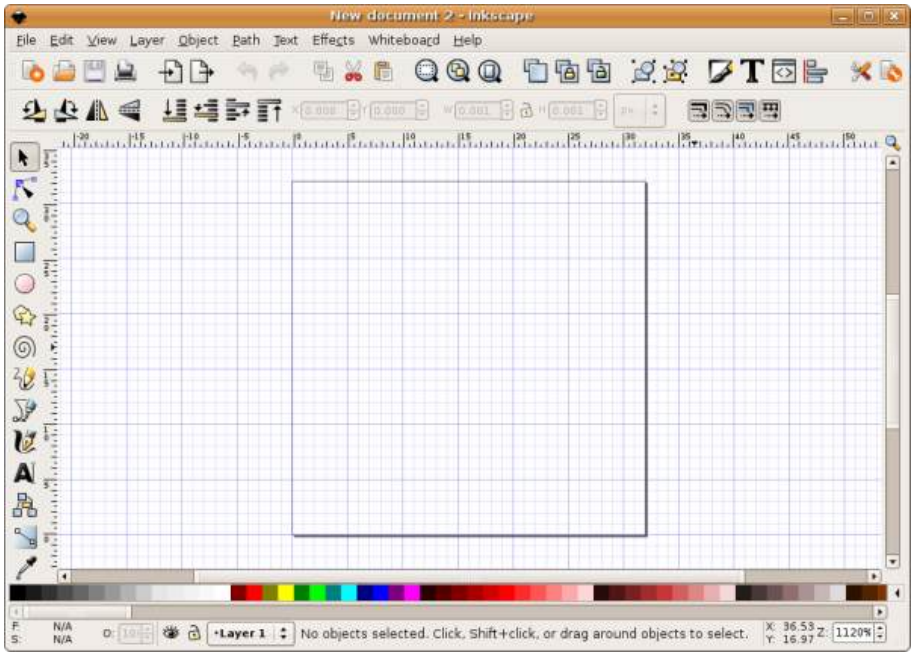
This tutorial was made with Ubuntu, but works with any operating system supported by Inkscape. In this case, only the way you open Inkscape and the general look and feel may be different.

Create New Icon File

First open Inscape and to create a new file. Click on the File menu and choose 'New' and then 'icon_32x32'.



A new blank file will then open with, in this example, a grid:



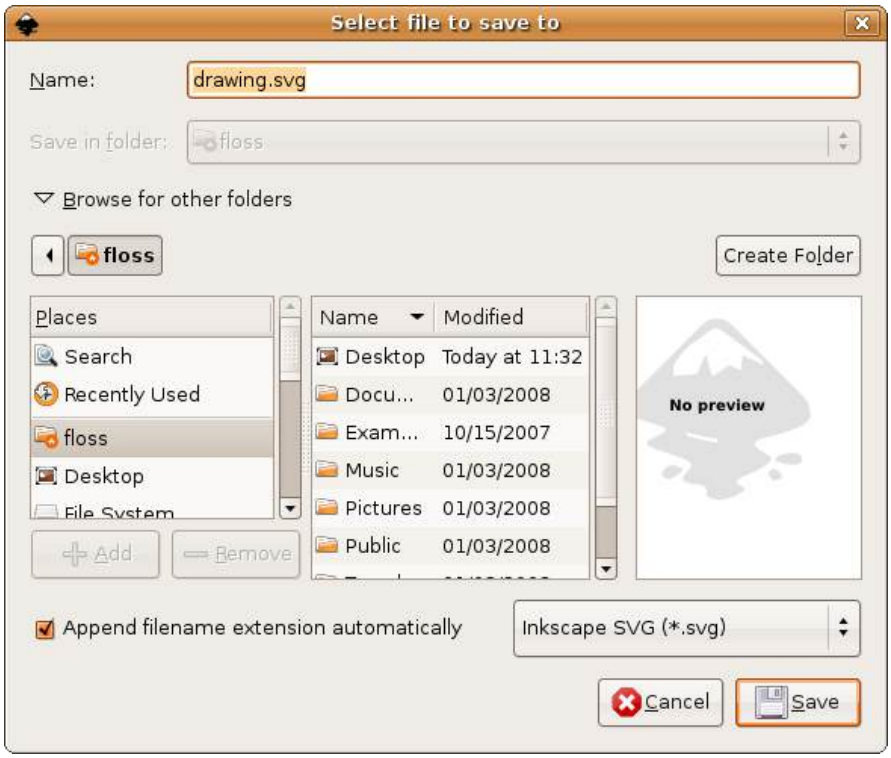
The grid is a 32x32 block grid, the same size as many of the icons you will find on your computer. The lines in this grid will not appear when the final image is exported to another type of file such as JPEG or GIF. The grid is just there to help you create the square icon within the standard 32x32 pixel dimensions. It's possible to add or remove the grid using the 'View>Grid' menu.

Save the File

Before you start work it is a good idea to first save the file. Even though the content of the file is empty it is good practice to always save the file before you start so that you can easily save the changes as you go. By

doing this you ensure yourself against losing your work if your computer crashes or turns off unintentionally. Save the file by clicking on 'File' and 'Save As...' :

This will pop open a dialog box. The style of your dialog may differ from the following :



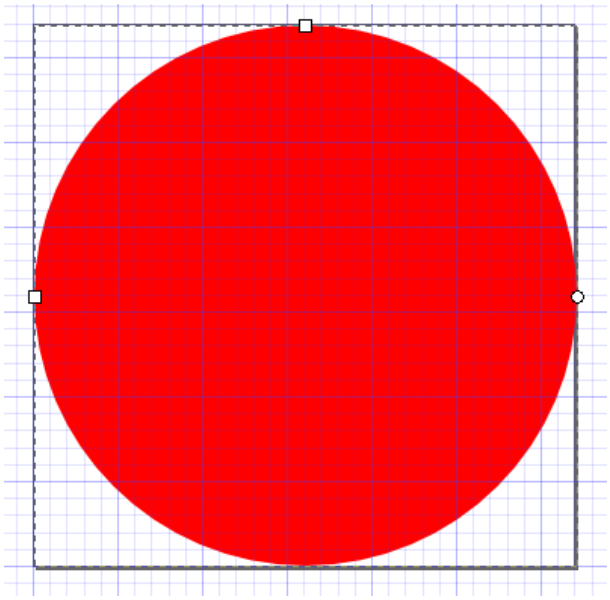
You can change the name of the file to anything to help you identify it (in the above example it is 'drawing.svg'). Make sure when you save it the file name has '.svg' at the end. You can also click through the folders on your computer using this dialog box until you find the folder where you wish to save the file. When you have the right location click 'Save'.

Start Work

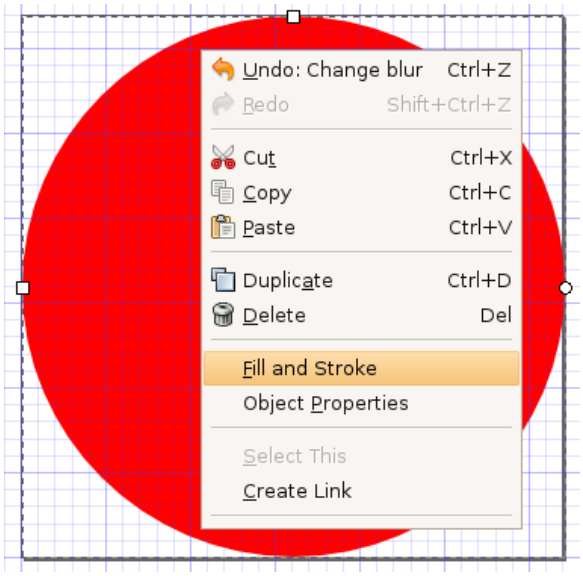
Now we will proceed to draw the icon. You can choose any of the tools for creating shapes and colors. Currently icons that look three dimensional are very popular but we will look at creating a very two dimensional icon. So lets make a simple radio icon. It will end up looking something like this:

Lets start with making the background circle. Choose the circle tool from the tool bar on the left of Inkscape:

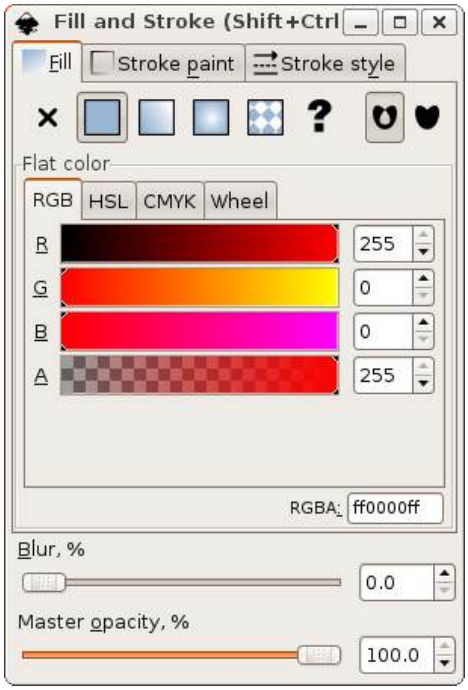
With this tool selected you need to click in one of the corners of the grid and drag to the diagonally opposite corner. It might take a bit of practice until you have this right. The end result should look something like this:



You may have a dark line around the outside of the circle. If you do we will look at how to remove this shortly. For now we will change the color of the circle to the desired color. I will choose the HTML color code '#ff7f00'. You may wish to choose another color. To change the color of the circle you must right-click on the circle and choose 'Fill and Stroke':



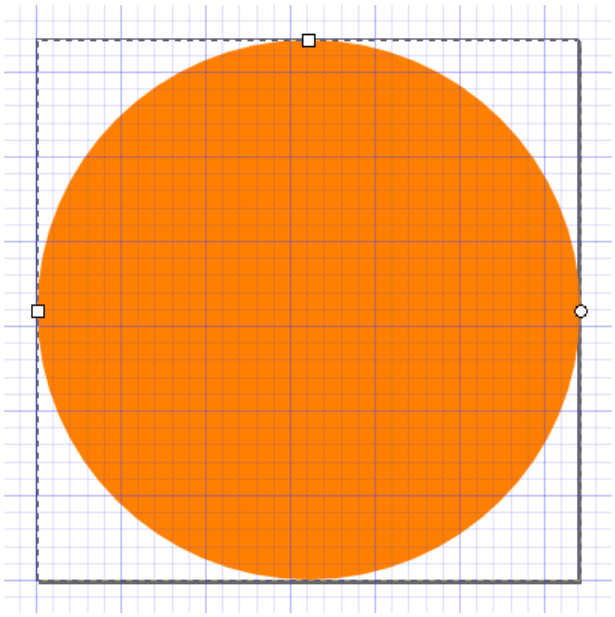
The following dialog should appear:



I know the HTML color code is the equivalent to the RGB code + 'ff'. So I will add 'ff7f00ff' to the RGBA box near the bottom :

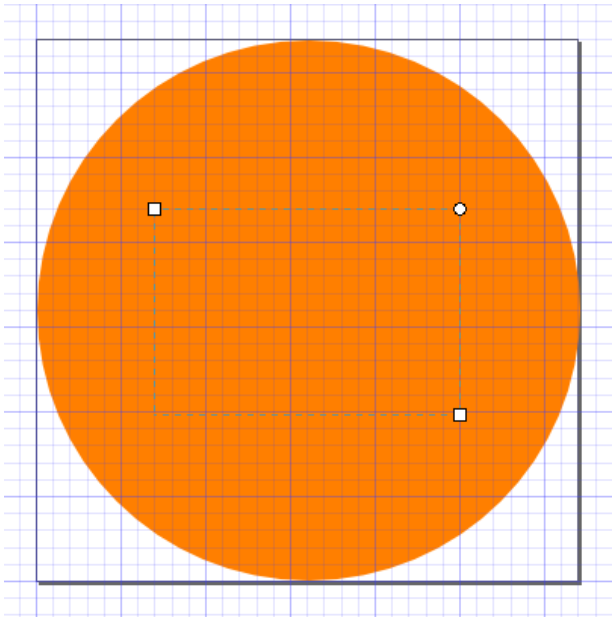
If you know your HTML color code then add it here. Otherwise you can use the RGB sliders to choose a color. As you change the values you will see the color of the circle change simultaneously. If you had a line around the outside of the circle when you created it you can now delete it by clicking on the 'Stroke style' tab and set the 'Width' box (at the top of the tab) to 0 (zero) px. When you are happy with the color of the circle just move the 'Fill and Stroke' dialog to the side (if you ahve enough room on your screen) or close it. We will use it again a little later.

My colored circle now looks like this :



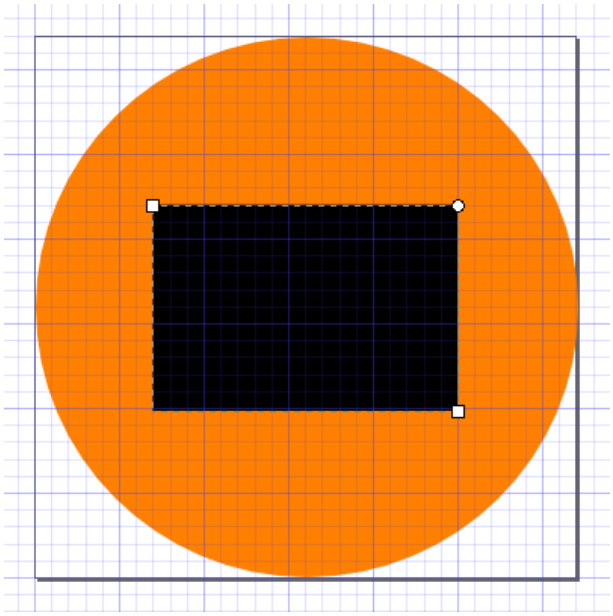
Now we wish to add the radio to the middle of our circle. We will first add a black box with rounded corners to the middle of the circle. Click on the 'Square and Rectangle' tool on the left :

Now click somewhere in the circle and drag the mouse in a diagonal towards the bottom of the page. You are now creating the square but you will not see anything happening on the screen so you have to guess the approximate distance you drag before you release the mouse. You should see something like this :



Now, return to the 'Fill and Stroke' dialog box and change the value of the RGBA box (In the 'Fill' tabe') to "000000ff" :

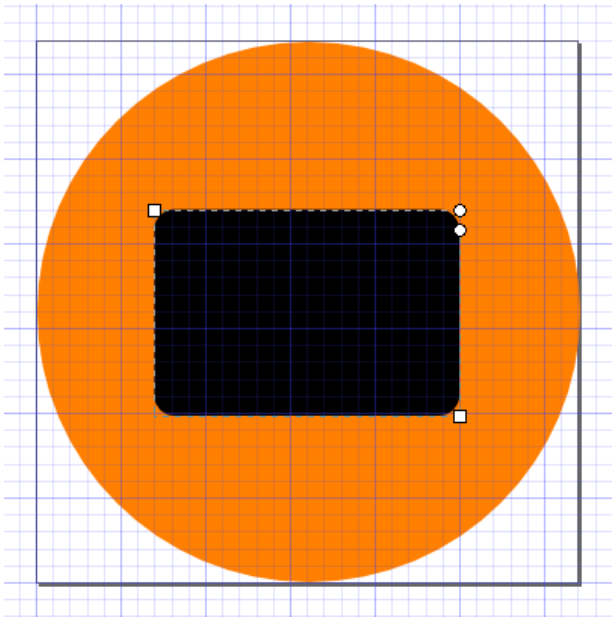
Now the square should appear black :



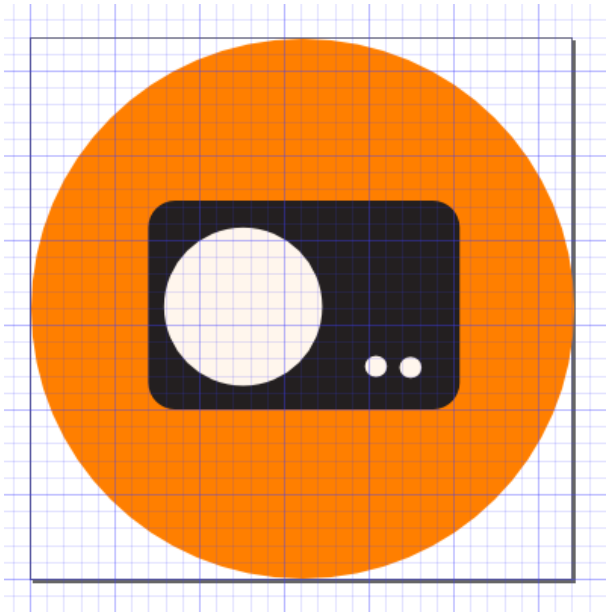
You could also have used the color sliders to choose a color.

Now we wish to make nice rounded corners on the black box. To do this mouse over the small round circle at the top right of the black box. It should go 'red' :

Click on the circle when it is red and drag it vertically down and you will all the corners of the black box change to rounded corners as you drag. When you are satisfied with the amount of 'rounding' release the mouse button :

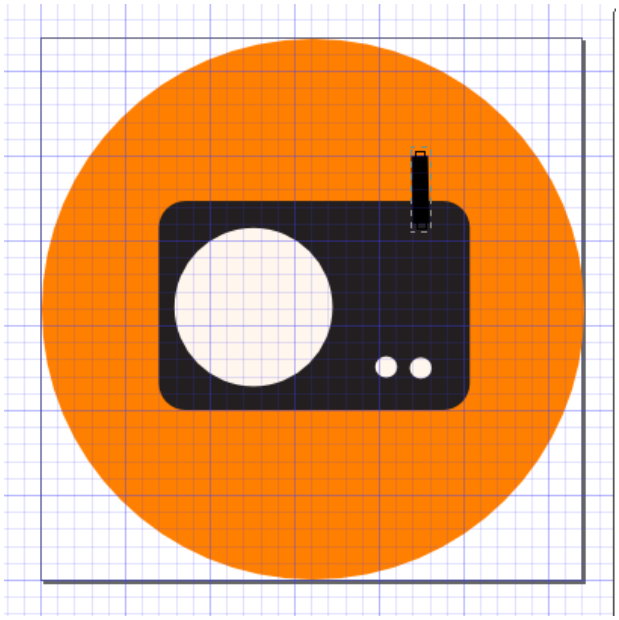


Now we need to create three white circles for the 'dial' and 'tuning buttons' of the radio. Do this using the same method as you used for creating the first large circle, except you want to try and create them in the right place in the black box.

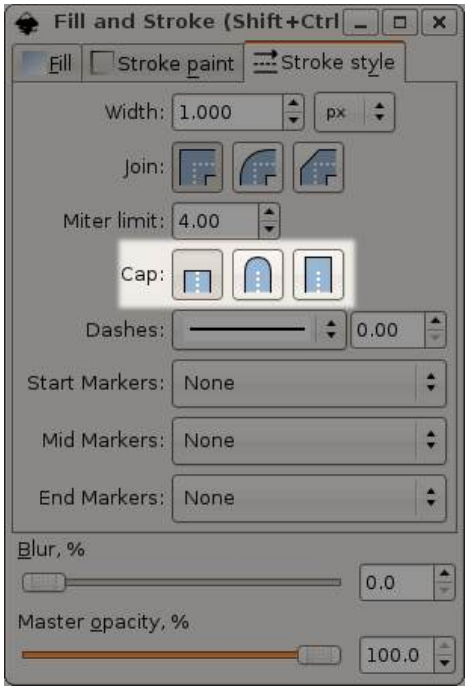


Now we want an antenna on the top right. We will use the pencil tool that you will also find on the left tool bar:

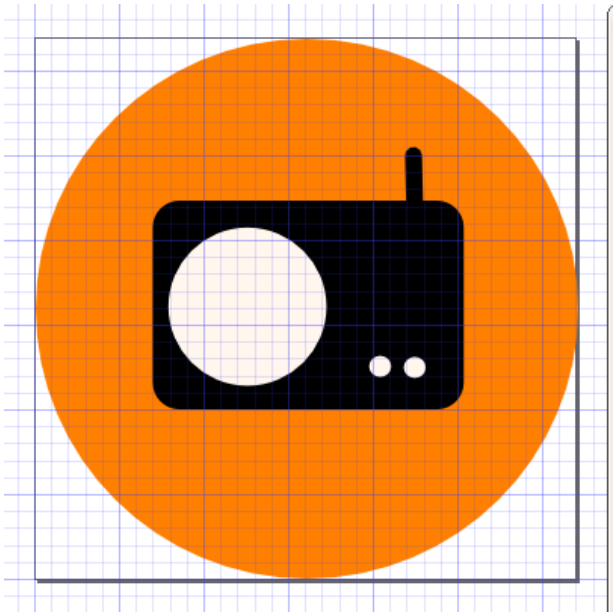
To draw a straight line click where you wish the line to start and then click again where you wish it to finish:



Now we wish to give the antenna a nice circular tip so we return to the 'Fill and Stroke' dialog and look at the 'Stroke style' tab. You will notice here a 'cap' section :



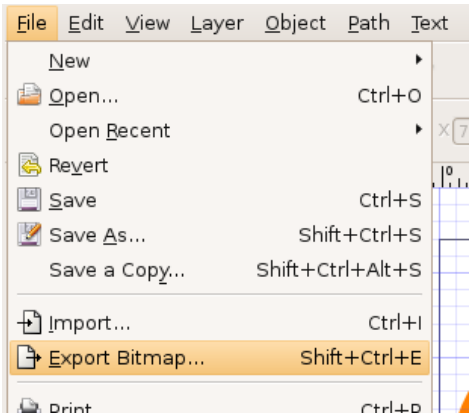
Click on the middle box in this section and your antenna will be instantly rounded :



Exporting to PNG

We will now export this to a PNG. You may wish to use another graphic file format however Inkscape will only support export to PNG. To convert the image to another format you will need to use an image software such as **GIMP**.

Choose 'File' and 'Export Bitmap...':



You will see a dialog appear like this:

If you just pressed 'Export' right now you would export a 32x32 PNG to the folder '/home/folder'. You can change any of these settings. To change the dimensions of the image click on the arrows next to the Width and Height boxes. To change the file name and the location you wish to export the file to you must click on the 'Browse' button.

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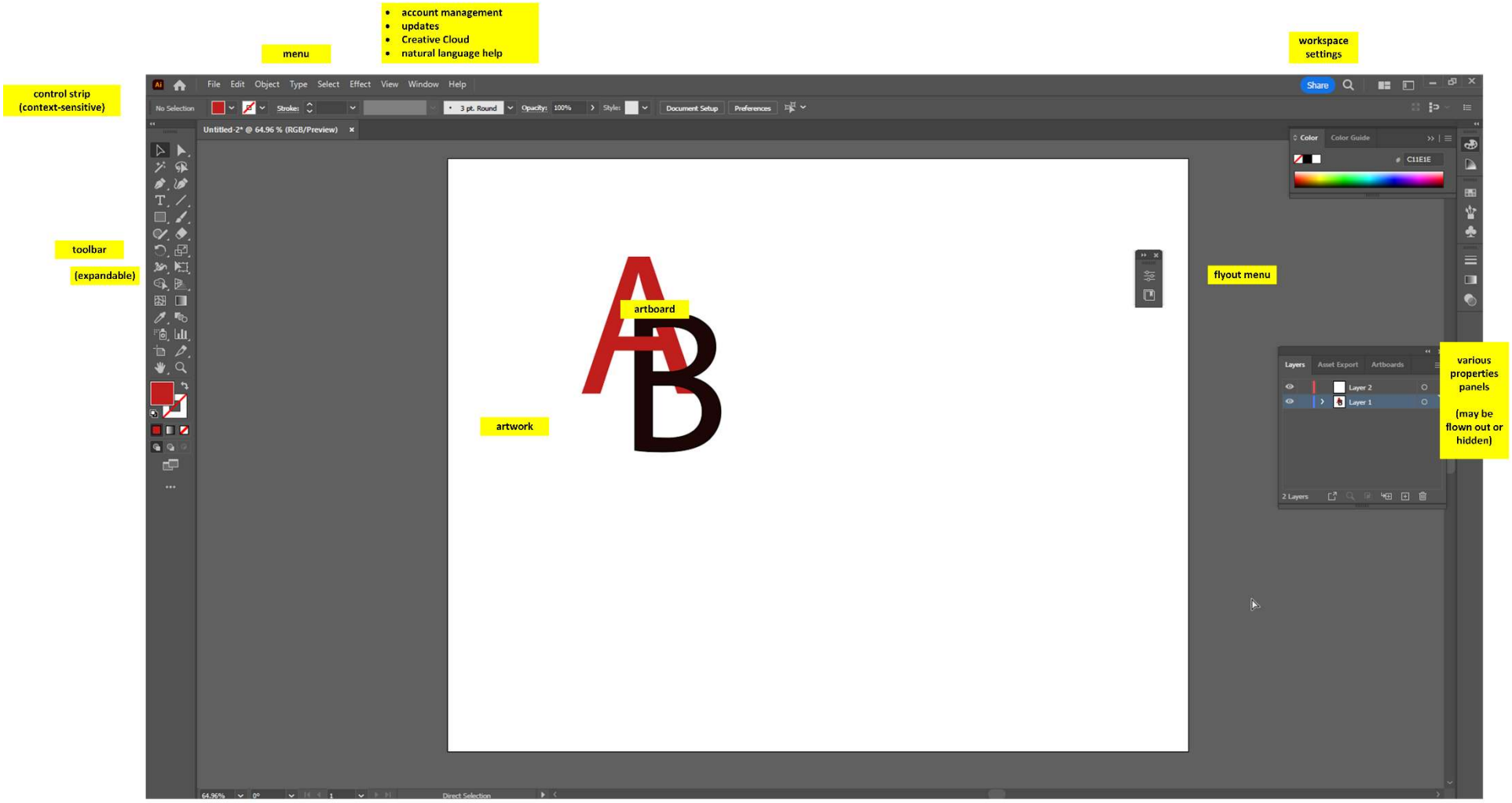
Intro to Adobe Illustrator 2023



Adobe Illustrator 2023 Graphical User Interface (GUI)

Intro to Adobe Illustrator Handout

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Downloadable Handout in Two File Formats (URLs)

These are updated just prior to (and often after) each training in many cases, so download the file just before the event (and maybe check after).

http://www.k-state.edu/ID/IntroAdobeIllustrator.pdf
http://www.k-state.edu/ID/IntroAdobeIllustrator.docx

You can do searches over the handout to see if there is some note about whatever feature you're interested in. You do not have to read this through in sequence although you can.

How to Approach the Two-Hour Training

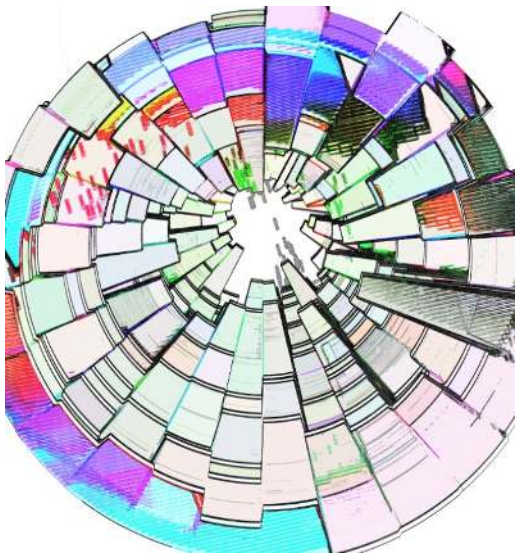
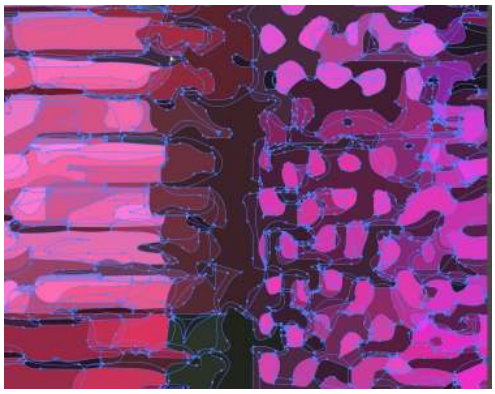
Blister Pack Lion

Two hours is a very short amount of time. This is only an introductory session which will showcase some basic aspects of a complex and powerful tool (that has been around since January 1987 and has evolved powerfully since then and has major elements of artificial intelligence / AI). The best way to learn this and other ~ tools is through experiential learning…by taking on projects and solving problems as you proceed. It helps to explore MS LinkedIn trainings (through the Manhattan Public Library)…and videos on YouTube…and other learning (based on however you like to learn).

This session will be run as a conversation, so unmute yourselves and speak up as questions arise…and / or post to the chat.

General Schedule

Hour 1		• General uses of Adobe Illustrator 2022 (logos, icons, badges; scalable designs; animations; and others) • Adobe Illustrator Graphical User Interface (GUI) • Vector visuals and scalability ; raster images and less scalability (based on spatial resolution / visual details)

		• Setting the proper type of workspace (based on your digital product) • Choosing an opening file type (and relevance of) • Setting up rulers and non-printing smart guides (with snap-to capabilities) • Born-digital seed (draw digitally), born-analog seed (for documents, diagrams, print photos, freehand drawings, and others) • Some walk-throughs of work with strokes (lines) and fills (and textures), curved lines, the width tool to enable changing stroke/line width manually (customized)
Hour 2		• Transcoding between raster – vector, vector to raster • Manual trace • Automated image trace • Various filters in Adobe Illustrator 2022 • Layers and how layers generally work • Masking (and various types of masks) • Applying gradients (linear, radial, etc.) • Or some selection of features and capabilities • Light troubleshooting

Preferences between Adobe Photoshop 2022 vs. Adobe Illustrator 2022

	Adobe Photoshop 2022	**Adobe Illustrator 2022**
Substance (but transcodable between the software programs)	Raster	Vector
Precision	Pixel, Coordinate Positioning, Colors	Stroke, Coordinate Positioning, Colors
Scalability	Limits of Pixels / Dots Per Inch	Unlimited (for all size use cases)
Drawing	Pen Tools, Paintbrush Tools, Shapes, and others	Pen, Pen for Curvature Drawing, Application of Paintbrush Types on Strokes, Shapes, Drawing with Shapes, and others
Logos, Glyphs	Suffices	More Typically
Editing	Layers, Filters, Others	Layers, Filters, Outlining View, Others
Auto Selection	By Color	By Variety of Attributes (Opacity, Color, Fill, Stroke, and Mixed Characteristics, among others)
Inputs, Raw Artworks	Raster	Vector to Maintain Editability, Multi-object Whiteboard as Better Analogy for Physical Workspaces; Vector Fonts and Font Families;
Animation		Puppet Warp, Maneuverable Objects and Characters
Printing	Pre-flighting, Printers Marks, Print Inks	Pre-flighting, Printers Marks, Print Inks, Pantone
XML		Can be transcoded to XML for integration into websites
Others		

Work Pipelines: In truth, both software programs may be part of a work pipeline, with handoffs back and forth. The either-or is a fallacious choice. However, practically speaking, the learning curves require investment, so most people are better at one vs. the other. The learning in each software will make acquiring the other easier, though.

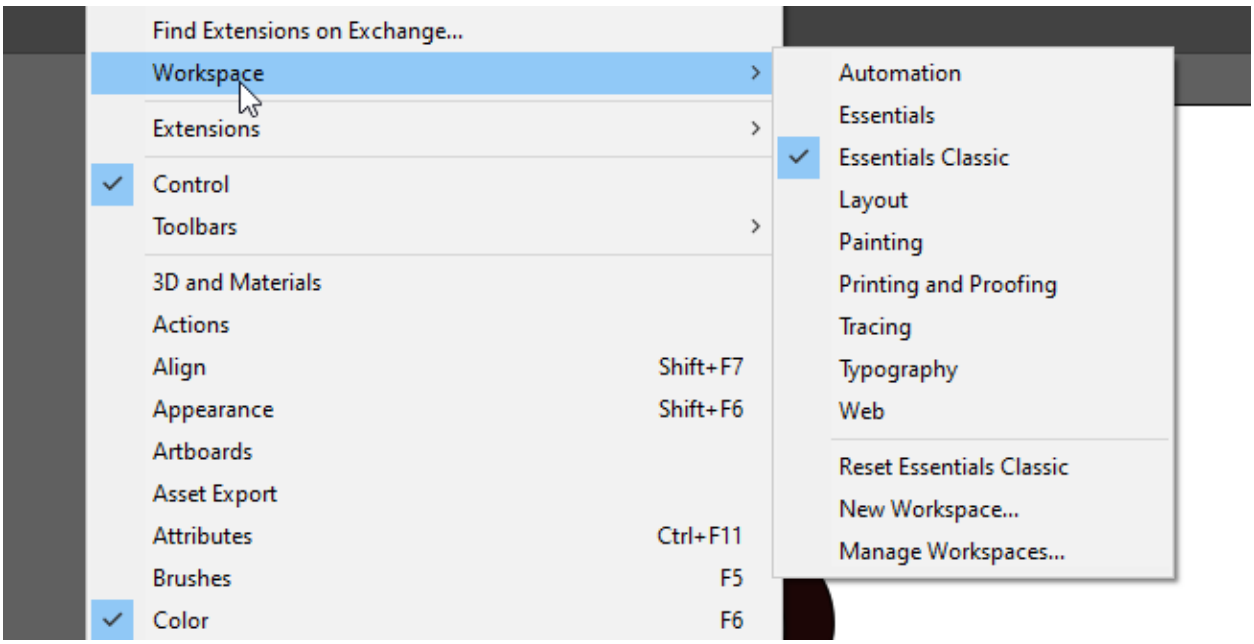
Preflighting for Publication / Deployment: A "pre-flight" enables the software to make sure that all local links to artworks and such are solid and integrates all contents into a publishable file with all the standards indicated in the digital file.

Your Preferences and Experiences: Your preferences matter. Your comfort level and familiarity matter.

Limited Use Case of the Presenter (Caveat): The presenter uses the software as part of instructional design work…so this is a limited use case. Don't let the limits of the trainer limit you!

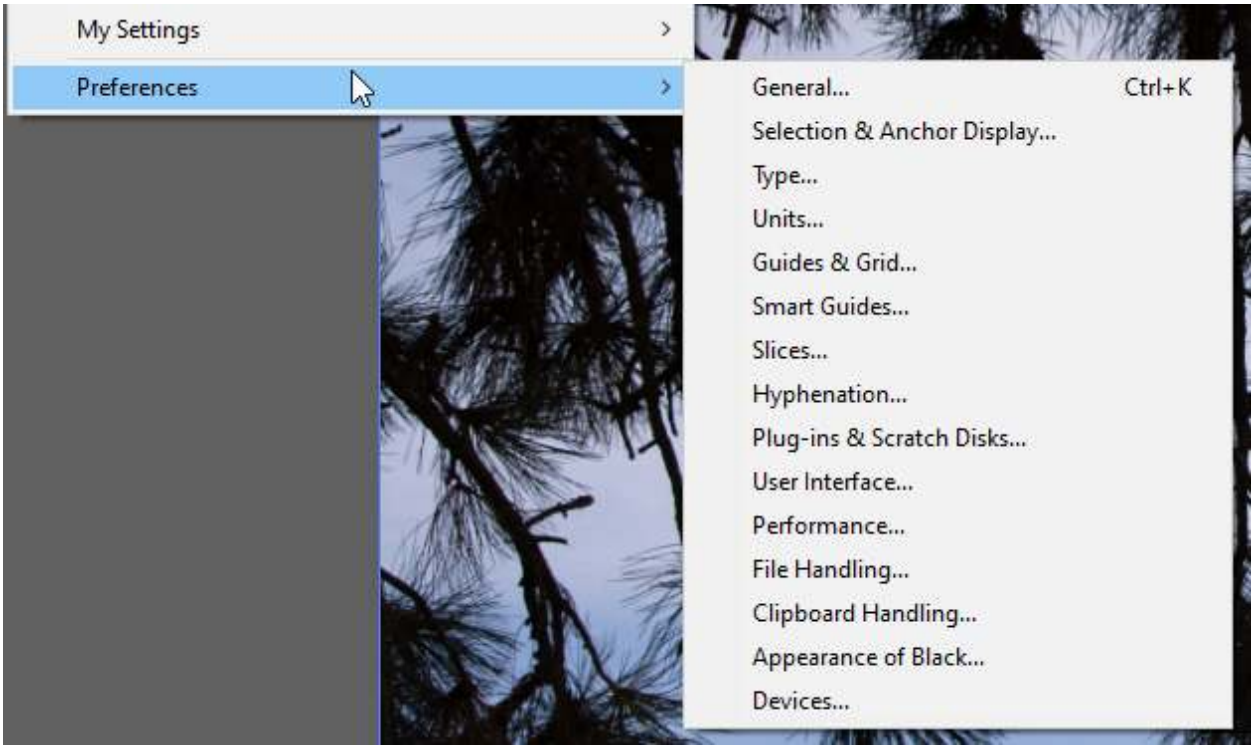
Workspaces

The respective workspaces are aligned to different needs of users creating different types of contents.



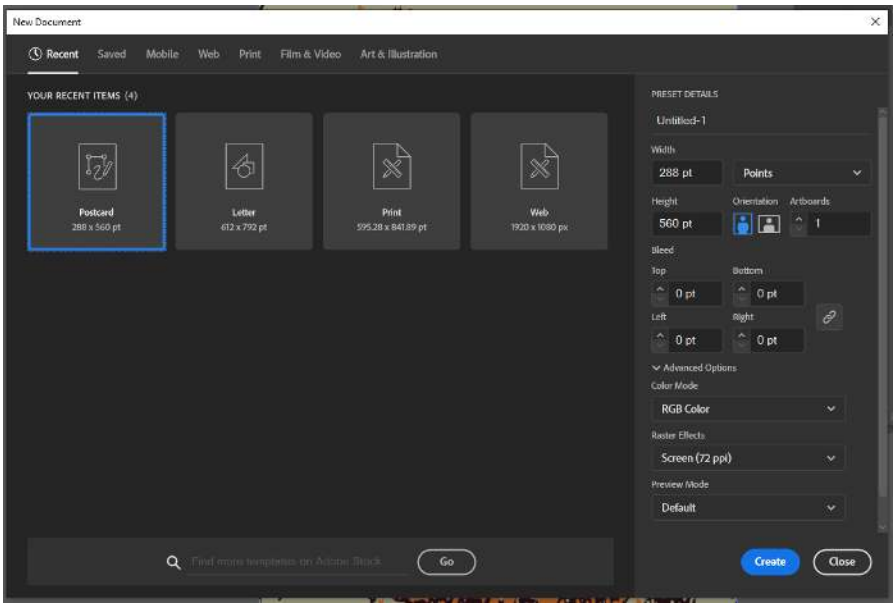
Also, the Edit -> Preferences enable the software setup in a way for optimal functioning for the person.

Also, see the importable / exportable "My Settings" in the Edit menu (above Preferences).



Preset Canvases Available

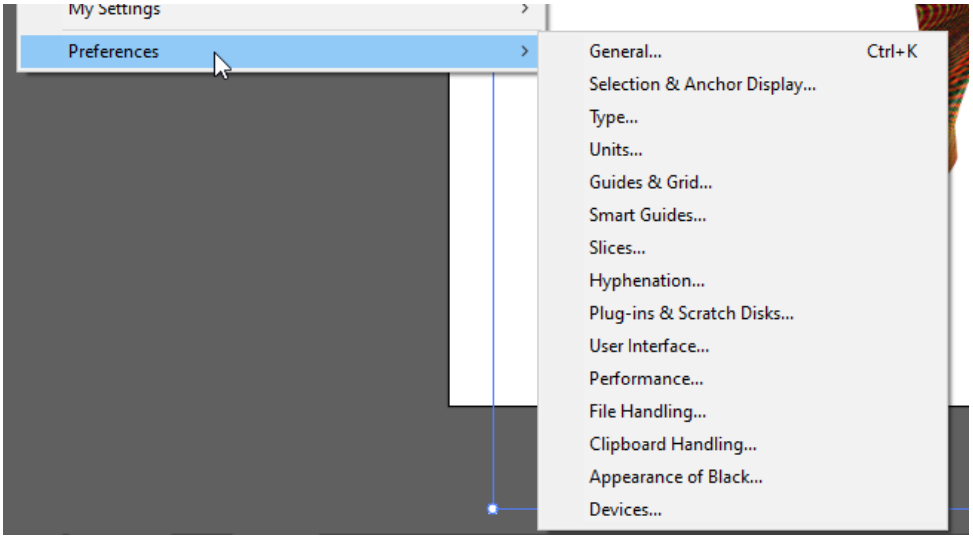
When starting a new file, various canvases are available including free preloaded types of contents for mobile, web, print, film and video (bottom thirds and others), and web.



A Different Conceptual Model

Assumption of knowledge of Adobe Photoshop as a basis? Or starting cold?

Make sure your scratch disk is sufficient for the various transforms and other work in Adobe Illustrator. (In Menu: Edit -> Preferences -> Plug-ins & Scratch Disks…)



Some Preliminary Understandings:

- A stroke is a line that moves through various anchor points. It can take on various attributes.
- Virtually any path imaginable is expressible in the 2D workspace of Illustrator.

- An open path is one which has a start point and an end point.

    - An open path can be closed onto itself by using the direct select tool to select both ends (anchor points) (hold down shift for multiple select). Then, go to Object -> Path -> Average -> Join, and define the type of join (vertical, horizontal, or both). Select both. Join. This makes the one open path a closed one (without adding a direct additional stroke).

- A closed path is one that does not have an open end. It is a closed "loop" of sorts.

- A compound path is a path of a certain type that can include holes (alpha) and can take applied colors (dragged on).

- Anchor points may be created anywhere…to enable dragging and moving.

- Every visual and piece of a visual is its own object. Every piece has its own number (in terms of backend management of contents in Illustrator).

    - Every piece has its own **reference point** (or origin point). The object rotates around its reference point.
- Illustrator is based on coordinates on a two-dimensional plane, with the x and y axes.

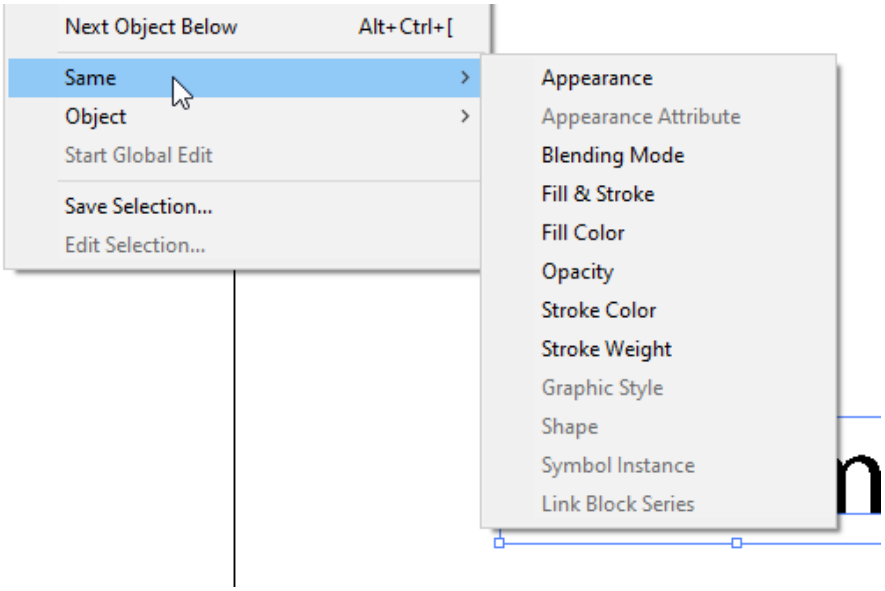
- All elements exist in a hierarchy, whether a flat or tall stack. The hierarchy is defined in the layers panel (in the properties panels to the right).
    - At the bottom of the layers hierarchy is a base layer, and it may be visible or invisible. (Digital visuals used to enable manual tracing—using a mouse or a stylus—are often placed at the bottom layer, used to inform to manual tracing, and then hidden or even deleted once the tracing is done.)
    - Any top layer is present in the visual (except for anything masked or clipped out).
- All forms or shapes are created with strokes (lines, edges) and fills (colors or transparency within the shapes). A polygon is a figure on a plane with a finite number of straight-line segments and potentially curved ones, too. The number of sides may be referred to as n-gon: 3-gon, 4-gon, and so on. The "n" refers to a number. "Poly" just means "several…" A polygon then is a shape with (at least) several sides.
    - All assets in Illustrator are eminently scalable, such as from a desktop image to a bus wrap or a building poster or aerial billboards and other visuals… The use of equations to express lines, shapes, colors, textures, and such, means that a vector graphic can be expanded exponentially without degradation of the visual (vs. a bitmapped or raster image).
- The main workspace may contain various art elements, artboards, and so on. A user can decide what is expressed: only what's on the artboard, or all elements in the workspace.
- All raster images can be vectorized, and all vector images can be rasterized. With each change, additional capabilities are possible (depending on what edits are being done). With each change, additional constraints are possible. (The strengths and weaknesses of the respective vector or raster forms can be seen.)
- The works in vector can be hyper precise.
    - It helps to understand basic geometry.
    - It helps to understand the degrees of a circle.
    - It helps to understand angles. (Yes, from middle school. It doesn't get more complex than this though.)

- Placing objects is somewhat different…so repeating objects may be placed around a circle such as numbers around a clock.
- It is possible to create tile patterning and space out the tiles in different ways (such as for textile and flooring and other such patterns). There are ways to lay out elements in grids that may be stretched or condensed.
    - Layout grids may be in various shapes: classic grid, circles, and others.
    - There can be custom grids.
- New elements may be added to a design. They may be removed.

- When working on an object, one can select the point of origin. When adding an effect, one can choose the point of origin (such as the fulcrum around which an effect is created).

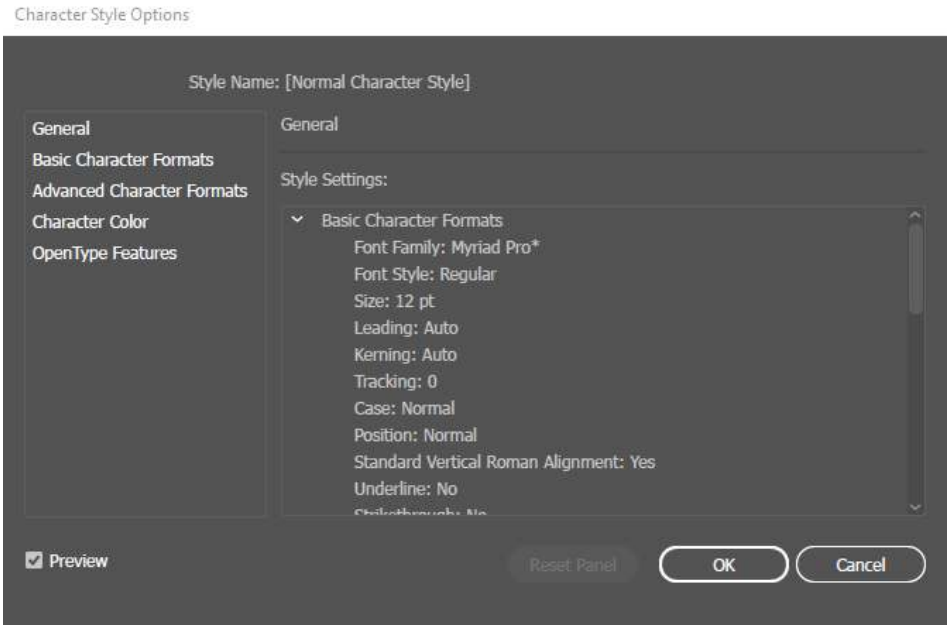
- Selection is powerful in Adobe Illustrator, too, such as selecting by color, by similar objects, by similar strokes, and so on.

- These tools are designed for fast work…so efficiencies are built into the software.

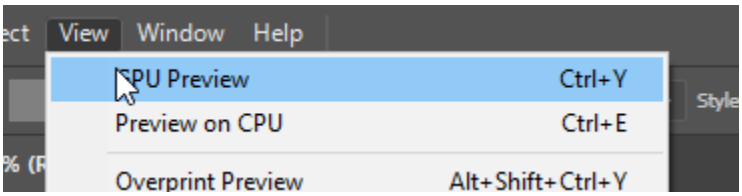


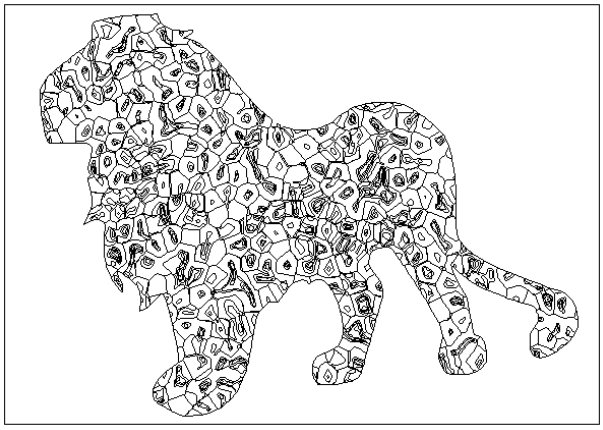
The “magic wand” tool enables selection by dimensions of different objects (color, line stroke feature, or some combinations of features). This enables a selection of objects in a combinatorial way.

- There are plenty of fonts that may be used and font effects. Typekit enables installing various font sets from the Creative Cloud. [Access this at the top right of the Creative Cloud window on the Web.]
  - To set a default font, start at the Window menu -> Type -> Character Styles -> Normal Paragraph Style [or Paragraph Styles]



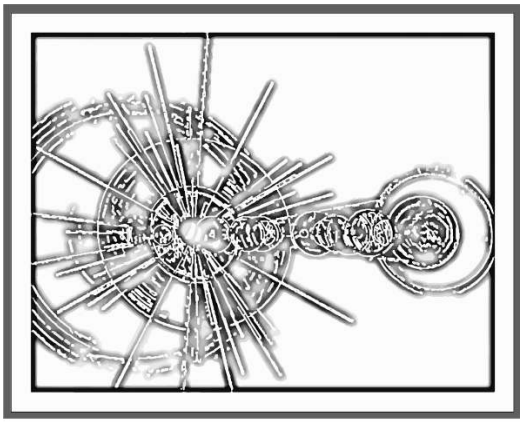
- There is a Preview feature (ctrl + y) to see the respective works in outlines (without any styles, without the GPU Preview). (example below of "View Outline" view) Another way to access this feature is through the menu: View -> Outline / View -> GPU Preview

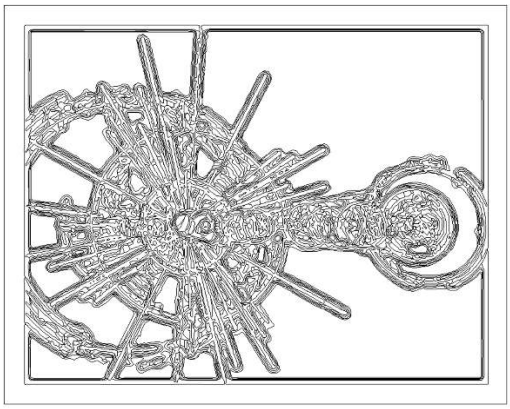




(above: outline view of the lion shape)

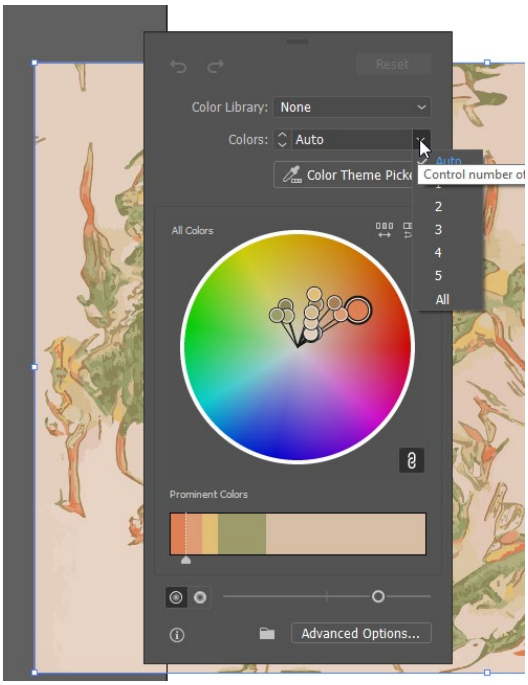
(to the right: GPU Preview at the top, outline view at the bottom)



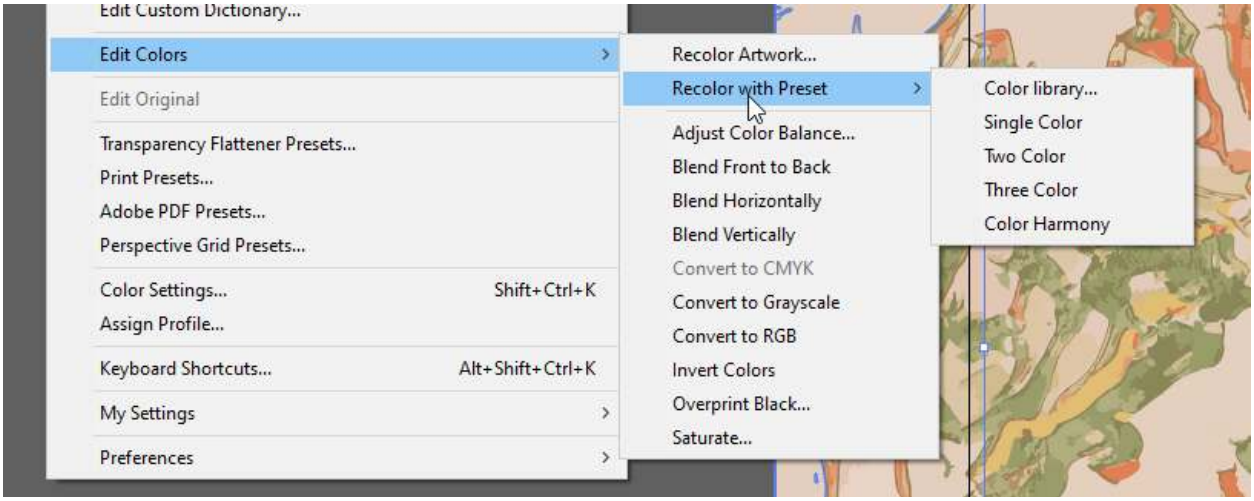


- It is possible to "recolor artwork" based on the built-in color wheel (in object bar above)…or through color themes from swatch sets or other artworks. It is also possible to select % of colors in the slider below.

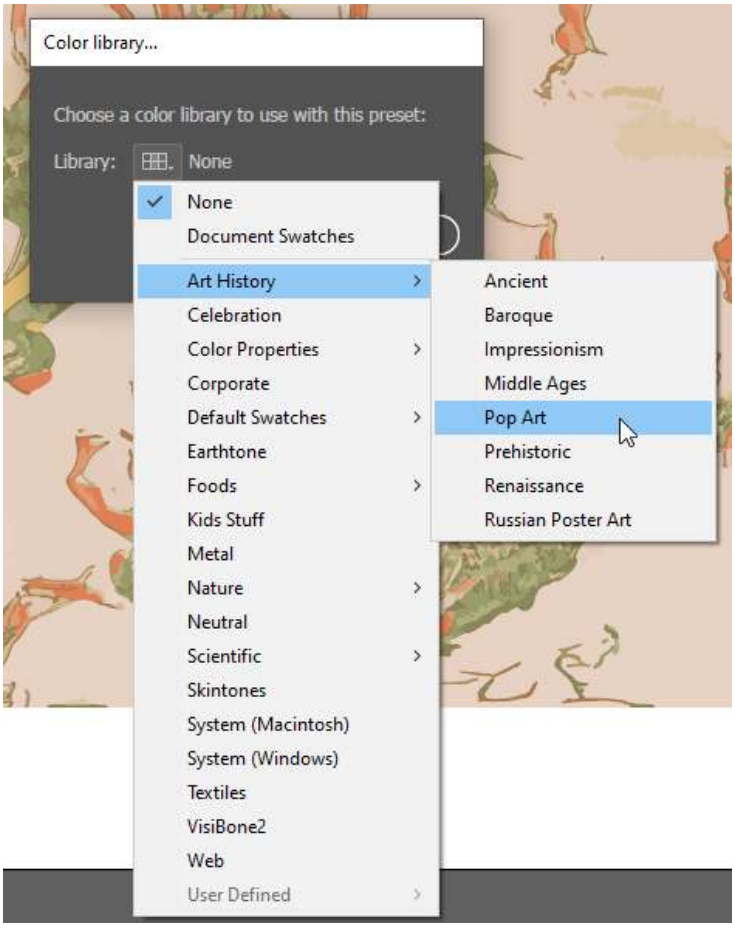
1. Highlight the vector graphic.
2. Go to Edit in the menu bar.
3. Select Edit Colors in the dropdown.
4. Select Recolor Artwork.
5. See the dropdown for the automatic reduction of colors.

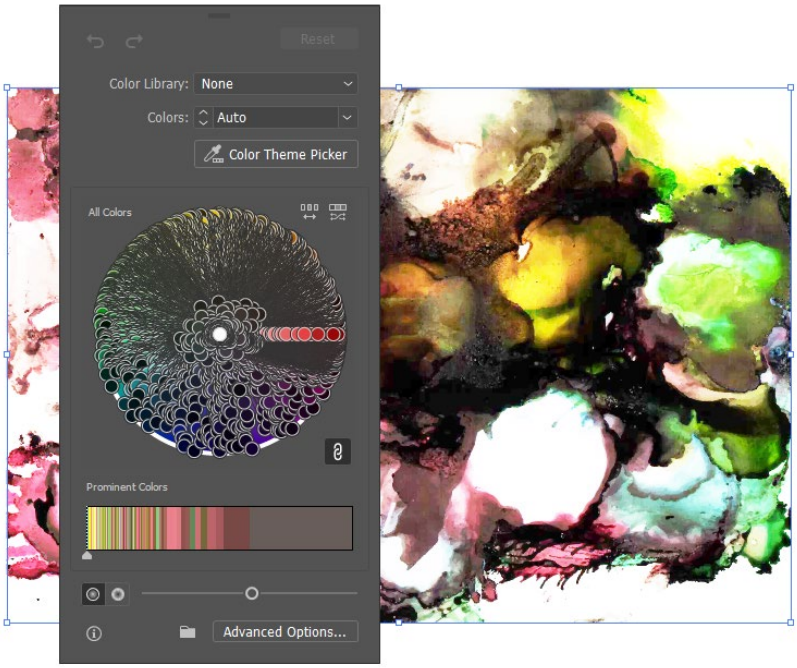


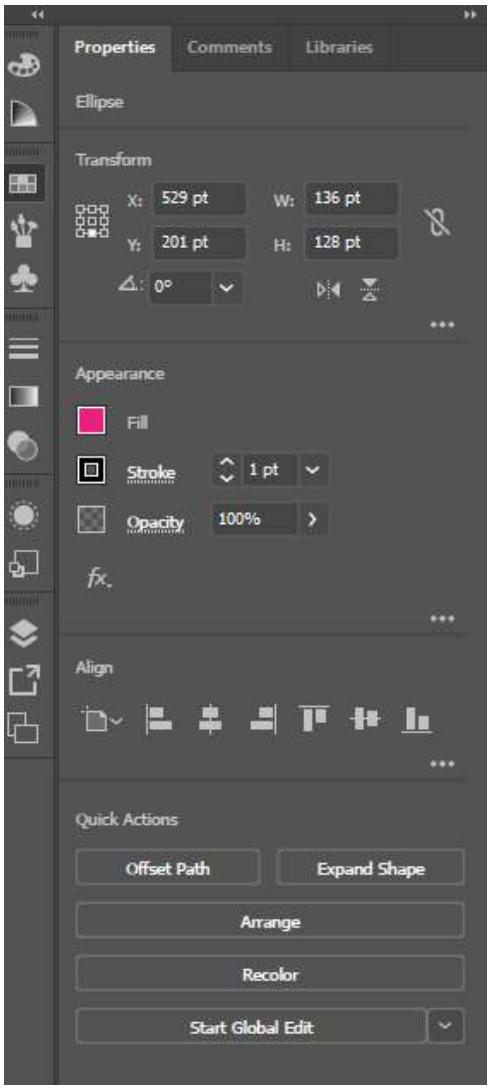
Recoloring may also be done by pre-sets (such as swatches from the built-in Color libraries…) or by color reductions.



Some examples of the color libraries may be seen in the dropdown illustrated here.

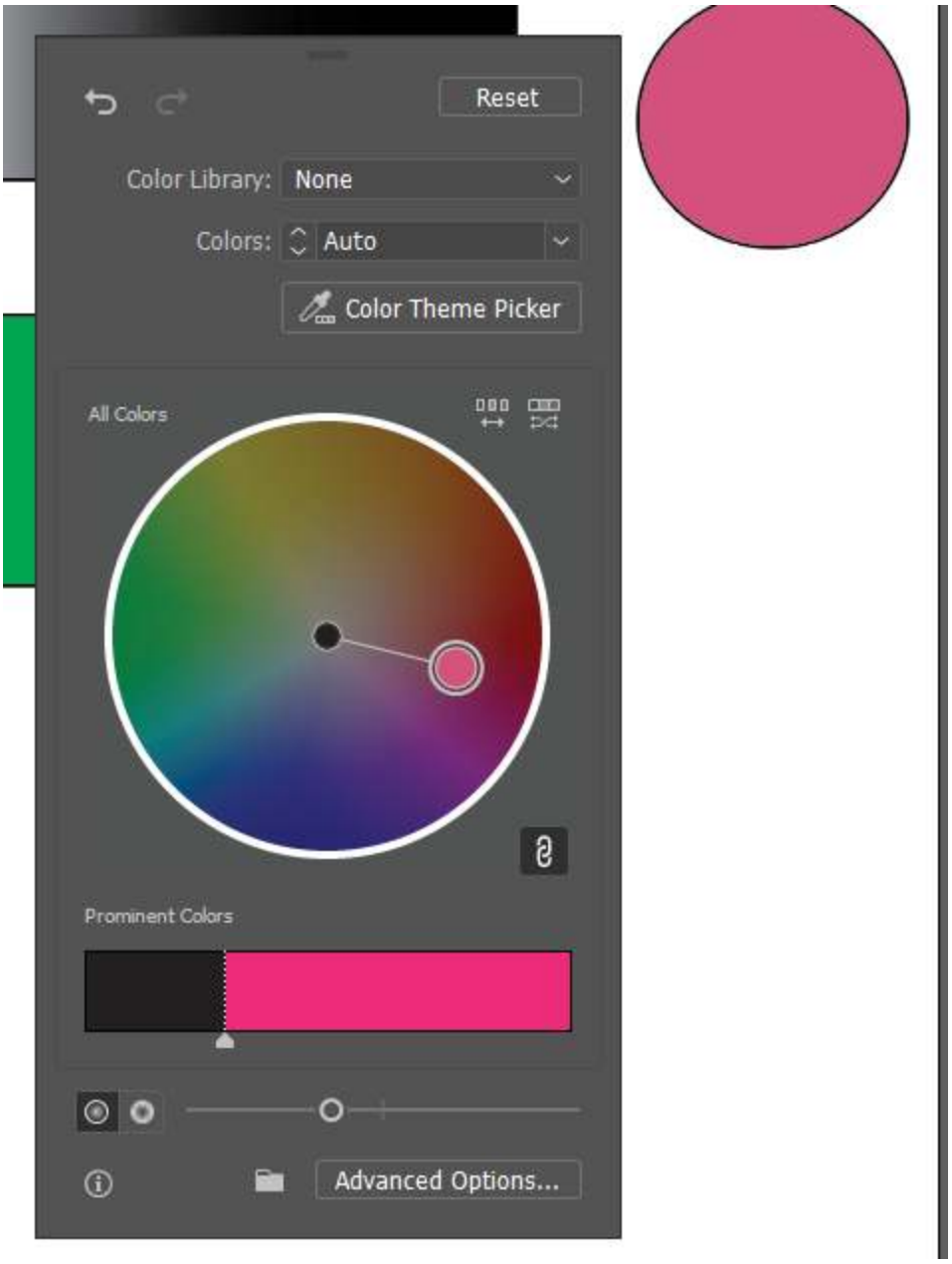

Color library...
Choose a color library to use with this preset:
Library:
None
None
Document Swatches
Art History
Celebration
Color Properties
Corporate
Default Swatches
Earthtone
Foods
Kids Stuff
Metal
Nature
Neutral
Scientific
Skintones
System (Macintosh)
System (Windows)
Textiles
VisiBone2
Web
User Defined
Ancient
Baroque
Impressionism
Middle Ages
Pop Art
Prehistoric
Renaissance
Russian Poster Art


Reset
Color Library:
None
Colors:
Auto
Color Theme Picker
All Colors
Prominent Colors
Advanced Options...



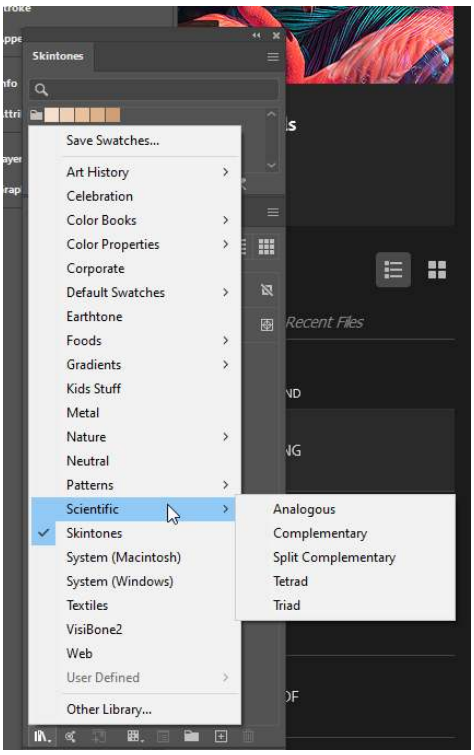
Or activate an object on the artboard. The Properties panel to the right has a Recolor button that opens a window.

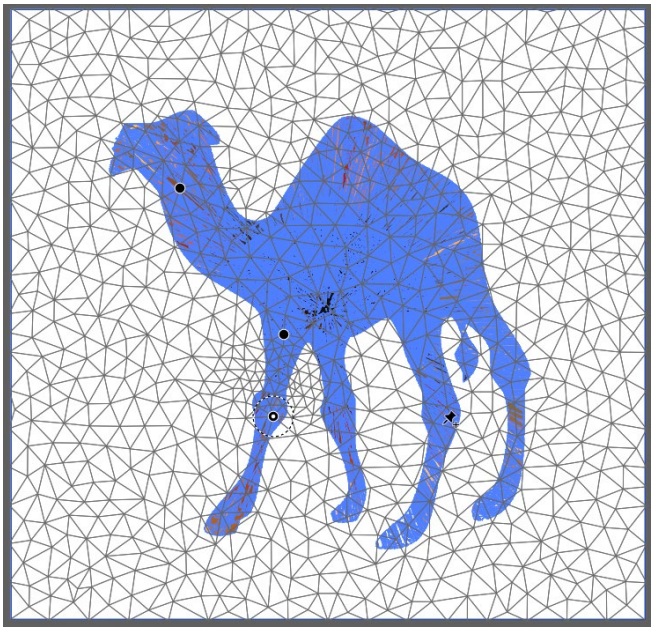
See the “Recolor” button near the bottom of the pane.



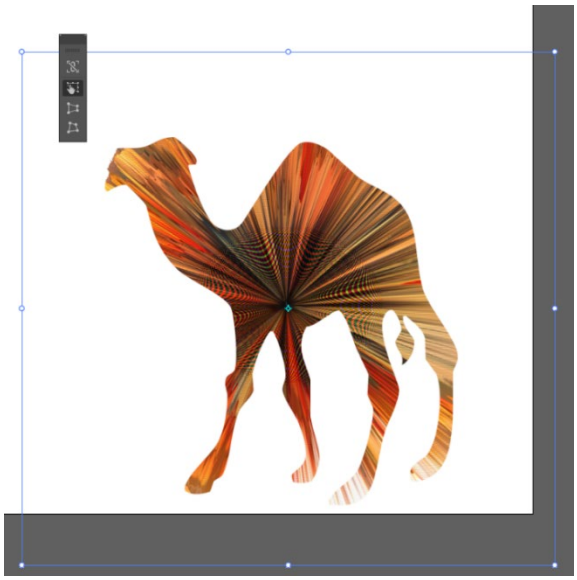
- Mosaic Tiles (a texture effect) can be used to create color groups from artwork.
- Collective colors may be used to create color swatch sets / color palettes.

- The library of colors has special palette sets. (See below.) The library link is at the bottom left of the swatches panel.
    - There are palette sets for skintones, scientific colors, and others.

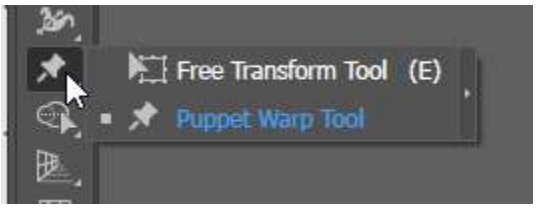




- There is a puppet warp tool in the free transform area that enables the "jointed" movement of characters (static and dynamic motion).
    - An image file cannot be puppet warped, so you have to run the Image Trace first and Expand…and then go to the Advanced toolbar to the left and the warp flyout…and pick Puppet Warp. Place pins to hold positions and to move positions. (See screenshots.)

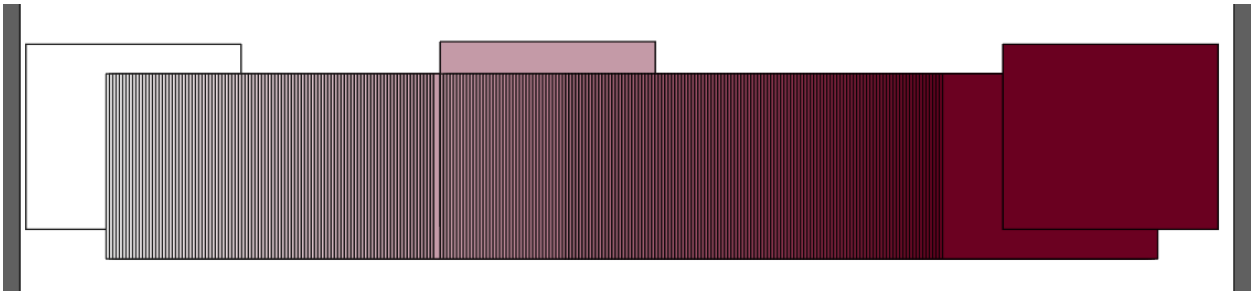


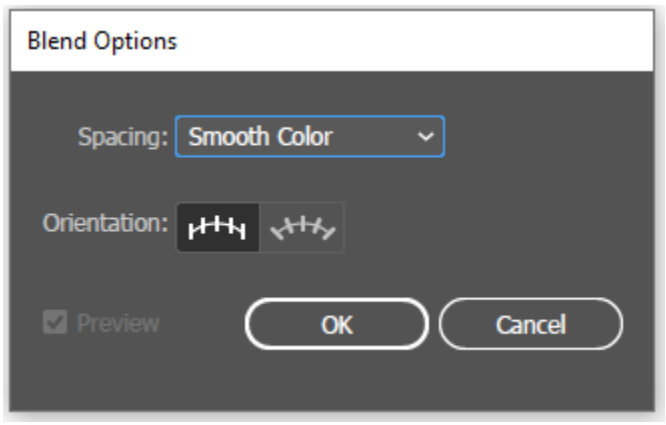
- There is a free transform tool, too, to enable reshaping of the image or objects.



Blend Mode

The “blend mode” tool enables the automated drawing of visuals in “in-between” states between a start and an end state. In the visual below, the start state is a white rectangle, and the end state is a maroon rectangle. The blend mode is “Smooth Color.”



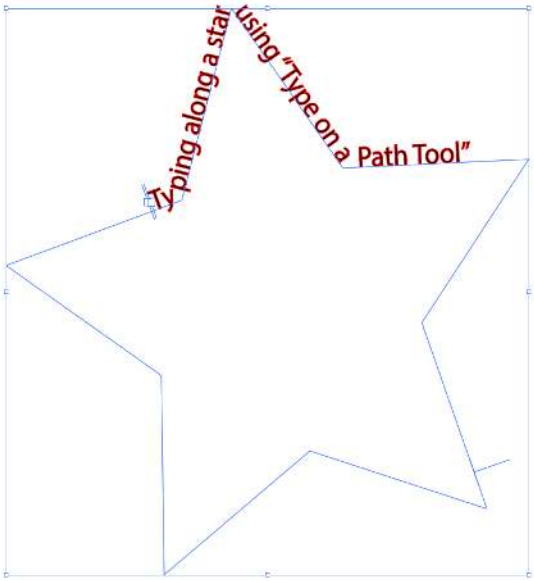


To access the blend mode, highlight the first and the last object in the sequence, go to Object in the menu -> Blend -> Blend Options -> and once the options are set, go back to Object -> Blend -> Make.

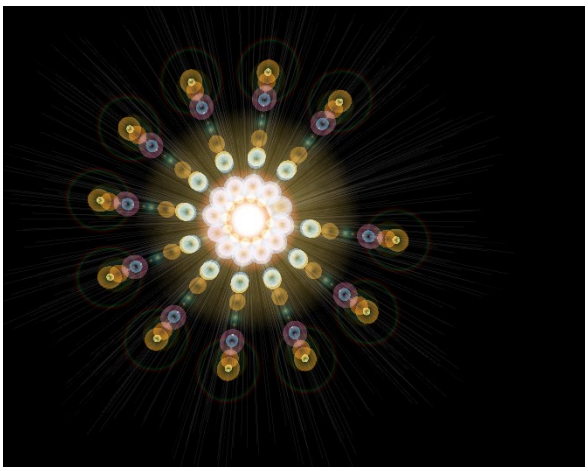
(The “make” will not be grayed out if the settings are correct. The keyboard shortcut for the “make” includes holding down Alt + Ctrl + B simultaneously.)

Typing along a Path

The type tool (in the Tools Menu) enables access to various fonts. The visual below illustrates the ability to write along a path, whether the path is part of a shape or is a line or is hand drawn, etc. The visual shows maroon text written along the edges of a star shape.



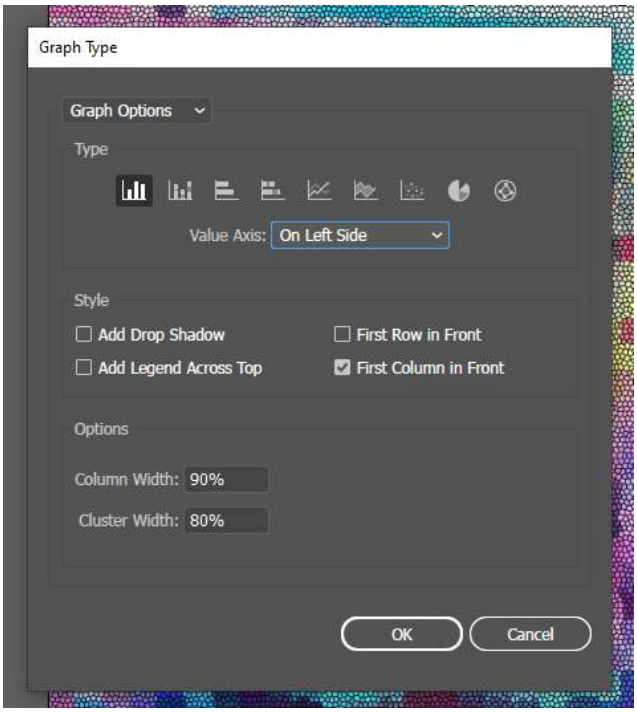
Distort and Transform along an Axis



It is possible to draw a visual, such as a lens flare, and make copies of that visual radially around a center using the Effect menu selection -> Distort & Transform -> Transform -> Rotate and Copies (settings in the window).

Visual Graph Tool for Data Visualizations

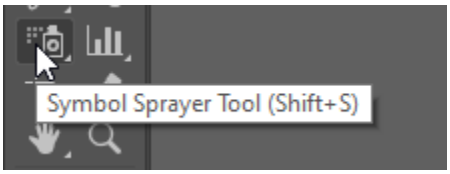
- There is a visual graph tool.
    - These enable the creation of data visualizations as part of the layout.



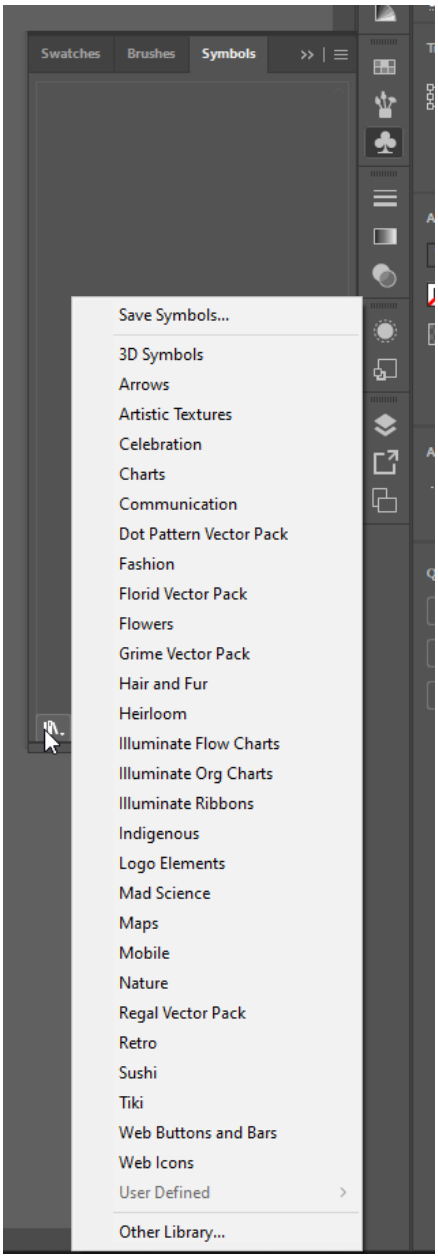
- There are resources in the Adobe Creative Cloud, with CC licensure (such as Adobe Fonts, templates, imagery, and others). The licensure has to be maintained for access.

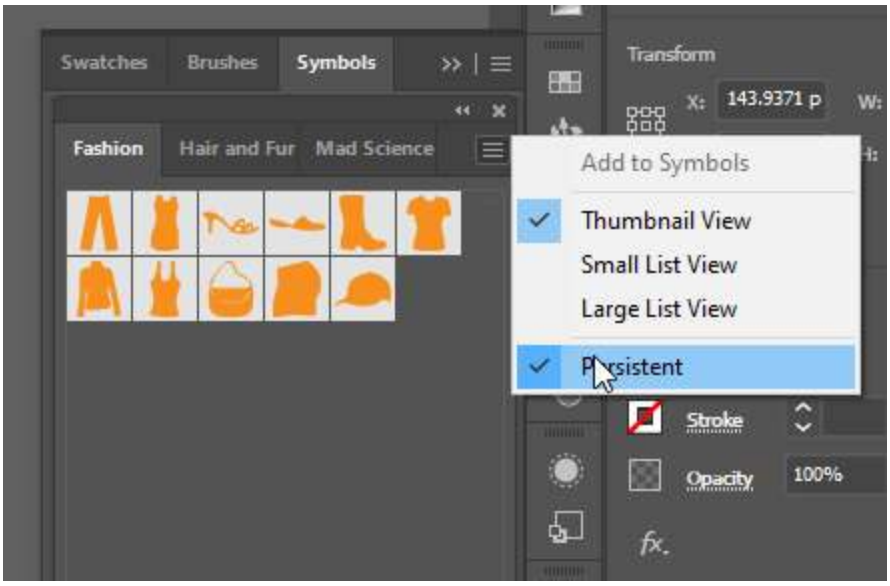
- Glyphs may also be used.

- Symbols: There is also a “symbol sprayer tool” and various symbols that may be selected from the toolbar to the left in the Essentials Classic and other workspaces.



Check out the library of symbols.



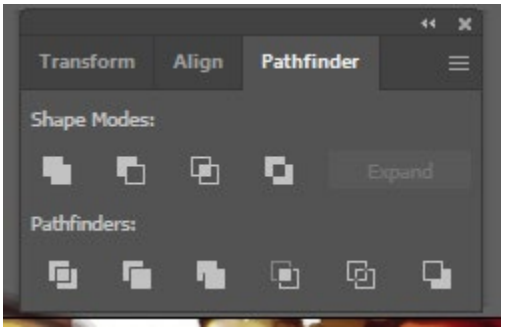


- Adobe Illustrator work may be collaborative, and it may be shared virtually via the cloud, or otherwise. The cloud-based versions are versioned, with a remembered history, so that the team can refer to an earlier version or return to an earlier version.
    - This sharing suggests that people need to work in ways that make handoffs easy:
        - clear informational labeling (of artifacts, of layers, of folders, of palettes, of color swatches, and others),
        - clear structuring of elements and artworks,
        - clear notations,
        - embedded artworks,
        - adherence to project standards (per stylebook),
        - editable project files (.ai),
        - and other aspects.

About Pathfinder to Enable Shape Interactions

Pathfinder enables the turning of lines and shapes into Paths…and Paths and Compound Paths enable more powerful editing of shapes and the creation of new shapes.

Go to Window -> Pathfinder to turn on.



There are various ways to combine shapes, including overlapping, intersecting, adding, subtracting, excluding, minus back, dividing, trim, cropping, outlining, and others.

Highlight the shapes, and then choose the various modes to select what you want.

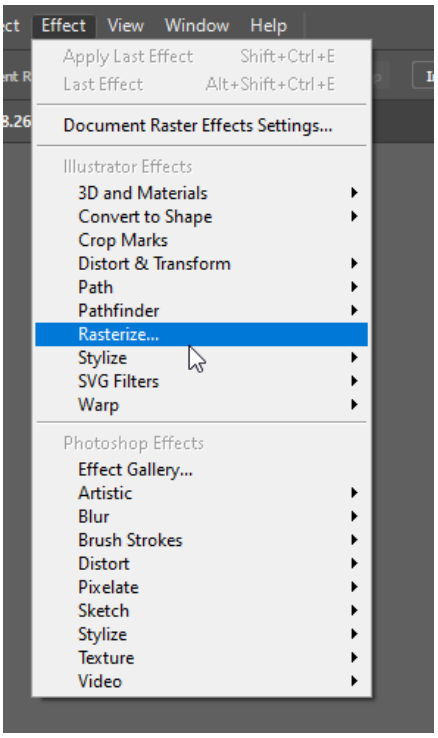
Light Refresher on Vector and Raster (before looking at Tracing)

Illustrator allows the drawing of strokes (lines), paths, and shapes as Bezier paths to vector graphics, which are resolution independent and therefore can scale up exponentially since the computer is drawing the image from mathematical equations, not pixels (picture elements, or squares with defined colors).

There is an interchange of raster to Illustrator, and vector to Photoshop, so each can handle the less native format to the tool.

	Common File Types (by file extensions)
Raster	.jpeg, .jpg, .png, .gif, .mpeg4
Vector	.svg, .eps, .pdf, and .ai (Adobe Illustrator file)
Other	.webp (less "lossy," lightweight image in Chrome, Firefox, Edge, and others; free converters available)

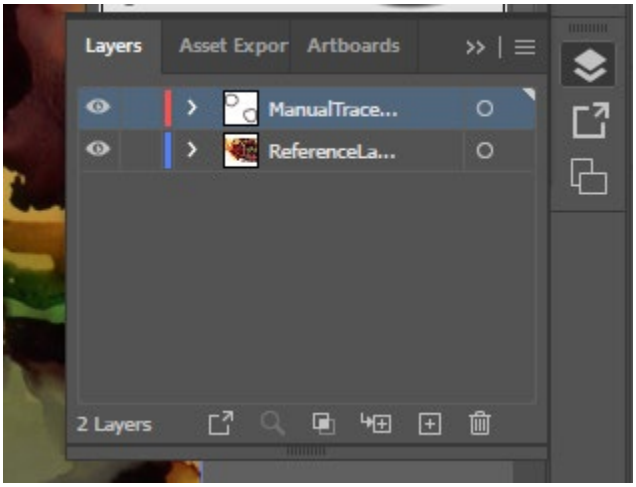
To rasterize an image in Illustrator, the path is Edit -> Rasterize. (screenshot below)



Small note: Complex .svgs can take a long time to open.

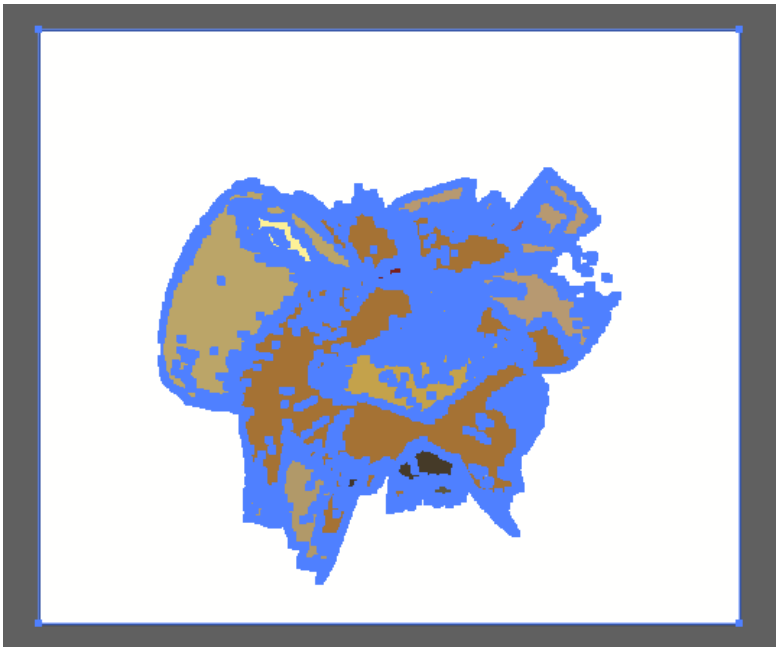
Manual (By Freehand) Image Trace

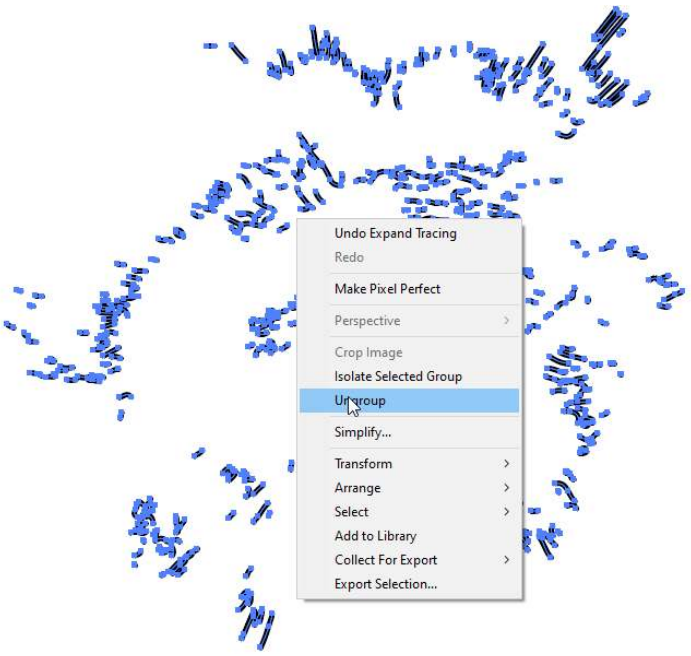
A manual image trace involves placing a bottom layer with a reference image…and setting up a new layer on top…and using a stylus or mouse to trace over the underlying image freehand. The paintbrush tool, the pencil tool, the pen tool, the curvature tool, and others may be used for the manual trace. Or, the trace layer may be created with various shapes as well, given the shapes editing capabilities of the Shapebuilder and the Pathfinder. Shapes may also be drawn freehand with the self-correcting Shaper Tool, which modifies a rough sketch into known shapes (circles/ellipses, squares, rectangles, triangles, and such). In the art space, all objects in the world may be redrawn using known basic or primary shapes.



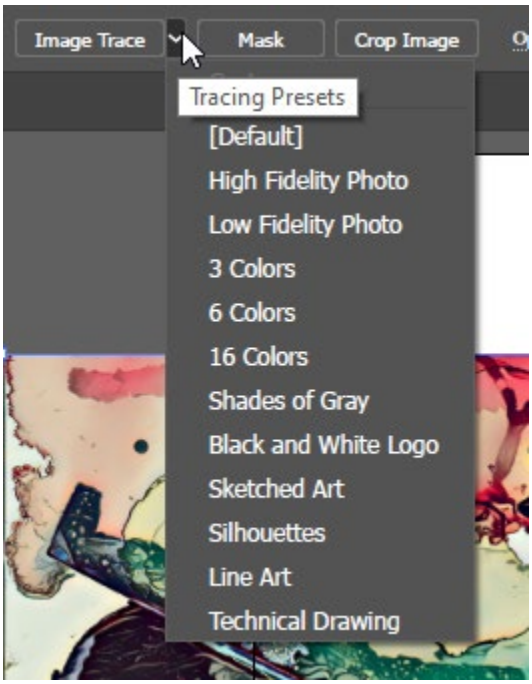
Automated Image Trace

- The “image trace” features enable raster images to be automatically traced to be turned into a scalable vector image.
    - After the image trace, click “expand.”
    - Right-click image to ungroup.
    - Each trace or segment is now editable.





The trace tool enables various types of traces. A visual may be treated in various ways based on the original seeding image. The user can decide how that image will be treated, such as a high fidelity photo, low fidelity photo, and so on. The colors of an image may be reduced with "3 colors, 6 colors, and 16 colors." Color data may be changed to "shades of gray." A visual may be rendered as a b/w logo. A visual may be changed into "sketched art, silhouettes, line art, technical drawing," and others. In most cases, the image trace is either a representation of the original (just the strokes identified out)…or is reductive (with less visual information after the process).

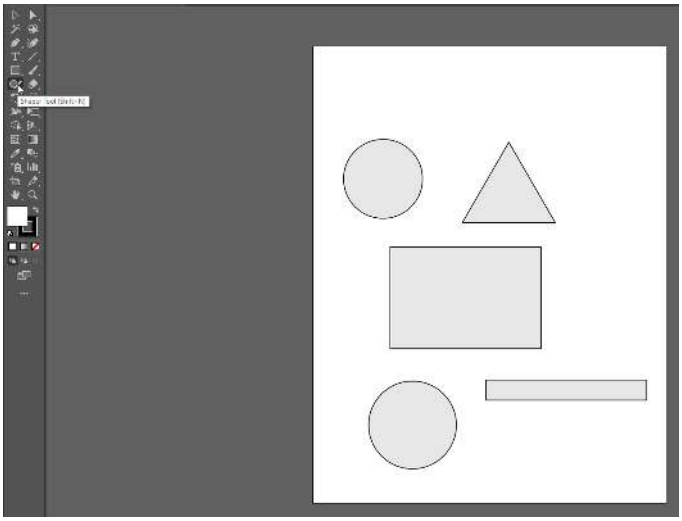


Traces can be **reductive**, such as by simplifying a visual to line art or reducing colors.

Neural filters, on the other hand, may be **additive**. Colorization adds artificial color. Style transfer can add color to a b/w piece. New lines may be added with style transfer.

Machine-corrected Freehand Sketching of Basic Shapes with Shaper Tool

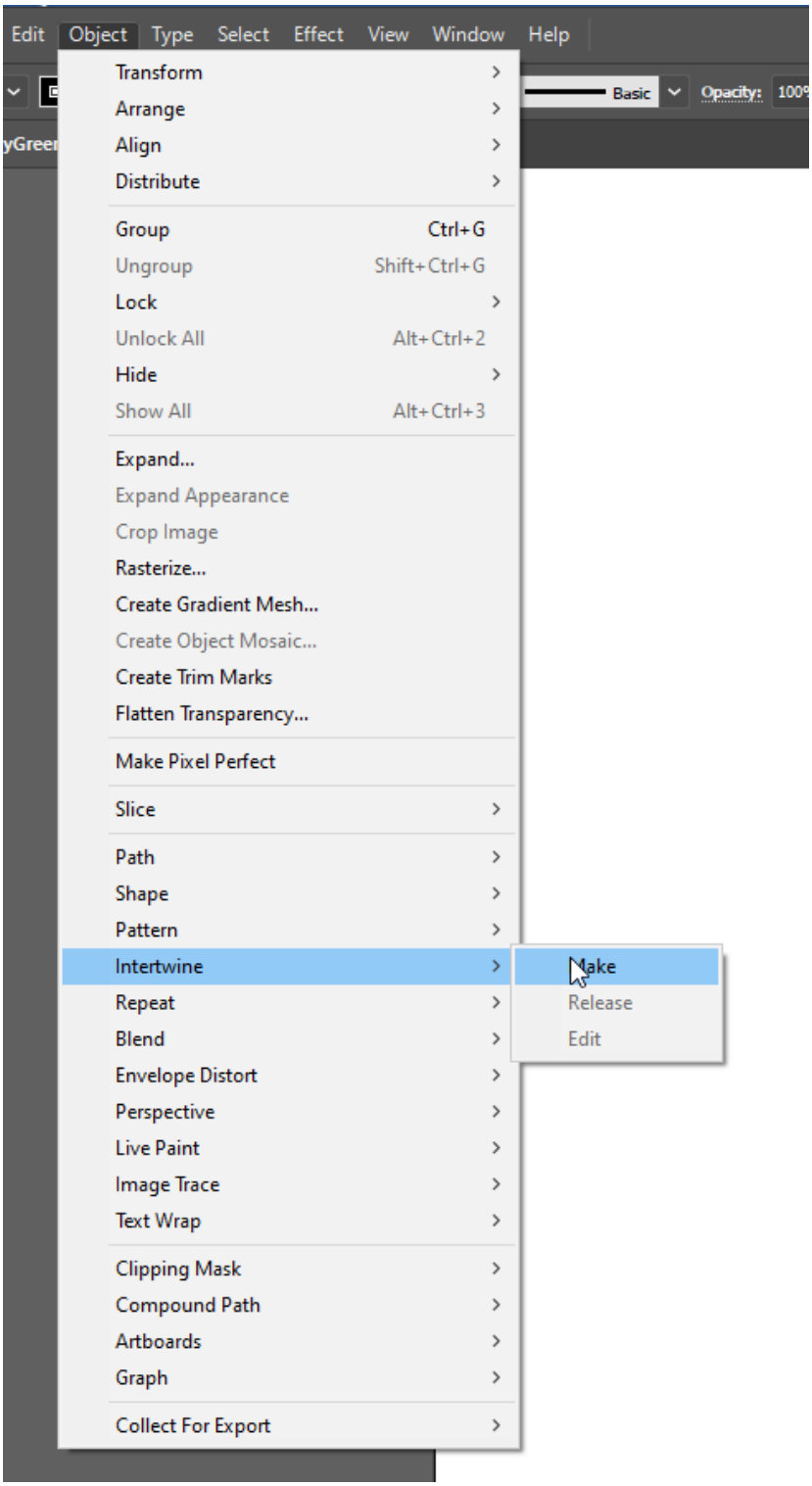
Click on the Shaper Tool in the toolbar to the left. Then draw without lifting the cursor from the artboard. The software will create the clean shapes even if the hand-made sketch is rough.



Creating Particulate Effects

Particulate matter—snowflakes, dust, rain, powder, flour, and others—may be created using filters (noise) and specialty brushes.

About the New Intertwine Feature in Illustrator 2023

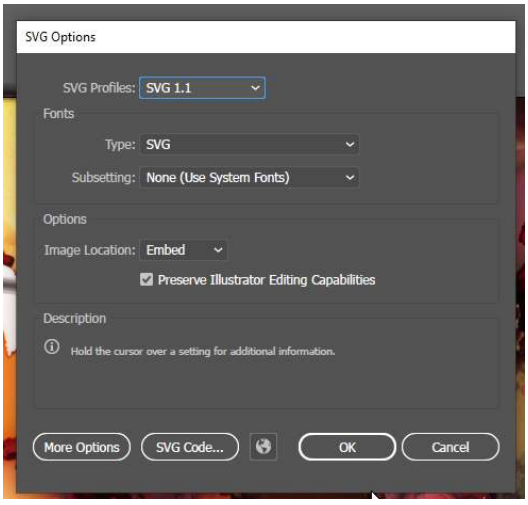


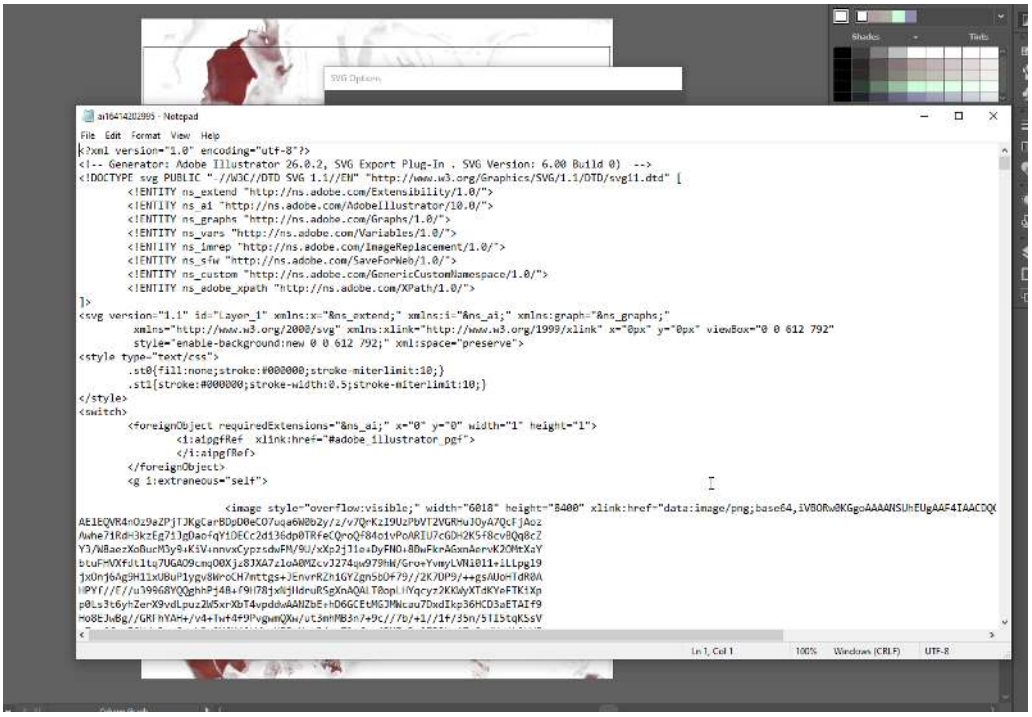
An example shows an “A” and a “B” intertwined.

SVGs as XML Code

Scalable vector graphics are often built into websites. SVGs can be turned into XML (extensible markup language) code. In Illustrator, one can open a text window with the XML code. (See below.)

When saving a file as a .svg, go to “SVG Code” button at the bottom.

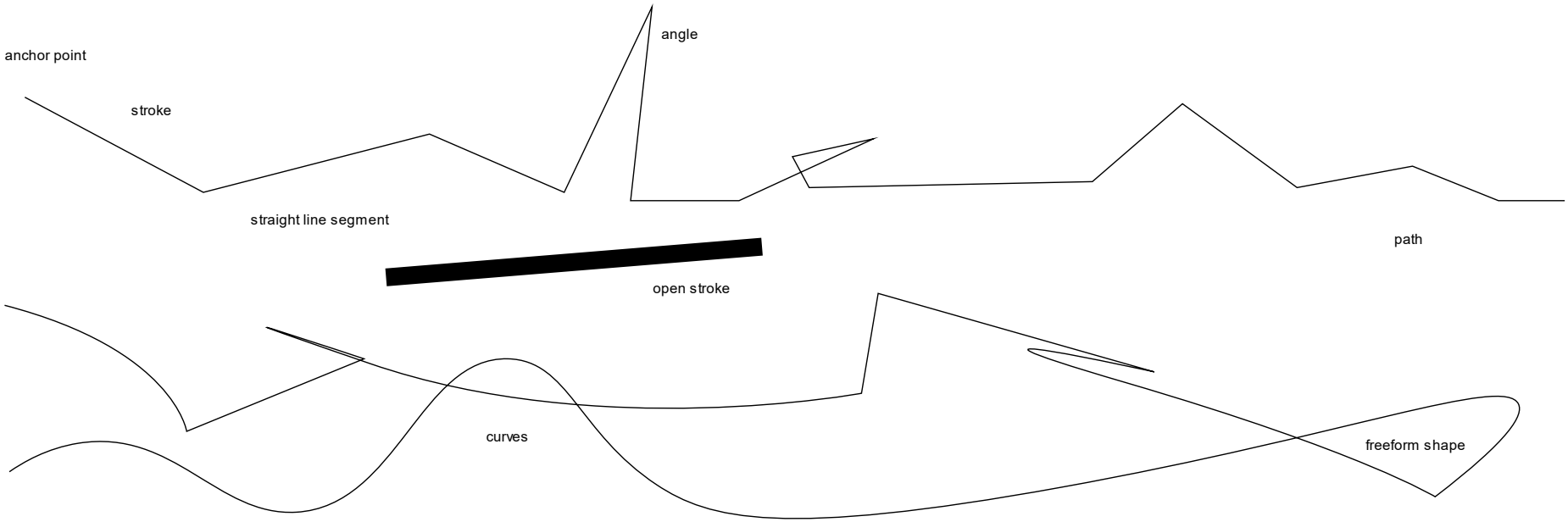


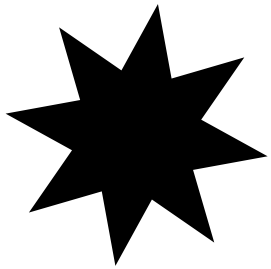


Turning Illustrator Files into Reusable Templates

- Files may be templated (.ait).
    - Templates may be reused again and again for some level of consistency.

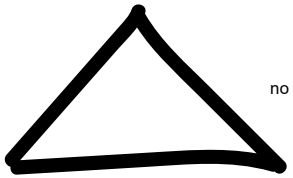
Some Basic “Getting Started” Terms





star polygon

rectangle polygon
(with solid blue fill and a heavy green stroke)



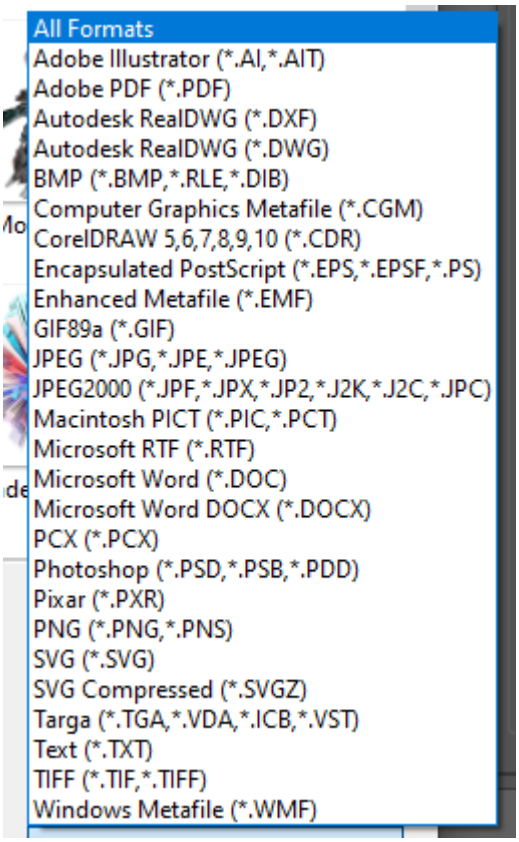
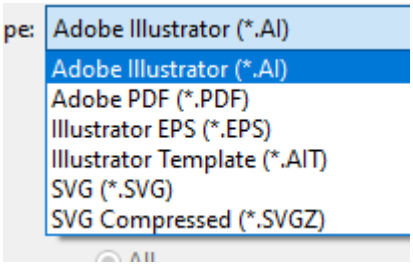
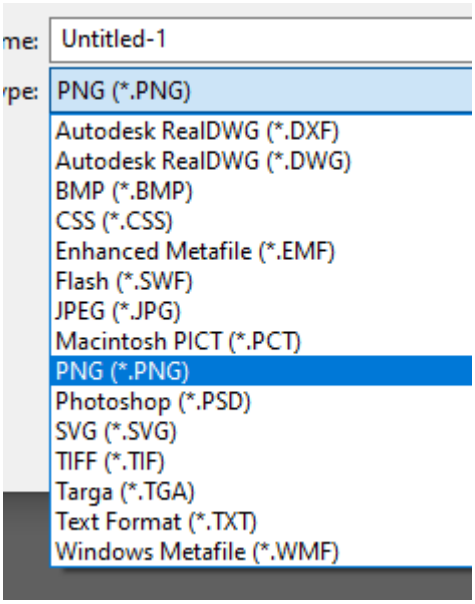
triangle (with the blob brush tool)

no fill (alpha channel)

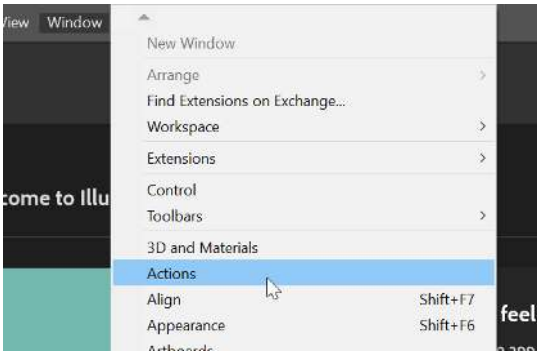
and more...

A Full Work Pipeline

Think about Adobe Illustrator 2022 as a tool for a full work pipeline. Various file types may be ingested and processed, and the same or other file types may be saved as, copied out, or exported.

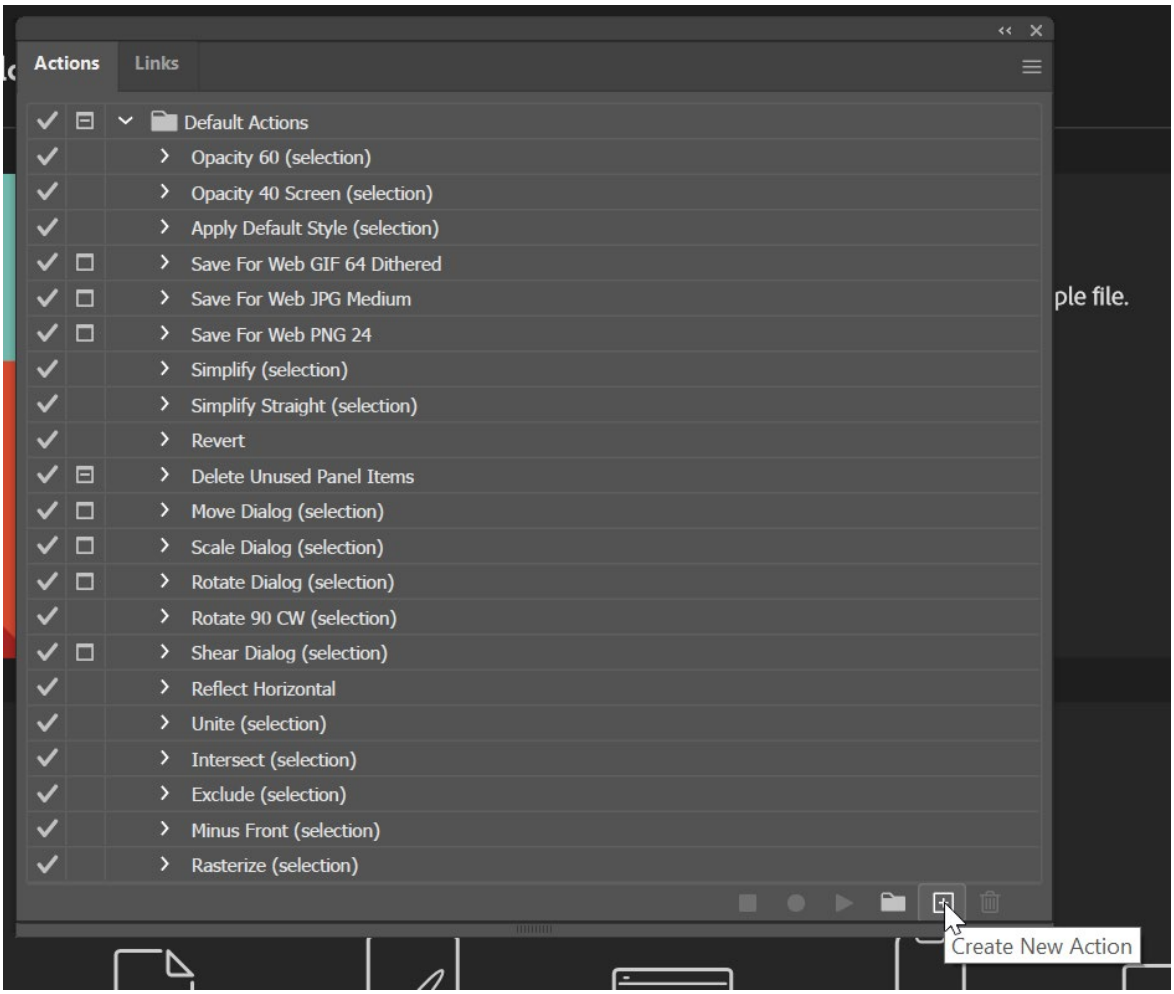
Inputs	Outputs
Place artwork (into an open artboard or workspace) Open an artboard (or set of artworks)  Note that the list above involves digital imagery, digital text, and project files.	Save as / save a copy  Export as  In general, the point is to export files in the most editable formats as possible.

There are optimized work sequences based on the capabilities of the respective software programs. It helps to pay attention as one is working to know where it is best to address particular transcoding and other work, for efficiencies.



Actions (automation): Those who want to batch-process a number of images in a folder or other recorded actions may set up using the following path: Window -> Action. These enable pre-recorded built-in macros/scripts (called "droplets" in Illustrator, as in drops of applets, I'm guessing)… (See left.)

But at the bottom right, there is a "plus" button which enables recording one's own. (See below.) [Always save a pristine master set of images before auto-processing.]



A Little (Unsolicited) Opinion Note: The software tool is all about control and precision, but there is creative room for serendipity, chance effects, and fortunate "accidents." Creativity requires the prior elements.

To Start

The pen tool is the main one for manually laying down lines. These can be straight lines. These can be curves (with the tangent handles).
Please check out various videos to see how this tool works. Try https://youtu.be/tXpnKlUMZQA and https://youtu.be/V1ibvMmKp7U.
As with anything, this feature will take practice.

The curvature tool enables controls for putting down anchor points in sequential steps and the moving of the cursor to emplace a curve around each ensuing anchor point.
Here is an excellent short video: https://youtu.be/MLg8xQgEIfk.

Various shapes can also be used to lay down lines. (These will include various types of unions and deletions and so on.) This is called "drawing by construction." This requires seeing a drawing as the interaction of various shapes…with extra parts of those shapes excised.

Various brushes come with Illustrator, too. As digital tools, they come with diverse effects.

The control strip across the top (under the menu bar) is context sensitive. It shows a lot of what may be done with the activated (highlighted) objects in a workspace.

Strokes (lines, edges) can be merged into shapes with the Shapebuilder tool. Shapes may be combined or subtracted for various new shapes based on prims (primitive or basic geometric and other shapes). (Highlight shapes first (by ctrl + a to select all or by dragging over the shapes to highlight them), and then click on the Shapebuilder tool in the toolbar…and then click to activate the various shapes and drag across to include new shapes into a melded one. The first clicked-on shape defines the color of the combined shape.
(If you want to delete sections, highlight the shapes, go to Shapebuilder, and hold down alt as you drag over the parts that you do not want.)
(If you want to remove a subsection, highlight the shapes, go to Shapebuilder, and drag over the section that you want to remove. Make sure the pieces are ungrouped. Drag out the piece using the general selection tool—not the direction selection one.)

Some Keyboard Shortcuts (to get started)

Hold down the spacebar brings up the temporary hand tool to enable dragging around the workspace (on the fly).

To copy something, generally, just highlight, hold down alt, and drag. The copy will be an exact visual of whatever it was copied from.

To copy a layer, go to the layers panel, grab, and drag to the "add a layer" icon below.

Ctrl + z undoes an action, in reverse order from the most recent. (The history button in the right panel can be accessed to go back as well.)

Ctrl + 0 (zero) zooms out of the zoomed-in view and resets to a full pasteboard view.

Ctrl and + zooms in.
Ctrl and - zooms out.

If selecting multiple objects, hold down the shift key and select objects by clicking on them.

Click: View + outline to see the shapes underlying the visuals (without the fills and other aspects that may get in the way of the underlying design).

And: There are various letter keyboard shortcuts that bring up frequently used tools.

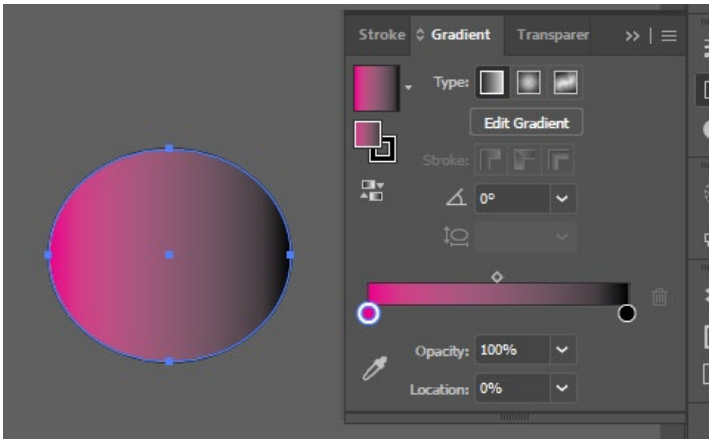
“a” goes right to the direct selection tool.
“p” goes to the pen tool.

Arrow keys can be used to “nudge” or move objects incrementally in particular directions.

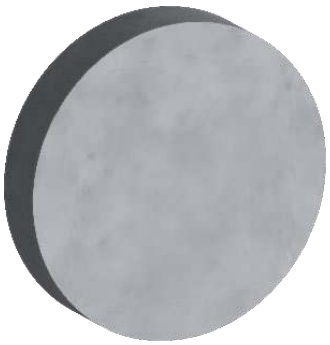
* On a Mac, the “ctrl” would be “cmd,” and the “alt” would be “option”.

Cool Visual Effects

Sophisticated gradients (types) in all shapes and tints



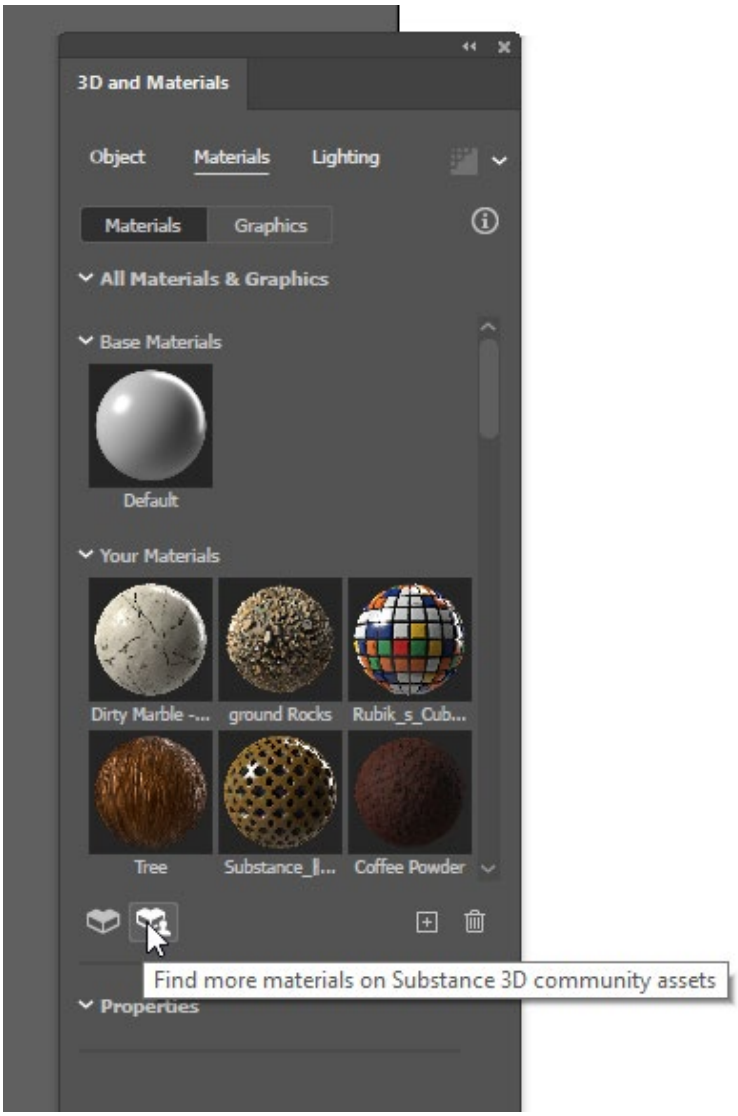
Making pattern files (tiles)



Envelope effects for reshaping text as an image (with “live” editable type)

3D and Materials (from Window -> … from Effect ->)
- May be based on a Plane, Extrude, Revolve, Inflate, and others
- May apply shadows
- May apply artificial lighting
- May move and rotate

- “Raytracing” (as in “light ray tracing”) involves the application of light reflectances and shadows with more sensitivities



- May apply materials on some shapes and surfaces (with free .spsm and .sbsar files available online) (see goldleaf to the right)
- May download free materials from the community and from Adobe
  - Just click the + to add to Illustrator in the 3D and Materials window (see below)
- At this time, multiple 3d effects / material effects cannot be applied to the same object. It’s a once-and-done situation (which explains why the pastries look awkward, in part).
- Also, the size of the object to which the texture is applied also affects the look-and-feel.

3D and materials applied:

A pink frosted doughnut anyone? (Eat fast—just a quick glance—without looking closely?)

Or a marbled cake-y glazed one?

The Substance materials do not wrap particularly intuitively on the various shape prims. These require touchup. (And even with touchup, mine do not look very appetizing, much less real.) I also exported 3d objects from Photoshop but could not find one that would “take” inside the 3d space in Illustrator.

Free transforms

Stylizing

Warping

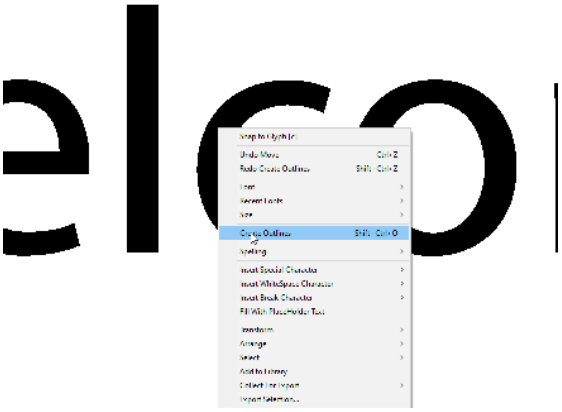
Art materials simulations

Brush strokes

Pixelation

Textures

Reshaping fonts (by turning these to outlines)



Next Steps and References

Illustrator Tutorials (by Adobe)
https://helpx.adobe.com/illustrator/tutorials.html

The LinkedIn (formerly Lynda.com) resources from the Manhattan Public Library also offer several excellent Adobe Illustrator trainings. This requires an active MPL library card, which is free for all residents.

Presenter:

Dr. Shalin Hai-Jew
Instructional Design, Research
ITS - K-State
Hale Library
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Updated: April 2023

INTRODUCTION TO

GRAPHIC *Designing*

WRITTEN BY

JERSEY DAVIS

INDEX

This eBook aims to provide a comprehensive guide to graphic design, covering both the fundamental principles and advanced techniques used by professional graphic designers. It also emphasizes the importance of ethical and responsible design practices in today's world. Whether you're a beginner looking to start a career in graphic design or an experienced designer seeking to enhance your skills, "Introduction to Graphic Design: A Comprehensive Guide" is your go-to resource for all things graphic design.

ABOUT THE BOOK

In a world where visual communication reigns supreme, the art of graphic design stands as the gatekeeper of creativity, message, and identity. "Design Unveiled" invites you to embark on a journey into the vibrant realm of graphic design, where you'll uncover the secrets to crafting visual excellence that captivates, informs, and inspires.

A COMPREHENSIVE GUIDE TO DISCOVER

- The Fundamentals of Graphic Design
- The Art of Composition and Layout
- The Power of Color and Typography
- Design Principles for Print and Digital Media
- Navigating the Digital Design Landscape
- Mastering Logo Design and Branding

Whether you're an aspiring designer looking to sharpen your skills or a seasoned professional seeking new horizons, "Design Unveiled" is your passport to unlocking the transformative potential of design.

Packed with practical insights, case studies, and real-world examples, this book is your trusted companion in the world of visual creativity.

Join us on this journey of design exploration, and let your creativity soar as you unveil the secrets of visual excellence!

-Jersey Davis
Author, Designer, Visionary

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INTRODUCTION

The World of Graphic Design :

Welcome to the exciting and visually stimulating world of graphic design! In this chapter, we'll embark on a journey to explore the multifaceted realm of graphic design, understanding its significance, history, and the pivotal role it plays in our modern, interconnected world. Whether you're a budding designer eager to dive into this creative field or someone simply curious about the magic that happens behind the visuals you encounter daily, you're in for an illuminating experience.

The Essence of Graphic Design:

Graphic design is the art and practice of visually communicating ideas, messages, and information through a combination of images, typography, colors, and layout. At its core, graphic design is about making information and concepts not only visually appealing but also easily comprehensible. It's the fusion of creativity and functionality, aesthetics and purpose.

Imagine a world without graphic design—it's hard, isn't it? From the packaging of your favorite cereal to the logo on your smartphone, from the websites you visit daily to the books you read, graphic design surrounds us. It's a silent but influential force that shapes our perceptions, conveys emotions, and facilitates communication.

The Evolution of Graphic Design :

Graphic design has a rich and evolving history. It's a journey that spans centuries, from the earliest cave paintings and hieroglyphics to the digital age of today. Let's take a quick tour through some key milestones:

Ancient Communication: Humans have been using visuals to communicate for thousands of years, whether it was through cave paintings, ancient manuscripts, or intricate tapestries.

The Printing Revolution: The invention of the printing press by Johannes Gutenberg in the 15th century transformed the spread of information, giving rise to typography and the concept of mass communication.

Art Nouveau and Art Deco: These design movements in the late 19th and early 20th centuries introduced new aesthetics, emphasizing organic shapes and intricate patterns (Art Nouveau) and streamlined, geometric designs (Art Deco).

Modernism and Bauhaus: The 20th century brought forth Modernism, characterized by simplicity and functionality. The Bauhaus school in Germany played a pivotal role in shaping modern design principles.

Digital Revolution: The advent of computers and software in the latter half of the 20th century revolutionized graphic design, making it more accessible and versatile than ever before.

CHAPTER 1

Getting Started with Graphic Design

This chapter is your initiation into the practical aspects of becoming a graphic designer. Whether you're an aspiring professional or someone looking to enhance their design skills, we'll cover the foundational knowledge and tools you need to embark on this creative journey.

What is Graphic Design?

Before diving into the technicalities, let's revisit the essence of graphic design. At its core, graphic design is about visually communicating ideas, information and emotion.

Essential Tools and Software

1.Computer: Graphic design is a digital endeavor, so a computer or laptop is your primary tool. Ensure it has the necessary hardware and software specifications to run graphic design applications smoothly.

2.Design Software: Familiarize yourself with industry-standard graphic design software like Adobe Creative Cloud, which includes Photoshop (for image editing), Illustrator (for vector graphics), and InDesign (for layout and print design). These tools offer vast creative possibilities and are widely used in the industry.

Essential Tools and Software

3.Tablet and Stylus: For digital illustration and fine-tuned control, consider investing in a graphic tablet and stylus. This setup mimics the experience of drawing on paper and is invaluable for artists.

4.Design Hardware: A high-resolution monitor, graphics card, and adequate storage are essential for handling large design files and ensuring accurate color representation.

5.Color Management: Calibrate your monitor to ensure accurate color representation. Color accuracy is crucial in graphic design, especially for print projects.

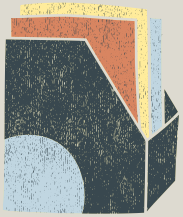
Setting Up Your Workspace

Your workspace plays a significant role in your productivity and creativity as a graphic designer. Here are some tips to create an optimal design environment:

1.Ergonomic Setup: Invest in an ergonomic chair and desk to maintain good posture during long design sessions.

2.Natural Lighting: If possible, position your workspace near a window to benefit from natural light. Adequate lighting reduces eye strain and enhances color perception.

3.Organization: Keep your workspace clutter-free and well-organized. A cluttered desk can lead to distractions and hinder your creative flow.



4.Inspiration Board: Create an inspiration board with printed designs, color swatches, and artwork that inspire you. This can serve as a visual reference and boost your creativity.

5.Backup and Storage: Set up a reliable backup system to protect your design files. Consider cloud storage solutions or external hard drives for redundancy.

As you embark on your graphic design journey, remember that it's a combination of creativity, technical skill, and a passion for visual storytelling. The tools and knowledge you've gained in this chapter are your foundation. In the chapters ahead, we'll delve deeper into the principles of design, helping you harness your creativity to create visually compelling and effective designs. So, roll up your sleeves, fire up your design software, and let's dive into the exciting world of graphic design!

CHAPTER 2

Understanding Design Principles :

Design is an art form deeply rooted in principles that guide every creative decision. To become a proficient graphic designer, it's crucial to grasp these fundamental design principles. In this chapter, we will delve into the core principles that underpin effective graphic design. Understanding and applying these principles will be the cornerstone of your design journey.

Balance: The Foundation of Composition

Balance is the distribution of visual weight in

a design. Achieving balance is vital because it ensures that your design feels stable and harmonious. There are two primary types of balance:

Symmetrical Balance: In symmetrical balance, elements on one side of a design mirror those on the other side, creating a sense of equilibrium. It's like folding a design in half, and both sides look identical. Symmetry often conveys a sense of formality and order.

Asymmetrical Balance: Asymmetrical balance involves distributing visual weight unevenly but in a way that still achieves equilibrium. It relies on contrast, scale, and color to create balance. Asymmetry often conveys dynamism and interest.

Understanding these design principles is fundamental to creating compelling and effective graphic designs. They provide a framework for making deliberate and informed decisions in your work. As you continue your journey as a graphic designer, remember that these principles are not rigid rules but tools that you can adapt and combine creatively to communicate your message and engage your audience effectively.

In the following chapters, we'll apply these principles to various design elements and projects, giving you practical insights and hands-on experience in the world of graphic design.

CHAPTER 3

Graphic Design Elements :

The world of graphic design is a playground of creativity where designers combine various elements to convey messages, tell stories, and create visual experiences. In this chapter, we'll explore the essential graphic design elements that form the building blocks of any design project. Understanding these elements and how they interact is key to crafting effective and visually appealing designs.

1. Shapes and Lines: The Basics of Form

Shapes and lines are fundamental elements in graphic design that help define the form and structure of your compositions.

2. Images and Illustrations: Visual Storytelling Images and illustrations are potent tools for conveying messages and evoking emotions.

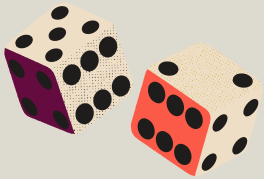
3. Icons and Symbols: Visual Communication in Miniature-Icons and symbols are simplified representations of objects, concepts, or actions. They are effective for conveying information quickly and universally.

4. Text and Typography: The Power of Words Text and typography are essential elements for delivering messages and conveying information.

5. Color: Evoke Emotions and Create Visual Impact-Color is a potent tool for setting the

tone, evoking emotions, and creating visual impact.

6. Texture and Patterns: Adding Depth and Interest-Texture and patterns can add depth, tactility, and visual interest to your designs.



As you continue your journey in graphic design, experimenting with these elements and learning how to combine them creatively will be your path to becoming a proficient and versatile designer. In the following chapters, we'll explore how to apply these elements in various design projects and contexts.

CHAPTER 4

Designing for Print :

Print design is a timeless and essential aspect of graphic design. Whether you're creating business cards, brochures, posters, or any other printed material, understanding the principles of designing for print is crucial. In this chapter, we'll delve into the key considerations, best practices, and techniques for creating stunning print designs.

1. Understanding the Medium - Print design differs significantly from digital design. In the print world, you're dealing with tangible materials, and the final product depends on

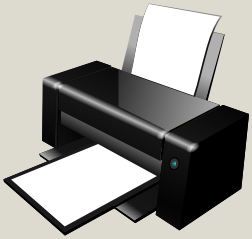
factors like paper type, ink, and printing methods.

2. Typography for Print - Typography plays a critical role in print design, impacting readability and aesthetics.

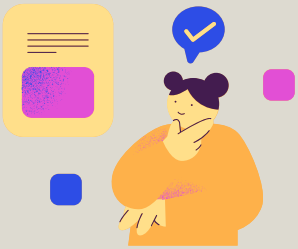
3. Layout and Grids - An organized layout is essential for print designs.

4. Color Considerations - Color is a vital aspect of print design.

6. Printing Techniques - Familiarize yourself with various printing techniques and finishes.



7. Print Proofing - Before mass production, always request a print proof. This is a sample of your design to ensure colors, layout, and overall quality meet your expectations.



Keep in mind that practice and collaboration with professional printers will further refine your print design expertise.

CHAPTER 5

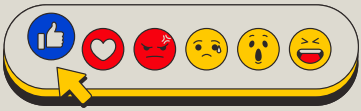
Web and Digital Design :

Web and digital design have become integral aspects of the modern world. As more of our daily activities and interactions move online, the role of web and digital designers has never been more crucial. In this chapter, we will explore the unique challenges, principles, and best practices that define web and digital design.

I. The Digital Landscape - Before diving into the specifics of digital design, it's essential to understand the digital landscape.

2. Principles of Web Design - Effective web design goes beyond aesthetics; it's about creating an optimal user experience.

3. Visual Design in Digital Media - Visual design remains a crucial aspect of digital design.



4. User Interface (UI) and User Experience (UX) Design - UI and UX design are closely intertwined and are central to digital design.

5. Mobile App Design - Designing for mobile apps requires specific considerations.

6. Prototyping and Testing - Prototyping allows you to create interactive models of

your designs to test usability and functionality.

7. Accessibility and Inclusivity - Digital designers must prioritize accessibility to ensure that their designs can be used by people of all abilities.

As you embark on your journey in digital design, keep in mind that successful designs are not just visually appealing but also functional and user-centric. By following best practices, staying up to date with emerging trends and technologies, and remaining open to user feedback, you'll be well-equipped to create engaging and effective digital experiences in an ever-changing digital landscape.

CHAPTER 6

Logo Design :

A logo is the face of a brand, a visual representation of its identity and values. Logo design is both an art and a science—a creative endeavor that requires a deep understanding of a brand's essence, its target audience, and the principles of graphic design. In this chapter, we'll explore the world of logo design, from its fundamental principles to the creative process and best practices.

The Significance of Logos :

Before delving into the design process, it's crucial to understand why logos are essential:

•Branding Identity: Logos encapsulate a brand's personality, mission, and values. They serve as a shorthand for everything a brand represents.

•Recognition: A well-designed logo is instantly recognizable and memorable, making it easier for consumers to remember and trust a brand.

•Differentiation: In a competitive market, logos help brands stand out. A unique and compelling

logo can set a brand apart from its competitors.

Types of Logos :

Logos come in various forms, each with its own strengths and applications :

•Wordmark: These logos consist of the brand's name, often stylized in a unique font. Examples include Google and Coca-Cola.

•Lettermark: Lettermarks use the initials or an acronym of the brand's name to create a distinctive logo. IBM and HBO are examples.

•Symbol or Icon: Symbols or icons are abstract representations of a brand and are

instantly recognizable without text. The Nike swoosh and Apple's apple are well-known examples.

•Combination Mark: Combination marks combine a wordmark with a symbol or icon. Adidas and Burger King use combination marks.

Logo design is a complex and rewarding field that requires a deep understanding of a brand's essence and the principles of graphic design. By embracing simplicity, relevance, and memorability, and by considering color, typography, and legal aspects, you can create logos that leave a lasting impression and effectively represent a brand's identity.

CHAPTER 7

Branding and Identity :

Branding is more than just a logo or a catchy slogan - it's the essence of a business or organization. It's the emotional connection that customers form with a brand, encompassing everything from visual elements to values and personality. In this chapter, we'll explore the multifaceted world of branding and identity, from its fundamental concepts to its practical applications.

What is Branding?

At its core, branding is the process of creating a unique and consistent identity for a

business, product, or organization. It's about shaping how the world perceives and interacts with that.

The Branding Process :

Creating a strong brand involves a systematic process :

•Research: Understand the business, its competitors, and the target audience. Identify opportunities for differentiation and alignment with values.

•Brand Strategy: Develop a brand strategy that outlines the brand's mission, vision, values, and positioning in the market.

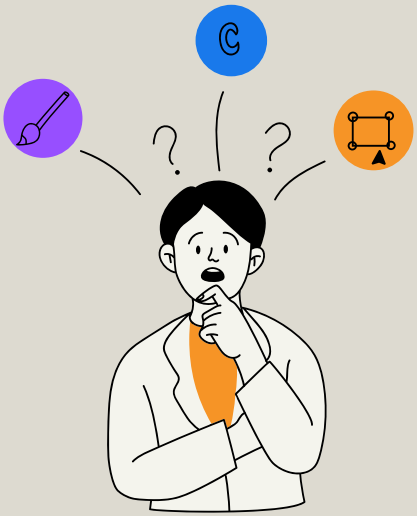
•Visual Identity: Create the visual elements that represent the brand, including the logo, color palette, typography, and imagery.

•Messaging: Develop a consistent brand voice and messaging that align with the brand's values and resonate with the target audience.

•Implementation: Roll out the brand across all touchpoints, including marketing materials, websites, social media, and physical spaces.

•Consistency: Ensure that all brand materials and messaging maintain consistency across different media and platforms.

Rebranding :



Rebranding involves refreshing or reinventing a brand's identity. It can be driven by smooth changing market conditions, mergers and acquisitions, or evolving brand values. Rebranding requires careful planning, as it can impact existing customer perceptions.

Branding and identity are fundamental aspects of any business or organization. A strong and consistent brand can build trust, set you apart from competitors, and resonate with your target audience.

CHAPTER 8

Illustration and Digital Art :

Illustration and digital art are vibrant and versatile forms of visual expression that have transcended traditional mediums. In this chapter, we'll explore the world of illustration and digital art, from the historical context and foundational principles to the modern digital tools and techniques that have revolutionized the field.

Traditional vs. Digital Illustration

Traditional illustration involves physical mediums like pencils, paints, and paper, while digital illustration relies on digital tools and software. Both have their merits, and many artists combine techniques from both worlds

.

•Traditional Illustration: Offers tactile and organic qualities, and it can be especially charming in its imperfections.

•Digital Illustration: Provides precision, speed, and a vast array of creative possibilities. It's the medium of choice for many contemporary illustrators.

Illustration Styles and Techniques :

Illustrators develop unique styles and techniques that set them apart. Styles can range from realistic to abstract, and techniques may involve line art, shading, pointillism, or digital brushes. Experimentation is key to discovering one's personal style.

Digital Tools and Software :

Digital artists leverage a variety of tools and software to create their work:

Graphic Tablets: Devices like Wacom tablets allow artists to draw directly onto a screen, simulating traditional drawing.

Digital Brushes: Software offers a wide array of digital brushes that mimic traditional mediums such as oil paint, watercolor, and charcoal.

Software Suites: Adobe Illustrator, Adobe Photoshop, Corel Painter, and Procreate are popular software choices for digital illustration.

Illustration and digital art are dynamic and ever-evolving forms of visual expression that offer limitless creative possibilities. Whether you're a seasoned digital artist or just starting your journey, understanding the historical context, embracing various styles and techniques, and staying abreast of digital tools and platforms will enrich your experience and allow you to create compelling and impactful visual art.

By understanding your target audience, maintaining brand consistency, creating compelling marketing collateral, embracing digital marketing, and continually optimizing your designs based on data, you can design for marketing success.

"There are three responses to a piece of design – yes, no, and WOW!
Wow is the one to aim for."

WRITTEN BY

JERSEY DAVIS

INTRODUCTION TO TYPOGRAPHY

GRAP2587 | Design for Digital Media 1

INTRO TO TYPOGRAPHY

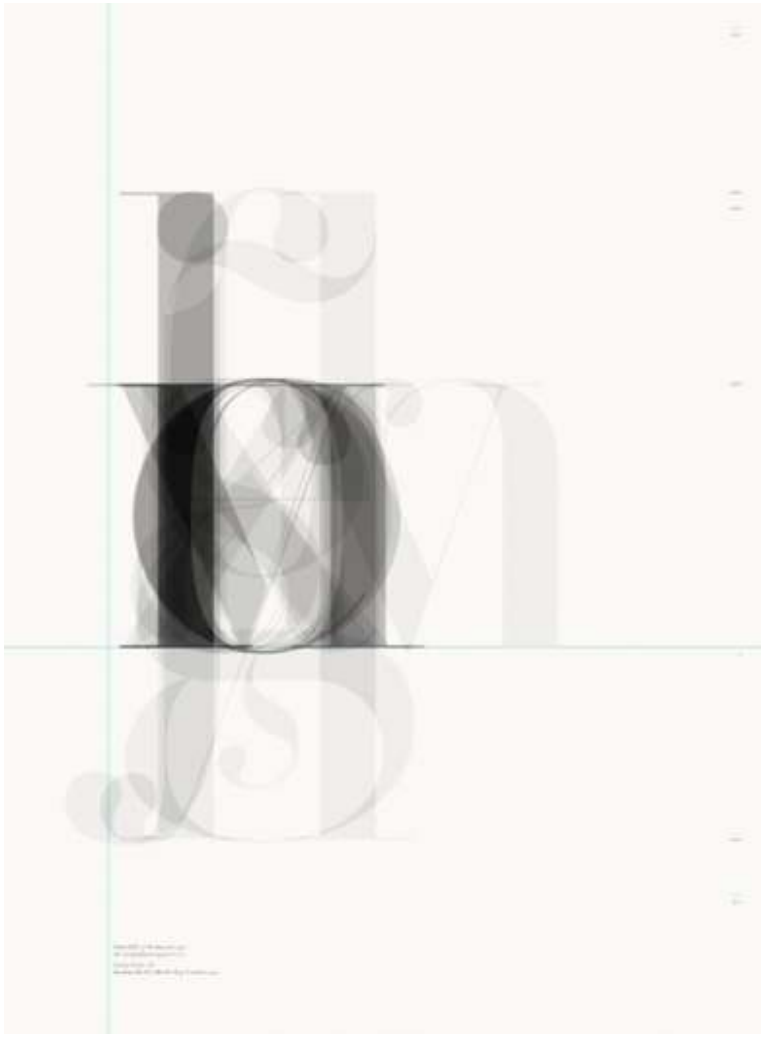
1. Understanding Typography
   - The Definition
   - Seeing Typography
   - Typography Practice
   - Font & Typeface
   - Formats

2. History of Typography
   - Sounds to Symbols
   - Evolution of Writing Systems
   - Gutenberg & Movable Type
   - Mechanization of Type
   - Photocomposition
   - The Digital Age

3. Typography in Use
   - In Communication Design
   - In Interactive Design
   - In Information Design
   - In Motion Design

4. Learning Typography
   - The Importance of Typography
   - All Principles Applied
   - How to Learn & Train Typography
   - Resources

THE DEFINITION



(Greek)

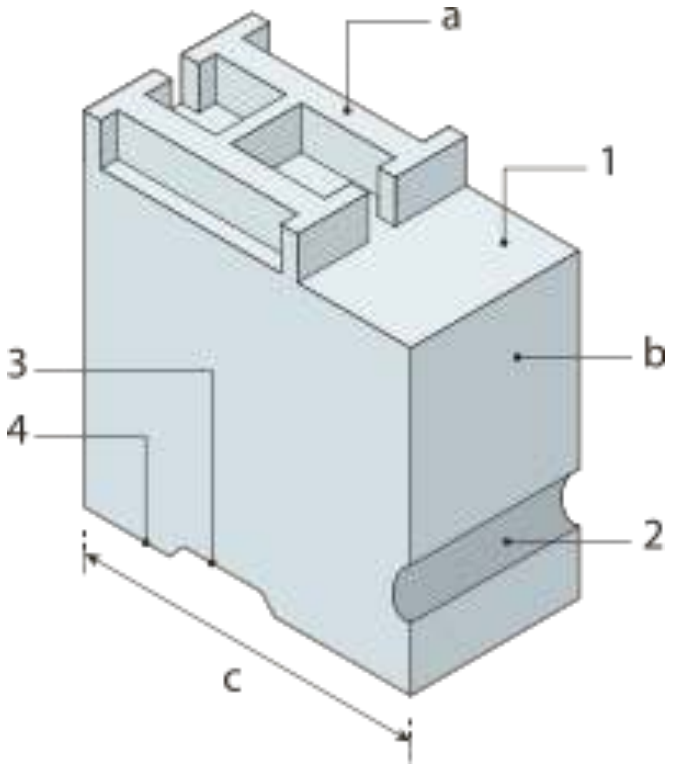
/ typos /: form

/ graphein /: to write

Typography is, simply put, the art & technique of choosing, setting & arranging type.

Good typography serves a purpose.

THE DEFINITION



SEEING TYPE

Type exists.

Type is everywhere.

Type is a fundamental part of human life & society.

SEEING TYPE

SEEING TYPE

SEEING TYPE

SEEING TYPE

SEEING TYPE

Individual letters, when arranged in a particular way, represent the sound of spoken language & visually express ideas in such a way that another person can understand them in the manner intended.

SEEING TYPE

SEEING TYPE

SEEING TYPE

TYPOGRAPHY PRACTICE

Then what exactly what typography is?

Typography engages with people on different levels:

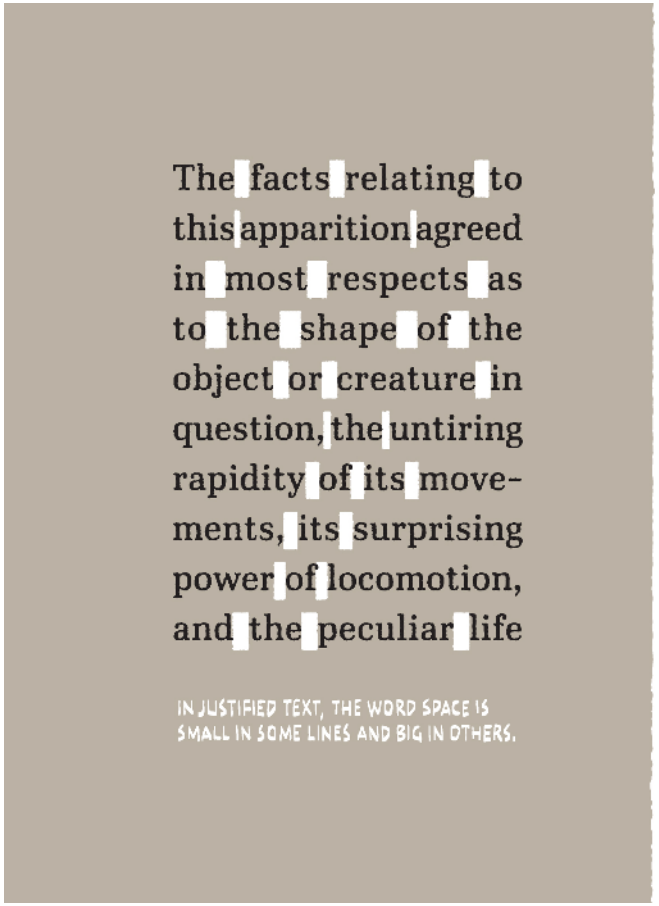
There are:

- The holistic view (macro-typography): type choice, visual impact, type choice, layout, compositions, hierarchy, whitespace, grid system (number of columns & rows), etc.
- The detailed view (micro-typography): type choice, sizes, spacing between lines, spacing between words, spacing between letters, etc.

MACRO / MICRO

macro-typography

MACRO / MICRO



When setting type, the typographer can change the word space independently from the rest of the font's built-in spacing. Unlike tracking, which should be adjusted cautiously in small increments, word space can absorb bigger changes since it's a relatively large space within the paragraph. Of course, the typographer still needs to be careful with the word space, but it's generally less delicate than letter space.

In a column of text, the typographer usually makes any word space adjustments to the whole thing at once. For example, a wide column may have a larger word space to give the words some breathing room. A narrow column may have a smaller word space to make the lines more compact. In a column of justified text, the word space can also be adjusted line by line. You've probably noticed this variation in word space in the narrow justified columns of a newspaper. Suddenly, there's a line with big gaps in it. In unjustified text, the word space stays consistent throughout the column. This will be more fully explained in the section on hyphenation and justification.

THE IMPORTANT THING IS HOW THE SIZE OF THE WORD SPACE RELATES TO THE SIZE OF THE LINE SPACE.

micro-typography

MACRO / MICRO

Stencil 90pt

Dancing Script 96pt

Comic Sans 90pt

Rosewood 96pt

macro-typography

MACRO / MICRO

Helvetica Neue 14pt / 18pt

Bacon ipsum dolor sit amet ball tip pork loin cow brisket drumstick bresaola short ribs beef jowl kielbasa pig spare ribs porchetta. Sausage pancetta t-bone chicken, swine filet mignon kevin. Boudin pork belly bacon, chuck pork loin venison ball tip prosciutto chicken leberkas jowl turducken salami swine ham. Hamburger sausage salami biltong bacon filet mignon ham hock. Tongue pork loin swine spare ribs strip steak pork chop shank biltong. Pig doner turducken filet mignon, fatback venison pork loin andouille tenderloin chicken.

Swine spare ribs frankfurter biltong. Rump brisket pork chop leberkas prosciutto hamburger shank sirloin, drumstick pastrami frankfurter jerky. Venison ball tip turducken t-bone sirloin andouille. Ham spare ribs bacon salami filet mignon pork capicola cow rump.

Source Sans Pro 14pt / 18pt

Bacon ipsum dolor sit amet ball tip pork loin cow brisket drumstick bresaola short ribs beef jowl kielbasa pig spare ribs porchetta. Sausage pancetta t-bone chicken, swine filet mignon kevin. Boudin pork belly bacon, chuck pork loin venison ball tip prosciutto chicken leberkas jowl turducken salami swine ham. Hamburger sausage salami biltong bacon filet mignon ham hock. Tongue pork loin swine spare ribs strip steak pork chop shank biltong. Pig doner turducken filet mignon, fatback venison pork loin andouille tenderloin chicken.

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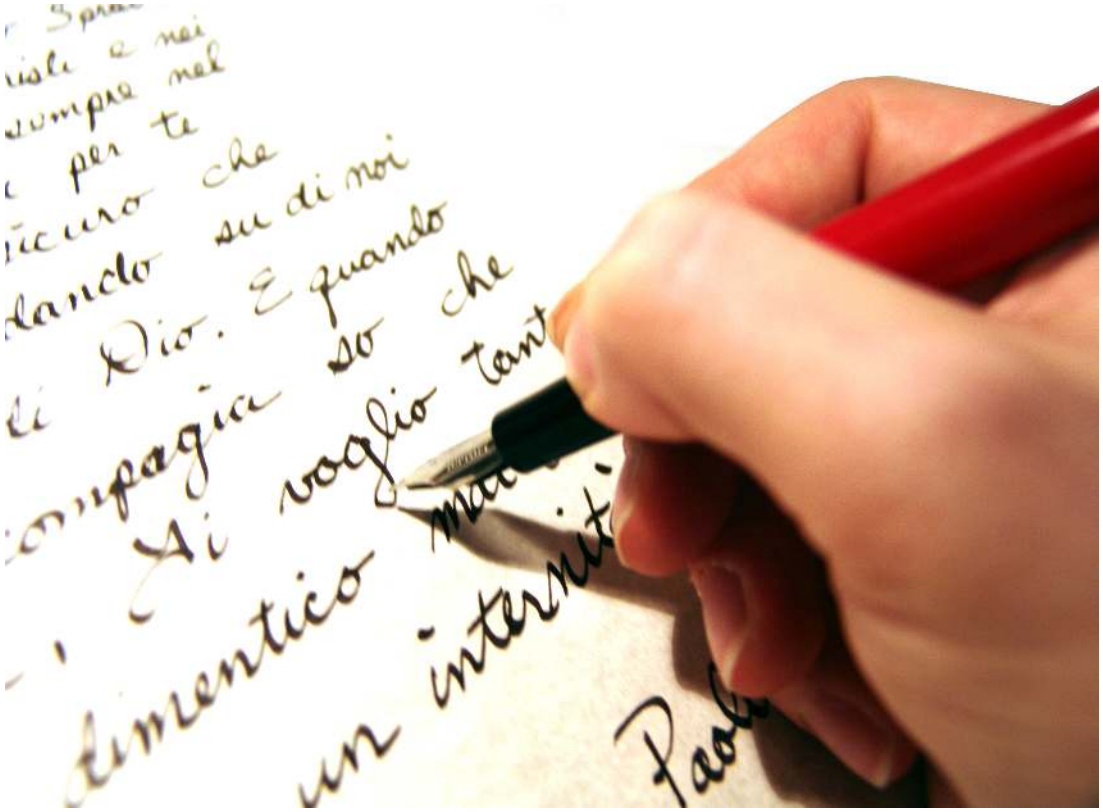
micro-typography

TYPOGRAPHY IS & ISN'T



Isn't!

TYPOGRAPHY IS & ISN'T

Is!

TYPOGRAPHY IS & ISN'T

Isn't!

TYPOGRAPHY IS & ISN'T

Isn't!

TYPEFACE / FONT / FAMILY

144 pt

NULLA PARI
Est laborum.

72 pt

IN VOLUPTATE VELIT ES
Consectetur adipisicing e

48 pt

OFFICIA DESERUNT MOLLIT ANIM ID
Fugiat nulla pariatur. Excepteur sint oc

FONT & TYPEFACE

Font: a set of letters, numbers, symbols, etc. of the same design. Examples:
Helvetica Neue Light is a Font

Typeface: a set of fonts that share common design features. Examples:
Helvetica Neue is a Typeface.

Font family: often used interchangeably with Typeface.

FONT & TYPEFACE

The physical embodiment of a collection of letters, numbers, symbols, etc. (whether it's a case of metal pieces or a computer file) is a font. When referring to the design of the collection (the way it looks) you call it a typeface.

FONT & TYPEFACE

Helvetica Neue Std

Helvetica Neue L T Std 85 Heavy

Helvetica Neue Std

Helvetica Neue L T Std 86 Heavy Italic

Helvetica Neue Std

Helvetica Neue L T Std 45 Light

Helvetica Neue Std

Helvetica Neue L T Std 46 Light Italic

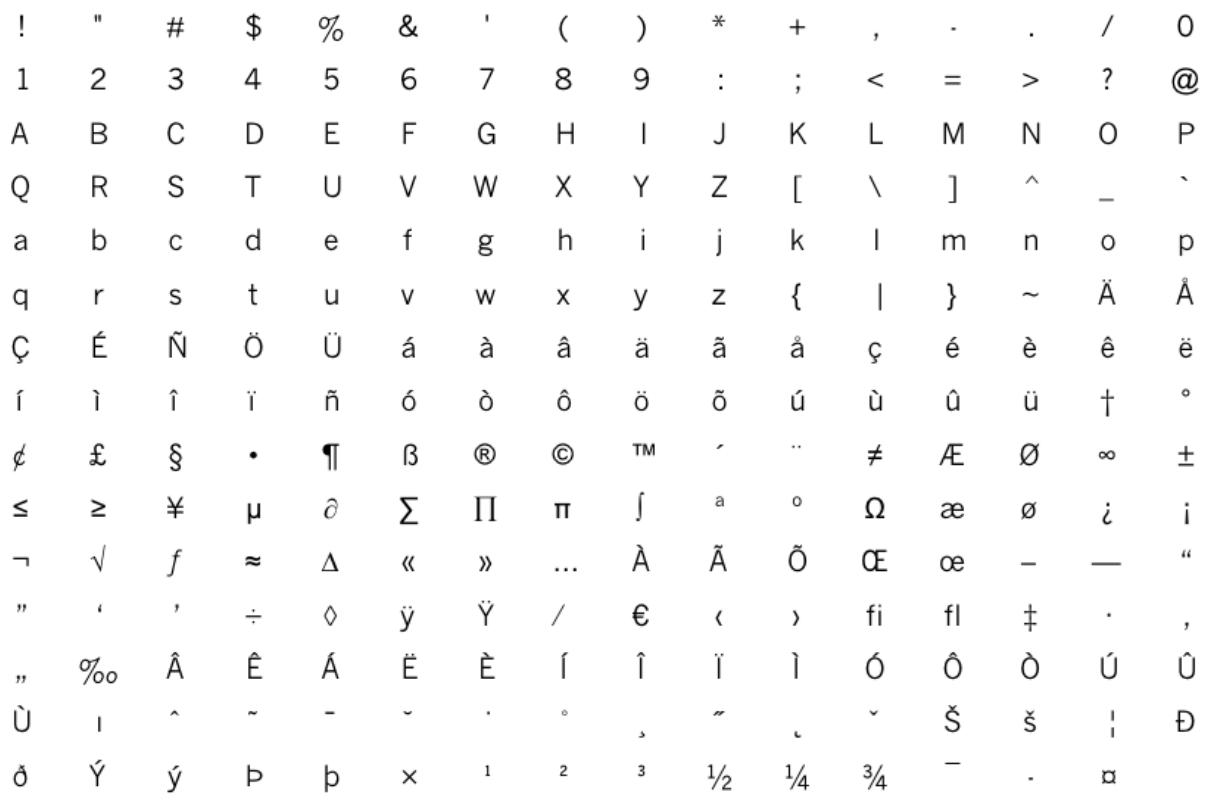
Helvetica Neue Std

Helvetica Neue L T Std 66 Medium Italic

Helvetica Neue Std

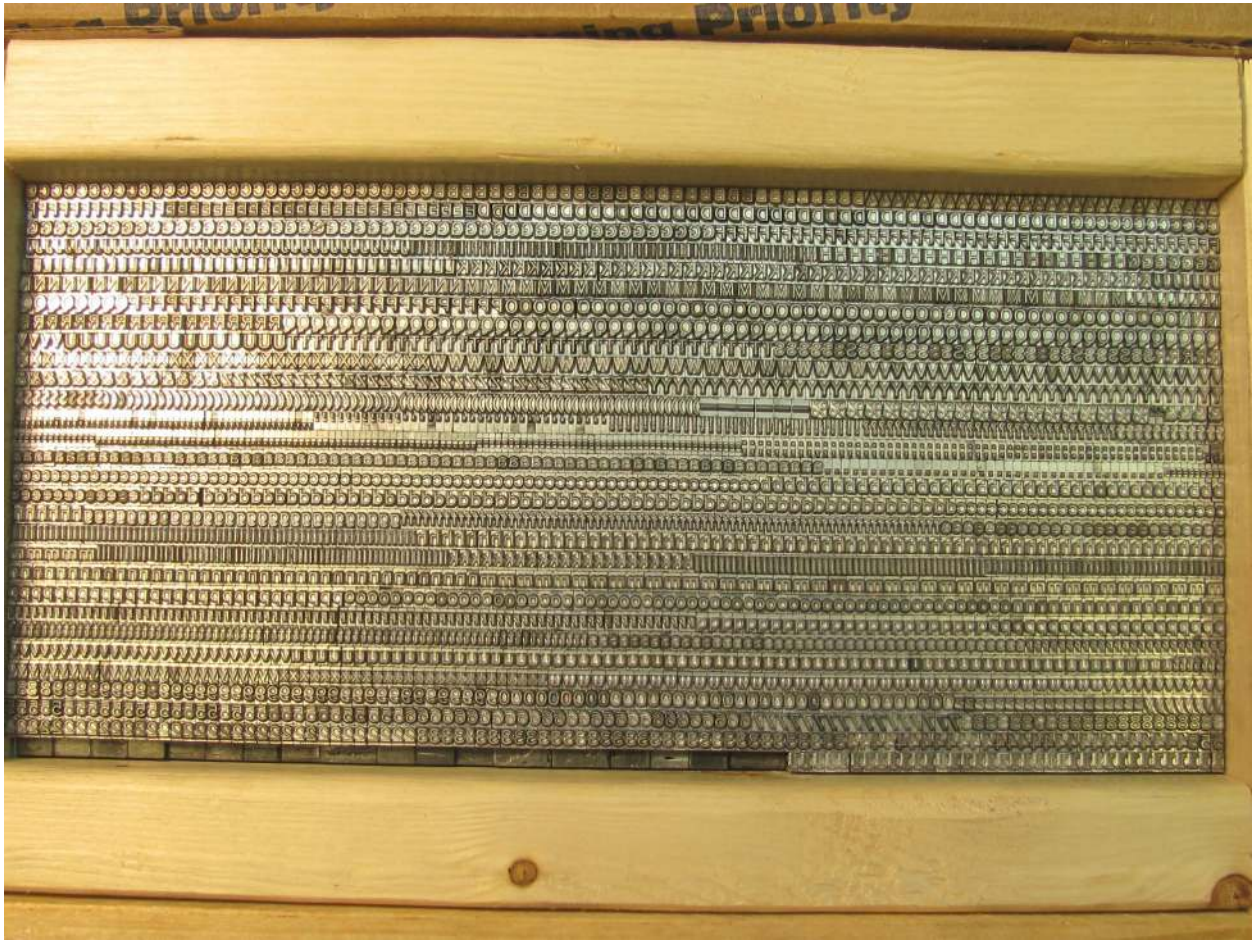
Helvetica Neue L T Std 65 Medium

FONT & TYPEFACE



typeface

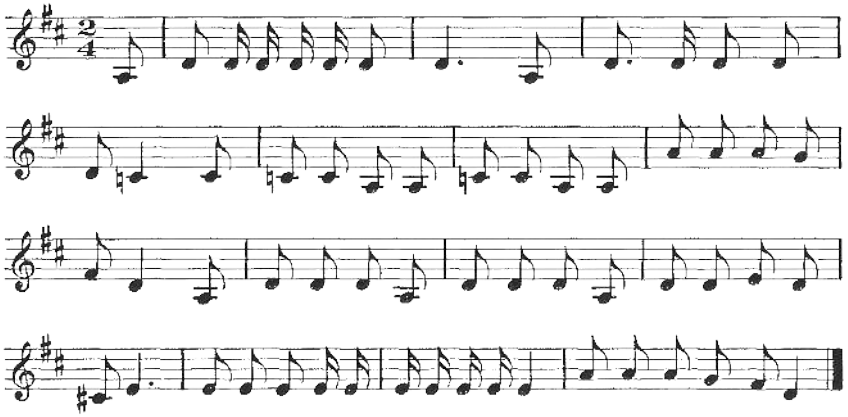
FONT & TYPEFACE



font

FONT & TYPEFACE

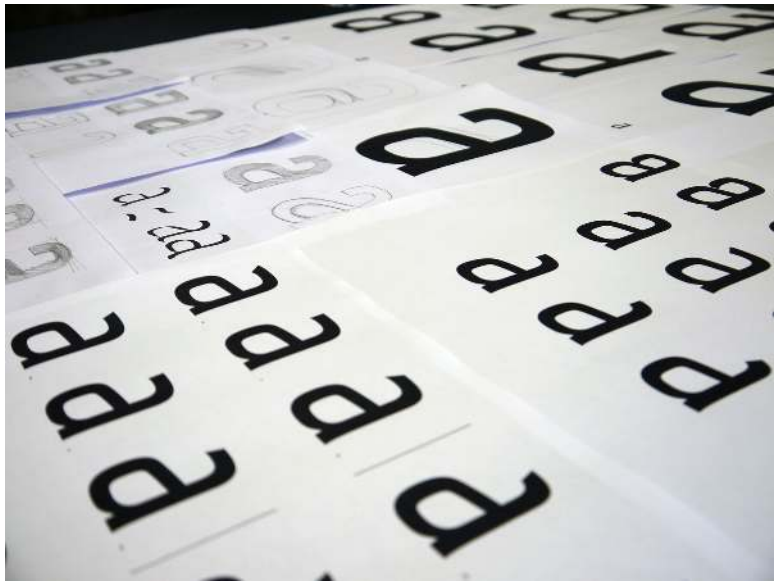
font (digital)

FONT & TYPEFACE

The way I relate the difference between typeface and font to my students is by comparing them to songs and MP3s, respectively (or songs and CDs, if you prefer a physical metaphor). – Nick Sherman

You talk about how much you like a tune, you don't say: "That's a great MP3". You say: "That's a great song". The MP3 is the delivery mechanism, not the creative work; just as in type a font is the delivery mechanism and a typeface is the creative work – Stephen Coles

MP3 AUDIO

TYPE DESIGN & DESIGNER



Type design: the art & craft of designing typefaces.

Type designer: designer of typefaces. Don't call them "font designers" (or even "font family designers". Ugh!).

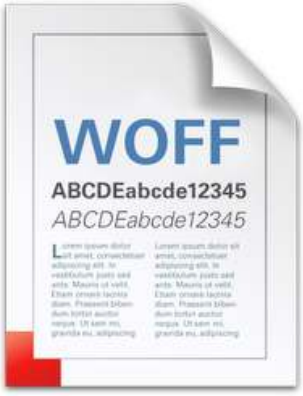
FORMATS

font.ttf

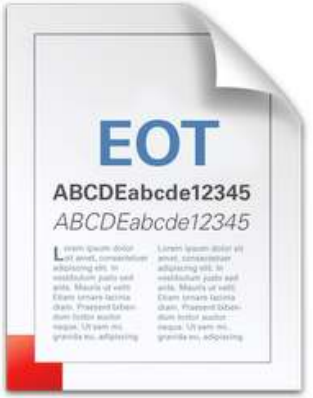
font.otf

Desktop font formats

Web font formats



font.woff



font.eot

INTRO TO TYPOGRAPHY

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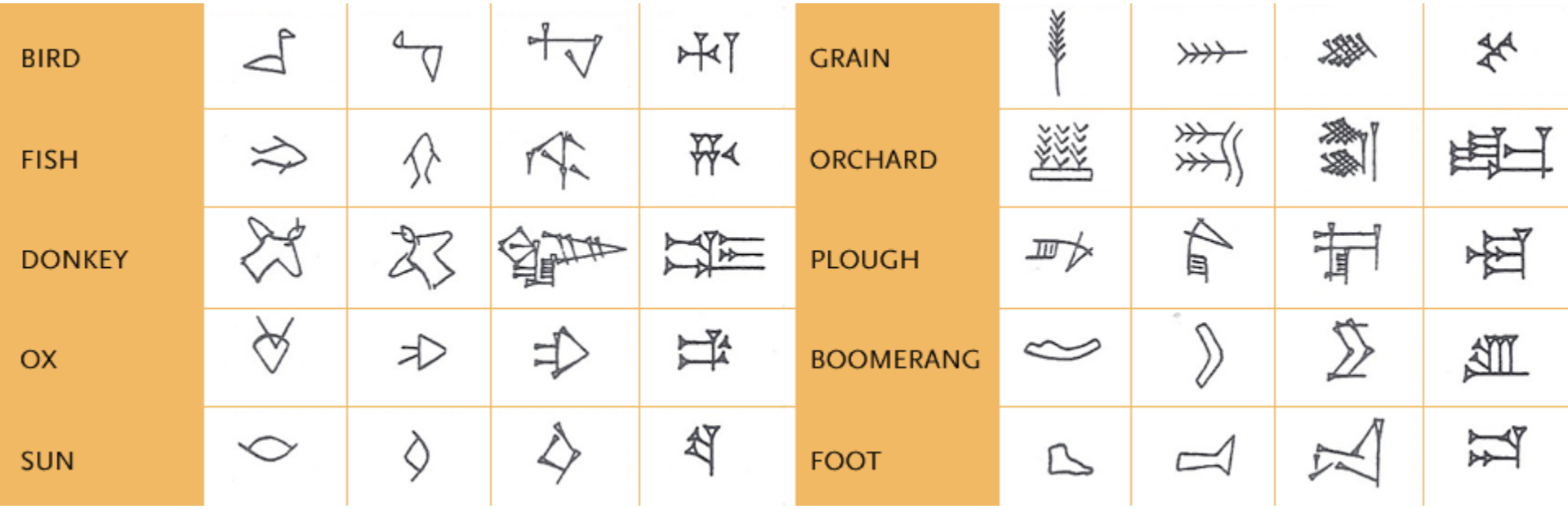
HISTORY OF TYPOGRAPHY

SOUNDS TO SYMBOLS

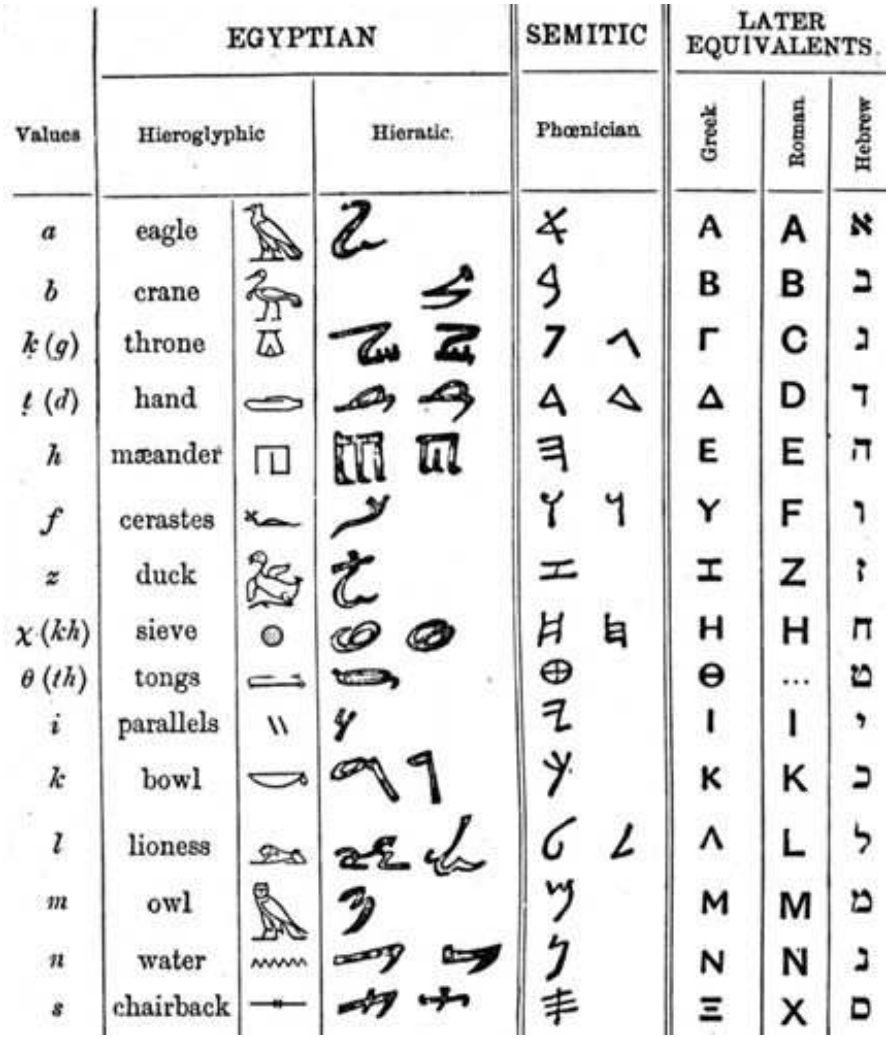
EVOLUTION OF WRITING SYSTEMS

Sumerians' Cuneiform – one of the earliest known systems of writing, c. 31st century BC to 1st century AD



EVOLUTION OF WRITING SYSTEMS

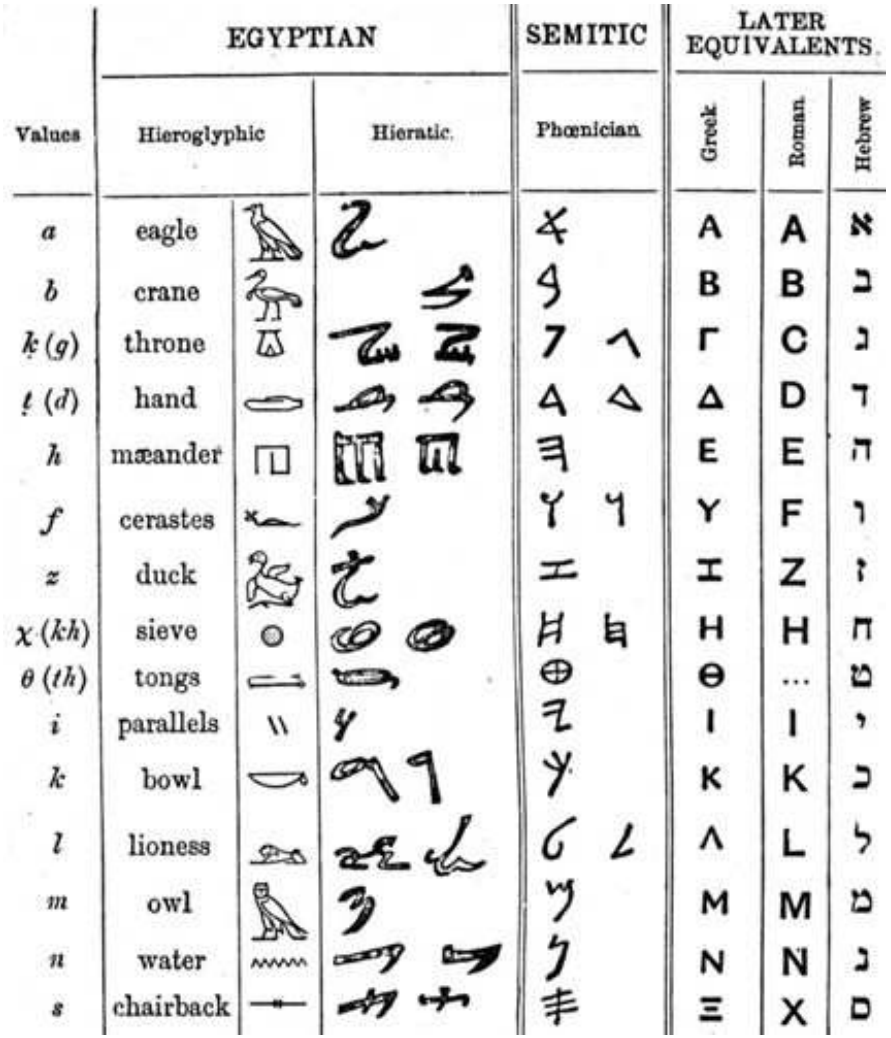
Evolution of Alphabets



Values	EGYPTIAN			SEMITIC	LATER EQUIVALENTS		
	Hieroglyphic		Hieratic.	Phœnician.	Greek.	Roman.	Hebrew.
a	eagle				A	A	א
b	crane				B	B	ב
ḳ (*g*)	throne				Γ	C	ג
ṯ (*d*)	hand				Δ	D	ד
h	mæander				E	E	ה
f	cerastes				Y	F	ו
z	duck				I	Z	ז
χ (*kh*)	sieve				H	H	ח
θ (*th*)	tongs				Θ	...	ט
i	parallels				I	I	י
k	bowl				K	K	כ
l	lioness				Λ	L	ל
m	owl				M	M	מ
n	water				N	N	נ
s	chairback				Ξ	X	ס

EVOLUTION OF WRITING SYSTEMS

Evolution of Alphabets



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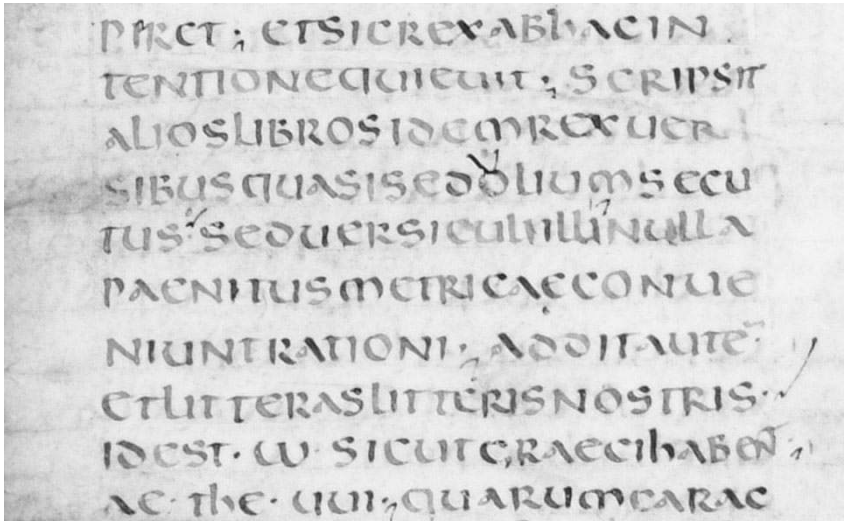
h	mæander				E	E	ה
f	cerastes				Y	F	ו
z	duck				I	Z	ז
χ (*kh*)	sieve				H	H	ח
θ (*th*)	tongs				Θ	...	ט
i	parallels				I	I	י
k	bowl				K	K	כ
l	lioness				Λ	L	ל
m	owl				M	M	מ
n	water				N	N	נ
s	chairback				Ξ	X	ס

EVOLUTION OF WRITING SYSTEMS

One of the most important contributions to early writing by the Romans: Trajan Columns

113 A.D.

EVOLUTION OF WRITING SYSTEMS

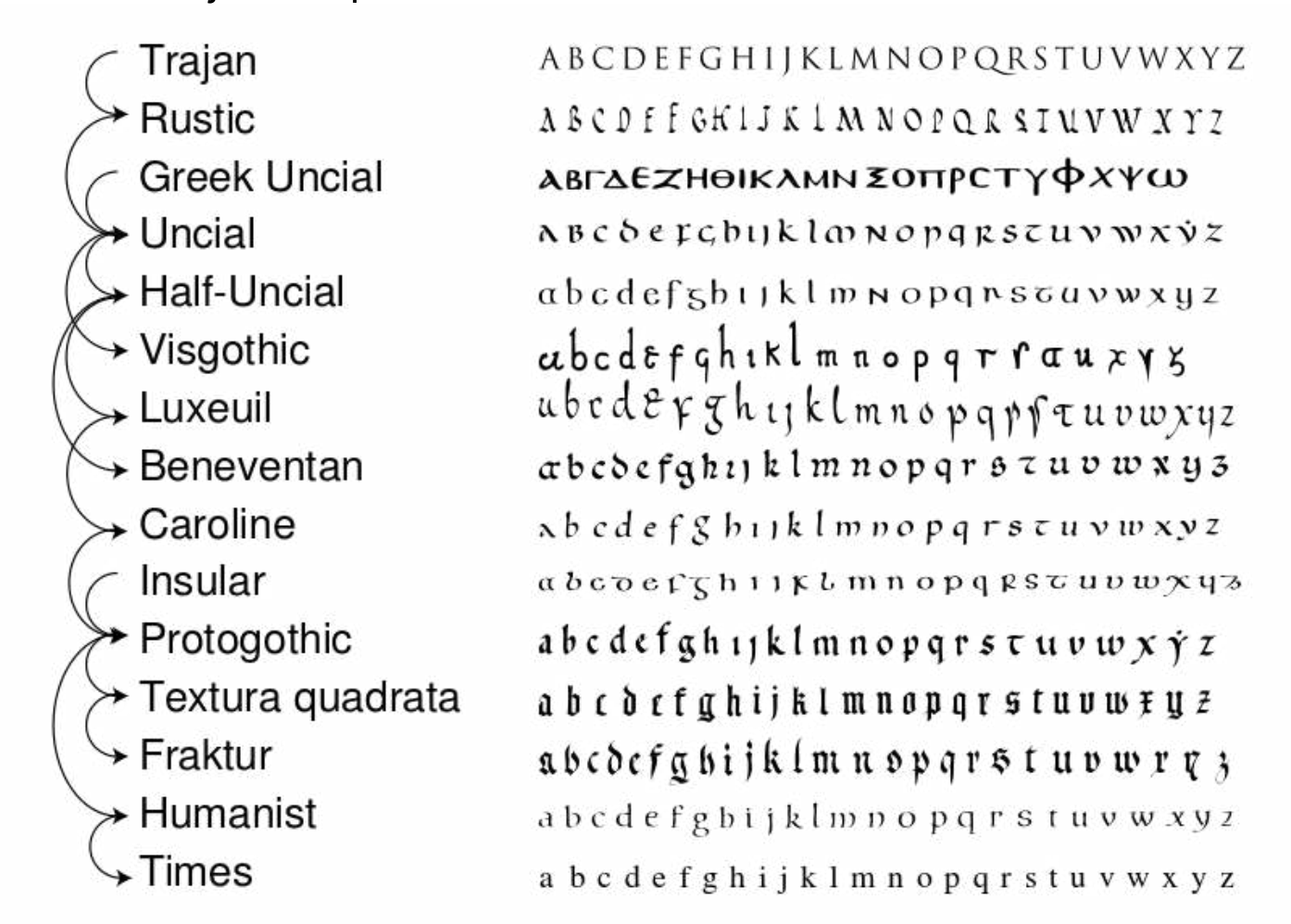


Uncial, France, 7th century.

Gothic Scripts

EVOLUTION OF WRITING SYSTEMS

17th century: the alphabet has 26 characters.



GUTENBERG & MOVABLE TYPE

Before printing:

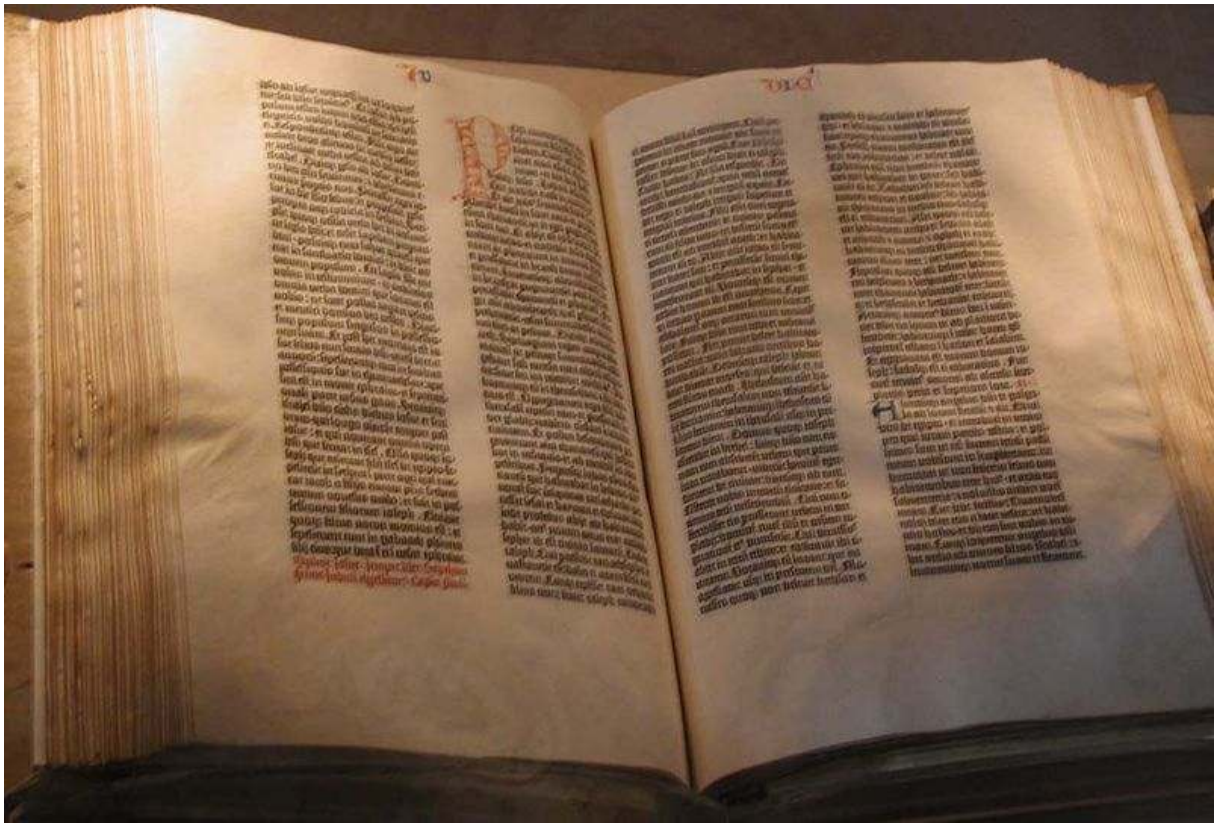
Scribes with breathtaking letterings.

GUTENBERG & MOVABLE TYPE

Johannes Gutenberg

& the first printed bible

1450 Germany



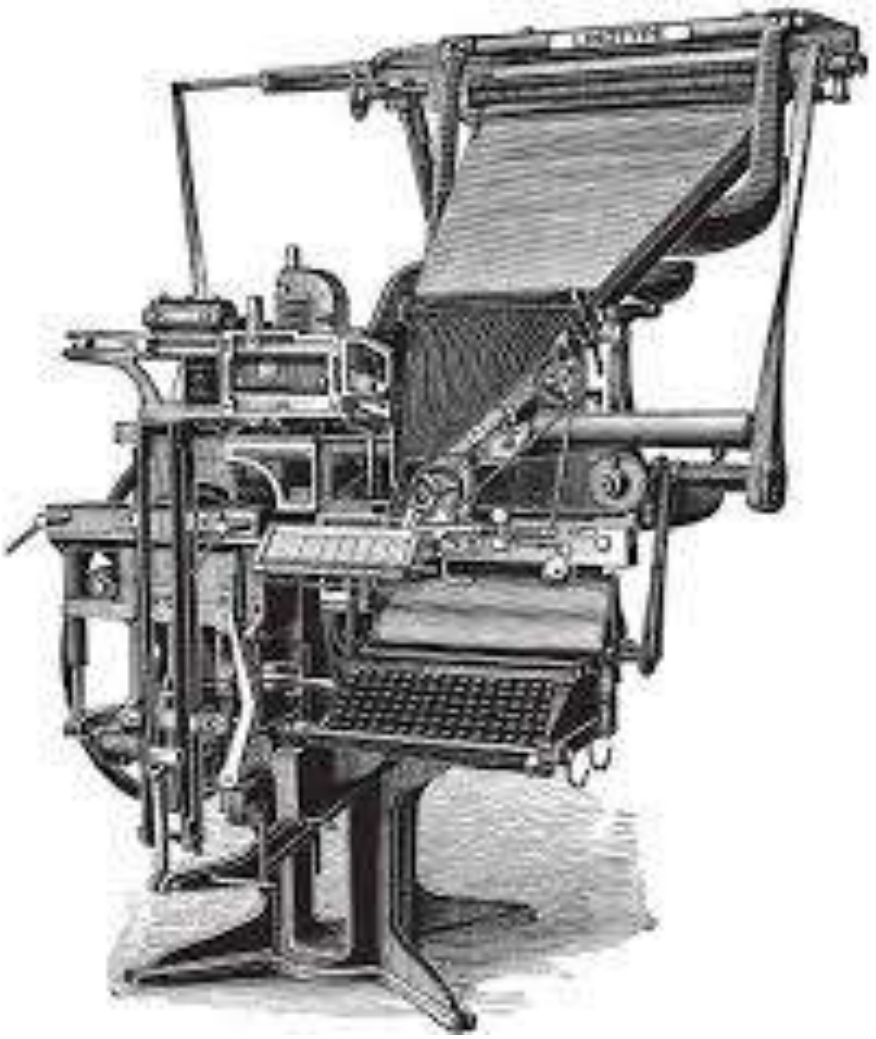
GUTENBERG & MOVABLE TYPE

MECHANIZATION OF TYPE

Typesetting machines: the result of Industrial Revolution in 19th & early 20th Century.

Linotype machine invented by Ottman Mergenthaler.

1880s



PHOTOTYPESETTING

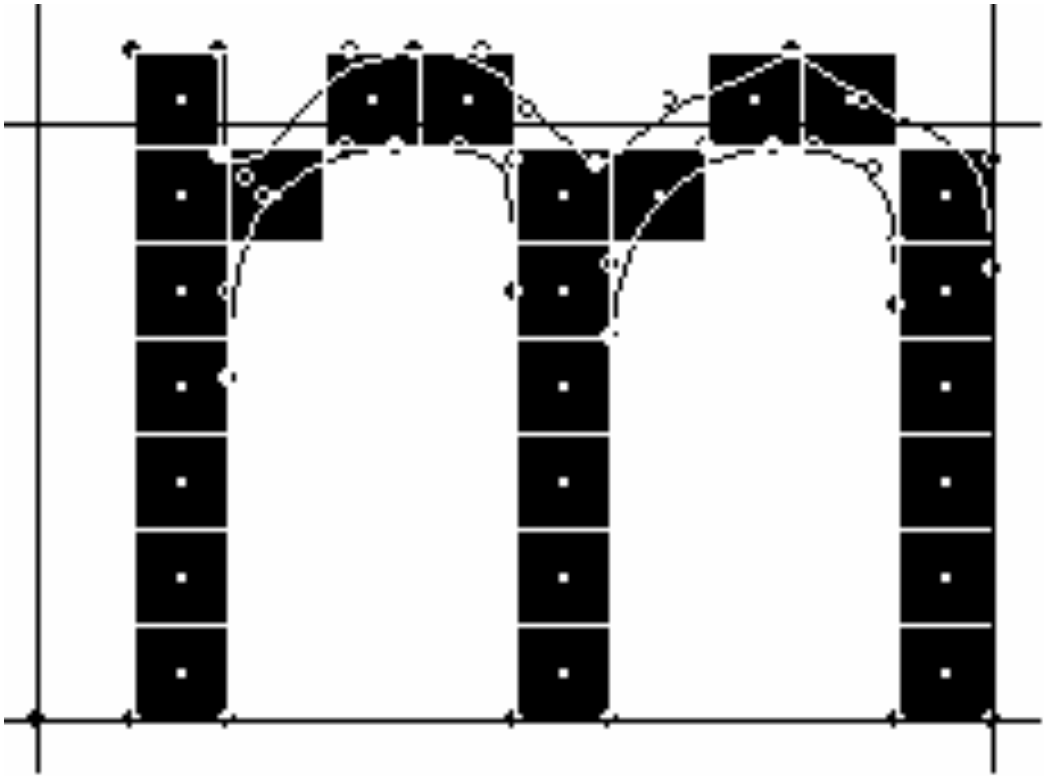
Applying photographic process to render type using “phototypesetters”

1960s – 1980s

THE DIGITAL AGE

Desktop publishing put typography into the digital world

1983 – 1984



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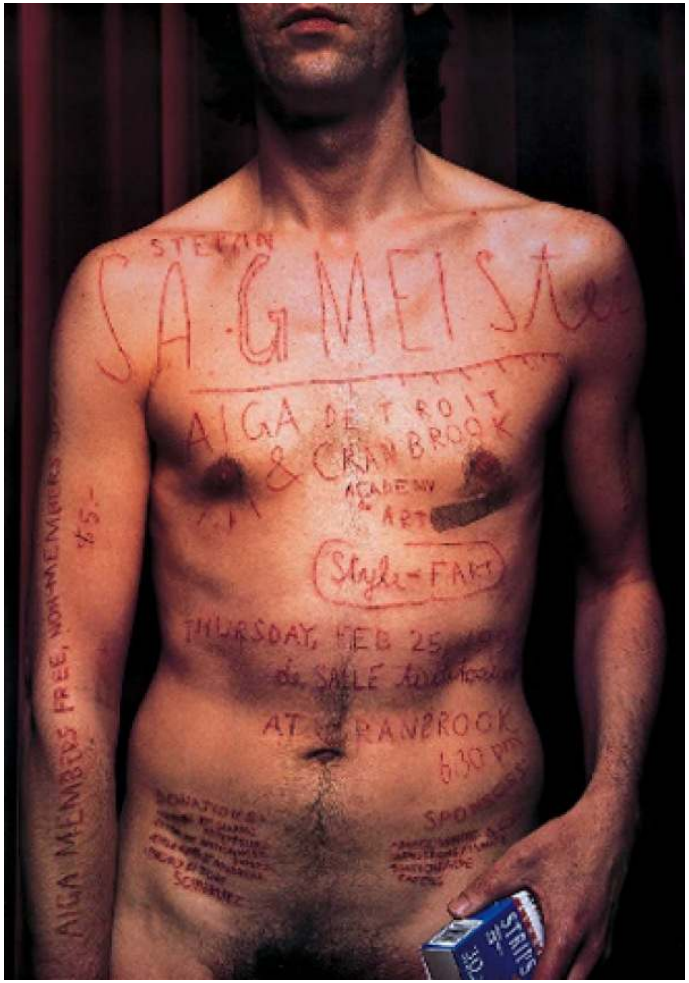
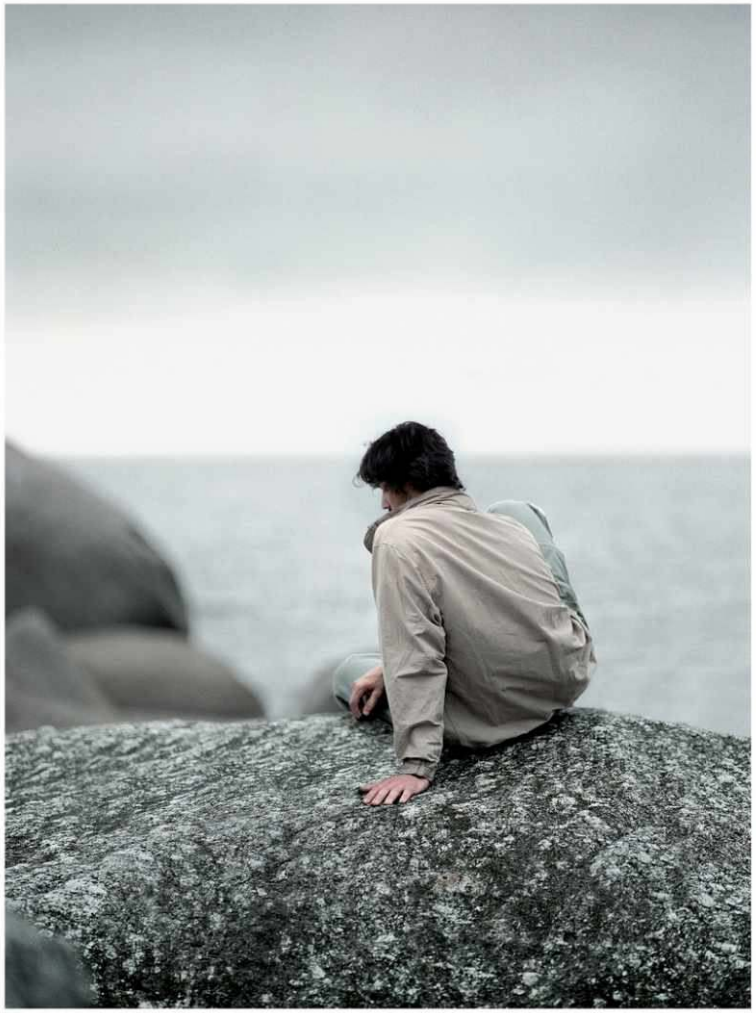
TYPOGRAPHY IN COMMUNICATION DESIGN

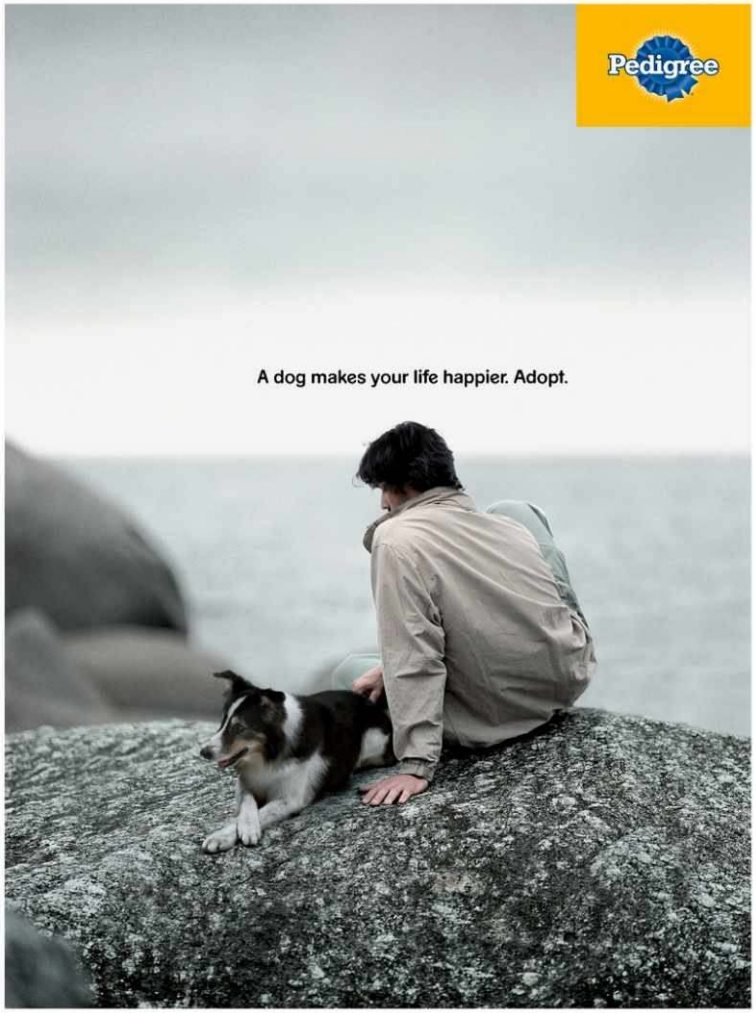


TYPOGRAPHY IN COMMUNICATION DESIGN

freedom

TYPOGRAPHY IN COMMUNICATION DESIGN

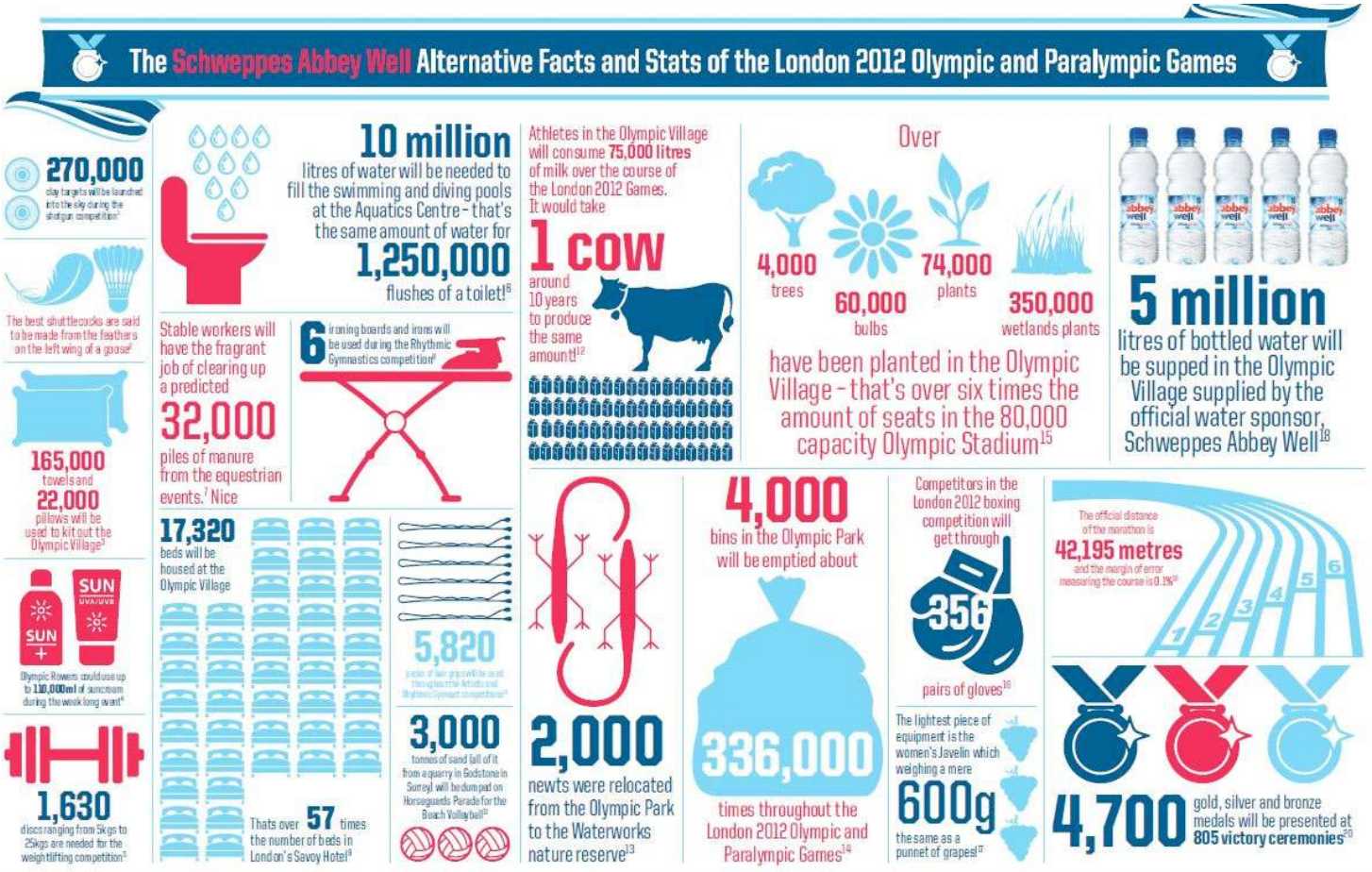




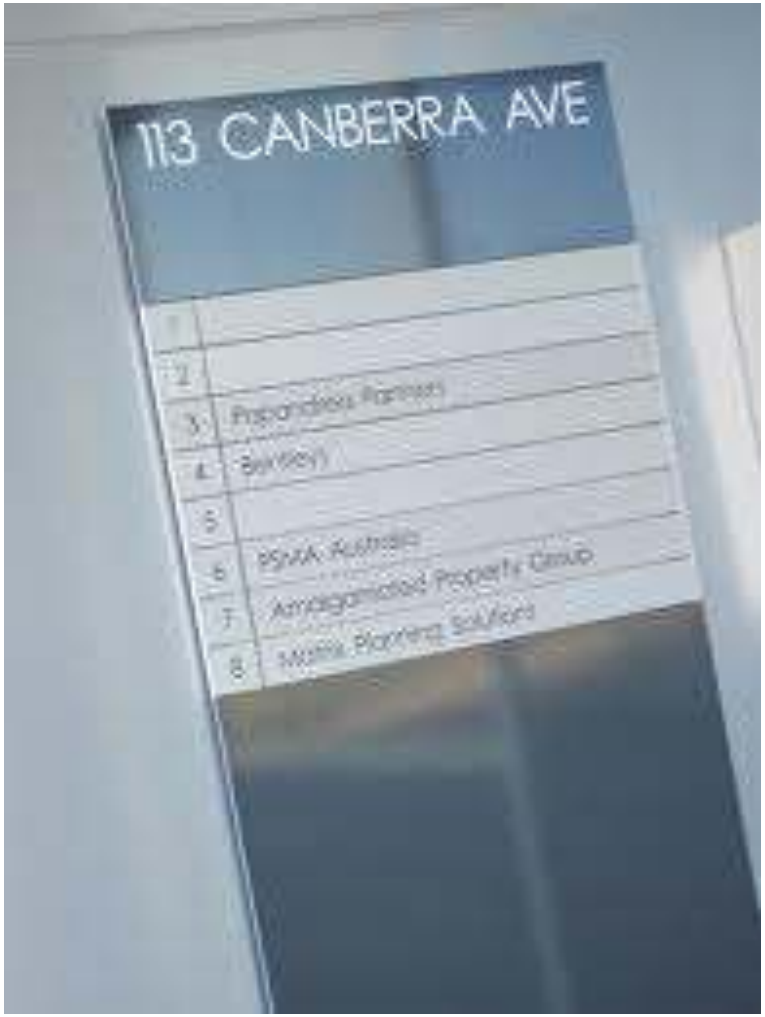
TYPOGRAPHY IN INTERACTIVE DESIGN

TYPOGRAPHY IN INTERACTIVE DESIGN

TYPOGRAPHY IN INFORMATION DESIGN



TYPOGRAPHY IN INFORMATION DESIGN



TYPOGRAPHY IN MOTION DESIGN

TYPOGRAPHY IN MOTION DESIGN

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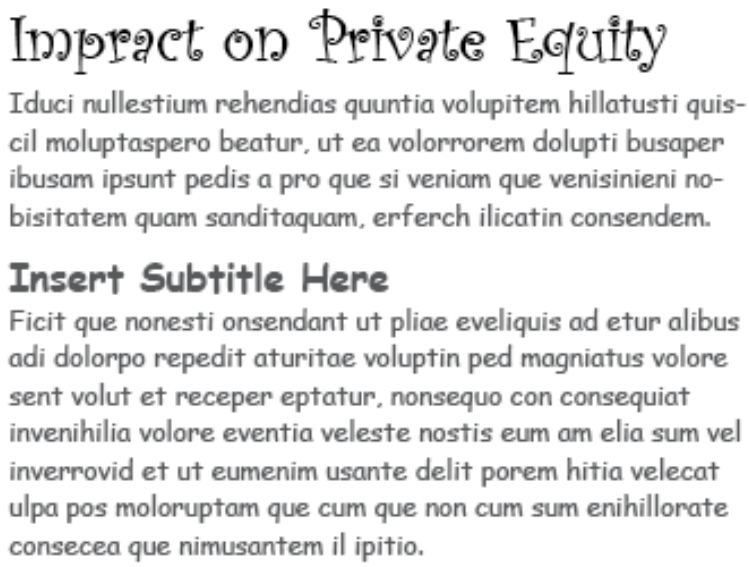
IMPORTANCE OF TYPOGRAPHY

Typography conveys emotions.

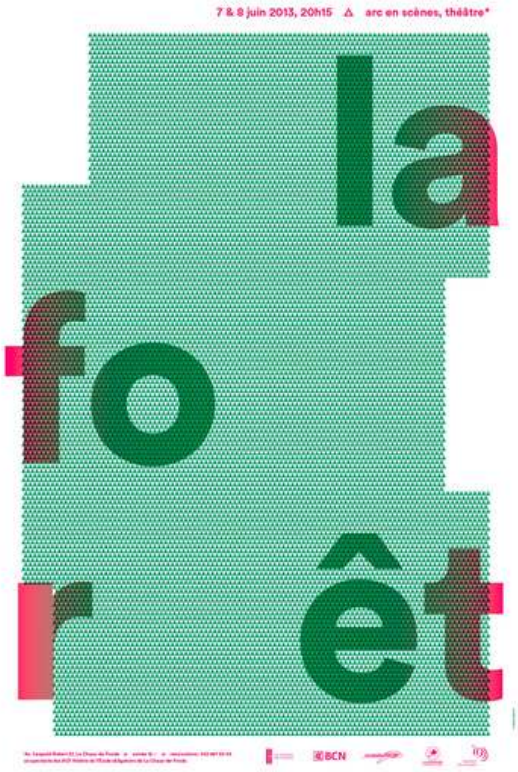


IMPORTANCE OF TYPOGRAPHY

Typography keeps people reading

IMPORTANCE OF TYPOGRAPHY

Typography is an essential part of design

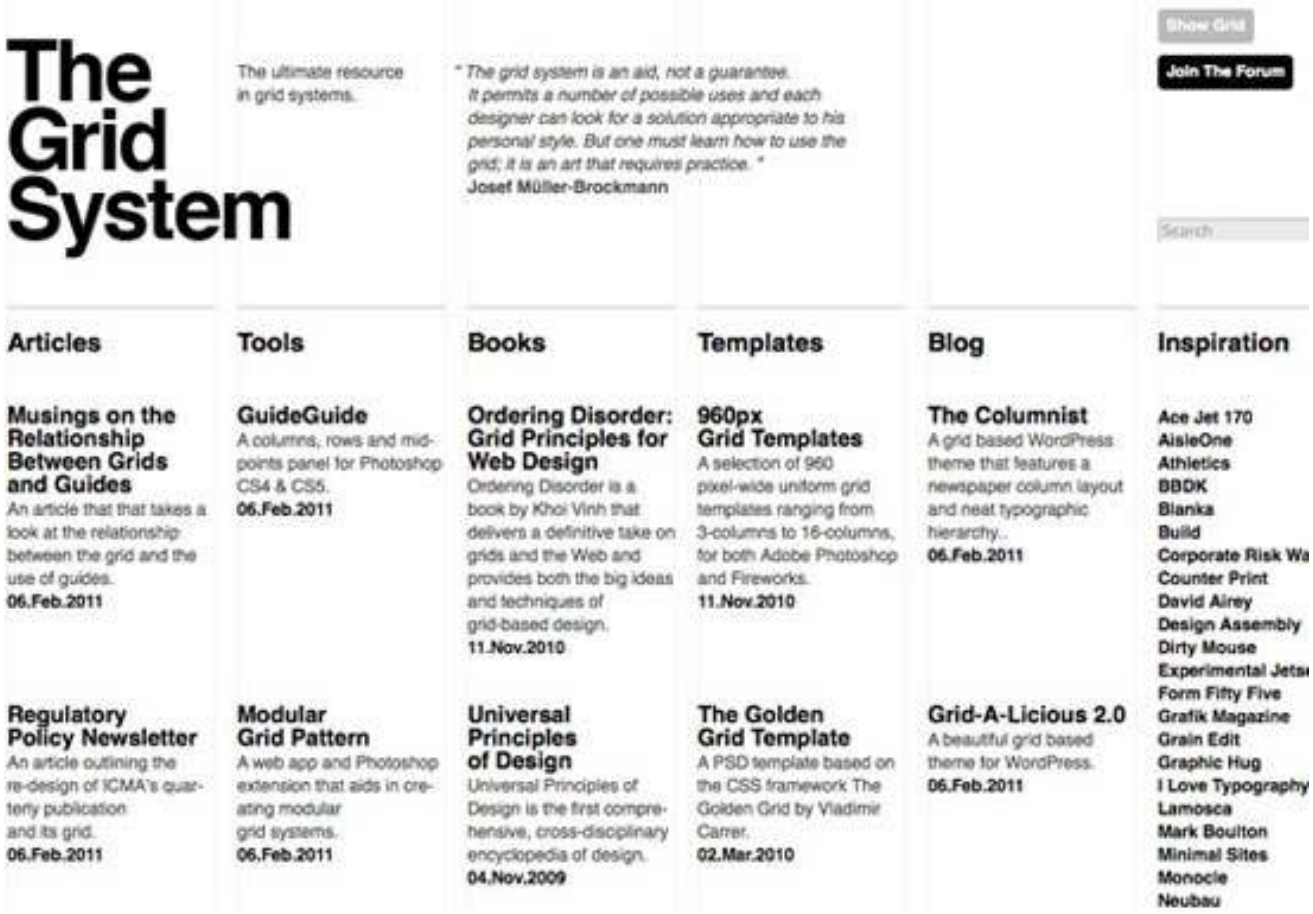


IMPORTANCE OF TYPOGRAPHY

APPLYING DESIGN PRINCIPLES

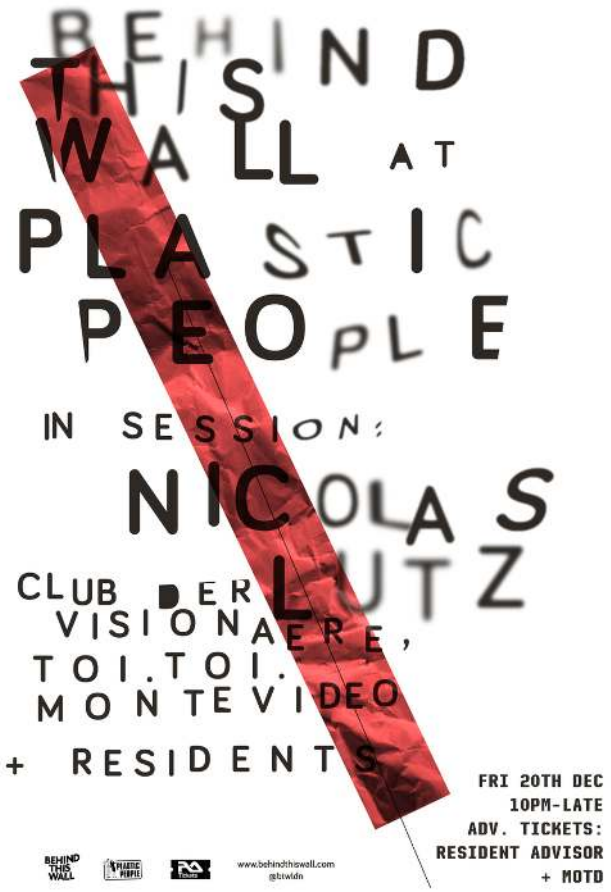
Proportion, Emphasis & Contrast

APPLYING DESIGN PRINCIPLES



Balance & Unity

APPLYING DESIGN PRINCIPLES

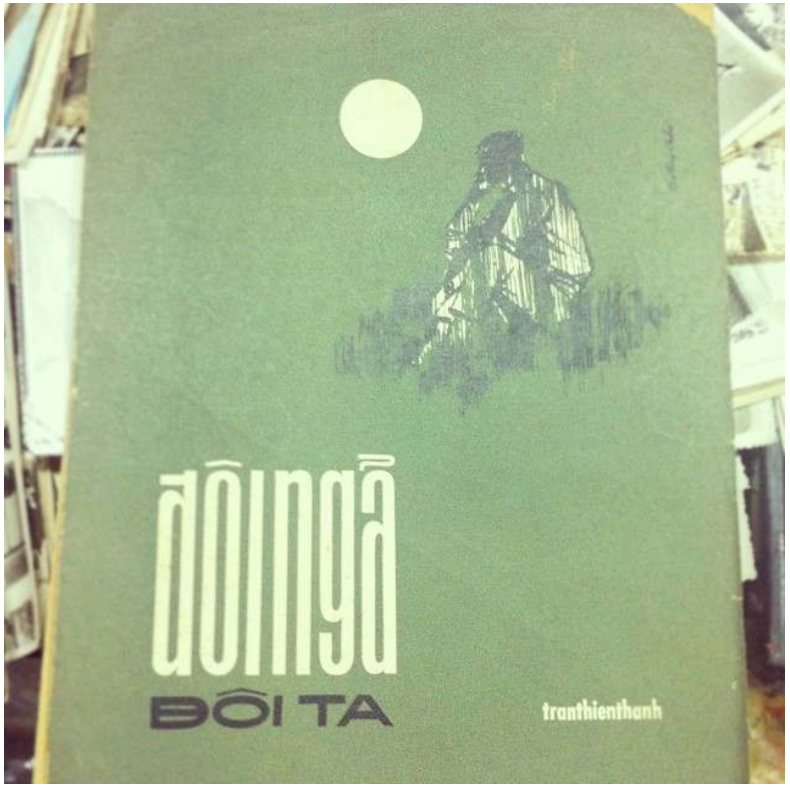


Movement & Rhythm

HOW TO LEARN TYPOGRAPHY

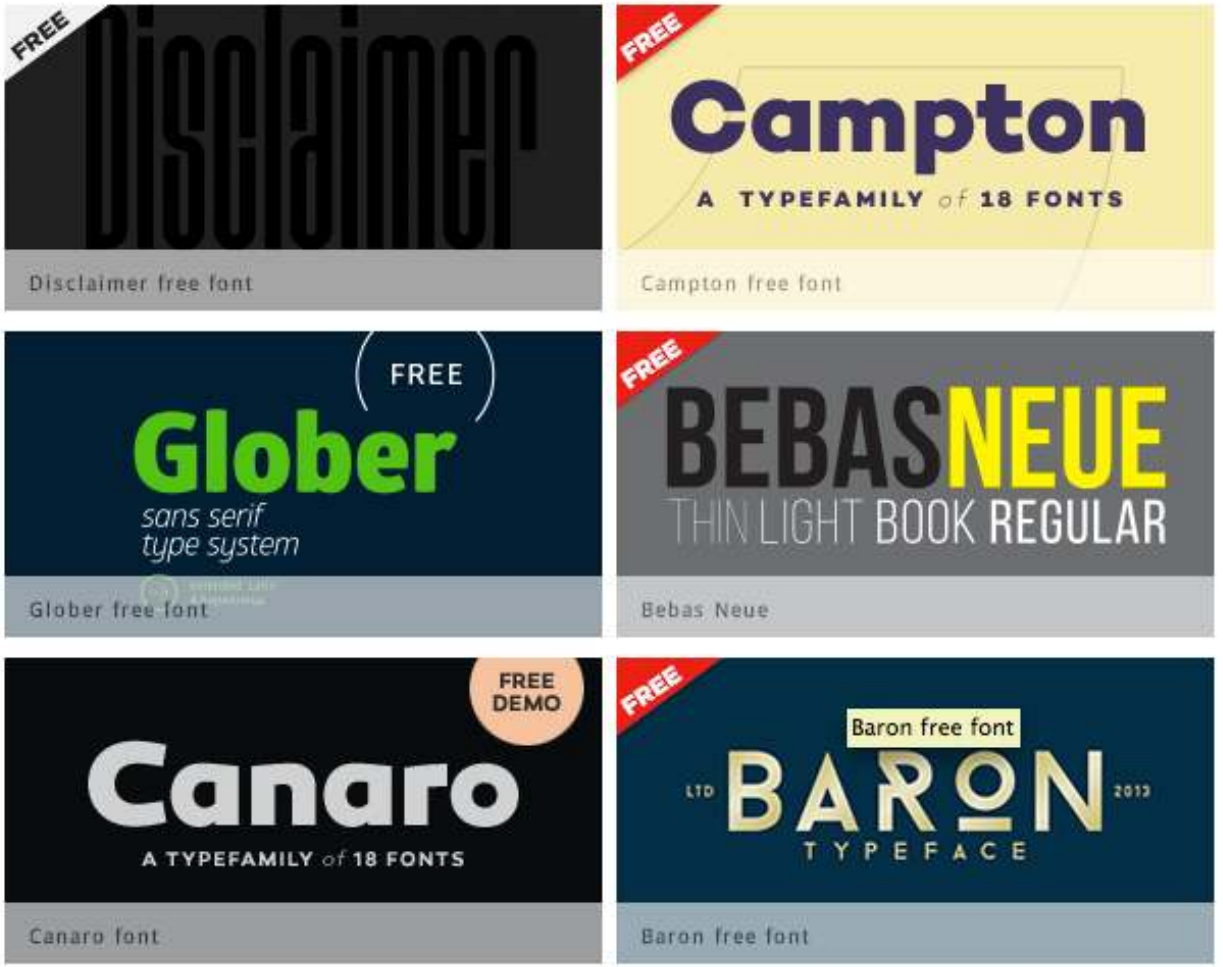
- Learn the principles
- Practice a lot
- Train your eyes
- Start noticing type
- Make friends with type
- Explore & experiment

HOW TO LEARN TYPOGRAPHY



Start noticing type

HOW TO LEARN TYPOGRAPHY



Make friends with type

HOW TO LEARN TYPOGRAPHY

Helvetica Neue 14pt / 18pt

Bacon ipsum dolor sit amet ball tip pork loin cow brisket drumstick bresaola short ribs beef jowl kielbasa pig spare ribs porchetta. Sausage pancetta t-bone chicken, swine filet mignon kevin. Boudin pork belly bacon, chuck pork loin venison ball tip prosciutto chicken leberkas jowl turducken salami swine ham. Hamburger sausage salami biltong bacon filet mignon ham hock. Tongue pork loin swine spare ribs strip steak pork chop shank biltong. Pig doner turducken filet mignon, fatback venison pork loin andouille tenderloin chicken.

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Source Sans Pro 14pt / 18pt

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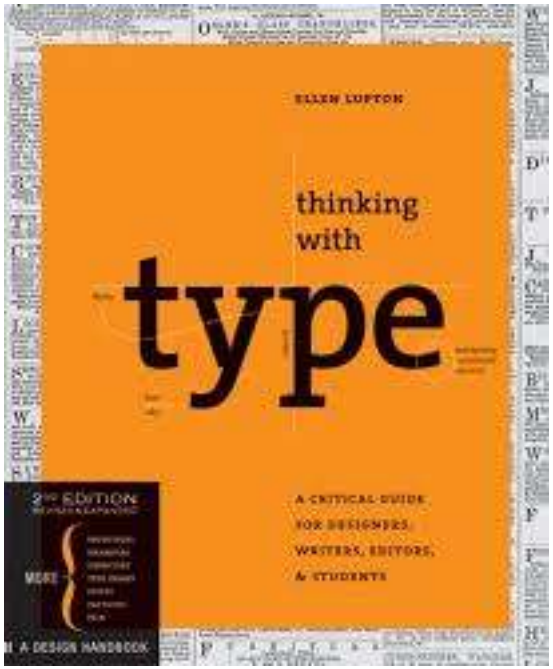
Explore & experiment

RESOURCES



Books on Type

Design & Typography Inspiration books

Any kind of book, actually. Even fictions.

RESOURCES

Resources & Inspirations

http://ilovetypography.com

http://typedia.com/

http://welovetypography.com/

http://incredibletypes.com/

http://fontsinuse.com/

http://typeverything.com/

Free Fonts

http://www.fontsquirrel.com

http://www.fontfabric.com

http://www.theleagueofmoveabletype.com/

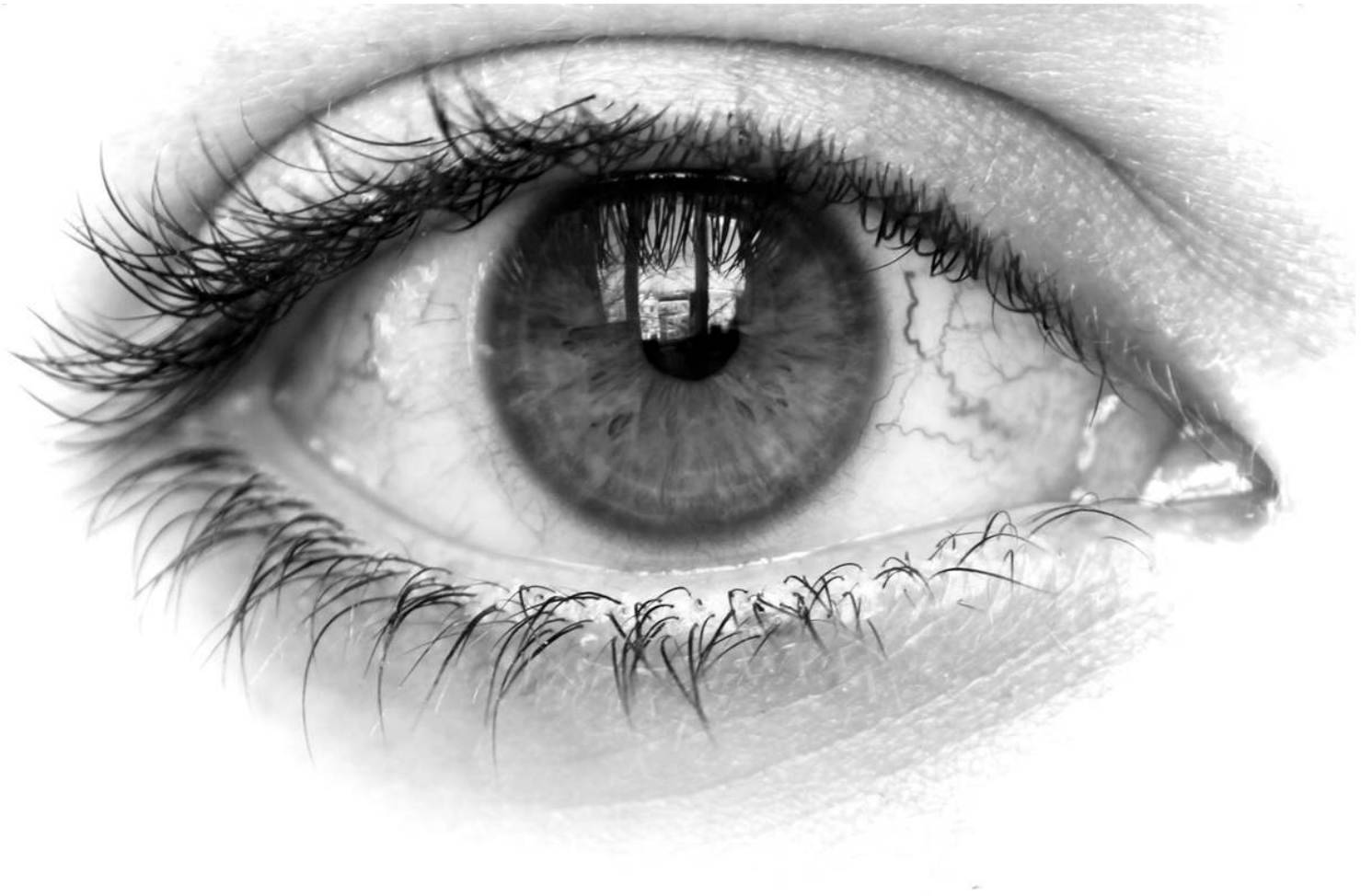
http://www.losttype.com/

http://www.exljbris.com/

RESOURCES

Any piece of type around you!

RESOURCES



And again, last but not least.

KNOW LOGO

DESIGNACTION COLLECTIVE

will design for change.

Design Action Collective. 2011. 1st edition.
All logos in this book were designed by Design Action Collective with the exception of the logos for HRC, the Black Panthers, the United Farm Workers, and Greenpeace.

DESIGNING LOGOS FOR SOCIAL CHANGE

WHAT IS A LOGO?

The term “logo” is often defined differently and used to describe different things, even within the field of graphic design. Design Action works primarily with social change organizations (grassroots groups, nonprofits, unions and sustainable businesses) so we use the term to describe:

A signature image and text combination that identifies an organization, campaign or project.

Logos are usually some combination of three elements: the symbol, the wordmark and the tagline.

Some may be made up of just the wordmark:

GroundWorks
CAMPAIGNS

Some logos may have the symbol and wordmark:

Or some may integrate the symbol and wordmark:

OTHER JARGON

When you start talking logo, you will soon start hearing a bunch of other terms that sound cool and professional. Most are thrown around loosely by people trying to sound cool and professional, and may not really help you understand the process. But when used properly, some of these terms can be useful shorthand, so here are a few common ones . . .

Identity System:

All the elements that your group uses in concert to communicate a cohesive and recognizable presence. This system includes visual elements such as your logo, your color palette and other print and web communications pieces, as well as writing style, tone, and key messaging elements.

Example: *IRN redid their identity with us in 2008. Their new identity system included shortening the name to International Rivers (and ditching the acronym), a new logo and tagline, a new color palette, and a style guide describing logo usage, color palette and fonts, and templates for their most common publications.*

Old Logo

New Logo

Brand:

How the world perceives your organization, based on how it has been presented through the use of the logo, identity, press coverage, interactions, etc. We can design your logo and identity system, but how your constituency perceives you (your "brand") will depend on that, *and* everything else you do.

Example: *Think Greenpeace*–their logo is just their name in a funky font. But their brand is one of the strongest in the environmental activist world.*

GREENPEACE

Signature:

Used synonymously with "logo."

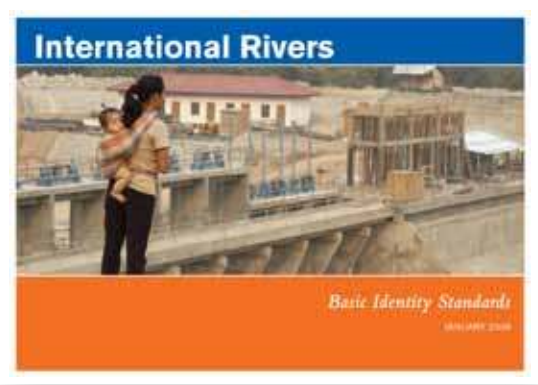
Mark:

Used to describe the "symbol" part of the logo. Also referred to as the "logo mark."

*Design Action did not design the Greenpeace logo

Style Guide:

A document describing such things as logo usage, color palette, fonts, images, etc., that have been approved for your organization. This Guide will allow future designers and people producing materials in-house the know-how to make your materials look consistent. Helping everyone adhere to a consistent presentation of your "identity" is part of the strategy for developing your "brand." Depending on how extensive the Guide is, you will see this document titled "Logo Usage," or "Organizational Identity," or "Brand Guidelines." Or sometimes you'll see "Logo Identity Brand Guidelines." As we said, there's a lot of jargon in this business.


International Rivers
Basic Identity Standards

INTERNATIONAL
RIVERS

RIVERS
RIVERS

Print Collateral:

All communications and outreach materials that your logo goes on, like brochures, factsheets, stationery, etc.

Business Suite:

Specifically, letterhead, business cards and envelopes.

ESLI
Environmental
Service
Learning Initiatve

ESLI
"WE NOT ONLY INHERIT THE EARTH FROM OUR ANCESTORS, WE BORROW IT FROM OUR CHILDREN."
Environmental Service Learning Initiative
ESLI YOUTH TO COPENHAGEN FOR CLIMATE TALKS!
CLIMATE JUSTICE
CLIMATE CHAOS
THE YOUTH ADVISORY BOARD IS HERE!
IN THIS ISSUE:
ESLI at SF Green Festival (page 2) School Updates (pages 3-7)

WHY DO YOU NEED A LOGO?

This is a “What, Why, and How” guide to logo design. But as a movement-servicing organization, we have to first suggest you ask “Why Not.” With organizational resources being so thinly distributed, and true revolutionary change coming through action, not just image, it is worth considering if spending time, money and energy on a logo process is appropriate for your organization.

For some organizations the answer will be obviously, yes.

But for others, particularly start-ups with limited funding, perhaps the logo is not the first thing that is needed. Perhaps building a strong reputation for successful campaigns or services is more important to your organization than the logo. Perhaps something simple and clean (or even homemade!) is what will give you “street-cred.” Perhaps an iconic image will emerge through your organizing efforts that will later be the basis for a logo, but right now what you need is some well-designed brochures and a website.

That being said, in this era of visual overload, investing some time and thought into developing a good logo can have tangible benefits . . .

Recognition

You do great work. People need to know it's you doing it.

Differentiation

There are others doing similar work. But they ain't you.

Visual Shorthand

"What type of organization is this? Ah, I can tell from the logo."

Credibility

"Is this a real organization or just someone with a website?"

Organizational Pride

You know who you are as a collection of people. This is who you are under one banner.

Personal Relationship

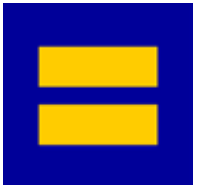
In a visual society, you need a face.

HOW TO DESIGN A GOOD LOGO

Your logo is not a hieroglyphic version of your mission statement.

There are no hard and fast rules in the ever-changing world of design. But here are some enduring concepts to think about when going into a logo design process.

1. Think "essence," not "literal." Avoid thinking of the logo as a montage of all your work or your vision. A more literal illustration that paints a full picture of what your organization does may be appropriate for a poster or a t-shirt or the cover of a report. But a good logo needs to be a visual metaphor – something symbolic that captures the essence of who you are, without spelling it out (unless the essence of who you are is literal and didactic and that's what you want to communicate).



*Design Action did not design these logos

2. Think about things like "Tone," "Mood" and "Association." A good logo engages emotions. People will shape their perceptions from the overall subtle combination of each aspect, and not just from the individual elements.

3. Identify your primary audience. Yes, the logo needs to work for everyone. Yes, your constituents are different from your staff and board, who, in turn, are different from your funders (and you have one department running a youth campaign). So you don't want to alienate any of your key audiences. But you may need to identify who your primary audience is.

4. It's not about you. Sorry, that may seem a bit harsh, but remember your logo is about how your organization is to be perceived by your target audience – which is presumably not you. If it's people very much like you, then great – you're in the focus group. But often that's not the case, and it's important to step back a bit from what you personally like, and consider what will best communicate your message and further your organizational goals.

5. Identify your decision-making group. Make sure all "stakeholders" are involved from the start. What you DON'T want is to go through a three-month logo design process that everyone is happy with, and then show it to your board chair, who could have reasons to hate it.

6. No Frankenstein logos! When reviewing logo concepts, remember the designers have developed them through a holistic process. There may be cases where swapping elements between logo options can work. But for the most part, mixing and matching design elements rarely works.

CASE STUDIES

ROAD MAP

Road Map provides capacity-building and strategic-planning support to activist and nonprofit organizations. The obvious concept behind the name is that they help organizations navigate their way to success. We played with lots of different navigation metaphors (maps, sextons, etc). Also footsteps, pathways, and roads. And some other off-the-wall ideas. Some were good ideas, but just not elegant. Some were overused (like the compass). We finally started honing in on the road, paring it down to its simplest and most elegant form. This works, not because the concept is so unusual (a road for an organization called Road Map), but because it captures the essence and tone of the organization. The calm, thoughtful process they offer to help organizations find a smooth and clear path through the bumps, is what Road Map is all about. >

CONCEPTS

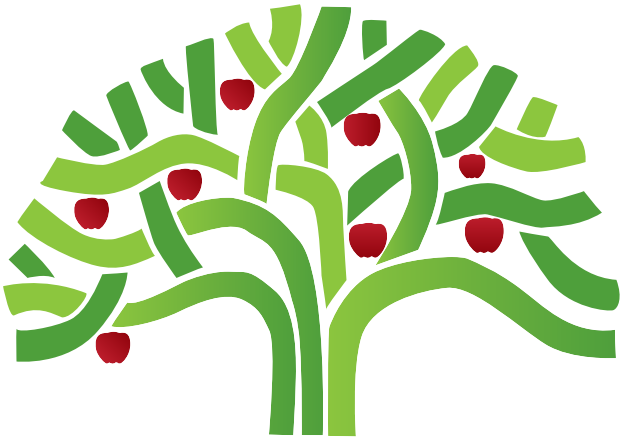
RoadMap
Strengthening Organizations. Advancing Social Justice.

HOPE COLLABORATIVE

Hope Collaborative, which addresses food and fitness disparities in Oakland neighborhoods, started out being attracted to literal representations. The cranes and the two towers are Oakland landmarks. The grains, eggplants and tomatoes are healthy foods.

But even though the rendering was nice, they ultimately made a better choice in going for a simpler and more clever logo. The tree is the City of Oakland logo. Traditionally an oak tree, this one has sprouted fruit! Get it? >

DESIGN DIRECTIONS



HOPE

COLLABORATIVE

Health For Oakland's
People & Environment

TEEN SOURCE

Teen Source provides sexual health information to young people, through frank and respectful communication. We started out working with their original tagline, and developed the logo based on the idea of the words “Teen” and “Source” speaking to each other about love issues. The logo needed to be hip, and appeal to younger people, so the use of very contemporary visual language and fresh typography and colors were appropriate. But then the idea of substituting an even “fresher” tagline and incorporating it into the logo emerged. And voila! it all came together. >

DESIGN DIRECTIONS

SEX + HEALTH
+ YOU
teensource.org

CAUSA JUSTA :: JUST CAUSE

When Just Cause and St. Peters Housing Association merged, they needed a logo that would speak to their two constituencies. Just Cause had a primarily African-American constituency, based in Oakland. St. Peters had a primarily Latino constituency based in San Francisco.

ORIGINAL LOGO

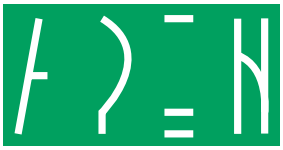
After selecting a new name that was in both English and Spanish, the logo process could start. Traditional black/brown unity images were strong. Urban, Oakland, San Francisco juxtapositions were also an option. But we kept working on paring the logo down to the most important elements. The Bay Bridge represents, literally, the bridge between the two cities. The name and tagline fill in the rest of the story. Bam. >

CONCEPTS AND DESIGN DIRECTIONS

CAUSA JUSTA
UNITY IS POWER ★ LA UNION HACE LA FUERZA
JUST CAUSE

ASIAN PACIFIC ENVIRONMENTAL NETWORK

APEN is one of the pioneering organizations in the Environmental Justice movement. The challenge here was to develop a new logo that clearly identified them as an Asian-focused organization which works in an urban environment with diverse Asian communities.



ORIGINAL LOGO

The Mother/Child says APEN's work is about people (not whales). The city skyline says urban. Paring down from versions that tried to show everything, we decided we could say "environment" through the color, and say "Asian" through the name and subtle details of the rendering style and font. Less is more, is almost always true when it comes to logos. >

CONCEPTS

DESIGN DIRECTIONS

APEN

ASIAN PACIFIC

ENVIRONMENTAL

NETWORK

COLORADO IMMIGRANT RIGHTS COALITION (CIRC)

CIRC is a state-wide coalition working to improve lives of immigrants and refugees in Colorado by advocating for workable, fair and humane immigration policies. As a cross-sector membership-based organization, they needed a logo that would show unity, yet also reflect the place they are working in – Colorado.

The concepts we worked with were landmarks of Colorado including the mountains, cityscape of Denver and the shape of the state. Unity. Diversity. Equality. CIRC as an umbrella organization. They were able to narrow it down to an abstract concept of the mountains and people, and make it a versatile, bilingual logo for print and their website. >

CONCEPTS

DESIGN DIRECTIONS

Colorado Immigrant Rights Coalition

PEACE ACTION: NO SOLDIER LEFT BEHIND

No Soldier Left Behind was a California Peace Action campaign that needed a recognizable identity. Time was short, so a concept was selected from a "moodboard" to keep rendering rounds to a minimum. This was an effective way to get to the symbolic image that best represented the campaign, as well as what the overall tone of the logo needed to be. >

MOODBOARD

DESIGN DIRECTIONS

No Soldier Left Behind

No Soldier Left Behind

No Soldier Left Behind

TYPOGRAPHY

no soldier left behind
Rebuild Iraq.
Rebuild America.

soldier left behind
Rebuild Iraq. Rebuild America.

no soldier left behind
Rebuild Iraq. *Rebuild America.*

COLOR

no soldier left behind
Rebuild Iraq. *Rebuild America.*

no soldier left behind
Rebuild Iraq. *Rebuild America.*

no soldier left behind
Rebuild Iraq. *Rebuild America.*

no soldier left behind
Rebuild Iraq. *Rebuild America.*

no soldier left behind

Rebuild Iraq. *Rebuild America.*

UMASS AMHERST WOMEN GENDER SEXUALITY STUDIES

The UMass Amherst Womens Studies Department is rooted in the feminist movement of the 1970s and their original logo reflected that legacy of militant, womens' rights-focused work. Their new logo needed to reflect the complexity and intersections of issues in the study of women, gender, and sexuality. The faculty decided early on that an abstract logo was the best way to go. It is impossible to be inclusive of the diversity of gender and representative of their intersectional analysis in a literal graphic. We explored ways to illustrate this approach to women, gender and sexuality studies through interlocking shapes and letters. We also created shapes representing the interruption of patterns of oppression. Their final logo achieved their goal of an updated look that would be relevant to students today and celebrates the department's work through intersecting lines and bold color. >

CONCEPTS

WOMEN,
GENDER,
SEXUALITY
STUDIES
UMass Amherst

ABOUT DESIGN ACTION

Design Action provides graphic design and visual communications services to nonprofit and social justice organizations. Through these services, we participate in building a broad progressive movement, and support ongoing struggles for social change.

Design Action Collective is an independent design and communications spin-off of Inkworks Press Collective. After more than a decade of offering graphic design services to nonprofit, grassroots, and activist organizations under the same roof as its print shop, Inkworks decided that both its print and design services would benefit from the creation of a new collective. Since then, Design Action has been providing web and “new media” services, and is expanding to include strategic communications and messaging.

Since the founding of Design Action, we have grown to be a multilingual, diverse nine-person shop. Our members have come to Design Action as activists and organizers within the social justice movement, and with the desire to offer our skills as designers to the progressive movement. Design Action is a union shop, affiliated with Communication Workers of America, AFL-CIO. We are also active members of the Network of Bay Area Worker Cooperatives, and the U.S. Federation of Worker Cooperatives.

www.designaction.org
510.452.1912

ABOUT INKWORKS PRESS

Inkworks is a collectively-owned union and certified-green printshop which provides high-quality, reasonably-priced printing for the progressive community committed to economic, social, and environmental justice.

Inkworks handles everything from design (with Design Action) to printing, finishing and mailing. We produce booklets, brochures, posters, postcards, flyers, letterheads, envelopes, business cards, short-run books (like this one), and more – in quantities from 50 to 50,000.

We are committed to high-quality, green practices, and thorough customer service. We are happy to answer any questions you may have. We are flexible and ready to work with you to keep your printing costs affordable.

contact us:
www.inkworkspress.org • (510) 845-7111
ESTIMATING REQUESTS: estimating@inkworkspress.org
FILES TO: inkworks@inkworkspress.org
Online orders are welcome at www.inkworksdirect.org

DESIGNACTION
COLLECTIVE

will design for change.

Logo Design Guide

Exploring the process and value of a visual identity

func-tion

Frank Maidens & Vivian Hui

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Version 1.0 – January 2016

Intro

What's this guide all about?

We spend a lot of time thinking about and working with logos. Logo design was a major part of our design educations, and represents a pillar of our studio service offering.

We believe logo designs have the power to define the perception of entire businesses. They can be the difference maker in a world saturated by similarity – directing our affection, loyalty, and hard-earned paychecks.

In the last ten years, there has been a major shift in the relationship between business and design. Thanks largely to the meteoric success of companies defined by strong branding and user-oriented products, more organizations have come to recognize the importance of visual communication. It's no longer a conversation reserved for a handful of highly trained specialists and a minority of progressive boardrooms.

As such, there is a growing number of people in our circles who show an interest in the curious craft of logo creation. They are freelancers, marketers, developers, educators, and entrepreneurs – smart, engaged people that want to learn more about the principles and benefits of good design.

Who are we writing for?

We imagine this guide will find itself in the hands *(aka download folders)* of various professionals working in the design space, or those whose work is related in some way to a design deliverable.

Specifically, this guide is intended for three primary audiences:

Creative Professionals with an interest in visual communication who want to explore another perspective of the design process, or include logo design as part of their service offering.

Clients & Entrepreneurs who want more tools to assess the quality of design work, or become more deeply involved in design conversations.

Project Managers, Marketers, Sales Professionals who want to better understand and communicate the value of design and the benefits of visual consistency when applied to brand collateral.

This guide is for: Creative Professionals

It's an exciting time to be a designer. The world has finally awoken to the opportunities created by the Voltron-like force of design and business working in tandem. More and more companies and business owners are interested in seeing how design strategy and targeted communication can help them achieve their goals; gone are the days of design as an afterthought.

As entrepreneurs incubate and bring their ideas to market, they create a steady stream of companies and products that require positioning in the competitive landscape. And as businesses mature and offerings change, those iterations often require updates to the existing visual representation. This constant demand for design work creates a well of opportunity that is difficult to ignore.

We hope that this guide will find its way into the reading lists of designers. Ideally, it will contribute to their understanding of the design process as it applies to logo design and encourage them to incorporate it into their service offering.

This guide is for: Clients & Entrepreneurs

It takes guts to own a business, or to own business decisions. There's a lot at stake, and a lot of responsibility. Savvy business people know their strengths and are able to step back from decisions they may not be fully suited to make. But that doesn't mean they should check out of the conversation completely.

This guide is meant to assist decision makers by elaborating our terminology and design philosophy. It aims to walk them through some important processes, and communicate the value of conceptual thinking paired with consistent execution.

We want to convince clients that design is worth the investment and provide them with some tools to better understand what's going on under the hood.

This guide is for: Marketers, Project Managers, Sales Professionals

A design education teaches someone how to create beautiful, meaningful things. However, a conventional design education doesn't always tell the whole story. It sometimes fails to express that a beautiful, meaningful design doesn't sell itself. If the client on the other side of the table is not convinced of its value, that design will likely never see the light of day.

It can be difficult to articulate the value of design. A designer may have an understanding of why something resonates, but struggle to express that understanding with brevity and confidence. This guide is partly aimed to address that issue. We want to express our perspective on the value of the design process and the positive effects of visual craft and consistency.

Our Vocabulary

There seems to be an ever-growing collection of words used when referring to design, so a quick recap of some general definitions will help us get on the same page.

What is a logo?

Logos have been used by people for thousands of years. Ancient merchants relied on markings to delineate ownership between similar-looking clay pots and burlap sacks. Likewise, owners of livestock would also *(and still do)* mark their animals as a means to prevent confusion and confrontation.

Eventually, trade guilds and trade unions started marking their products with symbols of authenticity. Since their products were typically of higher quality, those markings would come to suggest superiority and reliability to the end user. Consumer affinity and loyalty soon followed.

Logos now have a role representing businesses, organizations, and individuals, in addition to products. They are symbols that capture some key aspect of a brand's culture, character, value proposition, or aspirations.

In many cases, these marks are used to differentiate between products that may otherwise seem identical. They can also come to embody a service or group that has no other kind of visual representation. Logos have the power to make two

similar things feel completely different and are sometimes the only visual manifestation of a service, collective entity, or effort.

Wordmark

A wordmark is a kind of logo that relies solely on the use of letterforms. It has no additional graphic accents or icons. It is an efficient mark that usually offers the viewer less visual information to interpret.

Brand Icon

A brand icon is a symbol that can be used independently or in conjunction with a wordmark. These potent images can easily speak 1000 words *(whether simply and directly or with a powerful ambiguity).*

Combination Mark (Lockup)

This is a word we use when a wordmark and brand icon appear together. Combination marks can effectively encourage brand recognition, as they help the viewer make a visual association. This pairing can also be called a 'lockup' because they will often appear in the same configuration. Businesses may have a small set of different lockups for various design applications.

(Keep in mind that not all logos are going to fit neatly into these three categories. Sometimes a wordmark will incorporate some kind of non-typographic element – so is it still technically a wordmark? Not really... Which part of the combination mark is the 'logo?' Put simply, all of it. 'Logo' is a term that can casually refer to any wordmark, brand icon, or combination mark. When encountering a design that is hard to classify, just say logo!)

Logomark

This is a slightly fancier way of saying 'logo.' To us, there is no real difference in meaning. It's just a handy piece of honorific language that can be used to add a bit of reverence when referring to a particular logo design.

What is a visual identity?

The term 'visual identity' is used to describe the graphics, typography, and other visual elements that appear alongside a logo design in various design pieces.

Logos are seldom seen floating alone in empty space. Usually they live on posters and packaging, in apps and websites. These design environments need to be curated. Through the careful use of supporting typefaces, colours, photography, illustrations and patterns, a more dynamic and targetable communication is made possible. If a logo is the lead singer, a visual identity is the whole band – capable of creating rich layers of communication.

Visual identities also outline how to consistently display the logo and other supporting visual elements in various applications. They can standardize alignments, primary/secondary colour systems, typographic relationships, photographic treatments, and other art direction.

Typography

We use this word to refer to the style and treatment of letterforms. It's a term that can also describe how a particular font is used in a visual design.

Type Pairing

This phrase refers to the selection of two *(or more)* typefaces that appear together in a single context or application.

Art Direction

Art direction is a term we use when referring to the combined visual mood/tone of various curated elements in a design piece.

Creative Direction

We use this term to refer to high-level conceptual approaches to a design solution. Creative direction guides the communication strategy; art direction guides the execution and visual tonality of that idea.

Design Collateral

Design collateral refers to any piece of media that is a product of the design process and is used to represent or promote the associated brand.

Deliverable

Similar to design collateral, deliverable is a word used to describe the final product of a design process. It's the thing *(or package of things)* that is ultimately delivered to the client.

Branding

We use this term to refer to all the core visual identity elements: wordmarks, brand icons, combination marks, and alternate logo versions are all considered to be pieces of branding.

Brand

A brand is a little more difficult to describe. It's a word that refers to the full sensory experience of an organization *(or individual)* – not just the visual stuff. This includes social media presence, customer service experience, interior design, and even things like the choice of music in the elevator or the quality of paper towels in the washroom.

Another Thought

Why are identities important?

The final design of a logomark is important, but should not be the end goal of a branding project. Visual designers need to ensure a logo looks intentional when applied to various pieces of brand collateral. These collateral pieces will each have their own range of content requirements *(text content, image content, other visual elements)*. The appearance of this information needs to be considered if the logo *(and brand)* are to communicate the intended feeling successfully.

Without a clear definition of elements like supporting typefaces, colours, and image treatments, the logo is left to wander in a haphazard environment. Even high-quality logo designs cannot stand up against an ill-considered backdrop. Logomarks need a plan to ensure they sing, regardless of the context. Visual identities provide that plan.

Where to Start?

We believe that the creation of a visual identity system should be the end goal of any logo design project. Logos don't live in isolation. The designer needs to provide a plan for the accompanying visual elements that support a logo design in application.

Before the identity can be explored, the logo itself needs to be resolved to a functional degree. Logos are the cornerstone of visual identities. Their visual flavour and structure will influence design decisions made at the identity level, and in collateral application. As such, we feel the design process needs to begin with the logo to establish communication goals before moving forward into any supporting identity treatments.

Before starting work on a logo design, the stage must be set by a creative brief. A creative brief is a document created through conversation and collaboration with the client, and is the essential starting point of any visual design challenge. This is where our design process begins.

Section 1
The Creative Brief

The brief is an essential document as it describes the design problem and provides the means to measure the success of future creative work. It's the strategy roadmap for the duration of the project. The creative brief helps to calibrate the designer's mind to the task, and also holds the client accountable to an approved creative direction. It should be used when it comes time to review design concepts, and will help to diminish the effects of personal bias *(more on that later).*

This document also defines boundaries that are beneficial to the designer's creative process. Limitless opportunity can sometimes inflict a paralysis of choice. But knowing where not to step can clearly illustrate the path forward.

Essential elements

In general, the creative brief should cover the following bases:

Describe the company

Its history, personality, and values

> *This&That Inc. is a luxury design and lifestyle brand for the cat connoisseur. Founded in 2014, the company has grown into the destination for design-obsessed cat owners who are looking to invest in high-quality feline furniture and goods. This&That is the Chanel of cats.*

Describe the task at hand

The challenge, restrictions, and project deliverables

> *This&That requires a new logo and supporting identity system for the launch of their online store. In addition to digital applications, a 1-colour version of the mark will be required for laser engraving on wood and metal. An icon version of the logo must also be considered for use in various social media properties.*

Describe the target audience

Tailor design solutions to specific audiences for maximum effectiveness

> *The primary audience is 30–50 year old cat-obsessed professionals in North America who are willing to invest in high-quality, design-focused objects for their cats. These aficionados are tech-savvy and engaged in interior design trends, always seeking the latest and greatest; they are split 70% female and 30% male.*

Outline the main communication goals

The objectives for the work and priority of main messages

> *All deliverables should position the company as the high-end choice in feline goods, represent a commitment to animals and the environment, and reflect a dedication to quality and craft. Though This&That should project a feeling of luxury, communications should also be approachable and speak to the audience in an honest way.*

Assess the competitive landscape

Knowing other companies in the same space helps the designer intelligently position any new work

> *Local competitors include... Global competitors include... This&That is also indirectly competing with luxury furniture brands, such as...*

Provide a high-level creative direction/key concepts

A succinct roadmap for the conceptual foundation of the work, which can sometimes be achieved through a list of adjectives created with the client

> *All brand communications should feel: confident, refined, luxe, well crafted, and vibrant. Avoid cutesy, whimsical or traditional cat iconography (no whiskers, paw prints, cat ears, etc.)*

Steps to a successful brief

The appropriate path to a creative brief depends on the size of the project and the expectations of the client. The exact approach varies, but the general process stays the same:

1. **Ask questions**
   - IRL discovery meeting/interviews with stakeholders
   - Questionnaires/forms, user interviews or surveys
   - Learn about company, goals, values, audience
   - Understand how the company wants to be positioned visually

2. **Review answers**
   - Process the responses
   - Identify key points, eliminate redundancy
   - Distill into a single document
   - *(email, PDF, website, PPT, shared doc, wood carving, etc.)*

3. **Arrange and present the important points**
   - Use phrases, sentences, bullets, infographics
   - Reference visual examples, read between the lines
   - Submit for client review

4. **Revise and get approval**
   - Incorporate feedback, embrace clarity
   - Insist on client sign-off before design commencement

A rough framework

The quality of the brief is determined by the quality of the questions asked. Depending on the size of the project or expectations of the client, the creative brief can range from a simple email to an elaborate, multi-page document. Consider some of the following questions *(this is by no means an exhaustive list):*

Company Background

Who is the company? What is the product or service? What are the core values and what does the brand stand for?

Project Overview

What is the purpose of the project? Does the project face any specific challenges or restrictions? What are the project deliverables?

Content & Messaging

What is the project trying to achieve? What are the main communication objectives? What is the priority of key messages?

Target Audience(s)

Who are the primary, secondary, and incidental audiences? What is their current relationship with the product or service? Why should they care about the product or service?

Competition

Who else exists in the competitive landscape? What are their differentiating factors? How are they speaking to their audience with art direction and copywriting?

Voice & Tone

What are the desired adjectives to describe future communications? What are the goals for the art direction and visual design?

Crafting and delivering a creative brief

Good

Discovery:

- Prepare a Google doc questionnaire or Google form
- Have a discovery phone call to review answers or ask additional questions

Creative Brief:

- Email a basic creative brief for client approval, state the following:
  - the design challenge
  - communication goals
  - general audience profile
  - competitive landscape
  - 5 adjectives for creative direction

Awesome!

Discovery:

- Hold a multi-day discovery session to fully explore various discovery questions
- Conduct independent stakeholder interviews to further elaborate business/communication goals
- Send out user/customer surveys
- Interview members of the target audience
- Perform comprehensive competitive landscape research

Creative Brief:

- Prepare a PDF or web page to present the creative brief
- Consider including the following sections:
  - company profile
  - statement of the design challenge
  - communication objectives
  - list of key ideas to express through design
  - priority of messages
  - list of adjectives for creative direction
  - detailed audience profiles (primary/secondary)
  - visual mood boards
  - competitor analysis (local/global/indirect)

Another Thought

"I'm just not a fan of yellow."

A client's personal bias is a common issue encountered by designers. Whether it's a *(strange)* affinity for the colour brown, or the dismissal of an effective typeface because of one previous bad date, decisions made on unfounded gut feelings rarely improve a design solution.

Effective logos and successful identities are designed to resonate with a particular audience. Sometimes the client needs to be reminded that they aren't a part of that group. The creative brief helps keep the conversation centred on the expectations, visual/cultural vocabulary, and other sensitivities of the target audience. It doesn't matter that the client hates yellow if that colour communicates the right message to the intended audience.

Use the creative brief as a shield against personal bias when presenting and reflecting on the effectiveness of a design solution. Allow it to be the justification for the creative work and all parties will benefit from a stronger solution in the end.

Section 2

Logo Design

Designers don't sell logos. They sell a process. This process will ultimately deliver a solution to the client's problem, and should be tailored specifically to each client based on their needs and budget.

It may seem like a trivial adjustment to consider the process *(not the logo)* as the project deliverable. However, we find this shift helps the client recognize the time and effort required to create a simple, effective design.

The logo design process

1. **Problem Definition**

   Put simply, the designer needs to know what problem they are trying to solve. In terms of logo design, this could be a problem with a brand's position in the market, a lack of existing visual representation, or a logo that is illegible at small sizes. Whatever the case, everyone needs to fully understand the design challenge before any solutions can be tabled.

2. **Research**

   Before a designer can start their creative process, they need to study up on the competitive landscape, audience, and production techniques.

   A competitive landscape review is essential. It helps prevent the presentation of a design that may be embarrassingly similar to a competitor, or other tangential company or product. It also gives the designer a view of the visual language used in a particular space, allowing them to position the new mark intelligently.

   In addition to the competitive landscape, the designer must have some perspective on who will be looking at the logo, and which

group of people it is important to appeal to directly. Every human being has their own configuration of visual and cultural vocabularies. The designer's job is to use their powers of empathy to anticipate how a particular group will respond to certain forms and colours. Once the audience is defined, the designer can research the expectations and preferences of that audience *(through an assessment of visual design precedents, audience interviews/surveys, review of social media commentary, etc.)*, and build that knowledge into their design execution.

We also recommend a designer quickly research the opportunities and limitations of various production techniques that may be applied to the logo design. Are there opportunities for simple and meaningful animation? What are the visual effects of letterpress or screenprinting? Can the logo present a smaller, simpler version of itself in extreme contexts? Investigating various production methods may inspire a particular treatment or other conceptual approach.

Research can be an exhaustive step in the design process, but it ultimately leads to more successful work. Designers should treat research like investing: a greater contribution will yield a greater return.

3. Divergence

This is the sketching *(aka braindumping)* phase. Once the designer has calibrated their mind for the design challenge *(problem definition, communication goals, audience profile)*, they are ready to let loose with their ideas and creative intuition.

Most importantly, divergence is a time reserved for unedited exploration. We've encountered many junior designers who believe sketching equals 'drawing the solution' – they try to solve the whole problem in their head and then draw the final answer. We take a different view: sketching is a time for exploring how different key

ideas or messages can be presented visually. It's about seeking various components of a solution, not the final solution itself. Designers should use this stage to experiment without the fear of failure. The goal of this phase is to capture a dynamic range of ideas and expressions.

We recommend that designers stay off the computer and actually sketch during this phase – using pencils, pens, whiteboards, or whatever is most comfortable. Filtering ideas through cumbersome digital tools can interfere with the meandering of the mind, or force a designer's ideas into too much structure prematurely. Alternatively, using a pen or pencil is like a mainline to the brain.

It's also important to note that divergence can happen any time, and doesn't need to be so strictly or linearly defined. Designers may be inspired by a particular piece of research, or even during the creative brief phase. Just remember that design is the solution to a problem – so a solution that arrives before the problem is fully understood might not be the most successful answer.

4. Convergence

The goal of the convergence phase is to measure the effectiveness of the sketches by using the creative brief. What ideas stand out and what makes them successful?

Which ideas most directly communicate the key messages *(e.g. bold, established, traditional, innovative)?* Which ideas have the greatest chance to resonate with the intended audience? Are there certain components to a sketch that can be successfully combined with the components of another?

The designer needs to isolate the effective concepts and invest more time in their visual execution. This means rendering them in vector, choosing proper typefaces and type treatments, and conducting further exploration digitally with colouration, size and form.

5. Presentation

Creating beautiful, meaningful things is a wonderful ability. But unless the client understands the value and appropriateness of a piece of work, it will not see the light of day. Since the client's approval determines which idea goes live, the presentation itself can be as important as all the underlying design work.

Presentations are an opportunity to uncover aspects of a concept that aren't satisfying the requirements found in the creative brief. They can initiate conversations on key issues and provide opportunities to better understand the audience and the task at hand. These deeper levels of understanding lead to stronger work.

We never show up to a logo presentation without some supporting examples of logo application. There is limited value in seeing a logo design floating alone on screen. Logos are applied to things – they live in various printed and digital spaces, surrounded by other information. By taking the time to explore how the various logo concepts look in context, we give the client a more valuable perspective on the proposed logo designs. These unpolished example applications remind the client that a logo is part of a larger identity system, and needs to be evaluated in that way. Section 4 dives deeper into the intricacies and philosophy behind good visual identity presentations.

6. Revisions & Implementation

Collaboration between the client and the designer is an essential part of the logo design process. When approached correctly, it leads to stronger solutions.

How many rounds of revisions are appropriate? That depends on the design team and the scope of the project. But we suggest stating a limit to the number of rounds of revisions in your project agreement. Whether it's two or 20 rounds, it's necessary for all parties to agree on the number before the project commences, otherwise this phase of the design process could go on forever *(Note: if a client expects 20 rounds of revisions, GTFO).*

For identity design projects, we generally scope two rounds of revisions with any additional rounds billed hourly *(if required)*. Section 5 takes a closer look at the revision process.

What makes a good logo?

Good question! It's difficult to provide a standard for the assessment of logo design. No two design challenges are the same, and different audiences have their own set of subjective criteria. The same design viewed by two different people may evoke completely different reactions. Even so, we believe there are some core tenets of logo design. By adhering to these tenets, designers will be more likely to create a successful piece of communication that stands the test of time.

Good logos are: Simple

Generally speaking, a logo should be simple. As we mentioned previously, the natural state of a logo is in application – it is crafted to be used in various design pieces. Sometimes those pieces have limited space for the logo to appear, or present some other display-related challenge. Overly complicated designs, or logos with extraneous details, are not versatile and perform poorly at small sizes. If a logomark cannot scale down in size *(think favicons and mobile social icons)* and retain some legibility, it fails as a design. Additionally, if a logo cannot be replicated in a single colour, its potential for success in certain applications *(and therefore its value)* may be limited.

Simple things are attractive. Succinct visual designs deliver their messages with potency, confidence, and refinement. Brands want to benefit from the positive effects of focused

communication. It is for these reasons we believe a logo should be simple, first and foremost.

How can simplicity be achieved?

- choose one main concept to express in the logo *(recognizing the logo is only one facet of the identity, and doesn't need to say everything)*
- eliminate any superfluous visual elements that don't contribute directly to the expression of the main concept *(extra circles, underlines, hipster Xs, etc.)*
- learn to rely on the beauty of unaltered typography *(someone already spent years crafting those letterforms; maybe they don't need to be messed with?)*
- ensure all graphic/visual elements are sized generously enough to remain visible when the logo is reduced in size, or provide alternate versions with simplified solutions to be used at small scales
- be mindful of supporting/tagline text sizes
- make sure the logo design has a 1- or 2-colour fallback option in case certain production methods are required

Good logos are: Memorable

A good logo should be simple, but not without character. There needs to be some expression or identifiable characteristic the viewer can recognize and use to recall the mark later. After all, part of the logo's job is to help build affinity with a company, product, or service. If it is not memorable in some way, it does nothing to help the viewer recognize the associated brand and ultimately build a relationship.

A designer should ask: how is this logomark 'ownable?' The goal is to create a mark that feels like it belongs to the associated brand, and no one else. When viewed from the perspective of

ownership, a more tactile dimension is added and it seems easier to 'feel' specific visual details.

How can memorability be achieved?

- explore intentional colouration that helps the logo stand apart in the competitive landscape *(requires some competitive research)*
- craft a logo design that enables strong alignments, allowing it to appear confidently and consistently in various layouts
- don't be afraid to step outside of the well-worn channels of type selection *(maybe everything doesn't need to be in Gotham?)*
- use clever or meaningful graphic/visual elements that accent the name or key message
- aim for high visual craft, pleasing forms, smooth curves, and sensitive proportions

Good logos are: Meaningful

A meaningful logo design successfully represents a key characteristic, goal, or value of the brand. This means that each visual component of a logo *(like form, typography, and colour)* are chosen because they depict, or combine to depict, a facet of the brand.

A designer must also take the target audience's visual/cultural vocabulary into account when choosing visual components. The forms and colours of the logo must speak to the audience in a 'language' they understand.

How can a logo become more meaningful?

- typography should be selected to present or accent a desired communication objective *(friendly geometric sans serif type can feel approachable; condensed uppercase sans serif type can feel mature/confident)*

- specific colours or colour combinations should be used to define or underscore a key message *(if the logo needs to feel approachable, perhaps blue is more effective than red; if the logo needs to feel confident, perhaps red is better than green)*
- eliminate all decorative elements in a logo design: decoration is never meaningful
- look to the use of visual metaphor to communicate with poetic potency *(an apple can represent knowledge, a turtle can represent reliability, wings can represent freedom)*

Good logos are: Well crafted

Logomarks must be expertly crafted and beautiful to behold. As ambassadors for brands and representatives of the collective efforts of entire organizations, logos should be dressed to impress. Even though the audience may not be equipped to articulate the visual quality of a logo, poor execution still has a negative effect on viewers. Like bad karaoke, the audience can hear when something is off-key, even if they are not trained singers themselves.

What contributes to a well-crafted logo?

- ensure all kerning *(letter-spacing)* is specifically adjusted for an optimal presentation of text
- ensure that the wordmark uses proper typographic marks *(apostrophes instead of footmarks, optically sized trademark symbols, etc.)*
- avoid machine skewed type effects by using actual condensed, extended, italic, or bold typefaces when required
- pay attention to the appearance of curved lines *(making sure there are no unwanted bumps, wobbles, or nipples)*
- avoid creating awkward shapes or gaps when overlapping elements by giving each component room to breathe

Tackling the logo design process

Good

Research:

- Do quick internet searches on the list of competitors provided by the client; build a small reference folder of logo images and site bookmarks
- Look at the social properties of various competitors and review the audience of people who are engaged in conversations with those brands

Divergence:

- Set aside 1 or 2 afternoons for divergence and visual exploration
- Explore how different ideas can be expressed in sketches; stay off the computer

Convergence:

- Identify the most effective sketches and convert them to vector
- Experiment with stroke weight, colour, proportion, and positioning
- Choose meaningful typefaces that align with the creative brief

Presentation:

- Email a PDF presentation with notes or conduct a screen share to review the concepts
- See section 4 for suggested approaches and content of presentations

Revisions & Delivery:

- Include at least 1 round of revisions in the initial project scope
- Ask for the client's feedback in writing
- Make revisions to the selected concept based on client input

Awesome!

Research:

- Research local/global competitors; build image resource folders of their branding assets; take notes on their messaging strategy and other art direction
- Build a file that demonstrates the primary brand colour of various competitors
- Look at the social properties of various competitors and review the audience of people who are engaged in conversations with those brands
- Prepare a questionnaire and conduct user interviews to determine how people feel about a particular product or service

Divergence:

- Reserve at least 1 week for divergence and visual exploration
- Keep a sketchbook on hand at all times
- Explore how different ideas can be expressed in sketches
- Push past the initial layer of ideas; find other ways to represent key ideas; combine approaches to create new concepts
- Stay off the computer!

Convergence:

- Identify the most effective sketches and convert them to vector
- Experiment with stroke weight, colour, proportion, and positioning
- Isolate successful vector renderings and perform additional cycles of digital divergence; see how else that idea can be represented visually
- Apply meaningful type selections to each concept; explore a range of options

Tackling the logo design process

Good (cont.)

- If additional rounds of revisions are required, estimate and bill for the time required
- Prepare approved logo files for delivery – see section 5 for tips on building out the final Logo Pack deliverable

Awesome! (cont.)

Presentation:

- Meet with all stakeholders to present concepts; provide print outs or follow along screen presentation; provide a PDF after the presentation with slides and summary notes of each concept
- See section 4 for suggested approaches and content of presentations

Revisions & Delivery:

- Include at least 2 rounds of revisions in the initial project scope
- Ask for the client's feedback in writing or provide a written summary of revisions discussed in person
- Make revisions to the selected concept based on client input
- Continue to refine the mark if an additional round of work is required
- Prepare approved logo files for delivery – see section 5 for tips on building out the final Logo Pack deliverable

Another Thought

Music is a time machine

A metaphor is a form of communication that equates two things by highlighting some hidden or implicit similarity between them. "Music is a time machine" is a metaphor because 'music' does not literally 'transport someone back in time.' By claiming that music is a time machine, the phrase identifies points of comparison between both things and suggests a fundamental connection.

When metaphors are used in logo design, the viewer is encouraged to consider the similarities between the brand and the objects and ideas in the logomark. Twitter is a good example: a tiny songbird is used as a metaphor for mobile communication and community dialogue. By using a songbird as their brand icon, Twitter successfully associates their service with the idea of short, sweet messages that can be shared from essentially anywhere. That little bird does some very heavy lifting.

Keep in mind that metaphors don't need to be explicit illustrations, or actual pictures of something – the application can be much more subtle. Selection of a bold font for the wordmark can communicate stability, trust, or confidence. Using rounded geometric forms in a brand icon can be a metaphor for playfulness or youth. Rigid forms in the logomark can equate the brand character with a concept of maturity or precision.

The overt or subtle use of metaphor in a logomark is a sure way to add layers of meaning while increasing the potency and effectiveness of the design.

Section 3

Visual Identity Design

Logos are designed to be used. We believe that the relationship between the logo and supporting elements is the real design deliverable. The client doesn't need a great logo – the client needs a great logo that is brought to life by a beautiful system, one that helps communicate more dynamic and targetable brand messages through the thoughtful combination of visual elements.

A good identity design considers the requirements of logo application to various collateral pieces and provides a plan for consistency. This is achieved through the design of mock pieces, which are then presented alongside each logo concept. These example pieces can be further explored and revised once the client chooses an initial design direction to pursue.

In addition to a logo design, our identity presentations propose an art direction through typography, type pairing, colour, and image treatments.

Typography

Typography is a vessel for mood and emotion, and the selection of supporting type is the soul of every visual identity.

An effective supporting type system will identify which font size and style should be used to present various pieces of written information. Headlines, intro paragraphs, body paragraphs, emphasis text, and numeral treatments are a good starting point. Identities that more articulately describe the use of fonts will benefit from greater consistency across various collateral applications.

Type Pairing

Blending the use of two *(or more)* typefaces is similar to the blending of flavours: a more interesting and dynamic experience can be the result. And pairing two typefaces can help the designer satisfy the brief. For example, if the brief calls for the design to feel "established and approachable," perhaps that result could be achieved by combining a professional-feeling serif wordmark with a friendlier, humanist sans serif body typeface.

Colour

A meaningful colour palette will elevate any identity system. It has the power to symbolize an idea, evoke emotion, and relate directly to cultural stories and values. Colour is a core element of visual language that people process before they are consciously aware of it. Because colour is such a powerful element of visual communication, it's important to use it intentionally.

Primary, secondary, and supporting colour families should be selected with the audience's visual vocabulary and expectations in mind. Designers should also give the competitive landscape special attention to ensure the identity design is intelligently positioned in the group, either through harmony or contrast.

In addition to single, 2-colour, and full colour versions of the core logomarks, there are other special applications of colour that need to be considered. The identity should define digital palettes *(like links, button colours, and web typography)* and other targeted treatments that may be used by specific applications *(like the colour of uniform material, or Pantone specials only achievable in offset printing).*

Building a rich system

Type and colour are usually the pillars of a visual identity system. An art direction can be further enhanced by incorporating graphic elements such as photography, illustration, and pattern, and through specific, consistent usage of design principles.

Photography & Image Treatments

A consistent approach to photography, both in style and subject matter, creates an additional through line that audiences recognize and associate with the brand. Intentional image crops, standard colour treatments, and other filters/effects are important to consider when establishing an art direction. For example, if the tabled identity concept suggests portraits of people are an effective way to connect with the audience, make a plan for consistent lighting, cropping, background, and model styling/tonality if possible. This will result in a style of photography that is as memorable as the logomark itself.

Illustration

Illustration is a great way to capture abstract ideas in a visual form *(something that is often difficult with photography).* They can be informative while adding additional character to the brand voice. Like photography, illustrations also need a consistent approach to visual execution, visual tonality, and subject matter to maximize their effectiveness in an identity context.

Pattern

Just like a brand mascot or illustration style, the use of pattern in an identity system can become an iconic element that the audience recognizes and gravitates towards. Whether it's a single pattern used in various applications, or a family of different meaningful patterns, they can be a successful way to visually communicate an additional layer of brand character. Well executed patterns also provide a brand with more visual versatility, as they can be applied to specific visual environments that other identity elements may have trouble with *(like wrapping papers, textile designs, and backgrounds for text).*

Showing a logo design concept in context

Good

Sample Applications:
Consider these simple logo applications that don't require the design of additional layouts:

- presentation folder
- tshirt/uniform
- mug
- step & repeat pattern
- app splash screen
- truck graphic
- bag
- environmental graphic

See section 4 for more info about the full contents of a visual identity presentation

Awesome!

Sample Applications:
Consider these more complicated logo applications to help show the concept in context. These may take more effort to create since they involve more content and craft than just applying a logo to a space:

- business card
- responsive web design
- presentation deck
- brochure
- postcard
- corporate badges
- billboard
- print advertising

See section 4 for more info about the full contents of a visual identity presentation

Another Thought

You're just my type

Type selection is one of the fastest ways to make or break an entire visual identity system. Each typeface comes with a world of meaning and history baked into its humble glyphs. Good designers can anticipate how a typeface *(or type pairing)* will resonate with a specific audience, and can make intelligent selections that will deliver the intended meaning of the text.

Interpretation of type is always subjective, and is based on the visual and cultural vocabulary of the viewer. However, there are some common audience reactions that can be anticipated: serif typefaces tend to feel more traditional than sans serifs *(thanks to the history and chronology of type design),* uppercase text settings typically feel more mature/rigid than lowercase settings, and Comic Sans shouldn't be used on anything that wants to be taken seriously.

Beneath the more obvious layers of association is a world of typographic nuance and subtlety that is experienced on an emotional level. The designer's careful engineering of these characteristics should be explicitly noted in a visual identity presentation to support each concept's rationale. An example comment could sound like, "This rounded font used in a lowercase setting feels friendly to directly appeal to the desired 'approachable' quality defined in the creative brief." It's important for designers to clearly justify and articulate their type choices so that clients can better judge the effectiveness of those selections.

When selecting type for wordmark concepts and as supporting type pairings, consider the following:

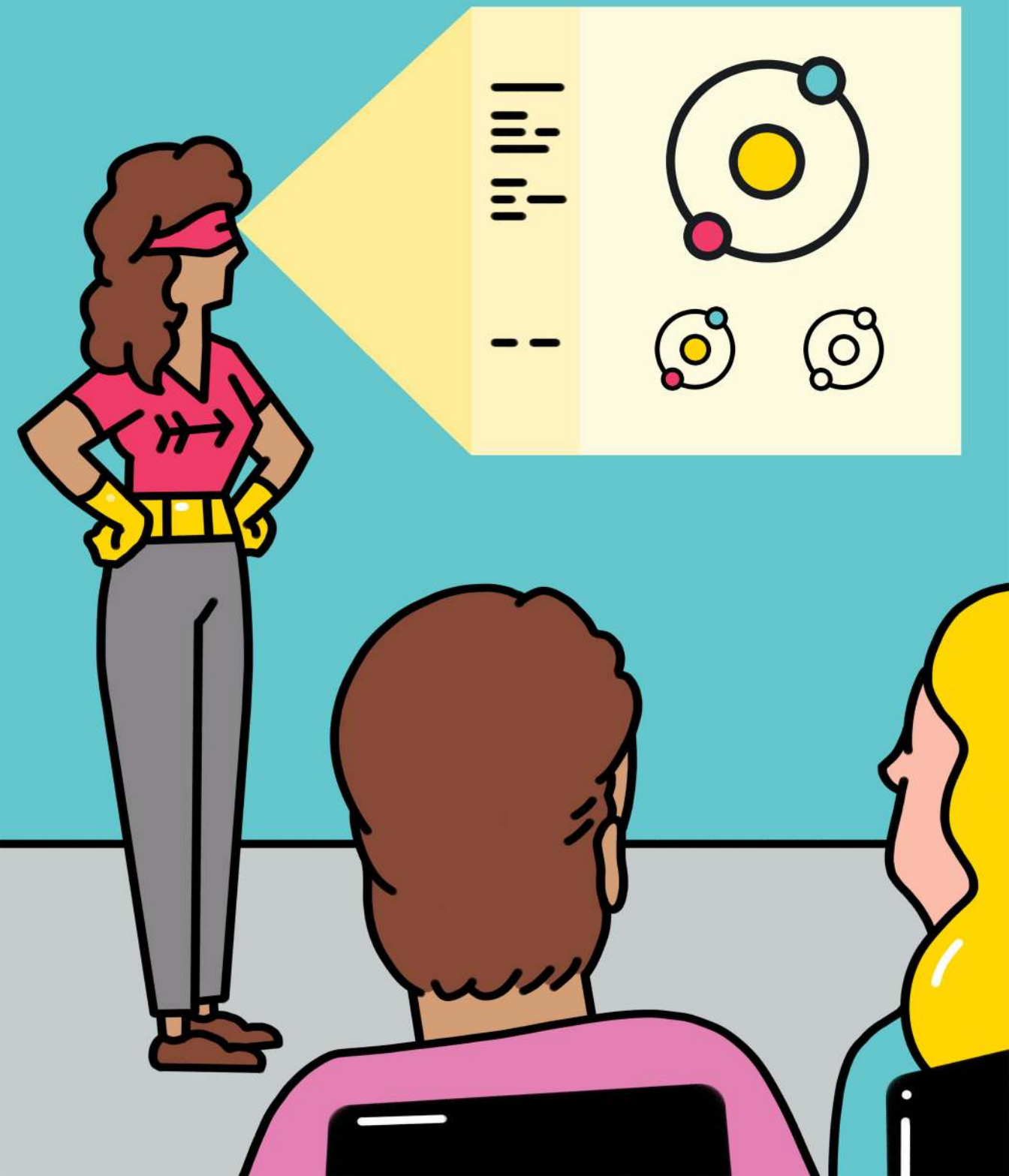
- how does the font feel visually *(based on its geometry/structural detail, overall weight, style)?*
- are those feelings meaningful to the key concepts in the creative brief?

Another Thought

You're just my type (cont.)

- what text setting *(lowercase, Title Case, camelCase, ALL CAPS, etc.)* or tracking/letter-spacing is suitable for the font in order to convey the right message to the audience?
- are there any existing cultural references to the typeface that should be weighed? *(e.g., Didot and other modern serif typefaces are dripping with 'fashion' references in our visual landscape, which may cause a mixed message if used incorrectly)*
- are the selected typefaces well crafted and technically suitable for the needs of the identity system? *(e.g., if print collateral appears in English and French, does the selected typeface provide the appropriate French accents?)*
- is the selected type complementary to the message or meaningfully opposed?
- is the selected type pairing visually and conceptually harmonious? *(whether the pairing is selected based on complementary or contrasting geometry or the history of the typefaces, it's important that the pairing is meaningful to the creative brief)*
- will the selected fonts provide enough versatility to satisfy all the needs of the visual identity system and deliver the right feeling and message across all touchpoints?

Section 4
Notes on Presenting



Good design does not sell itself. A thoughtful and articulate presentation of design work is essential if a successful concept is to become a real solution. The designer needs to capitalize on their intimacy with each concept and clearly explain how they solved the challenge.

Many successful designs are simple. Designers must reveal the intentionality behind these seemingly simple solutions to their clients. Clients need to be shown that a particular typeface, or some other visual treatment, has been chosen because it directly applies to an aspect of the creative brief.

We never share sketches with the client – they contain too many ideas for a targeted, effective presentation. Sketches also typically fail to demonstrate an articulate solution. Instead, we usually share our first round of logo convergence. These ideas are confidently defined, but still offer the client an opportunity to weigh in on the visual design early in the process.

We always table at least 3 logo/identity concepts. This allows us to explore multiple interpretations of the creative brief and can help the client confidently choose a design direction to pursue.

We don't include any concepts that make us nervous, or that we wouldn't want to see live in the world. Sometimes a client will ask us which option we prefer personally. We tell the client that we believe in all of the concepts presented, and that it doesn't really matter which we prefer – they should choose the concept they believe will resonate most powerfully with the audience defined by the brief. Research is extremely valuable, but designers will never have the same degree of audience

intimacy as the client. The client is in the unique position to select a design based on their deep understanding of their business, audience, and communication goals.

Tips for murdering a design presentation

1. **Review the brief**

   Every visual identity presentation should begin with a quick recap of the approved creative brief. This will calibrate all parties prior to the assessment of work. Design is the solution to a problem, and each design needs to resonate with a target audience. Referring to the brief before sharing concepts provides a refreshed perspective on the task at hand. And by encouraging all parties to view the work from the audience's perspective, personal bias is reduced.

2. **Review the competitive landscape**

   Occasionally, our presentations take a quick spin through the competitive landscape before showing design concepts. Sometimes competitive landscapes can be dominated by common themes or colours – we use this as an opportunity to show how specific concepts could be positioned against those existing designs. For clients that live in more saturated or fiercely competitive spaces, this simple positioning step provides a valuable vantage point for assessing the effectiveness of the following concepts.

3. **Highlight the details**

   Each concept needs to be thoroughly explained. Details that may seem obvious to visually-minded designers can be lost on a client who may not have as much experience reviewing and assessing visual nuances. We leave nothing to chance. We feel it's better to err on the side of thoroughness than to have some valuable strategy or detail go unnoticed. Additionally, each concept should be presented with as little personal bias as possible, allowing the client to assess how each solution meets the needs of the creative brief.

4. Set a comfortable pace

Pacing describes the rate at which concepts and slides are presented. This is highly variable and needs to be adjusted on a per case basis. Our general recommendation is to go slow. Designers spend hours staring at and refining visual concepts. They are intimate with the details, but can forget that clients are seeing the work for the first time. Clients need time to take in the details and assess the work from the perspective of the audience.

In addition to moving slowly between slides in a presentation, we also suggest putting pauses between each concept. Like a bite of pickled ginger between sushi pieces, blank screens between identity concepts can cleanse the visual palate and build anticipation for the next 'bite.'

5. Show each concept in a standardized way

Present all logos and example applications in a standard format. We strive to establish a consistent visual system to show the components of an identity so the client is better able to focus on the differences between each creative solution, rather than variations in the presentation itself.

In general, we include the following slides for each concept:

- **Primary logo design**
  The logo design should be shown alone on white space at a reasonable size *(not comically large)* so the client can appreciate the mark's details and communicative value.

- **Size reductions**
  Demonstrate that the logo can be reduced to very small sizes and still retain legibility – if it doesn't work small, it doesn't work at all.

- **1-colour version**
  Even if the client has not requested a 1-colour version, include one to demonstrate the versatility of the logo design, and its ability to anticipate any future production requirements.

- **Alternate versions / lockups**
  Show any secondary lockup configurations that may be required for specific deliverables, such as square social icons.

- **Colour palette**
  Isolating the colour swatches from the logo design helps the client fully consider and appreciate the palette so they can judge its effectiveness.

- **Type pairing**
  Choose a sample text setting of relevant content to outline the proposed fonts for headings, subheads, body text, etc. Provide licensing information and costs associated with each proposed typeface.

- **Basic applications**
  Choose example applications that are relevant to the client's business goals or audience expectations. Apply the logo and other identity elements to these applications to demonstrate the viability of each proposed identity system.

- **Summary**
  Include the main logo design, size and colour reductions, selected example application, and a short written summary of the approach. Ideally these summary pages should be printer-friendly *(so no solid black backgrounds).*

6. Summarize each concept

As mentioned above, it's important to include a summary slide at the end of each concept before moving to the next. This provides a concise picture of the proposed identity concept and gives the client an idea of the whole system. Each summary should include the logomark, size and colour reductions, a couple of important example collateral applications, and a short written description.

These summary slides can be pulled out of the presentation and delivered in a solo, printable deck for the client's convenience.

Recipes for a successful identity presentation

Good

Presentation Intro:

- Bullet point notes, audience recap, and key concepts from the creative brief
- List of biggest competitors as a reminder of how the work will be positioned

Logo & Visual Identity Presentation:

- Present at least 2 concepts to the client
- Each concept should include:
  - the logomark
  - size reductions of the logo
  - 1-colour version of the logo
  - an example icon treatment
  - basic type pairing
  - a simple example application to demonstrate the logo in context (e.g., tshirt design)
  - summary

Delivery Format:

- Prepare a single flat image for each concept
- Email the presentation with notes to elaborate each design solution or conduct a presentation via screen share
- Request client feedback and revisions by email

Awesome!

Presentation Intro:

- Full reiteration of the creative brief details, summarized in bullets points and diagrams
- Audience profiles and summary to remind everyone who the work is trying to reach
- In-depth competitive landscape analysis with visual examples and notes about each competitor – focus on the tone and voice, general art direction, and their approach to copywriting and logo design

Logo & Visual Identity Presentation:

- Present at least 3 concepts to the client
- Each concept should include:
  - the logomark
  - size reductions of the logo
  - 1-colour version of the logo
  - alternate lockup versions of the logo
  - colour palette in use (swatches and any associated mood images)
  - type pairing and licensing info
  - example icon treatments in context for various social media uses
  - a variety of more complicated sample applications to demonstrate the logo in context (e.g., business card design, print ad, example use in the nav of a web page with proposed type systems)
  - summary

Delivery Format:

- Prepare an annotated, multi-page PDF or presentation web site with all concepts
- Provide written rationales for each aspect of the concept (not just the logomark)

Recipes for a successful identity presentation

Awesome! (cont.)

- Schedule a meeting with all stakeholders to walk them through the designs on screen; provide print outs to reference during the presentation
- Capture first blush feedback and answer questions to help validate concepts
- Following the meeting, email the printer-friendly PDF presentation with summary slides of each concept, or leave a copy of the presentation on USB
- Schedule another meeting or call to discuss additional feedback and revisions
- Request client feedback and revisions by email or capture verbal feedback in writing for the client to review and confirm

Another Thought

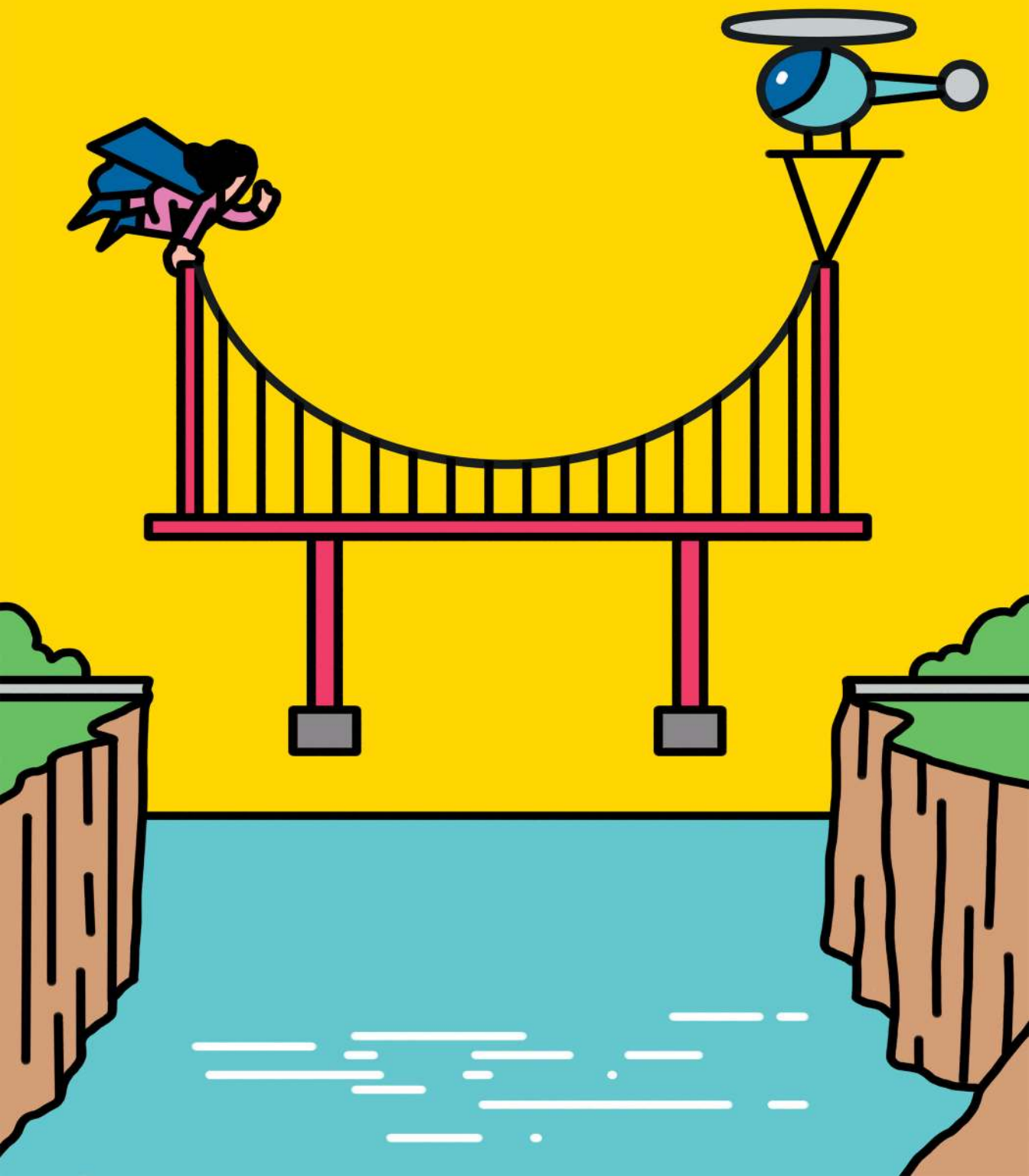
Presentations need to be designed

It might sound a little meta, but presentations are design pieces themselves. Beyond solving the initial design challenge of creating effective logo and visual identity concepts, designers face an additional design challenge of deciding how to best present *(and sell)* those concepts. In the case of a logo presentation, the client is the audience. Designers need to fully consider the best way to appeal to the client's sensitivities and tailor their presentation. How much time and patience do they have? What would comfort or confuse them? What are their expectations and what is their level of visual acuity or savviness?

In terms of pacing a design presentation, we approach the problem by considering what the client is expecting to see, based on our past interactions and discovery meetings. Designers make a living on being able to empathize with others and consider different user experiences from their perspective; crafting a presentation is no different.

By delivering a presentation within the comfort zone of the client, you are less likely to scare them off or make them nervous *(unless that is the desired effect, in which case, ignore this advice!).* Suppose we're presenting a logo redesign and throughout the discovery process the client has seemed reluctant to move away from the existing logomark. We'll take that into consideration when crafting our presentation. Our presentation may start with a logo concept that feels closest to home, reassuring the client that we're taking baby steps. From there, we would slowly migrate to concepts that feel like more of a departure.

Section 5
Revisions & Delivery



Revisions are a fact of life and a vital part of the design process. They provide opportunities to delve further into the most effective ideas.

Client feedback should be centred around the identity concept that most successfully addresses the requirements of the creative brief. Revision notes should be framed as issues *("this concept doesn't feel confident enough")* rather than solutions *("make the logo blue").* This allows the designer to explore diverse solutions to the problem, rather than implementing a revision that may not address the core issue.

Revisions to a concept should always be captured in writing so there is a list of approved changes for everyone to reference. Whether the client provides feedback via email, or the designer sends over meeting minutes of the requested changes for the client to approve, it's important that the revisions are written down. This helps ensure everyone is on the same page and makes it easy to track and delineate the rounds of revisions.

Once the final identity design is approved, the logo files should be packaged in a way that non-designers can easily use: think JPGS and transparent PNGS of all versions. Source artwork, fonts, and colour palette references should also be provided for anyone who needs to work with the visual identity in the future. Bonus points are awarded for more specialized file formats that may be requested down the line: SVG, PDF, or legacy working files.

Logo Pack FTW

Good

Revision Process:

- Include at least 1 round of revisions in the initial project scope
- Ask for the client's feedback in writing and make revisions to the selected concept based on input
- Continue to refine the mark if an additional round of work is required; estimate and bill for the time required
- Prepare approved logo files and visual identity design for delivery (the Logo Pack ZIP)

Logo Pack Contents:

- All logo files in full colour, 1-colour, and alternate lockups provided as AI for use in print
- All logo files in full colour, 1-colour, and alternate lockups provided as JPG, transparent PNG, and SVG for use in digital
- Collection of square logo JPGs or PNGs in different colours for use in social properties, app icons, etc.
- Master AI file of all logo lockups in all colours with organized/named colour swatches
- Fonts used in the logo and visual identity design (depending on license)
- PDF or web page colour reference sheet with swatches, cymk, and rgb/hex values

Awesome!

Revision Process:

- Ask for the client's feedback in writing or provide a written summary of changes discussed and make revisions to the selected concept based on input
- Continue to refine the mark if an additional round of work is required; estimate and bill for the time required
- Prepare approved logo files and visual identity design for delivery (the Logo Pack ZIP)

Logo Pack Contents:

- Table of contents readme.txt providing direction on when each logo file should be used
- All logo files in full colour, 1-colour, and alternate lockups provided as AI, EPS, PDF for use in print
- All logo files in full colour, 1-colour, and alternate lockups provided as JPG, transparent PNG, and SVG for use in digital
- Collection of square logo JPGs or PNGs in different colours for use in social properties, app icons, etc.
- Master AI file of all logo lockups in all colours with organized/named colour swatches
- PDF or web page style guide of typography
- Licensed fonts used in the logo and visual identity
- PDF or web page colour reference sheet with swatches, cymk, rgb/hex, and Pantone values
- PDF or web page of basic logo standards guide for alignments, clearspace, other usage details, and dos and don'ts
- Library of any additional art direction elements like pattern swatches, sample photography, PSDs of photo filters, or GIFs showing approved logo animations

Another Thought

"How much for a logo?"

We recommend designers look at every new project through the lens of process *(not delivery)* when building an estimate. It's important to remember that identity design is a multi-phased effort that takes time, even if the end deliverable is a simple graphic mark expressed within the boundaries of a simple visual system.

When it comes to quoting, the best a designer can do is spell out their process and decide on a price that is fair compensation for the time and skills required. This can be done by tallying hours required by each step of the process, or by considering the final solution and charging a fixed fee based on the value it delivers. Whichever the case, clients should understand that quality design requires skill, and that successful identities have the power to define businesses and drive prosperity.

Closing

TL;DR: design is a process that delivers a solution to a problem. When it comes to their audiences, most brands want to be taken seriously. Even if the underlying message is lighthearted or whimsical, good design can ensure that a brand feels credible and trustworthy.

Visual identities are more important than logos. The quality of an identity design directly represents the quality of a business, service, person, or product. Sometimes identity design is the only visual manifestation of a particular service or company.

Last but not least: identity systems allow brands to communicate with consistency. A sense of consistency is valuable because it promotes memorability and recall. It creates trust and builds affinity. Consistent visual communication facilitates brand loyalty and all the great things that follow: rewarding experiences, referrals, and repeat business.

Thanks for reading!
We hope you've discovered something useful about the creation of visual identities that can be incorporated into a future project. If you found this information valuable, please share it with a colleague.

We released this as a **pay what you want** resource so more creative professionals would be encouraged to examine what

goes on behind the scenes of a logo design. If you feel this guide is valuable to your workflow or business, consider making a contribution at *studiofunction.com/logo-design-guide/* or help us spread the love with a tweet.

More from our brains

If you have the energy for more words from us, we also wrote The Dev Pack: A guide to preparing design mockups for development. This resource recommends and outlines different ways to prepare visual designs for dev handoff *(beyond sending a PSD).* Grab it from *studiofunction.com/dev-pack-guide/* and let us know what you think!

About Studio Function

Our studio is built on three key ideas: We want the freedom to ask the right questions. We want to produce meaningful, functional, beautiful designs. And we want to partner with clients that are solving important problems.

Special thanks

All illustrations in this PDF are by Sam Island. Check out more of his work at *samislandart.com*. Copy editing and general good vibes provided by Caleb Sylvester.

Colophon

This PDF was created using Adobe InDesign. The text is set in *Crimson Text* by UofT engineering student Sebastian Kosch, available via Google Fonts. Headlines and other chunky bits are set in *Nimbus Sans Novus* by URW++.

Help us improve this guide

Our visual identity design process has been a work in progress over the last decade. We're always on the lookout for different approaches and would love to hear your feedback. Tweet @studiofunction or email us at info@studiofunction.com and let's chat.

The complete beginners guide to Adobe Illustrator

Get started

The complete beginners guide to Adobe Illustrator

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Useful shortcut keys — Go to page

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SECTION 3 - CREATE A PROJECT

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CHAPTER 3 - Colouring & Text

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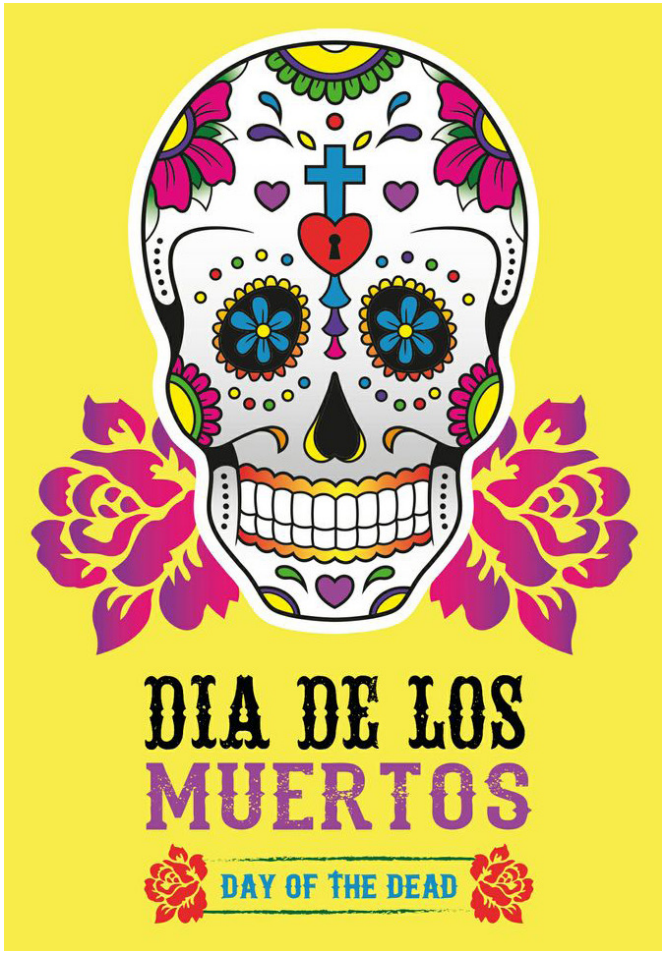
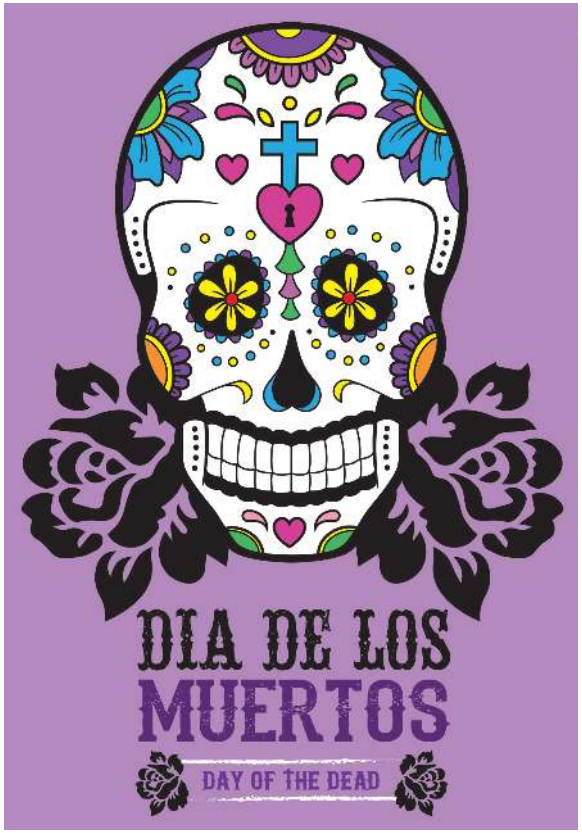
CHAPTER 4 - Finishing & Exporting

Ep 19 Exporting ready for print in Adobe Illustrator — Go to page

What are we going to create on this course

A vector illustration / Poster / T-shirt design





Watch speed art video

COURSE OVERVIEW

The complete beginners guide to Adobe Illustrator

Watch video

Overview

In this tutorial I am to going to introduce you to the Adobe Illustrator interface.

Download the project folder here

Download here

[Approx 270MB]

Project folder directory

01 - INTRODUCTION
- 1 - Introduction
- 3 - Artboards
- 4 - Vector basics
- 5 - Fill & Stroke effects
- 6 - Colour
- 7 - 10 handy tips

02 - ESSENTIAL PRACTICE
- 8 - Shapes
- 9 - Grouped vectors
- 10 - Drawing
- 11 - Blob brush tool
- 12 - Text

03 - TEST PROJECT

Chapter 1 - Setup
- 13 - Set up / Place drawing
- 14 - Image trace
- Assets

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- 15 - Trace drawing
- 16 - Compounding vectors

Chapter 3 - Colouring
- 17 - Colouring
- 18 - Adding type

Chapter 4 - Export
- 19 - Exporting
- PDF Examples

Useful Illustrator Short cut keys

V	Selection tool
A	Direct selection tool
S	Scale tool
P	Pen tool
G	Gradient tool
Z	Zoom tool
N	Pencil tool
B	Brush tool
R	Rotate tool
T	Type tool
F	Full screen mode
I	Eyedropper tool
Y	Magic wand tool

Shift C	Anchor tool
Shift B	Blob & brush tool
Shift E	Eraser tool
Shift T	Touch type tool

cmd (Ctrl PC) + C	Copy
cmd (Ctrl PC) + V	Paste
cmd (Ctrl PC) + 0	Zoom to fit document to workspace area
cmd (Ctrl PC) + 1	Zoom to 100%
cmd (Ctrl PC) + R	Show rulers
cmd (Ctrl PC) + ;	Show guides
cmd (Ctrl PC) + '	Show Grid
cmd (Ctrl PC) + Z	Undo
cmd (Ctrl PC) + G	Group selection
cmd (Ctrl PC) + A	Select all
cmd (Ctrl PC) + L	Create new layer
cmd (Ctrl PC) + X	Cut

cmd (Ctrl PC) + Shift + V	Paste in place
Tab	Toggle visibility of interface panels
Press & hold shift + click & drag	Scale object (with scale tool active)
Press & hold shift + click & drag	Snap rotate (with rotate tool active)
Press & hold shift + click & drag	Snap rotate anchor handles (with pen tool active)
Press & hold shift + click	Select multiple objects (with selection tool active)
Press & hold shift + click	Deselect objects (with selection tool active)
Press & hold shift + click	Draw straight & 45 degree lines (with pen tool active)
Press & hold spacebar	Maneuver around document
Press & hold alt + click & drag	Quick duplicate object (with selection tool active)
Press & hold alt + click	Zoom out (with zoom tool active)
Press up or down	Toggle points (with star shape tool active)
Press up or down	Toggle corners (with round rectangle shape tool active)
Press & hold cmd (Ctrl PC) + click and drag	Toggle depth of points (with star shape tool active)
Press ' [' to increase brush size, press '] ' to decrease brush size	Toggle brush size (with eraser tool active)
Press ' [' to increase brush size, press '] ' to decrease brush size	Toggle brush size (with eraser tool active)

SECTION 1 - INTRODUCTION

Episode 1:

Interface Introduction to Adobe Illustrator

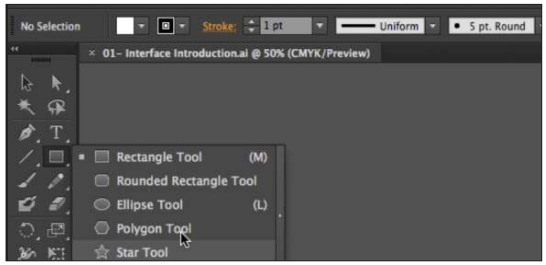
Watch video

Overview

In this tutorial I am to going to introduce you to the Adobe Illustrator interface.

Topics covered

- The tool panels 00:03:24
- The control panel 00:07:40
- Workspace panels 00:09:55
- The document area 00:13:30



Please note

In this video tutorial course we will be using Adobe Illustrator CC for mac. Almost all of the principles demonstrated and covered will apply to future and previous versions. Some differences may apply if you are using a previous or future version.

SECTION 1 - INTRODUCTION

Episode 2:

Panels & Workspaces in Adobe Illustrator

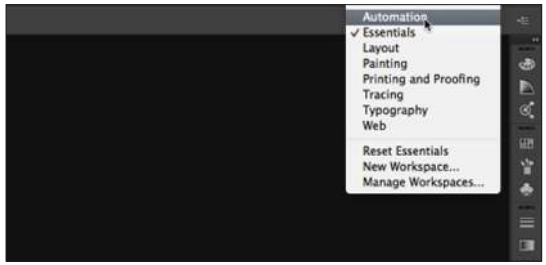
Watch video

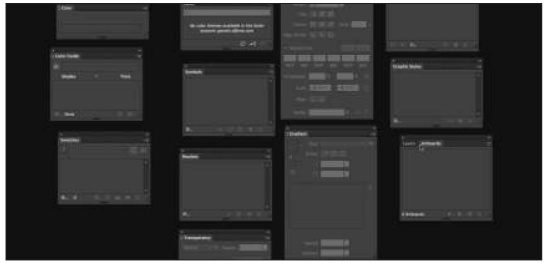
Overview

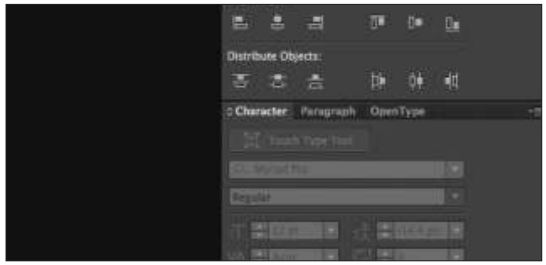
In this video I am going to show you how to customise the panel layout to create a more comfortable workspace. Also I will also be recommending my the workspace that I use that I find really effective.

Topics covered

- Panels review 00:00:00
- How to create a workspace 00:03:45







SECTION 1 - INTRODUCTION

Episode 3:

Artboards in Adobe Illustrator

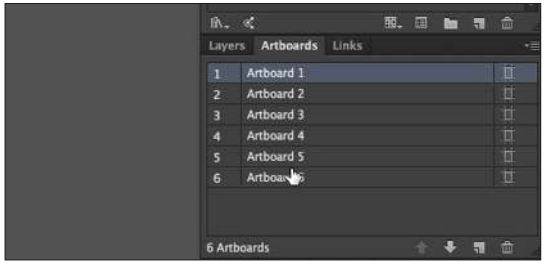
Watch video

Overview

In this video we are going to take a closer look at art boards in Adobe illustrator.

Topics covered

- The Artboard panel 00:02:14
- Artboard tool 00:05:40



SECTION 1 - INTRODUCTION

Episode 4:

Vector basics / Selection & Direct selection tool

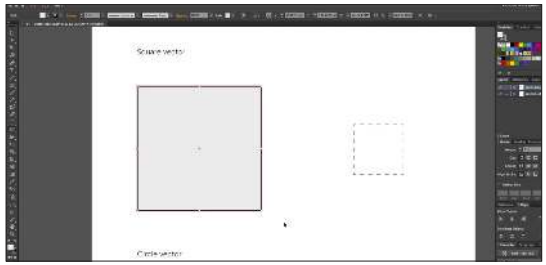
Watch video

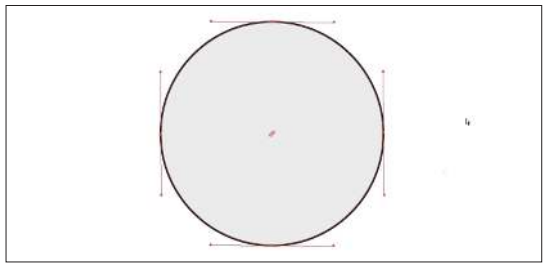
Overview

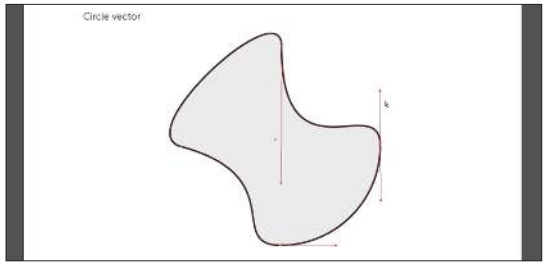
In this tutorial I am going to talk about how creative elements work and how we can begin to operate in illustrator. In this video you will get a good understand of the basics so we can move on later and start building our own creative elements.

Topics covered

- Vector overview 00:00:00
- The Selection tool 00:02:05
- The Scale tool 00:06:23
- The Rotate tool 00:07:47
- Multiplying 00:09:05
- The Direct selection tool 00:10:35
- Anchor points 00:12:20
- Anchor handles 00:15:00







SECTION 1 - INTRODUCTION

Episode 5:

Fill & Stroke effects in Adobe Illustrator

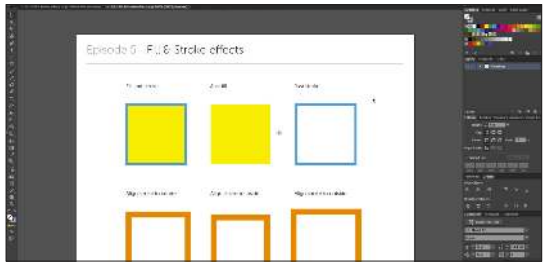
Watch video

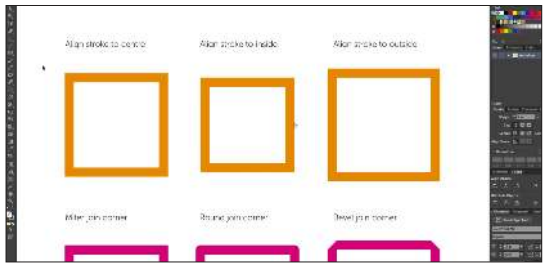
Overview

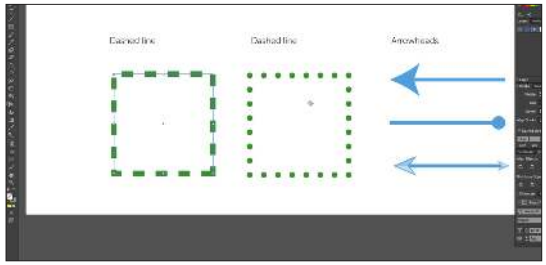
In this tutorial I am going to talk about Fill & Stroke effects in adobe Illustrator.

Topics covered

- Stroke effects 00:01:07
- Stroke align effects 00:05:20
- Stroke corner effects 00:08:15
- Stroke dash effects 00:08:51
- Arrowhead effects 00:09:50







SECTION 1 - INTRODUCTION

Episode 6:

Using Colour / Swatches / Pantone's / Gradients & more

Watch video

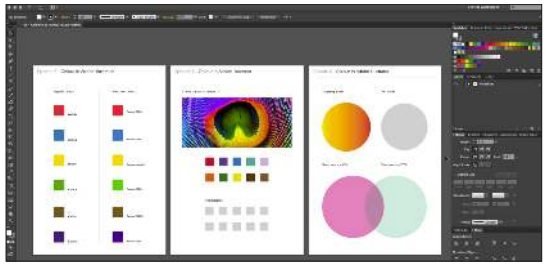
Overview

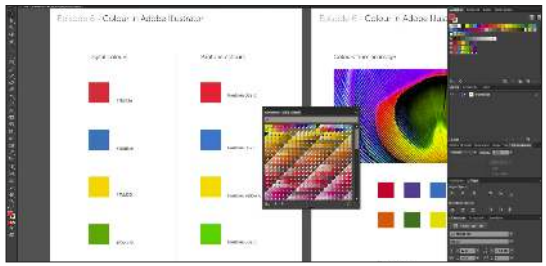
In this tutorial I am going to talk about colour and demonstrate some colour effects and tools in adobe illustrator.

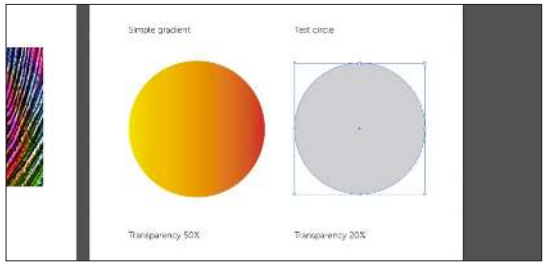
In illustrator there are many ways to use colour. It all really comes down to what exactly you're using the colour for.

Topics covered

- Swatches panel 00:07:57
- Pantones 00:9:55
- Picking colours from an image 00:11:10
- Gradient bar 00:13:40
- Transparency 00:16:12







SECTION 1 - INTRODUCTION

Episode 7:

10 Handy Tips / Things to know for Beginners

Watch video

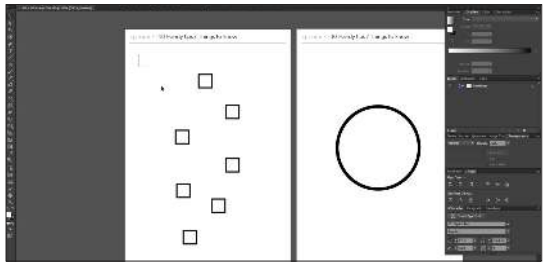
Overview

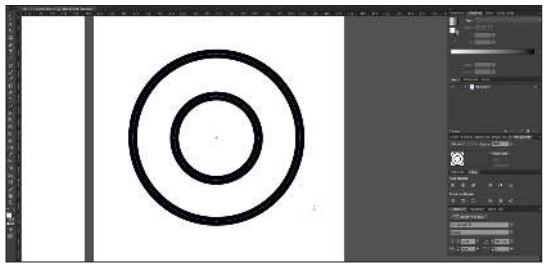
In this video I am going to cover 10 details in Adobe illustrator that you should know and be up to speed with before starting a project.

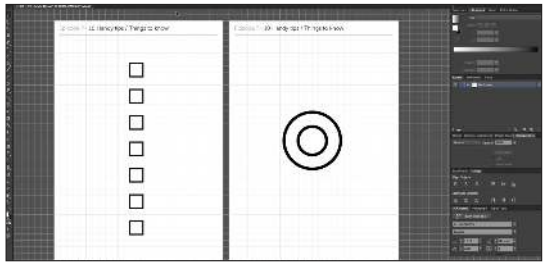
The things I am going to cover in the next few minutes are things that are going to improve your workflow and help you create your artwork.

Topics covered

Tip 1	Zooming	00:01:22
Tip 2	Manoeuvring	00:02:36
Tip 3	Duplicating objects	00:03:06
Tip 4	Select Multiple objects	00:03:33
Tip 5	Aligning	00:04:15
Tip 6	Rulers & guides	00:05:10
Tip 7	Smart guides	00:06:53
Tip 8	Scale strokes & Effects	00:08:33
Tip 9	Shift & Full screen	00:09:45
Tip 10	Grids	00:10:16







SECTION 2 - ESSENTIAL PRACTISE

Episode 8:

Creating shape vectors

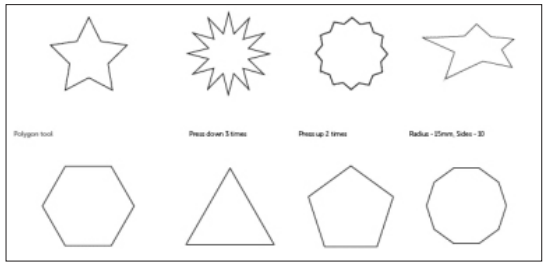
Watch video

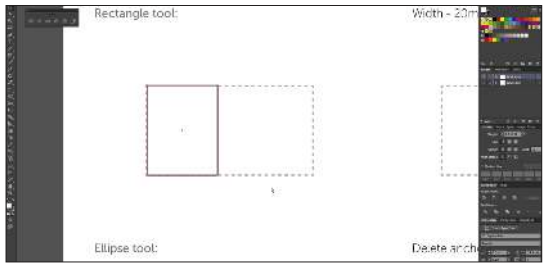
Overview

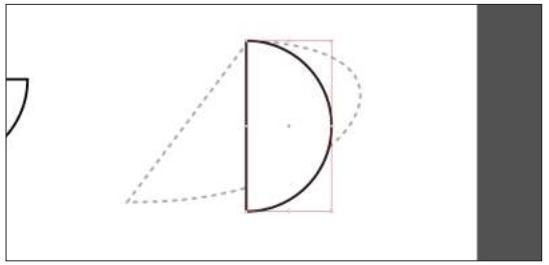
In this video we are going to be focusing on shape vectors. We are going to be creating a range of vector shapes then later look at how to manage multiple vectors in groups and then look at how to compound vectors.

Topics covered

- Rectangle shape tool 00:02:55
- Deleting anchor points 00:08:00
- Ellipse shape tool 00:08:52
- Free transform tool 00:10:49
- Star shape tool 00:11:57
- Free transform tool 00:14:10
- Polygon shape tool 00:15:36
- Rounded rectangle shape tool 00:17:08







SECTION 2 - ESSENTIAL PRACTISE

Episode 9:

Grouped vectors, Compounding vector shapes & Using the shape builder tool

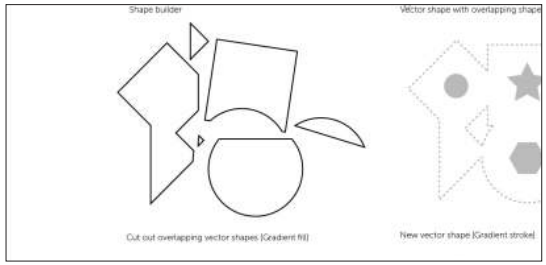
Watch video

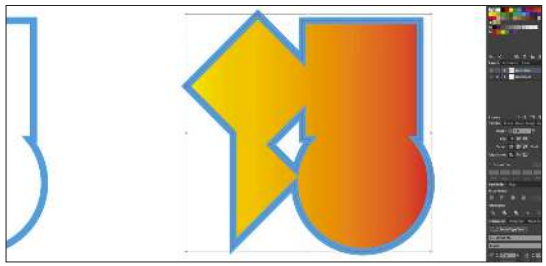
Overview

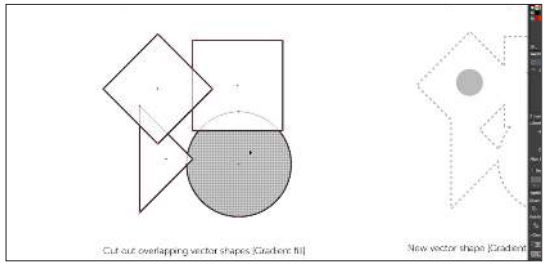
In this video we are going to be using some of the shapes created in episode 7 and learn how to create groups and compound vector shapes.

Topics covered

- Arranging objects 00:03:07
- Grouping vectors 00:05:20
- Compounding vectors 00:07:51
- Shape builder tool 00:09:52
- Cutting shapes out of vectors 00:11:39







SECTION 2 - ESSENTIAL PRACTISE

Episode 10:

Drawing with the Pen tool / Brush tool / Pencil tool & more

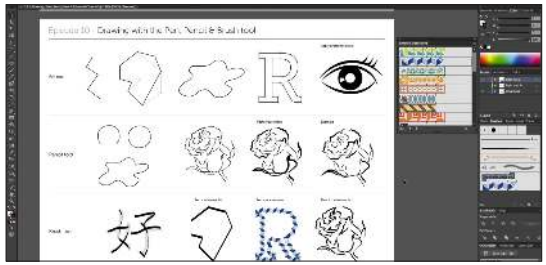
Watch video

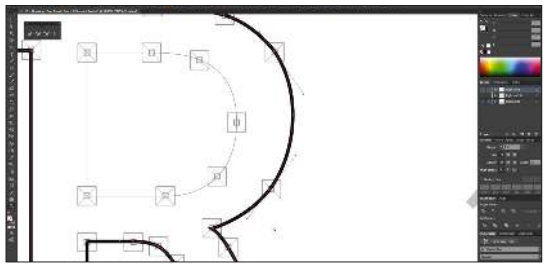
Overview

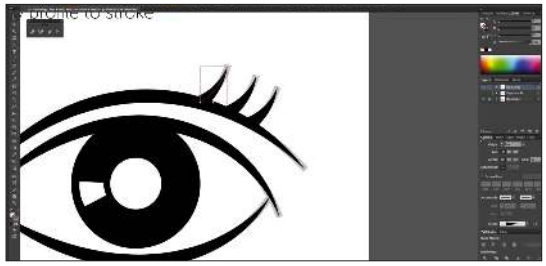
In this video we are going to be covering the drawing tools. Along the way i'll be demonstrating some techniques so you can get a good understanding of the possibilities you have in this programme and what tools you may want to use for any given task.

Topics covered

- Pen tool 00:04:18
- Click & drag technique 00:12:50
- Click & convert technique 00:15:26
- Pencil tool 00:32:40
- Brush tool 00:38:18







SECTION 2 - ESSENTIAL PRACTISE

Episode 11:

The Blob brush tool & Eraser tool in Adobe Illustrator

Watch video

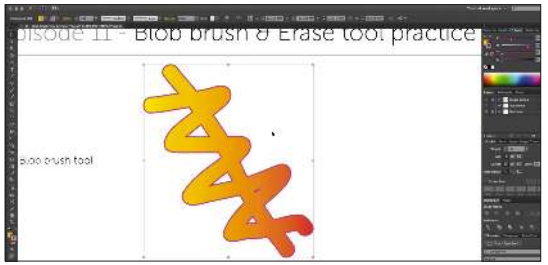
Overview

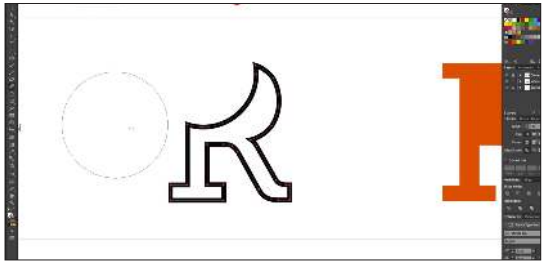
In this video we are going to be taking a look a the blob brush tool and the eraser tool. Now the blob brush tool allows us to draw similar to the brush tool by drawing freehand strokes.

Tho unlike the brush tool, the blob tool does not work along a vector path, the blob tool essentially creates and builds vector shapes.

Topics covered

- Blob brush tool 00:01:45
- Eraser tool 00:04:23
- Combination 00:05:40





SECTION 2 - ESSENTIAL PRACTISE

Episode 12:

Type tools in Adobe Illustrator

Watch video

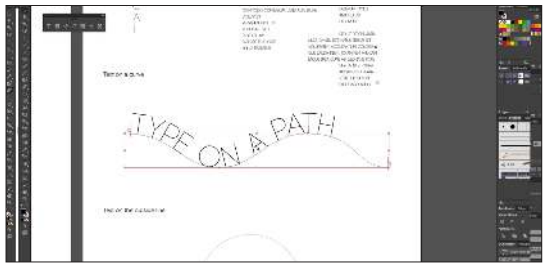
Overview

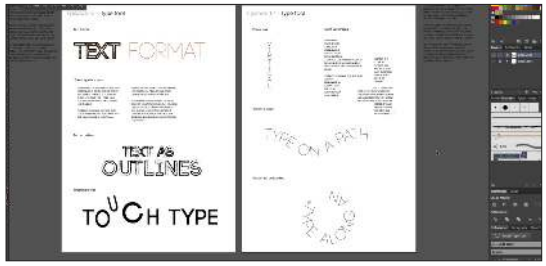
In this video we are going to be focusing on type in Adobe Illustrator. It would be pretty tedious and difficult to create brochures, magazines, newspapers or books in this program but illustrator does offer a range of tools to be creative with type which you may find useful for logos, Tshirt designs, posters or book covers.

In this tutorial we are going to be looking at the the Type tools on offer, and how we might use them.

Topics covered

- Type tool 00:01:50
- Type area box's 00:05:05
- Type as outlines 00:06:45
- Touch type tool 00:08:09
- Vertical type 00:09:37
- Area type tool 00:09:55
- Type on a path tool 00:11:00





SECTION 3 - CREATE A PROJECT / CHAPTER 1 - SETUP & IMAGE TRACE

Episode 13:

Setting up a document / Placing in a drawing / Sketch

Watch video

Overview

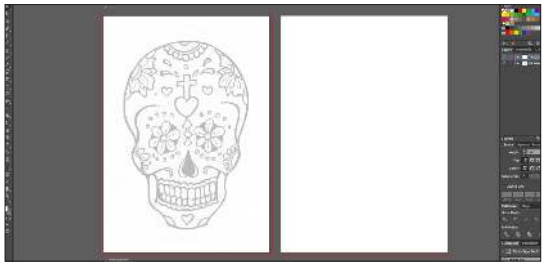
In this video we are going to begin our project by setting up a document in Adobe illustrator for our poster and Tshirt design. Then we will be placing in a previously scanned drawing, and arranging the layers in such a way to prepare us to begin tracing it.

I will be taking you through some of the details you need consider as we set up a new document.

Later In this project we are going to convert the scanned drawing into vector artwork by using the drawing tools in illustrator. Once converted into vector artwork we will be able to apply stroke and colour effects in future.

Topics covered

- Set up a new document 00:01:35
- Place in a sketch 00:05:20
- Apply transparency 00:06:30
- Prepare layers 00:07:15



SECTION 3 - CREATE A PROJECT / CHAPTER 1 - SETUP & IMAGE TRACE

Episode 14:

Image trace tool for sketches in Adobe Illustrator

Watch video

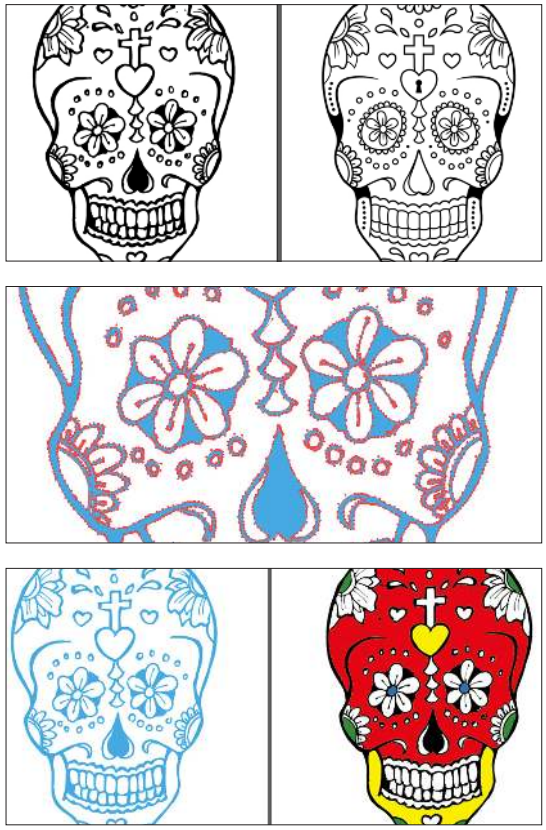
Overview

In this video I will be showcasing a particular tool called the image trace tool, known in earlier versions of illustrator as the live trace tool.

We will be taking a look at this tool, how it works, and later on I will be discussing its pro and cons and why we may, or may not want to use it.

Topics covered

- What is the image trace tool? 00:01:36
- Why use the image trace tool? 00:02:25
- How do we use the image trace tool? 00:04:45
- Why use drawing tools instead of image trace tool? 00:10:55



SECTION 3 - CREATE A PROJECT / CHAPTER 2 - DRAWING

Episode 15:

Tracing a hand drawn sketch & Converting to vector artwork

Watch video

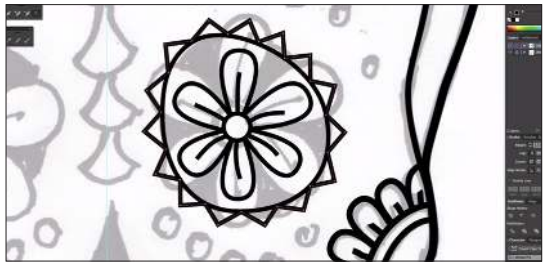
Overview

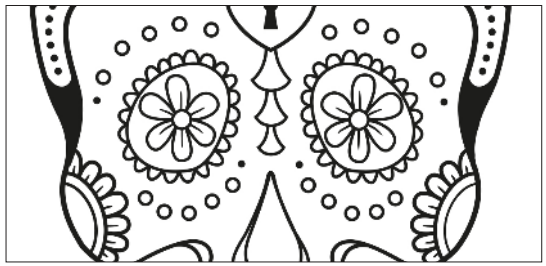
In this video, we are going to use the drawing tools in Adobe illustrator to trace a drawing. We are going to be using the pen tool, the brush tool, the blob brush tool and the Eraser tool to convert a scanned in sketch drawing into vector artwork.

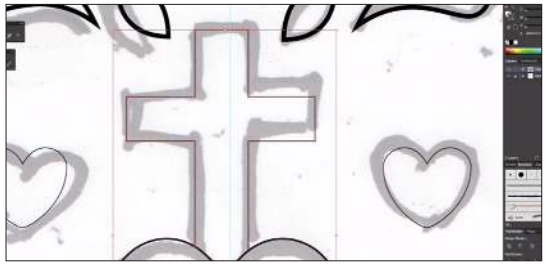
By the end of this tutorial you will be very familiar with the drawing tools and equipt with the knowledge to create your own artwork. I will also be sharing some of my tips and tricks.

Topics covered

- Commence tracing 00:03:20
- Create a custom brush 00:44:25







SECTION 3 - CREATE A PROJECT / CHAPTER 2 - DRAWING

Episode 16:

Compounding vector shapes & strokes / Pathfinder Tool

Watch video

Overview

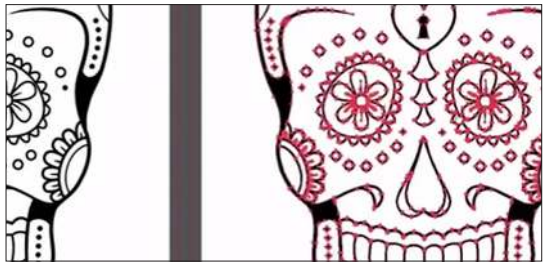
In this video I am going to demonstrate how we can take the vector drawing composition created in the previous video and compound it into a single flat vector.

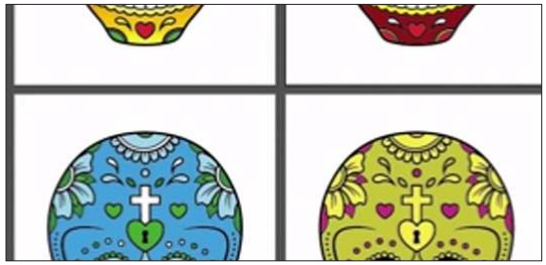
There are many good reasons to use this technique. The main reason why we are going to compound our vector is to help us add colour to the artwork in the next tutorial.

In this video I will be also be demonstrating how to use the compounded shape vector as a stencil.

Topics covered

- What does compounding a vector mean? 00:01:17
- Why compound vectors? 00:01:39
- How to compound vectors 00:02:30
- Using a compounded vector as a stencil 00:06:30





SECTION 3 - CREATE A PROJECT / CHAPTER 3 - COLOURING & ADDING TYPE

Episode 17:

Colouring a vector drawing in Adobe Illustrator

Watch video

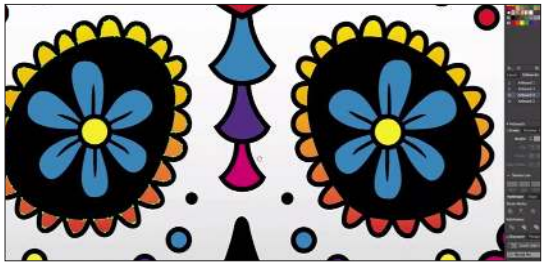
Overview

In this video we are going to carry on from where we left off in the previous video and add colour to our vector drawing. Now for this project we are creating artwork for both a printed poster and a T-shirt.

In this video we are going to deal with CMYK colors and pantones colours and look at the approach for both the Poster design and the T-shirt design. What you will soon discover is that there are various technicalities we need to keep in mind to provide the right artwork for the printers.

Topics covered

- Colouring poster design 00:02:45
- Colouring the T-Shirt design 00:13:55
- Using Pantone colours 00:14:40



SECTION 3 - CREATE A PROJECT / CHAPTER 3 - COLOURING & ADDING TYPE

Episode 18:

Adding type to a poster design in Adobe Illustrator

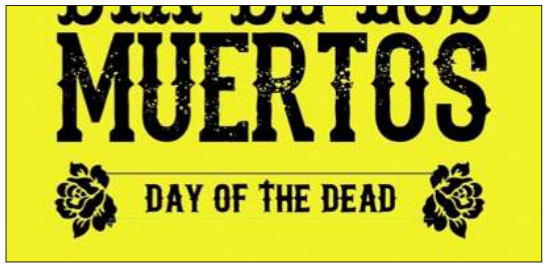
Watch video

Overview

In this video we are going to carry on from where we left off in the previous episode and using the type tools, continue to add some type elements to our poster and Tshirt design.

Topics covered

- Adding type 00:01:35
- Using Touch type tool 00:12:25



SECTION 3 - CREATE A PROJECT / CHAPTER 4 - FINISHING & EXPORTING

Episode 19:

Exporting ready for print in Adobe Illustrator

Watch video

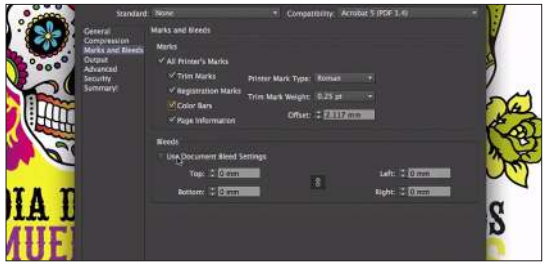
Overview

In this video we are going to prepare the artwork for print. Getting your artwork complete to your design is one thing, tho making sure the printer receives your artwork to print it exactly to your design is another.

Before I send my artwork to print, there are a |few things I need to prepare, review and be certain are correct before sending the artwork off to be printed.

Topics covered

- What is a PDF? 00:02:05
- Review checklist 00:04:34
- Export PDF 00:10:40



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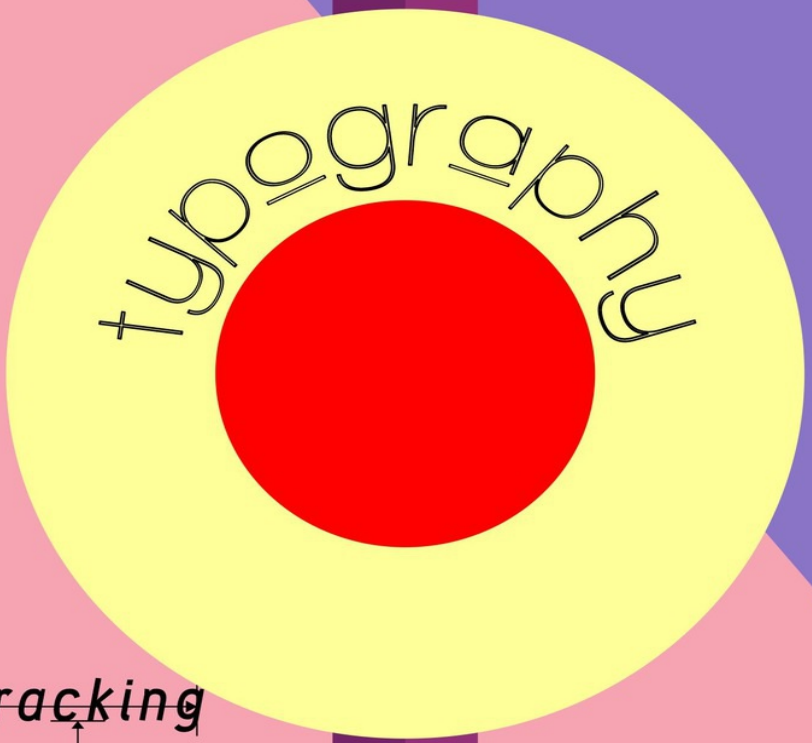
THE GRAPHIC DESIGN BOOK

A Comprehensive Guide for Beginners



FORM

Tracking
Leading
Kerning

Festus Atiba

The Graphic Design Book:

A Comprehensive Guide for Beginners

Festus Atiba © 2020 INTRODUCTION This guide provides information for individuals who are seeking to become expert graphic designers. The guide is a work of compilation aimed at introducing the beginner to what graphic design entails and how they can use the information to build a career in graphic design for themselves. It contains a lot of useful links to resources that would help the designer on this new journey to becoming a force in the creative industry. The guide is structured in sections to allow for easy learning and understanding.
SECTIONS
Section 1 "Graphic Design" ***Types of Graphic design, Specializations and what it entails to be a graphic designer.***
Section 2 "The Fundamentals and Elementsof Graphic Design" ***The concepts of 'Line', 'Shape', 'Form', 'Texture' and 'Balance'***
Section 3 "Layout and Composition" ***Key principles of 'Proximity', 'White Space', 'Alignment', 'Contrast', and 'Repetition'.***
Section 4 "Typography" ***Typefaces and Fonts, Kerning, Leading and Tracking How you can choose typography for your designs.***
"Colour" ***Colour Theory, Colour Wheel, Colour Models and How to adequately select a perfect colour scheme for your designs.***
Section 6 "Graphic Design Software" ***The prons and cons of different design software***
Section 7 Conclusion DICLAIMER The content of this book is

not fully original. Some of the texts, images, photos, illustrations and diagrams found in this work were obtained from blogs, open source websites and online educational platforms. Reasonable efforts have been made to credit these sources at the end of the book. Any liability either direct or indirect arising from the use and application of this book is hereby disclaimed.

GRAPHIC DESIGN

What is Graphic Design?

According to the American Institute of Graphic Arts (AIGA), graphic design is defined as **"the art and practice of planning and projecting ideas and experiences with visual and textual content."**

In simple terms, graphic design is the process of visual communication and problem-solving through the use of typography, photography, iconography and illustration. It communicates certain ideas or messages in a visual way. These visuals can be as simple as a road signs, business or corporate logo, to as complex as page layouts on a website or technical schematics.

Most times, visual designs around us are taken for granted. But hidden in every design from book covers to an exit sign lies a set of design ideas that subtly influence our perceptions, decisions and choices.

Most graphic design projects have commercial purposes like advertisements and promotions. Graphic design can be used by companies to promote and sell products through advertising, by websites to convey complicated information in a digestible way through infographics, or by businesses to develop an identity through branding, among other things. Among other things, it is also used purely as a means for artistic expression.

Design can aid in selling a product or idea. It is applied to products and elements of company identity such as logos, colors, packaging and text as part of branding. Branding has become increasingly more important in the range of services offered by graphic designers. Graphic designers often form part of a branding team.

Graphic design is applied in the entertainment industry in decoration, scenery and visual story telling. Other examples of design for entertainment purposes include novels, vinyl album covers, comic books, DVD covers, opening credits and closing credits in filmmaking, and programs and props on stage. This could also include artwork used for T-shirts and other items screen-printed for sale.

Information design from scientific journals to news reporting, the presentation of opinion and facts is often improved with graphics and thoughtful compositions of visual information. Newspapers, magazines, blogs, television and film documentaries may use graphic design. With the advent of the web, information designers with experience in interactive tools are increasingly used to illustrate the background to news stories. Information design can include data visualization, which involves using programs to interpret and form data into a visually compelling presentation, and can be tied in with information graphics (known as infographic).

To understand graphic design, it is important to know the fundamental elements and principles that make up a design. These elements will help you to decide what to include in your design and how create visually stimulating designs that resonates with the consumer again and again.

Types of Graphic Design

1. **Visual identity graphic design:** Designers that specialize in visual identity graphic design work with brands. They create brand identity assets like logos, business cards, Letterheads, typography, color palettes and image libraries that represent a brand's personality. Here, designers often develop a set of visual brand guidelines (style guides) that describe best practices and provide examples of visual branding applied across various media. These guidelines help to ensure brand consistency throughout future applications.

2. **Marketing & advertising graphic design**: This refers to designs created for marketing and advertising. It involves the creation of visual content to create awareness or promote more a product, service or brand. Designers can specialize in a specific type of media (vehicle wraps or magazine ads, for example) or create a broad assortment of collateral for print, digital, and beyond. Examples include: • Postcards and flyers, Magazine and newspaper

ads, Posters, banners and billboards, Infographics, Brochures (print and digital), Vehicle wraps, Signage and trade show displays, Email marketing templates, PowerPoint presentations, Menus, Social media ads, banners and graphics, Banner and retargeting ads, Images for websites and blogs.

3. **User interface graphic design:** A user interface (UI) is how a user interacts with a device or application. UI design is the process of designing interfaces to make them easy to use and provide a user-friendly experience. A UI includes all of the things a user interacts with—the screen, keyboard and mouse—but in the context of graphic design, UI design focuses on the user's visual experience and the design of on-screen graphic elements like buttons, menus, microinteractions, and more. It's a UI designer's job to balance aesthetic appeal with technical functionality.

4. **Publication graphic design:** Publications are long-form pieces that communicate with an audience through public distribution. They have traditionally been a print medium. However, there's recently been a significant rise in digital publishing with the advent of technology. Examples include Books, Newspapers, Newsletters, Directories, Annual reports, Magazines, Catalogs.

5. **Packaging graphic design:** Products require some form of packaging to protect and prepare them for storage, distribution, and sale. Packaging design can also communicate directly to consumers, which makes it an extremely valuable marketing tool. Every box, bottle and bag, every can, container, or canister is a chance tell the story of a brand. Packaging design involves specialization in a specific type of packaging (like labels or beverage cans) or a specific industry (like food or children's toys). It requires top-notch conceptual and problem-solving skills in addition to a strong working knowledge of print and industrial design. They must be flexible to meet the demands of clients, marketers and manufacturers and be aware of current trends.

6. **Motion graphic design (Animation):** Graphics that are in motion are referred to as motion graphics. This can include animation, audio, typography, imagery, video and other effects that are used in online media, television and film. It can be found across all digital platforms, which has created all sorts of new areas and opportunities. Examples of motion graphic

design: Title sequences and end credits, Advertisements, Animated logos, Trailers, Presentations, Promotional videos, Tutorial videos, Websites, Apps, Video games, Banners, GIFs. Motion graphics designers begin by developing storyboards and then bring their concepts to life with animation, video and traditional art. Depending on the industry, a strong working knowledge of marketing, coding and 3D modeling can be definite assets.

7. **Environmental graphic design:** Environmental graphic design is a multidisciplinary practice that merges graphic, architectural, interior, landscape and industrial design. Designers collaborate with people in any number of these fields to plan and implement their designs. Because of that, designers typically have education and experience in both graphic design and architecture. They must be familiar with industrial design concepts and able to read and sketch architectural plans. Environmental graphic design visually connects people to places to improve their overall experience by making spaces more memorable, interesting, informative or easier to navigate. Examples of environmental graphic design: Signage, Wall murals, Museum exhibitions, Office branding, Public transportation navigation, Retail store interiors, Stadium branding, Event and conference spaces.

Career

Specializations exist across different sectors and jobs and design is an exception. Graphic design is composed of many fields and specializations, ranging from print and web design to animation and motion graphics. Though it is often difficult to decide what type of graphic designer you are because of the overlapping boundaries, specialization still exists. Therefore, designers must adaptable and lifelong learners so they can change or add specializations throughout their careers. Many designers specialize in a single type; others focus on a set of related, similar types. This is mainly due to inconstancy of industry demand and changing technology.

Graphic design offers opportunities and options for individuals of almost any interest. About 20 years ago, graphic design was focused on print-related items like magazines, movie posters and advertisements. Now we’re living in the digital age, which has given birth to several new types of graphic design.

Today, there are many career opportunities in the design industry. These

opportunities majorly stem from advancement in technology used for design. Some of the most notable modern-day design careers include:

- **Graphic designer**
- **UI/UX Designer– User Interface/User experience designer**
- **Web designer**
- **Art director**
- **Creative director**

Other Design Careers
Web Designer: Website designer creates engaging and intuitive web pages for users. This includes overall layout, color scheme and navigation.
UI/UX Designer: They focus on ensuring a website or application is easy and satisfying to use. These designers emphasize value, usability, adoptability and desirability.

Motion graphics designer/Animator : They create motion graphics design —or animation. It involves bringing visual elements to life through special effects. They are used for TV shows, video games and movies.

Job Prospects

Demand for “traditional” graphic designers who work primarily for print publishers has fallen substantially, but that doesn’t tell the whole story. Employment of graphic designers in computer systems design services is projected to grow by 24 percent through 2028, according to the Bureau of Labor Statistics (BLS). Much of this demand is spurred by business’ and organizations’ increased need for digital graphics and imagery as they aim to increase their digital presence.

As you can see, once armed with the proper knowledge and training, there are several graphic design job options out there. You have the ability to cater your career to your personal skills and interests. Section 2
FUNDAMENTALS OF DESIGN

The fundamentals of design are the foundation of every visual medium, from fine art to modern web design. They're even present in seemingly unimportant details, like the fonts that make up most compositions. They help

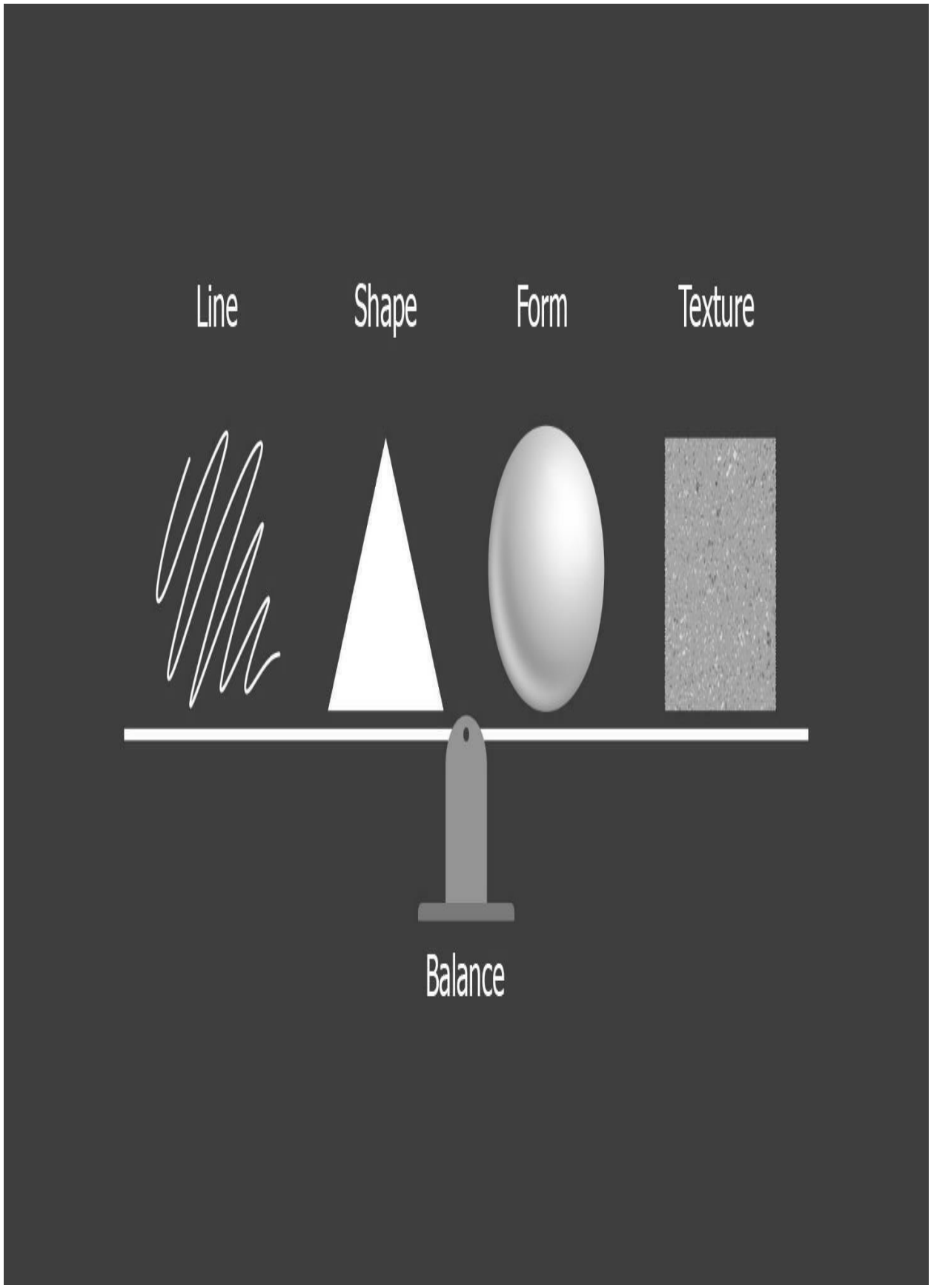
the designer to achieve effective composition, balance and aesthetically stable designs.

Artwerx
ARTISTS | MUSEUMS | FEATURED | ABOUT
Search by artist, style, etc...
GREEN WHEAT FIELDS, AUVERS
Vincent van Gogh
Dutch, 1853 - 1890
1890
Oil on canvas
28.5 x 36 inches
NATIONAL GALLERY OF ART
Collection of Mr. and Mrs. Paul Mellon

Basic Elements and Principles

The basic elements include: **line, shape, form, texture, and balance.** Others include: **Color, Size, and Space.** They might not seem like much on their own, but together, they're part of almost everything we see and create. Principles include: Balance, Contrast, Emphasis, Movement, Proportion, Rhythm

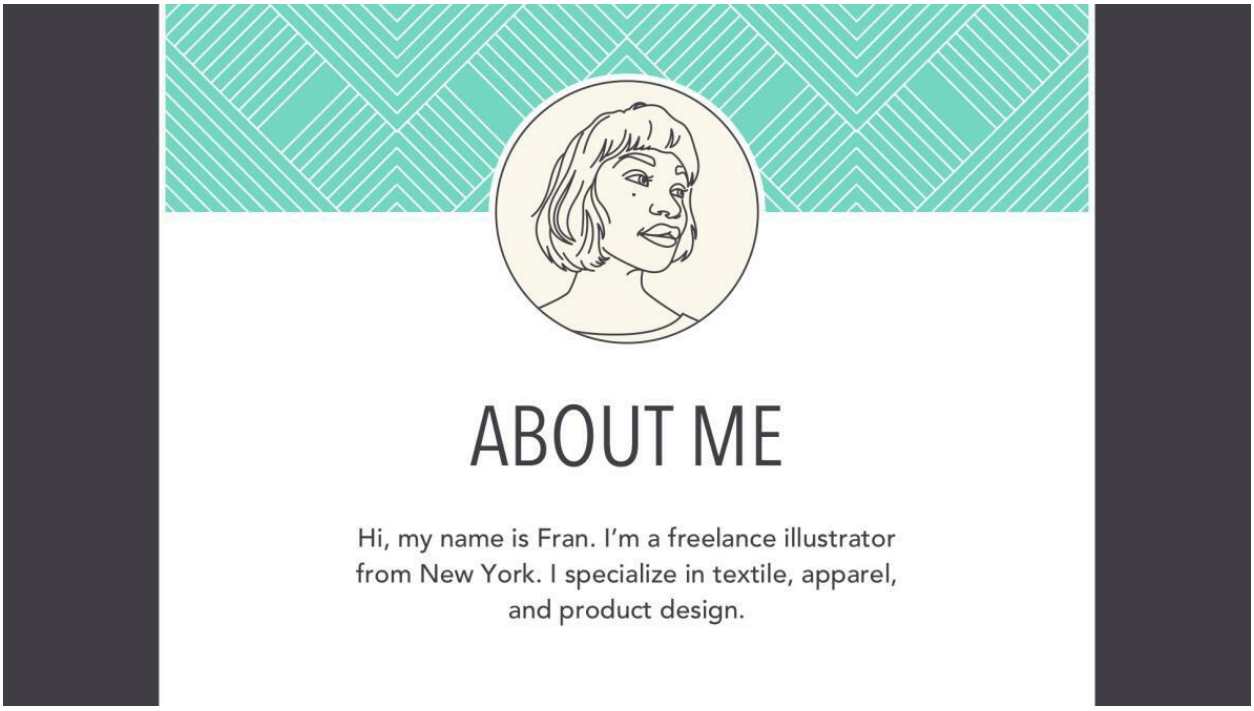
However, the designer is not limited in any way to any of these principles. Because **creativity has no boundaries.**


Line
Shape
Form
Texture
Balance

The fundamentals can be intimidating at first, especially if you don't consider yourself an artist. But keep an open mind—there's a lot they can teach you about working with different assets and creating simple visuals from scratch.

Line

A line is a shape that connects two or more points. It can be fat, thin, wavy, or jagged —the list goes on. Every possibility gives the line a slightly different feel.
Lines appear frequently in design; for example, in drawings and illustrations. They're also common in graphic elements, like textures, patterns, and backgrounds.

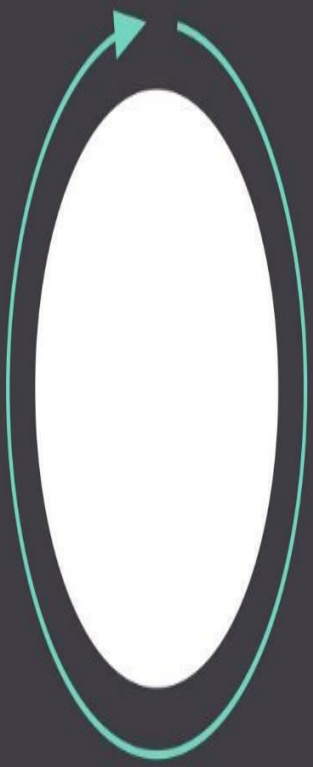


Lines can be used in simple compositions, too —for organization, emphasis, or just decoration. In the example below, lines have been used to create a flow chart that guides the reader's eye from one element to the next.
When working with lines, pay attention to things like weight, colour, texture, and style. These subtle qualities can have a big impact on the way your design is perceived.

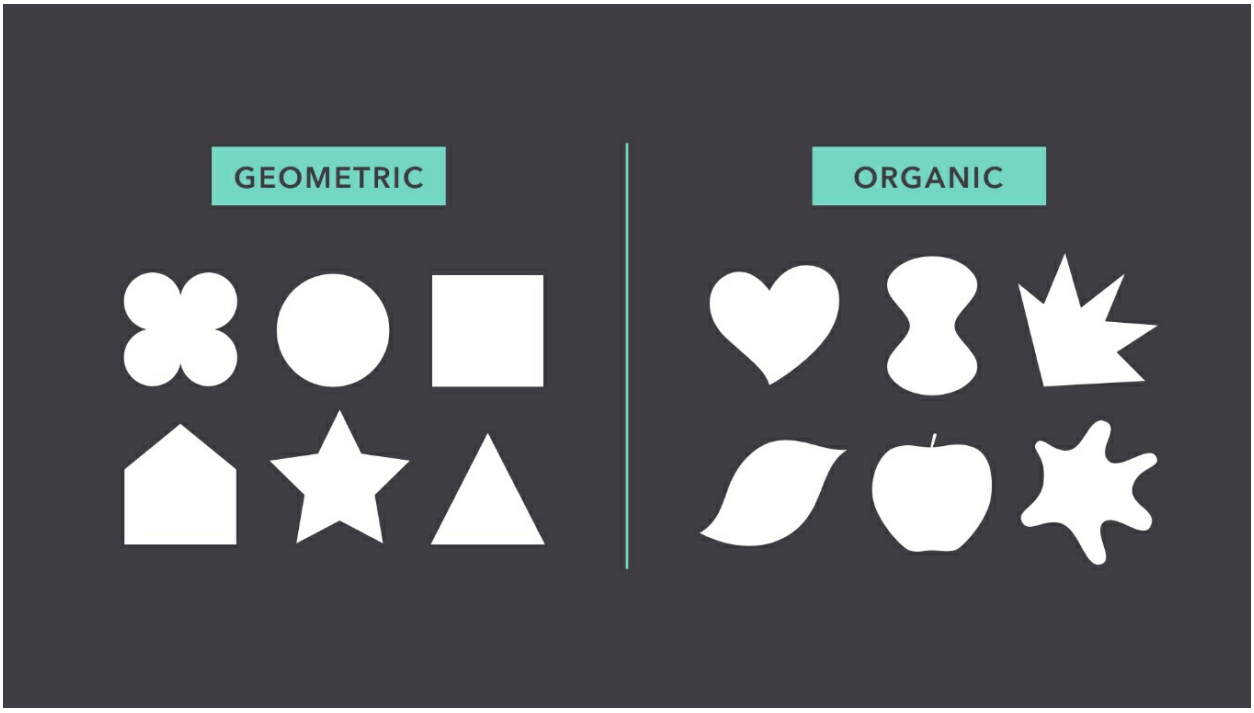
Look for places where lines are hiding in plain sight; for example, in text. Even here, experimenting with different line qualities can give you very different results.

NO PA

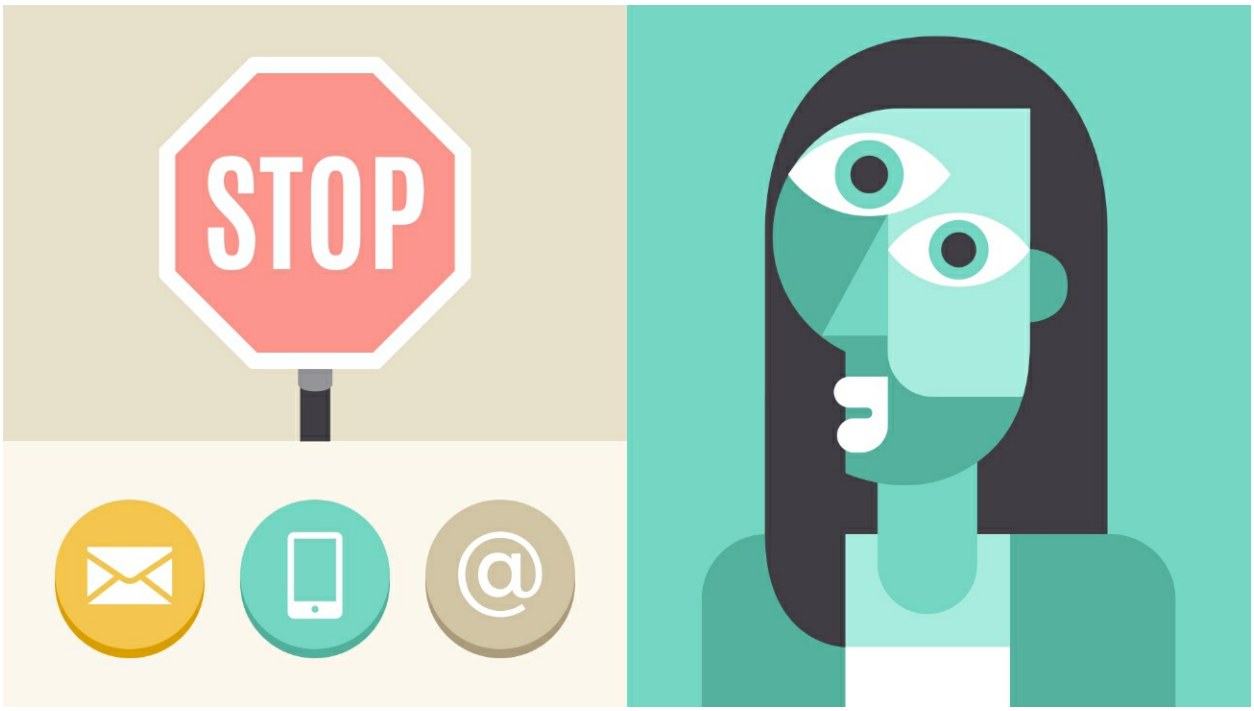
Shape
A shape is any two-dimensional area with a recognizable boundary. This includes circles, squares, triangles, and so on.



Shapes fall into two distinct categories: geometric (or regular) and organic (where the shapes are more free form).

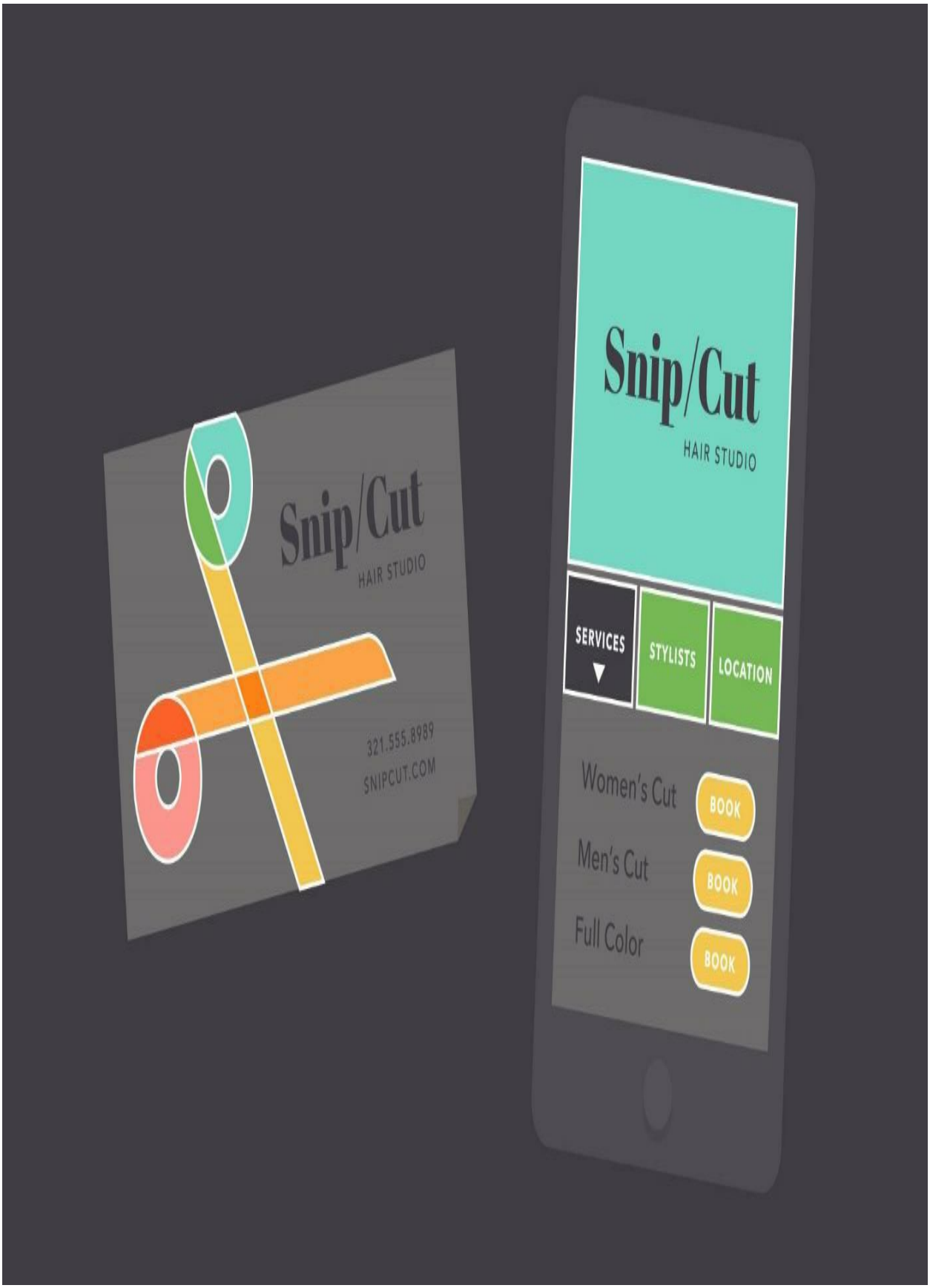


Shapes are a vital part of communicating ideas visually. They give images heft and make them recognizable. We understand street signs, symbols, and even abstract art largely because of shapes.



Shapes have a surprising number of uses in everyday design. They can help you organize or separate content, create simple illustrations, or just add interest to your work. See if you can spot the many examples in the image below.

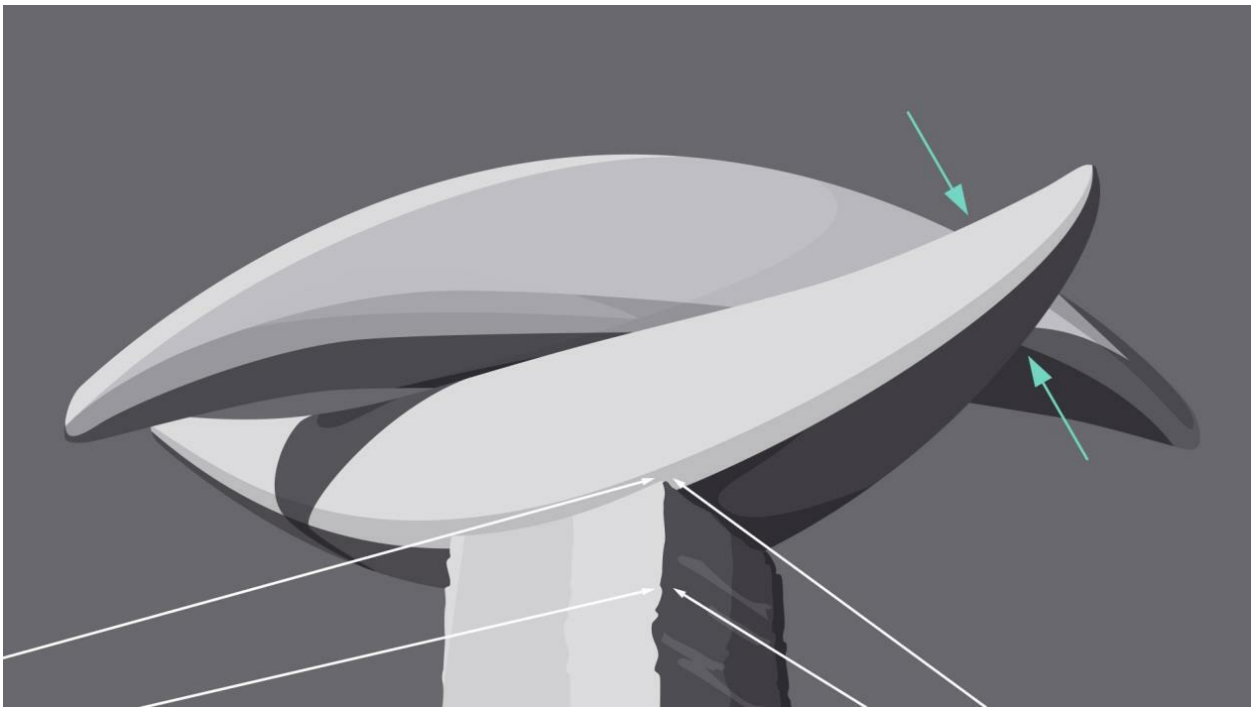
Shapes are important because they're the foundation of so many things. Learn to look for them in other designs, and soon you'll start seeing them everywhere.


Snip/Cut
HAIR STUDIO
321.555.8989
SNIPCUT.COM
Snip/Cut
HAIR STUDIO
SERVICES
STYLISTS
LOCATION
Women's Cut
BOOK
Men's Cut
BOOK
Full Color
BOOK

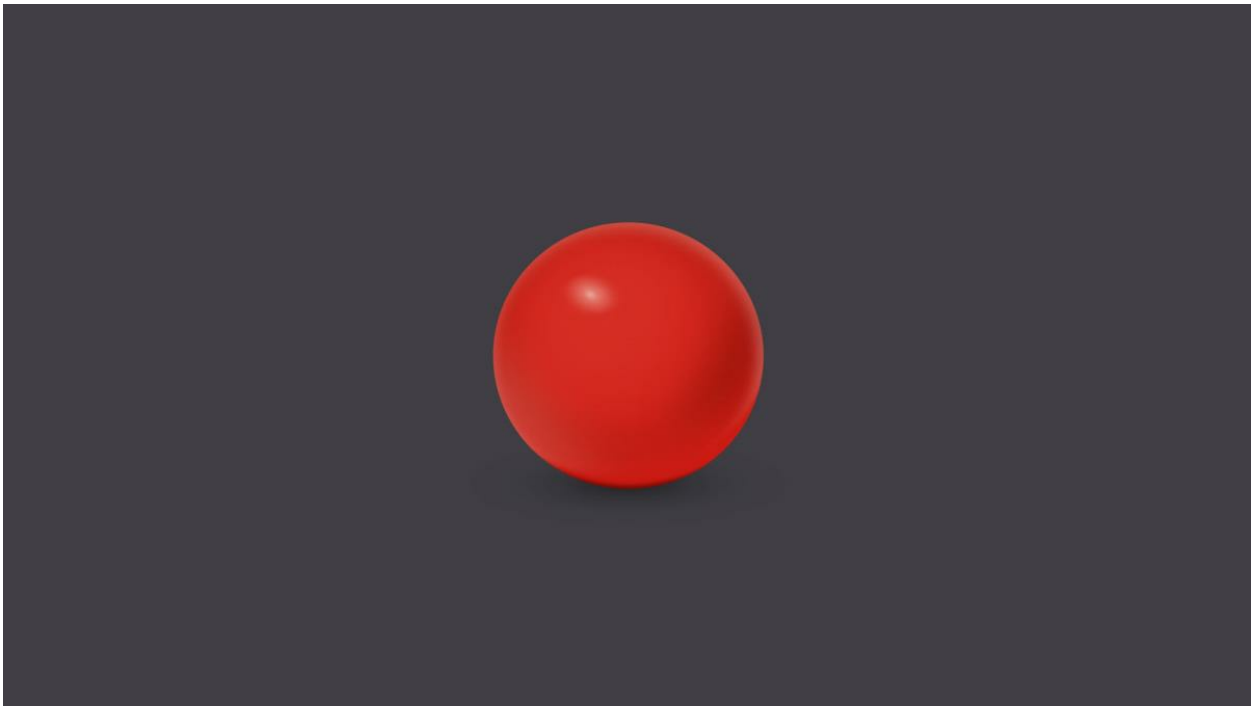
Form

When a shape becomes 3D, we call it a form. Forms make up a variety of things in the real world, including sculptures, architecture, and other three-dimensional objects.

However, forms don't have to be three-dimensional shapes. They can also be implied through illustration, using techniques like light, shadow, and perspective to create the illusion of depth.

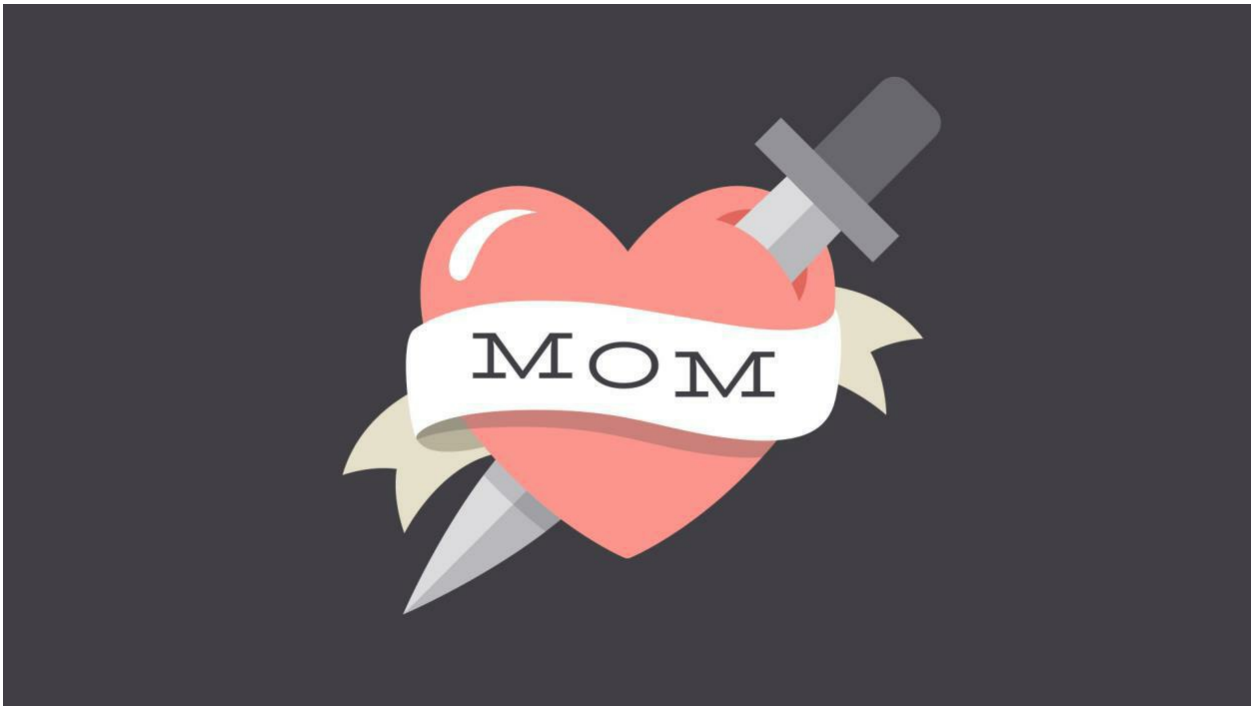


In two-dimensional design, form makes realism possible. Without it, renderings like the image below—a ball with highlights and shading—simply wouldn't be the same.



Even images that are less realistic use similar techniques to create dimension.

Below, the lighting and shading are stylized, but still hint at form and depth.



In everyday composition, the purpose of form is the same, but on a smaller scale. For example, a simple shadow can create the illusion of layers or give an object a sense of place.
Basic forms can bring a touch of realism to your work, which is a powerful tool when used in moderation.

Texture
Texture is the physical quality of a surface. Like form, it can be part of a three-dimensional object, as in the example below (a small prickly cactus in a shiny ceramic pot).

Or it can be implied through illustration, suggesting that it would have texture if it existed in real life. In design, texture adds depth and tactility to otherwise flat images. Objects can appear smooth, rough, hard, or soft, depending on the elements at play.

For beginners, textures make great background images and can add a lot of interest to your work. Look closely, and you may find texture in unexpected places, like distressed fonts and smooth, glossy icons.
Just be careful not to go overboard—too much texture in a single design can quickly become overwhelming.

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Balance

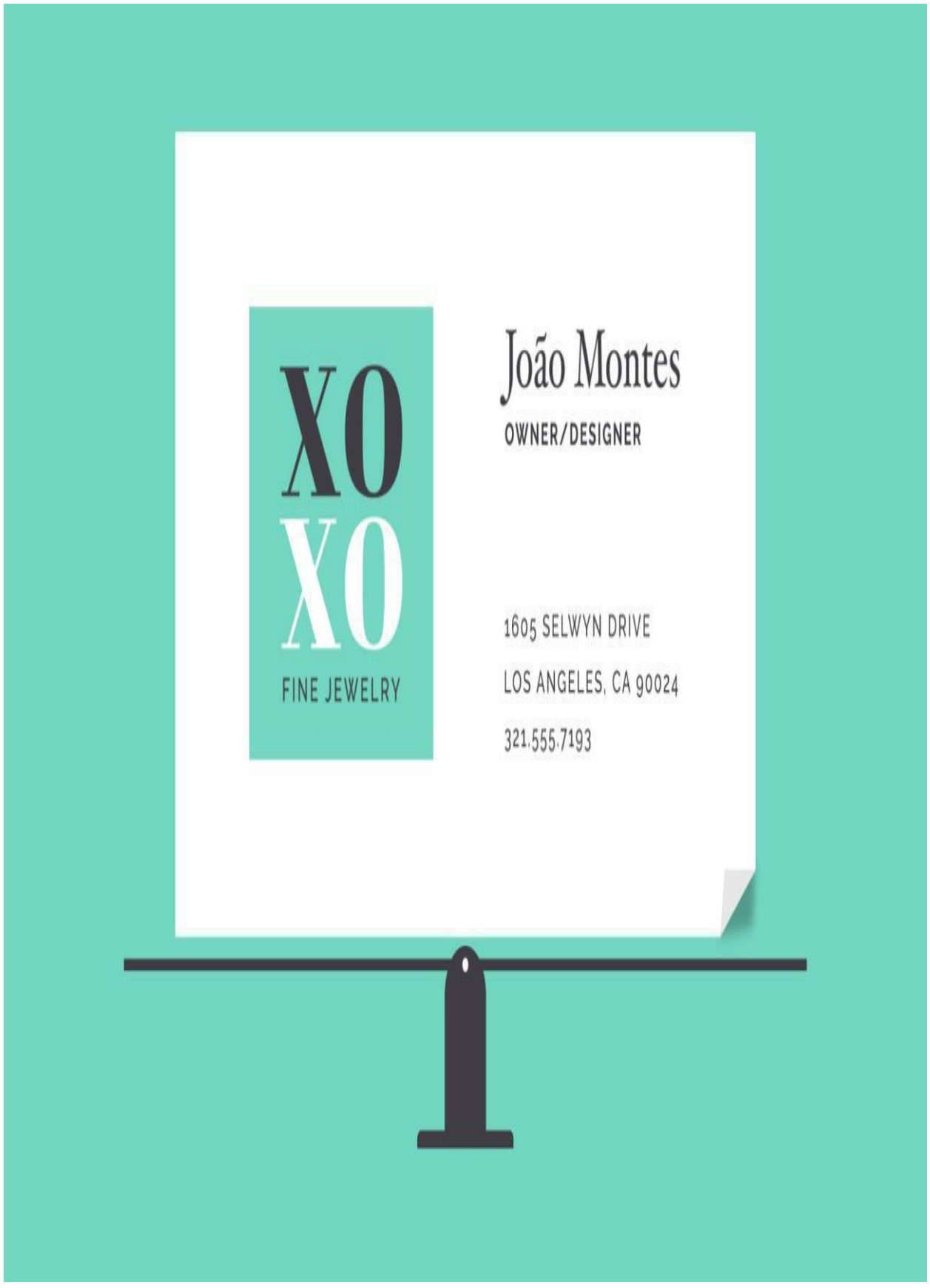
Balance is the equal distribution of visual weight (more specifically, how much any one element attracts the viewer's eye). Balance can be affected by many things, including colour, size, number, and negative space.

Mastering balance can be tricky for beginners because it does take some intuition. Luckily, the design world is full of examples that can help you understand its different iterations.

Symmetrical designs are the same or similar on both sides of an axis. They feel balanced because each side is effectively the same (if not identical).

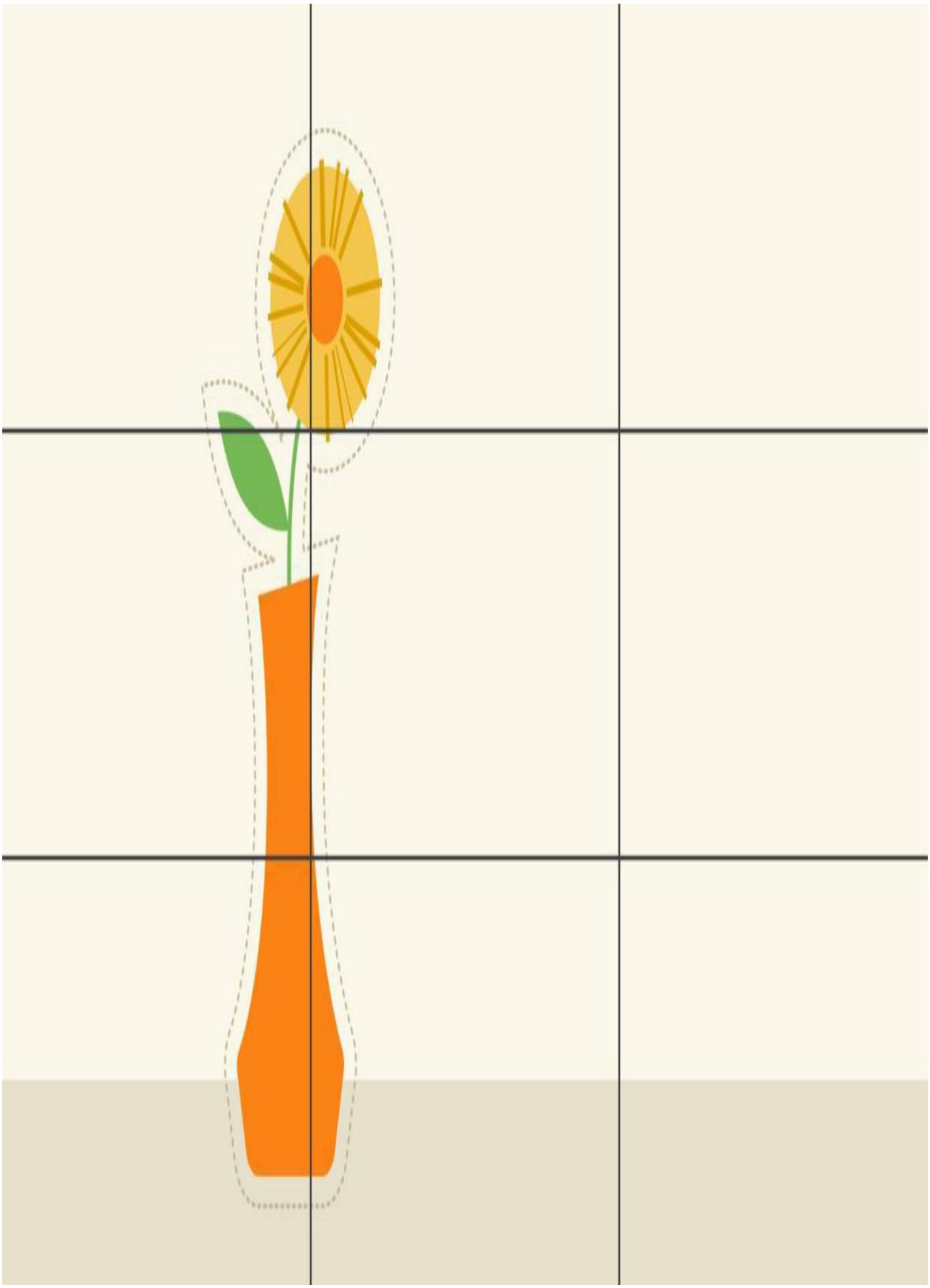
Asymmetrical designs are different, but the weight is still evenly distributed. The composition is balanced because it calls attention to the right things (in

this example, the person's name and company logo).

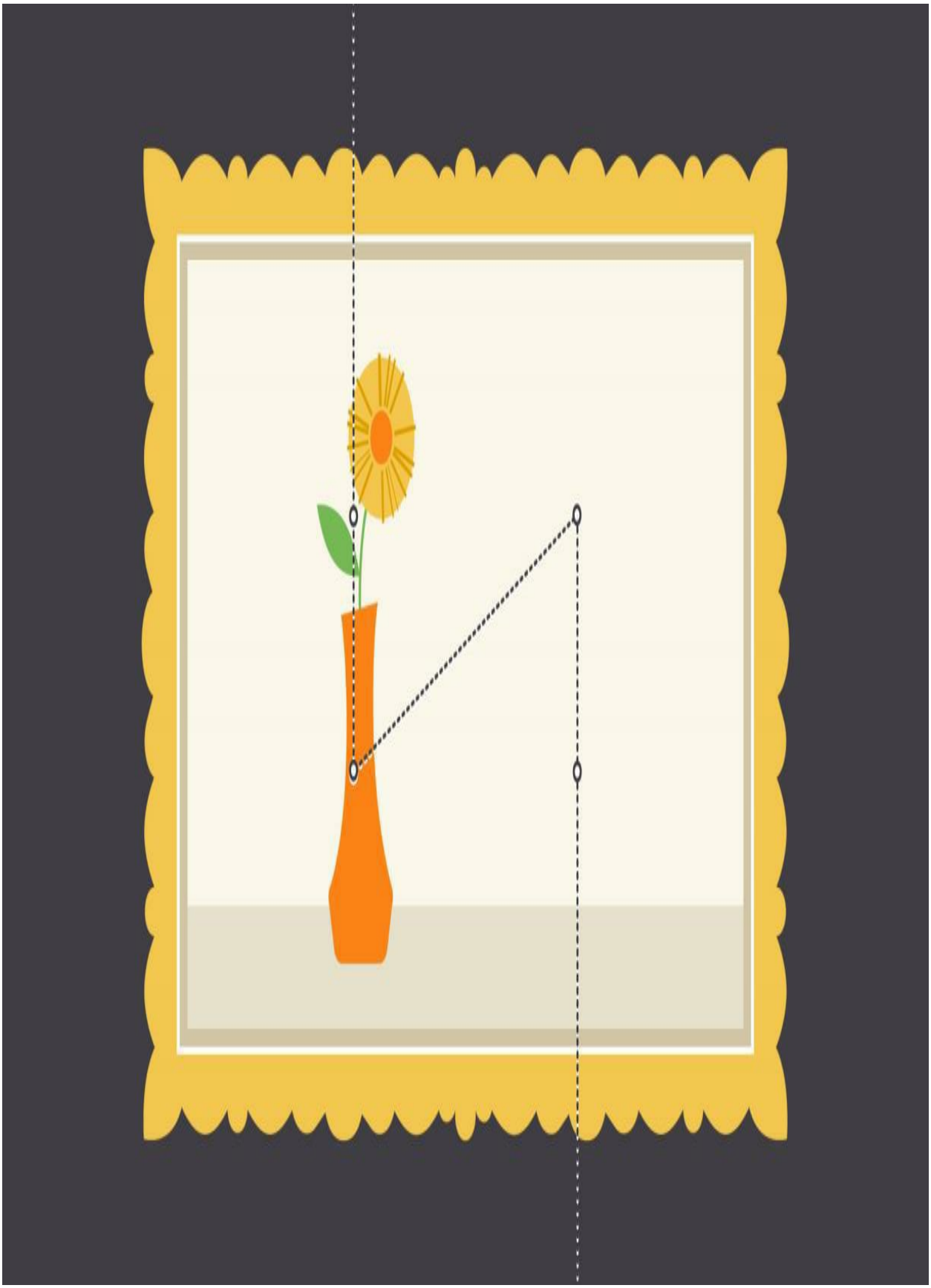

XO
XO
FINE JEWELRY
João Montes
OWNER/DESIGNER
1605 SELWYN DRIVE
LOS ANGELES, CA 90024
321.555.7193

The Rule of Thirds

Many people, including designers and photographers, use a strategy called the rule of thirds. This imagines your work area divided into a 3x3 grid. The focal point of the image is placed on or near one of the grid lines, creating visual balance with the rest of the space.

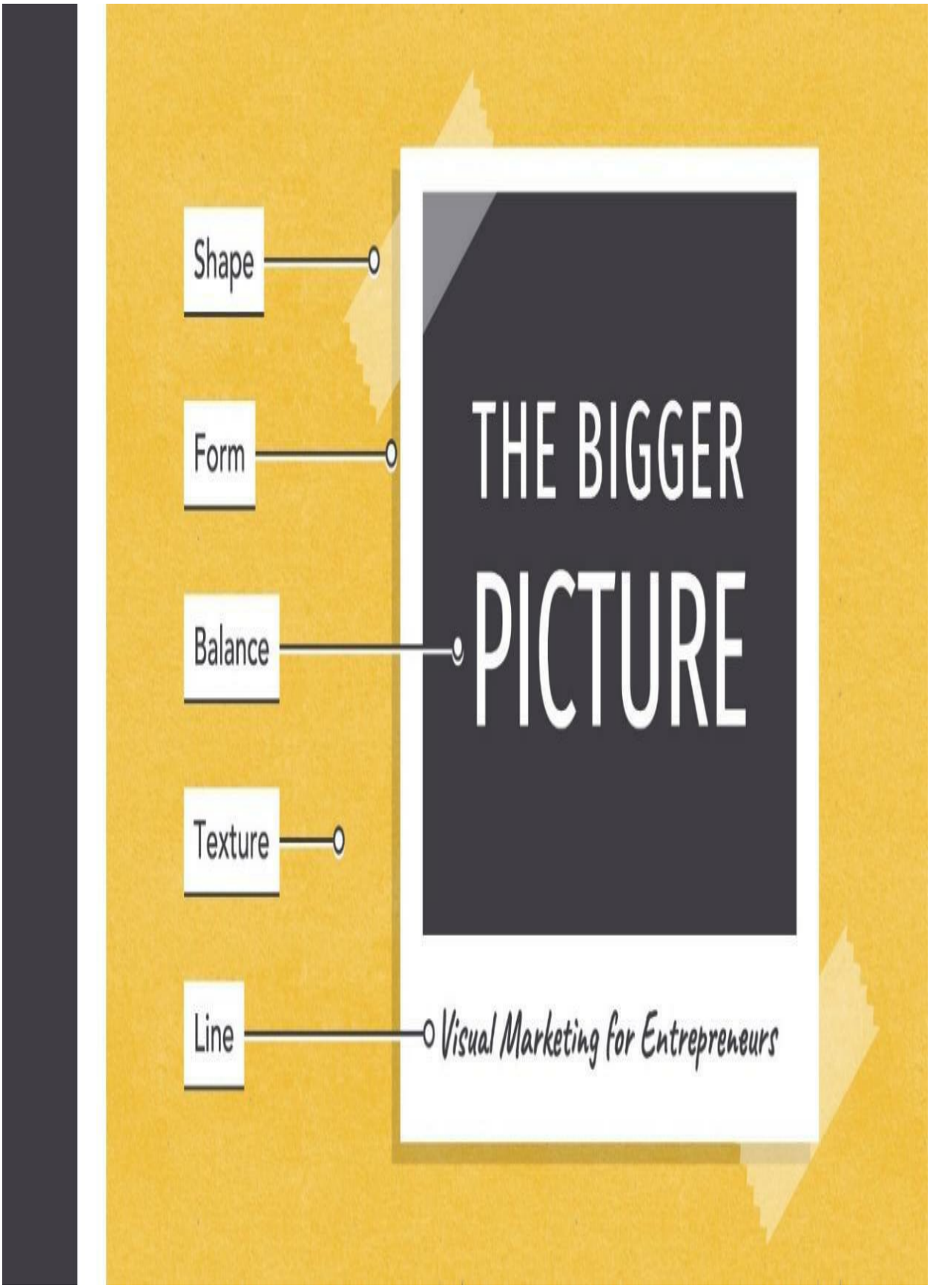


We find this type of composition appealing because, according to studies, the human eye naturally follows this path when scanning a design.

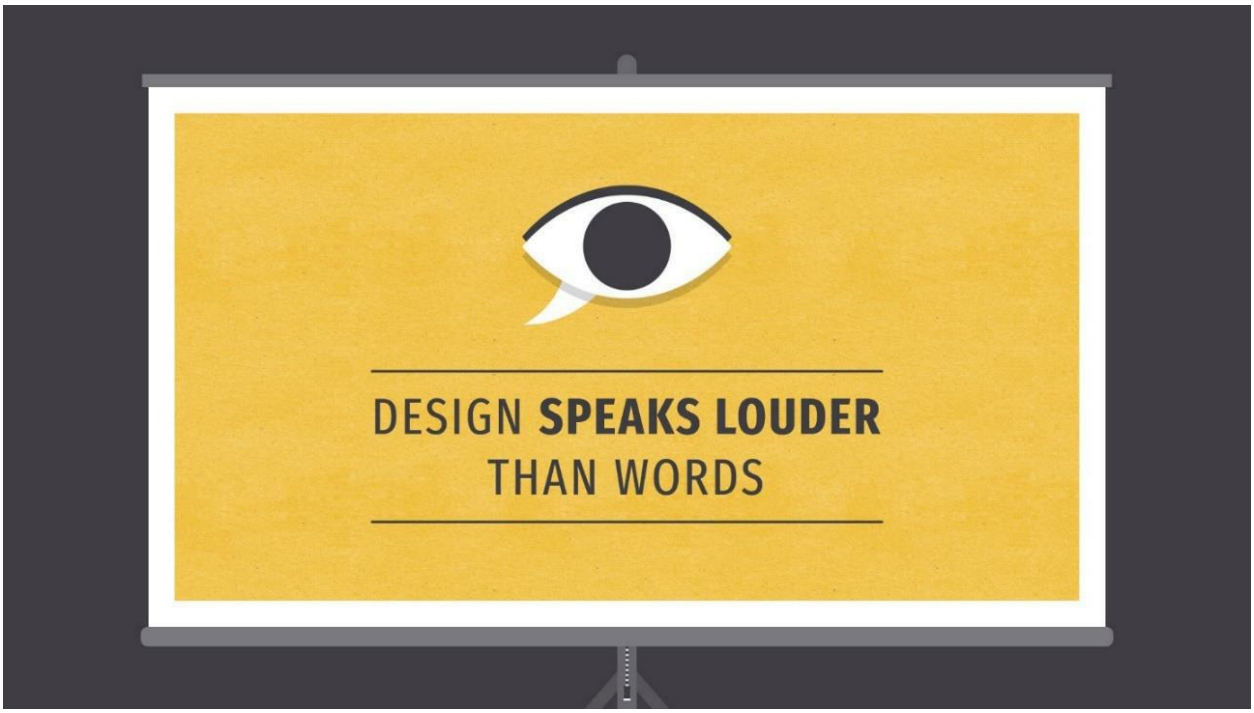


Putting it all together

The fundamentals of design are all about the bigger picture—in other words, learning to appreciate the many small details that make up every composition.


Shape
Form
Balance
Texture
Line
THE BIGGER
PICTURE
Visual Marketing for Entrepreneurs

This insight can be applied to almost any type of project, whether you're creating your own graphics or just looking for simple ways to enhance your work.



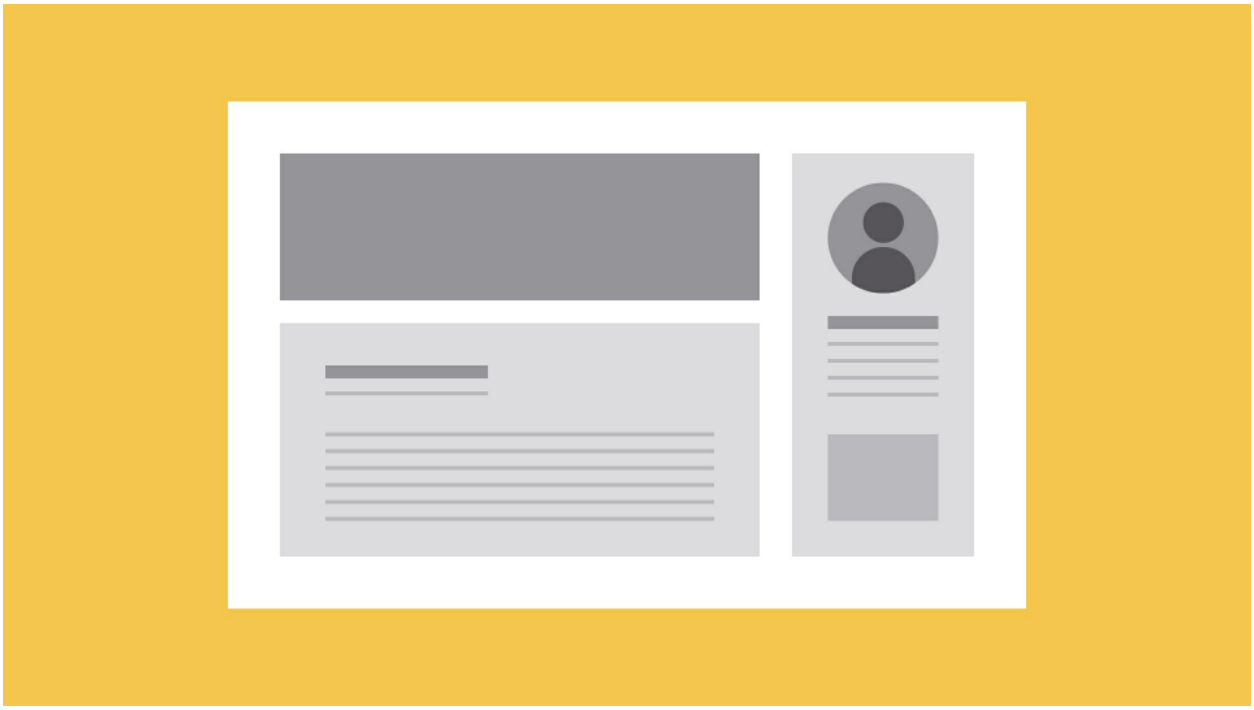
Understanding these basic fundamentals is very important to creating stunning designs that would be remembered for a long time. It is very important that every rule is followed for maximal effect in your design work.
Section 3

LAYOUT AND COMPOSITION

In any design, a balance has to be achieved between the visual impact and communication aspects. To achieve this, it is best to consider layout and composition well.

What is composition?

In many ways, layout and composition are the building blocks of design. They give your work structure and make it easier to navigate, from the margins on the sides to the content in between.



Why is composition so important? In short, it ’s the way your content is arranged. It doesn’t matter if you’re working with text, images, or elements in a graphic; without a thoughtful, well-composed layout, your work would basically fall apart.

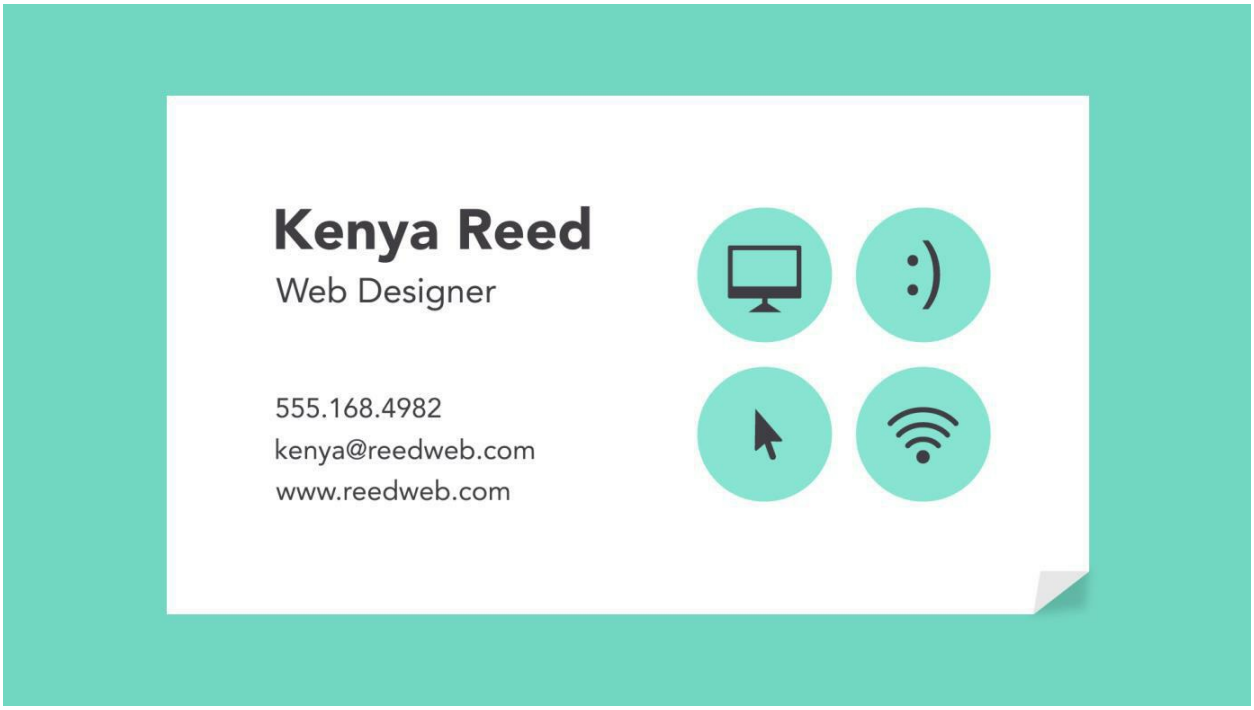
Five basic principles

The key to mastering layout and composition is to think like a designer. Luckily, it ’s easier than it sounds. There are five basic principles that can help you transform your work and sharpen your eye for design. Keep them in mind during your next project, and look for ways to apply them.

1. Proximity
2. White space
3. Alignment
4. Contrast
5. Repetition

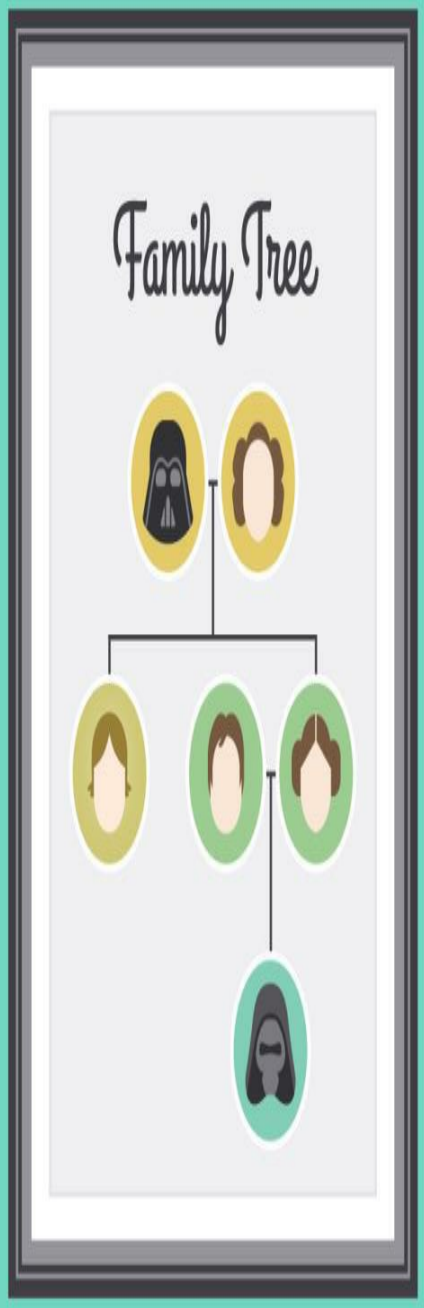
Proximity

Proximity is all about using visual space to show relationships in your content. In practice, it ’s pretty simple—all you have to do is make sure related items are grouped together (for instance, blocks of text or elements in a graphic, as in the example below).



Groups that are NOT related to each other should be separated to visually emphasize their lack of a relationship. All in all, this makes your work easier to understand at a glance, whether it’s purely text or something more visual.

Old-Fashioned Shortcake
INGREDIENTS
2 cups flour
1/2 teaspoon salt
2 tablespoons sugar
4 teaspoons baking powder
3 tablespoons shortening
3/4 cup milk
1 quart berries
Sift dry ingredients; mix in shortening. Add milk to make a soft dough, then smooth out lightly. Bake in greased layer cake tin for 20 to 25 minutes. Split, butter, and spread sweetened crushed berries or other fruit between layers.


Family Tree

White Space

White space is an important part of every composition. Now, this doesn ’t mean literal white space; it just means negative space, like the spaces between your content, between lines, and even the outer margins.

There ’s no one way to use white space correctly, but it’s good to understand its purpose. White space helps you define and separate different sections; it gives your content room to breathe. If your work ever starts to feel cluttered or uncomfortable, a little white space might be just what the doctor ordered.

"Keep your face to the sunshine and you cannot see a shadow."

– HELEN KELLER

Alignment

Alignment is something you deal with all the time, even if you don’t realize it. Whenever you type an email or create a document, the text is aligned automatically.

Alignment

Whenever you type an email
or create a document,
the text is aligned automatically.

When aligning objects by yourself (for instance, images or separate text boxes), getting it right can be tricky. The most important thing is to be consistent.

It might help to imagine your content arranged inside of a grid, just like the example below. Notice how there’s an invisible line centering each image to the text? Each grouping is also evenly spaced and aligned, with equal-sized margins.

SEE

no evil

HEAR

no evil

SPEAK

no evil

It’s this attention to detail that makes the composition easier to navigate. Without consistent alignment, your work could start to feel disorganized.

Contrast

Contrast simply means difference – that one item is different from another. Light and dark, big and small, bold and subtle. In layout and composition, contrast can help you do many things, like catch the reader’s eye, create emphasis, or call attention to something important.

To create contrast in the example below, we ’ve used colour, more than one style of text, and objects of differing sizes. This makes the design more dynamic and, therefore, more effective at communicating its message.

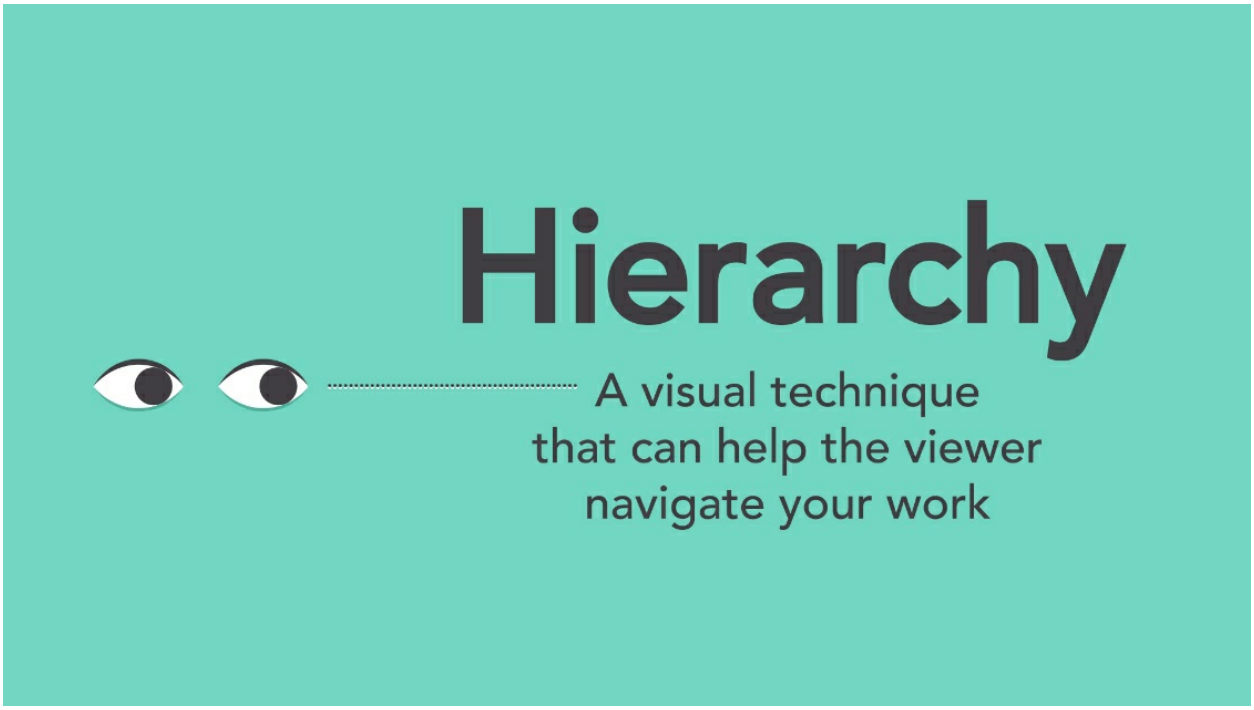
CAUTION

RADIATION

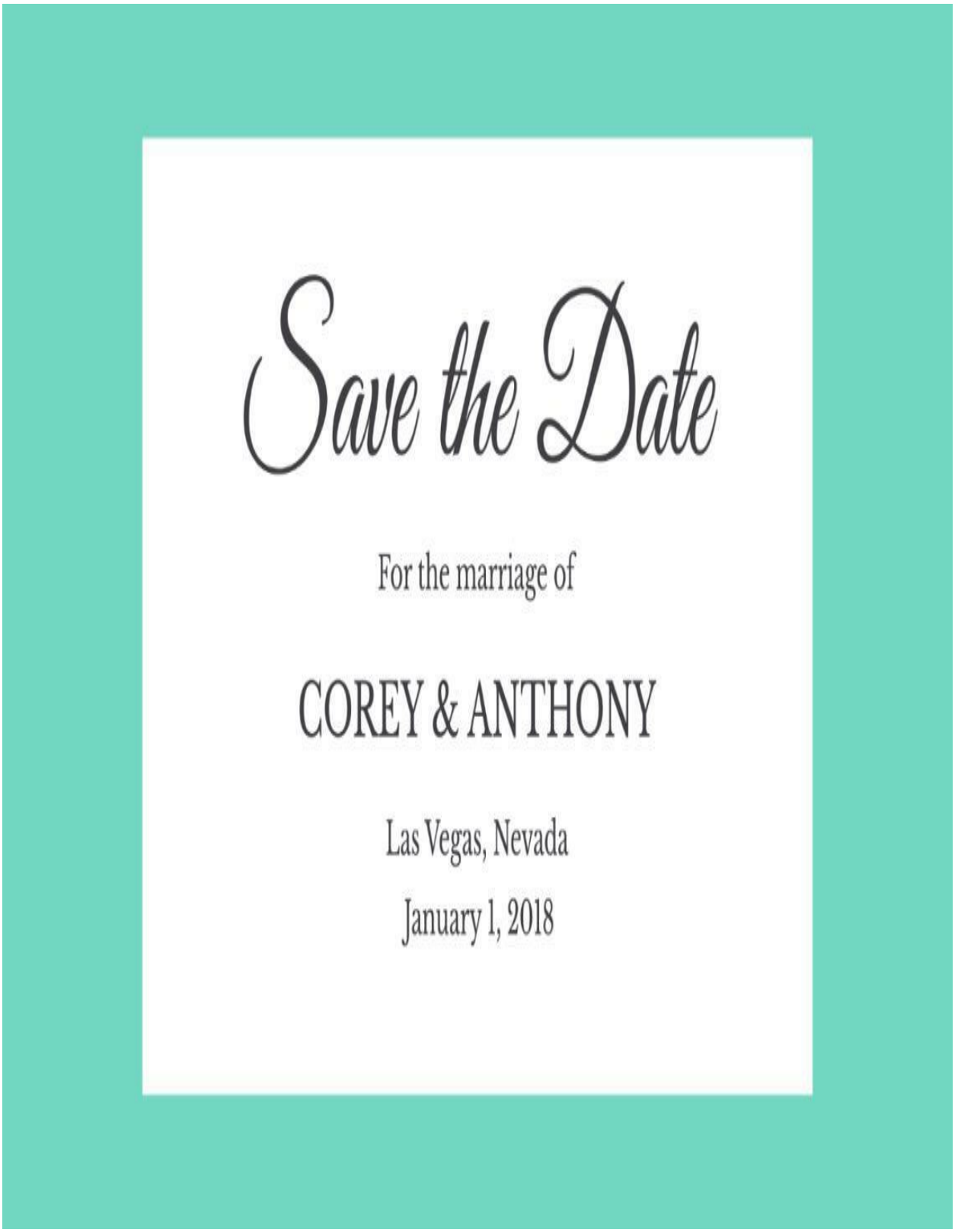
MAY CAUSE
SUPERPOWERS

Hierarchy

Contrast is also closely tied to hierarchy, which is a visual technique that can help the viewer know the importance of each element in your design in relation to the rest. It help to navigate your work. In other words, it shows them where to begin and where to go next using different levels of emphasis.

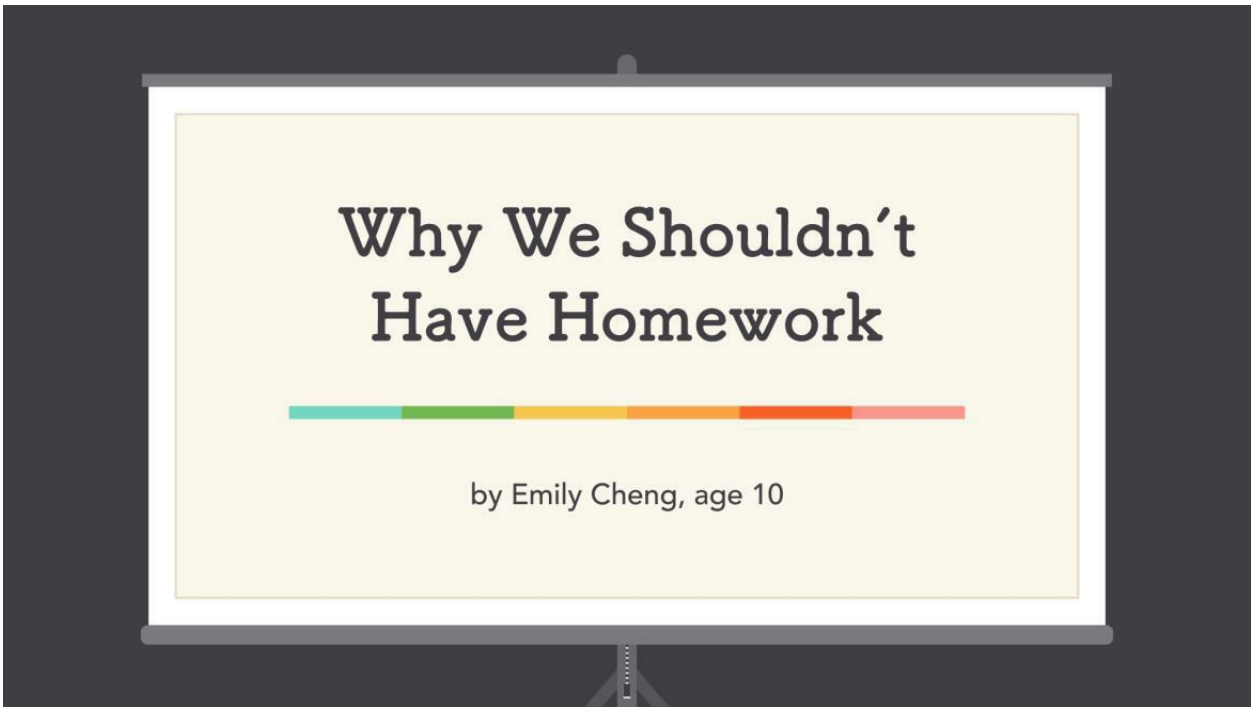


Establishing hierarchy is simple: Just decide which elements you want the reader to notice first, then make them stand out. High-level or important items are usually larger, bolder, or more eye-catching in some way. **Remember, when everything stands out, nothing stands out.**

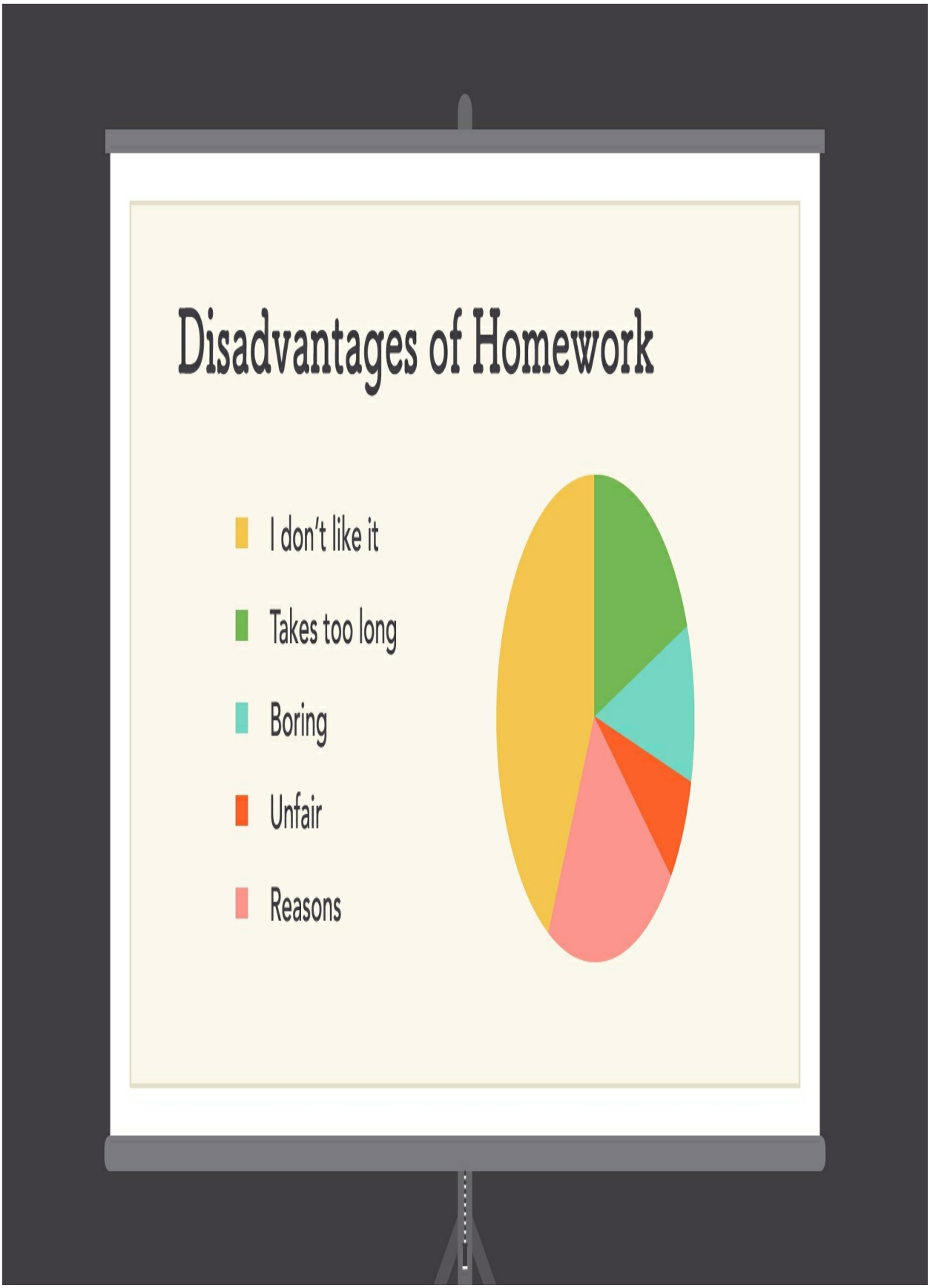

Save the Date
For the marriage of
COREY & ANTHONY
Las Vegas, Nevada
January 1, 2018

Repetition

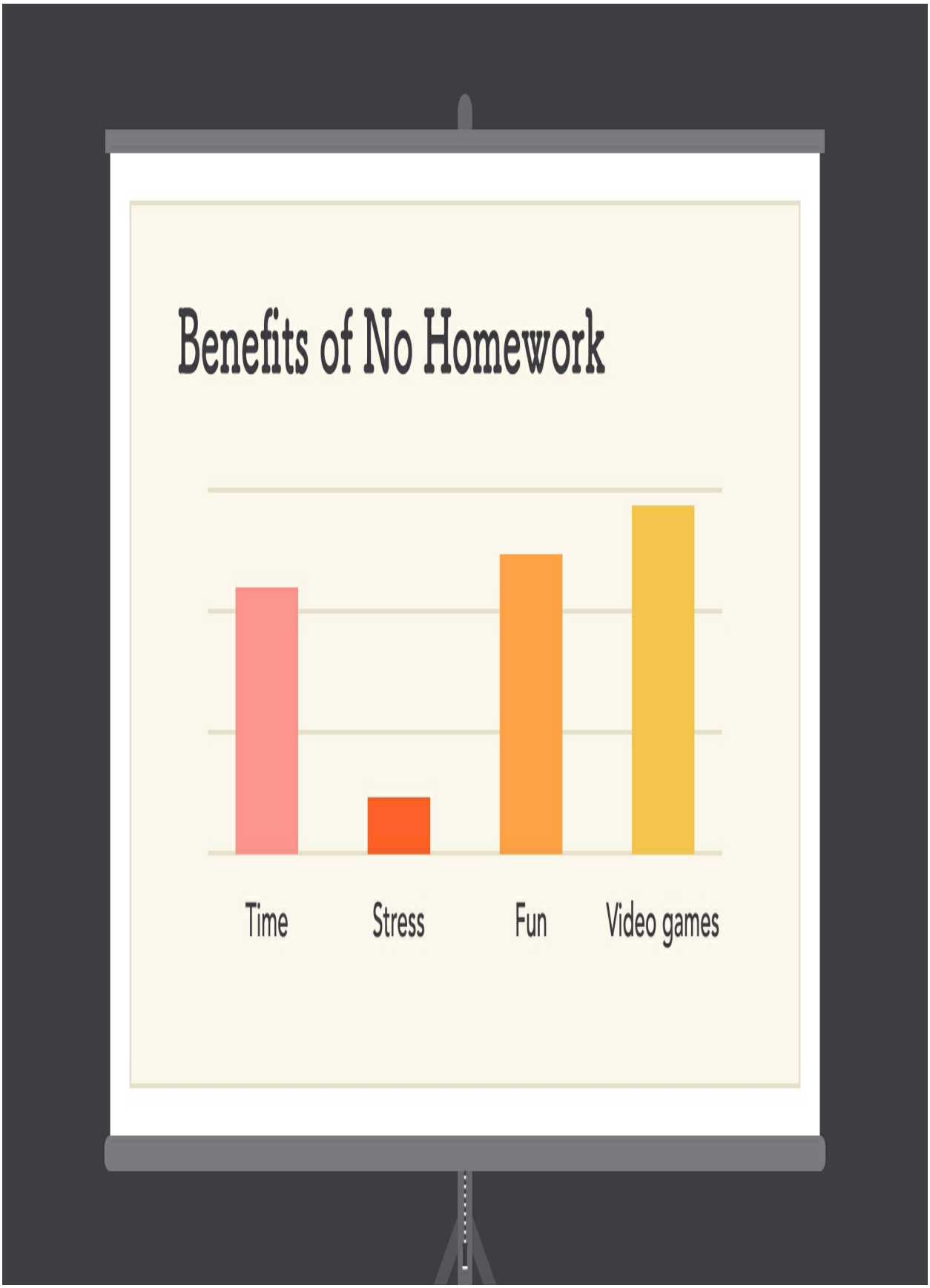
Repetition is a reminder that every project should have a consistent look and feel. This means finding ways to reinforce your design by repeating or echoing certain elements.



For instance, if you have a specific colour palette, look for ways to carry it through. If you ’ve chosen a special header style, use it every time.


Disadvantages of Homework
I don't like it
Takes too long
Boring
Unfair
Reasons

It’s not just for aesthetic reasons—being consistent can also make your work easier to read. When viewers know what to expect, they can relax and focus on the content.


Benefits of No Homework
Time
Stress
Fun
Video games

Putting it all together

You might say layout and composition are the unsung heroes of design. It ’s easy to overlook their role, but they’re part of everything you do.

The principles you just learned can help you elevate any project. All it takes is a little attention to detail and you can create beautiful, professional-looking compositions.

Section 4

TYPOGRAPHY

Today, typography used in designs convey different messages to the viewer: classical ones are for a strong personality, while more modern ones may convey clean, neutral look. Bold typefaces are used for making statements and attracting attention.

Typography is the art of arranging letters and text in a way that makes the copy legible, clear, and visually appealing to the viewer. Typography involves font style, appearance, and structure, which aims to elicit certain emotions and convey specific messages. In short, typography is what brings the text to life.

Why is Typography Important?

Typography is so much more than just choosing beautiful fonts: it is an important component of graphic design. Good typography will establish a strong visual hierarchy, provide a graphic balance to your design.

1. **Typography builds brand recognition**

Audiences associate a particular typeface featured on a design with a

particular type of brand. For example, typography in advertising often reflects a company's brand. This is especially true for logos of brands which are made with certain typefaces. E.g. Coca-Cola. Using unique, consistent typography can help to carry a brand forward because their designs (advertisement materials: flyers, banners etc.) are visually appealing to their audience.

2. Typography holds the attention of the readers
Good typography could be the difference between someone accepting a design. Good typography can be visually stimulating and memorable, and thus typography plays a huge role in the process of design.

3. Typography influences decision making
Typography has a profound effect on the way that users digest and perceive the information conveyed by the text. Eye-catching type is much more persuasive than weak fonts that don't reinforce the message of the text. This is true especially for brand designs that aim at attracting the attention of a potential buyer or customer.

The Different Elements of Typography

1. Typefaces and Fonts
There's some confusion surrounding the difference between typefaces and fonts, with many treating the two as synonymous. A typeface is a design style which comprises a myriad of characters of varying sizes and weight, whereas a font is a graphical representation of text character. Put simply, a typeface is a family of related fonts, while fonts refer to the weights, widths, and styles that constitute a typeface.

Choice of typeface or font is often used to draw attention to a particular design, when combined with efficient use of colour, shapes, and images.
The Difference between a Typeface and a Font
There are three basic kinds of typeface: **Serif, Sans-Serif, And Decorative/Display**.
• **Serif** comes from the Greek word meaning. A serif font has little feet attached to each letter – a classic style of font used in books- the little feet guide u-your eyes along lines of text.
• **Sans** is French word meaning without. So, san serif means without serif –

This type of fonts is modern and it is styled to look clean on digital screens. **Decorative/Display typeface** includes fonts that are neither serif nor sans-serif. These types of fonts are usually used for headlines and because of their differences, they are best used in small doses. Other types of fonts include: Monospaced, Script and Symbol.

Few Tips to help you get the perfect Font Pairing

To keep a design neat and organized, a good designer will never use more than three fonts—and keep decorative fonts to a minimum. Alternatively, pick fonts that are contrasting and complementary. Most graphic designers will pair serif fonts with sans-serif fonts, such as putting main body text in a serif font and putting your title in a sans-serif font, or vice-versa. You could also pick something that is round and sort of match it with something tall. Fonts that are opposite should attract each other to create harmony in your design. Usually, decorative fonts are used for digital designs aimed at attracting the attention of viewers- and they are used in such a way that it is visually stimulating. The easiest way to pair fonts is to use one font family. You can pick a font with different weights, like light, medium and bold, and you can create contrast with these fonts easily.

1. **Contrast**

Contrast is used to vary the shades of a Most designers create contrast by playing around with varying typefaces, colours, styles, and sizes to create designs that are unique and stylish enough to generate impact. Different types are used in fonts for emphasis

1. **Consistency**

Keeping your typefaces consistent is key to avoiding a confusing and messy design. When conveying information, it’s essential to stick to the same font style, so your viewers can understand what they’re reading, and begin to notice a pattern in your design.

4. White space

Often referred to as ‘negative space,’ white space is the space around text or graphics. It’s often overlooked and tends to go unnoticed by the user, but proper use of white space ensures the design is organized and neat. White space can even draw attention to the text and provides an overall aesthetically pleasing experience. White space often takes the form of margins, padding,

or just areas with no text or graphics.

5. Alignment
Alignment is the process of unifying and composing text, graphics, and images to ensure there is equal space, size, and distances between each element. Many UI designers create margins to ensure that their logo, header, and body of the text are aligned with each other. When aligning your user interface, it's good practice to pay attention to industry standards. For example, aligning your text to the right will seem counterintuitive for readers who read left to right.

6. Colour
Colour is one of the most exciting elements of typography. This is where designers can really get creative with their text. Text colour or font colour can make the text stand. Colour is used for its emotional effect in conveying the tone and nature of the message.

1. **Hierarchy**
Establishing hierarchy is one of the most vital principles of typography. Typographical hierarchy aims to create a clear distinction between textual information should be noticed and read first. In an age of short attention spans brought about by social media, designers are urged to be concise and create typefaces that allow users to consume the necessary information in short amounts of time.

Hierarchy can be created using sizing, colour, weight, contrast, and alignment. For example, if you have a text or character in red and larger than the others, it is a visual clue to the viewers that it's an important piece of the design. The most typical example of typographical hierarchy is size. Important information is usually larger and in standard text.

1. **Size**
All typefaces and fonts are not created equal. Some are fat and wide; some are thin and narrow. So, words set in different typefaces can take up a very different amount of space on the page.

The height of each character is known as its 'x-height' (quite simply because it's based on the 'x' character). When pairing different typefaces, it's

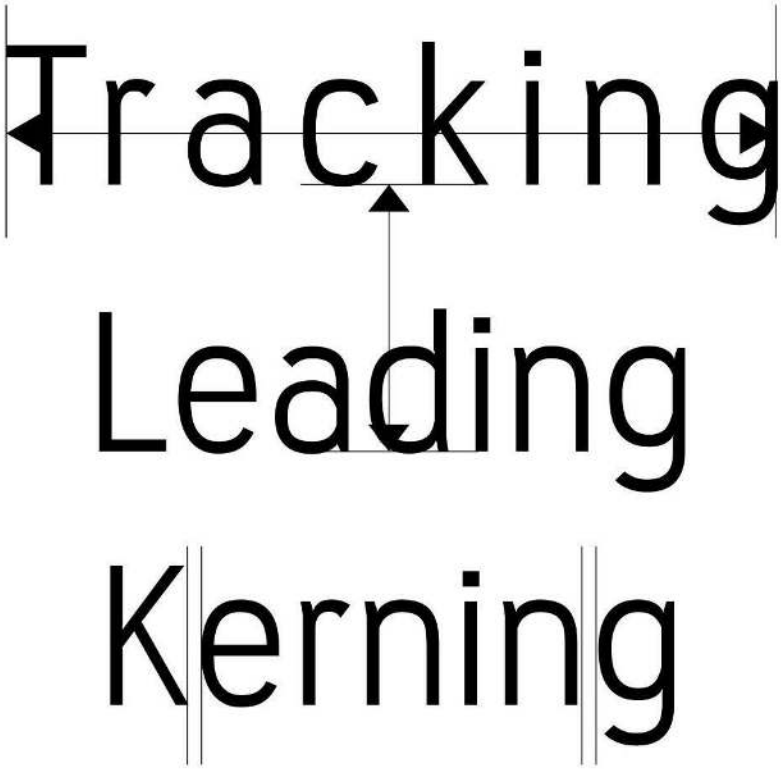
generally wise to use those that share a similar xheight.
The width of each character is known as the ‘set width’. This spans the body of the letter, plus the space that acts as a buffer between one letterform and the next.
The most common method used to measure type is the point system, which dates back to the 18th century. One point is 1/72 inch. 12 points make one pica, a unit used to measure column widths.

Type sizes can also be measured in inches, millimetres, or pixels.

Leading
Leading describes the vertical space between each line in a paragraph. It’s so named because, in the days of metal typesetting, strips of lead were used to separate lines of type. For legible body text that’s comfortable to read, a general rule is that your leading value should be anything between 1.25 and 1.5 times greater than the font size.

Tracking
Tracking is the process of adjusting the spacing of all characters in a word, and is applied evenly. It is simply described as the horizontal space between characters or words.

Kerning
Kerning is similar to tracking, but they are not the same thing. Kerning is the spacing between individual letters. Kerning is the process of adjusting the space between characters to create a harmonious pairing. For example, where an uppercase ‘A’ meets an uppercase ‘V’, their diagonal strokes are usually kerned so that the top left of the ‘V’ sits above the bottom right of the ‘A’. Like this together, VA.



How do you choose the right typeface or font for your design?
With so many different fonts and typefaces to choose from, it's easy to feel overwhelmed. The process of choosing typefaces or fonts depends on a few other things apart from creativity on the path of the designer. Thus, making the right choice depends on so much more than just seeing what looks nice. Here are a few key considerations based on the following principles:

1. Legibility and Readability
Legibility "refers to perception" (being able to see as determined by physical limitations of the eye) and readability "refers to comprehension" (understanding the meaning). Good graphic designers aim to achieve excellence in both.

Readability and Legibility are often used synonymously, typographically they are separate but related concepts. Legibility describes how easily individual characters can be distinguished from one another. Readability refers to how easy it is to read the text as a whole, as opposed to the individual character

recognition described by legibility.
The typeface chosen should be legible. That is, it should be read without effort. Sometimes legibility is simply a matter of type size; more often, however, it is a matter of typeface design. Case selection always influences legibility. In general, typefaces that are true to the basic letterforms are more legible than typefaces that have been condensed, expanded, embellished, or abstracted.

Don’t skimp on readability and legibility. Both of them are vital to the overall outlook of a design: There’s nothing worse than a design that looks good but is entirely illegible or readable. When deciding what typefaces or fonts to include in your design, set the style, aesthetic, and voice aside and reflect on whether the font is legible and readable. Can the text be legible without strain? Are the characters distinct enough?

2. Personality
One of the first things to consider is the type of design you are creating. Most importantly, it is imperative that the typography reflects the personality of the brand or product. A good starting point when faced with this challenge is to define the core traits of your brand, and start to gather typefaces that reflect these traits. From there, you can begin to notice a trend.

3. Tone
Are you creating it for a corporate organization? Does it reflect a certain age group – children, teenagers or adults? It is important to consider how the typeface or font harmonizes with the tone of the message. For example, if you want to convey serious or important information, choose a less stylized or decorative font that is both clearly legible and will limit distraction.

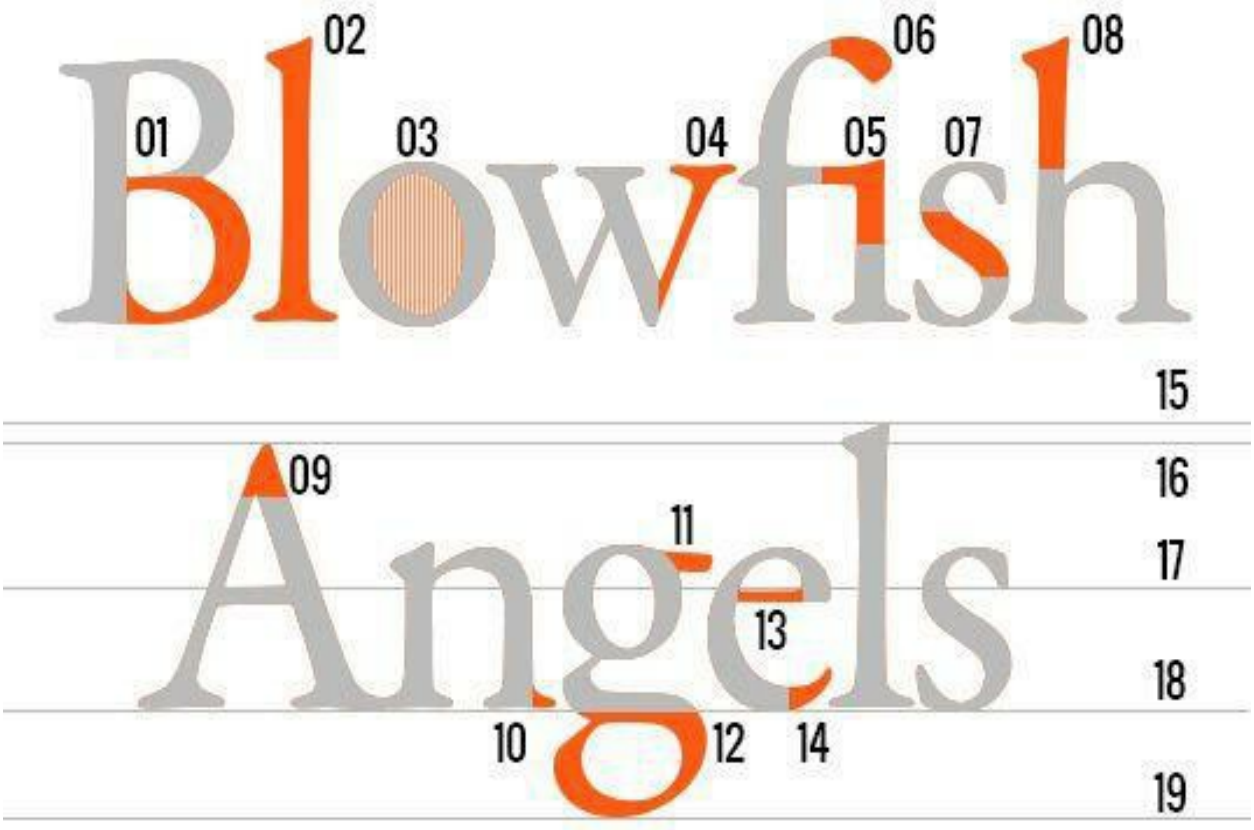
4. Be Creative
If you’re not sure where to start, try out your different typefaces and fonts on your own design and see what it looks like. This may be difficult at first but it would yield some positive results when applied with the other principles above. Also taking some time to see what other people are doing can prove useful. Open your eyes to the typography that you see around you. Can you notice similar patterns? Can you see good and bad examples of typography? Looking up how typography is used in designs online will give you some good ideas of what type of typefaces or fonts can be used and combined

together in designs.

Conclusion
Mastering typography is an important step in becoming a good graphic designer. Designers often use typefaces to set a theme and mood in a design. For example, in advertisement designs (using bold, large text to convey a particular message to the reader is really important).

If you're not sure where to start, why not head to your favourite online channels such as Instagram to see different kinds of designs and making a note of what, where, when and how typefaces and fonts are commonly used for various types of designs.

Font Description



1. Bowl; 2. Stem; 3. Counter; 4. Arm; 5. Ligature; 6. Terminal; 7. Spine; 8. Ascender; 9. Apex; 10. Serif; 11. Ear; 12. Descender; 13. Crossbar; 14. Finial; 15. Ascender height; 16. Cap height; 17. X-height; 18. Baseline; 19. Descender line

Section 5

COLOUR
The Power of Colour

Color is a deeply rooted form of communication —a universal language than can be understood 60,000 times faster than text alone. Colour plays a vital role in design and everyday life. It can draw your eye to an image. Sometimes it can trigger an emotional response. It can even communicate something important without using words at all.

However, choosing colour for any design (logo, flyer, pamphlet, brochure, magazine or book cover etc.) can be a challenging task. In creating designs that great, there is a whole science to colour and how it is used. This science is what enables us to know which colours to match together to create designs that look good.
So, what is the science that enables us to know which colours look good together and which ones don't? The answer is simple: **Colour theory.**

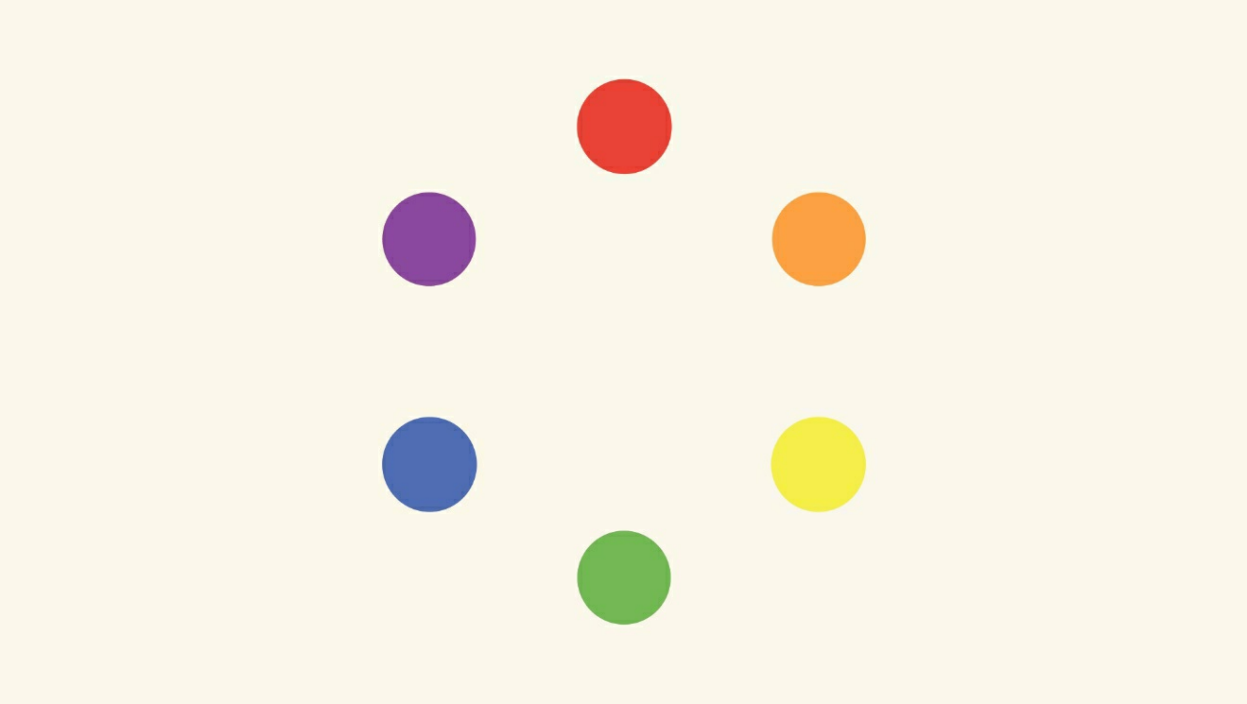
Colour Theory both the science and art of using colour. It explains how humans perceive colour; and the visual effects of how colours mix, match or

contrast with each other. Colour theory also involves the messages colours communicate; and the methods used to replicate colour.

Artists and designers have followed colour theory for centuries, but anyone can learn more about it. It can help you feel confident in many different situations, whether it's choosing colours for a design or putting together the perfect outfit. With a little insight, you'll be looking at colour in a whole new way. Remember that colour combinations for graphic design are not absolute, they can be combined in different ways. So do not be afraid to experiment.

Colour Basics

Let's start at the beginning with a refresher on the basics. Remember learning about primary and secondary colours in school? Then you already have some knowledge of colour theory. Secondary colours are created by combining two primary colours. Red and yellow make orange; yellow and blue make green; and blue and red make purple.



If we mix these colours together, we get even more in-between shades, like red-orange and yellowgreen. All together, they form what's called a **colour wheel**.

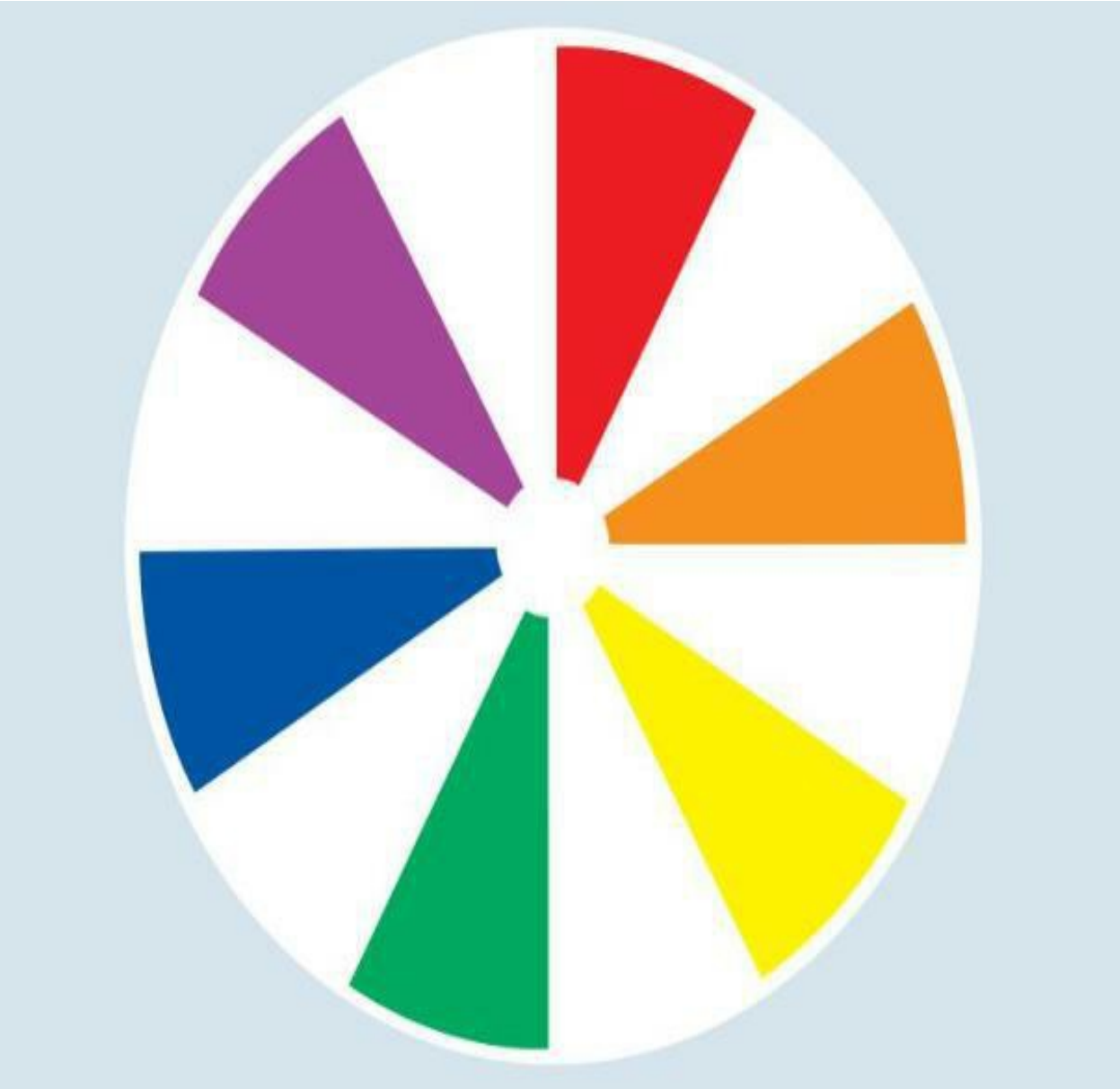
In colour theory, colours are organized on a colour wheel and grouped into 3 categories: primary colours, secondary colours and tertiary colours.

Colour Wheel

The colour wheel for graphic design is a circle with different coloured sections used to show the relationship between colours.

Primary Colours

The typical colour wheel includes the blue, red, and yellow primary colours. They are the building blocks for every other colours.



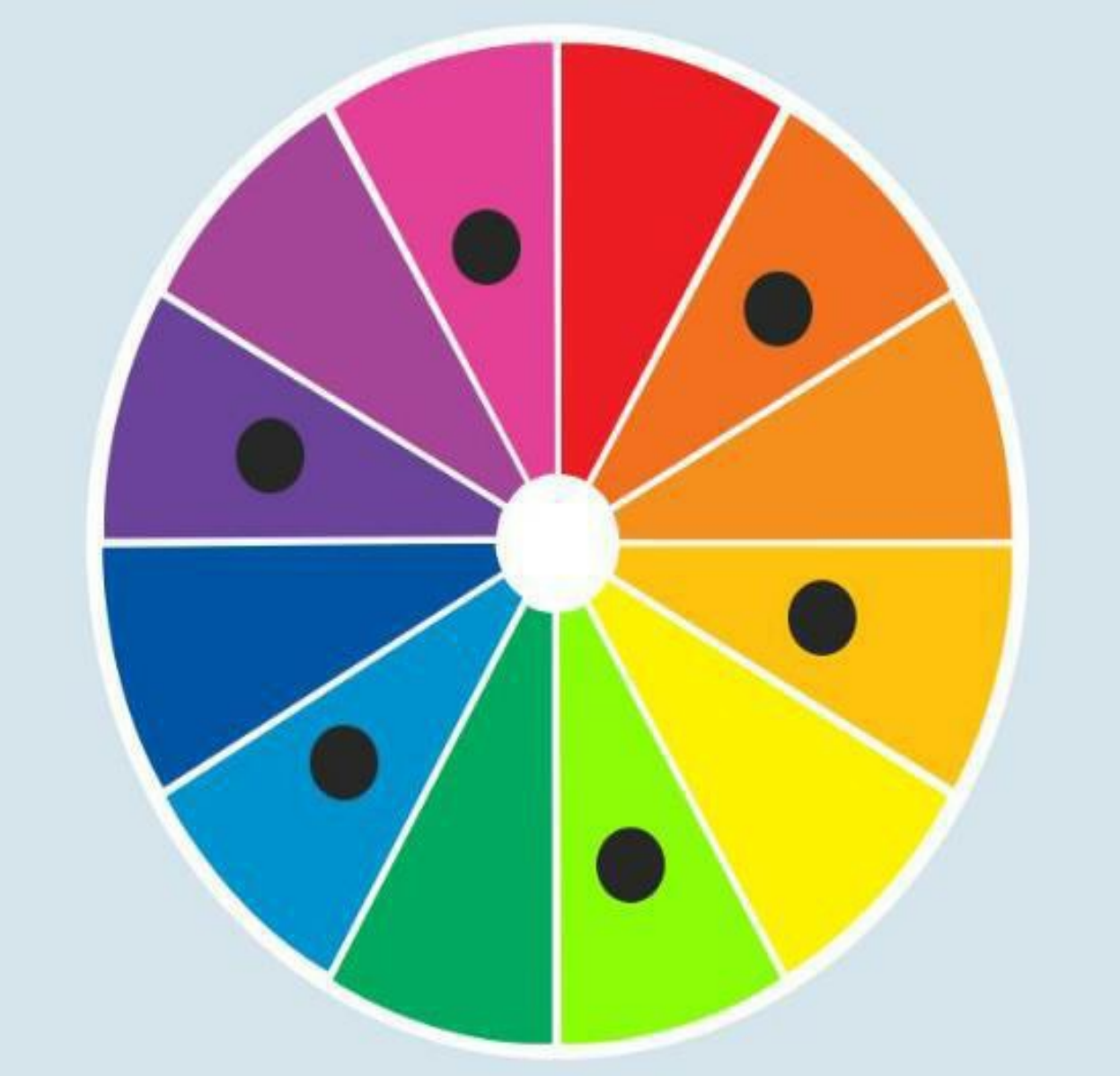
Secondary Colour

Secondary colours are colours that come from mixing two primary colours.

There are three secondary colours.

Here is the combination to create secondary colours:
Purple = Red + Blue,
Orange = Red + Yellow
Green = Yellow + blue.
Tertiary colours

Tertiary colours are colours made by combining a secondary colour with a primary colour. There are six colours considered tertiary. In the RYB colour wheel, these tertiary colours are red-orange, yellow-orange, yellow-green, blue-green, blue-violet, and red-violet.



Here is the combination to create tertiary colours:
Magenta (Red-Purple) = Red + Purple Violet (Blue-Purple) = Blue + Purple
Vermillion (Red-Orange) = Red + Orange Amber (Yellow-Orange) = Yellow + Orange
Teal (Blue-Green) = Blue + Green Chartreuse (Yellow-Green) = Yellow + Green Colour Terminologies - A Closer Look

Now that you know about the colour wheel, let's take it one step further with hue, saturation, brightness, value, tone, chroma, tint and shade. Describing colour is best done by these key terms. Let’s cover what exactly these terms

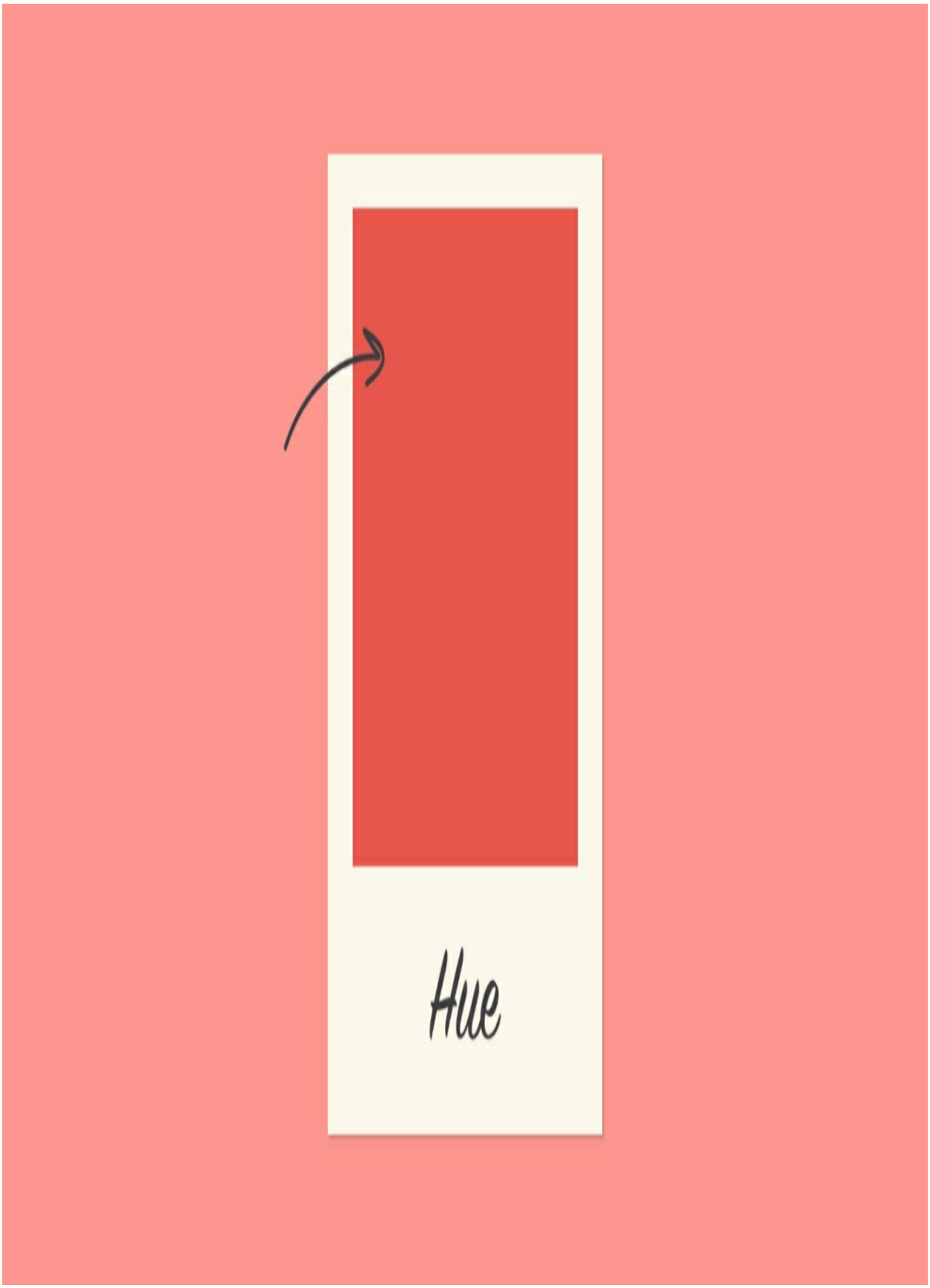
mean! These are terms you might not encounter in daily life, but they're the key to understanding more nuanced colours—like all those little paint chips at the home improvement store.

Hue

Hue is the easiest one; it's basically just another word for colour. In the example below, you might describe the hue as coral pink or light red, depending on your interpretation.

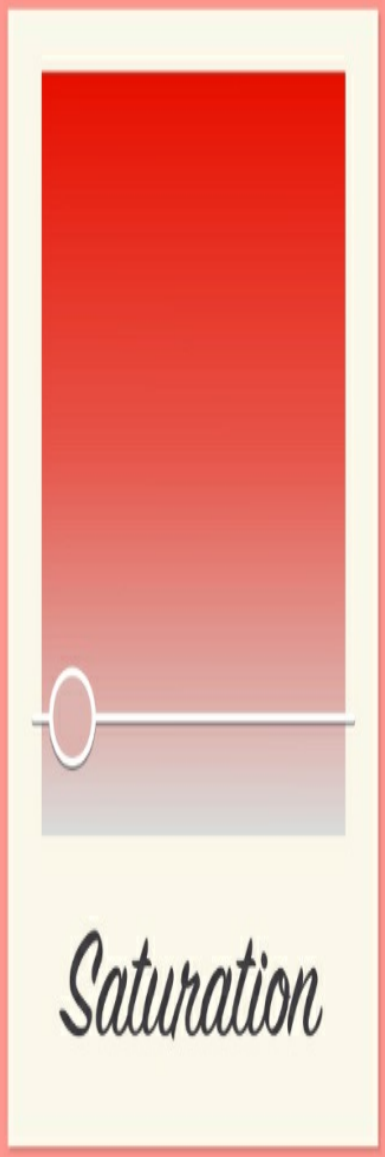
Hue is one of the main properties of colour and is the property of light by which the colour of an object is classified as red, blue, green, or yellow in the colour spectrum.

Green, orange, yellow, and blue — each of these is a hue, a colour or a shade. A rainbow shows the melting of one hue into another, from red to violet, and all the shades in between. The noun hue means both a colour and a shade of a colour. Green is a hue, and turquoise is a hue of both green and blue!


Hue

Saturation

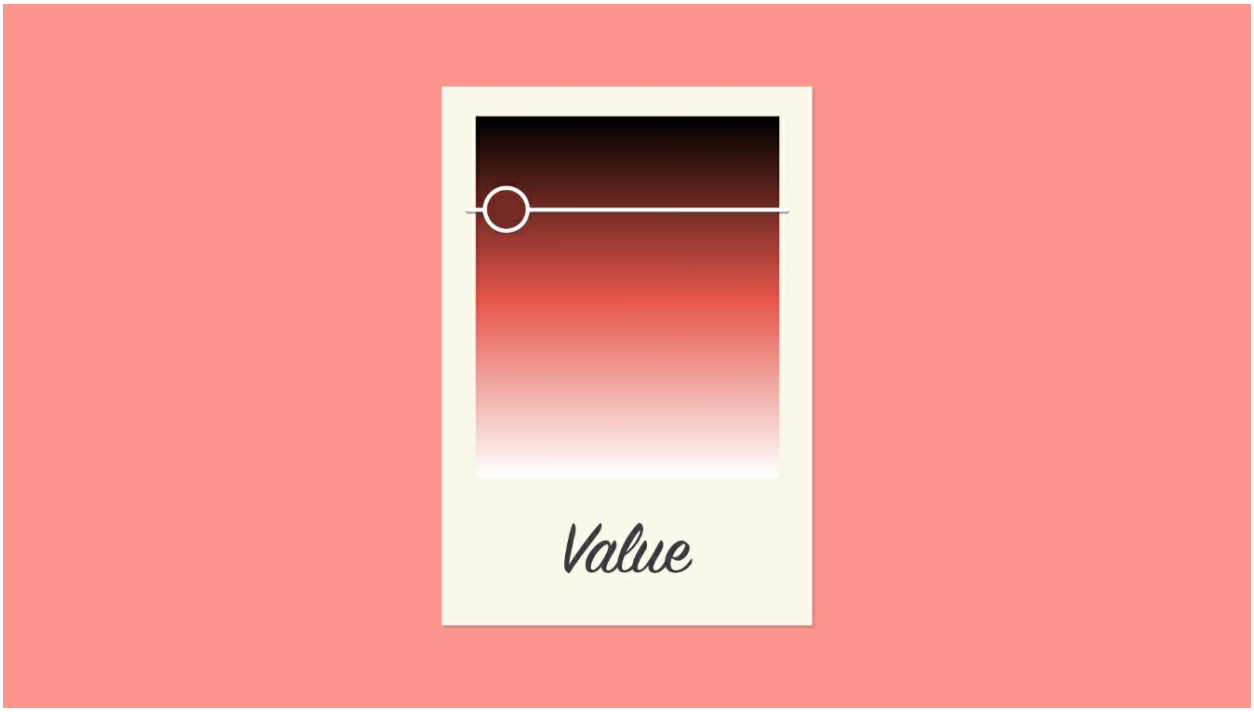
Saturation refers to intensity —in other words, whether the colour appears subtler or more vibrant. Saturation is the intensity of a hue from a gray tone with no saturation, to pure, vivid colour with high saturation.


Saturation

Highly saturated colours are brighter, richer, more colourful or deep. De-saturated or low saturation colours have less pigment and will come off muted or pastel.

Value

Value has to do with how dark or light the colour is, ranging from black to white. As you can see below, this gives us many different shades, from a deep reddish brown to a light pastel pink.

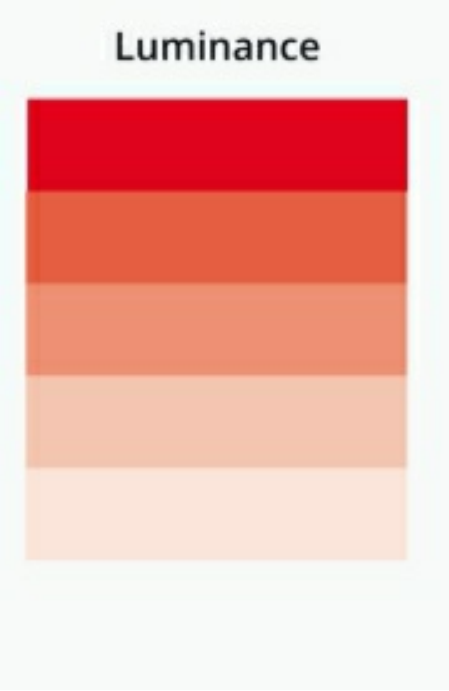


Value in the art will refer to the shadows and highlights and will give your work more dimension. It is especially important in black and white photos, design and illustration as it will separate objects from each other and their background.

Brightness

Brightness is simply both the hue of colour along with the value of the colour. It refers to its lightness and its ability to replicate light or reflection. This is also known as “luminance.”

Luminance is the amount of brightness or light in a colour.



Tone

Tone is a colour to which black and white (or grey) have been added. This darkens the original hue while making the colour appear more subtle and less intense.

Warm and Cool Colours

The colour wheel can also be divided into warm and cool colours. The warmth or coolness of a colour is also known as its colour temperature. The colour combinations found on a colour wheel often have a balance of warm and cool colours. According to colour psychology, different colour temperatures evoke different feelings. For example, warm colours are said to bring to mind coziness and energy, while cool colours are associated with serenity and isolation.

Warm colours are the colours from red through to yellow. These colours are said to bring to mind warmth, like the sun.
Cool colours are the colours from blue to green and purple. These colours are said to bring to mind coolness, like water.

Tint

A tint is a hue to which white has been added. For example, red + white = pink.

Shade

A shade is a hue to which black has been added. For example, red + black = burgundy.

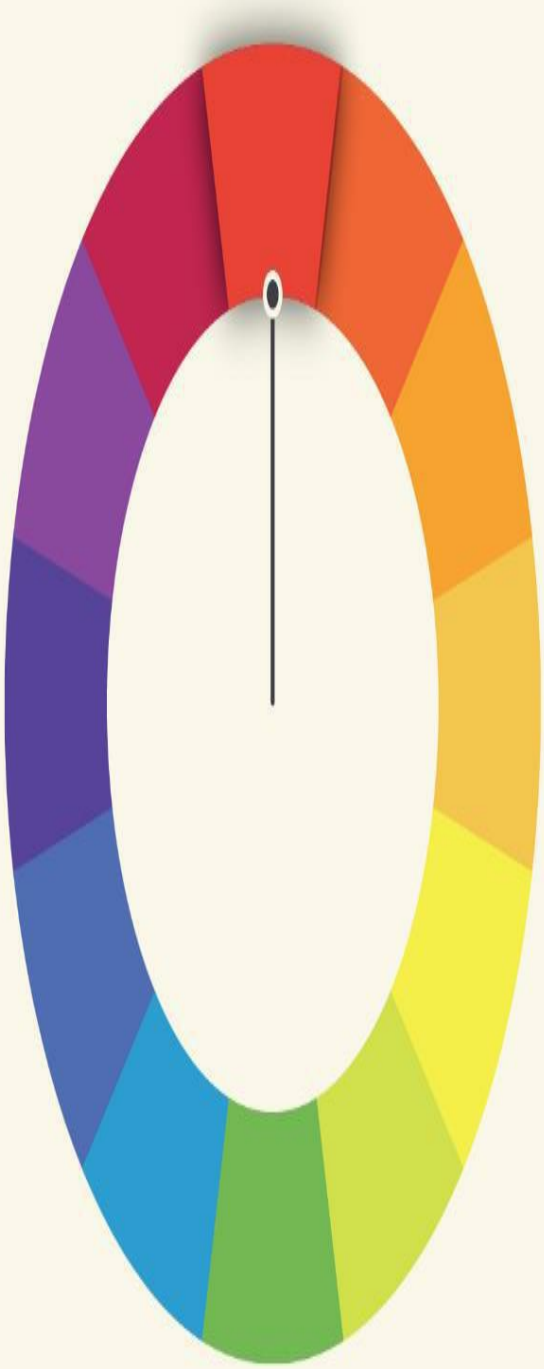
Creating Colour Schemes

So how do we put this all together to create professional-looking colour schemes? There are actually tried-and-true formulas based on something called colour harmony that can help.

Colour harmony uses the colour wheel to illustrate time-tested colour combinations. We'll explore some of the most common types of harmony below.

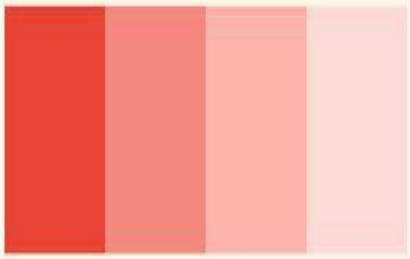
Monochromatic

The easiest formula for harmony is monochromatic because it only uses one colour or hue. To create a monochromatic colour scheme, pick a spot on the colour wheel, then use your knowledge of saturation and value to create variations.



Formula:

Monochromatic



The best thing about monochromatic colour schemes is that they're guaranteed to match. The colours suit each other perfectly because they're all from the same family.

Complementary

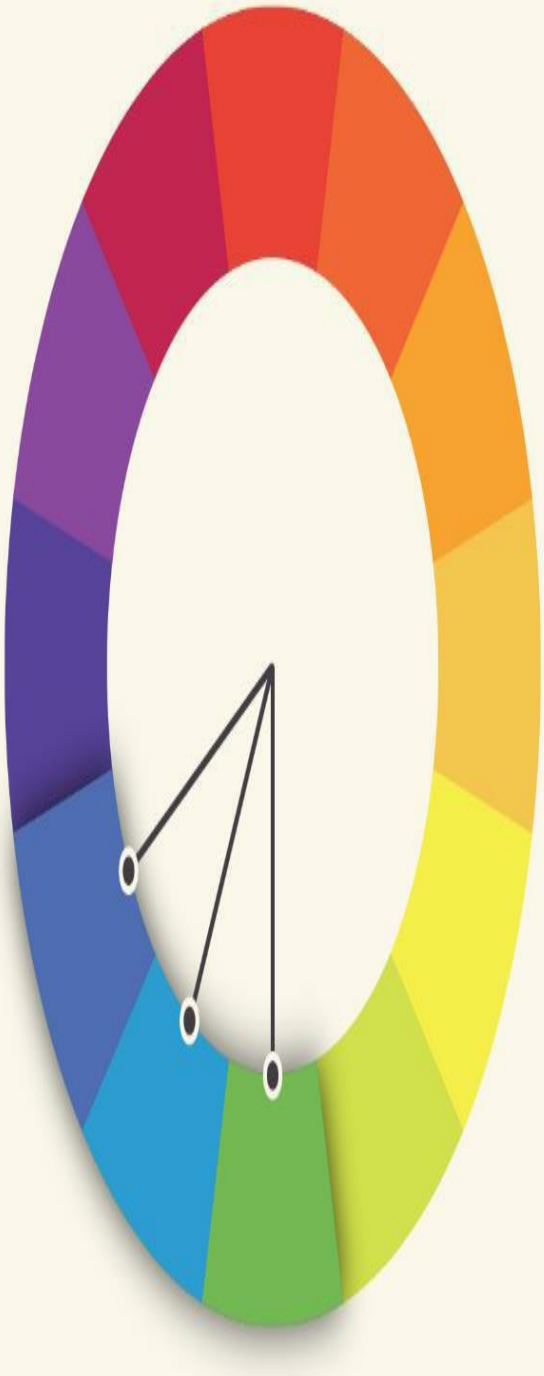
Complementary colours are opposite each other on the wheel; for instance, blue and orange or the classic red and green.

Complementary colours of graphic designs provide high contrast and high impact colour combination. When placed together or next to one another these colours will appear brighter and more vibrant.

To avoid complementary colour schemes that are too simplistic, add some variety by introducing lighter, darker, or de-saturated tones.

Analogous

An analogous colour scheme uses colours that are next to each other on the wheel, like reds and oranges or blues and greens.



Formula:

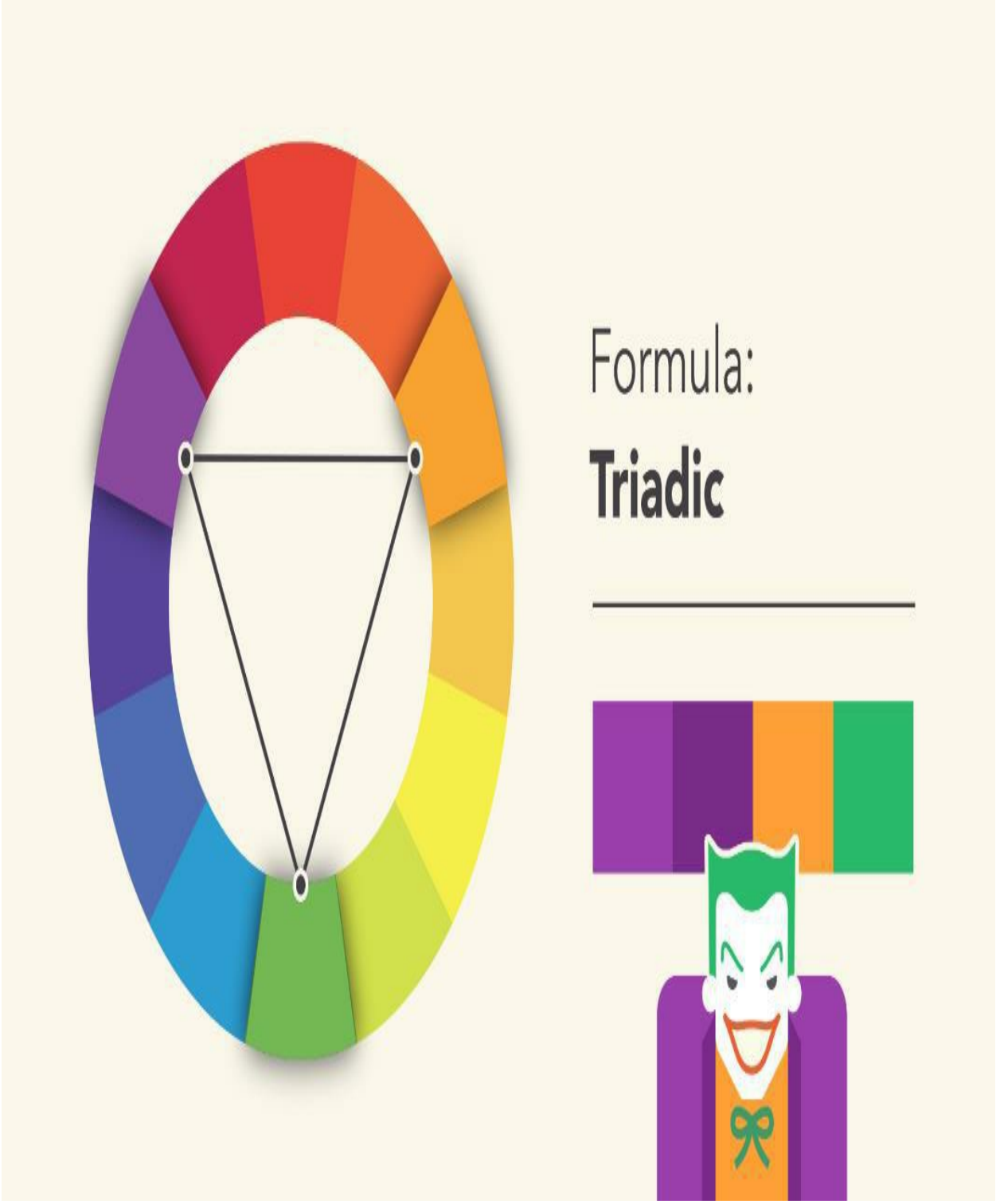
Analogous

Analogous colours are among the easiest to find on the colour wheel. Pick any colour at any point on the wheel. Now, look at any three colours directly to the left or right of the chosen colour. Together, those four are a group of analogous colours. Mixing colours that are adjacent to each other creates a colourful yet harmonious look. When using an analogous colourscheme it’s best to choose one main colour, using the other three as accent colours. This will help keep designs from becoming too chaotic or busy.

Split-Complementary

A split-complementary colour scheme uses the colours on either side of the complement. It involves the use of three colours. Start with one colour, find its complement and then use the two colours on either side of it.
This gives you the same level of contrast as a complementary colour scheme but more colours to work with (and potentially more interesting results).

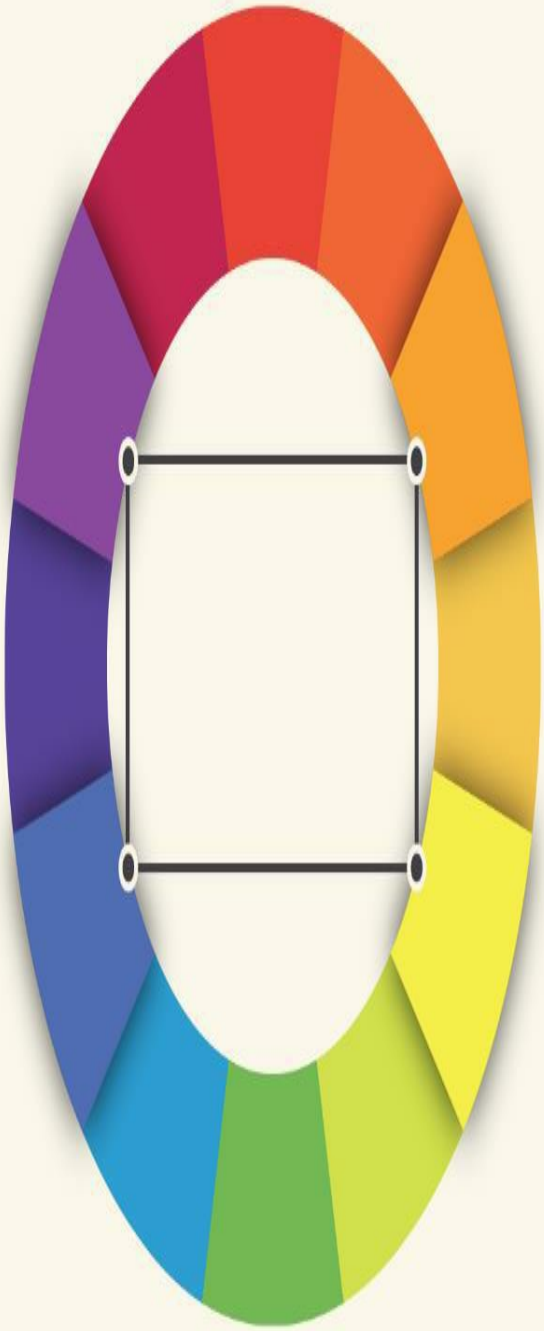
Triadic
A triadic colour scheme uses three colours that are evenly spaced, forming a perfect triangle on the wheel.



These combinations tend to be pretty striking—especially when they include primary or secondary colours—so be mindful when using them in your work.

Tetradic

Tetradic colour schemes form a rectangle on the wheel, using not one but two complementary colour pairs. This formula works best if you let one colour dominate while the others serve as an accent.



Formula:

Tetradic

Avoiding common mistakes

There are a few classic dos and don'ts when it comes to colour. For instance, have you ever seen colours that seem to vibrate when they're placed next to each other?

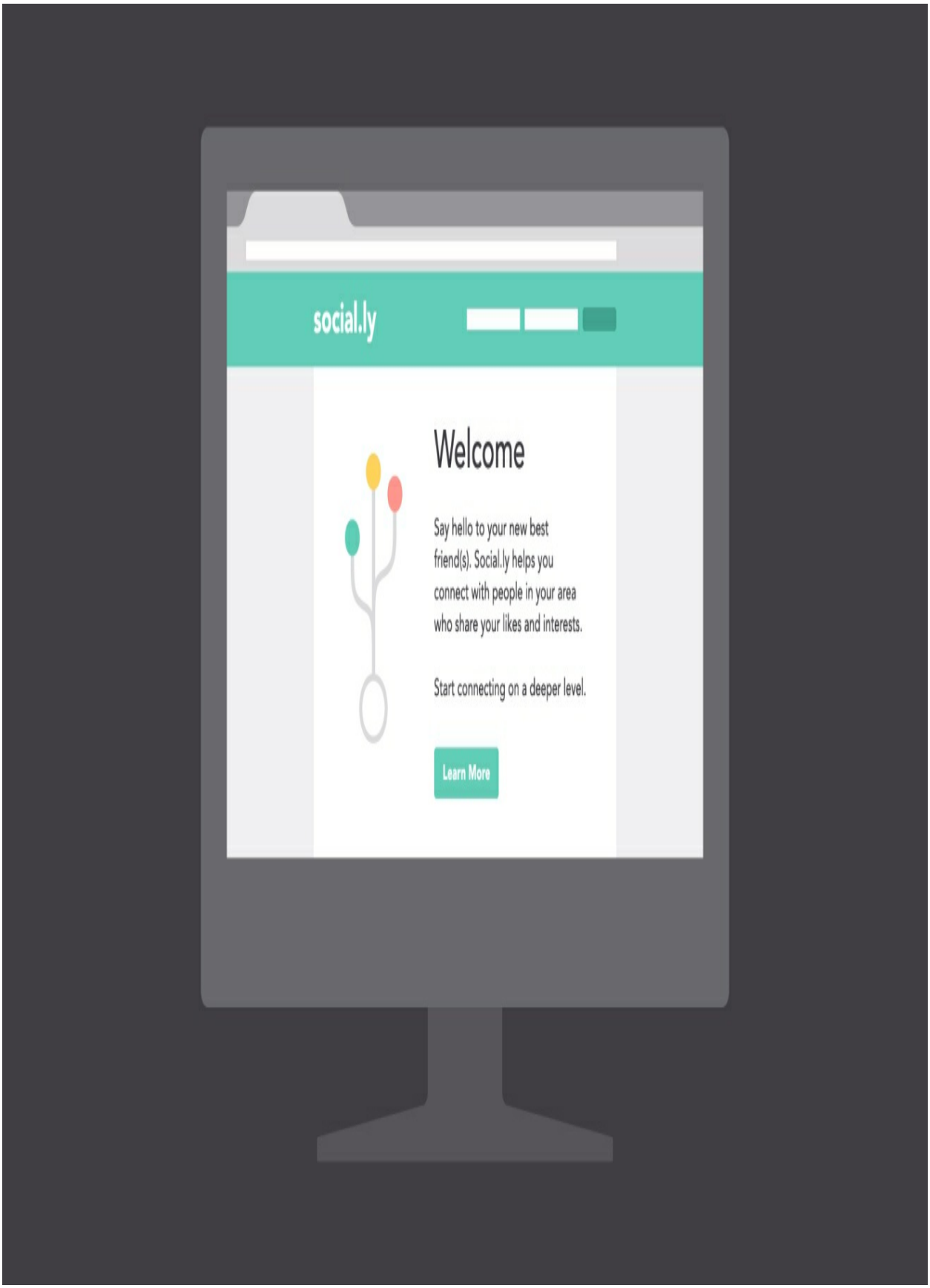
The solution is to tone it down —literally—and there's a simple way to do it. Start with one colour, and try adjusting its lightness, darkness, or saturation. Sometimes a little contrast is all your colour palette needs.

Readability is an important factor in any design. Your colours should be legible and easy on the eyes, especially when working with text.

Sometimes that means NOT using colour—at least not in every little detail.

social.ly
Welcome
Say hello to your new best friend(s). Social.ly helps you connect with people in your area who share your likes and interests.
Start connecting on a deeper level.
Learn More

Neutral colours like black, white, and gray can help you balance your design, so when you do use colour, it really stands out.


social.ly
Welcome
Say hello to your new best friend(s). Social.ly helps you connect with people in your area who share your likes and interests.
Start connecting on a deeper level.
Learn More

Choosing the right colours

Every colour sends a message. It's important to consider the tone of your project, and choose a colour palette that fits.

For example, bright colours tend to have a fun or modern vibe.

Snip/Cut
HAIR STUDIO
321.555.8989
SNIPCUT.COM

Desaturated colours often appear more serious or businesslike.

SANFORD
FINANCIAL

NINA YAZDANI

Senior Analyst

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www.sanfordfinancial.com

Sometimes it just depends on the context. With practice and creativity, there's no limit to what you can do.

Finding Inspiration

You can find ideas for colour schemes in all kinds of interesting places, from advertising and branding to famous works of art.

Using Web Resources

You can use a web resource to browse colour palettes or generate your own. Experienced designers often take inspiration from the world around them. There's nothing wrong with finding something you like and making it your own.

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Putting it all together

Everywhere you look, there's colour, colour, and more colour. It can be intimidating to use it in your work, but it doesn't have to be. Just keep experimenting, and remember what you've learned about colour theory. Soon, choosing great-looking colours will feel like second nature.

Online Resources for Colours

If you cannot seem to pick out colours suitable for your design on your own, you can seek help using online sources to create a nice design. Here are a few and they include:

1. Coolors (https://www.coolors.com)
2. Colour Hunt (https://colourhunt.co/)
3. COLOURLovers (https://www.colourlovers.com)
4. Paletton (https://paletton.com)
5. Google Colours (https://brandpalettes.com/google-colours)
6. Adobe Colours CC
7. Canva (https://www.canva.com/colours/colour-wheel)

Conclusion

Don't be afraid to play with the palette and create your own unique interpretation. That's what colour harmony is all about; the formulas are merely starting points to help guide and inspire you.

The Colour Mixing Model

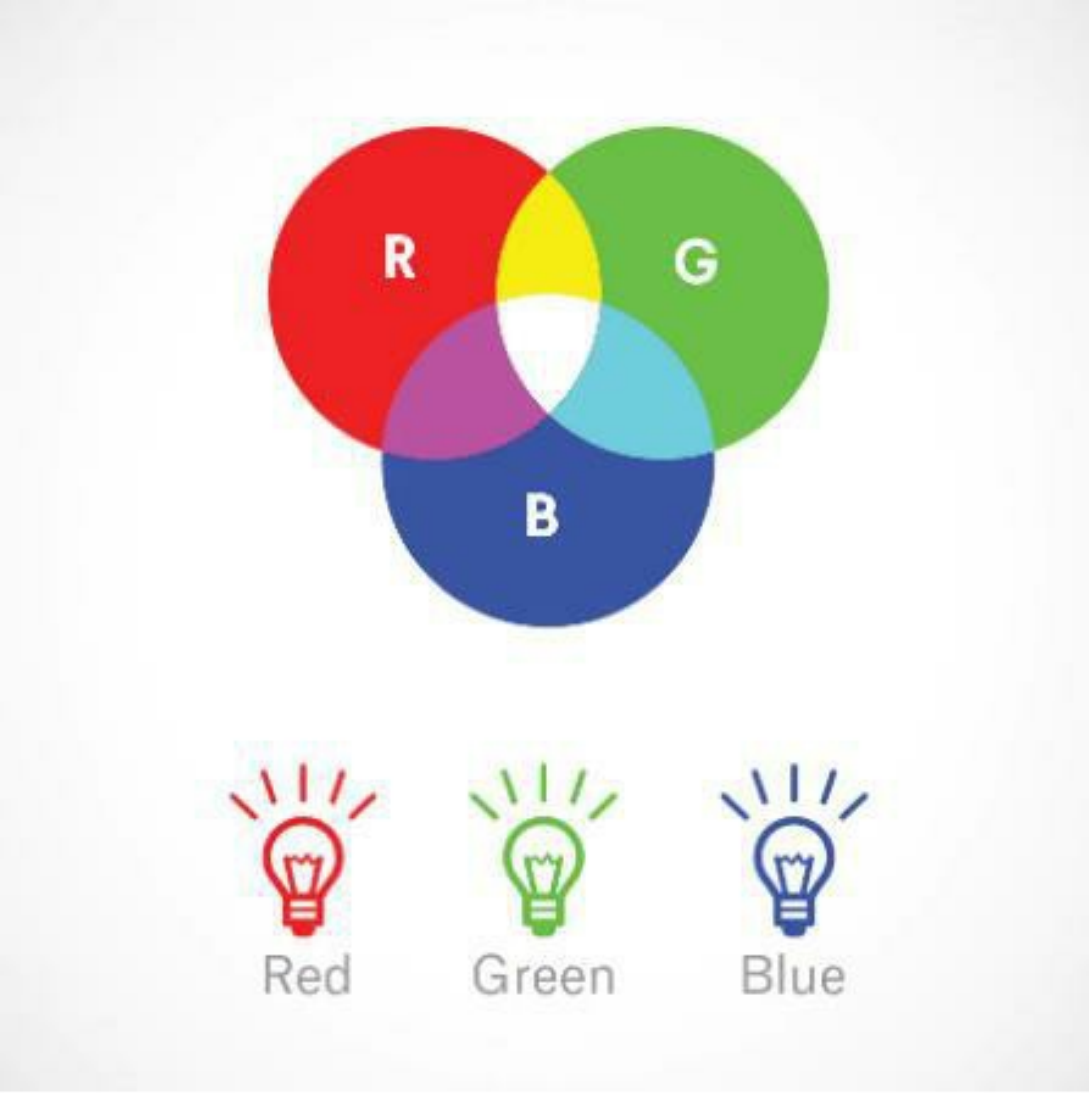
Humans see colours in light waves. The colour mixing model refers to the models that exist in graphic design?Two colour models exist in today’s world: The additive model (RGB) and the subtractive model (CMYK).

RGB: The Additive Colour Mixing Model

RGB stands for Red, Green and, Blue. RGB is known as the additive colour mixing model. Mixing light—or the additive colour mixing model—allows you to create colours by mixing red, green and blue light sources of various intensities. The more light you add, the brighter the colour mix becomes. If you mix all three colours of light, you get pure, white light.

Electronic displays: TVs, screens and projectors use red, green and blue (RGB) as their primary colours, and then mix them together to create other colours.

The RGB scale is 0 to 255. In RGB, black is R=0, B=0 and G=0, and white is R=255, B=255, and G=255.



Let’s say you have a very distinct brand with a bright yellow logo. If you post the logo on Facebook, Twitter or your website and don’t use the correct colour process, your logo will appear muddy instead of that bright yellow. That’s why, when working with files for any screen, use RGB, not CMYK.

CMYK: The Subtractive Colour Mixing Model

Any colour you see on a physical surface (paper, signage, packaging, etc.) uses the subtractive colour mixing model. Most people are more familiar with this colourmodel because it's what we learned in kindergarten when mixing finger paints. In this case, "subtractive" simply refers to the fact that you subtract the light from the paper by adding more colours.

Often you are printing on white papers. Adding colours block the white paper from coming through. The CMYK scale ranges from 0 to 100. When C=100, M=100, Y=100 and K=100 you get black colour. The opposite is white, which is C=0, M=0, Y=0 and K=0.

Why should you care?

Printing is very expensive especially when you are printing a full colour project. Therefore, it is important that colours are in the right model to avoid waste of money.

Since printing uses the subtractive colour mixing method, getting accurate colour reproduction can only be achieved by using CMYK. Using RGB will not only result in inaccurate colour.

GRAPHIC DESIGN SOFTWARE

Graphic Design Software

Presently, there are a variety of design software available for use. Some are free while others are not. Adobe products such as Photoshop, Illustrator and

InDesign are mainstays in the graphic design world. However, you will need to purchase these programs and pay a monthly subscription fee to be able to use its full features.If you are just beginning and don't want to commit to the high price tag Adobe products often carry, similar free open-source software such as GIMP can help you begin to master the basics.

Graphic designers are expected to be proficient in software programs for image-making, typography and layout. Nearly all of the popular and "industry standard" software programs used by graphic designers since the early 1990s are products of Adobe Systems Incorporated. Adobe Photoshop (a raster-based program for photo editing) and Adobe Illustrator (a vector-based program for drawing) are often used in the final stage. Also, some designers use CorelDraw. CorelDraw is a vector graphics editor software developed and marketed by Corel Corporation. You will need to purchase the software to be able to use it.

Raster images may be edited in Adobe Photoshop, logos and illustrations in Adobe Illustrator and CorelDraw, and the final product assembled in one of the major page layout programs, such as Adobe InDesign, Serif PagePlus and QuarkXpress.

Open source software used to edit the vector graphis is Inkscape. Primary file format used in Inkscape is Scalable Vector Graphics (SVG). You can import or export the file in any other vector format. Designers often use pre-designed raster images and vector graphics in their work from online design databases. Powerful open-source programs (which are free) are also used by both professionals and casual users for graphic design, these include Inkscape (for vector graphics), GIMP (for photo-editing and image manipulation), Krita (for painting), and Scribus (for page layout).

CONCLUSION

So far, I believe that you have learned a great deal from this book. We have looked at major concepts around the design niche. From colours to typography, we have really covered a lot.

However, it is noteworthy that most elements used in our design process are not entirely original. That is, they are not created by us, especially images, vectors and fonts.

Copyright is an important part of the design industry. As much as there are freely licensed resources, there are also resources that are premium which requires permission (purchase) before you can be allowed to use them. But due to the lack of strict regulations in the industry, most designers don't give it much thought.

As an emerging designer, it is important to acknowledge the work of other either by paying them or sending regards to them, depending on where you get such resources. Infringement or theft of intellectual property is a serious which needs to be taking seriously.

To this end, it is important that you check out whether the resources you are using for your design is freely licensed (open-source) or paid for.

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typography.

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Contents.

Covering everything from the basics, to all things complex.

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Technical Options.

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6

Glossary, Further Reading, References.

Typeface Design.

A typeface "is the collection of thousands of years of collective human social, technological and economic history, combined with the passion, skill, experience and personal history of its lone creator." (Earls, 2002)

1

An Introduction to Typeface Design

If you're a fan of type and lettering and feel up for a considerable yet potentially rewarding design challenge, then you might consider designing a typeface. Then, should you want a further challenge, you might also consider ways of digitising your work. There are a number of ways that you can approach this—some being far easier than others. Whatever approach you decide to take, a certain amount of patience and understanding will go a long way.

As an art form and a technical, engineering process, typeface design has, in recent years, grown exponentially. A process that has traditionally been the esoteric knowledge of the few, it continues to grow and diversify. Opinions regarding the expanding growth of typeface design have been met with conflicting views. The late English lettering artist and designer, Michael Harvey, expressed his dislike for the "confusing variety of styles" that add to the "over-crowded repertoire" of new typeface designs and the flourishing eccentricity and 'crumbling' standards of design. However, the argument about the choice or surplus of available fonts (depending upon your stance), is perhaps better summed up by Peter Bil'ak in his article "We don't need new fonts…".

In his piece, Bil'ak states:

"There are typefaces which haven't been made yet and which we need. Type that reacts to our present reality rather than being constrained by past conventions.... It is time to think about why we design type, not just how we design it."

Getting Started

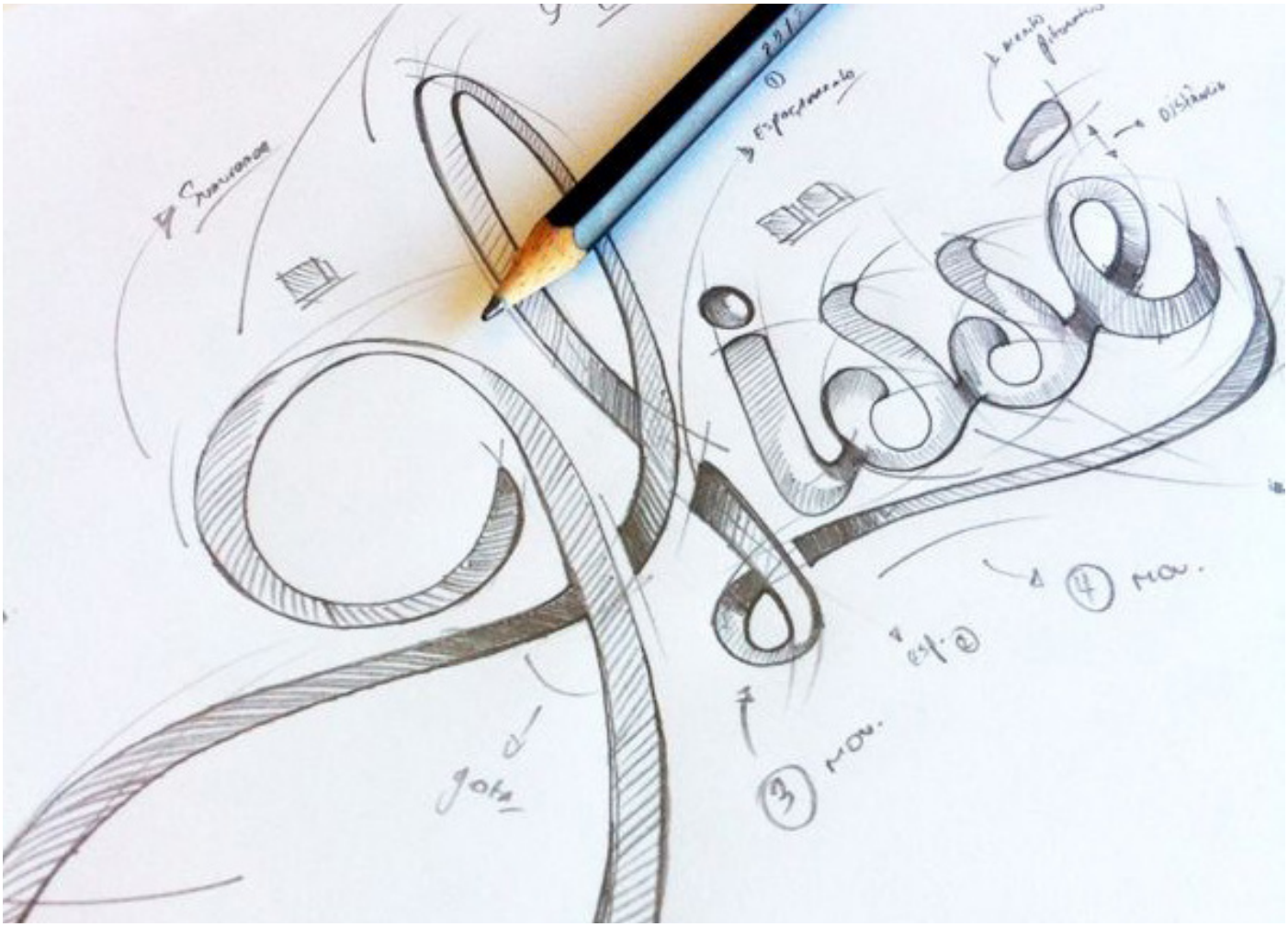
When you begin designing a new typeface, the Internet, along with old or out of print books, can provide a good starting point for your research. Don't feel obliged to sit at your desk during this process. Get yourself outside and look out for sources of potential inspiration and keep a visual record of the items that inspire you. Importantly, consider the function of the fonts that interest you. What is it that you think the designer set out to do with the type? Is it purely decorative or does it serve a particular function, such as assisting with way finding or traffic signs? With that in mind, consider what you want it to do. You should keep in mind the maxim, *"form follows function".*

Once you've established what the typeface is for, you can begin. If you're new to the process, you'll want to keep things simple. Typeface design can be a complex enough task without making it more difficult than it needs to be. Perhaps you should start with a simple display face. Depending upon the requirements of a brief, some designers spend months (even years!) perfecting their work, so it's a task that can easily grow in scale.

Some designers begin working by drawing letters, keeping sketchbooks and transferring these designs to the computer. Other designers work directly within a digital environment. For the latter, FontStruct is an ideal place to begin, being an easy to use, free, online resource, but we will address technological resources at a later point.

On finding inspiration, the French typeface designer Jean-Francois Porchez suggests we must stand on the shoulders of giants for inspiration and knowledge, stating:

"Each new typeface cannot live without its predecessors – all of them form one large family and are connected and influenced by each other. If Claude Garamond cut a splendid type, then it's because he studied Italian typefaces and tried his best to improve upon them…"

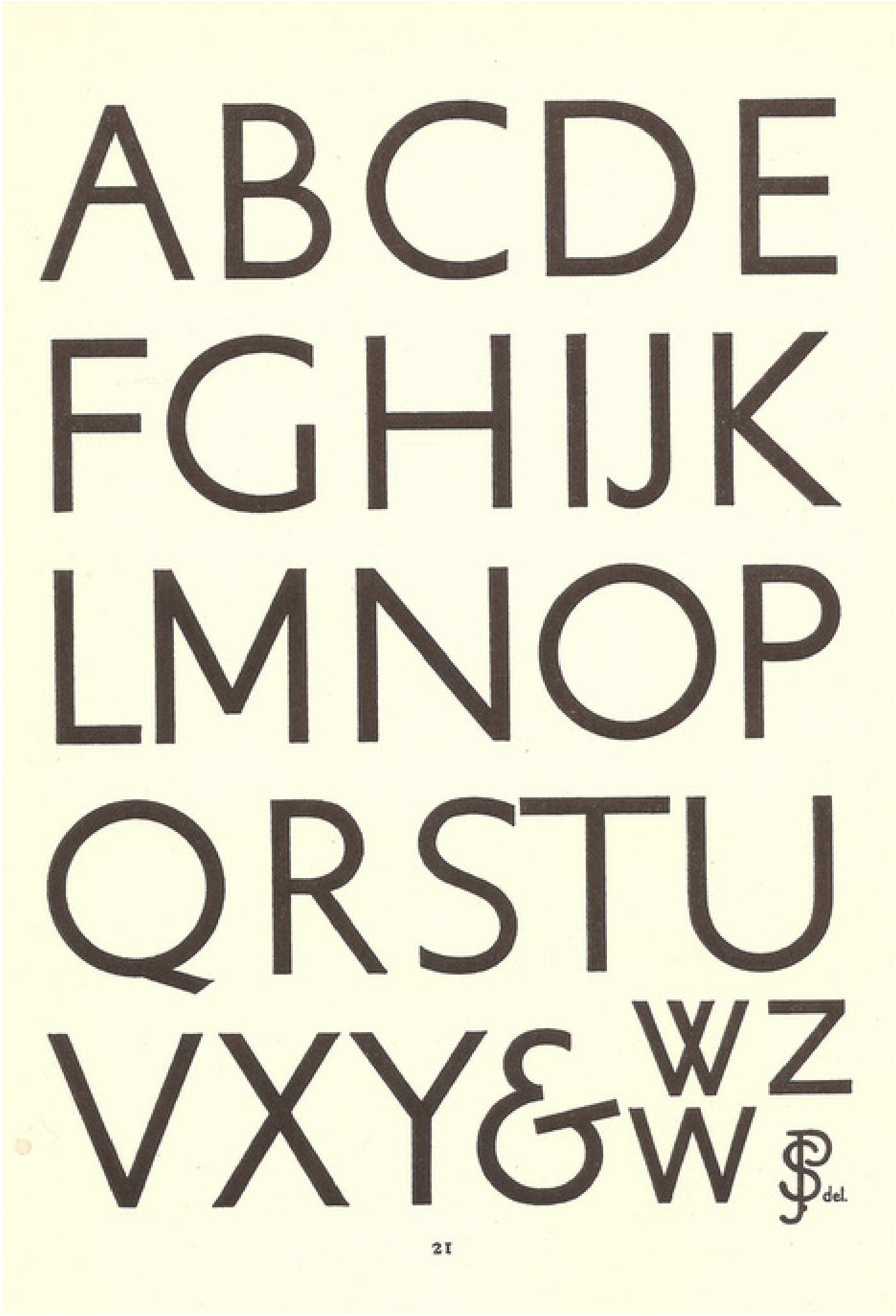


Egyptian for Carving.

A B C D E F G H I

J K L M N O P Q

R S T U V W X Y Z



21

Terminology & Anatomy.

"There are typefaces which haven't been made yet and which we need. Type that reacts to our present reality rather than being constrained by past conventions.... It is time to think about why we design type, not just how we design it." (Peter Bil'ak)

2

Even if you are an experienced calligrapher or lettering artist, making a typeface is an altogether different process, and before you start it's worth becoming familiar with some fundamental typographic concepts and terminology.

You have probably heard of the terms fonts and typefaces, and these two terms are often used interchangeably as there seems little harm in doing so. However, strictly speaking, there is a distinct difference in meaning between the two.

A typeface is essentially the design of an alphabet and its family, such as Gill Sans Regular, Bold and Italic in its various sizes (12 point, for example), whereas a font or "fount" (UK) is the physical collection of a typeface and the means by which it is rendered. For example, either the digital or metal version of Gill Sans Bold 14 point is a font.

Another simple example of these differences is stencil lettering and the means by which it is produced. The design of the letter shapes on the stencil is the typeface and the plastic stencil used to reproduce the design is essentially the font.

Where to Start?

A good place to start might be with what is known as a display face. Display faces are those used for headings and tend to be used at a larger size than body type. However, before proceeding, you should familiarise yourself with some of the fundamental concepts of typography along with its associated terminology.

Letterform Anatomy

Start by getting to know the differences between your cap heights and your x-heights and how to tell your serifs from your sans serifs. Whatever you choose to design, it will need to be consistently well styled and proportioned.

Familiarise yourself with the information contained within this chart on the following page:

Bowl
Ascender
Terminal
Stem
Stress
Ear
Bar
Loop
Serif
Cap Line
Cap Height
Stem
Leg
Crotch
Descender
Shoulder
Bowl
Juncture/
Joint
Spine
Counter
Tail
Italic
X Height
Vertex
X Line
Vortex
Stroke
Bracket Fillet
Swash

V

Types of Typefaces.

“No other design discipline requires so much learning and training as fontography, and by no other aspect can amateurs be so easily distinguished from professionals. To be font literate, a designer has to study the history and the principles of font design.” (Dmitry Kirsanov)

3

Classifications

Obviously type comes in thousands of shape variations, and as well as Serifs and Sans Serifs, there are specific classifications for these variations. One such classification system is that created by Maximilien Vox in 1954. Within the current version of the Vox System there are nine groups identified as follows:

1. Classicals

- Humanist
- Garalde
- Transitional

2. Moderns

- 2.1 Didone
- 2.2 Mechanistic
- 2.3 Lineal
- 2.3.1 Grotesque
- 2.3.2 Neo-grotesque
- 2.3.3 Geometric
- 2.3.4 Humanist

3. Calligraphics

- 3.1 Glyphic
- 3.2 Script
- 3.3 Graphic
- 3.4 Blackletter
- 3.5 Gaelic

Although this system encompasses most of the major categories of type, Cheng (2002) has noted that some of the many typefaces now in existence defy classification.

Humanist, Old Style or Venetian

Humanist typefaces echoed the letter shapes formed by broad nibbed pens.

Transitional

Sharing some of the characteristics of both the humanist and modern faces, transitional faces have medium contrast between the thick and thin parts of the letters.

Modern or Didone

These typefaces exhibit high contrasts between thicknesses. The serifs are referred to as hairlines due to their fineness.

Egyptian

Traditionally, Egyptian or slab serif typefaces were quite heavy with completely squared or bracketed serifs.

Examples for you to consider:

Technical Options.

“I do not think of type as something that should be readable. It should be beautiful.” (Ed Benguiat)

4

Having decided upon the purpose of your design and having invested a considerable amount of time drawing letters and trying out different variations and combinations, you may wish to attempt to digitise your typeface, effectively making a font.

An experienced designer will probably concur that this part of the process can be as involved—if not more so—than the initial designing of the typeface, particularly if you are striving to perfect the numerous intricacies of the working font.

Let us assume that, as a beginner, you are possibly striving to achieve a basic display face of some kind, at the very least. You might consider making a quirky or decorative typeface. Either way it might be wise to keep in mind the KISS principle of design, meaning 'keep it simple, stupid'. In other words, don't try too to achieve too much too soon.

There are several technological choices that you can make when approaching font production and these will largely be determined by your needs and your technical experience.

There is no single definitive design process, and typeface designers approach this somewhat esoteric process in different ways. Some designers work exclusively from screen, without sketching or drawing their designs onto paper, whereas some will scan their drawings and bring these either into font editing software or firstly into Illustrator to be worked on later within a font editor.

If you're striving to design a display face then you might consider starting out by using what is almost certainly the best beginners' option for getting quick results: Fontstruct.

Free and simple to use, Fontstruct allows you to use a variety of shaped building blocks in order to form a simple, modular typeface. Once complete, you can then share, download and use the fonts.

FontStruct allows for the design to take place within a web browser, meaning that there's no requirement to download and install any software. It is incredibly easy to use, and both building and editing are simple. Because of the modular method of construction there are no Bézier curves to manage, which is an easy way to begin.

Once you have completed your design, you can download your font and install it onto your computer. You can also decide whether or not to keep the font private or share it via the FontStruct website. The sharing of fonts within the application means that fellow designers can modify each other's designs, making this a truly social design experience.

If you want to create typeface that is altogether more sophisticated, then there are more powerful and complex editors such as Fontlab, and the more recently available, Glyphs. Fontlab has been the serious type designer's tool for a long time, but now has competition from applications such as Glyphs which according to its creator is a "smart and simple approach [that] helps you draw new typefaces, modify existing fonts, and sculpt your letterforms hassle-free." Unlike some of its competitors, Glyphs has a user friendly bank of tutorials and resources, including an excellent handbook that will help the novice get off to a good start and develop an understanding of many of the refining processes.
Unlike FontStruct, Glyphs allows you to draw and edit using Bézier curves, giving you the opportunity to create far more precise shapes.

An exciting new development is Prototypo. Prototypo appears to be a sophisticated browser based tool which is available to download in beta form.

Prototypo

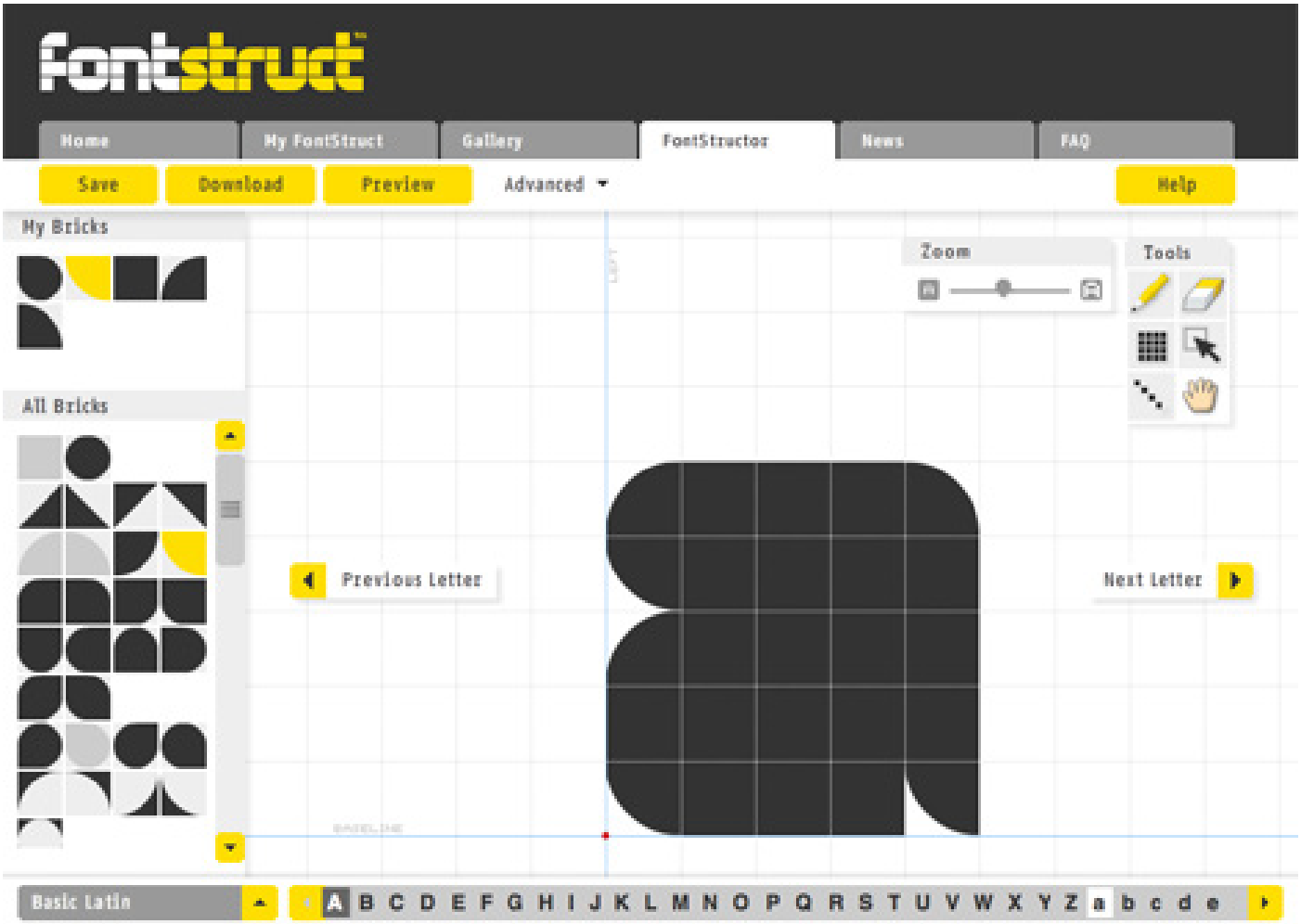
Of course, if you're familiar with Adobe Illustrator (and particularly its pen tool) you can design your letterforms with it. But in order to turn these into working fonts, you'd need to bring them into some font editing software and continue working on them to perfect their shaping and spacing. If you design a typeface using Illustrator you must remember to work in points as this is the true measure of type.

There are several other font editors; some are designed for use on PC and others for Mac such as RoboFont.

If you're looking for a way of digitising your handwriting, then there's MyScriptFont.com which is a free online tool and quite a fun way of generating a font with a personal touch.

Whatever approach you decide to explore, it is good to keep in mind that tools are simply tools and that type design and typography are steeped in centuries of convention. As designer and author David Jury says, *"Rules can be broken but never ignored."*

Tweaking
Tweaking
THICKNESS
CONTRAST
APERTURE
OPTIC THICKNESS
ROUNDNESS
BREAK PATH
AXIS
INKTRAP
File Edit Font View Help
Tweaking
Glyph list


fontstruct
Home
My FontStruct
Gallery
FontStructor
News
FAQ
Save
Download
Preview
Advanced
Help
My Bricks
All Bricks
Zoom
Tools
Previous Letter
Next Letter
Basic Latin

Understanding Typeface Design.

“Since typography is a communicaton method that utilizes a gathering of related subjects and methodologies that includes sociology, linguistics, psychology, aesthetics, and so much more – we aim to educate that there is no single approach within typography that applies to everything.”
(Shelley Gruendler)

We have touched upon some of the fundamental aspects of typeface design, along with a brief look at some of the type design tools available. To conclude, here are some final points to consider:

Read, read, read.

If you're really keen to develop an understanding of type design, be prepared to learn from those with experience. What has in the past been a fairly enigmatic discipline has become a more open and transparent activity with many people having a go at creating their own designs and sharing their knowledge. Scour books and investigate the many discussion threads in typography forums such as those at typedrawers.com. Specialist libraries are few, but places such as the St. Bride Library in London house an enormous collection of books, type specimens and print artefacts. However, if this is not an option, the reading and references provided below should get you off to a good start.

Draw, or at least doodle, regularly.

You don't have to have great handwriting or be an accomplished calligrapher, however, drawing does enable the immediate conveying of ideas as well as the development of sensitivity as to how shapes and forms interact in relation to each other. Use whatever you have available, be it scraps of paper, napkins or old packaging. Ideally, have a notebook or journal that you can keep with you and utilise as soon as inspiration strikes. A graphics tablet can be a good investment too, as this allows for greater freedom and control than a mouse.

Start with the basics.

Take time to develop sensitivity towards letter shapes. When designing letters, don't just consider the main form of the letter shapes as standalone designs. Instead, consider also the spatial relationship between other letters as well as the spaces inside of the letters (the counters and apertures).

Never steal someone else's design.

Tweaking someone else's design in an attempt to pass it off as your own is unethical and if it breaches copyright law, it's illegal. There are many observant type aficionados who are quick to spot a plagiarised font!

Enjoy it!

Leave the stress to the professionals and simply have fun learning about letters as you play with the possibilities. Good luck.

WILLIAM CASLON

{Baroque 1722 Baroque}

THE FIRST OF A FAMILY of english type founders, was born at Cradley, Worces tershire in 1692, and in 1716 start ed business in London as an engraver of gun locks and bar rels, and as a bookbinder's tool-cutter. Being thus brought into contact with printers,he was induced to fit up a type foundry, largely through the encouragement of printer William Bowyer.

ABCDEFGHI *ABCDEFGHI*
JLMNOPQR *JLMNOPQR*
STUVWXYZ *STUVWXYZ*
abcdefghij *abcdefghij*
klmnopqr *klmnopqr*
stuvwxyz *stuvwxyz*
1234567 *1234567*
89&?! *89&?!*

THE DISTINCTION AND legibility of his type secured him the patron age of the leading printers of the day in England and on the continent. The use of Caslon types, discontinued about the beginning of the 19th century, was revived about 1845 at the suggestion of Sir Henry Cole, and used for printing the Diary of Lady Willough by (a 17th century story) by the Chiswick Press. The head line on this page is "Caslon Old Face." He died on the 23rd of January 1766. His son, William Caslon (1720-1778), who had been partner with his father for some years, continued the business. The original printing of the American Declaration of Independence was set with Caslon type, and derivatives of these typefaces continue to be used as computer fonts.

Quoufque tandem abutere Catilina, patientia noftra? quamdiu nos etiam furor ifte tuus eludet? quern ad finem fefe effrenata ja&abit audacia?

QUOUFQUE TANDEM ABUTERE CATILINA, PATIENTIA NOFTRA? QUAMDIU NOS ETIAM FUROR IFTE TUUS ELUDET? QUERN AD FINEM FEFE EFFRENATA JA&ABIT AUDACIA?

Quoufque tandem abutere Catilina, patientia noftra? quamdiu nos etiam furor ifte tuus eludet? quern ad finem fefe effrenata ja&abit audacia?

History	*History*	History	History
Kolonien	*Kolonien*	Kolonien	Kolonien
Declaration	*Declaration*	Declaration	Declaration
Originality	*Originality*	Originality	Originality
Humanistic	*Humanistic*	Humanistic	Humanistic
Indepedence	*Indepedence*	Indepedence	Indepedence
Transparent	*Transparent*	Transparent	Transparent
Transitional	*Transitional*	Transitional	Transitional
Experimental	*Experimental*	Experimental	Experimental

Akzidenz Grotesk
(1898)

GROTESQUE

IN GERMAN, the word Akzidenz refers to any type of printing other than book and magazine work. Accidenz (sic) Grotesk was released by Berthold in Berlin in 1898, according to their own literature.

Akzidenz-Grotesk is a grotesque typeface originally released by the H. Berthold AG type foundry in 1898 under the name Accidenz-Grotesk. It was the first sans serif typeface to be widely used and influenced many later neo-grotesque typefaces. Max Miedinger at the Haas Foundry used it as a model for the typeface Neue Haas Grotesk, released in 1957 and renamed Helvetica in 1960. Miedinger sought to refine the typeface making it more even and unified. Two other releases from 1957, Adrian Frutiger's Univers and Bauer and Baum's Folio, take inspiration from Akzidenz-Grotesk.

24680

?!

abc defghijklmnopqrstuvwxyz

ABCDEFGHI
JLMNOPQR
STUVWXYZ
abcdefghijklmn
opqrstuvwxyz

The design of Akzidenz-Grotesk was theorized to be derived from Walbaum or Didot, as demonstrated by the similar font metrics when the serifs are removed. However, the font family also included fonts made by other foundries, such as the c. 1880 typeface Royal Grotesk Light from the Berlin foundry Ferdinand Theinhardt Schriftgiesserei, designed by Ferdinand Theinhardt for the scientific publications of the Royal Prussian Academy of Sciences in Berlin.

Glossary.

The following are some of the key terms that pertain to typeface design and typography.

Aperture

An opening of negative space within a letter such as that within the letter ‘U’ or that within the lower half of the letter ‘e’

Ascender

The part of a letter that extends above the x-height. Examples of letters with ascenders include ‘b’, ‘d’ and ‘h’

Baseline

The invisible line on which type sits

Body Type

Type that forms a main body of text.
(As opposed to a ‘display’ face)

Counter

A closed space, or partially closed space within a glyph such as that within the letter ‘d’ or the number 8

Cursive

Letters that join together as in scripted handwriting

Descender

The part of a letter that extends below the baseline. Examples of letters with ascenders include ‘p’ and ‘q’

Diacritics

Appearing either above or below a letter, diacritical marks indicate a change in pronunciation. Examples of diacritical marks include the acute (é) and a circumflex (ê)

Display Type

Type that is generally of a large size. Often this can be easily read at a distance or used in headings.

Fount (archaic UK) Font (US)

The physical means by which a typeface is applied or rendered. This normally takes the form of either a digital or metal set of letters and glyphs. Example: Gill Sans Light 10pt

Glyph

A symbol or character

Hinting

Part of the type design process whereby type is lined up against a pixel grid

Italic

A sloping type based on cursive handwriting.

Kerning

The adjustment of space between two letters

Legibility

“A measure of a type’s ability to be read under normal reading conditions.” (White, 2004)

Ligatures

A glyph that is formed by joining two or more glyphs

Majuscules

Capital or uppercase letters

Minuscules

Lowercase letters

Multiple Mastering

A technological facility that allows for the creation of a variety of font weights from one master

Opentype

A font file format that works across platforms. It generally has the suffix .otf

Pica (Py)

Consisting of 12 points, a pica is typographic measure expressed.

Point (Pt)

A single unit of a measure of type. There a 12 points to a pica.

Readability

“A measure of the type’s ability to attract and hold a reader’s interest. Increasing readability often causes a reduction in legibility…” (White, 2004)

Serif

A small stroke or addition to the end of letter or glyph stroke. These come in a variety of shapes and styles such as hairline, slab, bracketed and wedge.

Sans Serif

Type that lacks the addition of serifs.

Sidebearings

This term refers to the spaces either side of a glyph.

Swash

An added flourish or embellishment such as a lengthened tail or exaggerated, decorative serif.

True Type Font

or .ttf is a forerunner of the Opentype format.

Tracking

The adjustment of space between multiple letters or tracts of text.

Typeface

The design of a set of an alphabet, numbers, punctuation, etc.

Tail

A descending, often extended stroke the generally runs from left to right such as that of the letter ‘Q’

Terminal

The end of a stroke without a serif. Example of a stylised version is the ball terminal

Weight

Refers to the relative density and stroke widths of a given typeface, such as thin, regular, bold, extra bold, etc.

X-Height

The height of a given lowercase ‘x’

Further Reading.

BOOKS

Bringhurst, R., The Elements of Typographic Style

A seminal typographic text, Bringhurst's book is a thorough and expansive wealth of typographic information and is ideal for both the beginner as well as the advanced practitioner.

Cheng, K., Designing Type

Cheng's book is an incredibly useful resource. This is not a 'how to book' but is a great reference for understanding the structure and systems of type. Highly illustrated throughout.

Earls, D., Design Typefaces

Harvey, M., Lettering Design

Heller. S, Typography Sketchbooks

A compendium of typographic designers' sketchbooks.

White, A., Thinking in Type: The Practical Philosophy of Typography

Jury, D., About Face: Reviving the Rules of Typography

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Baseline Magazine

WEB

Blokland, E van., Value of Type

Briem, G., Notes on Type Design

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De Gregorio, J.P., Vector Drawing Mistakes

Shaw, P., Ten Simple Rules for Researching Letterforms

Stössinger, N., Sketching Out of My Comfort Zone: A Type Design Experiment

On drawing Type by TypeCooker.com

- Type Anatomy

FORUMS

http://typedrawers.com/ – "Typedrawers is a forum where fellow typophiles can gather to discuss typography."

GAMES

Kerntype - http://type.method.ac/

Shapetype - http://shape.method.ac/

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4 5 6
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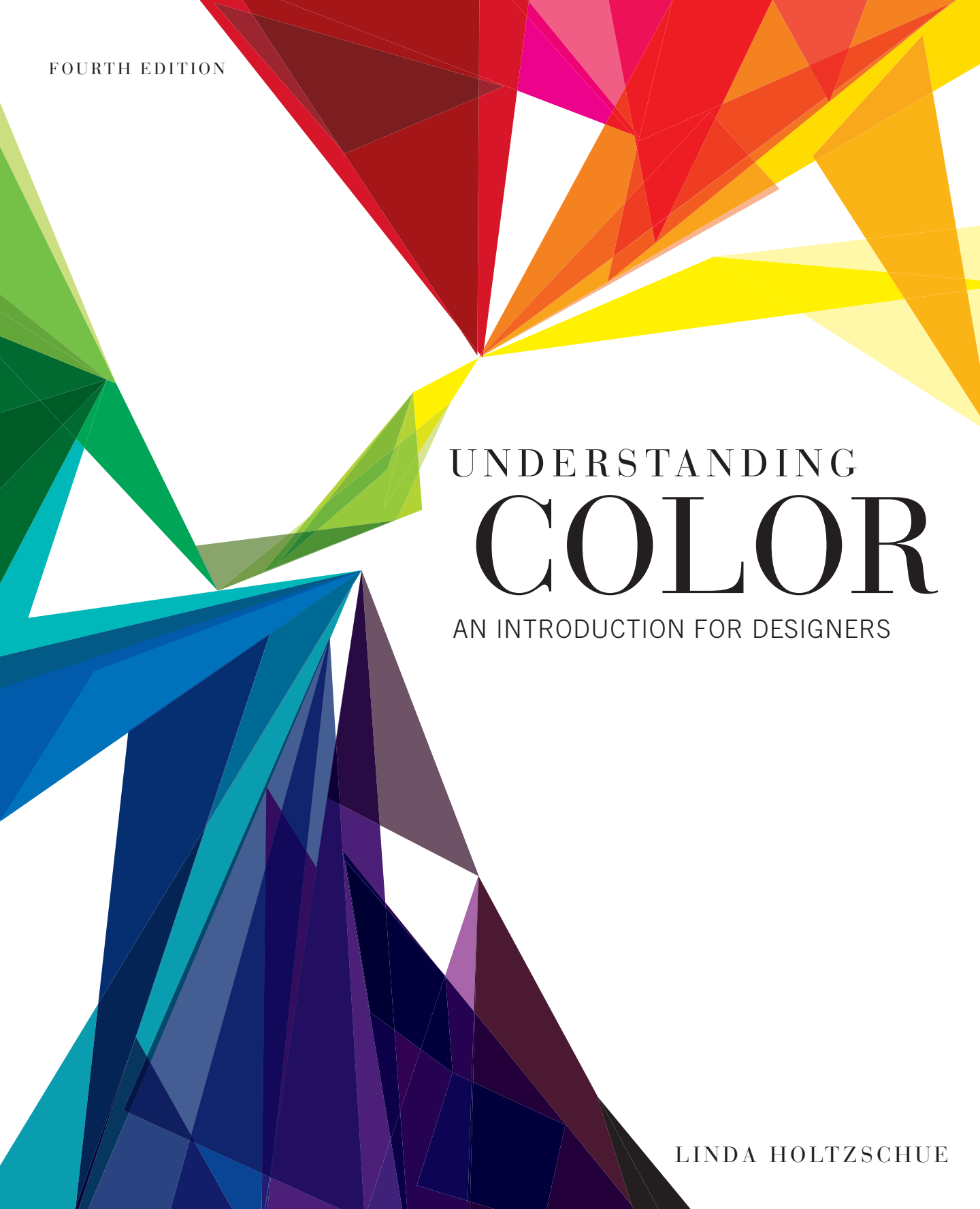

UNDERSTANDING COLOR

AN INTRODUCTION FOR DESIGNERS

LINDA HOLTZSCHUE

UNDERSTANDING COLOR

An Introduction for Designers
Fourth Edition

Linda Holtzschue

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DEDICATION

To my children Alison, Adam, and Sara, and the wonderful partners they have brought into our lives; to my grandchildren Amanda Carrico Schloss, Katherine Rose Schloss, and Daniel Elias Holtzschue; and most of all to my husband Karl, whose patience and support made this book possible.

ACKNOWLEDGMENTS

This fourth edition of *Understanding Color* would not have been possible without the generosity of friends and colleagues, both old and new. I was delighted that Jennifer Perman, who designed the second and third editions and created many of the illustrations, was available to design this edition as well, and that my editor Margaret Cummins, who cheered me on through the two last editions, continued in that role for this one. I could not have done this book without the contributions, guidance, and support of Steven King, Bob Stein, Laurent de Brunhoff, Phyllis Rose, Carin Goldberg, Ron Lubman, Emily Garner, Kenneth Charbonneau, Wilsonart, Mark Stevenson, Stephen Gerould, Dan Brammer, Jamie Drake, Adobe Systems, Farrow and Ball, David Setlow and Stark Carpets, Edward Fields Carpets, Tai Ping Carpets, Safavieh Carpets, The Color Association of the United States, Color Marketing Group, Elizabeth Eakins, Sanderson, First Editions, Erika Woelfel of Colwell Industries, X-Rite Corporation, Alison Holtzschue, Gwen Harris, my friends at Benjamin Moore, and anyone I may inadvertently have missed. Thank you, all.

PREFACE

More changes have taken place in the way that designers work with color in the last few decades than have occurred over the last few centuries. Technology has created a tectonic shift in the everyday experience of color. Colored light, once a limited area of interest to designers, is now the primary medium of the studio workplace. Color is a whole new world, and at times a very confusing one.

This is a book for everyone who uses color. It is written for design students and sign painters, architects and carpet salespeople, graphic designers and magicians. It is a road map to the relationships between colors, and even more, to the relationships between real colors and virtual colors. It is guide to using color freely, comfortably, and creatively without dependence on complicated theories or systems. This is a book about learning to see—in the new way as well as the old.

This book includes a workbook component that is available online at www.wiley.com/go/understandingcolor4e.

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Online Workbook (www.wiley.com/go/understandingcolor4e)

1 AN INTRODUCTION TO COLOR STUDY

The Experience of Color / Color Awareness / The Uses of Color / Color-Order Systems / Color Study

Color is essential for life.

—*Frank H. Mahnke*

Color is stimulating, calming, expressive, disturbing, impressional, cultural, exuberant, symbolic. It pervades every aspect of our lives, embellishes the ordinary, and gives beauty and drama to everyday objects. If black-and-white images bring us the news of the day, color writes the poetry.

The romance of color exists for everyone, but color has an additional aspect for design professionals. Forms, colors, and their arrangement are the foundation elements of design, and of these elements, color is arguably the most powerful weapon in the designer's arsenal. A skilled colorist understands what color is, how it is seen, why it changes, its suggestive power, and how to apply that knowledge to enhance the marketability of a product. Whether the product is a graphic design, an item of apparel, an interior, automobile, toaster, garden, or anything else, good coloring can determine its success or failure in the consumer marketplace.

The Experience of Color

Color is, first, a sensory event. Colors are true sensations, not abstractions or ideas. The beginning of every color experience is a physiological response to a stimulus of light. Colors

are experienced in two very different ways. The colors on a monitor screen are seen as direct light. The colors of the physical world—of printed pages, objects, and the environment—are seen as reflected light.

The perception of colored light is a straightforward experience: light reaches the eye directly from a light source. The experience of real-world color is a more complex event. Real-world colors are seen indirectly, as light reflected from a surface. For tangible objects and printed pages, light is the *cause* of color, colorants (like paints or dyes) are the *means* used to generate color, and the colors that are seen are the *effect*.

All colors, whether they are seen as direct or reflected light, are unstable. Every change in light or medium has the potential to change the way a color is perceived. The color of a carpet underfoot is very different from that of its image on a screen, and each of these is different from its illustration on a printed page. In addition, the same color will appear to be different colors depending on its placement relative to other colors.

Not only are colors themselves unstable, ideas about colors are unstable as well. The color that one person identifies as "true red" will be a bit different from another's idea of "true red." When colors are used as symbols, their meanings are equally mutable. A color used symbolically in one context may have another meaning entirely—and even be called by another name—when it appears in a different situation.

Most of the work of the design industries today is done in images of direct light, on a monitor, for products that will ultimately be produced as goods or printed pages. Are the screen image and the product the same color? Can they be the same color? Which one is the "true" color—the one on the screen, or the one that is the actual object? Is there such a thing as a "true" color at all?

Designers *use* color. Their concern is with effects, not with words, ideas, or causes. Understanding what is seen, and how and why it is seen—how colors work—is background knowledge that supports the *art* of color. Designers work with color every day in a comfort zone; a healthy mix of fact, common sense, and intuition. A skilled colorist exploits the instabilities of color and uses them to create interest and vitality in design.

We understand color in much the same way that we understand the shape of the earth. The earth is round, but we experience it as flat, and act on it according to that practical perception of flatness. Color is light alone, but it is experienced so directly and powerfully that we think of it as a physical entity. No matter what we may understand about the science of color, or what color technology is available, we believe our eyes. Color problems in the design industries are solved with the human eye. *Designers work with color from the evidence of their eyes.*

Color Awareness

Color is sensed by the eye, but the perception of color takes place in the mind, and not necessarily at a conscious level. Colors are understood in context. They are experienced at different levels of awareness depending on how and where they are seen. Colors may be perceived as an aspect of form, as light, or as surroundings. Colors permeate the environment, are an attribute of objects, and communicate without words.

Environmental color is all-encompassing. Both the natural world and man-made environments immerse us in colors, whether they are the cold whites of Antarctica, the lush greens of tropical forests, the accidental color compositions of urban streets, or the controlled-color environments of architecture, landscape design, interior design, or theater design.

Surrounding colors have a powerful impact on the human body and mind, but most of the time they are experienced with an astonishing lack of awareness. Environmental color is noticed only when it is a focus of attention, like a dazzling sunset or a freshly painted room. Someone who expresses a dislike for the color green may nevertheless take enormous pleasure in a garden, describing it as a blue or yellow garden, when in fact the surroundings are overwhelmingly green, with blue or yellow present as only a small part of the whole.

The colors of objects are perceived very directly. The separateness of an object allows the viewer to focus both eye and mind on a single entity and a single color idea. We are the most consciously aware of color when it is an attribute of a defined object: a blue dress, a red car, a yellow diamond.

Figure 1-1.
Environmental Color. Color surrounds us at all times, whether the setting is man-made or natural. *Photograph courtesy of Phyllis Rose Photography, New York & Key West.*

Figure 1-2.
Environmental color. The colors of the natural environment are complex and beautiful. *Photograph courtesy of Phyllis Rose Photography, New York & Key West.*

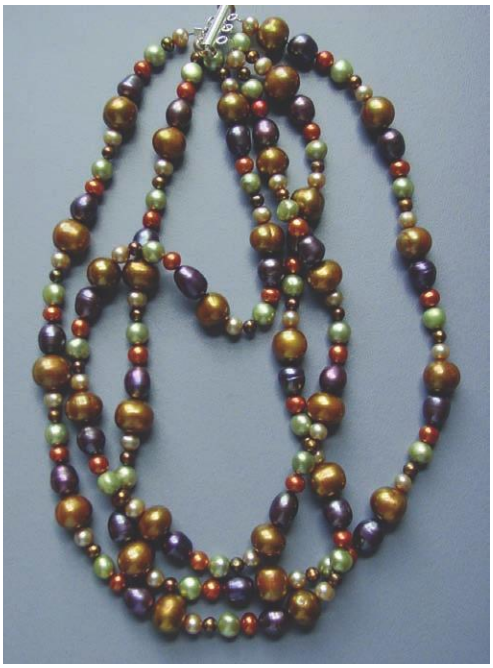


Figure 1-3. *Object Color.* Unexpected color in natural pearls adds surprise and individuality to a classic jewelry form. *Jewelry design by Suzy Arrington.*

Graphic colors are the colors of images: painted, drawn, printed, or on-screen. *Graphein*, the Greek root of the English word "graphic," means both "writing" and "drawing." Whether a graphic design is made of written words, illustrations, or both, its purpose is to communicate. It tells a story, sends a sales pitch or political message, even conveys emotion. Color in a graphic design is an integral part of the message. Graphic colors are experienced on many levels; conscious and unconscious, sensory and intellectual, at the same time.

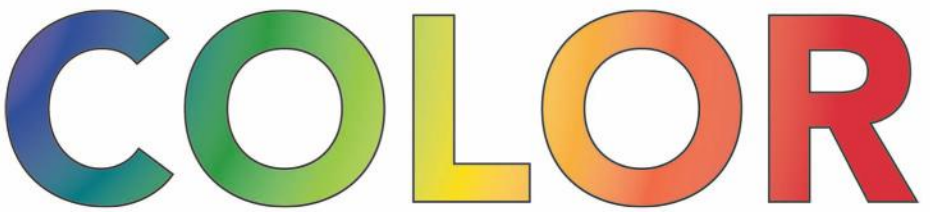


Figure 1-4. *Graphic Color.* Color adds meaning to the written word.

The Uses of Color

Color is recognized universally as a natural component of beauty. The Russian language word for *red* has the same root in Old Russian as the word for *beautiful*. But colors are far more than just beautiful; they are *useful.* Color can be used to communicate ideas and emotions, to manipulate perception, to create focus, to motivate and influence actions.

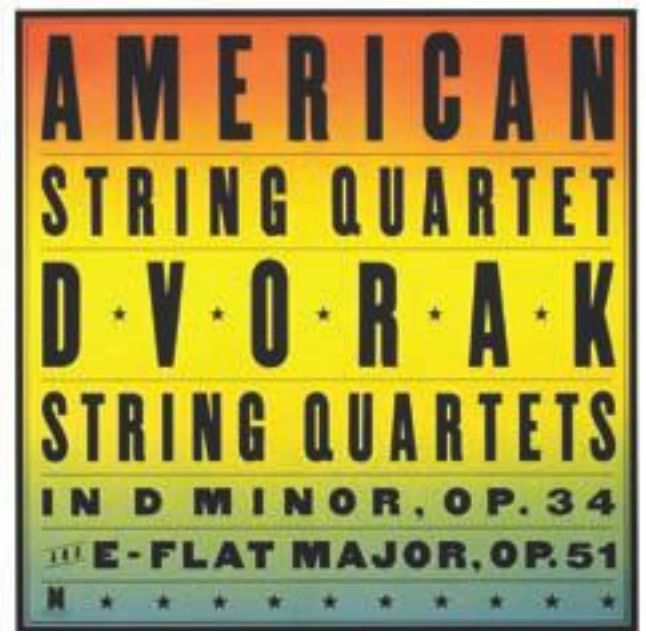


FIGURE 1-5.
Color Conveys Mood. An off-beat combination of colors perfectly expresses the edgy modernity of Dvorak's Piano Concerto in G Minor. *Image courtesy of Carin Goldberg Design.*

Color can be used as pure function, to increase or reduce available light. Light colors reflect light and increase the available light in a space; dark colors absorb light and reduce it. A room painted pale ivory will reflect more light—actually be lighter—than the same room painted dark red. When a room is dark, adding additional illumination alone will not necessarily solve the problem. If the walls are absorbing light, they will continue to do so. Illumination and color are equals in environmental space: it is the balance of the two that establishes the level of brightness.

Color can modify the perception of space, creating illusions of size, nearness, separation, or distance. Colors can be chosen to minimize or obscure objects and spaces, or to delineate

space, separating one area from another. Color can be used to create continuity between separated elements in design, or to establish emphasis or create focus in a composition.

Color can be a visual expression of mood or emotion. Intense colors and strong contrasts communicate action and drama. Gentle colors and soft contrasts convey serenity. Color can be also used to generate emotional response. Colors have physiological effects on the body. They can be chosen to stimulate, or to calm. They can be used to arouse a nonvisual sense, instill unconscious motivation, alter behavior or induce mood.

Colors can be a nonverbal language; communicating ideas without a single word. They can be used to represent a nation or institution, a product, or an idea. National flags are identified by color. IBM is Big Blue. Harvard is crimson. Colors can both symbolize and communicate social position. In ancient China the emperor alone wore yellow. Roman Catholic priests wear black; Tibetan priests, saffron yellow.

Figure 1-6.
Orange Alert.
Traffic cones warn drivers to take extra care.

Colors can be used to alert, or warn. A flashing red light evokes a different response than a green one. Violet warns of radiation danger.

Color identifies. It provides instant discrimination between objects of similar or identical form and size. The red file holds unpaid bills; the green file the paid ones. Color is associative. The ordinary things of everyday life are characterized by color associations. They can all be found in the Yellow Pages.

Figure 1-7.
Color Identifies.
An enormous filing system is made more manageable by tabs of different colors.

Color-Order Systems

One way to understand color is to organize it into a system; to hypothesize and illustrate a structured model of color *relationships*. The color-order model,

or color system, is a thread that runs from the earliest writings on color, to the absolute present, to tomorrow. These countless different and competing systems organize color in various ways, each convinced of its own rightness. But color is such an enormous topic that no single color-order system can be truly inclusive. In the early 1930s, The National Bureau of Standards tried to categorize and describe 10 million colors for scientific and industrial use. The result was a massive color-name volume and a breathtaking failure. The group "grayish-yellowish-pink," for example, included about thirty-five thousand samples.[1]

Today that effort might call for categorizing the more than 17 million colors available in design software, and even that number is not as great as the number of colors that can be distinguished by the human eye. More successful systems address a narrower range of issues. These systems fall into three groups:

Technical-scientific color-order systems
Commercial color-order systems
Intellectual-philosophical color-order systems

Technical-scientific systems are the province of science and industry. They measure color under specific and limited conditions. Most deal with the colors of light, not the colors of objects. One means of measuring the color of light is by determining the exact temperature in degrees Kelvin (K) of a piece of metal, called a blackbody, as it heats up. The color of a blackbody changes at specific temperatures, so that "color temperature" in the scientific sense refers to the point in degrees K at which the blackbody changes color as it heats, from yellow, to red, to blue, to white.

The International Commission on Illumination, known as the CIE, or Commission Internationale de l'Eclairage, has developed a color triangle that locates the color (in degrees K) of any light source. The CIE color triangle is mathematical and based on the human range of vision. It is said to be the most accurate of the color description models but is highly theoretical—so theoretical, in fact, that according to a CIE spokesman, it cannot be accurately illustrated. It is also not applicable to many technologies, including color printing and color monitors, so these must use other color models.

Another system, the Color Rendering Index (CRI), evaluates the way in which a given light source renders the colors of objects in comparison to a chart of eight standardized colors. Colors are judged in relation to daylight, and incandescent light is rated as best for illuminating the natural world because of its similarity to daylight.

Technical systems of color organization are not a part of the day-to-day work of the design studio. The artist or designer, for example, has a concept of "color temperature" that is something entirely different. Technical systems are, however, of value in maintaining quality control at the manufacturing level.

Commercial color-order systems are systematic arrangements of colors meant to assist the user in making selections from the colors available in a particular product line. There are two broad categories of commercial color systems: true *color systems* and *color collections*.

True color systems are based on a theory of colors and are published as a full range of colors arranged in a visually logical format. They attempt to illustrate all possible colors that can be produced within a specific medium, like paints or printing inks, and offer the option of adding colors beyond those illustrated. The Benjamin Moore® Color System, for example, includes over 3300 paint colors and offers a computer-matching option to create custom colors that do not appear in the printed collections.

One of the most widely known commercial color order systems is the PANTONE MATCHING SYSTEM®,[2] which provides a palette of standardized colors for a wide range of products ranging from printing inks to software, color films, plastics, markers, and color-related tools and products for use by designers and manufacturers. The most familiar are the swatch books of colors that can be printed using four-color (cyan, magenta, yellow, and black; or CMYK) process printing inks. The colors are arranged in a visually logical way, and each indicates the percentage of each of the four CMYK colors that mix to produce it. Bright red PANTONE® P57-8C, for example, is noted as C0, M100, Y66, K0. A dull red, PANTONE P50-8C, is made up of C0, M99, Y91, K47. Colors that do not appear on the charts can be produced by tweaking the percentages. Pantone offers similar charts for six-color printing inks, solid color inks, and a host of related color products.

Figure 1-8. *Commercial Color Order Systems.* Pantone offers swatch books of colors that can be printed using four-color process printing inks as well as swatch books for six-color, solid color, and specialty inks.

Color collections are narrower in scope. They offer a limited number of colors, often with a theme or point of view, to help the user in making a selection within a single product or group of related products. Color collections do not attempt to include all possible colors. Instead, they offer a color range intended to meet most needs within a specific target market or industry. A collection of historical paint colors, or Web or mail-order catalog color options, are color collections. The number of samples can vary widely. Coats and Clark, for example, offers more than 900 colors of sewing thread in its Global Colour Reference of Industrial Apparel Thread Shades. Old Village, a specialty paint company, offers only 25 paint colors, each painstakingly created to simulate an historic original, along with a very few colors in specialty products like buttermilk paint. There are countless numbers of these limited color collections available for use by artists, designers, and the general public.

Figure 1-9. *Color Collections.* A representative sampling of Farrow and Ball paint colors; part of a larger collection of colors based on historical prototypes. *© Farrow and Ball 2005. All rights reserved. No part of a colour card may be reproduced without prior permission of Farrow and Ball.*

Commercial color systems and collections are practical both in intent and in fact. They do not contribute to an understanding of color, but they were never intended to do so. Within their limitations they are invaluable as aids in specifying colors for design as well as other purposes.

Intellectual-philosophical systems explore the meaning and organization of color. Artists (famous and not-so-famous) have always written on color, but a fascination with the topic has never been limited to the world of art. Plato and Aristotle, Newton, Goethe, and Schopenhauer wrote about color, as did many other scientists and philosophers.

The late seventeenth and early eighteenth centuries saw a flowering of writing on color whose influence persists to the present day. Two threads of inquiry run through these studies: first, the search for a perfect color-order system; second, the search for laws of harmony in color combinations. The two paths have converged into the present-day field of study known as *color theory*.

Early writers made no distinction between the art of color and the science of color. The colors of light and the colors of objects were dealt with as a single discipline. The natural sciences developed quickly into separate fields of study: biology, mathematics, physics, chemistry, and later, psychology, among others, but even color theorists whose focus was the arts continued to approach color as an scientific discipline. The aesthetic and philosophical color-order systems they proposed were viewed as scientific information until well into the twentieth century. Today, writers on color still come from scientific, artistic, and philosophic disciplines, but color theory for artists and designers has been separated from science. Color theory has, at last, its own place in the arts.

Color Study

It makes no difference whether color is seen as light on a monitor screen or as an attribute of a physical object. No matter how color is seen, or in what medium it is rendered, or what other senses seem to respond to it, the experience of color is visual. Color study focuses first on *eye-training*: learning to distinguish in every color sample three objective attributes:

Hue, the name of the color.
Value, the relative darkness or lightness of a color.
Saturation or *chroma*, the relative brilliance or muted quality of a color.

These qualities are independent ideas only for purposes of study. Every color possesses these three attributes. No color can be described fully as only one idea. The emphasis of

one quality over the others—of hue-intensity, or lightness, or darkness—gives each single color its individual character. Other qualities of colors, like surface texture, or opacity, or translucency, are additional to these basics.

The second focus of color study is color *control*. The instability of colors cannot be eliminated, but it can be minimized. Designers deal with the instability of color in three principal ways: first, by anticipating the changes that occur when colors are seen under different lighting conditions; second, by adjusting for changes that take place when colors are given different placements relative to one another; and third, by adjusting for changes that take place when colors must be transposed from one medium to another—for example, in illustrating a piece of dyed fabric on a monitor screen.

Finally, color study provides guidelines that enable designers to create consistently effective color combinations. The ability to use color effectively is a skill that can be taught and strengthened. Color *competence* is the ability to problem-solve with color. It is the ability to predict and control, to the extent possible, color effects, and the ability to select and specify colors and combinations that enhance every product or page.

Many color courses taught today have their foundation in the teachings of Albert Munsell (1858–1918) and Josef Albers (1888–1976). Munsell's method follows a classic color-order tradition. It is a formal system based on orderly progressions of the three color qualities hue, value, and saturation. Albers took a more free-spirited approach. He believed that true understanding comes from an intuitive approach to color. He rejected the rigidity of color-order systems and stressed the power of eye-training exercises.

The Albers intuitive approach dominates American color education today. Detached from its background in color-order systems, it is not as accessible an approach to teaching color as it can be. The intuitive approach makes infinitely more sense when it follows an understanding of what color-order systems are about. By learning first to discriminate hue, value and saturation and mastering the concept of intervals, or steps between colors, students acquire skills that make the Albers exercises comprehensible from the start. Color-order systems and the Albers intuitive approach are not alternative ways to study color, nor are they competitive. The first leads seamlessly to the second, and together they embrace an understanding of color without limits or gaps.

Because no two individuals see color in exactly the same way, or share the exact idea of any given color, it is easy to conclude that color study in a class would be (at best) a waste of time or (at worst) a scheduled free-for-all. But there are great benefits from studying color in a group. For every problem assigned there are wrong answers, but there are also many different possible right answers. Twenty students may bring in twenty different, yet equally acceptable, solutions to the same color assignment.

Academic (nondesign) education tends to reward students who find the single "right" answer to a question or assignment. Some students who have been successful in traditional academic areas find this flexible standard of "rightness" very difficult. But the possibility of many right answers for each problem is extraordinarily liberating because it increases exponentially the chances of success. It is also extraordinarily instructive, because seeing a variety of responses, even "wrong" ones, sharpens developing critical skills.

The first step of color study is to enable the student to see colors without the interference of preconceived ideas. Students with a (self-assessed) "good sense of color" may feel inhibited, disoriented, or exasperated at the start. Just as mastering written music may at first slow down a natural musician, learning basic color skills may at first hamper an artist or designer. The inhibiting effect is short-term. There is a moment when learned material becomes reflexive, like the magical moment for children when learning to read ends and reading begins. The ability to work effectively with color is part of a designer's core competence. It is artistic empowerment. The designer who understands color has a competitive edge in every industry.

Endnotes

1 Sloane 1989, page 23.

2 PANTONE® and other Pantone trademarks are the property of and used with the permission of Pantone, LLC.

Chapter 1 Highlights

- Color is a sensory event; a physiological response to a stimulus of light. Color is also unstable. Every change in light or material has the potential to change the way a color is perceived and, in addition, individuals interpret the experience of color in different and personal ways.
- Colors are understood at different levels of awareness. Environmental color, whether natural or man-made, is all-encompassing. The separateness of an object allows the viewer to focus on a single entity and single color idea. Graphic colors are the colors of images: painted, drawn, printed, or on-screen.
- Colors on a monitor screen are seen as direct light. The colors of the physical world are seen as reflected light. Nearly all design today is done on a monitor screen in images of direct light for products that will be seen as reflected light.
- Color can be used as pure function, to increase or reduce available light. It can modify the perception of space, creating illusions of size, nearness, separation, or distance. Color can minimize or obscure objects and spaces and delineate space. It can be used to create continuity between separated elements in design, or to establish emphasis or create focus in a composition. Color can be used to express mood or emotion. Color can be used to alert, or to warn, or to provide discrimination between objects of similar form and size. It can be a nonverbal language; communicating ideas without words.
- A color-order model, or color system, is a structured model of color relationships. Technical-scientific systems measure color under limited conditions, and most deal with the colors of light. Commercial color-order systems are systematic arrangements of colors meant to assist the user in selecting colors from a limited palette. Intellectual-philosophical systems explore the meaning and organization of color.
- True color systems attempt to illustrate all colors and include the option of adding colors beyond those illustrated. Color collections offer a fixed and limited number of colors to help the user in making a selection within a single product or group of related products.
- Color study focuses first on learning to discriminate objective attributes of color: hue, value, and saturation (chroma). The second focus of color study is

color control. Color study also provides guidelines for creating effective color combinations.

- Many color courses are based on the writings of Albert Munsell (1858–1918) and Josef Albers (1888–1976). Munsell's system is based on formal progressions of hue, value, and saturation. Albers stressed the power of eye-training exercises.

2

A LITTLE LIGHT ON THE SUBJECT

Light / Additive Color: Mixing Light / Lamps / Lighting Level / Vision / The Illuminant Mode of Vision / The Object Mode of Vision / Subtractive Color: Colorants / Lamps and Color Rendition / Metamerism and Matching / Modifying Light: Surface / Transparent, Opaque, and Translucent / Iridescence / Luminosity / Indirect Light, Indirect Color / Modifying Light: Filters

Colors seen by candelight, Will not look the same by day.
—Elizabeth Barett Browning, "The Lady's 'Yes" 1844

Color is a visual experience, a sensation of light that cannot be verified by other senses: not by touch, taste, smell, or hearing. A colored *object* can be touched, but it is the object itself that is tangible, not its color. Color has no physical substance.

Colors are not only intangible, they also change constantly. Everyone has experienced buying an article of clothing and arriving home to find that it is a different color than it seemed to be in the store, or selecting a paint color and being astonished at the final result. Even the *idea* of a color is unstable. A number of people looking at the same thing often disagree about exactly what color it is.

The instability of color has a number of causes. The first of these is the way in which colors are generated by light, reflected from surfaces, and sensed by the human eye. The color of an object is no more permanent or absolute than the light under which it is seen.

Light

Only *light* generates color. Without light, no color exists. Light is *visible energy* that is emitted by a *light source*. A light source can be any one of a number of things: the sun, a luminous

panel, a neon sign, a light bulb, or a monitor screen. The eye is a sense organ that is adapted to receive light. The retina of the eye receives a stimulus—the energy signal—and transmits it to the brain, where it is identified as color.

Light sources emit this visible energy in pulses, or waves. All light travels at the same speed, but waves of light energy are emitted at different distances apart, or *frequencies*. The distance between the peaks of these energy emissions is called *wavelength*. Wavelengths of light are measured in nanometers (nm).

The human eye is able to sense wavelengths of light ranging from about 380 nm to about 720 nm. Individual wavelengths are sensed as discrete (separate) colors, or *hues*. The pure hues of light are defined by measurement. Red is the longest visible wavelength (720 nm), followed in order by orange, yellow, green, blue, indigo, and violet, the shortest visible wavelength (380 nm). ROYGBIV is an acronym for these wavelengths, which are the colors of the *visible spectrum*.

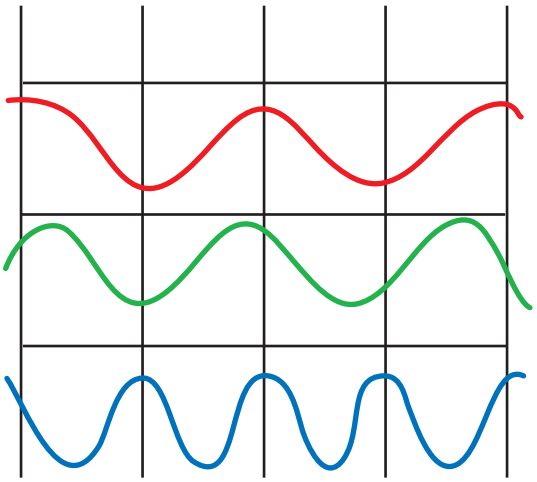


Figure 2-1.
Wavelength. Individual wavelengths of the visible spectrum are sensed as different colors.

Different light sources emit the various wavelengths (colors) at different levels of energy. One light source may give off a particular wavelength at such a low level of energy that it

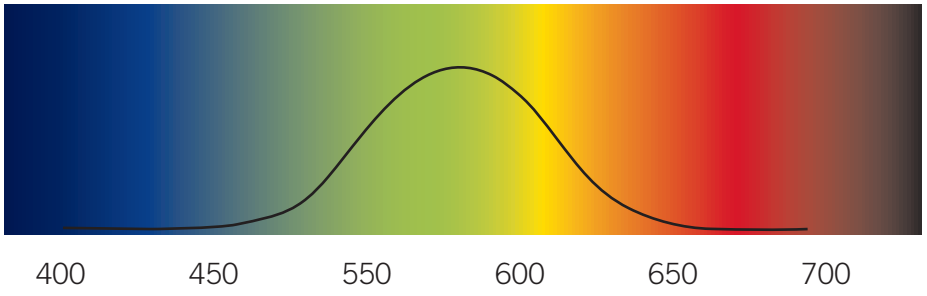


Figure 2-2 .
Visible Light. Human beings are able to sense light (visible energy) between wavelengths of about 380 nm to 720 nm.

is barely visible, while another emits it so strongly that it is seen as brilliant color. Although the color is the same, the intensity of the color experience is very different.

The human eye is most sensitive to light in the middle range of the visible spectrum and sees these colors, the yellow-green range, most easily. Yellow-green light can be sensed at a lower level of energy than other colors. If a light source emits all the wavelengths at roughly the same level of energy, the yellow-green range is sensed as brightest. There is visible light and color beyond the range of human vision. Some animals and insects can sense colors that humans cannot, and both ultraviolet and infrared, colors that lie beyond the ends of the visible spectrum, can be made visible to the human eye with special optical equipment.

Additive Color: Mixing Light

The sun is our fundamental light source. Sunlight is sensed as white, or colorless, but it is actually made up of a mixture of colors (wavelengths) that are emitted in a continuous band. The individual colors of sunlight can be seen when sunlight is passed through a prism. The glass of the prism bends, or *refracts*, each wavelength at a slightly different angle, so that each color emerges as a separate beam. Under the right set of atmospheric conditions water droplets will form natural prisms, and the component colors of sunlight can be seen as a rainbow.

Figure 2-3. *The Component Wavelengths of White Light.* The separate colors (wavelengths) that make up white light can be seen when light is passed through a prism. Each wavelength bends at a slightly different angle and emerges as a separate color. The individual colors of white light can be seen in nature when droplets of water in the atmosphere create a naturally occurring prism. *Photograph courtesy of Phyllis Rose Photography, New York & Key West.*

Other sources, like ordinary light bulbs, also produce

white light. A light source does not have to emit all of the visible wavelengths in order for white light to result. White light is produced as long as a source emits the red, green, and blue wavelengths in roughly equal proportions. *Red, green, and blue are the primary colors of light.* A source can emit additional wavelengths (colors) and the light it produces will remain white, but if any one of the three primaries is missing, the light is sensed as a color.

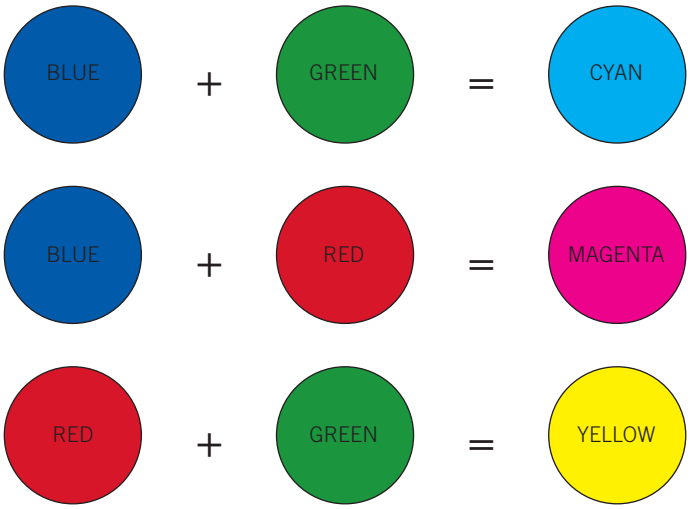


Figure 2-4. *Mixing Light.* The light primaries, red, green, and blue are mixed to form the secondaries cyan (blue-green), magenta (blue-red), and yellow.

Mixing two of the primary colors of light produces a new color. Blue and green wavelengths combined produce cyan (blue-green). Red and blue wavelengths combined produce magenta (a blue-red, or red-violet), and green and red wavelengths combined produce yellow. *Cyan, magenta, and yellow are the secondary colors of light.*

Wavelengths can be combined in unequal proportions to create additional colors. Two parts green light and one part red at equal levels of energy provide a yellow-green; two parts red and one part green produce orange. All hues, including violets and browns that are not found as wavelengths in the visible spectrum, can be produced in light by mixing the light primaries in different proportions.[1]

White or colored light seen as a result of a combination of wavelengths is called an *additive mixture* or *additive color.* "Additive" is a good description of the reality, because new colors of light are created as wavelengths are added to each other.

Lamps

Lamps are the principal man-made light sources. "Lamp" is the correct term for a light bulb. The fixture that holds the lamp is a *luminaire*. Technically, an ordinary table lamp is a "portable luminaire." There are three main families of light sources and, within these families, hundreds of different lamps. Each lamp produces light in a particular way and emits light of a specific color, quantity, and direction.

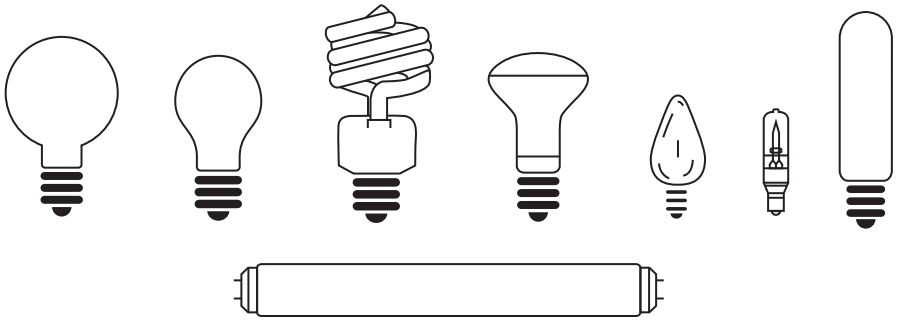


Figure 2-5. *Lamps.* Lamps are available in hundreds of different shapes and sizes.

Most lamps produce light that is white. They emit the three primary colors red, green, and blue, and most, or all, of the other colors. A catchall phrase for these lamps is *general light source*. General light sources provide *ambient light*, which is general area lighting. A lamp that is missing one or more of the primary colors gives off colored light. It is not a general light source. The lamps in neon signs are one example of a light source emitting a narrow range of wavelengths.

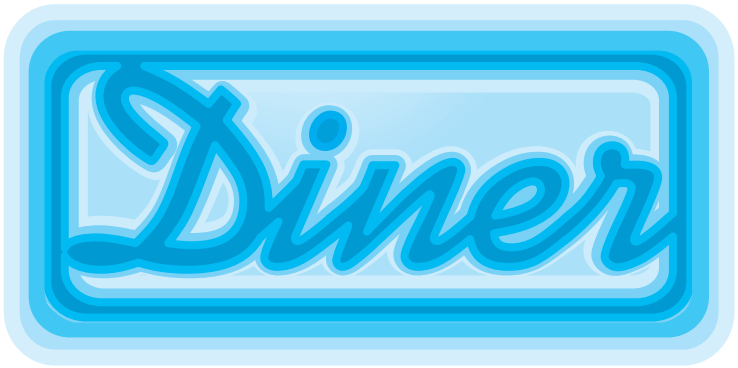


Figure 2-6. *Additive Color.* Neon signs are a familiar example of additive color, or color seen as direct light reaching the eye.

All white light sources, however, do not produce white light that is exactly the same. Each type of lamp emits wavelengths in a characteristic pattern called a *spectral distribution curve*, or *spectral reflectance curve*. The spectral distribution curve shows which wavelengths are actually present and the strength of each wavelength relative to the others for that particular type of lamp. The curve provides a visual profile of the color characteristics of any source. A lamp may emit one or two of the primaries at a higher level of energy than the others; or additional colors may be emitted strongly, or one or more may be missing entirely. If each wavelength leaving a lamp is imagined as a colored ribbon, some of the ribbons will be thicker, wider, and stronger than others.

Spectral distribution determines (and describes) the color quality of a light source: whether the light it produces is seen as neutral white ("pure" white, or red, green, and blue equally), warm white (stronger in the yellow-red range), or cool white (stronger in green or blue). Every lamp can be described technically by its spectral distribution. *Differences in spectral distribution determine how different kinds of lamps render (show, or display) the colors of objects.*

To refer to light sources as "true," or "natural," or "artificial," is deceptive. It leads to the misconception that there is "true" light and its opposite, "false" light. We think of natural light (sunlight) and artificial light (from man-made sources) as if they were different entities, but light is always visible energy. There are naturally occurring light sources and man-made ones. Every general light source produces light that can be differentiated from other sources in two ways. The first is by its spectral distribution. The second is by its *apparent* whiteness. Daylight is the standard of whiteness for man-made light sources, and, because response to sunlight is part of our genetic makeup, it also helps to determine whether light from a given source will be sensed as more or less natural.

About 40% of man-made interior lighting is used for domestic purposes.[2] The balance is used to illuminate public and commercial spaces. The most familiar man-made light sources are incandescent and fluorescent lamps. High-intensity discharge (HID) lamps, like the high pressure sodium lamps used in street lamps, or the mercury vapor lamps used as night lighting for ballfields, are also in common use.

Incandescent lamps, like the sun, produce light by burning. The light they emit is a small byproduct of heat—only about 5% of the energy used by an incandescent lamp results in light.

Like the sun, incandescent sources emit the visible wavelengths as a continuous spectrum. Candlelight, firelight, and incandescent lamplight are sensed as comforting because they emit light in the same way the sun does.

The apparent whiteness of an incandescent lamp depends on the temperature at which it burns, called its *color temperature.* Color temperature in lamps is measured in degrees Kelvin (K). A typical incandescent lamp burns at a relatively low temperature, around 2600-3000K. An incandescent lamp typically sheds a warm, yellow light because it gives off more energy in the red-orange-yellow range than in the blue or green. Lamps that burn hotter emit bluer light; very white light is hottest of all. A halogen lamp is a type of incandescent lamp with a gas inside the glass envelope that causes it to burn at a high temperature; as a result the light produced is a bluer white. The heat-light-color relationship is recognized in colloquial language; something that is "white-hot" is dangerously hotter than something "red-hot."

The color temperature of a lamp is used as a measure of whiteness for the color of *light* produced by lamps. *It does not help to predict how a light source will render the colors of objects*. Different lamps reach their color temperature in different ways, and spectral distribution information is not always available. Experienced designers use mockups in field conditions to make sure that the lamps they specify will deliver the right quantity and quality of light for each situation.

Fluorescent lamps produce light in a completely different way. The interior of the glass bulb is coated with phosphors, substances that emit light when they are bombarded with electrical energy. The color of a fluorescent lamp depends on the particular makeup of its phosphor coating. Fluorescent lamps do not burn, so they do not have an actual color temperature, but they are assigned an "apparent color temperature" to indicate their degree of whiteness.

Fluorescent lamps do not produce the continuous spectrum that is characteristic of incandescent light. Instead, they give off light as a series of separate bands of energy. Although fluorescent lamps are available with many different spectral distributions, a typical low-cost fluorescent lamp emits all wavelengths at similar levels of energy. Because of the eye's sensitivity to yellow-green, the white light emitted by an ordinary fluorescent lamp has as yellow-greenish cast.

Whether at home, at work, or at play, the color of light around them affects how people feel about their environment. Light from a source with a continuous spectrum—from a source that imitates sunlight—is sensed as the most comfortable, the most welcoming, and the most natural. Many lamps are marketed as "full-spectrum," as if this were some magical solution to a host of light and color problems. A lamp described by its manufacturer in this way tells the purchaser only that the lamp emits all wavelengths. It does not describe the strength of those wavelengths relative to each other, or tell whether or not the wavelengths are continuous, or describe the relative warm or coolness of the white light that is produced.

Present-day efforts to find environmentally responsible light sources have created an explosion of research into new sources of light. The current focus is on LED lamps, which produce light at low operating cost by combining the output of red, green, and blue light-emitting diodes. The light that results is white, strong, and energy efficient: excellent for limited uses like car headlamps, but problematic as a light source for interior environments because it contains only the three primary colors and does not have a continuous spectrum. LED lamps have been developed that include phosphors to increase their spectrum, but these lamps are not yet perfected, and even at their best will not produce a continuous spectrum. Research is ongoing in the effort to improve the color rendering of LED sources, but at the time of this writing LED lamplight, despite its environmental virtues, is less than ideal in its ability to render the colors of the real world.

Lighting Level

Lighting level refers to the *quantity* of available light, regardless of its color makeup. Lighting level describes the *total* amount of light coming from the source. The amount of light emitted by a lamp is unrelated to its spectral distribution. The same lamp type may give off more or less light, but its spectral distribution—the pattern of relative energy emitted at the different wavelengths—is identical for that lamp type no matter what quantity of light it gives off.

Too little available light makes it hard to see colors. Excessive and uncontrolled light falling on a surface can also impair color perception. *Glare* is an extreme, physically fatiguing level of general light. Glare obliterates color perception and can be temporarily blinding. Glare is corrected by adjusting the light source: by reducing the quantity of light, changing the direction from which it comes, or both.

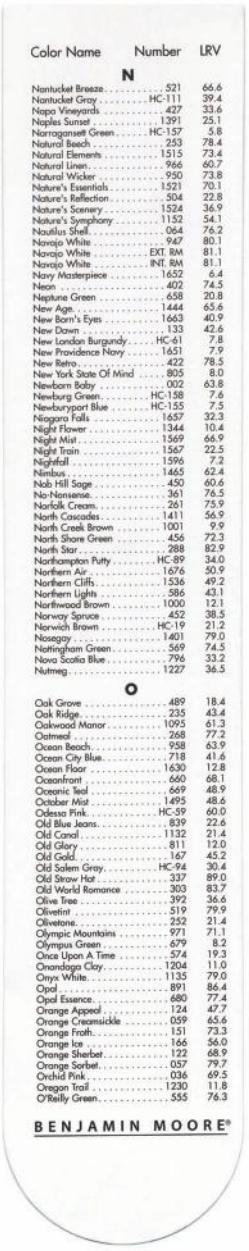
Reflectance, or *luminance*, is a measure of the amount of light falling on a surface that is reflected *back*. It is a measure of the total amount of light reflected, not the individual wavelengths (colors). Reflectance is so important to some products, like interior and exterior paints, that the percentage of light reflected back from each color, called its LRV (light-reflecting value), is part of the basic information the manufacturer provides.

The ability to perform most tasks depends on the ability to see the difference between dark and light, not on the ability to detect color. Black-and-white photographs and written text are perfectly understandable. The ability to see dark/light contrast is unaffected by lamp color. Only lighting level affects the ability to see dark and light differences. And although many people report different levels of comfort under a variety of light sources, human performance under general light sources has not been shown to be affected by lamp color, only by lighting levels.[3]

Vision

Vision is the sense that detects the environment and objects in it through the eyes. It is the principal way in which physical realities like objects and the environment are discerned and the only way in which color is perceived.

Color vision is experienced in two different ways: either as light directly from a light source, or as light reflected from an object. In the *illuminant mode of vision,* colors are experienced as direct light reaching the eye, like the colors of a monitor screen or a neon sign. In the *object mode of vision* colors are seen indirectly, as reflected light. The tangible things of the real world—objects and the environment—are seen in the object mode of vision.



Color Name	Number	LRV
N		
Nantucket Breeze	521	66.6
Nantucket Gray	HC-111	39.4
Napa Vineyards	427	33.6
Naples Sunset	1391	25.1
Narragansett Green	HC-157	5.8
Natural Beach	253	78.4
Natural Elements	1515	73.4
Natural Linen	966	60.7
Natural Wicker	950	73.8
Nature's Essentials	1521	70.1
Nature's Reflection	504	22.8
Nature's Scenery	1524	36.9
Nature's Symphony	1152	54.1
Nautilus Shell	064	76.2
Navajo White	947	80.1
Navajo White	EXT. RM	81.1
Navajo White	INT. RM	81.1
Navy Masterpiece	1652	6.4
Neon	402	74.5
Neptune Green	658	20.8
New Age	1444	65.6
New Born's Eyes	1663	40.9
New Dawn	133	42.6
New London Burgundy	HC-61	7.8
New Providence Navy	1651	7.9
New Retro	422	78.5
New York State Of Mind	805	8.0
Newborn Baby	002	63.8
Newburg Green	HC-158	7.6
Newburyport Blue	HC-155	7.5
Niagara Falls	1657	32.3
Night Flower	1344	10.4
Night Mist	1569	66.9
Night Train	1567	22.5
Nightfall	1596	7.2
Nimbus	1465	62.4
Nob Hill Sage	450	60.6
No-Nonsense	361	76.5
Norfolk Cream	261	75.9
North Cascades	1411	56.9
North Creek Brown	1001	9.9
North Shore Green	456	72.3
North Star	288	82.9
Northampton Putty	HC-89	34.0
Northern Air	1676	50.9
Northern Cliffs	1536	49.2
Northern Lights	586	43.1
Northwood Brown	1000	12.1
Norway Spruce	452	38.5
Norwich Brown	HC-19	21.2
Nosegay	1401	79.0
Nottingham Green	569	74.5
Nova Scotia Blue	796	33.2
Nutmeg	1227	36.5
O		
Oak Grove	489	18.4
Oak Ridge	235	43.4
Oakwood Manor	1095	61.3
Oatmeal	268	77.2
Ocean Beach	958	63.9
Ocean City Blue	718	41.6
Ocean Floor	1630	12.8
Oceanfront	660	68.1
Oceanic Teal	669	48.9
October Mist	1495	48.6
Odessa Pink	HC-59	60.0
Old Blue Jeans	839	22.6
Old Canal	1132	21.4
Old Glory	811	12.0
Old Gold	167	45.2
Old Salem Gray	HC-94	30.4
Old Straw Hat	337	89.0
Old World Romance	303	83.7
Olive Tree	392	36.6
Olivetint	519	79.9
Olivetone	252	21.4
Olympic Mountains	971	71.1
Olympus Green	679	8.2
Once Upon A Time	574	19.3
Onandaga Clay	1204	11.0
Onyx White	1135	79.0
Opal	891	86.4
Opal Essence	680	77.4
Orange Appeal	124	47.7
Orange Creamsicle	059	65.6
Orange Froth	151	73.3
Orange Ice	166	56.0
Orange Sherbet	122	68.9
Orange Sorbet	057	79.7
Orchid Pink	036	69.5
Oregon Trail	1230	11.8
O'Reilly Green	555	76.3

BENJAMIN MOORE®

Figure 2-7. Benjamin Moore provides an LRV for each of its paint colors.

Benjamin Moore LRVs are valid only for Benjamin Moore colors tinted by a Benjamin Moore retailer.

The Illuminant Mode of Vision

The *illuminant mode of vision* has two variables in the perception of color: the characteristics of the light source and those of the viewer. Wavelengths of light are measurable and can be established definitively, but each viewer's perception of color is to some extent individual. Viewers with normal color vision will vary slightly in their ability to sense colors, but more significantly, they interpret what they have sensed in different and personal ways. Each person's perception of color is to some extent a singular event.

Television and computer monitor displays, traffic lights, and neon signs are examples of the illuminant mode of vision. Colors seen in the illuminant mode of vision are additive colors. Additive colors are more stable than the colors of objects. Colors of light do not change when a source (like a monitor) is moved from a setting with one type of ambient lighting to a different one, or from darker to lighter surroundings. A measurable wavelength or mixture of wavelengths is constant unless there is a change in the pattern, or the strength, of wavelength emission from its source.

The Object Mode of Vision

In the *object mode of vision,* color is seen as light reflected from a surface. Physical objects and the environment are seen as reflected light. Color perception in the object mode of vision has *three* variables: the characteristics of the light source, the individual viewer's visual acuity for color and interpretation of it, and the light-modifying characteristics of the object. The three are so interdependent that if any one changes, the apparent color of the object will also change.

Light leaving a light source is the *incident beam*. When the incident beam reaches a surface, some or all of the light falling on the surface scatters, or reflects—in other words, bounces off it. The *reflected beam* is light that leaves a surface and reaches the eye.

Colors seen as reflected light are understood differently from colors seen as direct light. A classic misunderstanding of that difference took place in the 1970s when many fire engines were painted yellow-green instead of the traditional red. The apparent reasoning for this was that the sensitivity of the eye to yellow-green light made yellow-green the most visible, and

therefore the safest, color. Yellow-green fire engines disappeared very quickly and the traditional red returned, an outcome not of nostalgia but of practical reality. Fire engines are objects, not wavelengths of light. Very different principles of high visibility apply. Red fire engines are more visible than yellow-green ones because red is more likely than yellow-green to contrast sharply with the natural or man-made environment.

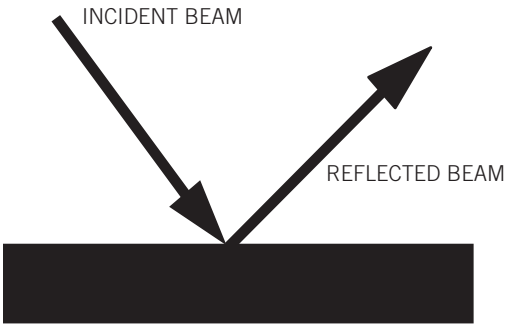


Figure 2-8. *Scattering, or Reflection.* Light reaching a surface is an incident beam. The reflected beam is light that leaves a surface and reaches the eye.

Subtractive Color: Colorants

Materials are the substances of the real world. They are the "stuff" of physical objects, the things that can be seen and touched. *Materials modify light*. Light reaching a material is modified in one of three ways:

Transmission: the material allows light to pass through, as through glass.

Absorption: the material soaks up light reaching it like a sponge, and the light is lost as visible. It can no longer be seen.

Reflection, or *scattering*: light reaching a material bounces off it, changing direction.

Colorants are special materials that modify light by *absorbing* some wavelengths and *reflecting* others. There are natural colorants, like chlorophyll, the green of plants, and hemoglobin, the red of blood. Manufactured colorants for products and print are derived from a wide variety of natural and synthetic substances. A colorant can be integrated into the substance of a material, like a color-through plastic, or applied to a surface as a coating. Colorants are also called color agents, dyes, pigments, and dyestuffs, depending on their makeup or end use.

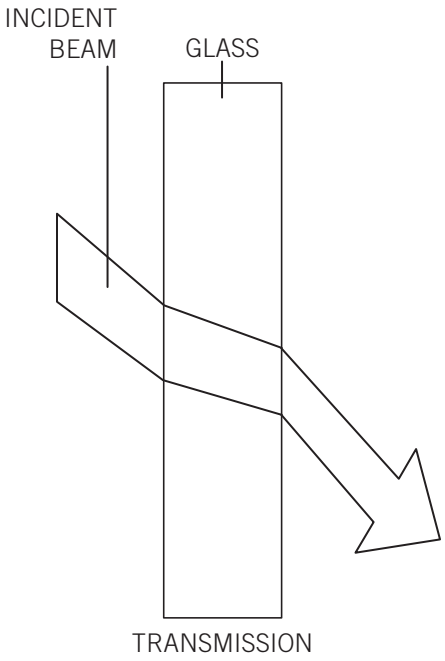


Figure 2-9.
Transmission. Window glass transmits light, allowing it to pass through with no perceptible change. Thin glass bends light so slightly that it retains its whiteness.

A white colorant reflects, or scatters, all wavelengths of light. A white object placed under a general light source reflects all of the wavelengths reaching it, and all (or most) of that reflected light reaches the eye. A black colorant *absorbs* all of the wavelengths of light. An black object placed under a general light source absorbs all wavelengths of light reaching it, and none (or very little) is reflected back to the eye.

Other colorants modify light *selectively.* A red colorant absorbs all of the wavelengths except the red. An object with a red colorant that is placed under a general light source absorbs all of the wavelengths of light reaching it except the red, which it reflects. Only the red wavelength reaches the eye, and the object is understood to be red.

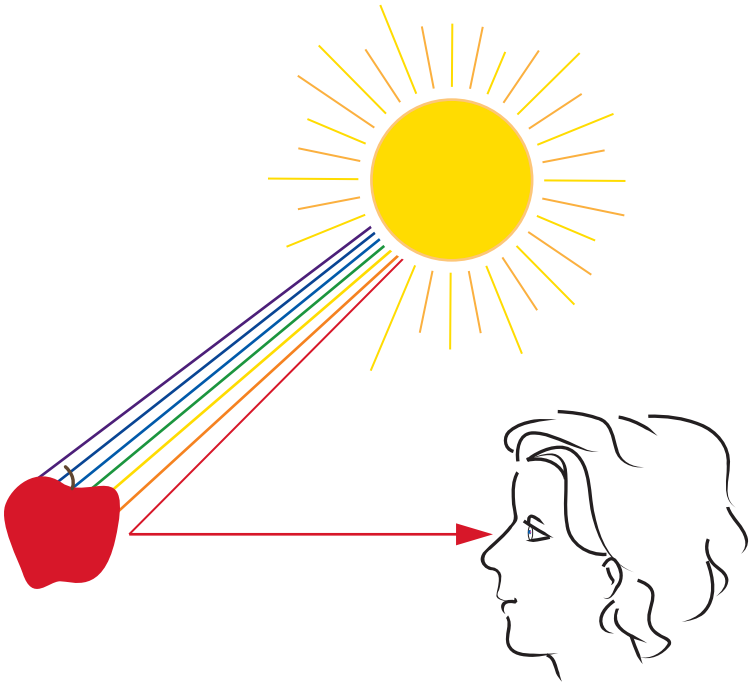


Figure 2-10.
Subtractive Color. The colorant of a red apple absorbs all the wavelengths except red. Only the reflected (red) wavelength reaches the eye, and the apple is seen as red.

In order for an object to be seen as a color, the wavelengths that its colorant reflects must be present in the light source. If a red object is placed under a source that lacks the red wavelength, all wavelengths reaching the object will absorbed. There is no red wavelength present to be reflected back to the eye. A red dress seen under green light is a black dress. In a parking lot illuminated by the light of yellow sodium lamps, red cars, green cars, and blue cars are indistinguishable from each other. Only yellow cars can be located by their color.

Colorants don't absorb and reflect individual wavelengths perfectly. They may absorb part of a wavelength and reflect part of it, or reflect more than one wavelength. So many possibilities exist that the range of visible colors is nearly infinite. Colors seen as the result of the absorption of light are *subtractive mixtures.* Colorants "subtract," or absorb, some wavelengths. The remaining wavelengths are reflected and reach the eye as color.

Lamps are compared by the power of emission at each wavelength. Colorants are compared by their ability to reflect each wavelength. An accurate comparison between color samples can be made only when the samples are compared under the same light source. The light source frequently used for critical color comparisons in a laboratory setting is a Macbeth lamp, which has spectral distribution similar to sunlight. One sample is designated as a *standard* and others are measured against it under the Macbeth lamp. The ability to measure color under controlled laboratory conditions has little practical application for the artist or designer, but it is important for quality control in manufacturing.

It is a common misconception that the "true color" of any object can be captured under the correct lamp. Among the most familiar light sources are A-lamps, or ordinary incandescent light bulbs, and cool fluorescent tubes. Each is different in spectral reflectance from the sun, from the Macbeth lamp, and from each other. A Macbeth lamp is a laboratory tool, not a readily available general light source. If true colors of materials existed, and were visible only under Macbeth lamps or in sunlight, they would be seen only in lighting laboratories and in the outdoors. Science can designate true colors of light as measurable wavelengths, but the concept that there are "true colors" of material objects is misleading. The color of an object is only as true as each viewer chooses to think of it.

Lamps and Color Rendition

General light sources vary widely in the way that they render the colors of objects. The same object will change in color under a variety of light sources because of differences in the spectral reflectance of each source.

Making color choices under one set of lighting conditions for use under a different set of lighting conditions can be a disaster. A woman wearing astonishingly bright red makeup has probably applied her cosmetics under standard fluorescent lighting. These lamps emit a relatively low level of red wavelength, so items with red colorants receive little red light to reflect. One way to compensate for this is to increase the amount of colorant—rouge or lipstick—to get a visibly red result under the fluorescent source. In daylight or incandescent light, the red is overwhelming.

Fluorescent lamps are the current standard for use in commercial applications. They provide high light output and low heat at reasonable operating cost, but vendors are aware of the weak spot in the color rendition of standard fluorescents. Grocers choose lamps with a stronger red component for use over meat counters to ensure that products appear fresh and healthy. High-end cosmetics counters provide separate light sources for sampling colors.

In residential interior design, considerations of color rendition usually take precedence over operating cost. Incandescent lamps are nearly always chosen over fluorescents as a general source. Lamps can be selected to enhance color schemes. Some manufacturers recommend specific lamps for different situations: one type for warm color schemes, another for cool ones. Conversely, lamps can be used to make color adjustments or corrections in completed interiors. Colors selected in daylight may be too warm or too cool for night lighting. Red-orange schemes can be muted by using a lamp with a weaker warm range; color schemes that are too blue or green can be muted by using lamps with stronger output in the red-orange-yellow range.

The near-universal specification of fluorescent lamping for commercial applications, including such crucial areas as studio work spaces, is often made without consideration of color rendering qualities. The poor color rendering that results from poor lamp choice

remains problematic for designers, manufacturers, retailers, and consumers because for most products, from foods and flowers to cosmetics and carpets, good color presentation is critical to sales.

Metamerism and Matching

Two objects that appear to match under one light source but not under another exhibit *metamerism*. The objects are called a *metameric pair*.

Materials differ in their ability to absorb colorants or accept them as coatings. The substance used as a blue colorant for wool may have little or no chemical relationship to one used to produce a similar blue in cotton. Still other substances are blue colorants for plastics, glass, paint, or ceramic dinnerware. Each colorant reacts to light in a very specific way. If the colorants of two things are different, they cannot be made to match under all light conditions. The underlying material, or substrate, can also at times influence color rendition.

Two red-dyed cottons may appear to match exactly under incandescent lighting. If their dyes are different, one may actually be more red-reflective than the other. When the two are placed under a fluorescent lamp (that is weaker in the red wavelength), the cotton with the more red-reflective dye appears brighter and "redder" than the other.

So-called "matching" wallpapers and fabrics, or silk shirts and wool skirts, or china sinks and enameled cast iron bathtubs, can never truly match because both their underlying materials and colorants are different. What is possible and practical is to reach an *acceptable match*, one that is pleasing to the eye. When different materials must reach an acceptable match, it is essential to compare them under lighting conditions that duplicate their end placement.

Some manufacturers and vendors control acceptable match in related goods by maintaining color laboratories. Sears is known for its care in ensuring that home products are color-compatible. A refrigerator purchased in Sears "Bisque" is likely to be an acceptable match to Sears "Bisque" dish towels or pots, and both are likely to be compatible with other nationally distributed products, like Wilsonart laminates, with the same color name.

Maintaining a laboratory for color and light problems is not realistic or necessary for graphic, textile, fashion, or interior designers, architects, small-scale manufacturers, or product designers. General Electric suggests that any pair of objects reaching acceptable match under both fluorescent and incandescent lamps will be acceptable under nearly all conditions.[4] This practical advice covers almost all color-compatibility situations, as long as the real limitations of color matching between two different materials are understood.

A sample submitted for color matching is a *standard.* There are times when the match to a standard must be stable under all light conditions. In order for this to happen, the new goods must be made of the same substrate (base material) and the same colorant used as in the original. A match that is perfect under any light conditions is possible only when the original standard and the new product are identical in all ways.

Only manufacturers who control the complete production and color process can offer perfect matches to a standard. But throughout the process and at its conclusion, the match is judged not only by laboratory testing but by the dyer, the designer, the manufacturer or the client: *ultimately*, *by human eyes alone.*

Modifying Light: Surface

Surface is the outermost layer of a thing, its "skin." Different surfaces—rough, smooth, or in between—have an impact on the way that colors are perceived. The word *value* refers to the relative lightness or darkness of a hue. *Only the perception of value is affected by surface texture.* Smooth surfaces reflect light very directly, so that a good deal of light reaches the eyes. Rough surfaces scatter light in many directions, and less light reaches the eyes. Surface texture has no effect on hue. A rough-skinned object and a smooth one that have the same colorant will be identical in hue under any kind of illumination, but the first will appear darker, the second lighter.

Light striking a surface reflects, or bounces off it. The position of a surface relative to the light source determines the angle at which the incident beam reaches it. The light that bounces off the surface, the reflected beam, does so at exactly the same angle as it reached it. This is the law of reflection: the angle of the incident beam determines the direction of the reflected beam.

The smoother the surface, the greater the amount of light that is reflected back directly. A *specular* surface is glossy, or mirror-like. Light leaving a specular surface is reflected so immediately, and so directionally, that most (or all) of it is seen as white light. When a specular surface is viewed from the *same* angle as the incident beam, only white light is seen: a specular reflection. When a specular surface is viewed from an angle that is *not* the same as the angle of the incident beam, some light reaching the underlying colorant can be seen. The surface still dazzles with white light, but color is seen as well—although darkened, because so much light has been deflected before reaching the colorant. The extreme reflectance of specular surfaces makes it difficult to interpret their color no matter what the viewing angle. Sequined garments are a perfect example: each sequin flashes with white light until the wearer moves and changes the viewing angle, when its color becomes suddenly, briefly, visible.

A *matte* surface is a smooth surface that is very slightly, even microscopically, roughened. Its roughness is too fine to be seen with the naked eye. A matte surface diffuses (spreads) light very evenly in many directions, so that light leaving the surface is constant from any angle of view. Colors on a matte surface, like an uncoated paper, have a flatness and uniformity under nearly all lighting conditions.

Textured surfaces are dynamic and lively. When incident light reaches a surface that has peaks and valleys of different orientations, the beams of light follow the law of reflection. Light scatters in random directions, dappling the surface with patches of light and shadow.

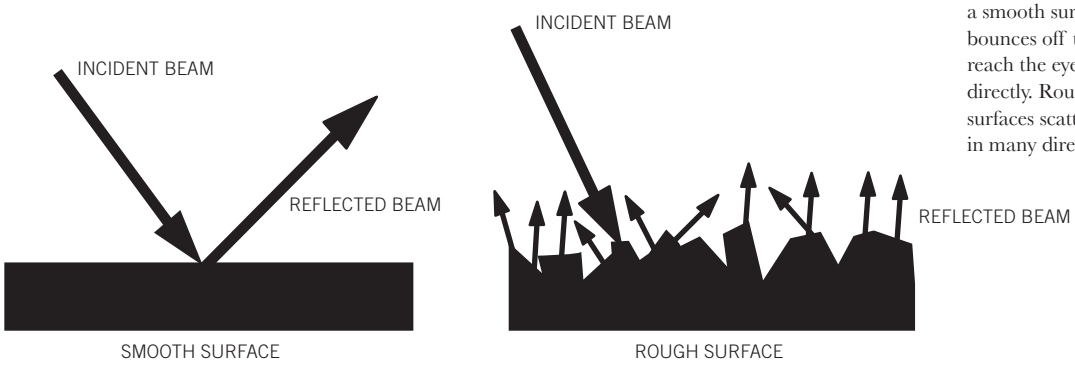


Figure 2-11. *Rough and Smooth.* Light reaching a smooth surface bounces off to reach the eye very directly. Rougher surfaces scatter light in many directions.

Texture is most apparent under point light sources, like sunlight or incandescent lamps. Light from a point source originates from a single location, or point, and the beams of light emitted are parallel. Fluorescent lamps are linear light sources. Linear light sources emit a broad-spread light in that is essentially non-directional. Light from a linear source does not reach the surface at an angle in the same way as a point light source. Reflected beams are less directional, so shadows are reduced or even absent. Even heavily textured surfaces tend to appear flat and uniform under fluorescent (or other linear) lighting. LED lamps are currently offered as both linear and point sources, but LED lamps are an emerging technology and their rendition of color and surface is difficult to evaluate at this time.

Figure 2-12.
Texture. Pattern can be created with yarn of only one color when it is woven as looped in some areas and as cut ends in others. *Image of carpet "Ardennes" courtesy of Edward Fields Carpet Makers.*

The sharper the angle of incident light, the more directional the reflected beam will be. *Raking light* describes light from a source that is positioned at an acute angle relative to a surface. Specular surfaces appear more glossy, and textured surfaces dramatically rougher, under raking light.

No matter what the ambient light, varying the textures of a surface allows designers to create an effect of two or more colors (or more accurately, lighter and darker variants of a single hue) using only one material. A piece of yarn, seen on its long side, is relatively smooth. Cut ends of the same yarn (a pile, or nap) reflect the identical wavelength but scatter light more widely and appear darker. Textiles made of a single yarn can be woven into intricate patterns by alternating areas of flat weave with areas of cut pile, or by contrasting them against areas of textured weave. Embossed metals and "frosted" glassware are other examples of surfaces that have been manipulated to create dark and light patterning within a single material and color.

Transparent, Opaque, and Translucent

A small amount of light is lost each time that light travels from a source to a surface and, when light reaches a surface, a very small amount reflects back immediately. The sum of this light loss can be so slight that as a practical matter it is unimportant. The light that remains is reflected, absorbed, transmitted, or a combination of these. If all of the light reaching an object is either reflected or absorbed, the object is *opaque*. If all (or nearly all) of the light reaching an object or material is transmitted, that object is *transparent*. Window glass is an example of transparent material.

When some of the light reaching an object or material is transmitted and some is reflected, the object is *translucent*. A translucent material can be white or a color, depending on its selective transmission and reflection of the various wavelengths. Translucent materials may allow a great deal of light to pass through (and be very translucent) or transmit very little light (and be barely translucent).

The terms "translucent" and "transparent" often are used interchangeably, but they do not mean the same thing. A truly transparent material is like window glass: for all practical purposes, it is invisible. A translucent material is detectably present, no matter how sheer it may be.

Iridescence

Iridescence is a attribute of surfaces on which the hue changes as the observer's angle of view changes. The changes from blue to green that are seen in a butterfly's wing as it flies, or the flashes of red, purple, and green in the black feathers of a grackle, or the brilliant and changing colors of soap bubbles and oil films are iridescence.

Iridescence is an optical phenomenon that occurs with reflected light. The color is produced by the structure of a surface that amplifies some wavelengths of light and suppresses others, depending on the angle of the light reaching it. The amplification of light makes iridescent color extremely vivid—the color that reaches the eyes may be reflected, but in the absence of a modifying colorant it is sensed as pure light. Because no colorant is involved—nothing that absorbs some wavelengths of light and reflects others—it is sometimes called structural color.

Iridescent textiles are brilliantly shimmery, seeming to be one color at one angle of view and a second color as the fabric moves. Iridescence in textiles is produced in a variety of ways. There are silk yarns with a molecular structure that creates iridescence, as well as synthetic yarns with similar properties. Most iridescent textiles, however, are made using special yarns and techniques of weaving. When the warp and filling are made from differently colored and light-reflective yarns; each color appears, vanishes, and reappears as the viewing angle shifts.

There are paints and inks with light-reflecting properties that create convincing iridescent effects on a page. As the observer's position changes, the color changes. An impression of iridescence is difficult to create on a screen, because light leaving a screen reaches the eye directly, no matter what the viewer's position or movements.

Luminosity

Luminosity is a word that appears often in color study. Its real meaning is the attribute of emitting light without heat. A luminous object is light-reflective, but it does not emit heat. The word luminous is used often to describe very light-reflecting colors and media with a great deal of light reflectance, like watercolor, dyes, or markers.

Indirect Light, Indirect Color

Indirect light occurs when light from a light source reaches a broad, light reflective plane that re-reflects it onto a second surface or object. In order for this to happen, the light source, the reflective surface, and the target surface or object must be positioned at similar angles to one another. Moonlight is a familiar form of indirect light. The moon is luminous: it reflects light but does not emit its own energy. Its surface reflects the light of the sun to the earth.

Each time light travels, some of it is lost through scattering. Moonlight is weaker than sunlight because much of the sun's light has been scattered and lost, first on its way from the sun to the moon, then again from the moon to the earth. Indirect lighting works in the same way as moonlight does. Light reaching a white surface is redirected to a target area. The indirectly lit area appears darker than it would under direct light, but no change in its apparent hue takes place.

Indirect *color* is a form of indirect light. Indirect color occurs when general light reaches a highly reflective color on a broad plane. Some of the general light—and a good deal of the strong color—will reflect onto any surface that is positioned to receive it.

Imagine sunlight reaching a wall that is covered with a highly reflective green paint. The green surface absorbs all wavelengths except the green, which is scattered onto a nearby chicken. If the chicken has white feathers, its surface reflects the green wavelength and the chicken appear to be green. If the chicken has red feathers, the green reaching it is a absorbed and the chicken appears to be a darker and duller red.

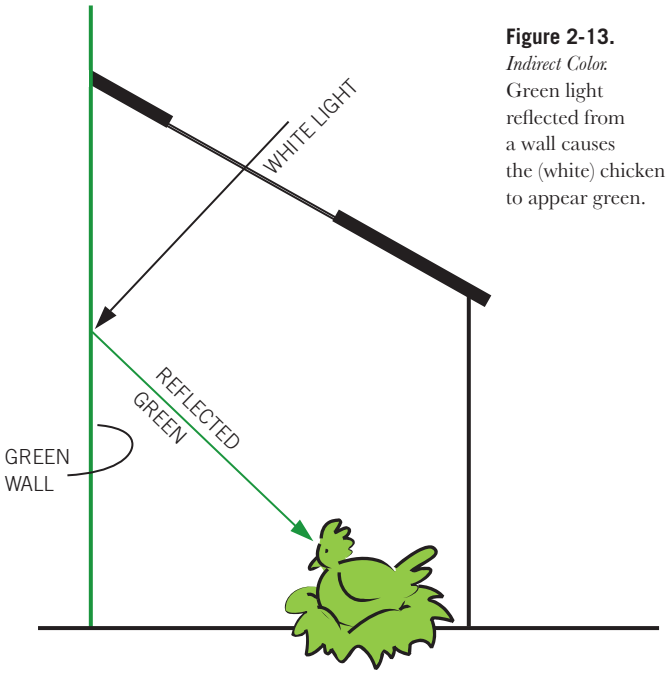


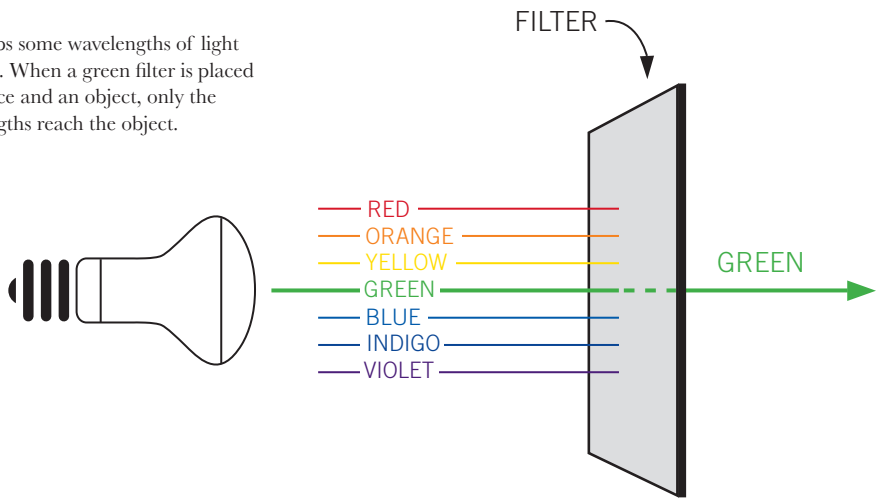
Figure 2-13.
Indirect Color.
Green light reflected from a wall causes the (white) chicken to appear green.

One way to describe the phenomenon of color reflected from one surface to another is *plane reflection.* The design applications most vulnerable to this are architecture and interior design, where planes of color on walls, floors, and ceilings interact with directional light sources to create potential conditions of light and color reflection. Although the strength of indirect color is diluted by scattered general light from the original source, plane reflection can cause substantial—and unexpected—distortions in architectural and interior design color programs.

Modifying Light: Filters

Filters are materials that transmit (pass through) some wavelengths of light and absorb others. A red filter placed between a light source and an object allows only the red wavelengths to pass through. Other wavelengths are absorbed. An object with a red colorant, seen through the filter, appears red. An object with a white colorant also appears red because a white

Figure 2-14.
Filters. A filter absorbs some wavelengths of light and transmits others. When a green filter is placed between a light source and an object, only the transmitted wavelengths reach the object.



surface reflects any wavelength reaching it. If the object reflects a color other than white or red, it will appear black (or near-black). The red filter transmits only the red wavelength. No other wavelength reaches the object, so none can be reflected back.

Filters are powerful modifiers of light, so they must be used with real understanding of their effects. *The New York Times* once published the story of a restaurateur who wanted to create a romantic atmosphere by bathing his establishment in warm, rose-red light. Instead of using lamps with a strong red wavelength (which would have provided a rosy white light, but allowed all colors to be seen), he installed red theatrical gels (filters), which blocked all the wavelengths except red. The gels were extremely effective: the vegetables and salads that arrived at the tables were black.

Endnotes

1 http://www.bway.net/-jscruggs/add..html
2 Fedstats/USA.gov/Department of Energy
3 General Electric Lighting Application Bulletin #205-42311.
4 Ibid.

Chapter 2 Highlights

- Only light generates color. Light is visible energy that is emitted by a light source. Light sources emit this energy in waves at different distances apart, or frequencies. Each wave is perceived as a separate color. The human eye is a sense organ that is adapted to receive wavelengths of light within a narrow range called the visible spectrum.

- White light is produced when a source emits red, green, and blue wavelengths in roughly equal proportions. Red, green, and blue are the primary colors of light. Cyan, magenta, and yellow are the secondary colors of light. Blue and green wavelengths combined produce cyan. Red and blue wavelengths combined produce magenta and green and red wavelengths combined produce yellow. White or colored light seen as a result of a combination of wavelengths is called an additive mixture or additive color.

- Sunlight is made up of all visible wavelengths emitted in a continuous band. Light emitted in a continuous band is sensed as the most natural. Lamps are the principal man-made light sources. Each type of lamp emits wavelengths in a pattern, or spectral reflectance (or distribution) curve, that shows which wavelengths are present and the strength of each relative to the others for that type of lamp. Differences in spectral distribution determine how different kinds of lamps render the colors of objects. A lamp marketed as "full-spectrum," does not tell the purchaser the strength of the wavelengths relative to each other or whether they are continuous.

- Incandescent lamps produce light by burning. The whiteness of an incandescent lamp depends on the temperature at which it burns, or color temperature. Fluorescent lamps contain phosphors that emit light when they are bombarded with electrical energy. The color of a fluorescent lamp depends on the makeup of its phosphor coating. Fluorescent lamps do not produce light as a continuous band. Present-day efforts to find environmentally responsible light sources are focused on LED lamps, which produce white light by combining the output of red, green, and blue light-emitting diodes.

- Lighting level refers to the quantity of available light, regardless of its color makeup. Reflectance, or luminance, is a measure of the amount of light falling on a surface that is reflected

back. The ability to perform most tasks depends on the ability to see the difference between dark and light, not on the ability to detect color.

- Colors are seen in one of two ways. The illuminant mode of vision has two variables: a light source and a viewer. Colors seen in the illuminant mode of vision are seen as direct light and are called additive colors. The object mode of vision has three variables: the light source, the viewer, and the light-modifying characteristics of an object. Colors in the object mode of vision are seen as light reflected from a surface material and are called subtractive colors.
- Materials are substances that modify light by transmission, by absorption, or by reflection. Colorants are materials that modify light selectively by absorbing some wavelengths and reflecting others. Colors seen as the result of the absorption and reflection of light are subtractive colors.
- If all light reaching a surface is transmitted, the object is transparent. If all light reaching a surface is reflected or absorbed, the object is opaque. If some of the light reaching a surface is transmitted and some is reflected, the object is translucent.
- In order for an object to be seen as a color, the wavelengths that its colorant reflects must be present in the light source. A true comparison between color samples can be made only when they are compared under the same light source. Two objects that appear to match under one light source but do not match under another are called a metameric pair.
- Light bounces off a surface at the same angle as it reaches it. A matte surface spreads light evenly, so that light leaving it is constant from any angle of view. Light reaching a surface that has peaks and valleys of different orientations scatters light in random directions, resulting in patterns of light and shadow, or texture.
- Iridescence is a attribute of surfaces on which the hue changes as the observer's angle of view changes. Indirect color occurs when general light reaches a highly reflective color on a broad plane. Both general and colored light are then re-reflected onto any surface positioned at the correct angle to receive them.

3

THE HUMAN ELEMENT

The Sensation of Color / Threshold / Intervals / The Perception of Color / Physiology: Responding to Light / Healing and Color / Synaesthesia / Psychology: Responding to Light / Naming Colors / Color as Language: From Name to Meaning / Impressional Color / Color as Words Alone

Color responses are more tied to man's emotions than to his intellect. In general, people do not respond to color with their minds.

—Deborah Sharpe

Of all the senses that connect us to the world—vision, hearing, taste, smell, and touch—vision is the most important. More than 80% of our sensory experiences are visual. We are drawn to light, and to color.

The instrument used in solving color problems in the design studio is the normal, unaided, human eye. For artists and designers, dyers and house painters, printers and carpet sellers, even when aided by tools of color technology, final decisions about color are made by human eyes alone.

The Sensation of Color

The experience of color begins with a *sensation.* A sensation is an actual, physical event. It is the body's response to a *stimulus*, something that is encountered from the outside world. Light, which is visible energy, is the stimulus for the sensation of sight. A stimulus is measurable: the color and quantity of light emitted by a light source can be measured.

Sensations are also measurable. An individual's ability to detect light is measured as *visual acuity*, or sharpness of vision. Visual acuity is the ability to sense patterns of light and dark

and to resolve detail. It is a measure of the weakest light stimulus that an individual can detect. The ability to see differences between dark and light is not the same as visual acuity for color. Someone who can discriminate very small differences between a dark gray and a slightly lighter one may not be able to detect a difference between two reds, one of which is slightly more red-violet than the other.

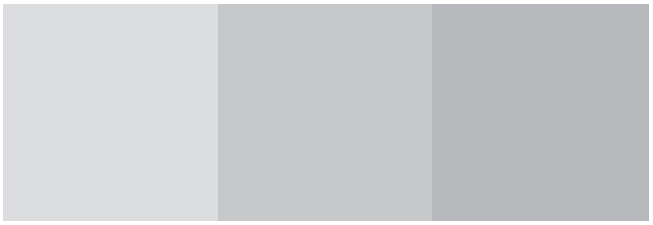
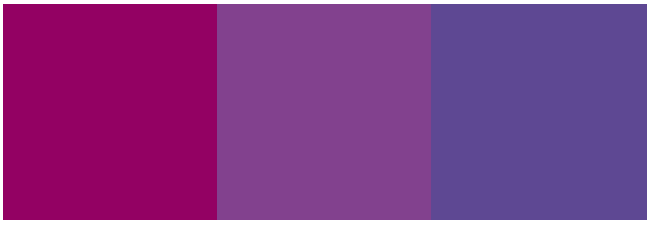


Figure 3-1.
Visual Acuity. Small differences between dark and light may be difficult for some people to see. Others with greater visual acuity can see a difference between very similar grays.

Visual acuity for *color* is the ability to detect differences between wavelengths (colors) of light. The strength and wavelength of each color of light can be separately measured using scientific instruments, but human beings do not see the spectrum as a progression of individual colors. The spectrum of light is sensed as a flowing and unbroken continuum, with each color blending into the next like a rainbow.

Color vision enables a viewer to discriminate small differences between hues in this continuous band, like the difference between a red and a red-orange, or a blue and a blue-violet. The human being is said to be able to distinguish about 150 individual colors (hues) of light, although this number is not scientifically established fact. It makes no difference whether the colors are seen as light directly from a light source or as light reflected from a surface. This number does not include the darker, lighter, or duller variations of each color. The multiples of the 150 colors with their variations mean that a person with normal color vision can distinguish millions of different colors.

Figure 3-2.
Visual Acuity for Color. Visual acuity for color is the ability to detect differences between similar hues.



Threshold

The *threshold* of vision is the point at which an individual can no longer detect a difference between two close samples. The threshold of *color* vision is the point at which a difference between two similar *hues* can no longer be discriminated. Individuals with normal color vision vary in the threshold of color acuity just as much as they do in their ability to see detail. Each person's vision is modified by factors like individual physiology, health, and age. Infants are believed to be able to detect differences between dark and light before they can see hue, and many older people experience a progressive loss of the ability to discriminate between blues, greens, and violets; thought to be caused by a gradual yellowing of the lens of the eye over time.

Intervals

One way to characterize differences between color samples is in *intervals*. An interval is a step of change between visual sensations. An individual's threshold establishes the *single interval:* the point at which a detectable middle step can no longer be inserted between two close colors. Single intervals take concentrated effort to see and are tiring to the eyes, but they play an important role in many special effects and illusions.

Many design situations call for intervals between three or more elements. *Parent-descendant color mixtures* are a fundamental interval relationship. They are illustrated as three colors arranged in a linear series, with a "parent" at each end and the "descendant," a middle step between the two, centered between them. *Parent-descendant color mixtures are the basis of many special effects and illusions and play a major role in color harmony.*

Figure 3-3. *Parent-Descendent Color Mixtures.* A middle interval can be established between any two parent colors, no matter how different.

Parent-descendant mixtures can be set up between colors having only hue difference, like red and blue; only value difference, like black and white; or only difference in saturation, like brilliant blue and gray-blue. Intervals can also be set up between color samples that contrast in more than one quality. A brilliant red-violet and a tint of gray-green have hue, value, and saturation contrast, but a middle step can still be found between them.

Even intervals occur when the middle step is *visually equidistant* between the two parents. Representative sets of even intervals might be black on one side, white on the other, and middle gray between them, or red on one side, yellow on the other, and orange between them. The important thing about even intervals is that the midpoint be just that: no more like one parent than the other.

Figure 3-4.

Even Intervals of Hue.

Even Intervals of Value.

Even Intervals of Saturation.

Intervals are not limited to the three steps of parent-descendant color mixtures. In a *series* of even intervals, each step is the visual midpoint between the samples on either side of it.

Figure 3-5.
Even Intervals in Series.

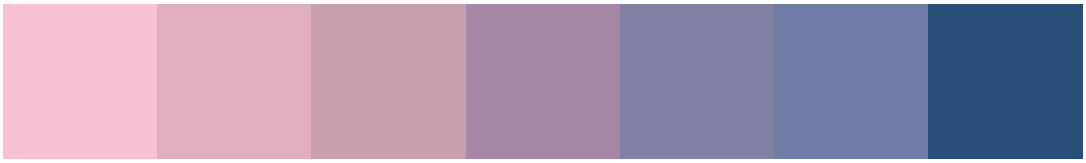


Figure 3-6.
Even and Uneven Intervals of Hue. In the upper set of squares, the center is the middle step between the two parents. In the lower sets of squares, the center is closer to one parent than the other. Do you agree?

Parent-descendant color mixtures occur so frequently in color study that the word "interval" alone is often taken to mean an equidistant step, but intervals can also be uneven. Tilting a middle color deliberately closer to one parent than the other is another technique used in creating special effects and illusions.

Creating order out of random information is a fundamental function of human intelligence. Things are categorized in order to control and understand the flow of information: large to small, A to Z, ascending numbers, dates,

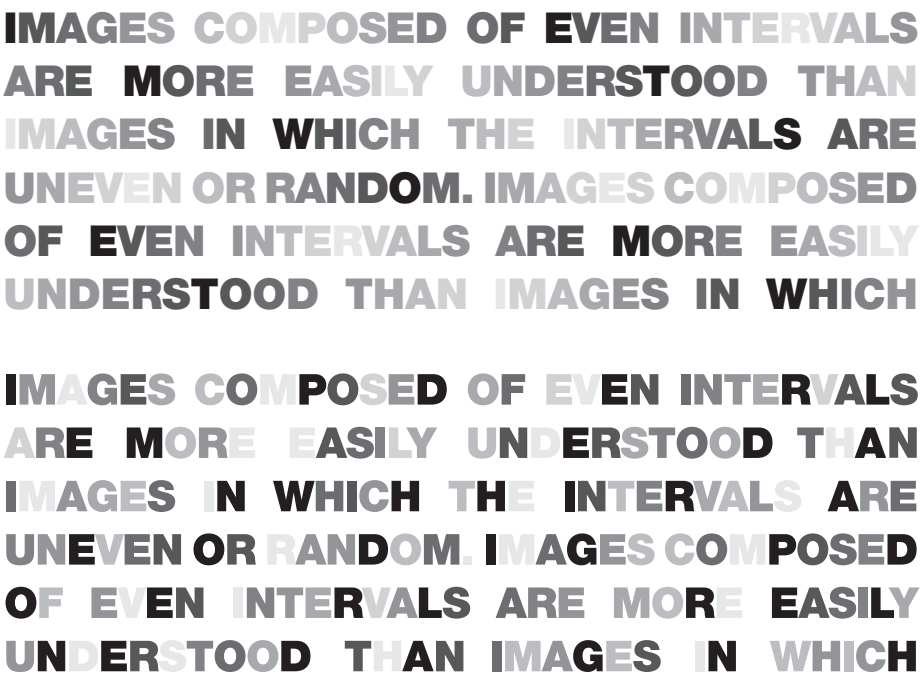


Figure 3-7. *Even and Uneven Intervals.* Images composed of even intervals are more quickly and easily understood than images in which intervals are uneven or random.

or sizes. Orderly information is easier for the brain to process; chaotic information is more difficult. Even intervals are orderly. They are easy for the eye and brain to process. Images composed of even intervals are more quickly and easily understood than images in which intervals are uneven or random.

Intervals are *visually logical* mixes. They are judged by eye alone. Members of a group rarely arrive at complete agreement as to the exact midpoint between two samples. The disagreement is about nuance, not about great differences, and can be attributed to individual differences in visual acuity and interpretation. A perfect interval, like perfect weather, is an opinion that includes a generous portion of fact.

A *gradient* is a series of progressive, even intervals so close that individual steps cannot be distinguished. It is a seamless transition between differences, whether from light to dark, or from one hue to another, or from a brilliant color to a dull one. Shading, for example, is a gradient of values, a "wash" of color from dark to light.

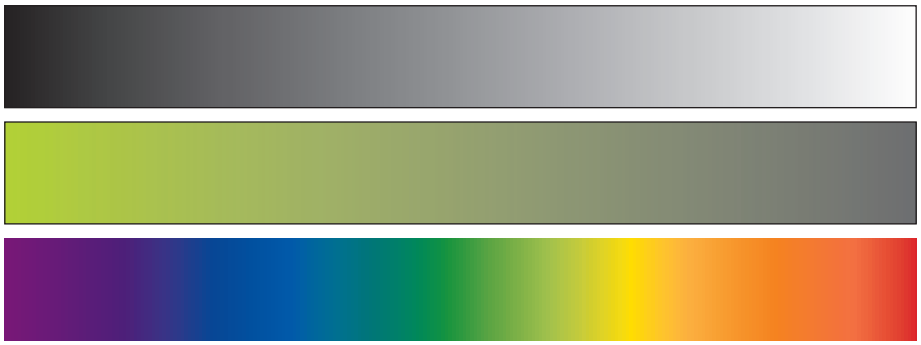


Figure 3-8. *Gradients.* A gradient is a progression of intervals in steps of change that are smaller than the human eye can distinguish. The transition can be in hue, value, saturation, or a combination of qualities.

The Perception of Color

A sensation alone—a touch, taste, smell, sight, or sound—is an incomplete event. The occurrence of a sensation is immediately followed by *perception.* Perception is the critical connection between human beings and their environment. It is cognitive, or *knowing*—the understanding and awareness of what has been sensed. Perception *decides* what has been sensed. It *recognizes* and *identifies* the sensation. It acts as a filter, separating useful and important information from competing stimuli in the environment.

When the brain receives a light stimulus it first interprets form as distinct from background by sensing patterns of light and dark. *Figure-ground separation*, or *pattern recognition*, is the first cognitive step in the process of perception. It identifies situations by forms and their arrangement.

Color plays an important, but secondary, role in recognition. A red file folder (unpaid bills) and a blue file folder (paid bills) seem at first to be identified by color. But both red and blue folders are identified first as file folders, and only secondly by color. The initial recognition is of form: this is a file folder, not a notepad, not a CD, not a book.

Recognition is based on learned information from a multitude of sources: individual experience, social and cultural traditions, environmental surroundings, formal teaching. The ability to recognize sensations develops with astonishing rapidity, beginning almost at birth. By adulthood, human beings have acquired and stored an immense database of recognized sensations. Everything seen is understood because its identity has been learned and the experience of it held in memory. New sensations, unless they are accompanied by additional information, are identified by referring to this stored information. Something new is recognized, correctly or incorrectly, because it is associated with some familiar thing that has similar characteristics.

Most perceptions occur unconsciously and at such high speed that they seem simultaneous with sensation.[1] What we think of as the sensory experience of color is always a fusion of sensation and perception. Unlike a sensation, *a perception cannot be measured. It can only be described.*

Understanding how we see, and how we process and respond to what we see, translates directly into design applications. A working knowledge of the fundamentals of perception provides information that helps the designer to direct, even at times to control, the way a design is received by its target audience. Physiology studies the body and its functioning. It is a measurable science that can quantify the body's physical responses to a stimulus of color. Psychology studies behavior; or how organisms perceive and react to situations when they are stimulated in different ways. Psychology can describe—but cannot precisely measure—the ways in which human beings recognize, interpret, and respond to the stimulus of color. Psychology deals with perception.

Physiology: Responding to Light

The nervous system is an information pathway from the outside world to the brain. It is made up of three kinds of cells: receptor cells, transmitter cells, and brain cells. Receptor cells receive information from the outside world (stimuli) and change it into a form of electrical energy that the brain can use. Transmitter cells carry these signals to the brain. Signals from each sense are received in separate, specific locations in the brain. The brain decodes each sensory event first by identifying which sense has been stimulated, then discriminates specific

qualities within that sense. In music, for example, it discriminates between notes; in vision, red from blue. Processing the information and generating a response to it is the final step.

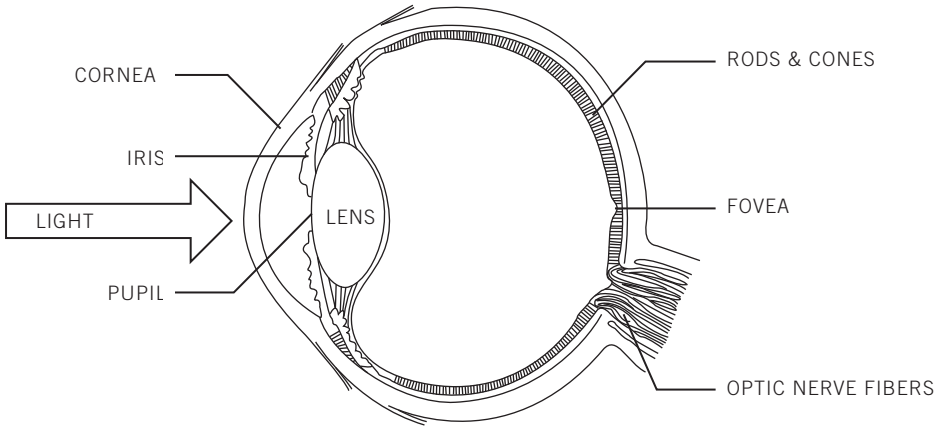


Figure 3-9.
The Human Eye.

The eye is a sense organ that detects light. Light enters the eye through the pupil and falls on the retina, the inside back of the eye. The retina is made up of two kinds of light-sensitive receptor cells, *rods* and *cones*. Both rods and cones connect to the optic nerve, which transmits the sensory message to the brain.

Rods and cones respond selectively to available light. Cones dominate vision when a great deal of light is present. Cones are responsible for color vision and for the ability to see detail. Objects appear more colorful and fine detail, like small print, is clearer when cones are dominant. Rods dominate vision in low light. Rods are responsible for peripheral (surrounding, less focused) vision. Colors appear muted, and fine detail is more difficult to see, when rods dominate.

The *visual field* is the extent of area that can be seen by the two eyes of a viewer standing in one position. The *fovea* is a tiny area at the back of the eye that is the center of the visual field. The fovea contains only cones. It is the most sensitive area of the retina, detecting

patterns of light and dark and color with the greatest clarity. Images and colors are seen less clearly when the light stimulus moves away from the fovea.

Both rods and cones are always at work. It is almost as if there are two separate systems, one for day and one for night. *Adaptation* is the involuntary response of the eye to the quantity of available light. The retina moves back and forth (adapts) quickly between rod and cone dominance as the amount of available light increases or decreases. Objects appear more colorful at higher lighting levels, when cones dominate. Color perception lessens in dim light when rods dominate: perception in low light is in shades of gray. A garden that is colorful by day loses its color gradually toward evening, but the contrast of dark foliage and light flowers persists as a different, but equally pleasing, image. Adaptation takes place under any lighting conditions. "Cones for color, rods for gray tones" is the same for sunlight or lamplight, fluorescent, incandescent, or any other kind of light. The sly old metaphor is literally true: all cats really *are* gray in the dark.

Lateral inhibition is an aspect of vision that increases the eye's ability to distinguish edges. When a pattern of light and dark contrast reaches the retina, the cells that receive the light part of the image inhibit the ability of the ones next to them to detect light. As a result, areas next to bright spots appear darker. The greater the quantity of light, the more lateral inhibition takes place: light areas appear lighter and dark areas darker.

The sensation of light is received in two areas of the brain: the cerebral cortex and the hypothalamus, or midbrain. The cerebral cortex is the center of cognitive activity. It receives information and processes it; recognizing, interpreting, and structuring a response to each stimulus. The midbrain controls the internal environment of the body. Sensations of light transmitted to the midbrain act as a biological stimulus to the central nervous system. The midbrain contains the center for regulation of blood pressure and body temperature. It also stimulates glands that control the production and release of hormones. When the brain is stimulated by a thought, mental image, or outside stimulus (like light), the midbrain triggers the release of hormones. A color stimulus has an effect on the strongest human needs and emotions—stress, hunger, thirst, and sex.

Sunlight, which contains all colors, is essential to human life. The human body is genetically

adapted to function at a normal level in response to the sun's pattern of energy emission. Changing the strength of a color stimulus causes a change in the body. Exposure to an elevated level of red stimulates hormone production and raises blood pressure, while exposure to an elevated level of blue has been shown to lower blood pressure and depress hormonal secretions.

The immediate biologic response of the body to a stimulus is *phasic arousal*. Phasic arousal is abrupt and lasts very briefly, like the surge of adrenalin that is experienced in a sudden and frightening situation. Phasic arousal requires a stimulus. *Tonic arousal* is the body's response over a prolonged period. The body has a norm for tonic arousal, and the brain continually directs the adjustment of hormone levels to keep it at that norm. Stimulation by a strong color causes phasic arousal—an immediate reaction—that can be physiologically measured, but the arousal is short term: *the duration of the effect is not continuous.*

Because exposure to color changes the body's hormonal balance, it can also cause changes in behavior. Colors can be chosen to stimulate, depress, or otherwise alter mood. In environmental design, overstimulation and understimulation have equally negative effects: human beings respond best to living spaces that have color, but not an overload of highly stimulating color. A graphic designer may choose a brilliant color to arouse short-term attention. Restaurant designers use red, in its many variations, to stimulate the appetite. The muted colors of funeral homes are meant to minimize emotional response. An extreme example of color used to modify behavior occurs with a color known as Baker-Miller pink (a bubble-gum pink). It has been hypothesized that exposure to Baker-Miller pink reduces aggressive behavior. The effects begin after a short period of exposure and last about for about half an hour—a perfect demonstration of the body's arousal pattern.

Colors can also be experienced without a stimulus of light. The brain alone, without a light stimulus, allows us to dream in color, or to imagine color with closed eyes. A headache or a blow on the head can trigger vivid images of blue stars. Color can be seen in the mind's eye.

Healing and Color

The eye is not the only organ that responds to light. Light is also absorbed through the skin.

The use of colored light to act on the body through the skin is a routine medical practice. The treatment of jaundiced infants with light is a standard (and effective) therapy, as is the treatment of psoriasis, a skin disease, by exposure to sunlight.

The use of color to heal has a long history. Ancient color therapies included (among other treatments) the application of colored *substances* to the skin. Some were helpful, but not because of their color. They were effective because the applied substance had medicinal properties.

Color therapy remains an active field of study. Most contemporary color therapists ascribe healing properties to different wavelengths, rather than to colored substances. Certain colors of light are thought to have affinities to different parts of the body. The medical establishment is skeptical of these claims, and the practice of medicine by color therapists is illegal in the United States.

Synaesthesia

Synaesthesia is a long-recognized but largely unexplained phenomenon in which one sense responds to the stimulation of another. There are reports of blind persons (and others) who are able to determine the colors of objects through touch. A woman reports a humming sound when she enters a certain red room; the sound stops as soon as she leaves the room. A man reports a strong taste of lemon when he listens to a particular piece of music. Research indicates that the sensory pathways to the brain are connected to each other in ways that can be demonstrated but are not yet understood. The idea that undiscovered connections exist between the senses seems less surprising when we remember what an ordinary thing it is to experience chills—ordinarily a reaction of the skin to change in temperature—from an emotional, musical, or visual experience.

Psychology: Responding to Light

Visual acuity for color, hormonal responses to color, adaptation, lateral inhibition, and synaesthesia are involuntary biologic responses of the body to a stimulus of light. The perception of color also includes involuntary *psychological* responses. The cerebral cortex, the reasoning part of the brain, identifies and organizes a response to each color stimulus that

is unconscious, but based on past learning. Stored information has a profound influence on color perception.

One of these responses is a kind of expectation called *memory color*. Memory color means that the viewer makes an unconscious assumption about the color of something, the "orange" of an orange, for example. A viewer influenced by memory color does not report the actual color experience. Instead, what is reported (or even illustrated) is a preconceived idea. A still-ripening Red Delicious apple may be more green than red, but it will be described as red. At times the sea can seem nearly maroon; even then an unthinking—or un*see*ing—artist will paint it as blue.

Memory color affects the perception of objects of familiar coloration. *Color constancy** is a second and equally powerful form of expectation. Color constancy means that the colors of familiar objects retain their identity no matter what the general lighting. The eye and brain together adapt to all general light sources as if they were the same. Colors seen in a daylit room may undergo dramatic changes at night under incandescent lighting, but the viewer is unaware of the differences. An image stored in memory overrides what is actually seen.

A second kind of color constancy occurs when close colors are perceived as being identical. In an all-white kitchen, the white of the refrigerator, the counters, the floor, the cabinets, and the paint may all be different, but the immediate cumulative effect is that they are the same. The various surfaces are categorized mentally as "white," and the concept of "whiteness" prevails over the differences that actually exist.

It is appropriate at times to look for fractional differences between colors (in a design situation, for example), but a constant concern with small color differences would be exhausting if it were applied to daily life. Memory color and color constancy screen out important color differences from ones that do not matter. Because they simplify and edit what is seen, they play a large and proper role in the visual comfort of living with color.

Naming Colors

Color is recognized universally as a particular kind of visual experience. When the brain receives visual information, it identifies it by name. In a widely-accepted study, researchers

* Color constancy is also called chromatic adaptation.

Brent Berlin and Paul Kay determined that ninety-eight languages had names for eleven basic colors. Simpler languages had fewer color names; more complex languages had more names. The languages studied gave names to colors in a consistent order of recognition: first, black and white (or dark and light), then red, followed by yellow and/or green, then blue, brown, orange, purple, and pink.

External evidence tells us that most people see the same thing when they see "red". We accept that something is red not by scientific measurement but by unspoken agreement, common language, and common experience. Green leaves are identified in each language as green, not as orange. Disagreements that arise about the names of colors are about variations within colors, not about broad categories. Something that is called red is simply more red than it is anything else. It may be a bluer red or a yellower red, but no one would ever call it blue or yellow.

Just as no one can know precisely what another person sees, no one can experience anyone else's *idea* about a color. Each individual holds in memory a personal "picture" of the meaning of each color name. Josef Albers observed, "If one says 'Red,' (the name of a color) and there are 50 people listening, it can be expected that there will be 50 reds in their minds. And one can be sure that all of these reds will be very different."[2] In addition, there are hundreds of different reds, and a surprising number of people assume that there is a fixed name for every one. If members of a group are asked to characterize a red object by its color name, the answers are more likely to be "fire-engine red," "cherry red," "spectrum red," or "lipstick red," than simply "red."

Color *study* requires only six names for colors: red, orange, yellow, green, blue, and violet. Each name represents a family of closely related hues. Restricting the names for colors to six words enables the observer to focus on the visual experience. Design and marketing professionals use (and need) romantic names for colors, like Venetian Red, Bermuda Blue, or Aztec Gold because the images that those words evoke play a large role in marketing. Both ways of naming colors are important in design as long as the critical difference between the two is recognized: the six hues of color study deal with eye training, color recognition, and color use; the countless color names of marketing are about product image and sales.

Color as Language: From Name to Meaning

Language is a collection of spoken words that convey ideas and feelings. Each word has the same meaning for those who speak the same language, although single words can have slightly different meanings for different groups of speakers. An American who orders "today's pudding" from a London menu may be surprised to be served a slice of chocolate cake. *Writing* is visible language. It is a visual code for spoken words: a means of preserving words and ideas and a way of controlling the way they are presented.

Dead languages are frozen in time. The meanings of words do not change. Living languages are continually evolving. Words change their meanings over time, or are lost entirely, or are replaced by new ones. Colors, too, can be a language. Colors (or color groups) can be visual codes that communicate ideas and feelings and influence the way ideas are presented. And like words, the meanings of colors can change, or be lost, or be replaced by new meanings over time. *Color is a living language.*

Culture is the social structure that establishes what is important to a group for people. For every individual the meaning of a color, or group of colors, is shaped by a hierarchy of outside forces: culture, spoken language, social status, setting, time, and individual life experience. *Semantics* is the study of the meaning of a word, passage of words, or other form of language—including the language of color. Awareness of cultural differences in the semantics of color is critical to the successful marketing of any product or image destined for the global market. Sometimes a color is only a color, but often it is more.

In *Color, Environment, and Human Response,* Frank H. Mahnke describes the experience of color as a pyramid with six levels of response:[2]

personal relationship
influence of fashions, styles and trends
cultural influences and mannerisms
conscious symbolism-association
collective unconscious
biological reactions to a color stimulus

Each step upward in Mahnke's pyramid represents a narrower interpretation of a color experience. The lowest level is the innate, unlearned response to color that is universal: the physiological response to a stimulus of light and its effects on the midbrain. The cognitive part of the brain becomes a part of the response at the second level, where each sensation is identified by name. "Collective unconscious" responses, like the association of the color red (and also the *word* for red) with blood, are also involuntary and cross-cultural.

At the third level colors become a language that is independent of words. Colors or color groups are used as symbols, or visual codes, for non-color ideas. Symbolic colors have formalized meanings for specific populations. White, the bridal color of the West, is identified as the color of mourning in India. Americans who see "red, white, and blue" interpret it as having an association with the United States, although the French, Chilean, and Yugoslavian flags (among others) employ the same colors.

Figure 3-10. *Color Symbolism.* Many national flags share the same configuration. Both configuration and colors are needed to identify each as a national symbol.

Colors or color groups that symbolize major societal concerns like nationhood, death, and marriage tend to maintain their meaning over long periods of time. These color symbols can be thought of as permanent for each population. Black as the color of mourning in the West, or white for mourning in India, for example, are meanings that are unlikely to change. "The East is Red" is both an anthem and a symbol of long-time political reality for Communist China.

Color symbols of lesser social importance tend to be less stable in meaning. Just as words change in meaning, often in politically and socially potent ways, so do the meanings of colors. A color symbol is sensitive to its time as well as to its audience. The Green Party, founded in Europe in the late 1970s, was perceived initially by many as a radical group bent on the destruction of modern society. Alliances of the Green Party with Communist and Socialist groups, known as Red-Green alliances, were viewed as threatening to social stability. In the

Figure 3-11.
Color Symbolism. The color green is recognized immediately as representing concern for the environment. *The Green Promise designation is Benjamin Moore's assurance that its environmentally friendly coatings meet and exceed the strictest industry standards, while also delivering the premium levels of performance you expect from Benjamin Moore.*

twenty-first century, with global warming and the deterioration of the environment everywhere apparent, green has not only lost any negative political meaning in the West, it has become an international rallying color of environmental politics and a powerful tool for advertisers. "Green" products carry the implicit message that using them will help to save the world.

Even with the same audience and within the same time period, however, no color is limited to a single meaning. *Colors are understood as codes for noncolor ideas only when they make sense in context. Colors may inform, but they do not define a situation.* Additional information is needed. The young woman in a long white dress may be a bride—or just a pretty young woman enjoying a summer dress. The green that represents environmental concern in one setting implies youth and inexperience in another. Shakespeare's aging Cleopatra speaks of her "salad days," when she was "green in judgment." New recruits are said to be green. At sea, one turns green from seasickness; in another circumstance, green with envy. Green, the color of Ireland, the "Emerald Isle," is also the color of Islam. And who can tell at a single glance if an orange and black scarf celebrates Halloween—or Princeton?

Figure 3-12.
Safety Colors. School buses and rain slickers are yellow as a reminder to exercise caution.

Some symbolic colors are so important in communicating ideas that their meanings that have been legislated. Color coding plays a central role in conveying safety information for many *vigilance tasks*, which

are situations that call for immediate response. The federally mandated OSHA[3] colors used in the United States, and in many cases internationally, communicate physical hazards and safety information. OSHA yellow indicates caution; orange, dangerous machinery or equipment; violet warns of radiation hazard. Red indicates fire-fighting equipment and is recognized world wide as an indicator of danger. Red says "STOP." Green indicates safety, or safety equipment. Safety colors and the situations in which they appear are closely linked in memory. They are processed very rapidly, so that little, if any, additional information is needed. An OSHA yellow rain slicker helps to protect the traffic officer. The same yellow as paint on a chair suggests nothing about safety at all.

When colors are seen with simultaneous and *conflicting* information, the areas of the brain that respond to color compete with other parts of the brain in structuring a response. The resulting delay and confusion in perception, called *Stroop interaction*, takes place because the relative importance of equal and competing streams of information must be sorted out.[4] The word "**BLUE**" written in **RED** will cause a hesitation in the steady flow of reading. An octagonal green traffic sign that said "STOP" would be misperceived—and delay response—with potentially fatal results.

Figure 3-13. *Stroop Interaction.* A delay in comprehension occurs when the different parts of the brain recieve conflicting information at the same time.

"Fashion" colors, the color trends of consumer marketing, are transitory and cyclical. They have a short-term influence on color preferences. Unconscious memory at the individual level, the "personal relationship" to color, is a stronger and more lasting determinant of color responses. When Deborah Sharpe states that "color responses are more tied to man's emotions

than to his intellect," the underlying message is that color responses are shaped by experiences and associations that are long-buried in memory. Aunt Agatha made you eat broccoli in her pink kitchen. You have visceral, and very negative, feelings about broccoli *and* pink.

Impressional Color

Impressional or *associative* colors evoke imagery without symbolic meaning. Grayed blue-greens may call to mind the icy cold of a winter sea, or a riot of brilliant greens a tropical forest. Steel gray suggests hi-tech; palest pastels a new baby. These responses can be personal *or* collective, and the slightest shift in hue can alter the imagery. Yellow tilted to orange may suggest warmth and richness, while yellow tilted to green suggests illness. Red is associated with passion of all kinds: "There are reds that are triumphal and there are reds which assassinate,"[4] but thoughts and images raised by a pale rose-red will be quite different from those of deep burgundy.

Color as Words Alone

Written and spoken *words* for colors communicate the same symbolic ideas as actual colors, but they are understood more indirectly. Words are processed as thought rather than as sensory experience. The *immediacy* of a color symbol diminishes when it is presented as words. Its *meaning* is unchanged. Reading the words "red, white, and blue" takes longer to process than the sight of a flag in full color. Color meanings of all kinds are inextricable from descriptive writing. A colorful personality is understood to be vital and interesting; a gray individual, not at all. No one has written more comprehensively, or with more eloquence and scholarship, on the imagery and symbolism of colors than Paul Theroux:

> Fox-red *tenné* in the language of chivalry shows a somewhat burnt tone. And garnet red with its low brilliance and medium saturation is a sort of "pigeon blood" or "Spanish wine." Cranberry red has a saucy sharpness to it, with a hint of yellow. Bluish red, popular in lipsticks, cardigans, and the aura of dramatic personalities....[5]

Does anyone doubt that celebrities have a bluish red aura? Theroux says it all.

Endnotes

1 Rodemann 1999, page 155.
2 Mahnke 1982, page 11.
3 OSHA United States Congress Occupational Safety and Health Act of 1971.
4 Varley 1980, page 132, quoting Leon Bakst.

Chapter 3 Highlights

- The experience of color begins with a sensation, a response to a stimulus of light. Both stimuli and sensation are measurable. A sensation is followed by perception; the understanding of what has been sensed. A perception can only be described. Most perceptions occur unconsciously and at such high speed that they seem simultaneous with sensation.

- Visual acuity for color is the ability to detect differences between wavelengths (colors) of light. The threshold of color vision is the point at which a difference between two similar hues can no longer be discriminated. An interval is a step of change between visual sensations. An individual's threshold establishes the single interval: the point at which a detectable middle step can no longer be inserted between two close colors.

- Parent-descendant color mixtures are illustrated as three colors in a linear series, with a "parent" at each end and a middle interval between them. Even intervals occur when the middle step is visually equidistant between the two parents. In a series of even intervals, each step is the midpoint between the samples on either side of it. Images composed of even intervals are more quickly and easily understood than those in which intervals are uneven or random. A gradient is a series of progressive, even intervals so close that individual steps cannot be distinguished.

- The eye is a sense organ that detects light. Light enters the eye through the pupil and falls on the retina, which is made up of light-sensitive receptor cells called rods and cones. These connect to the optic nerve, which transmits the message to the brain. Rods and cones respond selectively to available light. Cones dominate vision when a great deal of light is present. Cones are responsible for color vision and for the ability to see detail. The fovea is a tiny area at the back of the eye that contains only cones. It detects patterns of light and dark and color with the greatest clarity.

- The sensation of light is received in two areas of the brain. The cerebral cortex recognizes, interprets, and structures a response to each stimulus. The midbrain contains the center for regulation of blood pressure, body temperature, and glands that produce hormones. Sensations of light transmitted to the midbrain act as a biological stimulus to the central

nervous system. Changing the strength of a color stimulus can cause a change in the body.

- Adaptation is an involuntary response of the eye to the quantity of available light. The retina moves back and forth between rod and cone dominance as available light increases or decreases. Lateral inhibition increases the eye's ability to distinguish edges in patterns of light and dark. The greater the quantity of light, the more lateral inhibition takes place.

- The immediate response of the body to a stimulus is phasic arousal, which is abrupt and brief. Stimulation by color causes phasic arousal but the duration of the effect is not continuous. Tonic arousal is the body's response over a prolonged period. The human body is genetically adapted to function at a normal level in response to the sun's pattern of energy emission. It has a norm for tonic arousal and continually adjusts hormone levels to keep it at that norm. Exposure to color changes the body's hormonal balance and can cause changes in behavior.

- Colors can be experienced without a stimulus of light. The brain allows dreaming in color and imagining color with closed eyes. A headache or a blow on the head can trigger a color response. Light is also absorbed through the skin. The use of colored light to act on the body through the skin is a routine medical practice. Synaesthesia is a largely unexplained phenomenon in which one sense responds to the stimulation of another.

- Memory color is a psychological response to a stimulus of color. The viewer makes an unconscious assumption about the color of something and reports a preconceived idea rather than the actual color experience. Color constancy means that the colors of familiar objects retain their identity no matter what the general lighting. A second kind of color constancy occurs when close colors are perceived as being identical.

- Colors and color groups are used as symbols. Symbols for major societal concerns like nationhood tend to maintain their meaning over time and can be thought of as permanent for each population. Lesser symbolic meanings can change, be lost, or be replaced over time. Semantics is the study of the meaning of a words, passages of words, or other forms of language. Awareness of cultural

differences in the semantics of color is critical to the marketing of any product intended for the global market. Words for colors communicate the same symbolic ideas as colors, but less quickly. Words are processed as thought rather than as sensory experience.

- Color coding plays a central role in conveying safety information. Stroop interaction takes place when colors are seen at the same time as simultaneous and conflicting information. Areas of the brain that respond to color meaning compete with other parts of the brain in structuring a response.

4

THE VOCABULARY OF COLOR

Hue / The Artists' Spectrum / Primary, Secondary, and Intermediate Colors / Saturated Color / Other Spectrums, Other Primaries / Chromatic Scales / Cool and Warm Colors / Analogous Colors / Complementary Colors / Tertiary Colors: Chromatic Neutrals / Black, White, Gray / Value / Value and Image / Transposing Image/ Pure Hues and Value / Tints and Shades / Monochromatic Value Scales / Comparing Value in Different Hues / Saturation / Saturation: Diluting Pure Hues with Gray / Saturation: Diluting Pure Hues with the Complement / Tone

"When I use a word," Humpty Dumpty said, in a rather scornful tone,"
it means just what I choose it to mean—neither more nor less."

—Lewis Carroll

The words for colors in everyday life—red, green, blue, and so on—do not have precise meanings. Instead, each word is a code for a range of similar sensations. The same is true for words that describe colors, like brilliant, dull, dark, or light. Red is red, and never blue, and brilliant is never dull, but at the same time, these words in conversation can represent a number of different visual experiences.

When a word is used to identify a color sample, or as the verbal expression of a color not seen, it means something slightly different to each person. Mary and John see the same red-orange sample and respond to it in personal ways. Mary calls it red; John insists it is red-orange. Not everyone senses colors in exactly the same way, and even if they did, individuals do not *think* of colors in exactly the same way. The result is a stalemate; a difference of opinion; an inability to agree on what a particular sample *is*—and difficulty in communicating ideas about it. It makes no difference whether a color is seen as pure light, or as a solid object, or as a printed page. Each person interprets and names colors in his or her own way. Color is a *subjective* experience. One man's peach is always another man's melon.

The design process calls for a vocabulary that communicates color ideas with more precision, but no new words are needed for this. If a color cannot be precisely *identified* by

words, it can nevertheless be described. Colors have basic attributes that can be observed and named. Whether these are called qualities, aspects, dimensions, or something else, they all refer to the same thing: qualities that are present in colors of all kinds; colors of light or colors of ink, colors that are glossy or matte, transparent or opaque, chalky or clear. An observer with a trained eye can use these attributes to describe a color relative to another, similar one, and to communicate differences between them with reasonable precision. Color corrections can be directed in the same way.

The vocabulary of working with color also includes words that describe interrelationships between colors. Together, the terms of description and interrelationship make it possible to convey ideas about color using words whose meanings are clear, consistent, and objective.

Three qualities of color are already familiar:

Hue: the name of the color: red, orange, yellow, green, blue, or violet

Value: the relative lightness or darkness of a color

Saturation or *chroma*: the hue-intensity or brilliance of a sample, its dullness or vividness

Each word communicates an independent idea, but the ideas are independent only for the purposes of study. These attributes are present in every color, and all are needed to describe it. A color sample may have additional characteristics, like opacity or translucency, but it can always be described first in terms of its hue, value, and saturation.

Hue

Hue means the *name* of the color. In science, the colors of light, or *spectral colors*, can be established precisely by a measurement of wavelength, and the words "hue" and "color" are interchangeable. In everyday speech (including in this book), the word "color" is used in two different ways. It can mean (more precisely) the hue of something or (more generally) the complete visual experience of its hue, value, and saturation together. Only the context in which it is used tells which meaning is intended. "Hue," however, means *only* the name of the color.

Chroma is a synonym for hue. It is part of some familiar color words:

Chromatic: Having hue
Achromatic: Without hue
Polychromatic: Having many hues
Monochromatic: Having one hue only

It has been theorized that the average person can distinguish about 150 colors (hues) of light, and every one can be described using one or two of only six words. The only names ever needed to describe hue are:

RED ORANGE YELLOW GREEN BLUE VIOLET

A color is called by the name of its most obvious, or dominant, hue. Every color name represents a family of related hues. Nearly all color samples include more than one hue, but *one hue is most apparent* and others are present in smaller proportion. A sample may seem to be pure yellow until it is placed next to a different yellow sample. Suddenly, one yellow is seen to contain a bit of green, the other a fraction of orange. Both are *called* yellow because yellow predominates in each. Using the word "*contains*" helps to evaluate colors. "This yellow contains some orange" is perfectly descriptive. It identifies the principal hue and a second one that modifies it.

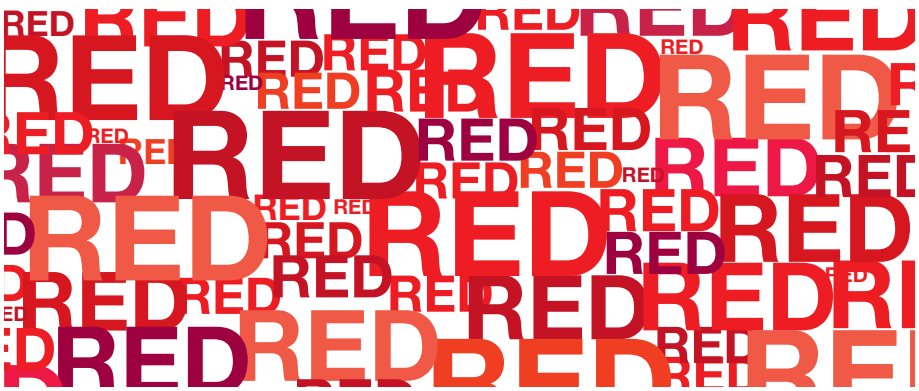
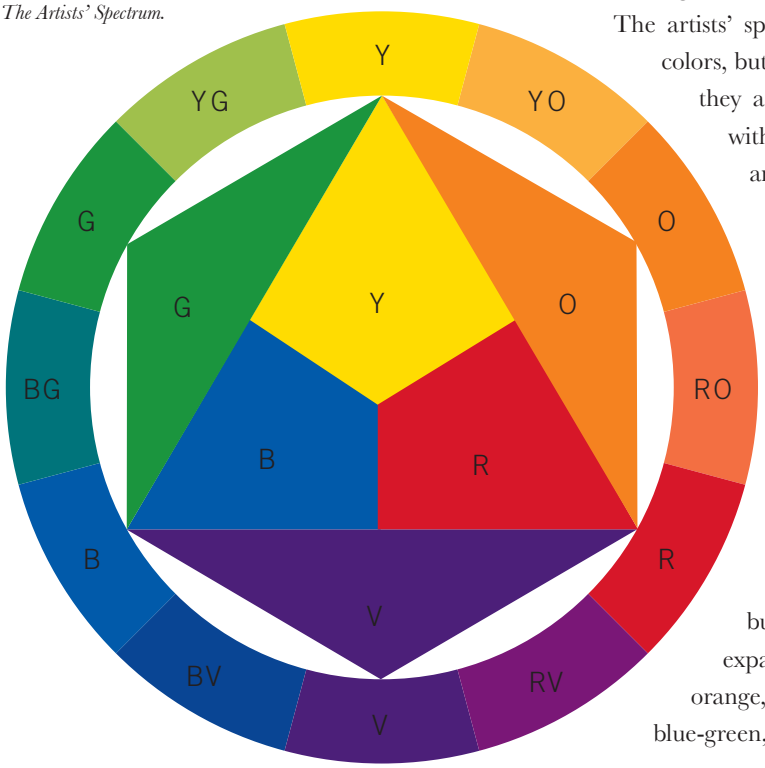


Figure 4-1.
Color Names. Color names are not absolute. Every color belongs to a enormous family of related hues.

The Artists' Spectrum

The *artists' spectrum* is a circle that illustrates hues in their natural (spectral) order. The spectrum of visible light (additive color) is linear, moving from short wavelengths of light (violet) to long ones (red), and the order of its colors is fixed. The artists' spectrum is also fixed in its order of colors, but it has six hues instead of seven, and they are presented as a continuous circle, with violet forming a bridge between red and blue. The artists' spectrum is also called the *color circle* or *color wheel*.

Figure 4-2.
The Artists' Spectrum.



There are too many hues in the range of human vision to include all of them in one circle, so the artists'spectrum is a sort of visual outline, or synopsis, of all visible hues. The basic spectrum is made up of six hues: red, orange, yellow, green, blue, and violet. An expanded spectrum has twelve hues, but *no new color names are introduced*. The expanded spectrum includes yellow-orange, red-orange, red-violet, blue-violet, blue-green, and yellow-green.

The artists' spectrum is limited to six or twelve hues only because this is a concise, easily illustrated figure. It can be expanded to any number of hues as long as the added colors are inserted at regular intervals in all hue ranges. There can be six, twelve, twenty-four, forty-eight, ninety-six, or more hues (but not thirty-seven or fifty-one). The only limits to the number of places on a color circle are the limits of human visual acuity and the practical problems of illustrating it.

Color circles can be illustrated in any medium. If the medium is a display of light on a screen, the colors will be sharp and clear; perhaps even measurable as true spectral hues. A subtractive medium like gouache or acrylic paint illustrates colors in a less measurable way. As long as the hues are in sequence and the intervals well-spaced, color circles can vary a good deal in appearance and be equally correct as representations of the visible hues.

Primary, Secondary, and Intermediate Colors

Red, yellow, and blue are the *primary colors* of the artists' spectrum. They are the simplest hues. They cannot be broken down visually into other colors or reduced into component parts. The primary colors are the most different from each other because they have no elements in common. All colors on the artists' spectrum are mixed *visually* from the primaries red, yellow, and blue. Green, orange, and violet are the *secondary colors* of the artists' spectrum. Each secondary color is an even interval, or visual midpoint, between two primary colors:

Green is the middle mix of blue and yellow.
Orange is the middle mix of red and yellow.
Violet is the middle mix of blue and red.

The secondary colors are less contrasting in hue than the primary colors. Each secondary color has one primary in common with each of the others. Orange and violet each contain red, orange and green each contain yellow, and green and violet each contain blue.

Yellow-orange, red-orange, red-violet, blue-violet, blue-green, and yellow-green are *intermediate colors.* They are the midpoints between the primary and secondary hues. Many times these colors are referred to, incorrectly, as tertiary colors.

Saturated Color

A *saturated color* is a hue in its strongest possible manifestation. The reddest red imaginable, or the bluest blue, are saturated colors. Saturated colors are also called *pure colors* or *full* colors. They are at *maximum chroma.* Saturated or pure colors can also be defined by what they do *not* contain. A saturated hue is made up of a single primary color or two primaries

in some mix or proportion, but never includes a third primary. A saturated hue does not contain black, white, or gray.

Saturated colors are not limited to the six or twelve "named" hues of the artists' spectrum. *As long as a color contains only one or two primaries and is undiluted by black, white, or grey, it is a saturated color.* Imagine the full range of visible hues as a circle with each color blending into the next, like a circular rainbow. Any hue, at any point on that circle, is a saturated color. The only limit to the number of saturated colors is the limit of human color vision.

Other Spectrums, Other Primaries

The artists' spectrum illustrates one color-order system. It is familiar, visually logical, easy to represent as a two-dimensional figure, and allows for theoretically unlimited expansion. Scientific and nonscientific disciplines use other spectrums to illustrate alternative color-order systems. Choosing one spectrum instead of another is exactly that: a choice. No spectrum is inherently more correct than another. The circles may vary in the names of colors, the number of colors illustrated and in the assignment of what might be called "prime points" on the wheel. Wilhelm Ostwald, for example, predicated an eight-hue spectrum that includes sea-green (blue-green) and leaf-green (yellow-green). Psychologists construct a four-hue spectrum with red, green, yellow, and blue as the primaries.

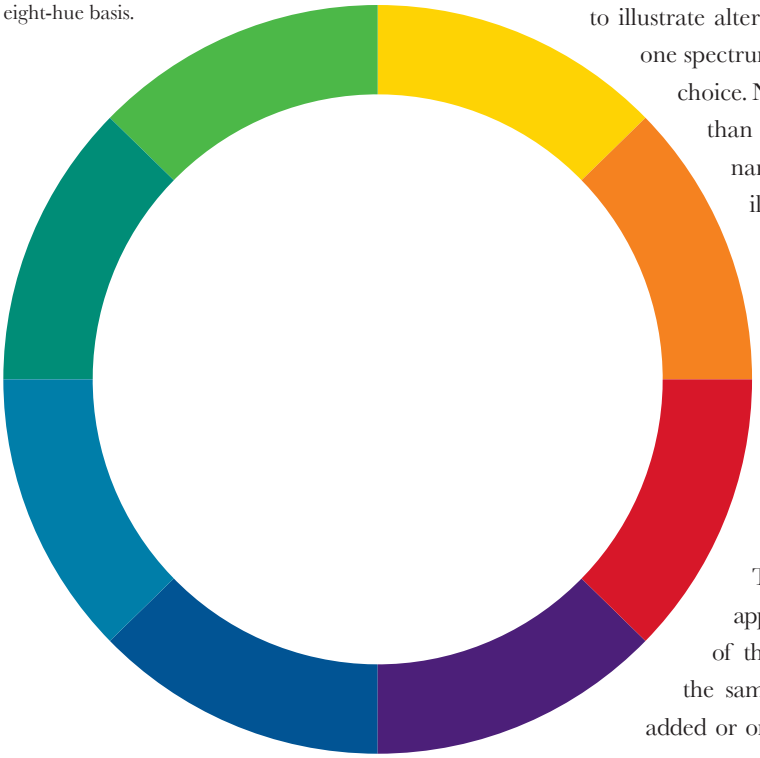


Figure 4-3.
Alternative Spectrums. Wilhelm Ostwald's spectrum has an eight-hue basis.

The different color circles may at first appear to conflict, but they are all variants of the same color-order idea. All recognize the same sequence of colors. Hues may be added or omitted. There may be a slight shift in

the placement of opposites. Munsell, for example, places blue-green, not green, opposite red. But no spectrum places red next to green, or orange next to blue. Arguments made for the number of colors included and their names are intellectual exercises. All color circles include the primary hues in some way, and all follow the same color order.

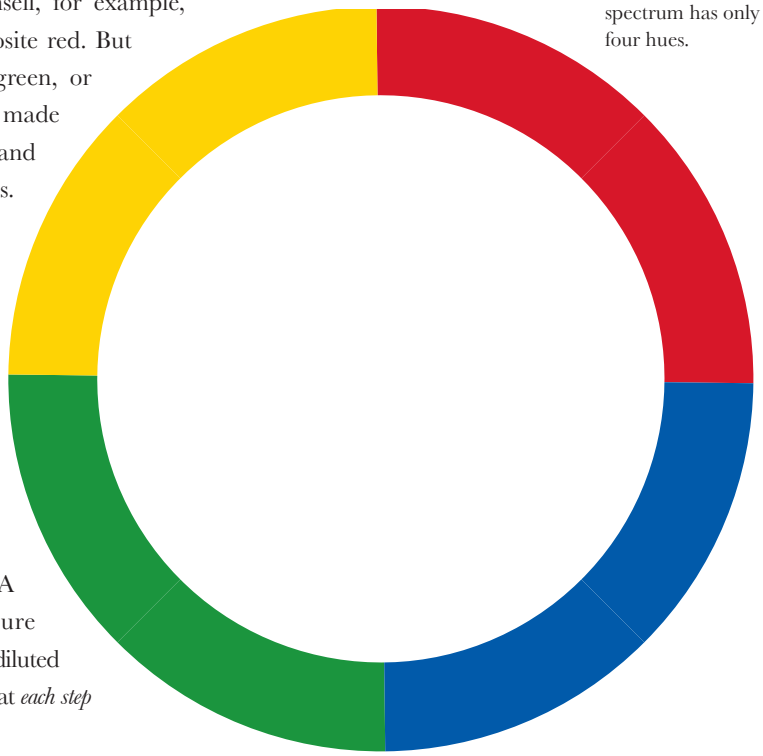


Figure 4-4.
Alternative Spectrums. The psychologist's spectrum has only four hues.

Chromatic Scales

A *chromatic scale* is any linear series of hues in spectrum order. A series of hues between blue and orange (blue-green-yellow-orange), for example, is a chromatic scale. A chromatic scale can illustrate pure (saturated) colors or more complex, diluted colors. Its defining characteristic is that *each step in the progression is a change in hue*.

Cool and Warm Colors

Cool and *warm* are words used to describe two opposing qualities of hue. Cool colors contain blue or green: blues, greens, violets, and steps between them. Warm colors are reds, oranges, yellows, and steps between them. Warmth or coolness in hues is sometimes referred to as color temperature.

The primary colors are weighted toward the warm. Only blue is cool, while both red and yellow are considered warm. As a result, the entire spectrum is more heavily "warm" than it is "cool." Blue is the polar extreme of cool, and orange, made of red and yellow, is the polar extreme of warm.

Warmth and coolness in colors are not absolute qualities. Any color, even a primary, can appear warmer or cooler relative to another color. There are cooler reds (closer to violet) and warmer ones (closer to orange). Violet and green are generally considered to be cool colors, but it is possible to describe one violet as warmer than another violet because it contains more red.

The terms "cool" and "warm" are helpful for describing families of colors or for comparing colors for warmth or coolness alone. They are less useful terms when hues need to be adjusted. Directing a color change toward a specific hue communicates more clearly. "This violet is too warm. Cool it off, add some blue. This red is too cold. Warm it up; bring it closer to orange."

Analogous Colors

Analogous colors are hues that are adjacent on the artists' spectrum. *Analogous color groupings contain two primary colors but never the third.* Analogous colors are among the most frequently used in design. Analogy affords the designer an opportunity to add colors to a design without changing the balance of an already established palette. A composition of red and orange is enriched, not changed, by the addition of red-orange.

Figure 4-5. *Analogous Hues.* Analogous hues contain two primaries, but never the third.

The traditional definition of analogous colors is "a group of colors consisting of a primary color, a secondary color, and any and all hues that lie between the two." In this time-honored definition, a primary hue dominates the grouping. Every color by this definition contains (visually) *at least* 50% of the same primary. Typical groups of this kind are blue, blue-violet, and violet (blue dominant); yellow, yellow-green, and green (yellow dominant); and red, red-orange, and orange (red dominant).

A more generous definition describes analogous colors as *a group of colors that are adjacent on the color wheel and lie between a primary and secondary color.* This broader definition includes color groups in which a primary or secondary color does not actually appear. Green, yellow-green, and yellow-yellow green are analogous, but no primary is present; red, red-red-violet, and red-violet are analogous but no secondary is present. As long as a group of hues is bounded by a primary and secondary color, and does not extend beyond either, the colors are analogous.

No matter how many hues are included within a primary-to-secondary range, the most successful analogous groupings are made up of even intervals of hue. This does not mean that the *placement* of colors in a composition must be in linear order. It means only that the overall composition has the best chance of success when the steps between colors are evenly spaced.

Analogy is not confined to pure colors. Colors that have been diluted in any way can also be analogous. *Analogy is a relationship between hues no matter what their value or saturation.*

Complementary Colors

Complementary colors are hues that are opposite one another on the artists' spectrum. Together, the two are called *complements* or a *complementary pair*. The basic complementary pairs of the artists' spectrum are:

Red and green
Yellow and violet
Blue and orange

In each of these pairs, one half is a primary color and the other half is the secondary that is a mixture of the remaining two primaries. *The three basic complementary pairs are different from each other in the exactly the same way as the primary colors. Neither half contains a hue in common with its opposite.*

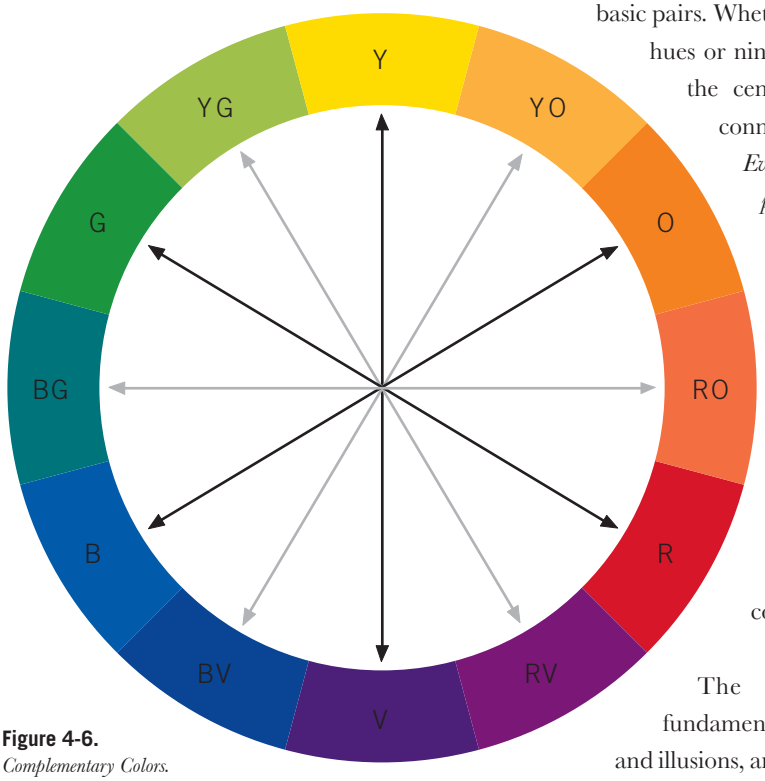


Figure 4-6.
Complementary Colors. Complements are opposite each other at any point on the artists' spectrum.

The complementary relationship is not limited to the three basic pairs. Whether a spectrum is illustrated with six hues or ninety-six, a straight line drawn across the center of the circle from any point connects a pair of complementary colors. *Every complementary pair includes the three primary colors in some mix or proportion.* These additional complementary pairs are less contrasting than the basic pairs because each color contains one primary in common with its opposite. The complementaries blue-green and red-orange, for example, each contain yellow. Yellow-green and red-violet each contain blue, and blue-violet and yellow-orange each contain red.

The complementary relationship is fundamental to color vision, to special effects and illusions, and to color harmony. Every color has an opposite that is its complement. The complementary relationship exists whether colors are pure hues or have been diluted in some way. *No matter what the value or saturation of a color, it maintains at all times a complementary relationship with its opposite.*

Tertiary Colors: Chromatic Neutrals

When a pair of complements is mixed together in a subtractive medium, the middle mix

absorbs (imperfectly, within the limits of the medium) all of the wavelengths of light, so that little or no hue reflects back to the eye. This effect can also be simulated on screen. Another way to characterize complements is to say that they are a pair of colors that, when mixed, produce a sample of no discernable hue. These are the *tertiary colors*.

Tertiary colors are an enormous, almost limitless class of colors. Tertiary means "of the third rank."[1] Tertiary colors are defined as "gray or brown, a mixture of two secondaries," which can be said more simply as "gray or brown, a mixture of three primaries." Tertiary colors are a sort of color soup, containing all possible hue ingredients but with no single hue apparent in the mixture (although it is almost always possible for a trained eye to identify the ingredients).

A color that has been dulled slightly by the addition of its complement is a muted hue, not a tertiary color. Red dulled by the addition of a little green is still red. Tertiary colors are sometimes called *a*chromatic neutrals, but it is more accurate to describe them as *chromatic* neutrals. They cannot be identified as hues, but neither are they a mix of black and white.

The word *brown* is used to describe many of the colors in this family. Brown is not a hue. We say "brown" instead of "tertiary color" because it is common usage and describes a family of familiar sensations. Brown typically has an orange or red overtone, rather than a cool one. Colors are not more or less brown, but browns can be more or less red, or orange, or yellow, or even more blue, green, or violet.

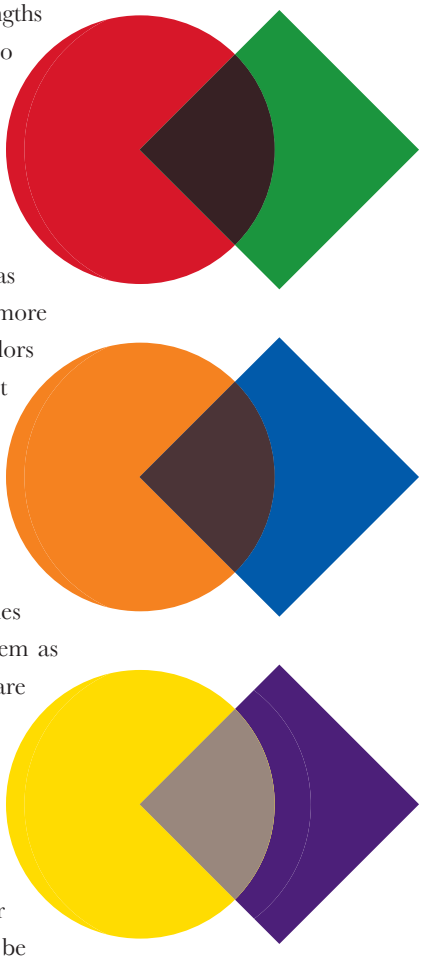


Figure 4-7.
Mixing Complements. When complementary colors are mixed to their midpoint, the result is a sample of no discernible hue.

Black, White, Gray

Black and white are *achromatic*. They are without hue. Absolute whites and blacks exist only in the medium of light. Light can be measurably white, and a total absence of light can be completely black. Subtractive media for black and white are not perfectly achromatic. Even the best quality colorants contain some suggestion of hue.

True grays, or mixtures of black and white, are also achromatic. Subtractive colorants for gray also contain a suggestion of hue. Grays are generally categorized as warm or cool, rather than as "orange" or "pink" grays, or "blue" or "green" ones. When warm and cool grays are placed next to each other, the presence of hue in each becomes immediately apparent. Even when a gray seems truly achromatic, placing it next to an area of color will cause it to lose its apparent neutrality and take on a suggestion of warmth or coolness.

Value

Value refers to relative light and dark in a sample. Hue is circular and continuous, but value is linear and progressive. A series of steps of value has a beginning and an end. *Value contrast exists whether or not hue is present.*

Value is first and most easily understood as a series of steps from black to white. White is the highest possible value. Middle gray, the midpoint between black and white, is a middle or medium value, neither dark nor light. Black is the lowest possible value. A *value scale* is a series of ever-doubling steps that move between the poles of dark and light. Each step in a value series is a middle interval between the two on either side of it: half as dark as the one before it, and twice as dark as the one following.

Figure 4-8.
Value Scale. A value scale moves from dark to light.

Value and Image

Only value contrast makes objects distinguishable from their background. Hue and saturation are not factors in the perception of image. The ability to discriminate objects from their background, or figure-ground separation, depends only on dark-light contrast. Black-and-white drawings, printed pages, and film images are completely understandable. People with a deficit in color vision are functional in a seeing world because "color-blind" really means "hue-blind."

The degree of contrast between light and dark areas determines the strength, or graphic

quality, of an image. Black and white, the polar extremes of value contrast, create the strongest images. Differences between form and their background may be further emphasized by contrasts of hue or saturation, but difference in value is the only factor in the ability to see a distinct edge between colors.

High-contrast images are not always desirable. Strong contrasts of dark and light induce lateral inhibition, and that means that the eye is working. Viewing extremely high-contrast images over a sustained period can cause eye fatigue. Superhighway signs in dark green and white, rather than black and white, provide good contrast at a slightly reduced level, decreasing the risk of eye fatigue—and accidents.

The closer in value an image is to its background, the harder it is to see. A light green frog on a light green leaf is invisible to predators, but resting on a dark green leaf, the frog is easy pickings. When there is no value contrast at all—like an igloo in a snowstorm—there is no image at all.

Figure 4-9. *Value and Image: Graphic Quality.* The strongest images contrast sharply in value with their background. Images are less distinct when the two are similar in value.

Line is an elongated area of color that has value contrast against its ground. There are thick lines and thinner ones, broken lines and lines of varying width, but all lines have the same two attributes: great length in relation to little width, and value contrast against the ground. Line has tremendous visual power. When blocks of color are similar in value they are difficult to make out, but the thinnest contrasting line between them creates an immediate separation.

Transposing Image

Value contrast makes objects distinguishable from their background. The placement of different values relative to one another *within* an image give it individual identity.

In order to transpose an image from one color to another, the number and placement of values within the two images must be identical. If there are five values present in the first image, there must be five in the second, and the placements of light, medium, and dark areas must be the same in both. Scrambling the placement of values within an image creates a new, different one.

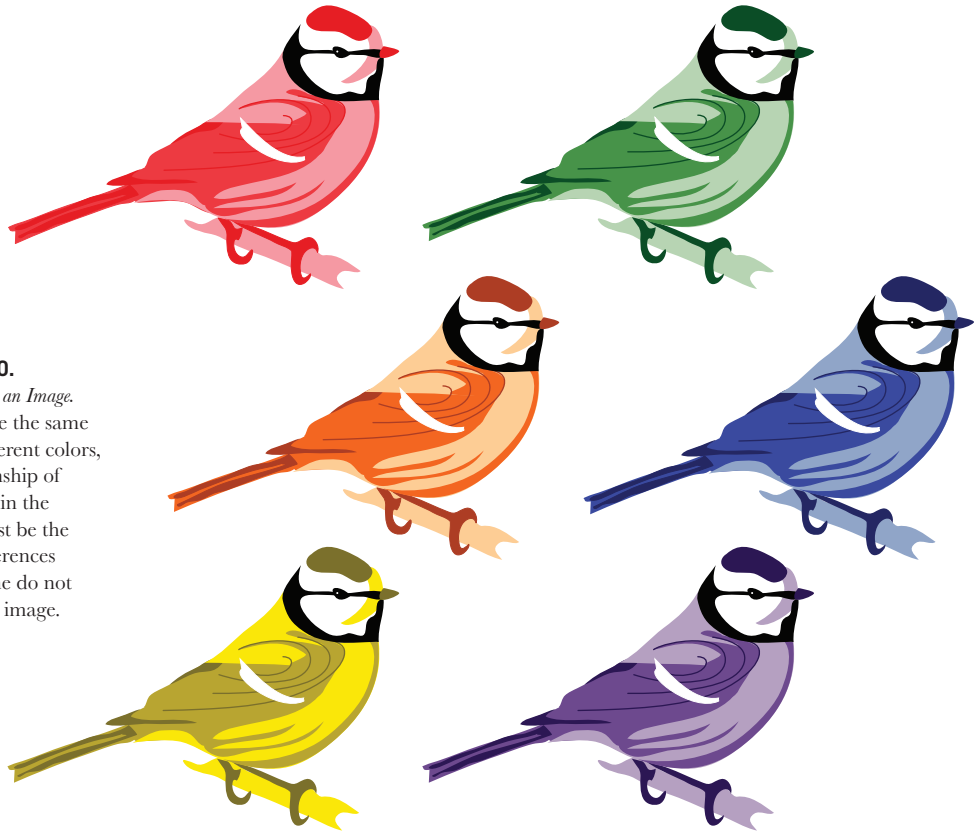


Figure 4-10.
Maintaining an Image. To illustrate the same bird in different colors, the relationship of values within the images must be the same. Differences in hue alone do not change the image.

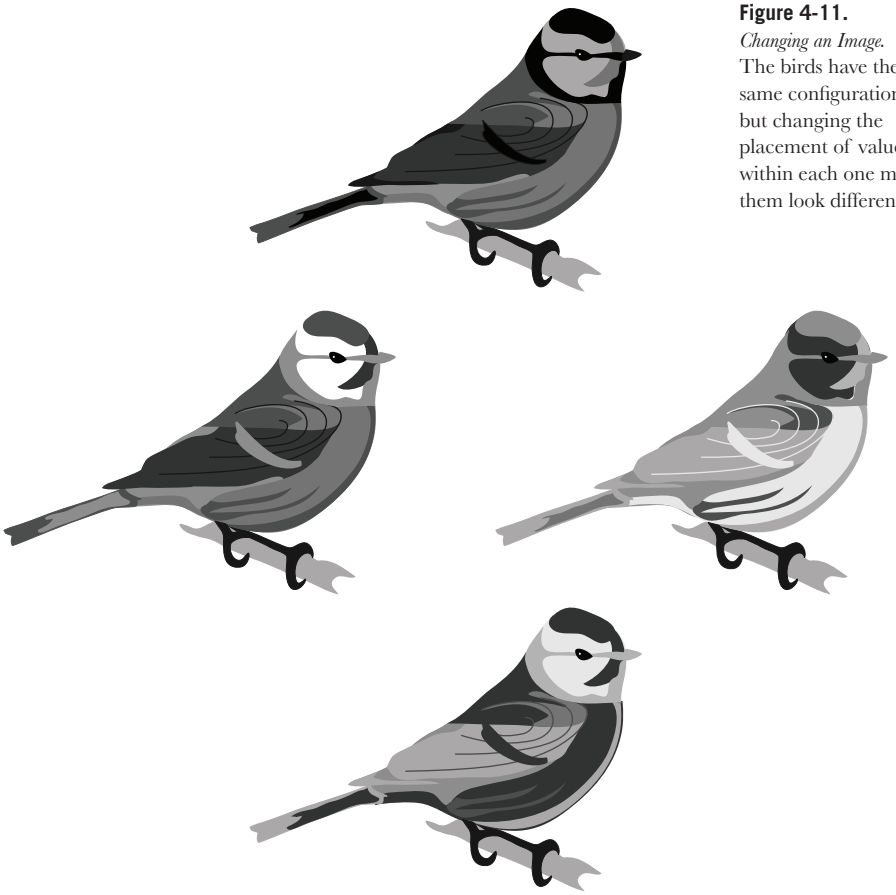


Figure 4-11.
Changing an Image.
The birds have the same configuration, but changing the placement of values within each one makes them look different.

Pure Hues and Value

Value is also associated with the idea of *luminosity*. A hue that is luminous reflects a great deal of light, appears light, and is high in value. A nonluminous hue absorbs light, is dark, and is low in value. It is immediately apparent that each of the six spectrum hues is at a different level of value. Yellow is lightest by far, and violet the darkest.

It is also apparent that red, orange, green, and blue are darker than yellow and lighter than violet. What is less apparent—but important to understand—is that *the saturated colors are*

not at evenly spaced steps of value. The value difference between yellow and green, for example, is much greater than the difference between blue and green. Blue and green are closer in value. The artists' spectrum illustrates colors at evenly spaced intervals of *hue*, but *not* at evenly spaced intervals of value.

An argument is sometimes made for expanding the spectrum with more intervals in the yellow-to-blue and yellow-to-red ranges than in the blue-to-red. Yellow is much lighter than blue, and the great difference in their values makes it possible to set up many perceptible steps between them. The same is true for yellow and red. Blue and red are closer in value, and the number of perceptible steps that can be established between them is fewer.

A spectrum illustrated in this way has many more steps in the yellow, orange, and green ranges than in the red, violet and blue. This creates an immediate problem in establishing complementary pairs. So many possible intervals can be illustrated between colors that contain yellow that they might actually end up opposite each other on a circle.

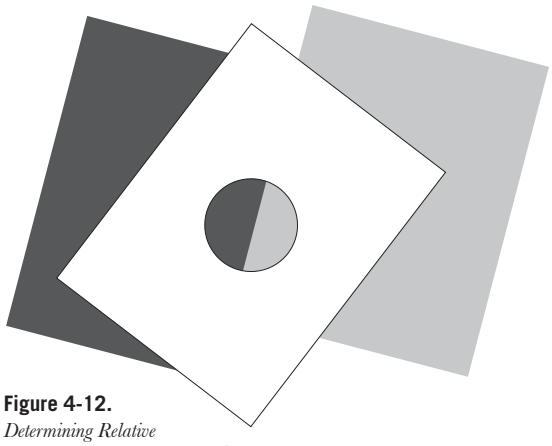


Figure 4-12. *Determining Relative Value.* A white frame that isolates samples is helpful in making value comparisons. Any opaque white paper with a small hole cut in it works well.

The purpose of the spectrum is to illustrate the full range of visible hues. No matter how many intervals are inserted between yellow-to-red and yellow-to-blue, *no new hue is introduced*. To say that there are more hues between yellow-to-red and yellow-to-blue than between red-to-blue misunderstands the nature and purpose of the spectrum.

Tints and Shades

Only occasionally are colors used at full saturation. In most situations, hues are diluted in one or more ways. The simplest way to dilute pure colors is to change their value by making them lighter or darker. A *tint* is a hue that has been made lighter. A *shade* is a hue has been made darker.

Tints are sometimes called hues with white added; shades are sometimes called hues with black added. "Added black" or "added white" is not meant as a recipe for paint mixing.

"Added white" is another way of saying "made lighter," and "added black" is another way of saying "made darker."

Tinting makes a hue more light-reflecting. A great deal of added white creates tints that are just barely identifiable as their original hues. A small amount of white results in a strong tint that is brilliant and light-reflecting, often creating a more intense color experience than that of the original saturated hues. Violet, the darkest of the pure hues, seems more chromatic when white is added. The same can sometimes be true of blue, green, or red. Strong tints are sometimes mistaken for saturated colors, but *no matter how hue-intense and brilliant a tint may be, it is a diluted hue, not a saturated color.*

Shades are reduced-hue experiences. Black absorbs all wavelengths of light, so adding black reduces light reflectance. Even slightly shaded hues are rarely mistaken for saturated colors. The range of shades is less familiar than tints, but it is just as extensive. Hue can be detected in a sample that appears at first to be completely black by placing it next to another, different black. Any hue that is present in either sample becomes instantly apparent.

Monochromatic Value Scales

A *monochromatic value scale* is a single hue illustrated as a full range of values in even steps, including both tints and shades. Monochromatic value scales are slightly more difficult to illustrate than black-gray-white scales. No one seems to have difficulty imagining (or illustrating) tints of pure colors. The dilution of any color with white makes it more light-reflecting and more visible. Understanding and illustrating shades can be more difficult. Shades of cool colors, like dark blue or dark green, are reasonably easy to

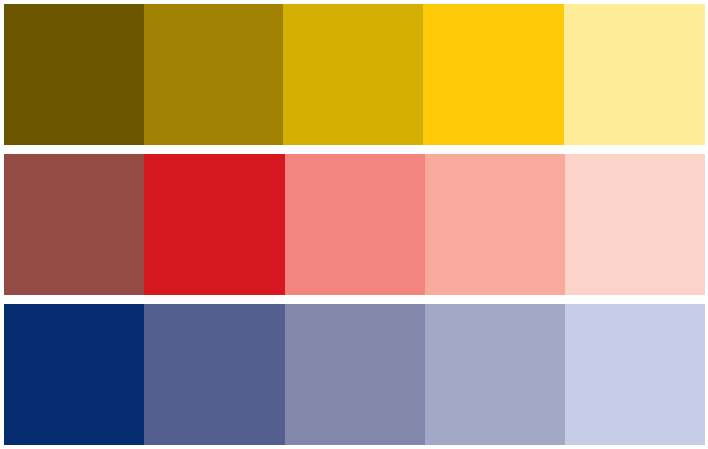


Figure 4-13. *Hue and Value.* Any hue can be illustrated as a full range of tints and shades, from near-white to near-black.

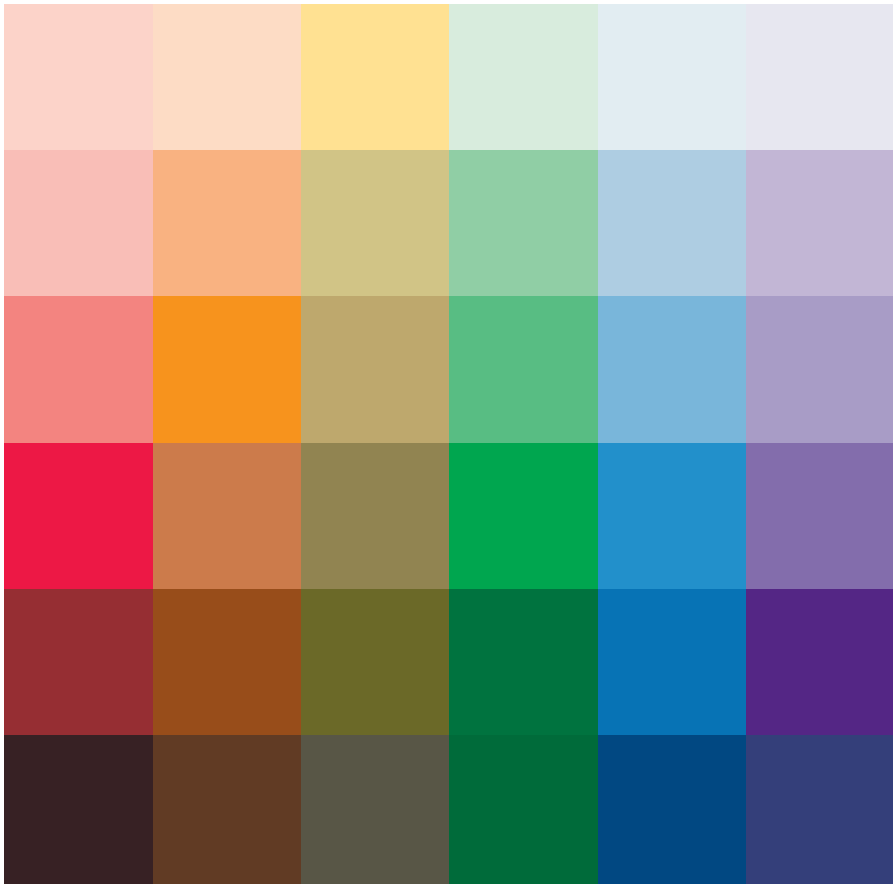


Figure 4-14. *Saturated Hues and Value.* A chart of seven steps of equal value in different hues illustrates how a single step of value may contain both tints and shades. On a limited chart such as this one, some saturated hues may not appear at all. *The accuracy of this image is limited by the constraints of CMYK printing.*

identify, but shades of warm colors can be problematic. Saturated yellow is so high in value that many people find it particularly difficult to associate yellows (and oranges) with their shades. It is hard to imagine the combination of yellow or orange with black because the essential nature of yellow, alone or as a component of orange, is so luminous and opposite to dark. But like all colors, yellow and orange can be illustrated as a full range of values, from near-white to near-black.

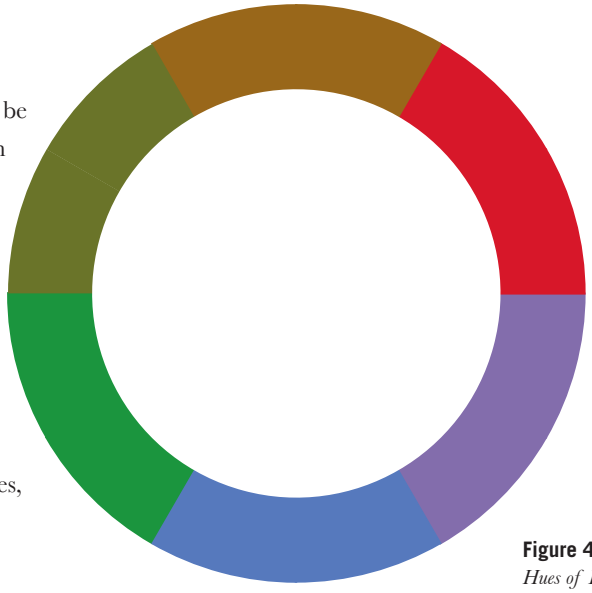


Figure 4-15. *Hues of Equal Value.* A spectrum of hues of equal value requires that some hues be saturated, some tints, and others shades. Is this yellow a pure hue, a tint, or a shade? What is the value of the violet?

Comparing Value in Different Hues

Deciding which of two gray samples is lighter or darker is not very difficult. Deciding which of two samples of a single hue is lighter or darker is also straightforward. Determining which of two *different* hues is lighter or darker than the other is difficult. It is more difficult when one hue is warm and the other is cool, and most difficult when the hues are complementary. Only saturated red and green are close to equal in value. In order for other complementary colors to be made equal in value to each other, one must be made darker or lighter. Adding a great deal of black to yellow can make it as dark as violet. Adding a great deal of white to violet can make it as light as yellow.

Hue is not a factor in "readability." Only value contrast creates an image. When two or more hues are used together, it is the value contrast between them, not the hue contrast, that determines the strength or weakness of the resulting image.

Figure 4-16. *Hue, Value, and Graphic Quality.* Contrast of dark and light, not hue, determines the strength of an image.

Saturation

The third descriptive quality of color is *saturation*, or *chroma*. Saturation and chroma refer to *hue intensity*, or the amount of pure color in a sample. A saturat*ed* color is a color at its fullest expression of hue. It is a color at *maximum chroma*. Saturat*ion* is a comparative term. It describes the contrast between dull and vivid. Saturation, like value, is linear and progressive. The beginning of a saturation scale is a color that is hue-intense. The end step is a color so muted that its hue can just be identified.

Saturation is a color quality that is distinct from value. Shades are already reduced in saturation because they contain black, so it is instinctive to think of muted colors as dark. But any hue or tint can be reduced in its chroma—made less vivid—*without* changing its value. A tint of red-orange can be muted to the color of light clay. The two contain the same hue and are equal in value, but one is a brilliant color and the other is muted.

Brilliant colors have a high level of saturation. A brilliant color may or may not *also* be a saturated color. Hues that contain a high proportion of yellow (yellow, yellow-orange, orange, and yellow-green) are good candidates for both brilliance and high saturation. Brilliant colors that do not contain yellow, or have a lower proportion of yellow, are more likely to be tints. Red-violet diluted with a little white is a brilliant color: a tint with a high level of saturation.

Muted (dull) colors are at a low level of saturation. Any color can be muted in steps from full saturation to very dull. As long as a color sample can be identified by its hue, it *is* a hue; just one that is muted. Dull orange is still orange. When a sample is so muted that its hue is no longer apparent, it has become a tertiary color: a sample that is chromatic but has no identifiable hue. Colors that lie at the threshold between muted hues and tertiary colors provide a classic situation for arguments set off by the way different individuals see and think about colors. Jane's idea of "burnt orange" may be John's idea of "brown."

Saturation: Diluting Hues with Gray

One way to change the saturation of a hue *without* changing its value is to dilute it with a gray of equal value. When a pure orange and a gray of equal value are the "parents" in

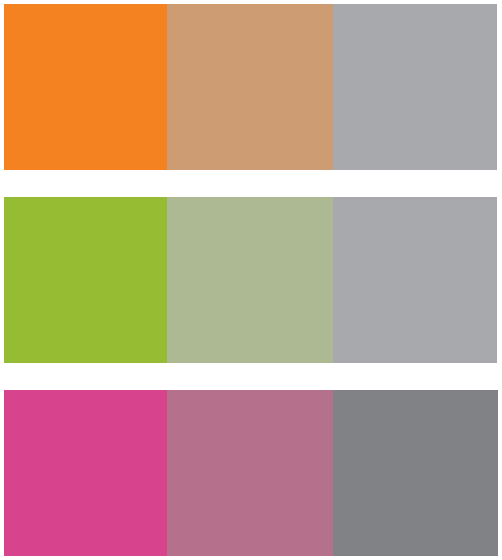


Figure 4-17.
Hues Diluted by a Gray of Equal Value.

a parent-descendant format, the middle interval, "gray-orange," is duller than the orange, more chromatic than the gray, and identical in value to both: neither lighter nor darker than either of its parents.

Saturation: Diluting Hues with the Complement

A second way to dilute pure color is to add some of its complement. Adding the complement to a color to reduce its saturation is a time-honored technique in subtractive media. When a pair of complements is arranged as a series of hue intervals moving toward each other, each step is reduced in saturation until the two reach a center, where each loses its identity. Complements mixed to this midpoint, where neither hue can be distinguished, become a tertiary color: a chromatic neutral.

Colors diluted in this way change in both saturation *and* value. A subtractive color absorbs more wavelengths of light as its complement is added. This effect is simulated on the screen, in colors of light, somewhat less successfully. Yellow and violet have extreme dark-light contrast, so a series of steps between them is a scale of light to dark as well as a scale of

progressively reduced chroma. As the yellow becomes more muted and darker, the violet becomes more muted and lighter. A series between orange and blue displays a similar dark-to-light value pattern, although the value differences are less extreme. Green and red are roughly equal in value, so each becomes darker as it moves toward the other; with the center mix darkest of all.

Theoretical gray is a concept used by color theorists to characterize a perfect tertiary color: one of no discernible hue. Theoretical gray (if it existed) would be created by the mixture of *any* pair of complementary colors. If theoretical gray could be illustrated, the middle mixes of violet and yellow, red and green, and blue and orange would be the same. Visual logic does not allow us to imagine the middle mix of different pairs of complements as the same. Whether a series of steps between complements is illustrated with paints, papers, light, or imagination, every pair of complementary colors moves to a center that is different from other pairs.

Figure 4-18. *Spectrum Hues Diluted by Their Complements.* Each pair reaches a different midpoint.

Some of the most interesting colors result from mixed complements that have been tinted to raise their value. The light neutrals of consumer goods, colors called beige, putty, natural, almond, and so on, are complementary mixes that have been diluted with white.

The natural world is much more a chromatic experience than a black and white one. There are at least 350 varieties of parrots, for example, but only 17 varieties of penguins—and not all of those are "plain" black and white. And although the exuberance of saturated color is easily found—tropical fish, birds, and flowers are examples—muted colors are by far the greatest part of our visual world. Soils and stones, forests, deserts, mountains, rivers, and seas, much of wildlife—all are hues muted by a complement. "Nature shows such mixed colors very elegantly,"[2] says Johannes Itten, as when green fruits ripen to red, or leaves turn from green to brilliant red in the fall.

Tone

There is no really satisfactory definition for "tone." *The Random House Dictionary of The English Language* offers three consecutive (and contradictory) definitions. First, it is defined as "pure color diluted by black or white," which we know as a tint or shade. A second definition states that tone is "one hue modified by another" (as in "this is a blue tone, that is a greener one"). The third meaning is given as "a hue muted by gray." Each definition means a modification of hue, but each means a different *kind* of modification of hue. The first means dilution by changing value, the second means dilution by changing hue, the third dilution by adding gray.

One word cannot mean variation in value, hue, and saturation interchangeably. If the word "tone" is to be used at all it is probably best to go with the most-used meaning: a color of reduced saturation. It seems clearest when used as a *verb.* To "tone down" a color means to mute it, to reduce its saturation. That phrase, at least, is familiar. No one ever says "tone up."

Endnotes

1 *Random House Dictionary of the English Language* 1967, page 1466.

2 Itten 1970, page 50.

Chapter 4 Highlights

- Names for colors are codes for similar sensations. A color is called by the name of its most obvious, or dominant, hue. Each name represents a family of closely related hues. The vocabulary of color includes words that describe interrelationships between colors as well as names for them.
- The basic qualities of colors are hue, value, and saturation, or chroma. Hue means the name of the color. The word "color" is also used more generally to mean the complete experience of hue, value, and saturation. Only six names are needed to describe hue: red, orange, yellow, green, blue, and violet. Black, white, and gray are achromatic.
- The artists' spectrum illustrates the full range of hues in spectral order. Red, yellow, and blue are the primary colors of the artists' spectrum. They are the simplest hues and cannot be broken down visually into other colors or reduced into component parts. Green, orange, and violet are the secondary colors of the artists' spectrum. Each secondary color is an even interval between two primary colors and has one primary in common with each of the others. An expanded spectrum includes the intermediate colors yellow-orange, red-orange, red-violet, blue-violet, blue-green, and yellow-green that are the midpoints between the primary and secondary hues. Many times these colors are referred to, incorrectly, as tertiary colors.
- A saturated color is a hue in its strongest possible manifestation. A saturated hue contains one or two primary colors but does not include a third primary or black, white, or gray.
- Cool colors contain blue or green. Warm colors contain reds, oranges, and yellows. Warmth and coolness in colors are relative qualities: a given color can be made to appear warmer or cooler relative to another.
- Analogous colors lie between primary and secondary colors. Analogous color groupings contain two primary colors but never include the third. Analogy is a relationship between hues no matter what their value or saturation.
- Complementary colors are hues that are opposite one another on the artists' spectrum. A line drawn across the center of a color circle from any point connects a pair of complementary colors. Each complementary pair includes the three primary colors in some mix or proportion. The basic pairs of the artists' spectrum are red and green, yellow and violet, and blue and orange. In each pair one half is a

primary color and its opposite is the secondary made of the remaining two primaries. The complementary relationship between hues is maintained no matter what their value or saturation.

- Tertiary colors are chromatic neutrals made of all three primaries. "Brown" is used to describe many of these colors. Brown is not a hue.

- Value refers to relative light and dark in a sample. White is the highest possible value. Black is the lowest possible value. Value contrast exists whether or not hue is present. A value scale is a series of steps between the poles of dark and light. A monochromatic value scale is a single hue illustrated as a full range of values.

- Only value contrast makes objects distinguishable from their background. The ability to discriminate objects from their background requires no hue. The degree of contrast between light and dark areas determines the strength of an image. The placement of different values relative to one another within an image give it individual identity. In order to transpose an image from one color to another, the placement of values within the two images must be the same.

- A tint is a hue that has been made lighter. Strong tints are sometimes mistaken for saturated colors. A shade is a hue has been made darker. Shades are reduced-hue experiences.

- Each of the six spectrum hues is at a different level of value. Saturated yellow is lightest and saturated violet darkest of the spectrum hues. Determining which of two different hues is lighter or darker than the other is difficult when one hue is warm and the other is cool, and most difficult when the hues are complementary.

- Saturation refers to hue intensity. It is a comparative term that describes the contrast between dull and vivid. The saturation of a hue can be reduced without changing its value by diluting it with a gray of equal value. Adding the complement to a color also reduces its saturation. Colors diluted in this way change in both saturation and value. A color that has been dulled by the addition of its complement but is still identifiable is a muted hue, not a tertiary color.

- Theoretical gray is a concept used by color theorists to characterize a perfect tertiary color: a chromatic sample of with no single discernible hue. If it existed, it would be created by the mixture of any pair of complementary colors. There is no satisfactory definition for "tone."

5

The Instability of Colors

The Instability of Colors / Color Composition / Ground and Carried Colors / Placement and Color Change / Equilibrium / Simultaneous Contrast / Afterimage and Contrast Reversal / Complementary Contrast / Ground Subtraction / Color and Area: Small, Medium, Large

The same color in two different contexts is not the same color.... This means that the identity of a color does not reside in the color itself but is established by relation.

—Rudolf Arnheim

Color is an experience of pure light, as insubstantial and unstable, fleeting and changeable, as a breath of air. Even the achromatic "colors"—grays, blacks, and whites—can alter when there is a change in where they are seen or how they are arranged. Unanticipated color changes can be costly as well as disconcerting. There are real implications for vendors when a color that has been selling well becomes unappealing when it is included as part of a new palette, or when companion products like wallpaper and fabric fail to sell because they do not coordinate when they move from the design studio to a retail setting. The ability to anticipate the sometimes startling changes that take place in colors when they are arranged in different ways, or placed in different settings, and to adjust for them as far as it is possible, is a critical skill for design professionals.

The Instability of Colors

Changes take place in colors for two very different reasons. The first lies in the relationship between colorants and light. Any change in ambient (general) lighting has the potential to alter the appearance of colors. *Changes in ambient lighting affect subtractive colors only*. Additive colors do not change if there is a change in room lighting. They are wavelengths of light that reach the eye directly from an independent source, like a monitor screen or a traffic

light. Screen images (or traffic lights) maintain their hues and hue relationships in daylight, under sodium lamps, incandescent lamps, or any other general lighting. Additive colors are stable under any ambient lighting as long as their source remains stable. They may be sensed as brighter overall in a dark area than in a well-lit one, but the hues, and their relationships to each other, do not change.

Arrangement is the second cause of color instability. Each time that two or more colors are used together the possibility exists that one or more will undergo a shift in hue, value, saturation, or some combination of these because of its placement relative to the others. Color shifts caused by a change in lighting conditions are difficult to manage because products and print media are moved from place to place. But the ways in which colors influence each other by placement are predictable and can be controlled by the designer. *Because placement affects both subtractive and additive colors, it has relevance to every step of the design process, from the first sketches, through computer rendering, to the finished product.*

The concept that every color is subject to being changed by its placement is perfectly expressed by Josef Albers' phrase "interaction of colors." No color is seen alone. Even an object that seems at first to be a single color is vulnerable to change from color interaction, because no color is seen without background. The Vermont barn that is red against green in summer is red against white in winter.

Understanding how colors affect and change each other is more than just a way of anticipating and preventing problems. There is a positive aspect to this instability. Two or three colors can be made to seem like three, or four, or even more, by manipulating their placement. In some kinds of printing, for example, the number of colors is a factor in pricing: the more colors, the higher the production cost. A skilled colorist can make two or three colors seem like many more by placing them strategically against each other. The interaction of colors offers opportunities to fine-tune color combinations in ways that add vitality and interest to color compositions.

Color Composition

A *composition* is something made up of individual parts that have been arranged in such a way that they are understood as a single, complete idea. An essay is made of separate

words; a song is made of single notes. A composition is understood as separate from its setting and from other things around it. A *design composition* is a planned arrangement of forms and colors meant to be sensed as a single visual idea. Colors used together create a *color composition: a group of colors meant to be sensed as a whole.*

A group of colors selected for use together is called (depending on the industry or design discipline) a palette, a colorway, a color story, or some other collective term. Selecting print, screen, or product colors really means creating a color composition, which can be a first step in the design process or a final one. Forms, colors, and their arrangement have equal importance in design. There are no rules about what must come first.

Figure 5-1.
Color Composition. A carpet design produced in different colorways offers a single pattern as different color compositions. *Carpet design by David Setlow. Image courtesy of Stark Carpet.*

Ground and Carried Colors

The background of a color composition is its *ground*. Different industries use different terms for the materials that are used as grounds. Colors printed on fabric or wall covering, for example, are said to be printed on a ground. The background of a carpet or a banner is called the *field*. The paper used in printing is called *stock*. A printer asks if printing will be done on white or colored stock, or on coated or uncoated stock. The monitor screen is also a ground; an empty picture plane waiting for images and colors.

No matter what word is used, "ground" means the background when color relationships are discussed. Colors laid on a ground are *carried colors*. The ground may be an accidental or unconsidered element—a blank white paper or an empty screen—but it is always a factor in the final composition. *Ground establishes the visual reference point for carried colors. It is a critical element in color compositions that is often overlooked.*

The ground is not necessarily the largest area in a composition. The area in a design that is ground is determined by the arrangement of forms, not by color or relative area. Visual cues determine which part of a composition is identified as image or pattern and which part is understood as background.

Negative space is the area within a composition that is not part of the image or pattern. It is the unfilled area around, and sometimes within, the design elements. Negative space is often, but not always, the same area as the ground.

Figure 5-2. *Ground and Area.* Ground is not necessarily the largest area in a composition.

Figure 5-3. *Shaky Ground.* Is a tiger black with yellow stripes, or yellow with black stripes?

In some kinds of patterning it can be difficult (or even impossible) to decide which part of a design is ground and which is carried color. A checkerboard has no identifiable ground; neither does the coat of a tiger. It is not necessary for ground to be a clearly defined area. *Colors will interact whether the ground is obvious or uncertain.*

Placement and Color Change

Three different kinds of color interaction cause apparent change in ground-and-carried-color situations: simultaneous contrast, complementary contrast, and ground subtraction.

All three serve to intensify the differences between colors. *All three are involuntary responses of the eyes, and all three occur in both subtractive and additive color compositions.*

Equilibrium

Equilibrium is a physiological state of rest that the eyes seek at all times. The eyes are at rest when the primary colors of light—red, green, and blue—are within the field of vision. The artists' primaries red, yellow, and blue reflect these wavelengths, as do the process printing primaries cyan, magenta, and yellow. The presence of any of these sets of primary colors in the visual field will bring the eyes to a state of equilibrium.

Figure 5-4. *Equilibrium.* The presence of three primary colors in any form allows the eyes to be at rest. *This is Elizabeth Eakins' carpet design "Sea Ranch"©1993.*

It is not necessary for the primaries to be present as individual colors for the eye to reach a state of rest. Any number of combinations and mixtures will allow the eyes to reach equilibrium: three primaries, or a pair of complements, or two secondary colors, or a hue diluted by its complement (a tertiary color). The three colors do not have to be equal in area. A green tree with one red apple is as effective in providing equilibrium as a red-and-green checkerboard.

Equilibrium is reached most easily when the primaries are mixed together into muted hues. The slightest dulling of a pure color makes it less stimulating to the eye. The popularity of "earth" colors, which are hues muted by the addition of their complement, may derive from the fact that they are genuinely, physically, restful. The eyes will always seek the most physiologically comfortable pathways in color perception.

Simultaneous Contrast

Simultaneous contrast is an involuntary response that takes place when the eyes are *not* at rest—when only a single hue is present in the field of vision. In this situation the eyes work to generate the missing complement, which appears as wash of hue in any nearby achromatic

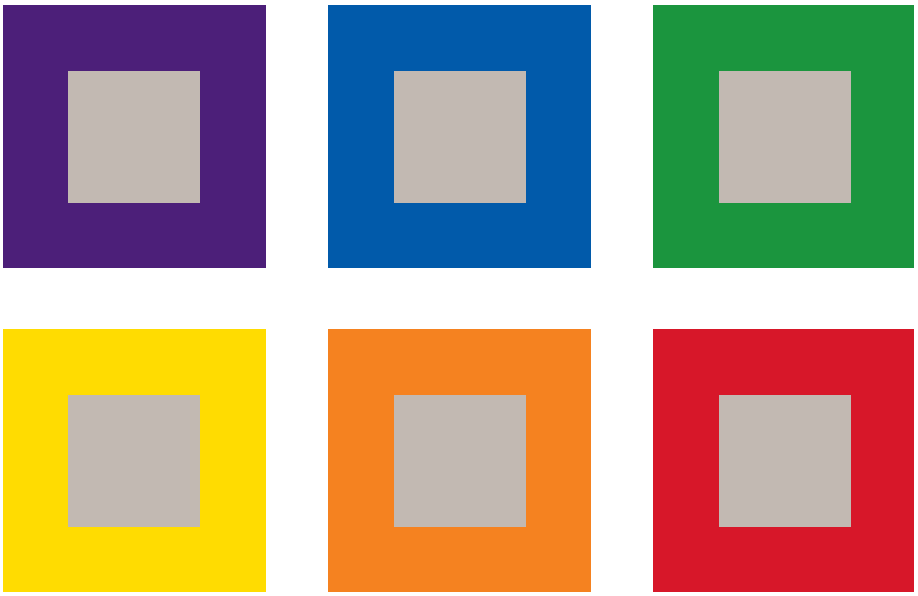


Figure 5-5.
Simultaneous Contrast. The strongest effects of simultaneous contrast occur when a neutral area is surrounded by a stimulating hue. All of the gray squares are the same. What hues can be seen in them?

area. If a single primary color is present, the missing secondary appears. If a secondary is present, the missing primary appears. *For any given color the eye spontaneously and simultaneously generates the missing complement.*

The effect of simultaneous contrast is most apparent when the stimulating hue is a saturated color or brilliant tint, but muted, tinted, or darkened hues will also cause it to take place. *Simultaneous contrast will occur to some extent whenever a single hue is placed on, or next to, an achromatic area.*

Colors and grays interact in real life exactly as they do in the classroom. If the possibility of simultaneous contrast is not anticipated, the results can be disastrous. A true story illustrates how costly this can be.

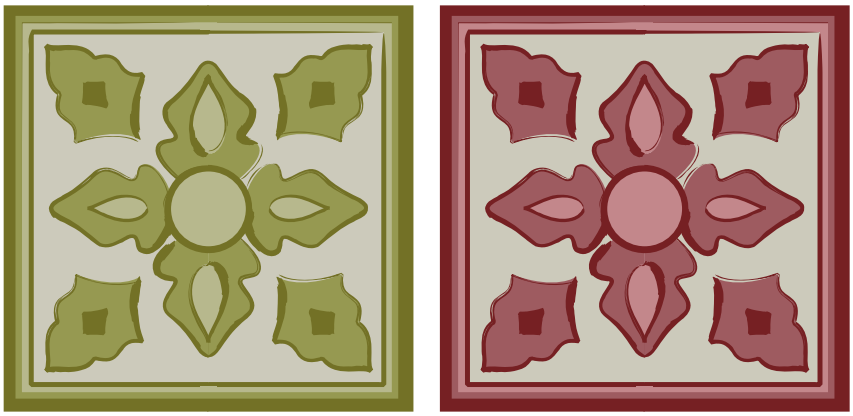


Figure 5-6.
Simultaneous Contrast. Muted colors have a reduced, but still definite, influence on neutrals. The same neutral ground shifts from cool to warm when the carried colors are changed.

An interior designer and his client selected a gray carpet for installation throughout an apartment. The client had been specific, insisting, "I hate blue, I only want gray." The apartment walls were painted "peach" and "terra cotta"—or, more accurately, tinted and muted variations of orange. When the carpet was installed, it was a pale, but definite, blue. The frantic designer brought a sample of the carpet back to the carpet showroom, insisting that the wrong goods had been installed. The sample was identical to the original selection. Simultaneous contrast was the villain. The orange-based wall colors caused the eye to generate the complementary blue, and the expanse of gray carpet was the perfect neutral field on which the complementary blue could be seen.

Simultaneous contrast is a factor in the selection of every neutral (including, and especially, variations of white) that is intended for use with a single hue or close family of hues. Fortunately, it is not difficult to anticipate and counteract unwanted effects. If a green textile is used with a white one, adding a slight green undertone to the white counteracts the red that the eye generates. Without that green, the white takes on a pink cast. A red textile calls for the opposite—a red undertone in the white to counteract the green that the eye supplies. If the designer in the example had used a gray carpet with a warm orange undertone, the problem would not have occurred.

Nearly all situations in which three primaries are present in the visual field allow the eye to be at rest, but compositions with blocks of very brilliant colors can be an exception. Vivid

hues used together can at times deliver such strong, separate, and contradictory stimuli that the eyes respond to each as if it were a single sensation. The struggle to maintain equilibrium means that the eyes must work, and work hard. The resulting eye fatigue can lead to genuine discomfort, like headache or blurred vision.

Afterimage and Contrast Reversal

Afterimage, or *successive contrast*, is exactly what it sounds like: an image that appears *after* a stimulating hue is taken away. Afterimage is caused by the same response of the eye as simultaneous contrast but takes place under different conditions. Simultaneous contrast can be induced by a brilliant color or a muted one, and the achromatic area that it affects must be adjacent. Afterimage requires a brilliant color stimulus and a nearby, but

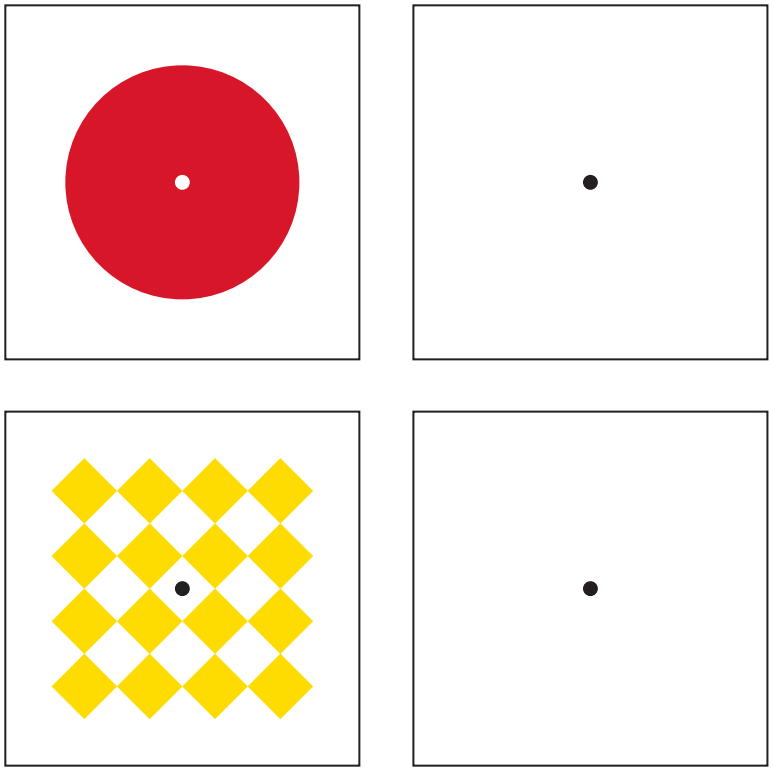


Figure 5-7.
Afterimage. Cover the lower half of the illustration with white paper. Stare at the red circle as long as possible without blinking. Blink firmly, then look immediately at the black dot in the center of the white square.

Next, cover the upper half of the illustration and repeat the exercise with the diamond design. What happens?

separate, blank white or light surface. The viewer stares hard at the color image, then blinks firmly and looks immediately at the white surface. The image appears ghost-like on the blank area—in complementary colors.

Contrast reversal is a variation of afterimage. In contrast reversal the "ghost" appears as a sort of double negative, with both complementary colors appearing, and in reversed positions. A stimulating pattern might be bright yellow diamonds on a white page. When the image is removed and a white field substituted, pale violet diamonds appear in place of the yellow ones, and the white spaces between them are faintly yellow.

Afterimage also occurs without hue. A black and white illustration viewed in this way will appear with the values reversed, like a photographic negative.

Complementary Contrast

Complementary contrast describes what happens when two colors with a complementary relationship—even the slightest complementary relationship—are used together. Simultaneous contrast takes place when only one hue is present. *Complementary contrast intensifies the difference between two hues that are already present, and already different.* Complementary contrast occurs with every form of color: saturated color, tint, shade, or muted hue.

A saturated color is seen at its maximum hue intensity when it is paired with its complement (or a near-complement). The difference in hue between the two is emphasized, but neither color undergoes any *change*. Red used with green remains red and green remains green; blue and orange together remain blue and orange, and so forth. This is true for saturated colors that are opposite at all points on the spectrum, not just the primary-secondary color

Figure 5-8. *Complementary Contrast.* Complementary colors are emphatically different when they are used together, but neither changes the other.

pairs. The intensity of any hue is enhanced by the presence of its complement. Red-violet is brilliant on its own; add a patch of yellow-green and it positively sings.

The second aspect of complementary contrast is its power to bring out undetected hue. It is with muted hues and their tints and shades that the hue-intensifying power of complementary contrast has its greatest impact. Complementary contrast does not require a brilliant stimulating color. A sample may seem so pale, or so dark, or so dull, that it seems achromatic, but when placed with its complement, or *a color that contains some of its complement*, previously unseen hue becomes suddenly visible. When both halves of a complementary pair are muted, *each* seems to become more hue-intense and more different from the other. No matter how slight, how muted, how "barely-there" the complementary contrast between two samples, the hue difference between them is intensified when they are placed together.

Goethe described the complements as "completing colors."[1] The basic complementary pairs, each made up of a primary and secondary color, are the foundations of complementary contrast. Colors do not have to be exact opposites for complementary contrast to occur.

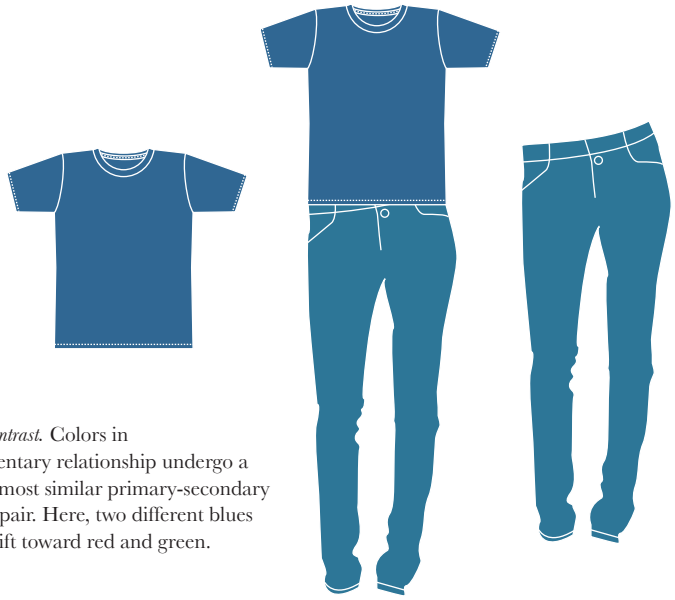


Figure 5-9.
Complementary Contrast. Colors in a part-complementary relationship undergo a shift toward the most similar primary-secondary complementary pair. Here, two different blues used together drift toward red and green.

They can be near-complements (or part-complements) like red-orange and green, or yellow-green and violet. *When colors other than the primary and secondary pairs are in a complementary (or part-complementary) relationship, they undergo a shift in hue toward the most similar primary-secondary pair.* Two "navy blue" samples, placed together, may suddenly appear greenish-navy and purplish-navy. The hue contrast that becomes visible is the red undertone of one and the green undertone of the other. *The eye seeks not only equilibrium, but also the simplest and most "completing" hue relationship.*

Ground Subtraction

Simultaneous contrast and complementary contrast are responses of the eyes to the complementary relationship. Each intensifies differences between samples that are already unlike. *Ground subtraction* is completely different. It takes place when a ground and its carried colors have *qualities in common—and also qualities that are different.* Whatever qualities that are shared by a ground and its carried colors are reduced; at the same time, differences between them are emphasized.

Figure 5-10. *Ground Subtraction of Value.* A middle gray appears darker on a light ground and lighter on a dark one.

For example, a middle gray placed on dark gray appears lighter: the darkness that is common to both is reduced and the remaining lightness intensified. The same middle gray placed on light gray appears darker. The lightness it shares with its ground is reduced, while the remaining darkness is emphasized. It is the "like-unlike" relationship between the ground and carried colors that generates change.

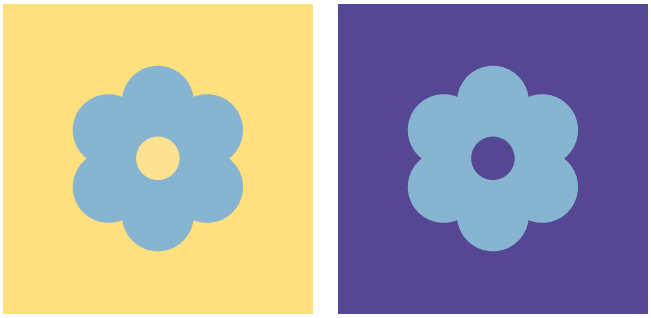


Figure 5-11. *Ground Subtraction of Value.* The same blue appears darker on yellow and lighter on deep violet.

The effect of altered value is the same when hue is present. A middle-value blue placed

on yellow-orange appears darker than the same blue placed on spectrum violet. *Any ground subtracts its own qualities from colors it carries. The more similarities a color has with its ground, the more apparent their differences will be.*

Figure 5-12. *Ground Subtraction: One Color as Two.* Colors appear to change when they are placed on grounds that share different aspects of the own qualities.

In theory, primary colors will not change in hue by placement, although they can be altered in apparent value. Secondary and intermediate colors (and all hues between them) will change, at times quite dramatically, when they are placed on grounds that share *different* aspects of their own qualities. Red-orange appears more yellow-orange by placing it on red (the common red is subtracted), or more red by placing it on yellow-orange (the common yellow is subtracted).

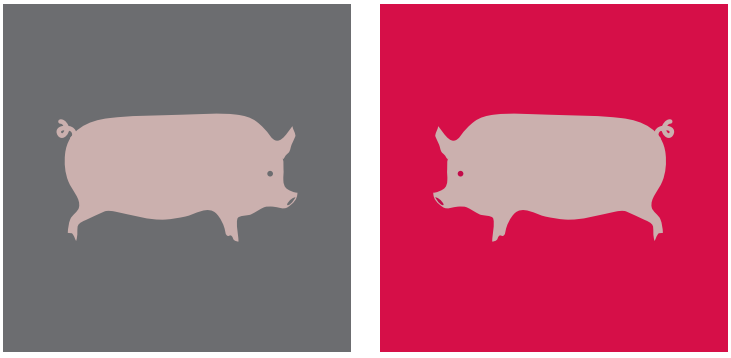


Figure 5-13. *Ground Subtraction of Hue.* A red-gray placed on gray appears more red. Placed on red, it appears much grayer.

The difference between the two carried colors seems even greater because both hue and value are affected. The red-orange appears lighter on the red; darker on the yellow-orange.

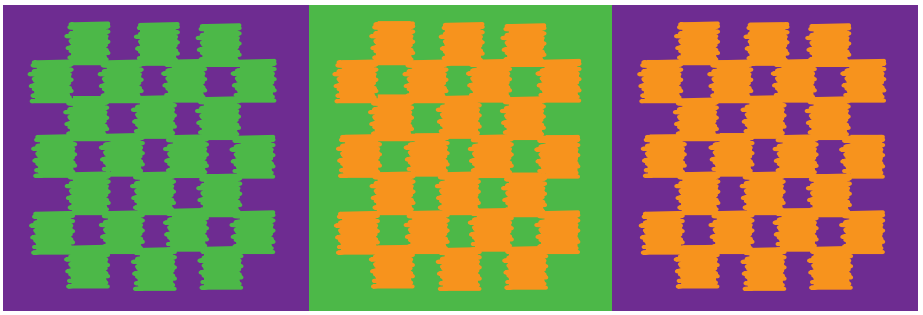


Figure 5-14. *Ground Subtraction of Hue.* How many colors are actually printed?

A muted color is more vivid on a grayed ground and more muted on a chromatic one. When a mix of gray and a brilliant hue is placed on each of its "parents," the mix seems very gray on the vivid ground; brighter and more chromatic on the gray one.

The more complex a color is—the more elements it contains—the more likely it is to be affected by colors around it. The changes that take place in complex colors are not necessarily more dramatic than those that take place with simpler colors. Change is simply more *likely*, because the more "ingredients" that are present, the greater the number of possibilities that it will have elements in common with (and also different from) its ground. A neutral "beige" mixed from blue, orange, and white can be made to appear duller or brighter, darker or lighter, more cool (or blue) or more warm (more orange) by placing it on different grounds.

Figure 5-15. *Ground Subtraction: Two Colors as One.* Different colors can be made to seem the same by placing them on grounds of opposing qualities.

Finally, color shifts can be extreme when *both* ground subtraction and complementary contrast are in play. If a brown square, made of a mixture of

blue and orange, is placed on an orange ground, the orange component that is common to both is reduced by ground subtraction. The blue is strengthened twice: first, as the color remaining after the orange is reduced, and again by complementary contrast against the orange ground. The same brown square, placed on a blue ground, looks very much more orange. The orange component is made stronger twice: first, as the color remaining after the blue is reduced, then by complementary contrast against the blue ground.

The principle of ground subtraction can be used in reverse to make different (but similar) colors appear to be identical. A green and a yellow-green can be made to look the same by placing them on grounds of *opposing* qualities. Yellow-green placed on yellow appears more green; the common yellow is reduced. Green placed on blue-green appears more yellow; the blue that is present in both is reduced.

Color and Area: Small, Medium, Large

A different kind of shift takes place in subtractive colors when a color that has been selected from a small sample, like a paint chip or fabric cutting, is applied to a large surface. The direction (placement in space) of a large color plane affects whether it will read as lighter or darker; or more muted or more chromatic, than it does as a small chip. Changes of this kind are not color interactions. They are differences seen in a single color, under the same lighting, but in different extents.

Typically, light reaches surfaces from above and at an angle. When light reaches a wall surface or a floor from the conventional direction, floors and walls appear lighter, and ceilings darker, than a small sample of color in the hand. The *quantity* of light reflected from a large surface is greater than the quantity of light reflected from a small sample. Charcoal gray, near-black on a paint chart, becomes a medium gray when applied to the exterior of a house. Carpet, medium beige in a small cutting, becomes lighter on the expanse of a floor. Conversely, a color selected for a ceiling must be made two or three steps lighter than a small sample to achieve the same effect, because less light that is directed conventionally—down from above—reaches a ceiling.

Colors also appear more *chromatic* on a larger plane. The presence of hue in chromatic neutrals (like "beige") is more visible in larger scale. Light and dark colors are equal in this.

An apparently neutral dark brown applied to a large wall is seen as both lighter and more chromatic — more red-brown, or more orange-brown — than in a small paint chip. A color experienced as "pale ivory" in a sample may read as true yellow on a large plane. A vivid color, mindlessly cheerful in a small dose, can be overwhelming as a painted wall. Adjusting a color selection to compensate for the difference between a small sample and the same color in a large area is an issue faced more in architecture and interior design than in other design fields.

Endnotes

1 Goethe 1971, page 55.

Chapter 5 Highlights

- Any color can alter when there is a change in how or where it is seen. Any change in general lighting has the potential to alter the appearance of subtractive colors, but does not change additive colors. Whenever colors are used together one or more may appear to change because of its placement relative to others. Changes in placement affects both subtractive and additive colors.
- A color composition is a group of colors meant to be sensed as a whole. The background of a color composition is its ground. Colors laid on a ground are carried colors. Ground establishes the visual reference point for carried colors. The area in a design that is the ground is determined by visual cues. Negative space is the area within a composition that is the unfilled area around or within the design elements. Negative space is often, but not always, the same area as the ground.
- The eyes constantly seek a state of rest, or equilibrium. Equilibrium is reached when the primary colors of light; red, green, and blue, are within the field of vision. Any number of combinations or mixtures of the primary colors will allow the eyes to reach equilibrium.
- Simultaneous contrast means that for any given color the eye spontaneously and simultaneously seeks its complement. Any adjoining achromatic area takes on aspects of that complement.
- Complementary contrast intensifies any complementary relationship that exists between a ground and a carried color.
- Ground subtraction takes place when a ground and carried colors contain elements in common as well as elements that are different. Qualities that are shared are reduced and differences between them are intensified.
- Different (but similar) colors can be made to seem identical by placing them on grounds of opposing qualities. The more complex a color, the more subject it is to change by placement.

6

Illusion and Impression

Optical Illusions / Color Illusions / The Illusion of Depth / Spatial Effects of Colors / Transparence Illusion / Fluting / Vibration / Vanishing Boundaries / Luminosity / Bezold Effect / Optical Mixes

Reality is an illusion, albeit a very persistent one.

—Albert Einstein

An optical illusion is a seemingly magical event that takes place when an image is misperceived or misinterpreted by a viewer. An illusion can only be *seen*. It cannot be confirmed by physical measurement or by any of the other senses. It is a purely visual experience that is mistaken for an objective reality.

Optical Illusions

The simplest illusions are initiated by involuntary responses of the eyes to overstimulation by something specific, like a single hue or extreme contrast. Afterimages and the color shifts caused by simultaneous contrast, complementary contrast, or ground subtraction are examples of this kind of direct response. Other aspects of vision, like lateral inhibition or involuntary eye movements, can also play a part. Traditionally, these have been considered to be illusions with a *physiological* basis. Illusions have traditionally been considered *cognitive* (knowing) when they take place because the observer makes unconscious assumptions or inferences about something when the actual image is different.

Seeing requires an elaborate interplay between the eyes and the brain. More complex illusions occur because an image is interpreted by a brain that is processing visual

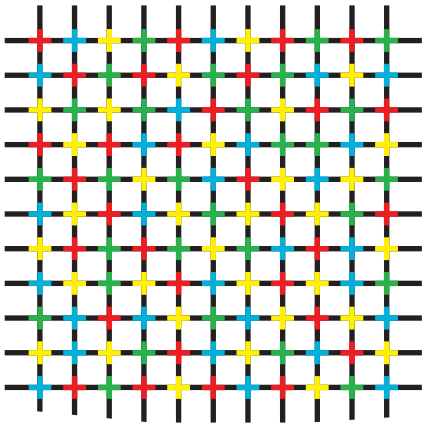


Figure 6-1. *The Importance of Arrangement.* In an illusion called neon color spreading, small areas of brilliant color are introduced into a repetitive pattern of black lines on a white ground, and the color seems to spread into the white space.

information—and processing it inaccurately. The current thinking among vision scientists is that *all* illusions have a physiological basis. All are caused by a disturbance to the flow of information between the eyes and the brain, although some engage more of the reasoning parts of the brain than others.

No matter how it originates or what it is called, every illusion takes place within a limited set of circumstances. Illusions depend on more than interaction between the human eye and brain. They depend equally on the *arrangement* of design elements: forms, lines, and colors.

Color Illusions

Color illusions occur when colors that have specific relationships are arranged in ways that mislead the eyes. Color illusions exemplify the current view: they are both a physiological response to and an interpretation (or, more accurately, a misinterpretation) of what is seen.

A true illusion fools the eye absolutely, and many effects that are called illusions do not meet this standard. Some, like an impression that printed colors are glowing on a page, might more accurately be called special effects, or even near-illusions. But no matter what a deceiving image is called—an illusion, a near-illusion, an impression, or a special effect—each has the same requirements: a set of colors with specific qualities and interrelationships, and that these colors be arranged in a particular way.

The Illusion of Depth

Whether a drawing is destined for print or a Web page, a great deal of graphic art is concerned with communicating three-dimensional objects—cars, soda bottles, articles of clothing, buildings—on the two-dimensional plane of a page or screen. An illustration of depth on a two-dimensional surface is not an illusion. Even the most skilled drawings that indicate depth are understood as *representing* three dimensions, without any expectation that

the depth will be seen as real. An illusion of three dimensions takes place only when a two-dimensional image is sensed by the observer as having actual depth. These illusions, made possible by technology, are a bridge between drawing and the real world.

Real-world depth perception depends first on binocular vision, or using two eyes. Only width and height are seen directly. Each eye receives two-dimensional information from a slightly different position and sends a slightly different image to the brain, which interprets the two messages as a three-dimensional situation.

Depth perception also depends on the assumptions of *perceptual constancy*. Perceptual constancy means that things are understood as remaining the same even as the eyes receive changing images. A familiar object seen as very small is understood to be farther away, not as reduced in size. Perceptual constancy provides important information, called *depth cues*, that complete the information provided by binocular vision.

Impressions of depth on a page or screen are created by using one or more of a set of drawing conventions called *pictorial depth cues*. Some important pictorial depth cues are overlap, linear perspective, shading, shadow, texture gradients, and atmospheric perspective.

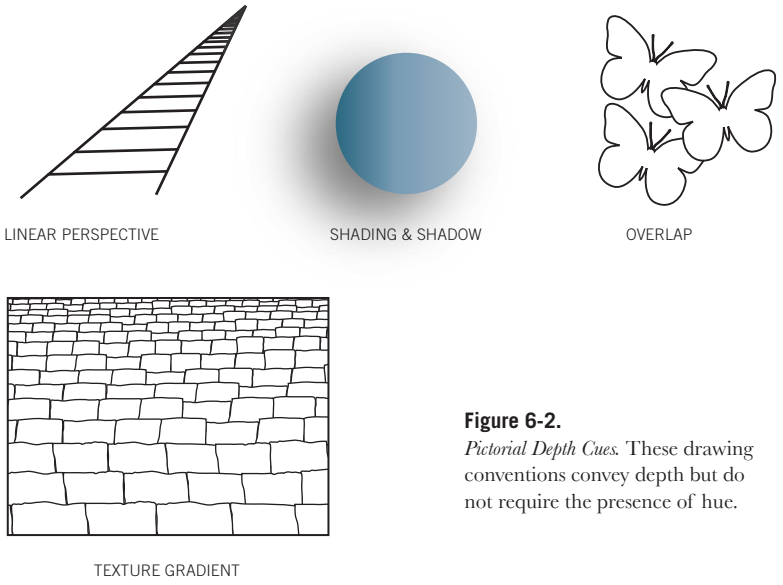


Figure 6-2.
Pictorial Depth Cues. These drawing conventions convey depth but do not require the presence of hue.

Atmospheric perspective (or *aerial perspective*) is the only pictorial depth cue that has a hue component. The farther away an object is in the real world, the more air and dust intrude between it and the observer, so that faraway objects appear less distinct. And, because the atmosphere scatters more of the blue wavelength of sunlight than the other colors, distant objects also appear slightly bluer than closer ones.

Figure 6-3. *Atmospheric Perspective.* Air and dust blur distant contours, making them increasingly indistinct. Scattering of the short wavelengths of sunlight by the atmosphere makes them appear slightly bluer. *Photograph by Gwen Harris.*

The more information—the more pictorial depth cues—provided by a drawing, the more convincingly three-dimensional objects can be represented. Unmoving images are limited to depth cues that represent three-dimensional space. They do not fool anyone into mistaking them as three-dimensional. It is always clear that the surface is flat. In order for an image to be perceived as three-dimensional on a two-dimensional plane, the additional depth cues of time and motion are needed.

Motion parallax (or *movement parallax*) is a real-world depth cue that includes motion and time. When a viewer is in rapid motion, near objects appear to move by in a blur while far objects seem to move slowly. Distant signs can be read from a moving train, but signs close to the track pass too quickly to be read. Objects are understood to be closer or farther away by a comparison of how fast they seem to move relative to the observer's position. Unlike a page, a screen image can include both motion and time. Motion parallax is a depth cue that can be simulated on-screen.

Motion graphics are drawings that include pictorial depth cues, motion, and time. Motion graphics can be magical. One additional element helps to complete the illusion of an alternate reality. *The most convincing illusions of depth result when a full array of depth cues is reinforced by the spatial effects of colors.*

Spatial Effects of Colors

One way to characterize the nature of colors is to think of them in terms of "near" or "far." Some colors have inherent qualities of "nearness" or "farness." Light blues seem to move away, and warm colors of any value close in. But in color compositions, the hue, value, and saturation of each form *in relation to its ground or surrounding colors* influences whether it is perceived as advancing or receding in space. Independent objects (and drawings of them) can be made to appear nearer or farther away by making changes in their hue, value, or saturation.

In general:

> *Hue*: Warm hues advance relative to cooler ones. Reds, yellows, and oranges appear to come forward relative to blues, greens, and violets.
>
> *Saturation*: Brilliant colors (saturated colors and strong tints) appear to come forward relative to muted colors or grays.
>
> *Value:* In a simple image of a figure against a ground, the figure will advance no matter whether it is lighter or darker than its background. A dark figure will come forward on a light ground and a light figure will come forward on a dark ground.

This interpretation of "near and far" is an aspect of visual processing called *figure-ground perception*. Figure-ground perception refers to the eye's ability to separate images into parts based on the contrast of dark and light: the basic process of seeing. The brain sorts visual information into dark and light areas and determines their edges, then translates this into a determination of which part of the image is figure (meaningful, seen as coming forward), and which is background (less meaningful, seen as receding.)

When figures of different value are laid against the *same* ground, however, the difference between each in relation to the ground determines which will advance and which will recede. *A figure with greater value contrast with the ground will advance and one closer in value to the ground will recede.* The greater the value contrast between a figure and its ground, the more sharply defined and nearer it seems. The depth cue of atmospheric perspective — that forms become less distinct as they move into the distance — signals that less contrasting figures are farther away.

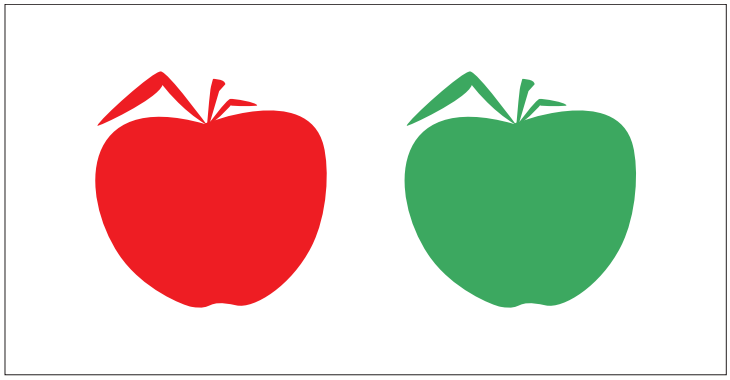


Figure 6-4.
Spatial Effects of Warm and Cool Hues. Warm hues advance relative to cooler ones.

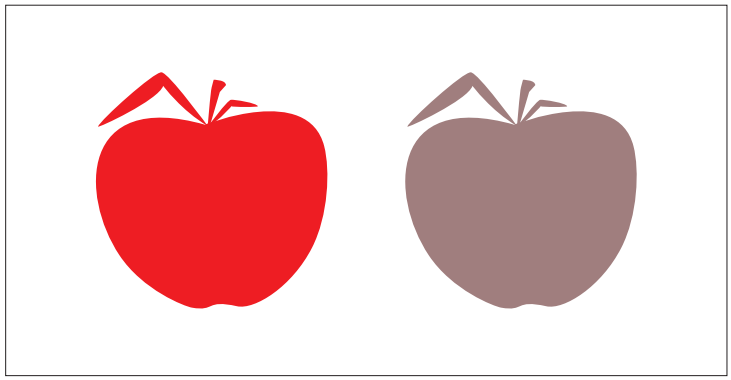


Figure 6-5.
Spatial Effects of Saturation. Brilliant colors come forward relative to muted ones.

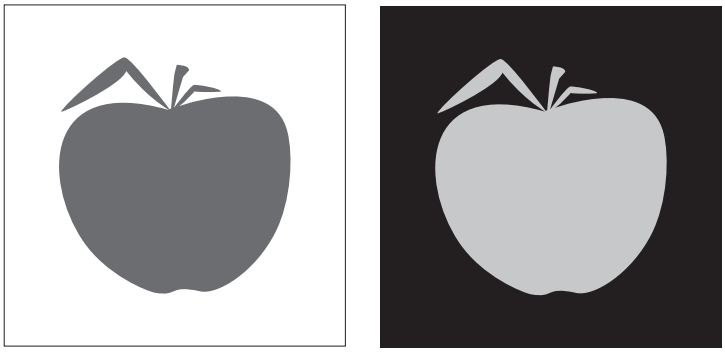


Figure 6-6.
Value and Figure-Ground Perception. A dark figure comes forward on a light ground; a light figure comes forward on a dark ground.

Dark or light color also influences how the size of a figure is perceived. The same object in a light color seems larger than it does in a darker color. A white bear would seem bigger than the black one—standing right in front of you, just ahead on the trail—in the same spot.

Color alone, however, does not create an effect of depth. It is a secondary indicator that supports one or more of the pictorial depth cues. The spatial effects of colors in a drawing can be reduced, or even reversed, by pictorial depth cues. Perspective and overlap, for example, are stronger indicators than color in conveying an impression of three dimensions.

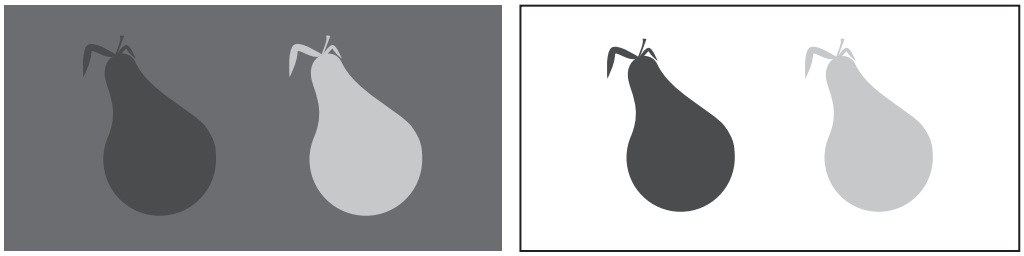


Figure 6-7. *Comparative Value, Ground, and Spatial Perception.* A figure that contrasts sharply in value with its ground advances relative to one that is closer in value to the ground.

The key to understanding whether a colors appears to advance or recede in relation to others lies in the phrase "*all other factors being equal*." Combinations of colors in which all factors are equal except one are rare. The *dominant* quality of a color—its coolness or warmth, high value, or brilliance relative to other colors present—determines whether it seems to move forward or back. In a gray world, yellow bounces forward. In a black world, gray glows with light.

Transparence Illusion

A *transparence illusion* is a three-dimensional illusion that takes place when two opaque colors and an interval between them are arranged in such a way that one color appears to be transparent and lying on top of the other.

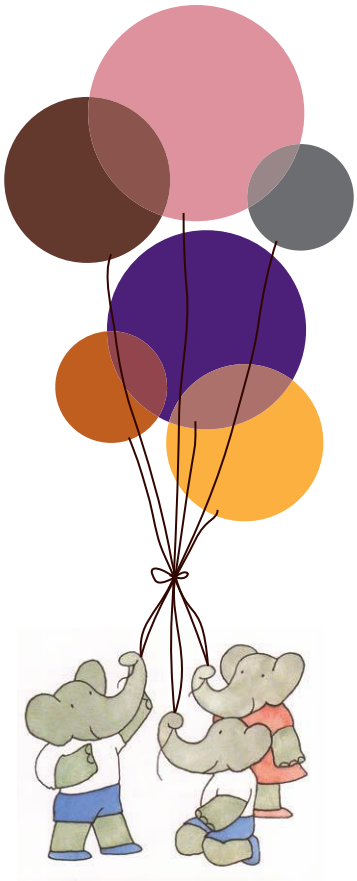


Figure 6-8.
Transparence Illusion.
When a number of colors and intervals between them are arranged in an overlapping way, the effect of transparency follows the guidelines of spatial effects. *Image courtesy of Laurent de Brunhoff. Laurent de Brunhof. Babar's Book of Color. New York: Harry N. Abrams, Inc. 2004*

A transparence illusion depends equally on two ideas: the pictorial depth cue of overlap and the intervals of a parent-descendant color series. A line drawing of overlapping forms suggests three dimensions; add three intervals of color and the suggestion becomes an illusion.

The guidelines of spatial effects determine which color in a transparence illusion will appear to be on top of the other. A color that is high in value, warm, or brilliant will appear to be on top when paired with a darker, cooler, or duller one. The top color will seem to come forward; the bottom color to recede.

The simplest transparence illusions are illustrated by arranging any two colors and an even interval between them in such a way that the middle interval fills any area where the two overlap. Changing the middle step between the parent colors to an uneven interval affects the apparent transparency of the top color. The more similar the color in the area of overlap is to the bottom (or receding) color, the more transparent the top color will appear.

Transparency in color can be a fact. Transparent liquids appear stronger in hue as their volume increases, a phenomenon Josef Albers describes as "volume color."[1] A strawberry milkshake is opaque, the same color in a gallon container as in a cup. Coffee is transparent. In a deep pot it is

Figure 6-9.
Transparence Illusion: Hue. If the middle color is an interval between warm and cool parents, the warmer parent color appears to be on top (transparent), and the cooler one underneath it.

Figure 6-10.
Transparence Illusion: Value. If the middle color is an interval of value between the parent colors, the lighter parent color appears to be on top (transparent), and the darker one underneath it.

Figure 6-11.
Transparency. If the middle color is an interval between chromatic and achromatic parents, the chromatic parent will appear to be on top (transparent) and the achromatic one beneath it.

very dark, but spilled in a saucer it is as light as tea. Some media, like watercolors or four-color process printing inks,, allow light to pass through a colorant and reflect back from white paper underneath. As layers are added, the effect of transparency decreases and the covered areas become deeper in hue. These are true transparencies, not illusions.

Certain hues suggest transparency. Cool hues, especially tints of blue and green, often seem transparent. Hues that are both warm and dark seem more dense and opaque. The medium makes a great difference in whether colors seem transparent or opaque.

Figure 6-12.
Transparence Illusion. Shifting the middle interval closer to one parent or the other alters the apparent degree of transparency of the top color.

Brilliance and transparency are more easily displayed on the screen than on paper, while deep and opaque colors are the reverse: more convincingly shown on paper, more difficult to render in the medium of light.

Fluting

Fluting is a three-dimensional illusion in which a series of vertical stripes of uniform width appear to have concavity, like the channels of a Doric column. Fluting occurs when stripes are arranged in a series of progressive steps of value. Each stripe is an even step of value between the darker and lighter stripe on either side of it. The edges of each stripe appear lighter where it abuts its darker neighbor and darker where it abuts its lighter neighbor. This creates an illusion of shading, the depth cue that give an impression of roundness.

Figure 6-13.
Fluting. Stripes in steps from dark to light create an illusion of concavity.

Figure 6-14.
Fluting. Colored stripes arranged in steps of value seem to have concavity.

Like the Three Bears' porridge, the width of the stripes has to be "just right." If the stripes are too wide, the eye cannot take in both edges at the same time. If they are too narrow, they appear to be lines instead of channels. Wide stripes sometimes give an illusion of convexity, but the illusion is still called fluting.

The effect of fluting is not confined to grays. Any hues can be used to create an illusion of fluting as long as they are arranged as a progression of values. When the same colors are configured as random values, the fluting disappears.

Figure 6-15. *Not Fluting.* Colors arranged in random values have no three-dimensional effect.

Vibration

Vibration is an effect that takes place when blocks of brilliant and complementary (or near-complementary) colors that are equal or very close in value are placed together. Combinations like a medium-value tint of blue and a saturated orange, or a tint of blue-green and a deep pink, will appear to shimmer where they meet, and edges between them are difficult to see.

Vibration is an uncomfortable visual experience. The impression of "shivering" color is the result of a potent combination of factors: the difficulty of finding the edge between forms of similar value, the conflict of trying to reach equilibrium in opposing and brilliant colors at the same time, and the effect of natural eye movements called *saccades.*

The eyes are always in search of visual resolution: they want to see clearly at all times. Saccades are involuntary eye movements that occur constantly and rapidly, but human beings are completely unaware of them. To stand and gaze at a painting may feel like a stationary event, but the eyes are constantly moving, finding first the more interesting parts, then putting them together to form a complete mental picture. Each saccade sends a "snapshot" of a small part of the image to photoreceptors (light receptors) in the brain. If the response time of a photoreceptor is different from the time that the area of the "snapshot" is moving across the retina, that small area of vision will blur. The moment is brief—focus returns quickly—but the rapidly alternating blur/focus sensation can be misinterpreted as motion

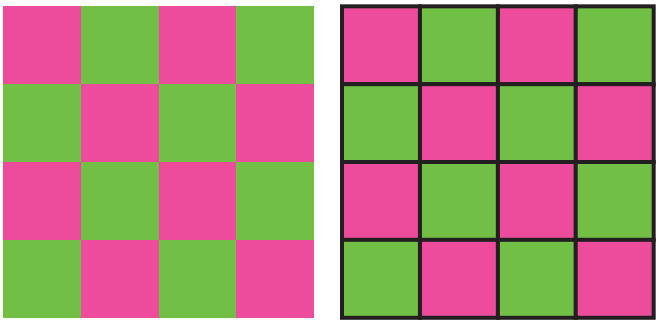


Figure 6-16. *Vibration.* A separating line lessens or eliminates vibration.

on the page. The more difficult it is to focus, the greater the impression of movement.

Palettes of brilliant and strongly contrasting hues have an inherent joyfulness, and these color groups are used constantly in many design fields. Vibration occurs only when adjacent colors are very different, very brilliant, and equal or close-to-equal in value. It can be eliminated by separating the colors so that they no longer touch, either by rearrangement or by adding a value-contrasting line between them.

Vanishing Boundaries

Vanishing boundaries occur when areas of *similar* hue and close value are placed next to (or on top of) one another. Vanishing boundaries are a "soft" effect—each form seems to merge into the other and lose its edges. Vanishing boundaries are not uncomfortable in the same way as vibration. The colors involved can be brilliant or muted, but their hues are similar, not opposed. The eyes do not need to seek equilibrium in two different ways at the same time.

Figure 6-17. *Vanishing boundaries.*

Luminosity

Figure 6-18. *Luminosity.* A gradient halo between a light area set into a darker field suggests glowing light.

A difficulty in finding edges also plays a part in another family of illusions: those in which subtractive colors seem to be luminous, or glowing on a page. Like all illusions and impressions, these effects require that colors with specific relationships be arranged in particular ways.

Effects of luminosity follow the guidelines of spatial effects: Lighter colors move forward relative to dark; more vivid colors move forward relative to dull ones. A glowing light is illustrated *without* hue by setting a relatively small, very light area (of any shape) into a large and darker field, with a "halo" of closely spaced, progressive intervals of value radiating out from the light area into the dark. When the halo is a gradient, or seamless transition from dark to light, the effect is of soft glow. When the halo is a series of intervals just at the threshold of vision, the effect is of a shimmering, light source.

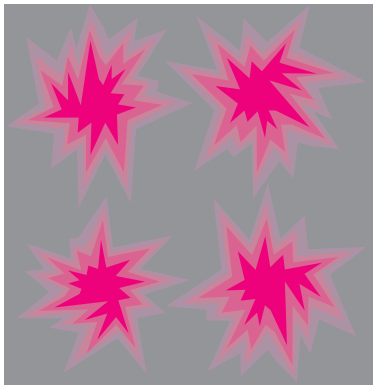


Figure 6-19. *Luminosity.* A gradient halo between warm and cool colors creates an impression that color is glowing on the page. *Photograph by Alison Holtzschue.*

Figure 6-20. *Shimmer.* A pattern of brilliant color separated from its field by close intervals of saturation seems to flicker on the page.

Dark and light areas arranged in this way *illustrate* glowing light, but a brilliant hue arranged in the same way has a much more dramatic impact: Illusions of luminous hue can be very

convincing. To achieve this effect a small area of brilliant color is set into a less chromatic field. The hue can be the same value as its field; it can also be lighter or darker. A "halo" of closely spaced, progressive intervals is set up between the two. If the halo is a gradient between hue and field, the effect is a soft glow. If the progression between the two is a series of intervals just at the threshold of vision, the color seems to shimmer on the page. Irregular drawing at the edges of the intervals reinforces this effect. But as long as a sequence of intervals is close and progressive, and moves from brilliant color to less brilliant color, a luminous effect will take place. Colors arranged in this way can bring subtractive color closer to the brilliance of colored light.

The colors seen on a monitor screen are already made of light, but the light is emitted from the screen overall, so there is no effect that colored light is emanating from a particular location. Colors on a screen must still be arranged in a specific way for an impression of glowing color to take place.

Bezold Effect

Bezold effect, or *spreading effect*, describes what happens when the value of an entire composition is altered by adding, removing, or changing one color only. Unlike most illusions and special effects, it is an effect of *line*. It takes place when internal design elements are outlined, or separated from the ground by dark or light line. When the forms are enclosed by dark line, all colors appear darker. When the forms are enclosed by light line, all colors appear lighter.

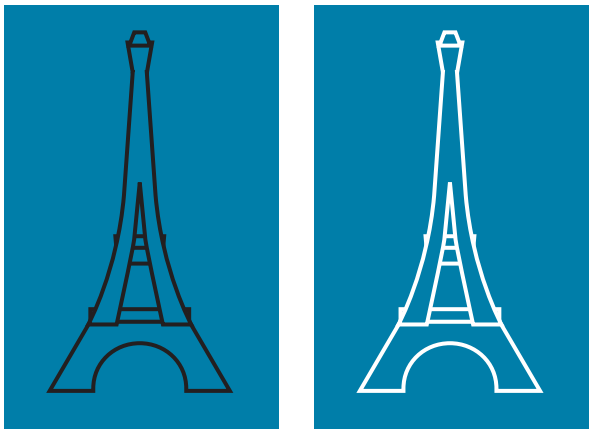


Figure 6-21.
Bezold Effect. When forms are enclosed by dark line, all colors appear darker. When the forms are enclosed by light line, all colors appear lighter.

The dark or light outline affects the apparent value of the entire composition. The presence of a dark or light outline changes only the perception of the overall value of a composition. Hue and saturation are not affected. Because every image, black and white or color, is created by the placement of different values within it, changing a *block* of color from dark to light (or the reverse) creates a new and different image—not a lighter or darker version of the original.

Optical Mixes

An *optical mix,* or *partitive color*, is created when two or more colors, in tiny masses or patches just at the threshold of vision, are used together to create a wholly new color. The color masses in an optical mix are so tightly configured that they are very difficult to distinguish as individual elements, but they do not entirely blend. Optical mixes are sometimes called *retinal mixes* because color is mixed in the eyes, not in a jar or on a palette.

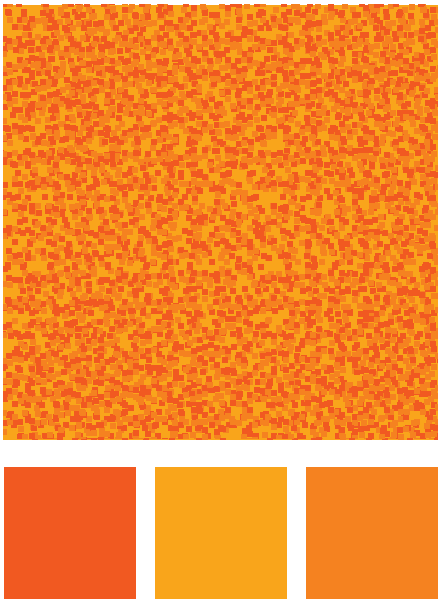


Figure 6-22. *Optical Mixing.* Optical mixes depend for their final effect on the color elements that are used together.

The success of an optical mix depends on the size of its color masses relative to the distance from which the whole will be ordinarily be seen. If the elements are too large, the colors read as separate patches. Too small, and the surface becomes as smooth and featureless as flat paint. Designing graphics for a telephone directory, for example, assumes a close viewing distance. Designing wallpaper presupposes a middle distance, highway billboards another viewpoint entirely: the "dot" in an optical mix on a billboard may be the size of a dinner plate when it is seen at close range.

Optical mixes have a vitality that flat color cannot offer. The hue, value, and saturation of each patch of color contrasts with its neighbors according to the guidelines of spatial

effects. Some patches advance and some recede; others create a "ground" of sorts, a matrix for the advancing and receding elements. An optical mix of brilliant analogous colors is dynamic and lively. Complementary colors mixed as dots just above the threshold of vision appear as a muted color or neutral, depending on the proportions of the colors. A blending of dark, light, and middle values create an impression of texture.

Optical mixing that is *below* the threshold of vision plays an enormous role in industry and design. The new colors created in this way are without texture or movement. Most color printing is achieved by the optical mixing of process colors. Dots of cyan, magenta, yellow, and black inks, too small to be seen without a magnifying glass, merge in the eye to create new colors. The colors on a monitor, too, can be described as a kind of optical mix. Thousands of tiny, individual dots of red, green, and blue light, additively mixed, create the hues, tints, and shades of the screen display.

Endnotes

1 Albers 1963, page 45.

Chapter 6 Highlights

- An optical illusion takes place when an image is misperceived or misinterpreted by a viewer. True illusions fool the viewer completely. Many effects that are called illusions are more accurately special effects or near-illusions.
- Simple illusions are initiated by involuntary responses of the eyes to overstimulation by something like a single hue or extreme contrast. More complex illusions occur because an image is misinterpreted. The current thinking among vision scientists is that all illusions are caused by a disturbance to the flow of information between the eyes and the brain. Color illusions occur when colors that have specific relationships are arranged in ways that mislead the eyes.
- An illustration of depth on a two-dimensional surface is not an illusion. An impression of depth in a drawing is created by conventions called pictorial depth cues: shading, shadow, linear perspective, overlap, texture gradient, and atmospheric perspective. Atmospheric perspective is the only pictorial depth cue that has a hue component. Distant objects appear grayer and slightly bluer than closer ones.
- Motion parallax is a depth cue that includes motion and time. When a viewer is in rapid motion, near objects appear to move by in a blur while far objects seem to move slowly. Motion graphics are drawings that include pictorial depth cues, motion, and time. Motion graphics can be true illusions. The most effective illusions of depth result when a full array of depth cues is reinforced by the spatial effects of colors.
- Certain hues suggest transparency. Cool hues often seem transparent. Warm colors containing red seem more dense and opaque. Some colors have spatial effects; inherent qualities of "nearness" or "farness." Light blues seem to move away and warmer and darker colors close in. Dark or light color also influences how the size of a figure is perceived. In general, color is a secondary indicator that supports one or more of the pictorial depth cues. Spatial effects of colors can be reduced or reversed by pictorial depth cues.
- In color compositions the hue, value, and saturation of each form in relation to its ground and to other colors present determine whether it is perceived as advancing or receding. In general, warm hues advance relative to cooler ones. Saturated colors and strong tints appear to come forward relative to muted ones.

- A dark figure will come forward on a light ground and a light figure will come forward on a dark ground. When figures of different value are laid against the same ground, figures with greater value contrast with the ground will advance and those closer in value to the ground will recede.
- A transparence illusion is a three-dimensional illusion that takes place when two opaque colors and an interval between them are arranged in such a way that one color appears to be transparent and lying on top of the other. The guidelines of spatial effects determine which color appears to be on top.
- Fluting is a three-dimensional illusion in which a series of vertical stripes of uniform width appear to have concavity. Vibration is an effect that takes place when blocks of brilliant and complementary (or near-complementary) colors that are close in value are placed together.. Vanishing boundaries occur when areas of similar hue and close value are placed next to or on top of one another. Each form seems to lose its edges and merge into the other.
- Effects of luminosity follow the guidelines of spatial effects. Glowing light is illustrated by setting a light area into a larger and darker field with a "halo" of closely spaced, progressive intervals of value radiating from the light area into the dark. A brilliant hue set into a less chromatic field has the same effect.
- Bezold effect takes place when internal design elements are separated from the ground by dark or light line. When forms are enclosed by dark line, all colors appear darker. When the forms are enclosed by light line, all colors appear lighter.
- An optical mix, or partitive color, is created when two or more colors, in tiny masses or patches just at the threshold of vision, are used together to create a wholly new color.

7

Color Theory: A Brief History

Setting the Stage / The Beginnings of Color Theory / Color and Controversy / The Scientific Model: Color Gets Organized / Color by Numbers / A New Perspective

The starting point is the study of color and its effects on men.

—Wassily Kandinsky Concerning the Spiritual in Art, 1912

Questions and ideas about color—what it is, what it means, how best to organize and display it, and especially, what makes color combinations harmonious—have a long history. The search for answers to these and other questions have produced an enormous library of writing known as *color theory*. Included within it are color-classification and color-order systems; color dictionaries, atlases, and encyclopedias; color as science, color as language, color as poetry, color as art. With the exception of the very earliest writers, there are underlying observations about color that are common to all. When these observations are distilled from the multitude of sources, color becomes a manageable field of study and guidelines—not laws—for achieving color harmony emerge that are consistent, easy to understand, and easy to apply.

Setting the Stage

The earliest known writers on color were Greek philosophers. Intrigued by the elusive nature of color, they sought to establish its place and meaning in the larger universe. To the ancient Greeks, beauty and harmony were aspects of mathematics. The Greek ideal—that beauty and harmony are the natural result of mathematical order, and are inextricable from each

other—is a premise that is still firmly in place. Beauty and mathematics conjoined have remained a basis for theories of color harmony from the earliest writings to the present.

Pythagorus (c. 569–490 BC) is credited with originating the concept of the "harmony of the spheres" a mathematical theory in which the planets are separated from each other by intervals corresponding to the harmonic lengths of strings, and therefore give rise to a beautiful musical sound. This ideal was extended to include forms and colors corresponding to the musical scale. Intended in its time to demonstrate the all-inclusiveness of the universe and an overriding wholeness in nature, this proposition still resonates today.

Aristotle (c. 384–322 BC) the most influential of the earliest writers on color, addressed it from both philosophic and scientific standpoints. Aristotle's premise that all colors derive from black and white, or darkness and light, was accepted as fact until well into the eighteenth century.

Renaissance writers, including Leonardo da Vinci (1452–1519) and others before and after him, wrote on aspects of color that ranged from the practicalities of mixing pigments to the philosophical and moral meanings of colors. But writers on color were few, and color remained a topic of narrow and specialized interest until the eighteenth century, when, as the result of experiments by Isaac Newton, it was swept into the mainstream of philosophic and scientific thought.

The Beginnings of Color Theory

During the eighteenth century in Europe, an historical period known as the Enlightenment (or Age of Reason), there was a fresh and vigorous search for rational, rather than mystical, explanations for all kinds of natural phenomena. People began to believe in the existence of irrefutable laws of nature. There was an assumption that there were natural laws for everything, including laws for combining colors, and that these laws, like laws of gravity, only awaited discovery. This search for absolutes determined by science was as rigid and uncompromising in its way as the demands of absolute faith that preceded it. Only the source of authority had changed, from God and his earthly representatives, the clergy, to reason and its earthly representatives, men.

The intellectual world of the eighteenth century was quite fluid. People didn't think of themselves as writers, biologists, or mathematicians but as "natural philosophers," "theologians," or "geometricians," all with wide-ranging and overlapping areas of interest. Philosophers and literary figures wrote confidently, if with dubious expertise, on all kinds of scientific and philosophical topics. Others, poets poised at the edge of the sciences, sought a rational basis for the nature of beauty itself and, as a corollary, color. In this way the "behavior" of colors could be explained and predicted, and the mystery of observed color phenomena mastered through an understanding of natural laws. The search for laws of color harmony was only one small part of the sweeping intellectual ferment of the time.

Two themes dominated eighteenth-, nineteenth-, and early twentieth-century color study. The first was the search for a comprehensive color-order system, including an appropriate format for visualizing it. Once in place, a color-order system could become a field in which the all-important search for laws of color harmony could take place. These treatises, dating from the late eighteenth century and continuing today, make up the collective body of knowledge known as *color theory*. Just as there are classics in literature, there are classics of color study. Two towering and very different figures dominate the beginnings of this discipline: Isaac Newton (1642–1727) and Johann Wolfgang von Goethe (1749–1832). Their observations are the foundations of modern color theory.

Isaac Newton, working at Cambridge in the late 1690s, first split sunlight into its component wavelengths by passing it through a prism. Newton observed that as each wavelength enters a prism it bends, or *refracts*. Because glass, the material of the prism, slows each wavelength down at a slightly different rate, each emerges as a separate beam of light: a distinctly different color. A rainbow is a naturally occurring demonstration of Newton's experiment. Droplets of water in the atmosphere act as tiny prisms, and sunlight is broken into visible colors.

Newton recombined the separated beams with a lens and reconstituted white light. From this he hypothesized the nature of light and the origins of perceived color. He published his results, entitled *Opticks*, in 1703. Newton's conclusion that light alone generates color remains a basis of modern physics.

When Newton observed that white light split in a prism yielded separate colors, he identified

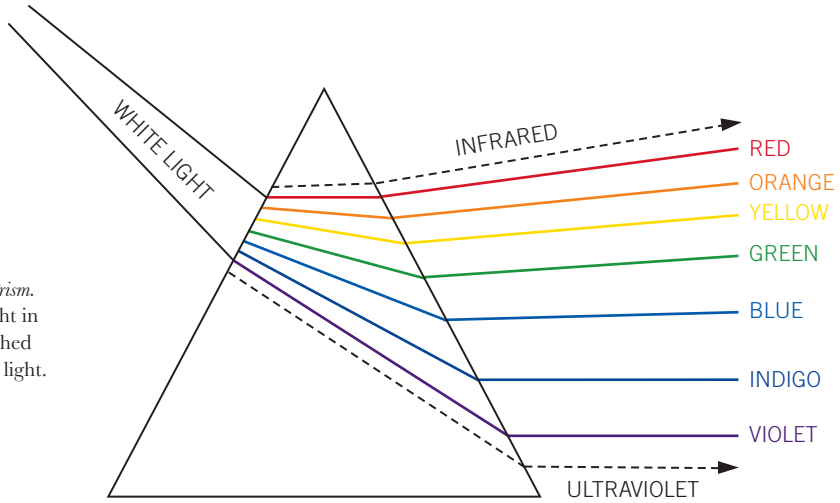


Figure 7-1. *Newton's Prism.* Newton split white light in a prism and distinguished seven colors of visible light.

seven spectral hues: red, orange, yellow, green, blue, indigo (blue-violet), and violet. Many people cannot detect indigo as a separate color of light between blue and violet. There are a number of possible explanations for Newton's choice of seven hues for the spectrum of light. Despite his genius, Newton was a product of the seventeenth century. He may have elected to include seven colors because that number corresponded to the musical notes of the diatonic scale. Mysticism was also a great part of Newton's time, and mystical properties were associated with the number seven. Perhaps he had unusual visual acuity in the blue-violet range. Whatever the reason or reasons, the seven hues of Newton's spectrum remain the standard of physical science in ongoing recognition of his discovery.

Although the spectrum of light is linear, Newton also originated the concept of colors as a *continuous* experience. He diagramed the seven hues as a circle, linking spectral red and violet. This first known illustration of colors as a closed circle made of arcs of individual color appeared in *Opticks*.

Newton's contemporaries viewed *Opticks* as a work on the nature of color, not on the nature of light. By the time of his death in 1727, interest in *Opticks* was widespread. The ideas in it generated tremendous controversy all over Europe.

During this same period artists and artisans were considering color from a wholly different direction. They were not concerned with the study of color as an abstract idea or visual experience. Instead, they sought to resolve the difficulties found in mixing two or more colored *substances* together to achieve a new (and predictable) color. Jacques Christophe LeBlon (1667–1741), a French printmaker, identified the primary nature of red, yellow, and blue while mixing pigments for printing. LeBlon's treatise, *Coloritto* (c. 1730) offers the first concept of three subtractive primary colors.[1] His work attracted a great deal of attention and acceptance. Unlike Newton's earlier theories, which addressed colors of light, and Goethe's later ones, which included ideas about perception and aesthetics, LeBlon's efforts and observations deal with practical realities. His work remains a foundation of present-day printing.

Moses Harris (1731–1785), an English engraver, used LeBlon's three primaries to produce the first known printed color circle (c. 1766). Harris, also addressing subtractive (artists' and printers') colors, was the first to publish them as an expanded circle of relationships—a true visualization of color organization. Harris believed that red, yellow, and blue were the most different from each other and should be placed at the greatest possible distances apart on the circle. To accomplish this he discarded Newton's indigo and created an expanded color circle based on equal intervals of color and multiples of three. This organization was later adopted by the man who is arguably the preeminent color theorist of the late eighteenth and early nineteenth centuries: Johann Wolfgang von Goethe.

Color and Controversy

Goethe was fascinated by color. He was familiar with Newton's theories of color but strongly opposed them, and bitterly resented the acceptance of Newton's ideas. He wrote, furiously, of Newton: "A great mathematician was possessed with an entirely false notion on the physical origin of color, yet, owing to his great authority as a geometer, the mistakes which he committed as an experimentalist long became sanctioned in the eyes of a world ever fettered in prejudices."[2]

Goethe spent a great deal of energy trying to prove that Newton was wrong, publishing his first treatise (of a lifelong series intended to refute Newton's hypotheses) in *Announcement for a Thesis in Color* in 1791.[3] Goethe viewed colors not as light, but as an entity of their

own, as *experienced reality*. His difficulties with Newton's ideas are evident in his own words. Newton's theory, he says, "does not help us to perceive more vividly the world around us," so that "even if we found a basic phenomenon, even then, the problem remains that we would not want to accept it as such," and, "Things which belong together according to our senses often lose their connections once we look into their causes." He scolded Newton sharply for his views: "The natural philosopher should leave the elementary phenomena in their eternal quietness and pomp."[4] Goethe's response to Newton is pragmatic. He is really saying, "What you say may or may not be true, but it is not useful in real life."

Like Newton, Goethe was both a genius and a child of the Enlightenment. Unlike Newton, however, he wrote with a sort of shotgun approach, aiming his considerable intellect at a topic, letting fly a lot of ideas, then turning without pause to fire again in a different direction. For today's readers Goethe's writing includes a lot of unintended humor, like discussion of the color sensibilities of earthworms and butterflies. Associations of color and beauty with morality were also a part of Goethe's treatises. There were sinful colors and chaste ones. "People of refinement have a disinclination to colors," declares Goethe, firmly—a bias that will appear repeatedly in the writings of later color theorists. He associated moral character not only with the choice of colors in clothing, but with skin color as well.[5]

Despite his freewheeling digressions, Goethe's observations were wide-ranging and seminal. He and his contemporary Otto Philip Runge (1770–1840), a German painter, shared in conceptualizing what are now called complementary colors. He called them, with enormous insight, "completing colors."[6] Goethe also reported extensively on the phenomena of simultaneous contrast and afterimage. He recognized that no pure color exists except in theory, and characterized the principal contrasts of color as polarity (contrast or opposition) and gradation (intervals). Although other artists and writers expanded Goethe's ideas and added new material, Goethe's observations were so wide-ranging and fundamental that almost every concept in modern color study can be found in his writing.

Goethe's most familiar contribution to color study is the six-hue color circle. Although Goethe believed that there were only two primary colors, blue and yellow, and that all colors derived from them, LeBlon's red-yellow-blue primary color basis prevailed, and the completed Goethe color circle reflects that convention. We know it today as the basic

artists' spectrum: equal arcs of six colors: red, orange, yellow, green, blue, and violet. Its elegant simplicity can be described as perfect visual logic.

Goethe's French contemporary Michel Eugene Chevreul (1786–1889), like Le Blon before him, addressed color from a practical viewpoint. As Master of the Gobelin Tapestry Works, Chevreul found difficulties with black dyes, which seemed to lose their depth or darkness when placed next to other colors. Chevreul accepted the three-primary-color theory. He observed and reported at length the phenomenon of simultaneous contrast, and his 1839 treatise *The Principles of Harmony and Contrast of Colors and Their Applications to the Arts (De La Loi du Contraste Simultane des Coleurs)* was a profound influence on the Impressionist movement in painting.

Goethe's six-hue spectrum of subtractive color remains the convention for artists; Newton's seven-hue model of the hues of visible light remains the scientist's spectrum. The battle between Newton and Goethe's color theories was a major schism in the history of ideas. It was unnecessary. Both theories are valid, but each describes a different reality:

Newton was looking at causes.
Goethe was looking at effects.

Probably because students who engage in the sciences are not usually the same ones who pursue the visual arts, the differences between the two ideas about color have, in the past, rarely come into conflict. Designers, like Goethe, work with color from the evidence of their senses. They deal with effects, not causes, and science traditionally has taken a back seat in the studio workplace. The advent of computer-generated design has changed the game. Designers today must understand both cause and effect and be able to work within—and between—both realities.

The Scientific Model: Color Gets Organized

Chevreul and Goethe were gifted observers and obsessive chroniclers. Most of the late nineteenth- and early twentieth-century color theorists who followed them worked instead from a scientific model, codifying their observations into rigidly formal systems. These theorists wrote on color as a discipline, as fact, as scientific truth. The stress was on rules, control, and order; the goal was to create a comprehensive color-order system and to find within it immutable laws of color harmony.

The earliest color-order systems displayed colors as two-dimensional charts: rectangular, triangular, and circular. The concept of color as three-dimensional model, conceptualized by earlier writers, was brought into widely-circulated realization by American Albert Munsell (1858–1918). In *A Grammar of Colors*, first published in 1921, Munsell proposed a "color tree" with infinite room for expansion. Munsell color space is constructed as progessive intervals of hue that rotate around a vertical axis of value from black to white. In Munsell's theory every possible color cannot be shown, but each has an assigned place on an alphanumeric (letter and number) scale. Munsell stated, "Naturally, every point (of color) has a defined number," so that "there can be no new color discovered for which a place and a symbol is not waiting."[7]

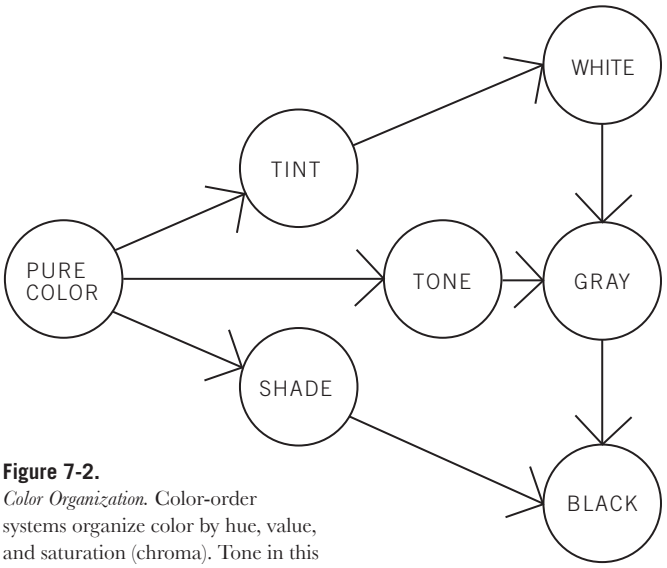


Figure 7-2.
Color Organization. Color-order systems organize color by hue, value, and saturation (chroma). Tone in this illustration represents grayness.

A saturated hue can be reduced in saturation in one of two ways. It can be diluted by the addition of achromatic gray, or by the addition of its complement. Each saturated hue at the perimeter of a Munsell color tree is diluted by a pure gray of equal value. Each hue moves in intervals of diminishing chroma toward the center axis until it arrives at an equal-value gray. Hue intensity is reduced at each step, but value remains uniform along each horizontal branch.

By contrast, complementary colors that dilute one another change in *both* saturation and value at each step as they move toward each other. Instead of reaching an achromatic gray where they meet, each pair of opposites arrives at a chromatic neutral that is visually logical for that pair alone. The complete series of intervals between the two reads as a natural progression. In the Munsell system each hue is placed opposite its complement,

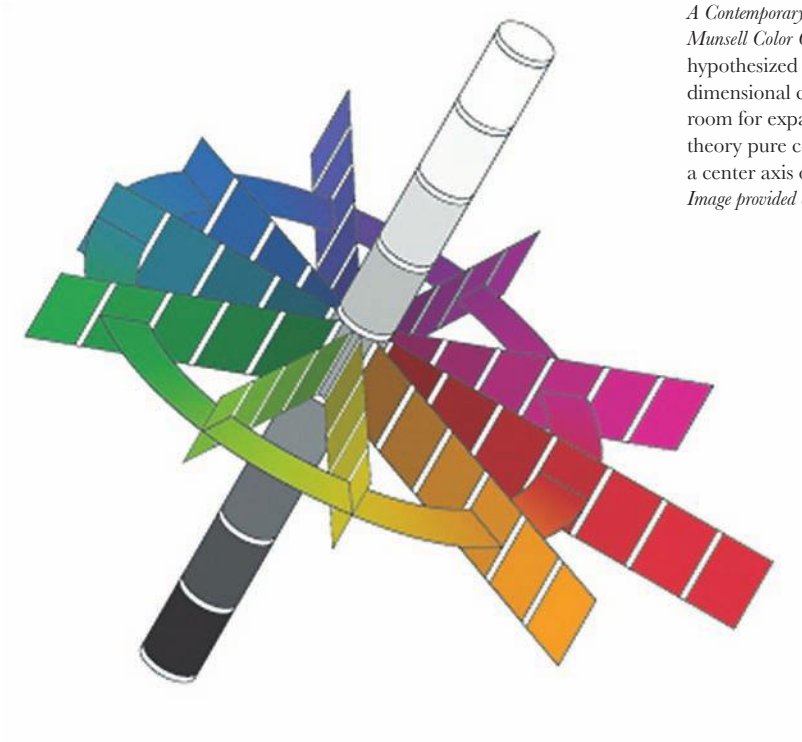


Figure 7-3.
A Contemporary Visualization of Munsell Color Order. Albert Munsell hypothesized color as a three-dimensional construct with infinite room for expansion. In Munsell's theory pure colors move toward a center axis of achromatic gray. *Image provided courtesy of X-Rite.*

but is independent of it. The entire range of muted hues that is created by the mixture of complements is absent. Munsell might have shown color space as *two* trees—a second one moving toward a center axis of tertiary colors in corresponding value intervals—and come closer to a comprehensive visualization of color space.

Color numbering systems have great value when they are used to communicate color information between individuals who have reference to the same set of standards. Equally useful for additive or subtractive colors, they are essential for specifying color mixes in

Web design, or charts for paints, inks, or countless other products. But color numbering is meaningless as an aid to *understanding* color. Letters, numbers, and formulae for colors are essential production aids that follow the creative process.

Like many early color theorists, Munsell's original work contains distracting associations of colors and morals. Following Goethe, Munsell associated color choices with character and morality, declaring that, "Quiet color is a mark of good taste," and, "If we wish our children to become well-bred, is it logical to begin by encouraging barbarous (i.e., colorful) tastes?"[8] Today, the Munsell system is widely used in industry and in classrooms, but without much of the original commentary.

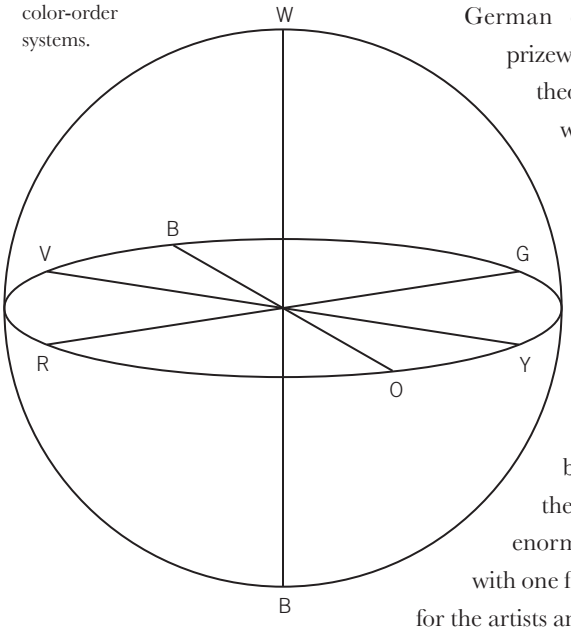


Figure 7-4. *Color as Three Dimensions.* A hypothetical color solid is a frequent theme in scholarly color-order systems.

German chemist Wilhelm Ostwald (1853–1932), a Nobel prizewinner, brought the conceptual color solid to full-blown theory in *Color Science*. Ostwald's later *The Color Primer*, with its eight-hue spectrum, became mandatory for color study in German schools and in many English ones. It was a strong influence on artists of the Bauhaus movement.

Wilhelm von Bezold (1837–1907) and Ludwig Von Helmholtz (1821–1894) contributed scientific fact to the growing body of color writing. The psychology and physiology of color vision, even color healing, became areas of increasing interest to scientists. By the early twentieth century, color study had become an enormous and wide-ranging topic, positioned uncomfortably with one foot in the sciences and the other in the arts. It remained for the artists and designers of the Bauhaus, a design group founded in 1919 by German architect Walter Gropius, to end its ambiguity.

The Bauhaus group brought the study of color to a level of attention not seen since Goethe's challenge to Newton. Feininger, Klee, Kandinsky, Itten, Albers, and Schlemmer, master-students of color and color theory, approached color from new directions with intelligence, wit, and energy.

Although inevitably some elements of the older, quasi-scientific style encroached on their writing, they made the definitive break between the study of color as science and the study of color as art and aesthetics. Light remained in the realm of physics; chemistry and engineering took over the nature of colorants; and psychology, physiology, and medicine became the arena for perception.

Johannes Itten (1888–1967) followed Goethe in exploring color as a series of contrast systems and opposing forces. He theorized seven contrasts of color based on perception alone: contrast of hue, value, saturation, warmth and coolness, complementary contrast, simultaneous contrast, and contrast of extension (area). Itten codified color harmonies as a series of chords based on the complementary relationship and diagramed them as geometric forms (*see* Figure 7-7), and although these chords are mathematically based, Itten's approach to color theory is notably less rigid than many that preceded it. Itten's focus was as much on individual perception as on mathematical relationships. Significantly, his major work is titled The *Art* of Color.

Color by Numbers

Color-order systems were the first concern of theorists because a formal system establishes a structured field in which to search for laws of color harmony. The primary focus of that search was on the relationship between hues, with value and saturation as secondary factors only lightly touched on. Each writer who sought laws of harmony concluded, in one way or another, that the basis of color harmony is order, and more specifically, that harmony lies in a balanced relationship between complementary hues. Goethe repeatedly characterizes color harmony as balance. Johannes Itten says "The concept of color harmony should be removed from the realm of subjective attitude into that of objective principle"[9]; Ostwald referred to "This basic law—Harmony = Order."[10] With his structured charts, Munsell could conclude, "What we call harmonious color is really balance."[11] There was a complete concurrence of opinion: balance between complementary colors was the first principle of color harmony.

The ancient ideal of mathematical balance was so much a part of the search for laws of harmony that hues were frequently associated with numbers or geometric forms. Two examples are representative of the mathematical-balance theories of harmony based on the complementary relationship. Schopenhauer theorized that equal light-reflectance

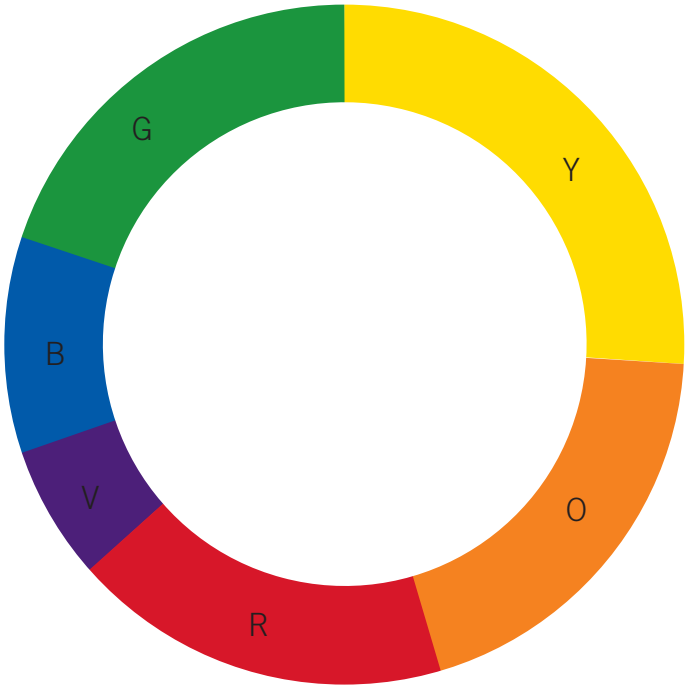


Figure 7-5. *Schopenhauer's Circle of Color Harmony.* Schopenhauer's harmonious color circle is made up of unequal arcs. Each complementary pair is meant to be equal in light-reflectance to each of the other two pairs.

in spectrum colors is inherently harmonious. Using Goethe's color circle as a basis, he suggested a scale of luminosity for each of the six pure hues. Each color is assigned a number representing its light-reflectance (or value) in relation to the others. Yellow, the most luminous, is assigned 9, the highest number. Violet, the darkest of the pure colors, is assigned 3. Red and green are equal in value, and blue and orange are placed relative to the others.

Red	Orange	Yellow	Green	Blue	Violet
6	8	9	6	4	3

The total of all of the numbers added together is 36, or 360 degrees: a full circle. When each pair of complements is added, the total is 12, or a 120° arc on a circle. Each pair of complementary colors thus occupies one-third of the color circle: a perfect mathematical balance of all colors.

red + green: 6 + 6 = 12
blue + orange: 4 + 8 = 12
yellow + violet: 9 + 3 = 12
36

While this color circle recognizes the disparities in value between the saturated colors, the conclusion it draws is deceptive. A patch of saturated violet that is three times the area of a patch of yellow does not necessarily reflect the same amount of light as the yellow. The light-reflectance of colors is not a function of their area. What it does illustrate, very effectively, is the way we *sense* value differences between pure colors.

Schopenhauer's theory can be illustrated as striped tee shirts. In order for each shirt to be harmonious, the red and green one must have equal-width stripes, the violet and yellow one must have violet stripes three times as wide as the yellow, and the blue and orange must have blue stripes twice as wide as the orange.*

Figure 7-6.
Harmonious Tee Shirts (according to Schopenhauer).

Johannes Itten superimposed geometric forms (squares, rectangles, triangles, and hexagons) over the artists' spectrum to demonstrate what he called "harmonious chords," calling them,

* On page 59 of *The Art of Color* (1961) and again in *The Elements of Color* (1970), Johannes Itten credits Goethe with this number series for light-reflectance in pure colors. This material does not appear in Goethe's *Color Theory* (1971), and Albers credits Schopenhauer with the creation of this theory, even referring to Goethe's "dismay" at the tampering with his color circle (Albers 1963, page 43). It's likely that Albers made the correct attribution.

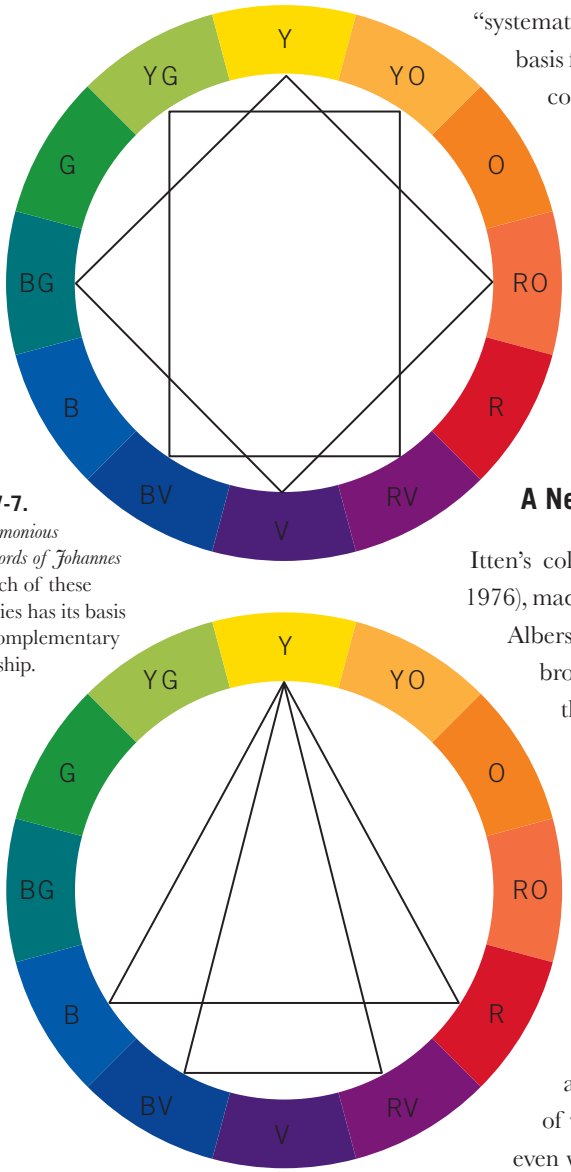


Figure 7-7. *The Harmonious Color Chords of Johannes Itten.* Each of these harmonies has its basis in the complementary relationship.

"systematic color relationships capable of serving as a basis for composition."[12] Each color chord illustrates complementary colors in some measurable proportion. The chords are described as dyads, triads, tetrads, and hexads, with the geometric points locating the "notes" of each chord. The rectangular and hexagonal constructs, called *split complements*, include intermediate colors; but *no chord strays from the complementary relationship*. Each is a variant of the same idea: the harmony of three primary colors presented as opposites.

A New Perspective

Itten's colleague at the Bauhaus, Josef Albers (1888–1976), made the final break with the color-order tradition. Albers fled Nazi Germany in the early 1930s and brought his teaching methods to Yale. He became the most influential name in color theory in the United States, but his 1963 book *Interaction of Colors* contained nothing like the usual charts or systems. Albers did not contribute to ideas of color order. He had a new role to play.

Josef Albers taught that true understanding of color comes from an intuitive approach to studio exercises. He stressed the instability and relativity of perceived colors and the power of visual training. At the same time, he taught that even within this unstable field, there are effects that can be predicted and controlled. In *Interaction of Colors* (1963), Albers casually discounts the generations of theory that

preceded him: "This book ... reverses this order and places practice before theory, which is, after all, the conclusion of practice."[13] Albers was not the first to recognize that the visual experience, more than conscious choice, determines how we perceive colors, but he was the first to assert the primacy of the visual experience over structure or intellectual considerations. For Albers, the visual experience, not theory, was paramount.

The late twentieth century saw the focus of color study move from philosophical inquiry to a greater interest in psychological and motivational effects of colors. There is an entire industry, for example, that is devoted to determining current and future consumer preferences in colors and color combinations. At the same time, color theorists continue to search for absolutes. There is an enduring assumption—or perhaps, a hope—that those elusive, timeless, and absolute laws for pleasing combinations of colors really do exist and simply await discovery.

Endnotes

1 Birren 1987, page 11 and Hope 1990, page 189.
2 Lecture 1941, page 3.
3 Goethe 1971, page 13.
4 Lecture 1941, pages 4–6.
5 Goethe 1971, pages 250–261.
6 Ibid. page 55.
7 Munsell 1969, page 10.
8 Ibid. page 41.
9 Itten 1969, page 19.
10 Ostwald 1969, page 65.
11 Munsell 1969, page 14.
12 Itten 1961, page 72.
13 Albers 1963, page 2.

Chapter 7 Highlights

- The Greek ideal that beauty and harmony are the natural result of mathematical order is a premise still firmly in place. Aristotle's (c. 384–322 BC) theory that all colors derive from black and white was a theory accepted as fact until well into the eighteenth century. In eighteenth century Europe a belief arose that natural laws for color existed and only awaited discovery. Treatises from that time to the present day make up the body of knowledge known as color theory. Color-order systems were a first concern of theorists because a formal system establishes a structured field in which to search for laws of color harmony.

- Isaac Newton (1642–1727) is the principal figure of early color theory. Newton passed sunlight through a prism so that each wavelength emerged separately, and from this hypothesized the origins of perceived color. His conclusion that light generates color is a basis of modern physics and the seven hues of his spectrum remain the standard of physical science. Newton also originated the concept of colors as a continuous experience. He diagramed the seven hues as a circle, linking spectral red and violet. Newton's contemporaries viewed his writing as a work on the nature of color, not on the nature of light.

- Jacques Christophe LeBlon (1667–1741) identified the primary nature of red, yellow, and blue pigments. Moses Harris (1731–1785) used LeBlon's three primaries to produce the first known printed color circle (c. 1766).

- Johann Wolfgang von Goethe (1749–1832) strongly opposed Newton's ideas and published extensively in an attempt to refute them. Goethe viewed colors not as light but as experienced reality. He and his contemporary Otto Philip Runge (1770–1840) shared in conceptualizing complementary colors. Almost every concept in modern color study can be found in Goethe's writings. His six-hue circle of subtractive color remains the convention for artists.

- Newton was looking at causes; Goethe was looking at effects. Designers today must be able to work with colors both of light and of substance; with additive and subtractive colors.

- Michel Eugene Chevreul (1786–1889) reported at length the phenomenon of simultaneous contrast in his 1839 treatise The Principles of Harmony and Contrast of Colors and Their Applications to the Arts.

- Most nineteenth- and early twentieth-century color theorists codified color observations into formal systems presented as scientific truth. Colors were presented as two-dimensional charts. An earlier concept of color as three-dimensional model was brought into widely-circulated realization by Albert Munsell (1858–1918) in *A Grammar of Colors* (1921). Munsell color space is constructed as intervals of hue rotating around a vertical axis of value from black to white. Every color cannot be shown, but each has an assigned place on an alphanumeric (letter and number) scale. The Munsell system places each hue opposite its complement, but independent of it. The entire range of muted hues that is created by the mixture of complements is absent.

- Wilhelm Ostwald (1853–1932) brought the conceptual color solid to full-blown theory in *Color Science*. His text *The Color Primer* (1923) became mandatory for color study in many European and English schools. Wilhelm von Bezold (1837–1907) and Ludwig Von Helmholtz (1821–1894) contributed scientific fact to the growing body of color writing.

- Color numbering systems are useful for specifying colors in Web design, charts for paints, inks, or other products. Color numbering is meaningless as an aid to understanding color.

- The artists of the Bauhaus, a design group founded in 1919, made the definitive break between the study of color as science and the study of color as art and aesthetics. Johannes Itten (1888–1967) theorized seven contrasts of color based on perception: contrast of hue, value, saturation, warmth and coolness, complementary contrast, simultaneous contrast, and contrast of extension (area).

- Historically, the primary focus of the search for color harmony has been on the relationship between hues. The bases of color harmony have been held to be order and the complementary relationship. Johannes Itten superimposed geometric forms over the artists' spectrum to demonstrate "harmonious chords," each made up of complementary colors in some proportion.

- Josef Albers' (1888–1976) book *Interaction of Colors* (1963) stressed the instability and relativity of perceived colors and the power of visual training.

8 Color Harmony

In Search of Beauty / Intervals and Harmony / Hue and Harmony / Value and Harmony / Saturation and Harmony / Major and Minor Themes / Some Harmonious Conclusions / On Beyond Harmony: Dissonant Colors / High-Impact Color / The X-tra Factor: Surface and Harmony

The enjoyment of colors, individually or in harmony, is experienced by the eye as an organ, and it communicates its pleasure to the rest of the man.

—Johann Wolfgang von Goethe

There is a theory in design that people will respect and care for surroundings and objects that they find beautiful, and will disdain and neglect, even damage, those that are unappealing. On some level, the pursuit of happiness always includes the pursuit of beauty.

In Search of Beauty

Beauty is the quality of an object or experience that gives pleasure to one or more of the senses. A color can be beautiful, or a sound, or a scent. Delight in the presence of beauty is as natural to the human condition as breathing.

Harmony is the happy condition that follows when two or more different things are sensed together as a single, pleasing experience. Harmony is perceived as complete, continuous, and natural. It is intuitive; a feeling that things are just as they should be. In a harmonious situation everything is in balance; everything *belongs*. Happy families live in harmony; barbershop quartets sing in harmony; a hermit lives in harmony with nature. Harmonious experiences are without gaps or surprises.

Color harmony occurs when two or more colors are sensed together as a single, pleasing,

collective impression. A single color can be beautiful, but it cannot be harmonious. Harmony requires a *grouping* of elements. A key characteristic of harmonious colorings is that they seem effortless and uncontrived. Each color seems natural in its relationship to the others. No color seems out of place. To paraphrase Goethe, the colors in harmonious compositions seem to "belong together according to our senses."[1]

A color does not have to be pleasing on its own to be used well. It is perfectly possible to dislike certain colors and still use them in harmonious ways. Someone who asserts that a certain color is "awful" is expressing a personal taste, like a preference for vanilla over chocolate, or jazz over opera. No color is inherently "bad." It is the relationship between the colors in a composition to each other that creates color harmony, not the colors themselves.

Johannes Itten defined color harmony simply, as "the joint effect of two or more colors."[2] But harmony, which implies beauty, is only one possible outcome of combining colors. Not all color combinations are intended to be harmonious. Harmony may be more pleasing than chaos, but it is not necessarily more interesting or exciting. A design concept may call for colors that are distinctly *un*beautiful: startling, visually aggressive, even disturbing. Dissonant combinations of color play a significant role in design.

A more comprehensive term for the force of colors used together is *color effects*. Color effects fall into two broad categories. The first is *color harmony*, the traditional idea of beauty or pleasingness in color combinations. The *"pleasing* joint effect of two or more colors" may be a closer definition of color harmony. The second is *visual impact*: the effect of color choices and combinations on the visual force of a design or image.

Successful color combinations are realized in terms of goal. Instead of thinking about combinations as harmonious or dissonant, they can be thought of as successful or unsuccessful. What was the colorist trying to achieve in choosing the colors? To create an image that has shock value, or high visibility, or a suggestion of luxury? To startle, excite, or disturb the viewer? To evoke association with a particular product or idea? *Color effect*s encompasses the central issue of color use:

What makes a group of colors work together to solve the problem at hand?

The old "laws" of harmony may seem antiquated now, but the observations that gave rise to them remain fresh and valid. They are immensely useful as a starting place for achieving color harmony. But no set of "laws" for color harmony is comprehensive, and no single factor determines it. All aspects of a color composition—hue, value, saturation, the spacing of intervals, and completeness—contribute to a harmonious effect. When all are considered, it is possible to generate color harmonies that transcend historical theory, individual taste, current trends, and cultural bias. Harmony can be found within a premise perfectly stated by Josef Albers: "What counts here—first and last—is not so-called knowledge, but vision—seeing."[3]

Intervals and Harmony

Human intelligence has been described as the search for order in all things. Putting things—any sorts of things—into logical order is irresistible. Babies demonstrate this long before speech, when they arrange blocks or toys in size order. Adults control information in exactly the same way, designating small, medium, large; or ascending or descending letters or numbers. Intervals, particularly even intervals, represent a kind of order in perception. Albers's "What counts here—first and last...is seeing" recognizes the natural human inclination to put things in order. Munsell speaks of "smooth sequences"[4] as harmonious. Even intervals make easy work for the eyes and brain. They are visually and intellectually comfortable. They do not challenge or disturb. *A series of visually logical steps between any colors is inherently pleasing.*

Creating intervals between unlikely elements responds to a recurring problem in design: how to achieve a good result when forced to work with colors that seem at first to be hopelessly incompatible. You have inherited Aunt Maude's buttercup yellow loveseat, but it will have to sit on your old dusty rose rug. Introducing colors that are a series of intervals between the two creates a visual bridge; a connectedness that responds to the human need for order. No matter how unlikely a color combination may seem at first, a series of intervals can establish a kind of order that the eye accepts as logical. *Creating a series of intervals between unrelated colors is a principal way in which they can be transformed into a harmonious grouping.*

At times a color composition seems too sparse, or "thin." A limited palette can be enriched by adding intervals between colors, which expands the palette without disturbing the original color concept. If colors are similar in value and an overall effect seems flat, adding steps of *value* (without changing hues) accomplishes this in a second way.

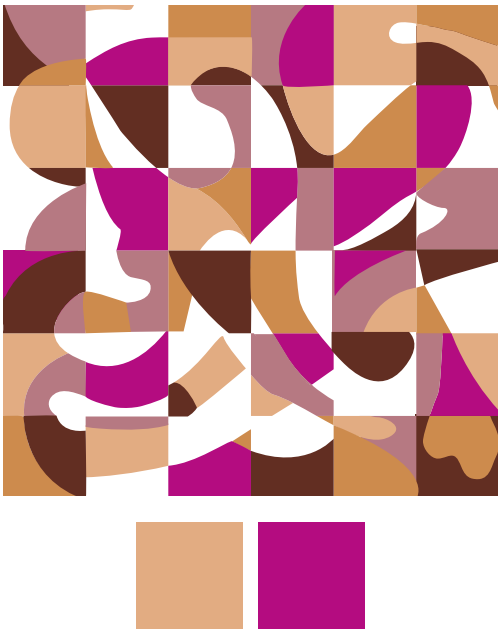


Figure 8-1. *Making it Work.* Creating intervals between two apparently incompatible colors turns them into a pleasing combination.

The *arrangement* of intervals is not necessarily important. That progressive intervals between colors are inherently harmonious is easily observed, but many times a linear progression does not make sense in a composition (*see* Figure 8-3). *Even intervals of any color quality (hue, value, saturation, or any mixture of these) within the same composition tend to be more pleasing than random intervals no matter how they are arranged.*

Hue and Harmony

Historically, the search for color harmony has focused on the relationship between hues, and more specifically, on the link between harmony and complementary colors. Complementary schemes are commonplace in every kind of product from fashion to automotive design. Red (or pink) and green, "peach" (orange) and blue, and "purple" (violet) and "gold' (yellow) are constantly recurring palettes.

Goethe's phrase "completing colors," is a reminder that the eye finds equilibrium in the presence of three primaries. Colors in a complementary relationship are *physiologically* satisfying. The eye is more comfortable at rest than at work, and comfort is pleasing. Albers's "seeing" again goes to the heart of the matter. The complementary-color theories of color harmony are supported by the visual experience at the most fundamental level.

But not every pleasing palette is made up of complements. Colorings are also harmonious when a single hue is used in a variety of values or saturations, as *monochromatic* schemes; or as *analogous* combinations, which contain two primaries but not the third. The menu of

possible harmonious hue combinations is a simpler proposition:

Any hues used together can be harmonious.

This does not mean that any hues used together *are* harmonious. It means only that there are no inherently bad hue combinations.

Value and Harmony

Although the principal function of value is as contrast that creates separation between figure and ground, traditional color theory offers three ideas about value and harmony:

Even intervals of value are harmonious
Middle values are harmonious
Equal values in different hues are harmonious

A range of values does not have to extend from the extremes of light to dark to be pleasing, nor does it have to be arranged in a linear progression. Intervals of value will be seen as harmonious as long as steps are equidistantly spaced. When the same image is illustrated using well-spaced steps of value and again with irregular steps, the even-interval version will almost invariably be chosen as preferable. This illustrates one relationship between

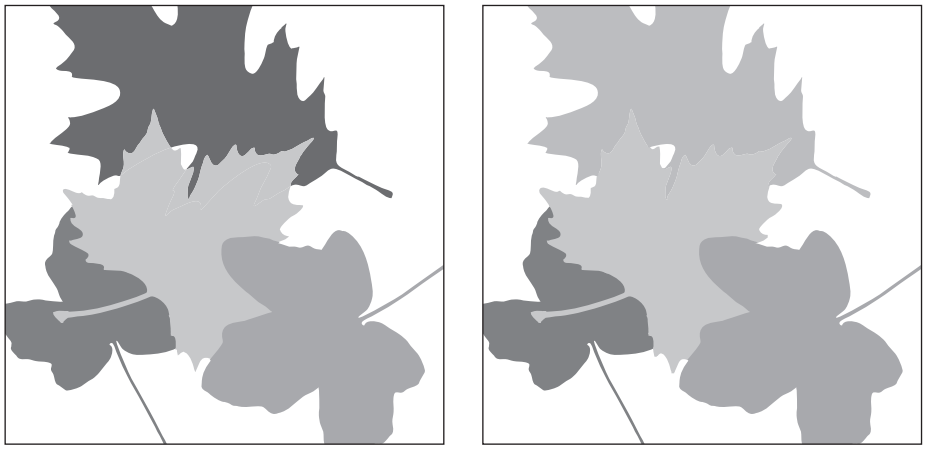


Figure 8-2. *The Harmony of Value in Even Intervals.* Which design is more appealing?

value and harmony and, at the same time, the larger idea that even intervals of any kind contribute to color harmony.

The second premise, that "middle values are harmonious," implies that hues at the extremes of light or dark are unpleasing. To examine this idea, consider first that only the darkest and lightest ends of the value scale are excluded. "Middle values" does not mean only a few samples. There is plenty of light-to-dark range in the middle values.

It is true that middle values are often selected as *preferable* over their much darker or lighter variations. Middle values are easy to see and easy to identify. Any viewer will select first those colors that can be discriminated from others with a minimum of effort. The idea that middle values are inherently harmonious tests the definition of "harmonious" and fails. It is more accurate to say that middle values are often *preferred* than to characterize them as "harmonious." Middle value is not a determinant of color harmony. All values, including extreme darks and lights, are equal in their potential to create harmonious palettes.

The final premise, that "equal values are harmonious," has two distinctly different aspects. Each depends for its success on the intent of the colorist. First, hues of close or equal value can be pleasing *when they are used as carried colors against a contrasting darker or lighter ground.* The image or pattern that emerges from the ground is flattened—it lacks the "forward and back" impression that is associated with the contrast of dark and light—but the presence of many colors offers a different kind of interest and liveliness.

Figure 8-3.
Equal Value Harmony. Many colors of the same or similar value have a pleasing effect when set against a value-contrasting ground.

Hues of close or equal value also create elegant harmonies *without* a contrasting ground when *no image is intended*. Some of the most beautiful stones, ceramic products, book endpapers, textiles, and wall finishes depend on the interplay of different hues of similar value. Similar-value color harmonies share some of the characteristics of

optical mixes, but large areas of color that are close in value do not meld into a single new color. Instead, they create surfaces on which color masses, soft-edged and elusive, float without creating any image or pattern. These richly multihued surfaces can be glorious, whether used alone or as backgrounds for value-contrasting images or patterns laid over them.

Figure 8-4.
Equal Value Harmony. Different hues of close or equal value create rich surface effects. They can be used alone or as backgrounds for value-contrasting patterns.

Saturation and Harmony

Considerations of saturation, or chroma, are an integral part of color-order systems, particularly the three-dimensional constructs. Like any series of steps, intervals between pure hues and their muted versions are pleasing, but the harmony of intervals is only one consideration in the relationship between hue-intensity and harmony. *Color compositions tend to be most successful when the overall level of saturation is relatively constant.*

A relatively constant level of saturation does not mean that all colors are at the same level of brightness. Elaborate compositions like Aubusson carpets have dozens of hues, each in many steps of saturation as well as many steps of value. Complex compositions that include different levels of saturation call for a studied balance between vivid and muted elements. Bright and dull elements are composed together to create a single, cumulative effect that is brighter or more muted.

Figure 8-5.
Muted and Bright. A reproduction William Morris wallpaper captures the muted beauty of the original nineteenth-century coloring. *Courtesy of Sanderson*. The contemporary pattern "Primavera" from Clarence House has all the exuberance of saturated color. *Courtesy of Clarence House.*

When a general level of saturation has been established, any atypical element is disruptive. A single pure color inserted into a muted palette will pop forward. A muted color appears grayed or dirty, and recedes when it is included in a composition of brilliant colors. It appears as a blot in the clean brightness around it.

Figure 8-6.
Uniform Saturation. A palette of pale, muted hues creates a soft background for strong black lettering. *Image courtesy of Carin Goldberg Design.*

The role of saturation in color harmony is distinct from its role in illustration. Muting a color in steps, or as a gradient, is a classic means of depicting three dimensions. Progressive intervals of saturation allow brighter colors to come forward gradually, creating a soft impression rather than a sharply graphic one. In a flower-printed textile, for example, movement from bright green to muted gray-green suggests leaves receding into gentle shadow.

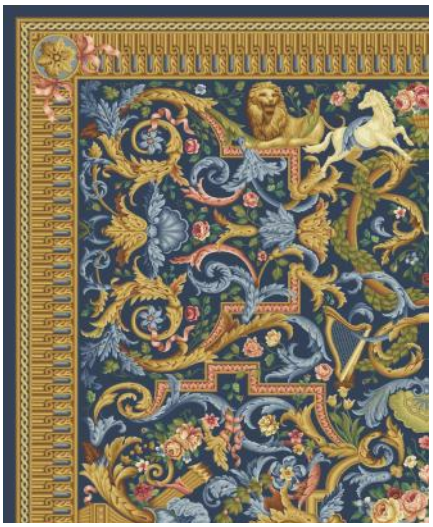


Figure 8-7.
Complex Harmony. This modern Savonnerie carpet contains over 60 different colors, but only three hues. *Carpet design by David Setlow. Image courtesy of Stark Carpet.*

It has been argued that muted colors are naturally more harmonious than saturated colors because the eye is at rest in the presence of muted color. This idea, too, fails the definition of harmony. Brilliant colors in combination are exciting and muted ones are calming, but neither is inherently more harmonious than the other. Only the relationship between colors creates color harmony, not the colors themselves. Harmonious compositions are possible with colors at any level of saturation.

Major and Minor Themes

Many complex colorings have an additional characteristic. There is a dominant hue family; most often a group of analogous colors; this principal theme is enlivened by smaller areas of complementary or near-complementary colors. A carpet may have dozens of yarns in the blue-to-green range. Some are brilliant and others muted; some dark, some light. The blue-to-green range is the major theme; the dominant hue family is blue-green. Small areas of the complementary red to red-orange support and reinforce the cool hues, emphasizing the blue-green by contrast and fulfilling the eye's need for equilibrium. Color compositions in which two or more hue families compete for *equal* attention are often less successful than those with major-minor hue relationships.

Some Harmonious Conclusions

A central feature of successful harmonies is completeness. The ground is often the largest single area in a composition, and that idea of completeness includes consideration of the color of the ground, even when it is simply white. White is no more an absolute than any hue. Every white has some undertone: yellow, green, blue, gray, violet, or some other color. Blacks and grays, too, carry undertones. There are green-blacks, blue-blacks, violet ones, brown ones. A well-chosen ground means the difference between a fully realized color harmony and a less satisfying one.

Much of the time colors for objects, print, and screen are chosen either to appeal to, or to attract the attention of, the widest possible audience. Following the guidelines for color harmony does not guarantee that a particular colorway will have universal appeal. There is always an element of personal bias in color preference. But there is no way to escape the conclusion that a great deal of what we find harmonious originates as involuntary responses of the eyes and mind. The brain has a built-in bias for certain kinds of combinations.

It is a little disconcerting to think that preferences in color combinations are predetermined by physiology and the unconscious, but that is only one part of the story. An impression of harmony is influenced by the eye's need for equilibrium, the comfort level of vision, the human need for logic in perception, and finally, by each individual's emotional response.

An instinct for what is harmonious can be trusted because the eyes dictate boundaries of comfort. We enjoy the accidental beauties of nature. In design, harmonious colorings are not accidental. They are deliberate. The designer's mind, hand, and eye creates each new palette, and the designer's *intent* determines whether that palette is harmonious—or otherwise.

It is good practice to avoid a reverence for authority when searching for color harmonies. The traditional principles have validity, but they are limited in scope. "Laws" of harmony are stifling to the creative spirit. No new idea ever grew from a strict adherence to rules.

What does make sense is to consider some observations about color harmony:

- No single factor determines color harmony.
- The complementary relationship between hues is a strong basis for harmony, but it is not the only basis. Any hues used together can be harmonious.
- Even intervals between colors contribute to harmony. Even intervals are pleasing whether they exist between hue, value, saturation, or any combination of these.
- Color compositions tend to be harmonious when the level of saturation is relatively constant.
- Compositions of many colors tend to be most successful when a dominant family of analogous hues is supported by smaller areas of their complements.

On Beyond Harmony: Dissonant Colors

If color harmony is the good child of design, its polar opposite is disharmony, or dissonance. Dissonant colorways are disturbing. Colors do not seem to belong with each other. If harmony conveys balance and order, disharmony communicates imbalance, unease, edginess, chaos; a sense of things missing or "off-kilter."

Dissonant colorings can be dynamic and exciting—not pleasing perhaps, but certainly a way to draw attention. When the guidelines of color harmony are deliberately ignored, the result may startle or repel, but it may also be memorable. Unpleasing colorways have their own strengths.

Figure 8-8. *Dissonant Color.* Dissonant color combinations have their own place in design. The coloring of this early-twentieth century textile was stylish in its time. "Stylish," however, does not necessarily mean "harmonious."

High-Impact Color

Some design problems call for colors or combinations that will draw instant attention. The strongest images are created by high value contrast alone, a graphic power that requires no hue. The addition (*not* the substitution) of brilliant color to an already powerful image does not change the strength of the image. Instead, it affects the amount of *time* it takes to capture the viewer's attention. Colors that are both hue-intense and light-reflecting, like a strong tint of red-violet, or a saturated yellow-green, have an eye-catching immediacy. Only a few saturated hues are high-impact colors. The range of violets, for example, is not light-reflecting enough to draw immediate attention. When a highly visible violet is called for, a strong tint is used. Working with high-impact colors is not necessarily an alternative to color harmony. Colorings can be both brilliant and harmonious.

Figure 8-9. *High Impact Color.* High-impact colors attract attention more quickly than softer ones.

Brilliant colors used together without some intervening value contrast are likely to vibrate, so although they draw immediate attention, they are poor candidates for good readability. Because these colors often contrast sharply with their surroundings, they are useful in communicating nonverbal warnings. The Occupational Safety and Health Administration (OSHA), for example, uses high-impact colors symbolically to alert for specific dangers — a tint of violet for radiation, vivid orange for hazardous situations.

Fluorescent colors, sometimes called neon, or "DayGlo" colors, are an extreme of high-impact color. They include a colorant that absorbs wavelengths of light from the UV range (non-visible light) of the spectrum and re-emits it as visible light. This added light-reflectance makes the colors extremely brilliant and attention-getting.

High impact color can also be used to *direct* attention. An area of brilliant color set into a more muted palette injects an element of surprise into a composition. It creates a point of focus that draws attention to itself and away from the composition as a whole.

Figure 8-10. *The Element of Surprise.* A patch of unrelated high-impact color draws attention to itself and away from the composition as a whole. *Image courtesy of artist Emily Garner.*

The X-tra Factor: Surface and Harmony

The natural world is a richly chromatic experience. Extraordinary brilliant and subtly muted colors coexist. Rarely is nature truly colorless. The colors of nature are also fragmented. Much of the time they are better described as optical mixes than as flat color.

Broken color, suggesting texture, invites a tactile response as well as a visual one. A textured surface (or the impression of one) engages more of the senses than an area of flat color. Fragmented color responds to the human need for connection to the natural world. One familiar example of broken surface color used to evoke this response exemplifies how human beings respond to it. Many medical facilities hang images of nature in waiting and treatment areas because they are calming to patients and their families. When medical facility art work is nonrepresentational, the colors are soft-edged and irregular, rather than hard-edged and geometric.

Flat color has its own purpose and place in design. Where broken color suggest nature, hard-edged, flat colors are dramatic and compelling. They have a discipline that responds

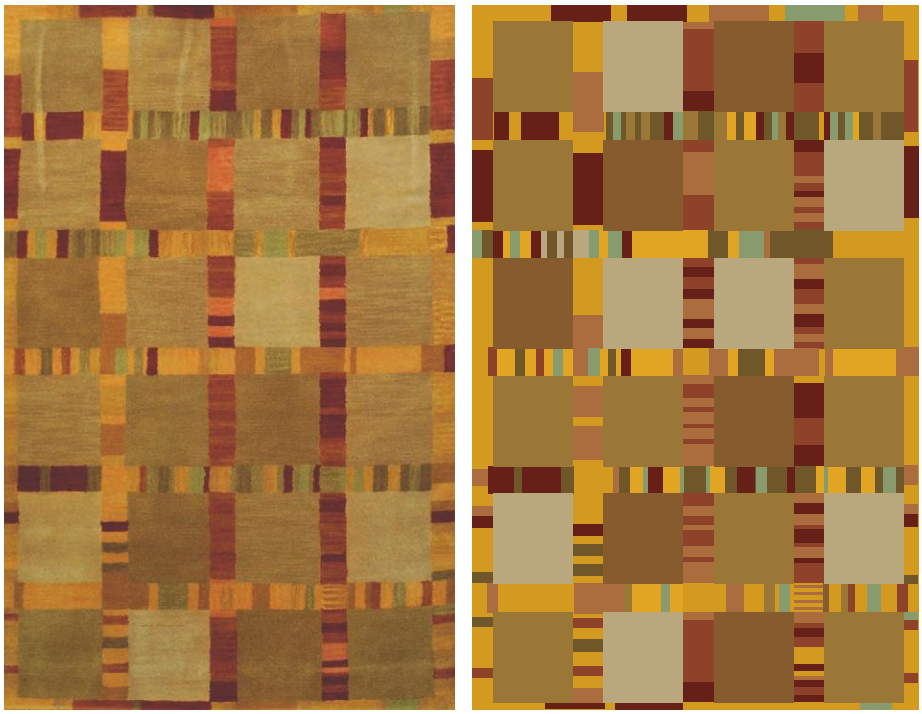


Figure 8-11.
The Appeal of Texture. The irregular colors of hand-dyed yarns give the carpet a warmth and appeal that seem to invite touch. The same pattern created in flat color lacks tactile appeal. *Image courtesy of Safavieh Cosmopolitan Collection CM413A © Safavieh 2005.*

to an entirely different human need: the need to control. The designer of a new electronic device is unlikely to specify coloring that suggests the texture of autumn leaves. A surface that is flat, sleek, and flawless offers an impression of precision. The decision to use flat or broken color is a small but meaningful side trip on the road to successful color choices.

Endnotes

1 Lecture 1941, pages 4–5.
2 Itten 1961, page 21.
3 Albers 1963, page 2.
4 Munsell 1992, page 36.

Chapter 8 Highlights

- Beauty is the quality of an object or experience that gives pleasure to one or more of the senses. Harmony occurs when two or more things are sensed together as a single, pleasing experience. Color harmony occurs when two or more colors are sensed together as a single, pleasing, collective impression. The relationship between colors in a composition to each other creates color harmony, not the colors themselves. Any colors can be used harmoniously. Not all color combinations are intended to be harmonious. Dissonant combinations of colors can be dynamic and exciting; a way to draw attention

- A comprehensive term for the force of colors used together is *color effects*. All aspects of a color composition, including the ground, contribute to color effects. Color effects fall into two categories. The first is color harmony. No set of "laws" for color harmony is comprehensive and no single factor determines it. The second is visual impact, or the effect of color choices and combinations on the power of an image. Color effects encompasses the central issue of color use: What makes a group of colors work together to solve the problem at hand?

- A series of even intervals between any colors is inherently pleasing. Creating a series of intervals between unrelated colors is a principal way in which they can be transformed into a harmonious grouping. A limited palette can be enriched by adding intervals between colors, which expands the palette without disturbing the original color concept.

- Colors in a complementary relationship are physiologically satisfying. The complementary color theory of color harmony is supported by the visual experience. The complementary relationship between hues is a strong basis for harmony, but it is not the only basis. Any hues used together can be harmonious. Monochromatic and analogous colorings are also harmonious. There are no inherently bad hue combinations.

- Even intervals of any color quality within the same composition tend to be more pleasing than random intervals no matter how they are arranged. All values, including extreme darks and lights, are equal in their potential to create harmonious palettes. Middle values are often preferred because they

are easier to discriminate than very light or very dark colors. Preference is unrelated to harmony.

- Hues of close or equal value can be pleasing when used as carried colors against a contrasting darker or lighter ground. Hues of close or equal value also create elegant harmonies without a contrasting ground when no image is intended.
- Color compositions tend to be most successful when the overall level of saturation is relatively constant. Complex compositions that include different levels of saturation call for a studied balance between vivid and muted elements. Harmonious compositions are possible with colors at any level of saturation. The role of saturation in color harmony is distinct from its role in illustration.
- Complex compositions made up of many colors tend to be most successful when there is a dominant family of hues supported by smaller areas of their complements. Color compositions in which two or more hue families compete for equal attention may be less successful than those with major–minor hue relationships.
- Brilliant colors that contrast sharply with their surroundings are useful in communicating nonverbal warnings. The addition of brilliant color to an already powerful image affects the amount of time it takes to capture the viewer's attention.
- Broken color, suggesting texture, invites a tactile response as well as a visual one. Hard-edged flat colors respond to the human need to control.

9 Tools of the Trade

It's the Real Thing: Color in Product and Print / Design Media / Artists' Media / Subtractive Mixing / Tinting Strength / Color Printing

The artist studies his materials and methods in order to gain the greatest possible control over his manipulations, so that he may bring out the best characteristics of his chosen technique, and express or convey his intentions....

—Ralph Mayer

Every manufactured object represents the culmination of a design/build process. Individuals or teams create design prototypes, but the final product—the real thing—is fabricated by others. The creative phase of product development ends at the moment when goods go into production.

Designers use renderings and models to visualize new products and graphics. Ideas for new forms and colors are sketched, modified, revised, and ultimately tweaked into final form for production. The finished illustration has a second and equally important use. Renderings are the means of *selling* design and color. They are the principal way in which new product and color ideas are communicated to potential buyers.

It's the Real Thing: Color in Product and Print

Objects and printed pages can be described in terms of their color qualities, but they are *experienced* in more complicated ways. The full impact of any object's color includes subtleties of texture, light-reflectance, and other, harder-to-define qualities. The blue of a ceramic tile and the blue of a velvet may be described using the same words, but one would never be mistaken for the other. It is difficult for people to think of the color of something as independent of its other attributes. Most of the time, the form, surface, and color of an object are understood as a single experience.

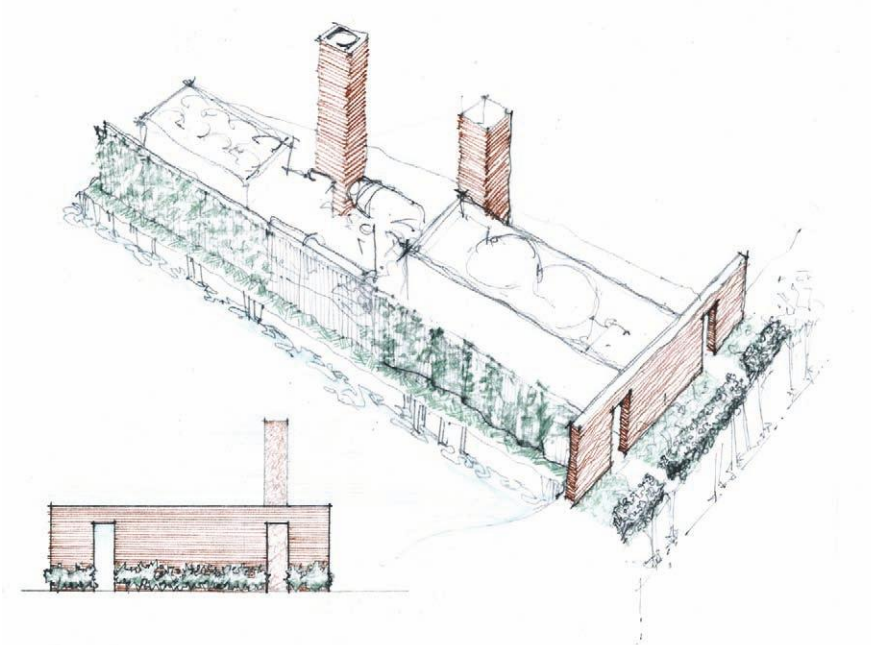


Figure 9-1.
Traditional Rendering: Architect Daniel Brammer's meticulous pencil drawing brings a structure to life on paper. *Image courtesy of Daniel Brammer for Cook+Fox Architects.*

Figure 9-2.
Traditional Rendering. A gouache rendering offers a vibrant image of a proposed new seaside veranda. *Image courtesy of Drake Design Associates.*

A surface scatters general light in a way that is characteristic of a particular thing and others like it. The pebbled yellow surface of a lemon is understood *as* the lemon. A lemon-shaped, lemon-colored form that is smooth and shiny is not identified in the same way. Two images printed on white paper using the same ink will read differently if the underlying paper surfaces are unlike. A matte paper scatters light in many directions; colors printed on it are diffused and softened. The same ink printed onto glossy paper seems sharper and more dramatic. Deep reds and golds printed onto velvet seem soft and lush; the same colors on a glossy vinyl seem sharp and aggressive.

Both natural and man-made materials are used in the manufacture of consumer goods. Natural materials like stone, wood, silk, linen, cotton, and wool have random gradations and shadings that many consumers find appealing as they are, without added color. Natural materials often have color added without losing the attributes that make them appealing in the first place. Wood can be stained to brilliant color without losing visible grain. Cottons, wools, linens, and silks can be dyed to any imaginable color without losing characteristic texture and light reflectance.

Figure 9-3.
Art Imitates Nature. Wilsonart laminates simulate the texture and colors of natural granite. *Image courtesy of ©Wilsonart International, Inc.*

Man-made materials in general need color added to make them attractive to consumers. The added color often simulates a natural material: petroleum- or cellulose-based materials textured and colored to simulate natural wool, linen, silk; wood, and stone are commonplace. Other synthetics may call for high color: a brilliant blue engineered stone, or a bright yellow vinyl floor tile.

Products are colored in three principal ways. Color can be introduced directly into the raw material before the start of the manufacturing process, chemically bonded to the surface of a material, or applied as a coating at a late stage of manufacture. Products are also colored using a combination of these means: a textile can be dyed, for example, then overprinted after dyeing.

Plastics, solid vinyl floorings, engineered stone, and solution-dyed man-made fibers are examples of products whose color is introduced into the base material at the earliest stage of manufacture. In solution dyeing of textiles, for example, the colorant is introduced into a chemical "soup" before it is extruded into fibers. There are limitations to introducing a colorant into any material. The colorant and base material must be chemically compatible, and not all colorants are compatible with all substances. The inclusion of color into the substance of a material offers some advantages: chipping or fading, for example, become lesser problems. But producing a separate base material for each colorway in a product line is costly, so the number of colors available in color-through products is often limited.

Dyes form a chemical bond with the substances to which they are applied. Textiles are the most familiar dyed products. Textiles can be dyed as fluff, before yarn is spun; or dyed as yarn; or piece-dyed after the fabric has been woven. Dyes are also used to introduce color into materials such as wood, paper, leather, fur, and even stone.

Figure 9-4.
Applied Color.
The floral design of this English Ironstone China plate is painted on at a late stage of manufacture.

Finally, color can be applied as a coating to the surface of a product, typically at a late stage of production. Coverage can be complete, like paint or ceramic glaze, or used in partial areas, as in a printed textile. The coating can be opaque, masking the underlying substrate completely, or it can be semi-transparent, allowing some of it to show through. Applied color enables a manufacturer to offer a greater number of color choices because the same underlying material is used for all colorways. Applied colors have their own limitations. They must adhere to the underlying material and perform well in normal use, without chipping, flaking, rubbing off, or bleeding.

Design Media

A *medium* is a means—something through which an idea, or a substance, or an image, is conveyed.[1] A *design medium* is a means of translating images, ideas, and colors from one

kind of visual experience to another. A three-dimensional reality (like a person or object) becomes a two-dimensional image through the medium of paint on paper, or light on a screen. A good design medium transmits the force of a whole visual idea. The best design media enable designers to create images that closely approximate the forms, textures, and colors of the end product.

A product moving through design development passes through many stages of visualization. It may begin as a marker sketch or a digital drawing, move into print, be a photograph or data on CD or DVD. It may appear on the Internet or on television, where it will be received by a wide variety of computer and television monitors, each with its own color-rendering qualities.

All fields of design share the same challenge: visual thinking and presentation must take place in (at least) one medium when the color of the actual product is achieved in an altogether different way. Each rendering must approximate as closely as possible not just color, but a sense of the whole product. Two very different kinds of media are used to achieve this goal. One is the traditional paints, inks, and markers of the artist-designer. The other is the near-universal medium of the present-day design studio: the on-screen imagery of digital design.

Artists' Media

Traditional artists' media produce subtractive color. They are substances made up of a liquid, paste, viscous, solid, or other inert base that contains a colorant. The colorant, a dye or pigment, selectively absorbs and reflects wavelengths of light. The base is a vehicle (or means) for transferring the colorant from one thing to another—from brush or pen to paper, for example. The base and colorant together make up the *medium*. In a good-quality artists' medium the colorant permeates the base evenly and transfers smoothly. There are literally hundreds of different subtractive media, among them poster paints, opaque and translucent watercolors, oil paints, acrylic paints, markers, crayons, colored pencils, dyes, and inks.

Dyes are colorants in solution. The colorant is fully dissolved in water, alcohol or some other solvent. Dyes penetrate the material they color and bond with it on the molecular level.

Some dyes are translucent, others are opaque. A translucent dye acts as a filter, allowing light to pass through it and reflect back from a white (or light) surface below. When light reflects from a white fabric back through a dye, or when a translucent ink is laid on a white paper ground, the effect is light-reflective and clear.

Pigments are finely ground particles of colorant suspended in a liquid or other vehicle. Pigments "sit" on a surface rather than bonding with it. They are typically more opaque and less brilliant than dyes. Not all artists' colors fall neatly into a dye or pigment classification. *Lake* colors, for example, are dyes that have been bonded to a finely ground material, usually a white clay, then suspended in a base. They have the brilliance of dyes but perform with the opacity of pigments.

Before the mid-nineteenth century most dyes were organic; derived from plant or animal material. Many were fugitive, oxidizing (fading or changing) rapidly on exposure to light or air. Pigments were made from ground earths; some, like lapis, were ground from semiprecious stones. The inorganic nature of pigments made them more durable than dyes. The accidental discovery in 1856 of coal-tar dyes precipitated a quest for ever more stable and brilliant synthetic colorants. Today's high-performance synthetic colorants have properties and durability unimaginable in the past. Because of their complexity, many of these products bridge the traditional distinctions between dyes and pigments.

Although the colorant alone determines the hue, the base of each medium has a part in overall color impression. A base is inert only in the sense that it does not modify light *selectively.* Dyes and pigments are selective, absorbing some wavelengths and reflecting others. A base modifies color by absorbing, scattering, or reflecting *general* light. Color qualities like translucency, opacity, and chalkiness depend many times on the light-modifying qualities of the base.

Watercolor and crayon are examples of media whose differences come largely from the modifying quality of the base. Water is a colorless substance that transmits light. It evaporates completely after it is applied, leaving the colorant essentially unmodified. A brilliant dye that is dissolved in water and applied to a white substrate (underlying material), renders colors that are light-reflective, clear and brilliant. Crayons carry the colorant in a wax base. The wax is dense, cloudy, somewhat shiny, and slightly translucent. Unlike water, the wax base of a crayon remains after it has been transferred to paper. Wax absorbs a bit of the light reaching it, scatters some light, and allows some light to reach the underlying

surface. No matter how brilliant the colorant in a wax crayon, the wax mutes the final color impression to some degree.

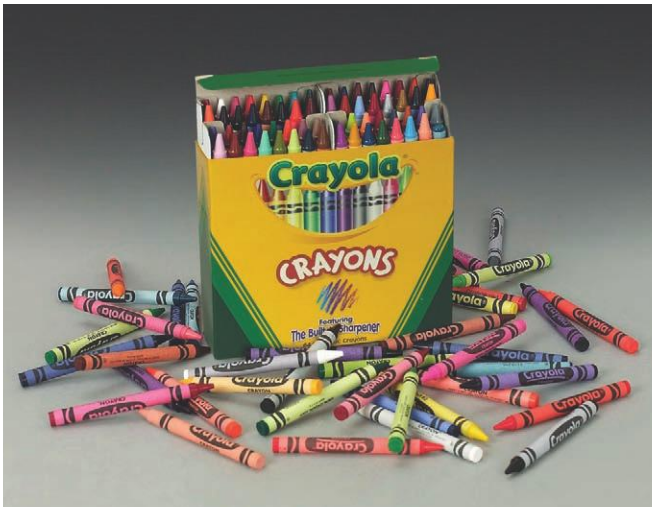
Some media have special properties and special uses. "Dayglo" colors, for example, include a substance that absorbs wavelengths of light from above the range of the visible spectrum and re-emits it at a lower range, as visible light. The added light-reflectance makes these colors useful for attention-getting purposes, like safety cones on the highway, because they are highly visible against any background. Metallic media are made up of finely ground particles of copper, zinc, or aluminum in different proportions that mimic the sheen and color of various metals. The particles are suspended in a base, usually a resin, that can be tinted with conventional colorants for further effect. Yellow can be added for gold, or muted orange for copper. The metal particles scatter light in many directions and the resulting colors are light-reflective but diffused, with a soft glow. High-gloss metallics are also available in many media.

Figure 9-5.
Artists' Media. The multitude of colors available in Crayola crayons has been an early inspiration to generations of artists and designers.

Image © 2004 Binney and Smith. All rights reserved. Crayola, the chevron design, and the serpentine design are registered trademarks, the smile design is a trademark of Binney and Smith.

Subtractive Mixing

No medium exists as a complete range of visible hues with all tint, shade, and chroma (saturation) variations. Instead, each has a limited number of colors that can be used alone or mixed together to extend its range. A medium may be available in a large number of colors, like oil paints or children's crayons, or have a only few very similar colors, like natural chalks. Different kinds of liquid or paste colors can be mixed together successfully only when the bases are compatible. A water-based acrylic paint, for example, will not mix with an oil-based paint.



Artists' colors are mixed in a visually logical way. Mixing a new hue follows the sequence of the artists' spectrum: two colors are mixed together to produce the hue that lies between

them. But two artists' colors that are mixed to make a third do not *dependably* produce the expected result.

A single medium may have a number of tubes or jars of what seem at first glance to be near-identical colors with different names. These colors share the same base, but each contains a different colorant, and each colorant reacts differently when mixed with others in the same medium. A red chosen at random and mixed with yellow will not necessarily make orange. *In order for two colors to produce a new hue, their colorants must reflect a wavelength in common.*

Most colorants do not reflect a single wavelength. Instead, they reflect light in a range of wavelengths, with one or a few wavelengths typically much stronger than others. The perceived color is the wavelength that is reflected most strongly, while others present may be reflected so weakly that they are not apparent.

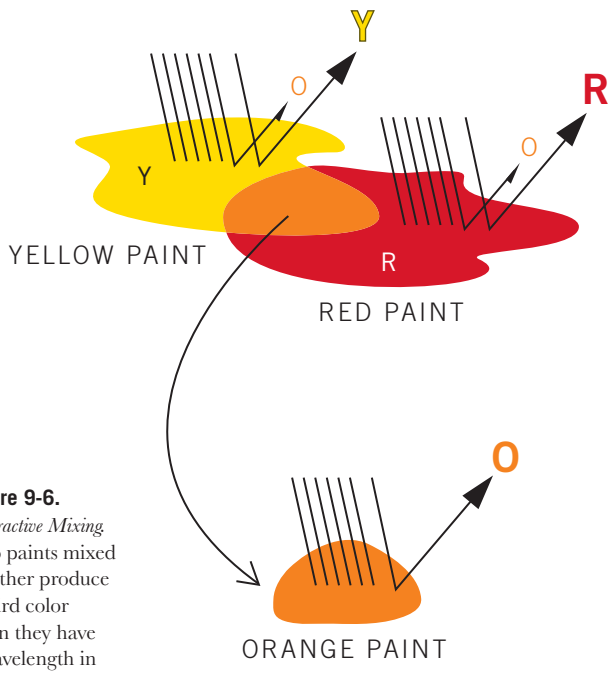


Figure 9-6.
Subtractive Mixing. Two paints mixed together produce a third color when they have a wavelength in common.

A red paint may absorb the yellow, green, blue, indigo, and violet wavelengths. It reflects a strong red wavelength and a weaker orange one. A yellow in the same medium may absorb red, green, blue, indigo, and violet wavelengths. It reflects a strong yellow wavelength and a weaker orange one. When the red and yellow paints are mixed, the mixture of the two absorbs all of the colors that each absorbed when used alone. The red in the new mixture absorbs the yellow wavelength, and the yellow absorbs the red. The only wavelength left to be reflected is orange, the *one wavelength that each of the two reflects in common.*

Paints that do not reflect a common wavelength mix to a kind of mud color. No matter how brilliant a red colorant may be,

for example, if that red also reflects violet and is mixed with a yellow that also reflects green, the resulting mixture will be drab. Mixed together, the two absorb all colors of light. Children's poster paints are a classic example of a medium with poor mixing qualities. The primary colors of "kindergarten" grade poster paints do not produce clean secondary colors. Blue and yellow together produce a sickly khaki; red and yellow a brownish orange, and blue and red a sort of purplish dirt color.

An experienced colorist mixes new colors based on the mixing affinities of specific tube or jar colors within a medium. The skills of color mixing are not the same as the ability to see differences, similarities, and intervals between colors. Each medium calls for its own set of technical skills, but all media demand the same color discrimination skills.

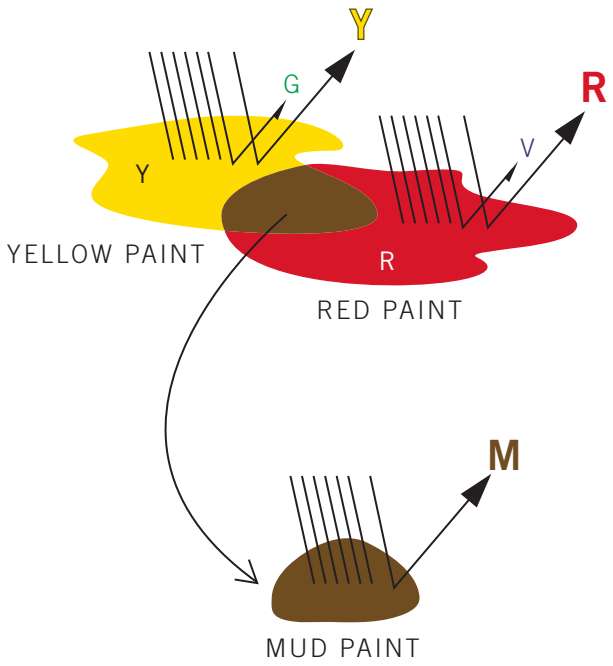


Figure 9-7. *Subtractive Mixing.* Two paints that do not reflect a wavelength in common mix to a muddy neutral.

Tinting Strength

Tinting strength refers to the relative *quantity* of a color needed to produce a perceptible difference when mixed into another color. Different tube or jar colors within the same medium will vary in tinting strength.

When colors are mixed, *quantity* does not necessarily produce a predictable result. Only tinting strength matters. Most yellows, for example, have little tinting strength. Adding one-half cup of yellow paint to one-half cup of green paint causes little change in the green. But a teaspoon of green added to a cup of yellow changes yellow to yellow-green at once. Colorants with great tinting strength are like garlic. A teaspoon of garlic in an apple pie will certainly be noticed. On the other hand, a teaspoon of apple in garlic soup does not make much of an impression.

Color Printing

Process colors are the principal medium of the printing and reprographic industries. The process color primaries are cyan (blue-green), magenta (red-blue), and yellow. Magenta is also called process red; cyan, process blue; and yellow, process yellow.

Figure 9-8.
Process Colors.

Process color, or CMYK (**C**yan, **M**agenta, **Y**ellow, and blac**K**) printing is the most familiar and universally available printing process. It is used in color xerography, computer color printers, and commercial color-press printing. The special colorants used in CMYK printing inks are also available in other media such as drawing inks, films, and markers, which is helpful to designers preparing art for CMYK printing. A design rendered by hand in process-color markers or drawing inks closely approximates the colors of the printed page.

Process colors are categorized as a subtractive medium, but they produce colors in a completely different way from artists' colors. Instead of selectively absorbing and *reflecting* wavelengths of light, process colors act as filters. Each process primary absorbs one wavelength of light and *transmits* the others.

Process magenta absorbs green and transmits red and blue.
Process yellow absorbs blue and transmits red and green.
Process cyan absorbs red and transmits blue and green.

When a process primary is laid on white paper, some wavelengths reaching it are absorbed and others are transmitted to the paper surface below. The white surface reflects the wavelength reaching it back through the ink "filter," and that color reaches the eye.

When process inks are mixed, they continue to absorb and transmit the same wavelengths. The secondary colors that result from mixing process primaries do not correspond to the colors that result from mixing artists' colors *or* the mixing of light.

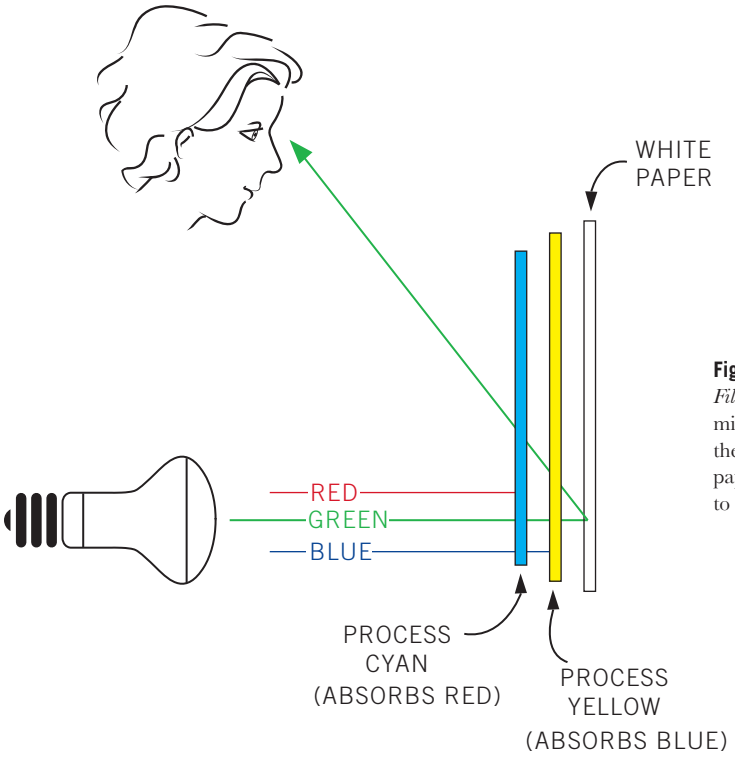


Figure 9-9. *Process Colors Act as Filters.* Here, cyan and yellow mixed absorb red and blue. Only the green wavelength reaches the paper surface and reflects back to the eye.

Magenta and yellow mixed absorb green and blue light (and transmit red), red results.

Yellow and cyan mixed absorb red and blue light (and transmit green), green results.

Magenta and cyan mixed absorb green and red (and transmit blue), blue results.

The three process primaries combined produce a medium gray tone, so black is needed to give sharpness and depth to a color image. The black also works to produce shades and muted colors. The brilliance of a pure color is reduced by adding black in 6% increments.[2]

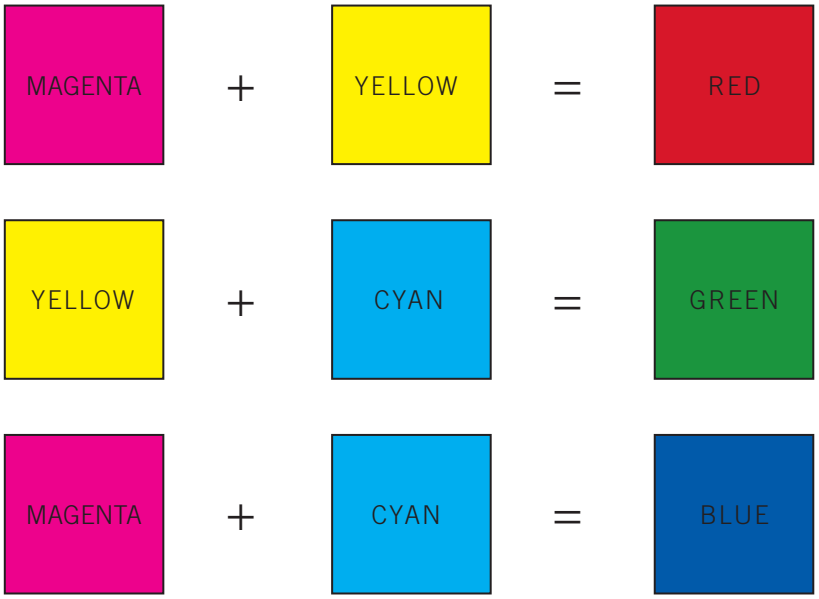


Figure 9-10.
Process Color Mixing

In CMYK printing the inks themselves are not actually mixed. Instead, each color is printed in tiny dots and as a separate step, called a *separation* or *color separation*. Printing the color dots in different densities enables the printer to achieve hundreds of colors using only four foundation inks. The dots are positioned at different angles to avoid a moire, or watermark, effect.

Figure 9-11.
Process Color Printing. Images in CMYK printing are made up of tiny dots. The dots are overlaid in different proportions to produce different colors.

Figure 9-12.
Color Separations. Each process color is printed as a separate step.

Most commercial printing is done on four-color presses using CMYK colors. Four-color process printing has some shortcomings. It does not produce clear colors in the orange range, many tints do not reproduce well, and gradations of color can be difficult to print (this book is printed in CMYK colors). Although excellent results can be obtained with CMYK colors using special printing presses called eight-color presses, only a few hundred

of these presses are available worldwide. Dealing with the constraints of CMYK color printing is an ongoing challenge for graphic designers.

Pantone[3], a world leader in color technology, has responded to the limitations of CMYK printing with the introduction of a six-color process called Hexachrome®, which adds vivid green and bright orange inks as well as brighteners to the standard CMYK colors. Hexachrome® printing requires a six-color printing press, but these presses are reasonably available; thousands are in use in the United States alone. Computer printers are also available in six colors, but most day-to-day color printing, including commercial color printing, computer printer output, and color Xerography, is still done in CMYK color with its attendant limitations.[4]

CMYK and Hexachrome® inks are not the only kind of printing inks. Some printing is done with specially formulated solid-color inks called *spot colors*. A spot color is mixed individually, in the same way as artists' colors, before being applied to a roller. Special inks like Dayglo or metallics are printed in this way. These colors offer much greater depth and clarity of hue than process colors and because they are specially mixed, the number of possible colors is nearly limitless. However, since each color must be mixed and printed separately, printing with more than one or two spot colors is costly.

Spot colors are difficult to translate into CMYK printing. CMYK printing has a limited color range, and spot colors do not. Some colors can be produced in either medium, but most spot colors converted to CMYK printing lose their depth and vibrancy. Hexachrome® offers, if not a limitless range, at least a greatly extended one, and, in addition, a result on the printed page that corresponds more closely to the brightness of a design developed on screen. A graphic design that has been printed in Hexachrome® will be closer in appearance to its screen original than any other currently available process.

Hand printing is associated more with fine art printing than with high-volume print production. In general, it employs subtractive colorants in the same way as other artists' media. True hand printing, like hand silk-screen or block printing, is used commercially in a small number of ways. Luxury goods like custom wallpapers or high-end home furnishings or fashion textiles are occasionally hand-silk-screen or hand-block printed, but the cost

of these techniques on a large scale is prohibitive. Some techniques in block printing and silk-screen have been adapted to machine printing and even to the use of CMYK process colors. Silk-screen in particular is in common use for printing onto fabric.

Figure 9-13. *Commercial Silk-screening.* Silk-screened wallpaper has a richness of surface not found in machine printing. *Image of "Compton Court II©" courtesy of First Editions Wallcoverings & Fabrics, Inc.*

Endnotes

1 Webster's 1987, page 620.

2 Warren 1996.

3 PANTONE® and other Pantone, Inc. trademarks are the property of Pantone, Inc.

4 Pantone, Inc., reprint from October 1998 edition of Electronic Publishing. PennWell, 1998.

Chapter 9 Highlights

- A design medium is a means of translating images, ideas, and colors from one kind of visual experience to another. Renderings and models are the principal way in which new product and color ideas are communicated to potential buyers. In all fields of design visual thinking and presentation takes place in (at least) one medium when the color of the product is achieved in an altogether different way.
- Most of the time, the form, surface, and color of an object are understood together as a single experience. Natural materials have random gradations and shadings that many consumers find attractive and can have color added without losing the attributes that make them appealing in the first place. Man-made materials in general need color added to make them attractive to consumers. Color can be introduced directly into the raw material before the start of the manufacturing process, chemically bonded to the surface of a material, applied as a coating at a late stage of manufacture, or colored using a combination of these means.
- Traditional artists' media are substances made up of an inert vehicle, or base, that contains (or carries) the colorant. Dyes are colorants in solution. Dyes penetrate the material they color and bond with it on the molecular level. Pigments are finely ground particles of colorant suspended in a base. Pigments "sit" on a surface rather than bonding with it. Many modern synthetic colorants bridge the traditional distinctions between dyes and pigments.
- Each medium has a limited number of colors that can be used alone or mixed together to extend its range. In order for two colors to produce a new hue, their colorants must reflect a wavelength in common. An experienced colorist mixes new colors based on the mixing affinities of specific tube or jar colors within each medium.
- Tinting strength is the relative quantity of a color needed to produce a perceptible difference when mixed into another color. Different colors within the same medium will vary in tinting strength.
- Process colors, or CMYK, are the principal medium of the printing and reprographic industries. The process

primaries are cyan (blue-green), magenta (red-blue), and yellow, and are used with black. Process magenta absorbs green and transmits red and blue. Process yellow absorbs blue and transmits red and green. Process cyan absorbs red and transmits blue and green. When a process primary is laid on white paper, some wavelengths reaching it are absorbed and others are transmitted to the paper surface below. The white surface reflects the wavelength reaching it back through the ink "filter," and that color reaches the eye.

- Process primaries mix into the secondary colors that do not correspond to the colors that result from mixing artists' colors or the mixing of light. Magenta and yellow mixed produce red. Yellow and cyan mixed produce green. Magenta and cyan mixed produce blue. The three process primaries combined produce a medium gray tone. Black is needed to give sharpness and depth to a color image.
- In CMYK printing each color is printed in tiny dots as a separate step called a color separation. Printing the dots in different densities enables the printer to achieve hundreds of colors using only four inks. Most color printing, including commercial color printing, computer printer output, and color Xerography, is done in CMYK color.
- Some printing is done with specially formulated solid-color inks called spot colors. A spot color is mixed individually in the same way as artists' colors. A six-color process called Hexachrome® adds vivid green and bright orange inks as well as brighteners to CMYK colors. Hand printing is associated more with fine art printing than with high-volume print production. In general, it employs subtractive colorants in the same way as other artists' media.

10

The Medium of Light

Then and Now / Images of Light / Lost in Translation / The Screen Display / Color Management / Color Display Modes / Presentation: Screen and Print / Color on the Web / Web Color Coding / Emerging Media

Now that we have the machines let us, instead of imitating former products and techniques, try to design goods that are characteristic of machine production—do not let us imitate former designs. Let us, with the help of these technical aids, produce the new.

—Gregor Paulsson, Design and Machinery 1919

Less than a generation ago the advent of computer-aided imagery created a revolution in the studio workplace. The new medium offered designers for the first time the opportunity to explore a variety of solutions without constraints of time or cost. Countless changes in the shape, size, and pattern of images and colors became possible without the time-consuming effort of reworking by hand. Visual ideas could be saved in memory or on a storage device at any point, so that discarded images were readily available for reconsideration or future use. Images could be sent between locations instantaneously, with absolute fidelity and without concern for loss or damage in transit. The speed and ease of design production meant that staffing needs could be reduced. Conversely, computer drawing gave freelance designers the tools to compete with larger organizations. The time, labor, material, and space-saving advantages of the digital medium translated directly into greater profitability for both designer and manufacturer.

Present-day imaging technologies and design software have capabilities unimaginable even a short time ago. Programs that offer an illusion of three-dimensional space and movement through it are commonplace. The same problem can be addressed by users in remote locations working simultaneously. For all of these reasons and more, the computer is now the near-exclusive means of drawing in the design industries.

Then and Now

The revolution in drawing medium paralleled a revolution in communication. The Internet, conceived in the 1960s as a means of communication for national defense, rapidly expanded into a new and seemingly limitless public broadcast medium. Made up of countless computers and their connections, the Internet swiftly became a series of interconnected networks that made it possible for information to be accessed and shared on a global basis.

The World Wide Web (the Web, or WWW) is an enormous sub-network of the Internet that offers text, images, motion, sound, and interactive communication to businesses, organizations, and individual users. The Web provides access to millions of information sites, or pages, that are linked in such a way that users can move between them with ease. Access to the Internet and the WWW is accomplished through paid providers called ISPs (Internet Service Providers) that enable individual access to the information superhighway.

By 2003, a not-so-distant "then," it had been estimated that 55% of United States households had Web-connected computers. The incidence of Internet use was weighted by factors like region, affluence, ethnicity, education, and age. By 2009 that estimate had risen to over 80% of households, with the number of hours spent on the Internet influenced by the same factors. There is no reason to think that the expansion of Internet access or the power of the World Wide Web has peaked. The phrase "computer-literacy" will, within the foreseeable future, become as meaningful as "literacy."

For information seekers, there is no longer any need to adhere to a broadcaster's schedule, or wait for a periodical to be published or a library to open. Information, whether about a consumer product, an academic or technical question, or just about current news or weather, is available 24 hours a day, 365 days a year. For vendors, the magnitude of the potential audience is breathtaking.

Images of Light

A *graphic* is something written or drawn. A graphic image can be as simple as a black and white drawing or text or as complex and richly colored as a Botticelli Venus. It may or

may not include drawings, text, photographs, and color, but no matter what it consists of, it is a means of communication.

Graphic design is the arrangement of forms and colors into a composition whose purpose is to deliver information, to draw attention, or both. Graphic design is often called, perhaps more accurately, *communication design*. Communication design is only one field in which screen drawing is the medium of choice. Architects, engineers, and interior designers; package designers, product designers — all use drawings to convey design ideas. Until the recent past, renderings were made in subtractive media like pencil, ink, gouache paint, watercolor, or marker. Hand-colored rendering are now a rarity. Every field of design employs drawing, and every field, with rare exceptions, employs drawing in light.

If screen drawings are graphic designs made in light, the color monitor is the designer's canvas: an empty picture plane awaiting the input of visual information. Among the principal uses of screen drawings are:

- The preparation of technical drawings
- The preparation of materials for printing, like books, periodicals, advertising flyers, and packaging
- The preparation of renderings for sales or production purposes
- The design of Web pages that must convey information about products; as online catalog
- The design of Web pages intended only for on-screen viewing and not associated with a physical product

The ways in which colors can be used to better communicate ideas — to improve the readability of text, or to create an impact, a special effect, or color harmony — are no different in drawing with light than in traditional media. What *is* different about designing in colors of light is that the colors must be selected not only with an eye to best resolving the immediate design problem, but also with consideration of its end use. If the drawing illustrates a product, can colors on the screen be made to "match" the product? Will the on-screen image end up as printed material, or will it be viewed only on a monitor? If the image is meant only for on-screen viewing, will it be seen on one screen, or on many? If on many screens, how many, whose, and what kind?

Josef Albers wrote about the interaction of colors in subtractive materials like paints and color papers. Designers today must navigate the interactions that exist between additive and subtractive colors; between light and substance. The monitor screen is a matrix in which three modes of color mixing can be in play at the same time: the red-yellow-blue-based colors of design thinking, the red-green-blue-based colors of working in light, and the cyan-magenta-yellow-based colors of process printing.

Figure 10-1. *Modes of Color Mixing.* Designers today must be familiar with (at least) three different modes of color mixing.

Today's technology demands of the user a constant adaptation to change. Hardware and software are in states of continual development. Changes take place so rapidly that nearly anything written about hardware or software is outdated before it goes into print. Future developments in the way that the colors of light are delivered and mixed on the screen will not affect the nature of the medium itself. Like every medium, light offers the designer a mix of advantages and disadvantages. Working successfully in color on a monitor screen means understanding the capabilities of the light medium: what it can do, and what it cannot.

Lost in Translation

The challenge of transposing a color idea from one medium to another is not new. Printed colors, for example, are not exactly the same as product colors; designers strive only to get the closest possible match. But even as technologies of imaging and the capabilities of software move forward at a dizzying pace, a crucial difference between computer renderings

and those produced by traditional media is often overlooked. The computer is the drawing instrument and the screen its blank page, but the *medium* is light. *Computer rendering means that much of the time the designer draws and paints in additive colors images of things that ultimately will be experienced as subtractive colors.*

Working in light can be an intoxicating experience. Colors are vibrant, even brilliant, and mutable at the click of a mouse. But colors seen as light are experienced differently from the colors of physical objects. The colors of an object are seen as scattered, or reflected light, and reflected light is inevitably reduced to some extent. Some light is lost as the light travels from the source to the surface, and more is lost as the light travels from the surface to the eye. An additive color that is the putative "match" of a subtractive color —a screen color that represents a printed or product color—will inevitably be sensed as more brilliant on the screen than in life or on a printed page.

Many screen images include representations of real-world objects. The things that most impart distinction and vitality to the colors of objects—variations in surface, variations that occur from the use of different colorants or changes in ambient lighting—do not occur on a screen. They can be simulated, but the complete experience of color, with its attendant sense of the *tactile* quality of an object, is absent in the medium of light. Although vision is the primary sense used to identify objects and surfaces, the sense of touch is closely associated with it, as every PLEASE DO NOT TOUCH sign in a museum reminds us. An image made in light is a vision-only, single-sensory experience.

Reflected colors also seem more natural. Human beings are predisposed to understand reflected color as "real." The surface of an object scatters light in distinctive ways that contribute to the way that colors are perceived. Colors on a monitor screen, unmodified by surface, have a uniformity of effect. The direction from which light originates matters as well. In nature, general (white) light that comes from an angle 45° above or behind the viewer is sensed as the most natural. Light emitted by a monitor screen reaches the eyes from a direct frontal position.

The brilliance of colored light can be used to advantage in illustrating certain kinds of surfaces. Additive color and the capabilities of drawing software combine to great effect

in the rendering of reflective materials. Screen drawings of materials like glass, polished metals, laminates, and high-gloss paints are particularly successful.

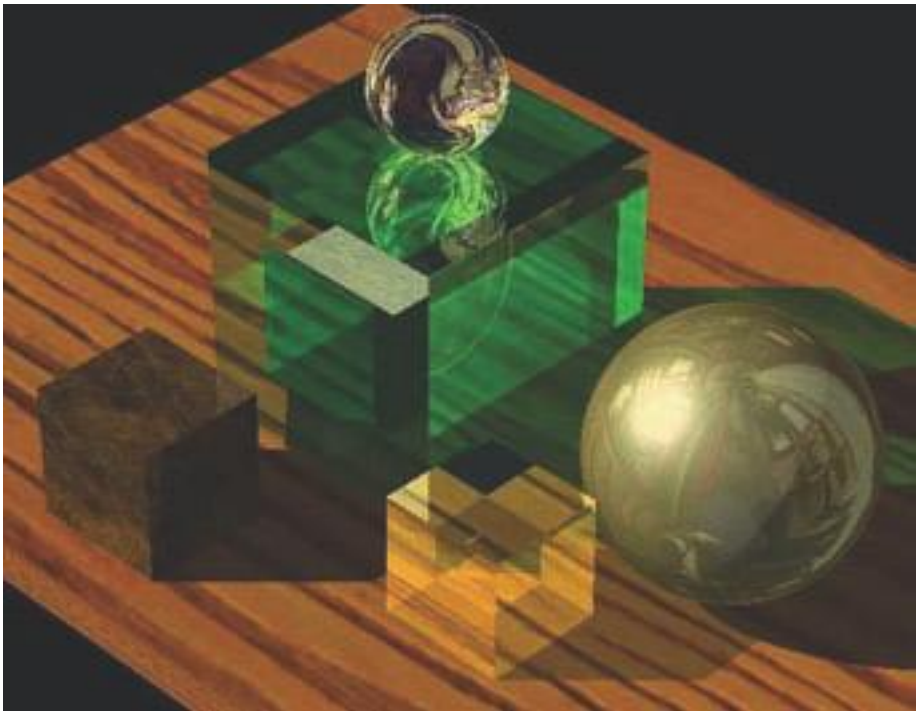


Figure 10-2.
Screen drawing. Digital drawing is perfectly suited to rendering reflective surfaces. *Digital drawing courtesy of Professor Ron Lubman, Fashion Institute of Technology.*

Materials with textured or matte surfaces like unpolished stone, natural woods, and textiles are more difficult to display. Pattern and color can be represented, but the way in which a textured surface reflects light is not easily conveyed. Textiles are particularly problematic. Color fields on a screen can be broken into areas of dark and light to suggest texture, but because the light that reaches the eye comes from only one direction, the characteristic flat brilliance of additive color persists.

Only computer drawing raises the issue of transposing colors from substance to light, and no other medium is in such widespread use. Good results can be achieved, but perfection is not possible. The best that can be achieved on a screen is a color that approximates its real-world counterpart. "Virtual" reality means "almost" reality.

The Screen Display

Successful color rendition on the screen begins with the selection of hardware and software that best meets a designer's particular needs. Every color image on a screen requires the interplay of platform, hardware, and software. Each platform—the combination of a particular type of computer and a particular operating system—has features that enable it to perform certain tasks better than others.

Software is designed to operate on a specific platform. The two platforms in most widespread use, PCs (personal computers) and Macs (Apple Macintosh) offer different capabilities. The display differences between competing products is small and is likely to continue to diminish. Mac and PC design programs are equally available, although Mac programs continue to dominate the design market, probably in response to designers' preference for their more intuitive operation.

Images on a monitor screen are created by the distribution (pattern) and strength (brightness) of light emitted at different wavelengths. The range of colors that a monitor displays is called its *gamut*, or *color gamut*. There are limitations to the color gamut of any monitor, but today's typical consumer-marketed monitor can display so many colors that its capability approaches the range of human color vision. *All monitors, however, display only additive color: wavelengths of light directly to the eye.*

A monitor screen is made up of individual elements called pixels ("picture elements"), which are the smallest units of the screen display. The pixels are arranged as a vertical and horizontal grid. Each pixel is made of up three components that can be triggered by an electrical impulse to produce the additive primaries, red, green, and blue. The voltage of the impulse applied to each pixel determines what color is displayed. Mixing the light emission *within* each pixel allows a full range of hues to be displayed. A pixel emitting both red and green light, for example, displays yellow, which is the mix of the two.

Each pixel on the screen can be altered in hue, value, and saturation. Pixels are measured as dpi, or "dots-per-inch" (or ppi, "pixels-per-inch"). The higher number of the dpi, the greater the screen's resolution (ability to show detail.) The number of ways in which a pixel can be *altered*—the total number of colors that a monitor is capable of displaying—is

determined by its *bit-depth*. A bit is the smallest element of electronic information that a computer can use. The more bit-depth a monitor has, the more pieces of information—the greater the number of colors—it can display.

The earliest monitors had only a 1-bit depth and displayed only black and white. The first generally marketed color monitors had an 8-bit depth, enabling them to display 256 colors. The 24-bit-depth monitor, now standard for consumers, can display 16,777,216 variations in color, but as a practical matter only about 20,000 are available for design purposes, a number that is adequate for most tasks. The rest are there—and are necessary—to make smooth gradations between colors.

A 24-bit-depth monitor is sometimes said to be capable of displaying "true color," implying that the range of its gamut is comparable to human vision (and also that there is such a thing as "true color" at all.) While the best monitors can display an enormous range of colors, none can display all of the colors within the range of human vision.

The first monitors in general use were cathode ray tube (CRT) monitors. These were the original television or computer monitors, bulky forms that occupied considerable space. The inside of the screen of a CRT monitor is coated with a grid of phosphors, substances that emit light when they are bombarded with electrical energy. Each pixel contains phosphors that emit each of the three primary colors of light (red, green, and blue) when it is excited by an electrical impulse. An electron gun at the back of the monitor bombards the phosphor grid with a constantly moving stream of electrical energy at a variety of frequencies and the phosphors glow in different colors, depending on the electrical stimulus.

The CRT monitor has been displaced by the LCD, or liquid crystal display. LCD monitors, which also employ pixels, were developed originally as flat screens for laptop computers. Today they are found as monitors for personal computers and televisions as well as laptops. LCD monitors are lightweight and take up only inches of depth on a work surface, a combination of features that makes larger screens practical. They are also more energy efficient.

Figure 10-3.
The LCD Monitor. The flat screen LCD monitor is used for all design tasks. *Image courtesy of The Stevenson Studio, www.thestevensonstudio.com ©2005*

An LCD monitor displays color when light passes through pixels that are filled with liquid crystal and red, green, and blue filters. Light, typically fluorescent light, is provided behind the liquid crystal matrix and the filters within each pixel bend (refract) the light to produce spectral colors. Since each pixel has three components (red, green, and blue) and each is represented by 8 bits, the total bit-depth of an individual pixel is 24 bits. By varying the voltage of electrical stimulus, the hue intensity of each pixel can be extended to 256 steps. Combining the pixels (256 X 256 X 256) allows a display of more than 16 million colors. Because colors on an LCD screen are seen as wavelengths of light, not as a glow emitted by phosphors, the colors produced are close to true spectral hues.

A limitation of early LCD monitors was their extremely restricted viewing angle. As soon as the viewer shifted away from the ideal angle, contrast was lost and colors diminished in clarity. The newest LCD monitors have greater sharpness and although the viewing angle issue persists, it is much improved. There is also a current trend toward the use of LEDs (light-emitting diodes) as the backlight source for LCD monitors. LED backlighting is brighter than fluorescent and has the advantage of enabling a wider color gamut than is available on present screens. Steven King, a pioneer and expert in the field of computer

color, predicts that within the near future all high-end LCD monitors will be produced with LED backlighting.

Monitors with up to 48-bit color are available for specialty purposes like professional image editing, high-end workstations, and medical imaging. These monitors use only the 24-bit-depth for colors; the additional bits are for special effects like animation or translucency. Very-high-end CRT monitors are still made for these purposes, but according to Paul Gagnon of DisplaySearch, a provider of display trends data, LCDs are supplanting these as well, and CRTs will disappear from the market within a very short time.

The LCD monitor, with its lighter weight, spectral color, lower energy consumption and rapidly advancing technology, has replaced the CRT, but that does not mean that there is a single all-purpose-perfect hardware or ideal software. Different manufacturers provide different quality levels and different capabilities. Every screen and mode of display has its own quirks, its own negative and positive aspects. With the rapid and constant developments taking place, selecting a monitor that is best for specific design tasks requires careful consideration of capabilities and features that are currently available.

Color Management

The difficulties inherent in displaying subtractive colors in an additive medium are so universally acknowledged that an entire industry is devoted to solving them. *Color management* tools are needed not only for colors on the screen that must correspond to product and print colors, but also for images that will be presented as light alone.

Hardware and software tools help to reduce the difficulties of making acceptable matches between screen colors and print or product colors. Color standards are a basic tool. A standard enables the designer to make constant comparisons against the same color at each step in the design process. Standards for color printing are most often quality-controlled printed colors on paper. A standard can also be a sample of actual product, like a metal finish, a wood finish, a yarn, or a lipstick. Standards are essential to maintaining a high level of consistency between the color of a product or printed page and its representation on-screen.

In order for screen colors to best represent the color of an actual product, the source of any image must also be standardized. This means that the color of the product and any photograph, transparency, scanned image, or rendering of it must conform to the same set of standards. Unless all of these elements can be cross-referenced to the same standard, even the most carefully prepared screen image is potentially unreliable as an indicator of product color.

A number of sophisticated color management devices are available for purposes of *calibration*. To calibrate a monitor means to adjust it so that specific combinations of red, green, and blue signals produce specific colors on the screen. The resulting screen colors can be calibrated to a set of equally standardized subtractive colors—print or product—so that the on-screen image is a reasonably accurate prediction of how colors will appear in their final form. The spectrophotometer of a color management tool measures and brings into synchrony the wavelengths of reflected color from the standard and those of emitted color from the screen.

At times the user must consider the end-use lighting condition of the standard before calibrating, because subtractive colors appear differently under different sources of ambient light. Colors on a monitor will also drift over time, even a relatively short period of time, so monitors that are used for design purposes need to be calibrated on a regular basis or before doing critical color work.

ColorMunki™ exemplifies a multipurpose color managment product. ColorMunki™ offers a program that can synchronize palettes to specific design programs, match screen colors to PANTONE® CMYK and spot color inks, create and name custom color palettes, calibrate colors for large screen projections. A colorimeter is used to calibrate the monitor with its software. A spectrophotometer captures the color of any material or image and creates an

Figure 10-4. *Color Management.* ColorMunki™ exemplifies the multipurpose calibration product. *Image provided courtesy of X-Rite.*

on-screen match; it even evaluates subtractive colors under different light sources. Versions are available for both Mac and PC platforms, and there is also a product for photography.

Calibration is also done to standardize colors between devices. A group of monitors can be calibrated to display like colors, but within limitations. Different platforms handle color slightly differently. Not all colors seen on a Mac, for example, will appear the same on a PC. Even within the same platform, in order for all viewers to see the same colors, their monitors must calibrated in the same way—standardized with each other as well as with the source of the color image. Monitors that share the same display capabilities can be calibrated identically so that a number of monitors display the same colors with a high level of consistency. Monitors that have different display capabilities cannot be calibrated in this way, but they can be calibrated to the same set of standards, so that a reasonable level of display consistency can be achieved. With care and good calibration, it is possible to display images that are reasonably close in color to a controlled audience of any size.

Reference standards and color management tools provide the designer with the means to make acceptable matches between additive and subtractive colors, particularly between screen and print colors. Not all users take advantage of these tools. In some studios a match between screen color and product is still made laboriously by matching each color individually by eye. Several design professionals interviewed confessed to shifting screen colors deliberately *away* from a standard in order to provide a livelier image for on-screen presentation. Even with the best tools, there are no absolutes. *No matter what standards and tools are employed, the ultimate decision on the quality of a match—and the best presentation—is made by the designer's eye.*

Color Display Modes

Bit-depth determines only how many colors a monitor is *capable* of displaying. The software determines how many colors are actually displayed and how colors are mixed on the screen. There are three different ways in which design programs display and mix colors. Generally, all offer a basic assortment of hues, a range of grays, and black and white presented as a box or circle called the color palette. Colors on the palette are mixed in one of three possible ways: the CMYK mode, the RGB mode, and the HSB mode. Each kind of display has a limited number of available colors and it is this range of colors that makes up its particular gamut.

Dithering is a capability of some graphics programs that extends the range of colors in software that has a limited color palette. Dithering takes advantage of the human eye's tendency to "mix" colors that are similar to each other. In a dithered image, colors that are not available in the gamut are approximated by pixels in a mixture of two or more similar colors from those that are available. Dithered images, particularly those with relatively few colors, often have a fragmented, grainy appearance. The number of available colors in the gamuts of most present-day software has made dithering less of a concern than in the past.

Figure 10-5. *Dithering.* Dithered images can have a fragmented, grainy appearance.

The *CMYK mode* of color display imitates the results of mixing process colors. Each color in the CMYK mode represents a color of process ink (cyan, magenta, yellow, black). The CMYK display mode facilitates working on the screen for print production.

Figure 10-6. *Color Display Modes.* Each color in the CMYK mode represents a color of process ink. *Adobe product screen shot reprinted with permission from Adobe Systems Incorporated. ©2010 Adobe Systems Incorporated. All rights reserved. Adobe and Photoshop is/are either [a] registered trademark[s] of Adobe Systems Incorporated in the United States and/or other countries.*

A bar display allows the user to select cyan, magenta, yellow, or black as a percentage of the screen display. When two of the CMY colors are mixed without black, clear colors and tints result. Three CMY colors mixed in equal percentages (with no black) make a middle gray that has a chromatic basis. Muted colors are achieved using this CMY gray mix and manipulating one or two colors until the desired effect is achieved.

Achromatic gray scales are made by manipulating percentages of black (K) alone. Black is used for making hues darker: one or two of the CMY colors plus black yields shades. The black-based gray that is mixed into a pure hue will make it darker, not more muted. There

is a difference in mixing colors between a CMYK light display and CMYK inks. In mixing inks, black is used in producing both shades and muted colors.

Varying the percentages of the three hues and black enables the designer to display on screen a full range of colors including most hues, values, and saturations. The CMYK display mode requires little new thinking about color mixing for designers who are familiar with process colors.

The *RGB mode* of screen display parallels the behavior of light. It is used to create images that will be viewed only as light display, such as Web pages or DVDs. In the RGB mode each of the light primaries, red, green, and blue, is displayed as a separate bar, called a channel. Each channel has an 8-bit depth so each primary can be displayed in 256 variations, and these multiply to the possibility of millions of variations.

The user selects each color for mixing in a range from no display (0%) to full display (100%). When 100% of each of the three light primaries is mixed, the result is white. When none (0%) of each primary color is mixed, the display is black. If 50% each of red, green, and blue is displayed, the result is a middle gray. Varying the relative percentages or red, green, and blue in RGB mixtures enables the user to view most hues, values and saturations.

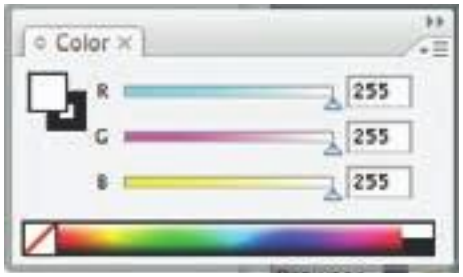


Figure 10-7.
Color Display Modes. The RGB mode of screen display is used to design for the medium of light. Colors are mixed exactly as light is mixed. *Adobe product screen shot reprinted with permission from Adobe Systems Incorporated. ©2010 Adobe Systems Incorporated. All rights reserved. Adobe and Photoshop is/are either [a] registered trademark[s] of Adobe Systems Incorporated in the United States and/or other countries.*

RGB mixing does not correspond to either of the two most familiar forms of subtractive mixing, artists' colors or process colors. It is not suitable for printing because the number of colors that it is possible to generate in the RGB mode is much greater than can be printed using the CMYK process and many of the colors that it displays cannot be printed. The RGB mode is most directly associated with working and viewing in the medium of light.

The *HSB mode* of display stands for hue, saturation, brightness (value). The HSB mode displays a color map. Next to the color map are three boxes, one each for hue, saturation,

and value. Each box has a range of numerical values. The user first selects a color from the map, then instructs each of the boxes to modify that selection in hue, saturation or value. Unlike the CMYK or RGB modes, it does not correspond to either additive or CMYK subtractive color mixing. Although colors are mixed on the screen in a way that is familiar to users of traditional media like paints, the HSB mode actually requires learning to mix light in a way that is associated only with digital design. Software for the HSB display mode is also marketed under different names: HSL (hue, saturation, lightness) and HSV (hue, saturation, value).

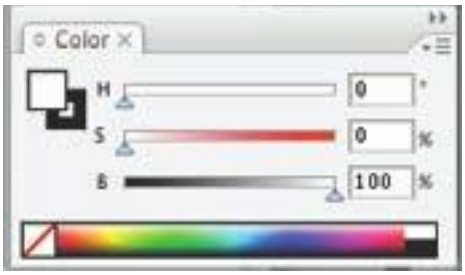


Figure 10-8. *Color Display Modes.* Colors in the HSB mode are mixed as separate steps—one each for hue, saturation, and brightness (value). *Adobe product screen shot reprinted with permission from Adobe Systems Incorporated. ©2010 Adobe Systems Incorporated. All rights reserved. Adobe and Photoshop is/are either [a] registered trademark[s] of Adobe Systems Incorporated in the United States and/or other countries.*

Presentation: Screen and Print

The level of color management needed during design development depends a great deal on the end use of a drawing. For example, the computer is superbly suited to the preparation of technical drawings, where pinpoint accuracy is critical. The degree of precision that is essential in plans drawn for constructing a ship, an airplane, or a building is now unimaginable except as digital drawing. In general, however, technical drawings use color only to make diagrammatic situations easier to read. These drawings may employ color, even require it, but accuracy in rendering a specific color—a certain red, or a particular blue, whether the drawing is presented on a screen or printed—is rarely essential to their purpose.

Renderings, on the other hand, call for close attention to color. Renderings have two distinct purposes. First, they are used to visualize new products and/or new colorings. The designer's rendering is the first, and sometimes the only, opportunity for a manufacturer to decide whether a new product or color offers a competitive edge, so it is critical to decision-making that its colors represent as closely as possible the colors of the finished goods.

The second purpose of renderings is to convey product design and color from the design studio to the manufacturing floor. The use of preproduction renderings is industrywide, ranging from the design of cars and appliances to textiles and building materials. A rendering can be sent electronically to a manufacturing location, where the colors in the image are

"matched" to actual materials. Alternatively, the selection of some material—yarn, metal, plastic, or stone—precedes the rendering and the artist begins work from a physical or samples. The factory uses the screen rendering as a color placement guide. In either case, the task of the renderer remains the same: to replicate in light, as closely as possible, the hue, value, saturation, and surface qualities of a subtractive material.

Figure 10-9.
Traditional Rendering. A gouache rendering of Edward Fields' carpet "Tropique" is shown to a client for approval, then sent from the salesroom to an overseas factory for fabrication. *Image courtesy of Edward Fields Carpet Makers.*

Figure 10-10.
Computer Rendering. Computer-generated drawings have overtaken hand-rendering for carpet designs. Edward Fields' carpet "Blythe Bridge" is rendered on the computer. *Image courtesy of Edward Fields Carpet Makers.*

Renderings may be printed or may be presented only on a screen. A good representation of real-life color on the screen is as vital for small-scale undertakings as for large ones. The architect or interior designer making a laptop presentation to a private client is just as much at the mercy of the screen's ability to display the colors of paints, textiles, and finish materials as the designer of a hotel complex making a PowerPoint presentation to a group of investors.

The most extensive efforts to manage the disparities between on-screen color and product have been made for the printing industry. When a screen drawing is done for printing purposes the drawing is made in light, but the final medium is printing ink. The great majority of printers, whether personal or commercial, produce continuous fields and shadings of color using process color inks, and these colors are a standard in graphic design software and many color management tools. Limiting the colors on the screen to the capabilities of CMYK printing helps to ensure that the screen design will be reproduced in print as closely as possible.

A *bitmap* is an image made of dots, or pixels, that correspond exactly to information in the computer memory. A computer printer separates the digital image into four bitmaps, one each for cyan, magenta, yellow, and black. Each bitmap duplicates the number and pattern of pixels of that color in the computer memory. The bitmaps are printed out as dots and overlaid in different densities in an effort to simulate as closely as possible the way in which an image is created on a commercial printing press. There are large-scale computer printers that use CMYK colors with additional bitmaps to extend the color range, adding grays and lighter cyan and magenta inks. There are also families of computer printers that do not use CMYK inks. Hexachrome® printers are available, although these are not in widespread use. Another type uses colored wax, which is more opaque than process inks and reflects light differently. Another, the dye-sublimation printer, uses CMYK inks on a plastic film that is fused at high heat onto paper. The heat causes the dyes to melt, or sublimate, onto the paper, which reduces the dot quality of typical ink printers. Dye sublimation prints appear more like photographs than typical computer prints but are costly to produce.

Color on the Web

Every Web page is a graphic design—a contained arrangement of forms and colors. Whether it is static or in motion, or accompanied by sound, its entire reason for being is to

communicate information. Whether or not a Web page will be successful in *attracting* viewers to that information depends in large part on the appeal of its page design and colors.

A much greater range of colors can be seen on-screen than can be printed. A screen image is potentially richer in color than any other means of presentation. Screen colors can be displayed with near-perfect consistency between monitors when they share the same platform, software, and are calibrated to the same set of standards. This kind of controlled situation might be found in a design studio, for example, including its remote locations.

The consumers who retrieve images from the Internet are not a controlled audience. They regularly experience color-display inconsistencies, although they are not necessarily aware of them in those terms. The consumer may be using a Mac, a PC, or some other platform. Home monitors vary in type and quality. The average person is unlikely to calibrate or use color standards. Since color depends on the display properties of the *receiving* monitor, there is no guarantee that the page designer's carefully planned colors will reach the home screen with any fidelity. Even viewers who do calibrate do not necessarily employ the same calibration software or set of standards. Realistically, the colors of a Web page cannot be assumed to arrive their destinations intact. They will be somewhat different on each screen they reach.

A major function of Web page design is to provide information about consumer goods and services. Many Web sites include on-screen catalogues, and these pages must convey as closely as possible the form, colors, and surface of goods and materials. No matter how well designed a product may be, success in the marketplace depends in great part on its color appeal. Consumers rate color as the single most important factor in making a decision to purchase. It is critical to sales that the screen display represent as closely as possible the colors of the product. If a drawing or photograph on a Web page has not been calibrated to a product or print standard before being broadcast, the number of opportunities for color shifts to take place between the product and its arrival on a home screen increases at each step.

Consumers encounter the same issue in reverse. As more and more items are purchased from online vendors, more decisions to buy are made based on images of light. A sweater

the purchaser identifies as blue on a screen is likely to be returned if it arrives in green. Careful Internet vendors remind potential purchasers that the color of the product may be different from the image on their screen.

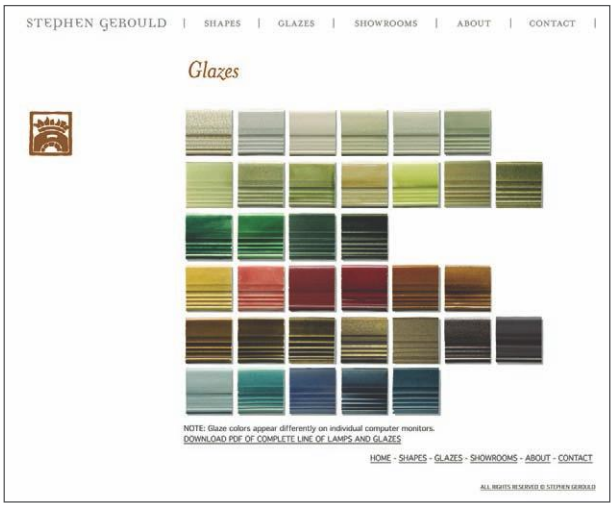


Figure 10-11.
Selling Color on the Web. Stephen Gerould reminds potential purchasers of his handmade ceramic lamps that the color of the product they receive may not be the same as the image on their screen. *Photography by Carolyn Schirmacher from the Website of Stephen Gerould Handmade Ceramic Lamps.*

When a Web page offers information only and does not represent a product, the page itself *is* the product. While it is likely that there will be a disparity in color between the designer's colors and the viewer's colors, only one step—from designer's screen to the receiving screen—separates the two. Because the image does not represent subtractive color, color differences between the sender's screen and receiver's screen are less likely to matter.

Web Color Coding

HTML (hypertext markup language) is the principal programming language of the Internet. It is the language in which Web pages are designed. Web design colors are specified by several widely used systems of coding.

Computer function is based on binary code. Colors in many widely used programs are specified in binary code by their relative proportions of red, green, and blue. Binary color coding identifies 256 steps, designated 0-255, of red, green, and blue (in that order) for

each pixel of the on-screen image. Each color is coded by its intensity within that pixel, with 0 representing no color and 255 representing maximum color. Black, for example, has no color and is 0,0,0. White has all three primaries equally and at full intensity and is coded as 255,255,255. Red is 255,0,0; green is 0,255,0, blue is 0,0,255. Binary codes for the secondary colors of light follow logically: yellow, the mix of red and green, is 255,255,0; cyan, made of green and blue, is 0,255,255; magenta, made of red and blue, is 255,0,255. Any color in the available gamut can be coded in this way. A color described as "orange-orange red," for example, is 255,102,0: full red, with enough green (because green and red mixed equally yield yellow) to make it into an orange.

Colors for Web graphics are also described using the *hexadecimal* system. The specification for each color in the hexadecimal system is based on a 16-symbol code consisting of the numbers 0 through 9 and the letters A through F. Each color is specified by two symbols as a percentage of each primary emitted in red, green, blue order. The full specification is called a hex triplet. No emission at a wavelength is indicated as 00; the maximum possible emission as FF. Black is therefore 00000, or no light emission; white is FFFFFF, or all primaries emitted equally. Red is indicated as FF0000; green as 00FF00, and blue as

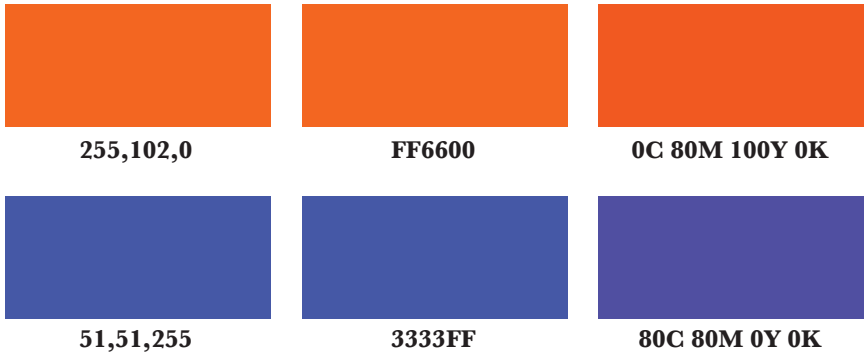


Figure 10-12.
Screen colors are coded in more than one way. A screen color can be specified in binary coding, hexadecimal coding, and by its closest CMYK print equivalent.

0000FF. Yellow is FFFF00—the secondary color mix of red and green, with no blue. Any color in the available gamut can also be coded in this way. The "orange-orange red" that is 255,102,0 in binary code is FF6600 in hexadecimal code.

Neither binary nor hexadecimal coding offers perfect color transitions. Middle codes may or may not be middle colors, and color intervals are not reliable by number. The designers eye is still necessary to make decisions about final colors and sequences.

Colors that have been specified in binary code can be converted to hexadecimal code, and vice versa, by using readily available charts or formulae. Both binary and hexadecimal coded colors are also specified by names that are generally acknowledged by most Web browsers. The names help Web designers to locate some colors more easily than by their hex triplet or binary designations, but the number of names includes only a small fraction of available colors.

Early color monitors had very limited bit-depth. A tool that helped designers working on material for Internet distribution was a group of RGB colors called the Web-safe (or Netscape) palette. These colors were (and still are) available online, in book form, and were an option in older design programs. The Web-safe palette identifies 216 colors in a full range of hues that are reasonably consistent for viewing on both Mac and PC platforms and monitors with a limited (256 colors) display range. Restricting colors to this palette gave designers some assurance that colors reaching the home computer would be displayed as a reasonable approximation of the original.

The colors in the Web-safe palette were selected for their probability of consistent transmission, not for design application. Many of the colors within each hue group are quite similar. For the greatest chance of success in reaching a wide variety of platforms and monitors, a designer

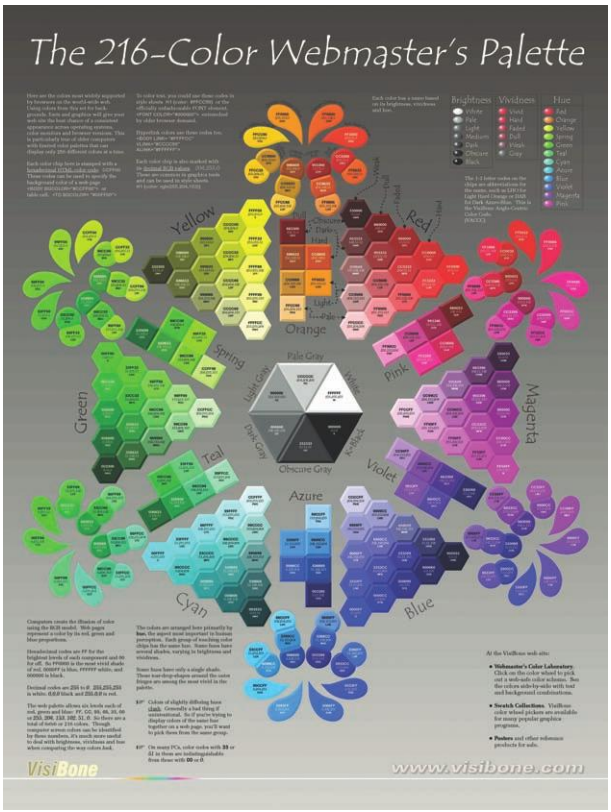


Figure 10-13. *The Web-Safe Palette.* VisiBone offers the Web-safe palette on line and in print as charts, books, folders, and cards. *©2005 VisiBone. (This figure does not use VisiBone's 8-Color printing process to match computer screen color.)*

working within the Web-safe palette had to avoid using colors with subtle differences within the same hue group, and the limited number of colors made working within the Web-safe palette less than satisfying. Advances in the capabilities of hardware and software and the proliferation of monitors with greater display capability have made the Web-safe palette less important than in the past. Many writers consider their use no longer necessary. This topic is covered admirably in a number of articles and books specific to Web design, notably those by Lynda Weinman and Bruce Heavin.

Outside the limits of the Web-safe palette lies an enormous range of available colors, each of which must be individually specified for each flat color situation. Selecting colors from this vast number can be overwhelming. Reducing a gamut to a smaller palette, a limited group of colors with preselected and designated relationships, helps to make color selection a more manageable process. Many of these preselected color palettes are on offer to designers, both as software and as free or for-purchase charts, each representing its own organization, set of preferences and point of view.

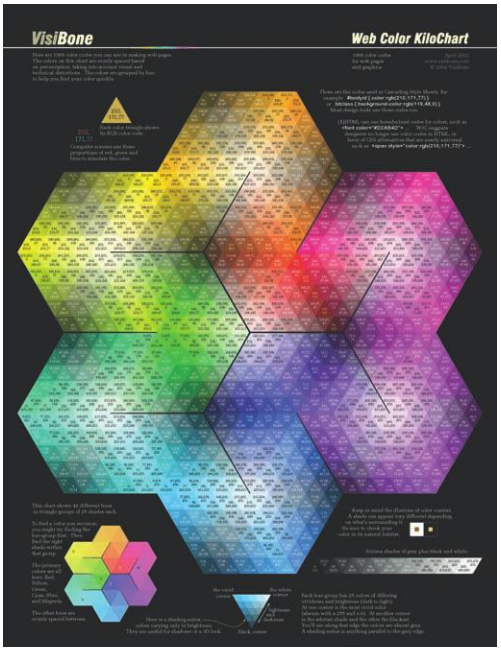


Figure 10-14.
Limited Palettes. Visibone offers a preselected palette of 1068 colors that are outside of the Web-safe palette with their binary code designations as a reference tool for Web page designers. *© 2010 VisiBone's 1068-Color reference chart. (This figure does not use VisiBone's 8-Color printing process to match computer screen color.)*

One of the problems associated with a fixed color palette is that colors needed to display a specific image may not be available in the palette, while many of the available colors may not be needed. A fixed palette containing mostly variations of pink, for example, would not be well-suited for images that do not contain pink. The present trend in software, which assumes a 24-bit depth monitor, is the use of an adaptive, or optimized color palette: one in which the available colors are based on how frequently they appear in the original source image. An image based on an optimized palette is much more likely to be closer to the original source image than any other approach, and an optimized palette eliminates dithering.

Emerging Media

One of the most exciting new technologies, not yet available in color, is the electronic paper display. The electronic paper display is a bridge between subtractive and additive media. It begins with a printing ink, or e-ink, that is made up of tiny capsules that contain both black and white particles. The particles are suspended in a clear fluid within the capsules. In one current formulation the white particles have a positive electrical charge; the black ones have a negative charge. The capsules are suspended in a medium that is then applied to a surface that is laminated to a layer of circuitry. When positive and negative charges are applied selectively, the capsules rotate so that either black or white becomes visible on the surface. The pattern of dark and light is established by the pattern of the circuitry and is controlled by a display driver.

A recent development in color display is four-color LCD technology, which adds a clear, strong yellow to the red-green-blue of each pixel. Advertised on the Web and in print as "four-primary-color" technology, it is said to expand the gamut of available screen colors enormously while also deepening blacks. Four-color technology is currently marketed only for television monitors but seems certain to enter the computer monitor market within the near future.

Endnotes

1 Rodemann 1999, page 170.

2 Developed by E-ink Corporation.

Chapter 10 Highlights

- Monitor screen images are created by the distribution (pattern) and strength (brightness) of light emitted at different wavelengths. Monitors display only additive color. There is no perfect way to translate an additive color to a subtractive color, or the reverse.
- Pixels are the smallest units of the screen display. Each pixel can be triggered to emit red, green, and blue light. Mixing light within each pixel allows each pixel to be altered in hue, value, and saturation.
- Each platform and monitor has its own color display capabilities. The range of colors that a monitor displays is its color gamut. Screen colors can be displayed with consistency between monitors only when they share the same platform, software, and are calibrated to the same set of standards. Consumers who retrieve images from the Internet constantly experience color-display inconsistencies.
- An LCD monitor displays color when light passes through pixels that are filled with liquid crystal and red, green, and blue filters. A early limitation of LCD monitors was their restricted viewing angle, now much improved.
- Color standards are samples that enable the designer to make constant comparisons against the same color at each step in the design process. Color management tools are needed for colors on the screen that must correspond to product and print colors, as well as for images that will be presented as light alone.
- To calibrate a monitor means to adjust it so that specific combinations of red, green, and blue signals produce specific colors on the screen. Screen colors can be calibrated to equally standardized subtractive colors. Calibration is also used to standardize colors between different devices.
- Software determines how many colors are displayed and how colors are mixed on the screen. The CMYK mode of color display imitates the results of mixing process colors. The RGB mode of color display parallels the behavior of light. The HSB mode of color display requires learning to mix light in a way that is associated only with digital design.
- A bitmap is an image made of pixels. A computer printer separates the digital image into four bitmaps, one each for cyan, magenta, yellow, and

black; each of which duplicates the number and pattern of pixels of that color in the computer memory.

- Colors are specified in binary code by their relative proportions of red, green, and blue. Colors for Web graphics are also described using the hexadecimal system, which specifies each color based on a 16-symbol code consisting of the numbers 0 through 9 and the letters A through F. Neither binary nor hexadecimal coding offers perfect color transitions. Middle codes may or may not be middle colors, and color intervals are not reliable by number. Colors that have been specified in binary code can be converted to hexadecimal code, and vice versa, by using charts or formulae.
- The Web-safe (or Netscape) palette, now considered obsolete by many, is composed of 216 colors in a range of hues that are reasonably consistent for viewing on both Mac and PC platforms and on monitors with a limited color display range. The trend in software suggests an adaptive, or optimized, palette in which the available colors are based on how frequently they appear in the original source image.
- A recent development in color display is four-color LCD technology, which adds yellow to the red-green-blue of each pixel. Another new technology, not yet available in color, is the electronic paper display. The electronic paper display is a bridge between subtractive and additive media.

11

The Business of Color

The Color Industries / Producing Color / Color Sampling / Color Forecasting / Color and Product Identity / Palettes, Color Cycles, and History /Traditional Colors / Influences on Palettes

The chief business of the American people is business.

—Calvin Coolidge

From a business point of view, design has a single purpose: products are made to be sold. A product sells because consumers find it more attractive than others like it. The decision to buy has as much to do with the purchaser's perception of a product as it does with the product itself. An item may be purchased because it seems comfortingly familiar, or because it makes a fresh and positive impression at the point of sale. No matter now well-functioning a product may be, to succeed it must be both attractive and memorable. A great part of success in sales is a result of good looks.

For consumers, color is the largest single factor in a decision of whether or not to make a purchase. Market research indicates that 90% of consumer purchases are the result of a deliberate search, and that only 10% of purchases are made on impulse. And of the planned purchases, 60% of the decision to buy involves color.[1] Large-scale purchases like automobiles are generally given some time and thought, but it has been estimated that in a supermarket aisle, the product packaging has about 20 seconds to attract a buyer.

Color is more than just a means of attraction. According to Color Marketing Group, a professional color research organization, the use of color expands readership: color advertisements are read up to 42% more than similar ones in black and white, and color

on a page both accelerates learning and increases comprehension. Colors increase brand recognition[2], which is crucial in a competitive market. Because colors have such a profound influence on sales, aesthetic decisions about color can become secondary to marketing ones. For designers, color means business. But what determines marketable colors? How are they selected, and by whom?

The Color Industries

Color not only *means* business; it *is* a business.The color industries are major, international, and widely varied. Manufacturers of consumer goods depend on them for support in product development and marketing, color technology, and environmental issues. Chemists and engineers create new colorants and the means to apply them. Testing laboratories determine that colorants are safe for the buying public. Research organizations explore consumer color preferences and analyze, chart, and publish observed color trends. Psychologists explore the potential for positive and negative reactions to colors by the buying public.

Advances or changes in one industry create a need for adaptation in others. Companies are increasingly interdisciplinary. Businesses that once specialized in a single field now extend their involvement to others. Pantone, for example, is so attuned to developing screen technologies and color marketing that it is involved in the development and manufacture of calibration tools and software, forecasting consumer color preferences, and a myriad of other color-related activities.

Producing Color

A foundation of the color industry is the manufacture of subtractive colorants: interior and exterior paints, furniture finishing materials, printing inks, ceramic glazes, automobile finishes, textile dyes, and colorants for numberless other materials. The manufacturing process is both art and a science; primarily, but not exclusively, the science of chemistry. Each color agent is a chemical mix with specific properties that make it compatible with the material it is meant to color. The necessity of working with legally permissible colorants adds a layer of complexity to the manufacturing process. Environmental concerns dictate that producers of every type or use of colorant stay current with constantly changing federal and local regulations.

Some colors may be difficult, or even impossible, to produce for particular products. CMYK printing, for example, lacks clear colors in the orange family. Bright, clear, reds are difficult to produce in ceramic glazes. Substrates also matter. Linen fiber, for example, does not dye to bright colors, while silk takes dyes brilliantly. Designers in every field are constrained by colors that are available to them. They must consider what is possible before selecting what may be desirable.

Color Sampling

Many times a single product is produced in a range of color choices. Some finishes are simply flat or glossy color; while others may display color shifts and special reflecting characteristics. In either case successful marketing requires that consumers be provided with a convenient and accurate way to make a selection. Colwell Industries, a century-old company, is the largest producer of color merchandising tools for decorative (and other) products. Colwell's processes exemplify the cutting-edge technology needed to create color samples that simulate the colors and finishes of real products. The charts and display materials that they produce range from interior and exterior architectural coatings to cars and cosmetics.

Colwell sampling charts are not conventionally printed, and only rarely are they available as samples of actual product.* Instead, paper grounds are coated with minute particles of metal, plastic, rubber or other substances to approximate the substrate of the real product. Car finish sample charts, for example, are made with a metal powder sprayed onto paper, followed by a coating of color lacquer. Colwell's chemists formulate colorants and finishes and its computers generate the mixing formulas, but even at this level of technology the human eye makes the final decision. Colwell President Bill Byers states it perfectly: "Color is an art, so a human eye is always needed."[3]

The care taken to duplicate color and finish does not attempt to address the additive-subtractive color issues of Internet shopping, but according to a Colwell spokesperson, metamerism is addressed in the laboratory. Although metamerism is unavoidable in some situations, Colwell strives for the closest sample-to-product match under a variety of light sources, a consideration that is increasingly important as new lamp types come into general use.

* Farrow and Ball, a British paint manufacturer, is a notable exception. It provides color charts made from actual paints.

Color Forecasting

A consumer economy relies on a constant demand for the new, including new colors. This demand originates at both ends of the market. Consumers want a fresh look, and producers make a conscious effort to step up sales by offering goods in new colors. Manufacturers depend for their survival (or at least for their profitability) on their ability to anticipate which new colors will be preferred by the purchasing public. And while this has always been true for the makers of short sales-life products like cosmetics and apparel, it is now true for all consumer goods. Even electronic devices, once available only in industrial black or gray, can now be purchased in "fashion" colors.

Nowhere do psychology, color, and marketing interact more closely than in the area of consumer color preferences. Hundreds of organizations and individuals provide (among other services) research and prediction of incoming color trends for target markets and target industries. These individuals and organizations provide manufacturers with an increasingly vital service called *color forecasting*.

Figure 11-1.
Forecast Colors.
The colors for youth, Fall and Winter 2011-2012, as forecast by The Color Association of the United States. *Image courtesy of the Color Association of the United States.*

Manufacturers and vendors rely on color forecasting to enhance their ability to compete for the consumer dollar. Forecasters provide information and guidance on the next wave of color demand, consulting on everything from hair colors and flowers to appliances and buildings. Organizations like The Color Association of the United States (CAUS), Color Marketing Group (CMG), and the Pantone Color Institute identify and predict color trends through a variety of methodologies: observation and reporting by individual members, meetings and workshops, scientific consumer testing, consumer surveys, and by analyzing how consumers respond to a number of cultural forces.

Regional factors are also considered. Not all colors sell in all climates and cultures. Color forecasts address national trends, but they also target narrower markets. Design legend Jack Lenor Larsen noted that "because color forecasting works (it is, after all, self-fulfilling) we shall see more of it but it will, necessarily, be more focused."[4]

Figure 11-2.
Forecast Colors. A collection of fashion colors for the urban market. *© 2010 by Color Marketing Group.*

Color forecasts are made for both short and long term, depending on the industry served. New palettes are arrived at either by a consensus of members or by smaller groups who meet on a regular basis to report their observations on "movement" in current colors: "toward a greener yellow," for example, as well as incoming and outgoing colors. The resulting palettes, published several times a year, reflect the focus of incoming color trends.

Color forecasting has its constraints. It must reflect public taste and direct it at the same time. People cannot be forced to like, or to buy, new colors. Individual consumers do not adopt all of the colors in a new palette. One reason for this, of course, is personal color preference, but a more determining factor is expense. It is costly to make color changes in big-ticket items like furniture and carpets. In home furnishings, completely new colors are purchased only when a household is undergoing a total renovation, or is a new household altogether. The same is true for apparel. Very few people discard all of their clothing from a previous year, and new items must work with the old. New colors and color combinations are most successful when they can easily be integrated into existing ones. New color preferences in the consumer market are not so much abrupt changes as they are movements from one palette into another.[5]

The advantage to manufacturers who follow the dictates of forecast colors is obvious: advance information about upcoming consumer color preferences means increased sales. The motto of Color Marketing Group is "Color sells...and the right colors sell better!" is followed by "CMG: Forecasting the Color of Profit."

Color forecasting benefits the consumer as well. When the fashion industry works within a similar seasonal palette, time-pressed shoppers find it easier to coordinate suits, ties, blouses, skirts, and cosmetics. When home-furnishings products are available in related colors, it is easy for the consumer to purchase compatible carpets, paints, wallpapers, building products, bed linens, china, and the multitude of other items that make a home.

Although color forecasting began in the United States, it is now a global phenomenon. American color forecasters consider international influences as a matter of routine and often seek out new ideas in exotic locales. Consumer color preferences in the Asian market are a current and lively area of study. Each year a European Hall of Prediction is held in Paris. London has its Colour Group, a panel with representatives from a variety of industries including paint, plastics, and automobiles.

Color and Product Identity

A second goal of color marketing is the attempt to establish a link in the public mind between a specific color and a specific product. Coca-Cola has its signature red; United

Parcel Service trucks (and uniforms) are brown. McDonald's arches are dependably golden. Color and form *together* can be extremely successful in establishing a color-product link. The scissors we like to use are orange. The chocolate-brown wrapper tells us that we are about to bite into a Hershey bar, not a Nestle's bar, whose blue-and-white wrapper suggests a dairy theme.

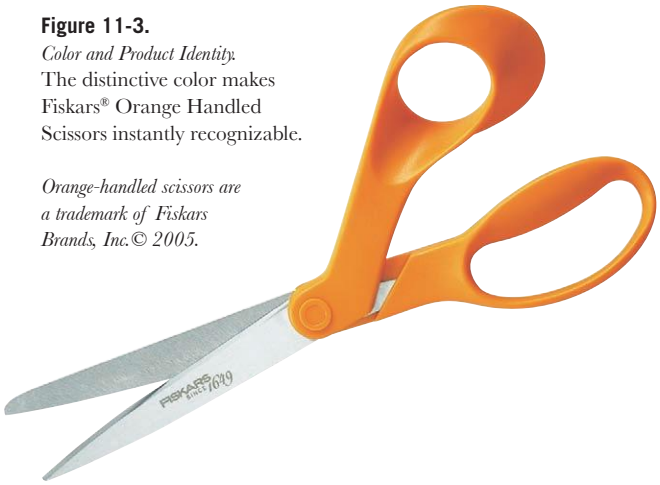


Figure 11-3.
Color and Product Identity.
The distinctive color makes Fiskars® Orange Handled Scissors instantly recognizable.

Orange-handled scissors are a trademark of Fiskars Brands, Inc.© 2005.

But color is a secondary identifier. Only rarely does a color alone identify a product. It becomes a leading visual cue to product identity when a form is both familiar and so generic that only the additional information of color creates an association.[6] A box is only a box, and unexciting to open, unless it bears the distinctive blue that signifies Tiffany and Company.

Some colors are so closely associated with a product that they have been trademarked. A trademark is a legal designation that protects a company's use of a specific image, color, or combination of these for a particular kind of product. It identifies the source of the product or service and distinguishes it from competing products of the same type. Owens Corning, for example, has trademarked a particular pink for insulation. Other manufacturers of insulation may not use it, but the color is available to makers of other products. Tiffany Blue, which resembles robin's egg blue, is a trademark owned by Tiffany & Co. The color is produced as a private custom color by Pantone as PANTONE number 1837, which is the year of Tiffany and Company's founding. Because it is a trademarked color, it is not included in any of the Pantone Matching System swatch books or Web color aids and its CMYK formula is closely held.

Trademarks are guarded fiercely by their owners. Disputes over trademarked colors have been so bitter and protracted that they have reached the United States Supreme Court. In one case the United States Supreme Court concluded that colors could constitute

trademarks because although colors do not automatically evoke a connection to any product by themselves, they could take on secondary meaning in the course of use, and in this way a color could function as a trademark by identifying the source of a particular product.[7] Despite this decision at the highest legal level, the trademarking of colors remains an area of uncertainty. Other attempts to trademark colors have been challenged and have met varied responses from other courts.

Palettes, Color Cycles, and History

Color forecasting attempts to predict the upcoming popularity of single colors; it also predicts *palettes*. A palette is a group of colors that is characteristic of something, like a culture, an artist, or a time period. The characteristic earth browns, black, and sand colors of antique African textiles are a palette in sharp contrast to present-day folk cloths from South and Central America, whose exuberant colors result from modern dyes, but whose origins lie in the dazzling feather cloaks of Pre-Columbian civilization. No one would confuse Vincent Van Gogh's palette of sharp yellows and acid greens with Monet's soft pastel hues. A palette can also be a collection of colors given an assigned meaning, like a spring apparel line marketed as Easter-egg pastels, or a fall palette of rusts and bronzed greens.

Palettes shift in response to outside forces like new technology, exposure to new cultures, politics, and fashions. Certain palettes even provide a clue to the age of objects. Aniline dyes, which reached both the southwestern United States and Japan in the later nineteenth century, replaced the natural colorants that preceded them, and as a result, the age of Navajo blankets and Japanese prints can be determined in part by their colors.

Color cycles are stages in the continually shifting consumer preference for certain palettes. A color cycle represents a prevalence of certain colors in the context of a particular time (and place). The candy-pastel colored buildings of Miami Beach, which saw it greatest development as a resort in the third decade of the twentieth century, exemplify one aspect of Art Deco design. The muted colors of natural dyes are characteristic of the textiles of the late nineteenth century anti-Industrial-Revolution designer William Morris. A new turquoise colorant came on the market after World War II; architectural elements in salmon and turquoise suggest that a structure was built in the 1950s. The psychedelic colors of the 1960s have as a subtext, for better or worse, a belief in "better living through chemistry."

Tracking the rise and fall of color cycles began in the twentieth century. At the outbreak of World War I in 1914, American manufacturers of textiles for women's apparel became aware that they would not have access to the fashion colors dictated by Paris, at that time the fashion capital of the world. The Textile Color Card Association of America was formed and, within a short time, produced a first book of samples called *The Standard Color Reference of America*. This palette of 106 named colors, dyed on silk ribbons, was produced specifically as a color marketing reference for the women's apparel textile industry. The colors and their names were inspired by nature, by university colors, and the colors of Armed Forces uniforms.[8]

The number of colors in the *Standard Reference* grew steadily over time. Colors initiated by the apparel industry trickled down to home furnishings and, later still, to hard goods like appliances and automobiles. The demand for new colors created by color stylists at the top

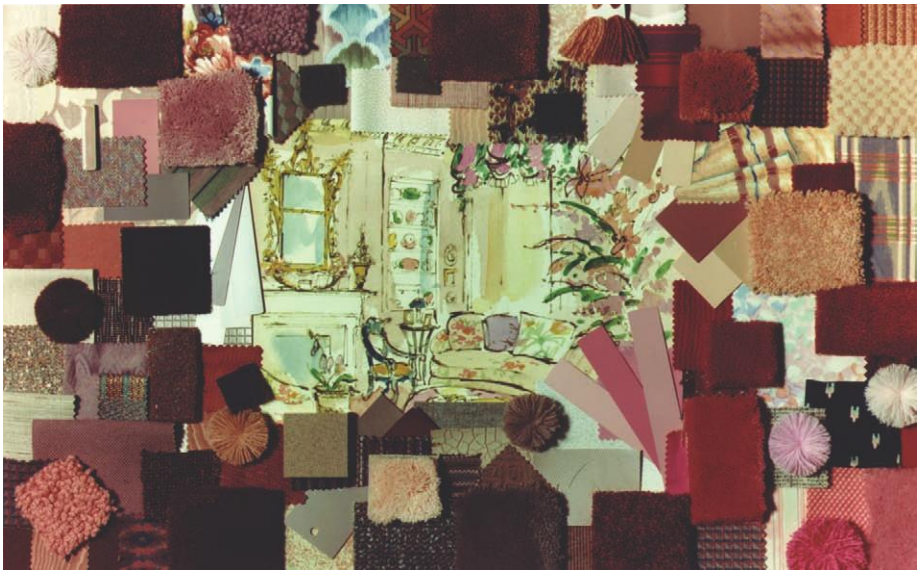


Figure 11-4. *Colors of the 1990s.* Kenneth Charbonneau's assemblage of colors from a multitude of manufacturing sources illustrates the colors of a decade. *Image courtesy of Kenneth Charbonneau.*

of the pyramid forced obsolescence of older goods. By the middle of the twentieth century, average color cycles were seven years or slightly longer, coinciding roughly with the decades. Fashion colors preceded other industries by about two years. Cycles tended to repeat in a

roughly predictable fashion, from brilliant, full-saturation, colors to more muted ones, to neutral palettes, then back again to strong hues.

The women's apparel industry remained the focus of interest until the end of World War II, when other consumer-oriented industries became aware of the marketing power and manufacturing advantages of a controlled palette. The Color Association of the United States (CAUS), an outgrowth of the Textile Color Card Association of America, began to publish palettes for other industries: man-made fibers in the1950s, menswear in the 1960s, home furnishing in the 1970s, and, in the 1980s, international and environmental colors.

CAUS still produces a color reference of dyed silk standards, but its intended audience is now American industry as a whole. *The Standard Color Reference of America*, revised approximately every ten years, is marketed as being "indispensable to anyone working in clothing, interiors, graphics, or with government specifications." Now in its tenth edition, it offers 198 named colors favored by the American public, from Electric Blue through Jade Green, Schiaparelli Pink to Tangerine, West Point Gray to Yale Blue.[9]

As color increasingly became recognized as a force in marketing, other industries seized the initiative in originating palettes. The apparel industry lost its place as the sole arbiter of fashionable colors. It is no longer possible to isolate a single industry as the originator of color trends, nor is the length of a color cycle dependable. Color influences are now global. Consumer response to certain colors arises from any number of sources: pop culture, movies, museum shows, travel, and Internet communications, as well as from new directions in technology and industry. These influences initiate the trends that precede each new color cycle. Cycles are shorter and more transient. Color cycles for clothing are now estimated by experts in the field at eight months, and colors for home furnishings and interiors for three to five years.[10]

Traditional Colors

Consumers are a complicated crowd, varying by age, gender, income, education, social status, cultural background and geography. No matter how successful a current color trend may be, consumers at all economic levels continue to demand products in "traditional" colorings. There are traditional colors for some items of apparel—bridal wear and hunting

gear are classic examples—but the market for traditional colors is particularly strong in home furnishings textiles.

Reproductions of period colors have not always been historically accurate, and they are not all accurate today. Colors oxidize (change after exposure to air and light) over time.

Figure 11-5.
Document Wallpaper. Arthur Sanderson and Sons wallpaper "Trellis" is available in the same colors as when it was introduced in 1864. *Image courtesy of Sanderson.*

Years of accumulated grime mute even the most durable colorants, and current levels of air pollution accelerate the deterioration. Not only are colors altered by soiling and oxidizing, the various colors in a composition alter at different rates. In a single textile or painted object, some colors change enormously over time and others hardly at all, so the original balance of the palette is lost. Many consumers think of the muted quality of faded colors *as* antique. Muted colors are often marketed as "traditional" in response to this common misconception. Sophisticated techniques of analysis now make it possible for the original colors of paintings, textiles, and decorative objects to be determined with greater accuracy. New methods of cleaning uncover close-to-original hues. Today there are wallpapers, textiles, and paints available that are near-perfect facsimiles of antique colors as they were when they were new. These reproductions are *document* (historically accurate) colors.

Present-day research and restoration approaches have caused a revolution in historic preservation. Building restoration, both interior and exterior, has taken on a new and controversial face as color accuracy supersedes the old conventions. When the Rose Room in the White House was refurbished during the Kennedy administration, there was a furor over the brilliance of its red-violet walls. The restored color duplicated the original, but it was shocking to a public used to thinking of period colors as subdued.

Influences on Palettes

From the beginning of time, new materials and new technologies — the *ability* to produce new colors — has been a powerful influences on palettes. Early man had only the tetrachrome palette; the red and yellow ochres, vine charcoal, and chalk that are still used by aboriginal cultures. The Egyptians, Minoans, Greeks, and Romans expanded the number of colorants in use until, by the Middle Ages, there was an enormous range of colors available for all of the arts — textiles, wall paintings, glass, ceramics, book illumination, and metals. Colorants in all cultures came from natural earths and organic materials. Earth colors were many and varied: blue from lapis and cobalt, orange from copper, red from iron, antimony yellow, cinnabar (sulfide of mercury, the prized and brightest red of the ancient world), and green from malachite. Others were animal or vegetable: murex purple from shellfish, red from madder, yellow from saffron, and blue from woad.

The colors and palettes of most cultures were determined by locally available materials, and

early palettes had a sort of ethnic identity. Despite the hardships of early travel and trade, a slow but steady exchange of colors and color influences took place between East and West. Early Crusaders brought turquoise to Europe from the Near East; the Portugese brought indigo from India. Returning travelers, soldiers, and explorers added colors to the Western color canon.

The expansion of Western influence into Africa and the Near and Far East in the nineteenth century accelerated the exchange of colors between cultures. Colors and combinations remained typical of certain groups, but were no longer confined to them. The exchange of colorants and palettes through conquest and trade continued steadily until the middle of the nineteenth century, when the most significant change in the ability to create colors came about entirely by accident.

In 1856, an eighteen-year-old English chemist's assistant named William Henry Perkin was attempting to formulate synthetic quinine from coal tars when he noticed that his rags were stained a deep purple. Perkin had stumbled upon coal tar dyes. Perkin's first dye color, called mauvine, or aniline purple, was followed quickly by additional colors. Aniline dyes were so well received that he was able to retire at thirty-five, a rich man.

Perkin's accidental discovery of aniline dyes laid the groundwork for today's synthetic (man-made) colorants. The first aniline colors were followed briefly by a return to natural dyes, then a rapidly expanding array of synthetics quickly took over. Man-made colors are the norm in today's market. Natural dyes are so rarely used that they are found almost entirely in luxury handcraft goods, like Oriental carpets and Harris tweeds, and in some historical reproduction items. Chemistry, not nature, dictates the colors of the new millennium.

Advances in chemistry are only one influence on palettes. Political, cultural, and social events also bring new colors and combinations to the public eye. When Worth, the Parisian designer to the Empress Josephine, began to design for the English aristocracy as well as the French, French colors first crossed the English Channel, then the Atlantic. Colors in the second decade of the twentieth century felt the impact of Diaghilev's dazzling costumes for the Ballets Russe in Paris (1909), while artist-designers Leon Bakst, Robert and Sonia Delaunay, and Matisse also infused brilliant color into the palette of the early century.

World War I ended the primacy of the French in dictating fashion colors. The inspiration for fashion colors after 1918 came from the United States. The 1920s saw a number of concurrent palettes, among them a palette of silver, black, and white inspired by the glamor of Hollywood and America's growing industrial might. The Great Depression of the 1930s brought browns, drabs, and colors associated with warmth and comfort. Art exhibitions inspired new palettes: a 1935 Van Gogh exhibition in the United States created a vogue for sunflower prints in browns, greens, and yellows. During the World War II, facilities turned to war production, and patriotic colors were popular.

High-speed communication and the globalization of commerce have blurred the lines between palettes that once were representative of cultures and eras. The time lapse between new color influences and the consumer awareness of them is now so short as to be effectively nonexistent. Traditionally Japanese colors and combinations appear as often in Milan as in Tokyo. So-called "retro" colors have their own audience. No single source inspires the colors of fashion or industry. There have never been richer or more varied sources of inspiration available to colorists, and there has never been a larger or more varied audience for those colors. In the fast-paced, freewheeling climate of twenty-first century design, any combination is desirable, any combination is acceptable.

Figure 11-6.
Inspiration Without Borders. The brilliant palette of traditional Chinese architecture has inspired the colorings of internationally marketed products like textiles and ceramics. *Photograph by Alison Holtzschue.*

Colorists are blessed with this multitude of sources, but they're also faced with conflicting demands. The market appeal of colors is crucial to sales, and therefore to the design process. Colorists who compete in the global marketplace must provide innovative colorings, up-to-the-minute fashion colorings, and classic, timeless colorings of equal marketability. Creating new palettes for today's consumer economy is a high-wire act—an exhilarating balance between art and design, technology and commerce.

Endnotes

1. Rodemann 1999, page 170.
2. http://www.colormarketing.org/uploadedFiles/Media/The Profit of Color!
3. www.themanufacturer.com/us/detail.html?contents_id=3405 ©2010 (**byers**)
4. Linton 1994, page x.
5. Rodemann 199, pages 171–172.
6. Sharpe 1974, pages 8–9, 31–32, 43–45, 75, 98.
7. *Qualitex Company v. Jacobson Products Company, Inc.*, 514 U.S. 159 (1995).
8. Linton 1994, page ix.
9. Promotional material CAUS New York 2005.
10. PCI Paint coating industry article *Techno Color* posted 9/01/2001 and interview with Erika Woelfel of Colwell Industries.

Chapter 11 Highlights

- Color is the largest single factor in consumers' decisions to make purchases. The color industries are major, international, and increasingly interdisciplinary.
- Producers of colorants must work within the constraints of increasingly strict environmental legislation. Makers of color samples consider, but cannot prevent, problems of metamerism.
- Color forecasting organizations attempt to predict consumer color preferences for upcoming periods of time. A second goal of color marketing is to establish in the public mind a link between a specific color and a specific product. Some colors are so closely associated with a manufacturer or product that they have been trademarked.
- A color cycle represents the prevalence of a certain palette in a certain time and place. So-called "traditional" colors, like bridal white, maintain their symbolic meanings over long periods of time. Document colors are accurate reproductions of colors of an earlier time and place.
- Palettes are influenced by the availability of colorants, advances in chemistry, cultural, social, and political factors.

GLOSSARY

Achromatic Having no discernible hue or color.

Acuity *See* Visual acuity.

Additive mixture Color seen as a result of light alone.

Additive primaries The three wavelengths of light that must be present to yield white light: red, green, and blue.

Admixture An artist's technique in which a single color is mixed into all (or most) colors in a composition.

Aerial perspective *See* atmospheric perspective.

Afterimage A "ghost" image that follows stimulation of the eye by a single color when its complement is not present in the field of vision.

Aniline dyes A family of alcohol-and coal tar-based dyes discovered in the 1830's; capable of producing brilliant colors but not as colorfast as the azo dyes that succeeded them.

Analogous colors Colors adjacent on a color spectrum, sometimes defined as hues limited to the range between a primary and secondary. A group of colors including any two primaries but never the third.

Artists' media A family of subtractive media that selectively absorb and reflect light. Artists' media are composed of a liquid, paste, viscous, solid, or other base into which pigments or dyes have been introduced to form a transferable colorant, such as paint, dye, crayon, or chalk.

Artists' primaries The simplest colors of artists' subtractive media, from which all other colors are derived: red, yellow, and blue. Colors that cannot be further reduced into component parts.

Artists' spectrum The full range of visible hues as organized by Goethe: red, orange, yellow, green, blue, and violet; expandable to include any and all hues in between them. Also called the color circle or color wheel.

ASTM The American Society for Testing and Materials, now ASTM International, which provides standards for more than 130 industries, including standards for paints, dyes, and other colorants.

Atmospheric perspective A pictorial depth cue in which drawn objects appear farther away because they are illustrated as more muted and more blue than nearer ones.

Azo dyes A family of petroleum-based dyes developed in the latter nineteenth century, with greater color fastness than the aniline dyes that preceded them.

Base A liquid, paste, viscous, wax, chalk, or other substance into which pigments, dyes, or other colorants may be introduced to create a medium such as oil paint, textile dye, or crayon; also at times called a binder. *See* vehicle.

Beauty The quality of an object or experience that gives pleasure to one or more of the senses.

Bezold effect An effect in which all colors in a composition appear lighter by the addition of light outline, or darker by the addition of dark outline; also an effect in which a colored ground appears lighter because of a linear design in light line or darker because of a linear design in dark line. Bezold effect is sometimes too broadly stated as "an effect that takes

place when changing one color composition appears to make all of the colors change."Also called assimilation effect and spreading effect.

Binary code colors A system for specifying colors for Web graphics by their relative proportions of red, green, and blue light emission.

Binder *See* base and vehicle.

Bit The smallest piece of digitally coded information.

Bitmap An image on the monitor made up of pixels that correspond exactly to the information in computer memory.

Brilliance The combined qualities of high light-reflectance and strong hue, typically found in saturated colors and strong tints.

Broken color A random distribution of patches or flecks of colors within a single area or field of vision.

Calibration The process of adjusting a monitor so that specific combinations of red, green, and blue signals produce specific colors on the screen.

Carried colors Colors in an image or design that are laid on the background.

Chroma A synonym of hue and color; the name of a color. Also, a term used to describe the relative presence of hue in a sample. A vivid color has high chroma; a muted color has lower chroma. *See* hue.

Chromatic Having hue or color.

Chromatic adaptation *See* successive contrast.

Chromotherapy The use of color for healing.

CIE The Commision Internationale de l'Eclairage; an organization that attempts to standardize color notation with regard to the colors of light.

CMYK display mode A mode of screen color display that imitates the results of mixing

process colors. Each color in the CMYK mode represents a color of process ink (Cyan, Magenta, Yellow, Black.)

Color A category of visual experience including hue, value, and saturation. Also, a synonym for hue and chroma; the name of a color. *See* hue.

Colorant A substance that reacts with light by absorbing some wavelengths and reflecting others, or by absorbing some wavelengths and transmitting others. *See* artists' media and process colors.

Color chord A group of theoretically harmonious hues, expressed geometrically as rectangles or triangles overlaid on the artist's spectrum.

Color coding The use of color to differentiate between similar objects or ideas.

Color constancy The perception that the colors of familiar objects remain the same no matter what the general lighting.

Color cycle A period of time or stage in consumer preference for certain palettes; the prevalence of certain colors in the context of a particular time.

Color display mode The way in which a user mixes colors on a monitor screen. *See* HSB display mode, RGB display mode, CMYK display mode.

Color forecasting A service that provides manufacturers and vendors with information and guidance on upcoming consumer interest in certain colors and palettes.

Color gamut *See* Gamut.

Colorimeter A device that measures the red, green, and blue wavelengths of emitted light; used to calibrate computers.

Color management The process of synchronizing colors between two or more devices or media.

Color memory The ability to retain color in memory; also the association of colors with images or events. Some experiments have indicated that the presence of color adds additional information to memory so that an image seen in color is more easily recalled than the same image seen without color.

Color notation A system of letters and numbers used to organize colors

Color Rendering Index A rating scale meant to assess the ability of a lamp to render the colors of objects. Lamps are rated based on the degree of color shift that occurs when an object placed under the test lamp is compared to the same object under a reference standard lamp.

Color solid A representation of colors organized by hue, value, and chroma (saturation) as a three-dimensional form.

Color temperature In lighting, the measurable temperature in degrees Kelvin of any given light source. In color theory and description, the relative warmth (red-yellow-orange cast) or coolness (blue or green cast) of a color.

Colorway A specific combination of colors for a product such as a textile or wall covering that is available in more than one combination of colors; as "this wallpaper is available in a red colorway, a green colorway, and a gray colorway."

Color wheel A synonym for spectrum. Also, a term sometimes used to mean a multi-color spinning top devised by nineteenth-century scientist James Maxwell to demonstrate responses of the human eye to colors in motion.

Complementary colors Colors that lie directly opposite one other on the artists' spectrum. Each pair of complements includes the three primary colors (red, yellow, and blue) in some proportion or mixture.

Complementary contrast An effect of intensified hue difference that takes place when colors used together contain even a partial complementary relationship.

Complements *See* complementary colors.

Composition A complete entity, something meant to be sensed as a whole.

Cones Light-sensitive receptor cells in the retina that respond to color and fine detail.

Contrast reversal A variation of afterimage in which the "ghost" image is seen as a negative of the original image and in complementary color.

CRI *See* Color Rendering Index

Design Used as a noun, art fused with function in an object that will be produced by others; as a verb, the process of originating applied art.

Design concept A broad solution to a design problem without resolution of details.

Design process Procedure for solving design problems.

Dilution Changing a pure or saturated hue by lightening, darkening, or muting by the addition of black, white, gray, or its complement.

Display mode The way in which a user mixes the colors of a monitor screen display. *See* color display mode.

Dithering A grainy or broken area of color caused by the insertion by the software of small areas of similar colors within an image to approximate a color that does not exist in the gamut. *See* gamut, hexadecimal colors, and Web-safe palette.

Divisionism *See* optical mix.

Document color Color or colors in a reproduction textile, wallpaper, paint, or other product that accurately reproduce the color or colors of that product as they were when originally produced.

Dye (dyestuff) A colorant that is fully dissolved in a vehicle, such as water or other liquid; a colorant in solution. Traditional dyes in general were organic, but most modern dyes are synthetic (man-made).

Equilibrium An involuntary, physiological state of rest that the eye seeks at all times. Equilibrium occurs when all three (additive or subtractive) primary colors are present within the field of vision.

Even intervals Visually equidistant steps between three or more colors.

Field In carpet and flag design, the term for the background upon which colors are laid. *See* ground.

Filter A material that transmits some wavelengths of light and absorbs others.

Fluting A three-dimensional illusion in which a series of vertical stripes of uniform width appear to have concavity, like the channels of a Doric column.

Fugitive Easily fading or deteriorating color.

Full color *See* saturated color.

Gamut The full range of colors available in software and seen as the light display of a color monitor. Also used, less frequently, to characterize the range of colors in subtractive media.

Glitter Sparkle; sharp light reflectiveness with an impression of movement.

Gloss A highly polished, light-reflective surface quality.

Gradient A series of progressive intervals of colors so close that individual steps cannot be distinguished.

Gray scale A value series made up of intervals of gray, which may or may not extend to include black and white.

Ground The background against which colors, forms, or shapes are laid.

Harmony In color, the pleasing joint effect of two or more colors used together in a single composition.

Hexadecimal colors Colors within a system of specifying colors for Web graphics based on a 16-symbol code consisting of the numbers 0 through 9 and the letters A through F. Each color is specified by two symbols as a percentage of red, green and blue.

HSB display mode The HSB display mode (Hue, Saturation, Brightness) mixes light on the monitor screen in a way that is associated only with digital design. Software for the HSB display mode is also marketed as HSL (Hue, Saturation, Lightness) and HSV (Hue, Saturation, Value.)

HTML Hypertext markup language; the principal programming language of the Internet

and the language in which Web pages are designed.

Hue The name of the color: red, orange, yellow, green, blue, or violet. Synonyms are chroma or color.

Hue intensity The saturation or purity of a color; its vivid versus dull quality. *See* saturation.

Illuminant mode of vision The presence of a viewer and light source only.

Image A representation or depiction of a person, animal, or object; also a form or shape seen against a ground. Images can vary from photographic likenesses to nonrepresentational forms that do not portray objects or natural appearances.

Incandescent light Light that results as a byproduct of burning.

Incident beam The beam of light leaving a light source.

Indirect color A secondary reflection of color that occurs when light from a general light source reaches a highly reflective color on a broad plane. The wavelength of color reflecting off the colored plane washes any surface that is positioned to receive it (that is, positioned at an angle equal to that of the incident beam leaving the general light source).

Intensity Sometimes used as a synonym for brilliance, or the strength of a hue. *See* hue intensity and light intensity.

Intermediate color A color on the spectrum between a primary and a secondary.

Interval A visual step between color samples. *See* even interval.

Iridescence An attribute of surfaces on which the hue changes as the observer's angle of view changes.

Lamp A device that emits light, or visible energy. The correct term for a light bulb.

Light Visible energy.

Light intensity The light-reflecting quality of a color. *See* luminosity and value.

Luminosity Literally, light emitted without heat. Used to describe the light-reflecting quality of a color. Luminous colors reflect light; non-luminous colors absorb light.

Luster Sheen; softened or diffused light reflectiveness on a surface.

Matte A smooth, dull, visually flat quality of surface.

Maximum chroma The strongest possible manifestation of a hue.

Medium The means by which something is transmitted. *See* artists' media.

Metameric pair Two objects that appear to match under one set of light conditions but do not match under a different set of light conditions.

Metamerism The phenomenon that occurs when two objects that appear to match under one set of light conditions do not match under a different set of light conditions.

Monochromatic Containing only one hue.

Monotone Color without variation. Used to describe two or more colors of very close hue, value, and saturation.

Motif A single, specific design element, an isolated and individual idea, form or shape.

Motion parallax A real-world depth cue that includes motion and time.

Motion graphics Drawings that include pictorial depth cues, motion, and time.

Negative space The area of a composition that is not occupied by motif or image. Negative space is often, but by no means always, the ground of a composition.

Object mode of vision The presence of a viewer, light source, and object.

Optical mix A new color that is seen as a result of the close juxtaposition of small areas of two or more other colors. Sometimes called divisionism. *See* partitive color.

Optimized palette A pre-selected, limited palette made up of colors that are found in the original source image.

Palette Literally, a board or plate upon which colors are mixed. Palette describes a group of colors used characteristically by an individual artist or designer; colors present in a specific design, group of designs, or body of work; or colors of a particular era.

Partitive color A new color that is seen as a result of the close juxtaposition of small areas of two or more other colors. *See* optical mix.

Pastel An apparel-industry term for colors diluted by white to high or middle value; also, clean tints with little or no muted quality.

Pattern A design composition formed by the reoccurrence of motifs. Pattern may be geometric or fluid, regular or random.

Physical spectrum The full range of visible colors of light as postulated by Newton: red, orange, yellow, green, blue, indigo (blue-violet), and violet.

Pigment A colorant that is finely ground and suspended as minute particles in a vehicle. Traditionally, pigments were inorganic (earth colors), but modern chemistry has blurred this distinction. Pigments in general are opaque.

Pixel One of the points of light that make up the picture on a monitor screen. The word derives from "picture element." The greater the number of pixels in a given area (the smaller and closer together), the higher the screen resolution.

Primary colors The simplest colors of the artists' spectrum; those that cannot be reduced or broken down into component colors: red, yellow, and blue. *See* artists' primaries, additive primaries, and process primaries).

Process colors A subtractive medium that selectively absorbs and transmits light. Yellow, cyan (blue-green), and magenta (red-violet) colorants that, when mixed or laid over one another, result in a wide range of colors for the printed page. Used with the addition of black in four-color (CMYK) printing.

Process primaries The simplest colors of process printing inks; those cannot be reduced or broken down into component colors: Cyan (blue-green), magenta (red-violet), and yellow. All hues in process printing are derived from these three inks. *See* artists' primaries,

additive primaries, and process colors.

Pure color *See* saturated color and maximum chroma.

Reflected beam The beam of light reaching an object that is reflected back.

Rendering Used as a noun, a final drawing; used as a verb, the process of drawing a final image.

Resolution A measure of the sharpness and clarity of an image and extent and clarity of detail that can be seen. In a monitor image this is determined principally by the number of dots (pixels) per square inch: the more pixels, the higher the resolution. Resolution is used to describe printers, monitors and scanners.

RGB display mode The RGB (Red, Green, Blue) mode of screen display parallels the results of mixing light. It is used to mix colors for images that will be viewed only as monitor screen light display, such as Websites or DVDs.

Rods Light-sensitive receptor cells in the retina that are responsible for peripheral, less focused vision.

Saturated color The most intense manifestation of a color imaginable; the "reddest" red or "bluest" blue. A fully saturated color contains one or two of the primary colors but never the third. Saturated colors are undiluted by black, white, or gray. Synonyms are pure color, full color, or hues at maximum chroma.

Saturation The degree of purity of a color; its hue intensity or vivid quality, as opposed to muted or dull quality.

Secondary colors Colors made up of two primary colors. *See* secondary colors of light, secondary colors of artist's media, and secondary colors of process printing.

Secondary colors of artist's media A group of colors, each resulting from the mixture of two subtractive primaries: orange (red and yellow), green (blue and yellow), violet (red and blue).

Secondary colors of light A group of colors of light (additive colors), each resulting from the mixture of two primary color wavelengths: cyan (blue and green), yellow (red and green), and magenta (blue and red).

Secondary colors of process printing A group of colors, each resulting from the mixture of two primary colors of process color ink: red (magenta and yellow), green (yellow and cyan), and blue (magenta and cyan).

Shade A pure color made darker, or with black added.

Simultaneous contrast A spontaneous color effect that results from a physiological response of the eye to stimulation by one color only. The eye, seeking the presence of all three primaries in the field of vision, generates a "wash" of its complement onto any adjacent neutral area.

Single interval The smallest difference between samples that a viewer can distinguish; established by the individual's threshold. *See* interval and threshold.

Sparkle Sharp light reflectiveness on a surface, with an appearance of movement. *See* glitter.

Spectral colors Scientifically defined colors of light; additive colors that are measurable by wavelength.

Spectral reflectance curve The pattern of relative energy that a specific lamp emits at the various wavelengths.

Spectrophotometer A color managment tool that provides a wavelength-by-wavelength analysis of additive or subtractive color.

Spectrum The full range of visible hues. *See* artists' spectrum and physical spectrum.

Spreading effect *See* Bezold effect.

Standard A sample against which a color is matched.

Subtractive mixture Color seen as the result of the absorption of light; the colors of objects and the environment.

Subtractive primaries The primary colors of artists' media, the artists' primaries: red, yellow, and blue *See* artists' primaries and process primaries.

Successive contrast *See* afterimage.

Surface The outermost layer of a two- or three-dimensional thing; its "face" or "skin."

Synaesthesia A largely unexplained phenomenon in which one sense responds to the stimulation of another.

Tertiary colors Colors made of any mixture of the three primaries; "brown," or chromatic neutrals.

Threshold That point in vision at which an individual can just distinguish between two close samples.

Tint A pure color plus white, or made lighter.

Tone A nonspecific word referring to some change in a hue. Most often used to mean a graying, or reduction in saturation (chroma).

Transition A change that takes place in steps (or intervals) from one form, color, size, surface, or other element to another, different one.

Transparence illusion An illusion in which opaque colors are made to appear transparent, created by drawing two colors as if they overlap and placing an interval between them in the area of overlap.

Value Relative light and dark, with or without the presence of hue. High-value samples are light; low-value samples are dark.

Vehicle A liquid, paste, viscous, wax, chalk, or other substance into which pigments, dyes or other colorants may be introduced to form a medium such as oil paint, textile dye, or crayon. Also referred to as a base or binder.

Visual acuity Sharpness of vision, the ability of the eye to see detail.

Wavelength A pulse of energy that a light source emits at specific distances apart. Within the range of visible energy, each single wavelength is perceived as a separate color.

Web-safe palette A palette of 216 selected colors that are reasonably consistent for viewing on any operating system and monitor. With advances in imaging this gamut is now considered by many to be an unnecessarily restricted number of colors. Web-safe

colors are non-dithering. *See* hexadecimal colors.

Weber-Fechner principle A mathematical series in which intervals between numbers are geometric (1, 2, 4, 8, 16, 32, 64, 128, etc.) in progression, each step twice the one before and half the one following. Weber-Fechner series are used to illustrate visual intervals, particularly intervals of value.

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What Design Can't Do

Essays on Design and Disillusion

Silvio Lorusso

Set Margins' #26

Design is broken. Young and not-so-young designers are becoming increasingly aware of this. Many feel impotent: they were told they had the tools to make the world a better place, but instead the world takes its toll on them. Beyond a haze of hype and bold claims lies a barren land of self-doubt and impostor syndrome. Although these 'feels' might be the Millennial norm, design culture reinforces them. In conferences we learn that "with great power comes great responsibility" but, when it comes to real-life clients, all they ask is to "make the logo bigger."

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Praise for *What Design Can't Do*

"Are you a bit depressed about design? This book might help you understand why. It may also make you laugh. With a lightness of touch, Silvio Lorusso provides an unflinching but well-reasoned discussion as to how design has become so 'bigged up' and what this actually means for its practitioners. After reading this book, design will never look the same to you."

– Guy Julier, author of *Economies of Design*

"What happens once design is a smokescreen and can no longer claim to be a blueprint for change? This is the question Silvio Lorusso puts on the table. How did form, no matter how cool and disruptive, become so futile and tired? Read this with caution: we can no longer design ourselves out of this painful realisation."

– Geert Lovink, author of *Stuck on the Platform*

"The disillusion of design is the disillusion of the world. This book is an essential read, not only for specialists. Because design affects us all, and because understanding where design fails helps us understand where design succeeds."

– Emanuele Quinz, author of *Strange Design*

"Italo-pessimist design critique at its best."

– Clara Balaguer, cultural worker
and grey literature circulator

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What Design Can't Do

Essays on
Design and Disillusion

Silvio Lorusso

Set Margins' #26

To my friends

"The challenge of modernity is to live without illusions and without becoming disillusioned."
– Antonio Gramsci, 1929–35[1]

"Until today, the history of design has remained a history of defeats, suffered by the high-flying aspirations of the designers in their battle against utilization by *Das Kapital.*"
– Hans Magnus Enzensberger, 1972[2]

1 Antonio Gramsci. *Letters from Prison*. New York: Columbia University Press, 1994.
2 Hans Magnus Enzensberger. "Remarks Concerning the New York Universitas Project." In *The Universitas Project: Solutions for a Post-Technological Society*, edited by Emilio Ambasz. New York: The Museum of Modern Art, 2006, p. 107.

Prologue:

Starter Pack

"[...] a fantastic paradox [concerns] modern industrial society and the way people live and act in it. It is the contradiction between the apparent omnipotence of humanity over its physical environment (the fact that technique is becoming more and more powerful, that physical conditions are increasingly controlled, that we are able to extract more and more energy from matter) and, on the other hand, the tremendous chaos and sense of impotence concerning the proper affairs of society, the human affairs, the way social systems work, etc."
– Paul Cardan (Cornelius Castoriadis), 1965[3]

"Soon chaos will be our common denominator, we carry it within us and we will find it simultaneously in a thousand places, everywhere chaos will be the future of order, order already no longer makes sense, it is nothing more than an empty mechanism and we wear ourselves out to perpetuate it so that it can consign us to the irreparable."
– Albert Caraco, 1982[4]

"In practice, of course a designer's life is as muddled, informal, and accident-prone, as most people's lives manage to be; not only behind the scenes, but sometimes in front of them."
– Norman Potter, 1969[5]

According to Victor Papanek, an early environmentalist, countercultural designer and outspoken critic of US consumerism, design is "the conscious and intuitive effort to impose meaningful order." Hence, tidying up a desk, curating a party playlist, structuring the chapters of a book... all of these activities can be understood as a design endeavour. It is distressing, however, to realise that very few human activities escape this definition. If this is the case, what specific kind of order are designers meant to

3 Cornelius Castoriadis, *The Crisis of Modern Society*, Solidarity pamphlet No. 23, 1965. https://libcom.org/article/crisis-modern-society-cornelius-castoriadis.

4 Albert Caraco. *Breviario del caos*. Milano: Adelphi, 1998, p. 93.

5 Norman Potter. *What Is a Designer? Things, Places, Messages*. London: Hyphen Press, 2006, p. 19.

A particularly niche graphic design starter pack.
Source unknown.

impose? And is it in any way more stable than the fragile, provisional order that we all attempt to carve, like a horde of industrious archivists in the Tower of Babel?

As every parent entering their kid's room knows, if design is about order, its precondition is not void, but chaos.[6] The book you are reading locates itself within chaos. As such, it's a weird design book. It explores the disorder that can't be contained, the mess that overflows the dams of what we consider arranged and designed, including our mental models and subjectivities. Imposing a meaningful order begins with drawing the line that separates what is subject to the design effort from what it is not: the former is what designers generally call 'problem.' Design is a magic circle that produces an orderly inside and a chaotic outside. Designers safeguard and rework the circle's shifting border,[7] and place ideas, things and people either within it or without it. But the line is porous: our attempts at ordering are inevitably artificial and their outcome necessarily unstable: entropy corrodes the negentropic islands that we call our projects. Order is always under siege. Chaos seeps in, corrupts the magic circle, erodes its contour. This collection of essays traces such erosion. Whereas what's inside design's magic circle can be defined, the chaos that surrounds it can only be described. Hence, *What Design Can't Do* is about describing more than prescribing.

Nowadays, chaos appears to be more powerful than order. Global-scale logistic systems seem as flawed as our tiny life projects. This is why today, more than ever, design

6 This is something that novelist Mary Shelley already pointed out in 1831: "Invention, it must be humbly admitted, does not consist in creating out of void, but out of chaos." The quote comes from a preface to her novel *Frankenstein; or The Modern Prometheus*. The plot seems to suggest that traces of chaos reside in the order, or even that order is nothing but provisionally patched chaos, like the body-corpse of the monster. See Maria Popova. "'Frankenstein' Author Mary Shelley on Creativity." *The Marginalian* (blog), June 25, 2018. https://www.themarginalian.org/2018/06/25/mary-shelley-creativity-franksenstein-1831/.

7 The general tendency is toward the border's extension: "And so we have been forced to expand the boundaries of the systems we deal with, trying to internalize those externalities." Horst W. J. Rittel and Melvin M. Webber. "Dilemmas in a General Theory of Planning." *Policy Sciences* 4, no. 2 (June 1, 1973): 155–69, p. 159.

is polarised. It feels either all-encompassing, infrastructural, planetary, big, baffling – or improvisational, ad-hoc, tiny, volatile. "Today's design culture is an expression of our intense prototype lives – wails Geert Lovink – [p]recarity as an open and free lifestyle is getting stuck in a never-ending series of failures. Projects either fall through or never get finished. Life feels like an endless row of proposals."[8]

Every problem is a wicked problem:[9] its resolution is temporary, its paradigm ever-shifting, its focus evolving. How can one provide yet another abstract definition of the design process when the circumstances of a messy reality are so imposing? Design is left with only one option: staring chaos in the eyes, waiving the somewhat reassuring notion of 'complexity.'[10] For chaos is not complexity: complexity is a field where various forms of expertise compete; chaos is the repressed that returns when the experts fail. If, as James Bridle argues, "complexity is not a condition to be tamed, but a lesson to be learned,"[11] chaos is a grievance that has nothing to teach.

Things do not only appear intricate: they feel meaningless, alien, even to those of us who have devoted themselves to the cause of order. Designers are torn between having to believe, for professional and vocational reasons, in the modern promise of a harmonic, fluid orderliness and being caught in an absurd, glitchy reality. They are the ideal type of a hyper-modern subjectivity – disillusioned evangelists who are losing faith.

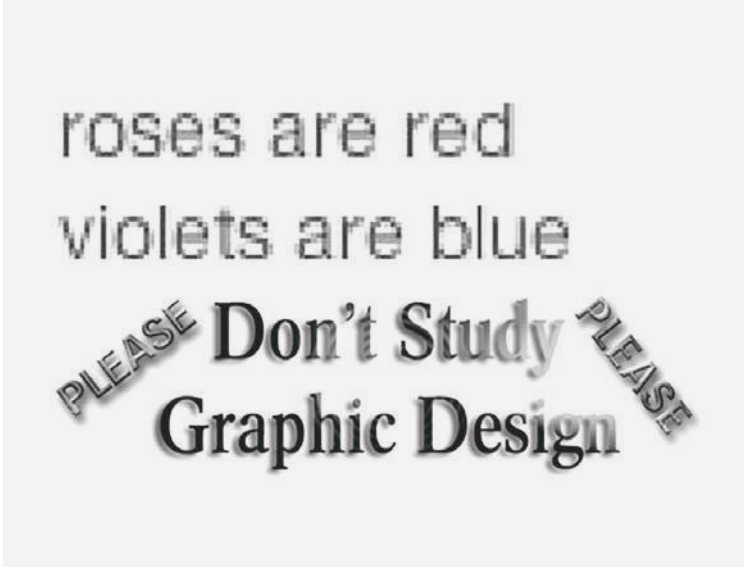
"Mess is the Law," declares architect Jeremy Till: "It has taken me this long to work out that maybe architecture

8 Geert Lovink. "Precarious by Design." In *Entreprecariat: Everyone Is an Entrepreneur, Nobody Is Safe*, by Silvio Lorusso. Eindhoven: Onomatopee, 2019. pp. 10-12.

9 A wicked problem is a problem that can't be unambiguously and definitively solved because its formulation is incomplete, shifting or even contradictory. The term was coined by design theorists Rittel and Webber, *op. cit*.

10 Hardly surprisingly many design papers and essays begin by paying an introductory tribute to complexity.

11 James Bridle. *New Dark Age: Technology and the End of the Future*. London: Verso, 2018, p. 138.



Critical Graphic Design (2015).
Source: https://hahahardcore.tumblr.com/post/107967560350.

is a mess; not an aesthetic mess but a much more complex social and institutional mess."[12] Writing in the '70s, architect Giancarlo De Carlo was more hopeful: "The mere sound of the word disorder generally provokes irrepressible neurosis, so it must be made clear that disorder does not mean the accumulation of a systematic dysfunction, but on the contrary the expression of a higher functionality capable of including and making manifest the complex interplay of all the variables involved in a spatial event [...] The salvation of the world – in all fields, from politics to aesthetics – is in disorder as the alternative of a constricting and overwhelming order that can no longer be shared."[13] In a similar vein, Henri Bergson argued that chaos is an order that we cannot see.[14] What if, instead, we think of order as a chaos that we try to ignore? Theodor Adorno believed that the task of art was to bring chaos into order.[15] Let's at least bring it into focus.

Talking of her book *Composing a Life*, anthropologist Mary Catherine Bateson wrote that the project "started from a disgruntled reflection on my own life as a sort of desperate improvisation in which I was constantly trying to make something coherent from conflicting elements to fit rapidly changing settings."[16] Here, I engage with a similar urgency, which I believe to be commonly felt. To do so, I explore the mechanisms that are put forward to maintain the illusion of order and the confidence in those who can bring it about. The goal is to shed light on the sense of disillusionment deriving from the distance between orderly expectations and a chaotic reality. A focus on this distance,

12 Jeremy Till. *Architecture Depends*. Cambridge, MA: MIT Press, 2013, p. XII.

13 Giancarlo De Carlo. *La piramide rovesciata: architettura oltre il '68*. Macerata: Quodlibet, 2018, pp. 112-4.

14 Quoted in Tomás Maldonado. *La speranza progettuale: ambiente e società*. Torino: Einaudi, 1997, p. 112.

15 Theodor W. Adorno. *Minima Moralia: Reflections from Damaged Life*. London: Verso, 1978, p. 222.

16 Quoted in Penelope Green. "Mary Catherine Bateson Dies at 81; Anthropologist on Lives of Women." *The New York Times*, January 14, 2021. https://www.nytimes.com/2021/01/14/books/mary-catherine-bateson-dead.html.

which might seem alien to the mission of design, is in fact at its core. After all, identifying a problem means nothing other than "knowing what distinguishes an observed condition from a desired condition."[17] The point is not to 'fix' disillusionment, but to understand its origin and the way it affects beliefs and behaviour. To do so, we must place ourselves *in the middle*, looking at how designers have to negotiate between management and execution, technics and humanities, autonomy and dependence, power and subjugation, bureaucracy and innovation, things and self.

* * *

What does the chaos surrounding the magic circle of design look like? Being unable to generalise, as chaos is always unique, I can only describe my own: a draconian series of InDesign paragraph styles, a lost Indexhibit site, a logo commissioned on Fiverr for a laugh, an absent-minded visit to the Dutch Design Week, a 404 error on a wrong jQuery URL in a static webpage, a bunch of riso-print zines, a student's expiring visa, a weak eduroam WiFi access point, a crowdfunded exhibition, a poster about the Anthropocene depicting mushrooms and bacteria, a Linmon-Lerberg IKEA desk travelling from one rented room to another, an unpaid internship report, a video essay featuring a North-American female voice-over, an Instagram ad followed by a @dank.lloyd.wright meme, some dusty mammoth Taschen volumes left at my dad's place, variable fonts, walking tote bags, people dressed like posters, hand-made protest signs, post-its, a dub DJ-set, Cinema 4D free assets, a Twitter hot take on the new CIA brand, a bachelor thesis on transhumanism, an urgent email from an obnoxious client, an alignment error on a 1200 print-run, two herniated discs, a MacBook Pro with Touch Bar, a pair of Lidl flip-flops, an expired

17 Rittel and Webber, *op. cit.*, p. 159.

Adobe Creative Cloud subscription, the daily advice of Stefan Sagmeister, a coffee-stained funding application, the e-flux spinner pattern, a witty rip-off of the MAGA hat, 10-page portfolios (10mb max), a Unity-based exploratory "videogame," a Marcel Breuer tubular chair (tried once in a corporate office), this very Markdown file edited in dark mode.

Someone who is familiar with design in the Global North (and probably elsewhere too) might have some of these items in their personal design starter pack. "Starter pack" is the name of a meme in which the defining features of a certain profession, subculture or fandom are displayed against a white background.[18] They can be items, tools, books or even habits. Often, clothes and accessories are included, showing that much of our professional identity inevitably conforms to a certain stereotype and signals it. Many different starter packs for the designer category exist: "the graphic design student" starter pack, the "pretentious designer" one, the "pissing off a graphic designer" version... Their implicit message being that there is no fundamental difference between tools, devices, literature or accessories. Everything is an appendix of identity, something that contributes to a sense of belonging, and in some lucky cases, to the accumulation of prestige. The starter pack meme highlights something else as well, namely that identity formation combines consumption with professional production. Or even that profession is, at least in part, a form of consumption.[19] A chaotic assemblage of designerly stuff floating on a white canvas is supposed to alchemically generate personal character and personalised meaning. Mieke Gerritzen and Geert Lovink speak of "an aesthetic ambiance around your personality, filled with seductive ideas, things and experiences."[20] But this form of

18 See https://knowyourmeme.com/memes/starter-packs.

19 I expand on this point in chapter 5.

20 Mieke Gerritzen and Geert Lovink. *Made in China, Designed in California, Criticised in Europe: Design Manifesto*. Amsterdam: BIS Publishers, 2020.

Found image. Source unknown.

aesthetic identification is fragile. Another recent memetic formula reads: "designer is not a personality." There is a growing feeling that both profession and consumption are insufficient means to build a solid, stable identity.

If we look specifically at graphic design, a dense, complex starter pack is the one built over three years in the Tumblr blog Critical Graphic Design.[21] There, one can scroll through a plethora of obscure inside jokes (some already outdated as the blog shut down in 2015), an obsession with avantgarde designers who are also cultural producers, especially from the Netherlands, the UK and the US (such as Experimental Jetset, Zak Keys, Michael Bierut), the parody of "criticality" as an attitude to display, some modified screenshots of the Photoshop interface, non-existing hyperstitional theory books.[22] Also noticeable is the mechanism of self-canonisation typical of small scenes, a fixation on ivy-league design schools such Yale (but pictures of Yale forklifts are shown instead), an ironic indulgence in amateur design, a sensibility towards precarity and the hardships of the job market, a few rants on the hypocrisy of political design, an acute awareness of consumerism and profession as two intersecting domains ("Everything is stuff," a book by Metahaven along with a Nike pair of sneakers or a Guy Fawkes mask). Finally, some traces of disillusion ("roses are red violets are blue please please don't study graphic design").

Design critic Francisco Laranjo lamented the lack of coherence of the blog, but it was exactly its schizophrenic

Originally, the subtitle of the book was "Amsterdam Design Manifesto", an apt choice that situates the specific design chaos the authors describe. Having lived in the Netherlands for several years, I get their perspective. This is why their books will appear frequently in the following pages. But, whereas Gerritzen and Lovink focus mainly on the contemporary state of design, I concentrate on the state of *designers*, who are the first to be redesigned by it.

21 https://criticalgraphicdesign.tumblr.com.

22 Hyperstition is a term coined by writer and philosopher Nick Land. A portmanteau of the words "hyper" and "superstition," it suggests that ideas can be pushed into the cultural arena where they reinforce themselves, functioning as memetic self-fulfilling prophecies. See Delphi Carstens. "Hyperstition." *orphan Drift Archive* (blog), 2010. https://www.orphandriftarchive.com/articles/hyperstition/.

polyvocality that gave Critical Graphic Design its edge.[23] Critique is no less messy than affirmation. Furthermore, what the anonymous group behind the blog was suggesting, already more than eight years ago, is that next to the visible lifestyle and professional items of design personality, there is a hidden starter pack,[24] one made of silent, sometimes unconscious factors: nightmarish bureaucratic procedures, financial troubles, rich families, gender biases, shitty clients, unpaid internships, dynamics of micro or macro-celebrity, generous funding or lack thereof, networks of friends, circles of gossip and so on. These threaten or sustain the project of doing projects: the professional life project of becoming a cultivated designer, a cultural professional, and more crucially, of remaining one.

A more recent meme could have easily been featured on Critical Graphic Design. Here, someone brags about her achievements: "My husband got a promotion, I'm pregnant and we just bought our first house." Unlike the accomplished high-school friend, the meme protagonist is busy joining vector points in Adobe Illustrator, an infamously tedious process that disgracefully hasn't been automated yet. There is much to unpack in this low-resolution image. First, a traditional idea of success and the good life (only the SUV is missing). Then, a vivid expression of personal disorientation and self-doubt. The meme also speaks of the trivialisation of skills and the drabness of the design profession, which is, for the most part, littered with repetitive tasks.

Finally, it seems to suggest that the exchange is happening between two women. This is no coincidence. For women it is structurally more difficult to get to the top as a designer and reach a position that is either authorial or

23 Francisco Laranjo. "Critical Graphic Design: Critical of What?" *Design Observer*, April 16, 2014. http://designobserver.com/feature/critical-graphic-design-critical-of-what/38416.

24 The hidden starter pack refers to the notion of hidden curriculum theorised by, among others, John Dewey, Ivan Illich and Paulo Freire. See https://en.wikipedia.org/wiki/Hidden_curriculum.

GRMMXI (2014).
Source: https://grmmxi.fi/post/75583597164.

authoritative.[25] Women are more likely to be stuck with the tiny, menial aspects of the job. After all, one tactic to protect oneself from the chaos of life is to narrow one's project down, to make it as small as the distance between two zoomed-in vector points. The 'tiny' design project, with its controllable Bézier curves, can act as shelter from the life project, which often seems chaotic and meaningless – quite the opposite of the baby-boomer family image painted by the meme's accomplished friend.

* * *

Italian cultural critic Tommaso Labranca once listed two opposite artificial hells of art and design practitioners. On the one hand, the aseptic horror of the white cube; on the other hand, "the nightmare of chaos experienced in the always temporary and shaky dwellings of the artistic underworld."[26] In 2018, Airbnb launched a series of design talks entitled *When Chaos Is Your Creative Director*.[27] The choice of topic makes a lot of sense for a company which deals with people dressing up their room and apartments, often small and Escheresque, into pleasant, generic habitats for tourists and business visitors alike.[28] The people at Airbnb explain: "[...] while the fog of chaos leaves some of us frozen, there are rare talents who can see clearly enough to activate and create." That's the hope: to be one of the lucky few who not only builds order, but protects it from seeping chaos.

They insist: "Because chaos is inevitable. Because we live in a world of political unrest, health crises, and

25 Ruber Pater reports that while most graphic design students are women, the designers who run studios are predominantly male. *Caps Lock: How Capitalism Took Hold of Graphic Design, and How to Escape from It*. Amsterdam: Valiz, 2021, p. 294.

26 Tommaso Labranca. *Vraghinaroda: Sopravvivendo a hipster situazionisti, santexuperine scalze e mistificatori deleuziani*. Milano: 20090, 2019, p. 84.

27 https://airbnb.design/designed-chaos/ and https://airbnb.design/seasontwo/.

28 Journalist Kyle Chayka dubbed this aesthetic style "airspace." See "How Silicon Valley Helps Spread the Same Sterile Aesthetic Across the World." *The Verge*, August 3, 2016. https://www.theverge.com/2016/8/3/12325104/airbnb-aesthetic-global-minimalism-startup-gentrification.

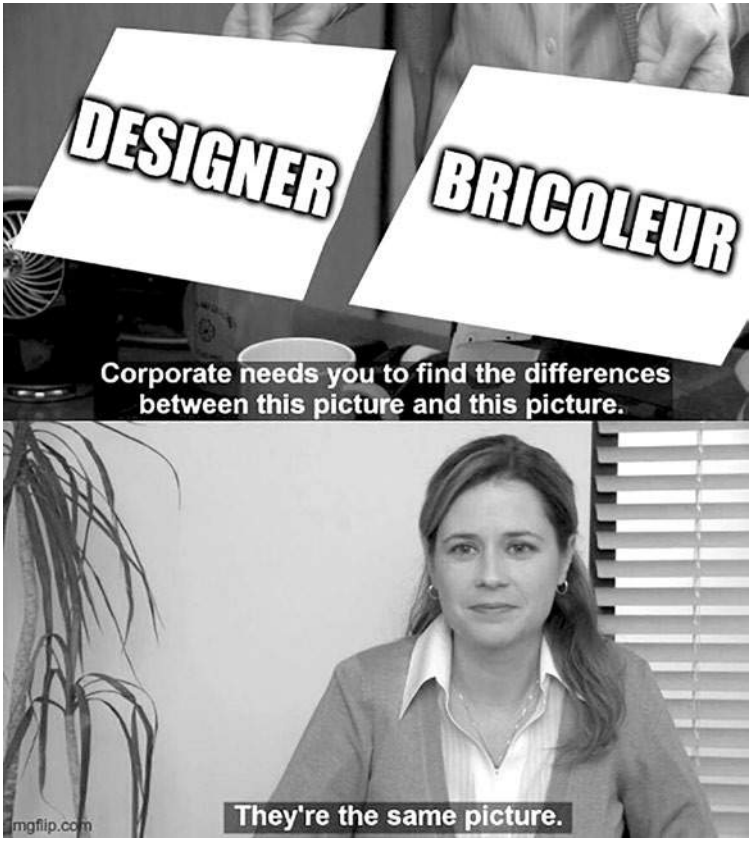
questionable ethical tech (*sic*), and design plays a critical role in addressing these global challenges." Even a behemoth company can't avoid admitting the inevitability of chaos, this blob-like material which is at the same time sublimely large and mundanely tiny, an agent whose indifference resists most organising efforts. Chaos precedes design and operates inside it: it is the manifestation of the Real beyond the designerly illusion of a stable and durable order. Given the chaotic qualities of design, one could appreciate the improvisational, almost absurdist illustration of designing provided by Enzo Mari, a legendary Italian designer who, growing up in the rubble of World War II, was first a vocal design utopianist, then a disillusioned realist, and finally a sort of Great Hater of the design world:

> We all design, every day, when we are forced to make our own decisions, even the seemingly trivial ones. For example, having to cook and finding in the fridge only a cup of yogurt and two onions.[29]

One way to look at design is by the capacity of its action, that is, the order it imposes. But design is also, more simply and fundamentally, yogurt and onions – what we are left with, the mess we're in.

29 Enzo Mari. *25 modi per piantare un chiodo: Sessant'anni di idee e progetti per difendere un sogno*. Milano: Mondadori, 2011, p. 5.

@dank.lloyd.wright (2022).



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Chapter 1.

In the Middle

"And just as it is only the burning awareness of what we cannot be that guarantees the truth of what we are, so it is only the lucid vision of what we cannot, or can not, do that gives consistency to our actions."
– Giorgio Agamben, 2011[30]

"To reach this goal – to feel what we know and know what we feel – is one of the tasks of our generation."
– László Moholy-Nagy, 1947[31]

A TERRIBLE LIFE DECISION

The state of design in 2023. Gone are the days of the Apple craze, of design as a positive force of change and economic growth, a golden age in which not only designers but also managers and politicians would jump on the smooth and colourful bandwagon of design. The period ranging from the mid-80s to the late 2000s was a promising one: design went hand in hand with creativity, which wasn't just a skill or a quality, but a full sociopolitical project, that of individual autonomy and cheerful reinvention, freedom of choice, agile making and breaking, self-design. During this time, management guru Tom Peters's conviction that "design is everything, it's how you live in the world" became the default, the same Tom Peters who imagined a "world where the timid goal of 'improvement' (and the tendency to tinker) has given way to... an unabashed commitment to destruction."[32]

30 Giorgio Agamben. *Nudities*. Stanford: Stanford University Press, 2011, p. 45.
31 László Moholy-Nagy. *Vision in Motion*. Chicago: P. Theobald, 1947, p. 11.
32 2015 McKinsey interview, https://www.youtube.com/watch?v=K3i7x54mOyo. Tom Peters. *Re-Imagine!* London: DK, 2006, p. 31.

Today, things are different. Creativity turned out to be not so emancipating: after the crisis of 2007 and 2008, we dwell in the debris of 'creative destruction.' What is constantly reinvented has no right to stability. A newer new is always impending. In fact, some time ago another keyword replaced design: *innovation*, a term which is itself now under scrutiny. Apple releases are still spectacular events, but the messianic aura around them has evaporated. The design field is still expanding, but it does so less boastfully than before. While change advances undaunted, a standstill is in place, an atmosphere of suspicion permeates the room. Needless to say, design is still employed to increase use value and exchange value (especially the latter), but there is a growing feeling that it has lost its transformative power. Or that design is not in control of this power. Or even that this power has *always* been out of control; as such, it is not power at all.

The "design culture turn" has run its course.[33] Design is not a buzzword anymore and today, more than ever, is ambiguously polarised. In this conjuncture, it is designers themselves, especially the young and not-so-young ones,[34] as well as those who inhabit the peripheries of the design citadels, who are starting to question the value and impact of such practice and its position within power structures. In a way, it is the very idea of design as an abstract and autonomous entity that is put into question: there is no Design, but designed artefacts, systems and processes, both material and immaterial; there are multiple influencing forces at play, and, in the middle of it all, there are

33 Guy Julier identifies a turn in design culture taking place between the '80s and the 2000s. This turn is linked to a new social and economic arrangement generated by neoliberal policies. *Economies of Design*. Los Angeles: Sage, 2017, p. 14.

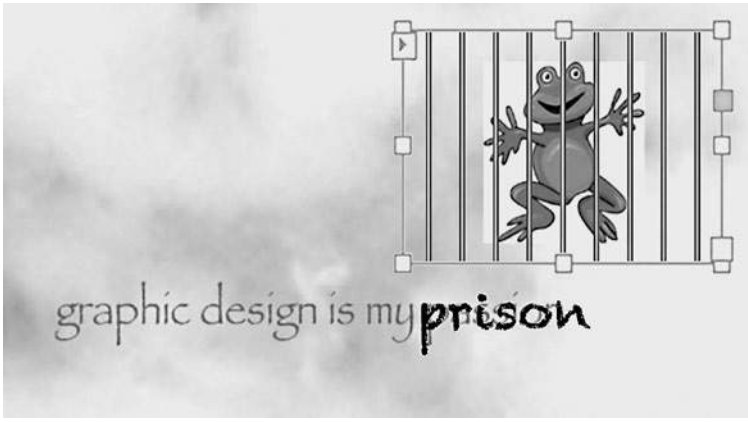
34 Perhaps, especially not-so-young ones: "The disgruntlement seems to go up the longer someone has been in the field: The more seasoned and experienced a UX person is, the more likely they are to be asking whether realizing user-centered values is even possible under capitalism." Jesse Garrett. "Ux Design Is More Successful Than Ever, but Its Leaders Are Losing Hope. Here's Why." *Fast Company*, June 3, 2021. https://www.fastcompany.com/90642462/ux-design-is-more-successful-than-ever-but-its-leaders-are-losing-hope.

designers, stuck between the grand project of modernity and the 'smol' tactics of everyday life.

While 'the power of design' mantra is ecumenically reiterated during conferences, the daily life of most designers is mundane, structured around those "trivial purposes" that the First Things First manifesto already lamented in 1964.[35] The cult of design heroes survives, but a sense of bitterness pervades the crowd, as the kind of hagiographic praise and unanimous approval that makes a hero doesn't match the times. Design heroes, even those who are alive and well, are the vestiges of a dying religion. It's a post-heroic age. Of course, a lot of work is yet to be done to eradicate the double myth of design as a force of good and the designer as the hero who governs it, but the sentiment is clear.

Let's zoom in on this sentiment. Can the field's 'sad passions' be revealing of something we don't fully comprehend? According to cultural theorist Raymond Williams, "one generation may train its successor, with reasonable success, in the social character or the general cultural pattern, but the new generation will have its own structure of feeling, which will not appear to have come 'from' anywhere."[36] What does it mean to inhabit design's current structure of feeling, its *Stimmung*? Can a sentiment not just be the object of analysis but also its medium, its propeller? Neither a theoretical inquiry, nor a critique, a how-to manual or an activist pamphlet, this book is a passionate diagnosis. Much of it deliberately insists on sense, feeling, perception. This is because perceptions matter as much as knowledge, and emotion as much as reason, or to put it another way, reason is one of the many forms that emotion takes. We now know that "the designer's own mindset/posture [is] an essential component of the design process"[37] and, if we are to trust László Moho-

35 Ken Garland. "First Things First: A Manifesto," 1964.
36 Raymond Williams. *The Long Revolution*. London: Penguin Books, 1965, p. 65.
37 Terry Irwin, Gideon Kossoff, and Cameron Tonkinwise. "Transition Design Provocation." *Design Philosophy Papers* 13, no. 1 (January 2, 2015): 3–11. https://doi.



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ly-Nagy, we must see designing not as a profession, but as an attitude.[38] Today, design is an attitude that wants to be considered a profession, but this attitude is perturbed by individual and social turmoil.

Constant redesign can be exhausting. Like in the Gartner Hype Cycle, after a "peak of inflated expectations," we reach a "trough of disillusionment."[39] The signs pop up in unexpected places: design is often the butt of the joke. Some designers are doubtful and disoriented. Others are anaesthetised and disappointed. Some of them are plain angry and resentful. Disillusionment is palpable. How many disillusioned designers are there? What's the exact percentage of chagrin in the design field? Clearly, there is no objective way to measure this. What is certain is that the "slope of enlightenment" is not in sight. Some designers get stuck: they feel unable to produce meaning with the instruments provided by the their field. "The truth is we are the most iconic, lazy, useful idiots of our era." This is how designer Baptiste Fluzin commented the call to arms to counter the rise of Donald Trump.[40] All in all, it might be a good time: perhaps designers are suddenly realising that their relationship with their discipline has always been a form of Stockholm syndrome.

Luckily for us, disillusion is not just *disillusionment*, a passive feeling of dismay and disappointment. It is also *disillusioning*, the active lifting of illusions, an engagement with reality without at least some of the old veils. Thus, disillusion is a pendulum oscillating between lucidity and dismay. Who is more prone to this sentiment? Design disillusion is the 'feel' of those who have access neither to the reassurance of the centre nor to the effervescence of the

org/10.1080/14487136.2015.1085688.

38 Moholy-Nagy, *op. cit*, p. 42.

39 Marcus Blosch and Jackie Fenn. "Understanding Gartner's Hype Cycles." *Gartner*, August 20, 2018. https://www.gartner.com/en/documents/3887767/understanding-gartner-s-hype-cycles.

40 Baptiste Fluzin. "Designers, Designers, Designers." *Tumblr* (blog), October 11, 2016. https://bfluzin.tumblr.com/post/152990139318.

margin, those who feel stuck in some sort of pseudo-cultural, semi-professional suburbia. Those who can't seem to attune themselves to design's display of good sentiments and prescription of appropriate behaviours. Despite running the risk of generating even more of it, one needs to think *with* disillusion to measure the space that separates expectations from reality.

Where is such disillusion expressed and how can it be probed? For obvious reasons, design pessimism mostly manifests informally, in ephemeral chats and conversations, in memes and tweets, often deleted shortly after. While it is becoming a genre in itself, the public anti-design critical outburst is still an exception. If we turn our gaze from the official disquisitions of museums, magazines and galleries which acrobatically rip design apart while reassembling it, to the oral environment of social media, we witness an outpouring of doubts, reality checks and self-deprecating humour. Just innocuous jokes, one might say. But what if we take those jokes seriously? A popular design publication suggested that jokes could bring down governments.[41] More humbly, I believe they could disclose something worthy of our attention. Let's look at some of them.

According to a series of memes, graphic design is: no longer my passion / my burden / my prison. A Twitter user admits: "Every day I think about what a terrible life decision being a designer has been." Another one rebukes: "design is so unimportant in the grand scheme of things and I'm sick of seeing people kid themselves into thinking their contributions as a designer are some form of visual activism." On a wall, painted red, we find the statement "design ruined my life," while a sticker on a trash can yells in caps: "graphic design is shit / coding is shit / all I want is revenge."[42] Facetiousness, for sure. Designers' humour, no

41 Metahaven. *Can Jokes Bring down Governments?* Moscow: Strelka Press, 2014.

42 To be fair, there are also positive expressions, such as "Every day I remind myself of how lucky I am that I get paid for drawing rectangles."

doubt. Nothing other than understatement and self-deprecation as a bonding mechanism. After all, that's what social media are for. More rigorous sources would surely reassure us. Among them, are a few surveys on designers' conditions. Lucienne Roberts, Rebecca Wright and Jessie Price, editors, together with social scientist Nikandre Kopcke, of the book *Graphic Designers Surveyed* report that "[a]lthough 83% of respondents said they would recommend a career in graphic design, only 55% of respondents expressed satisfaction with their career – and 23% were actively dissatisfied."[43] Stefanie Posavec, who designed the book, concludes that "[...] while designers are an independent, opinionated (and, dare I say it, mouthy) bunch when it comes to how we feel about our practice and our chosen field, this strong will doesn't always translate into higher wages or shorter hours."[44]

From Flavia Lunardi's 2018 survey on Italian graphic design studios we learn that although 86.9% of the respondents don't want to change job, 71% of them are not appeased by the recognition of the graphic designer's role in Italy.[45] In this case, like in those of the UK and US, it seems that the majority like their job and yet they are unhappy with it. Is this an instance of what Lauren Berlant called "cruel optimism?" The theorist used the term to refer to a situation in which "the object/scene that ignites a sense of possibility actually makes it impossible to attain the expansive transformation for which a person or a people risks striving [...]."[46] The object/scene being, in this situation, a designer's career.

According to the design census of 2019 carried out by the American Institute of Graphic Arts (9,429 participants), one out of three designers is dissatisfied, and 7% of

43 Lucienne Roberts, Rebecca Wright, and Jessie Price, eds. *Graphic Designers Surveyed*. London: GraphicDesign&, 2015, pp. 456. The survey spans the UK and US. In the overall sample, 85% of the respondents were under 40 and predominantly white, pp. 50-51.

44 *Ibidem*, p.18.

45 Flavia Lunardi. "Grafica Italia 2018." ISIA Urbino, 2018.

46 Lauren Berlant. *Cruel Optimism*. Durham, NC: Duke University Press, 2011, p. 2.

them are ready to call it quits. And this does not take into account the self-selecting bias of a professional association which plausibly includes practitioners who have obtained a certain degree of stability or success in their practice. What would be the dissatisfaction rate if designers at large were considered? Moreover, those surveys implicitly ask for a sort of definitive assessment of oneself, and therefore the respondent might be cautious when it comes to negative self-evaluation.

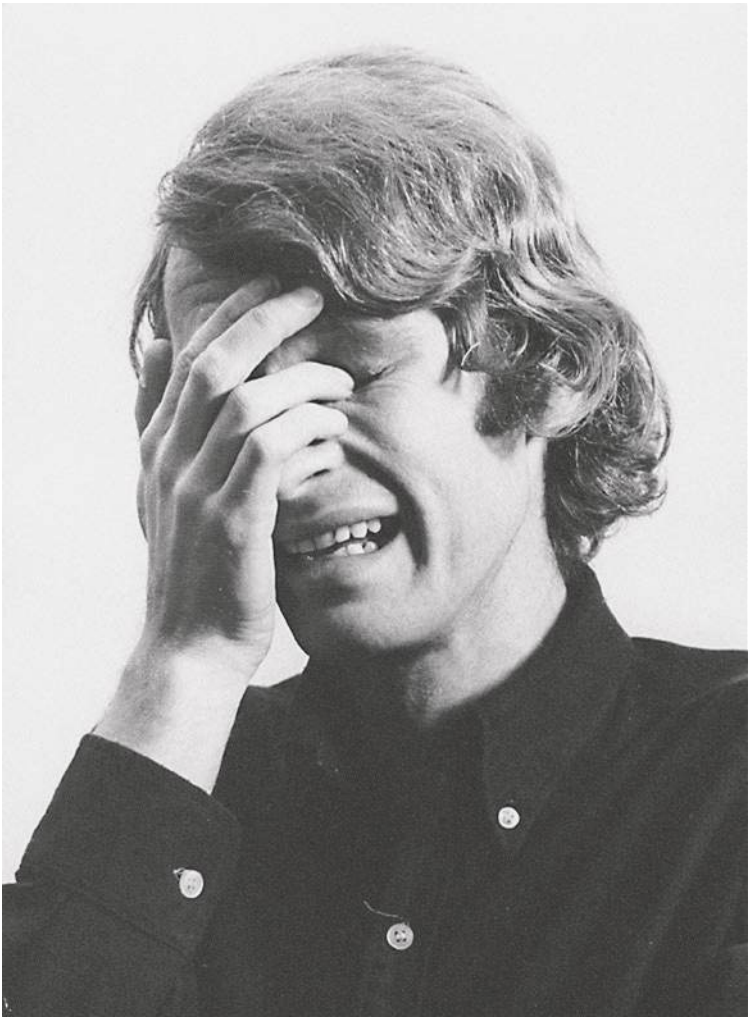
THE BIG SPLIT

Is this anything new? US sociologist C. Wright Mills diagnosed an uneasiness with the role of the designer already in the late 1950s. He highlighted two trends: first, the importance that distribution was gaining over production, thus that of status over subsistence; second, the subordination of activities to capitalism and nationalism. This is the nexus where designers operate:

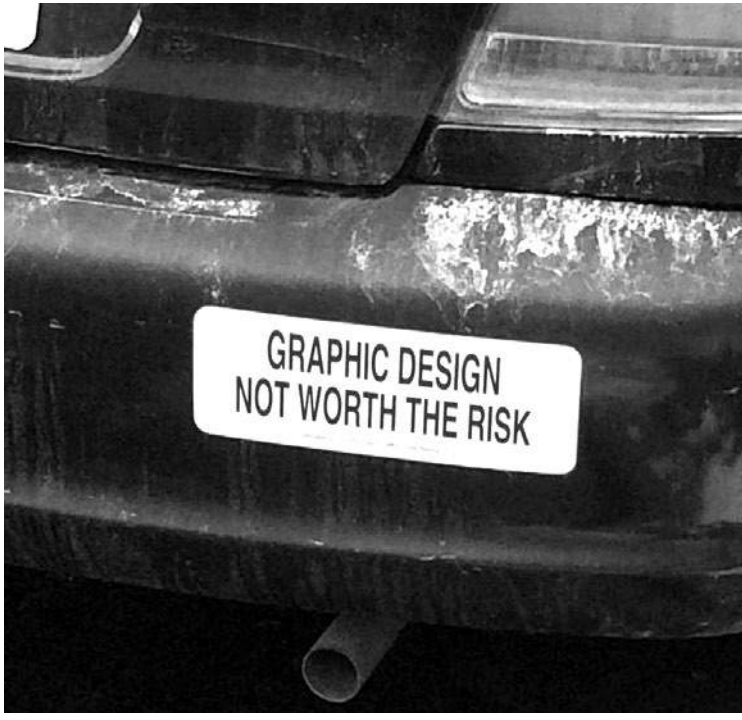
> Designers work at the intersection of these trends; their problems are among the key problems of the overdeveloped society. It is their dual investment in them that explains the big split among designers and their frequent guilt; the enriched muddle of ideals they variously profess and the insecurity they often feel about the practice of their craft; their often great disgust and their crippling frustration.[47]

According to Mills, the designer is a "man in the middle" who cannot fully understand their position without considering the cultural and economic conjuncture they're

47 C. Wright Mills. "Man in the Middle: The Designer." In *Power, Politics and People*, edited by Irving L. Horowitz, London: Oxford University Press, 1969, p. 374. This was part of Mills' talk at the 8th International Design Conference of Aspen in 1958. The same conference where, almost twenty years later, Saul Bass asked students "why do we have to assess capitalism? We're just trying to stage a design conference." See Alice Twemlow. "'A Guaranteed Communications Failure:' Consensus Meets Conflict at the International Design Conference in Aspen, 1970." In *Aspen Complex*, edited by Martin Beck. London: Sternberg Press, 2013.



I'm too sad to tell you, a mixed media artwork created by conceptual artist Bas Jan Ader between 1970 and 1971.



Bumper sticker by Freelance Studio, who found the original image on Twitter.

in. One suspects that his words caused some stir. Designers are not used to seeing themselves in the middle: they place themselves at the top of cultural and decision-making processes. They portray themselves as in charge of the plan. But then, they are frustrated by the realisation that the plan and the design are two very different things.[48] Mills's diagnosis, however, might not fully apply today. Whereas in the past the designer's crippling frustration derived from the awareness of being an "organization man," a cog in the cultural and economic machine, now, given the abundance of designers, the frustration has more to do with being a cog *outside* the machine, that is, being less and less able to shape its workings.[49] After all, Mills was speaking to a crowd of prominent representatives of the field many of whom, despite the "Aspen-style" informality,[50] were firmly embedded in the industrial and corporate world.

Mills explained that *White Collar,* one of his most popular books, was "about the new little man in the big world of the 20th century [...] for, in truth, who is not a little man?" Here, I propose to focus on the 'interscalar' gap between the personal tiny and the structural huge, to connect the vector points of individual life to the grand issues rooted in modernity and its crisis. The '60s white-collar worker was beginning to experience what would become the default torments of the no-collar professional worker, a category that the designer centrally represents: The "new little man [...] seems to have no firm roots, no sure loyalties to sustain his life and give it a center [...] Perhaps because he does not know where he is going, he is in a frantic hurry; perhaps because he does not know what frightens him, he is paralyzed with fear."

48 See chapter 4.

49 William H. Whyte. *The Organization Man*. Philadelphia: University of Pennsylvania Press, 2002.

50 *International Design Conference in Aspen: The First Decade*, 1961. https://www.youtube.com/watch?v=8MxCGKicSfg.

This is how the magazine *Industrial Design* reviewed Mills's talk:

> Generally when a speaker addresses members of a profession not his own, he tells them what they want to hear. He can do it obviously, by telling them how good they are; or subtly, by telling them how bad they are, then making it all right at the end by exhorting them to be better. In either case, since he tells them only what they tell each other, he contributes only the illusion of a fresh perspective. An exception is this paper read to the Design Conference in Aspen this summer by sociologist and author (*The Power Elite*) C. Wright Mills. Neither lullaby nor mock attack, it is a hard analysis of the designer in our society.[51]

Neither lullaby nor mock attack. We can take this as a methodological principle to carry out an inquiry that is self-conscious of its own participation in the design discourse, a discourse which mostly welcomes what it can digest in utilitarian terms, be it criticism or praise.

Since Aspen, design's identity crisis has only worsened. In the '70s, Tomás Maldonado detected a dimming of the "designerly hope," brought about by a youthful nihilism that was leading to a pre-emptive renunciation of action. Maldonado was thankful to the youth "for waking us up from our drowsiness and reminding us without euphemism that ours is not an Arcadian age, but an agonizingly convulsive one."[52] And yet, he firmly believed in the necessity of hope, hence his essay on the topic. Although I agree with Maldonado, this book takes a different approach: it examines hopelessness *in itself*, resisting the urge of immediately rejecting it, for it is in hopelessness that hope sprouts.

51 In Javier A. Treviño. "C. Wright Mills as Designer: Personal Practice and Two Public Talks." *The American Sociologist* 45, no. 4 (December 1, 2014): 335–60. https://doi.org/10.1007/s12108-014-9196-y.

52 Maldonado, *La speranza progettuale*. *op. cit.*, p. 10.

"Only for the sake of the hopeless ones have we been given hope," wrote Walter Benjamin.[53]

THE EVERYDAY DESIGNER

Maldonado had the young in mind, Mills the professional class. What's the subject of this book, then? The designer in the real world, I'm tempted to say. Still, what reality? The question is not an easy one and has serious theoretical implications. "When I talk about reality – wrote Ursula Franklin – I'm not trying to be a philosopher. I think of reality as the experience of ordinary people in everyday life."[54] With this in mind, who do we call a designer? What particularities do we include in this definition? The more we categorise, the more our categories quiver. Shall we consider designers only people who studied design? What do we do, then, with those self-taught practitioners who have achieved professional recognition? How could we leave out the "legions of designers who work in-house for companies or as freelancers?"[55] Shall we break down designers by the type of service they provide? That won't work. Graphic designers craft products, product designers conceive services, service designers do performances, performers strategically call their activity choreographic design, strategic designers print posters and fliers, etc. Specialisms have detached themselves from products to become a sort of shared attitude and a common set of cultural and methodological references. More than with the things that they do, designers identify with the sensitivity they adopt when they do those very things. To a certain extent, the contemporary designer is a designer without qualities.

Radical openness is our last resort: trust the person who calls themselves a designer to be one. Trust the designer who makes some of her income designing logos for friends,

53 Walter Benjamin. *Selected Writings Vol. 1*, 1913-1926. Cambridge, MA: Belknap, 2004.
54 Ursula M. Franklin. *The Real World of Technology*. Toronto: Anansi, 1999, p. 27.
55 Julier, *op. cit.*, p. 5.

Found image, photoshopped. In the original picture, taken in London, the sticker reads "Growth is shit, jobs are shit, all I want is revenge."

as well as the design *auteur* who sell his designs as collectible NFTs to pay rent; trust the business student who started calling themselves a designer after having learned about design thinking, trust the art director and the service designer, trust the critical designer, the user-centred designer, the design catalyst, the metadesigner, trust the designer who makes 'actually existing design.' This stance goes against notions of quality, excellence and even expertise, notions that have been mobilised by the field to acquire status. We need to relativise these notions to be able to see how, and if, they still lead to status. Excellence has always been the rhetorical instrument of a winning elite. But in order to understand elitist aspirations, we need to picture a kind of designer who is not so much part of an elite. This designer is not taken into account by design literature, which mostly focuses on promoting and strengthening the field. I will call this figure the *everyday designer*. While this kind of practitioner might have cultural and political aspirations, a good portion of their time is spent in mundane tasks. Perhaps the everyday designer has just launched a small boutique studio and struggles to find the clients who could appreciate their research capabilities. Perhaps they just had a kid. The everyday designer might be simply a student, taking an undergraduate or master's degree, either a technical or more artistic one. Or a recent graduate, trying to practice what has been preached to them. They might be sitting in a small agency thinking up a campaign to enrich the portfolio of a small bank. The everyday designer is a Photoshop whiz, he's toying with Figma or Sketchup, she's banging her head against CSS rules or the settings of a 3D printer. The everyday designer is the designer caught up in daily, menial tasks beyond creation, while daydreaming about their role and career. They're good enough designers – yet they might even be contemplating the idea of quitting.

What the everyday designer is not: the design star at the top of their field, the universally recognised expert,

Me in the middle of the night questioning my decision to become a designer

Found image. Source unknown.

the one everyone else looks at and emulates – in short, all those designers we hear tips and advice from, those who saturate the media sphere affirming their ideas and their role. It's not their fault: if asked, they answer. While the image of the design star is intrinsically iconic, that of the everyday designer is admittedly rather blurry, but its function is clear: to avoid exceptionalism. Or, more accurately, to isolate the exceptions that shape the rule and the deviations produced by the norm. In this sense, the everyday designer is seldom a resolved regularity, but more frequently a living discrepancy. That said, blurriness is part of the lens I adopt: many practitioners will readily recognise that they have acted as one or more of the 'personas' I just mentioned. Indeed, the everyday designer is not really a person, but a configuration. It's the personalisation of a semi-professional designing multitude at the edge of the design's field of vision.

In a similar vein, for his book *Economies of Design*, Guy Julier steered clear

> of the more senior, established end of the design career, not least because this is already much more represented through design media. These tend to focus on big success stories. We hear little of the designers who continue working the long hours for relatively low financial reward or those, indeed, who after a decade of so leave design altogether to do something else.[56]

Where is the everyday designer based? The answer is unavoidably partial. The practitioner I have in mind is generally active in the Global North: Europe and the US. This is of course a shortcoming, but I don't think it defeats my anti-exceptionalist purpose. A European designer is an exception, of course, but an exception that contains multiple normalities. Furthermore, Europe offers enough variety to highlight and compare stark differences in terms of

56 Julier, *op. cit.*, p. 55.

desires, aspirations and the possibility to realise them.[57] Finally, European design is not so European anymore. Go to design academies and you will find a substantial part of Chinese, Korean and Latin American students. Go to a final show in Amsterdam or Milan and you will see not only projects *on* the Middle East, but also projects made *by* Middle Eastern designers.

Most of my references and examples derive from graphic design. There are multiple reasons for this. First, a personal one: I trained as a graphic designer and therefore I'm more familiar with this context. Second, a methodological one: given the low access barrier of this field, the everyday designer is more likely to be a graphic designer than a more niche figure. Graphic design is a diluted practice, one in which professionals have to put increasing effort to assert their expertise to clients and society at large. Graphic design is where the necessity of radical openness becomes most obvious, since it's also where the pro/amateur binary abruptly collapses. Graphic design, one of those "rather inchoate professions" in and around design,[58] might even show the general trajectory of professions.

In 1995, design theorist Richard Buchanan proposed a model of design activities structured around four orders. In this model, graphic design is – controversially, one must say – the most limited one, as it deals particularly with symbols. The other three orders are industrial, interaction, and environmental design.[59] In contrast to this model, I propose to see graphic design as the widest of the orders, as it is more affected by technical, cultural and methodological democratisation: it is spread more diffusely

57 I think in particular of the gap between funding-rich countries like the Netherlands (where I used to reside) and funding-poor countries like Italy (where I come from). The designer is not just a "man in the middle" of cultural and economic trends, but also of cultural and economic contexts.

58 The expression is Mills's. Quoted in Treviño, *op. cit.*, p. 351.

59 Richard Buchanan. "Design Research and the New Learning." *Design Issues* 17, no. 4 (October 1, 2001): 3–23. https://doi.org/10.1162/07479360152681056.

across society. If we accept whoever professes themselves a designer to be one, we have to see design not just as an activity, but as a community of practice able to provide a sense of belonging. I would be tempted to speak of class, if it weren't for the abysmally different conditions among everyday designers. Many have argued that in a modern society everyone designs. While not denying this obvious fact, the lens chosen here is that of *consciousness*: everyday designers are conscious of their role and cultural context – they are self-conscious.[60] In this sense, the notion of everyday designer differs profoundly from the generalist claim that "everyone is a designer," a claim that can be either a defence of autonomy or a form of libertarian individualism. In other words, the everyday designer is explicitly a designer in a world where everybody designs.

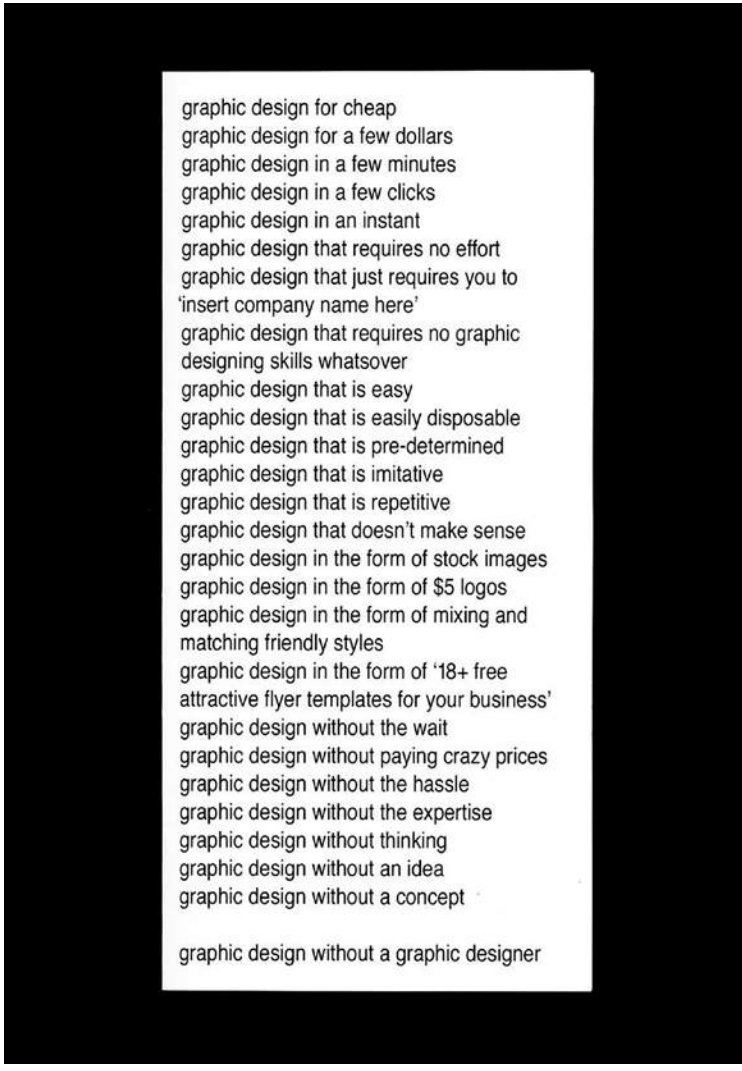
I'M NO EXPERT

It is true for me what was true for Karl Kraus: "I do not like to meddle in my private affairs."[61] However, as much as I'd like to avoid entering the stage, a few words on 'who speaks' are due. Shortly put, I consider myself an everyday designer who, thanks to his partial, intermittent, but hopefully lasting professional belonging to the field, has had enough time to write about his condition and the condition of his peers. This is my personal discrepancy, the exceptional normality I represent. While this work emerges from my own doubts and uncertainties, it doesn't derive from an urgency to critique or celebrate but, to use an apt expression by graphic designer Jan van Toorn, from a "passion for the real." Once again, it's neither lullaby nor mock attack.

This is a design book without much design in it. Or to be more precise, the design we encounter is not so much the portfolio material of designers, but the designed

60 This point is slightly controversial, as I sense that the professionals of design discourse are abandoning the profession as a matter of concern. See chapter 7.

61 Harry Zohn. *Karl Kraus*. New York: Ungar, 1971, p. 17.



Promotional (Mis)information by Adelia Lim (2018). A "collection of cheaply produced promotional materials that responds to the way graphic design is valued when design templates come into play."

environment that surrounds them, made of objects, services, but also institutions, protocols, values, myths. As discussed in the prologue, I'm more concerned with the 'mess we're in' than the projects we work on. Furthermore, this book is full of memes, a cultural form that exemplifies distributed agency and the attitude of the *bricoleur* – both aspects that challenge the compromised vocational image of the designer as planner. Memes also reflect the everyday designer's condition: disillusioned designers won't write papers to express their chagrin, they don't have time for that. They will make a meme instead.

For various reasons, several theorists have drawn a fairly clear line between professional designers and non-professional ones. Design historian Victor Margolin speaks of design with a capital 'd' and design with a small 'd,' the former being associated with the Industrial Revolution and the advent of mass communication, the latter being a broadly human activity.[62] Architect Ezio Manzini proposes a model in which "expert design" and "diffuse design" interact in various ways.[63] The everyday designer is located between these poles – not fully recognised as an expert, but also not a mere human being who plans ahead. The notion of the everyday designer sheds light on the fact that divides based on professionalism and expertise are nuanced, complicated and, more importantly, *dynamic*. The point is not only that these divides do not dignify practices that aren't traditionally recognised by the design field, but also that expertise and professionalism are categories in constant renegotiation. A professional skill might become a default one, and everyday designers might struggle to be considered expert. Furthermore, as with any profession, expert design is vampiristic: it sucks the vitality of diffuse

62 Steven Heller. "The Evolution of Design." *The Atlantic*, April 9, 2015. https://www.theatlantic.com/entertainment/archive/2015/04/a-more-inclusive-history-of-design/390069/.

63 Ezio Manzini. *Design, When Everybody Designs*. Cambridge, MA: MIT Press, 2015, p. 40.

design – the intelligence of the general intellect – to maintain its status. The crisis of the professions exacerbates vampirism: as expert designers lose prestige in the eyes of the public, they try to absorb diffuse design to reinstate their role and position.

LIFE PROJECT

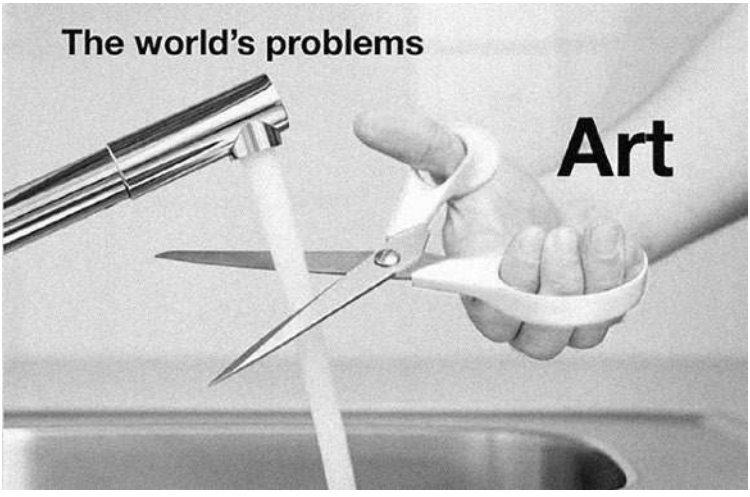
In *The Politics of the Artificial*, Margolin warns us that "if designers are going to realize the full potential of design thought, then they should also learn to analyze how the situations that frame design practice are themselves constructed."[64] Similarly, anthropologist Lucy Suchman is in favour of "ethnographic projects that articulate the cultural imaginaries and micropolitics that delineate design's promises and practices."[65] In these words, we hear the echo of Mills: the process of professional negotiation, the oscillating reputation of the everyday designer, the kernel of their vocations and ambitions should be made explicit.

The construction of ambition and vocation is strictly linked to the era in which design with a capital 'd' emerged: modernity. And when modernity becomes relentless and more flexible, in the phase that coincides with what we call neoliberalism, ambitions and vocations follow suit. Guy Julier explains that during the '80s, "the professional environment and aspirations of design mirrored the increased flexibility and speed afforded by the Big Bang for the financial sector."[66] In this regard, Ezio Manzini raises a relevant issue: "How and to what extent modernity's promise of designability of one's own life has been fulfilled is obviously an open question." This is the question we are tackling here, however not in general terms – that

64 Victor Margolin. *The Politics of the Artificial: Essays on Design and Design Studies*. Chicago: University of Chicago Press, 2018, p. 241.

65 Lucy Suchman. "Anthropological Relocations and the Limits of Design." *Annual Review of Anthropology* 40, no. 1 (2011).

66 Julier, *op. cit.*, p. 25.



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would be exceptionally pretentious – but within design itself. Thus, the question becomes: what is the horizon of designability of one's own design practice? The protagonist of Manzini's story is

> a subject immersed in his everyday life, taking part in various conversations; a node in various networks and an actor in various social forms. From his point of observation and action, he designs and co-designs his action on the world operating as a *bricoleur*: he looks for usable materials around him (products and services, but also ideas and knowledge) and, adapting and reinterpreting them, he uses them to compose his life project.[67]

This is a good characterisation of the everyday designer. When talking of life projects, the distinction between design and bricolage becomes blurry. The everyday designer is always working with a set of constraints that do not only belong to the internal logic of the design, but to its presence in the world. Everyday designers design their life project from the rubble of what is already there – their activities and career included. This process is dynamic and contingent, as Julier points out: "Designers' responsiveness to changing commercial contexts results in rapid reinventions of the professional field – a constant redesign of design itself."[68] Their practice is inevitably *compromising*.

Designers readjust. In Manzini's words, "in order to adapt to what is required of them, they must redesign themselves and their way of operating. But then this is what is required of everybody today."[69] If the modern project does not fulfil its promises of identity, administration and prosperity, the designer, being one of its poster children, becomes a hypermodern post-professional figure,

67 Manzini, *op. cit.*, p. 4.
68 Julier, *op. cit.*, p. 38.
69 Manzini, *op. cit.*, p. 2.

located between autonomy and dependence. And, like everyone else, caught in endless projects:

> The weak, depressed, self-critical, virtual self is essentially that endlessly adaptable subject required by the ceaseless innovation of production, the accelerated obsolescence of technologies, the constant overturning of social norms, and generalized flexibility. It is at the same time the most voracious consumer and, paradoxically, the *most productive self*, the one that will most eagerly and energetically throw itself into the slightest *project*, only to return later to its original larval state.[70]

The figure of the everyday designer shows us a dire situation, one where it is not individuals who direct projects, but projects that direct individuals.

CRACKED UP ON CAPITALISM

This reversal begins just after the phase of education. The leap from the *vita contemplativa* of the design school to the *vita activa* of independent or studio work is seldom devoid of trauma. Furthermore, a series of structural forces loom over the possibility of designerly gratification. According to a Gallup survey, 85% of workers worldwide do not feel engaged with their job.[71] From this perspective, the previous statistics about designer dissatisfaction acquires the opposite meaning. As dire as it might sound, one third of dissatisfied practitioners is not so bad. Let's consider again *Graphic Designers Surveyed*. The editors refer to an overall job satisfaction in the US of 48%, against the 78% of the UK. In this scenario, graphic designers living the States would be relatively more satisfied than those

70 The Invisible Committee. *The Coming Insurrection*. New York: Autonomedia, 2009, p. 31.

71 Jim Clifton. "The World's Broken Workplace." *Gallup*, June 13, 2017. https://news.gallup.com/opinion/chairman/212045/world-broken-workplace.aspx.

active in Great Britain.[72] But even in some affluent strongholds of design culture things are not so rosy: "[...] when asked about their satisfaction a year or two after graduating, a consistently high share of former design students – approximately 40%; twice as many compared with other studies – would not choose the same field of study again," reports Paola De Martin from Switzerland.[73]

All things considered, a certain degree of cynicism is to be expected. This is nothing new. The position of designers within capitalist and consumerist enterprises makes them pessimistic and frustrated.[74] But it's not only that. Issues related to the changing world of work need also to be taken into account. To begin with, there is the feeling of a decreasing return of investment from higher education, with the subsequent idea that a study loan spent on rising fees and rent might not be a good deal. 'ROI resentment' becomes particularly justified if we examine the income generated by the activity of designers: 44% of the *Graphic Designers Surveyed* are economically "just getting by."[75] The income disparity between various design specialisations might fuel such resentment. Furthermore, the work itself is often unstable and unpredictable, particularly when it comes to freelancing. Then, there are strong disparities between countries. For instance, the 15% unemployment rate in Spain against the 3.3% of the Netherlands. These disparities can fuel ambitions – and then curb them. There is, as well, the issue of sexism and the gender gap[76] (and yet women are surprisingly slightly

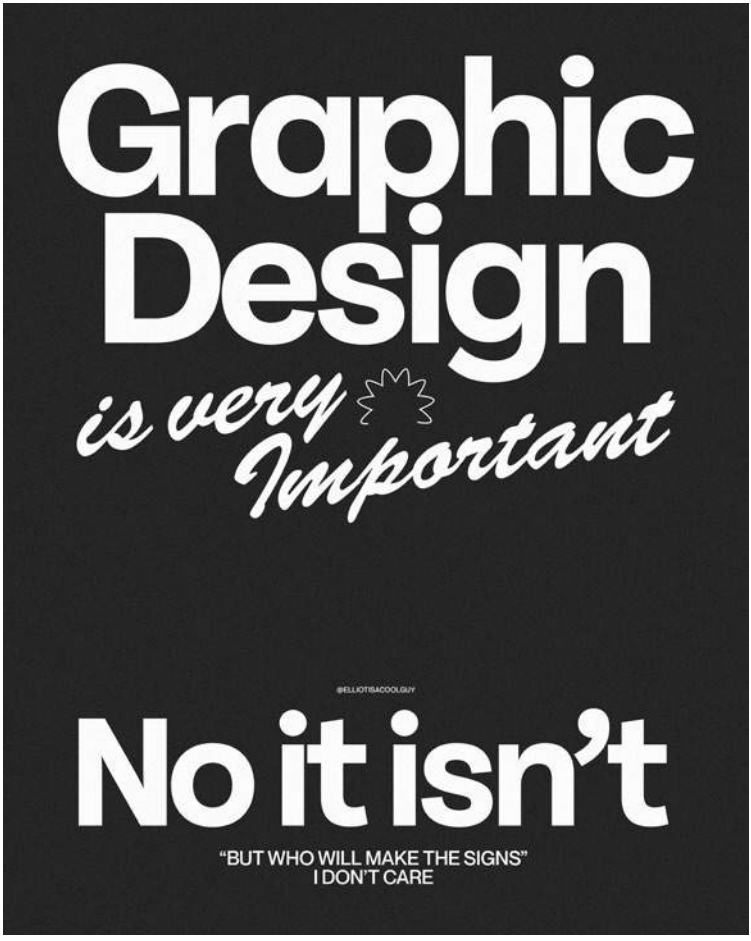
72 Roberts et al., *op. cit.*, p. 456.

73 Paola De Martin. "Breaking Class: Upward Climbers and the Swiss Nature of Design History." In *Design Struggles: Intersecting Histories, Pedagogies, and Perspectives*, edited by Claudia Mareis and Nina Paim. Amsterdam: Valiz, 2021, p. 66.

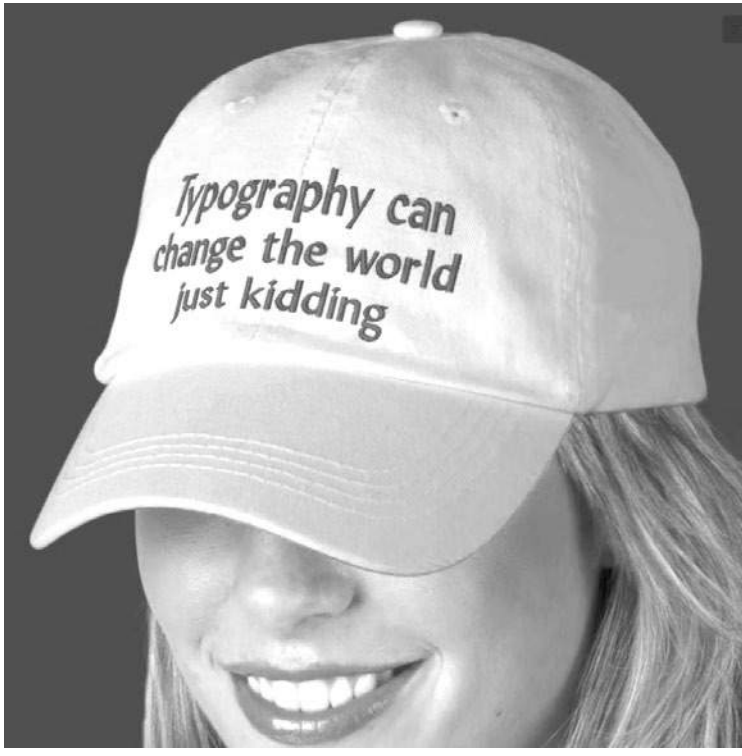
74 "Being at the centre of the sales curve, it is logical that many graphic designers develop a 'cynical relationship to both consumerism and capitalism', writes Boehnert. Designers are frustrated with the discipline, but see no other ways of making money." Pater, *op. cit.*, p. 208.

75 Roberts et al., *op. cit.*, p. 296.

76 *Ibidem*, p. 118.



Poster by Elliott Ulm (2021).



Print-on-demand baseball hat by Stephanie Davidson.

more satisfied with their career).[77] A widespread sense of futurelessness can also not be ignored. Design, with its fast-paced trends, is instrumental to it. The feeling is also related to the unfolding climate crisis, in comparison to which everything pales, any action feels inappropriate, abstract, irrelevant. How can one speak about petty things like professional recognition when we are facing extinction?[78]

A TikTok user summarises some of these problems in a 30-second rant:

> Hi guys, ever just wonder how much of a scam everything in our life is? When I first went to college I was like: oh my god, I need to find a purpose. What am I gonna do that's gonna give me such life fulfilment? And my answer to that was USER EXPERIENCE DESIGN. I really thought the ANSWER to my LIFE was UX DESIGN. I'M CRACKED UP ON CAPITALISM![79]

As authors like Guy Julier, Ruben Pater and Matthew Wizinsky aptly demonstrate, design is tightly (if not inextricably) connected to capitalism.[80] But not everything can be reduced to this relationship. Or, more precisely, not everything should be reduced to this relationship in its most general sense. The risk here is to perform Capital's job of abstracting lived practices and treat them as generic, 'lo-res' instances of exploitation. This usually leads to sacrosanct but also interchangeable calls for unionisation, petitions to 'pay your interns,' etc. We

77 *Ibidem*, p. 134.

78 Recently, MoMA design curator Paola Antonelli made the provocative claim that humans should design "an elegant ending" of their race. In such statements urgency meets powerlessness. Antonelli's provocation can be interpreted as an intelligent move to annex design powerlessness to her curatorial interests. Augusta Pownall. "'We Don't Have the Power to Stop Our Extinction' Says Paola Antonelli." *Dezeen*, February 22, 2019. https://www.dezeen.com/2019/02/22/paola-antonelli-extinction-milan-triennale-broken-nature-exhibition/.

79 @suwuuuuu, post deleted.

80 Julier, *op. cit*. Pater, *Caps Lock*, *op. cit*. Matthew Wizinsky. *Design After Capitalism: Transforming Design Today for an Equitable Tomorrow*. Cambridge, MA: MIT Press, 2022.

shouldn't treat the designer (or any other kind of practitioner, for that matter) as an economically-determined stick figure who doesn't possess any specificity other than being a 'cultural worker.' We shouldn't miss the *specific* manifestations of design's liaison with capitalism. This oversight might be strategically valid from the point of view of 'praxis,' but it is less so when it comes to comprehend the designer's condition. Instead, the question we need to ask is: how does design differ from other activities entrenched in capitalism? Let's begin with design culture's default optimistic attitude.

THE UNCHANGEABLE

All the bad news won't affect design's standard stance. The fact that optimism is intrinsic to design emerges clearly in Herbert Simon's definition: "Everyone designs who devises courses of action aimed at changing existing situations into *preferred* ones."[81] More than focusing on the state of things, design pays attention to how things ought to be. And, of course, they ought to be better. So, from the point of view of design, pessimism appears vulgar and easy. Optimism, instead, is praised. But optimism can also be easy and vulgar, it can be mesmerised by abstract possibilities at the expense of a callous reality. Marco Petroni argues that "[t]oo often in the world of design the harshness of the *real* is accepted only when translated or rendered imaginative."[82] This translation, which is what we call a project, is also a way of projecting oneself outside reality.

Ultimately, design is terrified by what it is unable to change. It must systematically repress any manifestation of its impotence. The unchangeable is design's taboo. In this sense, it simply mirrors a broader condition which Italian philosopher Giorgio Agamben describes as follows:

81 Herbert A. Simon. *The Sciences of the Artificial*. Cambridge, MA: MIT Press, 2019, p. 111. The italics are mine.

82 Marco Petroni. *Il progetto del reale: il design che non torna alla normalità*. Milano: Postmedia Books, 2020, p. 10.

> Separated from his impotentiality, deprived of the experience of what he can not do, today's man believes himself capable of everything, and so he repeats his jovial 'no problem,' and his irresponsible 'I can do it,' precisely when he should instead realize that he has been consigned in unheard of measure to forces and processes over which he has lost all his control. He has become blind not to his capacities but to his incapacities, not to what he can do but to what he cannot, or can not, do.[83]

Agamben associates such positive, affirmative alienation, such perceived surplus of potential, to flexibility, instability and to the confusion between role and vocation. Along these lines, we can interpret design disillusion as a return of the repressed, the revenge of a reality that perturbs, with its constraints and limitations, the optimistic project of establishing the preferable.

A GENEALOGY OF DESIGN DISILLUSION?

Is it possible to assemble a genealogy of design disillusion? It's no easy task. Should one begin with the Radical Design movement of the '60s and '70s? Or with Le Corbusier, who in 1923 wrote that "[e]ngineers are healthy and virile, active and useful, moral and joyful. Architects are disenchanted and idle, boastful or morose. That is because they will soon have nothing to do."[84] Or with Charles Baudelaire, who prophesied an Americanisation brought about by "the mechanical?"[85] Instead of listing such expressions of disenchantment one by one, we can highlight instead certain recurring themes: mechanisation and rationalisation;

83 Agamben, *op. cit.*, p. 44.

84 Le Corbusier quoted in Beatriz Colomina and Mark Wigley. *Are We Human?: The Archaeology of Design*. Zürich: Lars Müller, 2016, p. 114.

85 Baudelaire cited in Cornelius Castoriadis. *Postscript on Insignificance: Dialogues with Cornelius Castoriadis*. London: Bloomsbury, 2011, p. 25-6.

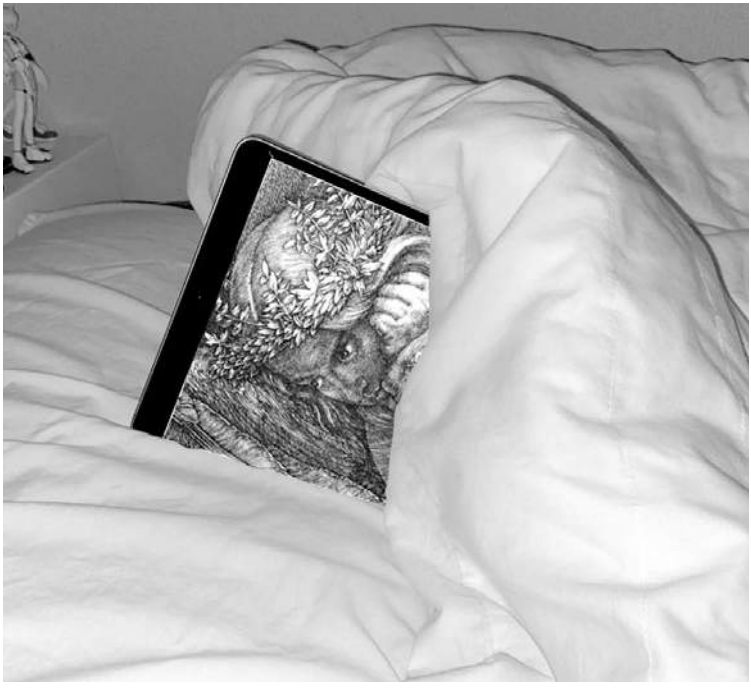
division of labour with its dulling effects; exploitation of natural and human resources; lack of meaning, autonomy and legitimacy.

These themes are often intertwined. This is the case for instance in John Ruskin's interpretation of the nature of Gothic architecture. Ruskin differentiated between "servile ornament" (typical of Greek and modern architecture) and "constitutional ornament" (typical of Gothic and medieval architecture). Servile ornament is about perfection. Here, the tasks assigned to tradespeople are so simple that they cannot be executed incorrectly, but they curb the labourers' inventiveness. On the other hand, constitutional ornament acknowledges a hierarchical relationship between the master and tradespeople, but allows the latter to invent, make mistakes and therefore be gratified with what they do. What Ruskin didn't like of servile ornament was basically the division of labour that such a style implied. This division of labour causes a 'mechanisation' of the labourers' work, so they lose autonomy, and therefore meaning in what they do. Nowadays, this sense of mechanisation feels bigger and more absurd: while there is room for imperfection, while the 'soul' seems to have found its place within work, no coherent whole emerges. There's no Gothic cathedral in sight. The 'whole' looks like a flawed socio-technical machine which is often jammed like an office printer.

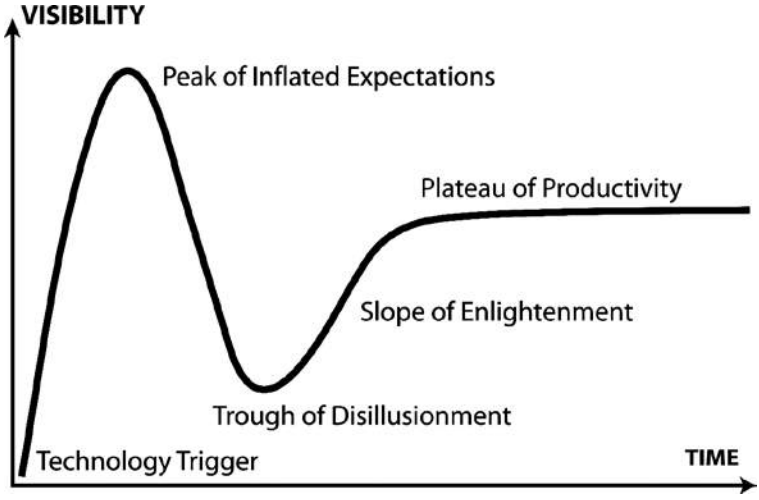
MELANCHOLY

A legendary engraving by Albrecht Dürer shows an allegory of melancholy, a 'humour' almost obsessively discussed in the philosophical treatises of the Renaissance.[86] The allegory shows a winged figure surrounded by an untidy accumulation of scientific equipment and tools of various trades: scales, an hourglass, a saw, some nails. Instead of employing them, the figure is cloistered, thoughtful, a cheek resting on a clenched fist, her sad eyes shining on a

86 Jacques Le Goff, ed. *L'uomo medievale*. Roma, Bari: Laterza, 2013, p. 305.



Melencolia MMXXIII by @young_agamben (2023).



Gartner Research's Hype Cycle diagram.

shadowed countenance.[87] These thoughtful eyes, staring at the distance, betray the realisation that whatever one could craft will always be imperfect. Some believe *Melencolia I* to be the spiritual portrait of Dürer himself:

> Despairing of the limits of human knowledge, she is paralyzed and unable to create, as the discarded and unused tools suggest. Ironically, this anguished representation of artistic impotence has proved a shining and enduring example of the power of Dürer's art.[88]

Melancholy is a disposition shared by scholars and artists. Dürer himself was divided between these two worlds, to the extent that his wife disapproved of his affinity with Italian humanists.[89] Dürer's engraving is a good representation of disillusioned everyday designers. Surrounded by a chaotic profusion of technological and methodological possibilities, they ponder their inability to achieve in their projects (their life being one of them) an ideal of perfection – or at least meaningfulness.[90] Too focused on thinking, they detach from a practice that has unrecognised limits and boundaries. As a result, practice itself becomes a process of introspection, a *recherche*. Christopher Lasch's theory of narcissism, summarised by Anthony Giddens, points towards something similar: "Exclusion of the majority from the arenas where the most consequential policies are forged and decisions taken forces a concentration upon the self; this

87 See Raymond Klibansky, Erwin Panofsky, and Fritz Saxl. *Saturn and Melancholy: Studies in the History of Natural Philosophy, Religion and Art*. Nendeln: Kraus Reprint, 1979, pp. 284-366.

88 The National Gallery of Washington. "Albrecht Dürer, Melencolia I, 1514." https://www.nga.gov/collection/highlights/durer-melencolia.html/.

89 Le Goff, *op. cit.*, p. 226.

90 "The idea behind Dürer's engraving, defined in terms of the history of types, might be that of Geometria surrendering to melancholy, or melancholy with a taste for geometry. [...] Geometria's workshop has changed from a cosmos of clearly ranged and purposefully employed tools into a chaos of unused things; their casual distribution reflects a psychological unconcern." Klibansky, Panofsky and Saxl, *op. cit.*, p. 317.

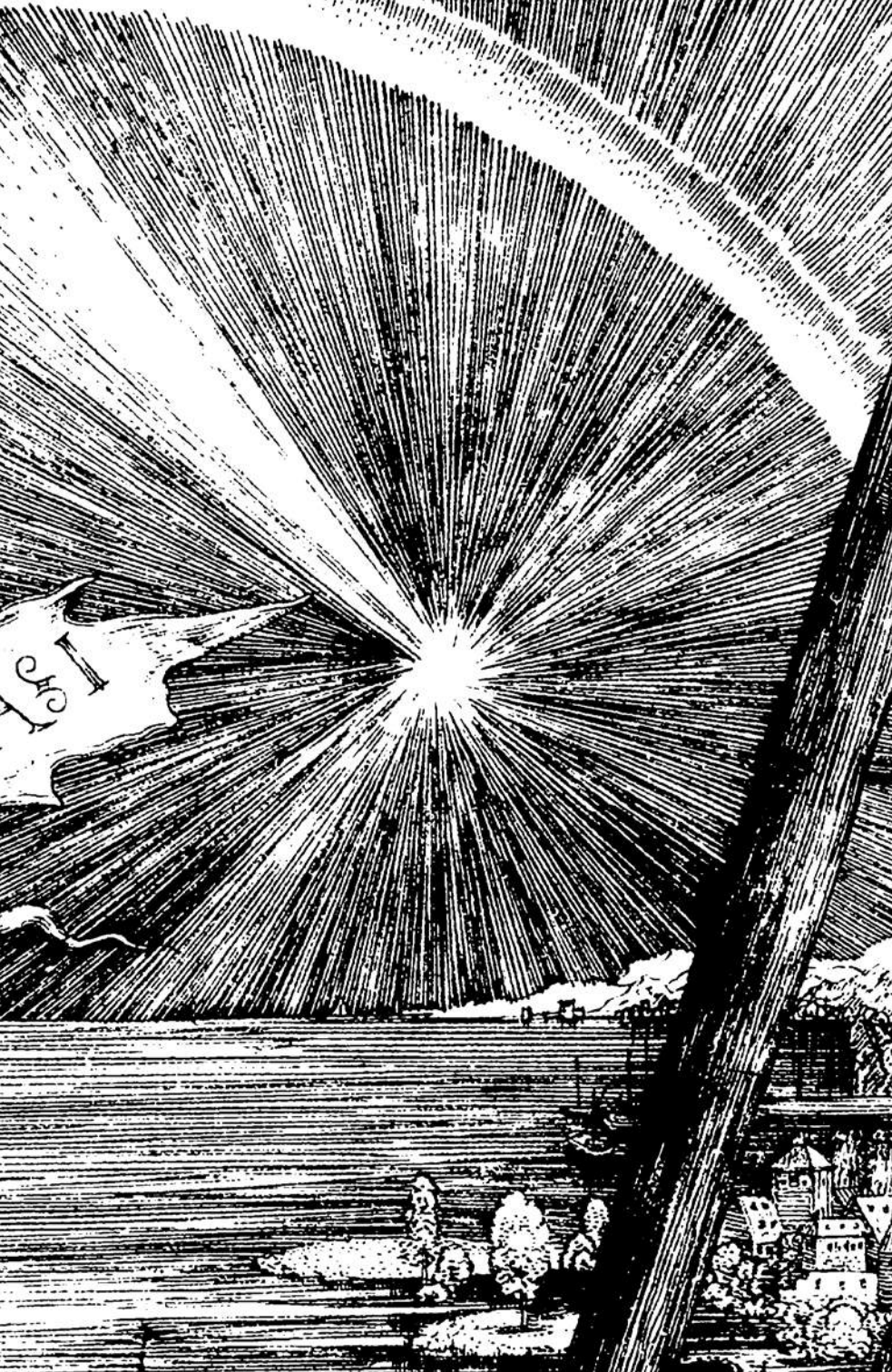
is a result of the powerlessness most people feel."[91] For designers, like for everyone else, the self becomes a refuge, but, more than everybody else, designers approach it as a project and a research area.

* * *

What Design Can't Do is divided in two parts. In the first part, entitled *Expectations,* I unpack the ambitions of design as a theoretical discipline, a field and a profession. In the second part, entitled *Reality,* I look at how these ambitions interact and clash with the activity of everyday designers as technologists, cultural professionals and students. In both parts I try to probe our extreme present by taking stock of an often obliterated past.

91 Anthony Giddens. *The Consequences of Modernity*. Cambridge, MA: Polity Press, 2015, p. 122-3.

Part I. Expectations



Chapter 2.

Everything Everyone All at Once: On Design Panism

"The secret ambition of design is to become invisible, to be taken up into the culture, absorbed into the background. The highest order of success in design is to achieve ubiquity, to become banal."
– Bruce Mau and Jennifer Leonard, 2004[92]

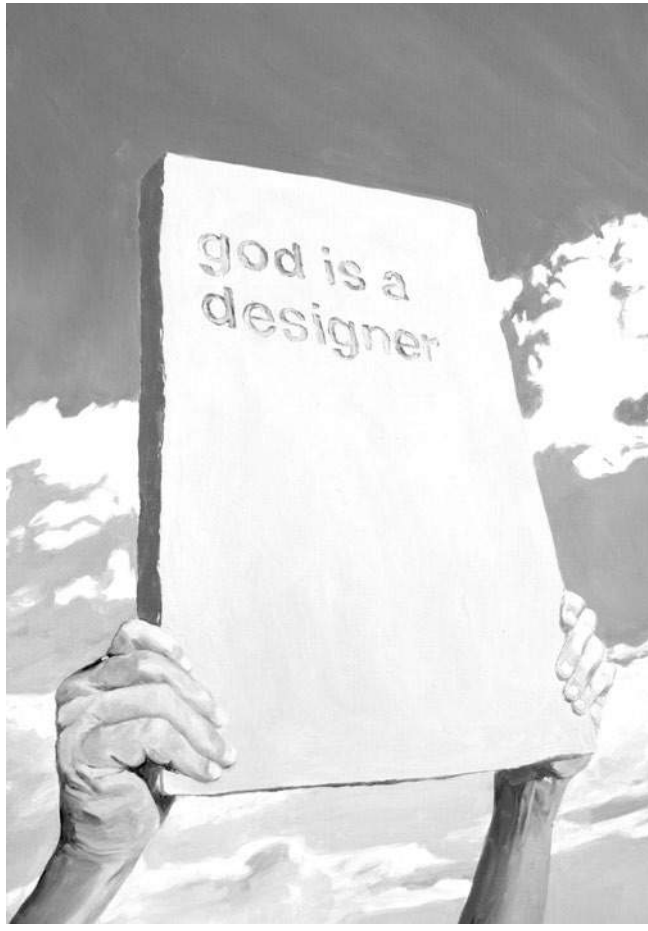
"Design has gone viral. The word *design* is everywhere. It pops up in every situation. It knows no limit."
– Beatriz Colomina and Mark Wigley, 2018[93]

ILUSIÓN

Design disillusion can go beyond lamentation, it can be more than chagrin, discontent, mere *disillusionment*. Hopefully, it can be a form of lucidity: one able to unveil a series of illusions, understood here not as false beliefs but as idiosyncratic, dysfunctional, or even contradictory notions. The Spanish term *ilusión* can refer to a reasonable hope and even joy! Design illusions include those fits of collective enthusiasm that influence and shape the design field. How did such illusions come about? How do they structure the minds of designers, their perception of the world and other people, as well as their relationship with them? Some of them seem now hard-coded into the design discipline – they've become platitudes. While others, maybe less explicit, blend with broader social configurations. In this part of the book, I will consider the illusions concerning the very essence of design; those that affect the way in which design positions itself among other domains; and, finally, those pertaining to design's

92 Bruce Mau, Jennifer Leonard, and Institute without Boundaries. *Massive Change*. London, New York: Phaidon, 2004, p. 4.
93 Colomina and Wigley, *op. cit.*, p. 46.

Mitch Goldstein (2022).



God is a Designer by Alix Gallet (2020), oil on canvas, 60x92cm.

capacity to act in the world. The first kind allows us to reconsider what design is, the second how design relates to other disciplines, the third what design can do. In three words: being, autonomy, power.

DESIGN HAUNTS

Two oft-heard assertions inform today's understanding of design: 'everyone is a designer' and 'everything is design.'[94] This all-embracing view of design can be named *design panism.* It is within the realm of design panism that designers (struggle to) articulate their role and position. But more than being a mere descriptor of a reality, design panism is an interpretative framework, a rhetorical instrument and a semi-conscious expansion agenda. Breaking down the conflicting meanings of these two common statements allows us to discern their consequences on the designer's identity and sense of realisation.

In a not-so-distant past, delimiting design was easy: whereas the mass-produced Moka pot hissing on the stove was unmistakeably design, the handmade ceramic cups used to drink its coffee weren't. Design began with the factory and ended with it: it was *industrial* design. Nowadays, things are more difficult: design escaped the assembly line to become a mentality, which is to say that our mentality has become industrialised.[95] The few handmade objects still surrounding us owe their aura to design. Design *haunts* them.

Whatever we make or encounter – a thing, an arrangement of things, a procedure to arrange such things – is haunted by design. Even if design is not there, it is already there. Thus, the ontological status of design is a ghostly one. As Bruce Mau and Jennifer Leonard put it, "for most of

94 Similar statements can be found in the field of art (Joseph Beuys's "everyone is an artist") and architecture (Hans Hollein's "*Alles Ist Architektur*"). For the latter see Hans Hollein. "Alles Ist Architektur." *Bau*, 1968. https://socks-studio.com/2013/08/13/hans-holleins-alles-ist-architektur-1968/.

95 Presumably, this industrialised mindset started to emerge in the West in the 16th century, with the appearance of the first mechanically reproduced books.

us, design is invisible. Until it fails."[96] But design is a peculiar ghost, one that craves the tangible world of the living. Design is like Slimer, the gluttonous spectre from *Ghostbusters*: one that enjoys ingurgitating as much food as it can.

Slimer's interactions with the material world are not seamless: they leave slime wherever they go. We can interpret such gelatinous secretions as design's ability to reconceptualise things within its mode of comprehension: suddenly, a pebble starts having a form and a function, it gets liable to a process of improvement, a process that is itself subject to a method. What has happened? Design has digested the thing: the pebble is now an *artefact*, a designed object.[97] This is the ultimate phase of the "extraordinary career" of the term 'design' noticed by Bruno Latour, a term that grew in "extension" (the typologies of products it is applied to) and "comprehension" (the parts of a thing design can comprehend):

> Today everyone with an iPhone knows that it would be absurd to distinguish what has been designed from what has been planned, calculated, arrayed, arranged, packed, packaged, defined, projected, tinkered, written down in code, disposed of and so on. From now on, "to design" could mean equally any or all of those verbs.[98]

But design's metabolism has gone further and left its muculent mark on what it couldn't process, namely, the thing's symbolic and ritual aura, its culture; substituted by design culture, with its own equalising symbols and rituals. The thing is apparently the same, but it is in fact completely different.

96 Mau et al., *op. cit.*, p. 2.

97 "The stone becomes an object only when it is given the function of a paperweight." Abraham A. Moles. "Objet et communication." *Communications* 13, no. 1 (1969): 1–21, p. 5.

98 Bruno Latour. "A Cautious Prometheus? A Few Steps Toward a Philosophy of Design (with Special Attention to Peter Sloterdijk)." Proceedings of the Annual International Conference of the Design History Society, edited by Fiona Hackne, Jonathan Glynne, and Viv Minto, 2–10. Falmouth: Universal Publishers, 2009.

ORANGES, PEAS AND ROSES

In 1963, Italian design polymath Bruno Munari amused himself by describing an orange, peas and a rose as industrial objects. Whereas the orange is "an almost perfect object," the rose is deemed completely useless and complicated. Munari's innocent divertissement, a sort of lesson in design thinking and perhaps a subtle critique of mass production, exemplifies the *actual* way in which design comes to reinterpret and thus change reality. Once such reinterpretation has happened it is very hard to think reality otherwise, that is, beyond functionality and efficiency.[99]

With such a mindset in place, the designer might well be in charge of everything. According to design curator Paola Antonelli, designers are "respectful, curious, generous, and hungry for other fields' bodies of knowledge and expertise, designers invade without colonizing. Who can we trust more? They should run the world."[100] Ruha Benjamin might disagree. Asked, during a workshop, to offer a definition of design, the sociologist suggested that "design is a colonizing project."[101] By that she meant that design is used to describe everything. *Description* is indeed the form that design's slimy digestion takes.

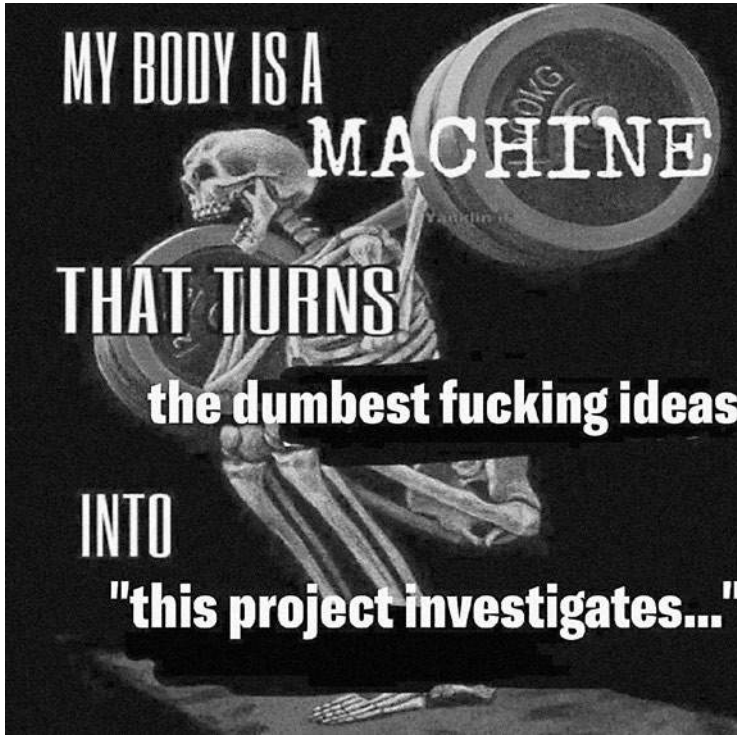
As Paul Rodgers and Craig Bremner declare, "design is neither a product nor a service. Design occurs in relationship to everyone and everything – it describes and shapes relationships."[102] Design proceeds through formalisation. Focusing on one of today's most successful design currents, design thinking, Benjamin points out its capacity to

99 Peas are described as "food pills of various diameters, packed in double valve cases, very elegant in form, colour, material, semi-transparent and easy to open." Munari, Bruno. 2010. *Good design*. Mantova: Corraini.

100 Paola Antonelli "Foreword." In Alexandra Midal. *Design by Accident: For a New History of Design*. Berlin: Sternberg Press, 2019.

101 Ruha Benjamin. *Race After Technology: Abolitionist Tools for the New Jim Code*. Cambridge, UK: Polity, 2020, p. 176.

102 Paul A. Rodgers and Craig Bremner. "The Design of Nothing: A Working Philosophy." In *Advancements in the Philosophy of Design*, edited by Pieter E. Vermaas and Stéphane Vial, 549–64. Design Research Foundations. Cham: Springer International Publishing, 2018. https://doi.org/10.1007/978-3-319-73302-9_25, p. 550.



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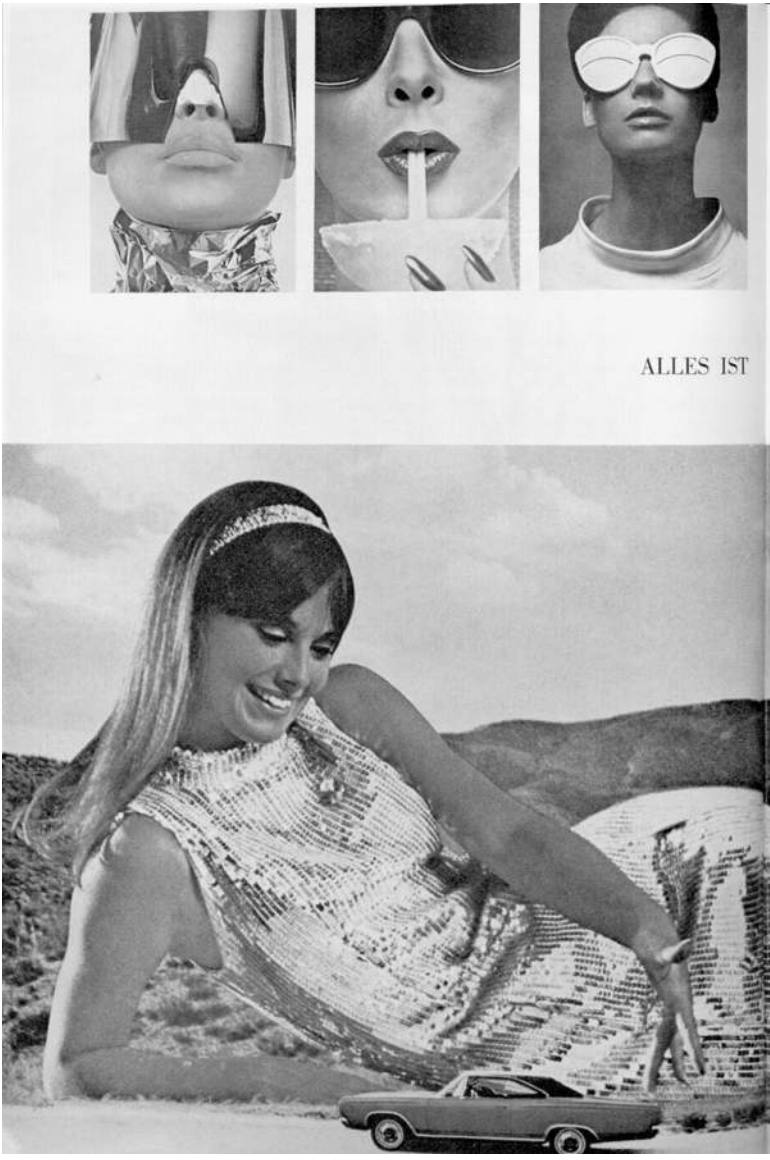
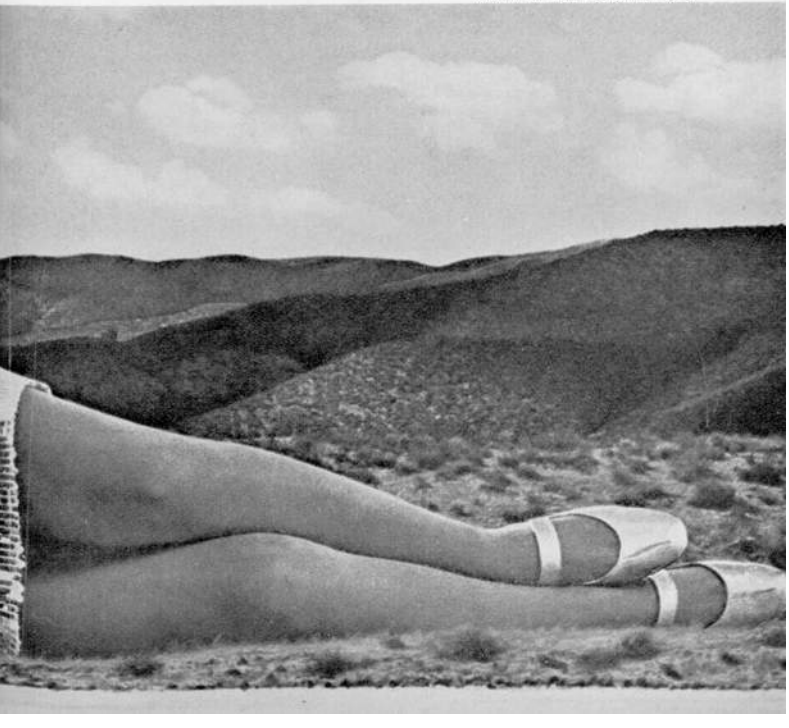


Spread from Hans Hollein's "Alles Ist Architektur," *Bau*, 1/2 1968.

ARCHITEKTUR

Aus einem Inserat der B. F. Goodrich Company.



encapsulate any form of activity, from the organisation of a protest to the user journey of a banking app (an uncommon etymology of the word 'design' links it to the Latin *designare*, which also means delimiting[103]). For Benjamin, the problem is that design thinking is forgetful, not unlike design in general: it neglects histories.[104]

> If one needs to "subvert" design, this implies that a dominant framework of design reigns – and I think one of the reasons why it *reigns* is that it has managed to fold any and everything under its agile wings.

Benjamin identifies a series of risks associated with design thinking, an "umbrella philosophy" that diminishes broader forms of human activity, erasing the genealogies from which they emerged in the first place, effacing tradition. It's a matter of hegemony: "Whether design-speak sets out to colonize human activity, it is enacting a monopoly over creative thought and praxis." Benjamin's concerns are specifically related to racial issues. From this vantage point she is able to see the way in which design depletes empowerment: "Maybe what we must demand is not liberatory *designs* but just plain old liberation. *Too retro*, perhaps?"[105]

THE LAST AVANTGARDE

A sense of disillusion partly derives from the realisation that design can be, like money, a general equivalent: something that severs the links between things and thus estranges them. It can devour contexts. Practitioners suspect that the umbrella is too small, that design is lacking the conceptual and practical means of encompassing

103 Tomás Maldonado. *Disegno industriale: un riesame*. Milano: Feltrinelli, 2008, p. 9.

104 Susan Stewart puts it in more systematic terms: "the excision of history from design thinking isolates the understanding that informs the design act from any understanding of the temporal trajectories in which it participates." "And So to Another Setting...". In *Design and the Question of History*, edited by Tony Fry and Clive Dilnot, 275–301, 2020. https://doi.org/10.5040/9781474245890.

105 Ruha Benjamin, *op. cit.*, pp. 174-180.

human activity, and that, by attempting to do so, it is actually making *tabula rasa*.[106] Design might be the last successful avantgarde: a deliberate repudiation of histories. Designers who come to terms with such awareness logically develop an impostor syndrome, an urge to resist design's reductive assimilation.

This can be the joint cause of many personal exoduses, dreamed of or executed: former designers decide to engage with the fullness of a certain human activity within its specific and historically-rich domain: farming, writing, cooking... all activities that resist reduction to "rural hacking," "content design," or "food design." More rarely, however, the act of design description provides an enrichment: the selectivity of design allows for the inclusion of forgotten voices, for the light-hearted reshuffling of austere practices, for a novel bridging of contexts. In these rare cases, design's 'ignorance' truly becomes its bliss.[107] At worst, design flattens a multiplicity of worlds into a one-dimensional, aseptic one; at best it nurtures them.[108]

It's hard to deny that "the logic of the world is the logic of

106 If we are to consider 'progressive' policy making as a form of design, we recognise a similar impetus for erasure, in brutally explicit terms: "There is a sense in which rapid economic progress is impossible without painful adjustments. Ancient philosophies have to be scrapped; old social institutions have to disintegrate; bonds of caste, creed, and race have to burst; and large numbers of persons who cannot keep up with progress have to have their expectations of a comfortable life frustrated." United Nations, Department of Social and Economic Affairs. "Measures for the Economic Development of Under-Developed Countries," 1951. http://digitallibrary.un.org/record/708544.

107 See chapter 3.

108 The field of creative coding provides a good example of non-flattening design. Processing, one of the main programming languages deliberately conceived with artists and designers in mind, is part of a rich history of experiences where design bridges diverse fields of knowledge. Emerging from the MIT 'lab' culture, in which Muriel Cooper, a graphic designer, had a leading role, Processing gave rise to a broad community of makers and thinkers who go beyond the drive towards efficiency of much computation culture. See Golan Levin and Tega Brain. *Code As Creative Medium: A Handbook for Computational Art and Design*. Cambridge, MA: MIT Press, 2021. For a counterargument based on Processing's blindness to its own materiality cf. Michael Murtaugh. "Torn at the Seams: Considering Computational Vernacular." In *Vernaculars Come to Matter*, edited by Cristina Cochior, Sofia Boshat-Thorez, and Manetta Berends, 93–110. Rotterdam: Everyday Technology Press, 2021.

the description of the world."[109] If so, the describing entity is inevitably revealed in the description, together with its limits. This is how the formalisation that design provides betrays design's own amorphousness: "design today feels like a vast formless body – or in the parlance of contemporary architecture, a 'blob' – able to absorb any blows delivered to it – lacking coherency and increasingly dispersed," observed Andrew Blauvelt in 2003.[110] This doesn't seem to be just an isolated impression: when a group of researchers employed data mining techniques to examine two decades' worth of conversations on the PhD-Design discussion list, an established platform for specialised exchange, they discovered that there had been minimal advancement in defining design.[111] Today, the doubtful designer asks: how can something so indefinite *define* the world? Recognising the reductive appropriation of pre-existing practices through design as a bureaucratic cultural instrument, they nevertheless cling to their authority, real or perceived, sensing that it is ineffective, provisional – if not illegitimate.

BUSY BEE

As we learn that design is everywhere (and thus nowhere), we are also convinced that everyone is a designer. As an activity, design is often placed, so to speak, in the mind of the beholder, framed as an integral component of human capabilities. Design is presented as a "universal human life-skill,"[112] it is "all that we do, almost all the time,"[113] until the whole human race is characterised as a

109 Francisco Varela. "Introduction." In Heinz von Foerster. *Observing Systems*. Systems Inquiry Series. Seaside CA: Intersystems Publications, 1984, p. XVI.

110 Andrew Blauvelt. "Towards Critical Autonomy or Can Graphic Design Save Itself?". *Emigre*, 2004.

111 Alethea Blackler, Levi Swann, Marianella Chamorro-Koc, Wathsala Anupama Mohotti, Thirunavukarasu Balasubramaniam and Richi Nayak. "Can We Define Design? Analyzing Twenty Years of Debate on a Large Email Discussion List." *She Ji: The Journal of Design, Economics, and Innovation* 7, no. 1 (March 1, 2021): 41–70. https://doi.org/10.1016/j.sheji.2020.11.004.

112 Rick Poynor. "First Things Next." In *Obey the Giant: Life in the Image World*. Boston, MA: Birkhauser Verlag AG, 2007, pp. 141-2.

113 Papanek, *op. cit.*, p. 3.

"designing species."[114] The list could continue endlessly.[115] In fact, "everyone is a designer" is a truism so common, both in and out of the design world, as to cause suspicion. Is this a marketing ploy to sell design books to a larger public?

To understand more clearly what this life-skill is about, we can consult an unlikely source, Karl Marx's *Das Kapital*. In the first volume, he famously compared an architect to a bee:

> A spider conducts operations that resemble those of a weaver, and a bee puts to shame many an architect in the construction of her cells. But what distinguishes the worst architect from the best of bees is this, that the architect raises his structure in imagination before he erects it in reality. At the end of every labour-process, we get a result that already existed in the imagination of the labourer at its commencement.[116]

Here, we gather that design is the exclusively human capacity to *prefigure*,[117] to imagine a structure before it is built.[118] Nobody can deny that humans have this imaginative ability, but is there any point lumping it together with a historically specific practice consolidated during the Industrial Revolution? In this process, not only is design (our contemporary, Western idea of design)

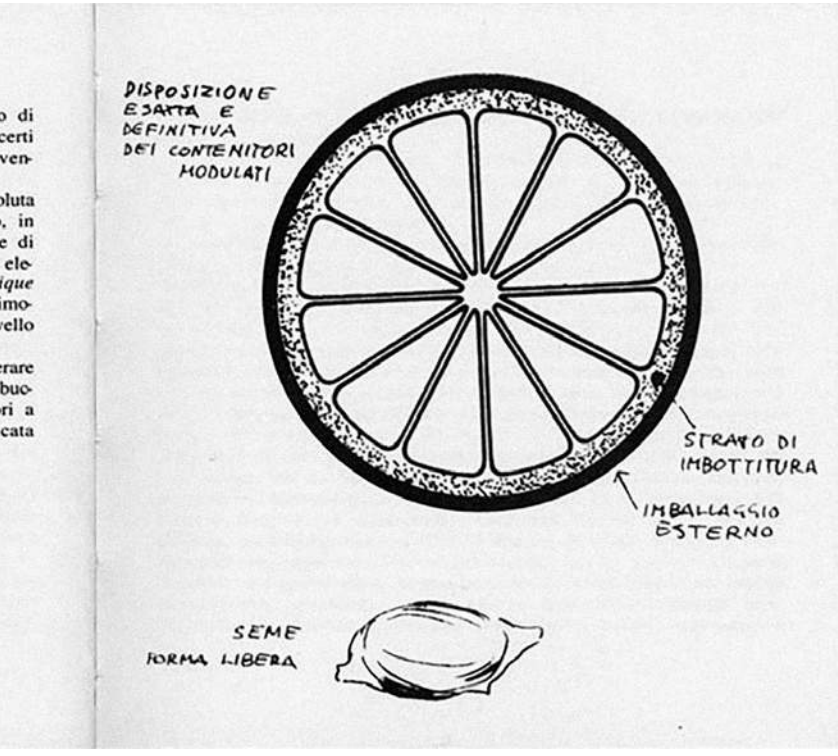
114 Victor Margolin interviewed by Max Bruinsma in 2015, https://vimeo.com/133073827.

115 On my blog, I assembled a chronology of "design panism", from 1962 to 2022, later collated in a zine made together with the students of the Information Design Department at Design Academy Eindhoven: https://networkcultures.org/entreprecariat/design-panism-a-timeline/.

116 Karl Marx. *Capital*. Vol. 1. Marx/Engels Internet Archive, 1999. https://www.marxists.org/archive/marx/works/1867-c1/index.htm.

117 Of course, we find earlier examples of this idea, for instance in the concept of *disegno interno* (inner design) of the Italian architect and painter Federico Zuccari. See *L'Idea de' pittori, scultori ed architetti*, 1607.

118 It seems, however, that animals like rats are also capable of prefiguration, imagining alternative paths before they take them. So, prefiguration might not be the fundamental difference between humans and other animals, but our capacity/necessity to shape our environment and externalise survival functions to it. See Thomas Hills. "Can Animals Imagine Things That Have Never Happened?" *Psychology Today*, October 22, 2019. https://www.psychologytoday.com/intl/blog/statistical-life/201910/can-animals-imagine-things-have-never-happened.



Spread from the book *Good Design* by Bruno Munari (1963).

naturalised, but also, as Beatriz Colomina and Mark Wigley remind us, the human reinvented.[119] In their writings, many architects and designers go back to the genesis of an archetypal artefact (a spear, a bridge, a nook) to justify their particular design philosophy, not unlike startup founders with their origin stories. A move that can be considered "an attempt to gain status for the profession by evoking a specious historical precedent," warns Papanek. Already in the '70s, he made the distinction clear: "In the beginning was Design, of course, but not industrial design."[120]

DESIGNED BY DESIGN

Proponents of an ontological idea of design like Tony Fry and Anne-Marie Willis add another element to design panism by suggesting that not just everything is design, and not only everyone is a designer, but also everyone is in turn *designed* by design, a point resonating with the formulation of Marshall McLuhan's friend Father John M. Culkin: "We become what we behold. We shape our tools and then our tools shape us."[121] In their view, design is a double movement: from the designer to the environment, but also, and more importantly, from the environment to the designer. Design is thus circular and bidirectional, one might say. Willis openly disassociates her idea of "ontological designing" from other forms of design panism, which can be found in "populist texts" such as Papanek's.[122] A lowbrow, demagogic idea of design panism corresponds to the perhaps wishful claim that anyone can, at least partially, design their own environment. On the contrary, highbrow design panism sees design as a sort of force field of intentionality investing both human

119 Colomina and Wigley, *op. cit.*

120 Papanek, *op. cit.*, p. 29.

121 To be fair, this aphorism should be formulated inversely: first, we are shaped by the designed world and then, perhaps, we design it back.

122 Anne-Marie Willis. "Ontological Designing." *Design Philosophy Papers* 4, no. 2 (June 1, 2006): 69–92. https://doi.org/10.2752/144871306X13966268131514.

and non-human agents. Here, 'everyone is a designer' and 'everything is design' are the same statement.

Both Willis' vertiginous conception of ontological designing and the apparently common-sense distinctions of Papanek and Margolin have their limits. The circularity of ontological designing clashes with the necessarily heuristic quality of design in practice, one that is based on a shared operational consensus: the belief that the agency lies in the designer, who is a synecdoche of the human. It is through this functional reduction that design operates, even when it supervises processes. Design intervenes by eschewing a full understanding of the world, in order to optimise its transformation. For a tree to grow healthily, it needs to be pruned. The alternative is apophenia: panic derived from the sense of total interrelation. The fact that Willis complains about designers' ability to comprehend ontological ramifications is, therefore, hardly surprising.

Anthropologist Lucy Suchman is careful when it comes to adopting the word 'design' in a broad sense. She suggests keeping "an eye to the tensions inherent in articulating projects in transformational change as 'small d' design, without reproducing the supremacy of Design with that initial capital letter." What is this supremacy about? She identifies four historical tendencies in the professional field of design: grandiosity, which is another word for universalism; progressivism, a standard development pattern "from them, to us, to a new us;" parochialism, that is, "the tendency to engage in conversations with each other, on behalf of everyone;" the predominance of ethics and values over politics.[123]

AUTHORITARIAN AND ANNOYING

Often, however, useful distinctions get lost in the fumes of design discourse. The result: a general idea of design

123 Lucy Suchman. "Keywords for Ethnography and Design: Design." *Society for Cultural Anthropology*, March 29, 2018. https://culanth.org/fieldsights/design.

activity where everything we do or is done to us is design. At this point, one could ask for specificity through precise definitions. As we have seen though, this doesn't work. So, let's keep it general to get a better understanding of the dysfunctions of design panism. If design is a universal human feature, why are there humans who call themselves designers? Why is the prefigurative function delegated to a specialised congregation of people? And what is then left to the others? The matter is in fact not new to debate. In 1994, Ivan Illich and Carl Mitcham gave a sermon against design:

> Contra the widely promoted belief that design is something all human beings do and have done throughout history, but now must do more consciously and thoroughly than ever before, design is something that has had a history. Its beginnings can be traced to the rise of modernity, and it will almost certainly come to an end with the modern project. Indeed, we have an obligation not so much to promote designing as to learn to live without it, to resist its seductions, and to turn away from its pervasive and corrupting influence.[124]

In 2009, Italian mass culture theorist and cyberpunk expert Antonio Caronia expressed his hostility to design:

> Personally, I would like to outlaw design. Design is one of the most authoritarian and annoying practices that modernity has created. Design is one of the most striking consequences at the level of properly human activities – which are activities of creation [and] have been killed by the separation between the activities of planning and realization. This is design.[125]

124 Carl Mitcham. "In Memoriam Ivan Illich: Critic of Professionalized Design." *Design Issues* 19, no. 4 (October 1, 2003): 26–30. https://doi.org/10.1162/074793603322545037, p. 29.

125 Caronia quoted in Salvatore Iaconesi and Oriana Persico. *Angel_F. Diario Di Una*

In theory, the question of design expertise could be put in very simple terms: the existence of design experts doesn't prevent non-designers to exercise the broadly human activity of designing. Norman Potter puts that succinctly: "Every human being is a designer. Many also earn their living by design [...]."[126] However, the self-evident recognition that there are, in every field, experts and non-experts, people who are paid for their expertise and people who aren't, conceals a fundamental predicament of modern society, that of *being run* by expertise. Most facets of everyday life – not just work – are organised, scrutinised and assessed by what US sociologist Anthony Giddens calls "expert systems."[127] Those systems prefigure and configure. Here, the dominance of prescriptive technologies fosters a culture of compliance.[128] Individual autonomy is thus reduced, a problem that the very Norman Potter, explicit in his leftist libertarianism, was very conscious of.

More balanced in his judgement, Gui Bonsiepe admits the necessity of expertise while, at the same time, warning against all-encompassing claims on designing:

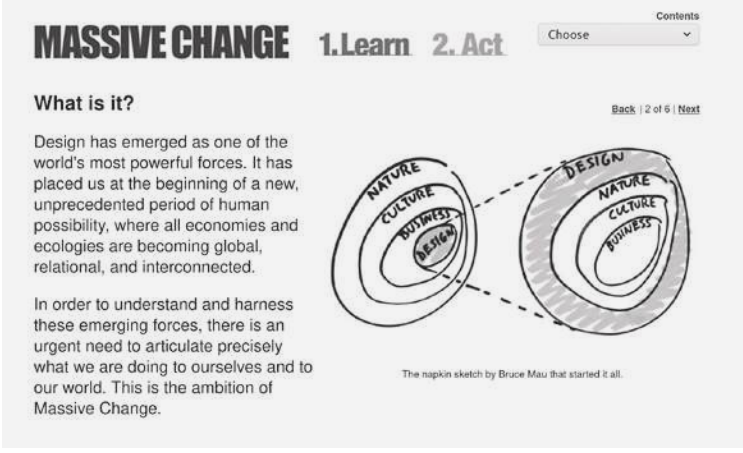
> There is a risk of falling into the trap of vague generalizations like "everything is design." Not everything is design, and not everyone is a designer. The term 'design' does refer to a potential to which everyone has access and which is manifest in everyday life in the invention of new social practices. Every one can become a designer in his special field, but the field that is the object of design activity always has to be

Intelligenza Artificiale. Roma: Castelvecchi, 2009, p. 276.

126 The extended quote: "Every human being is a designer. Many also earn their living by design – in every field that warrants pause, and careful consideration, between the conceiving of an action and a fashioning of the means to carry it out, and an estimation of its effects." Potter, *op. cit.*, p. 10.

127 "Systems of technical accomplishment or professional expertise that organise large areas of the material and social environments in which we live today." Anthony Giddens. *The Consequences of Modernity*. Cambridge, UK: Polity Press, 2015, p. 27.

128 Franklin, *op. cit*, pp. 18-9.



Screenshot of the massivechange.org website, including Bruce Mau's napkin sketch in which design becomes all-encompassing.

> identified [...] Design is a basic activity whose capillary ramifications penetrate every human activity. No occupation or profession can claim a monopoly on it.[129]

This idea is reminiscent of Antonio Gramsci's popular assertion that "all men are intellectuals, but not all men have in society the function of intellectuals." Here, the emphasis might seem on "all men," but it is actually on the intellectual's function. In fact, elsewhere Gramsci argues that

> The intellectual is a 'professional,' a skilled worker who knows how his own specialized 'machines' function. He has an 'apprenticeship' and a 'Taylor system' of his own. It is childish and illusory to attribute to everyone this acquired and not innate ability, just as it would be childish to believe that any unskilled worker can drive a train.[130]

"EVERYONE CAN DESIGN, EVEN DESIGNERS"

Whereas Illich and Mitcham were issuing a warning about the pervasiveness of design, critics like Caronia lamented the asymmetry that it creates between planners and executors. Today, such asymmetry appears to have evened up as design systems no longer offer just a singular product but an array of options. Intuitive graphic tools, customisation features, design methods, a do-it-yourself ethos... they all contribute to this shift.[131] The emergence of such design sandboxes could be understood as a democratisation of the ability to prefigure: designers design systems for non-designers to be able to design. This is the

129 Gui Bonsiepe quoted in Fernando Secomandi and Dirk Snelders. "Interface Design in Services: A Postphenomenological Approach." *Design Issues* 29, no. 1 (January 1, 2013): 3–13. https://doi.org/10.1162/DESI_a_00192,

130 Antonio Gramsci. *The Antonio Gramsci Reader: Selected Writings, 1916-1935*. Edited by David Forgacs. New York: New York University Press, 2000, pp. 389-90.

131 An example among many: IKEA distributes prefigurative ability offering online planning tools to its clients. See https://www.ikea.com/nl/en/planners/.

meaning of *Everyone is a Designer!*, Mieke Gerritzen and Geert Lovink's prosumerist manifesto written in 2000 and updated in 2010.[132] But while the designing ability, albeit limited, trickles down, a sense of powerless entrapment *trickles up*: designers, who are almost never in charge of the systems' system, lose their imagined *raison d'être*: their prefigurative know-how dissolves into the execution of someone or something else's high-level prefiguration: a manager, an engineer, an algorithm. With prefiguration sandboxes readily available, designers begin to perceive their relatively humble position.

'Everything is design' and 'everyone is a designer,' these ubiquitous truisms about the ubiquitousness of design, reflect the fact that designing agency has spilled out of the designer figure in various ways, to the point that common hierarchies of expertise can be overturned, as Cuban artist and designer Ernesto Oroza does when he quips that "everyone can design, even designers."[133] Design expands, while the designer shrinks. Designing capabilities are no longer just a feature of individuals but also of collective subjects like organisations.[134] When design starts to be conceived in terms of agency, it is reasonable to wonder whether it can be an impersonal agent in (or even for) itself. Design is then conceived as a force that surrounds designers, more than one that they exert: design designs them like it designs everyone else. More than being design's main subjects, designers are subjected to design: the designer becomes a condition more than a role. What kind of force is design? Tomás Maldonado speaks of a productive one: "Like all design activities that, in one way or another, intervene in the production-consumption

132 Mieke Gerritzen and Geert Lovink. *Everyone Is a Designer in the Age of Social Media*. Amsterdam: BIS, 2010 and *Everyone Is a Designer! Manifest for the Design Economy*. Amsterdam: BIS, 2001.

133 This is the title of a research unit led by Oroza within the Cité du design initiative, https://www.citedudesign.com/fr/a/everyone-can-design-even-designers-778.

134 "In a world in rapid and profound transformation, we are all designers. Here, 'all' obviously includes all of us, individuals but also organizations, businesses, public entities, voluntary associations, and cities, regions, and states." Manzini, *op. cit.*, p. 1.

relationship, industrial design acts as a true productive force. What's more, it is a productive force that contributes to the organization (and hence socialization) of the other productive forces with which it comes into contact."[135] In this sense, design is not just a dependent part of other, larger systems, but a system in itself, able to interact with other ones, affecting them.

The Massive Change program, initiated in 2004 by renowned architect and designer Bruce Mau with the Institute without Boundaries, shows the ambiguous relationship between design as a productive force and the designer. Mau deliberately speaks of design as "one of the world's most powerful forces." A 'napkin sketch' made in the back of a cab postulates that design is no longer a mere subset of business but the broadest superset, one that includes nature, culture and business. Far from being intimidated by this realisation, he places a generic "we" (designers? policy makers? humanity as a whole?) in charge of the power of design, in order to "minimize unintended consequences and maximize positive outcomes." The project shows an optimism, at times cautious but more often untamed, about the (broadly understood) designer's ability to harness the force of design: "We have an unprecedented capacity to plan and produce desired outcomes through good design." Before unleashing wildly cheerful future predictions, the book resulting from the project, a kaleidoscope of technical and scientific innovations seen from a globalist view, introduces a sense of foreboding reminiscent of the work of Paul Virilio: "Accidents, disasters, crises. When systems fail we become temporarily conscious of the extraordinary force of design, and the effects it generates."[136] The optimism of Mau lies in the belief that disasters happen only when design fails, and not when – and this is becoming increasingly clear – design acts as intended. Design

135 Maldonado, *Disegno industriale: un riesame*, *op. cit.*, p. 14.
136 Mau et al., *op. cit.*, p. 10.

disillusion reacts instead to the fact that design, one of the main features of modernity, is not akin to a "carefully controlled and well-driven motor car," but to a juggernaut.[137]

* * *

Design panism is plural. 'Everything is design' doesn't only imply that everything in our environment is artificial, but also that our very gaze has been artificialised. There is no nature for us, only culture. What culture, specifically? A reductive one that erases histories, or a respectful one that welcomes careful innovation? The game is still on, but the first option is broadly felt and generates impostor syndrome. Affirming that everyone is a designer can mean many things. Design panism can be a matter-of-fact recognition that humans have an innate prefigurative ability. It can also be driven by a professionalist propaganda which characterises designers as *the* experts of such general ability. Furthermore, design panism can be used to justify a particular design philosophy. But it can also lead to an admission of the cultural limitations of design as a mode of comprehension. Design panism may be in line with a demand for autonomy against the prescriptive and bureaucratic structure of modern life, as it is run by experts who detain, in various fields, the monopoly on planning and implementing. What level of prefiguration should a designer aim for? Who should be in charge of design as an impersonal force? These remain open questions.

137 Giddens, *op. cit.*, p. 53.

Chapter 3.

A Complex Relationship: On Synthesis and Autonomy

"Any discipline or generated system for the organization of reality faces the problem of having to exceed the scope of its object of inquiry, but since it, too, must be part of that object (if it is to be something as grand as reality), it must contain itself in a logical relationship to all it is trying to contain, which expands the initial problem of inclusion. There is, in other words, always *more* to and of reality."
– Lewis Gordon, 2011[138]

"As a generalist, [the designer] has made a specialty out of his lack of specialization."
– Gui Bonsiepe, 1974[139]

Design's ubiquity makes it hard to pinpoint. I'm reminded of a famous definition of deity offered by theologian Alain de Lille, a definition which almost sounds like a design brief: "God is an infinite sphere, whose centre is everywhere and whose circumference is nowhere."[140] With an indeterminate centre, design can easily reveal itself in between, and link, other disciplines and cultural domains. For instance, Walter Gropius, founder of the Bauhaus, understood design as "a new unity" of art and technics.[141] While they remain two distinct realms, design emerges amid them, producing a novel understanding of both.

As design becomes a cultural phenomenon under constant restructuring, it aspires, more and more consciously, to be the whole that exceeds the sum of its parts. That is, the very relationship between them. As design events

138 Lewis R. Gordon. "Shifting the Geography of Reason in an Age of Disciplinary Decadence." *Transmodernity* 1, no. 2 (2011). https://doi.org/10.5070/T412011810.
139 Gui Bonsiepe. "Design e Sottosviluppo." *Casabella*, 1974, p. 43.
140 Paolo Lucentini, ed. *Il libro dei ventiquattro filosofi*. Milano: Adelphi, 1999.
141 Herbert Bayer, Ilse Gropius and Walter Gropius, eds. *Bauhaus 1919-1928*. New York: The Museum of Modern Art, 1938. https://www.moma.org/documents/moma_catalogue_2735_300190238.pdf, p. 82.

show, any domain, any practice can be plugged into it: not just art and technics but also journalism, craftsmanship, obscure traditions, witchcraft... We can legitimately talk of design hyperconnectivity. Distinct phenomena appear to the design mindset as an assortment of equivalent nodes, with design itself acting as a *paranode,* "the conceptual space that lies beyond the borders of the node."[142]

BOTH FISH AND WATER

What are the consequences of hyperconnectivity on how designers view themselves? In 2006, Gui Bonsiepe called for an integration of the design perspective into higher education, lamenting the predilection for discursive results over projects, and insisting on the necessity for novel institutions:

> So far, design has tried to build bridges to the domain of the sciences, but not vice versa. We can speculate that, in the future, design may become a basic discipline for all scientific areas. But this Copernican turn in the university system might take generations, if not centuries. Only the creation of radically new universities can shorten this process.[143]

Here, Bonsiepe describes design's reasonable but not entirely successful attempt to build a bridge to another field. Victor Papanek had a more ambitious idea. In his seminal book *Design for the Real World,* he advocated a cross-disciplinary team where the designer acts as "the bridge between the disciplines." Notably, he saw the designer as a *synthesist*, a producer of wholes, one might say. For him, designers would have the epistemic last word, that is, the problem's formulation.[144]

142 Ulises A. Mejias. "The Limits of Networks as Models for Organizing the Social." *New Media & Society* 12, no. 4 (June 1, 2010): 603–17. https://doi.org/10.1177/1461444809341392, pp. 612-3.

143 Gui Bonsiepe. "Design and Democracy." *Design Issues* 22, no. 2 (2006): 27–34, pp. 28-9.

144 Papanek, *op. cit.*, pp. 187-8.

The US polymath Buckminster Fuller might have been the ultimate design synthesist: some of his most majestic plans, such as the 1959 proposal for a geodesic "Dome Over Manhattan" (a spherelike structure, unsurprisingly), reflect design's ambition to be an all-encompassing suprasystem.[145] However, it also manifests the limits of design's hyperconnectivity: as every problem is chased by another problem, the synthesist would end up enveloping "spaceship Earth" in its entirety. This, according to German philosopher Peter Sloterdijk, is simply not possible: while we live under modernity's vast dome, we tend to take for granted its life supports, which come invariably from an outside.[146] Italian theorist Raffaele Alberto Ventura employs an analogy close to Sloterdijk's ideas – the air conditioner:

> [It] consumes material resources to produce cold air, and in the meantime throws out hot air. That is, it solves one problem, but in doing so it creates another. Actually, more than solving it, it moves it from one subsystem to another.

Synthesis can never be complete. For Ventura, "every attempt to bring order to the world produces a certain amount of disorder."[147] Furthermore, one thing is to acknowledge the ambiguous planetary impact of modernity, and another one is to believe that designers can plan on that scale. Like technology, design is "both fish and water"[148] and it is crucial to specify whether we are referring to the former or the latter. Today, however, the passion for wholeness takes a different form: whereas Fuller's

145 A controversy exists regarding who actually designed this project as well as the other domes generally attributed to Fuller only. This issue of attribution casts a light on the politics of synthesis and its interactions with mechanisms of recognition. See https://archive.nytimes.com/cityroom.blogs.nytimes.com/2014/05/07/50-years-later-questions-over-who-designed-a-worlds-fair-dome/, and http://www.domerama.com/tc-howard-trying-to-understand-his-anonymity/.

146 Latour, *op. cit.*, p. 9.

147 Raffaele Alberto Ventura. *Radical choc: ascesa e caduta dei competenti*. Torino: Einaudi, 2020, p. 103.

148 Franklin, *op. cit.*, p. 6.

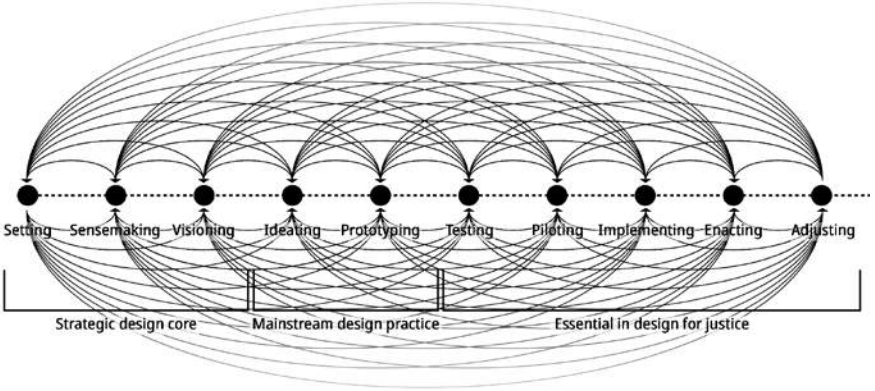


Diagram by Mauricio Mejía (2023) where each and every phase of the design process is linked to each and every other. An apt instance of the aesthetics of complexity. According to Mejía, "[m]ainstream design practices still focus on ideating and prototyping. The whole design process in complex situations includes a lot more activities."

craving for totality was rooted in an engineering mindset (a dream echoed by the '60s 'total design' agencies), contemporary design is an all-encompassing *cultural* sphere, one that emphasises comprehension, that is, understanding plus assimilation.

YOU KNOW NOTHING

To comprehend a wide range of phenomena, a cross-disciplinary approach is often invoked. But cross-disciplinarity is not exclusively a design virtue.[149] Cybernetics, for instance, was a successful bridge between scientific domains which built its own vocabulary pilfering terminology from disparate fields: homoeostasis, feedback, cognition...[150] We can also position the cross-disciplinary ethos within a prolonged criticism of specialisation. Inaugurated by Adam Smith,[151] it reached a peak with the '60s anti-bureaucratic counterculture. In 1973, sci-fi author and aeronautical engineer Robert Heinlein had one of his characters, Lazarus Long, assert that "specialization is for insects." Many designers and design theorists would agree, but only on the condition that design would provide the foundations of general activity. It is thus worth looking at the larger context of the quote:

> A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, *design* a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the

149 According to Wikipedia, "Cross-disciplinary knowledge is that which explains aspects of one discipline in terms of another." It is a form of translation. https://en.wikipedia.org/wiki/Academic_discipline. For an extensive list of the various forms of relations among disciplines, see Craig Bremner and Paul A. Rodgers. "Design Without Discipline." *Design Issues* 29 (July 1, 2013): 4–13. https://doi.org/10.1162/DESI_a_00217, pp. 11-12.

150 Klaus Krippendorff. "A Dictionary of Cybernetics." *Departmental Papers* (ASC), 1986. https://repository.upenn.edu/asc_papers/224. Of course, design itself has borrowed from cybernetics. As an example, the Studio at the Edge of the World recently proposed a practice of "second order design fiction," which draws from the idea of second order cybernetics. See https://www.thestudioattheedgeoftheworld.com/news.html.

151 Adam Smith. *The Wealth of Nations. Books 1-3*. London: Penguin, 2003.

> dying, take orders, give orders, cooperate, act alone, solve equations, analyse a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.[152]

The emphasis is mine. In this list, designing is not a prism of each and every human endeavour but simply one of its manifestations. Here, design does not provide total synthesis, it is not presented as a placeholder for human activity at large. Instead, it is humbly connected to a specific typology of artefact. *Design is a part, not the whole*.

Papanek believed that design could contribute a form of meta-knowledge deriving from the interpretation of various scientific and social languages. But this possibility rests on what we could call meta-ignorance. According to Don Norman, "the trouble with experts is that they know too much and they think the same way other experts think. Designers, on the other hand, know nothing, they ask stupid questions, and that's exactly why they're brilliant."[153] This is a pretty common idea: designers can be candid, devoid of preconceptions, free from the past. Their naivety being an asset, designers are able to elude the narrow precincts of expertise. However, designerly candidness is itself a preconception which simply displaces what is known, believed or assumed. The risk is evident: instead of developing a productive non-knowledge, designers might just turn a blind eye to their own biases. As a result, their 'unknown knows,' what they don't know that they know, might act in the world in ways they can't determine or even perceive. The past still casts its invisible shadow on the present. Given these premises, can there be actual design synthesis? Sure, but it would be shaped by the personal and cultural point of view of designers. Their "stupid

152 Robert A. Heinlein. *Time Enough for Love: The Lives of Lazarus Long*. New York: Ace, 2003, p. 248.

153 Don Norman. Opening speech at Koç University, October 4, 2012. https://www.youtube.com/watch?v=z_7Go53Zc-Y.

questions" will inevitably be sophisticated. Design can't be innocent.[154] Designers can frame problems, but as soon as these get slightly complex, 'designerly stupidity' becomes better at concealing rather than revealing.

THE EM DASH-SHAPED DESIGNER

In a scathing review of Papanek's bestseller for the Italian magazine Casabella, Gui Bonsiepe recognised the conservative and technocratic impulse implicit in his idea of synthesis:

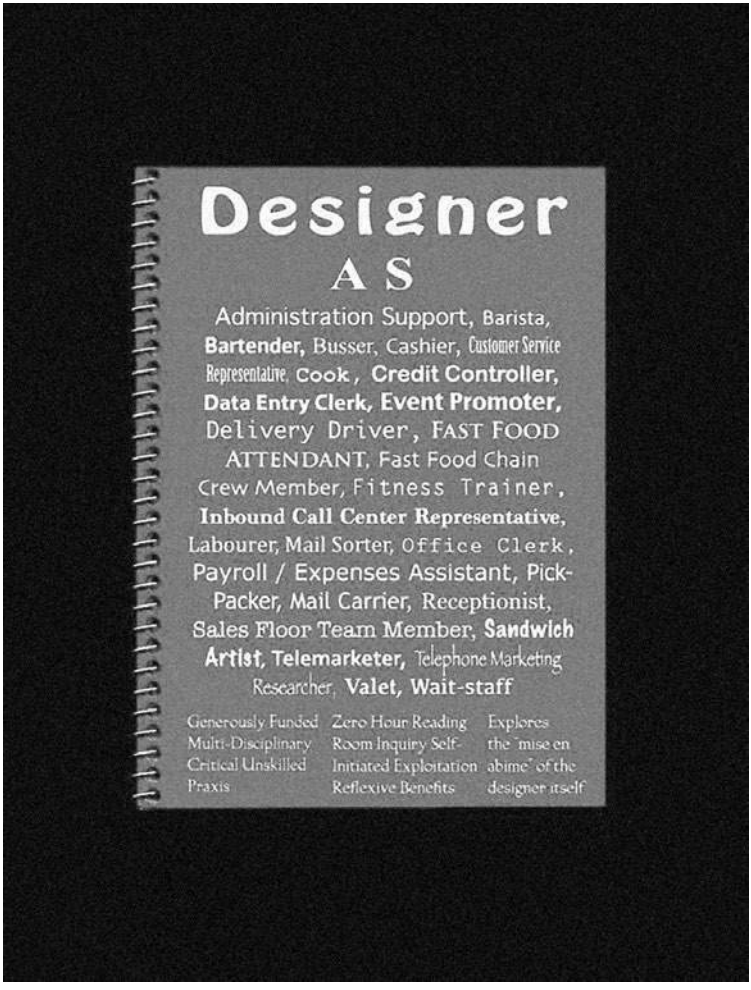
> [Papanek] tasks the designer with the role of a filler for the weaknesses of other specialists in the design team: that of an interpreter, of a generalist, essentially exercising, alongside his normal function which Papanek never discusses, a mediating function. The de-specialisation of the designer is ultimately reduced to a regression of the designer to the function of having-a-finger-in-every-pie.

Bonsiepe accused Papanek of turning design into "the only decisive and common-sense activity remaining to human beings." He was baffled by his conviction that designers should contribute to setting the goals for society at large, without negotiation or mediation: "The technologist throws the politician out [...] But for that you don't need a designer, you need a revolution."[155]

Specialisation or generalism? A typographic metaphor seems to offer a fruitful balance between the two. Circulating already in the late '70s in engineering circles, the idea of a T-shaped set of skills – meaning a deep, vertical specialisation accompanied by a horizontal breadth of

154 This is something Don Norman himself has recognised: "Designers who live in Western (including Western-influenced) societies inhabit a reality where everyone learns from the same books and universities and attends the same conferences. Consequently, everyone tends to think the same way: a dangerous thing." See "What Is Humanity-Centered Design?" *The Interaction Design Foundation*, 2022. https://www.interaction-design.org/literature/topics/humanity-centered-design.

155 Bonsiepe, "Design e sottosviluppo," *op. cit.*, pp. 43-4.



Critical Graphic Design (2015).

knowledge – was popularised by David Guest in 1991.[156] What the T-shaped professional knows is both broad and deep. Needless to say, the metaphor was followed by a plethora of variations, including not only letters but diacritics and pictograms. Following Bonsiepe, the shape deriving from the skill set of the designer, as portrayed by Papanek, would be purely horizontal – a hyphen, or perhaps an em dash.

For Arturo Escobar, design culture has contributed to manufacture a uniform, ever-expanding "One-World World." This, in turn, threatens to eradicate what's left of what he calls, following decolonial theorist Walter Mignolo, the pluriverse, which is where "things and beings *are* their relations [and] do not exist prior to them."[157] To sustain the pluriverse (whatever that is), designers might have to relinquish the ambition of a synthetic monopoly on ideas through a singular mode of comprehension. It wouldn't be a big loss, since most of them didn't actually hold such monopoly in the first place: rarely are designers given room to truly reframe the problem, that is, to take into account more of its relationships, to expand the magic circle. To maintain the wealth of pre-existing relationships – be they historical, political or cultural – design should resist the temptation to formalise, to abstract, to be *the* relation.

And yet, a solution is always a formalisation, one that mirrors the problem's formulation. Designers' discomfort with synthesis might have to do with the fact that designed artefacts can only go so far when it comes to incarnating different, if not opposite, worldviews. More than in use, a synthesis of disparate goals and desires is found in narrative. Stories, more than things, are able to keep different ideas together. They are ambiguous by design. This might explain the current centrality of storytelling and the

156 See https://en.wikipedia.org/wiki/T-shaped_skills.

157 Arturo Escobar. *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. London: Duke University Press, 2018, pp. 70-1.

humanistic ascendancy over the technical in design as an all-encompassing sphere.

Relinquishing an alleged synthetic monopoly might also have a positive side-effect. Forgoing design synthesis doesn't mean giving up generalism – it means giving up *genericity*, a vagueness that makes practitioners feel groundless. Authentic generalism offers instead a solid, fertile ground, one that doesn't rest on design alone, but requires an engagement with things, beings and practices on their own terms. Generalism is the opposite of specialisation, but it thrives on specificity.

A KNACK FOR MISCHIEF

"The discipline without the discipline of another discipline." This definition – courtesy of David Reinfurt – refers to graphic design, but it can be well applied to design in general.[158] Both meanings of the word discipline are mobilised here: field of study and self-restraint. (Graphic) design might lack the acquiescence that permeates more established branches of knowledge – it has a knack for mischief. What does it mean for a discipline to disobey? It means escaping its own semi-rigidly defined confines, its own "provisional territorialisation."[159] From this perspective, design's disobedience appears liberating. A case in point is the prestigious MIT Media Lab. Former director Joi Ito proudly spoke of the lab's *anti*disciplinarity:

> An antidisciplinary project isn't a sum of a bunch of disciplines but something entirely new – the word defies easy definition. But what it means to me is someone or something that doesn't fit within traditional academic discipline – a field of study with its own particular

158 David Reinfurt. *A *New* Program for Graphic Design*. Los Angeles: Inventory Press, 2019, p. 16.

159 Jacques Rancière. *Et tant pis pour les gens fatigués*. Paris: Éditions Amsterdam, 2009, p. 478.

words, frameworks, and methods.[160]

The target of Ito's criticism was the hyper-specialisation and gatekeeping of disciplinary experts who become incapable of communicating with the outer world and even among themselves. Furthermore, he rightly pointed out that reality is not organised around disciplines: they are artificial compartments of knowledge. If disciplines are the little dots on the map of knowledge, Ito explained, antidisciplinary space, the space of design, is the white area in between.

While antidisciplinarity is a clear attempt to provide a space for the "misfits," an emphasis on uniqueness and novelty might lead to the disregard of much of the slow, cementing work performed within the disciplines – the work of maintenance: "We shouldn't be doing something that someone else is doing. If someone else starts doing it, we should stop." Not only can design devour contexts, it can also ignore them.

Besides what is to be done, disciplines are normally busy with *how* things should be done, namely, methods. Nigel Cross recounts how design tried to pump up its reputation by presenting itself as a science.[161] This turned out to be an illusion: after endless debates, designers and theorists alike realised that design is not a science exactly because its processes can't be turned into a formula. Unlike science, which is based on a single method, design has *methods* that are plural, ad hoc and often non-replicable – to the extent that speaking of methods might already be a stretch. Donald Grant, writing in 1979, sententiously concluded:

> Most opinion among design methodologists and among designers holds that the act of designing itself is not and will not ever be a scientific activity; that is, that designing is itself a

160 Joi Ito. "Antidisciplinary." Joy Ito's Web (blog), October 2, 2014. https://joi.ito.com/weblog/2014/10/02/antidisciplinar.html.

161 Nigel Cross. "Designerly Ways of Knowing: Design Discipline versus Design Science." *Design Issues* 17, no. 3 (2001): 49–55.

> nonscientific or ascientific activity.[162]

Whereas the attempt to mimic the epistemology and methodology of science is doomed to fail, it is possible to turn design, a concrete set of "techniques of the artificial," into an object of study. This is what philosopher Donald Schön calls reflective practice.[163] In this way, design can finally achieve its status of discipline. Again Cross:

> Design as a discipline, therefore, can mean design studied on its own terms, and within its own rigorous culture. It can mean a science of design based on the reflective practice of design: design as a discipline, but not design as a science. This discipline seeks to develop domain-independent approaches to theory and research in design. The underlying axiom of this discipline is that there are forms of knowledge special to the awareness and ability of a designer, independent of the different professional domains of design practice.[164]

The design discipline becomes therefore the investigation of "designerly ways of knowing, thinking and acting." Such definition runs the risk of sounding circular, if not tautological. The idea of the "designerly," taken to its extreme, is that nothing precedes the designing act: it is this very act that forms its own understanding. Since a general design method akin to the scientific one cannot be established, design might then become whatever designers do.[165] If a designer were to propose using a pierced let-

162 Donald P. Grant. 'Design Methodology and Design Methods'. *Design Methods and Theories* 13, no. 1 (1979). See also Henrik Gedenryd, who adds that not only do designers disregard prescribed methodologies, but they also believe these simply don't work. "How Designers Work - Making Sense of Authentic Cognitive Activities." Lund University Cognitive Studies. Doctoral thesis, Lund University, 1998. http://lup.lub.lu.se/record/18828, p. 1.

163 Donald A. Schön. *The Reflective Practitioner: How Professionals Think in Action*. London, New York: Routledge, 2016.

164 Cross, *op. cit.*, p. 54.

165 This doesn't mean of course that certain replicable methods aren't employed by designers. For instance, the use of personas, paper prototyping or procedures like MoSCow are commonly used within design teams (See, for example, https://

tuce leaf as a face mask as protection from an aggressive virus, that would be a legitimate designerly approach.[166]

Schön's effort to systematise the design process is undoubtedly precious as it distils its underlying principle, namely, "a reflective conversation with the situation." But the form of this conversation is not, of course, set in stone. Given design's cultural expansion, it might be that a special form of knowledge, previously based on an intimate relationship with a set of materials, a technique or a category of products, is simply lost. The philosopher himself warns that, given the tendency "to think of policies, institutions, and behaviour itself, as objects of design, [we] risk ignoring or underestimating significant differences in media, contexts, goals, and bodies of knowledge specific to the professions."[167] When designers are integrated into a problem-solving chain, the validation of their results retrospectively includes their approach: if the product is good, the process must also be good. But things differ when the designerly approach becomes the focus, when the process itself is the product, when the production of meaning (what Manzini calls "sense-making") becomes prominent: the designerly attitude appears at the same time fully legitimate – and completely random.[168]

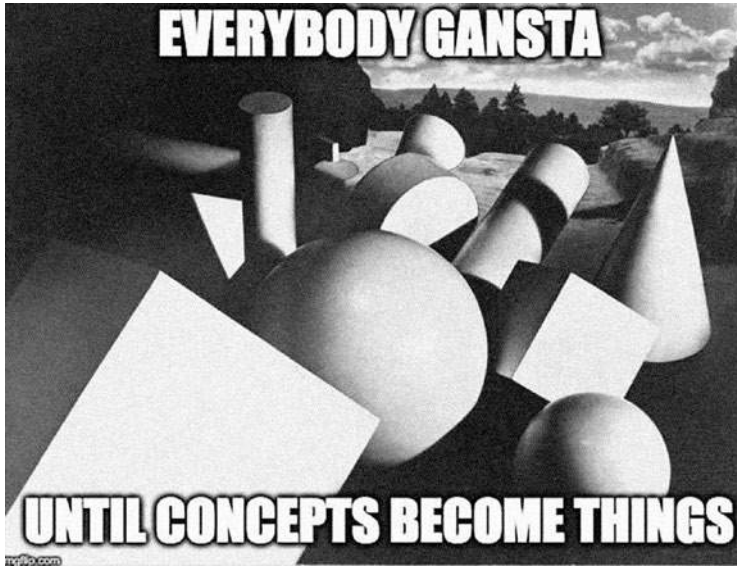
toolkits.dss.cloud/design/). But, as any designer will admit, those methods are limited or even detrimental to a genuine understanding of the issue at hand.

166 This is indeed what German-Namibian artist Max Siedentopf came up with. Of course, it was something between art project and provocation, but the very attention given to it by *design* platforms like Dezeen and the wave of indignation that followed (Siedentopf later apologised) signal the ambiguity of design methodologies: the way they lead to a result, as well as their very result, are dubious. See Amy Frearson. "Max Siedentopf Suggests Alternative Masks to Protect Against Coronavirus." *Dezeen*, February 17, 2020. https://www.dezeen.com/2020/02/17/alternative-coronavirus-masks-max-siedentopf/. During the first months of the Covid-19 pandemic, this was no isolated case. Architecture critic Kate Wagner called Coronagrifting the phenomenon of "cheap mockups of COVID-related design 'solutions.'" A phenomenon that seems to imply that the 'designerly' approach has some sort of immunity against serious scrutiny. "Coronagrifting: A Design Phenomenon." *McMansion Hell* (blog), May 23, 2020. https://mcmansionhell.com/post/618938984050147328/coronagrifting-a-design-phenomenon.

167 Schön, *op. cit.*, p. 77.

168 In the case of Coronagrifting, the product, being mostly an idea, coincides with the process, and it is deemed good, as long as it generates online engagement.

Rendering of the Dome Over Manhattan by Buckminster Fuller (1960). Black and white photograph on board with dome overlay, 12.75in x 18.38in.



Found image. Source unknown.

Design scholars Craig Bremner and Paul Rodgers react conservatively in the journal *Design Issues*. For them, the discipline of design is simply dissolving. This is the result of a three-fold crisis: professional, economic and technological. Bremner and Rodgers notice how "[p]ractice shifts from being 'discipline-based' to 'issue- or project-based.'" As a consequence, "design now finds itself in a position of not knowing what to project." Not unlike Ito, they recognise the non-disciplinary character of design, which is what they call undisciplinarity. Interestingly, they connect a disciplinary modus operandi to the character of the designer who adopts it.

> "Undisciplined" research straddles the ground and relationships between different idioms of distinct disciplinary practices. Here a multitude of disciplines "engage in a pile-up of jumbled ideas and perspectives. Undisciplinarity is as much a way of doing work as it is a departure from ways of doing work." It is an approach to creating and circulating culture that can go its own way without worrying about what histories-of-disciplines say is "proper" work. In other words, it is "undisciplined."[169]

Bremner and Rodgers commend the worry-free attitude of the undisciplined designer. But the chaotic quality of undisciplinarity might resemble pure relativism. In the pile-up of jumbled ideas and perspectives, one might feel dispirited. A tweet by Celeste Labedz aptly summarises the feeling: "Yeah, my work is interdisciplinary; I'm confused by *multiple* fields at the same time."[170] There may be a sense of being a Jack of all trades, master of none.[171] How can we prevent disorientation? The scholars' proposal is to hook up the act of designing to something stable, be

169 Bremner and Rodgers, "Design Without Discipline," *op. cit.*, p. 12.
170 https://twitter.com/celestelabedz/status/1391770429514190850.
171 This is, in fact, Bruce Archer's *positive* suggestion. Archer quoted in Potter, *op. cit.*, p. 165.

that either disciplinarity (hence, design as "disciplined and irresponsible") or responsibility ("undisciplined and responsible"). Otherwise, insecurity might prevail.[172]

According to Danah Abdulla, the undisciplinarity of design is connected to the specialisation of the designer, or lack thereof. She considers the following conversation:

> What do you do?
> I'm a designer.
> What kind? Graphic? Fashion? Furniture? Interior?

Pressured for a specific answer, the designer feels anxious and, instead of simply adopting a general, interdisciplinary label, ends up finding shelter in a specialised discipline. However – Abdulla argues – disciplines are constrictive, bureaucratic, myopic: they are mainly concerned with administering their own rules and regulations. This is what Lewis R. Gordon calls "disciplinary decadence."[173] Quoting Moholy-Nagy, Adbulla points out that the urge to specialise isn't new in design, where vocational education "breeds specialists with a rather narrow horizon." Finally, she argues for "border-thinking," which consists in locating the inquiry at the very edge of a system of thought in order to break disciplinary boundaries.[174]

However, it is important to realise that, in the real world of work, inhabiting disciplinary borders has become the default. The widespread diffusion of 'multi-hyphenated' professional profiles is proof.[175] The everyday designer might opt for a specialisation to signal their affiliation to a certain cultural group, but in practice, their work is likely to span various lines of work, if not various jobs! This is

172 Bremner and Rodgers, "Design Without Discipline," *op. cit,* p. 13.

173 Gordon, *op. cit.*, p. 229.

174 Danah Abdulla. "Disciplinary Disobedience." In *Design Struggles: Intersecting Histories, Pedagogies, and Perspectives*, edited by Claudia Mareis and Nina Paim. Amsterdam: Valiz, 2021.

175 See Nikki Shaner-Bradford. "What Do You Do? I'm a Podcaster-Vlogger-Model-Dj." *The Outline*, November 25, 2019. https://theoutline.com/post/8301/everyone-you-know-is-a-multi-hyphenate. Artist Brad Troemel refers to this phenomenon as "polyemployment."

why a designer could end up feeling a certain repulsion towards multiplicity: "I don't want to have to know every software and skill. I just want to be a graphic designer – not a motion designer, 3D artist, UX strategist."[176]

It seems, then, that undisciplinarity (which somehow includes antidisciplinarity), understood here as an implosion of disciplines from the perspective of those who practice them, is to be found either at the high end of the workforce (e.g. the art director) or at the bottom (the occasional web designer-barista). In any case, the incentive is primarily financial: "The evasion of normative procedures and professional identities by the design industry is in no small part the result of its constant adherence to 'following the money' by inventing new specialisms to exploit technological and business opportunities."[177] This shows that undisciplinarity, rather than a mission, is *a destiny* of the so-called creative class. This was already clear in the '90s to graphic designer Rudy VanderLans:

> In a sense, everything can be learned on the job, even critical thinking, exploration, introspection, offset printing, intellectual development, bookkeeping, French literary criticism, programming, [and] contract writing... It can all be learned as you slowly develop into the all-around professional you're *supposed* to be.[178]

The emphasis, which is mine, highlights the necessity of this condition. J. Dakota Brown associates the role of all-around professional to the "zany," a frenetic character type who does it all, but struggles to keep it all together.[179] Paradoxically, the contemporary creative profes-

176 "Design Thread 2 – Excess of Everything," 2022. https://www.designthreads.report/thread2.

177 Julier, *op. cit.*, 24.

178 VanderLans quoted in J. Dakota Brown. "American Graphic Design in the 1990s: Deindustrialization and the Death of the Author." *Post45*, January 10, 2019. https://post45.org/2019/01/american-graphic-design-in-the-1990s-deindustrialization-and-the-death-of-the-author.

179 Brown, "American Graphic Design in the 1990s", *op. cit*.

sional is an amateur, someone "who dabble[s] in a range of activities without dedicating or committing themselves to any one field."[180] For everyday designers, antidisciplinarity, rather than making it possible to escape the constriction of disciplines, means entering a frenzied state of insecurity.

GOOD DOUBT, BAD DOUBT

Designer Michael Rock maintains that such insecurity produces anxiety: "Designer anxiety is rooted in a fear that what we do is not respected, worthwhile, important. This anxiety is fuelled by a business world that, in general, neither respects nor considers design particularly worthwhile or important. [...] Thus design activity is fraught with a desperate quality."[181] Designer and educator Brian LaRossa agrees:

> Graphic designers are insecure. This is understandable; design lacks defensible boundaries. It is ubiquitous and absorbent, everywhere and everything. It is never itself, always its subject. It is diffused evenly across our lives to such a low concentration that we often doubt its worth.

LaRossa, saying something that can be said of design at large, goes on, describing precisely the idiosyncrasies – the 'magic,' one could say – of the design process: "Our inquiries compound, narrowing options until we're left with something real."[182] Designers are asked to 'trust the process,' to have faith in a liturgy that acquires meaning only in hindsight. LaRossa, however, is positive about persistent doubt: he sees it as a fundamental component of the design craft. But, for designers to merrily linger in a

180 Gerry Beegan and Paul Atkinson. "Professionalism, Amateurism and the Boundaries of Design." *Journal of Design History* 21, no. 4 (2008): 305–13, p. 309.

181 Michael Rock. "On Unprofessionalism." *2x4* (blog), August 5, 1994. https://2x4.org/ideas/1994/on-unprofessionalism/.

182 Brian LaRossa. "Design as a Third Area of General Education." *Medium* (blog), January 23, 2021. https://brianlarossa.medium.com/design-as-a-third-area-of-general-education-80bc59875e45.

state of doubt, they need not only self-confidence, but the confidence of their peers, be they fellow practitioners or clients. Desirable doubts require certainties.

How can we take advantage of this inherent and productive insecurity without sinking into 'bad doubt,' that is, crippling self-doubt? How can we maintain conviction in spite of indifference or even hostility without having to have a kind of *Fountainhead* me-against-the-world cockiness? Undisciplinarity requires legitimacy and recognition to be practised. In a time when a proper balance between specialisation and generalism is yet to be found, undisciplinarity requires a social act of faith. Clients, users and journalists have to be convinced that the designer's 'stupid questions,' awkward methods (often pretentiously called 'methodologies') and hardly explicable connections will spark innovation in the end and, therefore, generate value.

According to Umberto Eco, "[t]echnique is administered by the artist or by the technician, who acts like a shaman."[183] This is not just about design: any kind of expert knowledge is esoteric. Ettore Sottsass believed that "design begins where rational processes end and magic begins." At times, the magic reference goes beyond analogy. Here's how Kursat Ozenc describes the "design shaman":

> A shamanesque designer doesn't stop in just reacting (*sic*) what's asked of her. She knows the friction in the world, and entropy embedded in human kind to derail from what's right and important. With that in mind, she works hard to create myths around sustainability. She creates micro-visions for a harmonious future, mixes the spiritual with the functional without hesitation. At times of hardship, she infuses light superstitions for the greater good.[184]

183 Umberto Eco. "Critical Essay." In *The Universitas Project: Solutions for a Post-Technological Society*, edited by Emilio Ambasz. New York: The Museum of Modern Art, 2006, p. 130.

184 Kursat Ozenc. "The 3 Most Unconventional Designers of 21st Century – Starting

However, if the expert or the designer is denied the trust to exercise their magic, the spell breaks.[185] Thus, productive insecurity requires a safe space to be exercised. Societal trust makes insecurity productive; without it undisciplinarity becomes mere disorientation. Design schools and labs can be seen as spaces that allow for a suspension of disciplinary disbelief toward design methods and practices. But what happens outside these enlightened post-disciplinary institutions? Annelys de Vet, director of the Design Department of the Sandberg Instituut of Amsterdam for a decade, recognises that "although design students remain some of the most forward-thinking individuals, designers are occupying rather fragile positions in society. As a department, we hope to provide a space for young people to prepare for a practice within a profession that is increasingly hard to define."[186]

Here, de Vet speaks of profession rather than discipline. According to Donald Schön, a professional is someone who "claims extraordinary knowledge in matters of human importance, getting in return extraordinary rights and privileges."[187] This definition clearly shows why someone might want to be considered a professional: a profession is not just a job, but a prestigious social position: "the image of The Professional; respect, high pay, beautiful lovers."[188] So, we can say that designers are considered

with Design Shaman." *Ritual Design Lab* (blog), April 4, 2016. https://medium.com/ritual-design/a-new-breed-of-21st-century-designers-b73712963b4.

185 Here is a tangible example of a broken spell. Some years ago, I attended the XY summer school, organised by the soon-to-be Scuola Open Source in Castrignano De' Greci, in the deep south of Italy. There, some of the tutors insisted on the value of rituals and encouraged us to perform one every day. Journalists then came to visit. Later, in order to discredit the political party that indirectly funded the school, they wrote a vitriolic article where they labelled the whole event as 'shamanic' to denounce the waste of public money. Sergio Rame. "Gli sprechi di Vendola: un corso per formare gli sciamani (digitali)." *ilGiornale.it*, July 18, 2014. https://www.ilgiornale.it/news/politica/sprechi-vendola-corso-formare-sciamani-digitali-1038689.html.

186 Annelys de Vet, ed. *Design Dedication: Adaptive Mentalities in Design Education*. Amsterdam: Valiz + Sandberg Instituut Design Department, 2020, p. 31.

187 Schön, *op. cit.*, p. 4.

188 Rock, *op. cit*.

Poster designed by Herbert Bayer in 1923 to promote the first major exhibition of the Bauhaus. It includes Walter Gropius's new motto for the school: ""Art and Technics: A New Unity".

professionals when society recognises the added value that their particular mixture of disciplines provides.

The increasing difficulty to define the design profession can be situated within a more general crisis of the professions. Schön draws a trajectory of the social role of the professions characterised by triumphalism during the '60s and scepticism during the '70s and '80s. Given the current debate around the crisis of competence, it is fair to assume that such scepticism never went away. This timeline almost perfectly coincides with the parable of design recognition, with a jubilant culmination of good design in the '60s and the growing doubt and antagonism during the '70s (with the politically engaged Radical Design and the sensuous Anti-Design), followed by the emergence of deconstructionism and post-modernism in the '80s.[189]

OWNING COMPLEXITY

The traditional justification for an expansion of professional domains is the growing complexity of our world. It is a litany that has been repeated uninterruptedly at least since the '60s. Not surprisingly, the design field insists on complexity and anchors itself to it, often arguing that the conventional scientific approach fails to grasp it.[190] Complexity, from this point of view, is not just matter of fact, but also an instrument for professional lobbying. Here, an interesting reversal ensues. What worries and disorients the general public becomes a reassuring design trope: with the increase in complexity, the need for professional designers increases too. Guy Julier, with a hint

189 Along these lines, we can think of the popularity of Critical Design (which is itself critical of design's role in society) at the turn of the millennium as a semi-conscious attempt to professionalise disciplinary self-doubt.

190 "Design is a way of behaving that approaches complexity in a different way (and expecting a different type of outcome) than we have conventionally used. Designers work in a way (a conversation with the self via the medium of drawing) that allows them to deal with very complex, ill-defined and ambiguous situations that would probably be inaccessible using conventional approaches." Ranulph Glanville. "Designing Complexity." *Performance Improvement Quarterly* 20, no. 2 (June 2007): 75–96. https://doi.org/10.1111/j.1937-8327.2007.tb00442.x.

of controversy, remarks: "It has become an orthodoxy to talk of the growing complexity of design in our 'complex world.'"[191] Innovation experts like to speak about complexity, albeit in vague terms, such as when they state that "society is growing increasingly complex and will only be more complex in the future." Scholar Lee Vinsel asks:

> What does this claim even mean? Complex in what way? Increasingly complex with respect to what metric? I have asked many professional historians this question, and they believe this increasing complexity claim is unsupportable.[192]

On the other hand, 'complicating' an issue is increasingly deemed as a valuable activity, an attempt to look deeper. So, complexity is at the same time the illness and the remedy. The orthodoxy of complexity goes hand in hand with design's anthropological turn. David Graeber notices that "[a]nthropologists are drawn to areas of density. [Their interpretative tools] are best suited to wend our way through complex webs of meaning or signification."[193] Designers are increasingly attracted to such dense areas of meaning, because it is where their self-attributed role of synthesists can be, presumably, better performed.

But 'owning' complexity is not enough. The professional ambition of the design field clashes with the trend of deprofessionalisation, which can also be understood as a proletarianisation of the professions and, therefore, a proletarianisation of design. This structural crisis goes hand in hand with what Schön, writing in the '80s, calls an internal "crisis of confidence:"

> The crisis of confidence in the professions, and perhaps also the decline in professional self-image, seems to be rooted in a growing scepticism

191 Julier, *op. cit.*, p. 6.

192 Lee Vinsel. "Design Thinking Is Kind of Like Syphilis — It's Contagious and Rots Your Brains." *Medium* (blog), August 3, 2020. https://sts-news.medium.com/design-thinking-is-kind-of-like-syphilis-its-contagious-and-rots-your-brains-842ed078af29.

193 David Graeber. *The Utopia of Rules: On Technology, Stupidity, and the Secret Joys of Bureaucracy*. Brooklyn: Melville House, 2015, p. 51.

> about professional effectiveness in the larger sense, a sceptical reassessment of the professions' actual contribution to society's well-being through the delivery of competent services based on special knowledge.

The philosopher adds that scepticism involves the very knowledge that the professions purportedly possess. Again, the changes he highlights in the professional context uncannily resemble the current state of design discourse: "The situations of practice are not problems to be solved but problematic situations characterized by uncertainty, disorder, and indeterminacy." It turns out that complexity is not just a fog surrounding the professional ivory tower, but a haze invading its chambers. A need for synthesis emerges:

> What is called for, under these conditions, is not only the analytic techniques which have been traditional in operations research, but the active, synthetic skill of "designing a desirable future and inventing ways of bringing it about."[194]

Like other professions, design actively tries to affirm and preserve its place in society. But, unlike many of them, the design field is particularly subject to professional pluralism, the proliferation of conflicting views around the discipline's aims, methods and goals. Donald Schön's motivation to write his book was to shed light on the qualities of the reflective practitioner, concluding that "we are bound to an epistemology of practice which leaves us at a loss to explain, or even to describe, the competences to which we now give overriding importance."[195] While his attempt to clarify the tacit ways of knowing afforded by practice is laudable, this feeling of being at a loss is ineradicable in the design field. Furthermore, it becomes harder to convince non-professionals of the adequacy of design knowledge

194 Schön, *op. cit.*, p. 16.
195 *Ivi*, p. 20.

beyond the exploits of design stars, the micro-evolutionary changes in consumer products and services, or even the software skills of execution now available to the masses. In this setting, professional pluralism, whose most obvious manifestation in design is a proliferation of labels,[196] reinforces the crisis of confidence and the sense that design's professional knowledge is either common sense or a clunky 'methodocracy,' a generic set of procedures that the Methods Movement itself has abjured.

THE PLATONIC TURN

As a partial response to a disciplinary and professional impasse, various designers began to think of design as an autonomous sphere. What they advocated was the idea that design could be disconnected from its traditional service-based role to gain the freedom to decide upon its own societal concerns, instead of tackling the narrow ones of those who commissioned the work. The designer's privileged interlocutor would not be a client anymore, but the transcendental idea of society as a whole. The goal was – to apply Manzini's categories – to put problem-solving aside and fully devote designers' efforts to sense-making. Unsurprisingly, this claim has been put forward with particular energy in those design subfields, such as graphic and product design, that have been more impacted by the threefold crisis pointed out by Bremner and Rodgers: economic, professional and technological. Given the growing supply of trained designers and the aspirational allure surrounding the profession, product design is deeply affected by economic pressure.[197] Graphic design, on the other hand, has suffered a crisis of legitimacy due to the general availability of desktop publishing software, which

196 "Sometimes, one gets the impression that a designer aspiring to two minutes of fame feels obliged to invent a new label for setting her or himself apart from the rest of the professional service." Bonsiepe, "Design and Democracy," *op. cit.*, p. 27.

197 See Justin McGuirk. "Designs for Life Won't Make You a Living." *The Guardian*, April 18, 2011. https://www.theguardian.com/artanddesign/2011/apr/18/designs-milan-furniture-fair.

has lowered the access bar of certain design tasks and therefore its prices. Furthermore, graphic design has partially failed to reinvent itself for the web, giving way to the more systematic and service-oriented areas of user interface and user experience design (UI, UX).

From this perspective, the call for cultural and professional autonomy seems more like a retreat from a hostile marketplace and an indifferent public than the actual fulfilment of disciplinary independence. Whereas the traditional service-based path could likely lead to a relatively small income and little agency, the autonomous road would at least guarantee the possibility to set the terms of one's own activity, a positive self-narrative that fits the imperative to be in control of one's own life project.[198]

In 2002, Andrew Blauvelt invoked "critical autonomy" in graphic design. The Japanese-American designer and curator wasn't encouraging the withdrawal from society or fine-art romantic genius individualism. Critical autonomy meant that design could be busy "generating meaning on its own terms without undue reliance on commissions, prescriptive social functions, or specific media or styles." Not without a dose of "self-awareness and self-reflectivity," designers would develop "a capacity to manipulate the system of design for ends other than those imposed on the field from without and to question those conventions formed from within." Furthermore, Blauvelt pointed out that critical autonomy would confer coherence to the discipline of graphic design (thus, confirming its crisis), a coherence which was under the attack of stylistic pluralism and the exogenous terms and conditions under which it operates.

198 As identity is nowadays deeply rooted in one's job, the issue of a gratifying professional self-image becomes crucial. I tackled this issue in my previous book: "Many simply ignore or even reject the circumstances that determine their own story. In doing so they often create an idealized personal narrative to the detriment of the material reality that lies behind it. In other words, they claim to be artists, journalists or entrepreneurs regardless of the income these activities actually generate. In this sense, the issue of precarity seems to be linked not only to work but also to the need to build and maintain one's identity." *Entreprecariat*, op. cit., p. 55.

However, what Blauvelt, as well as most proponents of design's autonomy, overlooks is the actual possibility for (graphic) design to become autonomous. Can design turn itself into something different alone? Can it radicalise its sense-making, meaning-generating activity without having to rely on its socially attributed role? And can one even clearly identify a "system of design" to manipulate? J. Dakota Brown suggests another way forward:

> Rather than pinning our hopes on bootstrapping ourselves out of design's social function, we should ask why certain possibilities have consistently been foreclosed in the history of the practice.[199]

A radically different view on autonomy might help frame these questions. Arturo Escobar speculates that design can foster the autonomy of communities. His idea of autonomy is not a generic one, but it is strictly connected to movements of liberation in Latin America, what he calls *autonomìa*. The anthropologist derives this concept from Chilean biologist Francisco Varela's notion of autopoiesis: an autonomous system is one that "finds its way into the next moment by acting appropriately out of its own resources."[200] The key aspect of Varela's autopoiesis lies in its independence from the environment of a system's identity: its defining logic is internal and, while it responds to changes in the environment, it maintains an inner coherence.

This understanding of autonomy can be fruitfully applied to communities because it allows them to maintain their identity while changing: it's a fertile combination of tradition and innovation. It is what makes it possible to change "traditions traditionally."[201] But the idea of an

199 J. Dakota Brown. *The Power of Design as a Dream of Autonomy*. Chicago, IL: Green Lantern Press, 2019, p. 20.

200 Francisco Varela. *Ethical Know-How: Action, Wisdom, and Cognition*. Stanford: Stanford University Press, 1999, p. 11.

201 Escobar, *op. cit.*, p. 172.

autonomous community rests on the premise that identity and coherence are there to be preserved in the first place. Is this the case for design? Blauvelt argues that graphic design is fundamentally about difference. So, while the community the designer works for can hopefully operate 'autopoietically,' the designer might perceive their field as heteronomous, that is, fundamentally shaped by its environment to the extent that it can't preserve any organisational coherence. One can sense such *allopoiesis* as the technological pressures that affect the design field. Innovation is often what attacks the core of a discipline.[202] The reversal is ironic: whereas design as an endeavour is traditionally about reshaping the environment, as a culture it is mainly *shaped* by it. This might be the fundamental lesson we learn from the perspective of ontological design.

A PEDANTRY OF THE SPIRIT

Blauvelt's critical autonomy, one of the many attempts at freeing design practices from their service status, seems to be more a matter of wishful thinking than a viable option for the majority of designers. Looking at the ethical claims of graphic designers, J. Dakota Brown notices how "[the] inflation of design's autonomy and power entails a corresponding inflation of the autonomy and power of its practitioners."[203] Critical autonomy implies that the ethical and critical impetus to affect the world, doesn't happen via *use-artefacts* but by means of *conversation pieces*, that is, texts and objects that intervene by "making people think," "raising awareness," etc.[204] In this sense, we can

202 An example: according to Lorraine Wilde, "it's not clear that anything resembling the traditional role of the graphic designer is really necessary or desired in new media. If you surf the Web, you know that lots of visual 'things' have been produced without the participation of someone the profession would even call a graphic designer." "The Macramé of Resistance." *Emigre*, no. 47 (1998). Is the identity of graphic designers preserved while they make the jump to digital media? Is graphic design a tradition worth maintaining when it comes to laptops and smartphones?

203 Brown, *The Power of Design…*, *op. cit.*, p. 12.

204 Cf. Anthony Dunne and Fiona Raby. *Speculative Everything: Design, Fiction, and*

argue that design has attempted to keep up its status by radically transforming its internal principle of being a service. It's no coincidence that a recent anthology of Swiss writings on design is entitled *Not at Your Service*.[205] When it comes to the design field, autonomy derives from a sort of dysfunctional autopoiesis: an internal coherence, built from scratch, impedes a healthy relation with the environment. Critical autonomy signals the rise of the designer who focuses on ideas rather than things – the Platonic designer. Contra Bonsiepe's emphasis on practicality, Platonic designers return to the contemplative mode of academia as the access to problem-solving (as well as authentic problem-framing) is barred to them. They promote theories, foster reflection and criticise positions. Instead of transforming the environment through products and services, Platonic design claims to sway public opinion. But it has often little interest in verifying such claim.

Speculative designers Anthony Dunne and Fiona Raby blatantly celebrate this shift when they state that "once designers step away from industrial production and the marketplace we enter the realm of the unreal, the fictional, or what we prefer to think of as conceptual design – design about ideas."[206] But while the step away from industrial production might actually take place, there is no outside the market, at least the labour market, as the designer still has to sell their ideas. Conceptual design might be an immaterial commodity, but it is still a commodity that needs a buyer to complete its cycle, otherwise it is pure consumption. From this perspective, critical autonomy is perhaps nothing more than a concealment of heteronomy:

> Designers have attempted, through sheer will, to reinvent themselves as theorists and critics. But as long as they are at work, they are

Social Dreaming. Cambridge, MA: MIT Press, 2013, p. 21.

205 Björn Franke and Hansuli Matter, eds. *Not at Your Service: Manifestos for Design*. Zurich, Basel: Zurich University of the Arts, 2021.

206 Dunne and Raby, *op. cit.,* p. 11.

> constrained by social forces over which they have little control – and which they have, unfortunately, shown little interest in grasping.[207]

The elephant in the room is design's self-sufficiency at scale. Undoubtedly, a small minority of designers, assisted by the cultural structure of museums and enlightened state support are able to exercise critical autonomy, but for the rest, the ability to "manipulate the system of design" is a mere disciplinary smokescreen. Maldonado:

> One has to admit that industrial design, contrary to what its precursors imagined, is not an autonomous activity. Although its design choices may seem free, and perhaps sometimes they are, they are always choices made within the context of a system of very rigidly established priorities. Ultimately, it is this system of priorities that governs industrial design.[208]

In this chapter I have shown that looking at how design weaves its web of relationships might be the only way to nail down its essence, but design's intrinsic in-betweenness is not devoid of intricacies. Design professes a generalist attitude, but its cultural hyperconnectivity, together with the synthetic monopoly that it holds (or claims to hold), might reduce it to a generic and reductive pursuit. Furthermore, its undisciplined character, when it is not backed up by a social act of faith, can lead to disorientation and crippling self-doubt. Design is situated within a more general crisis of the professions, where the tacit contract that broadly defines a profession – legitimised expertise offered in exchange of prestige and recognition – vacillates. That of the profession is not an absolute, static category, but a relative one in constant renegotiation. Finally, through the decades, design has formulated a demand for autonomy: designers want to decide upon

207 Brown, *The Power of Design…*, *op. cit.*, p. 19.
208 Maldonado, *La speranza progettuale*, *op. cit.*, p. 12.

their own concerns, methods and outputs – they want to determine their own added value. However, design's "critical autonomy" might disguise the abdication of its specific social role, that is, contributing to the production of preferable conditions by transforming the artificial environment, or, more succinctly, solving problems. Instead of becoming masters of their own destiny, designers risk ending up retreating into a Platonic realm where their main concern is not artefacts any longer, but ideas and principles whose reception is uncertain. A pedantry of the spirit, in Norman Potter's formulation:

> For the social task we have fresh evidence all the time of man's fallibility, of his deepening technological commitment; of the nature of affluence divorced from the social or spiritual awareness. Yet there is a pedantry of the spirit in dwelling too much on these things.[209]

209 Potter, *op. cit.,* p. 45.

Chapter 4.

Flipping the Table:
On Power and Impotence

"But is there still room for design? Frankly, it is very small."
– Tomás Maldonado, 1970[210]

"In the way that we conduct ourselves as designers, we are as free as the marketplace allows us to be."
– Adrian Shaughnessy, 2005[211]

"[D]id it hurt? when they hired you for your design experience and then just ignored all of your advice and told you exactly what to design and where"
– @tiffanyton, 2021[212]

MASS-PRODUCED IMPACT

Conventionally, designers are thought of as changing the world through the things they design: products, services, systems, environments, tools, machines. By designing the artefacts that make up our artificial world, they contribute to make the world more and more artificial, and, it seems, inescapably so. At the beginning of the twentieth century, in the culminating phase of industrial production and mass communication, the link between cause (design) and effect (world changes) was particularly tangible since its scale was colossal. Armies of identical artefacts would enter the life of the masses for a relatively long time. The slightest design change would suddenly re-adjust the operations of this mechanically-reproduced battalion of things. As industrial designer Carroll Gantz explains, "[designers would] transform the process of design from an individual creative act that produced one artefact at a time into a complex industrial mass production system

210 Maldonado, *La speranza progettuale*, *op. cit.*, p. 76.
211 Adrian Shaughnessy. *How to Be a Graphic Designer Without Losing Your Soul*. New York: Princeton Architectural Press, 2005, p. 25.
212 https://twitter.com/tiffanyton/status/1433067196712833034.

that made millions of identical products."[213]

Designers could think of themselves as a commanders whose designerly 'orders' would be executed *en masse*, in line with traditional definitions of power, such as Carl Schmitt and Max Weber's, namely, "the possibility to find obedience to a command."[214] A modern perspective on production implied a modern understanding of design's power. "One design for all" wasn't just a technical necessity but a programmatic ambition. Consider, for instance, Henry Ford's faith in a universal car, the T model. In 1922 (the same year Weber offered his definition of power), Ford expressed his deliberate intention to "build a motor car for the great multitude."[215] As Hal Foster puts it, "the commodity was its own ideology, the Model T its own advertising: its chief attraction lay in its abundant sameness."[216]

The sun hasn't set on standardised production yet. Most of the things we purchase and use are still mass-produced. Furthermore, if we take digital services into account, we have to admit that an entirely new dimension of mass production has emerged. One where segments of this mass are relentlessly sieved by means of quick cycles of test and implementation, where an interface change on a platform used by the millions, instantly deployed, can have monumental consequences. But how common is it for a designer to work on such a level? Pretty rare. Only a handful of designers have access to the prototyping rooms of these massively used products.[217] The majority of them will likely

213 Gantz quoted in Wizinsky *op. cit.*, p. 59.

214 Max Weber. *Typen der Herrschaft*. Ditzingen: Reclam, 2019, p. 7. Weber uses the word *Herrschaft*, a term covering a vast semantic field including notions of authority, rule, dominion and power.

215 Henry Ford and Samuel Crowther. *My Life and Work*. Garden City, NY: Doubleday, Page & Co., 1922. https://college.cengage.com/history/primary_sources/us/henry_ford_discusses.htm.

216 Hal Foster. *Design and Crime*. London: Verso, 2002, p. 19.

217 Spotify is a case in point. In 2018 it was used by around 200 million monthly active users. At the time, it had only 200 designers. A ratio of one designer for one million users, more or less the population of Milan. See Shaun Bent, Marina Posniak, and Geerit Kaiser. "Reimagining Design Systems at Spotify." *Spotify Design*, September 2020. https://spotify.design/article/reimagining-design-systems-

work on projects whose impact is limited in time, space and reach. Big projects, perhaps, but hardly as gargantuan. This doesn't mean that their efforts have little or no effect, but that the effect doesn't match the modern *forma mentis* of power as vast scale standardisation. Mentions of the tremendous influence that designers possess will be met by most with some reasonable scepticism.

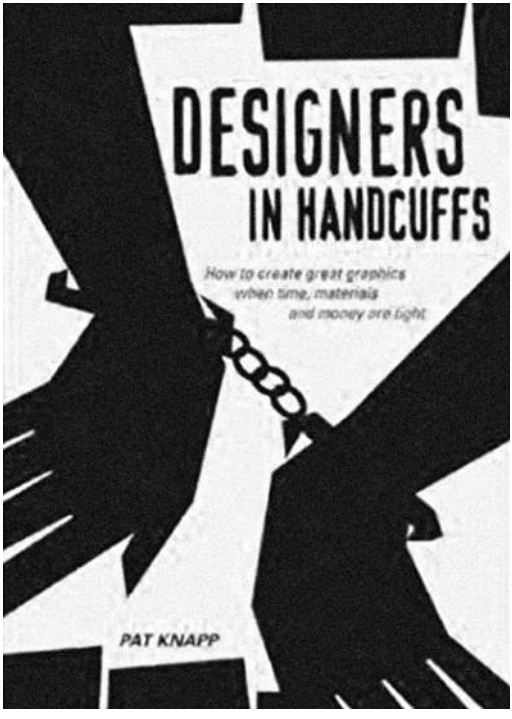
CAUSE AND EFFECT

The modern, 'scalar' idea of design power also tends to place too much emphasis on the single object, as if its transformative power weren't enhanced or hindered by the environment it inhabits. As if the object weren't caught in a network of interactions with other objects. A linear, almost mechanistic understanding of design power (designer > product > user)[218] masks the improvisational role that the object performs on the stage of everyday life. The leading role attributed to the isolated object reflects another leading role – that of the designer. A reassuring narrative of invention also plays its part: when we perceive a groundbreaking change in habit, we are tempted to look for a singular artefact that represents such change – a symbol. Then, we try to identify an individual who conceived it, possibly through a Eureka moment.

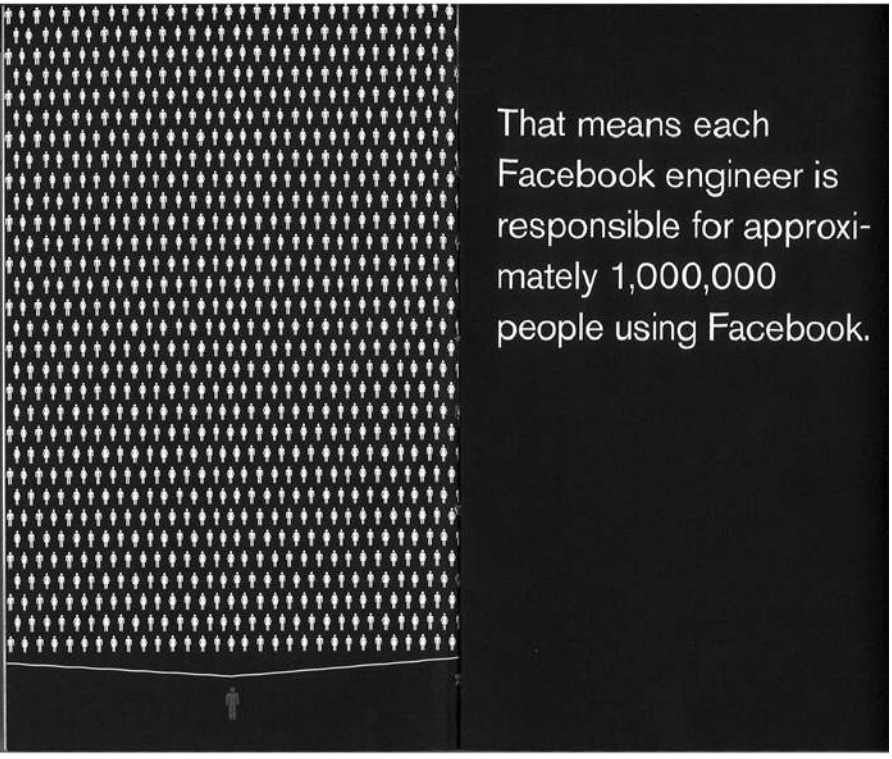
However, we shouldn't forget that while products and services affect the world, they are also affected by it, and not just after coming into existence. According to design historian Glenn Adamson, the commodity itself is a consequence of power more than a cause of it, as the commodity is shaped by power relationships *before* entering the everyday use stage. While discussing the formal similarity of two bookcases conceived by two politically antipodal designers

at-spotify and https://www.statista.com/statistics/367739/spotify-global-mau/.

218 The Greater Good Studio, for instance, proposes the following "theory of change:" "We believe that research changes design, design changes behavior, and behavior changes the world." See https://web.archive.org/web/20160918151937/http://www.greatergoodstudio.com:80/theoryofchange/.



Saul Bass-inspired cover of *Designers in Handcuffs*, one of the few books which candidly focuses on the limitation of doing a project. Designed by Lisa Buchanan.



Spread from Facebook's "Little Red Book," distributed internally to employees. Designed by Ben Barry and Tim Belonax. Texts by J Smith.

(the communist Enzo Mari and the Nazi sympathizer Ingvar Kamprad, founder of IKEA), he observes that "by the time something is designed, it is usually too late to determine its political effect. Commodities are principally the outcome of power relations, not the cause of them."[219]

Reflecting on how politics precede the use of a certain artefact, political theorist Langdon Winner mentions the infamous case of New York's Southern State Parkway bridges. These bridges, conceived by Robert Moses in the 1920s, were too low for a bus to pass. Plain bad design? Not really. Moses deliberately limited the access of disadvantaged citizens, mostly people of colour, who generally used buses to get into town.[220] This is an unusually blatant case of design wickedness (narratives of design power frequently take the form of a cautionary tale). Winner himself admits that it sounds almost conspiratorial. But from the perspective of power, it can be considered an exception more than a rule, the exception having to do not with racism and classism, which were and are both common, but instead with the coincidence of authority and execution, Robert Moses' *Herrschaft*. In fact, Winner highlights the influence that "the power broker" had over governors and presidents. How likely is it a for a designer to have full jurisdiction to, paraphrasing Winner again, embody their intention in physical form?

The symbol of the ouroboros, a serpent eating its own tail, aptly represents the relationship between design and people, since humans design things, but then those things design them back. "Design designs," as Tony Fry puts it. Design doesn't only shape use, but also meaning:

219 Glenn Adamson. "The Communist Designer, the Fascist Furniture Dealer, and the Politics of Design." *The Nation*, February 20, 2021. https://www.thenation.com/article/culture/enzo-mari-ikea-design/.

220 Langdon Winner. "Do Artifacts Have Politics?" *Daedalus* 109, no. 1 (1980): 121–36. For a more intricate account of Robert Moses' activity, see Thomas J. Campanella. "The True Measure of Robert Moses (and His Racist Bridges)." *Bloomberg*, July 9, 2017. https://www.bloomberg.com/news/articles/2017-07-09/robert-moses-and-his-racist-parkway-explained.

it transforms the way people make sense of the world, designers included.[221] However, this circularity might suggest that people are as active in the design of their environment as they are designed by it. Indeed, the environment's capacity to 'design' us is more vigorous than ours to design it. This is because we are born into a pre-designed normality, and not into a void:

> Complicating matters is the fact that we are firmly lodged in this loop before we have even developed any awareness of it, if we ever do. Just as we have no memory of how we acquired language and began to use and be shaped by it, we have no memory of how we began to use and be shaped by design.[222]

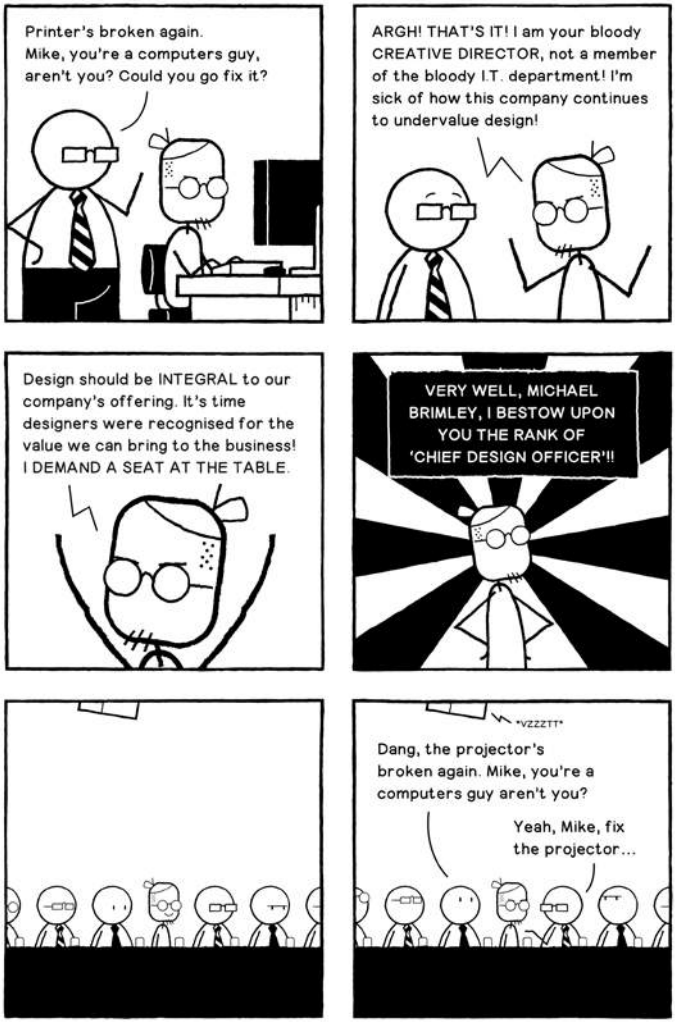
COGS AND NEEDLES

Dutch philosopher Koert van Mensvoort proposes an evolutionary reading of the development of a common product, the Gillette razor. His provocative account is radically 'object-oriented:'

> Obviously many designers and engineers have been involved in the creation of my razors over the years. No doubt these are all decent and friendly people – with good incomes too – but what more are these creators of the individual models than little cogs in the perpetuating Gillette Company? Calling them engineers and designers is arguably too much credit for the work they do, as they merely sketch the next razor model, of which one can already predict the 'innovative' new properties [...] It's not like they are in a position to think deeply on the meaning and origins of shaving, in order to

221 Fry quote in Willis, *op. cit.*, p. 86.

222 JP Harnett. "Ontological Design Has Become Influential in Design Academia – But What Is It?" *Eye on Design*, June 14, 2021. https://eyeondesign.aiga.org/ontological-design-is-popular-in-design-academia-but-what-is-it/.



Design Thinking! Comic. The strip humorously reflects on the designer's perceived authority.

> reinvent this ancient ritual. Like bees in a beehive their work is determined by the logic of the larger structure.[223]

As we have seen, it's not the first time that a designer is compared to a pollinating insect: Karl Marx deemed imagination the fundamental difference between the architect and the bee, the human and the animal.[224] Van Mensvoort reverses the axiom, showing that this faculty cannot be fully exercised within a highly crystallised or interrelated industry. In a perverse twist, the commodity is here the powerful agent shaping the work of designers. The thing makes us more than us making the thing, not only in a physical sense, with the assembly line determining the gestures of the worker, but also intellectually, with market forces setting objectives and constraints to be grappled with. The picture is bleak, probably too bleak. And yet, many designers feel trapped in such an "iron cage," to use Max Weber's notorious expression.

MoMA design curator Paola Antonelli is more optimistic. In a recent interview she states that:

> Design has a lot of power that is still untapped and unexplored. There are many different types of designers. They all have influence on our behavior. Some have fundamental, earth-shattering influence, like the designers behind apps and electronic appliances and the interfaces we use all the time. Facebook, Instagram, Twitter. That's design of the system but also the interface. Other designers might have less effect, but they all act under different pressure points under the great acupuncture system of human life.[225]

223 Koert van Mensvoort and Hendrik-Jan Grievink. *Next Nature: Nature Changes Along with Us*. Barcelona: Actar, 2015.

224 See chapter 2.

225 Paola Antonelli. "MoMA curator: '[Humanity] will become extinct. We need to design an elegant ending.'" Interview by Suzanne LaBarre. *Fast Company*, January 8, 2019. https://www.fastcompany.com/90280777/moma-curator-we-will-become-extinct-we-need-to-design-an-elegant-ending.

While acknowledging the sheer variety of existing designers (whose number, especially in the digital sector, is growing),[226] Antonelli adopts a poetic image that conflates them, with the risk of mitigating the immense power differential that designers can have. Furthermore, the acupunctural analogy doesn't say much about the specific power that design holds and the peculiar way in which such power is exercised.

TOO LOW IN THE POWER STRUCTURE

The case of "earth-shattering influence" of apps and digital devices deserves further scrutiny. Interdependence, complexity and hierarchy limit the designer's effort to impress their personal or professional agenda on the artefacts. Like a cobweb, the structures of production, distribution and consumption form a tense and yet mobile mesh of interrelations. This means that a radical intervention is unlikely, and even more unlikely is the possibility that the designer alone or in a team would take such action. The power of design is before everyone's eyes: design has indeed changed the face of our planet, our way of conceiving it and the way we exist within it. Design is indeed "one of the most powerful forces in our lives," as Alice Rawsthorn argues.[227] But the fact that design is powerful doesn't necessarily imply that designers are.

This seems to be true both for the everyday designer, be they a smalltown solo practitioner or a corporate employee, and the design maestro. In 2004, Enzo Mari published a paid advertisement in the architectural magazine Domus.

226 See Forrester. "The $162 Billion Design Industry Won't Stop Growing." *Forbes*, March 25, 2021. https://www.forbes.com/sites/forrester/2021/03/25/the-162-billion-design-industry-wont-stop-growing/.

227 Rawsthorn quoted in Steven Heller. "'Design Is One of the Most Powerful Forces in Our Lives.'" *The Atlantic*, March 13, 2014. https://www.theatlantic.com/entertainment/archive/2014/03/design-is-one-of-the-most-powerful-forces-in-our-lives/284388/. The British critic's statement is unconsciously but fittingly reminiscent of the opening line of Haug's 1971 book on commodity aesthetics: "Commodity aesthetics is one of the most powerful forces in capitalist society." Wolfgang Fritz Haug. *Critique of Commodity Aesthetics: Appearance, Sexuality, and Advertising in Capitalist Society*. Cambridge, UK: Polity Press, 1986, p. 10.

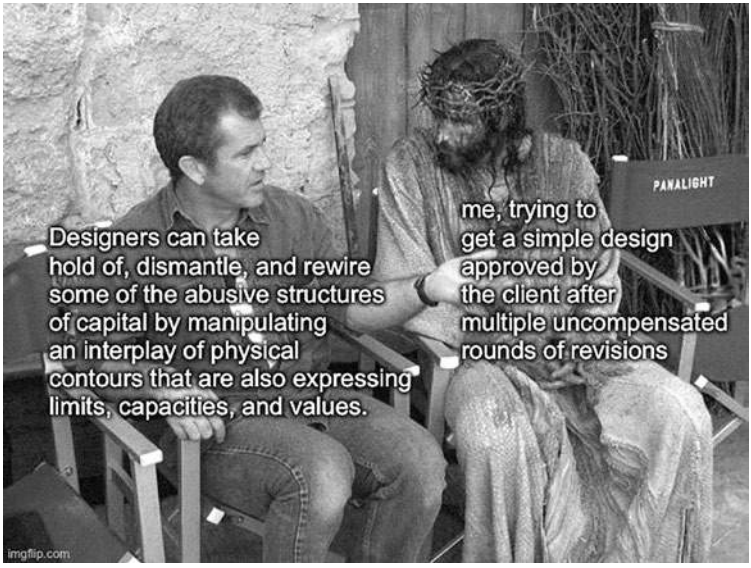
It read: "Highly experienced and qualified designer / seeks desperately / not only for himself / YOUNG ENTREPRENEUR." He did so to highlight the fact that in his line of work the one that truly allows things to happen is the entrepreneur.[228]

On more than one occasion, Don Norman raised doubts about Victor Papanek's oft-quoted claim that "There are professions more harmful than industrial design, but only a very few of them." According to Norman, the motto is "well-intended but wrong: [Papanek] gives too much credit to designers who are generally too low in the power structure to matter." He adds: "Designers then (and today) are servants of their masters: Their client or employers. Design is a tool of industry, and because industry does not pay for externalities, they think nothing of destroying the environment in mining for maters (*sic*), in manufacturing, and in disposal of poisonous stuff."[229] Norman is echoed by John Maeda, design-star who popularised the idea of a *design-led* company. In 2019 he made a sobering declaration: "In reality, design is not that important." Here's his reasoning: "We know that tech people are in charge of the world: They can approve pull requests and they push the code out and right or wrong happens." Designers, far from being the centre of the design process, are ancillary to it. Would it be good, then, to change this state of affairs? Not according to Maeda, who thinks companies led by designers might achieve resonance among a design-savvy crowd, but their technical structure might be brittle, leading eventually to failure.[230] Massimo Vignelli, interviewed together

228 Enzo Mari. "Inserzione a Pagamento." *Domus*, April 2004. It is worth pointing out that Mari still had a pretty romantic idea of the impact of design. From the ad: "A design entails a passionate alliance between two persons: a soldier of utopia (the designer) and a tiger from the real world (the entrepreneur). It is always the tiger, if he wishes, who can allow at least a fragment of utopia to be attained. Today tigers seem to be extinct." Here, Mari didn't consider the idea that the tiger of the real world, the entrepreneur, might be caged in economic imperatives.

229 Don Norman. Emails to the PhD-Design Mailing List, August 9, 2020 and March 18, 2021.

230 Katharine Schwab. "John Maeda: 'In Reality, Design Is Not That Important.'" *Fast*



@ethicaldesign69 (2021). The meme is a sarcastic response to a claim by Keller Easterling, represented on the left.

with his wife Lella by Barbara Radice, adds a tint of moralism to this socially determinist view – right after having bragged about their Rolls-Royce:

> [Transforming society] was always our utopia, the dream of Unimark, the reason we went to America. Until we realized that instead design cannot transform society but that it is society that determines design. There is nothing to be done. Every society has the design it deserves.[231]

So, what is the real chance of designers injecting their intentions into the design and therefore the world? First, Winner reminds us that the leeway for changing an artefact is greater when its typology is introduced for the first time: after that phase its malleability decreases and the artefact becomes stabilised. But not every designer can be a pioneer. Most will work with consolidated types of product, such as the razor. Second, the actual position of the designer in the hierarchy of an organisation matters: generally too low, according to Norman. And one can assume that the larger the structure, the more difficult it is to rise to the top.

THE HUMAN FACTOR

These accounts of design power, devoid of any trace of agency or free will, might sound depressingly deterministic. Design historian John Heskett offers an alternative point of view:

> Whether executed well or badly (on whatever basis this is judged) designs are not determined by technological processes, social structures, or economic systems, or any other objective source. They result from the decisions and choices of human beings. While the influence of context and circumstance may be considerable, the

Company, March 15, 2019. https://www.fastcompany.com/90320120/john-maeda-in-reality-design-is-not-that-important.

231 Massimo Vignelli and Lella Vignelli. "Emigranti di lusso." Interview by Barbara Radice. *Modo*, March 1981, p. 21. Thanks to Michele Galluzzo for this source.

> human factor is present in decisions taken at all levels of design practice. With choice comes responsibility. Choice implies alternatives in how ends can be achieved, for what purposes, and for whose advantage. It means design is not only about initial decisions or concepts by designers, but also about how these are implemented and by what means we can evaluate their effect and benefit.[232]

Heskett emphasises the multitude of choices that are made throughout the design process. These choices, as we have seen, are only partially made by designers. The "human factor" in these decisions is the very reason we should not consider this decision-making process a linear, logical one, but one that is affected by temperament, irrationality and chance. Out of the Brownian motion of all the particles involved (those who make choices) on the micro scale, a direction emerges on the macro scale. Some decisions, however, are more *decisive* than others. "Taken by whom?," Maldonado asks. He wonders about "the requirements needed to turn Design into Revolution; what power structures, existing or to come, must delegate to designers the responsibility for radically changing all the technical structures of the human environment, in a planetary-scale operation."[233] Anthony Masure, head of research at the Geneva University of Art and Design, has an answer:

> I'm not a pessimist, but I think designers have little power. I think a policy maker, for example, has a lot more impact on society. The question is why we have come to this. I think it's a problem, it's also the designers' fault. Most have relinquished power to decision-makers. Designers rarely place themselves in these decision-making

232 John Heskett. *Design: A Very Short Introduction*. Oxford: Oxford University Press, 2005, pp. 5-6.

233 Maldonado, *La speranza progettuale*, *op. cit*., p. 58.

> positions... In France, for example, we will find relatively few designers who create businesses. We will obviously find designers in companies, but few who carry projects from scratch, who initiate or who join as partners and not as executors.[234]

Masure tends to idealise the policy maker as the figure whose decisions matter the most. This is not uncommon. Here, we can obliquely recognise the synthetic urge to determine lower processes, designing being one among them. How can designers be authentic decision-makers? One option is to become entrepreneurs, that is, to hold the decision-making reins of a project in its entirety. In the '70s Victor Papanek used the case of designers as entrepreneurs as evidence to demystify the "myth of the designer's lack of control:" the bad products that they create while controlling every aspect of the design, production and marketing, were a proof, albeit negative, of design agency.[235] Both the designer as entrepreneur and the designer as policy maker betray an isolationist attitude: a technocratic image of decision-making that hides, once again, its chaotic, conflictual and plural dimension – its politics. Obviously, there is no such thing as full control. The sum of all decisions made by managers, engineers, clients, competitors, etc. provides only a fragile and provisional synthesis. Thus, synthesis is not so much the 'product' that the designer can offer but the emergent quality of a series of political forces clashing. Free will exists, but since it is plural, it leads to indeterminate results.

POSTPARTUM

In 1993, Adam Richardson recognised not only the designers' lack of control, but also their unwillingness to admit it. He noticed an identity crisis in industrial designers

234 Anthony Masure. Interview by Karl Pineau, August 2019. https://livre-ethique-numerique.designersethiques.org/content/interviews/interview-anthony_masure.html.

235 Papanek, *op. cit.*, p. 232-3.

Tweet by Billy Fleming (2021).

Screenshot of the website of the What Design Can Do initiative.

due to the fact that they "do not do what they generally say they do. That is, they have much less control over the process of product development than one might be led to believe by the common rhetoric. In addition, how users and cultures respond to the products which designers help create is not well understood." Richardson saw two specific limits in the way design impact is conceptualised. First, the designer's ability to shape the outcome of the design process, namely, the artefact. Second, the ability of an artefact to transforms culture *directly*, and in the way originally intended by the designer. Richardson's second point stands on semiotic grounds: inspired by Roland Barthes, he argued that every user would interpret the product differently, and shared culture would weigh on this interpretation far more than the designer's intention: "The culture-product interaction is a postpartum activity, so to speak, one that takes place after actual production." Referring to two distinct Barthesian "deaths" of the designer, Richardson showed how the designer's understanding of control is illusory:

> The issue here is that of control, or rather the illusion of having it when in fact it is in the hands of others. In the case of the first death, designers work as though they have control over the product once it enters the use-place. Although, as we have seen, this is not so. With the second death, designers are primarily passive spectators when it comes to initially deciding the function of products, and conveniently submerge this fact by stating, "I designed that."[236]

The first illusion, that of controlling meaning and interpretation is particularly present in the field of graphic design, where, for the most part, it is very difficult – if not impossible – to decipher the cultures that digest the

236 Adam Richardson. "The Death of the Designer." *Design Issues* 9, no. 2 (1993): 34. https://doi.org/10.2307/1511672, p. 41.

signs. As a result, cultural purposes acquire priority over cultural results. The quality of a project tends to align dangerously with its (good) intentions. This leads to a 'moralism of things,' exemplified by the irrelevance of poster competitions with social aims. Rather than 'educating the public,' these competitions seem to serve another function: to sanction the cultural and moral distance that separates designers and users.

Especially since the '90s, a novel idea of the role of design has gained momentum: the idea that design can be not just about objects or services hitting the market, but about projects acting as rhetorical devices, artefacts or scenarios meant to prove a point or mobilise conflicting worldviews. Such design outputs should make people think about and discuss relevant issues, perhaps even making them change their point of view. These products (be they speculative devices or fiction films) don't belong to the everyday life of the users but are divorced from it: they act like a commentary. This is what Bruce and Stephanie Tharp broadly call "discursive design."[237] The general framework they propose is more about the intention than the outcome of a design endeavour. Design critic Francisco Laranjo sheds lights on their emphasis on intentionality over result. Discussing Speculative Design, he highlights some key aspects of design equally applicable to the field of graphic, product and interaction design as "generation of discussion as an end in itself, reducing 'debate' to being mentioned in a press-release, a newspaper, a design blog."[238]

Such media insularity is a symptom of an excessively humanistic turn. Here, the textual production of

237 Bruce Tharp and Stephanie Tharp. "The 4 Fields of Industrial Design." *Core77*, January 5, 2009. https://www.core77.com/posts/12232/The-4-Fields-of-Industrial-Design-No-not-furniture-trans-consumer-electronics-n-toys-by-Bruce-M-Tharp-and-Stephanie-M-Tharp. See also Anthony Dunne and Fiona Raby. *Dunne & Raby*, 2009. http://dunneandraby.co.uk/content/bydandr/36/0. I shall return to these currents in chapter 6 and 7 to offer a theory of their popularity.

238 Francisco Laranjo. "We Don't Need Speculative Design Education, Just Better Design Education." *SpeculativeEdu*, November 17, 2020. https://speculativeedu.eu/interview-francisco-laranjo/.

speculative design is larger and more important than its artefactual counterpart, raising the suspicion that the artefact is not even necessary to the alleged generation of debate. For Daniel van der Velden, it is the very idea of debate that, in the age of platforms, vacillates:

> The structure that emerges out of the fusion of these technologies and platforms with each other and the world makes the idea of having a 'debate' seem quaintly inadequate, especially since most words lag behind the normlessness that is this structure's *de facto* governing force. In devouring the real by indexation, taxonomy, and optimization, technology platforms pre-own our critiques of them, meaning that our subjectivity and 'debates' are fully pre-included, pre-implied in their own dynamics and models.[239]

FLIPPING THE TABLE

In the last decade, designers who work with and for companies, especially in the UX field, have clamoured for a 'seat at the table.' They demand access to what they consider the true locus of decision-making. And yet, the idea of the seat at the table can be seen as a theory of power in itself, not dissimilar from that proposed by Carl Schmitt. According to the German jurist, power always relies on experts: a king is, to a certain extent, dependent on his counsellors.[240] Here, power coincide with *access to the locus of power*.

Clearly, asking for a seat implies its current absence. Design's preoccupation with decision-making can be read as a manifestation of its relative powerlessness: the one who has power doesn't ask for it. Not everyone agrees, though. According to Evan Osherow, designers are in the process of obtaining that coveted office chair:

239 Daniel van der Velden. "Lyrical Design." In de Vet, *op. cit.*, p. 58.
240 Carl Schmitt. *Dialogo sul potere*. Milano: Adelphi, 2012.

> We now have chief product officers. The McKinsey Design Index is a commonly known metric showing the business value of design. And the demand for UX designers is expected to continually grow.[241]

In other words, design is a factor of company growth, designers have pompous titles (and incomes) and the demand for them is expanding. This perspective is shared by Kate Aronowitz (Google Ventures): "We're making key strategic decisions and helping to shape the direction of companies. We got here because we proved design can solve big problems in a way that others cannot."[242] Daniel Burka questions these equivalences with rare frankness:

> Let's be honest. At some companies in 2020, designers make a lot of money, have important-sounding titles, and get invited to the leadership table. But even at huge companies with famous design teams, design loses many key battles when it comes to creating good user experiences. We have achieved $300,000 salaries but we have little of the power. What's the point in driving a BMW if we can't make good products for our users?[243]

It would seem that the designer's authority is recognised and remunerated as long as it aligns with the goals of the company. Put otherwise, the high bonuses and perks, more than a manifestation of design power, are a proof of the acquiescence or even complicity of designers with the cause of the company.

But what if power within an organisation doesn't

241 Evan Osherow. "Designers, Stop Asking for a 'Seat at the Table.'" *UX Collective* (blog), March 7, 2020. https://uxdesign.cc/designers-stop-asking-for-a-seat-at-the-table-4ab933d7037f.

242 Aronowitz quoted in Katharine Schwab. "Take The Survey: Do Designers Have A Seat At The Table?" *Fast Company*, April 25, 2018. https://www.fastcompany.com/90169455/take-the-survey-do-designers-have-a-seat-at-the-table.

243 Daniel Burka. "Mission Accomplished? The Hard Work of Design Is Still Ahead of Us." *Thinking Design* (blog), July 9, 2020. https://medium.com/thinking-design/mission-accomplished-the-hard-work-of-design-is-still-ahead-of-us-a4e47e5a0c8c.

coincide with the authority and prestige it gives? What if having power doesn't mean just having the possibility to take part in the conversations that lead to decision-making, but to *oppose* a decision altogether? An understanding of power as such is intrinsically conflictual, agonistic, if not antagonistic. Is power always confrontational? Is true power the power to veto? In a Twitter conversation, user @maetl quipped:

> In design leadership conversations for years, there was a thing about 'design needing a seat at the table' in orgs and governance – in hindsight this was all wrong, and the conversation should have been about flipping the table.[244]

TAKE HOLD, DISMANTLE, REWIRE

Claiming design's power through intricate formulations, some academics end up causing a sense of detachment among practitioners. Symptomatic of this approach is the latest book by urbanist Keller Easterling, entitled *Medium Design*. Here, the designing activity is extended to encompass activism, spatial organisation and politics. A quote from an article summarising the book's argument provoked a small social media stir:

> Designers can take hold of, dismantle, and rewire some of the abusive structures of capital by manipulating an interplay of physical contours that are also expressing limits, capacities, and values.[245]

Twitter meme account @ethicaldesign69,[246] who can be considered the most inventive public voice of design disillusion, matched the quote with "me trying to get a simple design approved by the client after multiple

244 https://twitter.com/maetl/status/1432190569854345216.

245 Keller Easterling. "On Political Temperament." *The Double Negative*, January 18, 2021. http://www.thedoublenegative.co.uk/2021/01/on-political-temperament-keller-easterling/.

246 At the time @neurotic_arsehole.

uncompensated rounds of revisions" in a gruesome template derived from the backstage of Mel Gibson's *The Passion of the Christ*. The meme went relatively viral.

Despite its abstruseness, Easterling's statement makes sense. Abusive capitalist structures can be curbed and mutated. Designers are indeed able to intervene. The issue here is not one of veracity but one of rhetoric: the grand statement that – if you're honest to yourself – can only make you feel pretty small. The register of the quote doesn't match the register of everyday life, except if your everyday life is mostly made of design conferences. The meme derives its strength from the distance that separates expectations from reality.

Another meme inverts Keller Easterling's point. Here, we see the Suez Canal blocked by a colossal container ship (representing Capital), while a small excavator (design) has been sent to unstuck it. The allegory could have served as an illustration for Kevin Rogan's review of Easterling's book: "The medium designer does not exist and never will exist; capital executes the plan and the designer must dance along as they always have." The medium designer is, according to Rogan, a theoretical fiction, one used to empower the author rather than the architectural field.[247]

Who's right? It seems that ideas about design's power are more a matter of attitude than analysis – a matter of optimism. Antonio Gramsci famously said: "I'm a pessimist because of intelligence, but an optimist because of will." Here, we have a peculiar reversal: infusing intelligence with optimism, as academics and practitioners often do, unwittingly spurs a pessimism of the will.

THE DESIGN AND THE PLAN

At this point, it is useful to make a distinction. According to Indian-American anthropologist Arjun Appadurai, the

247 Kevin Rogan. "Keller Easterling's Medium Design Ignores the Role of Power in Design." *The Architect's Newspaper*, February 22, 2021. https://www.archpaper.com/2021/02/keller-easterling-medium-design-review/.

categories of design and planning share many similarities: they're both "as old as humanity," they both are linked to a "universal inclination to utopias," they both relate to "the fear of disaster and dislocation." But they also differ – design is rooted in the explosion of industry and the growth of markets, while planning is linked to the goals and ambitions of the state. This is how Appadurai understands the separation between planning and design:

> Planning is more explicitly concerned with sustainability – both social and environmental – than design, and so it has a regulatory relationship to design, just as design has a regulatory relationship to fashion.[248]

Admittedly, it is not easy to make an essential distinction between designing and planning. It is tempting to assign a social value to the plan and a technical value to the design, but that would be a mistake, since any social decision is a technical one and "technology is society made durable."[249] Instead, we could simply posit a hierarchical relationship: the plan rules over the design. In such case, we need to add a political component, more evident in planning than in design. Our current masterplan is not even run by a human entity, but dictated by the highly dynamic machinic assemblage we call capitalism.[250] In this scenario, we can think of designing and planning in terms of strategy and tactics. The plan is not the cumulative result of various

248 Arjun Appadurai. *The Future as Cultural Fact: Essays on the Global Condition*. London: Verso, 2013, p. 266.

249 Bruno Latour. "Technology Is Society Made Durable." *The Sociological Review* 38, no. 1 (May 1, 1990): 103–31. https://doi.org/10.1111/j.1467-954X.1990.tb03350.x.

250 "In collecting goods and people from around the world, capitalism itself has the characteristic of an assemblage. However, it seems to me that capitalism also has characteristics of a machine, a contraption limited to the sum of its parts. This machine is not a total institution, which we spend our life inside; instead it translates across living arrangements, turning worlds into assets. But not just any translation can be accepted into capitalism. The gathering it sponsors is not open-ended. An army of technicians and managers stand by to remove offending parts – and they have the power of courts and guns. This does not mean that the machine has a static form." Anna Lowenhaupt Tsing. *The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins*. Princeton, NJ: Princeton University Press, 2021, p. 133.

designs, but the force able to select certain designs by excluding others. The plan consists of an impersonal strategy, while design is a personal tactic. According to Michel de Certeau, tactics are deployed by those who do not possess a full view of the strategic field.[251] In the case of design, it is not the view that is lacking, but the means to engage with the perceived field. This is the tragic state of design: seeing the field but not being able to modify it.

A relational understanding of design power can now be provided: designers might exert a certain influence over the *design* but they generally have little or no control over the *plan*. As the plan governs the design by structuring its field of action,[252] the plan is intrinsically unchangeable by the design; what is more: the design is a product and a function of the plan. Whereas design is a pawn, the plan is the chessboard. The plan, as Appadurai puts it, "limits the infinitude" of the design.[253] In order to change the plan, design cannot remain design. It has to enter the chaotic realm of agonism and antagonism. That of the plan is an "uneasy space" where action evolves into political action, be that in the form of activism, protest, revolt, etc. In other words, design needs to transubstantiate into a counter-plan, which will in turn generate new designs.

The designer who plots a counterplan is, while doing so, not a designer: they may be a citizen, a politician, a terrorist, an activist, a lobbyist, etc., namely, a political actor. They become a designer again when deploying the plan through design. Trying to get hold of the plan means

251 Langdon Winner mentions two types of choice when it comes to work with new technology. The first is a binary one: should it be developed in the first place? Yes or no. The second has to do with the way in which such technology should be developed. The former is normally out of the designers' scope, while the latter is within their purview: designers aren't asked *if* something should be done or not, they are asked *how* such a thing should be done. The first question is a matter of the plan, which can be in the hands of authorities like the firm or the state; the second relates to the design, which can be, to a certain extent, in the hands of the designer, unless the designer doesn't actively operate to obstruct the plan. Winner, *op. cit.*, p. 127.

252 Here, I'm borrowing Michel Foucault's notion of governmentality. For a basic introduction, see https://en.wikipedia.org/wiki/Governmentality.

253 Appadurai, *op. cit.*, p. 265.

setting aside one's profession. Maldonado was interested in how to change the world without changing one's job.[254] The relationship between the plan and the design suggests that to do so one needs to transcend the profession, as the profession is governed by the plan.

DAMNED IF YOU DO, DAMNED IF YOU DON'T

Designers rarely exert power as a direct form of command (the instructions they formulate as blueprints, prototypes, etc. are always prone to negotiation and to the final say of the client), so they often speak of their power indirectly, by means of its implication. The keyword is "responsibility." In fact, the most memorable aphoristic legacy from Spiderman's uncle, namely, that "with great power comes great responsibility," is frequently heard during public design events. Here, responsibility implies power. So, a rhetoric of responsibility can be understood as a form of 'power signalling,' or even "hero bait."[255] In the '70s, German philosopher Wolfgang Fritz Haug attributed an "exaggerated illusion of use-value" to commodities,[256] but the same can be said of the design field as a whole. Consider designer Mike Monteiro's *Ruined by Design.* Monteiro opens his book on design ethics with the following statement:

> Design is a craft with responsibility. The responsibility to help create a better world for all. Design is also a craft with a lot of blood on its hands. Every cigarette ad is on us. Every gun is on us. Every ballot that a voter cannot understand is on us.[257]

254 Maldonado, *La speranza progettuale, op. cit.*, p. 124.

255 "Even though the books that they promote are likely more nuanced, the claims have been reduced to a power promise, divorced from any limits or doubts. Let's call them hero bait – invitations to play the role of a saviour, carefully crafted for a designer who feels stuck at their desk doing non-meaningful work." Gijs de Boer. "Between Fox Traps and Hero Bait." *Design Drafts*, no. 1 (2023).

256 Haug, *op. cit.*, p. 36.

257 Mike Monteiro. *Ruined by Design: How Designers Destroyed the World, and What We Can Do to Fix It*. San Francisco, CA: Mule Design, 2019.

Critical Graphic Design (2015).

Who actually is this "us?" That's not very clear, since the author, following Jared Spools, argues that "[a]nyone who influences what the design becomes is the designer."[258] Such a generalisation avoids the fundamental question of the *specific* power that design has and justifies the examples of good and evil the author gives in the book: almost none of them involve designers *stricto sensu*.[259] Thus, his appeal to an ethical conduct becomes a generic one, one that masks hierarchies and relations of authority. It could even be read as a way to please the readership by granting them a potency that they (individually, as designers) don't have. In fact, the solution provided by Monteiro, when the company for which the designer works is crooked beyond repair, is to quit.

This is a not unusual example of self-aggrandisement by over-responsibilisation, an idealised narrative that sees design more as an individual moral framework than a practical philosophy. The logic is akin to that of ethical consumption, where the centrality of individual ethics compensates for design's subordination. Undoubtedly, the designer affects the world with their work and their choices, but this generally happens in unspectacular ways which wouldn't be worthy of the Spiderverse. Indeed, most designers concur that the quality of their work depends solely on the extent to which the client allows it.[260]

While studying schizophrenia, Gregory Bateson and his colleagues developed the concept of the double bind. This takes place when a figure of authority makes an injunction on the subject while implicitly expressing a secondary demand that contradicts the primary one: "damned if you do, damned if you don't."[261] In our case, the pri-

258 https://twitter.com/jmspool/status/836955987860914176.

259 Among the cases of virtue: a doctor halting the distribution of a painkiller with tremendous side effects; among those of wrongdoing, an engineer's complicity in the 2015 Volkswagen emissions scandal.

260 From *Graphic Designers Surveyed*, to the question "Is it true that your work is only as good as your client will let it be?" 62% of the respondents agreed, *op. cit.*, p. 288.

261 Gregory Bateson, Don D. Jackson, Jay Haley, and John Weakland. "Toward a

mary injunction is to design ethically, while the secondary one is dependent on daily requirements of the job that leave no room for any kind of ethical consideration. The result "leaves the subject torn both ways, so that whichever demand they try to meet, the other demand cannot be met. 'I must do it, but I can't do it' is a typical description of the double-bind experience."[262] On the one hand, designers are told that their work has a profound, terrible impact on society; on the other hand, society utilises them as "mere executors of strategic decisions made well before they were employed."[263] To understand design disillusion in this context, we can implement Expectation confirmation theory (ECT): our satisfaction with a product or a service is not absolute, but relative to what we expect from it: a product is good when it positively exceeds our expectations.[264] So, if our expectation as designers is to save the world, we get disappointed when all we are asked to do is make the logo bigger.

Chermayeff & Geismar – who have done branding for oil companies, banks and pharmaceutical companies – explain their view on corporate identity design as follows:

> When we create a great logo for an environmental organization, we do not see ourselves as saving the planet. In the same way, we cannot take responsibility for the 'evil' actions of corporations we brand.[265]

While this might sound like a dismissal of complicity, it could also be read as a sober understanding of design

Theory of Schizophrenia." *Behavioral Science* 1, no. 4 (1956): 251–64. https://doi.org/10.1002/bs.3830010402.

262 https://en.wikipedia.org/wiki/Double_bind.

263 Gerritzen and Lovink, *Made in China…*, *op. cit.*, p. 57.

264 See Richard L. Oliver. "Effect of Expectation and Disconfirmation on Postexposure Product Evaluations: An Alternative Interpretation." *Journal of Applied Psychology* 62, no. 4 (1977): 480–86. https://doi.org/10.1037/0021-9010.62.4.480.

265 Aaron Kenedi. "Marks Men: An Interview With Ivan Chermayeff, Tom Geismar, and Sagi Haviv of Chermayeff & Geism." *Print*, September 14, 2011. https://www.printmag.com/designer-interviews/marks-men-an-interview-with-ivan-chermayeff-tom-geismar-and-sagi-haviv-of-chermayeff-geismar/.

power. Is a studio that hasn't worked for any problematic organisations irreproachable? Or did they simply never receive an unethical brief? Does a rhetorical display of virtue conceal powerlessness and detachment? When there is no chance to 'sin,' the issue of responsibility can become an indirect form of self-congratulation.

One of the main issues of responsibility is that it is hard to pinpoint. Philosopher Vilém Flusser understood this well:

> The division of labor has as a result the impossibility to assign responsibility to any one of the people involved in the process. If a robot kills, who shall be held responsible: the designer of the robot, of the knife [held by the robot], or the man who has calculated the robot program? It is not possible to ascribe ethical responsibility to some error in design, in programming, or in production. And what about assigning the ethical responsibility to the industry that produced the robot, or the whole industrial complex, or finally the whole system of which that complex is part and parcel?

While he pointed out the complexity of conceiving a design ethics, he believed in its necessity: "unless some kind of ethics in industrial design be elaborated, total ethical irresponsibility will follow."[266]

DESIGN ETHICS? NO THANKS!

In a talk given at IxDA Budapest in 2020, Berlin-based designer Cade Diehm questions the design ethics frameworks that consider the ills of a technology as the direct result of decisions made by a designer or a design team. Designers who adopt such frameworks insist on a responsibility to assess their own motivations over what they choose to build.[267] The limit of this approach derives,

266 Vilém Flusser. "Ethics in Industrial Design?," March 20, 1991. Transcript of a talk given at the Eindhoven symposium.

267 https://vimeo.com/483988793.

according to Diehm, from its reliance on dubious notions of consent and inclusion. Emphasising the former means involving users in decision making, but informed consent, even when powered by a seamless user interface, "crumbles at scale:" there are simply too many intertwined systems with their own legal and technical complexities (such as terms of service and privacy settings) to take into account. Similarly, the notion of inclusion in a certain kind of technology doesn't take into consideration the actual political power to opt out of it.

Diehm gives the example of biometric data gathering. While the inclusive perspective focused on issues of representations (namely, the racial bias of facial recognition systems), the 2019 Hong Kong protests showed its failure: the only possible way for the protesters to avoid surveillance was a Luddite one: tear down the poles equipped with sensors and cameras. It appears, then, that agency and power are more concrete when expressed negatively: power, here, is the power to prevent, the agency of *not* doing something.

For Diehm, design ethics is a form of reductionism "that allows designers to escape the scrutiny of their work." It's a simplistic answer given to Flusser's complex questions. Not unlike Guy Julier's idea of design as a laxative,[268] Diehm sees technology as an "accelerant" which makes societal problems metastasise through quick fixes that, counterintuitively, *impede* change. Therefore, he invites us to reject the seductive image of design ethics, as it masks the ongoing trajectories that designers follow, consciously or not.[269]

268 Julier, *op. cit.*, p. 174.

269 Similarly, Susanne Bødker and Morten Kyng find the excessive focus on ethics within the field of Participatory Design troublesome. They write: "We find the issues of how researchers should behave, ethical issues, important. At the same time, we find it problematic that politics has been reduced to how researchers should behave and act fairly when involving users in projects. We would argue that politics is more than fairness and ethics, and we hypothesize that the tendency to avoid profound conflicts also makes it difficult for current PD to engage in some of the more controversial areas where researchers could really help people by taking side with them." See "Participatory Design That Matters –

BAD BOYS

In a time when modernity manifests its tragic shortcomings, designers' emphasis on responsibility and ethics is increasingly linked to design's culpability. Designers have exhumed the *confiteor:* "mea culpa," they utter. The international organisation What Design Can Do, known for its social design initiatives, exemplifies this attitude. "I've been a bad designer" is the motto of its No Waste Challenge. Here, a group of renowned designers and design theorists from around the globe confess that they have contributed to waste and pollution, like naughty kids apologising for their mischief.[270] Under the influence of capitalism, design has adopted a "new spirit" that enables it to capitalise on the criticism it receives: design actively assimilates accountability and converts it into a form of self-reflective sentimentality associated with a sense of responsibility.

There is a parallel with the world of tech. A counter-narrative, popularly dubbed *techlash*, is emerging around digital platforms. It stands in opposition to the messianic narrative of Big Tech amplified by the media throughout the '10s. Here, mistakes are acknowledged. Careers are built out of an anti-tech evangelism and boosted by public acts of atonement. The irony is that these negative evangelists are often the very same people at the top of the professional elite ladder, now repentant. If it's true that, as Beatriz Colomina and Mark Wigley point out, design has always been a pedagogic and moralising project, it now acquires the double role of the judge and the culprit.[271] A strange disassociation takes place, where design is simultaneously the good force and the evil one. This circularity is perfectly exemplified by a statement in an Arte documentary entitled, *ça va sans dire*, The Power of Design:

Facing the Big Issues." *ACM Transactions on Computer-Human Interaction* 25, no. 1 (February 13, 2018): 4:1-4:31. https://doi.org/10.1145/3152421, p. 9.

270 https://www.whatdesigncando.com/stories/havent-we-all-been-bad-designers/.

271 Colomina and Wigley, *op. cit.*, p. 77.

> Many designers today agree: that design is partly to blame for our current problems, and that the solutions to these problems should come from design.[272]

POWER USERS

The amplification of potency granted by technics complicates the question of power. As Carl Schmitt points out, power exceeds the one who holds it, whoever they may be. Power is an autonomous force – a Leviathan. What most people refer to when they speak of the power of design, is actually the power of technics, which has grown immensely since the beginning of modernity and is largely independent from the will of the people who make use of it. Because of the division of labour, the humblest of designers cannot help contributing, at least indirectly, to 'earth-shattering' endeavours. This is why we shouldn't confuse the autonomous, leviathanic power of technics within modernity (in Spinozian terms, *potentia*) with the indirect power held by the designer (*potestas*) which has to do with proximity to the locus of direct power.[273]

This chapter began with a military analogy: an army of commodities, shaped by the designer, that enter the market and drastically change the life of consumers. What if, when it comes to design power, it is still that way? Everyday designers are, after all, themselves consumers: reflex cameras, ergonomic mice, Pantone mugs, etc. Most of these items – in particular the Apple devices – provide both the tools of trade and a mythology of good design (which, as we have seen, is now in decline). What if the designers' multitude is defined more by their lifestyle-oriented equipment than the actual work they do with it? The MacBook can then be seen as the Ford T of the everyday designer: a product with a massive concrete and imaginative presence

272 Reinhild Dettmer-Finke (director). "The Power of Design." *Arte*, 2021.
273 Gilles Deleuze. *Spinoza: Practical Philosophy*. City Lights Books, 1988, p. 97.

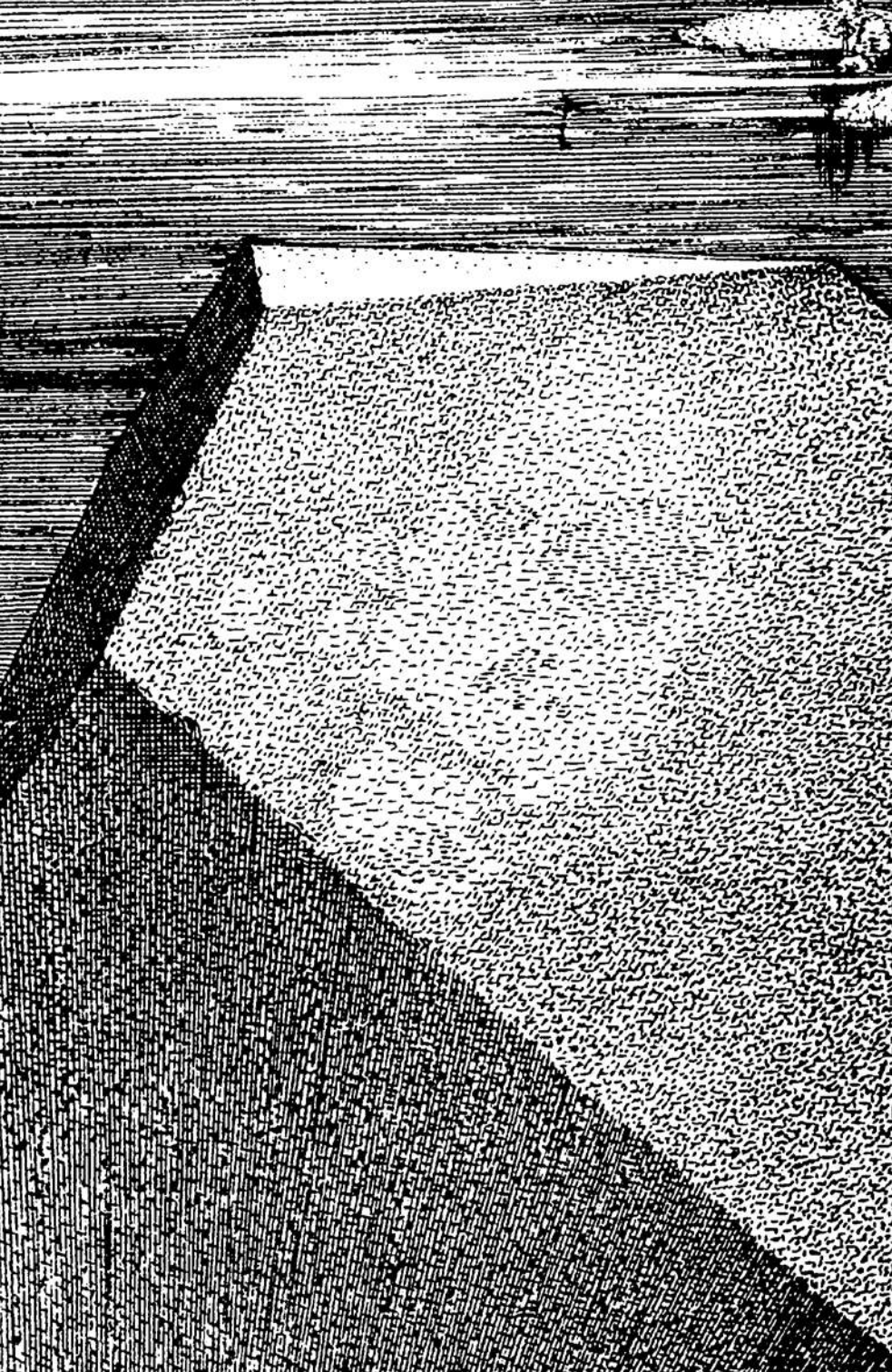
Poster by Valerio Nicoletti (2023) that shows the pyramid of power in design.

spreading the gospel of impact and creativity. From this point of view, design culture appears highly shaped by the dynamics of prosumerism (production as consumption). A depressing suspicion arises: for all the talk about design's role in society, the big-scale impact on the world left by designers as a demographic cohort might lie less in the design work they do than in the consumer choices they make; the laptop they buy counting more than the poster they design with it.

So, power becomes merely a brand idea. However, an idea that is frighteningly similar to the professional narrative of the power of design. Is design culture itself a product to sell? In an issue of *C Mag* on graphic design, curators Chris Lee and Ali S. Qadeer notice that one of the main ways in which design relates to power is by obfuscating its relationships.[274] We can argue that the general framework of professional discourse contributes to this obfuscation, concealing the power relationships in which designers themselves are enmeshed. In certain cases, then, quitting not only the job, but the profession, becomes the only viable solution. This would show that when designers have no power *as designers*, they might do as workers and perhaps citizens.

274 Chris Lee and Ali S. Qadeer. "Editorial." *C Mag*, 2019, p. 10.

Part II. Reality



Chapter 5.

Form Follows Format: On Semi-Automation and Cultural Professionalism

"I see graphic design as something that began in the 1920s and died in 2008. It doesn't exist anymore. And the reason it doesn't exist is because it's completely infiltrated everything."
– Michael Rock, 2021[275]

"Wouldn't it be better if websites just made themselves?"
– The Grid, 2014[276]

"What if – hear me out – WE are the Design Tools"
– @ktsuskin, 2019[277]

You are on the subway. Look at all these people staring at their phones. What are they doing? This woman is chatting, that guy is mindlessly playing Candy Crush, and then there are these kids crafting visual content for social media like Instagram, Twitter, TikTok. Using pretty powerful editing apps, they're creating compositions, correcting colours, tweaking typography, adding stickers and animations. In other words, they are *designing*. And not in a vague, generic sense: they are performing the very same activity as that of a graphic designer, that is, manipulating symbols and pushing pixels around – using just their thumbs.

Sure, "design is a job,"[278] but one could convincingly argue that today the bulk of the designing is performed by casual users – who don't even call it 'design'! The increasingly common availability of digital tools has been crucial in this respect: not only did it create a more subtle awareness of

275 Rock quoted in Jarrett Fuller. "Design Criticism Is Everywhere – Why Are We Still Looking For It?" *Eye on Design*, August 19, 2020. https://eyeondesign.aiga.org/design-criticism-is-everywhere-why-are-we-still-looking-for-it/.

276 https://www.youtube.com/watch?v=OXA4-5x31Vo. See also https://www.pagecloud.com/blog/what-happened-to-the-grid.

277 https://twitter.com/ktsuskin/status/1159851862201442304.

278 As the title of a Mike Monteiro book. *Design Is a Job*. New York: A Book Apart, 2012.

design culture in the layperson, but it also altered the perception of the designers' status and, in turn, their relationship to visual culture as a whole. When, in the late '80s, the desktop publishing revolution took place, the designers' reaction was mainly a defensive one against the new barbarians, that is, clueless non-professionals that were quickly catching up. "One of the great fears attached to the advent of the Macintosh and desktop publishing – explains Michael Rock – is that it has initiated others into our secret language; demystified our code of fonts, rags, picas, and leading."[279] This secret language was being converted into a function crystallised in software: (graphic) design had become semi-automated. With this shift, designers' authoritativeness was partially eroded: the semi-automated designer also became a semi-professional one. Speaking about programmers, Ellen Ullman brutally describes the effects of professional demystification: "If regular people, called 'users,' can understand the task accomplished by your program, you will be paid less and held in lower esteem."[280]

A minority of designers, however, were able to escape this fate by gaining the role of cultural producers who highlight and interpret visual culture, and in particular the emerging technical vernacular. Among them, prominent designers such as April Greiman, Zuzana Licko and Rudy VanderLans took up the role of "interpreter and mediator of the social meaning of technology"[281] and helped popularise the aesthetics of the new digital means of production among professionals. The Macintosh brought with it an apparent liberation from the material constraints of layout resulting in a Derridean ethos of non-linearity and deconstruction. The effect was often confusing and sometimes consciously uninviting. A "cult of the ugly" was

279 Michael Rock. "On Unprofessionalism." *2x4* (blog), August 5, 1994. https://2x4.org/ideas/1994/on-unprofessionalism/.

280 Ellen Ullman. *Life in Code: A Personal History of Technology*. New York: Picador, 2018, p. 8.

281 Emily McVarish. *Inflection Point*. Berkeley, CA: Emigre, 2017, p. 8.

Sponsored Instagram post for the Adobe Spark app.

formed, as design historian Steven Heller called it.[282] It was time to break the grid, let go of rules, defy tradition. Licko and VanderLans dedicated the 11th issue of their magazine *Emigre* to the relationship between graphic designers and the Macintosh. Analysing the issue, Emily McVarish argues that Emigre's vision of authorship didn't "stake its claim in self-defense against amateur incursions into graphic design's professional territory."[283] In fact, there was no need to do so: amateurs didn't threaten professionals because they were there to be *interpreted*.

Things didn't stop there. From the early 2000s to the present day, various practitioners, inspired by the internet, began adopting the presets, templates and defaults inscribed in desktop publishing and web browsers. Their intervention was minimal. By removing themselves from the typographical equation, designers would show that non-design was itself design, one that was taking place before anyone's actions and had its own aesthetic autonomy. Designer and writer Rob Giampietro dubbed this phenomenon "Default System Design:"

> Default Systems are machines for design creation, and they represent design publicly as an "automatic" art form, offering a release from the breathless pace at which design now runs, as clients ask for more, quicker, now. Default Systems are a number of trends present in current graphic design that exploit computer presets in an industry-wide fashion. They are a quasi-simplistic rule-set, often cribbing elements from the International Style in a kind of glossy pastiche, a cult of sameness driven by the laziness and comfort of the technology that enabled Emigre's rise, the Macintosh.[284]

282 Steven Heller. "Cult of the Ugly." *Eye Magazine*, 1993. https://www.eyemagazine.com/feature/article/cult-of-the-ugly.

283 McVarish, *op. cit.*, p. 16.

284 Rob Giampietro and Rudy VanderLans. "Default Systems in Graphic Design." *Emigre*,

Artwork for *The Life of Pablo* album by Kanye West. Here's how *The Verge* described it: "Designed (typed?) by the Belgian artist Peter De Potter, the post-modern cover features The Life of Pablo in all caps and a small family photo in the bottom corner."

Designers working with Default Systems would deliberately use 'neutral' fonts like Helvetica or default ones like Arial and Times New Roman. Or they would keep the default treatment of web hyperlinks. Or exaggerate ridge borders with CSS. Or indulge in the use of WordArt effects. To the untrained eye, their work is often indistinguishable from that of amateurs. More recently, they would make use of what Silvia Sfligiotti calls *auto-tune typography*, an approach that "transforms words into undifferentiated blocks that can be placed on a page without worrying too much [and] turns visual design into an automatic, a-critical activity."[285] In the context of the internet, terms like 'brutalism' are employed and sometimes nostalgia plays a role. Finally, appearing is a quirky, poetic web design avantgarde that sees HTML tags more as a semiotic text than a merely practical set of rules.[286]

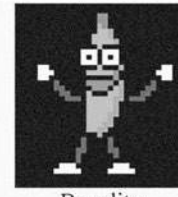
For the defaultists, "form follows format," as designer Hendrik-Jan Grievink put it.[287] That's how template culture comes about, where every new project actually derives from a long sequence of previous projects. When I open Word or InDesign, I run into a template since the page, only apparently blank, already contains a series of design choices registered in advance, such as the margins' width. Why waste time on design decisions when one can celebrate the aprioristic purity of the template? Perhaps, this is the most authentic embodiment of the passion for systems that permeates the modernist ideology. British designer Daniel Eatock epitomised this trend with his generic *Utilitarian Poster* that could be used for any type of event. Resembling a bureaucratic form, it includes boxes for the event's title, description, date, etc. In the 1952 dystopian novel *Player Piano*, Kurt Vonnegut depicts an almost

2003. https://linedandunlined.com/archive/default-systems-in-graphic-design/.

285 Silvia Sfligiotti. "This Is Auto-Tune Typography." *Medium* (blog), August 19, 2020. https://silviasfligiotti.medium.com/this-is-auto-tune-typography-3953e74cc2ac.

286 See https://brutalistwebsites.com/ and https://html.energy/.

287 Hendrik-Jan Grievink. "Template Culture." *Mediamatic*, 2009. http://www.mediamatic.net/en/page/116348/template-culture.

Hacking	Defaults
Hacking a Nintendo cartridge to make images	Using MS Paint to make images
	12 point Times New Roman
Net.Art 1.0	???
Anxiety	 Banality
"The Man is taking away our privacy... that's lame!"	"We willingly give up our own privacy (i.e. endlessly talking about ourselves on our Myspace profiles)... why?"
Empowering The People by subverting The Man's power	Being and critiquing The People by using the tools made by The Man
Rock & Roll attitude	Exuberant humility
Jodi's blogs	Tom Moody's blog
Sophisticated breaking of technology	Semi-naive, regular use of technology

Guthrie Lonergan (2007).

wholly automated society where it is sufficient to record the movements of a worker on a disk to ensure that the machine will endlessly repeat them with complete accuracy. As in Vonnegut's novel, Eatock has 'recorded' his design choices and, from the point of view of mere production, made himself obsolete.

MIDAS TOUCH

According to Giampietro, both '90s Ugly Design and 2000s Defaultism were rooted in the personal computer:

> Default Systems design claims, "This is how the computer works with minimal intervention." It also claims, "By keeping the designer from intervening, this design language is made available to all." So Default Systems look new, but they arise from the social concerns of the old.[288]

While most designers would immediately alter the setting to reinstate their personality into the work, the postmodernists would push them to their extreme. Defaultists, instead, would treat them as *objet trouvé*, as something that acquires value insofar as it is picked up by a charismatic figure – the designer. Both the maximalist approach of Ugly Design and the minimalist one of Defaultism spoke enthusiastically of the broad availability of a digitally-informed design language. "We are the primitives of a new technological era," the *Emigre* duo proclaimed. However, they ignored that such language was in the process of being made available to all *in the first place*. It was starting to populate the common imaginary ('photoshop' being not just a software, but a verb), regardless of the designer's role and intention. Both Ugly Design and Defaultism attempted to impress their mark on the design language of software. The expectation was one of control and recuperation but, in the meantime, graphic design was becoming an autonomous *Kulturtechnik*, a daily practice

288 Giampietro and VanderLans, *op. cit.*

that doesn't need the mediation of experts, akin to taking pictures with a mobile phone.[289]

Giampietro argued that the value of both postmodernists and defaultists's artefacts lays in a "critique of the conditions of their own making."[290] While doing so, though, they also foregrounded a position of cultural arbitrage held by the designer. "Ugliness is smart," opined Heller.[291] Their aesthetics of the technical is at the same time a form of subsumption: art and design elevate technology by injecting it with meaning. This interpretative web forms an invisible threshold around tools and objects. Both the 'template apologist' and the 'layout grammatologist' are cultural professionals. They possess a sort of Midas touch: through their interpretative work, they're capable of turning the mundane into culture. In this way, the cultural professional gains the legitimacy to operate 'unprofessionally,' that is, to venture into the sphere of the ordinary and make something extraordinary with it. The cultural professional is, therefore, a special kind of *amateur*, one who is paid for their *amour*. They're equidistant from the stiff flannel-suited *cadre* and the unpaid dabbler. That's a coveted position to be in, as design historian Chiara Alessi suggests: "designers today are proving to be agents more of the cultural industry than the manufacturing one."[292]

THE GRID

Whereas the Cult of the Ugly was about breaking the grid, and Default Systems are about exposing it, digital platforms have managed to bring the grid (understood here as a set of rules, formats, templates, presets) to a level

289 The concept was suggested to me by German artist Sebastian Schmieg, whom I thank.

290 Giampietro and VanderLans, *op. cit.*

291 Heller, *op. cit.*

292 Chiara Alessi. *Design senza designer*. Bari: Laterza, 2016, p. 98. Designers are not alone in their desire to be cultural professionals. Such desire is widespread and commodified under the mandate of being creative, which is propagandised, for instance, by Apple (MacBook *Pro*).

Ad for Ad World event (2021). Made with the Apple's default Notes app, it insists on the lack of any need for a designer.

of omnipresence and, thus, invisibility. Once platforms start mediating content dissemination, the type of content itself, be it modernist, postmodernist or defaultist, becomes irrelevant. According to Daniel van der Velden, "design has externalized its grid function (an element that used to 'stand for' its social ambitions and responsibilities) to the platform while it has itself become that platform's liquid contents."[293] The platform's grid is not just a set of visual proportions but a dynamic structure for organising information, a *medium*. One that generates the very tools that are meant to fill it: photo and video editors, cropping utilities, sorting algorithms, etc.

The defaultist and postmodernist 'authorial capture' of the grid (by means of breaking or highlighting it) was a response to the abstraction of the designer's role, now turned into a process. Today, some twenty years later, not a day goes by without someone wondering if the designer will be made obsolete by artificial intelligence.[294] According to an ultra-cited and now classic 2013 study, 47% of the jobs performed in the US are put at risk by computerisation.[295] The study includes a ranking of about 700 professions ordered by probability of automation in ascending order. Graphic design occupies the 161st place. Not so bad.

Giampietro argued that "[t]o view a computer through its default settings is to view it as it's been programmed to view itself, even to give it a kind of authority."[296] This

293 Van der Velden, "Lyrical Design," *op. cit.*, p. 61.

294 See, for instance, Rob Peart. "Automation Threatens to Make Graphic Designers Obsolete." *Eye on Design*, October 25, 2016. https://eyeondesign.aiga.org/automation-threatens-to-make-graphic-designers-obsolete/; Jon Gold. "Taking The Robots To Design School." *Jon Gold* (blog), May 25, 2016. https://web.archive.org/web/20160525193649/http://www.jon.gold/2016/05/robot-design-school/; Tone Bratteteig and Guri Verne. "Does AI Make PD Obsolete? Exploring Challenges from Artificial Intelligence to Participatory Design." In Proceedings of the 15th Participatory Design Conference - Volume 2, 1–5. PDC '18. New York, NY, USA: Association for Computing Machinery, 2018. https://doi.org/10.1145/3210604.3210646.

295 Carl Benedikt Frey and Michael A. Osborne. "The Future of Employment: How Susceptible Are Jobs to Computerisation?" *Technological Forecasting and Social Change* 114, no. C (2017): 254–80.

296 Giampietro and VanderLans, *op. cit.*

authoritative personification of the machine is the narrative that is driving today's frenzy about artificial intelligence applied to design. While the designer becomes a process, artificial intelligence is turned into a person. A case in point is The Grid, an AI-powered system to automatically design websites, announced in 2014 and never actually launched. The Grid had its own personified AI, a bot called Molly, portrayed as someone that is "quirky, but will never ghost you, never charge more, never miss a deadline, never cower to your demands for a bigger logo."[297] After raising a huge amount of money with crowdfunding, The Grid went silent. Currently, their website still promises the release of a third version after the demise of version 2, while a ticker tape reads "AI was the easy part."

Scholars point out that "artificial intelligence is a topic that is perceived between great euphoria and pure dystopia, and on the technical level between surprising functionality and frustrating technical failure."[298] This is not only true for the field of graphic design. Tools like IKEA Place, LiveHome3D or Planner 5D facilitate the work of interior design and are deliberately targeted at amateurs ("Use Planner 5D for your interior house design needs without any professional skills").[299] Most systems boast about their artificial intelligence capabilities, like Leaperr, "an Artificial Intelligence (AI) system that does the job of an interior designer."[300]

The lesson to be learned from The Grid and similar AI tools is not just that full automation, in the field of design as in many others, is difficult and perhaps unattainable, but that it nonetheless reinforces social expectations and

297 Ritupriya Basu. "Algorithms Are a Designer's New BFF – Here's Proof." *Eye on Design*, December 19, 2019. https://eyeondesign.aiga.org/algorithms-are-a-designers-new-bff-heres-proof/.

298 Klaus Neuburg, Sven Quadflieg and Simon Nestler. "Will Artificial Intelligence Make Designers Obsolete?" Berlin, 2020.

299 https://planner5d.com.

300 Harshajit Sarmah. "5 AI-Powered Home And Interior Designing Tools." *Analytics India Magazine*, July 26, 2019. https://analyticsindiamag.com/5-ai-powered-home-and-interior-designing-tools/.

desires. People want to believe that designers, like any other gatekeeper, are replaceable by a docile machine, as this has partly already happened with the personal computer. But, whereas the desktop publishing revolution offered static presets, AI makes them dynamic, updating them on the basis of new collected information – an ever-evolving defaultism.

The Duchampian gesture of 'signing' the system by letting it run without intervention is now paralleled by the designer's ambition to occupy the control room of AI in order to "set parameters and constraints, determine behaviour of automated systems," as Microsoft puts it.[301] The designer aims at becoming a systematist, because it is at the level of systems that synthesis can really happen.[302] Below, things are not so interesting: "what may be a delightfully stimulating plaything for the systems designer, may be the basis for a dehumanised work environment for the user."[303] If defaultists make themselves invisible by means of their lack of intervention, system designers would become invisible by defining the system's variables.

SEMI-AUTOMATION

That of controlling systems is clearly a target that can be met by a minority of designers. Generally, the discussion on AI replacement considers the designer class as a single, homogeneous entity. It might sound obvious, but this is not the case: for one system designer, there is already an army of designers affected by restricted uses of automation or caught in processes of semi-automation. As Ruben Pater explains,

> [...] graphic design is becoming increasingly automated itself. Internet banners and interface

301 Jasmine Oh. "Yes, AI Will Replace Designers." *Microsoft Design* (blog), August 22, 2019. https://medium.com/microsoft-design/yes-ai-will-replace-designers-9d90c6e34502.

302 See chapter 3.

303 Mike Cooley. *Architect or Bee? The Human Price of Technology.* Nottingham: Spokesman, 2016, p. 40.

Business card for Manuel Bürger made by OM Graphic, a copy shop in Mumbai, using Corel Draw 9.0 (2008).

Unused draft for a made-to-measure Variable Logo Service by Vera van de Seyp and Charlotte Rohde (2022). Here, van de Seyp, a researcher at MIT, and Rohde, a professor of typography at the Bauhaus Universität Weimar, mimic the graphic style of car repair shops.

> designs are already 'assembled' automatically based on different spaces and user data. A/B testing can update and optimize font styles, colours, and images. If your data profile says you are conservative, perhaps the system chooses the Trajan typeface. In this scenario many graphic design jobs will likely become obsolete within two decades.[304]

Fiverr.com is the biggest online marketplace for freelance services. Many among them revolve around graphic design. These gigs show that there is a fine line between automation and manual labour: some of the tasks performed by these remote designers can hardly be called creative: slide organising, vector tracing, detailed photo retouching... all assignments that show the vast gap, still extant, between fully automated processes and purely manual ones. Everyday designers of this kind temporarily reside within this gap. We want full automation but we predominantly deal with forms of semi-automation, and in some cases even "fauxtomation."[305] While AI-based text-to-image engines, chatbots and logo generators are given huge media attention, hordes of designers are still busy with tedious conversion tasks. Technological expectations clash with the harsh reality of routine work. On Upwork, another marketplace for freelancers, the description of a 'tasker' catches my attention: Belinda J., $25 per hour, claims to be a self-taught graphic designer, since all her previous jobs included tasks where she created graphics. The generalisation of graphic design offers the opportunity to create a gig out of its yet-to-be-automated processes.

Cases like Belinda's might seem distant from the reality

304 Pater, *op. cit.*, p. 215.

305 The term was coined by writer and film-maker Astra Taylor. It refers to activities that falsely appear fully automated but are, in fact, manually performed by humans. See "The Automation Charade." *Logic*, August 1, 2018. https://logicmag.io/failure/the-automation-charade/.

of the European or American hyper-scholarised designer, but, as Pater points out, it might not be like that for long:

> Sooner or later, designers too will be challenged by the same forces of global competition that laid off industrial workers. With the rise of algorhythmically produced design, and online job platforms for design services such as Fiverr and a growing designer class, the question is not if design will remain a viable profession in post-industrial societies under capitalism, but for how long.[306]

Not all of this is new. Already in 2006, Daniel van der Velden warned us that the competitive advantage of the European knowledge economy "will quickly become a thing of the past, if holding a mouse proves cheaper in Beijing than in the west of Holland."[307]

Some time ago, Fiverr introduced an online logo generator, based on pre-designed components automatically assembled on the basis of a set of parameters chosen by the user. Once those are entered, an animation appears while the data is being crunched: a flat-style designer working at their table. Here, the figure of the designer is an allegory of the design activity, a sort of skeuomorphism of work, akin to the floppy disk as the icon of saving. The designer as a person is used to illustrate a machinic design process. What brings together Defaultism and semi-automated remote work is that the designer role becomes a function governed by a set of instructions executable by either a human or a machine. And this might be true not just for the remote outsourced workers, but for entire fields like UX design:

> In 2023, being a 'designer' has been reduced to assembling prefab visual components and templated structures. Many of us have recognized early the similarities between using

306 Pater, *op. cit.*, p. 250.
307 Daniel van der Velden. "Research and Destroy." *Metropolis M*, 2006. http://indexgrafik.fr/daniel-van-der-velden-metahaven-research-and-destroy/.

> design systems in product design, and putting Lego pieces together to 'create' something that resembles a good product.[308]

In his essay, Giampietro was presciently speaking of systems, well before the actual widespread adoption of specific digital tools to apply and replicate parts of a company's interactive corporate identity, with frameworks such as Twitter Bootstrap and Google's Material Design. Giampietro understood that design systems are where management meets automation. Here, the design rules – the defaults – are written in code. The Lego analogy suggests that a default can be thought of as a decision crystallised in software and, therefore, design systems can be conceived as a form of automated decision making, a Taylorism of UX design. After all, "[t]he factory floor prefers interchangeable, replaceable parts."[309] Some examples of this kind of automation are more explicit: from logo generators that simply mix combinatorially pre-existing designs to more advanced ones that employ artificial intelligence. Neither should we forget reusable assets, which are becoming easier to employ within cloud-native software like Figma and Sketch. Something similar is happening in the field of illustration as well, where we have witnessed the rise of the so-called 'flat style.'[310] With its almost childlike simplicity, it is technically prone to semi-automated replication and assemblage.[311] Very popular on Big Tech platforms, this style has been accused of infantilising the user. Here, a

308 Corneliux. "Everyone Used to Be a Designer." *UX Planet*, February 13, 2023. https://uxplanet.org/everyone-used-to-be-a-designer-530aa762e415.

309 Garrett, *op. cit*.

310 Also called Corporate Memphis, Alegria art, big tech art, etc. See https://en.wikipedia.org/wiki/Corporate_Memphis and https://knowyourmeme.com/memes/subcultures/corporate-art-style.

311 In fact, there are websites (such as stubborn.fun) that allow a user to combine elements and generate flat style illustrations. See also Rachel Hawley. "Don't Worry, These Gangly-Armed Cartoons Are Here to Protect You From Big Tech." *Eye on Design*, August 21, 2019. https://eyeondesign.aiga.org/dont-worry-these-gangley-armed-cartoons-are-here-to-protect-you-from-big-tech/.

technical operation turns into a reflection of its cultural effects.

Despite all the hype around automation, there is and there will always be a surplus of manual work to be performed, which appears in many different places: from the design studio that needs to adjust the files for print or deployment, to the continuous training of AI models (what scholars Mary L. Gray and Siddharth Suri call "the paradox of automation's last mile"[312]), to the outsourcing of book typesetting to India, as it is still cheaper and quicker than using automated systems. On the one hand, technology makes things easier for designers, relieving them from the burden of dull tasks so they can devote themselves to their creative endeavour. On the other hand, however, technology redefines those very categories by shifting the threshold between creativity and dullness.[313] The extent of this process can be broad: certain operations that are at the core of some designers' practice, can be reduced to automated functions. At this point, designers can either cling to a fully but vaguely conceptual image of their role ('it's the idea that matters'), or hope for a higher meta-position, from which to govern those functions. Actually, though, the position is not occupied by the designer, but by the engineer. Pater briefly explains how this is happening in the field of online advertising, which "is no longer dictated by art directors, but by people with a PhD in mathematics and data science who spend all day thinking how to make people click on ads."[314] If engineers are the ones structuring the system, how do designers make ends meet? In a satirical vision of 2025, Francisco Laranjo offers two options:

> Designers earned a living in two ways: 1) operating their design bots to fulfil a design service;

312 Mary L. Gray and Siddharth Suri. *Ghost Work: How to Stop Silicon Valley from Building a New Global Underclass*. Boston: Houghton Mifflin Harcourt, 2019, pp. 38-63.

313 "The process by which these two dialectical opposites [creative and non-creative work] are united by the designer to produce a new whole is complex, and as yet an ill-defined and little-researched area." Cooley, *op. cit.*, p. 38.

314 Pater, *op. cit.*, p. 213.

> 2) feeding the centralised design bot network by informing the system about the rationale behind their decisions.[315]

Designers, apart from the few that have a say in the process of shaping systems, become either enlightened user-contributors of pseudo-intelligent automated systems, as in both defaultism and postmodernism, or they occupy a position that is missing from Laranjo's speculation – that of the in-house or outsourced clerk who executes tasks yet to be automated. In 2006, Ellen Lupton joked about how designers feared "that secretaries equipped with Times Roman and Microsoft Word would obliterate the design profession,"[316] but we shouldn't forget that for the majority of designers the job is largely clerical.

CHASING TECH

Given the risk of replacement, the tedium of repetitive tasks and the potential, albeit niche, designer's role as a systematist, technology appears at once as something to fear and something to master. But that's not all. Technology is also something to *chase*. Since the advent of personal computers, designers have found themselves 'catching up' with new tools, systems, devices and media. And the current excitement around AI reveals that by technology we mean, generally, *new* digital technology, and that we are somehow enchanted by it. Historian David Edgerton:

> In invoking 'technology,' the designer is blinding herself to the materiality of her world, her tools, her ambitions to transform the material [...] Invoking the concept of 'technology' in design, focused on the disembodied digital meaning of the term, far from putting 'technology' into design, removes most of it, leaving

315 Francisco Laranjo. "Ghosts of Designbots yet to Come." *Eye Magazine*, December 21, 2016. https://www.eyemagazine.com/blog/post/ghosts-of-designbots-yet-to-come.

316 Ellen Lupton. *D.I.Y. Design It Yourself*. New York: Princeton Architectural Press, 2006, p. 19.

> behind only digital simulacra. In other words, rather than bringing in the material to design, or rethinking it, the concept of 'technology' can be a way of getting rid of it. There is a radical difference between a maker culture focused on digital devices and one concerned with physical workshops stuffed with many tools for working many materials.[317]

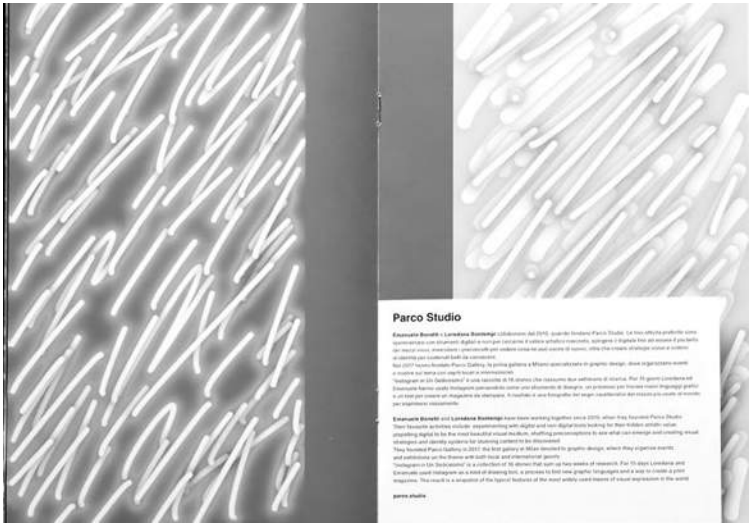
Edgerton's account is slightly over-dualistic: digital technology doesn't need to be disembodied, or better, it never is. But it is true that often the 'technology' that designers try to catch up with is a digital simulacrum. The design field, thanks to its porous boundaries, expands to include emergent technologies, but the medium's novelty is its only message. Various design schools and studios, for instance, felt the urge to equip themselves with virtual reality setups (an interest recently revived by Mark Zuckerberg's metaverse propaganda). Students and staff now 'experiment' with VR, but their work often acts like an ornament for the technology itself. They try to engage with the medium, but the medium prevails. As in the platform's conceptual grid, the work of designers becomes mere content.

LEARN TO CODE

Many designers and design theorists have tried to chase technological development by acquiring (or inviting others to acquire) the literacy needed to interact with computers on their own terms. This is how 'learn to code' became a sweeping injunction.[318] People debated (and still do) whether this practice should be within the designer's

317 David Edgerton, Hugo Palmarola, and Pedro Álvarez Caselli. "Some Problems with the Concept of 'Technology' in Design: Interview with David Edgerton." *Diseña*, no. 18 (January 29, 2021): 2–8. https://doi.org/10.7764/disena.18.Interview.2, pp. 3-4.

318 Designers weren't alone in developing this pressure. See Silvio Lorusso. "Learn to Code vs. Code to Learn: Creative Coding Beyond the Economic Imperative." In *Graphic Design in the Post-Digital Age*, edited by Demian Conrad, Rob van Leijsen, and David Héritier. Eindhoven: Onomatopee, 2021. https://silviolorusso.com/publication/learn-to-code-vs-code-to-learn/.



Instagram in un sedicesimo by Parco Studio, published by Corraini., "For 15 days Loredana and Emanuele used Instagram as a kind of drawing tool, a process to find new graphic languages and a way to create a print magazine. The result is a snapshot of the typical features of the most widely used means of visual expression in the world."

Mark Zuckerberg at the World Mobile Congress of Barcelona in 2016. The people in the audience wearing virtual reality look like extras promoting this technology.

domain. For some notable practitioners, such as John Maeda, the answer is a hard yes.[319] Other positions are more down to earth: sure, coding skills are a 'nice to have,' a skill that facilitates dialogue with developers, but in practice, the place of graphic, UI and even UX designers is the wireframe, the mockup, the clickable prototype designed in Illustrator or (goosebumps) Photoshop; and more recently in Figma, Sketch or Invision, with the occasional venture into CSS or SASS territory.

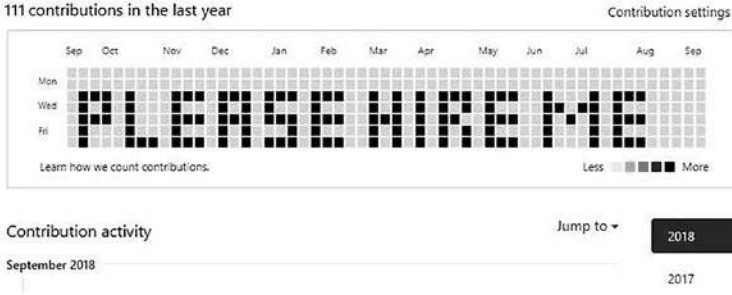
Coding, however, is not just one among many '21st century skills.' Coding is a battleground for authoritativeness and professional prestige, in a context where entire professions, such as graphic design, are perceived as something that it would be good to automate once and for all. If we consider the replacement of a cover image on Facebook or the adoption of a new template for our blog a design endeavour, graphic design suddenly acquires the same professional and cultural value of sending an email. So why not turn such a menial job into a button to press?

The battle is also clouded by the blurred meaning of code – are we talking of concocting a tiny Processing sketch or building a full Node.js application? In any case, code generally means a higher salary and a vantage point in a world that, as programmer and venture capitalist Marc Andreessen assured, is being eaten by software.[320] Coding emerges as a professional panacea for the long tail of everyday designers who are reasonably sensitive to the rhetoric of skill obsolescence and employability. It comes as no surprise, then, that US and UK students consider coding the third most useful skill to acquire (after networking and "idea generation").[321]

319 https://www.youtube.com/watch?v=JGGAitirfRY. See also Liz Stinson. "John Maeda: If You Want to Survive in Design, You Better Learn to Code." *Wired*, March 2017. https://www.wired.com/2017/03/john-maeda-want-survive-design-better-learn-code/.

320 Marc Andreessen. "Why Software Is Eating the World." *Andreessen Horowitz* (blog), August 20, 2011. https://a16z.com/2011/08/20/why-software-is-eating-the-world/.

321 Roberts et al., *op. cit.*, pp. 166-7.



Sentence composed by populating the contribution chart of Github, the most popular hosting service for software development.

Not only does code promise to upgrade individual practices, but also the activities of design's museum-educational complex. The result: a phantasmagoria of screens and interactive installations in exhibitions and final shows, often paired with the inadequacy to restructure cultural programs around computational ideas and approaches.[322] The edge afforded by coding, however, won't last long: while programmers and computer scientists possess an acknowledged and relatively arcane expertise with a correspondingly high salary, coding is increasingly perceived as semi-skilled labour. The programmer belongs to a profession, the coder to a workforce.

MANAGERIAL CAPTURE

In an episode of *Età dell'oro*, an Italian web series on the hardships of a group of creative workers from Milan, we see an art director instructing a graphic designer by perpetually repeating "two points to the left, no no, two points to the right." Here we see how graphic design can become micromanagement, turning the designer into a Kafkaesque, voice-controlled mouse. The scene seems like an aberration of the design practice, but might in fact be its essence. Design historian J. Dakota Brown retraces the evolution of the 'managerial capture' of typographers' trade knowledge and labour, from the theorisation of W. A. Dwiggins, who coined the term 'graphic design,' to Walter Paepke, who founded the International Design Conference in Aspen to convince people that design shouldn't be a mere appendix of production but a branch of management. In this sense, the whole history of design *stricto sensu* can be understood as a progressive disconnection of the brain from the hand, itself part of the larger phenomenon of the division of labour. The last act of Brown's historical pièce brings us to the present. Here, a bitterly

322 Some exceptions exist. Among them, the School for Poetic Computation in the US and Hackers and Designers in the Netherlands.

ironical reversal takes place: graphic designers, who were among the agents of printers' deskilling, are now themselves the victims of the deskilling caused by digital democratisation.[323]

According to Mieke Gerritzen and Geert Lovink, "after the democratization of the design discipline, the leading designer will become a software developer,"[324] but that ain't exactly easy, both technically and professionally, as it involves the risk of perverting one's acquired aptitude. Luckily, there is another way out of an unrewarding democratisation and it's called 'design thinking.' Albeit in decline,[325] design thinking is still the most successful managerial expression of design. At best, it's a methodology designed to fuel the innovative capacities of organisations; at worst, it's a form of self-help for organisations facilitated by designers.[326]

Either way, with its five easy pillars (Empathise, Define, Ideate, Prototype, and Test) design thinking is, unlike high-level programming, rather accessible. In fact, it aspires to become a democratic organisational instrument. One of its main proponents, Tim Brown, co-chair at the consulting firm IDEO, argued that "[design] has become too important to be left to designers."[327] This might be one of the few aims that design has actually achieved. In fact, as Beatriz Colomina and Mark Wigley

323 J. Dakota Brown. *Typography, Automation, and the Division of Labor: A Brief History*. Chicago, IL: Other Forms, 2019.

324 Gerritzen and Lovink, *Everyone Is a Designer in the Age of Social Media*, op. cit., p. 96.

325 Several articles deeply critical of design thinking have been published over the last few years. Among them, Rebecca Ackermann. "Design Thinking Was Supposed to Fix the World. Where Did It Go Wrong?" *MIT Technology Review*, February 9, 2023. https://www.technologyreview.com/2023/02/09/1067821/design-thinking-retrospective-what-went-wrong/. Vinsel, *op. cit*.

326 Self-help for organisations goes hand in hand with individual self-help: "In the Stanford d. school we attempt to bring students through a series of experiences that change their self-image so that they think of themselves as being more creative. We call this boosting their *creative confidence*." The emphasis is mine. Bernard Roth. *The Achievement Habit: Stop Wishing, Start Doing, and Take Command of Your Life*. New York, NY: Harper Business, 2015.

327 Tim Brown. *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation*. New York, NY: Harper Business, 2009, p. 8.

note, "design has gone viral [...] 'Design thinking' has become a dominant business model affecting everything from politics to education, personal relationships, research, communication, and philanthropy [...] Design has become dangerously successful."[328] New card sets of 'methods' and guides have popped up everywhere,[329] functioning not only as pedagogical tools that are easy to disseminate, but also as a form of guerilla promotion of a few design consulting firms.[330]

Design thinking is managing innovation, but what constitutes innovation, exactly? Design thinking helped designers realise that outputs were less important than the process, leading to what Anja Groten defined "workshopisation," that is, an intensification of design's performative dimension, its fetish for collaboration[331] and, subsequently, "a general disappointment in what workshops are actually capable of."[332] Similarly, Guy Julier speaks of a virtualist tendency in the field:

> [design's] emphasis on process and collaboration, its customer experience mapping or its frequent use of workshops, hackathons and jams makes it appear to contribute to a kind of virtualism where things are made real but not actual.[333]

Here, design gets close to the world of contemporary art, which is, according to Boris Groys, "basically art production

328 Colomina and Wigley, *op. cit.*, p. 46.

329 Among them, IDEO's *Field Guide to Human-Centered Design*, "[a] step-by-step guide that will get you solving problems like a designer". https://www.designkit.org/resources/1.html.

330 See Julier, *op. cit.*, p. 25. Another term that describes a similar phenomenon is UX Theatre, which "happens when designers are asked to *pretend* to do the work of design and aren't actually *permitted* to do the work of design." Tanya Snook. "UX Design Has a Dirty Secret." *Fast Company*, October 18, 2021. https://www.fastcompany.com/90686473/ux-design-has-a-dirty-secret.

331 Anja Groten. "Towards a Critical Collaborative Practice." In de Vet, *op. cit.*, p. 40.

332 Anja Groten. "The Workshop and Cultural Production." *Open!*, June 1, 2019. https://www.onlineopen.org/the-workshop-and-cultural-production.

333 Julier, *op. cit.*, p. 157.

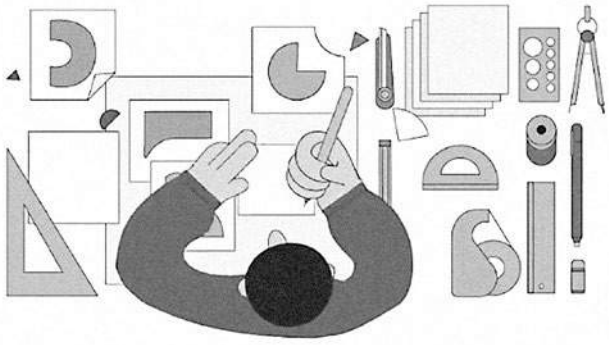
without a product."[334] Similarly, Paul Rodgers & Craig Bremner have argued that "design is largely a process of the management of its own spin, and management is the only product of the mature neo-liberal service economies."[335]

CULTURAL PROFESSIONALISM

One stratagem adopted by designers to reaffirm their expertise is to ridicule the lack of 'design common sense' of the amateur, who is guilty, for instance, of having aligned the text to the right, rendering it hard to read. The monopoly of common sense, however, has an expiration date: as the commandment that forbids the use of Comic Sans for something that is not comical became commonplace, so have the few precepts that are sufficient to create a decent poster. And if these, together with fashionable tricks, are then incorporated into commonly used software and templates – as happened for example with the Trend Generator by Trend List[336] – the monopoly might expire soon. Indeed, it might have expired already. We might be in the process of realising that, instead of being a set of rigorous structuring principles, graphic design might just be one composite style among many within the ocean of visual culture, an 'aesthetic' – call it *graphicdesigncore*.

When attacking non-professionals, graphic designers generally appeal to the notion of *quality.* They argue that the client should be 'educated' to recognise it. Perhaps, it is still possible to tie the concept of quality to some specialised activities like type design or the design of way-finding systems. One could argue about quality with some degree of objectivity in the case of large-scale projects, such as the identity of big museums or companies. However, quality acts first and foremost as a device for

334 Boris Groys. "Under the Gaze of Theory." *E-Flux*, May 2012. https://www.e-flux.com/journal/35/68389/under-the-gaze-of-theory/.
335 Rodgers and Bremner, *The Design of Nothing*, op. cit., p. 553.
336 https://apprecs.com/ios/767377188/trend-generator.



Screenshot from Fiverr.com. Animation showing their automated logo generator process. Manual labour is here depicted skeuomorphically.

professional legitimation, and this function, of course, can't be expressed too openly, as "every profession is a conspiracy against the laity."[337] Profession, then, has the double meaning of competence and declaration of a belief, namely, the belief in the expert. Such belief, however, seems to be fading due to a crisis of legitimisation that involves the whole "professional-managerial class."[338] The phenomenon is not new: as early as the '70s intellectuals like Luhmann and Habermas were debating the *Legitimationsprobleme.* The laity were becoming aware of the conspiracy.

According to Ruben Pater, "the [design] profession is a form of protection and exclusion, limiting the access to knowledge, the access to the means of production, and the influence over discourse."[339] Several critiques of professionalism, like this one, or even calls for deprofessionalisation, legitimately denounce how the design profession has disproportionately excluded marginalised groups. But, as already pointed out, a professional can be seen as someone who claims "extraordinary knowledge in matters of human importance," getting in return "extraordinary rights and privileges."[340] If the rights and privileges granted by a profession are extraordinary, it must mean that a profession is necessarily exclusive: it is an elite. But what kind? A useful distinction is that between "core elite" and "general elite." The designer is usually general elite because they mostly execute the directives of the core elite. Furthermore, this general elite resembles an army with increasingly approximate skills. Less a small clan than a swarm.[341] The anti-professional critique neglects the mechanisms of

337 George Bernard Shaw quoted in Rittel, *op. cit.*, p. 155.

338 Alex Press. "On the Origins of the Professional-Managerial Class: An Interview with Barbara Ehrenreich." *Dissent Magazine*, October 22, 2019. https://www.dissentmagazine.org/online_articles/on-the-origins-of-the-professional-managerial-class-an-interview-with-barbara-ehrenreich.

339 Pater, *op. cit.*, p. 334.

340 Schön, *op. cit.*, p. 4.

341 For a detailed overview of the analytical tools to categorise elites and professional cohorts, see Ventura, *Radical Choc*, *op. cit.*, p. 55-63.

deprofessionalisation that affect the design field *in the first place*. This is a problem, because higher education – which is where these calls generally originate – is bound to generate resentment and anger if it cannot guarantee its student body, which invests time and money in its institutions, the social and economic benefits of the profession. In this scenario, calling for a complete deprofessionalisation of design in the name of inclusivity is mere populism.

How can we solve design's *Legitimationsprobleme*? The magic words are culture and intelligence, as well as research and knowledge, which together form the ability to discern good from bad, to draw distinctions of both aesthetic and moral value. According to the Canadian designer Bruce Mau, who co-authored a massive book with Dutch architect-theorist Rem Koolhaas, "the only way to build real equity is to add value: to wrap intelligence and culture around the product. The apparent product, the object attached to the transaction, is not the actual product at all. The real product has become culture and intelligence."[342] In turn, the designer takes up the role of someone who produces culture intelligently. As the following excerpt shows, the invite for designers to become cultural producers was explicit:

> The true investment is the investment in design itself, as a discipline that conducts research and generates knowledge – knowledge that makes it possible to seriously participate in discussions that are not about design. Let this be knowledge that no one has asked for, in which the designer is without the handhold of an assignment, a framework of conditions, his deference, without anyone to pat him on the shoulder or upbraid him. Let the designer take on the debate with the institutions, the brand names or the political parties, without it all being about getting the

342 Mau quoted in Foster, *op. cit.*, p. 23.

Goya's *Saturn Devouring His Son* redrawn by Twitter user @clayhor using the Corporate Memphis illustration style (2020).

> job or having the job fail. Let designers do some serious reading and writing of their own. Let designers offer the surplus value, the uselessness and the authorship of their profession to the world, to politics, to society.[343]

This proposed role reminds us that, less enthusiastically put by Norman Potter, of designers as cultural generators: "obsessive characters who work in back rooms and produce ideas, often more useful to other designers than the public."[344]

DIGITAL VERNACULAR AND THE REVENGE OF THE AUTHOR

This emphasis on knowledge and culture often coincides with a partial renunciation of the not so safe grounds of technical expertise, as well as a renunciation of the 'technical' at large. By this I don't mean that designers stop using any technical equipment or technique, but that these retreat into invisibility, while the focus of the work becomes more humanistic – rather than a tool to act in the world, technics itself becomes the object of a humanistic perspective. At this point, a clarification of what, in this context, I mean by 'technical' is needed. Here, the technical character of a practice is relative to the extent of its adoption. For instance, the alphabet can be rightly considered a communication technology, but you wouldn't call a writer a 'communication technician.' This is to say that the explicit *technicity* of successful technologies lasts for a limited period, after which it fades into the background. A counterexample: today, unlike in the late eighties, the act of double clicking to open a computer folder is not seen as a technical act unless you are human-computer interaction engineer. When the technicity of a practice becomes implicit, such a practice becomes a cultural one – it has

343 Van der Velden, "Research and Destroy", *op. cit.*
344 Potter, *op. cit.*, p. 12.

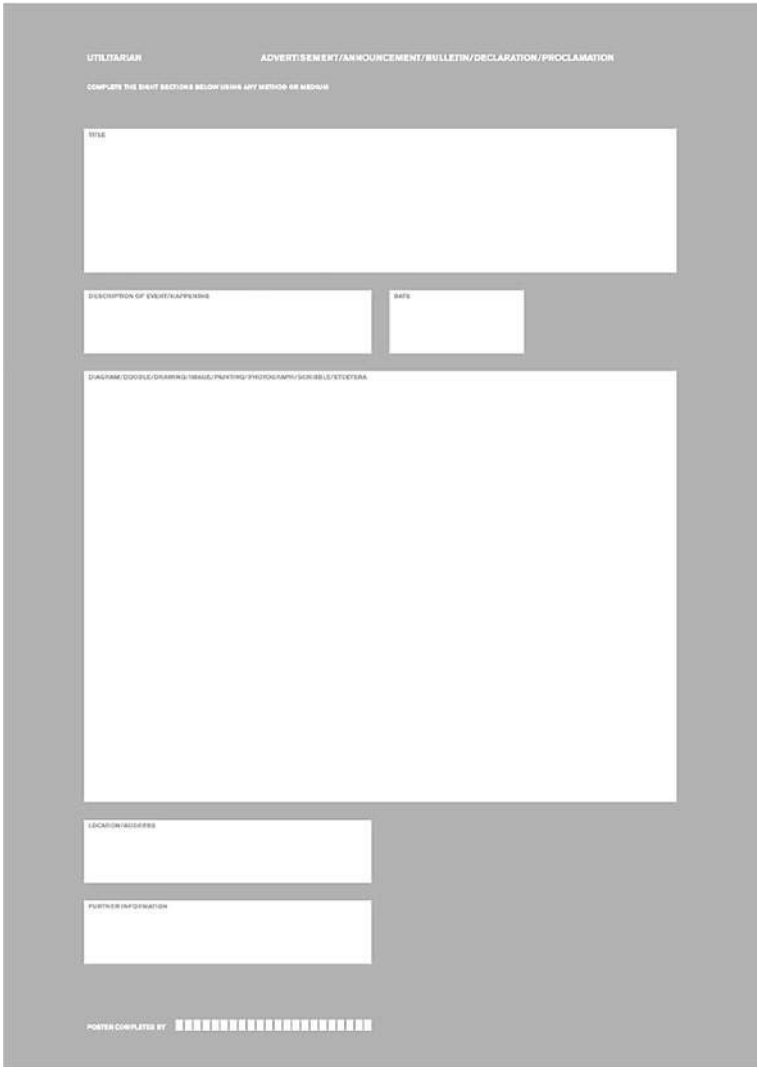
become a 'text' in Barthesian terms, to read and interpret. This is how the author avenges their death at the hands of technological democratisation.

In the '80s, art director Tibor Kalman and his design company M&CO began glorifying 'low,' vernacular visual cultures, following the analogue move of Robert Venturi, Denise Scott Brown and Steven Izenour in the field of architecture. It was time to learn. Not only from Las Vegas, but also from the roadside sign or the diner's menu. "Design without designers" became the motto, conceived by Kalman himself for an invitation to an event he organised. Who's motto, though? Of a group of hyper-professionalised graphic designers, of course. "We're interested in vernacular graphics – Kalman stated – because it's the purest and most honest and most direct form of communication. We will unabashedly steal from vernacular work."[345] But, if vernacular design is so pure and honest, why do we need professional graphic designers stealing from it? Because this gives the designer a new role, that is, to infuse the everyday with meaning and an ethos. What about the role of the vernacular, then? "[R]ather than being in opposition to modernism, [it] can be seen as a means of imprinting a sense of place onto modernity."[346] Given the diffusion of networked computers, vernacular recuperation of 'the Other' increasingly blends with digital default. In 2008 German graphic designer Manuel Bürger visited a copy shop in Mumbai to commission the design of his own business card, giving total freedom to the owner. Unsurprisingly, the card included several signifiers of graphic design's technicity such as a rainbow gradient and the clipart of a plotter machine.[347]

345 Kalman quoted in Rick Poynor. *No More Rules: Graphic Design and Postmodernism*. London: Laurence King, 2003, p. 81.

346 Beegan and Atkinson, *op. cit.*, p. 311.

347 See https://manuelbuerger.com/business-card. Today, we might turn up our noses at such a project, but we need to consider the fast pace at which our cultural sensibility evolved over the last decades. At the time, for instance, the notion of cultural appropriation or the term wasn't common knowledge, as it is now.



Utilitarian Poster by Daniel Eatock (1998). "A generic form/template silkscreened on newsprint paper, methodically guides the user through the steps of creating their own advertisement, and includes blanks in which to insert relevant information, such as titles of events, images, persons to contact, etc."

There are many present-day instances of the cultural recuperation of digital vernacular. A common one is the established design studio that uses the Comic Sans font in their social media posts and job announcements to signal a certain easy-going attitude. Here, Comics Sans does not appear so much as a manifestation of the bad taste of digital vernacular, but as an iconic residue of technological democratisation. Similarly, the infamous "graphic design is my passion" meme is less a mockery of amateurs than an embodiment of the materiality of design software, with its default fonts, standard visual effects and clipart libraries. How can a professional use what is generally deemed a non-professional font? Here's a little test for you: if you use Comic Sans and you can still be taken seriously, it means that you operate in a cultural sphere.

Cultural professionals, such as postmodernist and defaultist graphic designers, often take part in accelerating the mutation of technical into cultural. Their focus might be on practices that, until then, didn't feel the need to manifest self-awareness and didn't have any incentive to explicate their own theory and philosophy. Before being recuperated, the vernacular is always 'technical', as its culture *resides* in its technique, it's *Kulturtechnik*. After recuperation, the vernacular has become 'cultural' in a narrative, interpretive or moral sense: the technique is now the content of a story or a parable. Vernacular recuperation is, therefore, the production of narratives around techniques. Ultimately, there are two approaches towards the vernacular: ironic detachment and dignifying attachment, often confusedly combined (while Kalman spoke highly of the authenticity of vernacular design, critic Rick Poynor asked: "Was there not also an unspoken assumption in such borrowings, as visual ideas were transported from low culture (everyday designing) to high culture (professional design), that high was inherently superior?"[348]). They

348 Poynor, *No More Rules*, *op. cit.*, p. 84.

both, however, signal a distance. The ability to create and maintain such a distance is the distinctive trait of the cultural professional. What about autochthonous pride, then? As the shepherd doesn't call his sheep 'nature', in this case we cannot speak of vernacular, since there is no distance between the subject and the object.

Filipino designer Clara Balaguer and Dutch curator Nathalie Hartjes offer a temporal definition of the vernacular, that is, "[w]hat is truly of its time, often concocted through bucking propriety – for example, African-American vernacular English slang that thumbs its nose at canonical English. What is newest and thus not yet institutionalised/cannibalised. What has not yet been distilled and bottled into the formal. When a language formalises, this is when it can begin to alienate people." That being said, cannibalisation is not a destiny for Balaguer, who believes in the ability to "write about the vernacular *in* the vernacular": "[t]o understand how far you could go with that kind of popular working tool is to deeply listen to the literacy of the body popular. If you can understand where they're coming from, maybe you can learn to speak a similar language, opening a space of negotiation."[349]

GUILT AND LOSS

> In the end, the most potent critiques offered by designers using Default Systems seem to be linked to guilt and loss. Default Systems, and the formats that they include, comment not just on the mechanics of systems but on systemic thinking in general, and on the new life of man in the networked Global Village. The computer has changed design, but it has also changed our process of thinking and making. Formats and systems govern everything from our weaponry

349 Rhys Atkinson and Clara Balaguer. "Learning from the Vernacular." *Futuress*, June 16, 2021. https://futuress.org/stories/learning-from-the-vernacular/.

systems to our guidelines for citizenship.[350]

Guilt, fear, loss, ambition. Our relationship with technology is not just an operational one, but it's linked to a whole spectrum of sentiments: designers worry and get excited, they mourn the demise of a technique, and they celebrate the birth of new know-hows. The territory is murky and not without risks, such as that of deskilling, that is, the relative contraction and devaluation of specific skills. Here, the case of graphic design is emblematic: outside design circles, the graphic designer is considered a sort of technician who possesses a certain adroitness with desktop publishing applications. This perception determines the social value of the designer as a professional figure, and to some extent also their compensation. As happened with the service sector as a whole, some skills – some trades, even! – have been incorporated into general-purpose software. The Adobe suite has objectified some of the design know-how, it has turned practical knowledge into fixed capital. In the face of deskilling, how can design react? By replacing a specific competence – technical, if you will – that is no longer valuable, with an extended cultural competence. We can think of it as a process of *softskilling.* Paraphrasing design theorist Tomás Maldonado, the designer is increasingly intellectual and decreasingly technical. Unfortunately, however, the humanistic approach doesn't seem very lucrative: who will buy the narrative products of the cultural professional? I will try to answer this question in the last two chapters.

350 Giampietro and VanderLans, *op. cit*.

Chapter 6.

Kritikaoke: On Ornamental Politics and Identity as a Skill

"Formerly, like Kant and Hume, [artists] signed their letters 'Your most humble and obedient servant,' and undermined the foundations of throne and altar. Today they address heads of government by their first names, yet in every artistic activity they are subject to their illiterate masters."
– Max Horkheimer and Theodor W. Adorno, 1944[351]

"All that which today declares itself marginal, irrational, a revolt, anti-art, anti-design, etc. [...] all this obeys, willy-nilly, the same economics of the sign."
– Jean Baudrillard, 1972[352]

"My biography is more interesting than my art"
– *foundationClass collective at Documenta 15, 2022

CHARISMA

In 2017, I found myself in Berlin attending Re:Publica, an international conference on innovation meets politics meets branding meets tech. On the main stage, just after the vocal intervention of Russian chess master and activist Garry Kasparov, it was the turn of Dr Nelly Ben Hayoun, experience designer and "manufacturer of the impossible." Ben Hayoun is unanimously described, by the likes of Hans Ulrich Obrist and Michael Bierut, as a force of nature, "an inexhaustible source of renewable energy."[353] While the sheer scale of her design experiences for clients like NASA, MoMA or Airbnb implies the work of a team ("we work," "we believe"), Nelly Ben Hayoun Studios is clearly framed around a charismatic leader. Their productions are truly impressive, often including two dozen lines

351 Max Horkheimer and Theodor W. Adorno. *Dialectic of Enlightenment*. London: Verso, 1997, p. 105.

352 Jean Baudrillard. "Design and the Environment: Or, The Inflationary Curve of Political Economy." In *The Universitas Project*, *op. cit.*, p. 60.

353 Aimee Mclaughlin. "Nelly Ben Hayoun on Her Theory of Total Bombardment." *Creative Review*, June 27, 2018. https://www.creativereview.co.uk/nelly-ben-hayoun-on-her-theory-of-total-bombardment/.

of credits. Faced with such a vast and energetic orchestration of talent, any practitioner blanches.

Nelly Ben Hayoun was there, in prime time, to present the University of the Underground, a new postgraduate course made up of "dreamers of the day" with the goal of forming the "very hard working" critical thinkers and radical designers that our world is so much in need of these days. A school for the "the Willy Wonkas of modern times, the contemporary Joy Divisions, JG Ballards, Marie Curies and Rauschenbergs, action researchers and designers, mythologists and makers of new worlds!"[354] On stage, the French experience designer lived up to the endorsements: her performance was cheerfully chaotic, with a giggling audience and multiple plot twists (speaking of charismatic leadership, at a certain point there were *three* Nellys on stage).

The University of the Underground, hosted in Amsterdam by the prestigious Sandberg Instituut but established in London as well, is just one among the many experiments, large and small, in alternative education and pedagogy.[355] So, what makes the UUG a particularly fascinating case study? Besides its laudable commitment to tuition-free education and the ambitious plan to run its MA for 100 years, the bombastic branding, positioning and charismatic leadership of the University of the Underground – winking at grass-roots movements and do-it-yourself experiences but at the same time emphasising free will and personality – represents a good opportunity to reflect on the meaning of counterculture today and evaluate its role. How is it that institutions, both emergent and well-established, can neutralise, regurgitate and then even administer countercultural expressions?

354 http://nellyben.com/projects/experiences/the-university-of-the-underground/.

355 Among them, within the field of design, the Scuola Open Source in the south of Italy, the Parallel School and the nomadic Relearn sessions. Zooming in the Netherlands, Hackers and Designers and Open Set.

MYTHMAKERS, CREATIVE SOLDIERS, FUTURE PRESIDENTS

Within the UUG, chaos is considered "a method of public engagement." Rooted in Anthony Dunne and Fiona Raby's Critical Design, influenced by theatrical practices and inspired by Roland Barthes' idea of the mythologist, the school trains "creative soldiers" to infiltrate institutions in order to "engineer change" with the hope that – who knows – some of them might become presidents one day. "Manufacturing countercultures" and providing a "positive inspiration and disturbance" is the way to go. The set of references informing the culture of the school is maximalist and eclectic: punk, Artaud's theatre of cruelty, "pirate utopias,"[356] The Smiths, and so on. The school presents itself a bit like a teenage bedroom: a ludic space organised in such a way as to signal a sense of belonging to certain groups, to suggest different breeds of *coolness*.

Signalling plays a crucial role here. Describing the International Space Orchestra project, Ben Hayoun elucidates her understanding of counterculture, somehow derived from Pierre Bourdieu's model of the different forms of capital.[357] In a sense, it seems as though the University of the Underground has incorporated Bourdieu's analysis *a bit too well*. Its branding combines a critical/disruptive lexicon with an array of progressive cultural icons, a very diverse dream team of advisers, and a street aesthetics reminiscent of punk zines involving stencils, xerography and markers. This straightforward mobilisation of cultural capital both in its embodied and objectified state speaks of the current institutional landscape

356 See https://en.wikipedia.org/wiki/Pirate_utopia.

357 Pierre Bourdieu. "The Forms of Capital." In *Handbook of Theory and Research for the Sociology of Education*, edited by J. Richardson, 241–58. New York: Greenwood, 1986. https://www.marxists.org/reference/subject/philosophy/works/fr/bourdieu-forms-capital.htm.

Example of a faux-picket at Re:Publica 2017. Photo by Dennis Knake/Lemonbeat GmbH.

– which is not afraid of 'agitators' anymore, instead it welcomes them – and pitches the school to mildly progressive media outlets and their audience. These are the new "rules for radicals."[358]

Clearly, mobilising various forms of capital is both unavoidable and necessary. Yet, it seems that the UUG, like many other instances of 'radical change' in design, simply replicates traditional dynamics of accumulation, obfuscating them under the veil of "criticool" jargon.[359] Working with institutions? Boring... We "infiltrate" them. Designers? No thanks, Please call us "creative soldiers." A semblance of antagonism is the perfect accessory to the casual look of prestige. Paradoxically, the manifestation of a pseudo-antagonist social and cultural capital becomes a means of acquiring more of it in a pacified, institutionalised form, disguising direct and indirect economic conversions that happen elsewhere. 'Anti' is the precondition of 'into', and later its corollary.

Bourdieu himself recognised counterculture as an attempt to constitute a market with its own rules:

> Thus, what is nowadays called the 'counter-culture' may well be the product of the endeavour of new-style autodidacts to free themselves from the constraints of the scholastic market (to which the less confident old-style autodidacts continue to submit, al though it condemns their products in advance. They strive to do so by producing another market, with its own consecrating agencies, that is, like the high-society or intellectual markets, capable of challenging the pretension of the educational system to impose the principles of evaluation of competences and manners which reign in the scholastic market,

358 See Saul D. Alinsky. *Rules for Radicals: A Practical Primer for Realistic Radicals*. New York: Vintage Books, 1989.

359 Francisco Laranjo. "Critical Everything." *Modes of Criticism*, August 4, 2015. https://modesofcriticism.org/critical-everything/.

> or at least its most 'scholastic' sectors, on a perfectly unified market in cultural goods.[360]

What he didn't anticipate, however, is that the institutional market would be able to digest the countercultural one. Culture and social relationships become respectively cultural and social capital when they are used to compete, consciously or not, against other agents. Compete for what? Platforms of expression, attention, funding. And these are rarely redirected to those who reject the 'disruptive' role models institutionally tailored to the docile high-end creative professional. Shouldn't the goal of counterculture be to reveal the hidden automatisms behind the acquisition, mobilisation and conversion of, not only economic, but also social and cultural capital?

IRONIC ATTACHMENT

During the Re:Publica presentation, one candidate's application was showcased to convey the vibe of the school. His application consisted in a videoclip of himself playing an '80s song featuring several clichés of contemporary design discourse ("I want to change the world," "I'm process-led, concept-driven"). Ironically, his gig conveyed the idea that radical expression as an institutionalised practice is the new default. The performance was a parodic *mise-en-scène* of disposition, inspired by a common part in a common play, that of the creative mind addressing an organisation. The irony is in the juxtaposition of the solemn and enthusiastic ambition of changing the world to a dry inflection and a frivolous tune for entertainment and mindless consumption – enthusiastic engagement as muzak. Was the applicant trolling? Maybe. One thing is certain, though: on a meta-level of irony, what was a dig at enthusiastic commitment was then unironically used to actually commit.

360 Pierre Bourdieu. *Distinction: A Social Critique of the Judgement of Taste*. Cambridge, MA: Harvard University Press, 1984, p. 96.

The Indiani metropolitani, a post-hippie and art-oriented subcultural youth group which was part of the Italian '77 movement, used irony as a disorienting strategy for public protest, yelling slogans like "We demand to work harder and earn less!" Inspired by both *Grundrisse* and Dadaism, they were fascinated by the ambiguous nature of irony:

> What interests us is the sense of bitterness that irony leaves us with, its flattening action. Irony opens spaces, it unhinges, it reveals what cannot be hidden anymore [...]. Irony lacks flesh and blood, it is only partially a practice of liberation, as partial as is violence and its organization. Finally, irony is a frustrating "language that marks the space between our desires and the difficulty of their realization."[361]

Irony, then, was a way to inhabit the gap between expectations and reality. But our society is post-ironic, for it has learned to neutralise irony's subversive power by simply incorporating it. A global online marketplace can shamelessly launch a campaign that promotes unhealthy workaholism,[362] while H&M can successfully bring to market UNEMPLOYED hoodies.[363] As David Foster Wallace pointed out, irony, especially in its postmodern breed, moved away from its antagonistic origins to evolve into a mere advertising technique that, while pleasing the audience, acts as a protective shield against criticisms, because how can you ironically criticise something that is already ironic about itself?[364]

361 Patrick Cuninghame. "'A Laughter That Will Bury You All': Irony as Protest and Language as Struggle in the Italian 1977 Movement." *Libcom*, January 5, 2012. https://libcom.org/article/laughter-will-bury-you-all-irony-protest-and-language-struggle-italian-1977-movement.

362 See Silvio Lorusso. "We, the Doers: Fiverr's Entrepreneurial Populism and a 3-Day Workweek." *Entreprecariat* (blog), March 20, 2017. https://networkcultures.org/entreprecariat/we-the-doers/.

363 Brian O'Flynn. "Class Struggle: When Did 'Unemployed' Become a T-Shirt Slogan?" *The Guardian*, May 3, 2018, sec. Fashion. https://www.theguardian.com/fashion/2018/may/03/class-struggle-working-class-unemployed-become-t-shirt-slogans.

364 David Foster Wallace. "E Unibus Pluram: Television and U.S. Fiction." *Review of Contemporary Fiction* 13, no. 2 (1993). http://jsomers.net/DFW_TV.pdf.

Paraphrasing DFW, the applicant's videoclip manages simultaneously to make fun of itself, the design world, and the ones who are meant to evaluate his attitude, pleased by the fact that they *get the joke*. However, I'd like to offer another interpretation of the ironic stance of the videoclip, one that has to do with detachment. Commonly, irony functions as a means of coping with a feeling of powerlessness and irrelevance. When switched on, the 'ironic detachment' mode allows us to distance ourselves from collective and individual miseries. In exchange, we pay the toll of disengagement. What is the song, if not a disengaged take on impotent engagement?

TERRIBLE SITUATIONS, GREAT OPPORTUNITIES

During the Re:Publica talk, the notion of "performance of politics" caught my attention. The term indicated a technique to incite public engagement (this was before the expression 'performative activism' became derogatory). Tweaking it a bit, the notion can be used to identify one constitutive aspect of design's ambiguous value system. Design has long learned to abhor its commercial, utilitarian, wasteful and dehumanising nature. Key figures like Victor Papanek and Ken Garland vocally criticised the sheer amount of time and energy that designers spend on polishing the cogs of the capitalist machine. More recently, the aforementioned Anthony Dunne and Fiona Raby advocated a design that makes us think instead of making us buy (a questionable opposition, to be frank).[365] Designers have learned to jot down manifestos – and so have advertising agencies.[366] Of course, the UUG has

365 Anthony Dunne and Fiona Raby. "A/B." *Dunne & Raby*, 2009. https://dunneandraby.co.uk/content/projects/476/0.

366 See http://www.manifestoproject.it/ and Tara-Nicholle Nelson. "What Transformational Brands Do: Manifesto Marketing." *Ad Age*, April 25, 2017. https://adage.com/article/agency-viewpoint/transformational-brands-manifesto-marketing/308572. In 1978, Rem Koolhaas asked "how to write a manifesto in an age disgusted with them?" In 2023, the question becomes "how to write a

EMOTIONAL LABOR INVOICE

Marginalized Folks, Inc.

TO: Potential Ally
Privileged Folks, Ltd

SERVICE PROVIDER	SPECIALIZATION	DUE DATE
Marginalized Person You Know	**Existing in Oppression**	**DUE ON RECEIPT**

DESCRIPTION	#	UNIT PRICE	LINE TOTAL
Helped you understand your racism/sexism/transphobia/ableism/etc.		$ 100.00	
Endured your microaggression(s)		$ 200.00	
Taught you about microaggressions & structural oppression		$ 300.00	
Explained something about oppression you could've Googled		$ 125.00	
Clarified that you are not entitled to my time		$ 250.00	
Clarified that you are not entitled to my pain		$ 350.00	
Softened my reaction to spare your feelings		$ 500.00	
Made you feel like a "good" ally		$ 600.00	
Listened to "not all...." and similar derailments/fragility		$ 750.00	
Smiled when you apologized for not speaking up in the meeting		$ 1,000.00	
		Subtotal	
		Marginalization Tax Adjustment	$ 1,500.00
		Amount Due:	

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Emotional Labour Invoice by Anastasia Collins.

its own.[367] In the meantime, plenty of labels like Social Design, Critical Design, Speculative Design, etc. have followed one another and continue to do so. Each of these iterations has contributed to an increasingly urgent but also abstract focus on the big issues of our time. This focus is not just the mirror image of design hubris, but also a symbolic return of investment in the currency of confidence, for those who have spent years becoming designers.

The saga of design heroes fighting against gargantuan societal problems has been successfully packaged in events like What Design Can Do, a Dutch platform created to "demonstrate the power of design; to show that it can do more than make things pretty. To call on designers to stand up, take responsibility and consider the beneficial contribution that designers can make to society."[368] Every year a new challenge is launched, such as the "refugee challenge" or the "climate change challenge," so that a multidimensional geopolitical issue becomes, as the *Volkskrant* newspaper reported, a Dragon Den-like competition. Here, design positions itself as the "ultimate problem-solving discipline," superior to governments or NGOs.[369] Global tragedies become design opportunities. I mean, literally. Bruce Mau, author of the *Incomplete Manifesto for Growth*, reportedly stated that "a terrible situation is a great opportunity to use design thinking."[370] The same design thinking that can be sold to companies and corporations. The wet dream of a universal design language comes true in the paradigm of design thinking-as-consultancy.

Undoubtedly, individual designers are genuinely concerned with specific issues, whether small or large, and

manifesto in an age enthralled by them?"

367 https://universityoftheunderground.org/design-experiences-manifesto.

368 https://www.whatdesigncando.com/.

369 Ruben Pater. "Treating the Refugee Crisis as a Design Problem Is Problematic." *Dezeen*, April 21, 2016. https://www.dezeen.com/2016/04/21/ruben-pater-opinion-what-design-can-do-refugee-crisis-problematic-design/.

370 Bas van Lier. "If Things Are Bad, That's Good." *What Design Can Do*, March 28, 2017. https://www.whatdesigncando.com/stories/things-bad-thats-good/.

sympathise with or belong to particular social groups, but design as a field tends to target general problems concerning a category, a user group, a set of personas. Design is humanitarian, in the sense that it is not concerned with *some* human beings, but with Humanity. This is, at least partially, a form of professional propaganda aimed at policy makers to let them pour money into the creative industries machine, financing the next social design event or prize. An ecumenic notion of societal impact is appealing. Political and social engagement is thus performed, at least in some measure, to please the policy-making big Other. In doing so, design constructs an imperfect but highly artificial image of the world in order to offer a solutionist happy ending, which, as in many Messianic cults, is constantly postponed. Who will "interrupt the cycle of capitalism?" Designers, of course.[371]

Inevitably, this type of magical thinking influences education insomuch as students, explicitly or not, are encouraged to wield 'conspicuous morality' and use it to satisfy their ambition – they become *moral climbers*.[372] Progressiveness, together with social and political engagement, becomes a form of positional consumption and, as such, it is an added value to the project and to the designer. This sets the context of evaluation by teachers, funding bodies and stakeholders alike. Thus, the design museum-educational complex offers a dispositional *Grundkurs* where one is urged to feign a more or less standardised expression of critical and socially concerned thinking within safe and somehow predefined ethical boundaries. A sort of humblebrag of good intentions that doesn't hurt or upset anybody. Against this backdrop, the notion of performance of politics is less a form of deep engagement than a garnish

371 Diana Budds. "9 Ideas Shaping The Future Of Design, According To Ideo, Microsoft, Autodesk, MIT, And More." *Fast Company*, September 12, 2017. https://www.fastcompany.com/90139617/9-ideas-shaping-the-future-of-design-according-to-ideo-microsoft-autodesk-mit-and-more.

372 I owe this expression to Italian writer Guido Vitiello.

to apply to one's own projects and practice. The consequences, according to the anonymous admins of Instagram architectural meme account @dank.lloyd.wright, are dire: "The problem is that traditional power relations and patterns of extraction carry on as usual under a veneer of progressive language. The awareness of that disconnect doesn't lead to systemic change, but instead to feelings of disaffection and powerlessness."[373]

ORNAMENTAL POLITICS

> Anyone dealing with the Creative Industries Funds may notice that projects with a strong *social* undertone have been given preference in the last few years. Not only 'multifacetedness,' but politics itself becomes a survival strategy for designers – a necessary branding and important checkbox on our portfolios.[374]

Designers like Anastasia Kubrak, quoted above, realise that a way to facilitate their success in the cultural arena is to "design your struggle."[375] Being 'political' is a plus. However, the idea of politics that informs this attitude is reductive: it is more about what the work *says* than what it *does*. The statement, the manifesto, the invective is more positively scrutinised than the inner logic of a job or the social relations that it produces. Politics functions as a badge rather than a process, with the further risk of turning it into a fetish, a formal requirement, a norm. Politics acquires a decorative value, it becomes *ornamental politics*: the radical slogan, the activist posture, the glorious declaration are adopted as decorations of purely autonomous practices, that is, cut off from the murky waters

373 James Brillon. "Instagram Account Dank.Lloyd.Wright Aims to 'Amplify Narratives That Are Excluded from Architecture's Official Consensus.'" *Dezeen*, August 30, 2022. https://www.dezeen.com/2022/08/30/instagram-dank-lloyd-wright-interview/.

374 Anastasia Kubrak. "Under Unwritten Terms & Conditions." In de Vet, *op. cit.*, pp. 166-7.

375 Gerritzen and Lovink, *Everyone is a Designer in the Age of Social Media*, *op. cit.*, p. 103.

of micro and macro politics.[376] Often, but not always, the political is reduced to the equivalent of a Che Guevara pin on a Fjällräven Kånken backpack.[377] Certain symbolic forms in this very cultural arenas confirm this, such as the 'faux-picket,' that is, a bunch of picket signs, with innocuous or even meaningless slogans that function as temporary installations in festivals and conferences.[378]Are we facing a good-willed, institutionalised aestheticisation of politics? Or is this a fetishism of rebellion where expression matters more than design, and in which practitioners are urged to love the 'revolt' more than the world to which it might give rise?[379] Let's listen once again to Walter Benjamin, who "demanded that artists must not merely adopt political 'content,' but must revolutionize the means through which their work is produced and distributed."[380]

In 1998, Mr. Keedy wrote in *Emigre*: "Today's young designers don't worry about selling out, or having to work for 'the man,' a conceit almost no one can afford anymore. Now everyone wants to be 'the man.' What is left of an avant-garde in graphic design isn't about resistance, cultural critique, or experimenting with meaning. Now the avant-garde only consists of technological mastery: who is using the coolest bit of code or getting the most out of

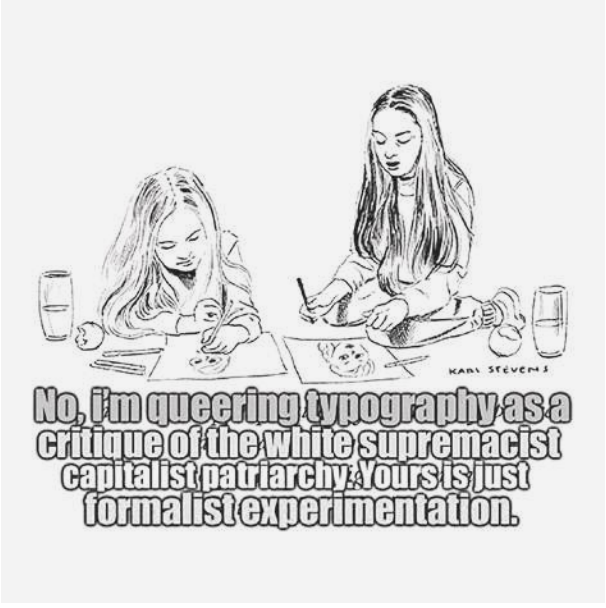
376 This is not to say that ornaments cannot be political. The problem, here, is with the reduction of politics to ornament. In short: "yes to political ornaments, no to ornamental politics." I thank Sofia Gonçalves for provoking this reflection.

377 See Schessa Garbutt. "Black Lives Matter Is Not a Design Challenge." *Design Toast* (blog), July 14, 2020. https://medium.com/design-toast/black-lives-matter-is-not-a-design-challenge-f6e452ff7821.

378 One faux-picket took place at the very Re:Publica event discussed above. Another one took place during the 2019 edition of Dutch Design Week. That said, even highly meaningful political statements can be made ornamental. This is the case, for instance, of a giant painting of a pedestrian section in downtown Washington, D.C. that reads "Black Lives Matter" in full caps. According to philosopher Olúfẹ́mi Táíwò, this is an example of "the elites' tactic of performing symbolic identity politics to pacify protestors without enacting material reforms." *Elite Capture: How the Powerful Took Over Identity Politics (and Everything Else)*. Chicago, IL: Haymarket Books, 2022.

379 Cf. See Maldonado, La speranza progettuale, *op. cit.*, p. 99.

380 Ellen Lupton. "Designer as Producer." In *Graphic Design: Now in Production*, edited by Ellen Lupton and Andrew Blauvelt. New York, 2012.



Meme by @ethicaldesign69 which mobilises the tension between the critical and the aesthetic disposition (2023).

their HTML this week."[381] In the museum-educational complex, quite the opposite is true: explicit cultural critique comes at the expense of "mastery," which is deemed as narrow-mindedness. The antidote? A broad understanding of politics in which a work that makes its politics explicit shouldn't be deemed superior to a 'formal' work which implicitly enacts its politics.

Manifestations of ornamental politics, virtue signalling, political performativity, etc. are generally interpreted through the lens of sincerity. If you take a political position for your own economic benefit, you're 'selling out' or you've been co-opted. This lens is not analytically useful: political purity (which is different from coherence) is a feature of martyrs. More interesting than looking at the veracity of the virtue that is signalled is to interpret the mechanisms of such signalling. If politics is an important checkbox on a designer's portfolio, it means that it is analogous to a skill. Therefore, a political position or, more precisely, the *manifestation of being politically positioned* in what is considered an acceptable or even correct way, becomes a form of competence. Within this line of thought, Raffaele Alberto Ventura concludes that political correctness "is not an innate merit but no less than an acquired skill."[382]

CRITICAL DISPOSITION

The aesthetic disposition was, according to Pierre Bourdieu, the traditional instrument of distinction between the masses and the elites. In 2018, Nicholas Holm argued that the aesthetic disposition was being replaced by a critical disposition. This is how the latter works:

> Like the aesthetic disposition, this new critical disposition is founded upon a distinct set of

381 Mr. Keedy. "Graphic Design in the Postmodern Era." *Emigre*, 1998. https://www.emigre.com/Essays/Magazine/GraphicDesigninthePostmodernEra.

382 Raffaele Alberto Ventura. "La cattiva notizia è che la cancel culture esiste eccome." *Wired Italia*, May 10, 2021. https://www.wired.it/play/cultura/2021/05/10/cancel-culture-esiste-debunker-politicamente-corretto/.

> rules for both interpreting and judging culture. However, whereas the aesthetic disposition attends to form and expression, the critical disposition encounters texts in terms of their politics: their entanglement in wider structures of power [...] In such a world view, what is thought to divide elites from the common people is no longer high culture, but a critical orientation towards the world that frequently manifests in terms of cultural interpretation.[383]

Mentioning magazines, blogs, podcasts such as *Buzzfeed*, *NPR*, and the *New Yorker*, Holm demonstrates that the critical disposition, which is often presented as a source of pride within design culture, is in fact a general phenomenon. Of course, the emergence of the critical disposition has a detrimental effect on the aesthetic one, hence formalist readings of an artefact are cast as naive. Within design culture, the aesthetic disposition derives from the notion of function (even when it opposes it!). If we consider functionalism as "appearance of use-value," we understand that it is nothing but a peculiar type of aesthetic judgement. However, when the critical disposition becomes pervasive, the designer obsessed with *gute Form* starts occupying a lower position in the cultural arena, hence all the attempts to catch up with the ever-growing inventory of critical-political concepts.

Designers display a critical disposition by expressing their ambition to raise awareness on a certain issue or problem. This impulse surely derives from the pedagogical role that designers aspire to occupy in society, but it's also linked to the larger scope of liberalism. According to artist Brad Troemel, "raising awareness is ultimately the process of making people aware that you are aware of

383 Nicholas Holm. "Critical Capital: Cultural Studies, the Critical Disposition and Critical Reading as Elite Practice." *Cultural Studies* 34, no. 1 (January 2, 2020): 143–66. https://doi.org/10.1080/09502386.2018.1549265, p. 10-7.

the conversation that needs to be had."[384] The issues one raises awareness of are always 'urgent'. By emphasising the urgency of the issues they raise awareness of, individuals also proclaim their own cultural significance. This becomes clear in social media campaigns. Similarly, Catherine Liu states that "[l]iberal members of the credentialed classes love to use the word *empower* when they talk about 'people,' but the use of that verb objectifies the recipients of their help while implying that the people have no access to power without them."[385] Thus, the empowering act creates a distinction, at least symbolic, between the powerful and the powerless. It's important to consider the economic aspect of raising awareness, as it is not expensive in terms of means. Issuing a statement requires less time and energy than building a product, hence the privileged position of graphic designers as critical actors. According to Glenn Adamson, "Graphic design in particular lends itself to gestures of protest, from punk album covers to handmade banners. But architecture and product design, where the big money is, have always been service businesses."[386]

ACTIVIST OR AUTHOR?

It's not uncommon for designers to present their work as a form of activism. To what extent is this fair to conventional activists who semi-anonymously take part in movements? Ann Thorpe, who proposed a set of criteria to identify what constitutes design activism, indicates that the practices of Critical Design are characterised by a certain detachment: "Critical design examples often skirt the twin issues of making specific claims for change and of clearly identifying how the change benefits a wronged or excluded group. Instead they show design's efforts to

384 https://www.patreon.com/posts/left-cant-meme-77324335.
385 Catherine Liu. *Virtue Hoarders: The Case Against the Professional Managerial Class*. Minneapolis, MN: University of Minnesota Press, 2021, p. 1.
386 Adamson, *op. cit*.

change or challenge cultural discourse."[387] Here we see, once again, that the arena of the self-anointed activist designer is the culture industry, more than society as a whole. In some educational contexts this orientation is explicit, such as the curriculum on auto-ethnography in design developed by Louise Schouwenberg, who considers design "a practice of cultural critique."[388] In an Instagram post, later deleted, design critic Alice Rawsthorn enthusiastically states that "[t]here are many examples of great design activism: from the simple physical gesture of raising and clenching a fist [...] to the colour-coded strategy [...] adopted by British suffragettes in the early 1900s." Design panism is here deployed according to a clear cultural strategy: by qualifying any form of activism as *design* activism, the cultural professional can write, curate and speak about anything.

From this perspective, politics and activism can be seen as a subset of a broader authorial approach, which has been long formulated and defended by designers in order to maintain, or more precisely gain, "critical autonomy."[389] Designer and writer Michael Rock recognises that for the designer-as-author there is no fundamental difference between any kind of self-directed:

> The general authorship rhetoric seems to include any work by a designer that is self-motivated, from artist books to political activism. But artist books easily fall within the realm and

387 Her criteria to consider design as activism: "It publicly reveals or frames a problem or challenging issue; it makes a contentious claim for change (it calls for change) based on that problem or issue; it works on behalf of a neglected, excluded or disadvantaged group; it disrupts routine practices, or systems of authority, which gives it the characteristic of being unconventional or unorthodox – outside traditional channels of change." Ann Thorpe. "Defining Design as Activism." In *Journal of Architectural Education*, 2011, p. 6.

388 Schouwenberg quoted in Jarrett Fuller. "The Auto-Ethnographic Turn in Design." *Design and Culture*, April 27, 2022, 1–3. https://doi.org/10.1080/17547075.2022.2061138. This turn is, of course, connected to autonomy. Schouwenberg continues: "We stressed the importance of designers formulating their own agendas and ideals and of basing theses on personal interests and artistic talents."

389 More on this in the next chapter.

> descriptive power of art criticism. Activist work may be neatly explicated using allusions to propaganda, graphic design, public relations and advertising.[390]

This lack of a 'client' is somewhat perplexing when it comes to activism, which is generally a communal activity in which a group has some specific needs and demands. In Thorpe's words: "Finally, most scholars of social movements note that activism is effective only as part of a broader movement or campaign."[391] However, for the critical designer-as-author, politics is the *content* of the work, more than the driving force of a horizontally negotiated praxis. We end up, again, in the domain of interpretation, where society is read as a text. What the critical designer, the radical designer, the activist designer, the designer as researcher, the speculative designer have in common is that they all manipulate their own content. Thorpe's conclusion is dispiriting: "In activist terms designers mainly talk to themselves."[392]

The example of Italian Anti-Design from the '60s and '70s helps shed light on the systemic reasons that lead designers to adopt a design practice consisting in activism as a form of cultural critique. Curator Emilio Ambasz described the approach of the 'anti-designers' as follows:

> Torn by the dilemma of having been trained as creators of objects, and yet being incapable of controlling either the significance or the ultimate uses of these objects, they find themselves unable to reconcile the conflicts between their social concerns and their professional practices. They have thus developed a rhetorical mode to cope with these contradictions.[393]

390 Michael Rock. "Designer as Author." *2x4* (blog), August 5, 1996. https://2x4.org/ideas/1996/designer-as-author/.

391 Thorpe, *op. cit.*, p. 6.

392 *Ibidem*, p. 14.

393 Ambasz quoted in Wizinsky, *op. cit.*, p. 177.

It is important to recognise that, as Matt Malpass points out, much of the anti-design activism happened during an economic downturn plagued by a high level of unemployment. Radical designers resorted to critical practices because of their lack of opportunities to operate as service providers: "these designers couldn't find work."[394]

THE IDENTITY TRAP

A particular case of political positioning within design has to do with politicising one's identity, understood here as the belonging to an oppressed or minoritarian group. It is not uncommon today to find practitioners who "identify as a mother," or use in their bios labels such as "queer designer," "black designer" or "liberatory designer." Whereas in the past one would find identification in the profession (for example, one's uncle or aunt would be characterised as 'the architect' or 'the doctor'), today we witness the professionalisation of identity. Clearly, the professional bio is one of the many means to manifest pride and a sense of affinity, especially in the face of the systemic exclusion that professional environments have enacted throughout history. That said, this development should be read against a broader shift in the role that identity plays in the labour market.

Some designers argue that their identity is inextricable from their practice. For instance, Dantley Davis, vice-president of digital product design at Nike, speaks of his racial identity as a sort of superpower: "I couldn't imagine my design practice without race; it helped me advance in so many ways. With it I see around corners that my peers can't see."[395] However, one of the risks concerning the professionalisation of identity has to do with the expectation that a person's work *must be* concerned with their identity,

394 Malpass quoted in Wizinsky, *op. cit.*, p. 174-5.

395 Davis quoted in Anne H. Berry, Kareem Collie, Penina Acayo Laker, Lesley-Ann Noel, Jennifer Rittner, and Kelly Walters, eds. *The Black Experience in Design: Identity, Expression & Reflection*. New York: Allworth Press, 2022, p. 68.

if such identity suffers or has suffered oppression, while hegemonic subjectivities are granted flexibility and autonomy of identity production. In 2016, E. Jane wrote a manifesto, entitled Nope, where they declared:

> I am not an identity artist just because I am a Black artist with multiple selves. [...] I am not grappling with notions of identity and representation in my art. [...] I am not asking who I am.[396]

More recently, Mohammed Al-Hawajri, argued that, for artists like him born in refugee camps in the Gaza Strip, making non-political art is *itself* a political gesture, since "a political reading is always projected unto their work, regardless of whether they intended it or not. That is how 'the other' sees them: as politicized bodies from a politically charged area."[397]

A recent meme format reminds us that a job, a hobby or even a pathology "is not a personality." In many cultural contexts we see the commingling of identity and personality. The political performance of identity can be a tool to claim recognition, equal opportunities, etc. – it is a form of strategic essentialism. At the same time, however, the prominence of personality can be seen as the ultimate outcome of the postmodern phenomenon of 'sampling' the self, that is, constructing an authentic self from bit and pieces of different cultural expressions, like music genres, artistic styles, and more recently the various -cores. Here, the root of the authenticity of that self lies in its very constructedness, one that reflects the micro-targeting of advertisers.[398] Already in the late '70s, social critic Chris-

396 https://e-janestudio.tumblr.com/post/132335744305/i-am-not-an-identity-artist-just-because-i-am-a.

397 https://thequestionoffunding.com/Documenta-Fifteen.

398 Some parallels can be found in the contemporary crafting of nonbinary identity. "Today, 'gender identity' references a core selfhood that requires no expression, no embodiment, and no commonality — in the case of some of the microidentities spreading on the internet — with genders as they are lived by others in the world. In this sense, contemporary gender identity is the apotheosis of the liberal Western fantasy of self-determining 'autological' selfhood, a regulatory ideal that gains meaning only in opposition to the 'genealogical' selfhood, overdetermined by social bonds, ascribed to racialized and indigenous peoples.

topher Lasch pointed out that personality had a role to play in the labour market: "In our time, the elimination of skills not only from manual work but from white-collar jobs as well has created conditions in which labour power takes the form of personality rather than strength or intelligence. Men and women alike have to project an attractive image and to become simultaneously role players and connoisseurs of their own performance."[399]

"Everything you make has you-ness – and part of what you will do at design school is develop this you-ness as you develop skills as a designer."[400] This is how US educator Mitch Goldstein speaks of a designer's personality and its role in their job. Goldstein might be more right than he thinks: the self – what he calls "you-ness" – doesn't just develop alongside professional skills: *the very self has become a skill*. And those who suspect that the self-as-skill is a reductive idea of individuality might find solace in the words of Horkheimer and Adorno: "only because individuals are none but mere intersections of universal tendencies is it possible to reabsorb them smoothly into the universal."[401]

CONDEMNED TO INDIVIDUALISATION

In 2011, Rob Giampietro noticed an emphasis on biography and a heightened self-awareness in design schools. He didn't consider this biographical focus as the result of a narcissistic leniency, but as one of the main burdens of the modern subject. As sociologist Ulrich Beck maintained, "people are condemned to individualization."[402] Being

[...] It is therefore difficult to imagine an identity more provincially Western and less decolonial than contemporary nonbinary identity." Kadji Amin. "We Are All Nonbinary." *Representations* 158, no. 1 (May 1, 2022): 106–19. https://doi.org/10.1525/rep.2022.158.11.106, p. 116.

399 Christopher Lasch. *The Culture of Narcissism: American Life in an Age of Diminishing Expectations*. New York, NY: W. W. Norton & Company, 1991, p. 92.

400 Mitch Goldstein. *How to Be a Design Student (and How to Teach Them)*. New York: Princeton Architectural Press, 2023.

401 Horkheimer and Adorno, *op. cit.*, p. 125.

402 Rob Giampietro. "School Days." In *Graphic Design: Now in Production*, edited by

proudly international, some schools value and encourage biographical expression as an interface to cultural difference, a badge of honour given their multicultural ethos. However, a biography perceived as uncommon – in geographical, class or bodily terms – can be exoticised and therefore 'othered' once again. Here, the unfamiliar biography is made valuable ('one's roots') not for its intrinsic value as the story of a unique life, but because of its scarcity. A work that mobilises an unfamiliar life story is framed as a cultural statement, while one that is rooted in a relatively ordinary biography might be deemed mere egotistic indulgence. Not everyone's 'becoming who they are' is validated in the same way. Either way, schools are in trouble because they struggle to discern the biographical from the personal, what relates to one's place in society from what is a feature of individual character, what is debatable from what should be unquestionable.

A keyword that points to such tensions is *position*. In the Netherlands, for instance, a frequent question asked in high-level design schools is "how do you position yourself?" The question is, of course, of a maieutic kind: it is meant to help students situate themselves within the issue they are investigating. The question works as an injunction because it forces the student to produce a self-image. The position can be the one of designer as mediator, as problem-solver, as activist, etc. Or more broadly as male, as Western, as able-bodied. Positional complexities are now at the heart of the design field's identity crisis. As schools rarely have the conceptual tools to address such complexities, this identity crisis (that, as we have seen, is somewhat intrinsic to the design field and now just feels more apparent) is shared with, if not offloaded on, the student-practitioner.

What's the institutional responsibility here? How can schools facilitate the generative crisis of the field without

Ellen Lupton and Andrew Blauvelt. New York, 2012. https://linedandunlined.com/archive/school-days/.

turning it into the identity crisis of individual students? Positional maieutics is a valid and useful means, but the dilemmas and wicked problems that it engenders should not be merely outsourced. Furthermore, schools shouldn't use those dilemmas either as formal or informal evaluation criteria. It is a risk for an educational organisation to engage with biography, especially in a time when individuality is forced upon individuals. Not every facet of biography should be scrutinised. And the ones that deserve attention, shouldn't be personalised. To manage this complexity, schools should become able to navigate intimacy, privacy and confidentiality. Most of all, they should avoid flattening a life story into a project-practice surface for evaluation or promotional purposes.

THE BUREAUCRACY OF IDENTITARIAN SKILLS

According to Fred Moten & Stefano Harney, "critical education only attempts to perfect professional education."[403] This statement distantly echoes Benjamin Bratton, who believes that "Institutional Critique is actually the last vestige of faith in the authority of art institutions."[404] As we have seen, there is a strong connection between a critical or political position, especially when rooted in one's identity, and the dynamics of professionalisation. That said, are there forms of profitability of the professionalised identity other than individual positioning? Created by Anastasia Collins, the Emotional Labour Invoice is a conversation piece that highlights the cost of all the invisible work which is traditionally carried out by marginalised folks ("endured your microaggressions(s)," "made you feel a 'good' ally," and so on). The fictional invoice, which can be read as a simple and yet powerful example

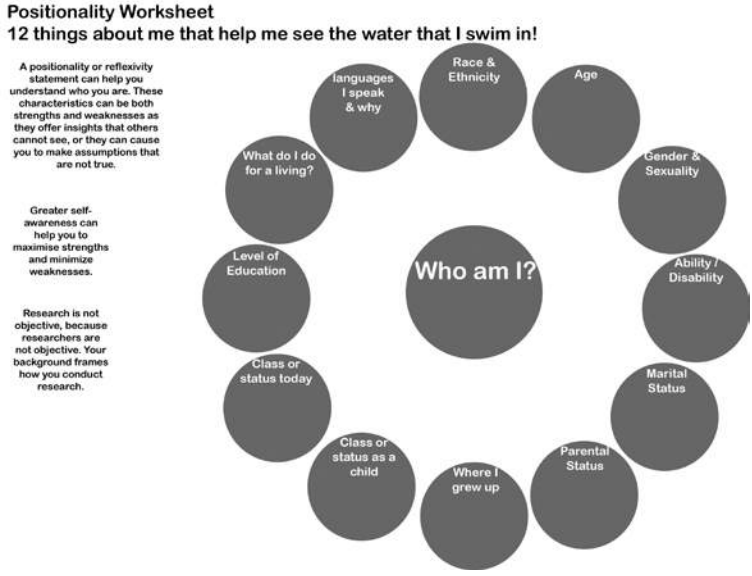
403 Stefano Harney and Fred Moten. *The Undercommons: Fugitive Planning & Black Study*. Wivenhoe: Minor Compositions, 2013, p. 32.

404 Benjamin Bratton. *Revenge of the Real: Politics for a Post-Pandemic World*. London: Verso, 2022, p. 10.

of speculative design, has a strong polemical value. But it can also be taken as an opportunity to reflect on the institutional monetisation of identitarian competence. Professionalised identity can be capitalised through its formalisation, that is, its bureaucratisation. Several universities now offer courses to obtain "diversity and inclusion" certificates.[405] This will soon be a requirement for a new managerial class profoundly embedded in institutions, who will produce their own frameworks and criteria to evaluate internationalisation, diversity and inclusion. We already see emerging a combination of design thinking methods with identity content, such as the Positionality Worksheet by design scholar Lesley-Ann Noel, a tool for self-reflection meant to help designers understand how features of their identity such as gender, level of education, class, etc. influence their practices.[406] While tools such as this represent an accessible way to introduce intersectional notions, they can easily become managerial tools in line with the reductionist achievements of design thinking, and therefore an enclosure of expertise. Furthermore, it cannot be denied that positionality itself is a pretty crude reflexive instrument, to the extent that it can be misleading. This is because it presents itself as multidimensional, but is in fact flat. While mapping their positionality, subjects do not acknowledge their *very act of positioning*, that is, in essence, a negotiation between their identity and the categories of an identitarian bureaucracy. In this way, positionality hides the acquisition of symbolic

405 Cornell, for instance. The program's fee is $3699.00. See https://ecornell.cornell.edu/certificates/leadership-and-strategic-management/diversity-equity-and-inclusion/.

406 Noel adds: "I've gotten positive and negative feedback in response to the wheel, and I have tried to take that in consideration when I facilitate positionality reflection activities. Some information that I provide when facilitating workshops: a) positionality conversations and reflections can be uncomfortable, b) the 12 elements of identity are suggestions and a starting point, there may be other reflection points needed in different contexts, c) there are tools to reflect on positionality. I created this one thinking of design practice, and at the time lived in California, so perhaps was thinking more internationally about designers in tech." Personal exchange via email.



The Positionality Worksheet by Lesley-Ann Noel (2019).

capital and the affiliations it weaves in the very moment it is performed. As a result, several interviewees of the Design Threads report "seem dissatisfied with mainstream attempts at inclusivity, specifically those in corporate contexts. Take for example: diverse representations without material inclusion/benefit, image-driven DEI efforts and other initiatives that end up resulting in tokenism instead of change in organizational structures."[407]

It's also worth noticing that a certain mode of exercising inclusivity *seamlessly* lends itself to managerial and bureaucratic approaches, to a sort of office politeness, even in settings, such as clubs, that one would expect to be adversarial or at least ambiguous. Attending a series of talks on club culture at the Circolo del Design in Turin, Valerio Mattioli, notices that,

> from safe spaces, clubs became sterile environments where to enact codes of behaviour as irreproachable as regulated by a kind of perfect woke etiquette – all agreeable in theory, yet all sort of polluted by a certain academically-flavoured pedantry. "Healthy" behaviours and "non-toxic" attitudes were being performed rather than internalised into a larger system of codes in the making. Words such as risk, conflict, desire, transformation were strictly banned, sacrificed in the name of an idyllic and sweetened pacification.[408]

STREET CRED

In educational institutions, the explicit attention to diversity and inclusion has another profitable function, that is,

407 See https://www.designthreads.report/.

408 Valerio Mattioli. "RIP Club Culture (finalmente)." *Not*, June 15, 2021. https://not.neroeditions.com/rip-club-culture-finalmente/. Before gasping at the use of the term 'woke' by an Italian writer who might not be fully synchronised with the nuances of a terminology in relentless evolution, I would invite the reader to finish the chapter.

to attract foreign students.[409] Commenting on the decision of the Willem De Kooning Academy of Rotterdam, a school that boasts an inclusive curriculum, to forcibly remove a pro-Palestine banner, Alina Lupu, a Romanian writer and artist, wrote:

> I guess it's safe to say, in the case where statements have gone missing or advocate neutrality after teaching decolonial theory, that your institution isn't woke, it's just instrumentalizing you for street cred. It's taking your hopes, your dreams, your aspirations, your country of origin as a marker for international reach, tolerance, and openness. And this used to be fine up until recently, up until spines started growing left and right within the art field. This used to be a symbiotic relationship. I scratch your back, you scratch mine. I give you my support, you give me yours [...] You take my image, you bend it and fold it and turn me into a worthy product. [...] In spite of the students asking on an open online stage, there was no instant materializing of general institutional accountability. But when faced with the sincerity of the demand, the institution laid itself bare as what it is, a neoliberal instrument which can be used to further one's career but isn't a real community, with ties that bind, but merely a casual, temporary, colorful, accumulation of individualities.[410]

Lebanese designer and researcher Imad Gebrayel notices that the deployment of performative criticality can be particularly appealing to those who are in need of an exit strategy: "A design program is sometimes a way out, an

409 In one of the schools where I used to teach, for example, the number of international students rose from 25% to 65% in the last five years.

410 Alina Lupu. "The Palestinian Conflict Rippling Across Dutch Art Educational Institutions." *The Office of Alina Lupu* (blog), 2022. https://theofficeofalinalupu.com/printed-matter/the-palestinian-conflict-rippling-across-dutch-art-educational-institutions/.

unclaimed, unlabeled, unofficial asylum-seeking process for individuals struggling with different forms of oppression, wanting an exit-route that isn't necessarily terminal."[411] These cases show something that will become more and more obvious in the future: the fact that the equilibrium of an institution can be disrupted not only by action, but also by inaction. It is becoming increasingly costly for self-proclaimed progressive institutions *not* to express solidarity for a marginalised group. This is why a pragmatic form of activism is one that makes the costs of 'doing the wrong thing' visible. Some people consider call-out culture a form of blackmail, but an economic understanding of institutional power shows that call-out culture is strategically rational, because such power is unevenly distributed.

At this point, we can draw a full circle of inclusivity. First, an intrinsically bureaucratic approach to identity is tested in autonomous spaces such as clubs. Then, it is used to attract students in institutions partly run by a section of the managerial class who sets the criteria for sanctioning this approach. Finally, the students themselves acquire familiarity with such knowledge to imbue their investment in education with meaning. As Brad Troemel puts it, "becoming bilingual in the language of inclusivity gives purpose to college degrees people paid much for."[412] Now, they are the bureaucrats – and they're looking for work. Holm invites us to have a realistic image of cultural environments: "No matter how much we might wish it to be otherwise, education – including that which we impart ourselves as teachers of cultural studies – is not simply a means to develop socially, politically and culturally productive awareness, but also a means of distinguishing sanctioned from unsanctioned forms of knowledge [...]". The use of specific terminologies, modes of inclusion and

411 Imad Gebrayel. "The Design Exit: Don't Look Behind!" In *Ununumimimdededesign*, edited by Joannette van der Veer. Eindhoven: Onomatopee, 2022.

412 https://www.patreon.com/posts/left-cant-meme-77324335.

approaches of strategic essentialism are turned into sanctioned forms of knowledge possessed by a selected cohort.

That said, not always is the sanctioning of 'correct' knowledge explicitly formalised. Speaking of her classes, theorist and educator bell hooks mentions the "gasp" that students made when one of them admitted they didn't know a prominent feminist author, something deemed "unthinkable and reprehensible."[413] The act of sanctioning happens tacitly, through gasps and perplexed facial expressions. It manifests, for instance, through the cringey feeling provoked by the non-native English speaker who misuses an outdated critical term.

KRITIKAOKE

Mattioli lamented the academic flavour of inclusive etiquette, suggesting that theory and critique, typical features of academia, aren't just methods of knowing and understanding, but tactical assets to position one's argument and oneself. As J. Dakota Brown explains, designers have been increasingly resorting to theory since the heydays of postmodernism:

> Postmodernists sought out intellectual justification for these new forms and methods, and they increasingly found it in "theory:" specifically, in contemporaneous writing on the postmodern condition and in the "cultural" and "linguistic" turns in the humanities. By the late 1980s, the critique of modernist principles had intensified under the flag of "deconstruction."[414]

In a context where praxis is not a necessary feature of a design practice, designers appear increasingly *condemned* to theory: what's left to them, to avoid succumbing to a submissive technical role is to indulge in sociology, anthropology, political theory... that is, to appear convincing by

413 bell hooks. *Teaching to Transgress: Education as the Practice of Freedom*. New York: Routledge, 1994, p. 113.

414 Brown, "American Graphic Design in the 1990s...", *op. cit.*

producing an allure of professionalism linked to their humanistic formation. Many project descriptions nowadays seem like AI-generated regurgitations of philosophy because designers (and artists alike) are busy practising what Italian essayist Tommaso Labranca used to called "kritikaoke."[415] Critical Design is probably the best known form of kritikaoke. Toke Riis Ebbesen neatly summarises its limits:

> Conceived basically as "useless," the value of critical design "ultimately lies in its ability to valuate: articulate, refuse, critique, spark, turn, transgress, formulate, transform etc" (Rosenbak, 5.15). However, circulating mainly in "art galleries, conference halls and academic publications" (Blythe, Yauner & Rodgers, 2015), useless critical design artefacts have been criticized for never entering everyday life (Bardzell & Bardzell, 2013). It has been argued that most critical design instead "reflect the fears, anxieties, desires, imaginaries, and ultimately, politics of an intellectual, liberal progressive white middle class" (Ansari & Hunt, 2015, 4). Removed from practical use, critical design may then become another echo chamber for designers, where they can safely repeat the slogans of design modernism without changing the world.[416]

The features of Critical Design's style of expression are very close to what Alix Rule & David Levine call International Art English:

> IAE has a distinctive lexicon: *aporia*, *radically*, *space*, *proposition*, *biopolitical*, *tension*, *transversal*, *autonomy*. An artist's work inevitably interrogates, questions, encodes, transforms, subverts, imbricates, displaces – though often it doesn't do these things so much as it serves to,

415 Labranca, *op. cit.*, p. 48.
416 Toke Riis Ebbesen. "Why Critical Design Is Useless: Criteria." Sursock Museum, Beirut, Lebanon, 2017.

> functions to, or seems to (or might seem to) do these things. IAE rebukes English for its lack of nouns: *Visual* becomes *visuality*, *global* becomes *globality*, *potential* becomes *potentiality*, *experience* becomes... *experiencability*.[417]

In this context, capitalism, as it is most broadly understood, has become the favourite scapegoat of the design cultural sphere. But the ritualistic vilification of capitalism is part and parcel of capitalism – that's its romantic mirror image. Baudrillard in 1972:

> Design is based from the start on the same rational abstraction as the economic system. There is no doubt that this rationality is virtually absurd, but it is so, in both cases, for the same reason. Their apparent contradiction is nothing but the logical outcome of their deep complicity. Designers complain that they are misunderstood and that their ideal is disfigured by the system! All puritans are hypocrites.[418]

At this point, we can briefly list the features of kritikaoke: a critical disposition functioning as a tool of distinction; the deployment of ornamental politics; an abstract notion of public that is to be made aware; the adoption of a critical-theoretical jargon – vaguely anticapitalist – adjacent to International Art English; a performance of identity as a professional skill that fits the criteria of an identitarian bureaucracy. What type of design does kritikaoke produce? To answer, we can resurrect Michael Rock's 2003 scrutiny of Dutch Design, which is probably the type of design that does kritikaoke best: "convoluted, challenging, intelligent, difficult, self-reflexive, coy, clever, often staggeringly beautiful [...]"[419]

417 Alix Rule and David Levine. "International Art English." *Triple Canopy*, 2013. https://canopycanopycanopy.com/contents/international_art_english.

418 Baudrillard, *op. cit.* p. 59.

419 Michael Rock. "Mad Dutch Disease." *2x4*, August 9, 2003. https://2x4.org/ideas/2003/mad-dutch-disease/.

An Artist Who Cannot Speak English Is No Artist by Mladen Stilinović (1992). Collection of Van Abbemuseum, Eindhoven. Today, one could imagine an International Art English update.

CLASH OF COUNTERCULTURES

To conclude, let us return to the University of the Underground. At a first glance, its most revolutionary aspect is the fact that it is tuition-free, a noble pursuit, given the rise of student expenses and the subsequent extortion known as student debt. The UUG's tuition fees are supposedly funded in part by philanthropic contributions and donations (80%) and in part by state funding (20%). Originally, the contributions from the government amounted to 50%. Generally, obtaining financial support entails a compromise regarding the way in which a project is presented to the funding bodies. Individual artists asking for grants are required to adopt the *néolangue* of the creative industries, and to detail their "competitive advantage." Antagonistic purity is rarely a good investment.

The UUG's conspicuous countercultural stance, however, clashed with a more traditional expression of counterculture: a group of students from the Sandberg Instituut penned an open letter to criticise (fiercely but politely) their own institution.[420] Their core concerns revolved primarily around the issue of corporately funded education: yes, the programs are tuition-free, but 80% of private and individual contributions opened the door to "direct privatisation." Other concerns included the lack of transparency regarding roles in the school and the smearing, so to speak, of the 'critical reputation' of the hosting institution. Finally, they rejected the UUG's countercultural branding, maintaining that it was improperly reminiscent of activist endeavours. Perhaps as a result of the open letter, much of the original countercultural jargon disappeared from the UUG website, as well as the characterisation of private contributions as philanthropy.

The concerns about countercultural branding and

420 *Medium*. "UUGH! Or: Issues Regarding University of the Underground," September 22, 2017. https://medium.com/@uugh/issues-regarding-the-university-of-the-underground-and-the-sandberg-instituut-fe58dbbf889b.

Sandberg's critical purity may be a cause of doubt: what does it mean nowadays to be an immaculately critical institution? Would that be effective anyway? If even Pepsi can adopt protest imagery, why wouldn't the UUG do the same? And yet, between the lines of the dispute between school management and students (and apparently some Sandberg teachers as well) we get a glimpse of what counterculture might be and perhaps has always been: a permanent distrust of opaque administration, especially when it presents itself as progressive; a constant tension against the ossification of certain power relationships, both implicit and explicit, against new elitist formations, especially when they pretend to speak for the masses. Maybe, then, counterculture is just crippling institutional self-doubt.

Chapter 7.

The School as Real World: On Aspirations and Compromise

"The starting point for organizing the program content of education or political action must be the present, existential, concrete situation, reflecting the aspirations of the people."
– Paulo Freire, 1968[421]

"In a period of 'diploma inflation' the disparity between the aspirations that the educational system produces and the opportunities it really offers is a structural reality which affects all the members of a school generation, but to a varying extent depending on the rarity of their qualifications and on their social origins."
– Pierre Bourdieu, 1984[422]

How do art and design schools produce their subjects, namely professional designers?[423] What values, criteria and mechanisms are tacitly or explicitly deployed in education? In this final chapter I explore the relationship between the school and the real world, the affinities between old and current waves of student protests, the targets of institutional critique, and the role that cultural capital plays in emerging professional models. Furthermore, I discuss how a broadly humanistic turn of design, which makes the designer an intellectual of technics more than a technical intellectual, is less a spontaneous evolution of the field than the logical outcome of design's unstable position within the technical domain. Finally, I lay out an ethos of compromise against the more commonly adopted one of autonomy, as the latter obscures various forms of dependency.

421 Paulo Freire. *Pedagogy of the Oppressed*. London: Penguin Books, 2017, p. 68.
422 Bourdieu, *Distinction*, *op. cit.*, p. 143.
423 I deliberately use the generic 'art and design school' formula because various institutions dedicated to the cultivation of design disciplines (academy, university, private school, master, summer school, etc.) share the issues I tackle here. The reasons why I insist a bit punctiliously on keeping art and design together will become obvious, I hope, throughout the chapter.

However, shouldn't education, being the inception of a practice, have been discussed at the beginning of this book? The odd placement of this chapter is due to the fact that we need to look at education from the point of view of work. From this perspective, education appears at the same time as the outcome of all the 'expectations' discussed above, and as a reality in itself, which belongs, at least indirectly, to the sphere of work. We can understand work as an activity, a system of relationships and, above all, a *myth*, that is, as a story we tell ourselves and one another.[424] Therefore, offering a realistic perspective of work means painting a picture which is not at all immune to idealisations and rhetorical veneers. One of these idealisations is the notion of 'real world,' that dimension that follows, in theory, the period of study. It is precisely this sequence that needs to be reconsidered.

Generally speaking, a school is conceived as a space protected from the disorienting brutality of the world of work. Traditionally, it was understood as the privileged place of *vita contemplativa*, as opposed to the context of labour, the domain of *vita activa*. There are many reasons why this is not the case, and some of them are specifically linked to the field of design. Years ago, Enzo Mari, a designer who rose to the status of "critical conscience of design," noted that, provided a design degree is considered on a par with a product, it is legitimate to say that the most impressive sector of this field, also in terms of turnover, is the "industry" of design schools. Thus, the structural reasons of the market interact *already at school* with the "hopes, strategies and naivety" of future designers.[425] Here, not only is the culture of design reproduced, but the subjects that will engage with it are generated.

However, there is another reason, more general and more alarming, to consider the school as an organ of the

424 See Lorusso, *Entreprecariat*, *op. cit.*

425 Mari, *25 modi per piantare un chiodo*, *op. cit.*, p. 122-3.

real world. In times of crisis, it provides less and less protection from the typical concerns of workers: students are not unfamiliar with steep rents, housing shortages, debts of various kinds, poorly paid jobs or internships that don't pay at all. Moreover, the instability of the labour market changes the meaning of the school experience: in a climate of uncertainty, you don't need to know who Gary Becker is to instinctively adopt the prism of *investment* when it comes to education. In short, the problems of school are hardly distinguishable from the problems of life.

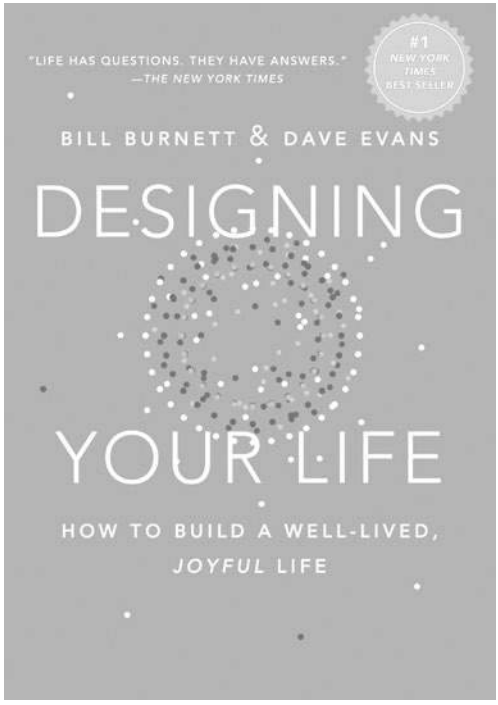
Although less protected than we imagine them to be, art and design schools remain, at best, a space that allows for critical thinking. While in the past, critique of design and through design has been primarily outward-looking (examining, for instance, dominant online platforms, pollution, and the ideologies that guide styles), today we are increasingly witnessing a self-referential critique that originates in institutions and points the finger at them, blaming their values and organisational models. The spectre of institutional critique, which until recently mainly haunted museums and art biennials, now also appears in design schools.

In the Netherlands, Germany and the UK, a network of unofficial Instagram pages has sprung up. These are anonymous accounts linked to the various academies that engage in a daily, all-round institutional critique: bureaucratic absurdities, sexism, bogus meritocracy, precarious contracts... all conveyed through the language of memes, inside jokes at once esoteric and accessible that are devoid of the pedantic seriousness typical of the movements. It is in these pages that, in the darkest months of the pandemic, students and staff felt the fervour of a community, much more than in the unctuous paternalism of official communications.[426]

426 See Silvio Lorusso and @wdka.teachermemes. "'May the Bridges We Burn Light the Way': Five Questions to a Dutch Design School's Meme Page." *Other Worlds*, no. 3 (July 26, 2021). https://buttondown.email/otherworlds/archive/ow-3-may-the-bridges-we-burn-light-the-way-five/. See also Tessel ten Zweege. "Calling Out Dutch

L' UNIVERSITA
E' UN GIARDINO
SBOCCIANO
MAO
le parole

DOVE
FIORI
CCUPATI
MOSKA
Maurizio



← "University is a garden where unemployed flowers blossom," DAMS, Bologna (1977). Photo by Enrico Scuro.

There are many books that mix elements of self-help with design culture. The circle closes: artistic expression becomes a tool for the realisation of one's own life and work project.

THE LOOOOONG '68

This is not the first time that the boundaries between the school and the real world have become so fragile and porous. The student protests of '68 and '77 were born out of a sense of alienation and precariousness, just as they are today. They were colossal events capable of breaking the spell that made the school a natural, and therefore immutable, phenomenon. In the eyes of Giancarlo De Carlo, who analysed the crisis of architecture faculties in Italy, students appeared "accidental and extraneous passengers in an institution that should be made for them and is justified only by their presence."[427] What then can be said about temporary faculty? "Sperm cells feverishly searching for an ovule that will give them a concrete configuration." Having followed the students' debate with keen interest, De Carlo reported in full the communiqués of the students from Turin:

> The current university does not meet the needs posed by society's existing labour demand. Our objective, however, is not simply to adapt to the needs of the demand for work. We believe that the University can and must provide those who attend it [...] with adequate professional preparation and critical tools.[428]

Not much has changed. Students then were aware of the "fate that would be reserved for them once they left the school, clueless in an indecipherable world," just as they are now.[429]

THE SCHOOL AS MARKET

As the communiqué suggests, both students and faculty conceive the school through either a materialist or

Art Institutions." *Futuress*, February 26, 2021. https://futuress.org/stories/calling-out-dutch-art-institutions/.

427 De Carlo, *op. cit.*, p. 46.

428 *Ibidem*, p. 54.

429 *Ibidem*, p. 73.

idealist lens. Materialists share a pragmatic focus on future employment, skills, market needs. Idealists, in turn, are divided between pessimists and optimists. Pessimist idealists argue that the school is a place of discipline and repression. They also lament the subjugation of education to the labour market. Schools, according to this point of view, are becoming a factory, only apparently egalitarian, for corporate executives, freelancers, workers or even the unemployed. Optimists see the school as a space for liberation through the exercise of critical thinking and the suspension of familiar preconceptions. On one side the pessimist Ivan Illich, on the other the optimist bell hooks.

Everybody is right, at least partially. How do we then merge their respective positions into a model? Claiming that art and design schools are an integral part of the real world is not enough. Nor is it correct to contrast professional training with the development of critical thinking. For example, are we really sure that a professionalising school cannot be liberating, or that emancipation is not, in some cases, a form of discipline? Pierre Bourdieu offers a useful synthesis of the issue. For the French sociologist, the school is first and foremost a *market* in which cultural capital is formed and exchanged, but above all legitimised and sanctioned.[430] Talking about culture as capital is crucial: it means emphasising that culture – criticality included – can be converted into economic capital, that is, money.

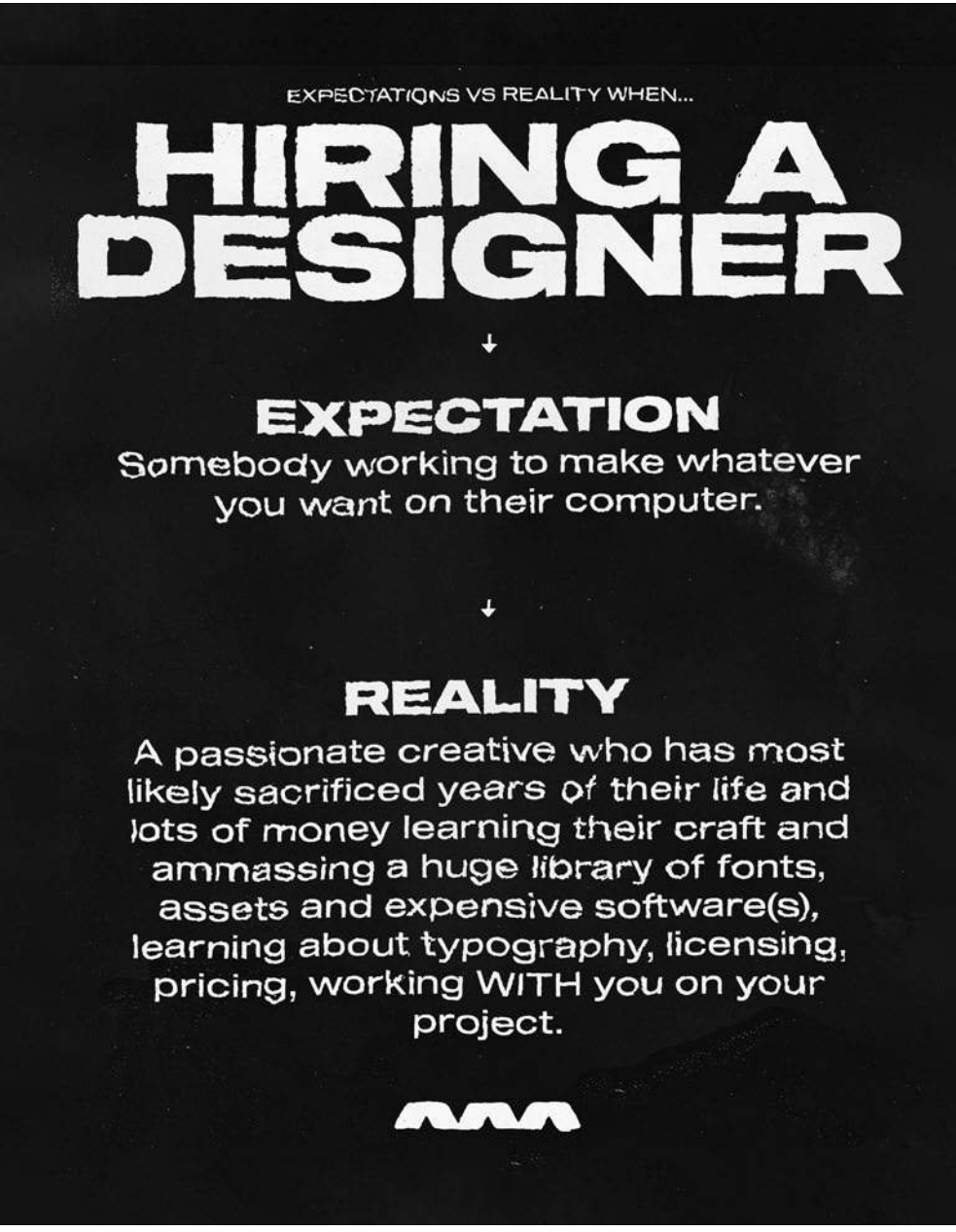
Scholastic capital, that is, the body of knowledge legitimated by the school, is a subset of cultural capital. At each level of education, the learner brings with them a varyingly substantial (sometimes even negative!) inheritance of cultural capital that is evaluated by the educational institution. No one enters school empty-handed. The art and design school is a special case because here, more than elsewhere, the student is encouraged to turn

430 Bourdieu, *Distinction*, *op. cit.*, pp. 80-1.

Babe, art school is just tiktok with a degree :)) by Content y Contenido (Ingeborg Kraft Fermin & Lucía Vives), billboard at the Rietveld art school of Amsterdam NL (2021).

László Moholy-Nagy photographed by Lucia Moholy in 1926. He is wearing a shirt under a coverall, a perfect outfit for the 'technical intellectual.'

→ Instagram post by Jack McArdle, Studio AAA (2022). The mere existence of this kind of propaganda shows that the expectations criticised here are themselves a reality.


EXPECTATIONS VS REALITY WHEN...
HIRING A DESIGNER
EXPECTATION
Somebody working to make whatever you want on their computer.
REALITY
A passionate creative who has most likely sacrificed years of their life and lots of money learning their craft and ammassing a huge library of fonts, assets and expensive software(s), learning about typography, licensing, pricing, working WITH you on your project.

their own cultural baggage, especially the most personal (interests, passions, hobbies, reading, ethical and political ideals, etc.) into a *practice*. Following Bourdieu, a design practice can be understood as the activity through which culture becomes profit and cultural consumers become cultural producers.

While the 'emancipatory school' reveals itself in the apparently autonomous ambitions of students and teachers, the 'conservative school' hides itself in the mechanism of legitimisation of these same ambitions. It can therefore happen that indignation, an emphatic manifestation of the critical spirit, becomes a legitimate or even prescribed value, as I have observed in some academies. However, this does not eliminate the mechanism of legitimisation *itself*, often tacit and made up of frictions, impediments, micro-censorship, which extend from the school to the labour market. The school will naturalise certain values, while hiding the way in which it legitimises them. One of these values is authenticity.

BECOMING YOURSELF

> Home was the place where I was forced to conform to someone else's image of who and what I should be. School was the place where I could forget that self and, through ideas, reinvent myself. [431]

The school of bell hooks, a black woman who grew up in a segregated, patriarchal environment, is the antithesis of home: a territory of reinvention outside the bastions of tradition.[432] Art and design schools go further and render reinvention a rediscovery. Here, the insistence is often on the authentic self, on what 'you and only you' can do or appreciate. By adopting a rhetoric of

431 hooks, *op. cit.*, p. 3.

432 Yet, as we have seen, the separation between home and school is blurred. One always carries to school, intentionally or not, part of the domestic burden. And the school environment may not necessarily be less conformist than the home one.

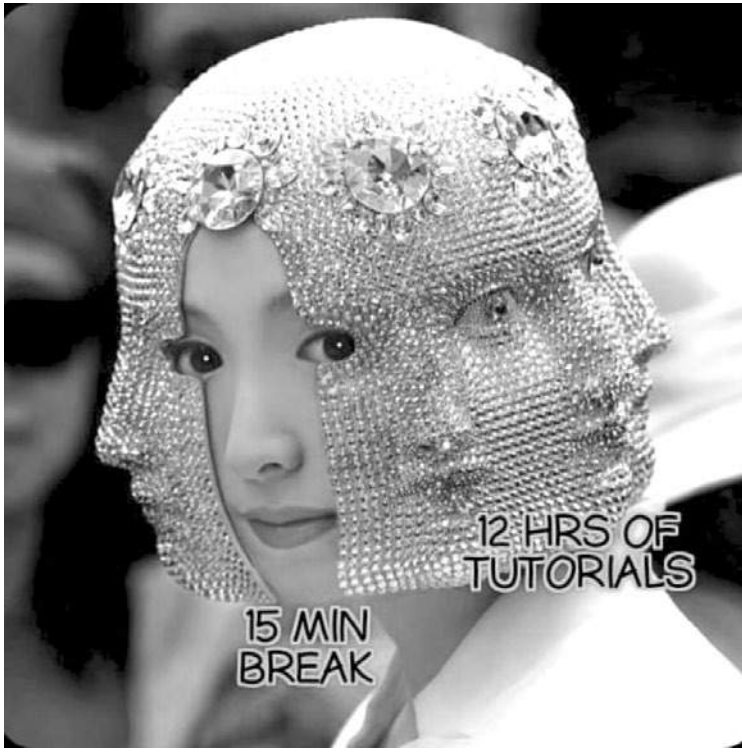
authenticity, design producers and consumers are promised to 'become who they are.'

This consideration allows us to address the age-old question of the relationship between art and design. Without wishing to resurrect the old distinctions (commissioned versus independent work, rationality versus intuition...), it is useful to consider the relationship between design of the self and design of things. Distinguishing the two is no easy task. After all, isn't design a reflexive practice in which, by designing the world, we redesign ourselves? And isn't art the making of artefacts that are partly independent of the person who creates them?

The things we design designs us back, no doubt, but there is an essential distinction between the design of the self and the design of things. This distinction is made evanescent by the constant praise of glorious designers who pour their personality into the things they conceive. The celebrated design figure is a pale copy of the romantic artist who survives in the promises of the school and the hopes of the students. It is, after all, a question of value. While the design dimension of the art and design school tends to value things and services, its artistic dimension – which is certainly not devoid of design – tends to value the expression of an autonomous and distinct identity. That said, what I'm suggesting here goes further: art and design schools are first and foremost laboratories for self-design, and only secondarily a context where things are designed.

Personal branding, human capital, identity politics: nowadays the design of the self is a project that is consciously (and often polemically) implemented. To what extent is it a liberating practice? In order to answer this question, it is necessary to clarify that self-design is *one of the key facts of modernity*, both an opportunity and a curse for anyone who inhabits an environment that is not entirely traditional. Since there are no predefined trajectories or 'careers,' we cannot refrain from designing

Chris Ashworth (2021).



@wdka.teachermemes (2021).

Me and the boys asking that one professor who worked at Herzog 15 years ago if he knows of any opportunities

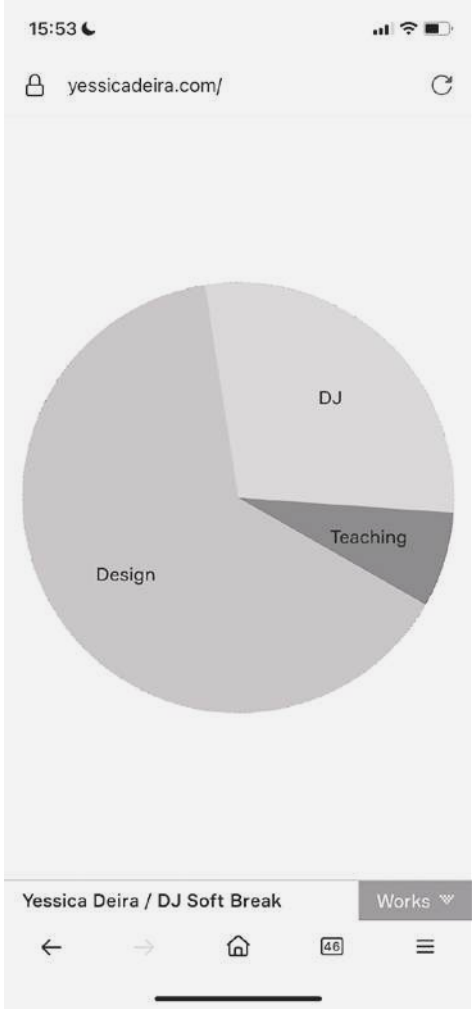
@b0ysfirm (2021). An allegorical manifestation of the hysteresis in architecture schools.

ourselves. Life appears to us as a more or less restricted gamut of risks and possibilities.[433] In this sense, the expressive drive that is generally associated with art is nothing but a reflection of the modern inevitability of self-design. No wonder then that creativity, a popular synonym of art as individual expression (and perhaps precisely for this reason frowned upon by designers), has acquired such primacy in recent decades.

Self-design carries a risk, though. In its extreme forms it is navel-gazing and pathologically self-reflexive – it's bad literature. By measuring itself obsessively against the ghost of identity, it is perturbed by an essence that does not always manifest. It is not difficult, then, to understand the disappointment of students who go to an art and design school believing they will grapple with a system of ideas, and instead find themselves placed in front of a mirror. The mirror image of the emancipatory school of bell hooks, self-design is an easy pedagogical shortcut, since it delegates the definition of content to students ('what do you hold dear?'). In a time devoid of consensual role models, the question 'who do I want to be?' is substituted by an incessant 'who am I?'. In this way, the school becomes a recruitment centre for subjectivities.

Identity is expressed through difference, via the distinctive device of taste. We should not think of taste as the snobbish prerogative of the *connoisseur*, but rather as a system of preferences (from TV series to political orientations) that actually constitute lifestyles. The art and design school, operating as a marketplace of differences, identifies and legitimises identity components and turns them into distinctive *skills* (like those, for example, of the mad genius) to be spent subsequently in the world of work, presenting some of them as authentic and therefore inalienable. This form of authenticity is, at least in part, a mechanism that

433 It is exactly from this consideration that design theorist Ezio Manzini describes a condition in which "everybody constantly has to design and redesign their existence, whether they wish it or not." Manzini, *op. cit.*, p. 1.



Online portfolio of designer Yessica Deira, designed by Justus Gelberg and Lukas Engelhardt. The pie chart is a playful visualisation of the multi-hyphenate condition.

is hidden from others and even from oneself. Through the self-hypnosis of authenticity, specific personality features can appear *true*. This is how the subject can become who they are. It is therefore possible to be inauthentic and sincere, or insincere and authentic. This mechanism makes it possible to explain, at least partly, the autobiographical turn of schools, already noted years ago by various observers.[434] However, with the encouragement of autobiography comes the omission of the fact that not all biographies are equally expendable, not all identities lend themselves to a favourable symbolic conversion. The mobilisation of identity and subjectivity is, thus, an integral part of the professionalisation effort. Let's look at it in detail.

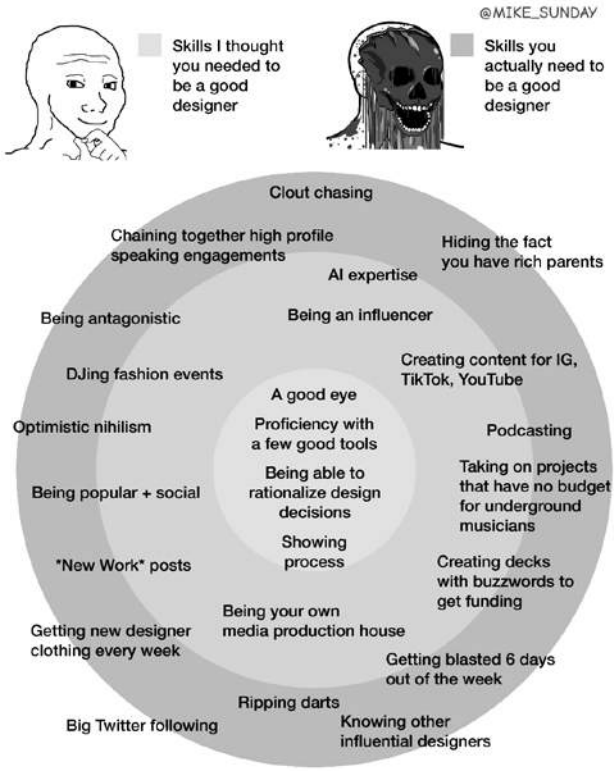
AMBIGUOUS ASPIRATIONS

Design is, like art, protean: its shape constantly changes. It is no coincidence that much of its specialised literature, rather than clarifying its nature, tries to direct its aims. Schools, as forges of currents and manifestos, actively participate in such an effort. This is of course reflected in the multiplicity of roles that the designer can perform: the charisma of the maestro, the kitsch of the star designer, the nostalgia of the craftsman, the pride of the expert and, increasingly, the struggle of the worker, can all converge within design.[435] One could say, paraphrasing Bourdieu, that since design is a fluid and confusing field, it allows for professional goals that are also fluid and confusing.[436] Its ambiguity gives rise to aspirations that are never fully

434 Among them, Rob Giampietro, who proposed an analogy between design school and creative writing courses. "School Days," *op. cit*.

435 Italian graphic designer and author Riccardo Falcinelli explains that none of these attitudes is truer than the others: "Everyone will choose for themselves what kind of designer or audience they want to be: they may prefer functionalism or expressive design; they may aspire to strictly informative or postmodern graphic design; they may believe in efficiency or practice performance. The important thing is you don't mistake these convictions for inescapable truths, but take them for what they are. It's the only way to not end up in the shallows of moralism, like those who believe there is only one right way to do things." *Critica portatile al visual design: da Gutenberg ai social network.* Torino: Einaudi, 2014, p. 301.

436 Bourdieu, *Distinction*, op. cit., p. 159.



Mike Sunday (2022). A humorous representation of the importance of cultural and symbolic capital for the development of a design practice.

realised nor completely frustrated, an ambiguity that is also useful to those ambitious youngsters who don't want to forgo either the 'real job' or certain hypothetical symbolic benefits.[437] It is tempting to replace the modernist diktat "less is more" with the vague "more or less"...

Design's intrinsic vagueness has, thus, a precise logic: aspirational ambiguity propels its growth. And the aspiring professional shouldn't be concerned with the recognition of their profession, since professionalisation, understood as the promotion of the designer's role by professional associations, seems to be an outdated problem. Aggie Toppins, a US designer and lecturer, recently argued that "the field's most pressing imperative today is no longer professionalisation; it's to make design practice a responsible part of building equitable, sustainable futures." As evidence of the success of professionalising efforts, Toppins reports that "graphic design is now an in-demand college major and it is more visible in popular media."[438] A statement that confirms Mari's thesis of the school as the main design industry.

It is critical to recognise, as Toppins does, that the design profession has historically been a means of excluding women, minorities and oppressed groups. But it's also important to remember that nowadays the professional issue is far from resolved. A study of 30,000 design graduates in the UK (design being the most popular undergraduate degree) showed that only a quarter of them end up working in high-skilled, well-paid jobs in the design field.[439]

MASS ELITISM

When it comes to design work, one cannot help but talk

437 De Martin, *op. cit.*, p. 71.

438 Aggie Toppins. "We Need Graphic Design Histories That Look Beyond the Profession." *Eye on Design*, June 10, 2021. https://eyeondesign.aiga.org/we-need-graphic-design-histories-that-look-beyond-the-profession/.

439 Sarah Dawood. "Design Most Popular University Choice – but Graduates Aren't Becoming Designers." *Design Week*, February 15, 2018. https://www.designweek.co.uk/issues/12-18-february-2018/design-popular-university-choice-graduates-arent-ending-designers/.

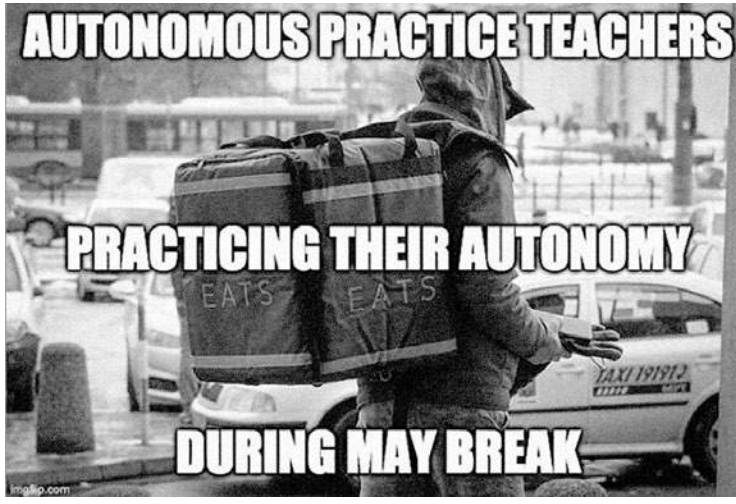
about technology. Democratisation, automation and disintermediation are keywords that are hard to escape. Ruben Pater sums up the long debate around the advent of desktop publishing in the following way:

> "By making our work so easy to do, we are devaluing our profession," warned design historian Steven Heller. "With everything so democratic, we can lose the elite status that gives us credibility." That hasn't been the case, as Ellen Lupton points out in her book on DIY design, seeing that "the field got bigger rather than smaller."[440]

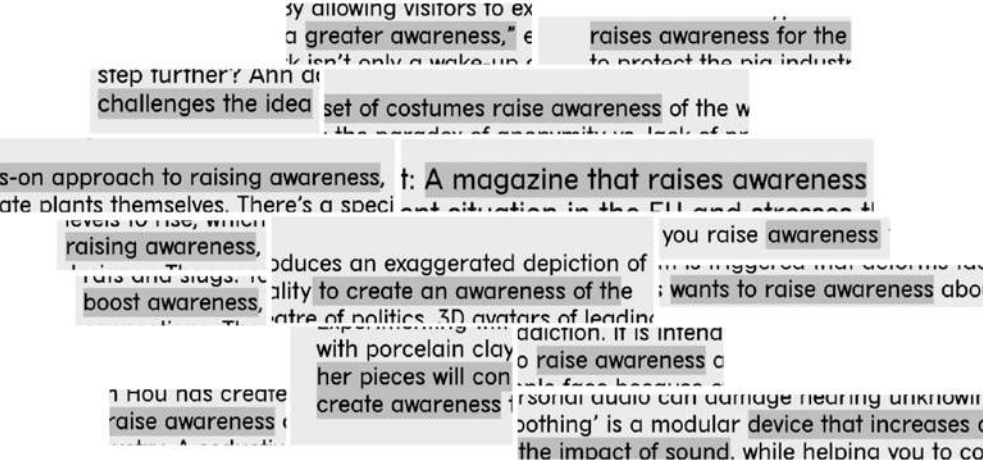
Now that the word 'elite' has become almost an insult, taking Ellen Lupton's side certainly feels more comfortable. However, the growth of the graphic design sector expected over the next ten years is lower than that of other career paths – 3% against an average of 8%.[441] Moreover – and this matters even more – Heller pointed out a problem of perception that translates into a decline in earnings and especially in symbolic profits. This is a reality that schools cannot take lightly. At a time when a significant portion of the student body does not have the opportunity to convert their substantial educational investment into a position of professional elite, a problem arises, and it is no small one.

For political scientist Jo Freeman, "an elite refers to a small group of people who have power over a larger group of which they are part, usually without direct responsibility to that larger group, and often without their knowledge or consent."[442] This definition, suited to a criminal organisation or a secret sect, doesn't really capture the essence of a profession's elite status. Let's recall, instead, Donald Schön's definition of a professional as someone who "claims extraordinary knowledge in matters of human importance, getting in return extraordinary rights and

440 Pater, *Caps Lock*, *op. cit.*, p. 329.
441 See https://www.bls.gov/ooh/arts-and-design/graphic-designers.htm#tab-5.
442 Freeman quoted in Táíwò, *op. cit.*, p. 22.



@wdka.teachermemes (2021).



"Awareness." Collage of graduate project descriptions from Design Academy Eindhoven. Afonso Matos (2022).

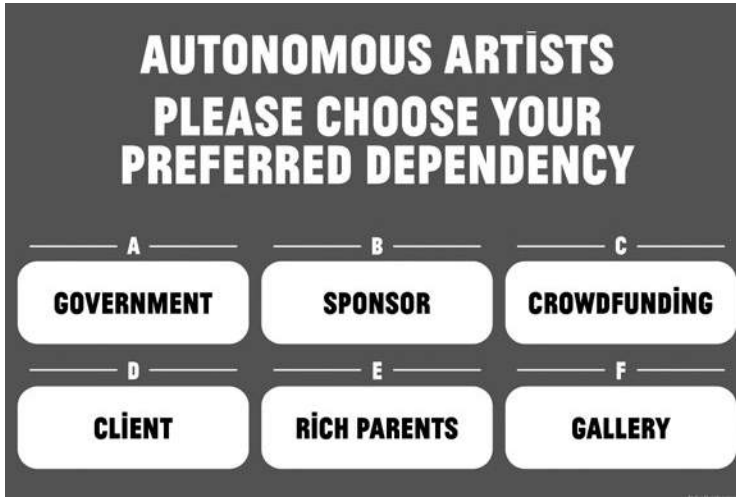


Illustration by Yuri Veerman (2021).

privileges."[443] This definition demonstrates that that of a professional is an inherently exclusive role (but not necessarily *excluding*, as it has been in the past). Here, the phrase 'professional elite' becomes almost tautological.

If designers' knowledge is not considered, at least in part, a form of expertise, they will hardly be recognised as professionals. This is the case of graphic design, a field whose know-how has lost its esoteric aura. The common perception is that, thanks to digital devices and the web, anyone can design a logo or a book. For some, the fact that graphic design is an actual university degree is even a source of amazement. My aim is not to debate whether this perception is right or wrong, but to point out that it does shape the general economic relationships between clients and designers, and thus the status of the designer. The effects are already evident in the wage gap between specialists in UX design, a field still considered esoteric, and those in graphic design, a practice now demystified.[444] The suspicion arises, therefore, that professionalisation is considered an outdated problem by those who have overcome this problem *individually*, that is, those who are considered to all effects professionals, perhaps members of a cultural elite, able to obtain both material and symbolic benefits from it.[445] They antagonise elitism, but while anti-elitism, as a practice, is the very 'emotional atmosphere' of the lower classes, as a discourse, it's the Trojan horse of the elites. As a practice, it is intuitive, matter-of-fact, material, lived; as a discourse, it is formal, theoretical, abstract, distinctive.

ECONOMIC RESPONSIBILITY

In 2022, I wholeheartedly signed Platform BK's open letter entitled "Graduates of art academies deserve more

443 Schön, *op. cit.*, p. 4.

444 $74,000 of UX Designers vs. $41,000 of graphic designers. Teo Yu Siang. "How to Change Your Career from Graphic Design to UX Design." *Interaction Design Foundation*, September 12, 2020. https://www.interaction-design.org/literature/article/how-to-change-your-career-from-graphic-design-to-ux-design.

445 Cf. Rock, "On Unprofessionalism", *op. cit*.

"This poster is worth the approximately €50,000 that my parents spent to get me to graduate from the D.A.M.S Cinema instead of buying a German luxury car." Poster by Giuseppe de Mattia (2021).

agency over their future."[446] Addressing the boards of Dutch art academies, the signatories argued that education should prepare students for "the unruly reality of the cultural sector's job market." The plan included four points: development of post-precarity courses; social engagement and self-organisation; insights into the world after the academy; student involvement in institutional developments. An earlier version of the letter called for "economically responsible art education." The question is: how to define economic responsibility? Up to this point, I have tried to show that an economic perspective does not only entail income and bills, nor mere entrepreneurial spirit, but that there are also inflections of symbolic capital, a gap between public perception and professional *proprioception*.

Within various art and design schools, both faculty and students are beginning to address the problem of the profession by taking the side of an exploited proletariat. Although, as the data show, there are very good reasons for adopting this stance, there is something self-defeating in it: by begrudgingly accepting the neo-worker condition, the legitimate professional aspirations of designers, which are not really working-class aspirations, are concealed. This is the reason why generic calls to unionisation, which re-emerge every five years or so, do not yield the desired results within the design professions. Again, we are in the presence of a double bind: the school educates students as professionals but demands of them a proletarian consciousness. A generic appeal to unionisation runs the risk of betraying design's professional promise,[447] not only by 'creating losers,' but by turning their resentment into material for courses and syllabi.[448]

446 https://www.platformbk.nl/en/graduates-of-art-academies-deserve-more-agency-over-their-future/.

447 See Silvio Lorusso. "No Problem: Design School as Promise." *Entreprecariat* (blog), December 7, 2020. https://networkcultures.org/entreprecariat/no-problem-design/.

448 I am aware of the weight of this remark, and I do not make it lightly, as the authors to whom I refer do not make it lightly either. See, in particular, Bourdieu,

ME, AN INTELLECTUAL

"The thing that pisses me off the most is the degradation of the intellectual role of the designer." This is what a designer friend told me once, as we listened to each other's anguished outpourings replete with VAT numbers, short-term contracts and late payments. In the introductory text of Ellen Lupton's aforementioned book (written together with her MICA students in Baltimore: it is a book born in a school), the historian likens the figure of the designer to the Gramscian "organic intellectual." Considering design "a *social* function, rather than as a profession or an academic discipline," Lupton suggests that

> These organic intellectuals could merge physical and mental labor, building "new modes of thought" out of acts of doing and making. Their skills would be both technical and theoretical.[449]

But what exactly is an intellectual? To this question, Tomás Maldonado, a complex and prolific figure in the field of design, has devoted an entire, wide-ranging volume which spans from Cervantes to Jonathan Swift, from Erasmus to Heidegger.[450] According to Maldonado, intellectuals are born *committed*. They "take a stand" and sign manifestos. Sometimes, however, they have a decorative function within the party they support and by which they are supported. Moreover, the moment they become modern, they cannot help but work on themselves:

> He expresses "the modern identity" as a way of being an intellectual that privileges "radical reflexivity." He is the intellectual "after Montaigne." With Montaigne "the self enters the scene." The intellectual is born who is turned in

Distinction, *op. cit.*, p. 143, and Ivan Illich. *Deschooling Society*. London: Calder & Boyars, 1971.

449 Lupton, *op. cit.*, p. 21.

450 Tomás Maldonado. *Che cos'è un intellettuale? Avventure e disavventure di un ruolo*. Milano: Feltrinelli, 2010.

> upon himself, but who offers himself to the gaze of others.[451]

The affinity with the prototype of the designer envisioned and promoted by the art and design schools is evident: this is an intellectual who takes a position, critiques and educates, who designs themselves and makes this work their own content to disseminate, their own voice. This type of intellectual meets the criteria of acting, but what about the making typical of designers, that is, their technical dimension?

THE TWO CULTURES

We now understand what an intellectual is. But how does this role relate to the designer? Maldonado offered a compelling definition: the designer is a *technical intellectual*.[452] This way, he attempted to abolish the false dichotomy that opposes humanistic culture to scientific culture. As Giovanni Anceschi explains,

> "Technical intellectual" is an expression that strikes us as an oxymoron, but it is such only according to the banal thinking and schizoid prejudice that continues to believe in the "two cultures."[453]

The attempt to "drive out this very concrete activity from the professionalist and practice-heavy middle-brow" is meritorious, yet things aren't that simple.[454] The difficulty lies in the fact that the distinction between technical and intellectual survives in the general perception of professional roles, and, broadly speaking, the designer is perceived more as a technician than an intellectual.

What kind of technician, though? It is true for many designers what Bourdieu argues about those who occupy

451 *Ibidem*, pp. 27-8.

452 Oddly, the word 'designer' never appears in his excursus on the role of the intellectual.

453 Giovanni Anceschi. *Tomás Maldonado intellettuale politecnico*. Milano: Edizioni del Verri, 2020, p. 35.

454 *Ivi*.

"Become who you are." Still from a TV ad for the Italian Accademia del lusso (2020).

the lower ranks of the ruling class. They "are relegated to the position of *technicians*, i.e., executants without economic, political or cultural power."[455] This is evident in the lamentations of the various design sub-sectors, not only in graphic design, which is in decline, but also in those on the rise such as interaction design, which is clamouring for its 'seat at the table.' There is a difference, then, between the technician and the *technologist*, a professional in their own right, recognised synthesis of the two cultures.

How, then, does one explain the intellectualistic, 'monocultural' propensity of contemporary design? Having failed to establish themselves authoritatively in the technical sphere, the designer attempts to occupy the humanistic sphere. This is a sensible move, albeit a largely unconscious one, consisting in the replacement of a technical elite (which is elite no longer) with an intellectual pseudo-elite, which stands as the 'consciousness' of technics. Schools can thus claim a formally autonomous role, which is the exercise of critical thinking and the production of awareness. Here, the focus is more on the why and less on the how, as Tibor Kalman once put it. In their latest book, Mieke Gerritzen and Geert Lovink frame this mainly European phenomenon through a scathing title: *Made in China, Designed in California, Criticized in Europe*.[456] Long gone are the days when"[t]he technical and commercial forms of vocational education became the desired goal through the constant ridiculing of the 'impractical' professor and the 'idealistic' intellectual [...]"[457] Today, the main object of ridicule is instead the 'hard realist.' But "[w]hen such 'derogatory' matters as arts, crafts, and machine technology are regarded as ballast and thrown out from liberal arts education, the student is left in a thin atmosphere of mere verbalism."[458]

455 Bourdieu, *Distinction*, *op. cit.*, p. 316.
456 Gerritzen and Lovink, *Made in China...*, *op. cit*.
457 Moholy-Nagy, *op. cit.*, p. 22.
458 *Ivi*.

THE PROBLEM WITH PROBLEMS

The shift from the technical intellectual to the intellectual *of technics* is most evident in Speculative Design, also called Critical Design.[459] This is the practice of imagining future scenarios in order to rethink – and thus transform – our present. The founding manifesto of the current, undersigned by Anthony Dunne and Fiona Raby in 2009, clarifies a number of principles.[460] Among them is a shift from an affirmative to a critical intention; design is no longer for production but for debate; provocation replaces innovation; and applications make way for implications. The result is a figure all in all similar to the traditional humanist intellectual. Indeed, Speculative Design is also sometimes called *design for debate*.

Every current has its excellences and its mediocrities. I am not interested here in determining the general validity of Speculative Design, but rather in giving reasons for its popularity, especially in pedagogical and academic circles. Notwithstanding some historical precedents acknowledged by speculative designers themselves, the novelty lies in defining and redefining problems through scenarios and prototypes, rather than devoting oneself to solving them. In doing so, Speculative Design solves *the problem of access to problems*. By creating fictions that are linked to reality and yet completely autonomous, it can disconnect from the concretely social dimension of technology, that is, what Galbraith used to call *technostructure*, which includes CEOs, policy-makers and corporate executives.[461] Moreover, on a pedagogical level such a practice lends itself to isolationism: a tenuous link to reality is enough to create a fictional world. A reality that for many designers is moreover second-rate: if problem solving is reduced to mere

459 See Dunne and Raby, *Speculative Everything*, *op. cit.*
460 Dunne and Raby, "A/B", *op. cit*.
461 John Kenneth Galbraith. *The New Industrial State*. Princeton: Princeton University Press, 2015.

execution, problem *framing* has the allure of cultural production, the traditional task of intellectuals. Hence, the shift from problems to solve to problematic situations to address.

This explains a growing scepticism toward problem solving. Such scepticism is not only about staging the 'wickedness' of complex problems, but also about evading the social obstacle that prevents one from participating in their resolution. This also explains the propensity to investigate colossal problems, since they lend themselves perfectly to the stunned gaze of the critic. By focusing on the highest systems, one can ignore those concrete, smaller systems from which they are excluded or over which they have no power.[462] In the words of Victor Papanek, "[i]t is also in the interest of the Establishment to provide science-fiction routes of escape for the young, lest they become aware of the harshness of that which is real."[463] Deploying images of preferable futures, these science-fiction routes inevitably tend to mobilise notions of ethics and politics. Once again, the question to ask is: how do ethics and politics *function* in the context of the art and design school?

IMPOTENT ETHICS

British historian Alison J. Clarke writes:

> Much present design discourse upholds a polarized model of design hatched in the broadly neo-Marxist 1970s paradigm that buoyed Papanek's vitriolic critique: [...] rehearsing a rhetoric that pits a morally and ethically virtuous design practice (sustainable, socially embedded, community-based, codesigned, etc.) against a model of designers as the handmaidens of a profit-driven corporate culture.[464]

462 See chapter 4.

463 Papanek, *op. cit.*, p. 283.

464 Alison J. Clarke. "Design for the Real World: Contesting the Origins of the Social in Design." In *Design Struggles: Intersecting Histories, Pedagogies, and*

This ethical binarism is not unrelated to the inaccessibility of problems. Generations of students and practitioners have passionately read Ken Garland's *First Things First* manifesto, which has been updated and republished several times in recent decades.[465] The original, explicitly anti-consumerist version indirectly lamented the reduction of the designer to a mere executor of a plan whose "trivial pursuits" are decided by others. Garland's consciousness-raising was thus not only ethical but also positional. Referring to the 2000 edition of the manifesto, J. Dakota Brown argues that "like earlier attempts to understand and to contest design's status as an 'institution of power,' the manifesto's critique was quickly rehabilitated into an apolitical affirmation of 'the power of design.'"[466] On closer inspection, then, the ethical afflatus of design, with its rhetoric of excessive responsibility, turns into self-aggrandisement. When designers and theorists proclaim that design has a great deal of responsibility, or even culpability, they are also saying: "we are important, influential, if not decisive." Examples of this excess are found both in canonical texts, such as Papanek's *Design for the Real World*, and in recent works, such as Monteiro's aforementioned book.[467]

Turning the attention to individual ethics (a fundamental component of self-design) is a way to mask one's own subordination. Undoubtedly, designers influence reality through their work and choices, but generally this is done in ways that are far from spectacular and hardly fit the cartoonish myth of a great power deriving from great responsibility. In fact, many designers agree that the success of a project depends largely on the goodwill of the client, understood as trust in the designer's competence.[468]

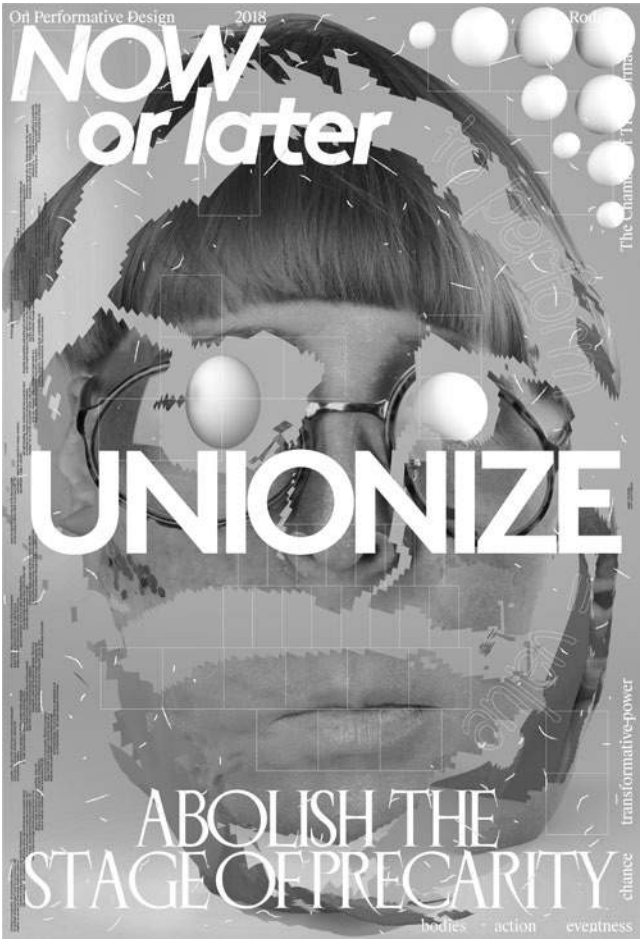
Perspectives, edited by Claudia Mareis and Nina Paim. Amsterdam: Valiz, 2021, p. 89.

465 Garland, *op. cit.*

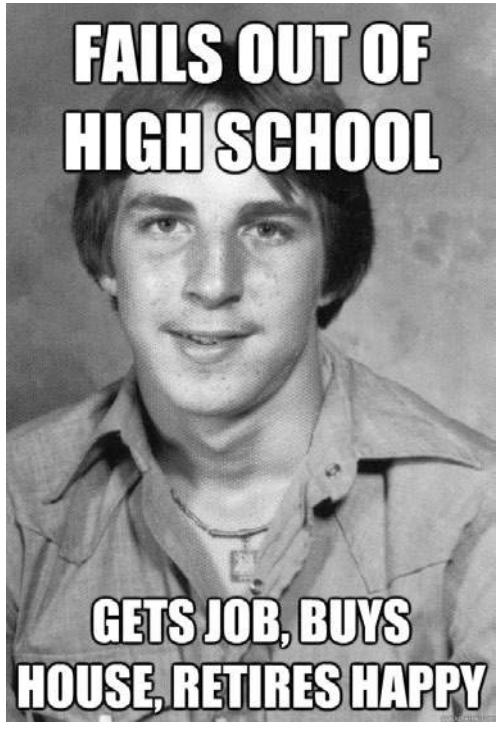
466 Brown, *The Power of Design…*, *op. cit.*, p. 2.

467 Monteiro, *op. cit*.

468 62% according to Graphic Designers Surveyed, *op. cit.*, p. 288.



This poster by the duo The Rodina (2018) stages the professional double bind: the workerist slogan clashes with the blown up portrait of a studio member, a more or less conscious manifestation of the performative activity required to create a professional and cultural aura out of one's individuality.



The many variations of the "Old Economy Steve" meme are a perfect instance of hysteresis.The memetic Steve doesn't understand the difficulties of the present because he looks at them through his own, allegedly easier, experience. Ironically, the actual Steve, now in his fifties, is a graphic designer who struggles to find work while dealing with chronic back pain. Thanks to J. Dakota Brown.

Music and Apocalypse, directed by Max Linz in 2019. The student protest appears here depowered and reduced to an ornamental motif. It also adopts the same generically political language of academia.

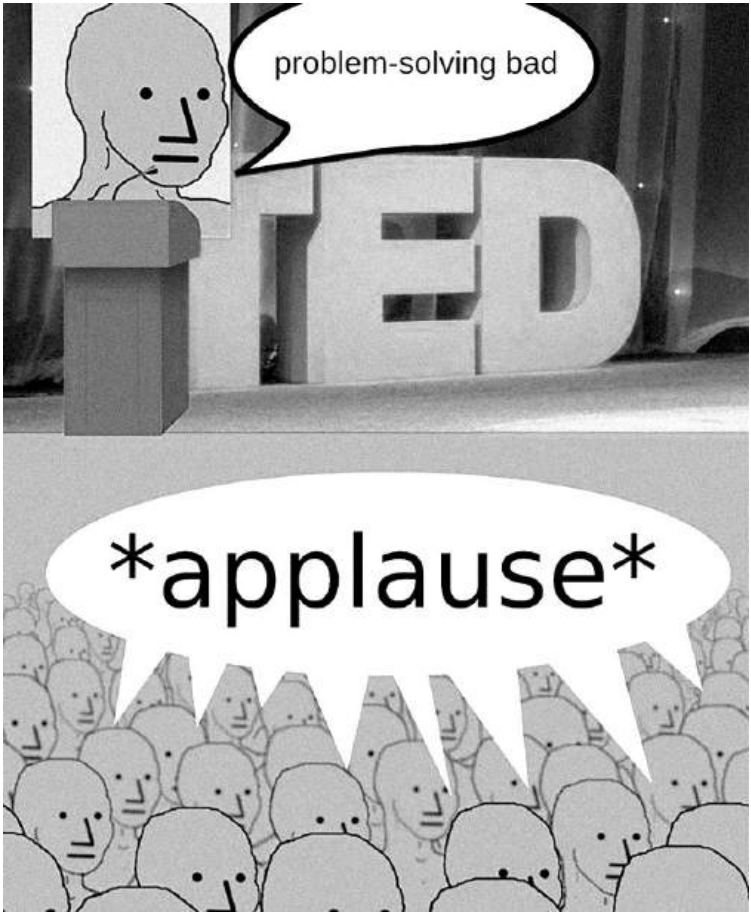
The evil entity that seduces, manipulates and deceives consumers, embodied from time to time by the organisation, the client or the designer themselves, is often a straw man. This abstract maleficence is the perfect target of a school that legitimises a distinctive lifestyle – *an attitude*, as Moholy-Nagy describes it. It is an attitude characterised by an affiliation with a kind of moral vanguard.[469] Within its ranks, what professional positioning – which is far from guaranteed – cannot provide, is instead provided by ethical positioning. If it's true that, as Tony Fry argues, design "either serves or subverts the status quo," it is legitimate to ask whether within certain instances of design education, the impetus to subvert the status quo is precisely what ultimately serves it.

POLITICAL ECONOMY

> Both in education and in business graphic design is often a case of the blind leading the blind. To make the classroom a perpetual forum for political and social issues for instance is wrong; and to see aesthetics as sociology, is grossly misleading. A student whose mind is cluttered with matters which have nothing directly to do with design; whose goal is to learn doing and making; who is thrown into the fray between learning how to use a computer, at the same time that he or she is learning design basics; and being overwhelmed with social problems and political issues is a bewildered student; this is not what he or she bargained for, nor, indeed, paid for.[470]

469 "Members of the PMC [Professional-Managerial Class] believe themselves to be virtuous vanguardists, floating above historical forms and conditions, transgressing boundaries and inventing new ways of being and seeing. It is hard to argue with them, because they do not accept debate as a meaningful form of the advancement of knowledge. For them, every conflict is moral, not intellectual or political." Liu, *op. cit.*, p. 33.

470 Paul Rand. "Confusion and Chaos: The Seduction of Contemporary Graphic Design." *AIGA Journal of Graphic Design* 10, no. 1 (1992). https://www.paulrand.design/writing/articles/1992-confusion-and-chaos-the-seduction-of-



Found image. Source unknown.

Here, Paul Rand bitterly laments the over-politicisation of graphic design education. It is Aggie Toppins again who reveals the backstory of this rant. Infuriated by the appointment of Sheila Levrant de Bretteville, a feminist designer associated with postmodernism, as director of Yale's graphic design course, Paul Rand abandoned his position in the department, convincing Armin Hoffman to do the same. Quoting Roger Kimball, Rand bemoaned the apparent predominance of "women's studies, black studies, gay studies" over supposedly "traditional curriculum and modes of intellectual inquiry."[471] It is as if the spirit of Jordan Peterson had retrospectively taken hold of him. Aggie Toppins' critique of this article deserves to be read in full.[472] Here, I will instead extract a grain of truth from this almost irredeemable material. The student bewildered by political issues is no less lost than the student thrown into an indecipherable world described by De Carlo in '68. Unfortunately, politics does not make the world a less complex place, and therein lies its value. As we have seen, the ethical disposition instead risks reducing the political to elementary binarisms and bombastic displays of outrage that serve to produce distinctive skills recognised, perhaps, only by designers themselves. It may be a good show, but do-goodism is not much of an educational offering and may even fuel conservatism.

How can we take advantage of political complexity in the educational context? By acknowledging, among other things, students' legitimate professional aspirations, including the less 'revolutionary' ones. The ornamental inclusion of blatantly political issues, the spectacle of indignation, and the ethical crossroads do not absolve schools from their professionalising shortcomings; on the contrary, they may obscure them. This is why a school that really wants to act

contemporary-graphic-design.html.

471 *Ivi.*

472 Aggie Toppins. "Good Nostalgia/Bad Nostalgia." *Design and Culture* 14, no. 1 (January 2, 2022): 5–29. https://doi.org/10.1080/17547075.2021.2010876.

upon so-called 'complexity' must rather extend the domain of what it understands as political, including the economic dimension of political manifestations. In other words, it must devote itself to *political economy*.

COMPROMISE

Hysteresis. Pierre Bourdieu uses a difficult word to describe an all-too-simple phenomenon, which consists of applying an outdated perspective to a mutated context. The sociologist writes:

> The hysteresis effect is proportionately greater for agents who are more remote from the educational system and who are poorly or only vaguely informed about the market in educational qualifications.[473]

The virulent diffusion of the "ok boomer" expression shows that the hysteresis effect has gone mainstream: the new generations are increasingly convinced that old approaches and perspectives don't work anymore in the mutated context of today. So-called boomers are like Don Quixote, immersed in obsolete world of ideas but – beware! – the boomer category is not strictly related to age. A boomer is whoever is tricked by the hysteresis effect – and the young can be tricked too. There are numerous informal accounts of this effect in both the art and design contexts. Joshua Citarella, a US artist, declares to his audience made up of students and emerging artists that "this education and career path is training you to become part of a professional class that, in reality, no longer exists. Institutions pay below market rates and no longer offer the protections that allowed artists to engage with complex ideas."[474] Roberto Arista, an Italian designer, reiterates the point by claiming that "[o]ur professionalizing studies were based on a model that was already crumbling in the

473 Bourdieu, *Distinction*, *op. cit.*, p. 147.
474 Citarella, Instagram story.

"We want to think." Slogan formulated in Italy by protesting students in '68.

nineties, and that in the 2000s was just a faded multiple of the original matrix."[475] The way the art and design school conceives self-design is out of sync. The glue that binds its dated ideals is the concept of autonomy. This manifests itself in various ways: romantic detachment from the things of the world, self-conviction of the social necessity of one's skills, critical repositioning within a context that limits access to problem-solving, a disconnect from the present and therefore from history.

In his text on critical autonomy, Andrew Blauvelt proposed that graphic design be understood as "a discipline capable of generating meaning out of its own intrinsic resources without reliance on commissions, functions, or specific materials or means."[476] As I have attempted to demonstrate, critical autonomy is more effect than cause; it is a dazzle, a nice story to tell in the classroom, at the museum or at a conference. This kind of autonomy is only apparent; it is in fact a symptomatic response to professional estrangement. What can this approach be replaced with? Let us take up Maldonado and Anceschi again:

> [...] the technical intellectual, while practising the stoic and realist art of design compromise is never a neutral manager, indifferent to the thing, the theme, the substance.[477]

Design compromise is the antithesis of critical autonomy. It is an admission of the fact that the designer is, after all, inevitably a *bricoleur* – a person who makes do with what they find, in the conditions in which they find themselves.[478] This doesn't mean that design compromise is docile or victimistic. Rather, it is a critique of compromise, an awareness of the reasons that guide awareness.

475 Roberto Arista. "Interfaces Are a Solid Object." *Progetto Grafico* 33 (2018).
476 Blauvelt, "Towards Critical Autonomy," *op. cit.*
477 Anceschi, *op. cit.*, p. 36.
478 Gaspare Caliri. "Cos'è il design-as(-a)-bricolage, la cultura del progetto attraverso le teorie della complessità." *cheFare*, January 19, 2021. https://www.che-fare.com/almanacco/cultura/design/cose-il-design-as-a-bricolage-la-cultura-del-progetto-attraverso-le-teorie-della-complessita/.

It is also historical compromise: the ability and courage to distinguish the past from the present. Honing the art of design compromise means recognising the *compromission* of the school as an integral part of the real world. It means overcoming professional proprioception to come to terms with others' perceptions of one's role, especially when these turn out to be unpleasant. An art and design school that wants to embrace an ethos of compromise must be idealist and materialist at the same time. To do so, it should be able to scrutinise the very ambitions that it produces and the mechanisms, both internal and external, that legitimise those ambitions. If such a school wants to become a space for self-realisation, it has to recognise the processes of professionalisation of the self, and overcome them.

Epilogue:

Ragequit

"Design is only possible where confidence and hope are united. Where there is resignation, that is, no belief in future prospects, there is no design."
– Gui Bonsiepe, 1992[479]

"[T]o turn a problem into a project is not to solve it."
– Hal Foster, 2002[480]

"Literature really changes things when it bumps up against its own impotence."
– Walter Siti, 2021[481]

According to G.H., the tortured protagonist of a quasi-mystical novel by Clarice Lispector, "[p]erhaps disillusionment is the fear of no longer belonging to a system."[482] In this book I've traced the porous boundaries of this system employing a two-fold method: on the one hand, I've revived old forms of disenchantment, some of which have been long forgotten; on the other hand, I've felt the pulse of the present by analysing memes and outbursts found in the informal arena of social media. This endeavour has led to the realisation that the larger the gap between expectations and reality, the more likely is the prospect of generating a dispirited, disillusioned practitioner. And from disillusionment to *ragequit* is but a short step. To avoid quitting in anger one should exercise realism, make a healthy compromise with reality, tame professional narcissism. This is what I call *disillusion,* the progressive

479 Gui Bonsiepe. *The Disobedience of Design*. Edited by Lara Penin. Radical Thinkers in Design. London: Bloomsbury, 2022, p. 237.
480 Foster, *op. cit.*, p. 121.
481 Walter Siti. *Contro l'impegno: Riflessioni sul Bene in letteratura*. Milano: Rizzoli, 2021, p. 263.
482 Clarice Lispector. *A paixão segundo G.H.* Rio de Janeiro: Rocco, 2009, p. 7.

tearing of disciplinary veils to develop a realist understanding of what a designer can do in the world.

What is to be done? Here is usually where the reader expects to be enlightened about the way forward, the path to take, the solution to all the discussed problems. This is where the *pars construens* goes. Let's tear off this last veil. Especially in design literature, conclusions have a rhetorical function: a happy end of sorts engineered to leave the reader energised. At the risk of disappointment, this is not gonna happen in this book. Will the reader be left "in the cynic's hall of mirrors" (as one anonymous reviewer put it), then? Hopefully not. Instead of proposing some half-baked alternatives for the sake of expectations, I want to defend pure criticism, as it is nowadays perceived as tantamount to cynicism. This is a stance that affects literature as a whole (see the triumph of manuals and how-to books) and design in particular, given its intrinsic focus on how things ought to be. Since, as we have seen, design is optimistic by default, I used the Gramscian pessimism of intelligence to criticise design's optimism of the will. The *fil rouge* running throughout this book is that it is not negativity and doubt that lead to cynicism but, instead, it is the false *pars construens*, the chimerical ambition, the appearance of constructiveness that do so. When structural problems (reality) are met with gestural solutions (expectations), disillusionment can only spread. To avoid that, it is crucial to "organise pessimism."[483]

HOW BOOKS CHANGE PEOPLE

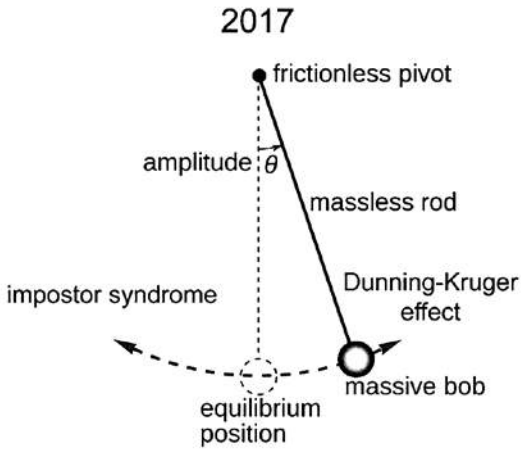
From this perspective, it becomes clear that the lack of *discussion* of alternatives is not the same as a lack of alternatives. The misunderstanding derives from our understanding of how books change people. Advocates of the

483 A book with a similar mission is the one written by Portuguese architectural critic Pedro Levi Bismarck, who borrows the expression "organised pessimism" from Walter Benjamin. *Arquitectura e «pessimismo»: Sobre uma condição política em arquitectura*. Porto: Punkto, 2020.

pars construens see change as linear, mechanistic: a book describes a solution which is then implemented by the reader. According to this view, a book without solutions, proposals or 'ways out' is an incomplete, crippled book. But arguing that criticism *per se* is not generative and that a case study is more 'real' than an analysis is misleading. What is really missing nowadays in design literature are texts that definitively place themselves outside of design optimism. In this regard, I would like to mention a person from whom the design field has learned a lot and still has a lot to learn: Salvatore Iaconesi, who together with Oriana Persico formed Art is Open Source. Iaconesi, who passed away last year, was a designer, a hacker and a theorist. He was deeply occupied with the question of tragedy within design practices. What the design field lacks, its great unspoken, is the tragic. Not the blockbuster representation of the tragedy of others (there is plenty of that), but the lived tragedy coming from the misery of everyday life. This story has not yet been sufficiently told. One cannot formulate a how-to for dealing with the tragic, one cannot develop a series of 'best practices' to fix tragedy, for tragedy is not a problem but a *dilemma*. How does a book act in the world? "The only way out of the tragedy is to *change state*," wrote Iaconesi.[484] A book really acts when it is able to transform the subject, and a subject can be transformed even by the least constructive book in the world – I am thinking of authors like Emil Cioran or Albert Caraco. Here, then, is the gap to be filled in the field of design: we have never been pessimists.

While discussing my ideas about disillusion, a friend called me a 'doomer.' Another one christened me 'blackpilled.' Finally, I was deemed hopeless. Can hope sprout from hopelessness? According to Maldonado, designing without hope is that "paradoxical behaviour that

484 Salvatore Iaconesi. "Nuovo Abitare, after COVID." *Counter Arts* (blog), July 15, 2021. https://medium.com/counterarts/nuovo-abitare-after-covid-e85e40e49861..



Federico Antonini (2021). The designer's activity as a pendulum between impostor syndrome and the Dunning-Kruger effect, that is, unfounded overconfidence.

→ *Saint Jerome in His Study*, Albrecht Dürer (1514), engraving.

stems from being compelled to design without having anything concrete to design. What's more: without having confidence, without the will."[485] Authentic hope, however, is not to be found *written* in the last pages of a book, but outside it, practised in the affinities and alliances built between the author and the readers, and among the readers themselves. This is why those rare obstinately hopeless books might be the most authentically hopeful ones. For Maldonado, hope is the functional belief in the designer's autonomy:

> When a designer – for example an architect – is persuaded that, as a designer, he can contribute to the transformation of society, he can only act in this sense to the extent that he believes in a relative innovative autonomy of his work. [The designer] will have to act even if this question will always remain open, that is, without knowing whether his autonomy will ultimately prove to be an illusion.[486]

As we see here, this belief in autonomy might prove itself illusory, but the issue is not a either-or one: illusions coexist with autonomy. In this book, I've highlighted the illusory aspects of the design field and described how they affect everyday designers' work and self-perception. Hopefully, this effort shows, in contrast, some of the real autonomy that designers can have, including the ability to disillusion themselves.

SIMPLIFYING ACCORDING TO TRUTH

In the previous chapters I have argued that complexity is not only the condition we inhabit, but also a lobbying instrument for design, a way to positions the designer as the figure who is most suited to dealing with it. The outcome of this situation is a paradoxical one: designers and

485 Maldonado, *La speranza progettuale, op. cit.*, p. 29.
486 *Ibidem*, p. 124.

design theorists have an incentive to increase complexity in order to reduce it. This way, any attempt at simplification is portrayed as a trivialisation of the issue at hand. The result is a sort of paralysing apophenia, where everything is linked to everything else. Here, drawing any type of boundary, proposing categories, employing models... all of this leads to suspicion.

In 2008, historian Russell Jacoby admitted that "the world is complex, but how did 'complication' turn from an undeniable reality into a desirable goal? How did the act of complicating become a virtue?" He noticed an emphasis on *problematising*, *contextualising*, *relativising*, etc.[487] These are valuable endeavours, but boundaries must be drawn, categories need to be produced, models need to be developed. If we don't do that, we will simply be trapped in other people's models, categories and boundaries. This doesn't mean that they should be crystallised once and for all. While designers formalise, they have to look at their formalisations as temporary arrangements. Here, as well, they should act as bricoleurs. The intransigent complexist insists that 'it's not that simple,' to which the cautious simplicist responds: 'exactly: sometimes complexity can be simplistic, and simplicity can be complex.'

That said, designers cannot simplify alone, they do not possess an all-embracing synthetic ability. Synthesis will always be concerted and involve negotiation and conflict: it's a political-epistemic act. But what should be the driving principle of this synthesis? Perhaps designers can embrace the motto of Italian writer and politician Leonardo Sciascia: "simplifying according to truth,"[488] where truth, that unfashionable notion, is the difficult part.

487 Jacoby quoted in Guido Vitiello. "I nuovi complessisti contro il 'pensiero unico'. Esercizi di atletica retorica." *Il Foglio*, April 5, 2022. https://www.ilfoglio.it/societa/2022/04/05/news/i-nuovi-complessisti-contro-il-pensiero-unico-esercizi-di-atletica-retorica-3876461/.

488 Sciascia quoted in Euclide Lo Giudice. "Sciascia su Sciascia." *Leonardo Sciascia Web*. https://www.amicisciascia.it/leonardo-sciascia/sciascia-su-sciascia.html.

COMPROMISSION

In his memoirs, Francesco Borromini includes a disclaimer to explain that when he served a particularly timid patron, it was better to curb his creative impetus and submit to their will.[489] Agreeing with and appreciating the humbleness of such a monumental figure as the Baroque architect means embracing compromise. For, when it comes to actually realising things, designing is neither about synthesis nor autonomy, neither order nor reason, but about compromise. We could speak of interdependence, but while interdependence acknowledges the constrictive and generative relationships that exist beyond autonomy, its essence is passive. Compromise, on the other hand, is an active and political endeavour: it implies mediation, if not *compromission*, that is, the adulteration of pure vision, with hands getting dirty in the messiness of reality. Stretching etymology a bit, compromission includes the word 'mission,' revealing an agency, an intention. As architect Marta Lonzi explains:

> My own experience tells me that there is nothing more exciting than to design accepting the limits that reality reveals and the constraints that the pre-existent imposes in a creative effort of great respect, above all, for the existential expectations that humans seek in order to be happy on this earth: at home, in a city, amid nature that makes them feel worthy of living this life, without upheavals or functional misery.[490]

After all, design is a compromise with the real, that is, a compromise with past things and, therefore, a negotiation of future ones. To design is both to compromise yourself and to compromise things. Design can undermine reality, while reality can jeopardise design. Against the

489 Marta Lonzi. *Autenticità e progetto*. Milano: Jaca Book, 2006, p. 12.
490 *Ibidem*, p. 42.

pseudo-scientific rationale of a project as the result of a logical, ineluctable process, we must opt for a political one, which does not necessarily correspond to activism, but it surely involves a compromise between conflicting forces.

THE PRACTICE OF PRACTICES

The first chapter of C. Wright Mills' *The Sociological Imagination* (1959) is entitled "The Promise." This chapter is not, as I suspected, about generic expectations, such as having a house, finding a job or building a career. What Mills talks about is "the promise of social science," ensured by a fundamental skill that a social scientist should muster. This skill is the sociological imagination, the ability to connect the personal and the societal, what Mills calls "the interplay of man and society, of biography and history, of self and world." It is about understanding personal troubles in the light of structural issues. A problem, for Mills, is an adequate formulation of these two scales. This is how he discusses it:

> The sociological imagination enables its possessor to understand the larger historical scene in terms of its meaning for the inner life and the external career of a variety of individuals. It enables him to take into account how individuals, in the welter of their daily experience, often become falsely conscious of their social positions. Within that welter, the framework of modern society is sought, and within that framework the psychologies of a variety of men and women are formulated. By such means the personal uneasiness of individuals is focused upon explicit troubles and the indifference of publics is transformed into involvement with public issues. The first fruit of this imagination – and the first lesson of the social science that embodies it – is the idea that the

> individual can understand his own experience and gauge his own fate only by locating himself within his period, that he can know his own chances in life only by becoming aware of those of all individuals in his circumstances. In many ways it is a terrible lesson; in many ways a magnificent one.[491]

Keeping in mind Mills' terribly magnificent lesson, what I attempted in this book was to turn the lens of sociological imagination to the *milieu* of design. Paraphrasing some of Mills' questions, this book asked:

> What varieties of [practitioners] now prevail in this field and in this period? And what varieties are coming to prevail? In what ways are they selected and formed, liberated and repressed, made sensitive and blunted?[492]

We have seen that part of this selection revolves around a promise of autonomy which implies self-direction. This self-direction is what we call a practice. What do we mean by that? 'Practice' is a term used mainly in the arts to define an artist's poietic activity. It involves the artist's concerns, their method, medium and even their theoretical and ethical ground. Through the decades, design has been considered a style, a craft, a thinking approach, etc. Now, with the notion of practice, we observe an extension of the understanding of the designer's activity as that of an increasingly generic cultural professional. What does this shift mean? Borrowing from programming, one could say that each *constant* of the discipline (methods, techniques, media and products, literacy, topics, ethical issues) is turned into a *variable*. Unavoidably, a degree of specificity is lost as the designer is encouraged to tweak all these variables. But if none of these is shared among peers, how can we call this a field? The liberating potential

491 C. Wright Mills. *The Sociological Imagination*. New York: Oxford University Press, 1959, p. 5.

492 *Ibidem*, p. 7.

of tweaking the very terms of one's work might also lead to isolation and individualisation. Educational institutions that are aware of this problem partially compensate the atomising drive with participatory and collaborative modes of interaction.

The variable-tweaking process that constitutes a design practice resembles micro-targeted advertising: as specific as to address one person only. It might be that 'at the intersection of' (an expression commonly found in designers' bios) there is no one else other than you. The autonomy of the self-directed professional can adumbrate precarity and insidiously replicate the much-despised design protagonism. In which case, it is not autonomy but wishful thinking. *Fauxtonomy*, if you will. The 'practice' model of the cultural professional is not dissimilar to the subject formation of the contemporary artist, which rarely offers more than a faint sense of belonging. Is this model, then, a weak form of professional, and therefore social, reproduction? The traditional medium-based or problem-based orientation was perhaps more solid for the simple fact that it had at least some fixed variables. The point, however, is not to choose one model over the other, but to raise a specific concern: what effects does the 'practice of practices' have on identification? Are design schools partially responsible for the disintegrating sense of belonging and tangible social isolation that many practitioners, often self-defined as outsiders, feel? By offering an abstract promise of autonomy, are schools uncritically abdicating their role to nurture a recognisable field?

ST. JEROME

The devaluation of skill-specificity is also worrisome because, in a society that cherishes work above all, craft is one of the few stable forms left of identity making: the mastery of a craft goes way beyond a professional title. A job well done for its own sake, the ample definition of

a craft offered by Richard Sennett,[493] can be an island of personal stability in a sea of impostor syndrome and self-doubt, punctuated by episodes of megalomania. Furthermore, crafts defy make-believe horizontalism by showing the positive side of hierarchy: a workshop master-apprentice relationship is not in itself an exploitative, abusive one.

Himself a craftsman, C. Wright Mills wrote that "[t]he most fundamental splits in contemporary life occur because of the break-up of the old unity of design, production and enjoyment." In craftsmanship, these three activities are conjoined. Mills also saw his research activity as a form of "intellectual craftsmanship," which we can understand as a mode of operating of the technical intellectual. As Treviño explains:

> "Social science," writes Mills "is the practice of a craft." A craft refers to the manual or mental processes through which workers freely employ their capacities and skills in creating the products of their enjoyment and enjoying the products of their creation. Mills uses the term "intellectual craftsmanship" in referring to a reflective style of work as well as "to the joyful experience of mastering the resistance of the materials with which one works."[494]

I started this book with Dürer's *Melencolia*, a brooding figure that expresses apathy and impotence, as the allegory of the disillusioned designer. I'd like to conclude it with another engraving, also by Dürer. Here, we see St. Jerome diligently occupied with the translation of the Bible into Latin. The scene exudes tranquillity: Jerome's study is well arranged, radiating brightness. The objects in the room are neatly organised, although not rigidly so. Rather than being burdensome, the saint's work is contemplative. He moves at a relaxed pace, undisturbed by the presence of a skull and

493 Richard Sennett. *The Craftsman*. London: Penguin, 2009.
494 Treviño, *op. cit.*, pp. 343-4.

an hourglass, symbols of mortality and the passage of time. These reminders do not provoke any sense of urgency or fear within him. Jerome has embraced the inevitability of death, and he focuses intently on his task without anxiety. A lion, the saint's loyal companion, lies quietly at the front of the sunlit room, symbolising the taming of human passions.[495] Jerome's craft seems to give him peace. Whereas *Melencolia* stands for disillusionment and chaos, *St. Jerome* represents a provisional yet fulfilling sense of order.

THE COMING FIELD

To find this provisional sense of order, to avoid succumbing to the multi-layered identity crises driven by fauxtonomy and futurelessness, it is time to put self-design aside and rebuild the design field around different, but not necessarily unprecedented, principles. This field should be a space inhabited by a series of connected communities of practice (where practice is *not* understood as in contemporary art, namely devoid of an authentic communal sense). It shouldn't shy away from problems. Instead, it should constantly redefine its own set of issues and concerns: functional problems, ethical problems, problems of method, access and inclusion. Unavoidably it will deal with complexity, but it won't try to tackle it in its entirety. Through the specialised knowledge it produces and the situated activity it performs, it will look at complexity without being blinded by its frightening, Cthulhu-like appearance.

The design field should be considered a political entity, but not because it regularly issues statements and manifestos. It is political because it is concerned with its own organisational politics, as well as the politics of the artefacts it circulates. Such a design field would be preoccupied with tangible, lower-case futures. The future rests in

495 See https://www.khanacademy.org/humanities/renaissance-reformation/northern/durer/a/durer-melancholia.

its surroundings, but also in the broader effects that interventions on these surroundings have. It thus moves within a gradient of multiple scales. A field such as this is not a 'scene:' its main engine isn't visibility. It might even unconsciously limit the exchange with the outside. But if it's too self-referential, it will become a club.

Nor is it a school: while learning takes place within it, scholastic hierarchies, both implicit and explicit, don't apply there. This doesn't mean that it rejects hierarchy completely: its structure is based on the healthy, reconfigurable hierarchies of apprenticeship, amateurship and curiosity. Design should be informal in nature, but it shouldn't fetishise informality: it should resist character normativity and protect its people from hurtful behaviour. It should be attentive to its flows of social, cultural and economic capital: generous with quoting, crediting and remunerating; mistrustful of impresarios and creative directors; rejectful of inner qualitative distinctions. All the work it needs is essential, interdependent work. Validation comes with effort, helpfulness and mutuality, more than with smartness, talent and bravado. It should believe in expertise, but not worship experts. It should provide an activity-based sense of belonging where people have roles and purposes, but these can be renegotiated. Here, biographical and cultural differences would be understood, but foregrounded only when necessary, so that a designer can blissfully forget themselves.

THE CROSSROAD

Right now, design culture is cohabited by two conflicting sets of values: a culture of smartness and one of contribution. Smart culture – the culture that has shaped my generation of designers – privileges the new idea, the concept that makes you go 'a-ha.' As such, it's based on detachment: worthy of attention is the thing that distances itself from the others, and by extension, the maker that stands out from the crowd. In this sense, it is tragically similar

to the operative logic of the neoliberal ethos: relentless reinvention. It is inattentive and uncompromising, as it ignores what's there and forgets what was there. Contribution culture, on the other hand, is humble, it *builds upon*. When a useful idea emerges, it helps it to flourish. Smart culture is uncompromising, intransigent and self-absorbed. Contribution culture is compromissory, tolerant and indulgent. It is realistic, but it doesn't ignore the reality of ideas. Whereas smart culture creates, contributory culture maintains. Smart culture is design, contribution culture is bricolage.

We see smart culture in the celebration of design heroes, in the throwaway solutions to epochal problems, in the pervasiveness of personal branding, in the distinctive performance of kritikaoke. We see contribution culture in the online syllabi put together collaboratively and semi-anonymously, in free and open-source tools, in students' translation of texts they find important into their mother tongue, in artefacts designed 'for and with' specific communities. Exploring design disillusion means venturing into the sorry shadow cast by smart culture, this kitschy relic of our shaky modernity. This book indicates the shadow's contour in the hope that other people, in other contexts, will dissipate it, together.

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For many years now, design schools have constituted a large part of my own 'real world.' Teaching in institutions located in various European countries has deeply influenced my understanding of the design field. These are the Lusófona University of Lisbon, the Royal Academy of Art in The Hague, the Sandberg Instituut of Amsterdam, the Design Academy Eindhoven, the Piet Zwart Institute in Rotterdam, the Burg Giebichenstein University of Art and Design in Halle, the Abadir Academy in Catania, the CFP Bauer in Milan.

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Over the years, I took notes on social media and tried to make sense of them publicly. In fact, *What Design Can't Do* can be seen as the byproduct of a never-ending Twitter thread.[496] I also discussed it on the Lurk instance of Mastodon. Many users engaged with this work and I want to collectively thank them.

A final thought goes to all the students that I got to know, both as a teacher and as a student myself.

Silvio Lorusso
Lisbon, September 2023

496 https://twitter.com/silvio_lorusso/status/1336248731033882624.

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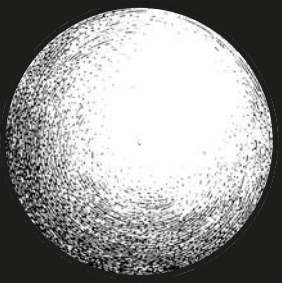
Silvio Lorusso is a writer, artist and designer based in Lisbon, Portugal. In 2018, he published his first book entitled *Entreprecariat: Everyone is an Entrepreneur. Nobody Is Safe*. He is an assistant professor and co-director of the Center for Other Worlds at the Lusófona University in Lisbon and a tutor at the Information Design Department of the Design Academy Eindhoven. Lorusso holds a Ph.D. in Design Sciences from the Iuav University of Venice. Paraphrasing the Italian architect Franco Raggi, he writes and talks about design, though actually doing very little of it, and believing in it less and less.



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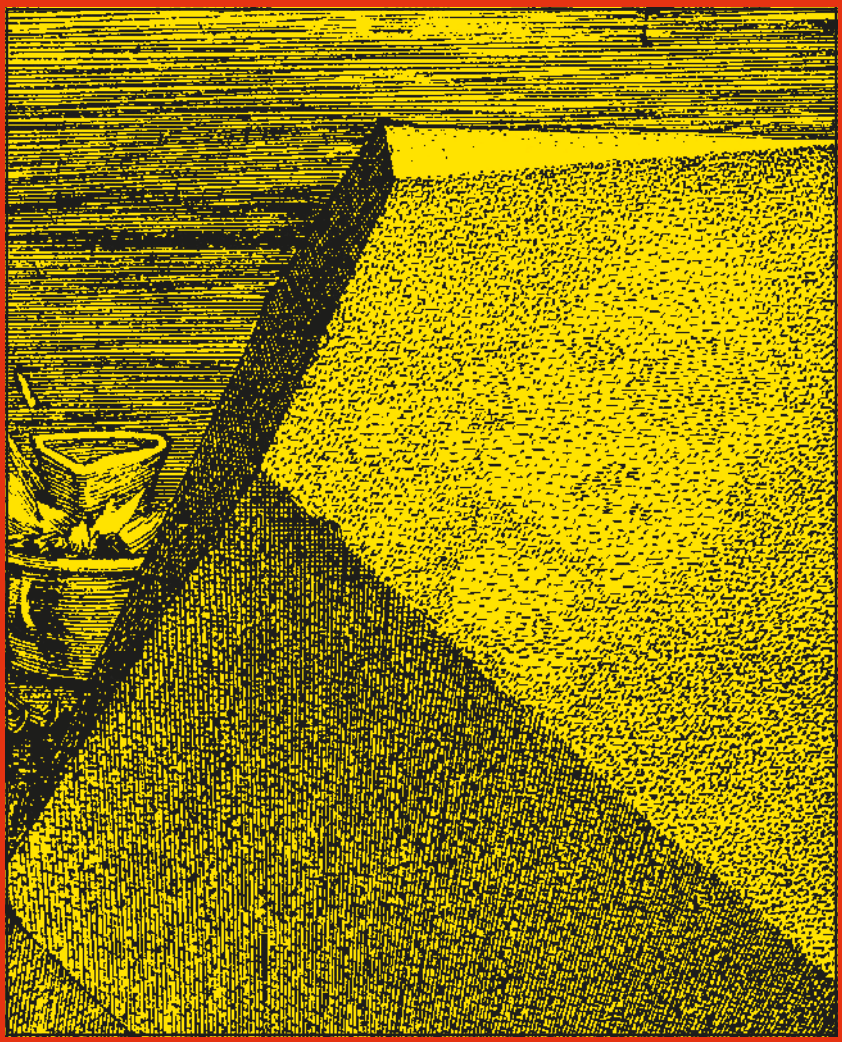

Design: what was once a promising field rooted in problem-solving has become a problem in itself. The skill set of designers appears shaky and insubstantial – their expertise is received with indifference, their know-how is trivialised by online services, their work is compromised by a series of unruly external factors. If you see yourself as a designer without qualities; if you feel cheated, disappointed or betrayed by design, this book is for you.